LEF/DEF 5.8 Language Reference

Product Version 5.8
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Cadence Design Systems, Inc., 555 River Oaks Parkway, San Jose, CA 95134, USA

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Preface

This manual is a language reference for users of the Cadence® Library Exchange Format (LEF) and Design Exchange Format (DEF) integrated circuit (IC) description languages.

LEF defines the elements of an IC process technology and associated library of cell models. DEF defines the elements of an IC design relevant to physical layout, including the netlist and design constraints. LEF and DEF inputs are in ASCII form.

This manual assumes that you are familiar with the development and design of integrated circuits.

This preface provides the following information:

- What's New on page 9
- Typographic and Syntax Conventions on page 9

What's New

For information on what is new or changed in LEF and DEF for version 5.8 see *What's New in LEF/DEF*.

Typographic and Syntax Conventions

This list describes the conventions used in this manual.

`text`

Words in `monospace` type indicate keywords that you must enter literally. These keywords represent language tokens. Note that keywords are case insensitive. They are shown in uppercase in this document, but a keyword like `LAYER` can also be `Layer` or `layer` in a LEF or DEF file.

`variable`

Words in *`italics`* indicate user-defined information for which you must substitute a name or a value.

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objRegExpr

An object name with the identifier *objRegExpr* represents a regular expression for the object name.

pt

Represents a point in the design. This value corresponds to a coordinate pair, such as x y. You must enclose a point within parentheses, with space between the parentheses and the coordinates. For example,

```
RECT ( 1000 2000 ) ( 1500 400 ).
```

|

Vertical bars separate possible choices for a single argument. They take precedence over any other character.

`[ ]`

Brackets denote optional arguments. When used with vertical bars, they enclose a list of choices from which you can choose one.

`{ } ...`

Braces followed by three dots indicate that you must specify the argument at least once, but you can specify it multiple times.

`{ }`

Braces used with vertical bars enclose a list of choices from which you must choose one.

`...`

Three dots indicate that you can repeat the previous argument. If they are used with brackets, you can specify zero or more arguments. If they are used with braces, you must specify at least one argument, but you can specify more.

`,...`

A comma and three dots together indicate that if you specify more than one argument, you must separate those arguments with commas.

`" "`

Quotation marks enclose string values. Write quotation marks within a string as `\"`. Write a backslash within a string as `\\`.

Any characters not included in the list above are required by the language and must be entered literally.

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LEF Syntax

This chapter contains information about the following topics:

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About Library Exchange Format Files

A Library Exchange Format (LEF) file contains library information for a class of designs. Library data includes layer, via, placement site type, and macro cell definitions. The LEF file is an ASCII representation using the syntax conventions described in "Typographic and Syntax Conventions" on page 9.

General Rules

Note the following information about creating LEF files:

- Identifiers like net names and cell names are limited to 2,048 characters.
- Distance is specified in microns.
- Distance precision is controlled by the `UNITS` statement.
- LEF statements end with a semicolon (`;`). You must leave a space between the last character in the statement and the semicolon.

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Character Information

For information, see Character Information on page 977 in the DEF Syntax chapter.

Name Escaping Semantics for LEF/DEF Files

For information, see Name Escaping Semantics for Identifiers on page 978 in the DEF Syntax chapter.

Managing LEF Files

You can define all of your library information in a single LEF file; however this creates a large file that can be complex and hard to manage. Instead, you can divide the information into two files, a “technology” LEF file and a “cell library” LEF file.

A technology LEF file contains all of the LEF technology information for a design, such as placement and routing design rules, and process information for layers. A technology LEF file can include any of the following LEF statements:

```
[VERSION statement]
[BUSBITCHARS statement]
[DIVIDERCHAR statement]
[UNITS statement]
[MANUFACTURINGGRID statement]
[USEMINSPACING statement]
[CLEARANCEMEASURE statement ;]
[PROPERTYDEFINITIONS statement]
[FIXEDMASK ;]
[LAYER (Nonrouting) statement
 | LAYER (Routing) statement] ...
[MAXVIASTACK statement]
[VIA statement] ...
[VIARULE statement] ...
[VIARULE GENERATE statement] ...
[NONDEFAULTRULE statement] ...
[SITE statement] ...
[BEGINEXT statement] ...
[END LIBRARY]
```

A cell library LEF file contains the macro and standard cell information for a design. A library LEF file can include any of the following statements:

```
[VERSION statement]
[BUSBITCHARS statement]
[DIVIDERCHAR statement]
[VIA statement] ...
[SITE statement]
```


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```
[MACRO statement
  [PIN statement] ...
  [OBS statement ...] ] ...
[BEGINEXT statement] ...
[END LIBRARY]
```

When reading in LEF files, always read in the technology LEF file first.

Order of LEF Statements

LEF files can contain the following statements. You can specify statements in any order; however, data must be defined before it is used. For example, the `UNITS` statement must be defined before any statements that use values that are dependent on `UNITS` values, `LAYER` statements must be defined before statements that use the layer names, and `VIA` statements must be defined before referencing them in other statements. If you specify statements in the following order, all data is defined before being used.

```
[VERSION statement]
[BUSBITCHARS statement]
[DIVIDERCHAR statement]
[UNITS statement]
[MANUFACTURINGGRID statement]
[USEMINSPACING statement]
[CLEARANCEMEASURE statement ;]
[PROPERTYDEFINITIONS statement]
[FIXEDMASK ;]
[  LAYER (Nonrouting) statement
 | LAYER (Routing) statement] ...
[MAXVIASTACK statement]
[VIARULE GENERATE statement] ...
[VIA statement] ...
[VIARULE statement] ...
[NONDEFAULTRULE statement] ...
[SITE statement] ...
[MACRO statement
  [PIN statement] ...
  [OBS statement ...]] ...
[BEGINEXT statement] ...
[END LIBRARY]
```

LEF Statement Definitions

The following definitions describe the syntax arguments for the statements that make up a LEF file. Statements are listed in alphabetical order, *not* in the order they should appear in a LEF file. For the correct order, see “Order of LEF Statements” on page 14.

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Bus Bit Characters

```
[BUSBITCHARS "delimiterPair" ;]
```

Specifies the pair of characters used to specify bus bits when LEF names are mapped to or from other databases. The characters must be enclosed in double quotation marks. For example:

```
BUSBITCHARS "[]" ;
```

If one of the bus bit characters appears in a LEF name as a regular character, you must use a backslash (`\`) before the character to prevent the LEF reader from interpreting the character as a bus bit delimiter.

If you do not specify the `BUSBITCHARS` statement in your LEF file, the default value is `"[]"`.

Clearance Measure

```
[CLEARANCEMEASURE {MAXXY | EUCLIDEAN} ;]
```

Defines the clearance spacing requirement that will be applied to all object spacing in the `SPACING` and `SPACINGTABLE` statements. If you do not specify a `CLEARANCEMEASURE` statement, euclidean distance is used by default.

`MAXXY`	Uses the largest x or y distances for spacing between objects.
`EUCLIDEAN`	Uses the euclidean distance for spacing between objects. That is, the square root of $x^2 + y^2$.

Divider Character

```
[DIVIDERCHAR "character" ;]
```

Specifies the character used to express hierarchy when LEF names are mapped to or from other databases. The character must be enclosed in double quotation marks. For example:

```
DIVIDERCHAR "/" ;
```

If the divider character appears in a LEF name as a regular character, you must use a backslash (`\`) before the character to prevent the LEF reader from interpreting the character as a hierarchy delimiter.

If you do not specify the `DIVIDERCHAR` statement in your LEF file, the default value is `"/"`.

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Extensions

```
[BEGINEXT "tag"
     extension

ENDEXT]
```

Adds customized syntax to the LEF file that can be ignored by tools that do not use that syntax. You can also use extensions to add new syntax not yet supported by your version of LEF/DEF, if you are using version 5.1 or later.

extension

Specifies the contents of the extension.

tag

Identifies the extension block. You must enclose *tag* in double quotation marks (" ").

Example 1-1 Extension Statement

```
BEGINEXT "1VSI Signature 1.0"
    CREATOR "company name"
    DATE "timestamp"
    REVISION "revision number"
ENDEXT
```

Here, *tag* is `1VSI Signature 1.0` and *extension* is all the remaining text before `ENDEXT`, including any newline characters. In this example, the extension string is as follows:

```
CREATOR "company name"
DATE "timestamp"
REVISION "revision number"
```

FIXEDMASK

```
[FIXEDMASK ;]
```

Does not allow mask shifting. All the LEF macro pin mask assignments must be kept fixed and cannot be shifted to a different mask. The LEF macro pin shapes should all have `MASK` assignments, if `FIXEDMASK` is present. In addition, macro OBS should all have `MASK` assignments unless they have the `SPACING` override. This statement should be included before the `LAYER` statements.

For example,

```
    ...
    MANUFACTURINGGRID 0.001 ;
```


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```
FIXEDMASK ;
LAYER xxx
```

Some technologies do not allow mask shifting for cells using multi-mask patterning. For example, the pin and routing shapes are all pre-colored and must not be shifted to other masks.

Layer (Cut)

Defines cut layers in the design. Each cut layer is defined by assigning it a name and design rules. You must define cut layers separately, with their own layer statements. For details, see the following chapter:

- LEF Syntax - Layer (Cut) on page 217

Layer (Implant)

Defines implant layers in the design. Each layer is defined by assigning it a name and simple spacing and width rules. For details, see the following chapter:

- LEF Syntax - Layer (Implant) on page 495

Layer (Masterslice)

Defines masterslice (non-routing) layers in the design. For details, see the following chapter:

- LEF Syntax - Layer (Masterslice) on page 525

Layer (Overlap)

Defines overlap layers in the design. For details, see the following chapter:

- LEF Syntax - Layer (Overlap) on page 607

Layer (Routing)

Defines routing layers in the design. Each layer is defined by assigning it a name and design rules. You must define routing layers separately, with their own layer statements. For details, see the following chapter:

- LEF Syntax - Layer (Routing) on page 609

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Library

Defining Library Properties to Create 32/28 nm and Smaller Nodes Rules

You can include library properties in your LEF file to create 32/28 nm and smaller nodes rules that currently are not supported by existing LEF syntax. The properties are specified inside the `PROPERTYDEFINITIONS` statements.

- The property names used for these rules all start with `LEF58_`.

All properties use the following syntax within the LEF `PROPERTYDEFINITIONS` statement:

```
PROPERTYDEFINITIONS
     LAYER propName STRING ["stringValue"] ;

END PROPERTYDEFINITIONS
```

The property definitions for the library properties are as follows:

```
PROPERTYDEFINITIONS
   LIBRARY LEF58_ANTENNACUMLAYERORDER STRING;
   LIBRARY LEF58_ANTENNADIFFONLYAREARATIOMODEL STRING;
   LIBRARY LEF58_ANTENNAMAXCUMAREA STRING ;
   LIBRARY LEF58_ANTENNAMAXGATEAREA STRING ;
   LIBRARY LEF58_ANTENNAMAXSIDETOVIAAREARATIO STRING ;
   LIBRARY LEF58_ANTENNAMODELGROUP STRING ;
   LIBRARY LEF58_CELLEDGESPACINGTABLE STRING ;
   LIBRARY LEF58_CELLLFORBIDDEN STRING ;
   LIBRARY LEF58_CELLLAYERSPACINGTABLE STRING ;
   LIBRARY LEF58_CELLLMAP STRING ;
   LIBRARY LEF58_CELLVARIANTS STRING ;
   LIBRARY LEF58_CONSTRAINTLENGTH STRING ;
   LIBRARY LEF58_CUTMASKTRACK STRING ;
   LIBRARY LEF58_EDGETYPEONOVERLAP STRING ;
   LIBRARY LEF58_FINFET STRING ;
   LIBRARY LEF58_GROUNDCONNECTIVITYLAYERS STRING ;
   LIBRARY LEF58_IMPLANTGROUP STRING ;
   LIBRARY LEF58_LAYERMASKSHIFT STRING ;
   LIBRARY LEF58_LINEENDTRACK STRING ;
   LIBRARY LEF58_MAXFLOATINGAREA STRING ;
   LIBRARY LEF58_MAXVIASTACK STRING ;
   LIBRARY LEF58_METALWIDTHTRACK STRING ;
   LIBRARY LEF58_METALWIDTHVIAMAP STRING ;
   LIBRARY LEF58_NOOVERLAP STRING ;
```


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```
   LIBRARY LEF58_OALAYERMAP STRING ;
   LIBRARY LEF58_OXIDECONVERSION STRING ;
   LIBRARY LEF58_PGVIATRACK STRING ;
   LIBRARY LEF58_POWERCONNECTIVITYLAYERS STRING ;
   LIBRARY LEF58_SITECOREOFFSET STRING ;
   LIBRARY LEF58_STACKVIALAYERRULE STRING ;
   LIBRARY LEF58_STACKVIARULE STRING ;
   LIBRARY LEF58_TAPDISTANCE STRING ;
   LIBRARY LEF58_TAPDISTANCERULE STRING ;
   LIBRARY LEF58_TRIMMETALTRACK STRING ;
   LIBRARY LEF58_VOLTAGEMODE STRING ;
END PROPERTYDEFINITIONS
```

Antenna Cumulative Layer Order Rule

You can define an antenna cumulative layer order rule by using the following `PROPERTYDEFINITIONS` statement:

```
PROPERTYDEFINITIONS
LIBRARY LEF58_ANTENNACUMLAYERORDER STRING
   "ANTENNACUMLAYERORDER {layerName}...
      [PARALLELLAYERORDER  firstMetalLayerName {parallelLayerName}...
            lastMetalLayerName]
      [ABOVEMETALONLYONTSV tsvLayerName]
      [{[OXIDE1 | OXIDE2 | OXIDE3 | OXIDE4 | OXIDE5 | ... |
            OXIDE32]{SUBLIST {layerName2}...}...}...]
      ;" ;
END PROPERTYDEFINITIONS
```

Where:

`ABOVEMETALONLYONTSV` *`tsvLayerName`*

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Specifies an antenna calculation method on *tsvLayerName*, which must be a cut layer of type TSV. The antenna calculation of a shape on the *tsvLayerName* is applied when the shape is connected to instance pins through the layers after *tsvLayerName* in `ANTENNACUMLAYERORDER` or through the first layer in `PARALLELLAYERORDER` where *tsvLayerName* appears first in *parallelLayerName*. For each instance pin, the antenna effect from different shapes on *tsvLayerName* is accumulated. That rule ratio is checked against the sum of each shape's calculated ratio although the shapes could be connected to the instance pin through different metal layers. This implies a constraint on the possible forms of the antenna rules on the *tsvLayerName* layer. The rule ratio cannot have dependence on the connected gate area or diffusion area.

`ANTENNACUMLAYERORDER {`*`layerName`*`}...`

Specifies the accumulated layer order to calculate the antenna cumulative ratio or cumulative area, using the layers with electrical connection in the given order. *layerName* must be a routing or cut layer or a masterslice layer with TS type property and any routing or cut layer must at most be defined in one statement. This is particularly useful to define the layer order with TSV along with front side and back side layers. Only one property can be defined on relevant layers.

`{SUBLIST {`*`layerName2`*`}...}...`

Specifies separate layer lists to calculate the antenna cumulative ratio or cumulative area individually. All layers defined in *layerName* must be specified once in *layerName2*. The relative layer order must be followed. If *layerName* contains "`layerA layerB layerC`" and `layerA` and `layerC` are defined in the same `SUBLIST`, `layerA` must be defined before `layerC`, but `layerB` can be left out of that `SUBLIST` and be defined on another `SUBLIST`. This means that *layerName* is broken into separate lists of *layerName2*.

`OXIDE1 | OXIDE2 | OXIDE3 | OXIDE4 | OXIDE5 | ... | OXIDE32`

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Specifies different sets of `SUBLIST` layers for different oxide models. If a set of `SUBLIST` layers is defined without an oxide model, it is applied to any oxide model without an explicit set of `SUBLIST` layers. The sum of all *`layerName2`* in all sets of `SUBLIST` layers for each oxide model must be equal to all layers in *`layerName`*. Any *`layerName2`* layer must still be defined in *`layerName`*.

`PARALLELLAYERORDER` *`firstMetalLayerName`* `{`*`parallelLayerName`*`}...` *`lastMetalLayerName`*

Specifies a separate parallel layer order connecting a bottom routing layer, *`firstMetalLayerName`*, to a top routing layer, *`lastMetalLayerName`*, through the intermediate layers in *`parallelLayerName`*. It defines an electric path on *`firstMetalLayerName`* and its above layers can be accumulated onto *`lastMetalLayerName`* through a connection composed of geometries from *`parallelLayerName`* layers. *`firstMetalLayerName`* and *`lastMetalLayerName`* must be routing layers defined in *`layerName`*. The sum of all layers in *`parallelLayerName`* and *`layerName2`* must be equal to all layers in *`layerName`*.

Antenna Cumulative Layer Order Rule Examples

- Consider the following example:

```
PROPERTYDEFINITIONS
LIBRARY LEF58_OXIDECONVERSION STRING "
    OXIDECONVERSION OXIDE1 TO OXIDE2
        NETCONNECTEDLAYER TSV CHECKBOTH ; " ;
LIBRARY LEF58_ANTENNACUMLAYERORDER STRING "
    ANTENNACUMLAYERORDER M0 V0 M1 V1 M2 V2 M3 V3 M4 V4 M5 V5
        M6 V6 M7 V7 M8 V8 M9 V9 M10 V10 TSV M11 B1 BV1 B2
        PARALLELLAYERORDER M10 TSV B1
        OXIDE1
        SUBLIST M0 M1 M2 M3 M4 M5 M6 M7 M8 M9 M10 M11 B1 B2
        SUBLIST V0 V1 V2 V3 V4 V5 V6 V7 V8 V9 V10 TSV BV1
        OXIDE2
        SUBLIST M0 M1 M2 M3 M4 M5 M6 M7 M8 M9
        SUBLIST M10 M11 B1 B2
        SUBLIST V0 V1 V2 V3 V4 V5 V6 V7 V8 V9
        SUBLIST TSV
```


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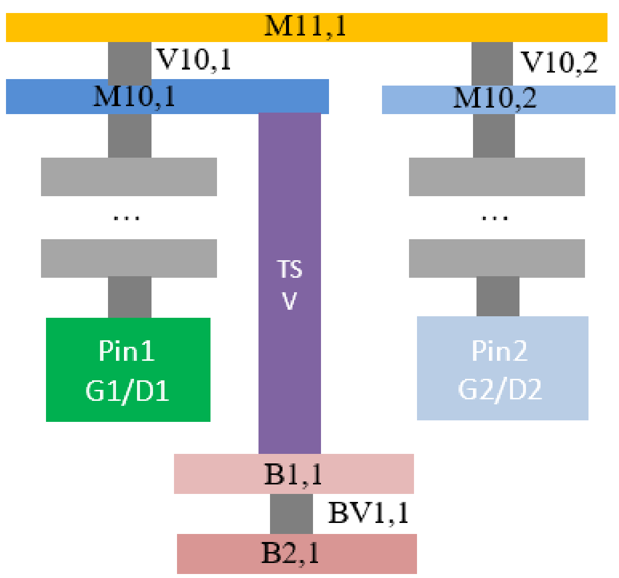
```
        SUBLIST V10 BV1 ; " ;
END PROPERTYDEFINITIONS
```

Here:

- The net is connected to `TSV`; `OXIDE1` and `OXIDE2` antenna rules are applied for the net with `CHECKBOTH`.
- The accumulated area ratio for `M10` layer can be calculated as `CAR[M10] = PAR[M10]`.
- The accumulated area ratio for `M11` layer can be calculated as `CAR[M11] = PAR[M11] + CAR[M10]`.
- The accumulated area ratio for `B1` layer can be calculated as `CAR[B1] = PAR[B1] + CAR[M11]`.
- The accumulated area ratio for `B2` layer can be calculated as `CAR[B2] = PAR[B2] + CAR[B1]`.
- The accumulated area ratio for `V10` layer can be calculated as `CAR[V10] = PAR[V10]`.
- The accumulated area ratio for `BV1` layer can be calculated as `CAR[BV1] = PAR[BV1] + CAR[V10]`.



With the above net connections, the following antenna checks are preformed:

```
Check M10 layer's antenna:
```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```
-    Metal M10,1 is electrically connected to pin1
     PAR(M10,1, G1) = Area(M10,1) / Area(G1)
     CAR(M10,1, G1) = PAR(M10,1, G1)
-    Metal M10,2 is electrically connected to pin2
     PAR(M10,2, G2) = Area(M10,2) / Area(G2)
     CAR(M10,2, G2) = PAR(M10,2, G2)

Check M11 layer's antenna:
-    Metal M11,1 is electrically connected to pin1 and pin2
     PAR(M11,1, G1) = Area(M11,1) / Area(G1+G2)
     CAR(M11,1, G1) = CAR(M10,1,G1) + PAR(M11,1, G1)
     PAR(M11,1, G2) = Area(M11,1) / Area(G1+G2)
     CAR(M11,1, G2) = CAR(M10,2,G2) + PAR(M11,1, G2)

Check B1 layer's antenna:
-    Metal B1,1 is electrically connected to pin1 and pin2
     PAR(B1,1, G1) = Area(B1,1) / Area(G1+G2)
     CAR(B1,1, G1) = CAR(M11,1, G1) + PAR(B1,1, G1)
     PAR(B1,1, G2) = Area(B1,1) / Area(G1+G2)
     CAR(B1,1, G2) = CAR(M11,1, G2) + PAR(B1,1, G2)
```

- Consider the following example:

```
PROPERTYDEFINITIONS
LIBRARY LEF_CDN_OXIDECONVERSION STRING "
    OXIDECONVERSION OXIDE1 TO OXIDE2
        NETCONNECTEDLAYER TSV CHECKBOTH ; " ;
LIBRARY LEF_CDN_ANTENNACUMLAYERORDER STRING "
    ANTENNACUMLAYERORDER M0 V0 M1 V1 M2 V2 M3 V3 M4 V4 M5 V5
        M6 V6 M7 V7 M8 V8 M9 V9 M10 V10 TSV M11 B1 BV1 B2
        PARALLELLAYERORDER M10 TSV B1
        ABOVEMETALONLYONTSV TSV
        OXIDE1
        SUBLIST M0 M1 M2 M3 M4 M5 M6 M7 M8 M9 M10 M11 B1 B2
        SUBLIST V0 V1 V2 V3 V4 V5 V6 V7 V8 V9 V10 TSV BV1
        OXIDE2
        SUBLIST M0 M1 M2 M3 M4 M5 M6 M7 M8 M9
        SUBLIST M10 M11 B1 B2
        SUBLIST V0 V1 V2 V3 V4 V5 V6 V7 V8 V9
        SUBLIST TSV
        SUBLIST V10 BV1 ; " ;
END PROPERTYDEFINITIONS
```


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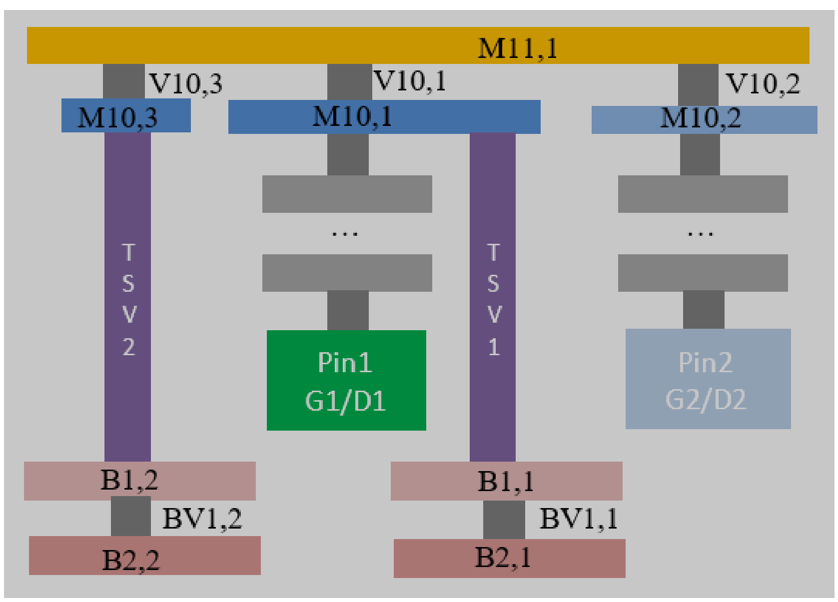
Here:

- The net is connected to two different vias, namely `TSV1` and `TSV2`, on `TSV`.
  - `TSV1` is newly electrically connected to `Pin1` from `M10,1`
  - `PAR(TSV1, G1) = Area(TSV1) / Area(G1)`
  - `TSV1` would not be first connected to `M11,1`, so no `TSV1` calculation for `Pin2` from `M11,1`
  - `TSV2` is newly electrically connected to `Pin1` and `Pin2` from `M11,1`
  - `PAR(TSV2, G1) = Area(TSV2) / Area(G1+G2)`
  - `PAR(TSV2, G2) = Area(TSV2) / Area(G1+G2)`
- `TSV` accumulated area ratio check for `Pin1`:
  - `CAR(TSV, G1) = PAR(TSV1, G1) + PAR(TSV2, G1)`
- `TSV` accumulated area ratio check for `Pin2`:
  - `CAR(TSV, G2) = PAR(TSV2, G2)`



Antenna Diff Only Area Ratio Model Rule

You can define an antenna diff only area ratio model rule by using the following `PROPERTYDEFINITIONS` statement:

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```
PROPERTYDEFINITIONS
LIBRARY LEF58_ANTENNADIFFONLYAREARATIOMODEL STRING
   "ANTENNADIFFONLYAREARATIOMODEL SKIPCHECK {ALL | OXIDE1 | OXIDE2
      | OXIDE3 | OXIDE4 | OXIDE5 | ... | OXIDE32}
      GATEAREA {ANY | OXIDE1 | OXIDE2 | OXIDE3 | OXIDE4
      | OXIDE5 | ... | OXIDE32}
      ;" ;
END PROPERTYDEFINITIONS
```

Where:

`ANTENNADIFFONLYAREARATIOMODEL SKIPCHECK {ALL | OXIDE1 | OXIDE2 | OXIDE3 | OXIDE4 | OXIDE5 | ... | OXIDE32} GATEAREA {ANY | OXIDE1 | OXIDE2 | OXIDE3 | OXIDE4 | OXIDE5 | ... | OXIDE32}`

Specifies that `ANTENNADIFFONLYAREARATIOMODEL` of the first given oxide should be skipped if the pin has `ANTENNAGATEAREA` defined on the second given oxide. If `ALL` is specified in `SKIPCHECK`, all given oxides will be skipped. If `ANY` is specified in `GATEAREA`, a pin having `ANTENNAGATEAREA` defined on any given oxides would skip the oxide model defined in `SKIPCHECK`.

Antenna Diff Only Area Ratio Model Rule Examples

- The following rule means that `ANTENNADIFFONLYAREARATIO` of all given oxides are not checked if the pin has `ANTENNAGATEAREA` defined on any given oxide.

```
PROPERTYDEFINITIONS
LIBRARY LEF58_ANTENNADIFFONLYAREARATIOMODEL STRING "
    ANTENNADIFFONLYAREARATIOMODEL SKIPCHECK ALL
        GATEAREA ANY ; " ;
END PROPERTYDEFINITIONS
```

- The following rule means that `ANTENNADIFFONLYAREARATIO` of all given oxides are not checked if the pin has `ANTENNAGATEAREA` defined on `OXIDE1`.

```
PROPERTYDEFINITIONS
LIBRARY LEF58_ANTENNADIFFONLYAREARATIOMODEL STRING "
    ANTENNADIFFONLYAREARATIOMODEL SKIPCHECK ALL
        GATEAREA OXIDE1 ; " ;
END PROPERTYDEFINITIONS
```

- The following rule means that `ANTENNADIFFONLYAREARATIO` of `OXIDE1` is not checked if the pin has `ANTENNAGATEAREA` defined on any given oxide.

```
PROPERTYDEFINITIONS
```


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```
    LIBRARY LEF58_ANTENNADIFFONLYAREARATIOMODEL STRING "
        ANTENNADIFFONLYAREARATIOMODEL SKIPCHECK OXIDE1
            GATEAREA ANY ; " ;
    END PROPERTYDEFINITIONS
```

Antenna Maximum Cumulative Area Rule

You can create an antenna maximum cumulative area rule to specify the maximum cumulative area of all layers to gate, which is not connected to the diffusion diode.

You can define an antenna maximum cumulative rule by using the following `PROPERTYDEFINITIONS` statement:

```
PROPERTYDEFINITIONS
LIBRARY LEF58_ANTENNAMAXCUMAREA STRING
   "ANTENNAMAXCUMAREA value [RANGE bottomLayer topLayer]
      [CUTLAYERONLY]
      ;" ;
END PROPERTYDEFINITIONS
```

Where:

`ANTENNAMAXCUMAREA` *`value`*

Specifies the maximum possible cumulative area of all layers, which is not connected to the diffusion diode, to be less than or equal to *`value`*. This would include floating metal that does not necessarily connect to a gate yet. The gate connection is done through a metal layer above the floating metal.

Type: Float, specified in microns

`CUTLAYERONLY` Specifies a maximum possible cumulative area only on cut layers.

`RANGE` *`bottomLayer topLayer`*

Specifies the maximum possible cumulative area between the given *`bottomLayer`* and *`topLayer`*.
Type: Float, specified in microns squared

Antenna Maximum Cumulative Area Rule Example

- The following rule means that the antenna is only measured on the cut layers, `VIA5 - VIA7`, on the given layer range.

  ```
  PROPERTYDEFINITIONS
  ```


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```
    LIBRARY LEF58_ANTENNAMAXCUMAREA STRING
        ANTENNAMAXCUMAREA 100.00 RANGE M5 M8 CUTLAYERONLY ; " ;
    END PROPERTYDEFINITIONS
```

Antenna Maximum Gate Area Rule

You can create an antenna maximum gate area rule to specify the maximum total area of all the antenna gate areas connected on any routing layer on the given antenna model.

You can define an antenna maximum gate area rule by using the following `PROPERTYDEFINITIONS` statement:

```
PROPERTYDEFINITIONS
LIBRARY LEF58_ANTENNAMAXGATEAREA STRING
   "ANTENNAMAXGATEAREA {OXIDE1 | OXIDE2 | OXIDE3 | OXIDE4 | OXIDE5
      | ... | OXIDE32} max_area
      ; " ;
END PROPERTYDEFINITIONS
```

Where:

```
ANTENNAMAXGATEAREA {OXIDE1 | OXIDE2 | OXIDE3 | OXIDE4 | OXIDE5
    | ... | OXIDE32} max_area
    ;
```

Specifies that the sum of all the antenna gate areas connected on any routing layer must be less than or equal to *max_area* on the given antenna model.

Type: Float, specified in microns squared

Antenna Maximum Side To Via Area Ratio Rule

You can create an antenna maximum side to via area ratio rule to specify the maximum possible ratio of cumulative side wall areas on all routing layers below a given cut layer to all the via areas on that cut layer.

You can define an antenna maximum side to via area ratio rule by using the following `PROPERTYDEFINITIONS` statement:

```
PROPERTYDEFINITIONS
LIBRARY LEF58_ANTENNAMAXSIDETOVIAAREARATIO STRING
   "ANTENNAMAXSIDETOVIAAREARATIO value
      RANGE bottomLayer topLayer
      ; " ;
END PROPERTYDEFINITIONS
```


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Where:

ANTENNAMAXSIDETOVIAAREARATIO *value* RANGE *bottomLayer topLayer*

Specifies that for each individual cut layer between *bottomLayer* and *topLayer*, which must be cut layers, if the wires on all the below routing layers are not connected to diffusion diodes, the maximum possible ratio of cumulative side wall areas on all below routing layers to all via areas on the cut layer must be less than or equal to *value*.

Type: Float

Antenna Maximum Side To Via Area Ratio Rule Example

- Consider the following example:

```
PROPERTYDEFINITIONS
LIBRARY LEF58_ANTENNAMAXSIDETOVIAAREARATIO STRING
"ANTENNAMAXSIDETOVIAAREARATIO ratio LAYER VIA9 VIA11
    ; " ;
END PROPERTYDEFINITIONS
```

Here:

One net has two pins – PIN1(G1) and PIN2(D2). There are floating wire segments as – M0,1 / M1,1 / M2,1 ... M9,1 / M10,1. See Illustration of the Antenna Maximum Side To Via Area Ratio Rule on page 29.

- Checking VIA9:
  - VIA9,1 is not connected to OD, and M0,1...M9,1 is a floating segment. *max_side_to_via_ratio*(VIA9,1, null) = *side_area*(M0,1 + M1,1 + M2,1 ...+ M9,1) / *Area*(VIA9,1) must be less than or equal to *ratio*.
  - VIA9,2 is connected to PIN2 OD, no floating segment is connected and no *max_side_to_via_ratio* check is needed.
- Checking VIA10:
  - VIA10,1 is not connected to OD, and M0,1…M10,1 is a floating segment. *max_side_to_via_ratio*(VIA10,1, G1) = *side_area* (M0,1 + M1,1 + M2,1 ...+ M9,1 + M10,1) / *Area*(VIA10,1) must be less than or equal to *ratio*.

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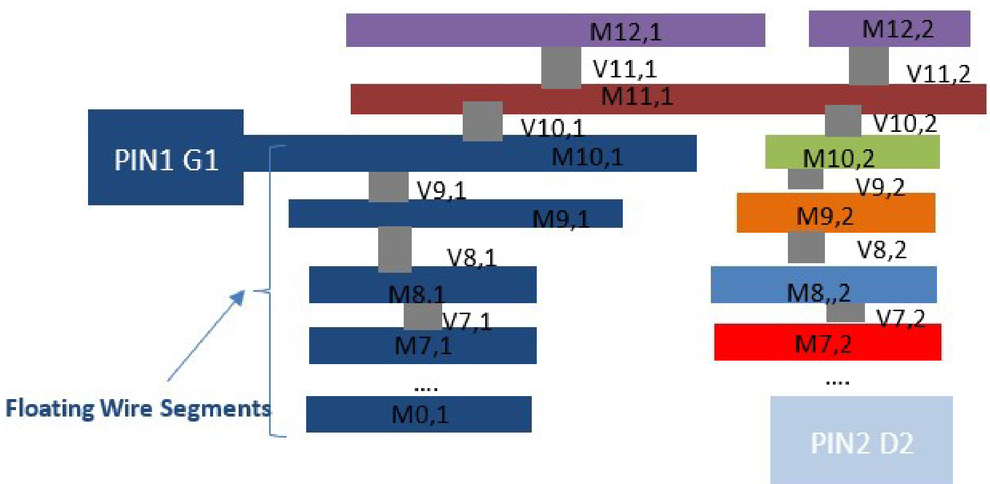
  - ❍ `VIA10,2` is connected to `PIN2 OD`, no floating segment is connected and no *`max_side_to_via_ratio`* check is needed.
- ❑ Checking `VIA11`:
  - ❍ `VIA11,1` and `VIA11,2` are connected to `PIN2 OD` and no *`max_side_to_via_ratio`* check is needed.

Figure 1-1 Illustration of the Antenna Maximum Side To Via Area Ratio Rule



Antenna Model Group Rule

You can define an antenna model group rule by using the following `PROPERTYDEFINITIONS` statement:

```
PROPERTYDEFINITIONS
LIBRARY LEF58_ANTENNAMMODELGROUP STRING
   "ANTENNAMODELGROUP OXIDEx MODEL {OXIDEy}...[GATEONLY]
      ;" ;
END PROPERTYDEFINITIONS
```

Where:

`ANTENNAMODELGROUP OXIDEx MODEL {OXIDEy}...`

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Specifies an antenna model group, which would be applied to all cut and routing layers, to combine all of the given `OXIDEy` in antenna consideration. `OXIDEx` would be considered when pins of all the `OXIDEy` models are connected and the individual `OXIDEy` models would be ignored. When some but not all of the `OXIDEy` models are connected, each of the individual `OXIDEy` models are checked separately and the `OXIDEx` model is ignored.

Type: String

`GATEONLY` Specifies that the antenna model group applies only to the pattern that metal is connected to a gate, and `ANTENNAAREARATIO`, `ANTENNACUMAREARATIO`, `ANTENNASIDEAREARATIO`, and `ANTENNACUMSIDEAREARATO` rules should be checked.

Antenna Model Group Rule Examples

- The following rule means that if two pins are electrically connected with pin1 having model `OXIDE1` and pin2 having model `OXIDE2`, `OXIDE3` by combining `OXIDE1` and `OXIDE2` would be checked while the individual `OXIDE1` and `OXIDE2`models would be ignored.

```
PROPERTYDEFINITIONS
LIBRARY LEF58_ANTENNAMODELGROUP STRING "
    ANTENNAMODELGROUP OXIDE3 MODEL OXIDE1 OXIDE2 ; " ;
END PROPERTYDEFINITIONS
```

- The following rule indicates that if two pins are electrically connected with pin1 having model `OXIDE1` and pin2 having model `OXIDE2`, `OXIDE1` and `OXIDE2` would be checked separately while `OXIDE4` would be ignored because not all of the models in the model group are connected.

```
PROPERTYDEFINITIONS
LIBRARY LEF58_ANTENNAMODELGROUP STRING "
    ANTENNAMODELGROUP OXIDE4 MODEL OXIDE1 OXIDE2 OXIDE3 ; " ;
END PROPERTYDEFINITIONS
```

- The following rule indicates that if three pins are electrically connected with pin1 having model `OXIDE1`, pin2 having model `OXIDE2` and pin3 having model `OXIDE3`, `OXIDE4` by combining `OXIDE1` and `OXIDE2` is checked along with `OXIDE3`, which does not belong to any model group. The individual `OXIDE1` and `OXIDE2`models are ignored.

```
PROPERTYDEFINITIONS
LIBRARY LEF58_ANTENNAMODELGROUP STRING "
    ANTENNAMODELGROUP OXIDE4 MODEL OXIDE1 OXIDE2 ; " ;
```


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```
END PROPERTYDEFINITIONS
```

- The following rule indicates that if two pins are electrically connected with pin1 having model `OXIDE1` and `OXIDE3` and pin2 having model `OXIDE2` and `OXIDE4`, `OXIDE5` by combining `OXIDE1` and `OXIDE2` and `OXIDE6` by combining `OXIDE3` and `OXIDE4` would only be checked separately.

```
PROPERTYDEFINITIONS
LIBRARY LEF58_ANTENNAMODELGROUP STRING "
     ANTENNAMODELGROUP OXIDE5 MODEL OXIDE1 OXIDE2 ;
     ANTENNAMODELGROUP OXIDE6 MODEL OXIDE3 OXIDE4 ; " ;
END PROPERTYDEFINITIONS
```

- The following illustration is an example of the rule below.

```
PROPERTYDEFINITIONS
LIBRARY LEF58_ANTENNAMODELGROUP STRING "
     ANTENNAMODELGROUP OXIDE3 MODEL OXIDE1 OXIDE2 ; " ;
END PROPERTYDEFINITIONS
```


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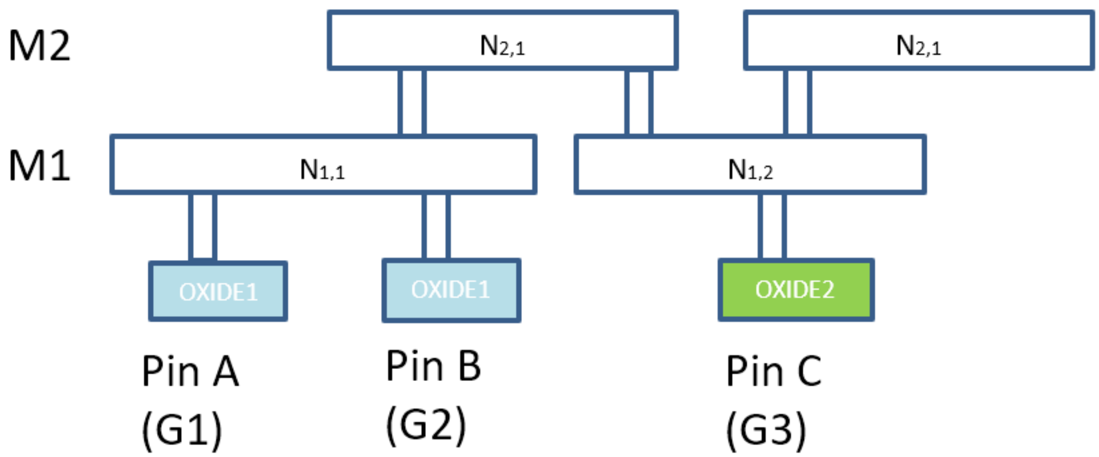
Figure 1-2 Illustration of the Antenna Model Group Rule



`OXIDE1` is defined on pins `A` and `B` and `OXIDE2` is defined on pin `C`. No pin should have the `OXIDE3` model.

Metal $N_{1,1}$ is electrically connected to pins `A` and `B` with model `OXIDE1`, so gate area (`G1+G2`) and model `OXIDE1 ANTENNA*` rules are applied.

Metal $N_{2,1}$ is electrically connected to pins `A` and `B` with model `OXIDE1` and `C` with model `OXIDE2`. Gate area of `G1+G2` for pins `A` and `B` of `OXIDE1` and `G3` for pin `C` of `OXIDE2` are combined as effective gate area for `OXIDE3 ANTENNA*` rules to be applied. Individual `OXIDE1` and `OXIDE2` models are no longer considered.

Gate Area

Layer	OXIDE1	OXIDE2	OXIDE3
M2	-	-	G1+G2+G3
M1	G1+G2	G3	-

Metal Area

Layer	OXIDE1	OXIDE2	OXIDE3
M2	-	-	N2,1
M1	N1,1	N1,2	-

Constraint Length Rule

You can create a constraint length rule to define the horizontal and vertical length of the concatenated constraint area. You can define a constraint length rule by using the following `PROPERTYDEFINITIONS` statement:

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```
PROPERTYDEFINITIONS
     LIBRARY LEF58_CONSTRAINTLENGTH STRING
     "CONSTRAINTLENGTH length {MAX|MIN} [HORIZONTAL|VERTICAL]
           [EXCEPTABUTTED] [WIDTH minWidth]
           CONSTRAINTAREATYPE {typeName | VERTICALCELLEDGESTACK
                 | HORIZONTALCELLEDGESTACK}
           ;" ;
END PROPERTYDEFINITIONS
```

Where:

`CONSTRAINTLENGTH` *`length`* `{MAX|MIN}`
    `CONSTRAINTAREATYPE` *`typeName`*

Specifies the horizontal and vertical length of the concatenated constraint area with type *`typeName`* defined in `CONSTRAINTAREATYPE` in `MACRO` must be less than *`length`* in `MAX` or greater than *`length`* in `MIN`.

Type: Float, specified in microns

`EXCEPTABUTTED` Specifies that the rule does not apply if there is an abutted shape from another cell in the orthogonal direction of the length constraint being checked.

`HORIZONTAL|VERTICAL` Specifies that the constraint length rule applies only on the given direction of either `HORIZONTAL` or `VERTICAL`.

`HORIZONTALCELLEDGESTACK`

Specifies that aligning any horizontal (top or bottom) cell edges would be subjected to this length constraint. `MAX` and `HORIZONTAL` must be defined for this predefined constraint type, which does not require `CONSTRAINTAREATYPE` to be defined in any `MACRO`.
Type: Float, specified in microns

`VERTICALCELLEDGESTACK`

Specifies that aligning any vertical (right or left) cell edges would be subjected to this length constraint. `MAX` and `VERTICAL` must be defined for this predefined constraint type, which does not require `CONSTRAINTAREATYPE` to be defined in any `MACRO`.
Type: Float, specified in microns

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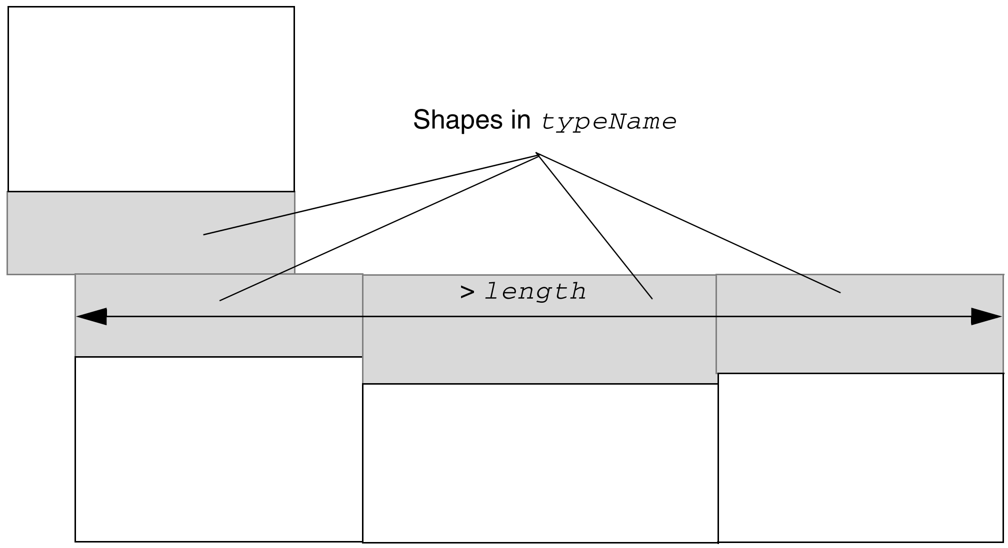
`WIDTH` *`minWidth`*	Specifies that the rule applies only to the portion of a shape with width greater than or equal to *`minWidth`*.

Figure 1-3 Illustration of Constraint Length Rule



It is not a violation because the left cell has a vertically abutted shape from another cell in the orthogonal direction of the checking length constraint.

Illustration of
`CONSTRAINTLENGTH` *`length`* `MAX HORIZONTAL`
`EXCEPTABUTTED CONSTRAINTAREATYPE` *`typeName`*

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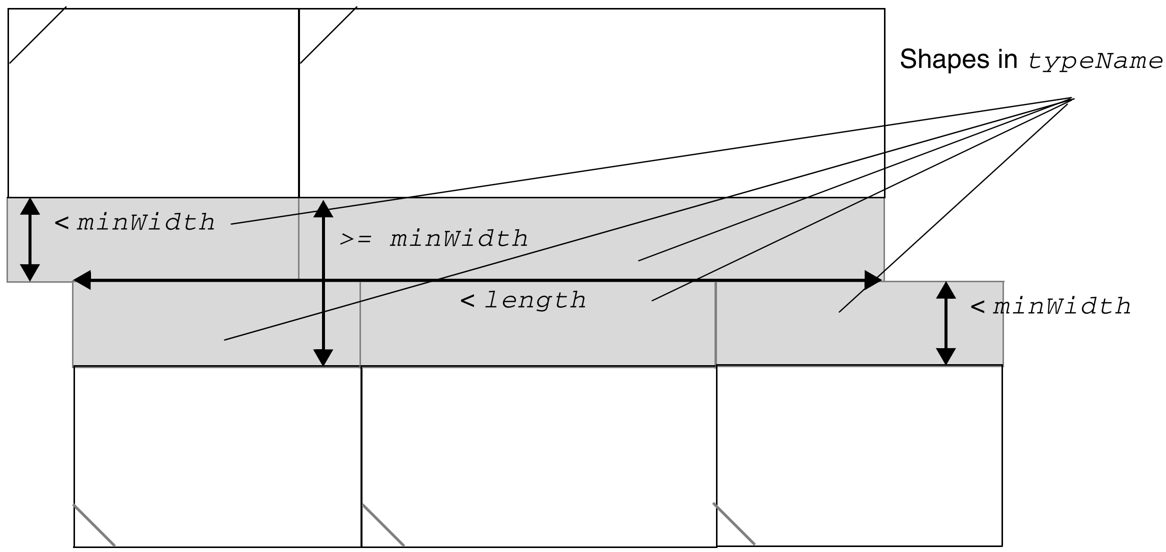
Figure 1-4 Illustration of Constraint Length Rule with WIDTH



Only the length from the portion of a shape with width >= *minWidth* needs to be checked.

Illustration of
```
CONSTRAINTLENGTH length MAX HORIZONTAL
    WIDTH minWidth CONSTRAINTAREATYPE typeName
```

Cell Edge Spacing Table Rule

You can create a cell edge spacing table rule to define a spacing table between two macro edges having different edge types.

This cell edge spacing table must be defined before reading in the `MACRO` definition with defined `EDGETYPE` in cell LEF files.

You can define a cell edge spacing table rule by using the following `PROPERTYDEFINITIONS` statement:

```
PROPERTYDEFINITIONS
LIBRARY LEF58_CELLEDGESPACINGTABLE STRING
    "CELLEDGESPACINGTABLE [NODEFAULT]
      {EDGETYPE {edgeType1 | DEFAULT}
      {edgeType2 | DEFAULT}
            [EXCEPTABUTTED | EXCEPTEXACTSPACING spacing]
            [EXCEPTNONFILLERINBETWEEN]
            [OPTIONAL | SOFT | EXACT]
```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```
            [UNDERPG | UNDERPGVIA] spacing} ...
      ;..." ;
END PROPERTYDEFINITIONS
```

Where:

`CELLEDGESPACINGTABLE`
`{EDGETYPE {edgeType1 | DEFAULT}`
`{edgeType2 | DEFAULT} spacing}...`

Defines a spacing table between a macro edge with type *edgeType1* and another macro edge with type *edgeType2*. The spacing is measured perpendicular from the edges. The `DEFAULT` keyword represents a macro edge without a type. If there is no spacing defined between the two types, it indicates that there is no spacing constraint (or 0.0 µm) between them.
Type: Float, specified in microns

`EXACT` — Specifies that it is a violation only when the spacing is exactly equal to *spacing*.

`EXCEPTABUTTED` — Specifies that two abutted macros of the given edge types are also allowed.

`EXCEPTEXACTSPACING` *spacing*

Specifies that the instances with spacing exactly equal to *spacing* can be exempted.

Type: Float, specified in microns

`EXCEPTNONFILLERINBETWEEN`

Specifies that the rule does not apply if there is, at least, one non-filler logic cell between the two constrained edges.

`NODEFAULT` — Specifies that `DEFAULT` should not be defined in the spacing table, and all the edges of any standard cell macros (with a `SITE` that is `CLASS CORE`) must have `EDGETYPE` property defined on both sides.

`OPTIONAL` — Specifies that the corresponding spacing constraints are optional. Users can use certain placement option to indicate whether those constraints should be observed or not.

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`SOFT`	Specifies that the corresponding spacing constraints are soft. This means that the tool has the freedom to honor the constraint with certain efforts. If the constraint will disturb the placement quality too much, the tool can choose to ignore the constraint.
`UNDERPG`	Specifies that the rule applies only when the bounding boxes on both instances are under a PG wire.
`UNDERPGVIA`	Specifies that the rule applies only when the overlapping segment of the edge constraint is under a PG via.

Cell Edge Spacing Table Rule Examples

- The following rule indicates that:
  - 0.1 µm is the spacing between a macro edge with type `GROUP1` and a macro edge with type `GROUP2`. Abutting the two macros is also allowed.
  - 0.2 µm is the optional spacing between two macro edges with type `GROUP1`; a certain placement option controls whether or not it is honored.
  - 0.3 µm is the spacing between a macro edge with type `GROUP2` and a macro edge without a type.
  - 0.4 µm is the soft spacing between two macro edges with type `GROUP2`, which would be ignored if placement quality is impacted too much.
  - 0.5 µm spacing is not allowed between a macro edge with type `GROUP1` and a macro edge without a type.
  - The rest of the combination of edges, `DEFAULT` to `DEFAULT`, do not have any spacing constraint, or 0.0 µm.

```
PROPERTYDEFINITIONS
  LIBRARY LEF58_CELLEDGESPACINGTABLE STRING
    "CELLEDGESPACINGTABLE
        EDGETYPE GROUP1 GROUP2 EXCEPTABUTTED 0.1
        EDGETYPE GROUP1 GROUP1 OPTIONAL 0.2
        EDGETYPE GROUP2 DEFAULT 0.3
        EDGETYPE GROUP2 GROUP2 SOFT 0.4
        EDGETYPE GROUP1 DEFAULT EXACT 0.5; ";
END PROPERTYDEFINITIONS
```

- The following rule illustrates the cell edge spacing rule for the given macro definition:

```
PROPERTYDEFINITIONS
  LIBRARY LEF58_CELLEDGESPACINGTABLE STRING
    "CELLEDGESPACINGTABLE
        EDGETYPE GROUP1 GROUP1 0.001 ; " ;
END PROPERTYDEFINITIONS
```


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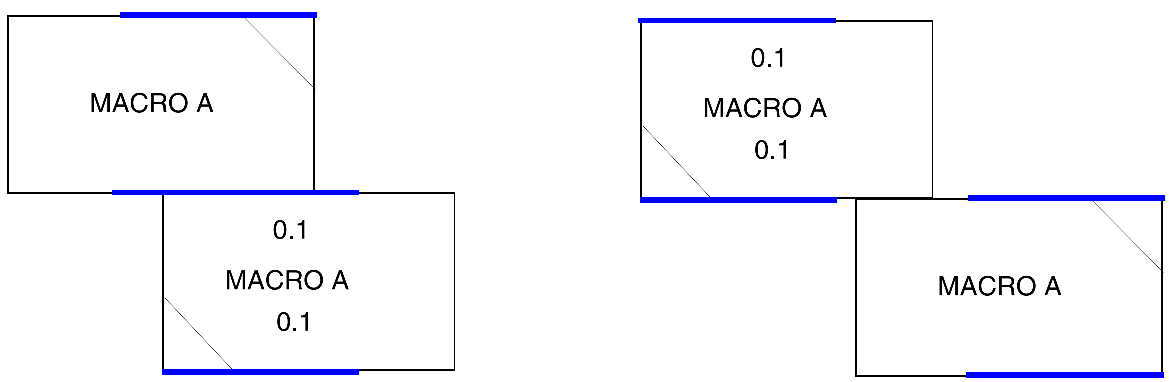
Figure 1-5 Illustration of Cell Edge Spacing Rule



a) Violation, the blue edges of GROUP1 should be 0.001 apart vertically or should not overlap horizontally

b) OK, the blue edges of GROUP1 do not overlap horizontally and there is no constraint on GROUP2 although edges of GROUP2 overlap

Example of

```
MACRO A
...
PROPERTY LEF58_EDGETYPE "
     EDGETYPE TOP GROUP1 RANGE 0 0.1
     EDGETYPE BOTTOM GROUP1 RANGE 0 0.1
     EDGETYPE TOP GROUP2
     EDGETYPE BOTTOM GROUP2 ; " ;
...
END A
```

- The following rule illustrates a cell edge spacing rule with `EXCEPTNONFILLERINBETWEEN`:

```
PROPERTYDEFINITIONS
  LIBRARY LEF58_CELLEDGESPACINGTABLE STRING
    "CELLEDGESPACINGTABLE EDGETYPE E1 E1
        EXCEPTNONFILLERINBETWEEN 1.5; " ;
END PROPERTYDEFINITIONS
```


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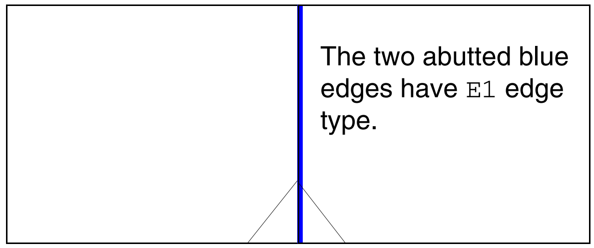
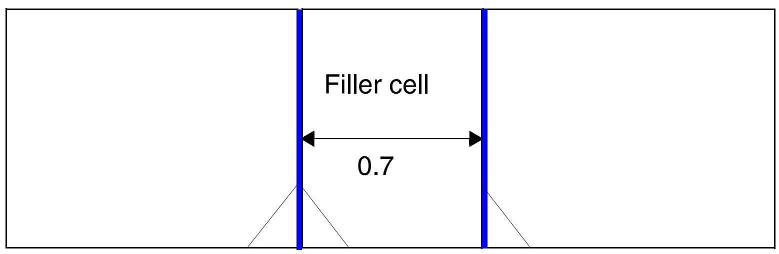
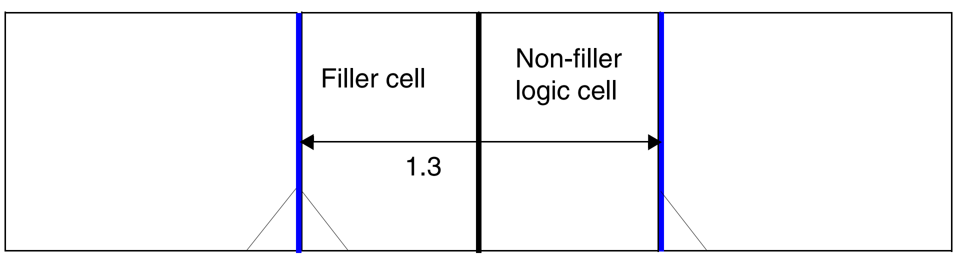
Figure 1-6 Illustration of Cell Edge Spacing Rule with `EXCEPTNONFILLERINBETWEEN`



a) Violation in abutted case.



b) Violation, having a filler cell is still a violation.



c) OK, there is a non-filler logic cell between the two blue `E1` edges.
It is a violation if `EXCEPTNONFILLERINBETWEEN` is omitted.

Cell Forbidden Rule

You can create a cell forbidden rule to specify the constraints on mixing cells of different types in the same design.

You can define a cell forbidden rule by using the following `PROPERTYDEFINITIONS` statement:

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```
PROPERTYDEFINITIONS
LIBRARY LEF_CDN_CELLFORBIDDEN STRING
"CELLFORBIDDEN codeName1 codeName2
     ;" ;
END PROPERTYDEFINITIONS
```

Where:

`CELLFORBIDDEN` *`codeName1 codeName2`*

Specifies that the cells of the first given cell type, *`codeName1`*, cannot be mixed with the cells of the second given cell type, *`codeName2`*, in the same design. The first given cell type cannot be the same as the second given cell type. *`codeName1`* and *`codeName2`* are given code names in `CELLMAP` in library and are defined in `CELLTYPE` in macro. If a cell has multiple types in `CELLTYPE` and the specified types are forbidden to be used together, this cell cannot be used.

Cell Layer Spacing Table Rule

You can create a cell layer spacing table rule to specify horizontal spacing between cell shapes on different layers.

You can define a cell layer spacing table rule by using the following `PROPERTYDEFINITIONS` statement:

```
PROPERTYDEFINITIONS
LIBRARY LEF58_CELLLAYERSPACINGTABLE STRING
    "CELLLAYERSPACINGTABLE
      [DEFAULTLAYER {notLayerName}...]
      {LAYER {DEFAULT|layerName} [WIDTH maxWidth]
      {DEFAULT|layerName2} [WIDTH maxWidth]
      [PRL minPrl maxPrl] [MERGED [HORIZONTAL|VERTICAL]]
      spacing [FORBIDDENSPACING minSpacing maxSpacing]} ...
      ;..." ;
END PROPERTYDEFINITIONS
```

Where:

```
CELLLAYERSPACINGTABLE
      [DEFAULTLAYER {notLayerName}...]
      {LAYER {DEFAULT|layerName} {DEFAULT|layerName2}
      spacing} ...
```


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Specifies the horizontal spacing between a cell shape in *layerName* and a cell shape in *layerName2* to be *spacing*. If DEFAULT is defined, DEFAULTLAYER must also be defined. DEFAULT means that a cell does not contain shapes on any given *notLayerName*. Spacing is measured to the boundary of such a DEFAULT cell. In addition, the spacing requirement should be considered on every half-row individually.

FORBIDDENSPACING *minSpacing maxSpacing*

Specifies that it is also considered a violation if the horizontal distance between the two cells is greater than or equal to *minSpacing* and less than or equal to *maxSpacing*.

Type: Float, specified in microns

MERGED [HORIZONTAL|VERTICAL]

Specifies that abutted shapes of the same layer in both given layers should be merged separately before spacing is applied. If HORIZONTAL or VERTICAL is defined, the merging happens only horizontally or vertically, respectively.

PRL *minPrl maxPrl*

Specifies that this spacing rule applies only between the two cells with vertical PRL greater than or equal to *minPrl* and less than or equal to *maxPrl*.

Type: Float, specified in microns

WIDTH *maxWidth*

Specifies that this spacing rule applies only on cells with horizontal width less than or equal to *maxWidth*. Abutted cell shapes are merged before checking against this width condition.

Type: Float, specified in microns

Cell Layer Spacing Table Rule Example

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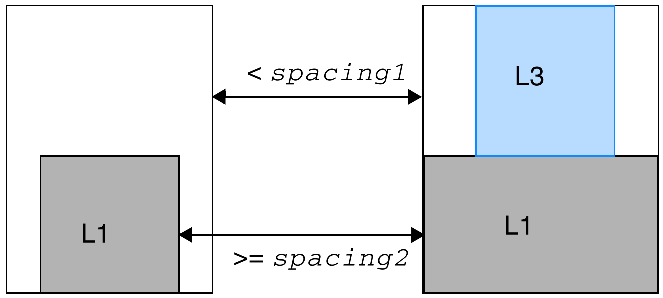
Figure 1-7 Illustration of Cell Layer Spacing Rule



Violation, the bottom half satisfies L1 to L1 *spacing2*, but the top half with no shape on the left to L3 on the right is treated as `DEFAULT` to `DEFAULT` and *spacing1* is not met.

Illustration of `CELLLAYERSPACINGTABLE DEFAULTLAYER L1 L2`
`LAYER DEFAULT DEFAULT` *spacing1*
`LAYER L1 L1` *spacing2* `;`

Cell Map Rule

You can create a cell map rule to specify the code name mapped to specific cell types.

You can define a cell map rule by using the following `PROPERTYDEFINITIONS` statement:

```
PROPERTYDEFINITIONS
LIBRARY LEF_CDN_CELLMAP STRING
"CELLMAP {REGULARSH|REGULARMH|PROTRUDEDMH|0.5MH}
     codeName
     ;" ;
END PROPERTYDEFINITIONS
```

Where:

`CELLMAP {REGULARSH|REGULARMH|PROTRUDEDMH|0.5MH}` *codeName*

Specifies that a cell type, `REGULARSH`, `REGULARMH`, `PROTRUDEDMH`, or `0.5MH`, is mapped to the given *codeName*. `REGULARSH` means regular single-height cells. `REGULARMH` means regular multi-height cells. `PROTRUDEDMH` means multi-height cells protruded on top and bottom by a half-row. `0.5MH` means multi-height cells with n.5 height.

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Cell Variants Rule

You can create a cell variants rule to specify the placement locations of electrically equivalent (EEQ) cells.

You can define a cell variants rule by using the following `PROPERTYDEFINITIONS` statement:

```
PROPERTYDEFINITIONS
LIBRARY LEF58_CELLVARIANTS STRING
"CELLVARIANTS variantTotalNum
     [STARTVARIANT startVariant [XFLIPSTARTVARIANT startVariant]]
     YFLIPMAP {flippedVariantNum siteVariantNum}...
     [XFLIPMAP {cellVariantNum siteVariantNum}... ]
     ;" ;
END PROPERTYDEFINITIONS
```

Where:

```
CELLVARIANTS variantTotalNum [STARTVARIANT startVariant]
     YFLIPMAP {flippedVariantNum siteVariantNum}...
```


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Specifies the total number of variants of electrically equivalent (EEQ) cells as *variantTotalNum*. By default, the original cell master MACRO without the EEQ construct will always have an implicit variant number of 1. If VARIANT is specified in that cell master, the variant number would become the given number. All variant numbers need not necessarily be defined in EEQ VARIANT. Any missing variant number means that there is no available EEQ cell placed on the corresponding site. Similarly, multiple EEQ cells can have the same variant number and tools have a choice to place any of them on the corresponding site.

The leftmost placement site of core area is labeled as variant 1 or *startVariant* if STARTVARIANT is defined. The subsequent sites range from 2 to *variantTotalNum* or *startVariant* % *variantTotalNum* + 1 to (*startVariant* + *variantTotalNum* - 2) % *variantTotalNum* + 1 and repeat back to 1 or *startVariant*. The non-Y-flipped instances should be placed such that the cell variant number matches the site variant number. Cells without any EEQ can be placed on any site.

YFLIPMAP defines which Y-flipped variant number, *flippedVariantNum*, should be placed on the site variant, *siteVariantNum*, when an instance is Y-flipped. There should be *variantTotalNum* pairs of variant values defined. The variant pair values should be symmetrical. That is, if *flippedVariantNum* of V1 is placed on *siteVariantNum* of V2, *flippedVariantNum* of V2 is placed on *siteVariantNum* of V1.
Type: Integer

Typically, this cell variant rule may vary for different cell libraries/ heights, and this library property would likely be defined in a cell LEF or side LEF file, instead of a tech LEF file.

XFLIPMAP {*cellVariantNum siteVariantNum*}...

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Specifies that the number of site variants, *variantTotalNum*, would be divided by two on each row. This means that *variantTotalNum* should be even and site variants would alternate between 1 and *variantTotalNum*/2 on each row. Only cell variants defined in *cellVariantNum* with a non-zero value in *siteVariantNum* could be placed on the MX rows, and they would be placed on the corresponding *siteVariantNum*. Only cell variants defined in *cellVariantNum* with a zero value in *siteVariantNum* could be placed on the R0 rows. There should be *variantTotalNum* pairs of variant values defined.

Type: Integer

XFLIPSTARTVARIANT *startVariant*

Specifies the leftmost placement site of the core area on the MX rows to be *startVariant*. STARTVARIANT would define the starting variant on the R0 rows. XFLIPSTARTVARIANT should only be defined along with XFLIPMAP.

Type: Integer

Cell Variants Rule Examples

- The following example illustrates the cell variant rule:

```
PROPERTYDEFINITIONS
LIBRARY LEF58_CELLVARIANTS STRING "
CELLVARIANTS 3 YFLIPMAP 1 1 2 3 3 2 ; " ;
END PROPERTYDEFINITIONS
...

MACRO A
...
END A

MACRO A2
PROPERTY LEF58_EEQ " EEQ A VARIANT 2 ; " ;
...
END A2

MACRO A3
```


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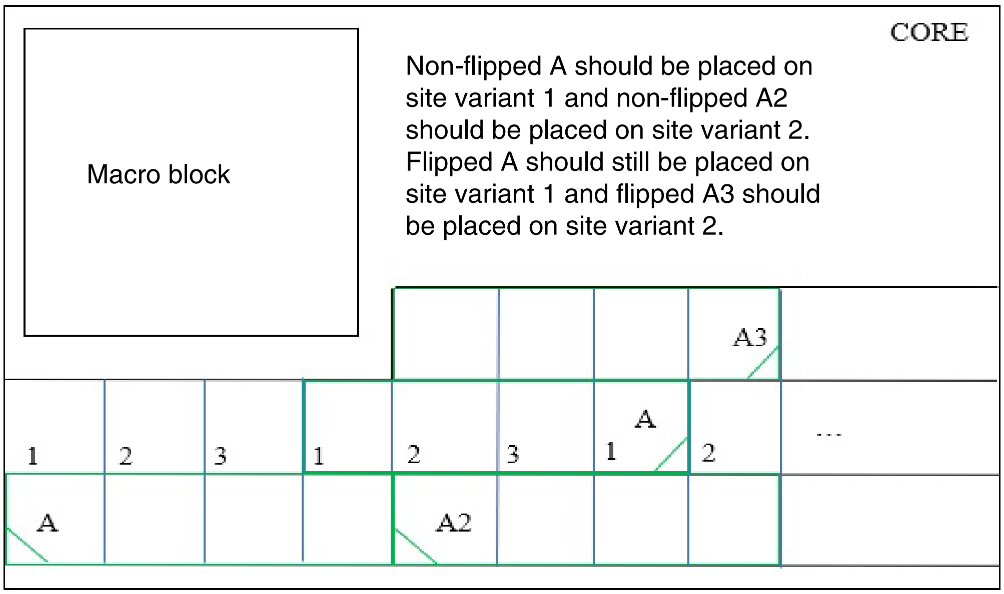
```
PROPERTY LEF58_EEQ " EEQ A VARIANT 3 ; " ;
...
END A3
```

Figure 1-8 Illustration of the Cell Variants Rule



- The following example illustrates a cell variant rule with `XFLIPMAP`:

```
LIBRARY LEF58_CELLVARIANTS STRING "
     CELLVARIANTS 4 STARTVARIANT 1 XFLIPSTARTVARIANT 2
     YFLIPMAP { 1 2 2 1 3 2 4 1 } XFLIPMAP { 1 0 2 0 3 1 4 2 } ; " ;
END PROPERTYDEFINITIONS
```


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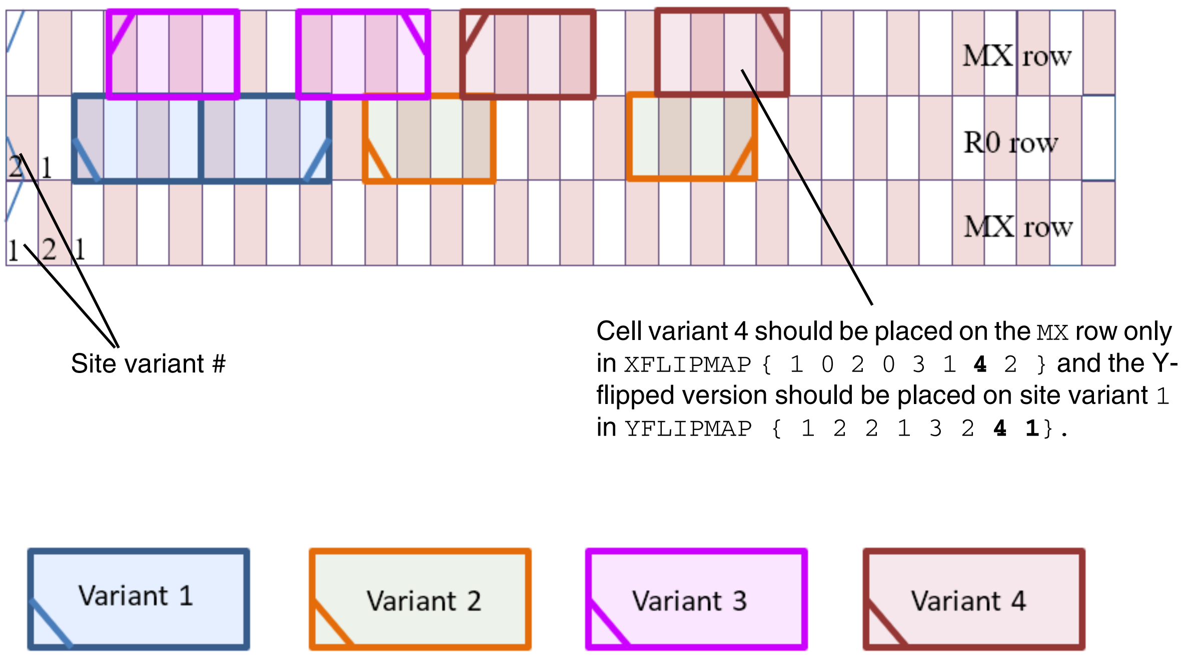
Figure 1-9 Illustration of the Cell Variants Rule with XFLIPMAP



Cut Mask Track Rule

You can create a cut mask track rule to specify a preferred cut mask vertical track for the mask of the cell master cuts.

You can define a cut mask rule by using the following `PROPERTYDEFINITIONS` statement:

```
PROPERTYDEFINITIONS
LIBRARY LEF58_CUTMASKTRACK STRING
"CUTMASKTRACK cutLayer MASK maskNum OFFSET offset PITCH pitch
     ;" ;
END PROPERTYDEFINITIONS
```

Where:

`CUTMASKTRACK` *`cutLayer`* `MASK` *`maskNum`* `OFFSET` *`offset`* `PITCH` *`pitch`*

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Specifies a preferred cut mask vertical track that the mask of the cell master cuts should follow. This construct is applied to all macros with `CLASS CORE`. *`cutLayer`* should be a cut layer with multiple masks using the `MASK` construct. *`maskNum`* defines the mask of the first track with *`offset`* from the cell origin of a cell master. Then, the subsequent track mask would be alternated with *`pitch`* being the track pitch. Cuts that are less than same-mask spacing apart should be clustered as a group, and the cut masks within a group are dependent on each other. However, the mask of an anchor cut in a group could be any mask. Once the anchor cut mask is picked, the mask of the rest of the cuts is determined. The cell architecture should have all of the center of cuts on the given tracks. The mask of the anchor cut should be picked to follow the track color.
Type: Floats, specified in microns

Cut Mask Track Rule Example

The following example indicates that the preferred cut mask vertical track for the mask of the cell master cuts is mask `1`:

```
PROPERTYDEFINITIONS
LIBRARY LEF58_CUTMASKTRACK STRING "
CUTMASKTRACK VIA1 MASK 1 OFFSET 0.15 PITCH 0.2
      ; ";
END PROPERTYDEFINITIONS
```


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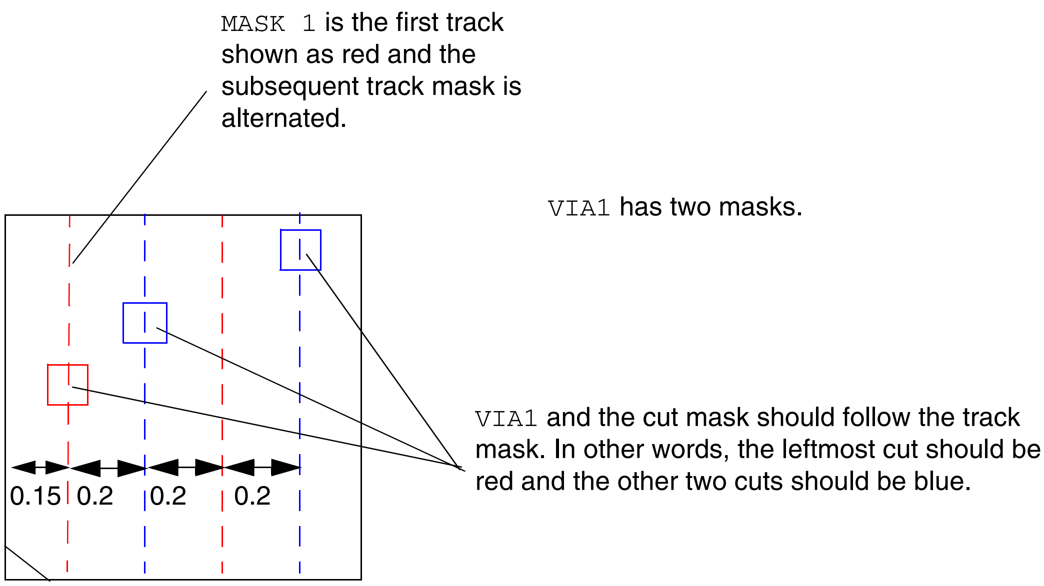
Figure 1-10 Illustration of the Cut Mask Track Rule



Edge Type on Overlap Rule

You can create an edge type on overlap rule to specify that edge type definitions reference cell boundaries on the `OVERLAP` layer.

You can define an edge type on overlap rule by using the following `PROPERTYDEFINITIONS` statement:

```
PROPERTYDEFINITIONS
LIBRARY LEF58_EDGETYPEONOVERLAP STRING
"EDGETYPEONOVERLAP
     ;" ;
END PROPERTYDEFINITIONS
```

Where:

`EDGETYPEONOVERLAP`	Specifies that `EDGETYPE` definitions reference the cell boundaries on the `OVERLAP` layer, where applicable, for all `MACRO`s. For `MACRO`s without an `OVERLAP` layer, the original cell boundaries are used as the reference by default.

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FinFET Rule

You can create a FinFET rule to specify the FinFET pitch to be pitch.

You can define a FinFET rule by using the following PROPERTYDEFINITIONS statement:

```
PROPERTYDEFINITIONS
LIBRARY LEF58_FINFET STRING
"FINFET
     PITCH pitch [OFFSET offset] {HORIZONTAL|VERTICAL}
         [NOCORECELL]
     ;" ;
END PROPERTYDEFINITIONS
```

Where:

```
FINFET
    PITCH pitch [OFFSET offset] {HORIZONTAL | VERTICAL}
```

Specifies the FinFET pitch to be pitch, which starts at the offset, if specified, or zero from the origin of the design. The HORIZONTAL/VERTICAL keywords mean that the FinFET pitch is used to define the legal Y/X values of the placement grid that all cells, blocks, and IOs must align to (specifically, the origin of every cell must be aligned to the legal Y/X values), in order to guarantee all the FinFETs inside are aligned properly. The X/Y value of the placement grid is derived from the standard cell site width and only applies to standard cells. The cell row height should typically be multiple of the FinFET pitch.
Type: Floats, specified in microns

NOCORECELL	Specifies that the FinFET grid does not apply to standard cells with MACRO with CLASS CORE.

FinFET Rule Example

The following example indicates that the FinFET y pitch is 0.108 μm:

```
PROPERTYDEFINITIONS
LIBRARY LEF58_FINFET STRING "
    FINFET
      PITCH 0.108 HORIZONTAL ; ";
END PROPERTYDEFINITIONS
```


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Ground Connectivity Layers Rule

You can create a ground connectivity layers rule to specify the routing layers between which ground nets must be completed.

You can define a ground connectivity layers rule by using the following `PROPERTYDEFINITIONS` statement:

```
PROPERTYDEFINITIONS
LIBRARY LEF58_GROUNDCONNECTIVITYLAYERS STRING
"GROUNDCONNECTIVITYLAYERS bottomLayer topLayer
     ;" ;
END PROPERTYDEFINITIONS
```

Where:

`GROUNDCONNECTIVITYLAYERS` *`bottomLayer topLayer`*

Specifies that ground nets must be completed between the given layers. *`bottomLayer`* and *`topLayer`* must be routing layers. It is a translated methodology to prevent certain DRC violations. A design can have multiple ground nets and not all of them need to be checked this way. A separate command is used to define which ground net should be checked in this methodology.

Implant Group Rule

You can create an implant group rule to specify rules that apply for all the shapes on an implant layer.

You can define an implant group rule by using the following `PROPERTYDEFINITIONS` statement:

```
PROPERTYDEFINITIONS
LIBRARY LEF58_IMPLANTGROUP STRING
   "IMPLANTGROUP groupName
       LAYER {layerName}... BASEDLAYER basedLayerName
      ;" ;
END PROPERTYDEFINITIONS
```

Where:

```
IMPLANTGROUP groupName
     LAYER {layerName}... BASEDLAYER basedLayerName
```


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Specifies that the combined shapes on all of the given layers in *layerName*, which must be an implant layer, must follow all of the rules on *basedLayerName*.

Implant Group Rule Example

```
PROPERTYDEFINITIONS
LIBRARY LEF58_IMPLANTGROUP STRING "
       IMPLANTGROUP GROUP1 LAYER VT1 VT2 BASEDLAYER VT1 ;
       IMPLANTGROUP GROUP2 LAYER VT1 VT3 BASEDLAYER VT1 ; " ;
END PROPERTYDEFINITIONS
```

In this example, the combined shapes on VT1 and VT2 and the combined shapes on VT1 and VT3 must follow the rules on VT1.

Layer Mask Shift Rule

You can create a layer mask shift rule to specify that mask could be shifted on the given layers.

You can define a layer mask shift rule by using the following PROPERTYDEFINITIONS statement:

```
PROPERTYDEFINITIONS
     LIBRARY LEF58_LAYERMASKSHIFT STRING
     "LAYERMASKSHIFT layer1 [layer2 ...]
          [MACROCLASS { RING | BLOCK | PAD | CORE | ENDCAP }...]
          ;" ;

END PROPERTYDEFINITIONS
```

Where:

```
LAYERMASKSHIFT layer1 [layer2 ...]
    [MACROCLASS { RING | BLOCK | PAD | CORE | ENDCAP }...]
```

Specifies that a mask could be shifted only on the given layers. If MACROCLASS is also specified, only the given layers on the MACRO with the given classes allow color shifting. If this LAYERMASKSHIFT construct is defined, none of the macros should define LAYERMASKSHIFT. In addition, LAYERMASKSHIFT is treated as an exemption to FIXEDMASK, which should be defined as a library property or on individual macros.

Type: String

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Layer Mask Shift Rule Examples

- The following example indicates that mask-shifting is allowed on LEF layer `M1` and `VIA1` but not on other layers:

```
FIXEDMASK ;

PROPERTYDEFINITIONS
LIBRARY LEF58_LAYERMASKSHIFT STRING "
   LAYERMASKSHIFT M1 VIA1 ; " ;
END PROPERTYDEFINITIONS
```

- The following example indicates that mask-shifting is allowed on the LEF layer `M1` and `VIA1` on `MACRO` with class `CORE` or `ENDCAP` but not on other layers and macros with other classes:

```
FIXEDMASK ;

PROPERTYDEFINITIONS
LIBRARY LEF58_LAYERMASKSHIFT STRING "
   LAYERMASKSHIFT M1 VIA1 MACROCLASS CORE ENDCAP ; " ;
END PROPERTYDEFINITIONS
```

Line End Track Rule

You can create a line-end track rule to specify conditions for aligning line ends to tracks on routing layers.

You can define a line-end track rule by using the following `PROPERTYDEFINITIONS` statement:

```
PROPERTYDEFINITIONS
     LIBRARY LEF_CDN_LAYERMASKSHIFT STRING
     "LINEENDTRACK layerName [MASK maskNum]
          COREOFFSET offset PITCH pitch
          TRACKDISTANCE distance
          ;" ;
END PROPERTYDEFINITIONS
```

Where:

```
LINEENDTRACK layerName [MASK maskNum]
      COREOFFSET offset PITCH pitch
      TRACKDISTANCE distance
```


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	Specifies that any line-end on *`layerName`*, which must be a routing layer, must align to certain tracks, which are *`pitch`* apart. There are two sets of tracks, which are apart by *`distance`*; the bottom line-end of a vertical wire or the left line-end of a horizontal wire aligns to one set of tracks while the top line-end of a vertical wire or right line-end of a horizontal wire aligns to the other set of tracks. *`offset`* specifies the starting coordinate of the track with respect to the origin of the core. In addition, `LINEENDTRACK` is mutually exclusive with `EOLTRACK`. If `LINEENDTRACK` is defined, `EOLTRACK` cannot be defined on any routing layer. *Type*: Float, specified in microns
`MASK` *`maskNum`*	*`maskNum`* defines the metal mask on which the line-end grids are applied.

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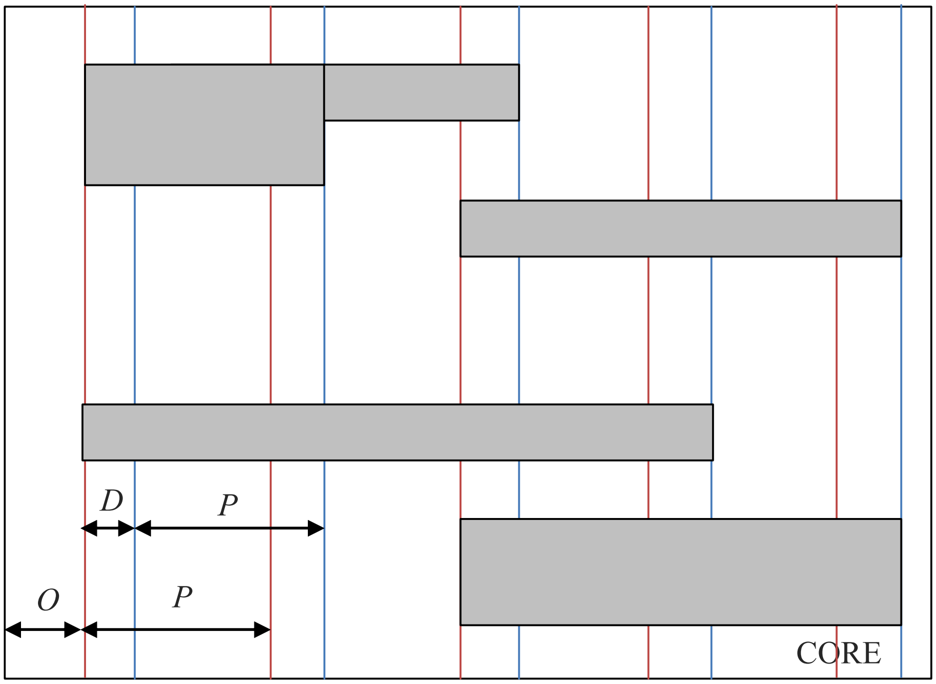
Figure 1-11 Illustration of Line End Track Rule



All left line-end edges must be aligned to the red tracks, and all right line-end edges must be aligned to the blue tracks.

Illustration of:

```
PROPERTY LEF_CDN_LINEENDTRACK "
LINEENDTRACK layerName  COREOFFSET O
PITCH P TRACKSDISTANCE D ; " ;
```

Maximum Floating Area Rule

Maximum area rules exist for floating metal shapes that are not connected to a diffusion or polysilicon gate. Similar to process antenna rules, maximum floating area rules apply only to the current layer and any lower layers (that is, all layers that have been fabricated up to the layer of interest). Maximum floating area rules can be used to avoid shorts between floating and non-floating metal wires that are caused by arcing due to charge buildup during processing steps.

You can create a global maximum floating area rule by using the following `PROPERTYDEFINITIONS` statement:

```
PROPERTYDEFINITIONS
LIBRARY LEF58_MAXFLOATINGAREA STRING
```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```
 "MAXFLOATINGAREA maxArea
  {SINGLELAYER | CONNECTED | ALLCONNECTED} minRoutingLayer maxRoutingLayer
   [[LAYERS minRoutingLayer maxRoutingLayer]
     SPACING minSpacing [PARSPACING minParSpacing minParallelLength ...]] ...
;" ;
END PROPERTYDEFINITIONS
```


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Where:

`ALLCONNECTED` *`minRoutingLayer maxRoutingLayer`*

Indicates that the rule applies to the connected area of floating metal shapes connected together on each layer between *`minRoutingLayer`* and *`maxRoutingLayer`* inclusive. The total connected area on the current and lower layers must be less than or equal to *`maxArea`*, or meet the minimum spacing to other grounded shapes on this layer and all lower-connected layer shapes. The names you specify for *`minRoutingLayer`* and *`maxRoutingLayer`* must be two previously defined LEF routing layers.

`CONNECTED` *`minRoutingLayer maxRoutingLayer`*

Indicates that the rule applies to the area of floating metal shapes connected together on each layer between *`minRoutingLayer`* and *`maxRoutingLayer`* inclusive. The connected area on a current layer must be less than or equal to *`maxArea`*, or meet the minimum spacing to other grounded shapes on this layer and all lower connected layer shapes. The names you specify for *`minRoutingLayer`* and *`maxRoutingLayer`* must be two previously defined LEF routing layers.

`LAYERS` *`minRoutingLayer maxRoutingLayer`*

Indicates that layers between *`minRoutingLayer`* and *`maxRoutingLayer`* inclusive must meet the specified `SPACING` rules. The names you specify for *`minRoutingLayer`* and *`maxRoutingLayer`* must be previously defined LEF routing layers, and can be the same layer.

`MAXFLOATINGAREA` *`maxArea`*

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Indicates that floating metal shapes must have a total area that is less than or equal to *maxArea*.
Type: Float, specified in units of microns squared

Floating metal is defined as metal on the current layer that cannot trace a path to a diffusion connection or polysilicon gate, using only same-layer or lower-layer metal connections. Grounded metal is defined as metal that can connect to a diffusion connection or polysilicon gate using only same-layer or lower-layer connections.

Note: The MAXFLOATINGAREA rule depends on the existing LEF MACRO PIN ANTENNADIFFAREA and ANTENNAGATEAREA statements to indicate which pins are connected to diffusion or polysilicon gates.

You can have two MAXFLOATINGAREA statements: one for SINGLELAYER, and one for CONNECTED or ALLCONNECTED.

PARSPACING *minParSpacing minParallelLength* ...

Indicates that floating metal that is greater than or equal to *minParSpacing* from grounded metal must have greater than or equal to *minParallelLength* at the minimum spacing distance. If more than one pair of values is given, the smallest spacing value that matches is used. (See Example 2.)
Type: Float, specified in microns (for both values)

The *minParSpacing* values must be defined in decreasing value, and be smaller than SPACING *minSpacing*. The intent of the PARSPACING rule is to spread out the charge build up on floating shapes to reduce the chance of spark; therefore, smaller spacing values require larger parallel lengths in order to be legal.

If you define PARSPACING for the same layer more than once (due to overlapping LAYERS layer ranges) the last PARSPACING values overwrite the previous values.

SINGLELAYER *minRoutingLayer maxRoutingLayer*

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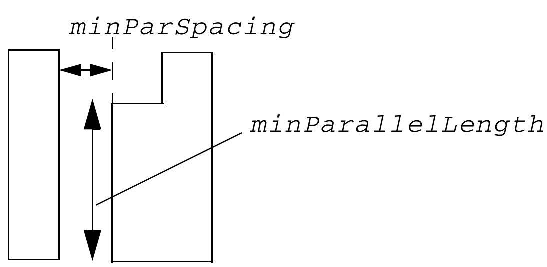
	Indicates the rule applies to each individual floating metal shape on any routing layer between *minRoutingLayer* and *maxRoutingLayer* inclusive. Each shape must have an area that is less than or equal to *maxArea*, or meet the minimum spacing to other grounded shapes. The names you specify for *minRoutingLayer* and *maxRoutingLayer* must be two previously defined LEF routing layers.
`SPACING` *minSpacing*	Indicates that if the current layer floating metal has an area that is greater than *maxArea*, the floating metal on the current layer (and all connected lower layers with `CONNECTED` or `ALLCONNECTED`) must be greater than or equal to *minSpacing* distance from grounded metal for all layers that have `MAXFLOATINGAREA` rules. If you define `SPACING` for the same layer more than once (due to overlapping `LAYERS` layer ranges) the last `SPACING` value overwrites the previous values. *Type:* Float, specified in microns

Definition of *minParSpacing* and *minParallelLength* for `MAXFLOATINGAREA` rule.



Maximum Floating Area Rule Examples

- **Example 1**

  Assume the following rule exists:

```
PROPERTYDEFINITIONS
  LIBRARY LEF58_MAXFLOATINGAREA STRING
    "MAXFLOATINGAREA 1000 SINGLELAYER m1 m6 SPACING 1.0 ;" ;
END PROPERTYDEFINITIONS
```

  Every shape on layers *m1* to *m6* with maximum floating area greater than 1000 μm^2 must have spacing of greater than or equal to 1.0 μm to any grounded metal.

- **Example 2**

  Assume the following rule exists:

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```
PROPERTYDEFINITIONS
  LIBRARY LEF58_MAXFLOATINGAREA STRING
    "MAXFLOATINGAREA 1000 CONNECTED m1 m6
      LAYERS m1 m1 SPACING 1.0 PARSPACING 0.5 0.8 0.2 2.0
      LAYERS m2 m6 SPACING 0.6 PARSPACING 0.3 0.9 ;" ;
END PROPERTYDEFINITIONS
```

For layer *m1*, any floating metal must be either greater than or equal to 1.0 µm distance from grounded metal, or:

- If it is greater than or equal to 0.5 µm distance away, there must be at least 0.8 µm of parallel length at the minimum spacing.
- If it is greater than or equal to 0.2 µm distance away, there must be at least 2.0 µm of parallel length at the minimum spacing.

Spacing that is less than 0.2 µm is not allowed.

Any floating *m2* through *m6* shapes with area that is greater than 1000 µm^2, must either be greater than or equal to 0.6 µm distance from grounded metal, or they must be greater than or equal to 0.3 µm distance away with greater than or equal to 0.9 µm of parallel length.

See Figure 1-12 on page 61 for different layout examples using this rule.

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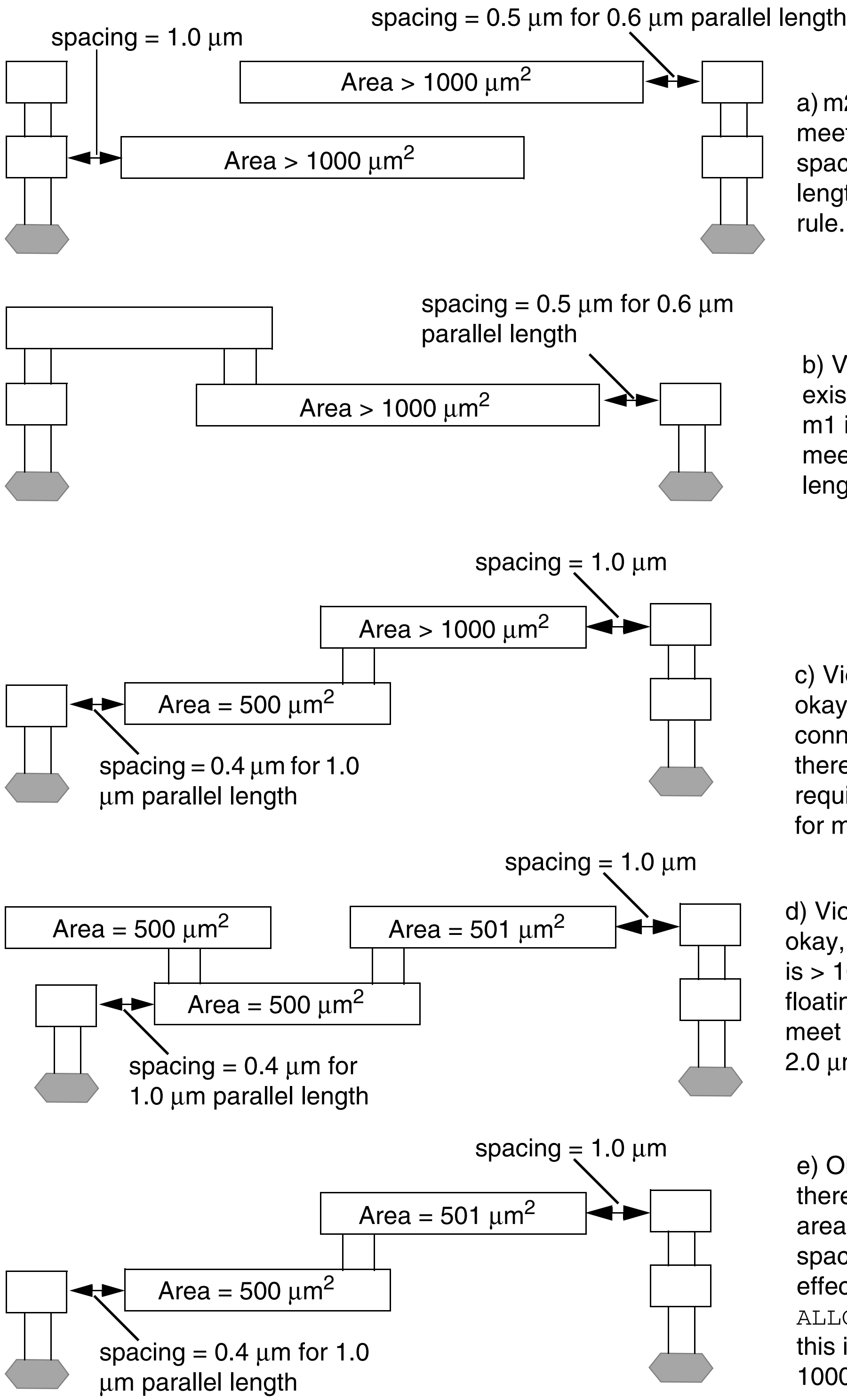
Figure 1-12 Maximum Floating Area Rule



a) m2 violation, m1 okay. m2 did not meet the spacing rule (0.5 µm spacing requires 0.9 µm parallel length). m1 met the 1.0 µm spacing rule.

b) Violation for m1. m2 does not exist when m1 is created; therefore m1 is still floating, and does not meet the required 0.8 µm parallel length for minSpacing >= 0.5 µm.

c) Violation for m1. m2 spacing is okay, but m2 floating area is connected to m1 floating area; therefore m1 must meet the required parallel length of 2.0 µm for minSpacing >= 0.2 µm.

d) Violation for m1. m2 spacing is okay, but total m2 floating area is > 1000 µm^2 connected to m1 floating area; therefore m1 must meet the required parallel length of 2.0 µm for minSpacing >= 0.2 µm.

e) Okay. m1 area < 1000 µm^2, therefore no spacing required. m2 area is <1000 µm^2; therefore no spacing required, and m2 has no effect on m1 spacing. If `ALLCONNECTED` was present, then this is a violation (m1 + m2 area > 1000 µm^2).

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Maximum Via Stack Rule

You can create a maximum via stack rule to require a series of stacked vias to all be multi-cut vias. A via is considered to be in a stack with other vias if the cuts of all the vias partially overlap (the boolean `AND` of the cut layer shapes from every via in the stack is not empty). A multi-cut via interrupts the stack, unless the `NOSINGLE` keyword is specified.

If multiple `MAXVIASTACK` statements are defined, all of them must be fulfilled individually.

You can define a maximum via stack rule using the following `PROPERTYDEFINITIONS` statement:

```
PROPERTYDEFINITIONS
     LIBRARY LEF58_MAXVIASTACK STRING
      "MAXVIASTACK maxStack [LAYER regionLayerName]
           [SAMENET] [NOSINGLE]
           [IGNOREONESAMEMETAL cutWithin
                  | IGNORESAMELAYER cutWithin]
           [MINMULTICUT numCut [ALLSAMEMETAL]]
           [WITHIN {LOWCUTSIZE | {cutLayerName within
                  [CENTERTOCENTER]}...}
              |CENTERCUTWITHIN cutLayerName
                  belowCutWithin aboveCutWithin]
           [EXCEPTPGNET
              | EXCEPTPGSHAPETYPE {RING | STRIPE | FOLLOWPIN
                  | IOWIRE | COREWIRE | BLOCKWIRE | PADRING
                  | BLOCKRING | FILLWIRE}...]
           [EXCEPTAREA individualArea sumArea
              | EXCEPTMAXAREA {metalLayer area}...
              | AREA {cutLayer area {ratio | ratio1 ratio2}}...]
           [RANGE bottomLayer topLayer]
           [EXCEPTCUTCLASSLIST {cutLayer cutClassName}...]
           [[NOVIALIST {viaName}...]...
              |[NOCUTCLASSLIST {cutLayer cutClassName}...]...]
           [EXCEPTENCLOSURE {cutLayer [ABOVE | BELOW] enclosure}...]
       ;" ;

END PROPERTYDEFINITIONS
```


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Where:

`ALLSAMEMETAL`

Specifies that multi-cut exemption applies only if all of the multi cuts on all the cut layers have the same metal on each of the metal layers sandwiching the cut layers. This construct must be used only along with `MAXVIASTACK 1`. For example, with `MAXVIASTACK 1` and `MINMULTICUT 3 ALLSAMEMETAL` defined for `VIA3` and `VIA4` cuts, at least 3 cuts must be on `VIA3` and `VIA4` individually and both `VIA3` and `VIA4` should have common metal on `METAL3`, `METAL4`, and `METAL5` to fully cover all cuts to be exempted from the rule.

`AREA {`*`cutLayer area`*` {`*`ratio`*` | `*`ratio1 ratio2`*`}}...`

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If only one ratio value is given, there are two slightly different usages.

- If *ratio* is defined as zero on one *cutLayer*, zero must be defined on all other *cutLayers*. This *ratio* construct is meaningless and is not used. Then, the area of the cut must be less than or equal to *area* to be counted as a stacked via. If the area of the cut is greater than *area*, the via should not be counted against the max stack rule as if it did not exist.
- If *ratio* is non-zero on one *cutLayer*, a non-zero value must be defined on all other *cutLayers*. This non-zero *ratio* usage can be used only when *maxStack* is `1` and all possible cut layers, except the first or the last layer, must be defined in *cutLayer* individually. This means that if the first or the last cut layer is missed, the cut on the missed layer cannot be a triggering cut but can be a neighbor cut. The triggering cut must have an area less than or equal to *area*. It is a violation if it is stacked with a below or above layer cut and the ratio of the area of the below or above cut to the area of the triggering cut is greater than *ratio*.

Type: Float, specified in microns squared, and double

If two values, *ratio1* and *ratio2*, are given, they must be positive values and *ratio2* must be larger than *ratio1*. *maxStack* can be any positive value greater than or equal to 1. If so, it is a violation when the ratio of the area of the below or above cut to the area of the triggering cut is greater than *ratio1* and less than or equal to *ratio2* for at least one pair of adjacent layer cuts.

Type: Float, specified in double

`CENTERCUTWITHIN` *cutLayerName belowCutWithin aboveCutWithin*

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Specifies that the maximum via stack rule applies only when all of the cuts below a cut on *cutLayerName* must be within *belowCutWithin* of that cut and all of the cuts above a cut on *cutLayerName* must be within *aboveCutWithin* of that cut. The maximum via stack rule must be defined such that having a cut on cutLayerName is the only way to violate the rule. For example, if a maximum via stack number of 2 is given between six metal layers (and a maximum of five stacked vias in between), CENTERCUTWITHIN cannot be defined on the second cut layer since the upper three vias could violate the rule without a second layer cut. CENTERCUTWITHIN can be defined on the third (middle) cut layer since three or more stacked vias would always have a cut on that layer. See Figure 1-19 on page 74.

Type: Float, specified in microns

CENTERTOCENTER
Specifies that the within distance between two cuts on adjacent cut layers is measured in the center-to-center style.

EXCEPTAREA *individualArea sumArea*

Specifies that the rule does not apply if the metal shape containing the stacked vias has an area greater than or equal to *individualArea* on any layer except the bottommost and topmost metal layers of the stacked vias or the union sum of those metal areas is greater than or equal to *sumArea*.

If any stacked via larger than the specified maximum number in the rule does not qualify for the exemptions in EXCEPTAREA, it is a violation even if there are additional stacked vias fulfilling the exemption in EXCEPTAREA.

Type: Float, specified in microns squared

EXCEPTCUTCLASSLIST {*cutLayer cutClassName*}...

Specifies that the rule is exempted for any vias belonging to the given cut class *cutClassName* on the given layer *cutLayer*. This means that those vias would interrupt a stack as if the vias did not exist for this max via stack rule.

EXCEPTENCLOSURE {*cutLayer* [ABOVE | BELOW] *enclosure*}...

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Specifies that the rule is exempted when all the vias on all the given layer *cutLayer* have enclosure greater than or equal to *enclosure* on all four sides, including corners, against both the above and the below metal layer. If ABOVE or BELOW is defined, the enclosure requirement is only against the above or below metal layer. This means that the rule is applied as long as one of the vias on the given layer *cutLayer* has enclosure less than *enclosure* on any side against either the above or the below metal layer.

Type: Float, specified in microns

EXCEPTMAXAREA {*metalLayer area*}...

Specifies that the rule is exempted if either the above or below metal area of a cut is less than *area* on any given one *metalLayer*.

Type: Float, specified in microns

EXCEPTPGNET

Specifies that the rule is exempted for power or ground net.

```
EXCEPTPGSHAPETYPE {RING | STRIPE | FOLLOWPIN | IOWIRE
     | COREWIRE | BLOCKWIRE | PADRING | BLOCKRING
     | FILLWIRE}...
```

Specifies that if all of the violated PG cuts belong to one or more of the specified list of shape types, it is not a violation. This means that even if one of the violated cuts does not belong to the specified list of shape types, it is still a violation.

IGNORESAMELAYER *cutWithin*

Specifies that any same-layer (same-net or different-net) cuts within *cutWithin* are considered multi-cuts, which interrupt the via stack.

Type: Float, specified in microns

IGNOREONESAMEMETAL *cutWithin*

Specifies that if the cuts within *cutWithin* have the same metal on at least one metal layer, they are considered multi-cuts, which interrupt the via stack. However, the bottommost stacked via can consider a same-metal cut only on the routing layer above, and the topmost stacked via can consider a same-metal cut only on the routing layer below.

Type: Float, specified in microns

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`LAYER` *`regionLayerName`*

Specifies that the rule applies only in the areas in *`regionLayerName`*, which must be a masterslice layer with the `TYPE REGION` property.

`MAXVIASTACK` *`maxStack`*

Specifies the maximum number of single-cut vias that are allowed on top of each other (that is, in one continuous stack).

Type: Integer

`MINMULTICUT` *`numCut`*

Specifies the minimum multi-cut number to be greater than or equal to *`numCut`* in order to be a multi-cut exemption to a stacked via violation with number of vias greater than or equal to *`maxStack`*. If `NOSINGLE` is specified, all of the vias must have a cut number greater than or equal to *`numCut`* to pass the rule.

By default, without `MINMULTICUT`, the multi-cut number is `2`. Therefore, *`numCut`* must be greater than `2`. However, *`numCut`* could be `0` and has a special meaning that the rule should be considered for each of the cuts individually. Multi-cut would not interrupt a stack.

Type: Integer

`[NOCUTCLASSLIST {`*`cutLayer cutClassName`*`}...]...`

Specifies that the given cut classes, which belong to the given *`cutClassName`* on the given cut layer *`cutLayer`*, are not allowed for stacking although the number of cut classes specified in `NOCUTCLASSLIST` is less than or equal to *`maxStack`*.

Type: String

`NOSINGLE`

Indicates that any single-cut via in a stack that is larger than *`maxStack`* is a violation, and multi-cut vias do not interrupt a stack. Therefore, any stack larger than *`maxStack`* must consist of all multi-cut vias.

`[NOVIALIST {`*`viaName`*`}...]...`

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Specifies that the given list of vias in `{`*`viaName`*`}...` is not allowed for stacking although the number of cut vias specified in `NOVIALIST` is less than or equal to *`maxStack`*. The via list should always start from a lowest cut layer and consecutively move up the cut layers, and they should be inside the metal layers in *`RANGE`* if defined. Wild card asterisk (*) is supported, which would map to all of the LEF predefined vias that match the wild card. If it is used alone to represent all of the vias on a certain cut layer, the via name on previous or following cut layers would have enough information to indicate what layer the asterisk is referring to.
Type: String

`RANGE` *`bottomLayer topLayer`*

Specifies a range of routing layers for which the maximum stacked via rule applies. If you do not specify a range, the *`maxStack`* value applies for all routing layers. The *`bottomLayer`* and *`topLayer`* values are routing layer names. The specified *`topLayer`* layer must be above the layer specified for *`bottomLayer`*.

`SAMENET` Specifies that the rule applies only to same-net vias.

`WITHIN {LOWCUTSIZE | {`*`cutLayerName within`*`}...}`

Specifies that an object is considered as a stacked via if the upper adjacent layer cut is within the minimum cut size of multiple `CUTCLASS` definitions or *`minWidth`* value in `WIDTH` on the lower cut layer in `LOWCUTSIZE` or within *`within`* distance of a cut on the lower layer in *`cutLayerName`*. Touching the within search window is considered as stacked.
Type: Float, specified in microns

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Maximum Via Stack Rule Examples

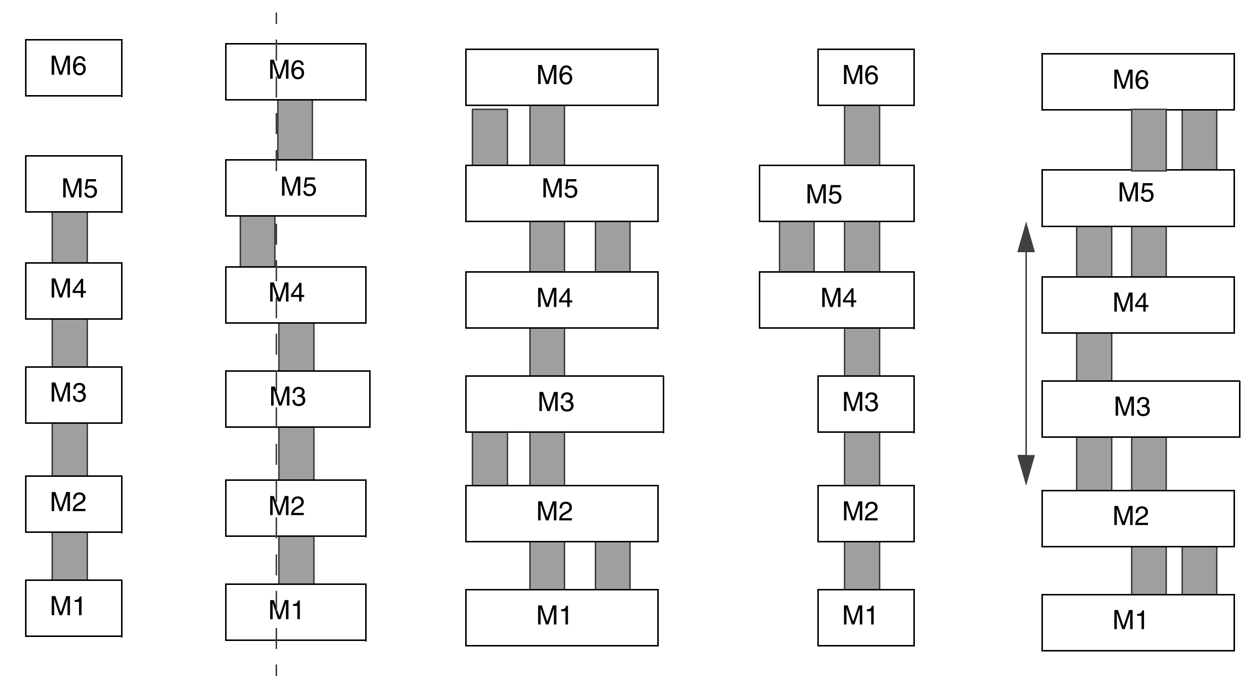
- If the following rule exists:

```
PROPERTYDEFINITIONS
  LIBRARY LEF58_MAXVIASTACK STRING
    "MAXVIASTACK 3 RANGE m1 m6  ;" ;
END PROPERTYDEFINITIONS
```

Only three single-cut vias can be stacked between layers *m1* and *m6*. See Figure 1-13 for different layout examples using this rule.

Figure 1-13 Max Via Stack Rule Examples

```
LIBRARY LEF58_MAXVIASTACK STRING "MAXVIASTACK 3 RANGE m1 m6 ;" ;
LIBRARY LEF58_MAXVIASTACK STRING "MAXVIASTACK 3 NOSINGLE RANGE m1 m6
;" ;
```



a) Violation. Stack = 4

b) Okay. Max-stack = 3

c) 1-cut max stack = 1, so default is okay. For `NOSINGLE`, max-stack = 5, so violation.

d) 1-cut max-stack =3, so default is okay. For `NOSINGLE`, max-stack = 5, so violation.

e) 1-cut max-stack =1, so default is okay. For `NOSINGLE`, max-stack = 3 (see arrow), so okay.

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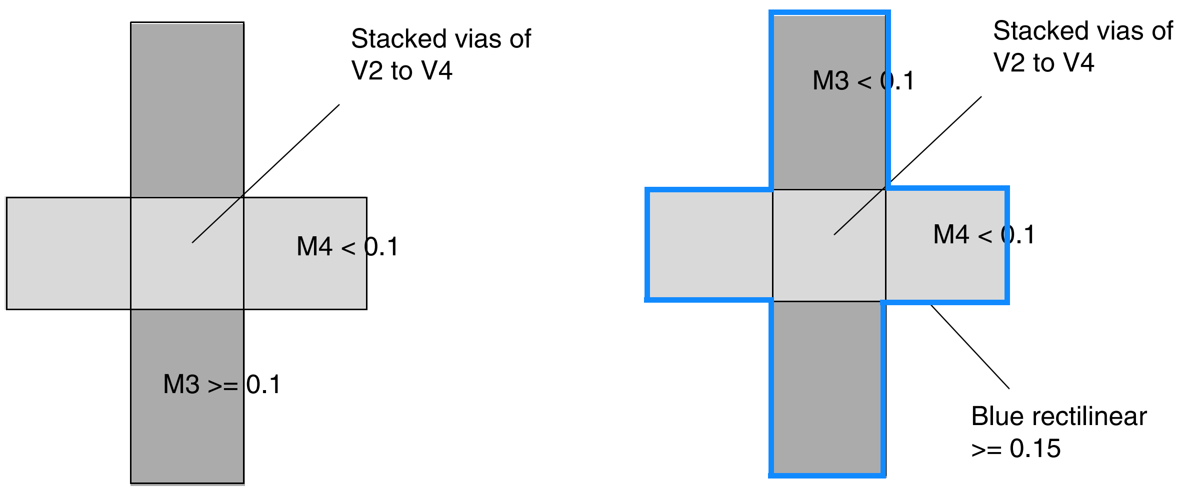
- The following rule applies if metal shapes containing the stacked vias have an area less than the individual area of 0.1 on any layer:

```
PROPERTYDEFINITIONS
  LIBRARY LEF58_MAXVIASTACK STRING
    "MAXVIASTACK 2 EXCEPTAREA 0.1 0.15
        RANGE M2 M6 ; " ;
END PROPERTYDEFINITIONS
```

Figure 1-14 Illustration of Max Via Stack Rule with EXCEPTAREA



a) OK, M3 has an area >= 0.1. Violation if EXCEPTAREA is omitted.

b) OK, the area of the blue rectilinear object >= 0.15. Violation if EXCEPTAREA is omitted.

Metal area of M2 (bottom most layer) or M5 (top most layer) of the stacked vias is irrelevant in EXCEPTAREA.

- The following property rule illustrates `WITHIN LOWCUTSIZE` with `V1` layer:

```
PROERPTY LEF58_CUTCLASS
    "CUTCLASS ...  WIDTH 0.06 ... ;
     CUTCLASS ...  WIDTH 0.12 ... ; " ;
```


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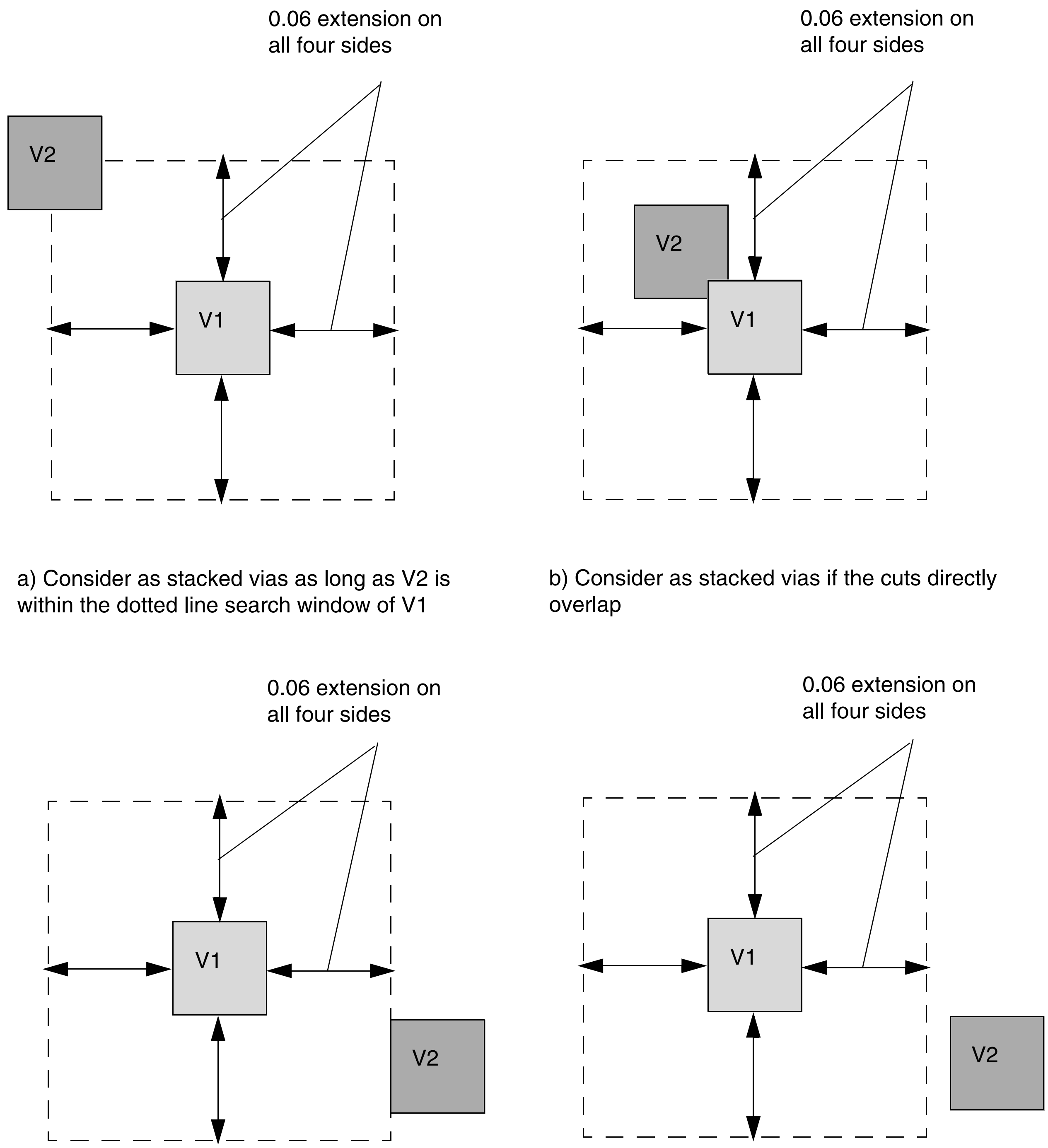
Figure 1-15 Illustration of Max Via Stack Rule with WITHIN LOWCUTSIZE



a) Consider as stacked vias as long as V2 is within the dotted line search window of V1

b) Consider as stacked vias if the cuts directly overlap

c) Consider as stacked vias when V2 is touching the dotted line search window of V1

d) Not considered as stacked vias since V2 is outside the dotted line search window of V1.

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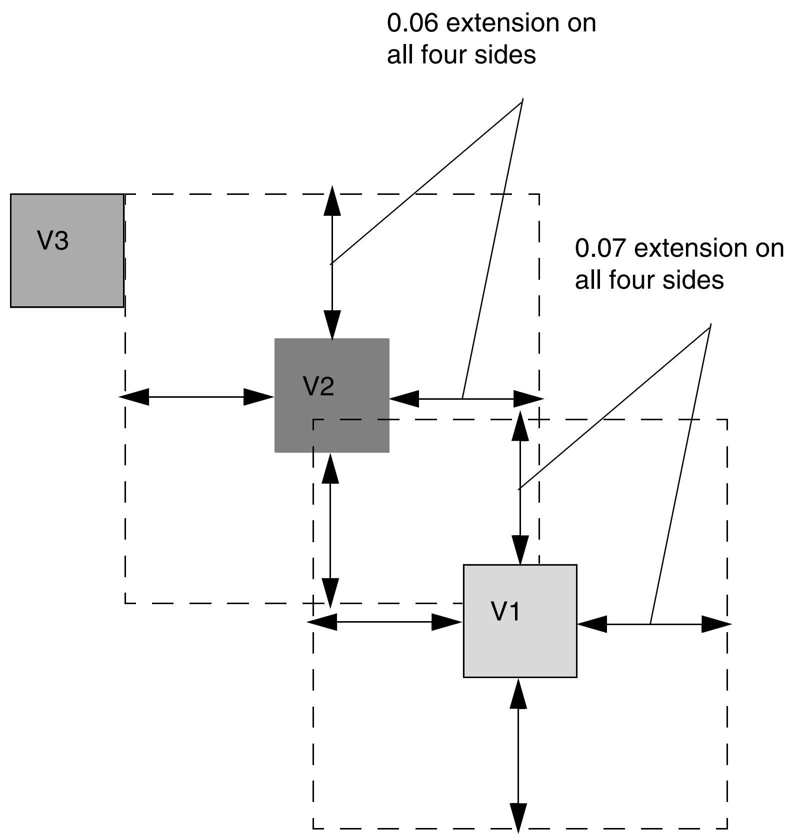
Figure 1-16 Illustration of Max Via Stack Rule with WITHIN



a) Consider as stacked vias from V1 to V3. V1 & V2 have their own specified within values and V3 touches V2 within search window

Illustration of WITHIN V1 0.07 V2 0.06

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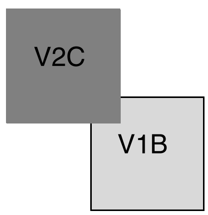
- The following property rule illustrates the max via stack rule with `NOVIALIST`:

```
PROPERTYDEFINITIONS
      LIBRARY LEF58_MAXVIASTACK STRING "
                MAXVIASTACK 3 NOVIALIST V1B V2C ; " ;
END PROPERTYDEFINITIONS
```

Figure 1-17 Illustration of Max Via Stack Rule with NOVIALIST

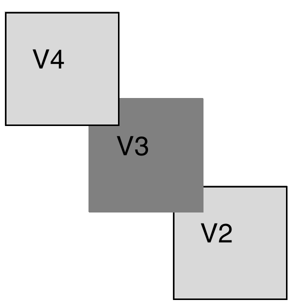


a) Violation, although three stacked vias are allowed, specific V1B and V2C combination is not allowed to be stacked

- The following property rule illustrates multiple max via stack rules:

```
PROPERTYDEFINITIONS
     LIBRARY LEF58_MAXVIASTACK STRING "
          MAXVIASTACK 2 RANGE M1 M5 ;
          MAXVIASTACK 3 RANGE M3 M7 ; " ;
END PROPERTYDEFINITIONS
```

Figure 1-18 Illustration of Multiple Max Via Stack Rules



a) Violation, the second statement passes, but failing the first statement alone is still a violation

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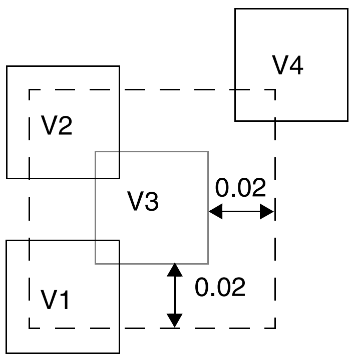
- The following example illustrates the max via stack rule with `CENTERCUTWITHIN`:

```
PROPERTYDEFINITIONS
LIBRARY LEF58_MAXVIASTACK STRING "
      MAXVIASTACK 3
            CENTERCUTWITHIN V3 0.000 0.020
            RANGE M1 M5 ; " ;
END PROPERTYDEFINITIONS
```

Figure 1-19 Illustration of Max Via Stack Rule with CENTERCUTWITHIN



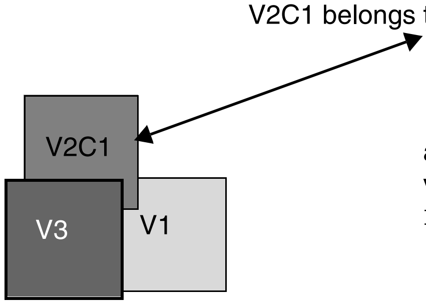
a) Violation. V1 and V2 overlap with V3 (within `0.000` of V3; V1 and V2 need not necessarily overlap) and V4 is within `0.020` of V3.

- The following property rule illustrates a max via stack rule with `EXCEPTCUTCLASSLIST`:

```
PROPERTYDEFINITIONS
     LIBRARY LEF58_MAXVIASTACK STRING "
          MAXVIASTACK 2
          EXCEPTCUTCLASSLIST V2 V2C; " ;
END PROPERTYDEFINITIONS
```

Figure 1-20 Illustration of Max Via Stack Rule with EXCEPTCUTCLASSLIST

V2C1 belongs to the cut class V2C on V2.



a) OK, V2C1 does not count against the max via stack rule. Violation if `EXCEPTCUTCLASSLIST` is omitted.

Metal Width Track Rule

You can create a metal width track rule to define the preferred routing direction for tracks on a specified layer.

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You can define a metal width track rule by using the following PROPERTYDEFINITIONS statement:

```
PROPERTYDEFINITIONS
LIBRARY LEF58_METALWIDTHTRACK STRING
"METALWIDTHTRACK routingLayer COREOFFSET offset
    {WIDTH width PITCH pitch [REPEAT repeat]}...
    ;" ;
END PROPERTYDEFINITIONS
```

Where:

`METALWIDTHTRACK` *`routingLayer`* `COREOFFSET` *`offset`*
`{WIDTH` *`width`* `PITCH` *`pitch`* `[REPEAT` *`repeat`*`]}...`

Specifies the preferred routing direction tracks on the layer *routingLayer*. *offset* specifies the starting coordinate of the track with respect to the origin of the core. The *width* and *pitch* set of values define the width and pitch of the subsequent tracks. *repeat* specifies how many times the given width and pitch should be repeated. Then, the entire sets of width and pitch would be repeated. Multiple METALWIDTHTRACK statements can be defined for different routing layers. For example:

```
METALWIDTHTRACK M1 ...
METALWIDTHTRACK M2 ...
```

However, one routing layer can have only one such statement. So, it would be illegal to have the following:

```
METALWIDTHTRACK M1 ...
METALWIDTHTRACK M1 ...
```

The METALWIDTHTRACK property is typically used to define a non-uniform track on a certain layer, which depend on the cell objects on the corresponding layer. In advanced nodes, all of the cell objects are typically required on the track on lower layers. Horizontal, non-uniform tracks may be created such that instances could be placed on any row such that the cell objects are still ensured on the track. Therefore, this library property is likely to be defined in a cell LEF or side LEF file, instead of in a tech LEF file.
Type: Float, specified in microns

Metal Width Track Rule Example

- The following example is an illustration of the metal track width rule:

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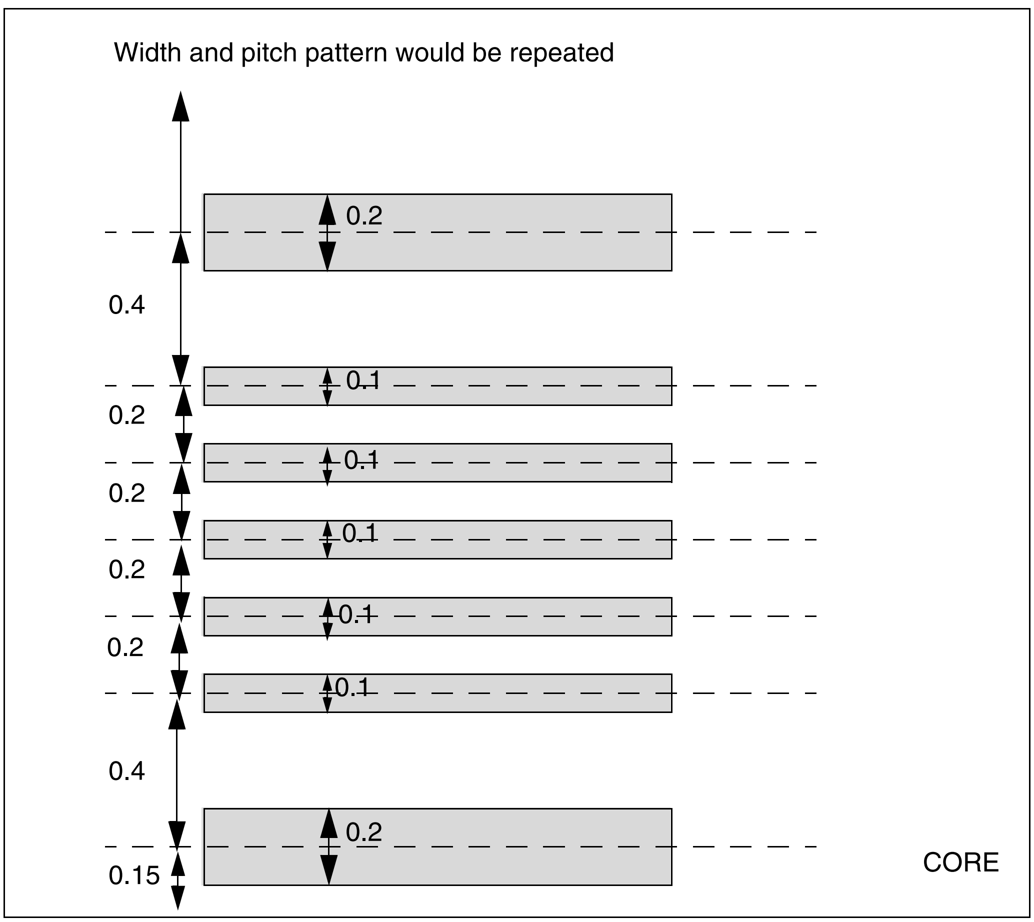
```
LAYER M1
    TYPE ROUTING ;
    DIRECTION HORIZONTAL ;
    ...
END M1
...
LIBRARY LEF58_METALWIDTHTRACK STRING
          "METALWIDTHTRACK M1 COREOFFSET 0.15
                WIDTH 0.2 PITCH 0.4
                WIDTH 0.1 PITCH 0.2 REPEAT 4
                WIDTH 0.1 PITCH 0.4 ; " ;
END PROPERTYDEFINITIONS
```

Figure 1-21 Illustration of Metal Track Width Rule



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Metal Width Via Map Rule

You can create a metal width via map rule to specify the vias to be used based on the given width values of the below and above metal layers.

You can define a metal width via map rule by using the following `PROPERTYDEFINITIONS` statement:

```
PROPERTYDEFINITIONS
     LIBRARY LEF58_METALWIDTHVIAMAP STRING
        "METALWIDTHVIAMAP
            [USEVIACUTCLASS]
            {VIA viaLayer
               {belowLayerWidth aboveLayerWidth
               |belowLayerLowWidth belowLayerHighWidth
                  aboveLayerLowWidth aboveLayerHighWidth}
               [TRACKS {ABOVE|BELOW} {trackNum}...]
               viaName [PGVIA]}...
            ;..." ;
END PROPERTYDEFINITIONS
```

Where:

`METALWIDTHVIAMAP`
`{VIA` *`viaLayer belowLayerWidth aboveLayerWidth viaName`*`}...`

Specifies the vias to be used based on the given width values of the below and above metal layers. *`viaName`* is used on cut layer *`viaLayer`* for below and above metal widths of *`belowLayerWidth`* and *`aboveLayerWidth`*. When multiple vias are defined for the same below and above metal width combination, the first defined via is preferred. When a DRC violation occurs, other defined vias are attempted.

Type: Float, specified in microns

`PGVIA`

Specifies that the given via is for PG net usage only. For a given below and above metal width combination, you can do one of the following:

- Define only one via to be used for both signal and PG nets. This via should be defined without `PGVIA`.
- Define multiple vias of which one is without `PGVIA`. In this case, the via without `PGVIA` would be used for signal nets, and the rest of the vias with `PGVIA` would be used for PG nets.

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`TRACKS {ABOVE|BELOW} {`*`trackNum`*`}...`

Specifies that the given via is applied on the given above or below metal tracks with track numbers in *trackNum*. `METALWIDTHVIAMAP` must be defined on the above or below metal layer, and *trackNum* refers to the track number in `METALWIDTHVIAMAP`. If `TRACKS` is defined on a certain width or width ranges on a certain cut layer, multiple statements with `TRACKS` on those width or width ranges on that cut layer must be defined. The sum of unique *trackNum* of all those statements must be equal to the total number of tracks on `METALWIDTHVIAMAP`.

`USEVIACUTCLASS`

Indicates that *viaName* is actually a cut class name on the given layer, *viaLayer*. Any cuts belonging to the given cut class could be used on the given width combination.

For example,

```
LIBRARY LEF58_ METALWIDTHVIAMAP STRING "
     METALWIDTHVIAMAP USEVIACUTCLASS
     VIA V1 0.05 0.08 V1_CUTCLASSA ; " ;

END PROPERTYDEFINITIONS
```

means that any `V1` cuts belonging to cut class `V1_CUTCLASSA` could be used for nets on below metal width of `0.05` and above metal width of `0.08` wires.

`VIA` *`viaLayer belowLayerLowWidth belowLayerHighWidth`*
`aboveLayerLowWidth aboveLayerHighWidth viaName...`

Specifies the vias to be used based on the given width range values of the below and above metal layers. That is the below and above metal width should be greater than or equal to *belowLayerLowWidth* and *aboveLayerLowWidth*, respectively, and less than or equal to *belowLayerHighWidth* and *aboveLayerHighWidth*, respectively, to use *viaName*.

Type: Float, specified in microns

Metal Width Via Map Rule Examples

- The following example indicates that `V1_SMALL` should be used on cut layer `V1` for below metal width of 0.05 and above metal width of 0.08 wires while `V1_LARGE` should be used for below metal width of 0.07 and above metal width of 0.08 wires:

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```
PROPERTYDEFINITIONS
LIBRARY LEF58_METALWIDTHVIAMAP STRING "
       METALWIDTHVIAMAP VIA V1 0.05 0.08 V1_SMALL
         VIA V1 0.07 0.08 V1_LARGE; " ;
END PROPERTYDEFINITIONS
```

- The following diagram illustrates the example below:

```
LAYER M1
    TYPE ROUTING ;
    DIRECTION HORIZONTAL ;
    ...
END M1
...
LIBRARY LEF58_METALWIDTHTRACK STRING "
    METALWIDTHTRACK M1 COREOFFSET 0.15
        WIDTH 0.2 PITCH 0.4
        WIDTH 0.1 PITCH 0.2 REPEAT 4
        WIDTH 0.1 PITCH 0.4 ; " ;
END PROPERTYDEFINITIONS

LIBRARY LEF58_METALWIDTHVIAMAP STRING "
    METALWIDTHVIAMAP
        VIA V1 0.0 3.0 0.0 3.0 TRACKS BELOW 1 3 5 VA
        VIA V1 0.0 3.0 0.0 3.0 TRACKS BELOW 2 4 6 VB ; " ;
END PROPERTYDEFINITIONS
```


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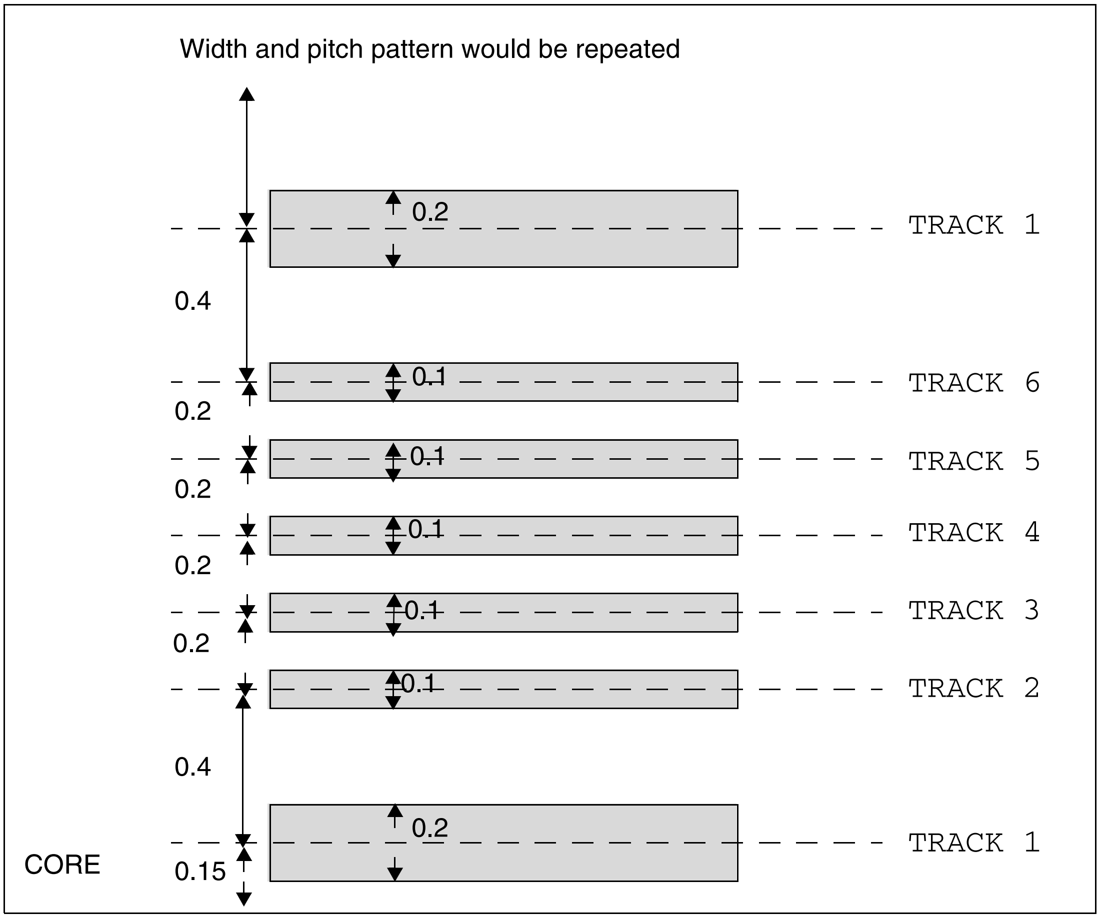
Figure 1-22 Illustration of Metal Width Via Map Rule



VA should be used on track 1, 3, and 5 on M1. VB should be used on track 2, 4, and 6 on M1.

No Overlap Rule

You can create a no overlap rule to specify shapes that cannot overlap on given masterslice layers.

You can define a no overlap rule by using the following `PROPERTYDEFINITIONS` statement:

```
PROPERTYDEFINITIONS
LIBRARY LEF58_NOOVERLAP STRING
"NOOVERLAP layerName1 layerName2 [WIDTH width]
    ;" ;
END PROPERTYDEFINITIONS
```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Where:

`NOOVERLAP` *`layerName1 layerName2`*

Specifies that the shapes on *`layerName1`* and *`layerName2`* must not overlap. Both the specified layers must be masterslice layers, with shapes defined only on cells.

Type: String and Float, specified in microns

`WIDTH` *`width`*

Specifies that only the shapes on *`layerName2`* with a width less than or equal to *`width`* are not allowed to overlap.

Type: Float, specified in microns

OpenAccess Layer Map Rule

You can create an OpenAccess layer map rule to define equivalence between LEF and OpenAccess layers and also between the geometries and shapes placed on the layers.

You can define an OpenAccess layer map rule by using the following `PROPERTYDEFINITIONS` statement:

```
PROPERTYDEFINITIONS
LIBRARY LEF58_OALAYERMAP STRING
   "OALAYERMAP oaLayer
{ [PURPOSE oaPurpose] BLOCKAGE layer
      | LAYER layer [MASK maskNum] }
      ;..." ;
END PROPERTYDEFINITIONS
```

Here, `LEF58_OALAYERMAP` can contain multiple mapping statements, separated by '`;`'. The property syntax allows two variants of the statement:

`OALAYERMAP` *`oaLayer`* `LAYER` *`layer`* `[MASK` *`maskNum`*`]`

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Specifies the equivalent OpenAccess layer name *oaLayer* for the LEF layer *layer* and also specifies the mapping between the shapes placed on the layers. If the MASK keyword is specified (allowed only for LEF multi-patterning layers with a MASK statement), the mapping will be mask-based; that is, the color mask *maskNum* of the LEF layer *layer* is mapped to the OpenAccess layer *oaLayer* and the LEF layer shapes with the color mask are converted into the OpenAccess layer shapes.

Note: Typically, mask-based mapping requires multiple OALAYERMAP statements per LEF layer, with one statement for each possible color mask.

Type: String

OALAYERMAP *oaLayer* [PURPOSE *oaPurpose*] BLOCKAGE *layer*

Maps the LEF/DEF blockage-type geometries on the LEF layer *layer* to regular shapes on the OpenAccess layer *oaLayer*. These blockage-type geometries are represented as LEF OBS statement geometries in the LEF file and DEF BLOCKAGES statement geometries in the DEF file.

If the PURPOSE keyword is specified in the statement, the OpenAccess shapes are set to the *oaPurpose* purpose in the conversion. Otherwise, they are set to the default drawing purpose.

Note: This variant of the OALAYERMAP statement does not specify the mapping between LEF and OpenAccess layers.

Type: String

OpenAccess Layer Map Rule Examples

- The following example indicates that the LEF layer M1 is mapped to OpenAccess layer Metal1:

```
PROPERTYDEFINITIONS
LIBRARY LEF58_OALAYERMAP STRING "
     OALAYERMAP Metal1 LAYER M1 ; " ;
END PROPERTYDEFINITIONS
```

- The following example indicates that the LEF MASK 1 shapes on M1 would go to OpenAccess layer Metal1A while MASK 2 shapes on M1 would go to OpenAccess layer Metal1B:

```
PROPERTYDEFINITIONS
```


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```
    LIBRARY LEF58_OALAYERMAP STRING "
         OALAYERMAP Metal1A LAYER M1 MASK 1 ;
         OALAYERMAP Metal1B LAYER M1 MASK 2 ; " ;
    END PROPERTYDEFINITIONS
```

Oxide Conversion Rule

You can define an oxide conversion rule by using the following `PROPERTYDEFINITIONS` statement:

```
PROPERTYDEFINITIONS
LIBRARY LEF_CDN_OXIDECONVERSION STRING
   "OXIDECONVERSION
        {OXIDE1 | OXIDE2 | OXIDE3 | OXIDE4 | OXIDE5 | ... | OXIDE32}
        TO
        {OXIDE1 | OXIDE2 | OXIDE3 | OXIDE4 | OXIDE5 | ... | OXIDE32}
        NETCONNECTEDLAYER layerName [WIREONLY] [CHECKBOTH]
      ; " ;
END PROPERTYDEFINITIONS
```

Where:

`CHECKBOTH`
: Specifies that the original first oxide model and the derived second oxide model should both be checked individually.

`OXIDECONVERSION`
`{OXIDE1 | OXIDE2 | OXIDE3 | OXIDE4 | OXIDE5 | ... | OXIDE32}`
`TO`
`{OXIDE1 | OXIDE2 | OXIDE3 | OXIDE4 | OXIDE5 | ... | OXIDE32}`
`NETCONNECTEDLAYER` *`layerName`*
: Specifies that the first specified oxide model can be converted to the second specified oxide model if a net, including any pins, wires and vias, is connected on *`layerName`*. This conversion should be done for all wires even if those wires do not connect to *`layerName`* yet. The first and second oxide models cannot be the same. The second oxide model is derived from the first oxide model. On `MACRO PIN`s, `ANTENNAGATEAREA` or `ANTENNADIFFAREA` cannot be defined on the second oxide model directly. Any missing rules on any layers in the second oxide model would assume the definition from the first oxide model.

`WIREONLY`
: Specifies that the conversion happens on wires connected to *`layerName`* only.

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PG Via Track Rule

You can use a PG via track rule to define vertical tracks for aligning all of the power and ground cell vias on a specified layer of a cell macro with the `ALIGNPGVIATOTRACK` property.

You can define a PG via track rule by using the the following `PROPERTYDEFINITIONS` statement:

```
PROPERTYDEFINITIONS
LIBRARY LEF58_PGVIATRACK STRING
   "PGVIATRACK cutLayer COREOFFSET offset PITCH pitch; " ;
END PROPERTYDEFINITIONS
```

Where:

`PGVIATRACK` *`cutLayer`* `COREOFFSET` *`offset`* `PITCH` *`pitch`*

Defines vertical tracks for aligning all the power and ground cell vias on the layer *`cutLayer`* of a cell macro with the `ALIGNPGVIATOTRACK` property. *`offset`* specifies the starting coordinate of the track with respect to the origin of the core. *`pitch`* specifies the pitch of the tracks. All the power and ground cell vias should be in multiples of *`pitch`* such that aligning any one of them to the track is sufficient.
Type: Float, specified in microns

PG Via Track Rule Example

The following example is an illustration of the PG via track rule:

```
    PROPERTYDEFINITIONS
    LIBRARY LEF58_PGVIATRACK STRING "
              PGVIATRACK VIA1 COREOFFSET 0.15
                   PITCH 0.2 ; " ;
    END PROPERTYDEFINITIONS
```


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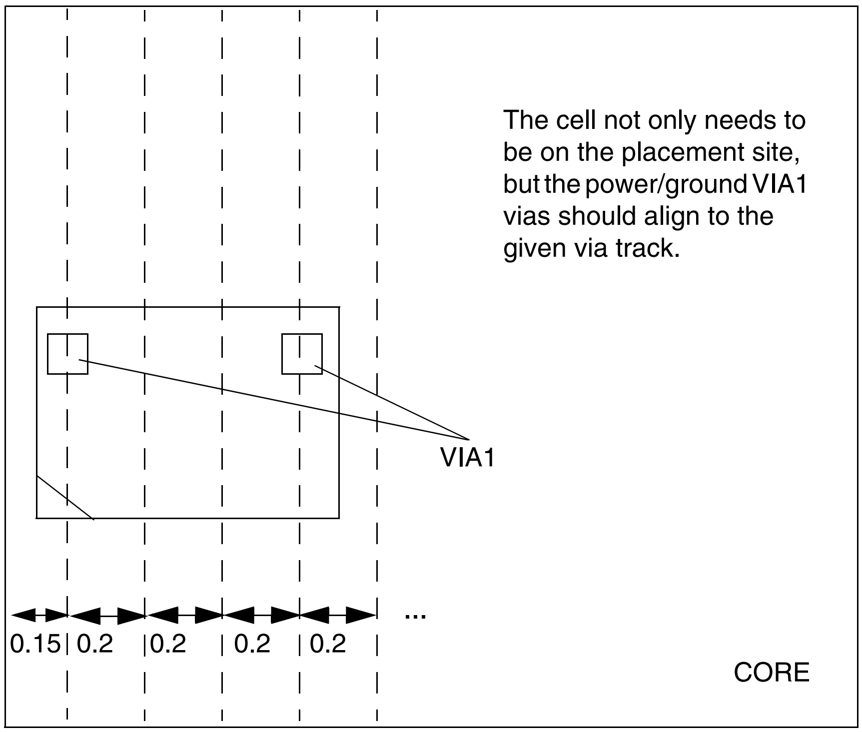
Figure 1-23 Illustration of the PG Via Track Rule



Power Connectivity Layer Rule

You can use a power connectivity layer rule to specify that power nets must be completed between the given layers.

You can define a power connectivity layer rule by using the following `PROPERTYDEFINITIONS` statement:

```
PROPERTYDEFINITIONS
LIBRARY LEF58_POWERCONNECTIVITYLAYERS STRING
"POWERCONNECTIVITYLAYERS bottomLayer topLayer
    ; " ;
END PROPERTYDEFINITIONS
```

Where:

`POWERCONNECTIVITYLAYERS` *`bottomLayer topLayer`*

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Specifies that power nets must be completed between the given layers. Both *bottomLayer* and *topLayer* must be routing layers. This is a translated methodology to prevent specific DRC violations. A design may have multiple power nets; however, not all need to be verified using this approach. A separate command is used to define which power net should be checked using this methodology.

Site Core Offset Rule

You can create a site core offset rule to specify the offset for the first row of a site.

You can define a site core offset rule by using the following PROPERTYDEFINITIONS statement:

```
PROPERTYDEFINITIONS
LIBRARY LEF58_SITECOREOFFSET STRING
"SITECOREOFFSET siteName offset
     ;" ;
END PROPERTYDEFINITIONS
```

Where:

SITECOREOFFSET *siteName offset*

Specifies that the first row of the given site, *siteName*, should start from *offset* with respect to the origin of the core.

Type: Float, specified in microns

Stack Via Layer Rule

You can use a stack via layer rule to define stack via rules on individual layers. These rules are combined to form a stack via rule.

You can define a stack via layer rule by using the following PROPERTYDEFINITIONS statement:

```
PROPERTYDEFINITIONS
LIBRARY LEF58_STACKVIALAYERRULE STRING
   "STACKVIALAYERRULE ruleName
      LAYER cutLayerName [CUTCLASS className]
         ROWCOL numCutRows numCutCols
            [XPITCH xPitch [MAXXPITCH maxXPitch]]
            [YPITCH yPitch [MAXYPITCH maxYPitch]]
            [BELOWMINLENGTH belowMinLength]
            [ABOVEMINLENGTH aboveMinLength]
```


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```
            [MAXCELLEXTENSION extensionPitch]
            [OFFGRIDCUT]
            [MAXXLINEEND maxLeftExtensiion maxRightExtension]
            [MAXYLINEEND maxBottomExtensiion maxTopExtension]
            [PREFERMASK maskNum]
            ; " ;
END PROPERTYDEFINITIONS
```

Where:

`ABOVEMINLENGTH` *`aboveMinLength`*

Specifies the minimum length of the default width wires on the above metal layer. This construct must be specified with `XPITCH` and/or `YPITCH`.
Type: Float, specified in microns

`BELOWMINLENGTH` *`belowMinLength`*

Specifies the minimum length of the default width wires on the below metal layer. This construct must be specified with `XPITCH` and/or `YPITCH`.
Type: Float, specified in microns

`MAXCELLEXTENSION` *`extensionPitch`*

Specifies the maximum number of pitch *`extensionPitch`* extended from the cell boundary of the pin along both directions from which the stacked vias must be enclosed. By default, the stacked vias must be inside the cell boundary if this construct is not specified.
Type: Integer

`MAXXLINEEND` *`maxLeftExtensiion maxRightExtension`*

Forms a boundary box of the pin shapes to which the stacked vias connect and expands to the left and right horizontally by *`maxLeftExtension`* and *`maxRightExtension`*, respectively. All of the via cuts on this layer must be enclosed by this expanded boundary box. If negative values are given, it would shrink the boundary box in the given direction. This construct should be mutually exclusive to `MAXCELLEXTENSION`.

Type: Float, specified in microns

`MAXXPITCH` *`maxXPitch`*

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Specifies the maximum number of pitch by which all the vertical minimum-default-width metal wires of the stacked vias could be apart. The pitch must be smaller than or equal to the given value.
Type: Integer

`MAXYLINEEND` *`maxBottomExtensiion maxTopExtension`*

Forms a boundary box of the pin shapes to which the stacked vias connect and expands to the bottom and top vertically by *`maxBottomExtension`* and *`maxTopExtension`*, respectively. All of the via cuts on this layer must be enclosed by this expanded boundary box. If negative values are given, it would shrink the boundary box in the given direction. This construct should be mutually exclusive to `MAXCELLEXTENSION`.

Type: Float, specified in microns

`MAXYPITCH` *`maxYPitch`*

Specifies the maximum number of pitch by which all the horizontal minimum-default-width metal wires of the stacked vias could be apart. The pitch must be smaller than or equal to the given value.
Type: Integer

`OFFGRIDCUT` Specifies that the cuts could be placed off grid to block fewer routing and the below metal shape would become an L shape. It is typically applied on a transition cut layer in which the cut size is larger than the default wire width on the below metal.

`PREFERMASK` *`maskNum`* Specifies a preferred mask, *`maskNum`*, for all the cuts on that layer. If there is no DRC for using that *`maskNum`*, it would be applied on all the cuts. The *`maskNum`* must be a positive integer, and most applications only support the values of `1`, `2`, or `3`.

Type: Integer

```
STACKVIALAYERRULE ruleName
   LAYER cutLayerName [CUTCLASS className]
      ROWCOL numCutRows numCutCols
```


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Specifies a stacked via rule on cut layer *`cutLayerName`* with *`className`* cuts. If `CUTCLASS` is not specified, the cut class with the smallest cut size would be used or the table lookup cut class would be applied based on the wire width if `METALWIDTHVIAMAP` is defined. These stacked via rules would be combined in `STACKVIARULE` to build stacked vias on top of some special instance pins. `ROWCOL` specifies the number of cut rows and columns of a multiple-cuts via of cut class *`className`* to use on the pin.
Type: String and Integer

Note that this rule is only for via creation. It is difficult to re-determine the cut dimension of a generated via, and the checker does not check for the correctness of the generated vias.

`XPITCH` *`xPitch`*

Specifies that all the individual cuts must be on the grid on both the top and bottom metal layers. The cuts are *`xPitch`* horizontally. There would be multiple vertical minimum-default-width metal wires, and each one of them would connect separately to a cut. In the case of a transition cut layer, the cut size would be larger than the default wire width on the below metal layer. It is typical to apply the `OFFGRIDCUT` construct such that the below metal shape becomes an L shape to block fewer tracks. If the common metal between two adjacent cut layers is vertical, *`xPitch`* on these two cut layers must be identical if they are defined. If neither `XPITCH` nor `YPITCH` is specified, the tool would automatically determine the proper pitch values. If zero is given in both of the pitches, a typical via with one large metal shape each on both the top and bottom layers would be created.
Type: Integer

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`YPITCH` *`yPitch`*	Specifies that all the individual cuts must be on the grid on both the top and bottom metal layers. The cuts are *`yPitch`* vertically. There would be multiple horizontal minimum-default-width metal wires, and each one of them would connect separately to a cut. In the case of a transition cut layer, the cut size would be larger than the default wire width on the below metal layer. It is typical to apply the `OFFGRIDCUT` construct such that the below metal shape becomes an L shape to block fewer tracks. If the common metal between two adjacent cut layers is horizontal, *`yPitch`* on these two cut layers must be identical if they are defined. If neither `XPITCH` nor `YPITCH` is specified, the tool would automatically determine the proper pitch values. If zero is given in both of the pitches, a typical via with one large metal shape each on both the top and bottom layers would be created. *Type:* Integer

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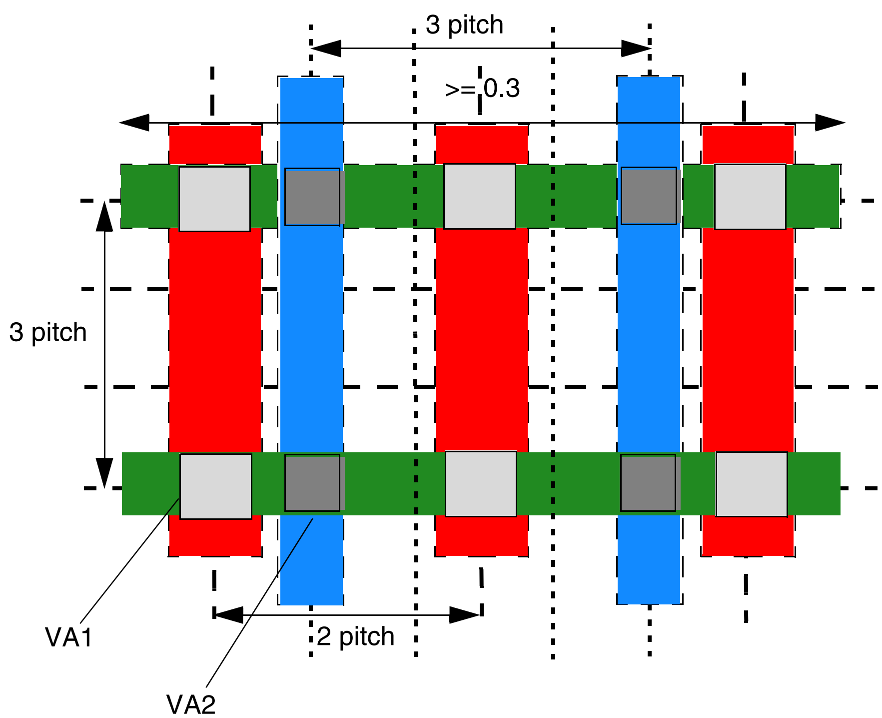
Stack Via Layer Rule Examples

- The following example is an illustration of the stack via layer rule:

```
PROPERTY LEF58_STACKVIALAYERRULE "
   STACKVIALAYERRULE SVLR1 LAYER VIA1 CUTCLASS VA1
      ROWCOL 2 3 XPITCH 2 YPITCH 3  ;
   STACKVIALAYERRULE SVLR2 LAYER VIA2 CUTCLASS VA2
      ROWCOL 2 2 XPITCH 3 YPITCH 3
      BELOWMINLENGTH 0.3 ; " ;
PROPERTY LEF58_STACKVIARULE "
   STACKVIARULE SVR1 SVLR1 SVLR2 ; " ;
```

Figure 1-24 Illustration of Stack Via Layer Rules



VIA1 would be one via with six cuts, three vertical red metal shapes and two horizontal green metal shapes.
VIA2 would be one via with four cuts, two vertical blue metal shapes and two horizontal green metal shapes.
Alternatively, multiple single-cut vias and multiple wires or patches could be used to represent them.

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Figure 1-25 Illustration of MAXCELLEXTENSION *extensionPitch*

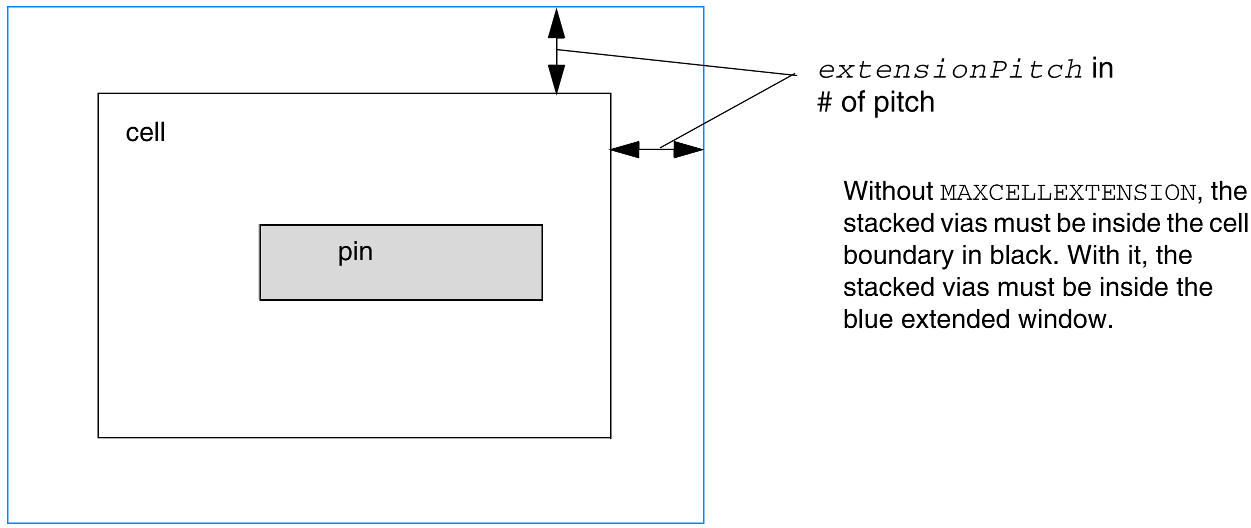

Figure 1-26 Illustration of OFFGRIDCUT

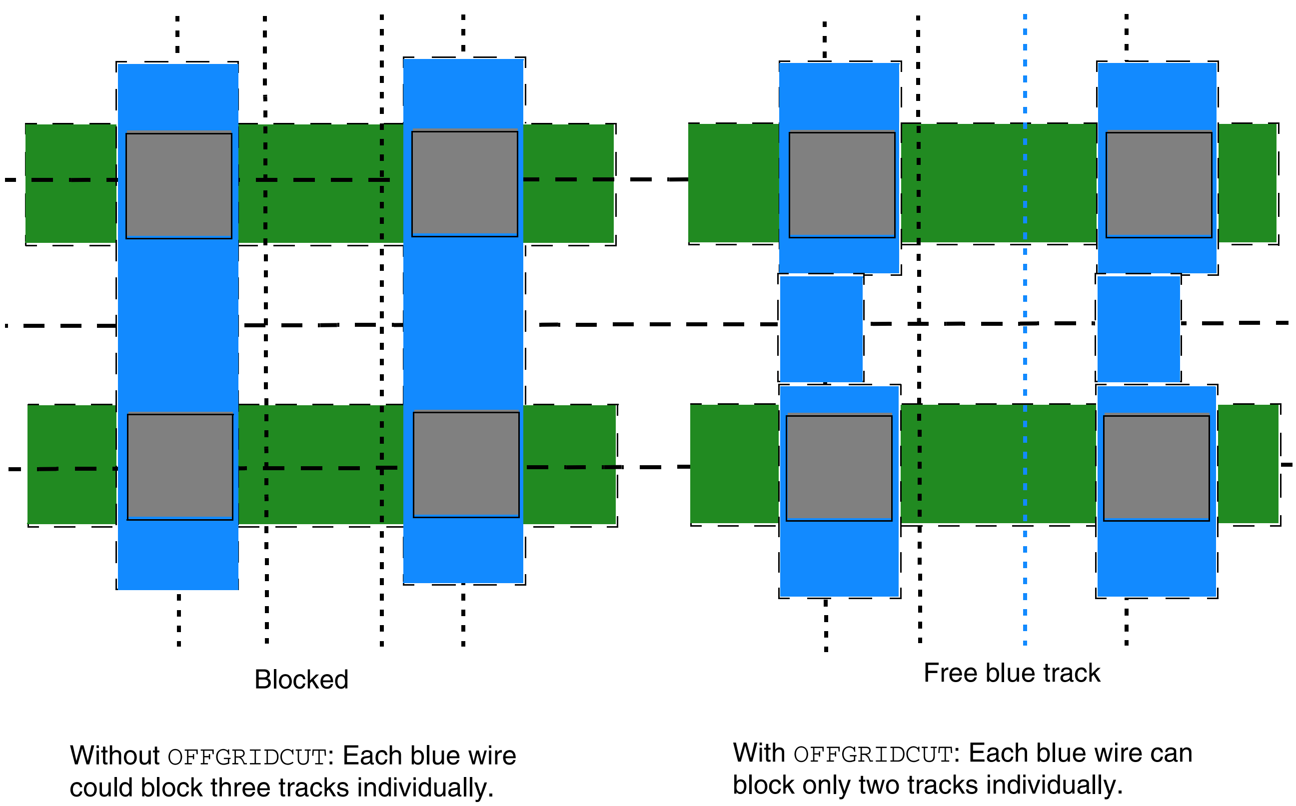

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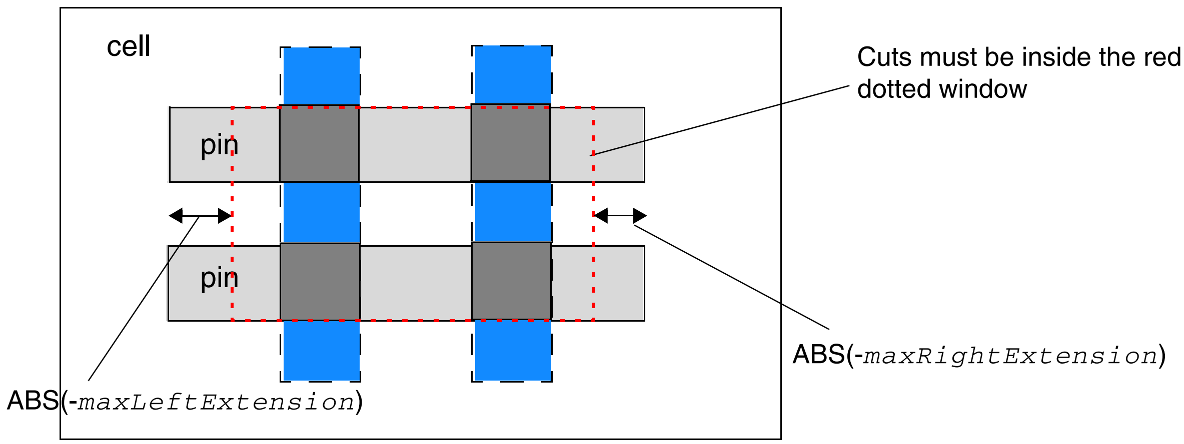
Figure 1-27 Illustration of MAXXLINEEND with Negative Values for *`maxLeftExtension`* **and** *`maxRightExtension`*



Stack Via Rule

You can create a stack via rule to specify a list of stack via rules for individual layers.

You can define a stack via rule by using the following `PROPERTYDEFINITIONS` statement:

```
PROPERTYDEFINITIONS
LIBRARY LEF58_STACKVIARULE STRING
"STACKVIARULE ruleName {stackLayerRuleName}...[EMFACTOR emFactor]
     ; " ;
END PROPERTYDEFINITIONS
```

Where:

`EMFACTOR` *`emFactor`*
: Specifies an EM-effective factor to indicate how well this stack via rule could prevent EM violations. The higher the specified number, the more effective is the rule in preventing EM violations.

  Type: Float

`STACKVIARULE` *`ruleName`* {*`stackLayerRuleName`*}...

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Specifies a stacked via rule, which consists of a list of stack via rules on individual layers in the given *stackLayerRuleName*. This list of stack via rules are the rule names specified with STACKVIALAYERRULE. *stackLayerRuleName* should correspond to individual layer rules on consecutive layers. It is illegal to define multiple stackLayerRuleName on the same layer. STACKVIALAYERRULE statements should be defined prior to STACKVIARULE in the tech LEF file.
Type: String

Tap Distance Rule

You can create a tap distance rule to specify distance between well tap cells.

You can define a tap distance rule by using the following PROPERTYDEFINITIONS statement:

```
PROPERTYDEFINITIONS
LIBRARY LEF58_TAPDISTANCE STRING
"TAPDISTANCE {distance | TAPDISTANCERULE ruleName
     |{HALFROW rowNum [LEFT|RIGHT]
         {distance | TAPDISTANCERULE ruleName}}...}
     TAPTYPE typeName
     ; " ;
END PROPERTYDEFINITIONS
```

Where:

```
TAPDISTANCE {distance | TAPDISTANCERULE ruleName
     |{HALFROW rowNum [LEFT|RIGHT]
         {distance | TAPDISTANCERULE ruleName}}...}
     TAPTYPE typeName
```


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Specifies that the effective distance of a well tap cell with type *`typeName`* should be *`distance`* on both the left and right sides of the cell, which the cell could cover. If `TAPDISTANCERULE` is specified, the distance is obtained from the tap distance rule *`ruleName`*. The distance is measured from the center of the cell. For example, the center-to-center distance between two well tap cells of the same type must be less than or equal to `2 * distance`.

Type: Float, specified in microns

`HALFROW` specifies that different distances could be defined on a half row basis, either in *`distance`* or in the distance defined in the tap distance rule *`ruleName`*. *`rowNum`* specifies the half row, with respect to the cell origin, on which the distance is defined.

`LEFT|RIGHT` specifies that different distances could be defined on the left or right sides on a half cell row basis. Left and right is referred with respect to the cell origin.

Tap Distance Rule Example

- The following example is an illustration of the tap distance rule:

```
PROPERTYDEFINITIONS

LIBRARY LEF58_TAPDISTANCE STRING "
     TAPDISTANCE HALFROW 1 2.4
          HALFROW 2 2.0 TAPTYPE TAP1
     TAPDISTANCE HALFROW 1 1.2 HALFROW 2 1.2
          HALFROW 3 LEFT 0 HALFROW 3 RIGHT 0.8
          HALFROW 4 LEFT 0 HALFROW 4 RIGHT 0.8
          TAPTYPE TAP2 ; " ;
END PROPERTYDEFINITIONS
```


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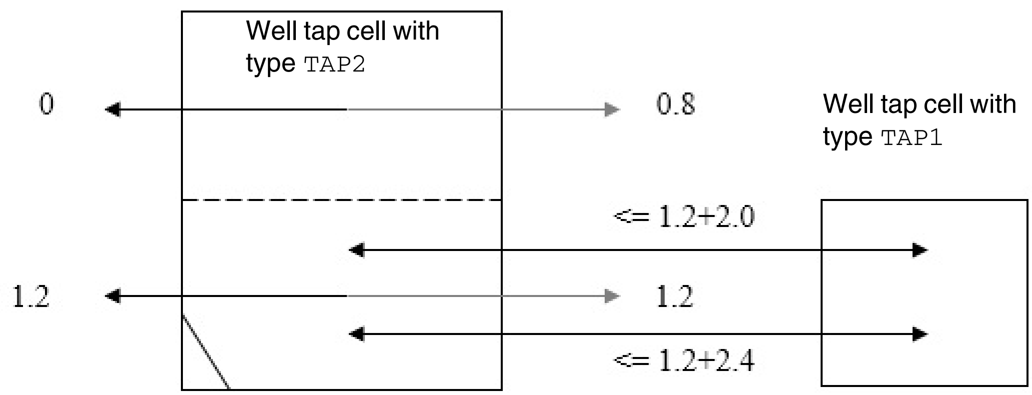
Figure 1-28 Illustration of Tap Distance Rule



Tap Distance Rule Property

You can create a tap distance rule property to specify effective tap distance.

You can define a tap distance rule property by using the following `PROPERTYDEFINITIONS` statement:

```
PROPERTYDEFINITIONS
LIBRARY LEF58_TAPDISTANCERULE STRING
"TAPDISTANCERULE ruleName DEFAULT distance
     [LAYER layerName layerDistance]...
     ; ";
END PROPERTYDEFINITIONS
```

Where:

`TAPDISTANCERULE` *`ruleName`* `DEFAULT` *`distance`*
`[LAYER` *`layerName layerDistance`*`]...`

Specifies that a rule with the name *`ruleName`* has an effective tap distance of *`distance`* by default and *`layerDistance`* when a tap cell overlaps with the areas on the layer *`layerName`*, which should be a `WELLDISTANCE` masterslice layer. The `TAPDISTANCE` library property could refer to this tap distance rule to determine the effective tap distance.

Type: Float, specified in microns

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Trim Metal Track Rule

You can create a trim metal track rule to define the trim metal tracks on a trim layer.

You can define a trim metal track rule by using the following `PROPERTYDEFINITIONS` statement:

```
PROPERTYDEFINITIONS
     LIBRARY LEF58_TRIMMETALTRACK STRING
     "TRIMMETALTRACK trimLayer [GROUP groupName]
          [ONTRACK [HALFINFILLER | HALFNONMERGED]
              [INCLUDELINEEND [TRIMCENTER]]
              [EXCEPTCOREEDGE]]
          [MASK maskNum
          |METALTRACKOFFSET trackNumber METALTRACKPITCH trackPitch]
          COREOFFSET offset PITCH pitch
          ;" ;
END PROPERTYDEFINITIONS
```

Where:

`EXCEPTCOREEDGE`	Specifies that the on-track requirement is not checked on the core edges.
`GROUP` *`groupName`*	Assigns a group name to the track definition. If any one of the `TRIMMETALTRACK` statements has `GROUP`, `GROUP` must be specified on all of the `TRIMMETALTRACK` statements. A separate command is used to define which group of `TRIMMETALTRACK` statements would be applied on a design. *Type*: String
`HALFNONMERGED`	Specifies that any non-merged trims could be on the half track as an exemption.
`INCLUDELINEEND`	Specifies that the line-end edges with default wire width or width lesser than or equal to the *`width`* value in `EXCEPTWIDTH` in the corresponding TRIMSHAPE definition if specified without a trim should also snap to the given hard trim grid. In other words, if a trim were to be added on that line-end edge, that imaginary trim should be on-grid.

`METALTRACKOFFSET` *`trackNumber`* `METALTRACKPITCH` *`trackPitch`*

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Specifies that the trim metal track is applied on the *trackNumber* metal track with respect to the core and all the tracks that are apart by a multiple of *trackPitch* track. If METALTRACKPITCH is defined on a certain given layer, all the other trim metal track statements on that layer must have METALTRACKPITCH with the same value. In addition, there should be at least *pitch* number of those statements. They should have different *trackNumber* statements and together they cover the track number of 1 to *trackPitch*. Then, multiple statements with the same *trackNumber* but with different *offset* could be defined.

Type: Integer

ONTRACK [HALFINFILLER]

Specifies the trim metal tracks are a hard constraint. It means that all of the inserted trims must be on a track and having an off-grid trim would be a violation by itself. HALFINFILLER means that trims on some special 1CPP filler cells could be on the half track as an exemption. The tool has a separate option to identify those special cells. This hard constraint does not apply to pre-defined trims in MACRO in the cell LEF. In addition, if one TRIMMETALTRACK statement defines ONTRACK on a *trimLayer*, all of the statements on that trim layer must also have ONTRACK.

TRIMCENTER

Specifies that line-end edges with default wire width or width lesser than or equal to the *width* value in EXCEPTWIDTH in the corresponding TRIMSHAPE definition if specified without a trim should snap to the center of the trim grid.

TRIMMETALTRACK *trimLayer* [MASK *maskNum*] COREOFFSET *offset*
PITCH *pitch*

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Specifies the trim metal tracks on layer *trimLayer*. The direction of the tracks should always be the orthogonal direction of the preferred routing direction of the metal layer defined in `TRIMMEDMETAL` on the layer *trimLayer* that the trim metal trims.
maskNum defines the metal track mask on which the trim grids are applied.
offset specifies the starting coordinate of the track with respect to the origin of the core with repeatable tracks that are *pitch* apart. Multiple statements could be defined on the same mask and/or different masks for different sets of tracks. Typically, the trim metal tracks depend on the cell objects, and this library property is likely to be defined in a cell LEF or side LEF file, instead of in a tech LEF file.

Type: Float, specified in microns

Trim Metal Track Rule Examples

- Consider the following example:

```
LAYER TM1
     TYPE MASTERSLICE ;
     MASK 2 ;
     PROPERTY LEF58_TYPE "TYPE TRIMMETAL ; " ;
     PROPERTY LEF58_TYPE "TYPE TRIMMEDMETAL M1 ; " ;
     ...
END TM1
...
LAYER M1
     TYPE ROUTING ;
     DIRECTION VERTICAL ;
     ...
END M1
...
PROPERTYDEFINITIONS
LIBRARY LEF58_TRIMMETALTRACK STRING "
            TRIMMETALTRACK TM1 MASK 1 COREOFFSET 0.01 PITCH 0.2 ;
            TRIMMETALTRACK TM1 MASK 1 COREOFFSET 0.12 PITCH 0.2 ;
            TRIMMETALTRACK TM1 MASK 2 COREOFFSET 0.08 PITCH 0.2 ;
            TRIMMETALTRACK TM1 MASK 2 COREOFFSET 0.19 PITCH 0.2 ;
END PROPERTYDEFINITIONS
```

The above example means that the trim metal layer of `TM1` has two sets of horizontal tracks on `MASK 1` with starting *y* locations of `0.01` and `0.12` with respect to the origin of the core with repeatable pitch of `0.2`, and it has another two sets of horizontal tracks on `MASK 2` with starting *y* locations of `0.08` and `0.19` with respect to the origin of the core with repeatable pitch of `0.2`.

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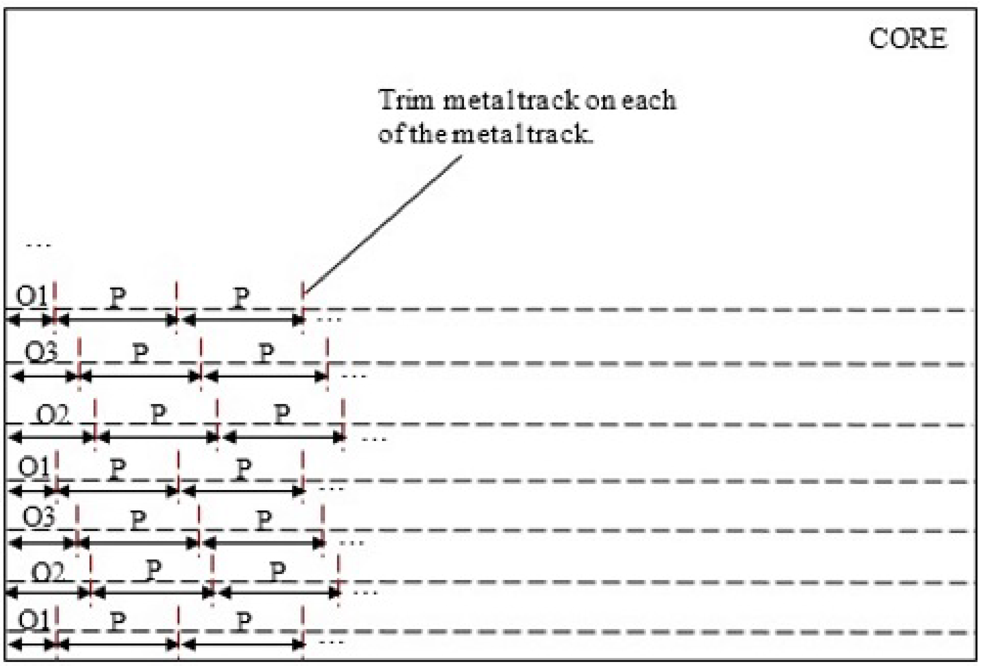
- The following example is an illustration of the trim metal track rule with `METALTRACKOFFSET` and `METALTRACKPITCH`:

```
TRIMMETALTRACK TM1 METALTRACKOFFSET 1 METALTRACKPITCH 3
    COREOFFSET O1 PITCH P ;
TRIMMETALTRACK TM1 METALTRACKOFFSET 2 METALTRACKPITCH 3
    COREOFFSET O2 PITCH P ;
TRIMMETALTRACK TM1 METALTRACKOFFSET 3 METALTRACKPITCH 3
    COREOFFSET O3 PITCH P ;
```

Figure 1-29 Illustration of Trim Metal Track Rule



Voltage Mode Rule

You can create a voltage mode rule to define how voltage should be computed in `VOLTAGESPACING` spacing.

You can define a voltage mode rule by using the following `PROPERTYDEFINITIONS` statement:

```
PROPERTYDEFINITIONS
LIBRARY LEF58_VOLTAGEMODE STRING
"VOLTAGEMODE ABSOLUTE
     ; " ;
END PROPERTYDEFINITIONS
```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Where:

`VOLTAGEMODE ABSOLUTE`

Specifies that the absolute maximum voltage of a net is used to define which `VOLTAGESPACING` spacing value should be applied on that net. The voltage of neighboring nets is irrelevant.

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Macro

```
MACRO macroName
     [CLASS
       { COVER [BUMP]
       | RING
       | BLOCK [BLACKBOX | SOFT]
       | PAD [INPUT | OUTPUT |INOUT | POWER | SPACER | AREAIO]
       | CORE [FEEDTHRU | TIEHIGH | TIELOW | SPACER | ANTENNACELL | WELLTAP]
       | ENDCAP {PRE | POST | TOPLEFT | TOPRIGHT | BOTTOMLEFT | BOTTOMRIGHT}
       }
     ;]
     [FIXEDMASK ;]
     [FOREIGN foreignCellName [pt [orient]] ;] ...
     [ORIGIN pt ;]
     [EEQ macroName ;]
     [SIZE width BY height ;]
     [SYMMETRY {X | Y | R90} ... ;]
     [SITE siteName [sitePattern] ;] ...
     [PIN statement] ...
     [OBS statement] ...
     [DENSITY statement] ...
     [PROPERTY propName propVal ;] ...
     [PROPERTY LEF58_ACCESSAREA
        "ACCESSAREA {LAYER layerName RECT pt pt [CUTCLASS className]
        [EXCEPTEXTRACUT]}...
        ; " ;]
     [PROPERTY LEF58_ALIGNPGVIATOTRACK
        "ALIGNPGVIATOTRACK
        ; " ;]
     [PROPERTY LEF58_ALLPINSCONNECTED
        "ALLPINSCONNECTED
        ; " ;]
     [PROPERTY LEF58_BOUNDARYSPACINGRANGE
        "BOUNDARYSPACINGRANGE minSpacing maxSpacing LAYER layerName
        ; " ;]
     [PROPERTY LEF58_CELLTYPE
        "CELLTYPE {codeName}...
        ; " ;]
     [PROPERTY LEF58_CLASS
       "CLASS
         {COVER [BUMP | FILL]
         | RING
         | BLOCK [BLACKBOX | SOFT]
         | PAD [INPUT | OUTPUT | INOUT | POWER | SPACER | AREAIO]
         | CORE [FEEDTHRU | TIEHIGH | TIELOW | SPACER
               | ANTENNACELL | WELLTAP [TAPWALL] [TAPTYPE typeName] | TSV]
         | ENDCAP {PRE | POST | TOPLEFT | TOPRIGHT | BOTTOMLEFT
                   | BOTTOMRIGHT
                   | TOPEDGE | BOTTOMEDGE | LEFTEDGE | RIGHTEDGE
                   | LEFTEVENSITEEDGE | LEFTODDSITEEDGE
```


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```
| RIGHTEVENSITEEDGE | RIGHTODDSITEEDGE
| LEFTTOPEDGE | RIGHTTOPEDGE
| LEFTBOTTOMEDGE | RIGHTBOTTOMEDGE
| LEFTTOPEVENSITEEDGE | LEFTTOPODDSITEEDGE
| RIGHTTOPEVENSITEEDGE | RIGHTTOPODDSITEEDGE
| LEFTBOTTOMEVENSITEEDGE
| LEFTBOTTOMODDSITEEDGE
| RIGHTBOTTOMEVENSITEEDGE
| RIGHTBOTTOMODDSITEEDGE
| LEFTTOPCORNER | RIGHTTOPCORNER
| LEFTBOTTOMCORNER | RIGHTBOTTOMCORNER
| LEFTTOPEVENSITECORNER | LEFTTOPODDSITECORNER
| RIGHTTOPEVENSITECORNER
| RIGHTTOPODDSITECORNER
| LEFTBOTTOMEVENSITECORNER
| LEFTBOTTOMODDSITECORNER
| RIGHTBOTTOMEVENSITECORNER
| RIGHTBOTTOMODDSITECORNER
| LEFTEDGETOPBORDER
| LEFTEDGEBOTTOMBORDER
| RIGHTEDGETOPBORDER
| RIGHTEDGEBOTTOMBORDER
| LEFTTOPEDGENEIGHBOR
| RIGHTTOPEDGENEIGHBOR
| LEFTBOTTOMEDGENEIGHBOR
| RIGHTBOTTOMEDGENEIGHBOR
| LEFTTOPCORNERNEIGHBOR
| RIGHTTOPCORNERNEIGHBOR
| LEFTBOTTOMCORNERNEIGHBOR
| RIGHTBOTTOMCORNERNEIGHBOR
| LEFTCORNERTOPBORDER
| RIGHTCORNERTOPBORDER
| LEFTCORNERBOTTOMBORDER
| RIGHTCORNERBOTTOMBORDER
| TOPCORNEREDGENEIGHBOR
| BOTTOMCORNEREDGENEIGHBOR
| TSVTOPEDGE | TSVBOTTOMEDGE
| TSVLEFTEDGE | TSVRIGHTEDGE
| TSVLEFTTOPCORNER | TSVLEFTBOTTOMCORNER
| TSVRIGHTTOPCORNER | TSVRIGHTBOTTOMCORNER
| TSVLEFTTOPEDGE | TSVLEFTBOTTOMEDGE
| TSVRIGHTTOPEDGE | TSVRIGHTBOTTOMEDGE
| TSVLEFTTOPEDGENEIGHBOR
| TSVRIGHTTOPEDGENEIGHBOR
| TSVLEFTBOTTOMEDGENEIGHBOR
| TSVRIGHTBOTTOMEDGENEIGHBOR
| TSVLEFTTOPCORNERNEIGHBOR
| TSVRIGHTTOPCORNERNEIGHBOR
| TSVLEFTBOTTOMCORNERNEIGHBOR
| TSVRIGHTBOTTOMCORNERNEIGHBOR
| REGULAR}
```


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```
                [TAPWALL] [TAPTYPE typeName]
      }
    ; " ;]
  [PROPERTY LEF58_CONSTRAINTAREATYPE
     "CONSTRAINTAREATYPE typeName {RECT pt pt}...
      ;..." ;]
  [PROPERTY LEF58_EDGETYPE
     "EDGETYPE {RIGHT | LEFT | TOP | BOTTOM}
      edgeType | BOTHSOURCE | SOURCEDRAIN | DRAINSOURCE | BOTHDRAIN
      | BOTHFLOATING
      [CELLROW cellRow [RANGE xLow xHigh] | HALFROW halfRow
      | RANGE xLow xHigh]
      ;..." ;]
  [PROPERTY LEF58_EEQ
     "EEQ macroName [VARIANT num]
      ;... " ;]
  [PROPERTY LEF58_FOREIGN
     "FOREIGN foreignCellName [pt [orient]]
           [COMPORIENT foreignOrientName {compOrient}...]...
      ;... " ;]
  [PROPERTY LEF58_LAYERMASKSHIFT
     "LAYERMASKSHIFT layer1 [layer2 ...]
      ;... " ;]
  [PROPERTY LEF58_OBSPARTIAL
    "OBSPARTIAL {LAYER layerName
      WIDTH width SPACING spacing RECT pt pt}...
    ; " ;]
  [PROPERTY LEF58_OBSSPACING
    "OBSSPACING {FULLDRC | MIN | spacing} [LAYER layerName]...
    ; " ;]
  [PROPERTY LEF58_PLACEWITHIN
    "PLACEWITHIN within PRL prl {WITH|WITHOUT} LAYER {layerName}...
    ; " ;]
  [PROPERTY LEF58_REQ
     "REQ macroName
      ; " ;]
  [PROPERTY LEF58_TAPCENTEROFFSET
    "TAPCENTEROFFSET
      {offset | {HALFROW rowNum [LEFT|RIGHT] offset}...}
    ; " ;]
  [PROPERTY LEF58_VARIANT
     "VARIANT num
     ; " ;]

END macroName
```

Defines macros in the design.

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Note: The keywords must be specified in the given order. For example if `ORIGIN` and `SITE` are both defined, `ORIGIN` must be specified first.

`CLASS`

Specifies the macro type. If you do not specify `CLASS`, the macro is considered a `CORE` macro, and a warning prints when the LEF file is read in. You can specify macros of the following types:

`COVER` Macro with data that is fixed to the floorplan and cannot change, such as power routing (ring pins) around the core. The placers understand that `CLASS COVER` cells have no active devices (such as diffusion or polysilicon), so the `MACRO SIZE` statement does not affect the placers, and you do not need an artificial `OVERLAP` layer. However, any pin or obstruction geometry in the `COVER` cells can affect the pin access checks done by the placers.

A cover macro can be of the following sub-class:

`BUMP`—A physical-only cell that has bump geometries and pins. Typically a bump cell has geometries only on the top-most “bump” metal layer, although it might contain a via and pin to the metal layer below.

`RING` Large macro that has an internal power mesh, and only exposes power-pin shapes that form a ring along the macro boundary. When power stripes are added across the macro, they connect to each side of the ring-pin but do not go inside the ring. The `CLASS RING` macro can also be used for power-switch cells that are abutted together to form a power-ring around a power-domain. In that case, their power-pins have the same effect of interrupting power stripes as the ring-pins in a single block `RING` macro.

`BLOCK` Predefined macro used in hierarchical design.

A block macro can have one of the following sub-classes:

`BLACKBOX`—A block that sometimes only contains a `SIZE` statement that estimates its total area. A blackbox can optionally contain pins, but in many cases, the pin names are taken from a Verilog description and do not need to match the LEF `MACRO` pin names.

`SOFT`—A cell that also contains a version of the sub-block that is not fully implemented. Normally, a soft block LEF can still have certain parts of it modified (for example, the aspect ratio, or pin locations) because the sub-block is not yet fully implemented. Any changes should be passed to the sub-block implementation. In contrast, a `BLACKBOX` has no sub-block implementation available.

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`PAD` I/O pad. A pad can be one of the following types: `INPUT`, `OUTPUT`, `INOUT`, `POWER`, or `SPACER`, for I/O rows; `INPUT`, `OUTPUT`, `INOUT`, or `POWER`, for I/O corner pads; `AREAIO` for area I/O driver cells that do not have the bump built in as part of the macro (and therefore require routing to a `CLASS COVER BUMP` macro for a connection to the IC package).
Note: Unlike `CORE SPACER`, `PAD SPACER` can contain logic pins.

For an example of a macro pad cell, see Example 1-2 on page 107.

`CORE` A standard cell used in the core area. `CORE` macros should always contain a `SITE` definition so that standard cell placers can correctly align the `CORE` macro to the standard cell rows.

A core macro can be one of the following types:

`FEEDTHRU`—Used for connecting to another cell.

`TIEHIGH`,`TIELOW`—Used for connecting unused I/O terminals to the power or ground bus. The software does not rely on this sub-class. A tie-cell has to be `CLASS CORE`, but the software does not consider the sub-class to determine its type. The dotlib representation of the cell's output pin is considered, and based on the function on that pin, it is determined whether it is a tiehigh, or tielow.

`SPACER`—Sometimes called a filler cell, this cell is used to fill in space between regular core cells. The `SPACER` sub-class needs to be cells with no logic-pins. Thus even with the sub-class defined, a cell will not be considered `SPACER` (also called `FILLER`) unless it has no logic/signal pins. A filler can only have Power and Ground pins. The instances of these cells will be marked by the insertion command to be of type 'Physical'.

`ANTENNACELL`—Used for solving process antenna violations. This cell has a single input to a diode to bleed off charge that builds up during manufacturing.

`WELLTAP`—Standard cell that connects N and P diffusion wells to the correct power or ground wire. The `WELLTAP` cells provide a tap for the N and P wells to the power/ground wires.

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`ENDCAP` A macro placed at the ends of core rows (to connect with power wiring).

If the library includes only one corner I/O macro, then appropriate `SYMMETRY` must be included in its macro description. An `ENDCAP` macro can be one of the following types:
`PRE`—A left-end macro
`POST`—A right-end macro
`TOPLEFT`—A top left I/O corner cell
`TOPRIGHT`—A top right I/O corner cell
`BOTTOMLEFT` —A bottom left I/O corner cell
`BOTTOMRIGHT`—A bottom right I/O corner cell

The `ENDCAP` sub-class is required. The `PRE` and `POST` are `CORE` area cells, whereas the other four are `PAD CLASS`.

Example 1-2 Macro Pad Cell

The following example defines a power pad cell that illustrates when to use the CLASS CORE keywords on power ports. For the VDD pin, there are two ports: one to connect to the interior core power ring, and one to complete the I/O power ring. Figure 1-30 on page 109 illustrates this pad cell.

```
MACRO PAD_0
    CLASS PAD ;
    FOREIGN PAD_0 0.000 0.000 ;
    ORIGIN 0.000 0.000 ;
    SIZE 100.000 BY 300.000 ;
    SYMMETRY X Y R90 ;
    SITE PAD_SITE ;

# Define pin VDD with SHAPE ABUTMENT because there are no obstructions
# to block a straight connection to the pad rings. The port without
# CLASS CORE is used for completing the I/O power ring.

    PIN VDD
        DIRECTION INOUT ;
        USE POWER ;
        SHAPE ABUTMENT ;
        PORT
           LAYER metal2 ;
             RECT 0.000 250.000 100.000 260.000 ;
           LAYER metal3 ;
             RECT 0.000 250.000 100.000 260.000 ;
    END
```


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```
# Define VDD port with PORT CLASS CORE to indicate that the port connects
# to the core area instead of to the pad ring.

      PORT
           CLASS CORE ;
           LAYER metal2 ;
             RECT 0.000 290.000 100.000 300.000 ;
           LAYER metal3 ;
             RECT 0.000 290.000 100.000 300.000 ;
      END
    END VDD

# Define pins VCC and GND with SHAPE FEEDTHRU because these pins
# cannot make a straight connection to the pad rings due to obstructions.
    PIN VCC
          DIRECTION INOUT ;
          USE POWER ;
          SHAPE FEEDTHRU ;
          PORT
            LAYER metal2 ;
              RECT 0.000 150.000 20.000 160.000 ;
              RECT 20.000 145.000 80.000 155.000 ;
              RECT 80.000 150.000 100.000 160.000 ;
            LAYER metal3 ;
              RECT 0.000 150.000 20.000 160.000 ;
              RECT 20.000 145.000 80.000 155.000 ;
              RECT 80.000 150.000 100.000 160.000 ;
          END
    END VCC
    PIN GND
        DIRECTION INOUT ;
        USE GROUND ;
        SHAPE FEEDTHRU ;
        PORT
           LAYER metal2 ;
             RECT 0.000 50.000 20.000 60.000 ;
             RECT 80.000 50.000 100.000 60.000 ;
        END
    END GND
    OBS
```


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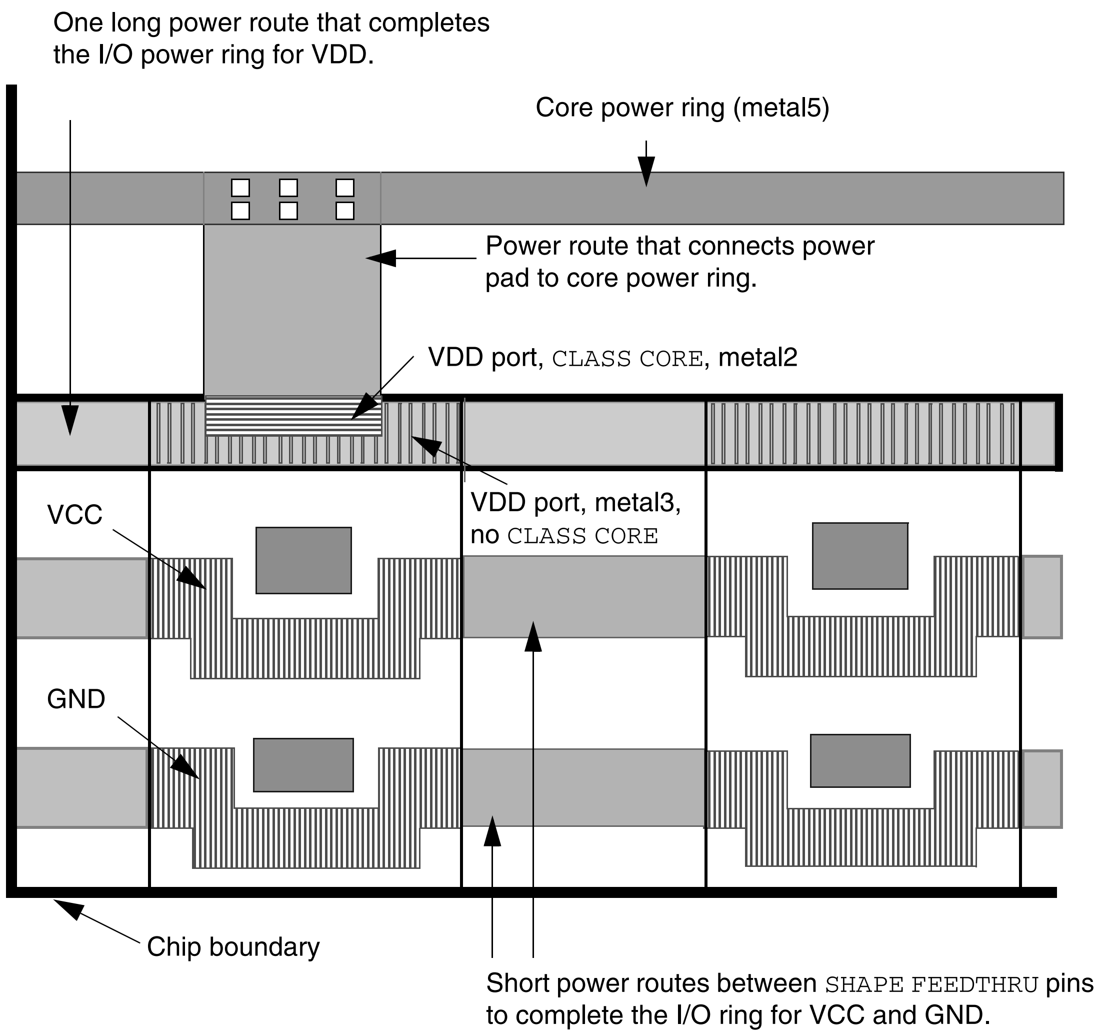
```
        LAYER metal1 ;
            RECT 0.000 0.000 100.000 300.000 ;
        LAYER metal2 ;
            RECT 25.000 50.000 75.000 60.000 ;
            RECT 30.500 157.000 70.500167.000 ;
    END
END PAD_0
```

Figure 1-30 Power Pad Cell



`DENSITY` *`statement`*

Specifies the metal density for large macros.

The `DENSITY` rectangles on a layer should not overlap, and should cover the entire area of the macro. You can choose the size of the rectangles based on the uniformity of the

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density of the block. If the density is uniform, a single rectangle can be used. If the density is not very uniform, the size of the rectangles can be specified to be 10 to 20 percent of the density window size, so that any error due to non-uniform density inside each rectangle area is small.

For example, if the metal density rule is for a 100 µm x 100 µm window, the density rectangles can be 10x10 µm squares. Any non-uniformity will have little impact on the density calculation accuracy.

If two adjacent rectangles have the same or similar density, they can be merged into one larger rectangle, with one average density value. The choice between accuracy and abstraction is left to the abstract generator.

The `DENSITY` syntax is defined as follows:

```
[DENSITY
   {LAYER layerName ;
      {RECT x1 y1 x2 y2 densityValue ;} ...
   } ...
END] ...
```

densityValue	Specifies the density for the rectangle, as a percentage. For example, `50.0` indicates that the rectangle has a density of 50 percent on *layerName*. *Type:* Float *Value:* `0` to `100`
layerName	Specifies the layer on which to create the rectangle.
x1 y1 x2 y2	Specifies the coordinates of a rectangle. *Type:* Float, specified in microns

Example 1-3 Macro Density

The following statement specifies the density for macro `testMacro`:

```
MACRO testMacro
    CLASS ...
    PIN ...
    OBS ...
    DENSITY
        LAYER metal1 ;
        RECT 0 0 100 100 45.5 ; #rect from (0,0) to (100,100), density of 45.5%
        RECT 100 0 200 100 42.2 ; #rect from (100,0) to (200, 100), density of 42.2%
    END
    ...
END testMacro
```


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`EEQ` *`macroName`*

Specifies that the macro being defined should be electrically equivalent to the previously defined *`macroName`*. EEQ macros include devices such as OR-gates or inverters that have several implementations with different shapes, geometries, and orientations.

Electrically equivalent macros have the following requirements:

- Corresponding pins must have corresponding functionality.
- Pins must be defined in the same order.
- For each group of corresponding pins (one from each macro), pin function and electrical characteristics must be the same.
- The EEQ *`macroName`* specified must refer to a macro in the list of LEF files read in at the same time. If the EEQ *`macroName`* referenced is already electrically equivalent to other model macros, all referenced macros are considered electrically equivalent.

`FIXEDMASK`

Indicates that the specified macro does not allow mask-shifting. All the LEF PIN MASK assignments must be kept fixed and cannot be shifted to a different mask to optimize routing density. All the LEF PIN shapes should have `MASK` assignments, if the `FIXEDMASK` statement is present. In addition, macro OBS should all have `MASK` assignments unless they have the `SPACING` override.

For example,

```
MACRO my_block
    CLASS BLOCK ;
    FIXEDMASK ;
    ...
```

`FOREIGN` *`foreignCellName`* `[`*`pt`* `[`*`orient`*`]]`

Specifies the foreign (GDSII) structure name to use when placing an instance of the macro. The optional *`pt`* coordinate specifies the macro origin (lower left corner when the macro is in north orientation) offset from the foreign origin. The `FOREIGN` statement has a default offset value of `0 0`, if *`pt`* is not specified.

The optional *`orient`* value specifies the orientation of the foreign cell when the macro is in north orientation. The default *`orient`* value is N (North).

Example 1-4 Foreign Statements

The following examples show two variations of the `FOREIGN` statement. The negative offset specifies that the GDSII structure should be above and to the right of the macro lower left corner.

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```
MACRO ABC ...
FOREIGN ABC -2 -3 ;
```

The positive offset specifies that the GDSII structure should be below and to the left of the macro lower left corner.

```
MACRO EFG ...
FOREIGN EFG 2 3 ;
```

MACRO *macroName*

Specifies the name of the library macro.

OBS *statement*

Defines obstructions on the macro. Obstruction geometries are specified using layer geometries syntax. See "Macro Obstruction Statement" on page 163 for syntax information.

ORIGIN *pt*

Specifies how to find the origin of the macro to align with a DEF COMPONENT placement point. If there is no ORIGIN statement, the DEF placement point for a North-oriented macro is aligned with 0, 0 in the macro. If ORIGIN is given in the macro, the macro is shifted by the ORIGIN x, y values first, before aligning with the DEF placement point. For example, if the ORIGIN is 0, -1, then macro geometry at 0, 1 are shifted to 0, 0, and then aligned to the DEF placement point.

PIN *statement*

Defines pins for the macro. See "Macro Pin Statement" on page 166 for syntax information.

PROPERTY *propName propVal*

Specifies a numerical or string value for a macro property defined in the PROPERTYDEFINITIONS statement. The *propName* you specify must match the *propName* listed in the PROPERTYDEFINITIONS statement.

SITE *siteName* [*sitePattern*]

Specifies the site associated with the macro. Normal row-based standard cells only have a single SITE *siteName* statement, without a *sitePattern*. The *sitePattern* syntax indicates that the cell is a gate-array cell, rather than a row-based standard cell. Gate-array standard cells can have multiple SITE statements, each with a *sitePattern*.

The *sitePattern* syntax is defined as follows:

[*xOrigin yOrigin siteOrient* [*stepPattern*]]

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Where:

xOrigin yOrigin	Specifies the origin of the site inside the macro. *Type:* Float, specified in microns
siteOrient	Specifies the orientation of the site at that location. *Value:* `N`, `S`, `E`, `W`, `FN`, `FS`, `FE`, or `FW`

Note: Legal placement locations for macros with site patterns must match the site pattern inside the macro to the site pattern in the design rows.

If the site is repeated, you can specify a *stepPattern* that defines the repeating pattern. The *stepPattern* syntax is defined as follows:

`[DO` *xCount* `BY` *yCount* `STEP` *xStep yStep*`]`

xCount yCount	Specifies the number of sites to add in the x and y directions. You must specify values that are greater than or equal to `0` (zero). *Type:* Integer
xStep yStep	Specifies the spacing between sites in the x and y directions. *Type:* Float, specified in microns

Example 1-5 Macro Site

The following statement defines a macro that uses the sites created in Example 1-17 on page 191:

```
MACRO myTest
    CLASS CORE ;
    SIZE 10.0 BY 14.0 ;   #Uses 2 F and 1 L site, is F + L wide, and double height
    SYMMETRY X ;          #Can flip about the X axis
    SITE Fsite 0 0 N ;    #The lower left Fsite at 0,0
    SITE Fsite 0 7.0 FS ; #The flipped south Fsite above the first Fsite at 0,7
    SITE Lsite 4.0 0 N ;  #The Lsite to the right of the first Fsite at 4,0
    ...
    PIN ... ;
END myTest
```

Figure 1-31 on page 114 illustrates the placement results of this definition.

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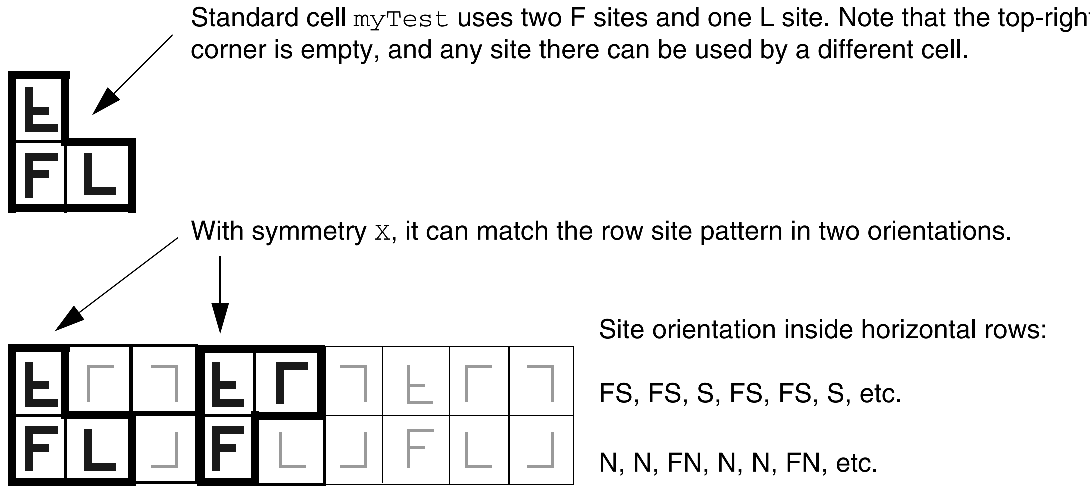
Figure 1-31



The following statement includes the gate-array site pattern syntax. It uses two F sites in a row with N (North) orientation.

```
MACRO myTest
    CLASS CORE ;
    SIZE 8.0 BY 7.0 ; #Width = 2 * Fsite width, height = Fsite height
    SITE Fsite 0 0 N DO 2 BY 1 STEP 4.0 0 ; #Xstep = 4.0 = Fsite width
    ...
END myTest
```

This definition produces a cell with the sites shown in Figure 1-32 on page 114.

Figure 1-32

`SIZE` *`width`* `BY` *`height`*

Specifies a placement bounding rectangle, in microns, for the macro. The bounding rectangle always stretches from (0, 0) to the point defined by `SIZE`. For example, given `SIZE` 10 `BY` 40, the bounding rectangle reaches from (0, 0) after adjustment due to the `ORIGIN` statement, to (100, 400).

Placers assume the placement bounding rectangle cannot overlap placement bounding rectangles of other macros, unless `OBS OVERLAP` shapes are used to create a non-rectangular area.

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After placement, a DEF `COMPONENTS` placement *`pt`* indicates where the lower-left corner of the placement bounding rectangle is placed after any possible rotations or flips. The bounding rectangle width and height should be a multiple of the placement grid to allow for abutting cells.

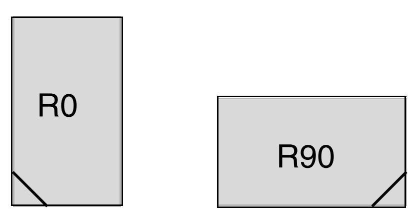
For blocks, the placement bounding rectangle typically contains all pin and blockage geometries, but this is not required. For example, typical standard cells have pins that lie outside the bounding rectangle, such as power pins that are shared with cells in the next row above them.

`SYMMETRY {X | Y | R90}`

Specifies which macro orientations should be attempted by the placer before matching to the site of the underlying rows. In general, most standard cell macros should have symmetry `X Y`. N (North) is always a legal candidate. For each type of symmetry defined, additional orientations become legal candidates. For more information on defining symmetry, see "Defining Symmetry" on page 153.

Possible orientations include:

`X`	N and FS orientations should be tried.
`Y`	N and FN orientations should be tried.
`X Y`	N, FN, FS, and S orientations should all be tried.
`R90`	Only applicable for non-standard cells.



Note: If you do not specify a `SYMMETRY` statement, only N orientation is tried.

For corner I/O pads, if the library includes `BOTTOMLEFT`, `BOTTOMRIGHT`, `TOPLEFT`, and `TOPRIGHT` I/O corner cells, then they are placed in North orientation (no flipping). However, if the library includes only one type of corner I/O, then `SYMMETRY` in x and y are required to create the rows for all four of them.

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Defining Macro Properties to Create 32/28 nm and Smaller Nodes Rules

You can include macro layer properties in your LEF file to create 32/28 nm and smaller nodes rules that currently are not supported by existing LEF syntax. The properties are specified inside the Macro statements where they can be seen with other rules.

Before you can reference them, properties must be defined at the beginning of the LEF file in the `PROPERTYDEFINITIONS` statement, immediately before the first `Macro` statement.

- Properties belong to the `MACRO`/`PIN` object and have a type of `STRING`.
- The property names used for these rules all start with `LEF58_`.

All properties use the following syntax within the LEF `PROPERTYDEFINITIONS` statement:

```
PROPERTYDEFINITIONS
     MACRO propName STRING ["stringValue"] ;
     PIN propName STRING ["stringValue"] ;
END PROPERTYDEFINITIONS
```

The property definitions for the macro properties are as follows:

```
PROPERTYDEFINITIONS
     MACRO LEF58_ACCESSAREA STRING ;
     MACRO LEF58_ALIGNPGVIATOTRACK STRING ;
     MACRO LEF58_ALLPINSCONNECTED STRING ;
     MACRO LEF58_BOUNDARYSPACINGRANGE STRING ;
     MACRO LEF58_CELLTYPE STRING ;
     MACRO LEF58_CLASS STRING ;
     MACRO LEF58_CONSTRAINTAREATYPE STRING ;
     MACRO LEF58_EDGETYPE STRING ;
     MACRO LEF58_EEQ STRING ;
     MACRO LEF58_FOREIGN STRING ;
     MACRO LEF58_LAYERMASKSHIFT STRING ;
     MACRO LEF58_OBSPARTIAL STRING ;
     MACRO LEF58_PLACEWITHIN STRING ;
     MACRO LEF58_REQ STRING ;
     MACRO LEF58_TAPCENTEROFFSET STRING ;
     MACRO LEF58_VARIANT STRING ;
     PIN LEF58_ACCESSAREA STRING ;
     PIN LEF58_ANTENNADIFFAREA STRING ;
     PIN LEF58_BOTTOMTOPLAYER STRING ;
     PIN LEF58_MUSTJOINALLPORTS STRING ;
     PIN LEF58_SHORTEDPINGROUP STRING ;
```


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```
     PIN LEF58_VIAINMUSTJOIN STRING ;
     PIN LEF58_VIAINPINONLY STRING ;
END PROPERTYDEFINITIONS
```

Access Area Rule

The access area rule can be used on the macro to specify a via access area on the specified layer.

You can specify the access area rule by using the following property definition:

```
PROPERTY LEF58_ACCESSAREA
        "ACCESSAREA {LAYER layerName RECT pt pt [CUTCLASS className]
        [EXCEPTEXTRACUT]}...
        ; " ;
```

Where:

`ACCESSAREA {LAYER` *`layerName`* `RECT` *`pt pt`* `}...`

Specifies a via access area on *layerName*, which must be a cut layer. All the cuts of a via connected to any pin shape on the bottom metal layer that overlaps the given area in `RECT` must be inside the given area. If there are multiple via access areas overlapping a pin shape, the entire size of the cuts of a via connected to the pin need to be inside one of the areas. Different via access areas should not overlap or be abutted. For a `MUSTJOINALLPORTS` pin, the given area must overlap with all of the pin shapes on the bottom metal layer. Then, the cuts on each of the individual pin shapes must be inside the given area and must be connected by a same-metal wire on the above metal layer as well. This construct can only be defined on standard cells with `MACRO CLASS` of `CORE`.

Type: Float, specified in microns

`CUTCLASS` *`className`*

Specifies the via access area for the given *className*, which is connected to the pin shapes. Without `CUTCLASS`, the via access area would be applied to vias of any cut class. Multiple via access areas could be defined for different cut classes. If a cut class does not have a corresponding via access area defined, the vias of that cut class could be connected to the pins without any restriction.

Type: String

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`EXCEPTEXTRACUT` Specifies that the via access area is not applied to vias with multiple cuts connected to a single pin shape. If a via pillar with multiple wires on the above metal layer of the pin shape layer is used, this via access area does not need to be honored. For example, if a via pillar with 2x2 cuts is connected to a `MUSTJOINALLPORTS` with two pin shapes, this via access area could be ignored.

Area Access Rule Examples

- The following example means that the cuts of `VIA1` of `VA` must be inside `1.0 1.0 2.0 2.0` and the cuts of `VIA1` of the other cut class must be inside `1.5 1.5 2.2 2.2` when connecting to a `M1` (below metal layer) pin shape that overlaps with the given area.

```
PROPERTY LEF58_ACCESSAREA "
     ACCESSAREA LAYER VIA1 RECT 1.0 1.0 2.0 2.0 CUTCLASS VA
          LAYER VIA1 RECT 1.5 1.5 2.2 2.2 ; " ;
```

- The following example means that the cuts of `VIA1` of `VA` must be inside `1.0 1.0 2.0 2.0` when connect to a `M1` (below metal layer) pin shape that overlaps with the given area and the cuts of `VIA1` of the other cut class does not have any restriction.

```
PROPERTY LEF58_ACCESSAREA "
     ACCESSAREA LAYER VIA1 RECT 1.0 1.0 2.0 2.0 CUTCLASS VA ; " ;
```

- The diagram below illustrates the following access area rule:

```
PROPERTY LEF58_ACCESSAREA "
     ACCESSAREA LAYER VIA1 RECT 1.0 1.0 3.5 1.8
          EXCEPTEXTRACUT
          LAYER VIA1 RECT 6.0 0.5 7.0 1.0
          LAYER VIA1 RECT 6.0 1.8 7.0 2.5 ; " ;
```


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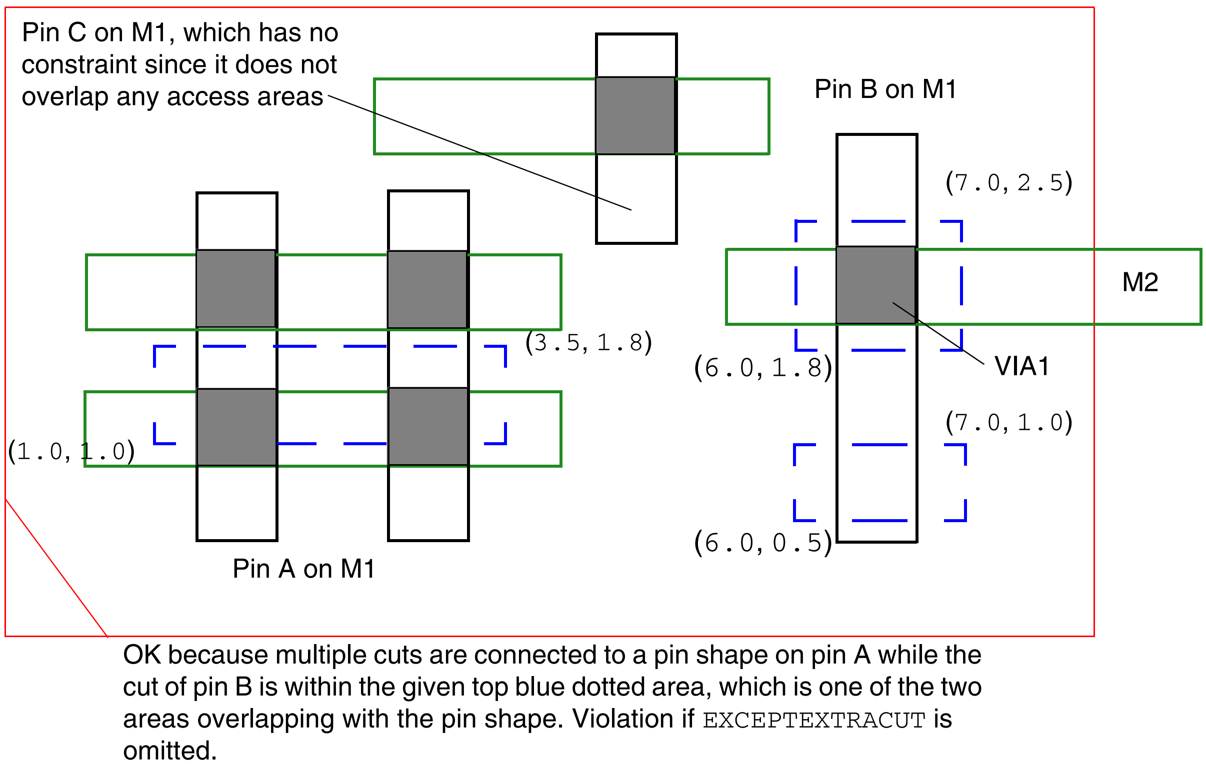
Figure 1-33 Illustration of the Access Area Rule



Align PG Via to Track Rule

The align PG via to track attribute can be used on the macro to align the power or ground cell vias to the given tracks defined in the PG via track rule in `PGTRACKVIA`.

You can specify the align PG via to track attribute by using the following property definition:

```
PROPERTY LEF58_ALIGNPGVIATOTRACK
     "ALIGNPGVIATOTRACK
        ;..." ;
```

Where:

`ALIGNPGVIATOTRACK`	Specifies that all the power or ground cell vias of this macro should be aligned to the tracks defined in `PGVIATRACK` in the library property in a tech LEF file.

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All Pins Connected Rule

The all pins connected rule can be used to specify that all of the pins in the macro are strongly connected internally.

You can specify the all pins connected rule by using the following property definition:

```
PROPERTY LEF58_ALLPINSCONNECTED
        "ALLPINSCONNECTED
        ; " ;
```

Where:

`ALLPINSCONNECTED`	Specifies that all of the pins in this macro are strongly connected internally. All of the pins should only have one port. This construct could be defined only in a `MACRO` with `CLASS COVER`.

Boundary Spacing Range Rule

The boundary spacing range rule can be used to define spacing rules for shapes on left or right cell boundaries in the specified CPODE layer.

You can specify the boundary spacing range rule by using the following property definition:

```
PROPERTY LEF58_BOUNDARYSPACINGRANGE
        "BOUNDARYSPACINGRANGE minSpacing maxSpacing LAYER layerName
        ; " ;
```

Where:

`BOUNDARYSPACINGRANGE` *`minSpacing maxSpacing`* `LAYER` *`layerName`*

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Specifies that shapes on *layerName*, which must be a CPODE masterslice layer, located on the left or right boundaries of a cell, must have other CPODE neighbor shape(s) to be positioned more than *minSpacing* and up to *maxSpacing* away—measured horizontally in the direction away from the cell. These neighboring shapes must also completely cover the boundary CPODE shape. Essentially, any CPODE shape at the cell's left or right edge needs a neighbor shape greater than *minSpacing* and less than or equal to *maxSpacing* on that respective side. In addition, any CPODE neighbor shape at a distance less than or equal to *minSpacing* is also a violation.

Type: Float, specified in microns

Cell Type Rule

The cell type rule can be used to specify the type of a cell.

You can specify the cell type rule by using the following property definition:

```
PROPERTY LEF58_CELLTYPE
        "CELLTYPE {codeName}...
        ; " ;
```

Where:

CELLTYPE {*codeName*}...

Specifies one or more cell type of a cell. *codeName* is the given code name in CELLMAP in library. Typically, a cell has one cell type. For a complex cell, multiple cell types can be defined.

Class Rule

You can use the class rule to define a macro as a special cover macro for metal filling purpose.

You can create class rule by using the following property definition:

```
PROPERTY LEF58_CLASS
     "CLASS
          {COVER [BUMP | FILL]
```


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```
| RING
| BLOCK [BLACKBOX | SOFT]
| PAD [INPUT | OUTPUT | INOUT | POWER | SPACER | AREAIO]
| CORE [FEEDTHRU | TIEHIGH | TIELOW | SPACER | ANTENNACELL
    | WELLTAP [TAPWALL] [TAPTYPE typeName] | TSV]
| ENDCAP {PRE | POST | TOPLEFT | TOPRIGHT | BOTTOMLEFT
    | BOTTOMRIGHT
    | TOPEDGE | BOTTOMEDGE | LEFTEDGE | RIGHTEDGE
    | LEFTEVENSITEEDGE | LEFTODDSITEEDGE
    | RIGHTEVENSITEEDGE | RIGHTODDSITEEDGE
    | LEFTTOPEDGE | RIGHTTOPEDGE
    | LEFTBOTTOMEDGE | RIGHTBOTTOMEDGE
    | LEFTTOPEVENSITEEDGE | LEFTTOPODDSITEEDGE
    | RIGHTTOPEVENSITEEDGE | RIGHTTOPODDSITEEDGE
    | LEFTBOTTOMEVENSITEEDGE | LEFTBOTTOMODDSITEEDGE
    | RIGHTBOTTOMEVENSITEEDGE | RIGHTBOTTOMODDSITEEDGE
    | LEFTTOPCORNER | RIGHTTOPCORNER
    | LEFTBOTTOMCORNER | RIGHTBOTTOMCORNER
    | LEFTTOPEVENSITECORNER | LEFTTOPODDSITECORNER
    | RIGHTTOPEVENSITECORNER | RIGHTTOPODDSITECORNER
    | LEFTBOTTOMEVENSITECORNER
    | LEFTBOTTOMODDSITECORNER
    | RIGHTBOTTOMEVENSITECORNER
    | RIGHTBOTTOMODDSITECORNER
    | LEFTEDGETOPBORDER
    | LEFTEDGEBOTTOMBORDER
    | RIGHTEDGETOPBORDER
    | RIGHTEDGEBOTTOMBORDER
    | LEFTTOPEDGENEIGHBOR
    | RIGHTTOPEDGENEIGHBOR
    | LEFTBOTTOMEDGENEIGHBOR
    | RIGHTBOTTOMEDGENEIGHBOR}
    | LEFTTOPCORNERNEIGHBOR
    | RIGHTTOPCORNERNEIGHBOR
    | LEFTBOTTOMCORNERNEIGHBOR
    | RIGHTBOTTOMCORNERNEIGHBOR
    | LEFTCORNERTOPBORDER
    | RIGHTCORNERTOPBORDER
    | LEFTCORNERBOTTOMBORDER
    | RIGHTCORNERBOTTOMBORDER
    | TOPCORNEREDGENEIGHBOR
    | BOTTOMCORNEREDGENEIGHBOR
    | TSVTOPEDGE | TSVBOTTOMEDGE
    | TSVLEFTEDGE | TSVRIGHTEDGE
    | TSVLEFTTOPCORNER | TSVLEFTBOTTOMCORNER
    | TSVRIGHTTOPCORNER | TSVRIGHTBOTTOMCORNER
    | TSVLEFTTOPEDGE | TSVLEFTBOTTOMEDGE
    | TSVRIGHTTOPEDGE | TSVRIGHTBOTTOMEDGE
    | TSVLEFTTOPEDGENEIGHBOR
    | TSVRIGHTTOPEDGENEIGHBOR
    | TSVLEFTBOTTOMEDGENEIGHBOR
```


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```
            | TSVRIGHTBOTTOMEDGENEIGHBOR
            | TSVLEFTTOPCORNERNEIGHBOR
            | TSVRIGHTTOPCORNERNEIGHBOR
            | TSVLEFTBOTTOMCORNERNEIGHBOR
            | TSVRIGHTBOTTOMCORNERNEIGHBOR
            | REGULAR}
            [TAPWALL] [TAPTYPE typeName]
        }
    ; " ;
```

Where:

All other keywords are same as the existing macro `CLASS` syntax.

`FILL`

Specifies that the macro is a special cover macro for metal filling purpose.

`LEFTCORNERTOPBORDER | RIGHTCORNERTOPBORDER`
`| LEFTCORNERBOTTOMBORDER | RIGHTCORNERBOTTOMBORDER`

Specifies a left or right edge endcap cell, which is on top or bottom a corner cell. For example, `LEFTCORNERTOPBORDER` would be placed on the left edge and on top of a corner cell.

`LEFTEDGETOPBORDER | LEFTEDGEBOTTOMBORDER | RIGHTEDGETOPBORDER | RIGHTEDGEBOTTOMBORDER`

Specifies a left or right edge endcap cell, which is right next to the `LEFTTOPEDGE`/`LEFTBOTTOMEDGE` or `RIGHTTOPEDGE`/`RIGHTBOTTOMEDGE` cell.

```
LEFTEVENSITEEDGE | LEFTODDSITEEDGE
    | RIGHTEVENSITEEDGE | RIGHTODDSITEEDGE
    | LEFTTOPEVENSITEEDGE | LEFTTOPODDSITEEDGE
    | RIGHTTOPEVENSITEEDGE | RIGHTTOPODDSITEEDGE
    | LEFTBOTTOMEVENSITEEDGE | LEFTBOTTOMODDSITEEDGE
    | RIGHTBOTTOMEVENSITEEDGE | RIGHTBOTTOMODDSITEEDGE
    | LEFTTOPEVENSITECORNER | LEFTTOPODDSITECORNER
    | RIGHTTOPEVENSITECORNER | RIGHTTOPODDSITECORNER
    | LEFTBOTTOMEVENSITECORNER
    | LEFTBOTTOMODDSITECORNER
    | RIGHTBOTTOMEVENSITECORNER
    | RIGHTBOTTOMODDSITECORNER
```


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Specifies that the macro with *EVENSITE* or *ODDSITE* endcap type has even or odd site. These are used to always form even number of sites horizontally, including the endcap cells.

LEFTTOPCORNERNEIGHBOR | RIGHTTOPCORNERNEIGHBOR
| LEFTBOTTOMCORNERNEIGHBOR | RIGHTBOTTOMCORNERNEIGHBOR

Specifies a top or bottom edge endcap cell, which is right next to the LEFTTOPCORNER/RIGHTTOPCORNER or LEFTBOTTOMCORNER/RIGHTBOTTOMCORNER cell. When all four cells are specified, they are placed in R0 (or MX) orient, abutting the CORNER cells on the same row. If only one of the four cells, for example RIGHTTOPCORNERNEIGHBOR, is specified, it will be flipped and rotated to be used in all four locations, provided the row of the location needing the bottom edge is flipped (MX). If both top and bottom rows are in R0 orient, the same cell cannot be used in place of LEFT/ RIGHTBOTTOMCORNERNEIGHBOR, as that would result in shorting power to ground. If two cells, TOP and BOTTOM, are provided, they can be flipped around the Y axis to be used in both the LEFT and RIGHT locations.

LEFTTOPEDGENEIGHBOR | RIGHTTOPEDGENEIGHBOR | LEFTBOTTOMEDGENEIGHBOR
| LEFTBOTTOMEDGENEIGHBOR

Specifies a top or bottom edge endcap cell, which is right next to the LEFTTOPEDGE/LEFTBOTTOMEDGE or RIGHTTOPEDGE/ RIGHTBOTTOMEDGE cell. When all four types of *NEIGHBOR are defined, they are placed in the R0 or MX orientation. If only LEFTTOPEDGENEIGHBOR and LEFTBOTTOMEDGENEIGHBOR are defined, they can be used by y-flipping for RIGHTTOPEDGENEIGHBOR and RIGHTBOTTOMEDGENEIGHBOR. However, LEFTTOPEDGENEIGHBOR and RIGHTTOPEDGENEIGHBOR cannot be x-flipped to replace LEFTBOTTOMEDGENEIGHBOR and LEFTBOTTOMEDGENEIGHBOR because the power/ground pins cannot be matched.

REGULAR

Specifies that this is an end cap cell without reference to any particular edge, corner, or location.

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`TAPTYPE` *`typeName`*
Identifies different well tap cells with `CLASS WELLTAP` or end cap cells with `CLASS ENDCAP`, which would also serve tap cell purpose, by tagging them with the name, *`typeName`*. Other LEF constructs or properties can refer to certain well tap cells by *`typeName`*, and define well tap specific information on them.
Type: String

`TAPWALL`
Specifies a special well tap cell or a special `ENDCAP` cell that is also serving tap wall purpose, which could be used to break OD diffusion and aligned vertically to form a tap wall.

`TOPCORNEREDGENEIGHBOR | BOTTOMCORNEREDGENEIGHBOR`

Specifies a top or bottom edge endcap cell as `TOPCORNEREDGENEIGHBOR` or `BOTTOMCORNEREDGENEIGHBOR`, which is between a corner cell and an edge cell.

```
TOPEDGE | BOTTOMEDGE | LEFTEDGE | RIGHTEDGE
      | LEFTTOPEDGE | RIGHTTOPEDGE
      | LEFTBOTTOMEDGE | RIGHTBOTTOMEDGE
      | LEFTTOPCORNER | RIGHTTOPCORNER
      | LEFTBOTTOMCORNER | RIGHTBOTTOMCORNER
```

Specifies that the macro has nwell resided on certain edge or corner of the endcap cells. For example, the `TOPEDGE` keyword indicates that the endcap cell has n-well on the top edge, while `LEFTTOPEDGE` indicates that nwell exists on both top and left edges. The `LEFTTOPCORNER` keyword indicates that nwell only exists on the top left corner.

`TSV`
Specifies a standard cell containing a TSV, which connects the front and back side layers. Typically, the standard cell has a pin on the front side layer and another pin on the back side layer.

```
TSVTOPEDGE | TSVBOTTOMEDGE | TSVLEFTEDGE | TSVRIGHTEDGE
      | TSVLEFTTOPCORNER | TSVLEFTBOTTOMCORNER
      | TSVRIGHTTOPCORNER | TSVRIGHTBOTTOMCORNER
      | TSVLEFTTOPEDGE | TSVLEFTBOTTOMEDGE
      | TSVRIGHTTOPEDGE | TSVRIGHTBOTTOMEDGE
```


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Specifies special endcap cells around TSV with a keepout zone. `TSVTOPEDGE`, `TSVBOTTOMEDGE`, `TSVLEFTEDGE` and `TSVRIGHTEDGE` are added on the corresponding edges around TSV. The rest of TSV*EDGE are added on the corners. TSV*CORNER are special end-corner cells when they are next to a regular blockage or block.

`TSVLEFTTOPEDGENEIGHBOR | TSVRIGHTTOPEDGENEIGHBOR`
`| TSVLEFTBOTTOMEDGENEIGHBOR | TSVRIGHTBOTTOMEDGENEIGHBOR`

Specifies a top- or bottom-edge endcap cell, which is right next to the `TSVLEFTTOPEDGE`/`TSVLEFTBOTTOMEDGE` or `TSVRIGHTTOPEDGE`/`TSVRIGHTBOTTOMEDGE` cell. When all four types of TSV*EDGENEIGHBOR cells are defined, they are placed in the R0 or MX orientation. If only `TSVLEFTTOPEDGENEIGHBOR` and `TSVLEFTBOTTOMEDGENEIGHBOR` are defined, they can be y-flipped for `TSVRIGHTTOPEDGENEIGHBOR` and `TSVRIGHTBOTTOMEDGENEIGHBOR`. However, `TSVLEFTTOPEDGENEIGHBOR` and `TSVRIGHTTOPEDGENEIGHBOR` cannot be x-flipped to replace `TSVLEFTBOTTOMEDGENEIGHBOR` and `TSVRIGHTBOTTOMEDGENEIGHBOR` because the power/ground pins cannot be matched.

`TSVLEFTTOPCORNERNEIGHBOR | TSVRIGHTTOPCORNERNEIGHBOR`
`| TSVLEFTBOTTOMCORNERNEIGHBOR | TSVRIGHTBOTTOMCORNERNEIGHBOR`

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Specifies a top- or bottom-edge endcap cell, which is right next to the TSVLEFTTOPCORNER/TSVRIGHTTOPCORNER or TSVLEFTBOTTOMCORNER/TSVRIGHTBOTTOMCORNER cell. When all four types of TSV*CORNERNEIGHBOR cells are defined, they are placed in the R0 or MX orientation. If only one of the four cells, such as TSVRIGHTTOPCORNERNEIGHBOR, is specified, it will be flipped and rotated to be used in all four locations, provided the row of the location needing the bottom edge is flipped (MX). If both the top and bottom rows are in R0 orient, the same cell cannot be used in place of TSVLEFTBOTTOMCORNERNEIGHBOR/ TSVRIGHTBOTTOMCORNERNEIGHBOR, as that would result in shorting power to ground. If only TSVLEFTTOPCORNERNEIGHBOR and TSVLEFTBOTTOMCORNERNEIGHBOR are provided, they can be y-flipped for TSVRIGHTTOPCORNERNEIGHBOR and TSVRIGHTBOTTOMCORNERNEIGHBOR.

Class Rule Examples

- The following example indicates that macro A is a cover macro for metal filling purpose:

```
MACRO A
    PROPERTY LEF58_CLASS “CLASS COVER FILL ; ” ;
...
END A
```

- The following example shows four different set of configurations for class rule:

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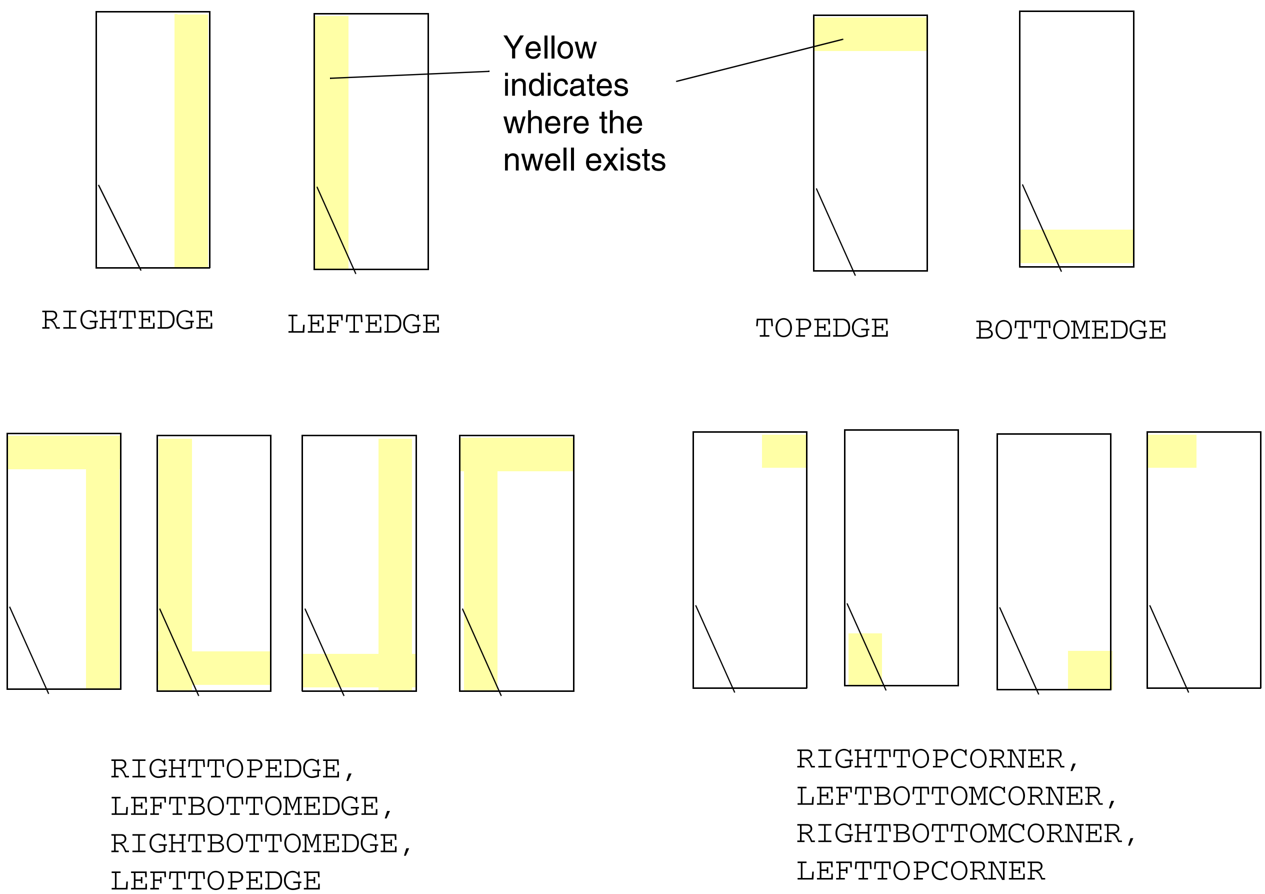
Figure 1-34 Illustration of Class Rule



A library can have cells in all of these configurations, but not likely nor necessarily. At minimum, one configuration from each of the 4 sets must be specified. It is expected that the available configuration from `TOPEDGE` or `BOTTOMEDGE` would have a list of cells of different size (similar to fillers), in order to reduce number of cells used to fill up the row.

- The following example shows the use of `RIGHTEDGE (RE)`, `TOPEDGE (TE)`, `LEFTOPEDGE (LTE)` and `LEFTTOPCORNER (LTC)` endcap cells:
  - `RIGHTEDGE` in N and FS orientation is used in start of rows
  - `RIGHTEDGE` in FN and S orientation is used in end of rows
  - `TOPEDGE` in N orientation is used in bottom row of the core or top row right above a block macro (the first row should have N orientation; otherwise, the bottom FS row would be wasted)

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- `TOPEDGE` in FS orientation is used in top row of the core or bottom row right below a block macro
- `LEFTTOPEDGE` is used at the 4 outer corners in N (left-bottom corner), FN (right-bottom), S (right-top) and FS (left-top) orientation
- `LEFTTOPCORNER` is used at the 4 inner corners similarly. Hence, all the cells must have `SYMMETRY X Y`.

Figure 1-35 Illustration of Class Rule

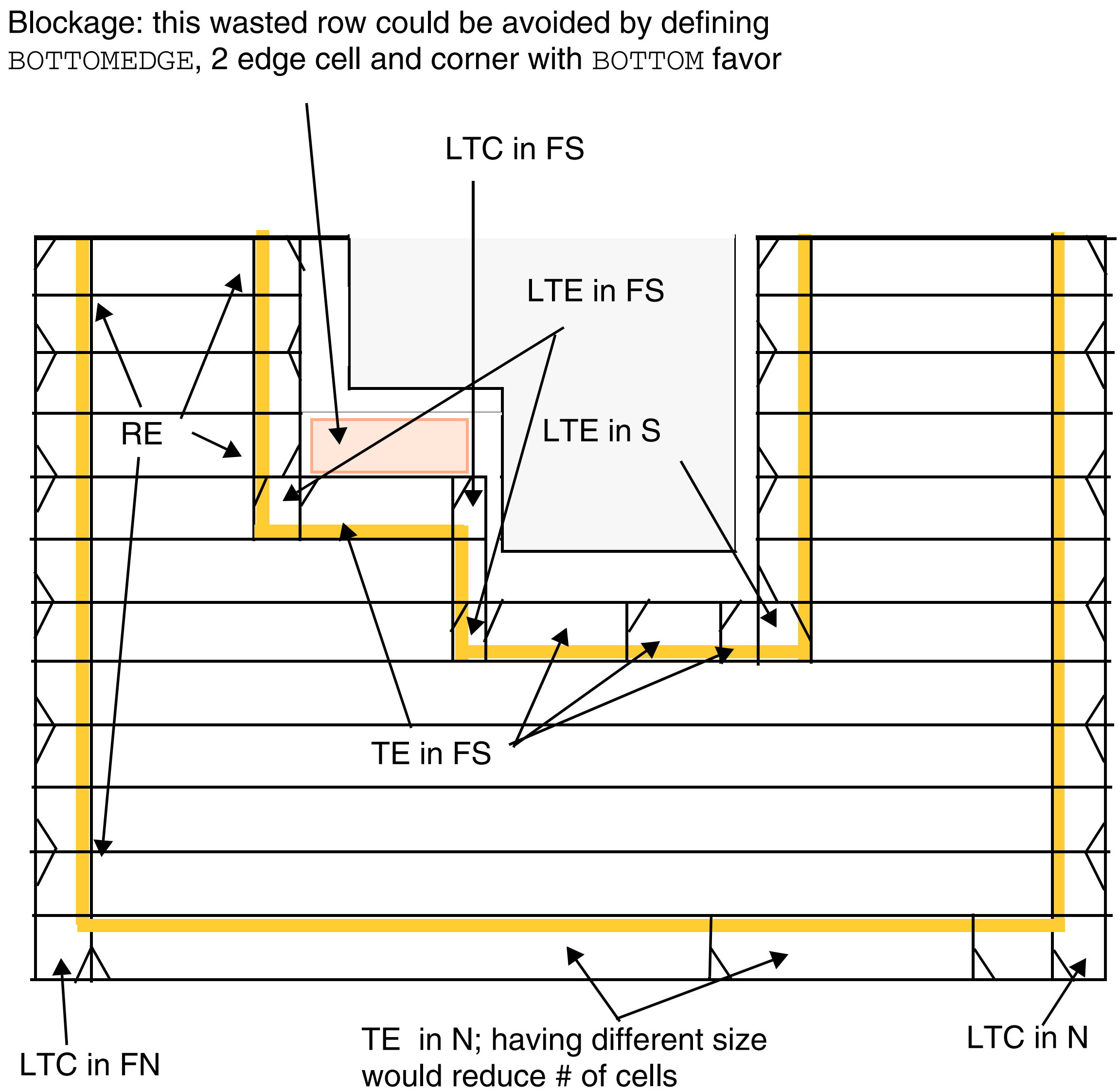

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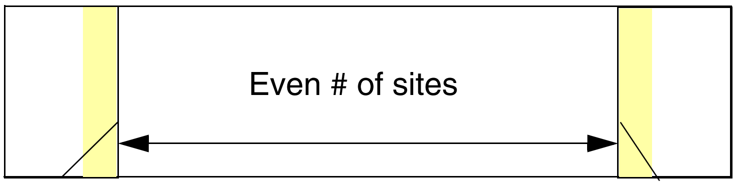
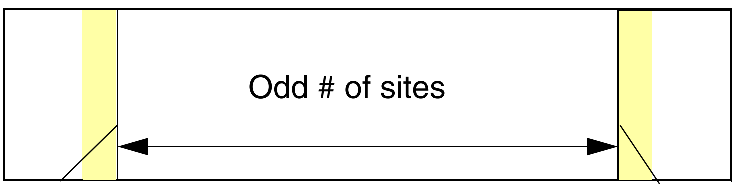
Figure 1-36 Illustration of Class Rule



The above endcap cells must have `LEFTEVENSITEEDGE CLASS`



One of the endcap cells must have `LEFTODDSITEEDGE CLASS` & another must have `LEFTEVENSITEEDGE CLASS` such that the # of sites including the endcap cells would be even

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Figure 1-37 Illustration of `LEFTEDGETOPBORDER`**,** `LEFTEDGEBOTTOMBORDER`**,** `RIGHTEDGETOPBORDER` **and** `RIGHTEDGEBOTTOMBORDER`

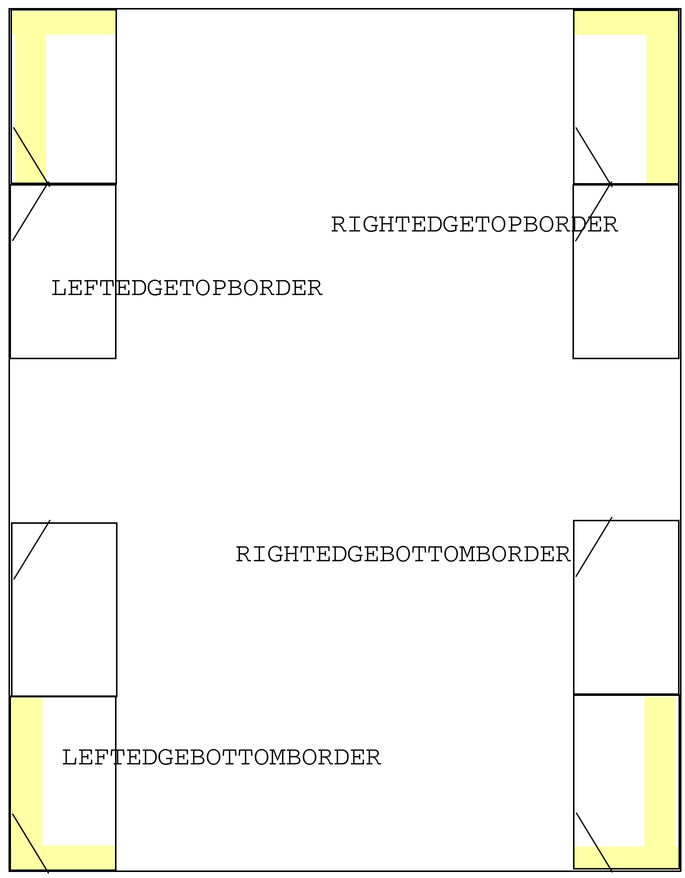

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Figure 1-38 Illustration of `LEFTTOPEDGENEIGHBOR`**,**`RIGHTTOPEDGENEIGHBOR`**,** `LEFTBOTTOMEDGENEIGHBOR`**,** `RIGHTBOTTOMEDGENEIGHBOR`, `LEFTTOPCORNERNEIGHBOR`, `RIGHTTOPCORNERNEIGHBOR`, `LEFTBOTTOMCORNERNEIGHBOR` **and** `RIGHTBOTTOMCORNERNEIGHBOR`

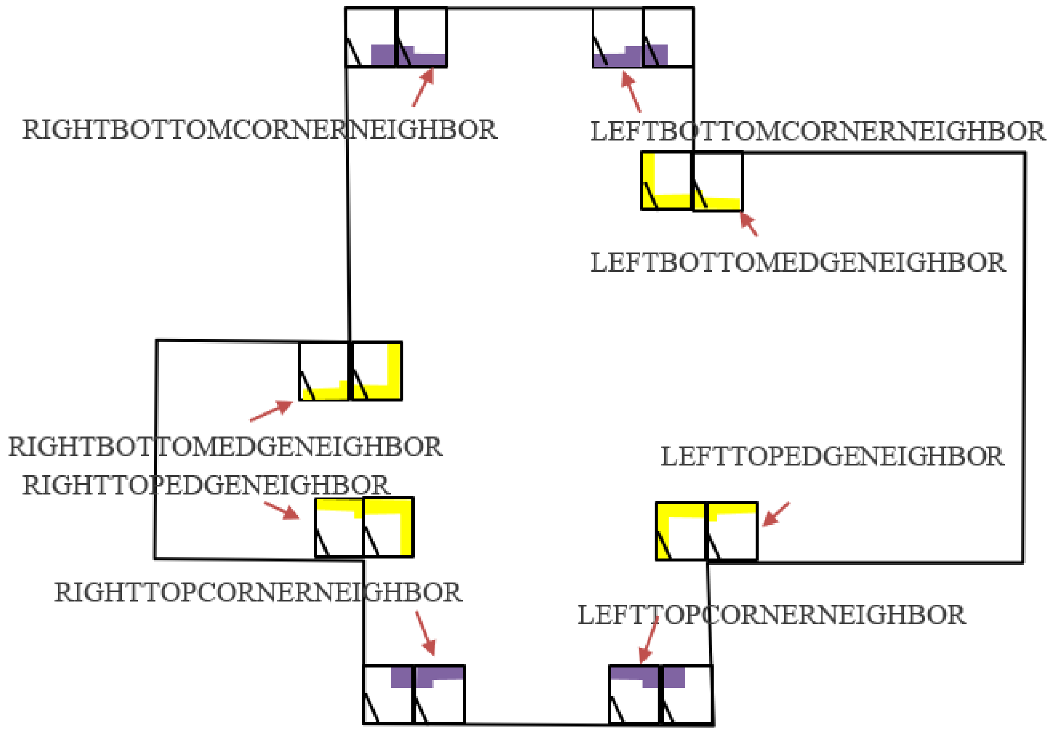

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Figure 1-39 Illustration of `LEFTCORNERTOPBORDER`**,** `RIGHTCORNERTOPBORDER`**,** `LEFTCORNERBOTTOMBORDER`**,** `RIGHTCORNERBOTTOMBORDER`**,** `TOPCORNEREDGENEIGHBOR`**, and** `BOTTOMCORNEREDGENEIGHBOR`

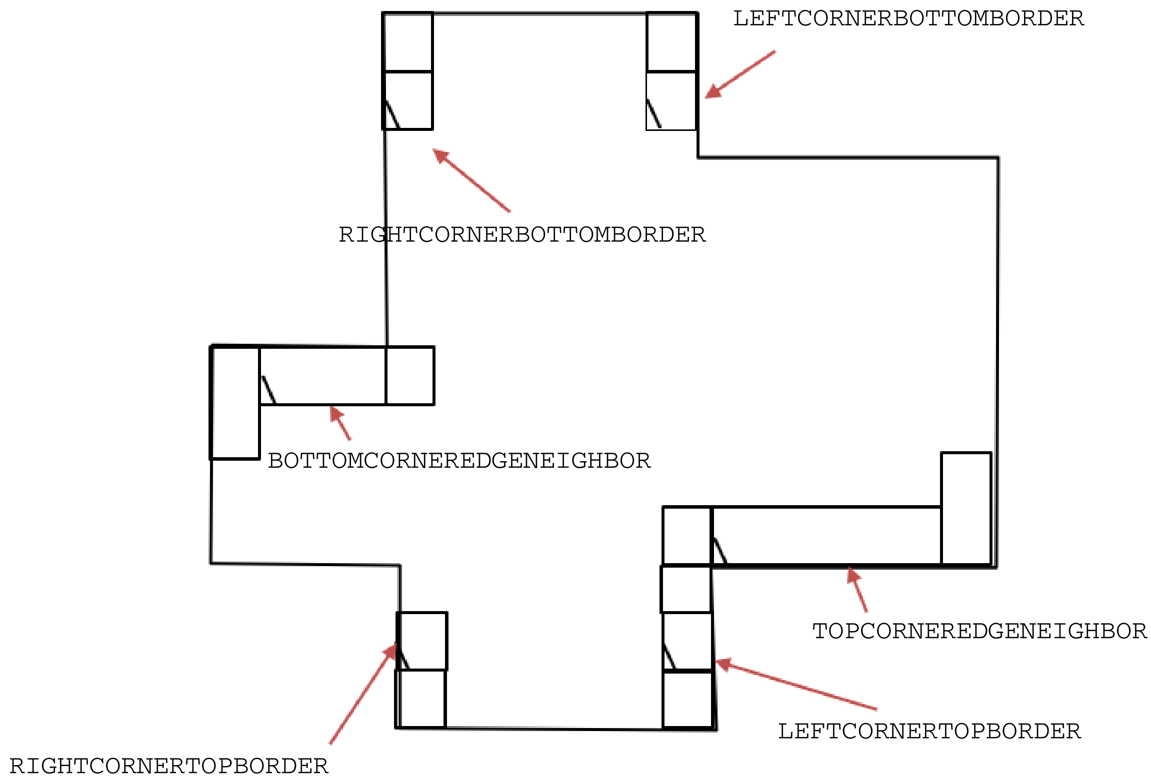

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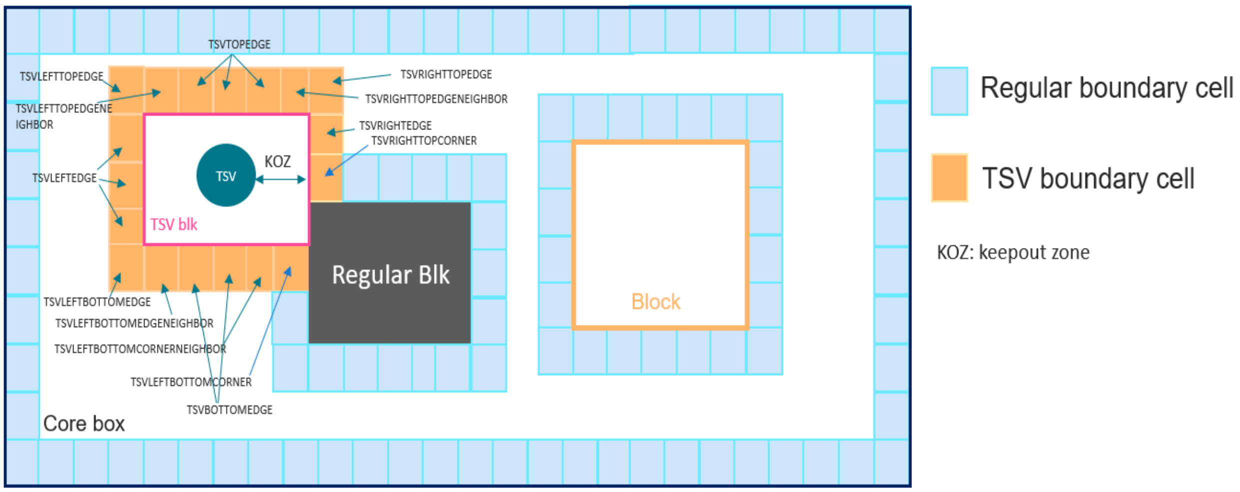
Figure 1-40 Illustration of TSV* End Cap Cells



Constraint Area Type Rule

The constraint area type rule can be used to define a list of constraint rectangular areas.

You can create a constraint area type rule by using the following property definition:

```
PROPERTY LEF58_CONSTRAINTAREATYPE
        "“CONSTRAINTAREATYPE typeName {RECT pt pt}...
         ;...” ;
```

Where:

`CONSTRAINTAREATYPE` *`typeName`* `{RECT` *`pt pt`*`}...`

Defines a list of constraint rectangular areas in `RECT` with respect to the cell origin with type name, *`typeName`*. Then, rules such as length limit, could be defined on those areas in separate LEF constructs, such as `CONSTRAINTLENGTH`.

Type: Float, specified in microns

Edge Type Rule

The edge type rule can be used to define edge types for right and left edges.

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You can create an edge type rule by using the following property definition:

```
PROPERTY LEF58_EDGETYPE
     "EDGETYPE {RIGHT | LEFT | TOP | BOTTOM}
         edgeType | BOTHSOURCE | SOURCEDRAIN | DRAINSOURCE | BOTHDRAIN
         | BOTHFLOATING
         [CELLROW cellRow [RANGE xLow xHigh] | HALFROW halfRow
         | RANGE xLow xHigh]
         ;..." ;
```

Where：

CELLROW *cellRow* [RANGE *xLow xHigh*]

Defines the cell row, *cellRow*, on which the *edgeType* is defined for multiple-height cells. Acceptable values for *cellRow* range from 1 to *n*, where *n* corresponds to the number of cell heights in the cell. By default, *edgeType* applies to the entire edge, encompassing all cell rows. Specifying 1 will apply *edgeType* to only the bottommost cell row.

RANGE can be defined only on TOP or BOTTOM edges to label a portion of the edge as the specified type.

Type: Float, specified in microns

EDGETYPE { RIGHT | LEFT | TOP | BOTTOM } *edgeType*
|BOTHSOURCE | SOURCEDRAIN | DRAINSOURCE | BOTHDRAIN | BOTHFLOATING

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Defines an edge type, *edgeType*, on the specified edge (RIGHT, LEFT, TOP, or BOTTOM) such that a spacing can be defined in CELLEDGESPACINGTABLE by referring to the types. For a macro, you can specify multiple statements on one edge.

BOTHSOURCE, SOURCEDRAIN, DRAINSOURCE, BOTHDRAIN, and BOTHFLOATING are special keywords that can be defined only on the LEFT or RIGHT edges:

- BOTHSOURCE indicates that the entire portion of that edge is a source.
- BOTHDRAIN indicates that the entire portion of that edge is a drain.
- BOTHFLOATING indicates that the entire portion of that edge is floating.
- SOURCEDRAIN indicates that the top portion is a source and the bottom portion is a drain.
- DRAINSOURCE indicates that the top portion is a drain and the bottom portion is a source.

Edges of those special types may or may not have corresponding spacing defined in CELLEDGESPACINGTABLE. They are used for ensuring a common edge type be defined in different cell libraries such that spacing can be defined in some combinations on those edge types in CELLEDGESPACINGTABLE.

See Figure 1-43 on page 140.

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`HALFROW` *`halfRow`*	Defines the edge type per half row, which can be defined only on the `LEFT` or `RIGHT` edges. For a single-height cell, the *`halfRow`* value of `1` would apply the given edge type on the bottom half of the cell while a value of `2` would apply the given edge type on the top half of the cell. Multiple height cells would have the same meaning and values of `3` or above would be used to represent the second and higher rows. For example, a double-height cell can have *`halfRow`* values from `1` to `4`, where the *`halfRow`* value of `1` would apply the given edge type on the bottom half of the first row while a value of `4` would apply the given edge type on the top half of the second row. See Figure 1-42 on page 139. *Type*: Integer
`RANGE` *`xLow xHigh`*	Defines a partial range of an edge that the type should be labeled. This can only be defined on `TOP` or `BOTTOM` edges. *Type*: Float, specified in microns

You can define multiple `EDGETYPE` statements on an edge, including single height cells, such that different type of constraints can be defined on an edge.

Edge Type Rule Examples

- The following statement indicates that edge type `GROUP1` contains the right edge of `MACRO1` and `GROUP2` contains the left edge of `MACRO1` and right edge of `MACRO2`:

```
MACRO MACRO1
…
PROPERTY LEF58_EDGETYPE "
    EDGETYPE RIGHT GROUP1 ;
    EDGETYPE LEFT GROUP2 ; " ;
END MACRO1

MACRO MACRO2
…
PROPERTY LEF58_EDGETYPE "
    EDGETYPE RIGHT GROUP2 ; " ;
END MACRO2
```


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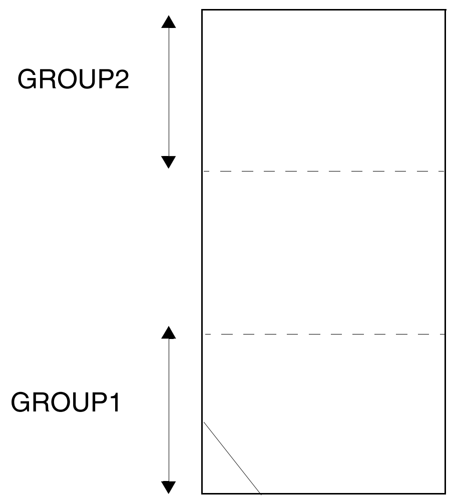
- The following example indicates that `GROUP1` and `GROUP2` are defined on the first and last (third) row, respectively, on the left edge for the triple-height cell:

```
PROPERTY LEF58_EDGETYPE "
    EDGETYPE LEFT GROUP1 CELLROW 1 ;
    EDGETYPE LEFT GROUP2 CELLROW 3 ; " ;
```

Figure 1-41 Illustration of Macro with Cell Row



- The following example indicates that `GROUP1` and `GROUP2` are defined on the first and last (fourth) half row, respectively, on the right edge of the double-height cell:

```
PROPERTY LEF58_EDGETYPE "
    EDGETYPE RIGHT GROUP1 HALFROW 1 ;
    EDGETYPE RIGHT GROUP2 HALFROW 4 ; " ;
```


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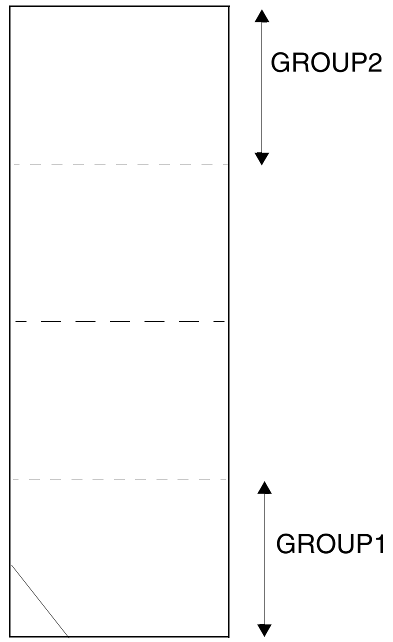
Figure 1-42 Illustration of Macro with Half Row



- The following example illustrates the edge type rule with special edge type keywords:

```
MACRO A
...
   PROPERTY LEF58_EDGETYPE
      "EDGETYPE LEFT DRAINSOURCE ;
       EDGETYPE RIGHT SOURCEDRAIN ; " ;
...
END A
MACRO B
...
   PROPERTY LEF58_EDGETYPE "
      EDGETYPE LEFT BOTHDRAIN ;
      EDGETYPE RIGHT BOTHSOURCE ; " ;
...
END B
```


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Figure 1-43 Illustration of Macro with special edge type keywords

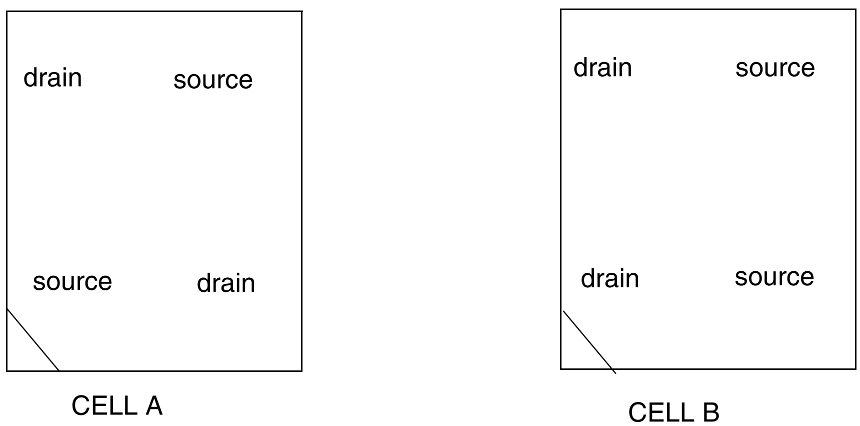
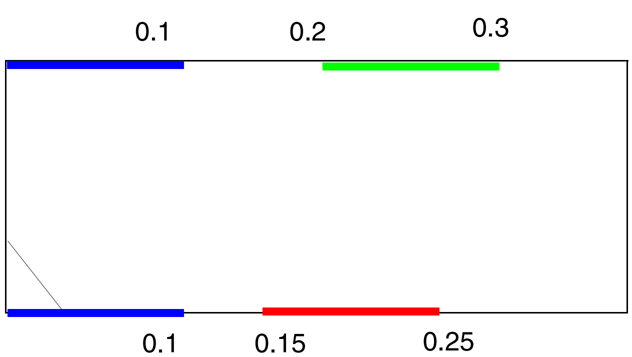


Figure 1-44 Illustration of Macro with Range

```
MACRO A
...
PROPERTY LEF58_EDGETYPE "
        EDGETYPE TOP GROUP1 RANGE 0 0.1 ;
        EDGETYPE BOTTOM GROUP1 RANGE 0 0.1 ;
        EDGETYPE TOP GROUP2 RANGE 0.2 0.3 ;
         EDGETYPE BOTTOM GROUP3 RANGE 0.15 0.25 ; " ;
...
END A
```



Blue line indicates GROUP1 edge
Green line indicates GROUP2 edge
Red line indicates GROUP3 edge

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EEQ Rule

You can use the EEQ rule to specify the electrically equivalent (EEQ) variant for macros with class `CORE` standard cells.

You can create an EEQ rule by using the following property definition:

```
PROPERTY LEF58_EEQ
          "EEQ macroName [VARIANT num]
          ;... " ;
```

Where:

All other keywords are same as the existing macro `EEQ` syntax.

`VARIANT` *`num`*

Specifies the EEQ variant number as *`num`*. This should be defined consistently with `CELLVARIANTS`, which indicates the usage. This construct should be applied only on `MACRO` with `CLASS CORE` standard cells.
Type: Integer

Foreign Rule

You can use the foreign rule to indicate that the foreign structure of the specified foreign orient should be used if a component of the macro matches one of the given component orientations.

You can create foreign rule by using the following property definition:

```
PROPERTY LEF58_FOREIGN
          "FOREIGN foreignCellName [pt [orient]]
               [COMPORIENT foreignOrientName {compOrient}...]...
          ;... " ;
```

Where：

All other keywords are same as the existing macro `FOREIGN` syntax.

`COMPORIENT` *`foreignOrientName`* {*`compOrient`*}...

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Specifies that the foreign structure of *foreignOrientName* should be used if a component of the macro matches one of the given component orientations specified using *compOrient*. Otherwise, all the other components will use *foreignCellName*.
Values: N, S, E, W, FN, FS, FE, or FN

FOREIGN Rule Examples

The following example indicates that the foreign structure of foreign orient `EFG` should be applied if the instance is placed with orientation of `S` or `FS`, and all the other instances use the foreign orient `ABC`.

```
PROPERTY LEF58_FOREIGN
     "FOREIGN ABC COMPORIENT EFG S FS ; " ;
```

Layer Mask Shift Rule

You can create a layer mask shift rule to specify that the mask could be shifted on the given layers.

You can define a layer mask shift rule by using the following property definition:

```
PROPERTY LEF58_LAYERMASKSHIFT
     "LAYERMASKSHIFT layer1 [layer2 ...]
          ;" ;
```

Where:

`LAYERMASKSHIFT` *`layer1`* `[`*`layer2`* `...]`

Specifies that mask could be shifted on the given layers only on the given `MACRO`. `LAYERMASKSHIFT` should not be specified as a library property if any macro has this `LAYERMASKSHIFT` construct. In addition, `FIXEDMASK` must be defined either as a library property or on this macro. This `LAYERMASKSHIFT` construct then overrides `FIXEDMASK` and allows mask shifting on the specified layers on this macro.

Type: String

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OBS Partial Rule

You can use the OBS partial rule to indicate a partially routing blockage.

You can create a OBS partial rule by using the following property definition:

```
PROPERTY LEF58_OBSPARTIAL
     "OBSPARTIAL {LAYER layerName
     WIDTH width SPACING spacing RECT pt pt}...
     ; " ;
```

Where：

`OBSPARTIAL {LAYER layerName`
`WIDTH width SPACING spacing RECT pt pt }...`

Specifies a partially routing blockage. Any wires in the given RECT area on *layerName* must have minimum width of *width* and spacing of *spacing*. In addition, wires cannot change layer within that area.
Type: Float, specified in microns

OBS Partial Rule Example

```
MACRO A
...
PROPERTY LEF58_OBSPARTIAL
   "OBSPARTIAL LAYER M2 WIDTH 0.2 SPACING 0.2
       RECT -1.0 -1.0 10.0 8.0 ; " ;
```


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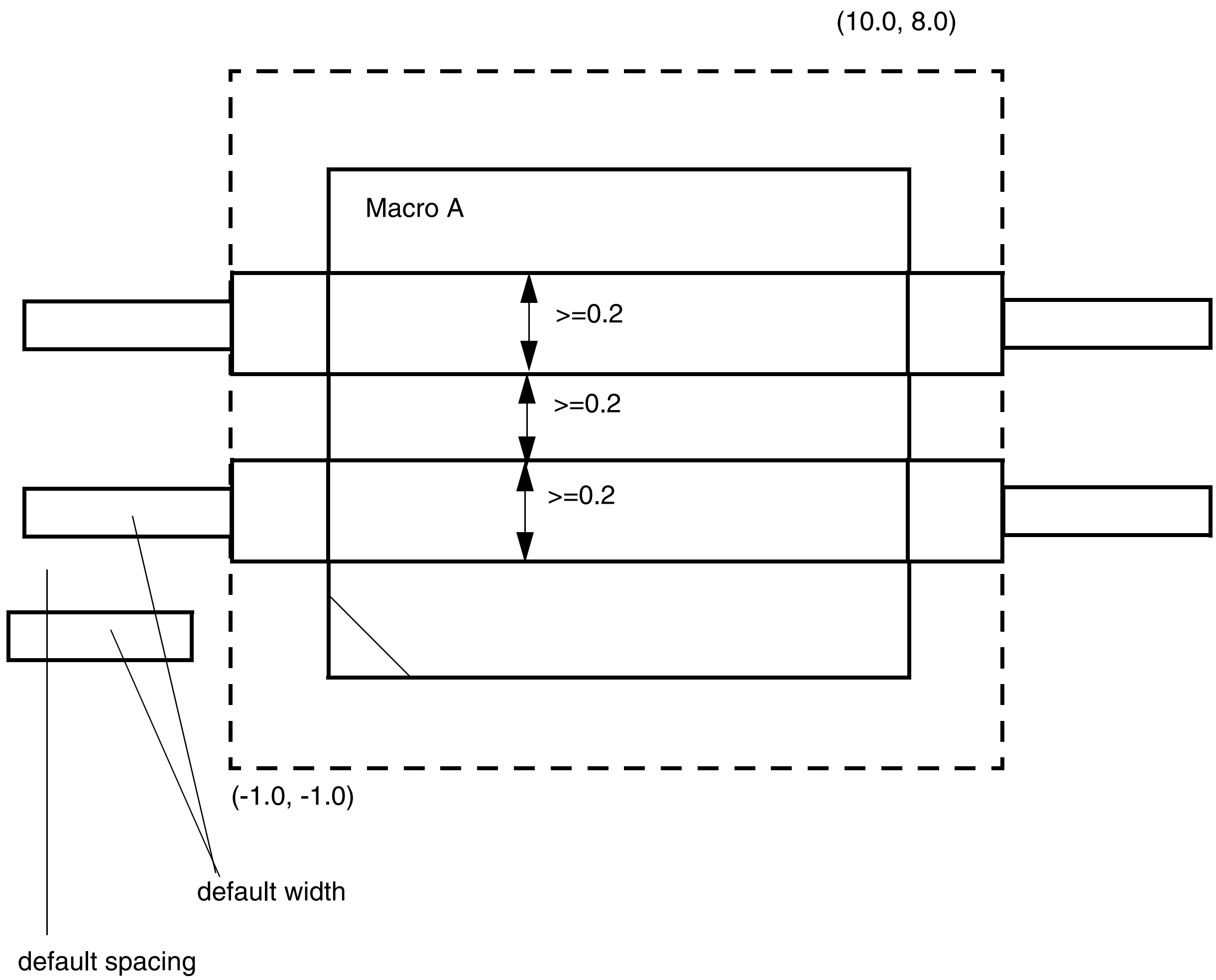
Figure 1-45 Illustration of OBS Partial Rule



All of the wires in the dotted area on layer M2 must
have minimum width of 0.2 and spacing of 0.2.

OBS Spacing Rule

You can use the OBS spacing rule to specify spacing for the OBS of the macro.

You can create a OBS spacing rule by using the following property definition:

```
PROPERTY LEF58_OBSSPACING
       "OBSSPACING {FULLDRC | MIN | spacing} [LAYER layer]...
       ; " ;
```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Where:

`FULLDRC`
: Specifies that any OBS shapes without `SPACING` or `DESIGNRULEWIDTH` should be treated as a real geometry so all design rule check (DRC) rules will be checked for these shapes. By default, blocks and pad cells (MACROs with a `CLASS` of `BLOCK`, `RING` or `PAD`) treat OBS shapes as "abstract" blockages to prevent routing over the block, and only simple DRC spacing checks are done to the OBS shapes. Note that standard cells (`CLASS CORE` or `ENDCAP`) always treat OBS shapes as real geometry with full DRC checks. Therefore, `FULLDRC` should normally only be used on one or two top routing layers of a block that has very sparse routing to make room for the top-level to route across the block. Note that this may cause extra cross-coupling between the block and top-level routing, so `FULLDRC` should be used carefully.

`LAYER` *layer*
: Specifies that the given spacing would be applied on OBS of this macro only on the given *layer*.

`OBSSPACING {MIN |` *spacing*`}`
: Specifies that OBS of this macro would either use min spacing if `MIN` is given or the given *spacing* value. This is equivalent to using `OBS SPACING` value for all the OBS shapes in this LEF MACRO. If `LAYER` is specified, the given spacing is applied on OBS of this macro only on the given *layer*. If `LAYER` is not specified, the spacing value applies to all the OBS shapes. Note that this construct must be used with caution because the defined spacing would always be used even on a neighbor wide wire, which may require a larger spacing.
Type: Float, specified in microns

OBS Spacing Rule Examples

- The following example indicates that all of the OBS on M3 would use 0.2 um while the rest of OBS uses min spacing.

```
PROPERTY LEF58_OBSSPACING "
      OBSSPACING MIN ;
      OBSSPACING 0.2 LAYER M3 ; " ;
```


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- The following example indicates that all of the OBS on M1 and M2 of this macro would use min spacing while M3 OBS would use 0.2um.

```
PROPERTY LEF58_OBSSPACING "
      OBSSPACING MIN LAYER M1 LAYER M2 ;
      OBSSPACING 0.2 LAYER M3 ;" ;
```

- The following example indicates that all of the OBS on M7 and M8 are real geometries that get full DRC checks, while the OBS on other layers are “abstract” blockages that only need min spacing checked.

```
PROPERTY LEF58_OBSSPACING "
      OBSSPACING MIN ;
      OBSSPACING FULLDRC LAYER M7 LAYER M8 ;" ;
```

Place Within Rule

You can use the OBS spacing rule to specify place within conditions for a filler cell.

You can create a place within rule by using the following property definition:

```
PROPERTY LEF58_PLACEWITHIN
       "PLACEWITHIN within PRL prl {WITH|WITHOUT} LAYER {layerName}...
       ; " ;
```

Where：

```
PLACEWITHIN within PRL prl {WITH|WITHOUT} LAYER {layerName}...
```


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Specifies that this filler cell must be placed under certain within conditions, and these conditions should be fulfilled horizontally on the same row.

`WITH` means that this cell must be placed horizontally within *`within`* on both sides of another cell with shapes on any given layer in *`layerName`* and should have a PRL with that cell greater than or equal to *`prl`*. Abutment with such a neighboring cell would fulfill the condition on that side.

`WITHOUT` means that this cell must be placed horizontally in a location without a neighboring cell with shapes on any given layer in *`layerName`* within *`within`* and having a PRL with that cell greater than or equal to *`prl`*. This means that either the spacing to such a neighboring cell should be greater than or equal to *`within`* or the PRL to that cell should be less than *`prl`*. Abutment with such a neighboring cell is allowed on only one side, and the other side needs to fulfill the within and PRL conditions. In other words, abutment on both sides would be a violation.

This cell should contain shapes on a layer with type `ABUTFILLER`. *`layerName`* should be a layer with type `ABUTLOGIC`.

Type: Float, specified in microns

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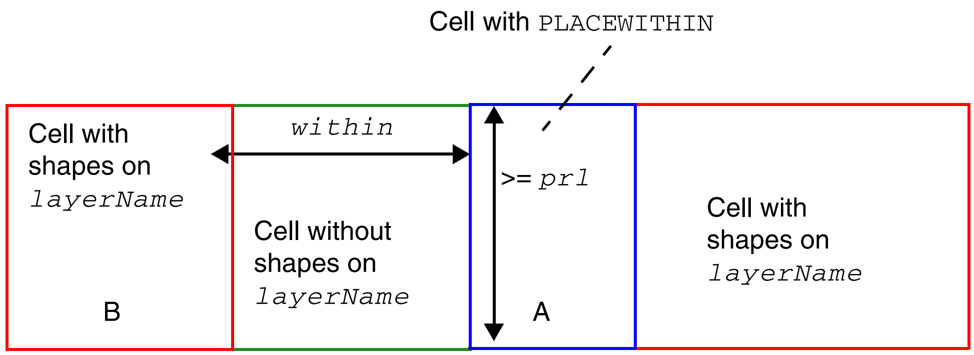
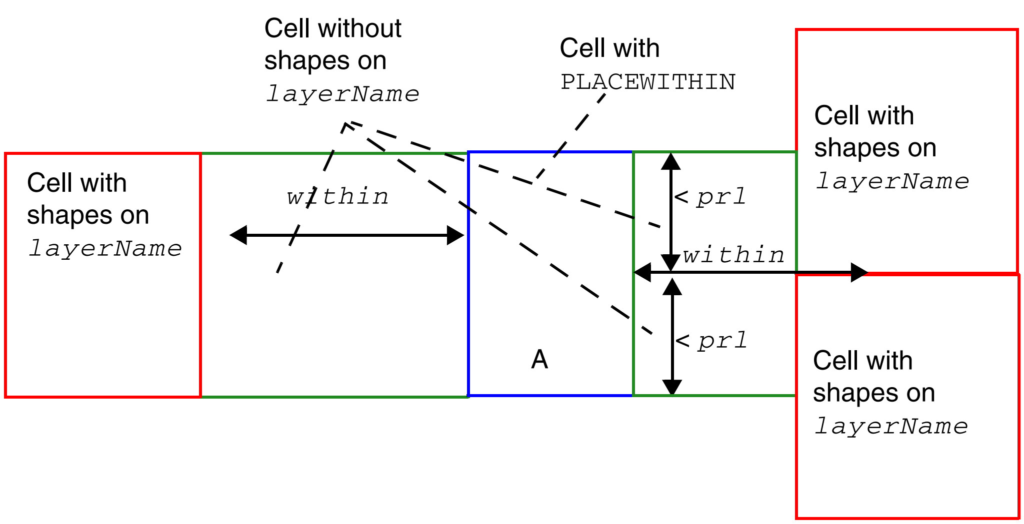
Figure 1-46 Illustration of the Place Within Rule Using WITH

Illustration of `PLACEWITHIN` *`within`* `PRL` *`prl`* `WITH LAYER {`*`layerName`*`}...`



Cell A is fine because it has a proper neighbor cell B on the left and abuts to another proper neighbor cell on the right.



Cell A has issues on both sides. The proper neighbor cell on the left is too far away and the proper neighbor cells on the right have PRL < *`prl`* individually.

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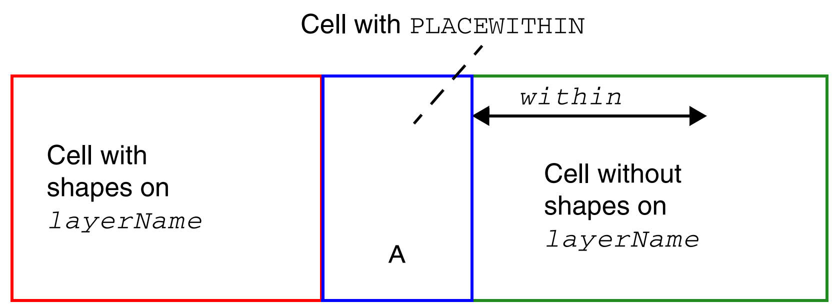
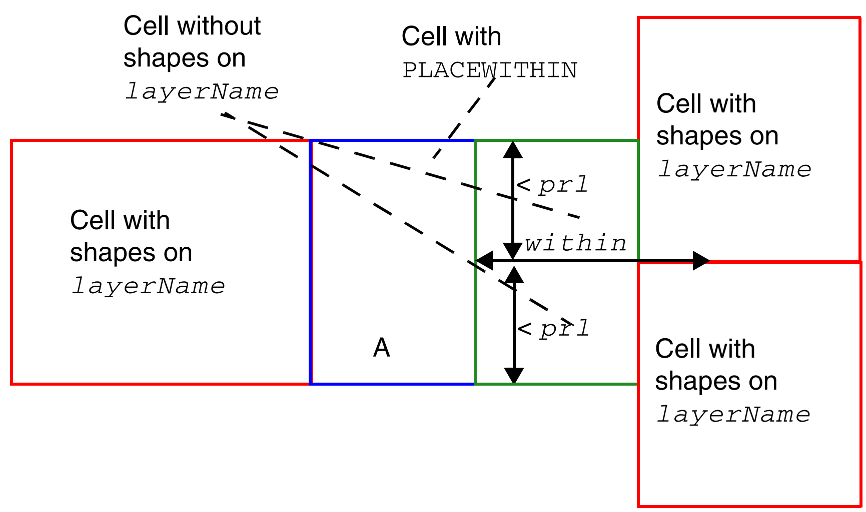
Figure 1-47 Illustration of the Place Within Rule Using WITHOUT

Illustration of `PLACEWITHIN` *`within`* `PRL` *`prl`* `WITHOUT LAYER {`*`layerName`*`}...`



Cell A is fine because it does not have any neighbor cells with shapes on *`layerName`* within *`within`* on the right and abutment on only the left side is allowed.



Cell A is fine. Although it has two neighbor cells with shapes on *`layerName`* within *`within`* on the right, each of them has an individual PRL < *`prl`* to pass the condition.

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REQ Rule

You can define a REQ rule by using the following property definition:

```
PROPERTY LEF58_REQ
        "REQ macroName
         ; " ;
```

Where:

`REQ macroName`	Specifies that the macro being defined should be a router-preferred equivalent (REQ) to the previously defined *`macroName`*. REQ cells can be placed at the same site to achieve a different routing pattern depending on the design context. This is different from EEQ cells, which provide cell placement flexibility at different sites. In addition, timing variation of REQ cells can be different from EEQ cells. This construct can only be defined on macros with the class CORE standard cells and should be mutually exclusive with EEQ. *Type*: String

Tap Center Offset Rule

You can use the tap center offset rule to specify an offset value from the cell center from which the tap distance should be measured.

You can create a tap center offset rule by using the following property definition:

```
PROPERTY LEF58_TAPCENTEROFFSET
       "TAPCENTEROFFSET
         {offset | {HALFROW rowNum [LEFT|RIGHT] offset}...}
       ; " ;
```

Where:

```
TAPCENTEROFFSET
    {offset | {HALFROW rowNum [LEFT|RIGHT] offset}...}
```


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Specifies an offset value from the cell center from which the tap distance should be measured. A positive value moves the starting point away from the center by the given amount. A negative value moves the starting point behind the center or away from the center in the other direction.
Type: Float, specified in microns

`HALFROW` specifies that the offset could be defined on a half row basis. *`rowNum`* specifies the half row, with respect to the cell origin, on which the offset is defined.

`LEFT|RIGHT` specifies that the offset could be defined on the left or right sides on a half cell row basis. Left and right is referred with respect to the cell origin.

Tap Center Offset Rule Example

```
MACRO A
PROPERTY LEF58_TAPCENTEROFFSET "
     TAPCENTEROFFSET HALFROW 1 2.4
          HALFROW 2 LEFT -1.2 HALFROW 2 RIGHT 3.6 ; " ;
...
END A
```


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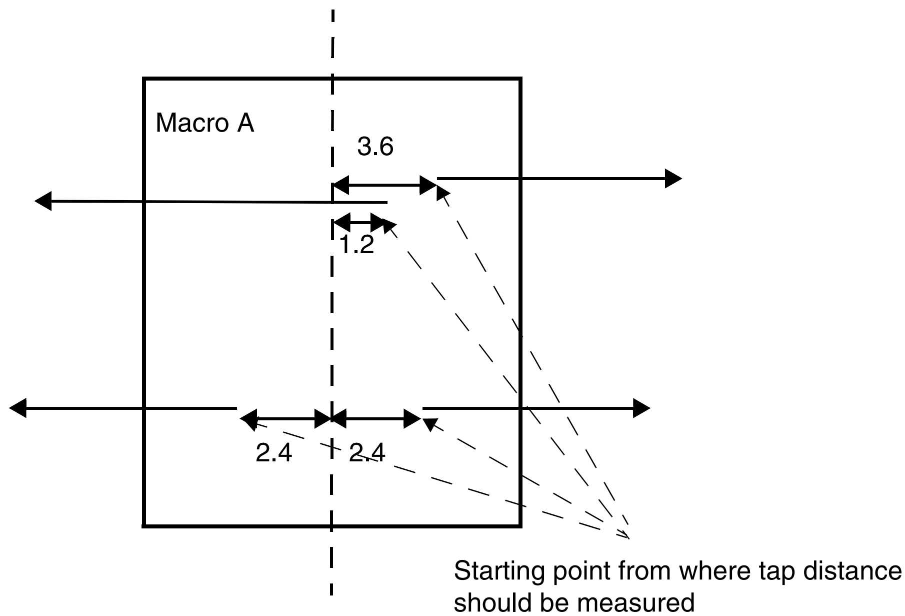
Figure 1-48 Illustration of Tap Center Offset Rule



Variant Rule

You can use the variant rule to specify the variant number of a cell.

You can create a variant rule by using the following property definition:

```
PROPERTY LEF58_VARIANT
        "VARIANT num
       ; " ;
```

Where:

`VARIANT num`	Specifies the variant number of this cell as *num*. This should be defined consistently with `CELLVARIANTS`, which indicates the usage. This construct should be applied only on `MACRO` with `CLASS CORE` standard cells. It is meant for non-EEQ cell master only as EEQ cells can define the variant number in the EEQ construct. *Type*: Integer

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Defining Cover Macros

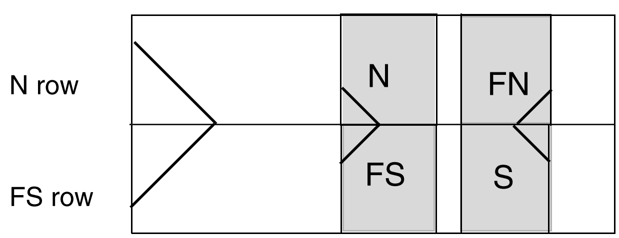
If you define a cover macro with its actual size, some place-and-route tools cannot place the rest of the cells in your design because it uses the cell boundary to check for overlaps. You can resolve this in two ways:

- The easiest way to support a cover macro is to define the cover macro with a small size, for example, 1 by 1.
- If you want to define the cover macro with its actual size, create an overlap layer with the non-routing `LAYER TYPE OVERLAP` statement. You define this overlap layer (cover macro) with the macro obstruction (`OBS`) statement.

Defining Symmetry

Symmetry statements specify legal orientations for sites and macros. Figure 1-49 on page 153 illustrates the normal orientations for single-height, flipped and abutted rows with standard cells and sites.

Figure 1-49 Normal Orientations for Single-Height Rows



The following examples describe typical combinations of orientations for standard cells. Applications typically create only N (or FS for flipped) row orientations for horizontal standard cell rows; therefore, the examples describe these two rows.

Example 1-6 Single-Height Cells

Single-height cells for flipped and abutted rows should have `SITE` symmetry `Y` and `MACRO` symmetry `X Y`. These specifications allow N and FN macros in N rows, and FS and S macros in FS rows, see Figure 1-50 on page 154. These symmetries work with flipped and abutted rows, as well as rows that are not flipped and abutted, so if the rows are all N orientation, the cells all have N or FN orientation. The extra `MACRO` symmetry of `X` is not required in this case, but causes no problems.

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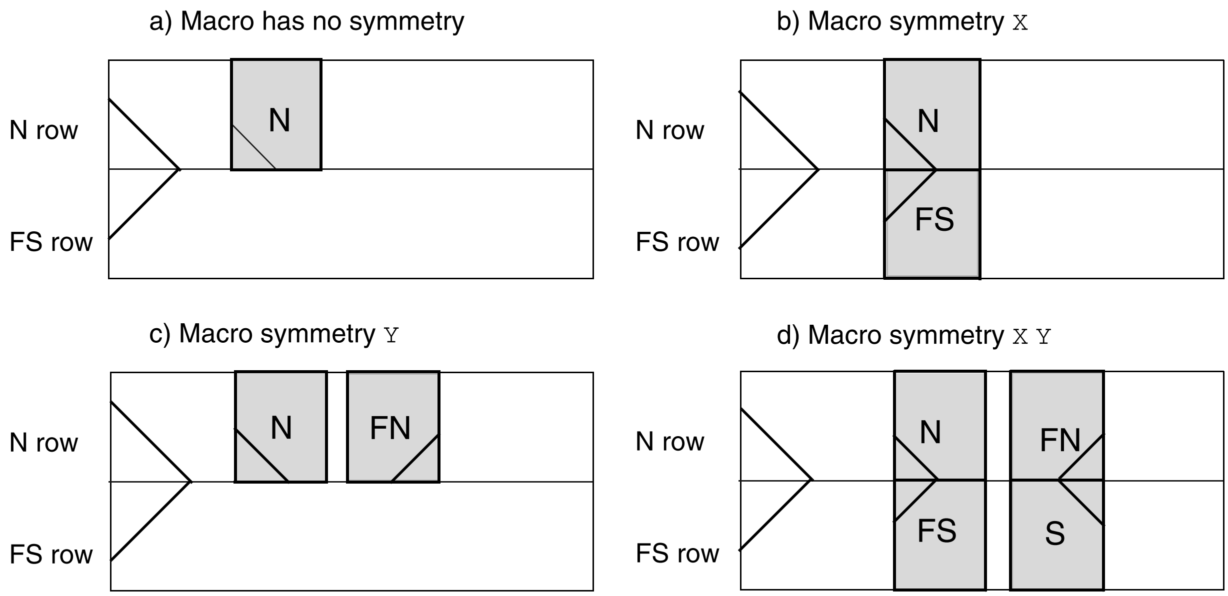
Figure 1-50 Legal Placements for Row Sites with Symmetry Y



Example 1-7 Double-Height Cells

Double-height cells that are intended to align with flipped and abutted single-height rows should have `SITE` symmetry `X Y` and `MACRO` symmetry `X Y`. These symmetries allow all four cell orientations (N, FN, FS, and S) to fit inside the double-height row (see Figure 1-51 on page 155). Usually, double-height rows are just N orientation rows that are abutted and aligned with a pair of single-height flipped and abutted rows.

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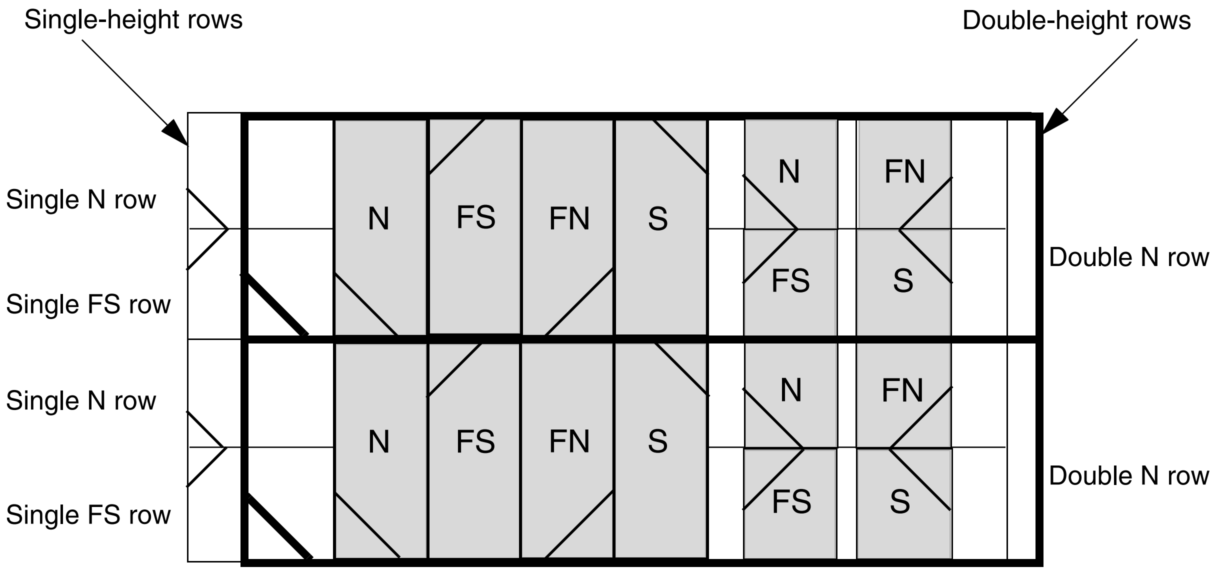
Figure 1-51 Legal Placements for Single-Height Row Sites with Symmetry Y and Double-Height Row Sites with Symmetry X Y



Note: The single-height rows are shifted slightly to the left of the double-height rows in the above figure for illustration purposes. In a real design, they should be aligned.

Example 1-8 Special Orientations

Some single-height cells have special orientation needs. For example, the design requires flipped and abutted rows, but only N and FS orientations are allowed because of the special layout of well taps on the right side of a group of cells that borrow from the left side of the next cell. That is, you cannot place an N and FN cell against each other in one row because only N cells are allowed in an N row. In this case, the `SITE` symmetry should not be defined, and the `MACRO` symmetry should be `X`. A `MACRO` symmetry of `X Y` can also be defined because the Y-flipped macros (FN and S orientations) do not match the N or FS rows. See Figure 1-52 on page 156 for the different combinations when the SITE has no symmetry.

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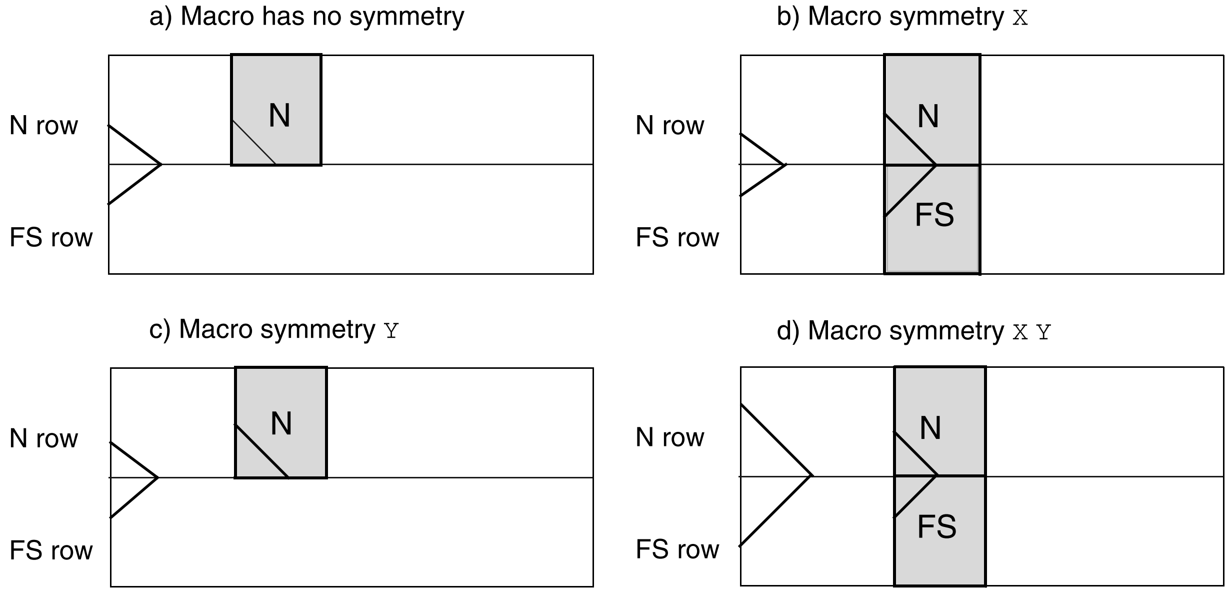
Figure 1-52 Legal Placements for Row Sites with No Symmetry



Example 1-9 Vertical Rows

Vertical rows use N or FN row and site orientations. The flipped, abutted vertical row orientation is N and FN, rather than the horizontal row orientation of N and FS. Otherwise, the meaning of the site symmetries and macro symmetries is the same as those for horizontal rows.

Single-height sites are normally given symmetry `X`, and single-height cells are normally given symmetry `X Y`. Example d in Figure 1-53 on page 157 shows the legal placement for a site with symmetry `X`, and the typical standard cell `MACRO` symmetry `X Y`.

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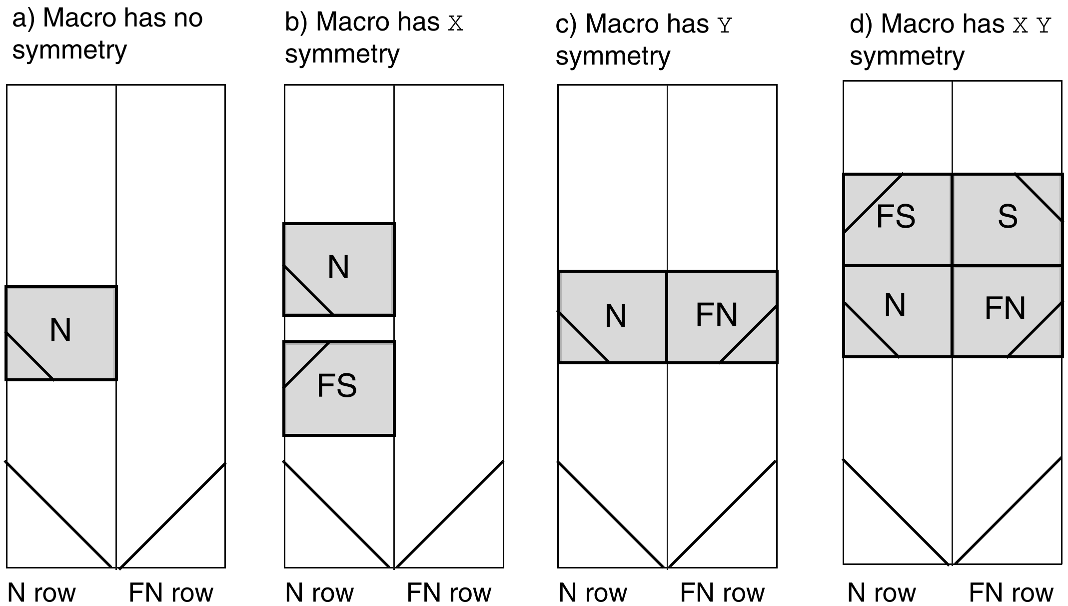
Figure 1-53 Legal Placements for Vertical Row Sites With Symmetry X



Layer Geometries

```
{ LAYER layerName
    [EXCEPTPGNET]
    [SPACING minSpacing | DESIGNRULEWIDTH value] ;
  [WIDTH width ;]
  { PATH [MASK maskNum] pt ... ;
  | PATH [MASK maskNum] ITERATE pt ... stepPattern ;
  | RECT [MASK maskNum] pt pt ;
  | RECT [MASK maskNum] ITERATE pt pt stepPattern ;
  | POLYGON [MASK maskNum] pt pt pt pt ... ;
  | POLYGON [MASK maskNum] ITERATE pt pt pt pt ... stepPattern ;
  } ...
| VIA [MASK viaMaskNum] pt viaName ;
| VIA ITERATE [MASK viaMaskNum] pt viaName stepPattern ;
} ...
```

Used in the macro obstruction (`OBS`) and pin port (`PIN`) statements to define layer geometries in the design.

`DESIGNRULEWIDTH` *`value`*

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	Specifies the effective design rule width. If specified, the obstruction or pin is treated as a shape of this width for all spacing checks. If you specify `DESIGNRULEWIDTH`, you cannot specify the `SPACING` argument. As a lot of spacing rules in advanced nodes no longer just rely on wire width, `DESIGNRULEWIDTH` is not allowed for 20nm and below nodes. *Type:* Float, specified in microns
`EXCEPTPGNET`	Indicates that the obstruction shapes block signal routing, but do *not* block power or ground routing. This can be used to block signal routes that might cause noise, but allow connections to power and ground pins.
`ITERATE`	Creates an array of the `PATH`, `RECT`, `POLYGON`, or `VIA` geometry, as specified by the given step pattern. `ITERATE` specifications simplify the definitions of cover macros. The syntax for *`stepPattern`* is defined as follows: `DO` *`numX`* `BY` *`numY`* `STEP` *`spaceX spaceY`* *`numX`* — Specifies the number of columns of points. *`numY`* — Specifies the number of rows of points. *`spaceX spaceY`* — Specifies the spacing, in distance units, between the columns and rows of points.
`LAYER` *`layerName`*	Specifies the layer on which to place the geometry. *`layerName`* can be any type of layers.

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`MASK` *`maskNum`*	Specifies which mask from double- or triple-patterning to use for this shape. The *`maskNum`* variable must be a positive integer. Most applications only support values of 1, 2, or 3. Shapes without any defined mask have no mask set (they are considered uncolored). The uncolored PIN shapes can be assigned to an arbitrary mask as long as they do not have a spacing conflict with neighbor objects. The meaning of uncolored OBS shapes depends on the cell. For standard cell MACROs (with a `SITE` that is `CLASS CORE`), the uncolored OBS shapes are considered to be real metal shapes that can be assigned to any mask as long as no mask spacing conflicts occur. For other MACRO types, uncolored OBS shapes are assumed to be abstractions that may be any mask, so other shapes must be spaced far enough away to avoid a violation to any mask shape at that location.
`MASK` *`viaMaskNum`*	Specifies which mask for double- or triple-patterning lithography to be applied to via shapes on each layer. The *`viaMaskNum`* is a hex-encoded 3 digit value of the form: *`<topMaskNum><cutMaskNum><bottomMaskNum>`* For example, MASK 113 means the top metal and cut layer *`maskNum`* is `1`, and the bottom metal layer *`maskNum`* is `3`. A value of 0 means the shape on that layer has no mask assignment (is uncolored), so 013 means the top layer is uncolored. If either the first or second digit is missing, they are assumed to be 0, so 013 and 13 means the same thing. Most applications only support *`maskNum`* values of 0, 1, 2, or 3 for double or triple patterning.

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The *topMaskNum* and *bottomMaskNum* variables specify which mask the corresponding metal shape belongs to. The via-master metal mask values have no effect. For the cut-layer, the *cutMaskNum* defines the mask for the bottommost, and then the leftmost cut. For multi-cut vias, the via-instance cut masks are derived from the via-master cut mask values. The via-master must have a mask defined for all of the cut shapes and every via-master cut mask is "shifted" (1 to 2, 2 to 1 for two mask layers, and 1 to 2, 2 to 3, 3 to 1 for three mask layers) so the lower-left cut matches the *cutMaskNum* value. See Example 1-11 on page 162 .

Similarly, for the metal layer, the *topMaskNum*/*bottomMaskNum* would define the mask for the bottom-most, then leftmost metal shape. For multiple disjoint metal shapes, the via-instance metal masks are derived from the via-master metal mask values. The via-master must have a mask defined for all of the metal shapes, and every via-master metal mask is “shifted” (1->2, 2->1 for two mask layers, 1->2, 2->3, 3->1 for three mask layers) so the lower-left cut matches the *topMaskNum*/*bottomMaskNum* value.

Shapes without any defined mask that need to be assigned, can be assigned to an arbitrary choice of mask by applications.

`PATH` *pt*

Creates a path between the specified points, such as *pt1 pt2 pt3*. The path automatically extends the length by half of the current *width* on both endpoints to form a rectangle. (A previous `WIDTH` statement is required.) The line between each pair of points must be parallel to the x or y axis (45-degree angles are not allowed).

You can also specify a path with a single coordinate, in which case a square whose side is equal to the current *width* is placed with its center at *pt*.

`POLYGON` *pt pt pt pt*

Specifies a sequence of at least three points to generate a polygon geometry. Most fab DRC rules require each polygon edge to be parallel to the x or y axis, or at a 45-degree angle. However, odd-angles are allowed sometimes on a few layers, such as bump layers. Each `POLYGON` statement defines a polygon generated by connecting each successive point, and then by connecting the first and last points.

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`RECT` *pt pt* — Specifies a rectangle, where the two points specified are opposite corners of the rectangle. There is no functional difference between a geometry specified using `PATH` and a geometry specified using `RECT`.

`SPACING` *minSpacing* — Specifies the minimum spacing allowed between this particular `OBS` and any other shape. While the syntax is shared for both `OBS` and `PIN`, it is only intended to be used with OBS shapes. The *minSpacing* value overrides all other normal LAYER-based spacing rules, including wide-wire spacing rules, end-of-line rules, parallel run-length rules, etc. An `OBS` with `SPACING` is not "seen" by any other DRC check, except the simple check for *minSpacing* to any other routing shape on the same layer.

One common application is to put an `OBS SPACING 0` shape on top of some `PIN` shapes to restrict the access of a router to other parts of the `PIN` without the `OBS` shape. This is sometimes needed for cells with large drive strengths to avoid electromigration problems by restricting the router to connect only to the middle of the output pin.

The *minSpacing* value cannot be larger than the maximum spacing defined in the `SPACING` or `SPACINGTABLE` for that layer. Tools may change larger values to the maximum spacing value with a warning.

`VIA` *pt viaName* — Specifies the via to place, and the placement location.

`WIDTH` *width* — Specifies the width that the `PATH` statements use. If you do not specify *width*, the default width for that layer is used. When you specify a width, that width remains in effect until the next `WIDTH` or `LAYER` statement. When another `LAYER` statement is given, the `WIDTH` is automatically reset to the default width for that layer.

Example 1-10 Layer Geometries

The following example shows how to define a set of geometries, first by using `ITERATE` statements, then by using individual `PATH`, `VIA` and `RECT` statements.

The following two sets of statements are equivalent:

```
PATH ITERATE 532.0 534 1999.2 534
    DO 1 BY 2 STEP 0 1446 ;
VIA ITERATE 470.4 475 VIABIGPOWER12
```


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```
    DO 2 BY 2 STEP 1590.4 1565 ;
RECT ITERATE 24.1 1.5 43.5 16.5
    DO 2 BY 1 STEP 20.0 0 ;
PATH 532.0 534 1999.2 534 ;
PATH 532.0 1980 1999.2 1980 ;
VIA 470.4 475 VIABIGPOWER12 ;
VIA 2060.8 475 VIABIGPOWER12;
VIA 470.4 2040 VIABIGPOWER12;
VIA 2060.8 2040 VIABIGPOWER12;
RECT 24.1 1.5 43.5 16.5 ;
RECT 44.1 1.5 63.5 16.5 ;
```

Example 1-11 Layer Geometries - multi-mask patterns

The following example shows how to use multi-mask patterns:

```
LAYER M1 ;
    RECT MASK  2 10 10 11 11 ;
LAYER M2 ;
    RECT 10 10 11 11 ;
VIA 5 5 VIA1_1 ;
VIA MASK 031 15 15 VIA1_2 ;
```

This indicates that the:

- `M1` rect shape belongs to `MASK 2`
- `M2` rect shape has no mask set
- `VIA1_1` via has no mask set (all the metal and cut shapes have no mask)
- `VIA1_2` via will have:
    - No mask set for the top metal shape (*`topMaskNum`* is 0 in the 031 value)
    - `MASK 1` for the bottom metal shape (*`botMaskNum`* is 1 in the 031 value)
    - The bottommost, and then the leftmost cut of the via-instance is `MASK 3`. The mask for the other cuts of the via-instance are derived from the via-master by "shifting" the via-master cut masks to match. So if the via-master's bottom-left cut is `MASK 1`, then the via-master cuts on `MASK 1` become `MASK 3` for the via-instance, and similarly cuts on 2 to 1, and cuts on 3 to 2. See Figure 1-54 on page 163.

```
LAYER M1 ;
    RECT MASK  2 10 10 11 11 ;
LAYER M2 ;
```


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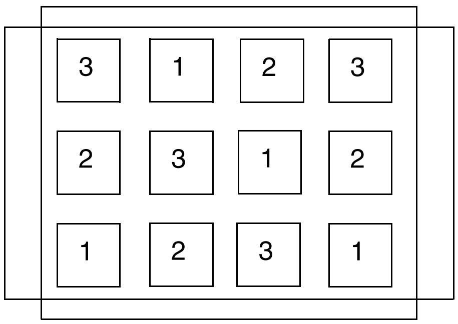
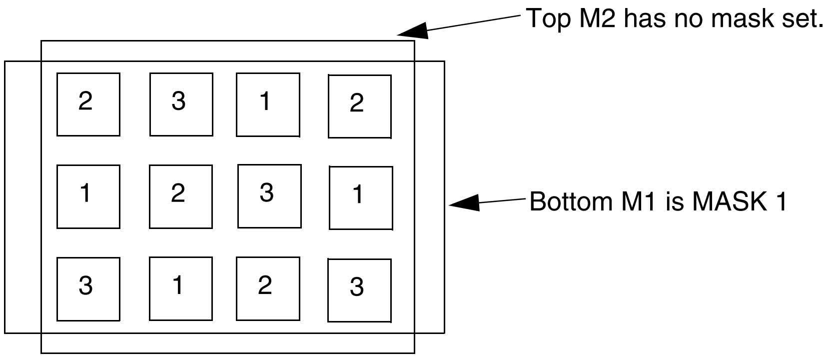
```
        RECT 10 10 11 11 ;
    VIA 5 5 VIA1_1 ;
    VIA MASK 031 15 15 VIA1_2 ;
```

Figure 1-54 Via-master multi-mask patterns



Via-master cut masks for VIA1_2.



Masks for via-instance: `VIA MASK 031 15 15 VIA1_2` ;
Bottom `M1` is `MASK 1`.Top `M2` has no mask set. Lower-left cut is `MASK 3`, and other via-instance cut shape masks are derived from via-master cut-masks as shown above by "shifting" the via-master masks to match: 1->3, 2->1, 3->2.

Macro Obstruction Statement

```
[OBS
     {layerGeometries}...
END]
```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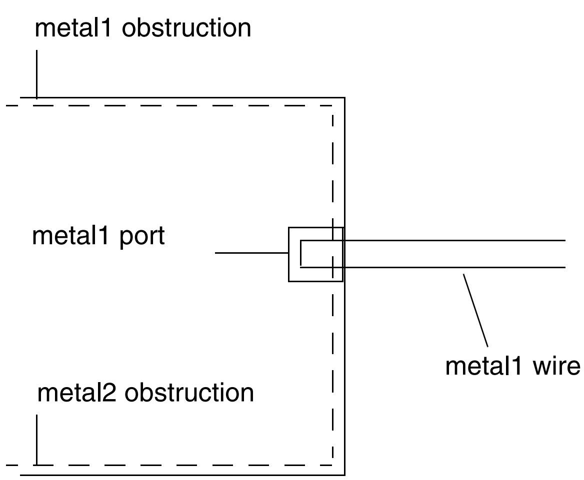
Defines a set of obstructions (also called blockages) on the macro. You specify obstruction geometries using the layer geometry syntax. See "Layer Geometries" on page 157 for syntax information.

Normally, obstructions block routing, except for when a pin port overlaps an obstruction (a port geometry overrules an obstruction). For example, you can define a large rectangle for a *metal1* obstruction and have *metal1* port in the middle of the obstruction. The port can still be accessed by a via, if the via is entirely inside the port.

In Figure 1-55 on page 164, the router can only access the *metal1* port from the right. If the *metal2* obstruction did not exist, the router could connect to the port with a *metal12* via, as long as the *metal1* part of the via fit entirely inside the *metal1* port.

Figure 1-55



Routing can also connect to such a port on the same layer if the routing does not cross any obstruction by more than a distance of the total of minimum width plus minimum spacing before reaching the pin. This is because the port geometry is known to be "real," and any obstruction less than a distance of minimum width plus minimum spacing away from the port is not a real obstruction. If the pin is more than minimum width plus minimum spacing away from the obstruction edge, the router can only route to the pin from the layer above or below using a via (see Figure 1-56 on page 165).

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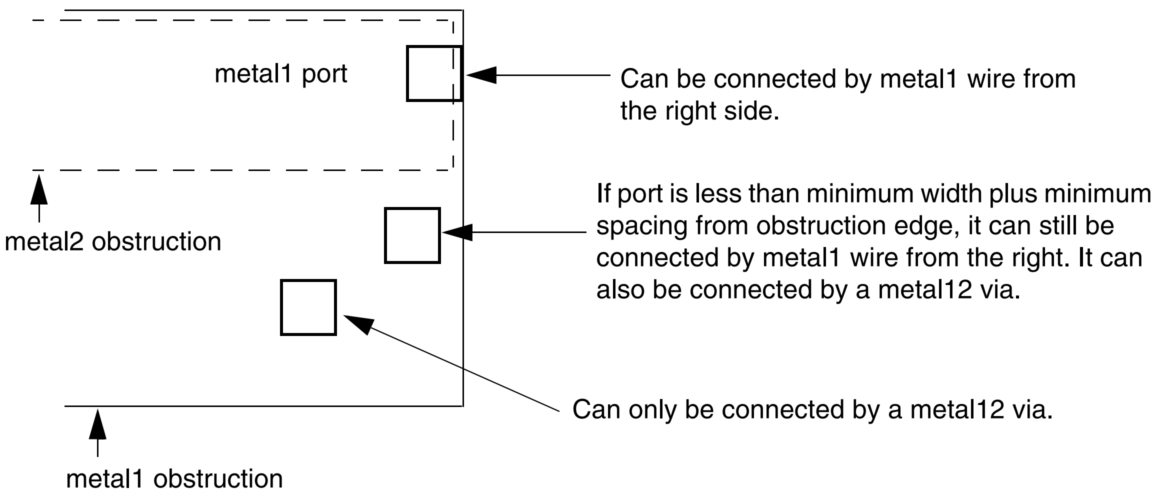
Figure 1-56



If a port is on the edge of the obstruction, a wire can be routed to the port without violations. Pins that are partially covered with obstructions or in apparent violation with nearby obstructions can limit routing options. Even though the violations are not real, the router assumes they are. In these cases, extend each obstruction to cover the pin. The router then accesses the pin as described above.

Benefits of Combining Obstructions

Significant routing time can be saved if obstructions are simplified. Especially in *metal1*, construct obstructions so that free tracks on the layer are accessible to the router. If most of the routing resource is obstructed, simplify the obstruction modeling by combining small obstructions into a single large obstruction. For example, use the bounding box of all *metal1* objects in the cell, rather than many small obstructions, as the bounding box of the obstruction.

You must be sure to model via obstructions over the rest of the cell properly. A single, large *cut12* obstruction over the rest of the cell can do this in some cases, as when *metal1* resource exists within the cell outside the power buses.

Rectilinear Blocks

Normally, footprint descriptions in LEF are rectangular. However, it is possible to describe rectilinear footprints using an overlap layer. The overlap layer is defined specifically for this purpose and does not contain any routing.

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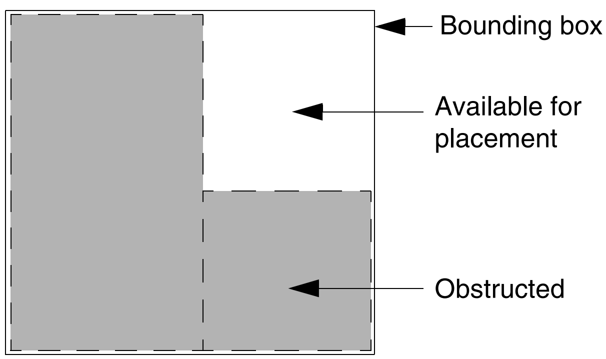
Describe a rectilinear footprint by setting the `SIZE` of the macro as a whole to a rectangular bounding box, then defining obstructions within the bounding box on the overlap layer. The obstructions on the overlap layer indicate areas within the bounding box which no other macro should overlap. The obstructions should completely cover the rectilinear shape of the macro, but not the portion of the bounding box that might overlap with other macros during placement.

Note: Specify the overlaps for the macro using the `OBS` statement. To do this, specify a layer of type `OVERLAP` and then give the overlap geometries, as shown in Figure 1-57 on page 166.

Figure 1-57




```
OBS
     LAYER OVERLAP ;
     RECT 0 0 500 1000 ;
     RECT 500 0 1000 500 ;

END
```

Macro Pin Statement

```
[PIN pinName
     [TAPERRULE ruleName ;]
     [DIRECTION {INPUT | OUTPUT [TRISTATE] | INOUT | FEEDTHRU} ;]
     [USE { SIGNAL | ANALOG | POWER | GROUND | CLOCK } ;]
     [NETEXPR "netExprPropName defaultNetName" ;]
     [SUPPLYSENSITIVITY powerPinName ;]
     [GROUNDSENSITIVITY groundPinName ;]
     [SHAPE {ABUTMENT | RING | FEEDTHRU} ;]
     [MUSTJOIN pinName ;]
     {PORT
       [CLASS {NONE | CORE | BUMP} ;]
       {layerGeometries} ...
      END} ...
     [PROPERTY propName propVal ;] ...
     [PROPERTY LEF58_ACCESSAREA
        "ACCESSAREA {LAYER layerName RECT pt pt [CUTCLASS className]
        [EXCEPTEXTRACUT]}...
        ; " ;]
```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```
     [PROPERTY LEF58_ANTENNADIFFAREA
        "ANTENNADIFFAREA [OXIDE1 | OXIDE2 | OXIDE3 | OXIDE4
              | OXIDE5 | ... | OXIDE32] [BACKSIDE]
              [PROTECTOR] value [LAYER layerName]
        ; ..." ;]
     [PROPERTY LEF58_BOTTOMTOPLAYER
        "BOTTOMTOPLAYER layerName
        ; " ;]
     [PROPERTY LEF58_MUSTJOINALLPORTS
        "MUSTJOINALLPORTS [ANYUPPERLAYERCONNECTION]
        ; " ;]
     [PROPERTY LEF58_SHORTEDPINGROUP
        "SHORTEDPINGROUP groupName
        ; " ;]
     [PROPERTY LEF58_VIAINMUSTJOIN
        "VIAINMUSTJOIN
        ; " ;]
     [PROPERTY LEF58_VIAINPINONLY
        "VIAINPINONLY
        ; " ;]
     [ANTENNAPARTIALMETALAREA value [LAYER layerName] ;] ...
     [ANTENNAPARTIALMETALSIDEAREA value [LAYER layerName] ;] ...
     [ANTENNAPARTIALCUTAREA value [LAYER layerName] ;] ...
     [ANTENNADIFFAREA value [LAYER layerName] ;] ...
     [ANTENNAMODEL {OXIDE1 | OXIDE2 | OXIDE3 | OXIDE4} ;] ...
     [ANTENNAGATEAREA value [LAYER layerName] ;] ...
     [ANTENNAMAXAREACAR value LAYER layerName ;] ...
     [ANTENNAMAXSIDEAREACAR value LAYER layerName ;] ...
     [ANTENNAMAXCUTCAR value LAYER layerName ;] ...
END pinName]
```

Defines pins for the macro. `PIN` statements must be included in the LEF specification for each macro. All pins, including VDD and VSS, must be specified. The first pin listed becomes the first pin in the database. List the pins in the following order:

- Netlist pins, including inout pins, output pins, and input pins
- Power and ground pins
- Mustjoin pins

`ANTENNADIFFAREA` *`value`* `[LAYER` *`layerName`*`]`

Specifies the diffusion (diode) area, in micron-squared units, to which the pin is connected on a layer. If you do not specify a layer name, the value applies to all layers. For more information on process antenna calculation, see Appendix C, “Calculating and Fixing Process Antenna Violations.”

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ANTENNAGATEAREA *value* [LAYER *layerName*]

Specifies the gate area, in micron-squared units, to which the pin is connected on a layer. If you do not specify a layer name, the value applies to all layers. For more information on process antenna calculation, see Appendix C, "Calculating and Fixing Process Antenna Violations."

ANTENNAMAXAREACAR *value* LAYER *layerName*

For hierarchical process antenna effect calculation, specifies the maximum cumulative area ratio value on the specified *layerName*, using the metal area at or below the current pin layer, excluding the pin area itself. This is used to calculate the actual cumulative antenna ratio on the pin layer, or the layer above it.

For more information on process antenna calculation, see Appendix C, "Calculating and Fixing Process Antenna Violations."

ANTENNAMAXCUTCAR *value* LAYER *layerName*

For hierarchical process antenna effect calculation, specifies the maximum cumulative antenna ratio value on the specified *layerName*, using the cut area at or below the current pin layer, excluding the pin area itself. This is used to calculate the actual cumulative antenna ratio for the cuts above the pin layer.

For more information on process antenna calculation, see Appendix C, "Calculating and Fixing Process Antenna Violations."

ANTENNAMAXSIDEAREACAR *value* LAYER *layerName*

For hierarchical process antenna effect calculation, specifies the maximum cumulative antenna ratio value on the specified *layerName*, using the metal side wall area at or below the current pin layer, excluding the pin area itself. This is used to calculate the actual cumulative antenna ratio on the pin layer or the layer above it.

For more information on process antenna calculation, see Appendix C, "Calculating and Fixing Process Antenna Violations."

ANTENNAMODEL {OXIDE1 | OXIDE2 | OXIDE3 | OXIDE4}

Specifies the oxide model for the pin. If you specify an ANTENNAMODEL statement, the value affects all ANTENNAGATEAREA and ANTENNA*CAR statements for the pin that follow it until you specify another ANTENNAMODEL statement. The ANTENNAMODEL statement does not affect ANTENNAPARTIAL*AREA and ANTENNADIFFAREA statements because they refer to the total metal, cut, or diffusion area connected to the pin, and do not vary with each oxide model.
Default: OXIDE1, for a new PIN statement

Because LEF is often used incrementally, if an ANTENNA statement occurs twice for the same oxide model, the last value specified is used.

For most standard cells, there is only one value for the ANTENNAPARTIAL*AREA and ANTENNADIFFAREA values per pin; however, for a block with six routing layers, it is

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possible to have six different ANTENNAPARTIAL*AREA values and six different ANTENNAPINDIFFAREA values per pin. It is also possible to have six different ANTENNAPINGATEAREA and ANTENNAPINMAX*CAR values for each oxide model on each pin.

Example 1-12 Pin Antenna Model

The following example describes oxide model information for pins IN1 and IN2.

```
MACRO GATE1
    PIN IN1
        ANTENNADIFFAREA 1.0 ;             #not affected by ANTENNAMODEL
        ...
        ANTENNAMODEL OXIDE1 ;        #OXIDE1 not required, but is good
                                     #practice
        ANTENNAGATEAREA 1.0 ;             #OXIDE1 gate area
        ANTENNAMAXAREACAR 50.0 LAYER m1 ; #metal1 CAR value
        ...
        ANTENNAMODEL OXIDE2 ;             #OXIDE2 starts here
        ANTENNAGATEAREA 3.0 ;             #OXIDE2 gate area
        ...
    PIN IN2
        ANTENNADIFFAREA 2.0 ;             #not affected by ANTENNAMODEL
        ANTENNAPARTIALMETALAREA 2.0 LAYER m1 ;
        ...
        #no OXIDE1 specified for this pin
        ANTENNAMODEL OXIDE2 ;
        ANTENNAGATEAREA 1.0 ;
        ...
```

ANTENNAPARTIALCUTAREA *value* [LAYER *layerName*]

Specifies the partial cut area above the current pin layer and inside the macro cell on the layer. For a hierarchical design, only the cut layer above the I/O pin layer is needed for partial antenna ratio calculation. If you do not specify a layer name, the value applies to all layers.

For more information on process antenna calculation, see Appendix C, “Calculating and Fixing Process Antenna Violations.”

ANTENNAPARTIALMETALAREA *value* [LAYER *layerName*]

Specifies the partial metal area connected directly to the I/O pin and the inside of the macro cell on the layer. For a hierarchical design, only the same metal layer as the I/O

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pin, or the layer above it, is needed for partial antenna ratio calculation. If you do not specify a layer name, the value applies to all layers.

For more information on process antenna calculation, see Appendix C, "Calculating and Fixing Process Antenna Violations."

Note: Metal area is calculated by adding the pin's geometric metal area and the `ANTENNAPARTIALMETALAREA` value.

`ANTENNAPARTIALMETALSIDEAREA` *`value`* `[LAYER` *`layerName`*`]`

Specifies the partial metal side wall area connected directly to the I/O pin and the inside of the macro cell on the layer. For a hierarchical design, only the same metal layer as the I/O pin or the layer above is needed for partial antenna ratio calculation. If you do not specify a layer name, the value applies to all layers.

For more information on process antenna calculation, see Appendix C, "Calculating and Fixing Process Antenna Violations."

`DIRECTION {INPUT | OUTPUT [TRISTATE] | INOUT | FEEDTHRU}`
Specifies the pin type. Most current tools do not usually use this keyword. Typically, pin directions are defined by timing library data, and not from LEF.
Default: `INPUT`
Value: Specify one of the following:

`INPUT`	Pin that accepts signals coming into the cell.
`OUTPUT [TRISTATE]`	Pin that drives signals out of the cell. The optional `TRISTATE` argument indicates tristate output pins for ECL designs.
`INOUT`	Pin that can accept signals going either in or out of the cell.
`FEEDTHRU`	Pin that goes completely across the cell.

`GROUNDSENSITIVITY` *`groundPinName`*

Specifies that if this pin is connected to a tie-low connection (such as `1'b0` in Verilog), it should connect to the same net to which *`groundPinName`* is connected.

`groundPinName` must match a pin on this macro that has a `USE GROUND` attribute. The ground pin definition can follow later in this `MACRO` statement; it does not have to be defined before this pin definition. For an example, see Example 1-13 on page 171.

Note: `GROUNDSENSITIVITY` is useful only when there is more than one ground pin in the macro. By default, if there is only one `USE GROUND` pin, then the tie-low connections are already implicitly defined (that is, tie-low connections are connected to the same net as the one ground pin).

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`MUSTJOIN` *`pinName`*

Specifies the name of another pin in the cell that must be connected with the pin being defined. `MUSTJOIN` pins provide connectivity that must be made by the router. In the LEF file, each pin referred to must be defined before the referring pin. The remaining `MUSTJOIN` pins in the set do not need to be defined contiguously.

Note: `MUSTJOIN` pin names are never written to the DEF file; they are only used by routers to add extra connection points during routing.

`MUSTJOIN` pins have the following restrictions:

- A set of `MUSTJOIN` pins cannot have more than one schematic pin.
- Nonschematic `MUSTJOIN` pins must be defined after all other pins.

Schematic and nonschematic `MUSTJOIN` pins are handled in slightly different ways. For schematic `MUSTJOIN` pins, the pins are added to the pin set for the (unique) net associated with the ring for each component instance of the macro. The net is routed in the usual manner, and routing data for the `MUSTJOIN` pins are included in routing data for the net.

The mustjoin routing is not necessarily performed before the rest of the net. Timing relations should not be given for `MUSTJOIN` pins, and internal mustjoin routing is modeled as lumped capacitance at the schematic pin.

Nonschematic `MUSTJOIN` pin sets get routed in the usual manner. However, when the DEF file is outputted, routing data is reported in the `NETS` section of the file as follows:

```
MUSTJOIN compName pinName + regularWiring ;
```

Here, *`compName`* is the component and *`pinName`* is an arbitrary pin in the set. You can also use the preceding to input prewiring for the `MUSTJOIN` pin, using `FIXED` or `COVER`.

`NETEXPR "`*`netExprPropName defaultNetName`*`"`

Specifies a net expression property name (such as `power1` or `power2`) and a default net name. If *`netExprPropName`* matches a net expression property in the netlist (such as in Verilog, VHDL, or OpenAccess), then the property is evaluated, and the software identifies a net to which to connect this pin. If this property does not exist, *`defaultNetName`* is used for the net name.

`netExprPropName` must be a simple identifier in order to be compatible with other languages, such as Verilog and CDL. Therefore, it can only contain alphanumeric characters, and the first character cannot be a number. For example, `power2` is a legal name, but `2power` is not. You cannot use characters such as `$` and `!`. The *`defaultName`* can be any legal DEF net name.

Example 1-13 Net Expression and Supply Sensitivity

The following statement defines sensitivity and net expression values for four pins on the macro `myMac`:

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```
MACRO myMac
    ...
    PIN in1
        ...
        SUPPLYSENSITIVITY vddpin1 ; #If in1 is 1'b1, use net connected to vddpin1.
                                   #Note that no GROUNDSENSITIVITY is needed
                                   #because only one ground pin exists.
                                   #Therefore, 1'b0 implicitly means net from
                                   #pin gndpin.
        ...
    END in1

    PIN vddpin1
        ...
        NETEXPR "power1 VDD1" ; #If power1 net expression is defined in the
                                #netlist, use it to find the net connection. If
                                #not, use net VDD1.
        ...
    END vddpin1
    PIN vddpin2
        ...
        NETEXPR "power2 VDD2" ; #If power2 net expression is defined in the
                                #netlist, use it to find the net connection.If
                                #not, use net VDD2.
        ...
    END vddpin2
    PIN gndpin
        ...
        NETEXPR "gnd1 GND" ; #If gnd1 net expression is defined in the
                             #netlist, use  it to find the net connection. If
                             #not, use net GND.
        ...
    END gndpin
    ...
END myMac
```

`PIN` *`pinName`*

Specifies the name for the library pin.

`PORT`

Starts a pin port statement that defines a collection of geometries that are electrically equivalent points (strongly connected). A pin can have multiple ports. Each `PORT` of the same `PIN` is considered weakly connected to the other `PORT`s, and should already be connected inside the `MACRO` (often through a resistive path).

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Strongly connected shapes (that is, multiple shapes of one PORT) indicate that a signal router is allowed to connect to one shape of the PORT, and continue routing from another shape of the same PORT.

Weakly connected shapes (that is, separate PORTs of the same PIN) are assumed to be connected through resistive paths inside the MACRO that should not be used by routers. The signal router should connect to one or the other PORT, but not both.

Power routers should connect to every PORT statement, if there is more than one for a given PIN. For example, if a block has several PORTs on the boundary for the VSS PIN, each PORT should be connected by the power router.

The syntax for describing pin port statements is defined as follows:

```
{PORT
  [CLASS {NONE | CORE | BUMP} ;]
  {layerGeometries} ...
END} ...
```

CLASS {NONE | CORE | BUMP}

Specifies the port type.
Default: NONE

A port can be one of the following:

BUMP—Specifies the port is a bump connection point. A bump port should only be connected by routing to a bump (normally a MACRO CLASS COVER BUMP cell).

CORE—Specifies the port is a core ring connection point. A core port is used only on power and ground I/O drivers used around the periphery. The core port indicates which power or ground port to connect to a core ring for the chip (inside the I/O pads).

NONE—Specifies the port is a default port that is connected by normal "default" routing. NONE is the default value if no PORT CLASS statement is specified.

layerGeometries

Defines port geometries for the pin. You specify port geometries using layer geometries syntax. See "Layer Geometries" on page 157 for syntax information.

PROPERTY *propName propVal*

Specifies a numerical or string value for a pin property defined in the PROPERTYDEFINITIONS statement. The *propName* you specify must match the *propName* listed in the PROPERTYDEFINITIONS statement.

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`SHAPE`

Specifies a pin with special connection requirements because of its shape.
Value: Specify one of the following:

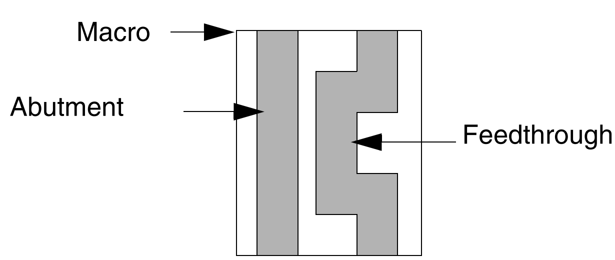
`ABUTMENT`	Pin that goes straight through cells with a regular shape and connects to pins on adjoining cells without routing.
`RING`	Pin on a large block that forms a ring around the block to allow connection to any point on the ring. Cover macro special pins also typically have shape `RING`.
`FEEDTHRU`	Pin with an irregular shape with a jog or neck within the cell.

Figure 1-58 on page 174 shows an example of an abutment and a feedthrough pin.

Note: When you define feedthrough and abutment pins for use with power routing, you must do the following:

- Feedthrough pin widths must be the same on both edges and consistent with the routing width used with the power route commands.
- Feedthrough pin centers on both edges must align for successful routing.
- Power pins in fork shapes must be represented in two ports and be defined as a feedthrough shape. In most other cases, power pin geometries do not represent more than one port.
- An abutment pin must have at least one geometric rectangle with layer and width consistent with the values specified in the power route commands.

Figure 1-58



`SUPPLYSENSITIVITY` *`powerPinName`*

Specifies that if this pin is connected to a tie-high connection (such as `1'b1` in Verilog), it should connect to the same net to which *`powerPinName`* is connected.

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powerPinName must match a pin on this macro that has a USE POWER attribute. The power pin definition can follow later in this MACRO statement; it does not have to be defined before this pin definition. For an example, see Example 1-13 on page 171.

Note: SUPPLYSENSITIVITY is useful only when there is more than one power pin in the macro. By default, if there is only one USE POWER pin, then the tie-high connections are already implicitly defined (that is, tie-high connections are connected to the same net as the one power pin).

TAPERRULE *ruleName*

Specifies the non-default rule to use when tapering wires to the pin.

USE {ANALOG | CLOCK | GROUND | POWER | SIGNAL}

Specifies how the pin is used. Pin use is required for timing analysis.
Default: SIGNAL
Value: Specify one of the following:

ANALOG	Pin is used for analog connectivity.
CLOCK	Pin is used for clock net connectivity.
GROUND	Pin is used for connectivity to the chip-level ground distribution network.
POWER	Pin is used for connectivity to the chip-level power distribution network.
SIGNAL	Pin is used for regular net connectivity.

Access Area Rule

The access area rule can be used to specify a via access area on the specified cut layer. You can specify the access area rule by using the following property definition:

```
PROPERTY LEF58_ACCESSAREA
        "ACCESSAREA {LAYER layerName RECT pt pt [CUTCLASS className]
        [EXCEPTEXTRACUT]}...
        ; " ;
```

Where:

ACCESSAREA {LAYER *layerName* RECT *pt pt* }...

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	Specifies a via access area on *layerName*, which must be a cut layer. All the cuts of a via connected to any pin shape on the bottom metal layer that overlaps the given area in RECT must be inside the given area. If there are multiple via access areas overlapping a pin shape, all the cuts of a via connected to the pin need to be inside one of the areas. Different via access areas should not overlap or be abutted. For a MUSTJOINALLPORTS pin, the given area must overlap with all of the pin shapes on the bottom metal layer. Then, the cuts on each of the individual pin shapes must be inside the given area and must be connected by a same-metal wire on the above metal layer as well. This construct can only be defined on standard cells with MACRO CLASS of CORE. If a pin has ACCESSAREA, the ACCESSAREA defined in MACRO has no meaning on that pin. In other words, the ACCESSAREA defined in MACRO is applied only on pins without their own ACCESSAREA. Actually, tools are not likely to support overlap of pin-based and macro/cell-based access areas. *Type*: Float, specified in microns
CUTCLASS *className*	Specifies the via access area for the given *className*, which is connected to the pin shapes. Without CUTCLASS, the via access area would be applied to vias of any cut class. Multiple via access areas could be defined for different cut classes. If a cut class does not have a corresponding via access area defined, the vias of that cut class could be connected to the pins without any restriction. *Type*: String
EXCEPTEXTRACUT	Specifies that the via access area is not applied to vias with multiple cuts connected to a single pin shape. If a via pillar with multiple wires on the above metal layer of the pin shape layer is used, this via access area does not need to be honored. For example, if a via pillar with 2x2 cuts is connected to a MUSTJOINALLPORTS with two pin shapes, this via access area could be ignored.

Access Area Rule Example

- The diagram below illustrates a case where the access area rule is defined for the pin as well as macro:

```
MACRO Z
...
PROPERTY LEF58_ACCESSAREA "
```


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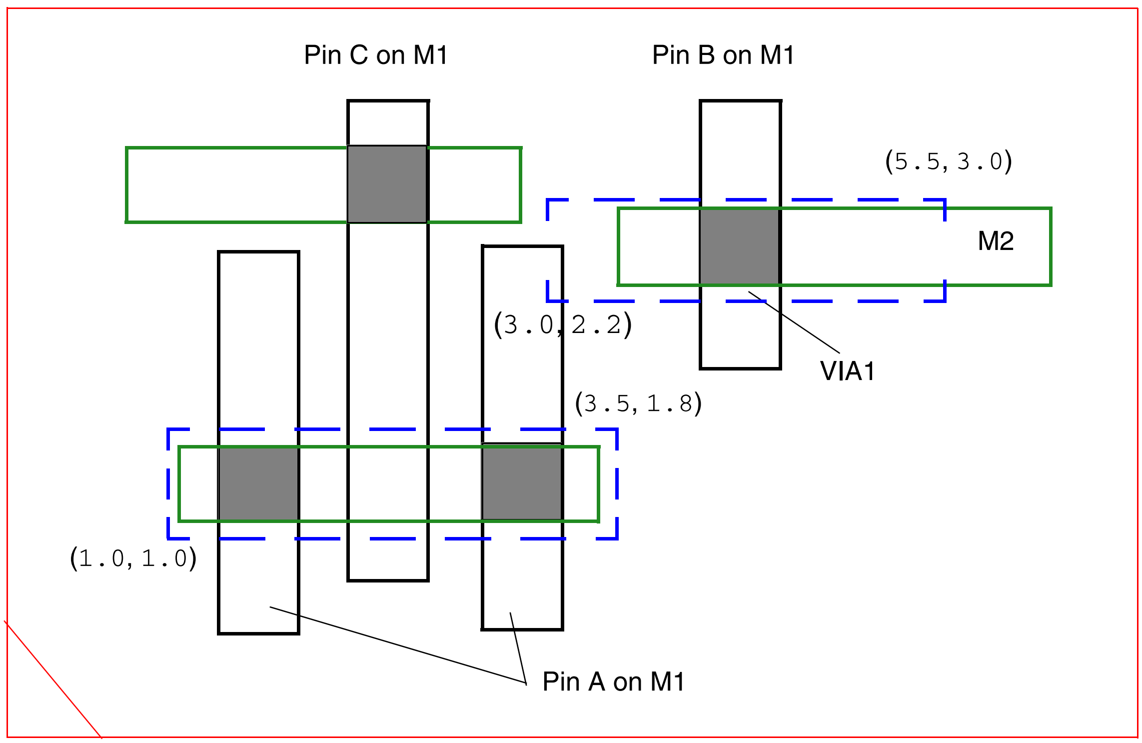
```
ACCESSAREA LAYER VIA1 RECT 3.0 2.2 5.5 3.0 ; " ;
PIN A
...
PROPERTY LEF58_ACCESSAREA "
ACCESSAREA LAYER VIA1 RECT 1.0 1.0 3.5 1.8 ; " ;
...
END A
...
END Z
```

Figure 1-59 Illustration of the Access Area Rule on a Pin



OK because both the cuts of pin `A` are within the access area of the pin and the access area of macro `Z` is irrelevant on pin `A`. The cut of pin `B` is within access area of macro `Z`. Pin `C` does not have any constraint because the access area of pin `A` is irrelevant, and it does not overlap with the access area of macro `Z`.

Antenna Diff Area Rule

The antenna diff area rule can be used to specify the diffusion area to which the pin is connected on a layer.

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You can specify the antenna diff area rule by using the following property definition:

```
PROPERTY LEF58_ANTENNADIFFAREA
        "ANTENNADIFFAREA [OXIDE1 | OXIDE2 | OXIDE3 | OXIDE4
              | OXIDE5 | ... | OXIDE32] [BACKSIDE]
              [PROTECTOR] value [LAYER layerName]
        ; ..." ;
```

Where:

`ANTENNADIFFAREA value [LAYER` *`layerName`*`]`

Specifies the diffusion (diode) area, in micron-squared units, to which the pin is connected on a layer. If you do not specify a layer name, the value applies to all layers.

`BACKSIDE`

Specifies that the diffusion area applies only for the metal on the layers with `BACKSIDE`. On a design with `BACKSIDE` layers but without `BACKSIDE` or `LAYER` specified in `ANTENNADIFFAREA`, the diffusion area applies only on the front side layers.

`OXIDE1 | OXIDE2 | OXIDE3 | OXIDE4 | OXIDE5 | ... | OXIDE32`

Specifies that the rule applies to the specified oxide model only. If an oxide model is defined on one macro pin, all the pins with `ANTENNADIFFAREA` should also be defined with an oxide model or `PROTECTOR`.

`PROTECTOR`

Specifies the diffusion area for protected devices. If the metal is electrically connected to a pin defined with protector diffusion, `ANTENNADIFFPROTECTORAREARATIO` would only be checked. Both gate and non-protector diffusion would be ignored for other `ANTENNA*` rules.

Bottom Top Layer Rule

You can specify the bottom top layer rule by using the following property definition:

```
PROPERTY LEF58_BOTTOMTOPLAYER
        "BOTTOMTOPLAYER layerName
        ; " ;
```

Where:

`BOTTOMTOPLAYER` *`layerName`*

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Specifies that all wires connected to the pin must be on the specified layer, *layerName*, or on the cut and metal layers above it if *layerName* is a front-side layer. If *layerName* refers to a back-side layer, the wires must be on *layerName* or the cut and metal layers below it. *layerName* must be a routing layer, and some of the pin shapes must belong to this layer.

Type: String

Must Join All Ports Rule

You can use the must join all ports rule to specify that all ports must be connected individually.

You can create a must join all ports rule by using the following property definition:

```
PROPERTY LEF58_MUSTJOINALLPORTS
     "MUSTJOINALLPORTS [ANYUPPERLAYERCONNECTION]
     ; " ;
```

Where:

ANYUPPERLAYERCONNECTION

Specifies that the must join port shapes must be connected by at least one wire from the layer immediately above the topmost pin layer of the port shape. For example, if all port shapes are on M1, at least one M2 wire and part of VIA1 must be used to connect them; using M1 wires only is not sufficient. As back-side layers are reversed, "upper" actually refers to the lower adjacent layer. So, if all port shapes are on BM1, at least one BM2 wire must be used to connect them.

MUSTJOINALLPORTS

Specifies that all of the ports must be connected individually. If there are multiple shapes per ports, at least one of the shapes of the port needs to be connected. However, tools typically would connect to only one of the shapes.

For example:

```
PIN A
...
     PROPERTY LEF58_MUSTJOINALLPORTS "
          MUSTJOINALLPORTS ; " ;
     PORT
          LAYER M1 ;
               RECT 1 1 2 2 ;
```


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```
                RECT 4 4 5 5 ;
        END
        PORT
            LAYER M2 ;
                RECT 2 2 3 3 ;
        END
...
END A
```

means that the router must connect to at least one of the two M1 shapes and to the M2 shape individually. However, tools typically would connect to exactly one of the two M1 shapes and to the M2 shape individually.

Shorted Pin Group Rule

You can create a shorted pin group rule by using the following property definition:

```
PROPERTY LEF58_SHORTEDPINGROUP
    "SHORTEDPINGROUP groupName
    ; " ;
```

Where:

`SHORTEDPINGROUP` *`groupName`*

Specifies that the pins with the same *`groupName`* are physically shorted/connected internally in a macro. If `SHORTEDPINGROUP` is defined, the macro should have at least two pins with the same *`groupName`*.

Via In Must Join Rule

You can create a via in must join rule by using the following property definition:

```
PROPERTY LEF58_VIAINMUSTJOIN
    "VIAINMUSTJOIN
    ; " ;
```

Where:

`VIAINMUSTJOIN`

Specifies that two stacked vias must be used to connect to the must-join pins and all of the cuts of these two stacked vias must be within the boundary box of the must-join pins.

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For example, if the must-join pins have multiple `M1` shapes, the cuts of the stacked vias to `M3` must be within the boundary box formed by the `M1` pin shapes.

Via In Pin Only Rule

You can use the via in pin only rule to specify that vias must be dropped inside the original pin shapes to connect to the pin.

You can create a via in pin only rule by using the following property definition:

```
PROPERTY LEF58_VIAINPINONLY
     "VIAINPINONLY
     ; " ;
```

Where:

`VIAINPINONLY`

Specifies that at least one cut of a via must be dropped inside the original pin shapes to connect to the pin, and a planar connection to the pin is not allowed. In some advanced nodes, the pin shapes can be extended for metal alignment purpose. For a single-cut via, it means that the cut would be in the original pin shapes. For a multiple-cut via, particularly with the via pillar having multiple fingers, only one cut needs to be in the original pin shapes and the rest of the via pillar structure can be outside the original pin shapes.
To support negative enclosure, for a square cut with size *CS* greater than the minimum width *W* on the below or above metal layer of a pin shape, the following method is used to determine whether the cut is inside the pin shape. From the cut center, extend *CS*/2 on any two opposite sides and *W*/2 on the other two opposite sides, if one of the two expanded sizes (one for extending *CS*/2 horizontally and another for extending *CS*/2 vertically) is completely covered by the pin metal shape, the cut is considered to be inside the pin shape.

Manufacturing Grid

```
[MANUFACTURINGGRID value ;]
```

Defines the manufacturing grid for the design. The manufacturing grid is used for geometry alignment. When specified, shapes and cells are placed in locations that snap to the manufacturing grid.

`value`

Specifies the value for the manufacturing grid, in microns. *`value`* must be a positive number.
Type: Float

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Maximum Via Stack

```
[MAXVIASTACK value [RANGE bottomLayer topLayer] ;]
```

Specifies the maximum number of single-cut stacked vias that are allowed on top of each other (that is, in one continuous stack). A via is considered to be in a stack with another via if the cut of the first via overlaps any part of the cut of the second via. A double-cut or larger via interrupts the stack. For example, a via stack consisting of single *via12*, single *via23*, double-cut *via34*, and single *via45* has a single-cut stack of height 2 for *via12* and *via23*, and a single-cut stack of height 1 for *via45* because the full stack is broken up by double-cut *via34*.

The `MAXVIASTACK` statement should follow the `LAYER` statements in the LEF file; however, it is not attached to any particular layer. You can specify only one `MAXVIASTACK` statement in a LEF file.

`RANGE` *`bottomLayer topLayer`*

Specifies a range of layers for which the maximum stacked via rule applies. If you do not specify a range, the value applies for all layers.

`value`

Specifies the maximum allowed number of single-cut stacked vias.
Type: Integer

Example 1-14 Maximum Via Stack Statement

The following `MAXVIASTACK` statement specifies that only four stacked vias are allowed on top of each other. This rule applies to all layers.

```
LAYER metal9
    ...
END LAYER

MAXVIASTACK 4 ;
```

If you specify the following statement instead, the stacked via limit applies only to layers *metal1* through *metal7*.

```
MAXVIASTACK 4 RANGE m1 m7 ;
```

Nondefault Rule

```
[NONDEFAULTRULE ruleName
     [HARDSPACING ;]
     {LAYER layerName
        WIDTH width ;
        [DIAGWIDTH diagWidth ;]
```


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```
        [SPACING minSpacing ;]
        [WIREEXTENSION value ;]
    END layerName} ...
    [VIA viaStatement] ...
    [USEVIA viaName ;] ...
    [USEVIARULE viaRuleName ;] ...
    [MINCUTS cutLayerName numCuts ;] ...
    [PROPERTY propName propValue ;] ...
    [PROPERTY LEF58_USEVIACUTCLASS
    "USEVIACUTCLASS cutLayerName className
        [ROWCOL numCutRows numCutCols]
        ;... " ;]
END ruleName]
```

Defines the wiring width, design rule spacing, and via size for regular (signal) nets. You do not need to define cut layers for the non-default rule.

Some tools have limits on the total number of non-default rules they can store. This limit can be as low as 30; however most tools that support 90 nanometer rules (that is, LEF 5.5 and newer) can handle at least 255.

Note: Use the `VIA` statement to define vias for non-default wiring.

`DIAGWIDTH` *`diagWidth`*

Specifies the diagonal width for *`layerName`*, when 45-degree routing is used.
Default: The minimum width value (`WIDTH` *`minWidth`*)
Type: Float, specified in microns

`HARDSPACING`

Specifies that any spacing values that exceed the LEF `LAYER` spacing requirements are "hard" rules instead of "soft" rules. By default, routers treat extra spacing requirements as soft rules that are high cost to violate, but not real spacing violations. However, in certain situations, the extra spacing should be treated as a hard, or real, spacing violation, such as when the route will be modified with a post-process that replaces some of the extra space with metal.

`LAYER` *`layerName`* `...` `END` *`layerName`*

Specifies the layer for the various width and spacing values. This layer must be a routing layer. Every routing layer must have a `WIDTH` keyword and value specified. All other keywords are optional.

`MINCUTS` *`cutLayerName numCuts`*

Specifies the minimum number of cuts allowed for any via using the specified cut layer. Routers should only use vias (generated or predefined fixed vias) that have at least

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numCuts cuts in the via.
Type: (*numCuts*) Positive integer

NONDEFAULTRULE *ruleName*

Specifies a name for the new routing rule. The name DEFAULT is reserved for the default routing rule used by most nets. The default routing rule is constructed automatically from the LEF LAYER statement WIDTH, DIAGWIDTH, SPACING, and WIREEXTENSION values, from the LEF VIA statement (any vias with the DEFAULT keyword), and from the LEF VIARULE statement (any via rules with the DEFAULT keyword). If you specify DEFAULT for *ruleName*, the automatic creation is overridden, and the default routing rule is defined directly from this rule definition.

PROPERTY *propName propValue*

Specifies a numerical or string value for a non-default rule property defined in the PROPERTYDEFINITIONS statement. The *propName* you specify must match the *propName* listed in the PROPERTYDEFINITIONS statement.

SPACING *minSpacing*

Specifies the recommended minimum spacing for *layerName* of routes using this NONDEFAULTRULE to other geometries. If the spacing is given, it must be at least as large as the foundry minimum spacing rules defined in the LAYER definitions. Routers should attempt to meet this recommended spacing rule; however, the spacing rule can be relaxed to the foundry spacing rules along some parts of the wire if the routing is very congested, or if it is difficult to reach a pin.

Adding extra space to a non-default rule allows a designer to reduce cross-coupling capacitance and noise, but a clean route with no actual foundry spacing violations will still be allowed, unless the HARDSPACING statement is specified.
Type: Float, specified in microns

USEVIA *viaName*

Specifies a previously defined via from the LEF VIA statement, or a previously defined NONDEFAULTRULE via to use with this routing rule.
Applications may treat USEVIA as a soft constraint by putting the given vias on the list of target vias to be used, and they may have a separate option to specify a hard constraint for vias.

USEVIARULE *viaRuleName*

Specifies a previously defined VIARULE GENERATE rule to use with this routing rule. You cannot specify a rule from a VIARULE without a GENERATE keyword.

VIA *viaStatement*

Defines a new via. You define non-default vias using the same syntax as default vias. For syntax information, see "Via" on page 197. All vias, default and non-default, must have

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unique via names. If you define more than one via for a rule, the router chooses which via to use.

Note: Defining a new via is no longer recommended, and is likely to become obsolete. Instead, vias should be predefined in a LEF `VIA` statement, then added to the non-default rule using the `USEVIA` keyword.

`WIDTH` *`width`*

Specifies the required minimum width for *`layerName`*.
Type: Float, specified in microns

`WIREEXTENSION` *`value`*

Specifies the distance by which wires are extended at vias. Enter `0` (zero) to specify no extension. Values other than `0` must be greater than or equal to half of the routing width for the layer, as defined in the non-default rule.
Default: Wires are extended half of the routing width
Type: Float, specified in microns

Note: The `WIREEXTENSION` statement only extends wires and not vias. For 65nm and below, `WIREEXTENSION` is no longer recommended because it may generate some advance rule violations if wires and vias have different widths. See Illustration of WIREEXTENSION on page 674.

Example 1-15 Nondefault Rule Statement

Assume two default via rules were defined:

```
VIARULE via12rule GENERATE DEFAULT
    LAYER metal1 ;
    ...
END via12rule
VIARULE via23rule GENERATE DEFAULT
    LAYER metal2 ;
    ...
END via23rule
```

- Assuming the minimum width is 1.0 µm, the following non-default rule creates a 1.5x minimum width wire using default spacing:

```
NONDEFAULTRULE wide1_5x
    LAYER metal1
        WIDTH 1.5 ; #metal1 has a 1.5 um width
    END metal1
    LAYER metal2
        WIDTH 1.5 ;
```


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```
    END metal2
    LAYER metal3
        WIDTH 1.5 ;
    END metal3
END wide1_5x
```

Note: If there were no default via rules, then a `VIA`, `USEVIA`, or `USEVIARULE` keyword would be required. Because there are none defined, the default via rules are implicitly inherited for this non-default rule; therefore, `via12rule` and `via23rule` would be used for this routing rule.

- The following non-default rule creates a 3x minimum width wire using default spacing with at least two-cut vias:

```
NONDEFAULTRULE wide3x
    LAYER metal1
        WIDTH 3.0 ; #metal1 has 3.0 um width
    END metal1
    LAYER metal2
        WIDTH 3.0 ;
    END metal2
    LAYER metal3
        WIDTH 3.0 ;
    END metal3
    #viarule12 and viarule23 are used implicitly
    MINCUTS cut12 2 ; #at least two-cut vias are required for cut12
    MINCUTS cut23 2 ;
END wide3x
```

- The following non-default rule creates an “analog” rule with its own special vias, and with hard extra spacing:

```
NONDEFAULTRULE analog_rule
    HARDSPACING ;     #do not let any other signal close to this one
    LAYER metal1
        WIDTH 1.5 ;   #metal1 has 1.5 um width
        SPACING 3.0 ; #extra spacing of 3.0 um
    END metal1
    LAYER metal2
        WIDTH 1.5
        SPACING 3.0
    END metal2
    LAYER metal3
        WIDTH 1.5
```


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```
        SPACING 3.0
    END metal3
    #Use predefined "analog vias"
    #The DEFAULT VIARULES will not be inherited.
    USEVIA via12_fixed_analog_via ;
    USEVIA via_23_fixed_analog_via ;
END analog_rule
```

Use Via Cut Class Rule

You can use the use via cut class only rule to specify the cut class to be used with the routing rule.

You can create a use via cut class rule by using the following property definition:

```
PROPERTY LEF58_USEVIACUTCLASS
     USEVIACUTCLASS cutLayerName className
        [ROWCOL numCutRows numCutCols]
        ;... " ;
```

Where:

`USEVIACUTCLASS` *`cutLayerName className`*
`[ROWCOL` *`numCutRows numCutCols`*`]`

Specifies *`className`* cuts on *`cutLayerName`*, which must be a cut layer, to be used with this routing rule. `ROWCOL` specifies the number of cut rows and columns of a multiple-cut via of the cut class *`className`* to be used with this routing rule. By default, a single cut via or the given cut number defined in `MINCUTS` is used.
Applications may treat `USEVIACUTCLASS` as a hard constraint and apply the given cut class vias on the non-default rule.

Property Definitions

```
[PROPERTYDEFINITIONS
     [objectType propName propType [RANGE min max]
        [value | "stringValue"]
     ;] ...
END PROPERTYDEFINITIONS]
```

Lists all properties used in the LEF file. You must define properties in the `PROPERTYDEFINITIONS` statement before you can refer to them in other sections of the LEF file.

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objectType

Specifies the object type being defined. You can define properties for the following object types:

LAYER

LIBRARY

MACRO

NONDEFAULTRULE

PIN

VIA

VIARULE

propName

Specifies a unique property name for the object type.

propType

Specifies the property type for the object type. You can specify one of the following property types:

INTEGER

REAL

STRING

RANGE *min max*

Limits real number and integer property values to a specified range. That is, the value must be greater than or equal to *min* and less than or equal to *max*.

value | "*stringValue*"

Assigns a numeric value or a name to a LIBRARY object type.

Note: Assign values to other properties in the section of the LEF file that describes the object to which the property applies.

Example 1-16 Property Definitions Statement

The following example shows library, via, and macro property definitions.

```
PROPERTYDEFINITIONS
```


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```
    LIBRARY versionNum INTEGER 12;
    LIBRARY title STRING "Cadence96";
    VIA count INTEGER RANGE 1 100;
    MACRO weight REAL RANGE 1.0 100.0;
    MACRO type STRING;
END PROPERTYDEFINITIONS
```

Site

```
SITE siteName
     CLASS {PAD | CORE} ;
     [SYMMETRY {X | Y | R90} ... ;]
     [ROWPATTERN {previousSiteName siteOrient} ... ;]
     SIZE width BY height ;
END siteName
```

Defines a placement site in the design. A placement site gives the placement grid for a family of macros, such as I/O, core, block, analog, digital, short, tall, and so forth. `SITE` definitions can be used in DEF `ROW` statements.

`CLASS {PAD | CORE}` Specifies whether the site is an I/O pad site or a core site.

`ROWPATTERN {`*`previousSiteName siteOrient`*`}`

Specifies a set of previously defined sites and their orientations that together form *`siteName`*.

The previously defined sites are abutted horizontally or vertically, depending on which of the following conditions is met:

- The height of each previously defined site is the same as the height specified for *`siteName`*, and the sum of the widths of the previously defined sites is equal to the width specified for *`siteName`*. In this case, the previously defined sites are abutted horizontally.
- The width of each previously defined site is the same as the width specified for *`siteName`*, and the sum of the heights of the previously defined sites is equal to the height specified for *`siteName`*. In this case, the previously defined sites are abutted vertically.

`previousSiteName`

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Specifies the name of a previously defined site. The sites in *previousSiteName* have to be basic sites with no pattern.

siteOrient

Specifies the orientation for the previously defined site. This value must be one of `N`, `S`, `E`, `W`, `FN`, `FS`, `FE`, and `FW`. For more information on orientations, see "Specifying Orientation" in the DEF `COMPONENT` section.

`SITE` *siteName* Specifies the name for the placement site.

`SIZE` *width* `BY` *height*

Specifies the dimensions of the site in normal (or north) orientation, in microns.

`SYMMETRY {X | Y | R90}`

Indicates which site orientations are equivalent. The sites in a given row all have the same orientation as the row. Generally, site symmetry should be used to control the flipping allowed inside the rows. For more information on defining symmetry, see "Defining Symmetry" on page 153.

Possible orientations include:

`X` Site is symmetric about the x axis. This means that N and FS sites are equivalent, and FN and S sites are equivalent. A macro with an orientation of N matches N or FS rows.

`Y` Site is symmetric about the y axis. This means that N and FN sites are equivalent, and FS and S sites are equivalent. A macro with an orientation of N matches N or FN rows.

`X Y` Site is symmetric about the x and y axis. This means that N, FN, FS, and S sites are equivalent. A macro with orientation N matches N, FN, FS, or S rows.

`R90` Site is symmetric when rotated 90 degrees. Typically, this value is not used.

Note: Typically, a site for single-height standard cells uses symmetry `Y`, and a site for double-height standard cells uses symmetry `X Y`.

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Site Row Pattern Examples

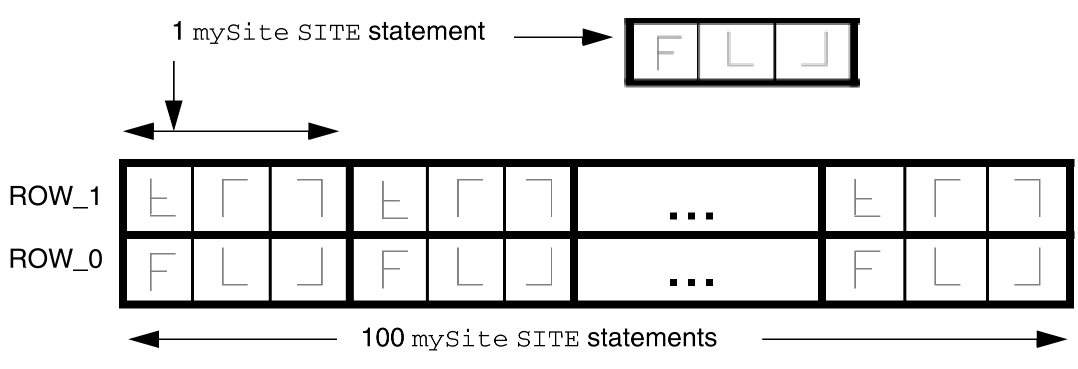
Example 1-17 Horizontal Pattern

The following example defines three sites: `Fsite`; `Lsite`; and `mySite`, which consists of a pattern of `Fsite` and `Lsite` sites:

```
SITE Fsite
    CLASS CORE ;
    SIZE 4.0 BY 7.0 ; #4.0 um width, 7.0 um height
END Fsite
SITE Lsite
    CLASS CORE ;
    SIZE 6.0 BY 7.0 ; #6.0 um width, 7.0 um height
END Lsite
SITE mySite
    ROWPATTERN Fsite N Lsite N Lsite FN ; #Pattern of F + L + flipped L
    SIZE 16.0 BY 7.0 ;                    #Width = width(F + L + L)
END mySite
```

The figure below illustrates some DEF rows made up of `mySite` sites.

Figure 1-60 Site Row Pattern Example




```
ROW ROW_0 mySite 1000 1000 N DO 100 BY 1 STEP 1600 0 ;
ROW ROW_1 mySite 1000 1700 FS DO 100 BY 1 STEP 1600 0 ;
```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Example 1-18 Vertical Pattern

The following example defines three sites: `SFsite`; `TFsite`; and `MySiteTST`, which consists of a pattern of `SFsite` and `TFsite` sites:

```
SITE SFsite
    CLASS CORE ;
    SIZE 4.0 BY 13.0 ; #4.0 um width, 13.0 um height
END SFsite
SITE TFsite
    CLASS CORE ;
    SIZE 4.0 BY 16.0 ; #4.0 um width, 16.0 um height
END TFsite
SITE MySiteTST
    ROWPATTERN TFsite N SFsite FS TFsite N ;
    SIZE 4.0 BY 45.0 ; #Height = height(TF + SF + TF)
END MySiteTST
```

The figure below illustrates some DEF rows made up of `MySiteTST` sites.

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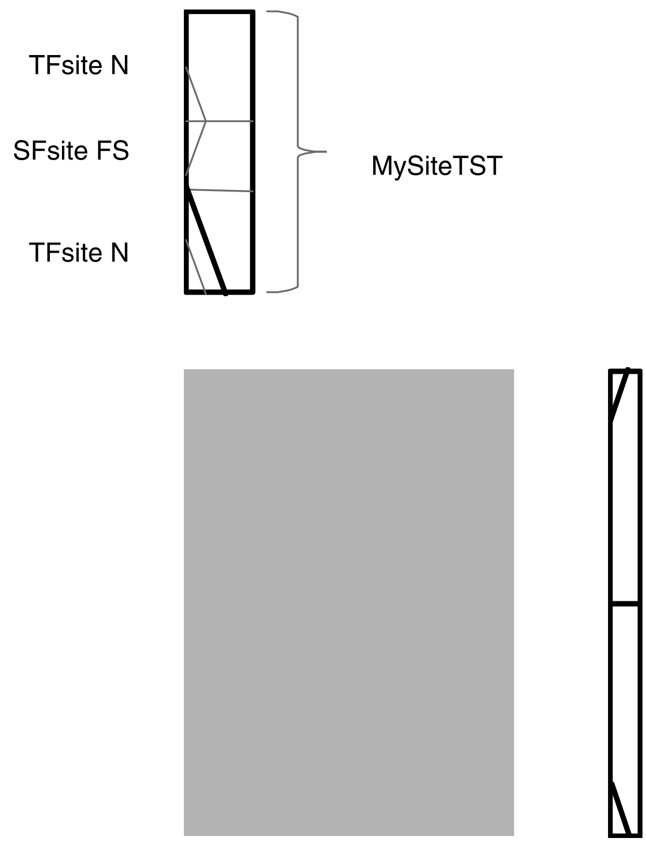
Figure 1-61 Vertical Pattern Example




```
ROW ROW_0 TFsite 1000 1000 N  DO 100 BY 1 STEP 40 0 ;
ROW ROW_1 SFsite 1000 1160 FS DO 100 BY 1 STEP 40 0 ;
ROW ROW_2 TFsite 1000 1290 N  DO 100 BY 1 STEP 40 0 ;
ROW ROW_3 TFsite 1000 1450 FS DO 100 BY 1 STEP 40 0 ;
ROW ROW_4 SFsite 1000 1610 N  DO 100 BY 1 STEP 40 0 ;
. . .
ROW ROW_T_0 MySiteTST 1000 1000 N  DO 100 BY 1 STEP 40 0 ;
ROW ROW_T_1 MySiteTST 1000 1450 FS DO 100 BY 1 STEP 40 0 ;
```

Units

```
[UNITS
     [TIME NANOSECONDS convertFactor ;]
     [CAPACITANCE PICOFARADS convertFactor ;]
     [RESISTANCE OHMS convertFactor ;]
     [POWER MILLIWATTS convertFactor ;]
     [CURRENT MILLIAMPS convertFactor ;]
     [VOLTAGE VOLTS convertFactor ;]
     [DATABASE MICRONS LEFconvertFactor ;]
     [FREQUENCY MEGAHERTZ convertFactor ;]
```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```
END UNITS]
```

Defines the units of measure in LEF. The values tell you how to interpret the numbers found in the LEF file. Units are fixed with a *convertFactor* for all unit types, except database units and capacitance. For more information, see "Convert Factors" on page 195. Currently, other values for *convertFactor* appearing in the UNITS statement are ignored.

The UNITS statement is optional and, when used, must precede the LAYER statements.

CAPACITANCE PICOFARADS *convertFactor*

Interprets one LEF capacitance unit as 1 picofarad.

CURRENT MILLIAMPS *convertFactor*

Interprets one LEF current unit as 1 milliamp.

DATABASE MICRONS *LEFconvertFactor*

Interprets one LEF distance unit as multiplied when converted into database units.

If you omit the DATABASE MICRONS statement, a default value of 100 is recorded as the *LEFconvertFactor* in the database. In this case, one micron would equal 100 database units.

FREQUENCY MEGAHERTZ *convertFactor*

Interprets one LEF frequency unit as 1 megahertz.

POWER MILLIWATTS *convertFactor*

Interprets one LEF power unit as 1 milliwatt.

RESISTANCE OHMS *convertFactor*

Interprets one LEF resistance unit as 1 ohm.

TIME NANOSECONDS *convertFactor*

Interprets one LEF time unit as 1 nanosecond.

VOLTAGE VOLTS *convertFactor*

Interprets one LEF voltage unit as 1 volt.

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Database Units Information

Database precision is relative to Standard International (SI) units. LEF values are converted to integer values in the library database as follows.

SI unit	Database precision
1 nanosecond	= 1,000 DBUs
1 picofarad	= 1,000,000 DBUs
1 ohm	= 10,000 DBUs
1 milliwatt	= 10,000 DBUs
1 milliamp	= 10,000 DBUs
1 volt	= 1,000 DBUs

Convert Factors

LEF supports values of 100, 200, 400, 800, 1000, 2000, 4000, 8000, 10,000, and 20,000 for *`LEFconvertFactor`*. The following table illustrates the conversion of LEF distance units into database units.

LEFconvertFactor	LEF	Database Units
100	1 micron	100
200	1 micron	200
400	1 micron	400
800	1 micron	800
1000	1 micron	1000
2000	1 micron	2000
4000	1 micron	4000
8000	1 micron	8000
10,000	1 micron	10,000
20,000	1 micron	20,000

The DEF database precision cannot be more precise than the LEF database precision. This means the DEF convert factor must always be less than or equal to the LEF convert factor.

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The LEF convert factor must also be an integer multiple of the DEF convert factor so no round-off of DEF database unit values is required (e.g., a LEF convert factor of 1000 allows DEF convert factors of 100, 200, 1000, but not 400, 800). The following table shows the valid pairings of the LEF convert factor and the corresponding DEF convert factor.

LEFconvertFactor	Legal DEFconvertFactors
100	100
200	100, 200
400	100, 200, 400
800	100, 200, 400, 800
1000	100, 200, 1000
2000	100, 200, 400, 1000, 2000
4000	100, 200, 400, 800, 1000, 2000, 4000
8000	100, 200, 400, 800, 1000, 2000, 4000, 8000
10,000	100, 200, 400, 1000, 2000, 10,000
20,000	100, 200, 400, 800, 1000, 2000, 4000, 10,000, 20,000

An incremental LEF should have the same value as a previous LEF. An error message warns you if an incremental LEF has a different value than what is recorded in the database.

Use Min Spacing

```
[USEMINSPACING OBS { ON | OFF } ;]
```

Defines how minimum spacing is calculated for obstruction (blockage) geometries.

`OBS {ON | OFF}`
: Specifies how to calculate minimum spacing for obstruction geometries (`MACRO OBS` shapes).
Default: `ON`

`OFF`
: Spacing is computed to `MACRO OBS` shapes as if they were actual routing shapes. A wide `OBS` shape would use wide wire spacing rules, and a thin `OBS` shapes would use thin wire spacing rules.

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`ON`	Spacing is computed as if the `MACRO OBS` shapes were min-width wires. Some LEF models abstract many min-width wires as a single large `OBS` shape; therefore using wide wire spacing would be too conservative.

Note: `OFF` is the recommended value to specify because it is a better abstract model for the various wide wire spacing rules that are more common at process nodes of 130nm and smaller. Certain older style LEF abstracts use `ON`, but it can have unexpected side effects (such as hidden DRC errors) if the abstracts are not created very carefully. You cannot mix both types of LEF abstracts at the same time.

Version

```
VERSION number ;
```

Specifies which version of the LEF syntax is being used. *`number`* is a string of the form *`major.minor`*`[.`*`subMinor`*`]`, such as `5.8`.

Note: Many applications default to the latest version of LEF/DEF supported by the application (which depends on how old the application is). The latest version as described by this document is 5.8. However, a default value of `5.8` is not formally part of the language definition; therefore, you cannot be sure that all applications use this default value. Also, because the default value varies with the latest version, you should not depend on this.

Via

```
VIA viaName [DEFAULT]
     { VIARULE viaRuleName ;
          CUTSIZE xSize ySize ;
          LAYERS botMetalLayer cutLayer topMetalLayer ;
          CUTSPACING xCutSpacing yCutSpacing ;
          ENCLOSURE xBotEnc yBotEnc xTopEnc yTopEnc ;
          [ROWCOL numCutRows numCutCols ;]
          [ORIGIN xOffset yOffset ;]
          [OFFSET xBotOffset yBotOffset xTopOffset yTopOffset ;]
          [PATTERN cutPattern ;]
     }
     | {[RESISTANCE resistValue ;]
        {LAYER layerName ;
          { RECT [MASK maskNum] pt pt ;
          | POLYGON [MASK maskNum] pt pt pt ...;} ...
```


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```
        } ...
      }
    [PROPERTY propName propVal ;] ...
END viaName
```

Defines two types of vias: fixed vias and generated vias. All vias consist of shapes on three layers: a cut layer and two routing (or masterslice) layers that connect through that cut layer.

A LEF fixed via is defined using rectangles or polygons, and does not use a `VIARULE`. The fixed via name must mean the same via in all associated LEF and DEF files.

A LEF generated via is defined using `VIARULE` parameters to indicate that it was derived from a `VIARULE GENERATE` statement. The parameterized via name must mean the same via in all LEF files.

Note that unlike LEF, a DEF parameterized VIA can have its name changed as long as the parameters are kept the same (see DEF Vias section), but LEF VIAs are assumed to be “common library data” that can be used by other LEF files, such as LEF MACROs, so all LEF VIA definition names should be defined only once in all the LEF files that are loaded.

If a LEF MACRO block needs its own locally defined VIA definition that is not already defined, it should use a unique via name. One way to do so is to use the MACRO name in the VIA definition name. For example, `MACRO my_block1` might have a unique via named `VIA my_block1_via1` inside the same LEF file.

`CUTSIZE` *`xSize ySize`*

Specifies the required width (*`xSize`*) and height (*`ySize`*) of the cut layer rectangles.
Type: Float, specified in microns

`CUTSPACING` *`xCutSpacing yCutSpacing`*

Specifies the required x and y spacing between cuts. The spacing is measured from one cut edge to the next cut edge.
Type: Float, specified in microns

`DEFAULT`

Identifies the via as the default via between the defined layers. Default vias are used for default routing by the signal routers.

If you define more than one default via for a layer pair, the router chooses which via to use. You must define default vias between *metal1* and masterslice layers if there are pins on the masterslice layers.

All vias consist of shapes on three layers: a cut layer and two routing (or masterslice) layers that connect through that cut layer. There should be at least one `RECT` or `POLYGON` on each of the three layers.

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`ENCLOSURE` *`xBotEnc yBotEnc xTopEnc yTopEnc`*

Specifies the required x and y enclosure values for the bottom and top metal layers. The enclosure measures the distance from the cut array edge to the metal edge that encloses the cut array.
Type: Float, specified in microns

Note: It is legal to specify a negative number, as long as the resulting metal size is positive.

`LAYER` *`layerName`*

Specifies the layer on which to create the rectangles that make up the via. All vias consist of shapes on three layers: a cut layer and two routing (or masterslice) layers that connect through that cut layer. There should be at least one `RECT` or `POLYGON` on each of the three layers.

`LAYERS` *`botMetalLayer cutLayer topMetalLayer`*

Specifies the required names of the bottom routing (or masterslice) layer, cut layer, and top routing (or masterslice) layer. These layer names must be previously defined in layer definitions, and must match the layer names defined in the specified LEF *`viaRuleName`*.

`MASK` *`maskNum`*

Specifies which mask for double- or triple-patterning lithography is to be applied to the shapes defined in `RECT` or `POLYGON` of the via master. The *`maskNum`* variable must be a positive integer. Most applications only support values of 1, 2, or 3. For a fixed via made up of `RECT` or `POLYGON` statements, the cut-shapes should either be all colored or not colored at all. It is an error to have partially colored cuts for one via. Uncolored cut shapes should be automatically colored if the layer is a multi-mask layer.

The metal shapes with a shape per layer of the via-master do not need colors because the via instance has the mask color, but some readers will color them as `mask 1` for internal consistency (see Figure 1-67 on page 209). So a writer may write out `MASK 1` for the metal shapes even if they are read in with no `MASK` value.

`OFFSET` *`xBotOffset yBotOffset xTopOffset yTopOffset`*

Specifies the x and y offset for the bottom and top metal layers. By default, the `0,0` origin of the via is the center of the cut array, and the enclosing metal rectangles. These values allow each metal layer to be offset independently. After the non-shifted via is computed, the metal layer rectangles are offset by adding the appropriate values—the x/y *`BotOffset`* values to the metal layer below the cut layer, and the x/ y *`TopOffset`* values to the metal layer above the cut layer. These offsets are in addition to any offset caused by the `ORIGIN` values.

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Default: `0`, for all values
Type: Float, specified in microns

`ORIGIN` *`xOffset yOffset`*

Specifies the x and y offset for all of the via shapes. By default, the `0,0` origin of the via is the center of the cut array, and the enclosing metal rectangles. After the non-shifted via is computed, all cut and metal rectangles are offset by adding these values.
Default: `0`, for both values
Type: Float, specified in microns

`PATTERN` *`cutPattern`*

Specifies the cut pattern encoded as an ASCII string. This parameter is only required when some of the cuts are missing from the array of cuts, and defaults to "all cuts are present," if not specified.

For information on and examples of via cut patterns, see Creating Via Cut Patterns.

The *`cutPattern`* syntax uses "_" as a separator, and is defined as follows:

`numRows_rowDefinition`
  `[`*`_numRows_rowDefinition`*`] ...`

`numRows`	Specifies a hexadecimal number that indicates how many times to repeat the following row definition. This number can be more than one digit.
`rowDefinition`	Defines one row of cuts, from left to right.

The *`rowDefinition`* syntax is defined as follows:

`{[R`*`repeatNumber`*`]`*`hexDigitCutPattern`*`} ...`

`hexDigitCutPattern`	
	Specifies a single hexadecimal digit that encodes a 4-bit binary value, in which `1` indicates a cut is present, and `0` indicates a cut is not present.
`repeatNumber`	Specifies a single hexadecimal digit that indicates how many times to repeat *`hexDigitCutPattern`*.

For parameterized vias (with `+ VIARULE` ...), the *`cutPattern`* has an optional suffix added to allow three types of mask color patterns. The default mask color pattern (no suffix) is a checker-board defined as an alternating pattern starting with `MASK 1` at the bottom left. Then the mask cycles left-to-right, and from bottom-to-top, as shown in Figure 1-65 on page 206.The other two patterns supported are alternating rows, and alternating columns, see Figure 1-66 on page 207.

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The optional suffixes are:

<cut_pattern>**_MR** alternating rows

<cut_pattern>**_MC** alternating columns

`POLYGON` *`pt pt pt`*

Specifies a sequence of at least three points to generate a polygon geometry. Most fab DRC rules require each polygon edge to be parallel to the x or y axis, or at a 45-degree angle. However, odd-angles are allowed sometimes on a few layers, such as bump layers. Each `POLYGON` statement defines a polygon generated by connecting each successive point, and then by connecting the first and last points.
The *`pt`* syntax corresponds to an x y coordinate pair, such as `-0.2 1.0`.
Type: Float, specified in microns

Example 1-19 Via Rules

The following via rule describes a non-shifted via (that is, a via with no `OFFSET` or `ORIGIN` parameters). There are two rows and three columns of via cuts. Figure 1-62 on page 202 illustrates this via rule.

```
VIA myVia
    VIARULE myViaRule ;
    CUTSIZE 20 20 ;             #xCutSize yCutSize
    LAYERS metal1 cut12 metal2 ;
    CUTSPACING 30 30 ;           #xCutSpacing yCutSpacing
    ENCLOSURE 20 50 50 20 ;      #xBotEnc yBotEnc xTopEnc yTopEnc
    ROWCOL 2 3 ;
END myVia
```


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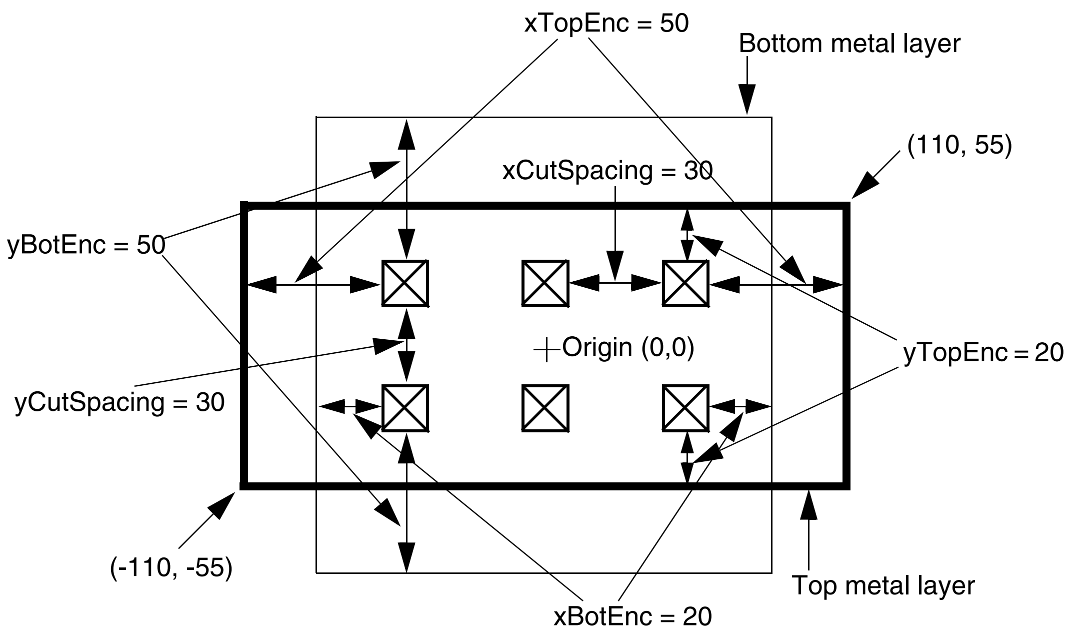
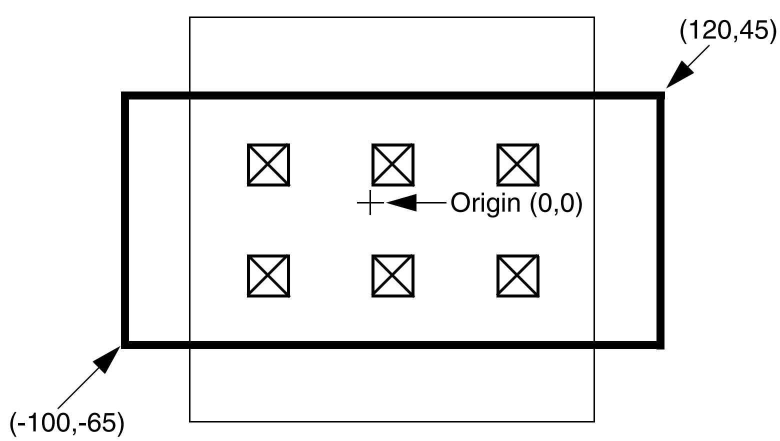
Figure 1-62 Via Rule



The same via rule with the following `ORIGIN` parameter shifts all of the metal and cut rectangles by 10 in the x direction, and by -10 in the y direction (see Figure 1-63 on page 202):

```
ORIGIN 10 -10 ;
```

Figure 1-63 Via Rule With Origin



If the same via rule contains the following `ORIGIN` and `OFFSET` parameters, all of the rectangles shift by 10, -10. In addition, the top layer metal rectangle shifts by 20, -20, which means that the top metal shifts by a total of 30, -30.

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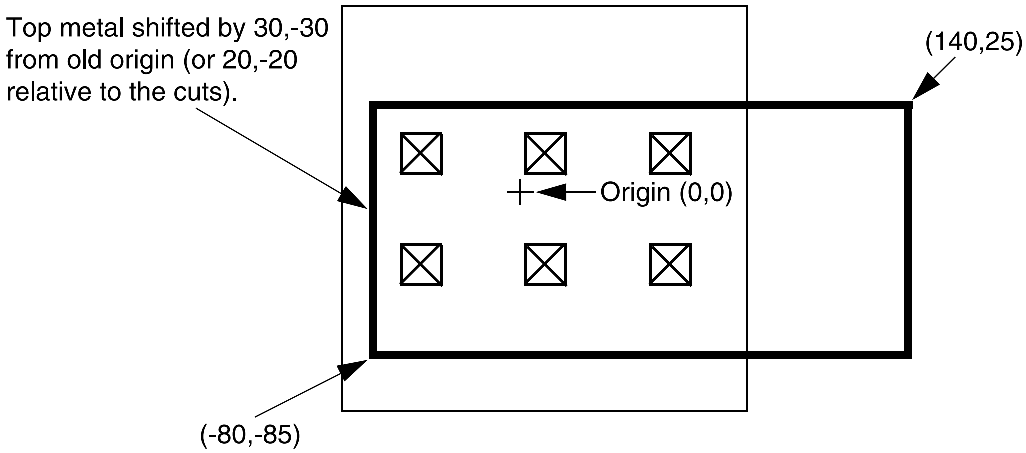
```
    ORIGIN 10 -10 ;
    OFFSET 0 0 20 -20 ;
```

Figure 1-64 Via Rule With Origin and Offset



Example 1-20 Via Polygon

The following via definition creates a polygon geometry used by X-routing applications:

```
VIA myVia23
    LAYER metal2 ;
    POLYGON -2.1 -1.0 -0.1 1.0 2.1 1.0 0.1 -1.0 ;
    LAYER cut23 ;
    RECT -0.4 -0.4 0.4 0.4 ;
    LAYER metal3 ;
    POLYGON -0.1 -1.0 -2.1 1.0 0.1 1.0 2.1 -1.0 ;
END myVia23
```

`PROPERTY` *`propName propVal`*

Specifies a numerical or string value for a via property defined in the `PROPERTYDEFINITIONS` statement. The *`propName`* you specify must match the *`propName`* listed in the `PROPERTYDEFINITIONS` statement.

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`RECT` *`pt pt`*

Specify the corners of a rectangular shape in the via. The *`pt`* syntax corresponds to an x y coordinate pair, such as `-0.4 -4.0`. For vias used only in macros or pins, reference locations and rectangle coordinates must be consistent.
Type: Float, specified in microns

`RESISTANCE` *`resistValue`*

Specifies the lumped resistance for the via. This is not a resistance per via-cut value; it is the total resistance of the via. By default, via resistance is computed from the via `LAYER RESISTANCE` value; however, you can override that value with this value. *`resistValue`* is ignored if a via rule is specified, because only the `VIARULE` definition or a cut layer `RESISTANCE` value gives the resistance for generated vias.
Type: Float, specified in ohms

Note: A `RESISTANCE` value attached to an individual via is no longer recommended.

`ROWCOL` *`numCutRows numCutCols`*

Specifies the number of cut rows and columns that make up the via array.
Default: `1`, for both values
Type: Positive integer, for both values

`viaName`

Specifies the name for the via.

`VIARULE` *`viaRulename`*

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Specifies the name of the LEF `VIARULE` that produced this via. This indicates that the via is the result of automatic via generation, and that the via name is only used locally inside this LEF file. The geometry and parameters are maintained, but the name can be freely changed by applications that use this via when writing out LEF and DEF files.

`viaRuleName` must be specified before you define any of the other parameters, and must refer to a previously defined `VIARULE GENERATE` rule name. It cannot refer to a `VIARULE` without a `GENERATE` keyword.

Specifying the reserved via rule name of `DEFAULT` indicates that the via should use a previously defined `VIARULE GENERATE` rule with the `DEFAULT` keyword that exists for this routing-cut-routing (or masterslice-cut-masterslice) layer combination. This makes it possible for an IP block user to use existing via rules from the normal LEF technology section instead of requiring it to locally create its own via rules for just one LEF file.

Example 1-21 Generated Via Rule

The following via definition defines a generated via that is used only in this LEF file.

```
VIA myBlockVia
    VIARULE DEFAULT ;                  #Use existing VIARULE GENERATE rule with
                                        #the DEFAULT keyword
        CUTSIZE 0.1 0.1 ;               #Cut is 0.1 x 0.1 um
        LAYERS metal1 via12 metal2 ;    #Bottom metal, cut, and top metal layers
        CUTSPACING 0.1 0.1 ;            #Space between cut edges is 0.1 um
        ENCLOSURE 0.05 0.01 0.01 0.05 ; #metal1 enclosure is 0.05 in x, 0.01 in y
                                        #metal2 enclosure is 0.01 in x, 0.05 in y
        ROWCOL 1 2 ;                    #1 row, 2 columns = 2 cuts
END myBlockVia
```

Example 1-22 Parameterized via cut-mask pattern

The following example shows a `VIARULE` parameterized via:

```
    VIA  myParamVia1
        VIARULE myGenVia1          CUTSIZE 0.4 0.4
        LAYERS M1 VIA1 M2          CUTSPACING 0.4 0.4
        ENCLOSURE 0.4 0 0 0.4      ROWCOL 3 4          #3 rows, 4 columns
        PATTERN 2_F_1_D;                               #1 cut in top row is missing
```


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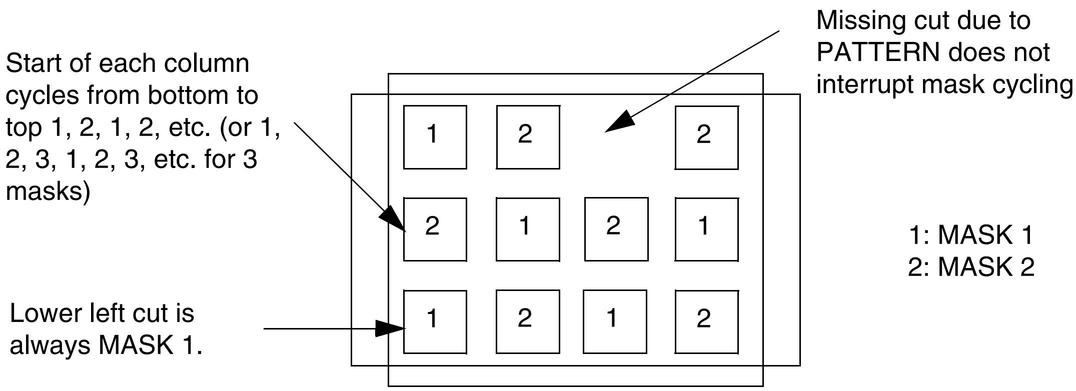
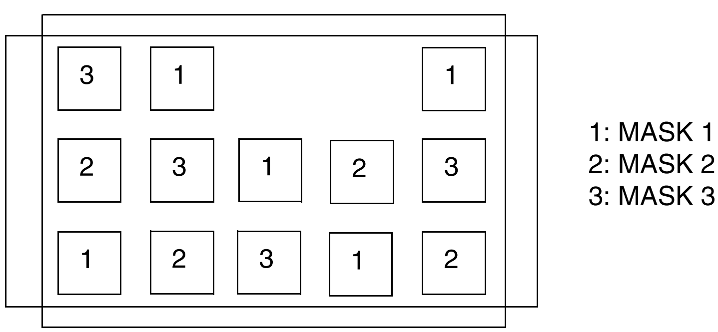
Example of a parameterized via checker-board cut-mask pattern for a 3-mask layer with 2 missing cuts. For parameterized vias (with `VIARULE ...`), the mask of the cuts are pre-defined as an alternating pattern starting with `MASK 1` at the bottom left. The mask cycles from left-to-right and bottom-to-top are shown.

Figure 1-65 Parameterized via cut-mask pattern using PATTERN



Assuming that the layer VIA1 is a two-mask cut-layer, then the mask color for each cut-shape is a checker-board as shown above. The default checker-board pattern is: the bottom left cut starts at 1, and then goes from left to right, 1, 2, 1, 2. The next row starts at 2, and goes 2, 1, 2, 1, etc. The missing cuts due to `PATTERN` statement do not change the mask assignments.



Example of a parameterized via checker-board cut-mask pattern for a 3-mask layer with 2 missing cuts. For parameterized vias (with `VIARULE ...`), the mask of the cuts are pre-defined as an alternating pattern starting with `MASK 1` at the bottom left. The mask cycles from left-to-right and bottom-to-top are shown.

- The following examples show a parameterized via with cut-mask patterns for a 3-mask layer and 2-mask layer using _MC and _MR suffixes:

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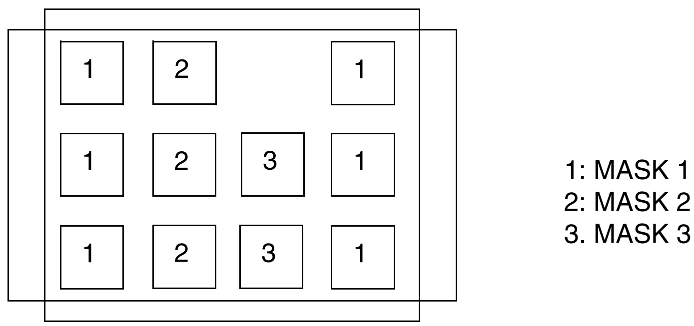
Figure 1-66 Parameterized via cut-mask pattern using Suffixes

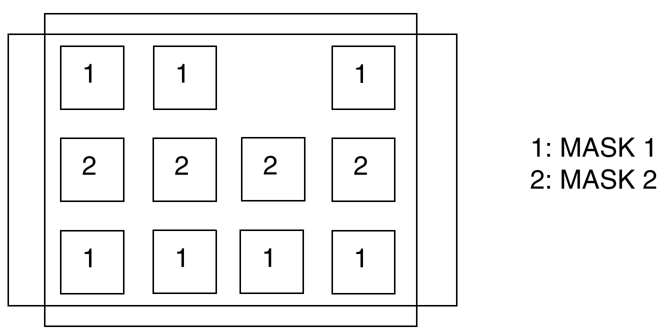
Mask color pattern for a parameterized via with 3-mask cut-layer:

```
VIA  myParamVia1
  VIARULE myGenVia1          CUTSIZE 0.4 0.4
  LAYERS M1 VIA1 M2          CUTSPACING 0.4 0.4
  ENCLOSURE 0.4 0 0 0.4      ROWCOL 3 4
  PATTERN 2_F_1_D_MC;        #alternating mask color columns due to _MC suffix
```



For a VIARULE parameterized via like:

```
VIA  myParamVia1
  VIARULE myGenVia1          CUTSIZE 0.4 0.4
  LAYERS M1 VIA1 M2          CUTSPACING 0.4 0.4
  ENCLOSURE 0.4 0 0 0.4      ROWCOL 3 4
  PATTERN 2_F_1_D _MR ;      #alternating mask color rows due to _MR suffix
```



Example 1-23 Fixed-via with pre-colored cut shapes

The following example shows a fixed-via with pre-colored cut shapes:

```
VIA myVia1
    LAYER m1 ;
    RECT -0.4 -0.2 1.2 0.2 ;           #no mask, some readers will set to mask 1
    LAYER via1 ;
    RECT MASK 1 -0.2 -0.2 0.2 0.2 ;    #first cut on mask 1
```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```
        RECT MASK 2  0.6 -0.2 1.0 0.2 ;    #second cut on mask 2
        LAYER m2 ;
        RECT -0.2 -0.4 1.0 0.4 ;           #no mask, some readers will set to mask 1
    END myVia1
```

For a fixed via made up of `RECT` or `POLYGON` statements, the cut shapes must all be colored or not colored at all. If the cuts are not colored, they will be automatically colored in a checkerboard pattern as described above for parameterized vias. Each via-cut with the same lower-left Y value is considered one row, and each via in one row is a new column. For common "array" style vias with no missing cuts, this coloring is a good one. For vias that do not have a row and column structure, or are missing cuts this coloring may not be good (see Figure 1-67 on page 209). If the metal layers having only one shape per layer are not colored, some applications will color them to `MASK 1` for internal consistency, even though the via-master metal shape colors are not really used by LEF/DEF via instances. For multiple disjoint metal shapes, it is highly recommended to provide proper color.

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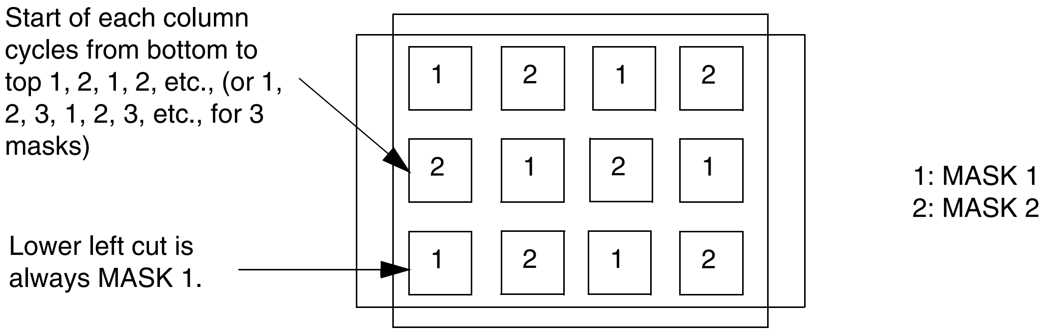
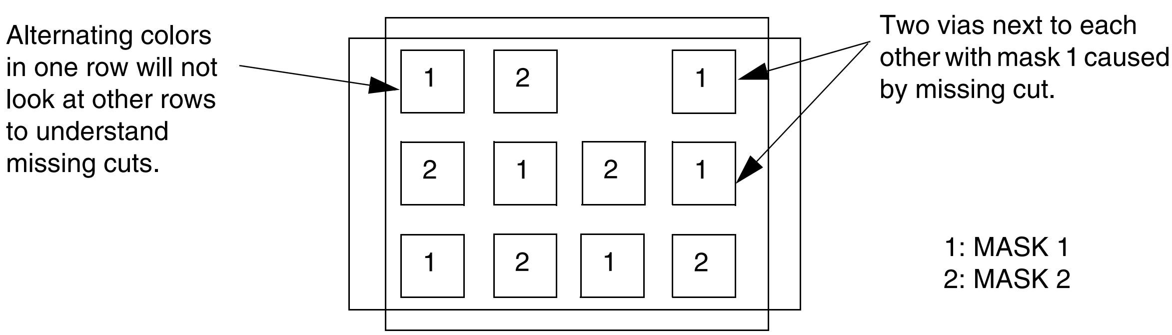
Figure 1-67 Fixed-via with pre-colored cut shapes



Uncolored fixed vias (from RECT/POLYGON statements) will get similar coloring as a parameterized via. Each cut with the same Y value is one row. The cuts inside one row alternate. This works well for cut-arrays without missing cuts as shown above.



Unlike parameterized vias, there is no real row/column structure required in a fixed via, so the automatic coloring will not work well for a fixed via with missing cuts (or cuts that are not aligned). This type of via must be pre-colored to be useful.

See the MACRO Layer Geometries statement to see how a via-instance uses these via-master mask values.

Via Rule

```
VIARULE viaRuleName
     LAYER layerName ;
        DIRECTION {HORIZONTAL | VERTICAL} ;
        [WIDTH minWidth TO maxWidth ;]
     LAYER layerName ;
        DIRECTION {HORIZONTAL | VERTICAL} ;
        [WIDTH minWidth TO maxWidth ;]
     {VIA viaName ;} ...
     [PROPERTY propName propVal ;] ...
```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`END` *`viaRuleName`*

Defines which vias to use at the intersection of special wires of the same net.

Note: You should only use `VIARULE GENERATE` statements to create a via for the intersection of two special wires. In earlier versions of LEF, `VIARULE GENERATE` was not complete enough to cover all situations. In those cases, a fixed `VIARULE` (without a `GENERATE` keyword) was sometimes used. This is no longer required.

`DIRECTION {HORIZONTAL | VERTICAL}`

Specifies the wire direction. If you specify a `WIDTH` range, the rule applies to wires of the specified `DIRECTION` that fall within the range. Otherwise, the rule applies to all wires of the specified `DIRECTION` on the layer.

`LAYER` *`layerName`*

Specifies the routing or masterslice layers for the top or bottom of the via.

`PROPERTY` *`propName propVal`*

Specifies a numerical or string value for a via rules property defined in the `PROPERTYDEFINITIONS` statement. The *`propName`* you specify must match the *`propName`* listed in the `PROPERTYDEFINITIONS` statement.

`VIA` *`viaName`*

Specifies a previously defined via to test for the current via rule. The first via in the list that can be placed at the location without design rule violations is selected. The vias must all have exactly three layers in them. The three layers must include the same routing or masterslice layers as listed in the `LAYER` statements of the `VIARULE`, and a cut layer that is between the two routing or masterslice layers.

`VIARULE` *`viaRuleName`*

Specifies the name to identify the via rule.

`WIDTH` *`minWidth`* `TO` *`maxWidth`*

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Specifies a wire width range. If the widths of two intersecting special wires fall within the wire width range, the `VIARULE` is used. To fall within the range, the widths must be greater than or equal to *`minWidth`* and less than or equal to *`maxWidth`*.

Note: `WIDTH` is defined by wire direction, not by layer. If you specify a `WIDTH` range, the rule applies to wires of the specified `DIRECTION` that fall within the range.

Example 1-24 Via Rule Statement

In the following example, whenever a *metal1* wire with a width between 0.5 and 1.0 intersects a *metal2* wire with a width between 1.0 and 2.0, the via generation code attempts to put a *via12_1* at the intersection first. If the *via12_1* causes a DRC violation, a *via12_2* is then tried. If both fail, the default behavior from a `VIARULE GENERATE` statement for *metal1* and *metal2* is used.

```
VIARULE viaRule1
    LAYER metal1 ;
        DIRECTION HORIZONTAL ;
        WIDTH 0.5 TO 1.0 ;
    LAYER metal2 ;
        DIRECTION VERTICAL ;
        WIDTH 1.0 TO 2.0 ;
    VIA via12_1 ;
    VIA via12_2 ;
END viaRule1
```

Via Rule Generate

```
VIARULE viaRuleName GENERATE [DEFAULT]
     LAYER routingLayerName ;
        ENCLOSURE overhang1 overhang2 ;
        [WIDTH minWidth TO maxWidth ;]
     LAYER routingLayerName ;
        ENCLOSURE overhang1 overhang2 ;
        [WIDTH minWidth TO maxWidth ;]
     LAYER cutLayerName ;
        RECT pt pt ;
        SPACING xSpacing BY ySpacing ;
        [RESISTANCE resistancePerCut ;]

END viaRuleName
```


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Defines formulas for generating via arrays. You can use the `VIARULE GENERATE` statement to cover special wiring that is not explicitly defined in the `VIARULE` statement.

Rather than specifying a list of vias for the situation, you can create a formula to specify how to generate the cut layer geometries.

Note: Any vias created automatically from a `VIARULE GENERATE` rule that appear in the DEF `NETS` or `SPECIALNETS` sections must also appear in the DEF `VIA` section.

`DEFAULT`

Specifies that the via rule can be used to generate vias for the default routing rule. There can only be one `VIARULE GENERATE DEFAULT` for a given routing-cut-routing (or masterslice-cut-masterslice) layer combination.

Example 1-25 Via Rule Generate Default

The following example defines a rule for generating vias for the default routing or masterslice rule:

```
VIARULE via12 GENERATE DEFAULT
    LAYER m1 ;
    ENCLOSURE 0.03 0.01 ; #2 sides need >= 0.03, 2 other sides need >= 0.01
    LAYER m2 ;
    ENCLOSURE 0.05 0.01 ; #2 sides need >= 0.05, 2 other sides need >= 0.01
    LAYER cut12 ;
    RECT -0.1 -0.1 0.1 0.1 ; # cut is .20 by .20
    SPACING 0.40 BY 0.40 ;   #center-to-center spacing
    RESISTANCE 20 ;          #ohms per cut
END via12
```

`ENCLOSURE` *`overhang1 overhang2`*

Specifies that the via must be covered by metal on two opposite sides by at least *`overhang1`*, and on the other two sides by at least *`overhang2`* (see Figure 1-68 on page 213). The via generation code then chooses the direction of overhang that best maximizes the number of cuts that can fit in the via.

Note: If there are also `ENCLOSURE` rules for the cut layer that apply to a given via, the via generation code will choose the `ENCLOSURE` rule with values that match the `ENCLOSURE` in `VIARULE GENERATE`. If there is no such match, the via generation code will ignore the `ENCLOSURE` in `VIARULE GENERATE` and choose which `ENCLOSURE` rule is best in `LAYER ENCLOSURE` values that apply to the same width via being generated. This means that only `ENCLOSURE` statements in `LAYER CUT` are honored, and one of them will be used.

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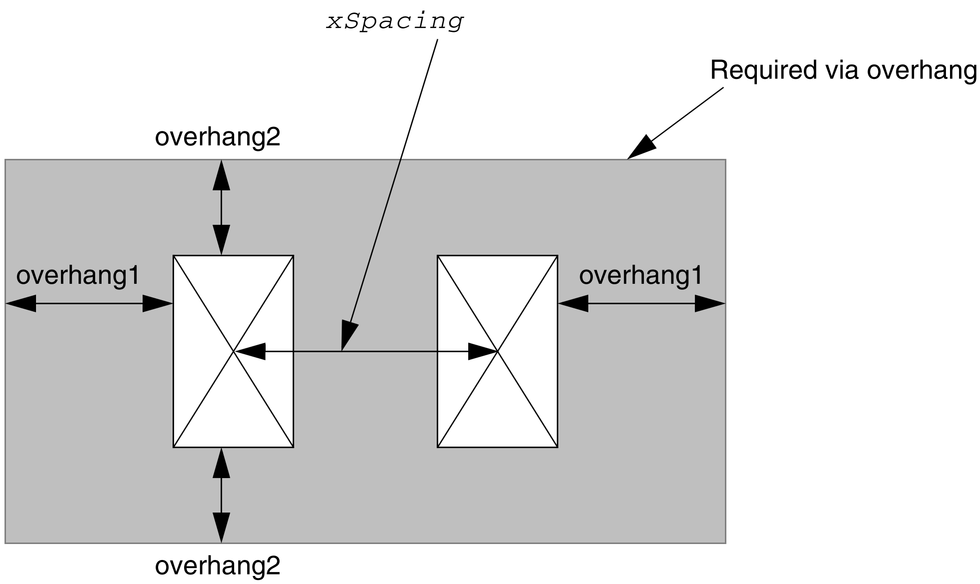
For example, `VIARULE GENERATE ENCLOSURE 0.2 0.0` combined with a `LAYER CUT` rule of `ENCLOSURE 0.2 0.0`, `ENCLOSURE 0.1 0.1` and `ENCLOSURE 0.15 0.15 WIDTH 0.5`, would mean that any via inside a wire with width that is greater than or equal to 0.5 wide, `0.15 0.15` enclosure values are used. Otherwise, `0.2 0.0` enclosure values are used. See the `LAYER CUT ENCLOSURE` statement for more information on handling multiple enclosure rule.
Type: Float, specified in microns

Figure 1-68 Overhang



Example 1-26 Via Rule Generate Enclosure

The following example describes a formula for generating via cuts:

```
VIARULE via12 GENERATE
    LAYER m1 ;
        ENCLOSURE 0.05 0.01 ; #2 sides must be >=0.05, 2 other sides must be >=0.01
        WIDTH 0.2 TO 100.0 ;  #for m1, between 0.2 to 100 microns wide
    LAYER m2 ;
        ENCLOSURE 0.05 0.01 ; #2 sides must be >=0.05, 2 other sides must be >=0.01
        WIDTH 0.2 TO 100.0 ;  #for m2, between 0.2 to 100 microns wide
    LAYER cut12
    RECT -0.07 -0.07 0.07 0.07 ; #cut is .14 by .14
    SPACING 0.30 BY 0.30 ;       #center-to-center spacing
END via12
```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The cut layer SPACING ADJACENTCUTS statement can override the VIARULE cut layer SPACING statements. For example, assume the following cut layer information is also defined in the LEF file:

```
LAYER cut12
    ...
    SPACING 0.20 ADJACENTCUTS 3 WITHIN 0.22 ;
    ...
```

The 0.20 µm edge-to-edge spacing in the ADJACENTCUTS statement is larger than the VIARULE GENERATE example spacing of 0.16 (0.30 – 0.14). Whenever the VIARULE GENERATE rule creates a via that is larger than 2x2 cuts (that is, 2x3, 3x2, 3x3 and so on), the 0.20 spacing from the ADJACENTCUTS statement is used instead.

Note: The spacing in VIARULE GENERATE is center-to-center spacing, whereas the spacing in ADJACENTCUTS is edge-to-edge.

GENERATE

Defines a formula for generating the appropriate via.

LAYER *cutLayerName*

Specifies the cut layer for the generated via.

LAYER *routingLayerName*

Specifies the routing (or masterslice) layers for the top and bottom of the via.

RECT *pt pt*

Specifies the location of the lower left contact cut rectangle.

RESISTANCE *resistancePerCut*

Specifies the resistance of the cut layer, given as the resistance per contact cut.
Default: The resistance value in the LAYER (Cut) statement
Type: Float

SPACING *xSpacing* BY *ySpacing*

Defines center-to-center spacing in the x and y dimensions to create an array of contact cuts.The number of cuts of an array in each direction is the most that can fit within the bounds of the intersection formed by the two special wires. Cuts are only generated where they do not violate stacked or adjacent via design rules.

Note: This value can be overridden by the SPACING ADJACENTCUTS value in the cut layer statement.

VIARULE *viaRuleName*

Specifies the name for the rule.

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The name `DEFAULT` is reserved and should not be used for any via rule name. In the LEF and DEF `VIA` definitions that use generated via parameters, the reserved `DEFAULT` name indicates the via rule with the `DEFAULT` keyword.

`WIDTH` *`minWidth`* `TO` *`maxWidth`*

Specifies a wire width range to use for this `VIARULE`. This `VIARULE` can be used for wires with a width greater than or equal to (>=) *`minWidth`*, and less than or equal to (<=) *`maxWidth`* for the given routing (or masterslice) layer. If no `WIDTH` statement is specified, the `VIARULE` can be used for all wire widths on the given routing (or masterslice) layer.

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2

LEF Syntax - Layer (Cut)

This chapter contains information about the following topics:

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Layer (Cut)

```
LAYER layerName
     TYPE CUT ;
     [MASK maskNum ;]
     [SPACING cutSpacing
        [CENTERTOCENTER]
        [SAMENET]
        [  LAYER secondLayerName [STACK]
         | ADJACENTCUTS {2 | 3 | 4} WITHIN cutWithin [EXCEPTSAMEPGNET]
         | PARALLELOVERLAP
         | AREA cutArea
        ]
     ;] ...
     [SPACINGTABLE ORTHOGONAL
           {WITHIN cutWithin SPACING orthoSpacing} ... ;]
     [ARRAYSPACING [LONGARRAY] [WIDTH viaWidth] CUTSPACING cutSpacing
        {ARRAYCUTS arrayCuts SPACING arraySpacing} ... ;]
     [WIDTH minWidth ;]
     [ENCLOSURE [ABOVE | BELOW] overhang1 overhang2
       [ WIDTH minWidth [EXCEPTEXTRACUT cutWithin]
       | LENGTH minLength]
     ;] ...
     [PREFERENCLOSURE [ABOVE | BELOW] overhang1 overhang2 [WIDTH minWidth] ;] ...
     [RESISTANCE resistancePerCut ;]
     [PROPERTY propName propVal ;] ...
     [ACCURRENTDENSITY {PEAK | AVERAGE | RMS}
       { value
       | FREQUENCY freq_1 freq_2 ... ;
           [CUTAREA cutArea_1 cutArea_2 ... ;]
           TABLEENTRIES
            v_freq_1_cutArea_1 v_freq_1_cutArea_2 ...
             v_freq_2_cutArea_1 v_freq_2_cutArea_2 ...
             ...
       } ;]
     [DCCURRENTDENSITY AVERAGE
       { value
       | CUTAREA cutArea_1 cutArea_2 ... ;
           TABLEENTRIES value_1 value_2 ...
       } ;]
     [ANTENNAMODEL {OXIDE1 | OXIDE2 | OXIDE3 | OXIDE4 | OXIDE5 | ...
       | OXIDE32}
     ;] ...
     [ANTENNAAREARATIO value ;] ...
     [ANTENNADIFFAREARATIO {value | PWL ( ( d1 r1 ) ( d2 r2 ) ...)} ;] ...
     [ANTENNACUMAREARATIO value  ...
     [ANTENNACUMDIFFAREARATIO {value | PWL ( ( d1 r1 ) ( d2 r2 ) ...)} ;] ...
     [ANTENNAAREAFACTOR value [DIFFUSEONLY] ;] ...
     [ANTENNACUMROUTINGPLUSCUT ;]
     [ANTENNAGATEPLUSDIFF plusDiffFactor ;]
```


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```
[ANTENNAAREAMINUSDIFF minusDiffFactor ;]
[ANTENNAAREADIFFREDUCEPWL
   ( ( diffArea1 diffAreaFactor1 ) ( diffArea2 diffAreaFactor2 ) ...) ; ]
[PROPERTY LEF58_TYPE
  "TYPE [TSV [LAYER bottomLayer topLayer | MULTITIERS]
      | PASSIVATION
      | MIMCAP | HIGHR | SPECIALCUT LAYER bottomLayer topLayer
      | SUBSTRATE | STACKEDMIMCAP bottomLayer topLayer]
      ;" ;]
[PROPERTY LEF58_ADJACENTFOURCUTS
  "ADJACENTFOURCUTS cutSpacing [ONECUTSETONLY]
      CUTCLASS {className [TO ALL] | ALL}
      {ABOVE | BELOW}
      PARALLEL parLength WITHIN parWithin [MERGEDCUTS]
      [HORIZONTAL|VERTICAL]
      [EXCEPTDIFFMETAL [NOEXTRACUT]]
      [MINCORNER | MININNERCORNER | MAXINNERCORNER]
      [EXCEPTMAXPRL maxPrl]
      [EXCEPTMINLENGTH minLength]
      [WITHINCUT within | CUTWITHINONLY within]
      [EXCEPTOTHERWIDTHWITHCUT width]
      [METALWIDTH maxWidth]
      ; " ;]
[PROPERTY LEF58_ANTENNACUMAREARATIO
  "ANTENNACUMAREARATIO {OXIDE1 | OXIDE2 | OXIDE3 | OXIDE4 | OXIDE5 | ...
    | OXIDE32} value
    [GATEEXPONENT index] [GATEAREA area]
    [DIFFONLY]
    ;" ;]
[PROPERTY LEF58_ANTENNADIFFGATEPWL
  "ANTENNADIFFGATEPWL {OXIDE1 | OXIDE2 | OXIDE3 | OXIDE4 | OXIDE5 | ...
    | OXIDE32}
    ((gateArea1 effectiveGateArea1) (gateArea2 effectiveGateArea2) ... )
    ;" ;]
[PROPERTY LEF58_ANTENNADIFFONLYAREARATIO
  "ANTENNADIFFONLYAREARATIO {OXIDE1 | OXIDE2 | OXIDE3 | OXIDE4
      | OXIDE5 | ... | OXIDE32}  value
    ;..." ;]
[PROPERTY LEF58_ANTENNADIFFPROTECTORAREARATIO
  "ANTENNADIFFPROTECTORAREARATIO {OXIDE1 | OXIDE2 | OXIDE3
      | OXIDE4 | OXIDE5 | ...| OXIDE32}  value
    ;..." ;]
[PROPERTY LEF58_ANTENNAGATEPWL
  "ANTENNAGATEPWL {OXIDE1 | OXIDE2 | OXIDE3 | OXIDE4 | OXIDE5 | ...
    | OXIDE32}
    ((gateArea1 effectiveGateArea1) (gateArea2 effectiveGateArea2) ... )
    ;" ;]
[PROPERTY LEF58_ANTENNAGATEPLUSDIFF
  "ANTENNAGATEPLUSDIFF {OXIDE1 | OXIDE2 | OXIDE3 | OXIDE4 | OXIDE5 | ...
    | OXIDE32}
    {plusDiffFactor |
```


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```
    PWL ((diffArea1 plusDiffProtect1) (diffArea1 plusDiffProtect2) ...)}
    ;" ;]
 [PROPERTY LEF58_ARRAYSPACING
   "ARRAYSPACING [CUTCLASS className] [PARALLELOVERLAP]
      [LONGARRAY] [WIDTH viaWidth]
      [WITHIN within ARRAYWIDTH arrayWidth] CUTSPACING cutSpacing
      {ARRAYCUTS arrayCuts SPACING arraySpacing} ...
      ;..." ;]
 [PROPERTY LEF58_BACKSIDE
   "BACKSIDE ;" ;]
 [PROPERTY LEF58_CUTCLASS
   "CUTCLASS className WIDTH viaWidth [LENGTH viaLength] [CUTS numCut]
       [ORIENT {HORIZONTAL | VERTICAL}]
       ; " ;]
 [PROPERTY LEF58_CUTONCENTERLINE
   "CUTONCENTERLINE width CUTCLASS className [ABOVE | BELOW]
      [EXTRACUT cutWithin WIDTH exactWidth]
      ; " ;]
 [PROPERTY LEF58_DIRECTIONALSPACING
   "DIRECTIONALSPACING cutSpacing {HORIZONTAL | VERTICAL}
      PRL prl
      CUTCLASS className1 TO className2
      [PARALLEL parLength WITHIN parWithin
          [EXCLUDECUTEDGE]
          [CUTS numCuts]
          [VIAGROUP groupName]]
      ; " ;]
 [PROPERTY LEF58_ENCLOSURE
   "ENCLOSURE [CUTCLASS className][ABOVE | BELOW]
      [MINCORNER]
      {EOL eolWidth [HORIZONTAL | VERTICAL]
          [MINLENGTH minLength]
          [EOLONLY] [SHORTEDGEONEOL] eolOverhang otherOverhang
          [SIDESPACING spacing EXTENSION backwardExt forwardExt
          |ENDSPACING spacing EXTENSION extension
          ]
     |{overhang1 overhang2
        |[OFFCENTERLINE] END overhang1 SIDE overhang2
        |HORIZONTAL overhang1 VERTICAL overhang2}
        [JOGLENGTHONLY length [INCLUDELSHAPE] ]
        [HOLLOW {HORIZONTAL|VERTICAL} length]
        [ WIDTH minWidth
            [INCLUDEABUTTED]
            [EXCEPTEXTRACUT cutWithin
                 [PRL | NOSHAREDEDGE | EXACTPRL prl]]
        | LENGTH minLength
        | EXTRACUT [EXTRAONLY [PRL prl]]
        | REDUNDANTCUT cutWithin
        | PARALLEL parLength [parLength2] WITHIN parWithin [parWithin2]
            [BELOWENCLOSURE belowEnclosure
                 [ALLSIDES enclosure1 enclosure2]
```


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```
              |ABOVEENCLOSURE aboveEnclosure]
          | CONCAVECORNERS numCorner
          | OTHERWITHINWIDTH width WITHIN within
          [ OTHERSIDE otherEnclosure]]
        }
        ;..." ;]
  [PROPERTY LEF58_ENCLOSUREEDGE
    "ENCLOSUREEDGE [CUTCLASS className][ABOVE | BELOW] overhang
        {OPPOSITE
             {[EXCEPTEOL eolWidth] [NOCONCAVECORNER within]
                  [CUTTOBELOWSPACING spacing
                      [ABOVEMETAL extension]]
             | WRONGDIRECTION
             }
        |[INCLUDECORNER]
          {WIDTH [BOTHWIRE] minWidth [maxWidth]
          |SPANLENGTH minSpanLength [maxSpanLength]}
             PARALLEL parLength
             {WITHIN parWithin | WITHIN minWithin maxWithin}
             [EXCEPTEXTRACUT [cutWithin]]
             [EXCEPTTWOEDGES [exceptWithin]]
        |CONVEXCORNERS convexLength adjacentLength
             PARALLEL parWithin LENGTH length
        }
    ;..." ;]
  [PROPERTY LEF58_ENCLOSURETABLE
    "ENCLOSURETABLE [CUTCLASS className] [MASK maskNum]
        [OTHERMASK otherMaskNum]
            [USEMAXWIDTH | USEMINWIDTH | WITHINFIRSTWIDTH]
        [DEFAULT {[ABOVE| BELOW]
            overhang1 overhang2 overhang3 overhang4}...]
        {WIDTH width {[ABOVE| BELOW]
            overhang1 overhang2 overhang3 overhang4 [MINSUM]
            [MUSTUSE]
            [MAXLENGTH length | LENGTH length]
            [LAYER trimLayer OVERLAP {1|2}
                  [TRIMLENGTHOUTERMETAL trimLength]
            | OTHERWITHINWIDTH width [LAYER layerName]
                  WITHIN within [EXCEPTOVERLAP]
                  [OTHERSIDE otherEnclosure]]
            | LINEENDGAPONLY
            | ROWBOUNDARY within OTHERSIDE otherEnclosure
                  [EXCEPTINSIDECUT]]
            [EXACTZERO]
            [INCLUDEABUTTED]}...
        | OTHERWIDTH otherWidth [PARALLEL parLength WITHIN parWithin]
            {{ABOVE| BELOW}
            overhang1 overhang2 overhang3 overhang4 [MINSUM]}...}...
        ; " ;]
  [PROPERTY LEF58_ENCLOSURETOJOINT
    "ENCLOSURETOJOINT [CUTCLASS className] [ABOVE | BELOW]
```


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```
        toOneJointOverhang [toBothJointOverhang]
        [EOLMINLENGTH minLength]
        JOINTWIDTH jointWidth JOINTLENGTH spanLength
        ; " ;]
  [PROPERTY LEF58_ENCLOSUREWIDTH
    "ENCLOSUREWIDTH {VIAOVERLAPONLY | USEMINWIDTH}
        ;..." ;]
  [PROPERTY LEF58_EOLENCLOSURE
    "EOLENCLOSURE eolWidth [MINEOLWIDTH minEolWidth]
        [HORIZONTAL | VERTICAL]
        [INCLUDEEDGEALIGNED]
        [EQUALRECTWIDTH] [CUTCLASS className] [ABOVE | BELOW]
        {{LONGEDGEONLY | SHORTEDGEONLY} overhang
        |overhang
              [exactOverhang
              |PARALLELEDGE parSpace EXTENSION backwardExt forwardExt
                    [MINLENGTH minLength]
              |MINLENGTH minLength
              |ALLSIDES
              ]
        }
        ; " ;]
  [PROPERTY LEF58_EOLSPACING
    "EOLSPACING cutSpacing1 cutSpacing2
        [CUTCLASS className1 [{TO className2 cutSpacing1 cutSpacing2}...]]
        ENDWIDTH eolWidth PRL prl
        ENCLOSURE smallerOverhang equalOverhang
        EXTENSION sideExt backwardExt SPANLENGTH spanLength
        ; " ;]
  [PROPERTY LEF58_FORBIDDENSPACING
    "FORBIDDENSPACING CUTCLASS className minSpacing maxSpacing
        [SAMEMASK]
        [SHORTEDGEONLY | LONGEDGEONLY | HORIZONTAL | VERTICAL]
        [PRL {prl TO prlClassName}...]
        ; " ;]
  [PROPERTY LEF58_KEEPOUTZONE
    "KEEPOUTZONE CUTCLASS className1 [TO className2]
        [SAMEMASK][SAMEMETAL | DIFFMETAL]
        [EXCEPTEXACTALIGNED [SIDE | END] spacing]
        {EXTENSION sideExtension forwardExtension |
        ENDEXTENSION endSideExtension endForwardExtension
        SIDEEXTENSION sideSideExtension sideForwardExtension |
        HORIZONTALEXTENSION horzSideExtension horzForwardExtension
        VERTICALEXTENSION vertSideExtension vertForwardExtension}
        SPIRALEXTENSION extension [CORNERONLY]
        ;" ;]
  [PROPERTY LEF58_MANUFACTURINGGRID
    "MANUFACTURINGGRID value
    ; " ;]
  [PROPERTY LEF58_MAXSPACING
    "MAXSPACING spacing [CUTCLASS className]
```


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```
  ;" ;]
[PROPERTY LEF58_MINENCLOSURE
  "MINENCLOSURE CUTCLASS className [ABOVE | BELOW]
       overhang1 overhang2
      ; " ;]
[PROPERTY LEF58_NOCUTCLASS
  "NOCUTCLASS className [EXCEPTMULTICUTS]
  ;" ;]
[PROPERTY LEF58_NOMETALSPACING
  "NOMETALSPACING spacing BOTHSPACING bothSpacing [PRL prl]
      [SAMEMASK] [CUTCLASS className]
      NEIGHBORMETAL metalPrl metalWithin
         noMetalPrl {noMetalWithin | noMetalLowWithin noMetalHighWithin}
      [SAMESIDE]
      [ABOVE | BELOW]
      [EXCEPTLINEENDSPACING [INCLUDELESSTHAN] lineEndSpacing]
       ;" ;]
[PROPERTY LEF58_ONEDARRAY
  "ONEDARRAY CUTCLASS className CUTSPACING cutSpacing
      ARRAYCUTS arrayCuts SPACING spacing
      LAYER secondLayerName SPACING interLayerSpacing
      ENCLOSURE overhang1 overhang2
      ; " ;]
[PROPERTY LEF58_ORTHOGONALSPACING
  "ORTHOGONALSPACING cutSpacing
      [CUTCLASS className [TO ALL]| CUTCLASSLIST {classNameList}...]
      {ABOVE | BELOW}
      PARALLEL parLength WITHIN parWithin WIDTH width
      [METALPRL metalPrl METALWITHIN metalWithin]
       [HORIZONTAL | VERTICAL]
      ; " ;]
[PROPERTY LEF58_OPPOSITEOVERLAPCUTSPACING
  "OPPOSITEOVERLAPCUTSPACING spacing PRL prl
      LAYER secondLayerName
      ; " ;]
[PROPERTY LEF58_PRLTWOSIDESPACING
  "PRLTWOSIDESSPACING prlSpacing nonPrlSpacing
      CUTCLASS className1 TO className2
      PRL prl1 prl2 prl3 prl4
      ; " ;]
[PROPERTY LEF58_REGION
  "REGION regionLayerName BASEDLAYER cutLayerName
  ; " ;]
[PROPERTY LEF58_SAMEMETALALIGNEDCUTS
  "SAMEMETALALIGNEDCUTS numCuts
      CUTCLASS {className | ALL}
      | {CUTCLASSLIST {classNameList}...}...}
      [ABOVE | BELOW] WIDTH width SPACING spacing
      [METALMASK metalMaskNum]
      [EXCEPTENCLOSURE exceptEnclosure {ABOVE|BELOW}]
      [PARALLEL parLength WITHIN parWithin
```


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```
        | METALWITHIN metalWithin [WITHCUT]
        | CUTWITHIN cutWithin
        |NOVIAOVERLAPMETAL within]
        [CENTERTOCENTER | EDGETOEDGE]
        ; " ;]
  [PROPERTY LEF58_SPACING
    "SPACING cutSpacing
       [SAMEMASK
       |MAXXY
       |[CENTERTOCENTER]
         [SAMENET | SAMEMETAL | SAMEVIA]
         [LAYER secondLayerName
              [STACK
              | ORTHOGONALSPACING orthogonalSpacing]
              | CUTCLASS className
                  [SHORTEDGEONLY [PRL prl]
                  |LONGEDGEONLY
                  |CONCAVECORNER
                       [WIDTH width ENCLOSURE enclosure
                           EDGELENGTH edgeLength
                       | PARALLEL parLength WITHIN parWithin
                           ENCLOSURE enclosure
                       | EDGELENGTH edgeLength
                           ENCLOSURE edgeEnclosure adjEnclosure]
                       | WRONGDIRECTIONONLY xSize ySize
                           [LAYER adjacentCutLayer]]
                  |EXTENSION extension
                  |NONEOLCONVEXCORNER eolWidth
                       [MINLENGTH minLength]
                  |ABOVEWIDTH width [ENCLOSURE enclosure]
                  |MASKOVERLAP
                  |WRONGDIRECTION
                       [EXCEPTRECTANGLE | OVERLAPONLY]
                  |DIFFCONCAVECORNER ]]]
         | ADJACENTCUTS {2 | 3 | 4}
              [EXACTALIGNED exactAlignedCut]
              [TWOCUTS twoCuts [TWOCUTSSPACING twoCutsSpacing]
                  [SAMECUT]
              | EACHCUTS numCuts [EXCEPTEVENNUMCUT]]
              WITHIN {cutWithin | cutWithin1 cutWithin2}
              [EXCEPTSAMEPGNET]
              [EXCEPTALLWITHIN exceptAllWithin]
              [ENCLOSURE [ABOVE|BELOW] enclosure
                  [ANYCUT] [EXCEPTSAMEMETAL]
                  [METALMASK metalMaskNum]
                  [METALWIDTH maxWidth]
                  [EXCEPTLINEENDGAP]]
              [CUTCLASS className [TO {ALL | VIAGROUP groupName
                  [CUTCLASSLIST {classNameList}...]}]
              |VIAGROUP groupName TO CUTCLASS {className|ALL}]
              [NOPRL [HORIZONTAL|VERTICAL] | SIDEPARALLELOVERLAP
```


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```
              |PRL prl [HORIZONTAL|VERTICAL] [SAMESIDE]]
              [SAMEMASK]
         | PARALLELOVERLAP [EXCEPTSAMENET | EXCEPTSAMEMETAL
              |EXCEPTSAMEMETALOVERLAP | EXCEPTSAMEVIA]
         | PARALLELWITHIN within [EXCEPTSAMENET]
              [CUTCLASS className
                   [LONGEDGEONLY
                   | ENCLOSURE enclosure {ABOVE | BELOW}
                       PARALLEL parLength WITHIN parWithin]]
         | SAMEMETALSHAREDEDGE parwithin [ABOVE][CUTCLASS className]
              [EXCEPTTWOEDGES] [EXCEPTSAMEVIA numCut]
         | AREA cutArea]
   ;..." ;]
 [PROPERTY LEF58_SPACINGTABLE
   "SPACINGTABLE CENTERSPACING
      LAYER secondLayerName
      [OVERLAPMETALSPACING metalLayerName WIDTH width
          {className1 TO className2 spacing}...]
      CUTCLASS {{className1 | ALL} }...
          {{className2 | ALL} {cutSpacing}...}...
   ; " ;]
 [PROPERTY LEF58_SPACINGTABLE
   "SPACINGTABLE
      [ORTHOGONAL
         {WITHIN cutWithin SPACING orthoSpacing} ... ;
      |[DEFAULT defaultCutSpacing]
         [SAMEMASK]
         [SAMENET | SAMEMETAL | SAMEVIA]
         [LAYER secondLayerName
               [NOSTACK]
               [NONZEROENCLOSURE
               | PRLFORALIGNEDCUT
                 {{className1 | ALL} TO {className2 | ALL} }...
               | EXCEPTENCLOSURE exceptEnclosure [ABOVE|BELOW]]]
         [CENTERTOCENTER
               {{className1 | ALL}| TO {className2 | ALL}}...]
         [CENTERANDEDGE [NOPRL]
               {{className1 | ALL}| TO {className2 | ALL}}...]
         [PRLSPACING spacing PRL prl]
         [PRL {prl|USEDEFAULT} [ HORIZONTAL| VERTICAL][MAXXY]
               [{{className1 | ALL} TO {className2 | ALL} ccPrl
                      [ HORIZONTAL| VERTICAL] }...] ]
         [PRLTWOSIDES
              {prl1 prl2 prl3 prl4 [WITHIN within]
                     className1 TO className2 spacing}...]
         [ENDEXTENSION extension [{TO className classExtension}...]
               [SIDEEXTENSION {TO className classExtension}...] ]
         [EXACTALIGNEDSPACING [HORIZONTAL | VERTICAL]
               {{className | ALL} exactAlignedSpacing}...]
         |EXACTALIGNEDWIREWIDTHSPACING
               {ABOVE|BELOW}
```


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```
                [HORIZONTAL| VERTICAL]
                {{className|ALL} width exactAlignedSpacing}...]...
          [NONOPPOSITEENCLOSURESPACING
                {className nonOppositeEnclosureSpacing}...]
          [OPPOSITEENCLOSURERESIZESPACING
                {className resize1 resize2
                     oppositeEnclosureResizeSpacing}...
                [EDGEALIGNED [HORIZONTAL|VERTICAL]]]
          CUTCLASS { {className1 | ALL} [SIDE | END]}...
            {{className2 | ALL} [SIDE | END] {-|cutSpacing1}
              {-|cutSpacing2}...}...;
      ]
       ; " ;]
   [PROPERTY LEF58_VIACLUSTER
     "VIACLUSTER CUTCLASS className
         CUTS perpendicularNumCut
         [DIAGONAL diagonalNumCut]
             [NOPRLSPACING diagonalSpacing bendSpacing WITHIN within
             EXTENSION sideExtension edgeExtension]]
         WITHIN minWithin maxWithin
         ; " ;]
   [PROPERTY LEF58_VIAGROUP
     "VIAGROUP groupName CUTS numCut CUTCLASS {{className}... | All}
         [MIXEDCUTCLASSES] [SAMEMASK]
         {{[PRLTWOSIDES prl1 prl2 prl3 prl4] SPACING minSpacing maxSpacing
           | PRL prl PRLSPACING prlSpacing [HORIZONTAL|VERTICAL]}
               [ALLOWBEND [NOHORIZONTAL|NOVERTICAL]]
         | ALIGNEDSPACING {HORIZONTAL | VERTICAL}
               minSpacing maxSpacing
         | LGROUP HORIZONTALSPACING horizontalSpacing PRL verticalPrl
               VERTICALSPACING verticalSpacing PRL horizontalPrl}
         ; " ;]
   [PROPERTY LEF58_VIAGROUPCLUSTER
     "VIAGROUPCLUSTER groupName CUTS numCut
         CUTCLASS {className | ALL}
         TWOCUTS verticalSpacing horziontalPrl
         TWOCUTSCLUSTER horziontalSpacing verticalPrl
         ; " ;]
   [PROPERTY LEF58_VIAGROUPOVERRIDESPACING
     "VIAGROUPOVERRIDESPACING
         ; " ;]
   [PROPERTY LEF58_VIAGROUPSPACING
     "VIAGROUPSPACING spacing [MAXSPACING]
         [SAMEMASK | DIFFMASK]
         [{HORIZONTAL| VERTICAL} [PRL prl] [TWOSIDES]]
         {VIAGROUP groupName | VIAGROUPCLUSTER groupName}
         CUTS numCut
         [LAYER cutLayerName] CUTCLASS {className | All}
         [TOVIAGROUP]
         [MERGEDCUTS | NOMERGEDCUTS]
         [PARALLEL parLength WITHIN parWithin
```


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```
              OTHERCUTCLASS otherClassName]
         ; " ;]
   [PROPERTY LEF58_VOLTAGESPACING
     "VOLTAGESPACING {voltage spacing}...
          ;..." ;]

END layerName
```

Defines cut layers in the design. Each cut layer is defined by assigning it a name and design rules. You must define cut layers separately, with their own layer statements.

You must define layers in process order from bottom to top. For example:

```
poly        masterslice
cut01       cut
metal1      routing
cut12       cut
metal2      routing
cut23       cut
metal3      routing
```

ACCURRENTDENSITY

Specifies how much AC current a cut of a certain area can handle at a certain frequency. For an example using the ACCURRENTDENSITY syntax, see Example 6-1 on page 626.

The ACCURRENTDENSITY syntax is defined as follows:

```
{PEAK | AVERAGE | RMS}
{ value
| FREQUENCY freq_1 freq_2 ... ;
    [CUTAREA cutArea_1 cutArea_2 ...  ; ]
    TABLEENTRIES
    v_freq_1_cutArea_1 v_freq_1_cutArea_2 ...
    v_freq_2_cutArea_1 v_freq_2_cutArea_2 ...
    ...
} ;
```

PEAK	Specifies the peak limit of the layer.
AVERAGE	Specifies the average limit of the layer.
RMS	Specifies the root mean square limit of the layer.
value	Specifies a maximum current limit for the layer in milliamps per square micron (mA/μm^2). *Type:* Float

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`FREQUENCY`	Specifies frequency values, in megahertz. You can specify more than one frequency. If you specify multiple frequency values, the values must be specified in ascending order. If you specify only one frequency value, there is no frequency dependency, and the table entries are assumed to apply to all frequencies. *Type:* Float
`CUTAREA`	Specifies cut area values, in square microns (μm^2). You can specify more than one cut area. If you specify multiple cut area values, the values must be specified in ascending order. If you specify only one cut area value, there is no cut area dependency, and the table entries are assumed to apply to all cut areas. *Type:* Float
`TABLEENTRIES`	Defines the maximum current for each frequency and cut area pair specified in the `FREQUENCY` and `CUTAREA` statements, in $mA/\mu m^2$. The pairings define each cut area for the first frequency in the `FREQUENCY` statement, then the cut areas for the second frequency, and so on. The final value for a given cut area and frequency is computed from a linear interpolation of the table values. *Type:* Float

`ANTENNAAREADIFFREDUCEPWL ( (` *`diffArea1 diffAreaFactor1`* `)`
`(` *`diffArea2 diffAreaFactor2`* `) ...)`

Indicates that the cut_area is multiplied by a *`diffAreaFactor`* computed from a piece-wise linear interpolation, based on the diffusion area attached to the cut.

The *`diffArea`* values are floats, specified in microns squared. The *`diffArea`* values should start with `0` and monotonically increase in value to the maximum size *`diffArea`* possible. The *`diffAreaFactor`* values are floats with no units. The *`diffAreaFactor`* values are normally between `0.0` and `1.0`. If no statement rule is defined, the *`diffMetalReduceFactor`* value in the PAR(m_i) equation defaults to `1.0`.

For more information on the PAR(m_i) equation and process antenna models, see Appendix C, “Calculating and Fixing Process Antenna Violations.”

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`ANTENNAAREAFACTOR` *`value`* `[DIFFUSEONLY]`

Specifies the multiply factor for the antenna metal area calculation. `DIFFUSEONLY` specifies that the current antenna factor should only be used when the corresponding layer is connected to the diffusion.
Default: `1.0`
Type: Float

For more information on process antenna calculation, see Appendix C, "Calculating and Fixing Process Antenna Violations."

Note: If you specify a value that is greater than 1.0, the computed areas will be larger, and violations will occur more frequently.

`ANTENNAAREAMINUSDIFF` *`minusDiffFactor`*

Indicates that the antenna ratio cut_area should subtract the diffusion area connected to it. This means that the ratio is calculated as:

ratio = (*cutFactor* x cut_area - *minusDiffFactor* x diff_area)/gate_area

If the resulting value is less than 0, it should be truncated to 0. For example, if a *via2* shape has a final ratio that is less than 0 because it connects to a diffusion shape, then the cumulative check for *metal3* (or *via3*) above the *via2* shape adds a cumulative value of 0 from the *via2* layer. (See Example 1 in Cut Layer Process Antenna Models, in Appendix C, "Calculating and Fixing Process Antenna Violations."
Type: Float
Default: `0.0`

`ANTENNAAREARATIO` *`value`*

Specifies the maximum legal antenna ratio, using the area of the metal wire that is not connected to the diffusion diode. For more information on process antenna calculation, see Appendix C, "Calculating and Fixing Process Antenna Violations."
Type: Float

`ANTENNACUMAREARATIO` *`value`*

Specifies the cumulative antenna ratio, using the area of the metal wire that is not connected to the diffusion diode.
`GATEEXPONENT` *`index`* specifies an exponential cumulative antenna ratio, using the electrical connected gateArea into the rule ratio formula as $value * gateArea^{index}$.
`GATEAREA` *`area`* specifies that the rule applies only if the gate areas are less than *`area`*.
For more information on process antenna calculation, see Appendix C, "Calculating and Fixing Process Antenna Violations."
Type: Float

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ANTENNACUMDIFFAREARATIO {*value* | PWL ((*d1 r1*) (*d2 r2*)...)}

Specifies the cumulative antenna ratio, using the area of the metal wire that is connected to the diffusion diode. You can supply an explicit ratio *value* or specify piece-wise linear format (PWL), in which case the cumulative ratio is calculated using linear interpolation of the diffusion area and ratio input values. The diffusion input values must be specified in ascending order.
Type: Float

For more information on process antenna calculation, see Appendix C, "Calculating and Fixing Process Antenna Violations."

ANTENNACUMROUTINGPLUSCUT

Indicates that cumulative ratio rules (that is, ANTENNACUMAREARATIO, and ANTENNACUMDIFFAREARATIO) accumulate with the previous routing layer instead of the previous cut layer. Use this to combine metal and cut area ratios into one rule.

For more information on process antenna models, see Appendix C, "Calculating and Fixing Process Antenna Violations."

ANTENNADIFFAREARATIO {*value* | PWL ((*d1 r1*) (*d2 r2*)...)}

Specifies the antenna ratio, using the area of the metal wire connected to the diffusion diode. You can supply an explicit ratio *value* or specify piece-wise linear format (PWL), in which case the ratio is calculated using linear interpolation of the diffusion area and ratio input values. The diffusion input values must be specified in ascending order.
Type: Float

For more information on process antenna calculation, see Appendix C, "Calculating and Fixing Process Antenna Violations."

ANTENNAGATEPLUSDIFF *plusDiffFactor*

Indicates the antenna ratio gate area includes the diffusion area multiplied by *plusDiffFactor*. This means that the ratio is calculated as:

ratio = cut_area / (gate_area + *plusDiffFactor* x diff_area)

The ratio rules without "DIFF" (the ANTENNAAREARATIO, ANTENNACUMAREARATIO, ANTENNASIDEAREARATIO, and ANTENNACUMSIDEAREARATIO statements), are unnecessary for this layer if ANTENNAGATEPLUSDIFF is defined because a zero diffusion area is already accounted for by the ANTENNADIFF*RATIO statements.
Type: Float
Default: 0.0

For more information on process antenna models, see Appendix C, "Calculating and Fixing Process Antenna Violations."

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`ANTENNAMODEL {OXIDE1 | OXIDE2 | OXIDE3 | OXIDE4 | OXIDE5 | ... | OXIDE32}`

Specifies the oxide model for the layer. If you specify an `ANTENNAMODEL` statement, that value affects all `ANTENNA*` statements for the layer that follow it until you specify another `ANTENNAMODEL` statement.

Default: `OXIDE1`, for a new `LAYER` statement

Because LEF is sometimes used incrementally, if an `ANTENNA` statement occurs twice for the same oxide model, the last value specified is used. For any given `ANTENNA` keyword, only one value or PWL table is stored for each oxide metal on a given layer.

For an example using the `ANTENNAMODEL` syntax, see Example 6-2 on page 631.

`ARRAYSPACING`

Specifies array spacing rules to use on the cut layer. An array spacing rule is intended for large vias of size 3x3 or larger.

The `ARRAYSPACING` syntax is defined as follows:

```
[ARRAYSPACING [LONGARRAY]
  [WIDTH viaWidth] CUTSPACING cutSpacing
  {ARRAYCUTS arrayCuts
      SPACING arraySpacing} ... ;
]
```

`CUTSPACING` *`cutSpacing`*

Specifies the edge-of-cut to edge-of-cut spacing inside one cut array.

`ARRAYCUTS` *`arrayCuts`* `SPACING` *`arraySpacing`*

Indicates that a large via array with a size greater than or equal to *`arrayCuts`* x *`arrayCuts`* in both dimensions must use N x N cut arrays (where N = *`arrayCuts`*) separated from other cut arrays by a distance of greater than or equal to *`arraySpacing`*.

For example, if *`arrayCuts`* = 3, then 2x3 and 2x4 arrays do not need to follow the array spacing rule. However, 3x3 and 3x4 arrays must follow the rule (3x4 is legal, if the `LONGARRAY` keyword is specified), while 4x4 or 4x5 arrays are violations, unless an *`arrayCuts`* = 4 rule is specified. (See Array Spacing Rule Example 1).

If you specify multiple `{ARRAYCUTS ...}` statements, the *`arrayCuts`* values must be specified in increasing order. (See Array Spacing Rule Example 3.)

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Specifying more than one `ARRAYCUTS` statement creates multiple choices for via array generation.

For example, you can define an *arrayCuts* = 4 rule with *arraySpacing* = 1.0, and an *arrayCuts* = 5 rule with *arraySpacing* = 1.5. Either rule is legal, and the application should choose which rule to use (presumably based on which rule produces the most via cuts in the given via area).

`LONGARRAY` Indicates that the via can use N x M cut arrays, where N = *arrayCuts*, and M can be any value, including one that is larger than N. (See Array Spacing Rule Example 2.)

`WIDTH` *viaWidth*

Indicates that the array spacing rules only apply if the via metal width is greater than or equal to *viaWidth*. (See Array Spacing Rule Example 1.)

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Example 2-1 Array Spacing Rules

- **Array Spacing Rule Example 1**

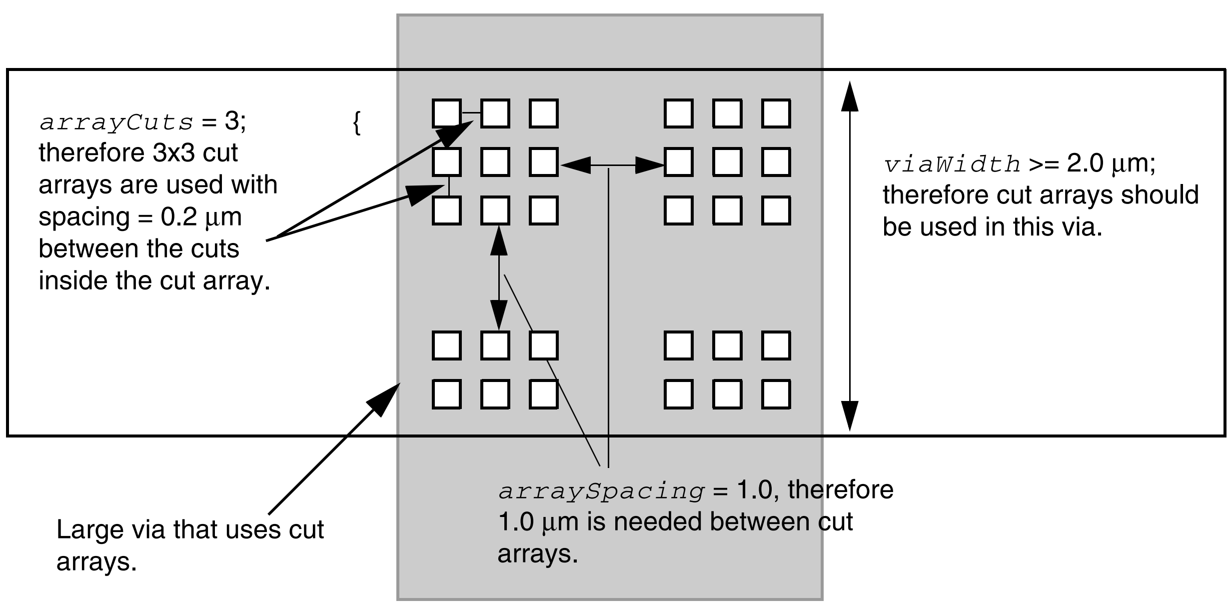
Assume the following array spacing rule exists:

```
ARRAYSPACING WIDTH 2.0 CUTSPACING 0.2 ARRAYCUTS 3 SPACING 1.0 ;
```

Any via with a metal width greater than or equal to 2.0 µm should use the cut spacing of 0.2 µm between cuts inside 3x3 cut arrays, and the cut arrays should be spaced apart by a distance of greater than or equal to 1.0 µm from other cut arrays. This creates the via shown in Figure 2-1 on page 234.

An array of 3x4 or 3x5 cuts spaced 0.2 µm apart is a violation, unless the `LONGARRAY` keyword is specified. This is because the 3x3 sub-array, inside 3x4 or 3x5 cut array, does not meet 1.0 µm spacing from other cut arrays. Also, any larger array, such as 4x4 or 4x5 cuts, is a violation because the 3x3 sub-array inside 4x4 or 4x5 cut array requires 1.0 µm spacing from other cut arrays.

Figure 2-1 Via Created With Array Spacing Width Rule



- **Array Spacing Rule Example 2**

The following array spacing rule is the same as Example 1, except the `LONGARRAY` keyword is present and the `WIDTH` keyword is not specified, so it creates the via shown in Figure 2-2 on page 235:

```
ARRAYSPACING LONGARRAY CUTSPACING 0.2 ARRAYCUTS 3 SPACING 1.0 ;
```

An array of 2x2, 2x3, or 2xM cuts ignores this rule.

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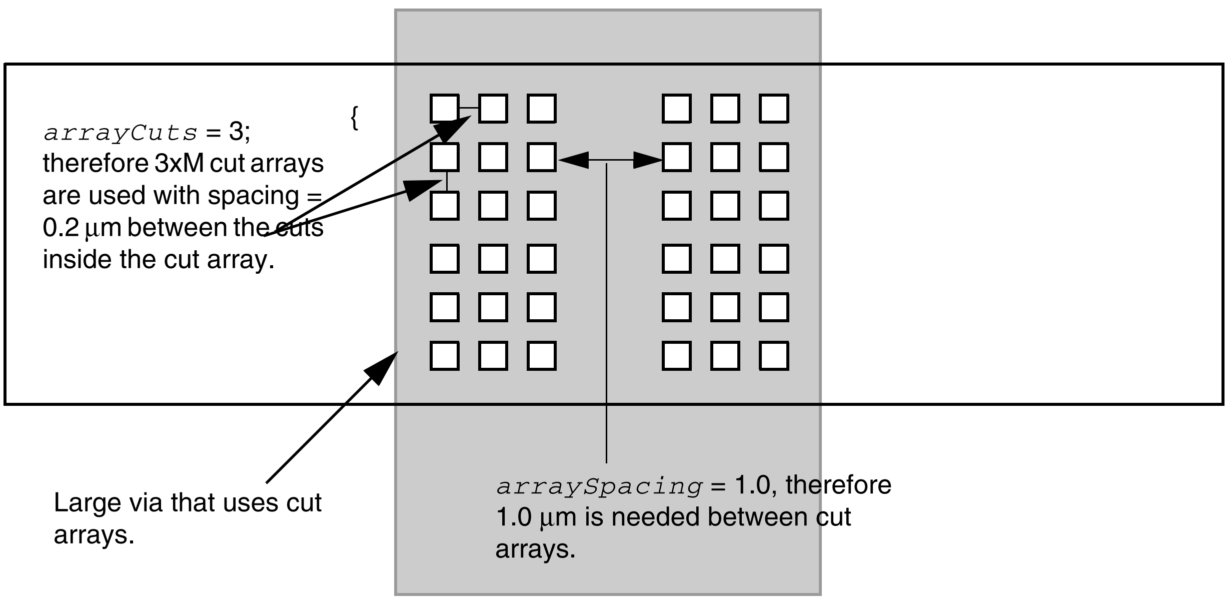
An array of 3x3 or 3xM must have 1.0 µm spacing from other cut arrays and 0.2 µm spacing between the cuts.

An array of 4x4 or 4xM is a violation because the array does not have 1.0 µm space from the 3xM sub-array inside the 4xM array.

Figure 2-2 Via Created With Array Spacing Long Array Rule



- **Array Spacing Rule Example 3**

  Assume the following multiple array spacing rules exist:

```
ARRAYSPACING LONGARRAY CUTSPACING 0.2
     ARRAYCUTS 3 SPACING 1.0
     ARRAYCUTS 4 SPACING 1.5
     ARRAYCUTS 5 SPACING 2.0 ;
```

  The application can choose between 3xM cut arrays with 1.0 µm spacing, 4xM cut arrays with 1.5 µm spacing, or 5xM cut arrays with 2.0 µm spacing, using 0.2 cut-to-cut spacing inside each cut array. No `WIDTH` value indicates that any via with more than three via cuts in both dimensions (that is, 3x3 and 3x4, but not 2x4) must follow these rules.

`DCCURRENTDENSITY`

Specifies how much DC current a via cut of a certain area can handle in units of milliamps per square micron (mA/µm^2). For an example using the `DCCURRENTDENSITY` syntax, see Example 6-3 on page 633.

The `DCCURRENTDENSITY` syntax is defined as follows:

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```
AVERAGE
{ value
| CUTAREA cutArea_1 cutArea_2 ... ;
   TABLEENTRIES value_1 value_2 ...
} ;
```

AVERAGE	Specifies the average limit for the layer.
value	Specifies a current limit for the layer in mA/μm^2. *Type:* Float
CUTAREA	Specifies cut area values, in square microns. You can specify more than one cut area value. If you specify multiple cut area values, the values must be specified in ascending order. *Type:* Float
TABLEENTRIES	Specifies the maximum current density for each specified cut area, in mA/μm^2. The final value for a specific cut area is computed from a linear interpolation of the table values. *Type:* Float

ENCLOSURE

Specifies an enclosure rule for the cut layer.

The ENCLOSURE syntax is described as follows:

```
[ENCLOSURE
 [ABOVE | BELOW] overhang1 overhang2
 [ WIDTH minWidth [EXCEPTEXTRACUT cutWithin]
 | LENGTH minLength]
;]
```

ENCLOSURE [ABOVE | BELOW] *overhang1 overhang2*

Indicates that any rectangle from this cut layer requires the routing layers to overhang by *overhang1* on two opposite sides, and by *overhang2* on the other two opposite sides. (See Figure 2-3 on page 238.)
Type: Float, specified in microns

If you specify BELOW, the overhang is required on the routing layers below this cut layer. If you specify ABOVE, the overhang is required on the routing layers above this cut layer. If you specify neither, the rule applies to both adjacent routing layers.

WIDTH *minWidth*

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Indicates that the enclosure rule only applies when the width of the routing layer is greater than or equal to *minWidth*. If you do not specify a minimum width, the enclosure rule applies to all widths (as if *minWidth* equaled 0).
Type: Float, specified in microns

If you specify multiple enclosure rules with the same width (or with no width), then there are several legal enclosure rules for this width, and the application only needs to meet one of the rules. If you specify multiple enclosure rules with different *minWidth* values, the largest *minWidth* rule that is still less than or equal to the wire width applies.

For example, if you specify enclosure rules for 0.0 µm, 1.0 µm, and 2.0 µm widths, then a 0.5 µm wire must meet a 0.0 rule, a 1.5 µm wire must meet a 1.0 rule, and a 2.0 µm wire must meet a 2.0 rule. (See Example 2-2 on page 238.)

EXCEPTEXTRACUT *cutWithin*

Indicates that if there is another via cut having same metal shapes on both metal layers less than or equal to *cutWithin* distance away, this ENCLOSURE with WIDTH rule is ignored and the ENCLOSURE rules for minimum width wires (that is, no WIDTH keyword) are applied to the via cuts instead. (See Example 2-3 on page 239.)
Type: Float, specified in microns

LENGTH *minLength*

Indicates that the enclosure rule only applies if the total length of the longest opposite-side overhangs is greater than or equal to *minLength*. The total length of the overhang is measured at the via cut center (see illustration F in Figure 2-5 on page 242).
Type: Float, specified in microns

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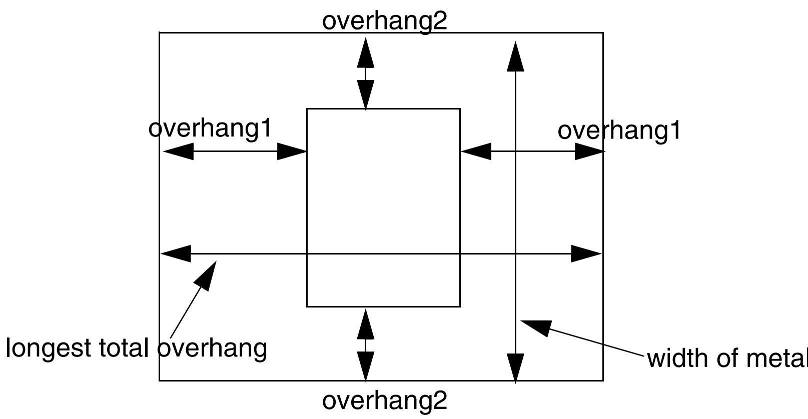
Figure 2-3 Enclosure Rule



Example 2-2 Enclosure Rules

- The following definition describes a cut layer that has different enclosure rules for *m1* below than for *m2* above.

```
LAYER via12
TYPE CUT ;
WIDTH 0.20 ;                    #cuts .20 x .20 squares
ENCLOSURE BELOW .03 .01 ;      #m1: 0.03 on two opposite sides, 0.01 on other
ENCLOSURE ABOVE .05 .01 ;      #m2: 0.05 on two opposite sides, 0.01 on other
RESISTANCE 10.0 ;               #10.0 ohms per cut
...
END via12
```

- The following definition describes a cut layer that requires extra enclosure if the metal width is wider:

```
LAYER via23
TYPE CUT ;
WIDTH 0.20 ;                  #cuts .20 x .20 squares
SPACING 0.15                  #via23 edge-to-edge spacing is 0.15
ENCLOSURE .05 .01 ;           #m2, m3: 0.05 on two opposite sides, 0.01 on
                              #other sides
ENCLOSURE .02 .02 WIDTH 1.0 ; #m2 needs 0.02 on all sides if m2 width >=1.0
                              #m3 needs 0.02 on all sides if m3 width >=1.0
ENCLOSURE .05 .05 WIDTH 2.0 ; #m2 needs 0.05 on all sides if m2 width >=2.0
                              #m3 needs 0.05 on all sides if m3 width >=2.0
...
END via23
```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- The following definition describes a cut layer that requires an overhang of .07 µm on all sides of *metal3*, and an overhang of .09 µm on all sides of *metal4*, if the widths of *metal3* and *metal4* are greater than or equal to 1.0 µm:

```
LAYER via34
TYPE CUT ;
WIDTH 0.25 ;                        #cuts .25 x .25 squares
ENCLOSURE .05 .01 ;                 #minimum width enclosure rule
ENCLOSURE BELOW .07 .07 WIDTH 1.0 ;  #m3 needs .07 on all sides if m3 width >=1.0
ENCLOSURE ABOVE .09 .09 WIDTH 1.0 ; #m4 needs .09 on all sides if m4 width >=1.0
...
END via34
```

Example 2-3 Enclosure Rule With Width and ExceptExtraCut

The following definition describes a cut layer that requires an enclosure of either .05 µm on opposite sides and 0.0 µm on the other two sides, or 0.04 µm on opposites sides and 0.01 µm on the other two sides. It also requires an enclosure of 0.03 µm in all directions if the wire width is greater than or equal to 0.03 µm, unless there is an extra cut (redundant cut) within 0.2 µm.

```
LAYER via34
TYPE CUT ;
WIDTH 0.10                      #cuts .10 x .10 squares
SPACING 0.10 ;                  #minimum edge-to-edge spacing is 0.10
ENCLOSURE 0.0 0.05 ;            #overhang 0.0 0.05
ENCLOSURE 0.01 0.04 ;           #or, overhang 0.01 0.04
#if width >= 0.3, need 0.03 0.03, unless extra cut across wire within 0.2µm
ENCLOSURE 0.03 0.03 WIDTH 0.3 EXCEPTEXTRACUT 0.2 ;
...
END via34
```


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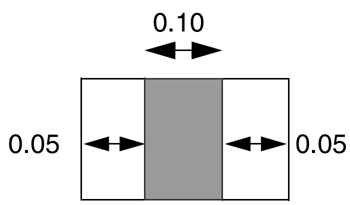
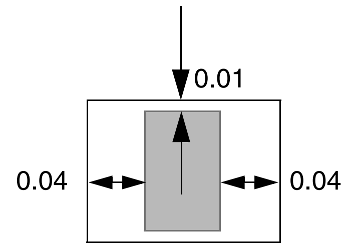
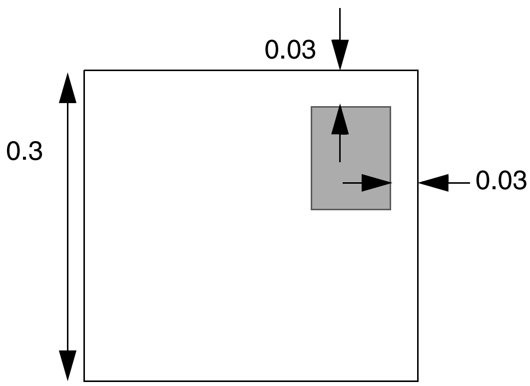
Figure 2-4 Illustrations of Enclosure Rule With Width and ExceptExtraCut



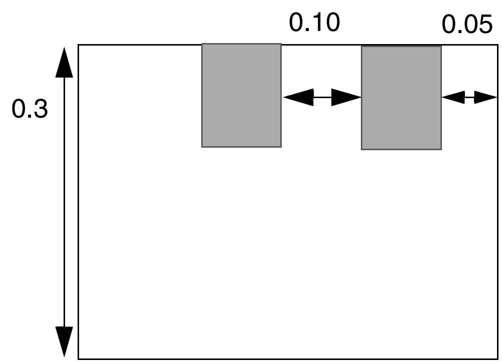
a) Okay; has 0.0 and 0.05 overhang.



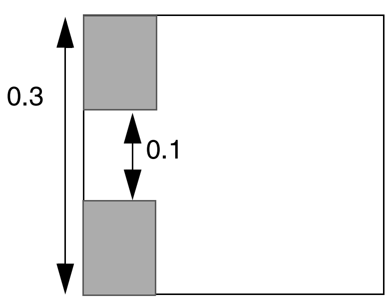
b) Okay; has 0.01 and 0.04 overhang.



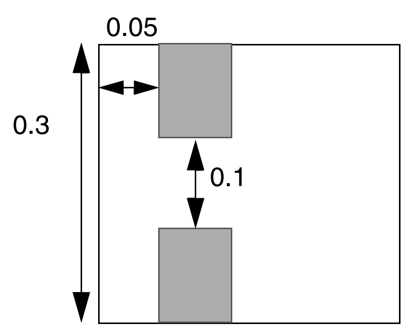
c) Okay; meets wide-wire enclosure rule of 0.03 0.03.



d) Okay; extra cut is <= 0.2 away; therefore, use min-width rule, and both cuts meet min-width enclosure rule of 0.0 and 0.5.



e) Violation. Extra cut is <= 0.2 away; therefore, use min-width rule, but cannot meet either 0.0 0.05 or 0.01 0.04 enclosure rules.



f) Okay. Extra cut is <= 0.2 away; therefore use min-width rule, and both cuts meet the min-width enclosure rule of 0.0 0.05.

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Example 2-4 Enclosure Rule With Length and Width

The following definition describes a cut layer that requires an enclosure of .05 μm on opposite sides and 0.0 μm on the other two sides, as long as the total length enclosure on any two opposite sides is greater than or equal to 0.7 μm. Otherwise, it requires 0.05 μm on all sides if the total enclosure length is less than or equal to 0.7 μm. It also requires 0.10 μm on all sides if the metal layer has a width that is greater than or equal to 1.0 μm. (Figure 2-5 on page 242 illustrates examples of violations and acceptable vias for the three `ENCLOSURE` rules.)

```
LAYER via34
TYPE CUT ;
WIDTH 0.20                          #cuts .20 x .20 squares
SPACING 0.20 ;                      #via34 edge-to-edge spacing is 0.20
ENCLOSURE 0.05 0.0 LENGTH 0.7 ;     #overhang 0.05 0.0 if total overhang >= 0.7
ENCLOSURE 0.05 0.05 ;               #or, overhang 0.05 on all sides
ENCLOSURE 0.10 0.10 WIDTH 1.0 ;     #if width >= 1.0, always need 0.10
...
END via34
```


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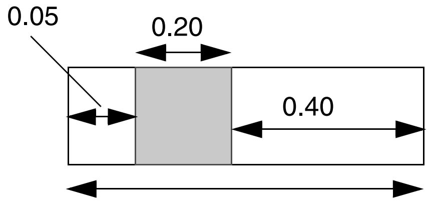
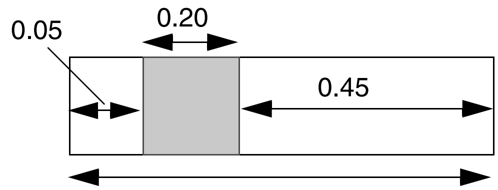
Figure 2-5 Illustrations of Enclosure Rule With Length and Width



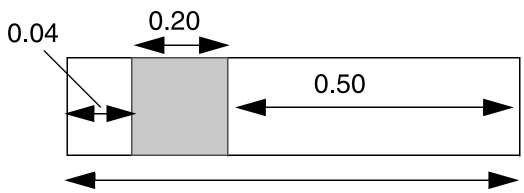
Longest overhang length = 0.65

a) Violation. Longest overhang length < 0.70, and did not meet second rule of 0.05 on all sides.



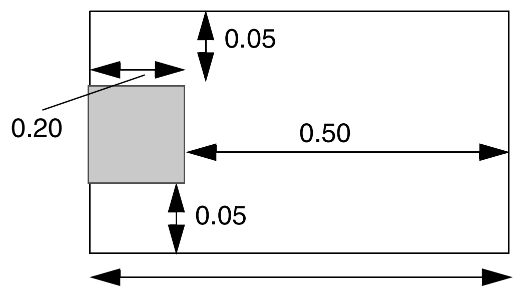
Longest overhang length = 0.70

b) Okay. Longest overhang length >= 0.70, and has 0.05 on opposite sides, and 0.0 on other sides.



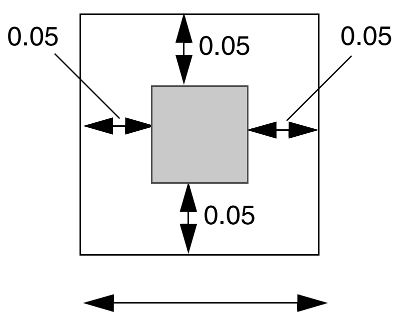
Longest overhang length = 0.74

c) Violation. Longest overhang length >= 0.70, but does not have 0.05 on opposite sides, and did not meet second rule of 0.05 on all sides.



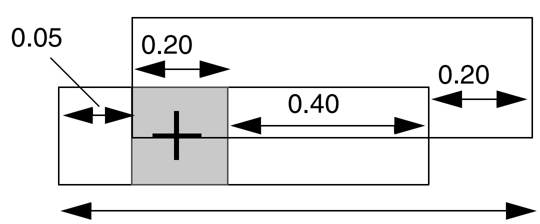
Longest overhang length = 0.70

d) Okay. Longest overhang length >= 0.70, and has 0.05 on opposite sides, and 0.0 on other sides.



Longest overhang length = 0.40

e) Okay. Total length < 0.7; therefore first rule fails, but second rule for 0.05 on all sides is met.



Longest overhang length = 0.85

f) Okay. Overhang length >= 0.70. (The center of the via cut is where the total overhang length is measured.)

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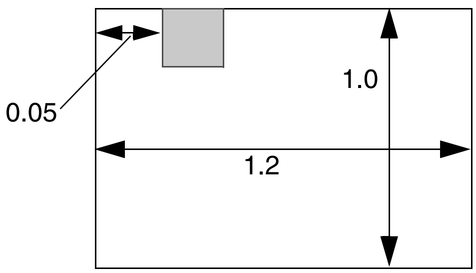

g) Violation. Meets first rule, but width >= 1.0;
therefore must meet third rule: 0.10 on all sides.

`LAYER` *`LayerName`*

Specifies the name for the layer. This name is used in later references to the layer.

`MASK` *`maskNum`*

Specifies how many masks for double- or triple-patterning will be used for this layer. The *`maskNum`* variable must be an integer greater than or equal to 2. Most applications support values of 2 or 3 only.

`PREFERENCLOSURE [ABOVE | BELOW]` *`overhang1 overhang2`* `[WIDTH` *`minWidth`*`]`

Specifies preferred enclosure rules that can improve manufacturing yield, instead of enclosure rules that absolutely must be met (see the `ENCLOSURE` keyword). Applications should use the `PREFERENCLOSURE` rule when it has little or no impact on density and routability.

`PROPERTY` *`propName propVal`*

Specifies a numerical or string value for a layer property defined in the `PROPERTYDEFINITIONS` statement. The *`propName`* you specify must match the *`propName`* listed in the `PROPERTYDEFINITIONS` statement.

`RESISTANCE` *`resistancePerCut`*

Specifies the resistance per cut on this layer. LEF vias without their own specific resistance value, or DEF vias from a `VIARULE` without a resistance per cut value, can use this resistance value.

Via resistance is computed using *`resistancePerCut`* and Kirchoff's law for typical parallel resistance calculation. For example, if R =10 ohms per cut, and the via has one cut, then R =10 ohms. If the via has two cuts, then R = (1/2) * 10 = 5 ohms.

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SPACING

Specifies the minimum spacing allowed between via cuts on the same net or different nets. For via cuts on the same net, this value can be overridden by a spacing with the SAMENET keyword. (See Example 2-5 on page 246.)

The SPACING syntax is defined as follows:

```
[SPACING cutSpacing
  [CENTERTOCENTER]
  [SAMENET]
  [ LAYER secondLayerName [STACK]
  | ADJACENTCUTS {2 | 3 | 4} WITHIN cutWithin
     [EXCEPTSAMEPGNET]
  | PARALLELOVERLAP
  | AREA cutArea]
;] ...
```

cutSpacing — Specifies the default minimum spacing between via cuts, in microns.
Type: Float

CENTERTOCENTER

Computes the *cutSpacing* or *cutWithin* distances from cut-center to cut-center, instead of from cut-edge to cut-edge (the default behavior). (See Spacing Rule Example 4.)

SAMENET — Indicates that the *cutSpacing* value only applies to same-net cuts. The SAMENET *cutSpacing* value should be smaller than the normal SPACING *cutSpacing* value that applies to different-net cuts.

LAYER *secondLayerName*

Applies the spacing rule between objects on the cut layer and objects on *2ndLayerName*. The second layer must be a cut or routing layer already defined in the LEF file, or the next routing layer declared in the LEF file. This allows "one layer look ahead," which is needed in some technologies. (See Spacing Rule Example 1.)

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`STACK`
: Indicates that same-net cuts on two different layers can be stacked if they are aligned. If the cuts are not the same size, the smaller cut must be completely covered by the larger cut, to be considered legal. If both cuts are the same size, the centers of the cuts must be aligned, to be legal; otherwise, the cuts must have *`cutSpacing`* between them. If *`cutSpacing`* is 0.0, the same-net cut vias can be placed anywhere legally, including slightly overlap case. (See Spacing Rule Example 7.)

  Most applications only allow spacing checks and `STACK` checking if *`secondLayerName`* is the cut layer below the current cut layer.

`ADJACENTCUTS {2 |3 | 4} WITHIN` *`cutWithin`*

  Applies the spacing rule only when the cut has two, three, or four via cuts that are less than *`cutWithin`* distance, in microns, from each other. You can specify only one `ADJACENTCUTS` statement per cut layer. For more information, see "Adjacent Via Cuts."
  Type: Float (*`distance`*)

`EXCEPTSAMEPGNET`

  Indicates that the `ADJACENTCUTS` rule does *not* apply between cuts, if they are on the same net, and are on a power or ground net. (See Spacing Rule Example 5.)

`PARALLELOVERLAP`

  Indicates that cuts on different metal shapes that have a parallel edge overlap greater than 0 require *`cutSpacing`* distance between them.

  Only one `PARALLELOVERLAP` spacing value is allowed per cut layer. The rule does not apply to cuts that share the same metal shapes above or below that cover the overlap area between the cuts. (See Spacing Rule Example 8.)

`AREA` *`cutArea`*

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Indicates that any cut with an area greater than or equal to *cutArea* requires edge-to-edge spacing greater than or equal to *cutSpacing* to all other cuts. (See Spacing Rule Example 6.)

A SPACING statement should already exist that applies to all cuts. Only cuts that have area greater than or equal to *cutArea* require extra spacing; therefore, *cutSpacing* for this keyword must be greater than the default spacing.

If you include CENTERTOCENTER, the *cutSpacing* values are computed from cut-center to cut-center, instead of from cut-edge to cut-edge.
Type: Float, specified in microns squared

Example 2-5 Spacing Rule Examples

- **Spacing Rule Example 1**

The following spacing rule defines the cut spacing required between a cut and the routing immediately above the cut. The spacing only applies to “outside edges” of the routing shape, and does not apply to a routing shape already overlapping the cut shape.

```
LAYER cut12
    SPACING 0.10 ;              #normal min cut-to-cut spacing
    SPACING 0.15 LAYER metal2 ; #spacing from cut to routing edge above
    ...
END cut12
LAYER metal2
    ...
```


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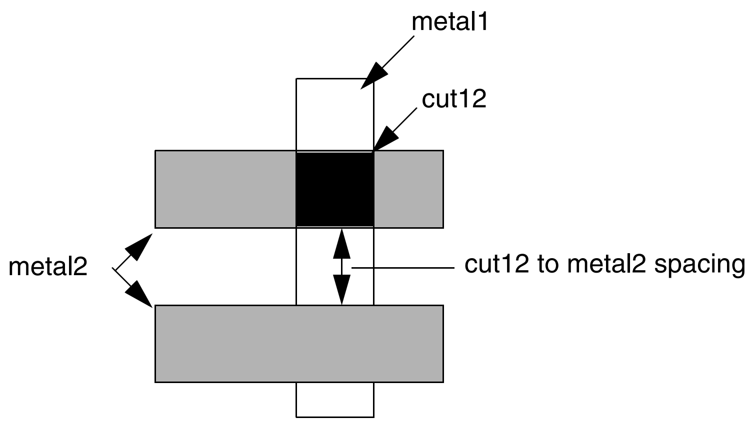
```
END metal2
```



The "`SPACING 0.15 LAYER metal2 ;`" rule only applies to outside edges; therefore, no violations between cut12 and the top metal2 shape will occur. Only the spacing to the bottom metal2 shape is checked.

- **Spacing Rule Example 2**

The following spacing rule specifies that extra space is needed for any via with more than three adjacent cuts, which happens if one via has more than 2x2 cuts (see Figure 2-6 on page 248). A cut that is within .25 µm of three other cuts requires spacing that is greater than or equal to 0.22 µm.

```
LAYER CUT12
    SPACING 0.20 ;                           #default cut spacing
    SPACING 0.22 ADJACENTCUTS 3 WITHIN 0.25 ;
    ...
END CUT12
```

Adjacent Via Cuts

A cut is considered adjacent if it is within *`distance`* of another cut in any direction (including a 45-degree angle). Figure 2-6 on page 248 illustrates adjacent via cuts for 2x2, 2x3, and 3x3 vias, for typical spacing values (that is, the diagonal spacing is greater than the `ADJACENTCUTS` distance value). For three adjacent cuts, the `ADJACENTCUTS` rule allows tight cut spacing on 1x*n* vias and 2x2 vias, but requires larger cut spacing on 2x3, 2x4 and 3x*n* vias. For four adjacent cuts, the rule allows tight cut spacing on 2x*n* vias, but it requires larger cut spacing on 3x*n* vias.

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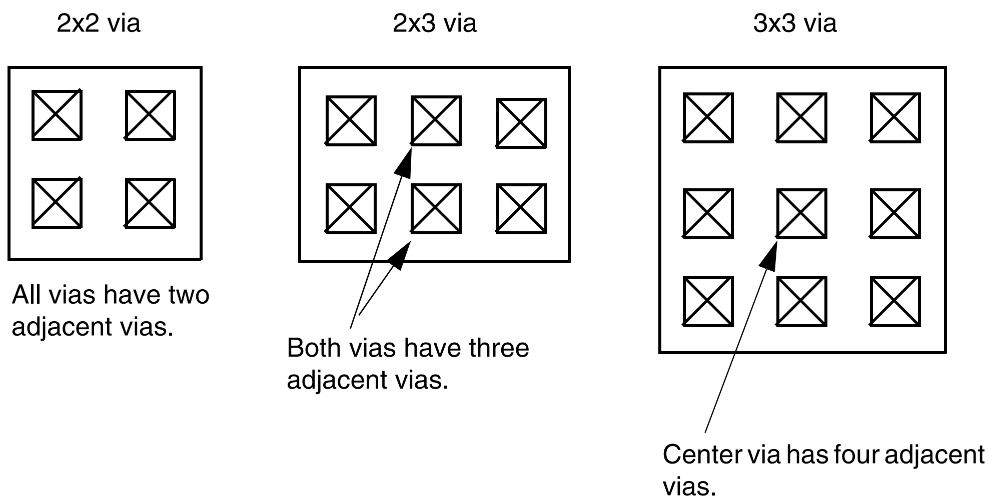
The `ADJACENTCUTS` rule overrides the cut-to-cut spacing used in `VIARULE GENERATE` statements for large vias if the `ADJACENTCUTS` spacing value is larger than the `VIARULE` spacing value.

Figure 2-6



- **Spacing Rule Example 3**

  The following spacing rule specifies that extra space is required for any via with 3x3 cuts or more (that is, a cut with four or more adjacent cuts – see Figure 2-6 on page 248). A cut that is within .25 µm of four other cuts requires spacing that is greater than or equal to 0.22 µm.

```
LAYER CUT12
    SPACING 0.20 ;                          #default cut spacing
    SPACING 0.22 ADJACENTCUTS 4 WITHIN 0.25 ;
    ...
END CUT12
```

- **Spacing Rule Example 4**

  The following spacing rule indicates that center-to-center spacing of greater than or equal to 0.30 µm is required if the center-to-center spacing to three or more cuts is less than 0.30 µm. This is equivalent to saying a cut can have only two other cuts with center-to-center spacing that is less than 0.30 µm.

```
SPACING 0.30 CENTERTOCENTER ADJACENTCUTS 3 WITHIN 0.30 ;
```

- **Spacing Rule Example 5**

  Figure 2-7 on page 249 illustrates the following spacing rule:

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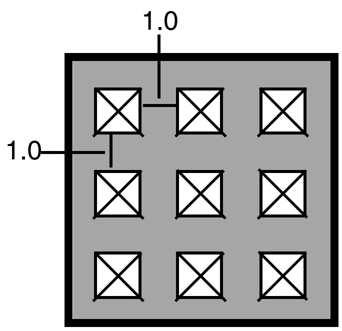
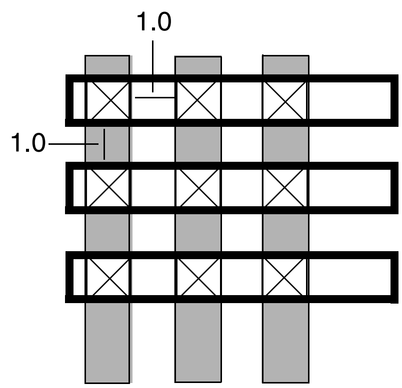
```
SPACING 1.0 ;
SPACING 1.2 ADJACENTCUTS 2 WITHIN 1.5 EXCEPTSAMEPGNET ;
```

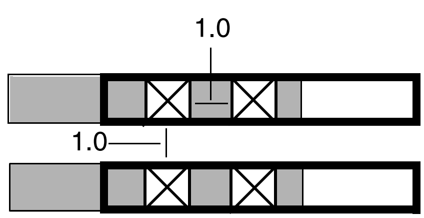
Figure 2-7 Except Same PG Net Rule



a) Allowed if cuts are on power or ground nets. Violation if cuts are on signal net.



b) Allowed if cuts are on the same power or ground nets. Violation if cuts are on signal nets.



c) Allowed if cuts are on the same power or ground nets. Violation if cuts are on signal nets, or different nets.

- ### Spacing Rule Example 6

The following spacing rule indicates that normal cuts require 0.10 µm edge-to-edge spacing, and cuts with an area greater than or equal to 0.02 µm^2 require 0.12 µm edge-to-edge spacing to all other cuts:

```
SPACING 1.0 ;
SPACING 0.12 AREA 0.02 ;
```

- ### Spacing Rule Example 7

The following spacing rule indicates *cut23* cuts must be 0.20 µm from *cut12* cuts unless they are exactly aligned:

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```
LAYER cut23 ;
SPACING 0.20 SAMENET LAYER cut12 STACK ;
```

- **Spacing Rule Example 8**

  Figure 2-8 on page 251 illustrates the following spacing rule:

```
SPACING 1.0 ;
SPACING 1.5 PARALLELOVERLAP ;
```


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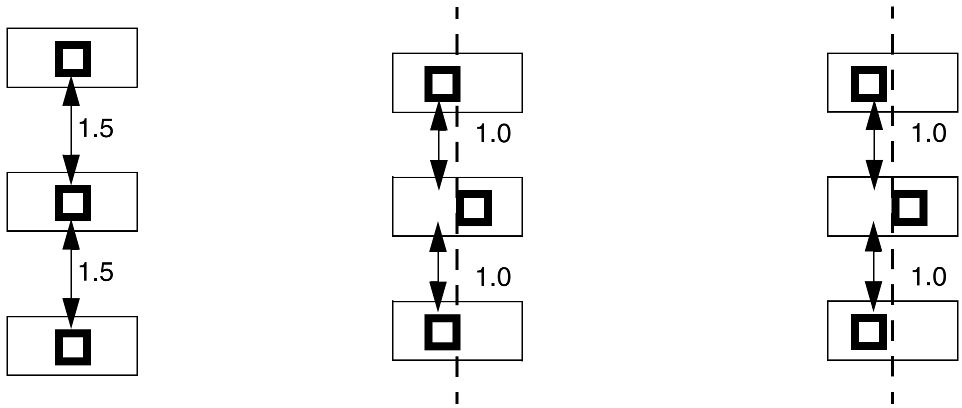
Figure 2-8 Parallel Overlap Rule



a) Okay. Cuts have parallel overlap > 0; therefore `PARALLELOVERLAP` rule of 1.5 applies.

b) Okay. Cuts have parallel overlap = 0; therefore `PARALLELOVERLAP` rule does not apply, and only 1.0 spacing is needed.

c) Okay. Cuts have no parallel overlap; therefore `PARALLELOVERLAP` rule does not apply, and only 1.0 spacing is needed.

d) Okay. Cuts overlap, but share the same metal above or below; therefore `PARALLELOVERLAP` rule does not apply, and only 1.0 spacing is needed.

e) Violation. Cuts must have above or below shared metal to cover the projected area between the cuts.

`SPACINGTABLE`

Specifies spacing tables to use on the cut layer.

The `SPACINGTABLE` syntax is defined as follows:

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```
SPACINGTABLE ORTHOGONAL
 {WITHIN cutWithin SPACING orthoSpacing}...
;]
```

`WITHIN` *`cutWithin`* `SPACING` *`orthoSpacing`*

Indicates that if two cuts have parallel overlap that is greater than `0`, and they are less than *`cutWithin`* distance from each other, any other cuts in an orthogonal direction must have greater than or equal to *`orthoSpacing`*. (See Example 2-5 on page 246, and Figure 2-9 on page 253.)
Type: Float, specified in microns (for both values)

Example 2-6 Spacing Table Orthogonal Rule

The following example shows how a spacing table orthogonal rule is defined:

```
SPACING 0.10                  #min spacing for all cuts
SPACINGTABLE ORTHOGONAL
    WITHIN 0.15 SPACING 0.11
    WITHIN 0.13 SPACING 0.13
    WITHIN 0.11 SPACING 0.15 ;
```


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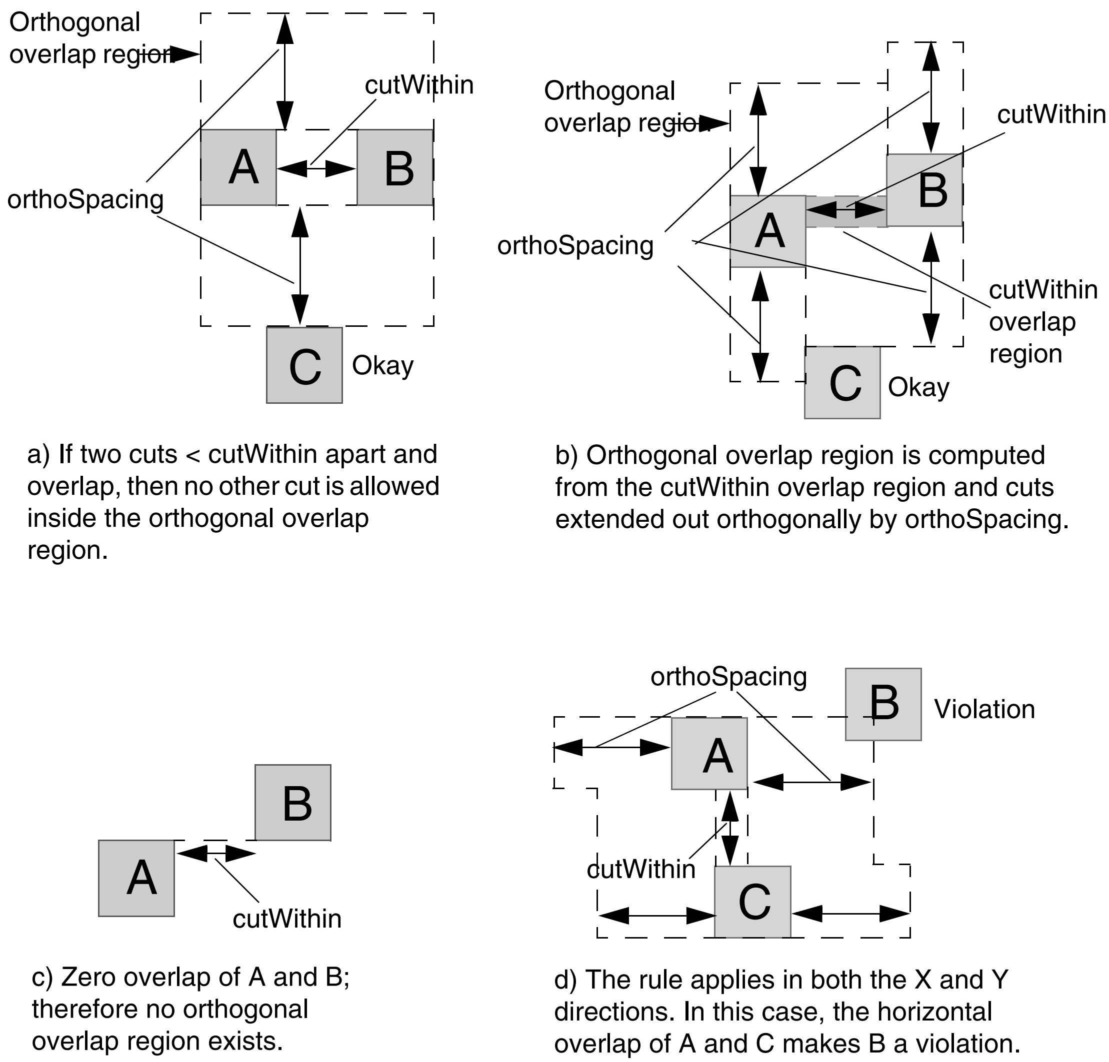
Figure 2-9 Spacing Table Orthogonal Overlap Regions



`TYPE CUT`

Specifies that the layer is for contact-cuts. The layer is later referenced in vias, and in rules for generating vias.

`WIDTH` *`minWidth`*

Specifies the minimum width of a cut. In most technologies, this is also the only legal size of a cut.
Type: Float, specified in microns

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Defining Cut Layer Properties to Create 32/28 nm and Smaller Nodes Rules

You can include cut layer properties in your LEF file to create 32/28 nm and smaller nodes rules that currently are not supported by existing LEF syntax. The properties are specified inside the `LAYER CUT` statements where they can be seen with other rules.

Before you can reference them, properties must be defined at the beginning of the LEF file in the `PROPERTYDEFINITIONS` statement, immediately before the first `LAYER` statement.

- Properties belong to the `LAYER` object and have a type of `STRING`.
- The property names used for these rules all start with `LEF58_`.

All properties use the following syntax within the LEF `PROPERTYDEFINITIONS` statement:

```
PROPERTYDEFINITIONS
     LAYER propName STRING ["stringValue"] ;

END PROPERTYDEFINITIONS
```

The property definitions for the cut layer properties are as follows:

```
PROPERTYDEFINITIONS
   LAYER LEF58_TYPE STRING ;
   LAYER LEF58_ADJACENTFOURCUTS STRING ;
   LAYER LEF58_ANTENNACUMAREARATIO STRING ;
   LAYER LEF58_ANTENNADIFFGATEPWL STRING ;
   LAYER LEF58_ANTENNADIFFONLYAREARATIO STRING ;
   LAYER LEF58_ANTENNADIFFPROTECTORAREARATIO STRING ;
   LAYER LEF58_ANTENNAGATEPWL
   LAYER LEF58_ANTENNAGATEPLUSDIFF
   LAYER LEF58_ARRAYSPACING STRING ;
   LAYER LEF58_BACKSIDE STRING ;
   LAYER LEF58_CUTCLASS STRING ;
   LAYER LEF58_CUTONCENTERLINE STRING ;
   LAYER LEF58_DIRECTIONALSPACING STRING ;
   LAYER LEF58_ENCLOSURE STRING ;
   LAYER LEF58_ENCLOSUREEDGE STRING ;
   LAYER LEF58_ENCLOSURETABLE STRING ;
   LAYER LEF58_ENCLOSURETOJOINT STRING ;
   LAYER LEF58_ENCLOSUREWIDTH STRING ;
   LAYER LEF58_EOLENCLOSURE STRING ;
   LAYER LEF58_EOLSPACING STRING ;
   LAYER LEF58_FORBIDDENSPACING STRING ;
```


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```
   LAYER LEF58_KEEPOUTZONE STRING ;
   LAYER LEF58_MANUFACTURINGGRID STRING ;
   LAYER LEF58_MAXSPACING STRING ;
   LAYER LEF58_MINENCLOSURE STRING ;
   LAYER LEF58_NOCUTCLASS STRING ;
   LAYER LEF58_NOMETALSPACING STRING ;
   LAYER LEF58_ONEDARRAY STRING ;
   LAYER LEF58_REGION STRING ;
   LAYER LEF58_SAMEMETALALIGNEDCUTS STRING ;
   LAYER LEF58_SPACING STRING ;
   LAYER LEF58_SPACINGTABLE STRING ;
   LAYER LEF58_VIACLUSTER STRING ;
   LAYER LEF58_VIAGROUP STRING ;
   LAYER LEF58_VIAGROUPCLUSTER STRING ;
   LAYER LEF58_VIAGROUPOVERRIDESPACING STRING ;
   LAYER LEF58_VIAGROUPSPACING STRING ;
   LAYER LEF58_VOLTAGESPACING STRING ;
END PROPERTYDEFINITIONS
```

Type Rule

A type rule can be used to further classify a cut layer.

You can create a type rule by using the following property definition:

```
TYPE CUT;
     PROPERTY LEF58_TYPE
       "TYPE [TSV [LAYER bottomLayer topLayer | MULTITIERS]
          | PASSIVATION
          | MIMCAP | HIGHR | SPECIALCUT LAYER bottomLayer topLayer
          | SUBSTRATE | STACKEDMIMCAP bottomLayer topLayer]
          ;" ;
```

Where:

`HIGHR`	Specifies that the layer is a special cut layer that is used to connect to a high resistance cut layer with type `HIGHR`.

`LAYER` *`bottomLayer topLayer`*

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Specifies a special `TSV` cut layer with cut vias that cross multiple layers from *`bottomLayer`* to *`topLayer`*. *`bottomLayer`* and *`topLayer`* must be routing layers, including the `BACKSIDE` routing layer.

Type: String

`MIMCAP` Indicates a mimcap layer, which is a cut layer that is not suitable for routing.

For example, the following rule indicates that layer VA is a mimcap cut layer:

```
LAYER VA
   TYPE CUT ;
   PROPERTY LEF58_TYPE "TYPE MIMCAP ;" ;
END VA
```


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`MULTITIERS` Specifies a TSV cut layer between two tiers in a style similar to 3D IC, but within a single design, by using a new design methodology.

The following properties can be defined on a `TSV MULTITIERS` layer:

```
PROPERTY LEF58_CUTCLASS
     "CUTCLASS className WIDTH viaWidth [LENGTH viaLength]
     [ORIENT {HORIZONTAL | VERTICAL}]
     ; " ;
```

The definitions of all keywords are the same as in the regular cut layer.

```
PROPERTY LEF58_ENCLOSURE
     "ENCLOSURE [CUTCLASS className] [ABOVE | BELOW]
     {overhang1 overhang2 | END overhang1 SIDE overhang2}
     ; " ;
```

The definitions of all keywords are the same as in the regular cut layer.

```
PROPERTY LEF58_SPACING
     "SPACING spacing
     ; " ;
```

where

`SPACING` *`spacing`*

Specifies that the cuts on this `TSV MULTITIERS` layer must have spacing greater than *`spacing`*. If `CUTCLASS` is defined on this layer, the spacing is applied on all the cuts on this layer.

Type: Float, specified in microns

`PASSIVATION` Indicates that the cut layer is a passivation cut layer.

`SPECIALCUT LAYER` *`bottomLayer topLayer`*

Specifies a special cut layer that is used to define connectivity. `LAYER` indicates the bottom and top layers to which this special cut layer is used to connect. Either *`bottomLayer`* or *`topLayer`* must be a routing layer. The other layer could be a masterslice or routing layer.

Type: String

`STACKEDMIMCAP` *`bottomLayer topLayer`*

Specifies stacked MIMCAP cut layers, each connected to the specified *`bottomLayer`* and *`topLayer`* routing layers.

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`SUBSTRATE`

Specifies a substrate layer, which typically exists between a back-side layer and a front-side layer.

The following property could be applied on this `SUBSTRATE` layer.

```
PROPERTY LEF_CDN_MAXSPACING
"MAXSPACING spacing [HORIZONTAL|VERTICAL]
    ; " ;
```

Here:

`MAXSPACING` *`spacing`* `[HORIZONTAL|VERTICAL]`

Specifies the maximum spacing between any two cuts on this layer. `VERTICAL|HORIZONTAL` restricts the checking in the given direction only when cut edges have a parallel run length (PRL) greater than `0`. For example, `VERTICAL` means that any horizontal cut edge should have another horizontal edge from another cut with PRL greater than `0` within *`spacing`*. Tools may only support `VERTICAL` at this moment.

Type: Float, specified in microns

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`TSV` Indicates that the cut layer is a through-silicon via (TSV) cut layer.

- The TSV cut layer allows all `ANTENNA*` properties in a regular cut layer to be defined.
- You can specify an `OVERLAP` layer name as the second layer to TSV cut layer `SPACING` rules, to indicate that the TSV (through-silicon-vias) cut must be some minimum spacing from any diffusion, poly, or well shapes. The syntax is as follows:

  [SPACING *spacing* LAYER *secondLayerName* ;]

  The *secondLayerName* can be a previously defined `OVERLAP` layer name, rather than a routing or cut layer name. However, if `LAYER` is specified in `TYPE` in a `TSV` layer, *secondLayerName* could be any routing or cut layer name between the given bottom and top layers.

  Note that in many cases standard cells are designed with a small amount of diffusion, poly, or well shapes sticking outside the cell boundary (the `LEF SIZE` statement), so that must be considered.

  For example, if 5 µm spacing is needed from the TSV-cut to any poly, diffusion, or well shape, and the standard cells allow up to 0.1 µm of diffusion, poly, or well shapes outside the cell boundary, then the rule should be specified as:

  ```
  SPACING 5.1 LAYER OVERLAP ;
  ```

Adjacent Four Cuts Rule

An adjacent four cuts rule can be used to describe a cut spacing rule for two sets of two cuts with `PRL` greater than zero when few other conditions are met.

You can create an adjacent four cuts rule by using the following property definition:

```
PROPERTY LEF58_ADJACENTFOURCUTS
       "ADJACENTFOURCUTS cutSpacing [ONECUTSETONLY]
          CUTCLASS {className [TO ALL] | ALL}
          {ABOVE | BELOW}
          PARALLEL parLength WITHIN parWithin [MERGEDCUTS]
          [HORIZONTAL|VERTICAL]
          [EXCEPTDIFFMETAL [NOEXTRACUT]]
          [MINCORNER | MININNERCORNER | MAXINNERCORNER]
          [EXCEPTMAXPRL maxPrl]
          [EXCEPTMINLENGTH minLength]
```


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```
[WITHINCUT within | CUTWITHINONLY within]
[EXCEPTOTHERWIDTHWITHCUT width]
[METALWIDTH maxWidth]
;" ;
```

Where:

`ADJACENTFOURCUTS cutSpacing CUTCLASS {className [TO ALL] | ALL}`
`{ABOVE | BELOW}`
`PARALLEL parLength WITHIN parWithin`

Specifies that if there are two sets of two cuts of cut class *className* with spacing less than *cutSpacing* and PRL greater than 0 of cuts within each sets, and each of the cuts has a neighbor cut of the other set within *parWithin* having parallel run length greater than *parLength*, it is a violation to have a wire that does not overlap or contain the cuts on the above or below metal layer in ABOVE or BELOW in a maximum bounding box formed by the facing cut edges.

If TO ALL is specified, the rule is triggered if at least one cut belongs to the given cut class, *className*; the other cuts could belong to any cut class.

If ALL is specified in CUTCLASS, any cut combination of four cuts could trigger the rule.

Type: Float, specified in microns

`CUTWITHINONLY within`

Specifies that a violation occurs only if there is a same-layer neighbor cut within *within* of the search window. The presence of a metal neighbor overlapping with the search window is irrelevant. However, if METALWIDTH is specified, the width of the metal containing the neighbor cut should still be checked. If the metal width exceeds *maxWidth*, the rule still does not apply.

Type: Float, specified in microns

`EXCEPTDIFFMETAL` Specifies that the adjacent four cut rule is exempted if both sets of the two cuts have different metal on the given above or below metal layer. In other words, the rule applies only if at least one set of the two cuts has the same metal on the given above or below metal layer.

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EXCEPTMAXPRL *maxPrl*

Specifies that the rule does not apply if the middle metal neighbor has a parallel run length overlap of less than *maxPrl* with the window.
Type: Float, specified in microns

EXCEPTMINLENGTH *minLength*

Specifies that the rule does not apply if the middle metal neighbor has a length greater than or equal to *minLength*.
Type: Float, specified in microns

EXCEPTOTHERWIDTHWITHCUT *width*

Specifies that the rule does not apply if there is a wire on the other layer (the below/above layer when ABOVE/BELOW is defined) with width greater than or equal to *width* and a cut on the specified layer in a search window formed by the maximum bounding box of the inner cut corners.
Type: Float, specified in microns

HORIZONTAL|VERTICAL Specifies that *cutSpacing* and *parLength* should be only along the given direction while *parWithin* should be only along the orthogonal direction when HORIZONTAL|VERTICAL is defined. For example, if HORIZONTAL is specified, *cutSpacing* and *parLength* should be only along the horizontal direction while *parWithin* should be only along the vertical direction.

MAXINNERCORNER Specifies a different way to form the metal searching window by using the maximum bounding box of the inner cut corners.

MERGEDCUT Specifies a different way to interpret the PARALLEL and WITHIN conditions. The two sets of two cuts with spacing less than *spacing* and a parallel run length (PRL) greater than 0 should be merged inside a set individually. If the merged cuts have PRL greater than *parLength* and within *parWithin*, the rule is triggered.

METALWIDTH *maxWidth*

Specifies that the rule applies only for the metal neighbor with width less than or equal to *maxWidth*.

Type: Float, specified in microns

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`MINCORNER`	Specifies a different way to form the metal searching window. If neighboring cuts, ones that are not in the same set, have PRL greater than `0`, the cut center is used to form the window. Otherwise, the inner corner is used. The window is still the maximum bounding box of all of the cut locations.
`MININNERCORNER`	Specifies a different way to form the metal searching window by using the minimum bounding box of the inner cut corners.
`NOEXTRACUT`	Specifies that the except different metal, `EXCEPTDIFFMETAL`, cannot be applied if there is an extra cut of any cut class inside any set of the two sets of two cuts, which triggers the rule.
`ONECUTSETONLY`	Specifies that the adjacent four cut rule also applies when at least one of the two sets of cuts has a spacing less than *`cutSpacing`* and PRL greater than `0`. This means the other set can have spacing greater than or equal to *`cutSpacing`*, particularly when negative *`parLength`* is specified. In such cases, the second set of cuts should have a maximum spacing of less than *`cutSpacing`* plus twice the absolute value of -*`parLength`* and two cut sizes (*`cutSpacing`* + 2 * abs(-*`parLength`*) + 2 cut sizes).
`WITHINCUT` *`within`*	Specifies that the rule applies even if the middle metal neighbor does not overlap with the metal search window, but the wire contains an above or below cut within *`within`* of the window. *Type*: Float, specified in microns

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Adjacent Four Cuts Rule Examples

Figure 2-10 Illustration of Adjacent Four Cuts Rule

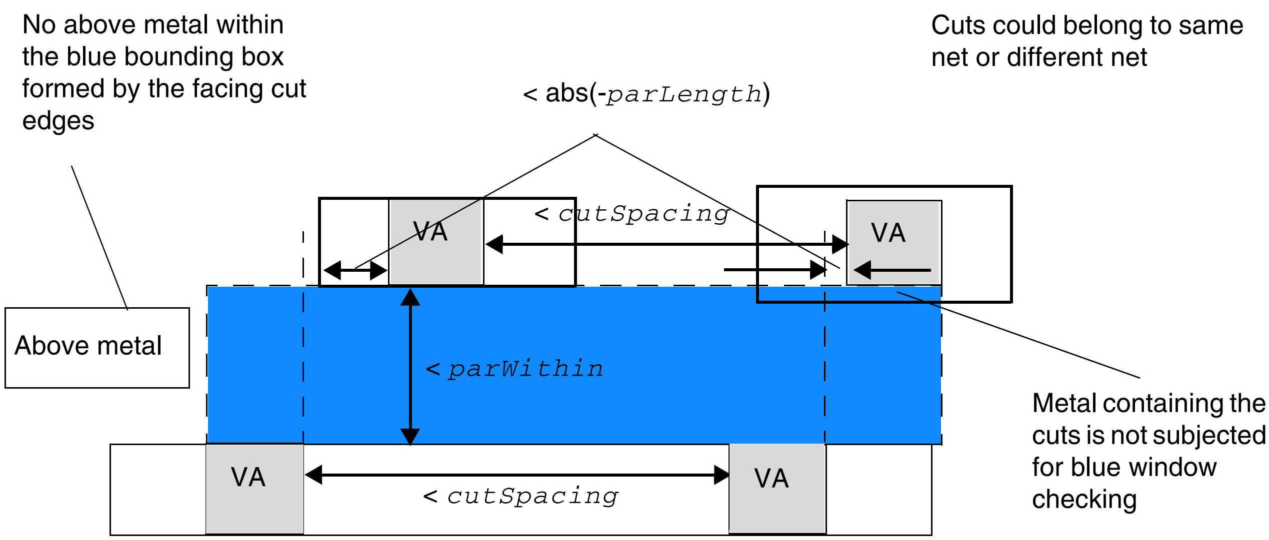

Illustration of `ADJACENTFOURCUTS` *`cutSpacing`*
`CUTCLASS VA ABOVE`
`PARALLEL -`*`parLength`* `WITHIN` *`parWithin`* `HORIZONTAL`

If `VERTICAL` is defined, *`cutSpacing`* and *`parLength`* should be vertical while *`parWithin`* should be horizontal. If `HORIZONTAL` or `VERTICAL` is not defined, both cases should be checked.

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Figure 2-11 Illustration of Adjacent Four Cuts Rule with EXCEPTDIFFMETAL

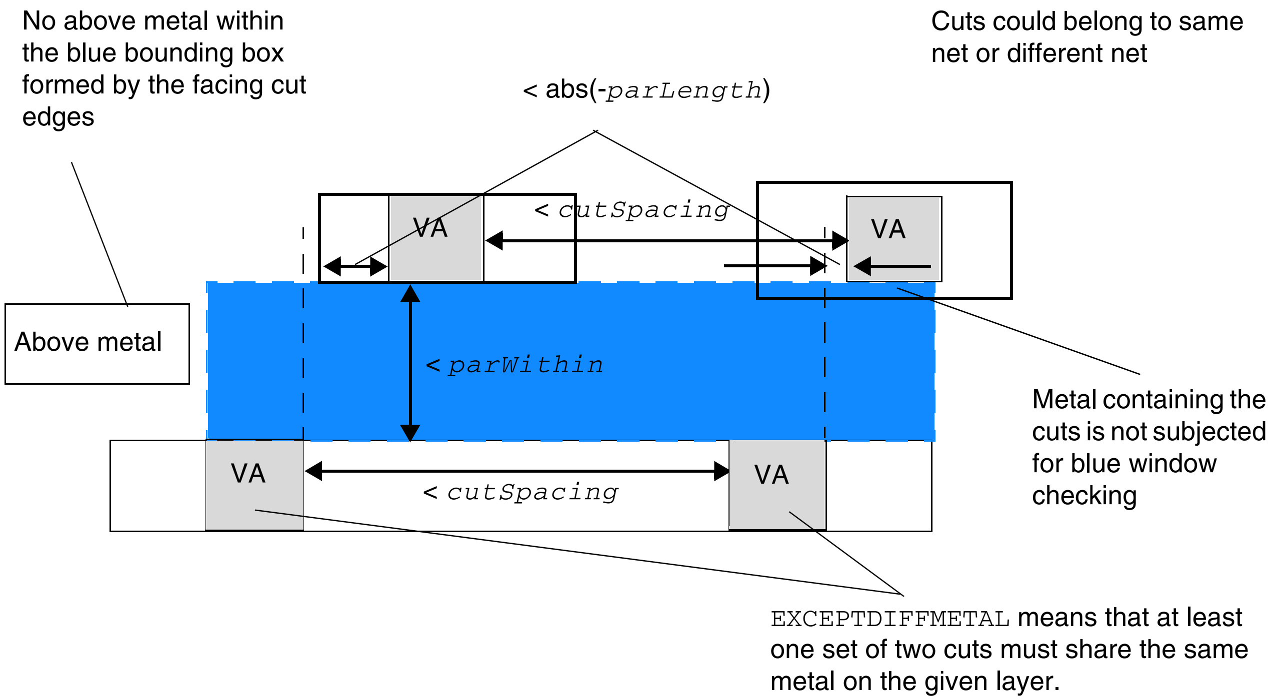

Illustration of `ADJACENTFOURCUTS` *`cutSpacing`*
`CUTCLASS VA ABOVE`
`PARALLEL` *`-parLength`* `WITHIN` *`parWithin`* `HORIZONTAL`
`EXCEPTDIFFMETAL`

If `VERTICAL` is defined, *`cutSpacing`* and *`parLength`* should be vertical while *`parWithin`* should be horizontal. If `HORIZONTAL` or `VERTICAL` is not defined, both cases should be checked.

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Figure 2-12 Illustration of Adjacent Four Cuts Rule with MERGEDCUTS

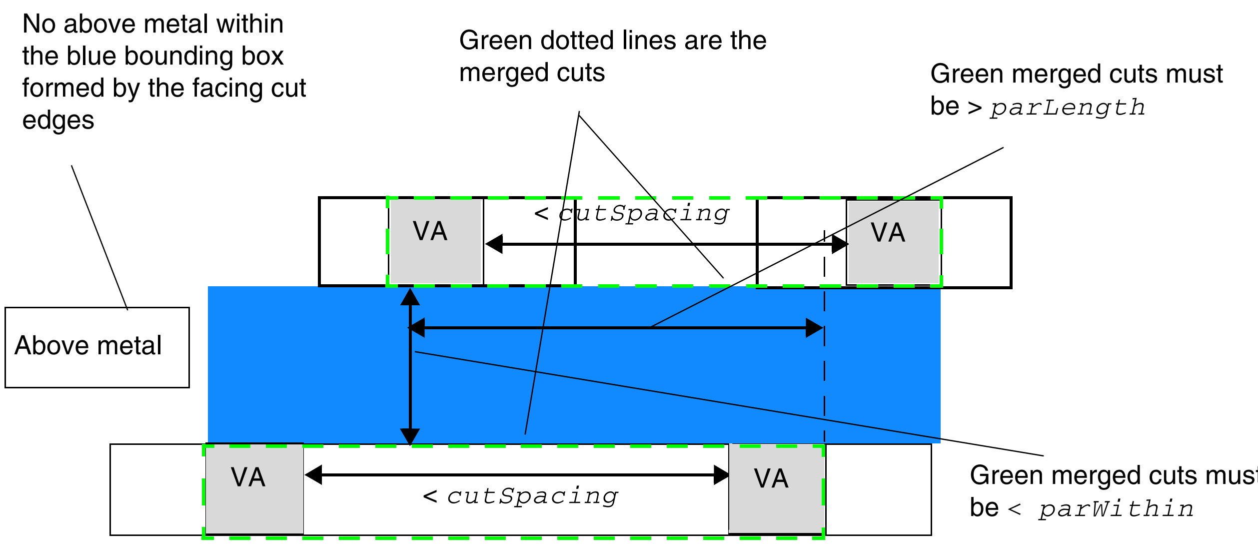

Illustration of `ADJACENTFOURCUTS` *`cutSpacing`*
`CUTCLASS VA ABOVE`
`PARALLEL` *`parLength`* `WITHIN` *`parWithin`*
`MERGEDCUTS`

Figure 2-13 Illustration of Adjacent Four Cuts Rule with MINCORNER

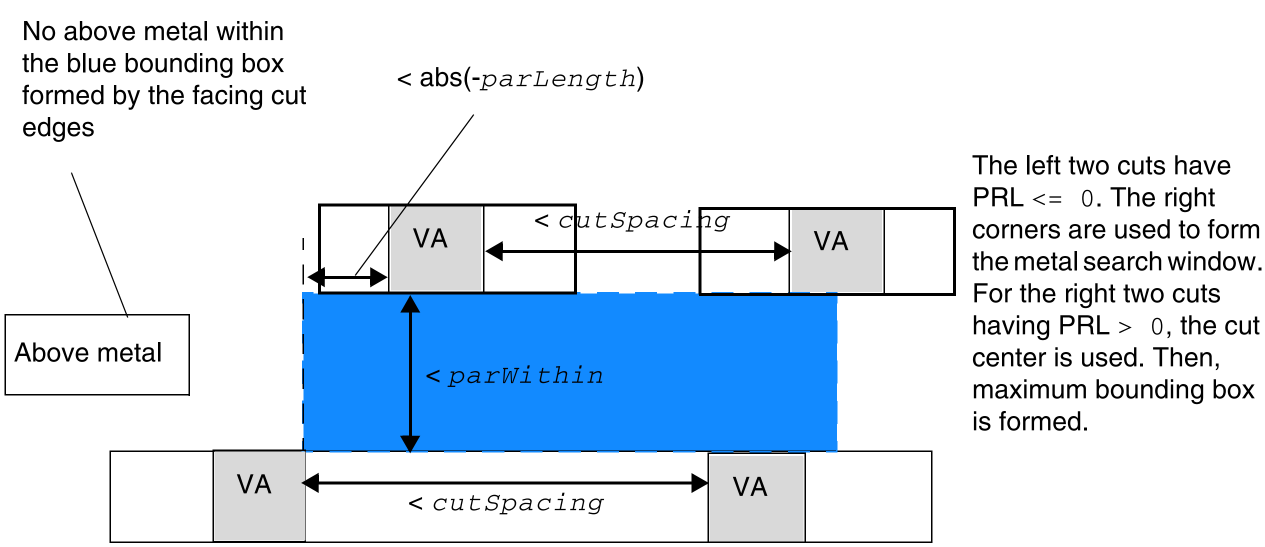

Illustration of `ADJACENTFOURCUTS` *`cutSpacing`*
`CUTCLASS VA ABOVE`
`PARALLEL -`*`parLength`* `WITHIN` *`parWithin`* `HORIZONTAL`
`MINCORNER`

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Figure 2-14 Illustration of Adjacent Four Cuts Rule with MININNERCORNER

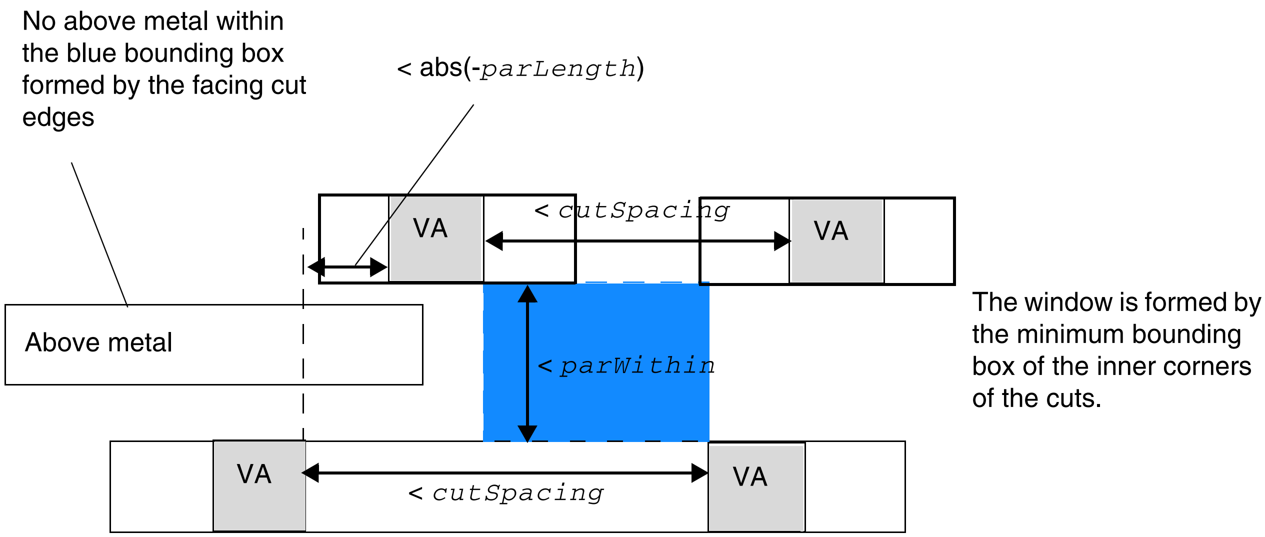

Illustration of `ADJACENTFOURCUTS` *`cutSpacing`*
`CUTCLASS VA ABOVE`
`PARALLEL -`*`parLength`*` WITHIN `*`parWithin`*` HORIZONTAL`
`MININNERCORNER`

Figure 2-15 Illustration of Adjacent Four Cuts Rule with MAXINNERCORNER

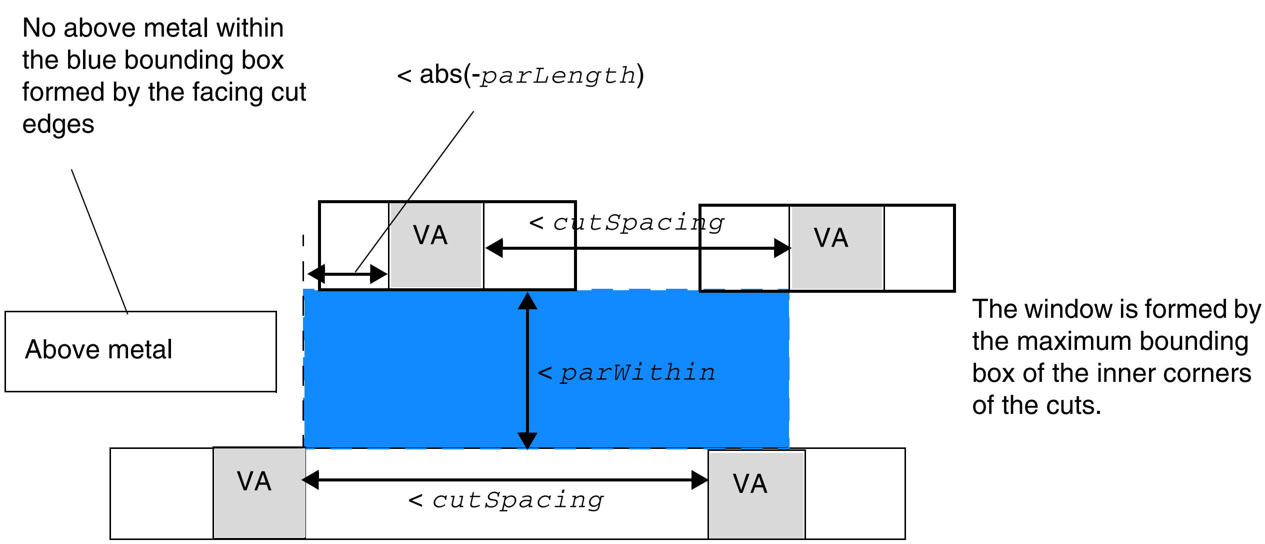

Illustration of `ADJACENTFOURCUTS` *`cutSpacing`*
`CUTCLASS VA ABOVE`
`PARALLEL -`*`parLength`*` WITHIN `*`parWithin`*` HORIZONTAL`
`MAXINNERCORNER`

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Figure 2-16 Illustration of Adjacent Four Cuts Rule with EXCEPTMAXPRL

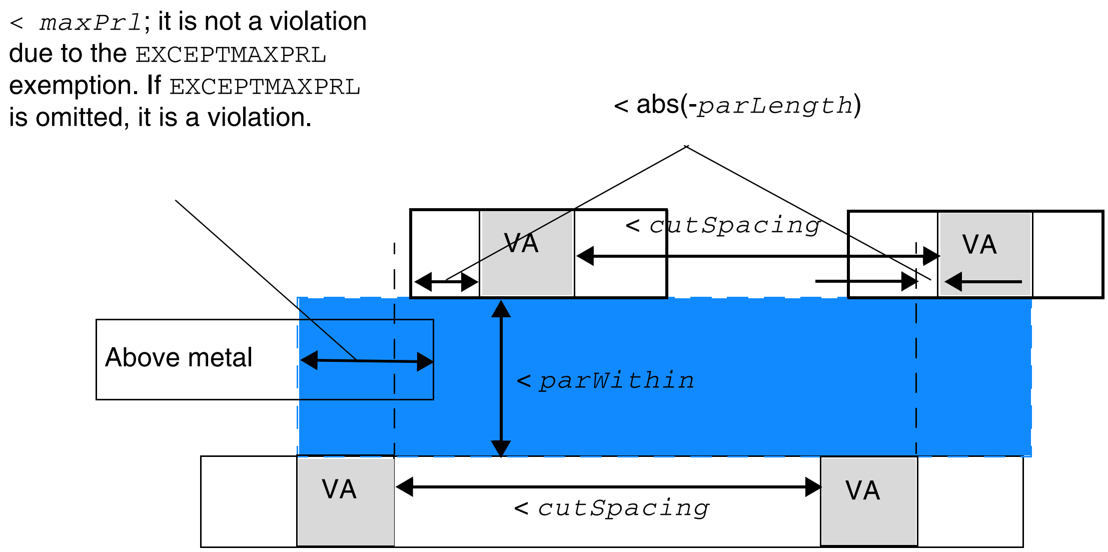

Illustration of `ADJACENTFOURCUTS` *`cutSpacing`*
`CUTCLASS VA ABOVE`
`PARALLEL` *`-parLength`* `WITHIN` *`parWithin`*
`EXCEPTMAXPRL` *`maxPrl`*

Figure 2-17 Illustration of Adjacent Four Cuts Rule with EXCEPTMINLENGTH

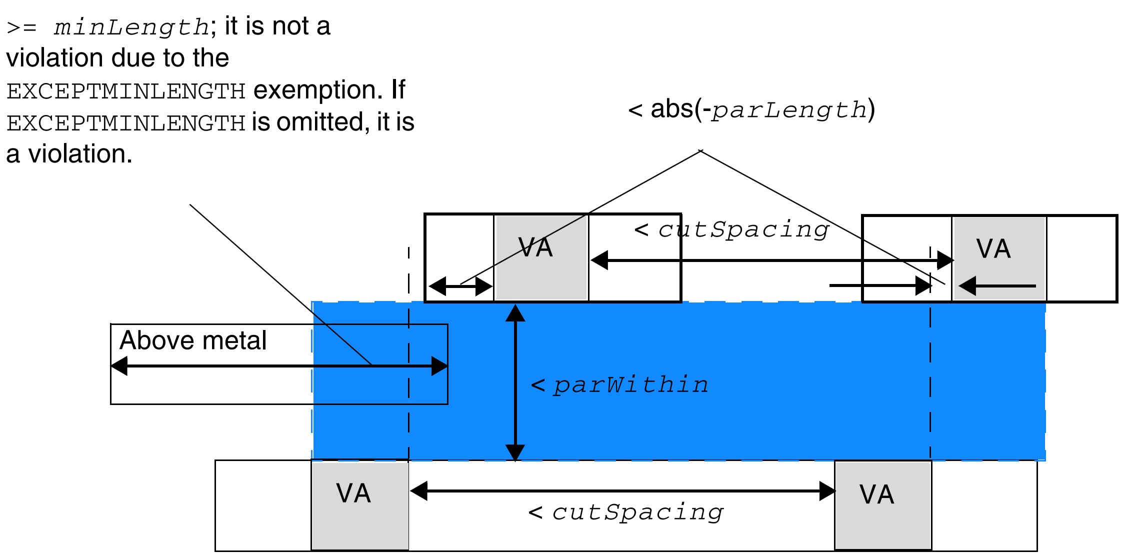

Illustration of `ADJACENTFOURCUTS` *`cutSpacing`*
`CUTCLASS VA ABOVE`
`PARALLEL` *`-parLength`* `WITHIN` *`parWithin`*
`EXCEPTMINLENGTH` *`minLength`*

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Figure 2-18 Illustration of Adjacent Four Cuts Rule with WITHINCUT

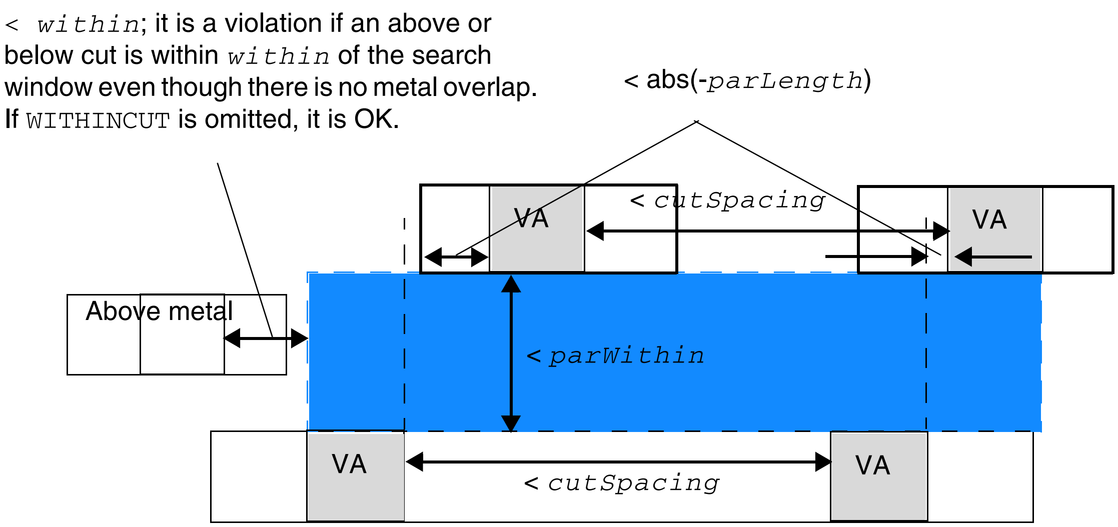

Illustration of `ADJACENTFOURCUTS` *`cutSpacing`*
`CUTCLASS VA ABOVE`
`PARALLEL` *`-parLength`* `WITHIN` *`parWithin`*
`WITHINCUT` *`within`*

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Figure 2-19 Illustration of Adjacent Four Cuts Rule with EXCEPTOTHERWIDTHWITHCUT

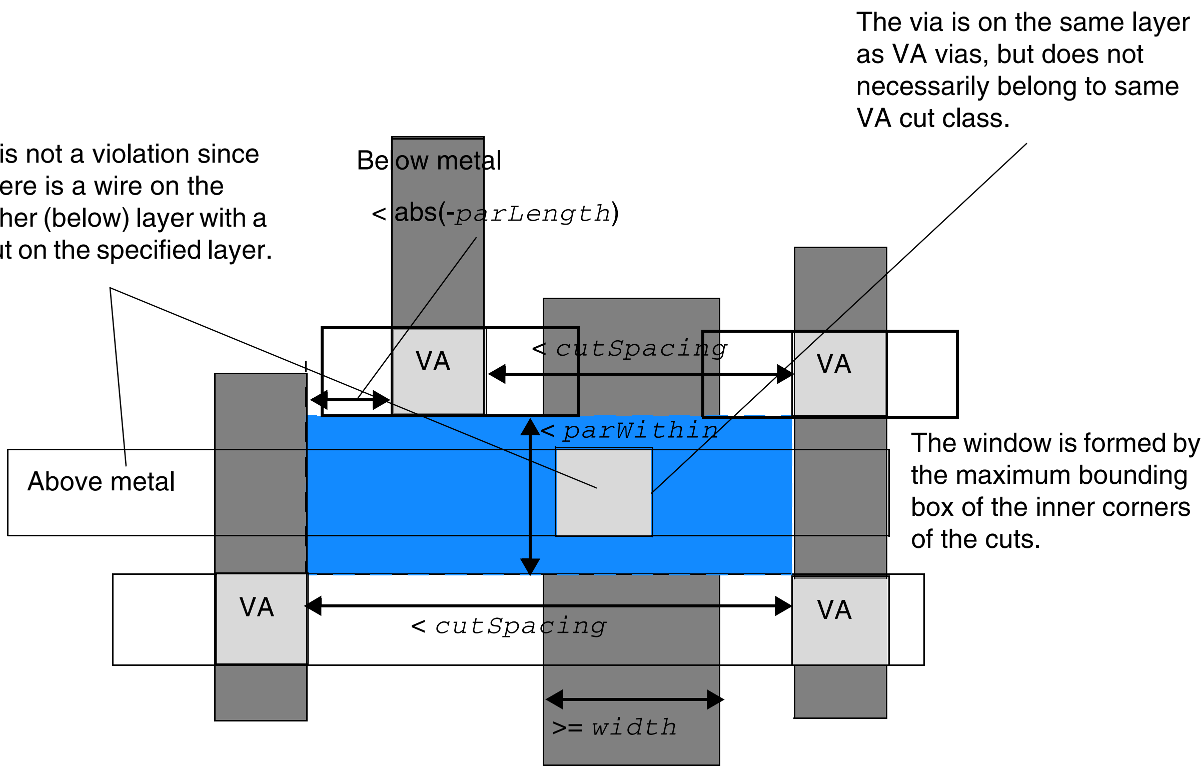

Illustration of `ADJACENTFOURCUTS` *`cutSpacing`*
`CUTCLASS VA ABOVE`
`PARALLEL -`*`parLength`* `WITHIN` *`parWithin`*
`HORIZONTAL EXCEPTOTHERWIDTHWITHCUT` *`width`*

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Figure 2-20 Illustration of Adjacent Four Cuts Rule with METALWIDTH

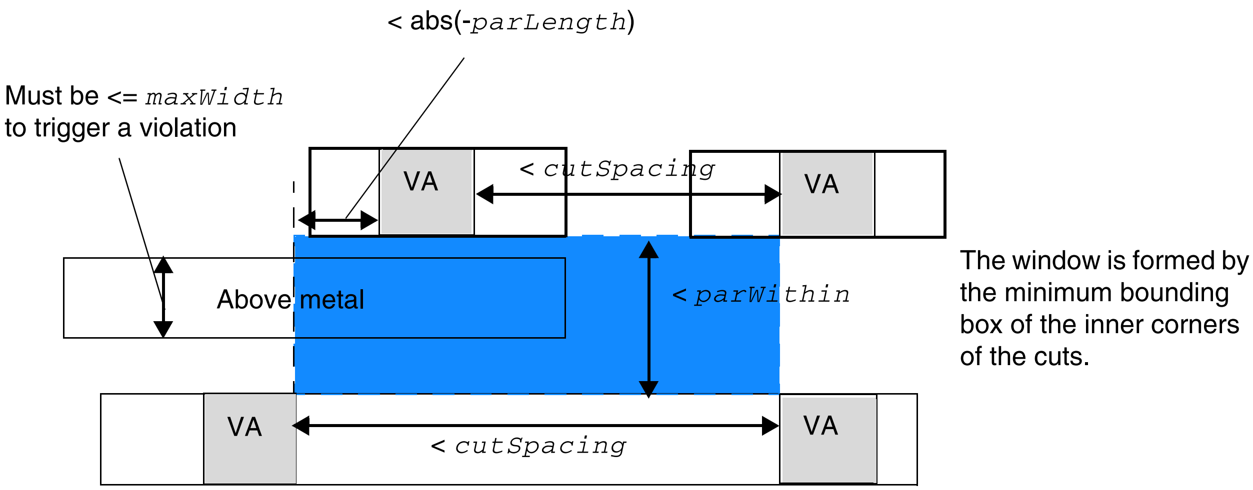

Illustration of `ADJACENTFOURCUTS` *`cutSpacing`*
`CUTCLASS VA ABOVE`
`PARALLEL` *`-parLength`* `WITHIN` *`parWithin`*
`HORIZONTAL METALWIDTH` *`maxWidth`*

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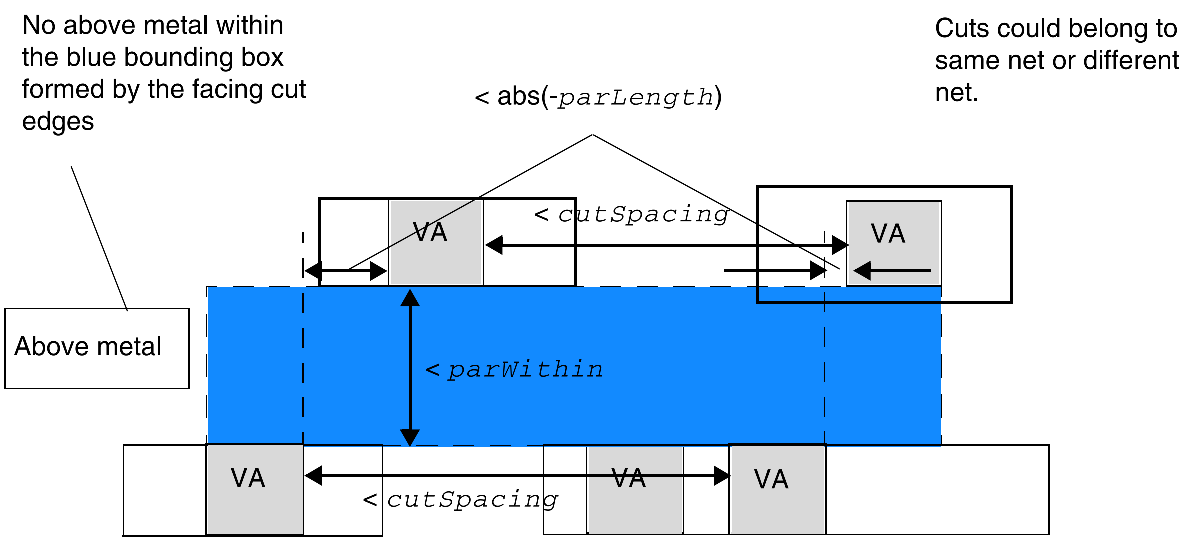
Figure 2-21 Illustration of Adjacent Four Cuts Rule with NOEXTRACUT



Although the outer four cuts have different metal individually, the `EXCEPTDIFFMETAL` exemption is not applied because there is an extra cut on the bottom with `NOEXTRACUT`.

Illustration of `ADJACENTFOURCUTS` *`cutSpacing`*
`CUTCLASS VA ABOVE`
`PARALLEL` *`-parLength`* `WITHIN` *`parWithin`* `HORIZONTAL`
`EXCEPTDIFFMETAL NOEXTRACUT`

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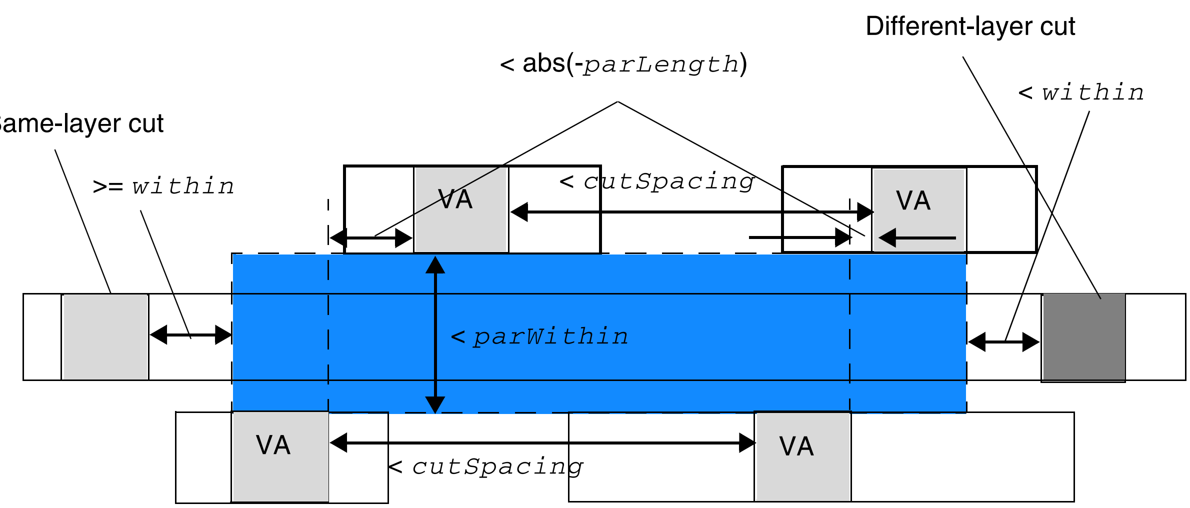
Figure 2-22 Illustration of Adjacent Four Cuts Rule with CUTWITHINONLY



It is OK. Having a different-layer neighbor cut within `within` of the blue search window on the right or on either an above or a below metal passing through the search window is irrelevant. As long as there is no same-layer neighbor cut within `within` of the search window, it is fine. This is the case here since the same-layer neighbor cut is located outside `within` of the search window on the left.

Illustration of `ADJACENTFOURCUTS` *`cutSpacing`*

```
CUTCLASS VA
PARALLEL -parLength WITHIN parWithin HORIZONTAL
CUTWITHINONLY within
```

Antenna Cumulative Area Ratio Rule

The antenna cumulative area ratio rule is used to specify the cumulative antenna ratio, using the area of the metal wire that is not connected to the diffusion diode.

You can create an antenna cumulative area ratio rule by using the following property definition:

```
PROPERTY LEF58_ANTENNACUMAREARATIO
       "ANTENNACUMAREARATIO {OXIDE1 | OXIDE2 | OXIDE3 | OXIDE4 | OXIDE5 | ...
        | OXIDE32} value
        [GATEEXPONENT index] [GATEAREA area]
        [DIFFONLY]
        ;" ;
```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Where:

`ANTENNACUMAREARATIO {OXIDE1 | OXIDE2 | OXIDE3 | OXIDE4 | OXIDE5 | ...| OXIDE32}` *`value`*

Specifies the cumulative antenna ratio, using the area of the metal wire that is not connected to the diffusion diode.

Type: Float

`DIFFONLY` Specifies the cumulative antenna ratio, using the area of the metal wire connected to a normal diffusion device, which is defined in `ANTENNADIFFAREA` on a macro pin. This check is skipped if the area is connected to the gate.

`GATEAREA` *`area`* Specifies that the rule applies only if the gate areas are less than *`area`*.

Type: Float

`GATEEXPONENT` *`index`*

Specifies an exponential cumulative antenna ratio, using the electrical connected gateArea into the rule ratio formula as *`value * gateArea`*$^{\mathrm{index}}$.

Type: Float

Antenna Diff Gate PWL Rule

The antenna diff gate PWL rule is used to specify that the antenna gate connected to the diffusion diode can be used to define a piece-wise linear (PWL) table on the given antenna model that is indexed by the real gate area, and returns an “effective gate area” interpolated from the table.

You can create an antenna diff gate PWL rule by using the following property definition:

```
PROPERTY LEF58_ANTENNADIFFGATEPWL
     "ANTENNADIFFGATEPWL {OXIDE1 | OXIDE2 | OXIDE3 | OXIDE4 | OXIDE5 | ...
         | OXIDE32}
         ((gateArea1 effectiveGateArea1) (gateArea2 effectiveGateArea2) ... )
         ;" ;
```


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Where:

```
ANTENNADIFFGATEPWL {OXIDE1 | OXIDE2 | OXIDE3 | OXIDE4 |
    OXIDE5 | ... | OXIDE32}
    ((gateArea1 effectiveGateArea1) (gateArea2
      effectiveGateArea2)... )
```

Specifies that the antenna gate connected to the diffusion diode can be used to define a piece-wise linear (PWL) table on the given antenna model that is indexed by the real gate area, and returns an “effective gate area” interpolated from the table.
Type: Float, specified in microns

effectiveGateArea indicates the effective gate area interpolated from the PWL table. If this table it not defined, the real gate area is used.

gateArea indicates the real gate area.

Note that if ANTENNADIFFGATEPWL is defined, the ANTENNAGATEPWL rule is applied only to the antenna gate that is not connected to the diffusion diode. If ANTENNADIFFGATEPWL is not defined, the ANTENNAGATEPWL rule is applied to all the antenna gates.

Antenna Diff Only Area Ratio Rule

The antenna diff only area rule is used to specify a diffusion-area-based antenna ratio.

You can create an antenna diff only area rule by using the following property definition:

```
PROPERTY LEF58_ANTENNADIFFONLYAREARATIO
      “ANTENNADIFFONLYAREARATIO {OXIDE1 | OXIDE2 | OXIDE3 | OXIDE4
         | OXIDE5 | ... | OXIDE32}  value
       ;..." ;
```

Where:

```
ANTENNADIFFONLYAREARATIO {OXIDE1 | OXIDE2 | OXIDE3 | OXIDE4
     | OXIDE5 | ... | OXIDE32} value
```


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Specifies a diffusion-area-based antenna ratio, using the area of the metal wire connected to a diffusion device, which is defined in `ANTENNADIFFAREA` on a macro pin. This ratio is associated with the given `OXIDE` model.

Type: Float, specified in microns

The metal connected to a pin with a protector diffusion device would only be checked against this given ratio. This includes the case when the metal also connects to gates or non-protector diffusions. Other `ANTENNA*` rules would be ignored.

The partial area ratio (PAR) is calculated for diffusion and gate area separately as:

$\mathrm{PAR}(N_{i,j}, D_k)$ = Drawn area $(N_{i,j})$ / Total area of diffusion area below $N_{i,j}$

$\mathrm{PAR}(N_{i,j}, G_k)$ = Drawn area $(N_{i,j})$ / Total area of gate area below $N_{i,j}$

The following two rule checks are preformed individually:

- The PAR for node $N_{i,j}$ with respect to diffusion D_k by dividing the drawn area of the node by the area of diffusions that electrically connected to it must be less than or equal to the given value in `ANTENNADIFFONLYAREARATIO`.
- The PAR for node $Ni_{,j}$ with respect to gate G_k by dividing the drawn area of the node by the area of gates that electrically connected to it must be less than or equal to the given value in `ANTENNAAREARATIO`.

Antenna Diff Protector Area Ratio Rule

The antenna diff protector area rule is used to specify a protector diffusion-area-based antenna ratio.

You can create an antenna diff protector area rule by using the following property definition:

```
PROPERTY LEF58_ANTENNADIFFPROTECTORAREARATIO
       "ANTENNADIFFPROTECTORAREARATIO {OXIDE1 | OXIDE2 | OXIDE3
          | OXIDE4 | OXIDE5 | ... | OXIDE32}  value
        ;..." ;
```


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Where:

```
ANTENNADIFFPROTECTORAREARATIO {OXIDE1 | OXIDE2 | OXIDE3
      | OXIDE4 | OXIDE5 | ... | OXIDE32} value
```

Specifies a protector diffusion-area-based antenna ratio, using the area of the metal wire connected to a protector diffusion device, which is defined in ANTENNADIFFAREA with PROTECTOR on a macro pin. This ratio is associated with the given OXIDE model.

Type: Float, specified in microns

The metal connected to a pin with a protector diffusion device would only be checked against this given ratio. This includes the case when the metal also connects to gates or non-protector diffusions. Other ANTENNA* rules would be ignored.

The partial area ratio (PAR) is calculated for protector diffusion as:

$PAR(N_{i,j}, D_k)$ = Drawn area $(N_{i,j})$ / Total area of protector diffusion area below $N_{i,j}$

The PAR for node $N_{i,j}$ with respect to protector diffusion D_k by dividing the drawn area of the node by the area of protector diffusions that are electrically connected to it must be less than or equal to the given value in ANTENNADIFFPROTECTORAREARATIO.

Antenna Gate PWL Rule

The antenna gate PWL rule can be used to define a PWL (piece-wise linear) table on the given antenna model that is indexed by the real gate area, and returns an “effective gate area” interpolated from the table.

You can create an antenna gate PWL rule by using the following property definition:

```
PROPERTY LEF58_ANTENNAGATEPWL
     "ANTENNAGATEPWL {OXIDE1 | OXIDE2 | OXIDE3 | OXIDE4 | OXIDE5 | ...
        | OXIDE32}
        ((gateArea1 effectiveGateArea1) (gateArea2 effectiveGateArea2) ... )
        ;" ;
```


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Where:

effectiveGateArea

Indicates the effective gate area interpolated from the PWL table. If this table it not defined, the real gate area is used.
Type: Float, specified in microns

gateArea Indicates the real gate area.
Type: Float, specified in microns

Antenna Gate Plus Diffusion Rule

The antenna gate plus diffusion rule can be used to represent the protection provided by the diffusion area that is added to the gate area value in the PAR (partial antenna ratio) equation on the given antenna model, as given below, which can be considered as the “additional effective gate-area”.

PAR = {(*cutFactor* * cut) * *diffReducePWL*(diff) - (*minusDiffFactor* * diff)} / {*gatePWL*(gate) + *plusDiffProtect*(diff)}

You can create an antenna gate plus diffusion rule by using the following property definition:

```
PROPERTY LEF58_ANTENNAGATEPLUSDIFF
     "ANTENNAGATEPLUSDIFF {OXIDE1 | OXIDE2 | OXIDE3 | OXIDE4 |OXIDE5 | ...
        | OXIDE32}
        {plusDiffFactor |
        PWL ((diffArea1 plusDiffProtect1) (diffArea1 plusDiffProtect2) ...)}
        ;" ;
```

Where:

All the other keywords are the same as the existing LEF cut layer `ANTENNAGATEPLUSDIFF` syntax.

gatePWL(gate) Indicates the gate area as defined in the PWL table.

plusDiffProtect(diff)

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Indicates the diffusion area that is added to the gate area value in the PAR (partial antenna ratio) equation.

If *plusDiffFactor* is specified, then:

plusDiffProtect = *plusDiffFactor* * *diffArea*

If the PWL table is defined, then:

plusDiffProtect = interpolated PWL value indexed by *diffArea*.

If PAR is greater than *cutGateRatio* (from ANTENNAAREARATIO), and no diffusion is connected then there will be a violation.

If PAR is greater than *cutGateDiffRatio* (from ANTENNADIFFAREARATIO), then there will be a violation. This is checked even if the diff_area = 0 (no diffusion is connected).

Antenna Gate Plus Diffusion Rule Examples

The following example shows a rule for layer via1:

cut_area <= 100 * (funcGate(gate_area) + funcDiff(diff_area))

funcGate = 1 for gate_area < 1, and equals 2 for gate_area >= 1

funcDiff = 10 for diff_area < 2, and equals 50 for diff_area >= 2.

Then,

100 >= cut_area / (funcGate(gate) + funcDiff(diff))

This fits the PAR formula above using *gatePWL*(gate) and *plusDiffProtect*(diff) values.

Then the LEF file will have the following:

```
#Max-ratio = 100 for all diff-area values because the diff-protect value is
#accounted for inside the PAR equation. Note, there is no need for ANTENNAAREARATIO
#which is only checked for cut with no diff connected, while ANTENNADIFFAREARATIO
#is checked for diff and no diff connected.
ANTENNAMODEL OXIDE1 ;
ANTENNADIFFAREARATIO  ( 0 100 ) ( 10000.0 100 ) ;

#This models funcGate(gate_area) above: 0 to 0.99 is 1, above 1.0 is 2
```


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```
PROPERTY LEF58_ANTENNAGATEPWL OXIDE1
    "ANTENNAGATEPWL ( ( 0  1 ) ( 0.99 1 ) ( 1.0 2 ) ( 10000.0 2 ) ) ;" ;

#This models funcDiff(diff_area) above: 0 to 1.99 is 10, above 1.0 is 50
PROPERTY LEF58_ ANTENNAGATEPLUSDIFF
    "ANTENNAGATEPLUSDIFF OXIDE1
       (( 0  10 ) ( 1.99 10 ) ( 2.0 50 ) ( 10000.0 50 )) ;" ;
```

Array Spacing Rule

You can use array spacing rules to require extra space between cut arrays, and between each cut array inside one large via. This rule only applies to large vias with many cuts; it does not apply to cuts for smaller vias.

When a cut layer has multiple cut classes defined, at the most one `ARRAYSPACING` property statement can be specified per cut class.

You can create an array spacing rule by using the following property definition:

```
PROPERTY LEF58_ARRAYSPACING
     "ARRAYSPACING [CUTCLASS className] [PARALLELOVERLAP]
           [LONGARRAY] [WIDTH viaWidth]
           [WITHIN within ARRAYWIDTH arrayWidth] CUTSPACING cutSpacing
           {ARRAYCUTS arrayCuts SPACING arraySpacing} ...
           ;..." ;
```

All other keywords are the same as the existing LEF cut layer `ARRAYSPACING` syntax.

Where:

`CUTCLASS` *`className`*

Defines the array spacing rule for a specific cut class (*`className`*). If `CUTCLASS` is defined in cut layer, then `CUTCLASS` must be specified in the `ARRAYSPACING` statement. Specify individual rules with the `CUTCLASS` keyword for each cut class, if needed.

`PARALLELOVERLAP`

Indicates that the array spacing rule applies only when there is a parallel edge overlap greater than `0`.

`WITHIN` *`within`* `ARRAYWIDTH` *`arrayWidth`*

Indicates a specific way to check cut spacing of a via array. With `WITHIN`, only one `ARRAYCUTS` statement should be defined, and *`cutSpacing`* and *`arraySpacing`* should have the same value. Any same-net or different-net cuts within *`within`* would be considered as a via array. Then, a rectilinear bounding box of all of the cuts of the via

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array is formed. Any cuts completely inside a portion that has width less than or equal to *arrayWidth* is removed. The remaining cuts must have spacing of *cutSpacing* (or *arraySpacing*).
Type: Float, specified in microns

Array Spacing Rule Examples

- The following array spacing rule indicates that any `AY_array` via with a metal width greater than or equal to `1.0` µm should use cut spacing of `0.10` µm between cuts inside 3x3 cut arrays. The cut arrays should be at a distance greater than `0.30` µm from other cut arrays with a parallel edge overlap greater than `0`.

```
PROPERTY LEF58_ARRAYSPACING
    "ARRAYSPACING CUTCLASS AY_array PARALLELOVERLAP WIDTH 1.0 CUTSPACING 0.10
    ARRAYCUTS 3 SPACING 0.30 ;" ;
```

- The next example illustrates the following array spacing rule:

```
PROPERTY LEF58_ARRAYSPACING  "
     ARRAYSPACING WITHIN 0.2 ARRAYWIDTH 0.06
          CUTSPACING 0.05
          ARRAYCUTS 3 SPACING 0.05 ; " ;
```


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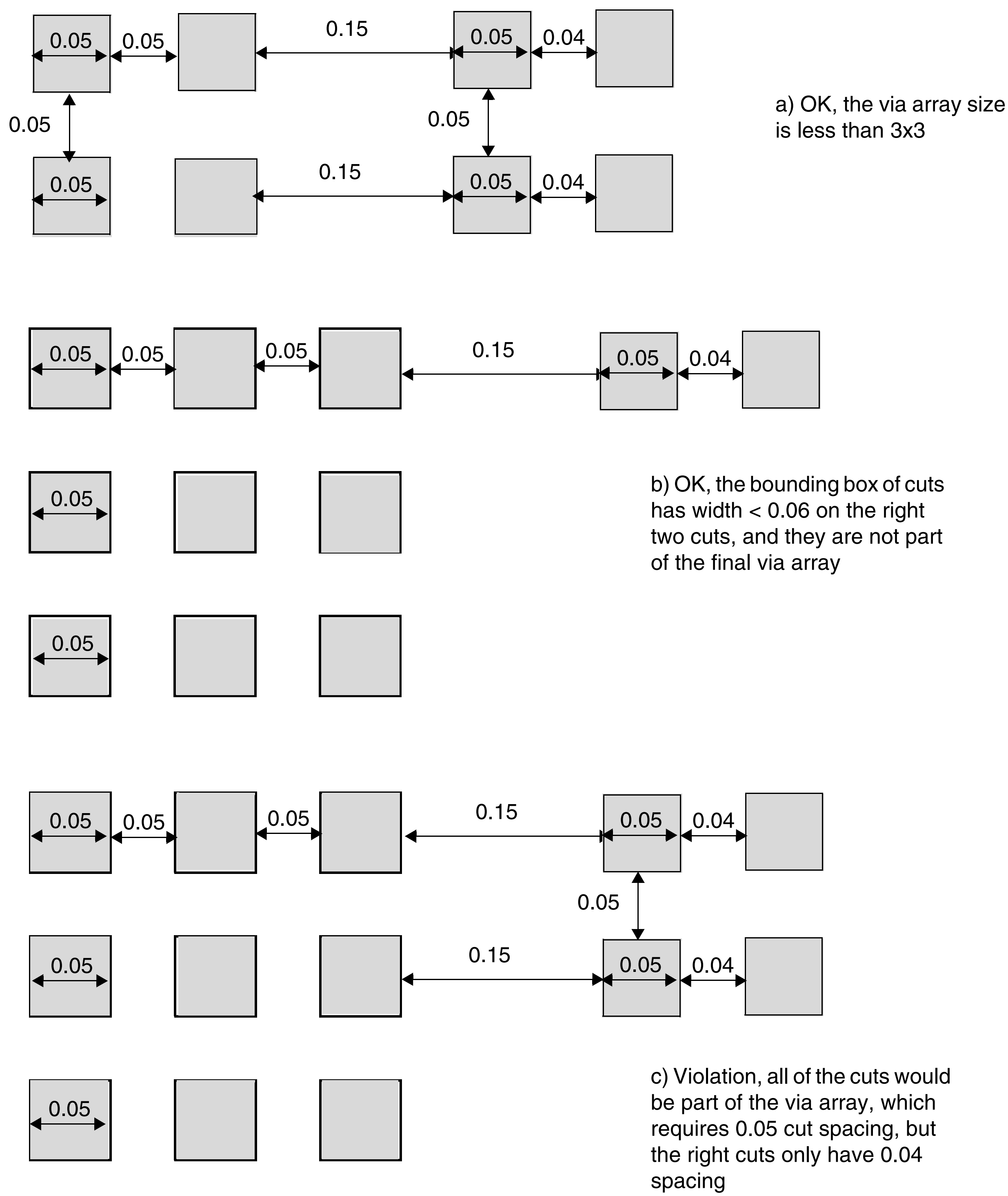
Figure 2-23 Illustration of Array Spacing Rule



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Backside Rule

A backside rule can be used to specify that cut layer is used on the underside of the die.

You can create a backside rule by using the following property definition:

```
PROPERTY LEF58_BACKSIDE
     “BACKSIDE ;” ;
```

Where:

`BACKSIDE`

Indicates that the cut layer is a backside cut layer. Only a regular cut layer or a passivation cut layer can be a backside layer; a TSV cut layer cannot be a backside layer.

Cut Class Rule

Cut class rules can be used to define the cut classes to which different types of vias can belong.

You can create a cut class rule by using the following property definition:

```
PROPERTY LEF58_CUTCLASS
     "CUTCLASS className WIDTH viaWidth [LENGTH viaLength] [CUTS numCut]
           [ORIENT {HORIZONTAL | VERTICAL}]
           ; " ;
```

Where：

`CUTCLASS` *`className`*

Specifies the name of the cut class. This name is used in later references to the cut class.

`CUTS` *`numCut`*

Defines the equivalent number of cuts of the cut class for calculation of resistance value. Resistance value for this cut class vias would be (*Resistance of the cut layer*)/*numCut*.
Default: 1
Type: Integer

Figure 2-24 illustrates three via sizes. Via `Vx1` contains one cut, via `Vx2` contains 2 cuts, and via `Vx4` contains four cuts. Using the above equation, Figure 2-24 shows how many cuts of each cut type are required to meet the listed minimum cut rule.

The resistance value is determined using the following equation:

cut layer resistance / numCut

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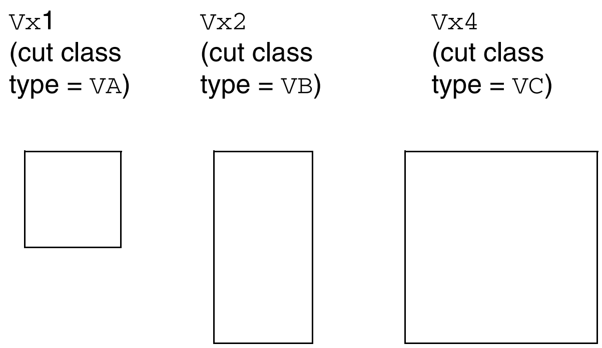
Figure 2-24 Illustration of Via Sizes



`LENGTH` *`viaLength`*

Specifies the cut length for this cut class. Any vias with cut lengths of *`viaLength`* belong to this class. `LENGTH` is only used for rectangular rather than square cuts, and *`viaLength`* must be greater than *`viaWidth`*.
Type: Float, specified in microns

`ORIENT {HORIZONTAL | VERTICAL}`

Specifies that the given horizontal or vertical direction is the only legal orientation of a cut via in this cut class. This keyword should be specified only for a rectangular cut class.

`WIDTH` *`viaWidth`*

Specifies the cut width for this cut class. Any vias with cut widths of *`viaWidth`* belong to this class.
Type: Float, specified in microns

Cut Class Rule Examples

- The following cut class rule indicates that any cut vias with a square dimension of `0.15` µm belong to cut class `VA`:

```
PROPERTY LEF58_CUTCLASS "CUTCLASS VA WIDTH 0.15 ;" ;
```

- The following cut class rule indicates that any cut vias with a rectangular dimension of `0.15` µm and `0.35` µm belong to cut class `VB`. This cut class uses `2` cuts; therefore, for a minimum cut rule requirement of two cuts, one via of this cut class is required to meet the rule. If the resistance value of a cut layer is 10 ohms, the resistance value of vias of this cut class is $1/2 \times 10 = 5$ ohms.

```
PROPERTY LEF58_CUTCLASS "CUTCLASS VB WIDTH 0.15 LENGTH 0.35 CUTS 2 ;" ;
```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- The following cut class rule indicates that any cut vias with a square dimension of `0.20` µm belong to cut class `VC`. This cut class uses `4` cuts; therefore, for a minimum cut rule requirement of `4` cuts, one via of this cut class is required to meet the rule.

  ```
  PROPERTY LEF58_CUTCLASS "CUTCLASS VC WIDTH 0.20 CUTS 4 ;" ;
  ```

Cut On Center Line Rule

You can create a cut on center line rule to specify that the center of a via cut must be placed on the center lines of the wires on both metal layers if the wire width is less than the specified width.

You can create a cut on center line rule by using the following property definition:

```
PROPERTY LEF58_CUTONCENTERLINE
       "CUTONCENTERLINE width CUTCLASS className [ABOVE | BELOW]
          [EXTRACUT cutWithin WIDTH exactWidth]
          ;..." ;
```

Where:

`ABOVE	BELOW`	Specifies that the cut on center line rule will be triggered based on the above or below metal wire width alone, if `ABOVE` or `BELOW` is specified.
`CUTCLASS` *`className`*	Defines the cut on line center rule for a specific cut class, *`className`*. This `CUTONLINECENTER` keyword can only be defined on a cut layer with cut class definition.
`CUTONCENTERLINE` *`width`*	
	Specifies that the center of a via cut must be placed on the centerlines of the wires on both metal layers if the wire width is less than *`width`*. *Type:* Float, specified in microns
`EXTRACUT` *`cutWithin`* `WIDTH` *`exactWidth`*	
	Specifies that for perfectly aligned via cuts within *`cutWithin`* in a wire with width equal to *`exactWidth`*, having the center of the bounding box of the cuts on the center lines of the wire can also meet the rule. *Type*: Float, specified in microns

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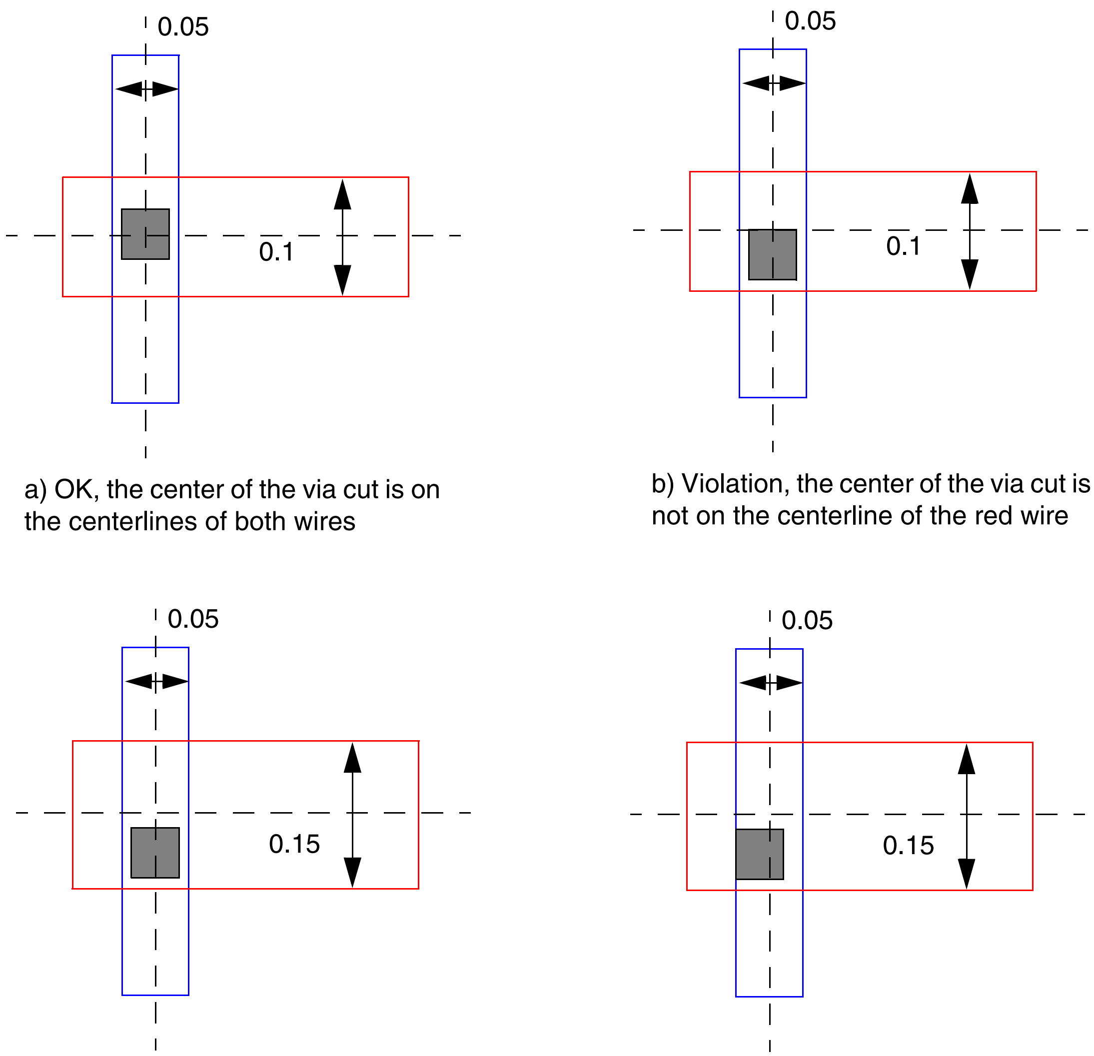
Cut On Center Line Rule Examples

- The following cut on center line rule indicates that the center of a `VA` via cut must be on the centerlines of the wires on both metal layers with width less than 0.13 µm:

```
PROPERTY LEF58_CUTONCENTERLINE
    "CUTONCENTERLINE 0.13 CUTCLASS VA ;" ;
```

Figure 2-25 Illustration of the Cut on Center Line Rule



a) OK, the center of the via cut is on the centerlines of both wires

b) Violation, the center of the via cut is not on the centerline of the red wire

c) OK, the width of the red wire is 0.15 (>= 0.13), the centerline rule does not apply to that wire

d) Violation, the centerline rule still needs to be applied on the blue wire

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Directional Spacing Rule

You can create a directional spacing rule by using the following property definition:

```
PROPERTY LEF58_DIRECTIONALSPACING
     "DIRECTIONALSPACING cutSpacing {HORIZONTAL | VERTICAL}
          PRL prl
          CUTCLASS className1 TO className2
          [PARALLEL parLength WITHIN parWithin
               [EXCLUDECUTEDGE]
               [CUTS numCuts]
               [VIAGROUP groupName]]
          ; " ;
```

Where:

`CUTS` *`numCuts`*

Specifies that the directional cut spacing rule applies only if the number of neighboring cuts of the cut class *`className2`* in a search window is greater than or equal to *`numCuts`*.
Type: Integer

`DIRECTIONALSPACING` *`cutSpacing`* `{HORIZONTAL | VERTICAL}`
`PRL` *`prl`*
`CUTCLASS` *`className1`* `TO` *`className2`*

Specifies a horizontal or vertical spacing of *`cutSpacing`* between two cuts, one belonging to *`className1`* and the other belonging to *`className2`*, and the cuts have a parallel run length greater than *`prl`* vertically when `HORIZONTAL` is specified and horizontally when `VERTICAL` is specified.
Type: Float, specified in microns

`EXCLUDECUTEDGE`

Specifies that there are two separate neighboring cut search windows for the cuts of the cut class *`className2`* formed by *`parLength`* and *`parWithin`* by excluding the cut edge of a via of the cut class *`className1`*. This means that the windows are formed from the cut corners. Each of the windows would be checked for a neighboring cut individually, and the rule will be triggered if any one of the windows fulfill the neighboring cut condition.
Type: Float, specified in microns

`PARALLEL` *`parLength`* `WITHIN` *`parWithin`*

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Specifies that the directional cut spacing rule applies only if a cut edge in the orthogonal direction of the spacing of a via of cut class *className1* has a neighbor cut of cut class *className2* within *parWithin* having parallel run length greater than *parLength* on the same side.

Type: Float, specified in microns

`VIAGROUP` *groupName*

Specifies that the directional cut spacing rule applies only if the neighbor via of cut class *className2* belongs to the via group *groupName*. If the max rectangle formed by the cut corners of two vias in the via group *groupName* overlaps with the window formed by *parLength* and *parWithin*, it is also considered as a neighbor cut.

Type: Float, specified in microns

Directional Spacing Rule Examples

- The following diagram illustrates the directional spacing rules given below:

```
PROPERTY LEF58_VIAGROUP "
    VIAGROUP VGA CUTS 3 CUTCLASS VA
        PRLTWOSIDES -0.01 -0.03 -0.01 -0.03 WITHIN 0.02 0.04 ; " ;
PROPERTY LEF58_DIRECTIONALSPACING  "
    DIRECTIONALSPACING 0.07 VERTICAL PRL -0.02
        CUTCLASS VB TO VA PARALLEL -0.04 WITHIN 0.05 ;
    DIRECTIONALSPACING 0.08 VERTICAL PRL -0.02
        CUTCLASS VB TO VA
        PARALLEL -0.04 WITHIN 0.05 VIAGROUP VGA ; " ;
```


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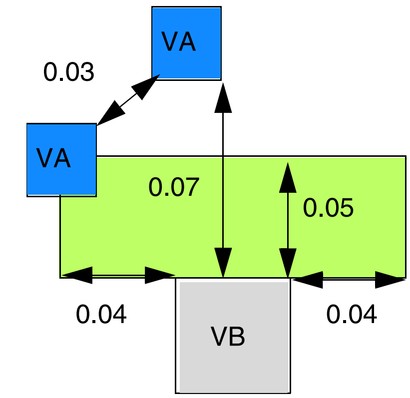
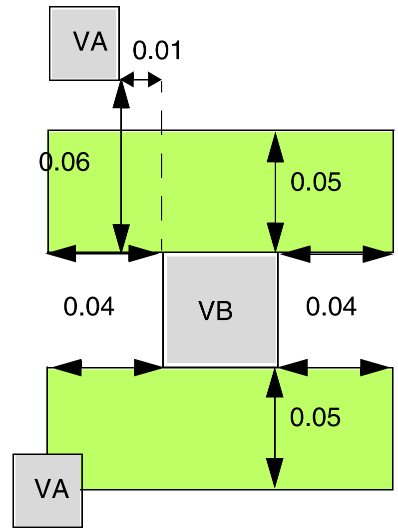
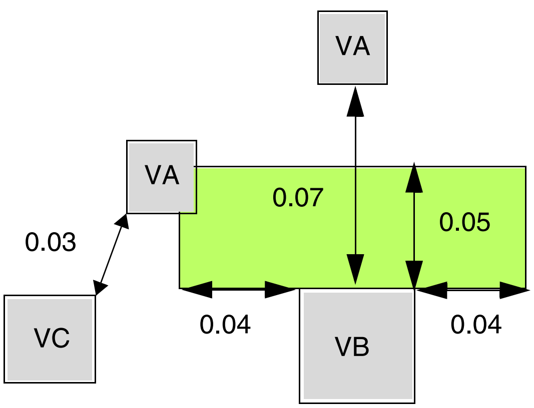
Figure 2-26 Illustration of the Directional Spacing Rule



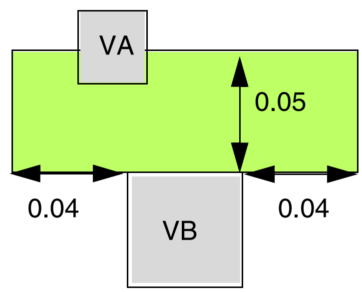
a) Violation. The first statement is met, but the second statement fails, assuming that the two blue VAs belong to via group VGA.



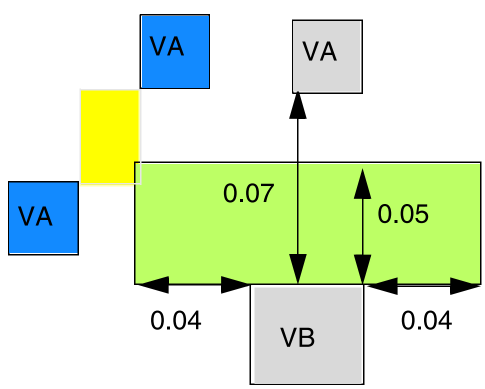
b) OK. The neighbor cut and the cut on which the spacing is applied must be on the same side.



c) OK. The bottom VA does not belong to the via group VGA, and the first statement is met.



d) Violation. If the neighbor cut of VA overlaps with VB (within 0.02 in `DIRECTIONWITHIN`), cut spacing of 0.07 is applied and fails.



e) Violation, assuming that the two blue VA belong to the via group VGA. Although the individual blue cuts do not overlap with the green window, the max yellow rectangle formed by their corners overlap with the green window to be counted as a neighbor cut.

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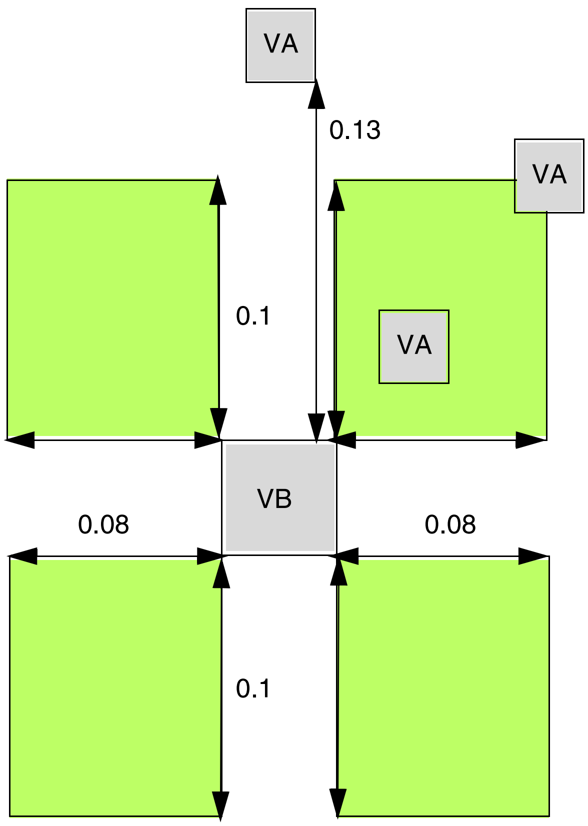
- The following diagram illustrates the directional spacing rule with `EXCLUDECUTEDGE` given below:

```
PROPERTY LEF58_DIRECTIONALSPACING  "
    DIRECTIONALSPACING 0.15 VERTICAL PRL 0
        CUTCLASS VB TO VA PARALLEL -0.08 WITHIN 0.01 ;
        EXCLUDECUTEDGE CUTS 2 ; " ;
```

Figure 2-27 Illustration of the Directional Spacing Rule with EXCLUDECUTEDGE



a) Violation. `EXCLUDECUTEDGE` means that the four green search windows are formed from the cut corners and each of the windows would be checked for neighboring cuts of `VA` individually. As the upper right window has two `VA` cuts (`>= 2` in `CUTS`) and there is another `VA` cut on top of `VB` with spacing of `0.13` (`< 0.15`), it is a violation.

Enclosure Rule

An enclosure rule can be used to prohibit via cuts from sharing the same wire edge.

You can create an enclosure rule by using the following property definition:

```
PROPERTY LEF58_ENCLOSURE
     "ENCLOSURE [CUTCLASS className][ABOVE | BELOW]
     [MINCORNER]
        {EOL eolWidth [HORIZONTAL | VERTICAL]
```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```
          [MINLENGTH minLength] [EOLONLY]
          [SHORTEDGEONEOL] eolOverhang otherOverhang
          [SIDESPACING spacing EXTENSION backwardExt forwardExt
          |ENDSPACING spacing EXTENSION extension
          ]
   |{overhang1 overhang2
     |[OFFCENTERLINE] END overhang1 SIDE overhang2
     | HORIZONTAL overhang1 VERTICAL overhang2}
     [JOGLENGTHONLY length [INCLUDELSHAPE] ]
     [HOLLOW {HORIZONTAL|VERTICAL} length]
     [WIDTH minWidth
        [INCLUDEABUTTED]
        [EXCEPTEXTRACUT cutWithin [PRL | NOSHAREDEDGE | EXACTPRL prl]]
     |LENGTH minLength
     |EXTRACUT [EXTRAONLY [PRL prl]]
     |REDUNDANTCUT cutWithin
     |PARALLEL parLength [parLength2] WITHIN parWithin [parWithin2]
          [BELOWENCLOSURE belowEnclosure
               [ALLSIDES enclosure1 enclosure2]
          |ABOVEENCLOSURE aboveEnclosure]
     |CONCAVECORNERS numCorner
     |OTHERWITHINWIDTH width WITHIN within
     [OTHERSIDE otherEnclosure]]
     }
     ;..." ;
```

Where:

All other keywords are the same as the existing LEF cut layer `ENCLOSURE` syntax.

`ABOVEENCLOSURE` *aboveEnclosure*

Specifies that the enclosure rule applies only if the enclosure on the above metal layer is less than *aboveEnclosure* on the entire wire width or the edge containing the cut on either side perpendicular to the side having neighbors, or the wire direction containing the cut on the below metal layer. This means that `BELOW` must be defined along with it.
Type: Float, specified in microns

`ALLSIDES` *enclosure1 enclosure2*

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Specifies that the rule is exempted if the below metal enclosure is greater than or equal to the minimum of the specified values, *enclosure1* and *enclosure2*, on all four sides including corners and greater than or equal to the maximum of *enclosure1* and *enclosure2* on any two opposite sides of the cut.

This exemption works independently from *belowEnclosure*. In other words, if the conditions on ALLSIDES are met or enclosure is greater than or equal to *belowEnclosure* on two opposite sides in the direction perpendicular to the cut side having neighbor, the rule is exempted.
Type: Float, specified in microns

BELOWENCLOSURE *belowEnclosure*

Specifies that the enclosure rule only applies if the enclosure on the below metal layer is less than *belowEnclosure* on the entire wire width or the edge containing the cut on either side perpendicular to the side having neighbors, or the wire direction containing the cut on the above metal layer. This means that the ABOVE keyword must be defined along with it.
Type: Float, specified in microns

CONCAVECORNERS *numCorner*

Specifies a search window by extending *overhang1* on four sides of the cut, *overhang1* must be equal to *overhang2*. If the search window finds the number of concave corners of the metal enclosing the cut to be greater than the *numCorner*, then it is a violation. If the metal corner collides with a corner of the search window, then it does not count. If the metal corner aligns with a side edge of the search window, then it will count.
Type: Integer

CUTCLASS *className*

Defines the enclosure rule for a specific cut class (*className*). If CUTCLASS is defined in cut layer, then CUTCLASS must be specified in ENCLOSURE.

Specify individual rules with the CUTCLASS keyword for each cut class, if needed.

END *overhang1* SIDE *overhang2*

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Specifies that for rectangular cut vias, *overhang1* applies to the end edges, and *overhang2* applies to the side edges. You must use this syntax only with cut class having rectangular cut vias.

ENDSPACING *spacing* EXTENSION *extension*

Specifies the enclosure rule only applies if there is no different-net neighbors on the EOL edge with *spacing* away and extension of *extension*.
Type: Float, specified in microns

EOL *eolWidth* [MINLENGTH *minLength*] [EOLONLY]
eolOverhang otherOverhang

Specifies the enclosure values for a cut near an end-of-line edge with length less than *eolWidth*. The first overhang, *eolOverhang*, must be applied to the EOL edge and it's opposite side while the second overhang, *otherOverhang*, must be applied to the other two sides. If EOLONLY is also specified, it should be specified for all ENCLOSURE EOL statements, and for an EOL edge, only one of the ENCLOSURE EOL statements should be observed while other ENCLOSURE statements without EOL keyword are ignored. Otherwise, fulfilling one of the ENCLOSURE with or without EOL keyword is allowed.
Type: Float, specified in microns

MINLENGTH specifies that an edge is EOL only if the end-of-line length is greater than or equal to *minLength* along both the sides. In the other words, if the end-of-line length is less than *minLength* along any one side, the rule does not apply.
Type: Float, specified in microns

EXCEPTEXTRACUT *cutWithin* [PRL | NOSHAREDEDGE | EXACTPRL *prl*]

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Indicates that if there is another via cut having same metal shapes on both metal layers less than or equal to *cutWithin* distance away, then the ENCLOSURE with WIDTH rule is ignored, and the ENCLOSURE rules for minimum width wires (that is, no WIDTH keyword) are applied to the via cuts instead. If the NOSHAREDEDGE keyword is specified, the via cuts cannot share the same failing wire edge. (See Figure 2-32 on page 302)

If the PRL keyword is used, the exemption will only be applied if there are neighbor cuts with common parallel run length greater than 0 on the opposite edges for all of the failing edges of a cut.

If the EXACTPRL keyword is used, the exemption will be applied only if the cuts have parallel run length exactly equal to *prl*.

If you have more than one ENCLOSURE statement for a given WIDTH, only one of the ENCLOSURE statements for that WIDTH needs to be met.
Type: Float, specified in microns (for all values)

EXTRACUT

Indicates that the enclosure rule only applies when there are two or more cuts having same metal shapes on both metal layers. If you have multiple ENCLOSURE statements (some with EXTRACUT) defined, only one of the rules need to be met.

EXTRAONLY

Specifies that vias with two or more cuts having the same metal shapes on both metal layers must follow this given enclosure statement, but not other ENCLOSURE without WIDTH statements. The ENCLOSURE WIDTH keyword will still be used on the corresponding wire width.

HOLLOW {HORIZONTAL|VERTICAL} *length*

Specifies that part of a via does not require any metal/ enclosure. Take the center point of a cut and extend *length* horizontally or vertically, based on whether HORIZONTAL or VERTICAL is specified, and that portion does not require any metal/enclosure. See Figure 2-46 on page 317
Type: Float, specified in microns

HORIZONTAL | VERTICAL

Specifies that the rule applies only if the EOL edge is in the given direction.

HORIZONTAL *overhang1* VERTICAL *overhang2*

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Specifies that *overhang1* must be applied on the left/right opposite sides horizontally, and *overhang2* must be applied to bottom/top opposite sides vertically.
Type: Float, specified in microns

`INCLUDEABUTTED`
Specifies that if a cut is abutted to the wide wire with width greater than or equal to *minWidth*, the corresponding overhang values will still be applied on the cut.

`JOGLENGTHONLY` *length* `[INCLUDELSHAPE]`

Specifies that the enclosure rule applies only if a cut overlaps with a wrong way jog with minimum wrong way width and maximum edge length less than *length* sandwiched by two right way wires with projected `PRL` less than `0` in the wrong way direction in a 'Z' shape. If `INCLUDELSHAPE` is specified, the enclosure rule also applies if a cut overlaps with a wrong way jog with minimum wrong way width and length less than *length* connected to a right way wire in an 'L' shape.
Type: Float, specified in microns

`MINCORNER`
Specifies that the enclosure in the corners only requires to cover the minimum enclosure value instead of MAXXY style of the two enclosure values.

`OFFCENTERLINE`
Specifies that the enclosure rule applies only for a via cut not colliding with the center line of the metal wires with width exactly equal to *width*. This keyword must be used along with `CUTCLASS` for rectangular cut vias and `WIDTH` for certain width.

`OTHERSIDE` *otherEnclosure*

Specifies the enclosure on the opposite side of a cut edge, which faces the wire fulfilling the `OTHERWITHINWIDTH` condition, to be *otherEnclosure*. This means that *otherEnclosure* is only checked for that opposite cut edge, while the remaining three edges must fulfill *overhang1* and *overhang2*. See Figure 2-48 on page 318
Type: Float, specified in microns

`OTHERWITHINWIDTH` *width* `WITHIN` *within*

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Specifies that the enclosure rule applies only if there is a neighboring wire within *`within`* of the cut and completely covering a cut edge in the non-preferred routing layer direction having a width greater than *`width`* on the other layer on which the enclosure applies and that has the `RECTONLY` construct. This means that `ABOVE` or `BELOW` must be specified. In addition, this enclosure rule is not applied on wide wires, which should be subjected to `ENCLOSUREWIDTH` statements as usual.
Type: Float, specified in microns

`PARALLEL` *`parLength`* `[`*`parLength2`*`] WITHIN` *`parWithin`* `[`*`parWithin2`*`]`

Specifies that the enclosure values must be fulfilled, but not other `ENCLOSURE` statements if a cut has a neighbor wire within (less than) *`parWithin`* having common parallel run length to the cut greater than or equal to *`parLength`* on one and only one side. The neighbor wire should have an edge in the preferred routing direction facing the cut to trigger the rule. In other words, that wire edge should not have a parallel run length greater than or equal to `0` with the orthogonal cut edge. If *`parLength`* is a negative value, the two edges will be checked in `WITHIN` style. In addition, the enclosure on a wide wire in `ENCLOSURE WIDTH` is usually larger, but this enclosure with a neighbor could be even larger. Hence, this enclosure rule should always be checked.

If a second set of length and within variables, *`parLength2`* and *`parWithin2`*, is specified, then they must be defined together. Here *`parLength2`* must be smaller than *`parLength`* and *`parWithin2`* must be larger than *`parWithin`*. Then, it will also be a violation if there is a neighbor that is greater than or equal to *`parWithin`* and less than *`parWithin2`* having a common parallel run length to the cut less than *`parLength`* and greater than or equal to *`parLength2`* on any one (or both) side(s). In other words, it is only legal if there are neighbors on both the sides based on *`parWithin`* and *`parLength`*, or there is no neighbor on either side based on *`parWithin2`* and *`parLength2`*.
Type: Float, specified in microns

`PRL` *`prl`*

Specifies that the enclosure rule must be applied for vias with two or more cuts if the cuts have parallel run length exactly equal to *`prl`*.
Type: Float, specified in microns

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`REDUNDANTCUT` *`cutWithin`*

Specifies the enclosure on redundant cuts, which share the same metal shapes on both metal layers to a cut within *`cutWithin`* distance that fulfills a `ENCLOSURE` statement without `REDUNDANTCUT` keyword. This includes `ENCLOSURE` statement with `EXTRACUT` keyword. For example, when one of the cuts fulfills the `ENCLOSURE` statement using the `EXTRACUT` keyword, and is outside of a wide object, that triggers a separate `ENCLOSURE` statement with the `WIDTH` keyword, the neighbor cut within *`cutWithin`* distance away can merely fulfill the `ENCLOSURE` statement using the `REDUNDANTCUT` keyword, and it does not necessarily need to fulfill the `ENCLOSURE` statement with the `WIDTH` keyword.

If a separate `ENCLOSURE` statement with the `WIDTH` keyword is used, the cut in a wide object with width greater than or equal to *`minWidth`* should fulfill the corresponding overhang values in order for the redundant cuts to follow the overhang values in the `ENCLOSURE` statement with the `REDUNDANTCUT` keyword. You can specify only one `ENCLOSURE` statement with the `REDUNDANTCUT` keyword.
Type: Float, specified in microns

`SIDESPACING` *`spacing`* `EXTENSION` *`backwardExt forwardExt`*

Specifies the enclosure rule only applies if there is no different-net neighbors on the side (non-EOL) edges with spacing away and extensions from the EOL edge by *`backwardExt`* going backward and *`forwardExt`* going forward.
Type: Float, specified in microns

`SHORTEDGEONEOL`

Specifies that the *`eolOverhang`* value must be applied to the end/short edges of a rectangular cut via while the *`otherOverhang`* value must be applied to the other opposite sides. This keyword must be used along with `CUTCLASS` for rectangular cut vias.

Enclosure Rule Examples

- The following enclosure rule specifies that `VC` vias should have `0.10` µm overhang on all four sides of the routing layers:

```
PROPERTY LEF58_ENCLOSURE
"ENCLOSURE CUTCLASS VC 0.10 0.10 ;" ;
```


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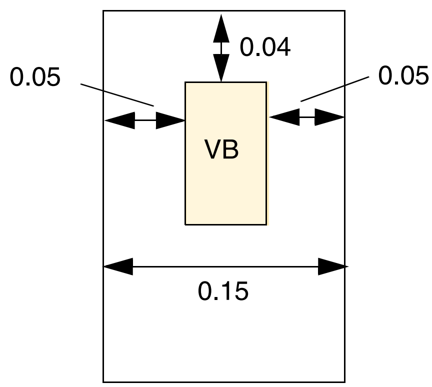
- The following enclosure rule specifies that the end/short edge must have 0.05 µm overhang to a EOL edge:

```
PROPERTY LEF58_ENCLOSURE
"ENCLOSURE CUTCLASS VB EOL 0.2
     SHORTEDGEONEOL 0.05 0.04 ;" ;
```

Figure 2-28 Illustration of Enclosure Rule With SHORTEDGEONEOL



a) Violation, the end/short edge must have 0.05 overhang to a EOL edge

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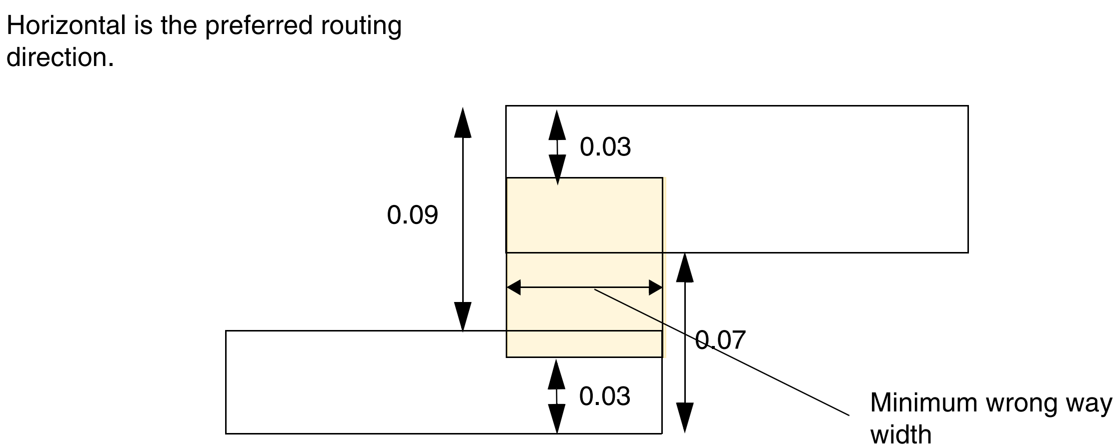
- The following enclosure rule illustrates `JOGLENGTHONLY`:

```
ENCLOSURE 0.00 0.04 ;
PROPERTY LEF58_ENCLOSURE "
   ENCLOSURE 0 0.03 JOGLENGTHONLY 0.1 ; " ;
```

Figure 2-29 Illustration of the Enclosure Rule with JOGLENGTHONLY



a) OK, the cut is inside a wrong way jog such that 0.03 enclosure is sufficient.

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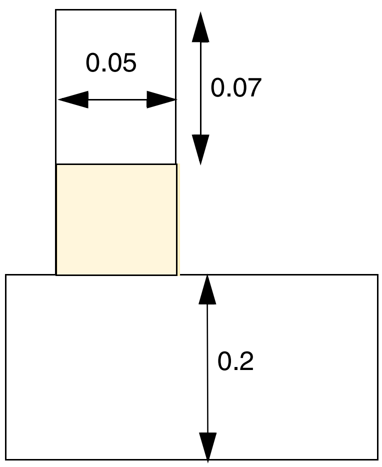
- The following enclosure rule illustrates `INCLUDEABUTTED`:

```
ENCLOSURE 0.00 0.04 ;
PROPERTY LEF58_ENCLOSURE
     "ENCLOSURE 0.05 0.05
         WIDTH 0.2 INCLUDEABUTTED ; " ;
```

Figure 2-30 Illustration of the Enclosure Rule with INCLUDEABUTTED



a) Violation, the cut is abutted to wide wire of 0.2. OK, if INCLUDEABUTTED is omitted.

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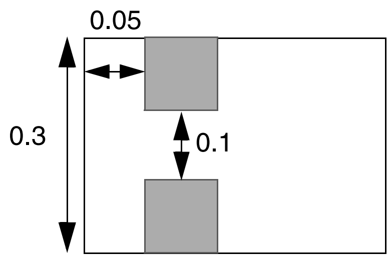
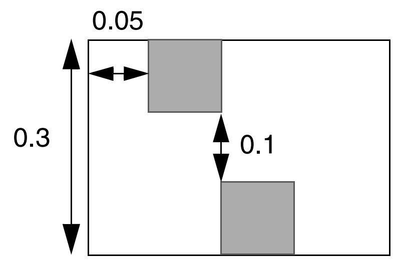
- Figure 2-31 illustrates the following enclosure rules:

```
ENCLOSURE 0.0 0.05 ;             #overhang 0.0 0.05
ENCLOSURE 0.02 0.02 ;           #or, overhang 0.02 0.02
PROPERTY LEF58_ENCLOSURE
    "ENCLOSURE 0.03 0.03 WIDTH 0.3 EXCEPTEXTRACUT 0.2 PRL ;" ;
```

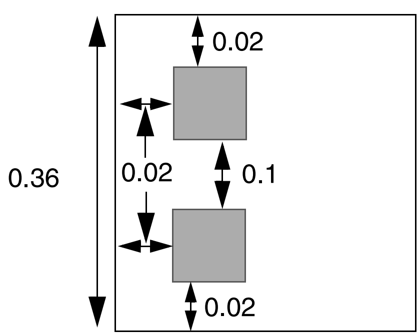
Figure 2-31 Illustrations of the Enclosure Rule With PRL



a) OK. Extra-cut is <= 0.2 away, with PRL > 0, so use min-width rule, both cuts meet the 0.0 0.05 enclosure rule of 0.0 0.05



b) Violation. Extra-cut is <= 0.2 away, but no PRL, so wide-wire enclosure of 0.03 0.03 is required



c) Violation, the left 0.02 failing edges do not have a neighbor cut on the opposite right edges

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- Figure 2-32 illustrates the following enclosure rules:

```
ENCLOSURE 0.0 0.05 ;          #overhang 0.0 0.05
ENCLOSURE 0.01 0.04 ;         #or overhang 0.01 0.04
PROPERTY LEF58_ENCLOSURE
    "ENCLOSURE 0.03 0.03 WIDTH 0.3 EXCEPTEXTRACUT 0.2 ;" ;
```


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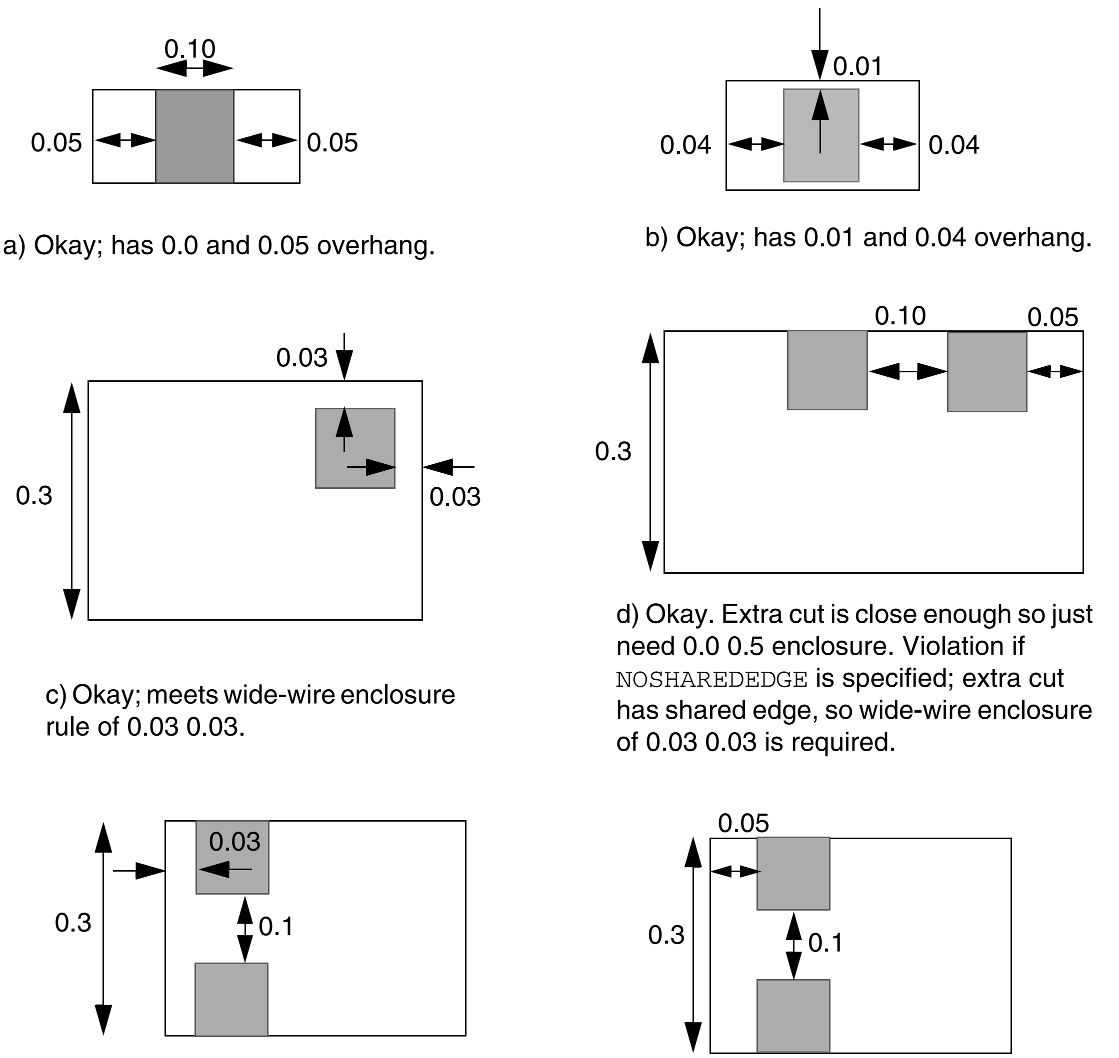
Figure 2-32 Illustration of the Enclosure Rule With ExceptExtraCut and NoSharedEdge



a) Okay; has 0.0 and 0.05 overhang.

b) Okay; has 0.01 and 0.04 overhang.

c) Okay; meets wide-wire enclosure rule of 0.03 0.03.

d) Okay. Extra cut is close enough so just need 0.0 0.5 enclosure. Violation if `NOSHAREDEDGE` is specified; extra cut has shared edge, so wide-wire enclosure of 0.03 0.03 is required.

e) Violation. Extra cut is <= 0.2 away; therefore, use minimum width rule, but cannot meet either 0.0 0.05 or 0.01 0.04 enclosure rules.

f) Okay. Okay for `NOSHAREDEDGE` also. Extra cut is <= 0.2 away, there is no shared edge; so use minimum width rule; both cuts meet the minimum width enclosure rule of 0.0 0.05.

- The following enclosure rule specifies that `VB` rectangular cut vias should have `0.10` µm overhang on the end edges, and no overhang on the side edges on the routing layers. (See Figure 2-33 .)

```
PROPERTY LEF58_ENCLOSURE
    "ENCLOSURE CUTCLASS VB END 0.10 SIDE 0.000 ;" ;
```


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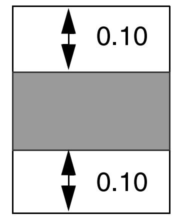
Figure 2-33 Illustration of the Enclosure Rule With CutClass

a) Okay; has 0.0 and 0.10 overhang on proper sides.

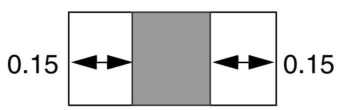


b) Violation; overhang on wrong sides.

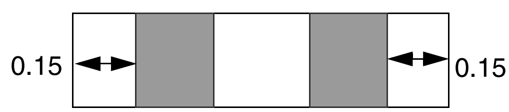
- The following enclosure rule specifies that `VA` vias with two or more same-metal cuts should have `0.15` µm overhang on any two opposite sides of the routing layers. (See Figure 2-34).

```
PROPERTY LEF58_ENCLOSURE
   "ENCLOSURE CUTCLASS VA 0.000 0.20 ;
    ENCLOSURE CUTCLASS VA 0.000 0.15 EXTRACUT;" ;
```

Figure 2-34 Illustration of the Enclosure Rule With CutClass



a) Violation; only one cut.



b) Okay; has 0.15 overhang on two opposite sides with two or more same metal cuts.

- The following enclosure rule indicates that a `VA` cut via must at least have either `0.10` µm, `0.12` µm or `0.0` µm, or `0.20` µm enclosures. For redundant cuts having same metal shapes on both metal layers, one of the `VA` cuts should at least have either `0.10` µm, `0.12` µm or `0.0` µm, or `0.20` µm enclosures, and the rest of the `VA` cuts within `0.15` µm distance from it, should have at least `0.0` µm, `0.05` µm enclosures.

```
PROPERTY LEF58_ENCLOSURE
   "ENCLOSURE CUTCLASS VA 0.10 0.12 ;
    ENCLOSURE CUTCLASS VA 0 0 0.20 ;
    ENCLOSURE CUTCLASS VA 0.0 0.05 REDUNDANTCUT 0.15 ;" ;
```

- Figure 2-35 illustrates `ENCLOSURE` rule with `REDUNDANTCUT`:

```
ENCLOSURE 0.03 0.03 WIDTH 0.3 ;
PROPERTY LEF58_ENCLOSURE
   "ENCLOSURE 0.0 0.02 EXTRACUT ;
    ENCLOSURE 0.0 0.01 REDUNDANTCUT 0.1 ; " ;
```


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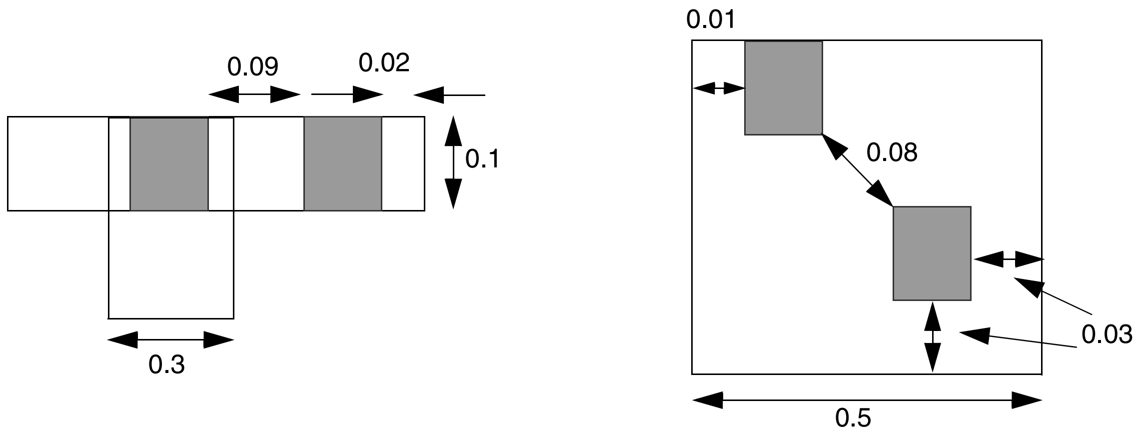
Figure 2-35 Illustrations of the Enclosure Rule With REDUNDANTCUT



a) OK, the right (and left) cut(s) fulfills the EXTRACUT enclosure, which would exempt the left cut from the WIDTH enclosure, and can merely follow and meet the REDUNDANTCUT enclosure

b) OK, the bottom cut fulfills the WIDTH enclosure, which would exempt the top cut from the WIDTH enclosure, and can merely follow and meet the REDUNDANTCUT enclosure

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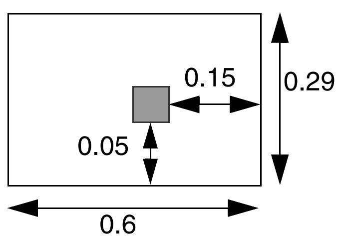
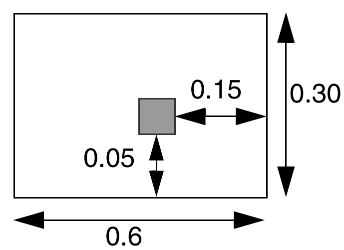
- The following enclosure rule indicates that the overhang of a via cut must be 0.15 µm on an EOL edge with length less than 0.30 µm and it's opposite side, while the overhang must be 0.05 on the other two sides. If no EOL edge is involved, a via cut should have 0.10 µm overhang on all four sides of the routing layers:

```
ENCLOSURE 0.10 0.10 ;
PROPERTY LEF58_ENCLOSURE
    "ENCLOSURE EOL 0.30 0.15 0.05 ; " ;
```

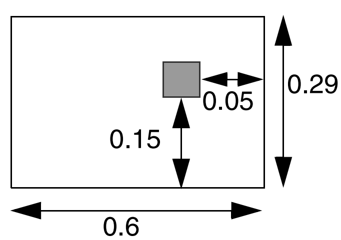
Figure 2-36 Illustration of the Enclosure Rule With Enclosure EOL



a) OK, ENCLOSURE EOL rule must be applied and met



b) Violation, no EOL edge, and ENCLOSURE without EOL must be applied



c) Violation, at least, 0.15 overhang must be on the EOL edge

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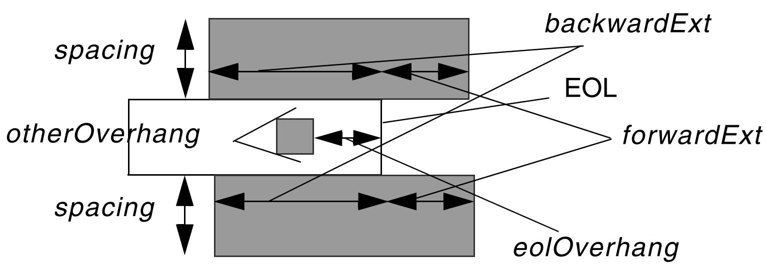
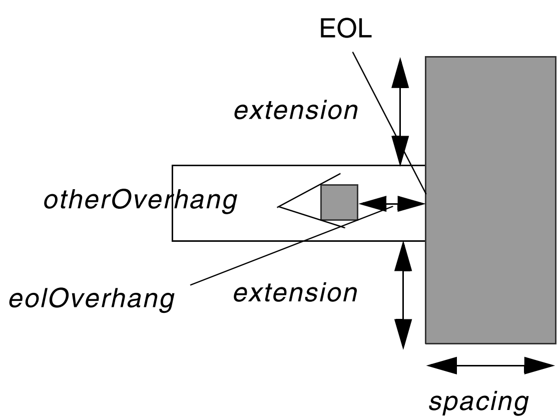
Figure 2-37 Definition of SIDESPACING and ENDSPACING

Definition of SIDESPACING *spacing* EXTENSION *backwardExt forwardExt*



There should have no neighbor wires on the 2 checking regions (in gray) in order to apply the given enclosure.

Definition of ENDSPACING *spacing* EXTENSION *extension*



There should have no neighbor wires on the checking region (in gray) in order to apply the given enclosure.

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- The following enclosure rule indicates that 0.15 µm and 0.05 µm can be the enclosure values of a cut on a EOL edge and the other sides if there is no neighbor wires within 0.04 µm with extension of 0.03 on the side edges or within 0.08 µm with extension of 0.06 on the EOL edge:

```
ENCLOSURE 0.1 0.1 ;
PROPERTY LEF58_ENCLOSURE
    "ENCLOSURE EOL 0.2 0.15 0.05 SIDESPACING 0.04 EXTENSION 0.03 0.03 ;
    ENCLOSURE EOL 0.2 0.15 0.05 ENDSPACING 0.08 EXTENSION 0.06 ; " ;
```


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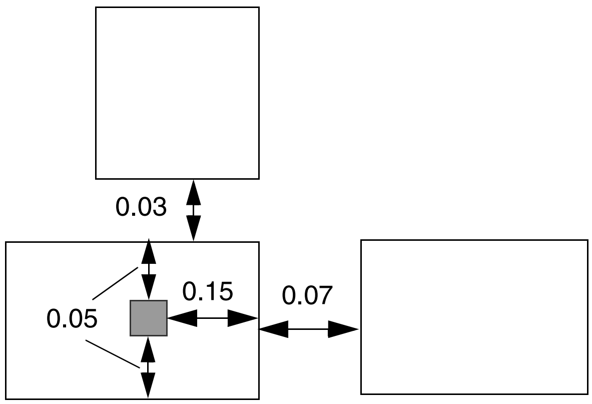
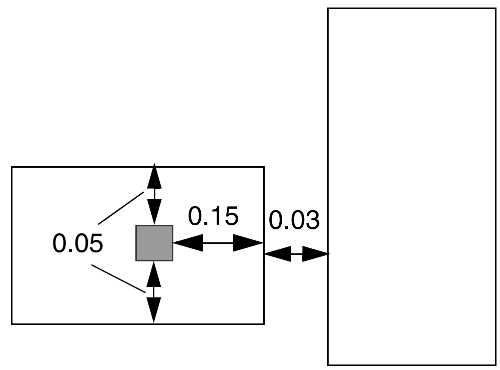
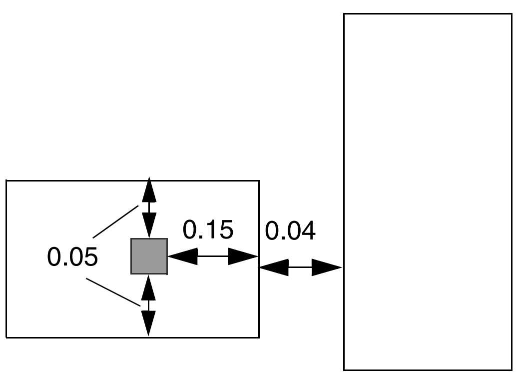
Figure 2-38 Illustration of the Enclosure Rule With SIDESPACING and ENDSPACING



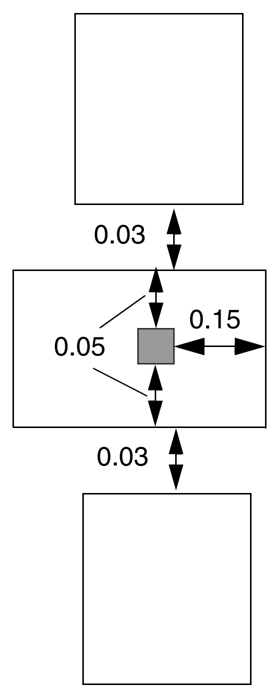
a) Violation, it has neighbor wires on orthogonal edges. 0.10 and 0.10 enclosure should be applied, and fails.



b) Violation, a neighbor wire with 0.03 (< =0.03 extension) in the corner would fail both region checking. 0.10 and 0.10 enclosure should be applied, and fails.



c) OK, the wire is just outside 0.04 from the top/ bottom edges. 0.15 and 0.05 enclosure should be applied, and is met.



d) OK, as long as the wires are not neighbors of orthogonal edges, the 0.15 and 0.05 enclosure can be applied.

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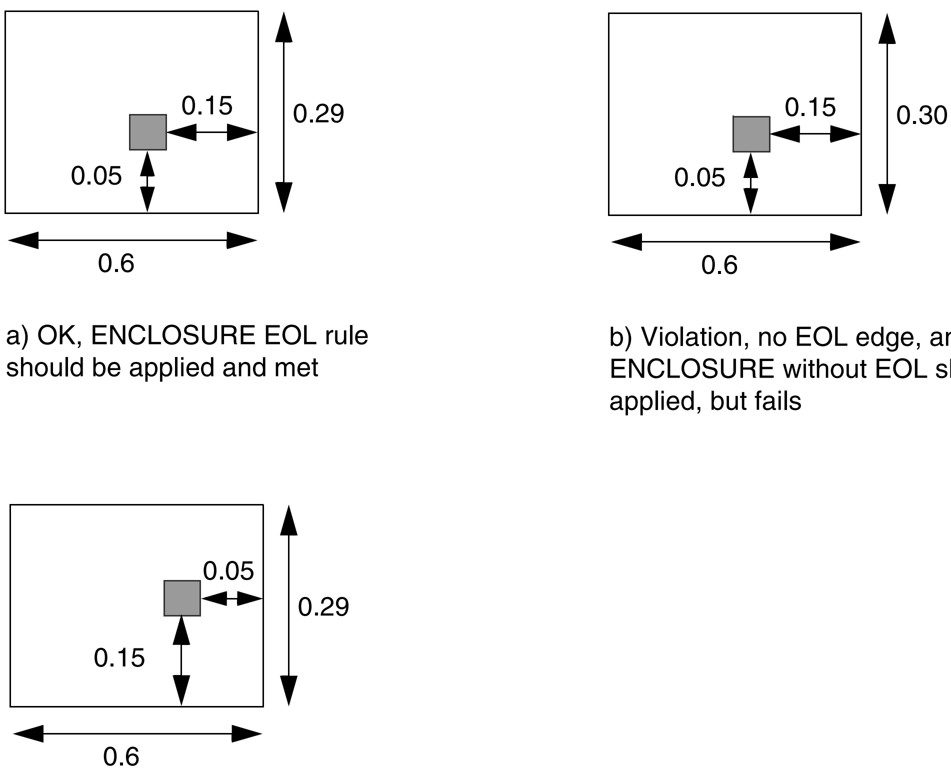
- The following enclosure rule indicates that the overhang of a via cut must be 0.15 µm on an EOL edge with length less than 0.30 µm and the opposite side while the overhang must be 0.05 µm on the other two sides. If no EOL edge is involved, a via cut should have 0.10 µm overhang on all four sides of the routing layers:

```
PROPERTY LEF58_ENCLOSURE
    "ENCLOSURE EOL 0.30 EOLONLY 0.15 0.05 ; " ;
```

Figure 2-39 Illustration of the Enclosure Rule With EOLONLY

```
ENCLOSURE 0.10 0.10 ;
PROPERTY LEF58_ENCLOSURE
    "ENCLOSURE EOL 0.30 EOLONLY 0.15 0.05 ; " ;
```



a) OK, ENCLOSURE EOL rule should be applied and met

b) Violation, no EOL edge, and ENCLOSURE without EOL should be applied, but fails

c) Violation, at least, 0.15 overhang must on the EOL edge

- Figure 2-40 illustrates the following enclosure rules:

```
ENCLOSURE 0.05 0.00 ;
ENCLOSURE 0.06 0.00 WIDTH 0.1 ;
PROPERTY LEF58_ENCLOSURE
```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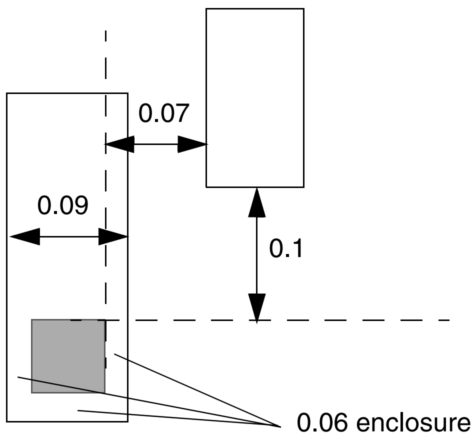
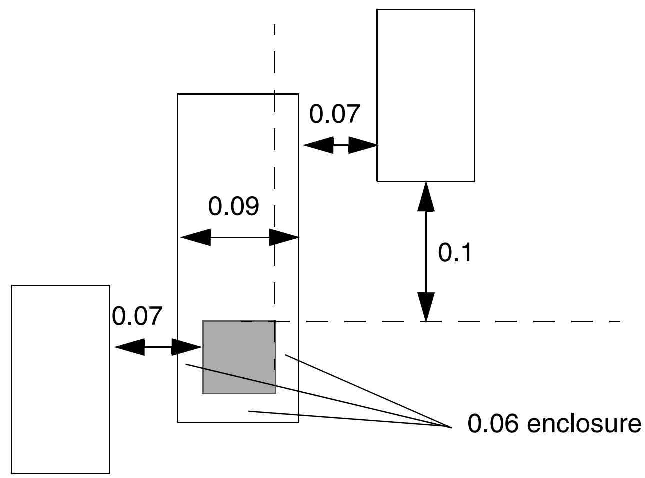
```
"ENCLOSURE 0.09 0.09 PARALLEL -0.11 WITHIN 0.08 ;" ;
```

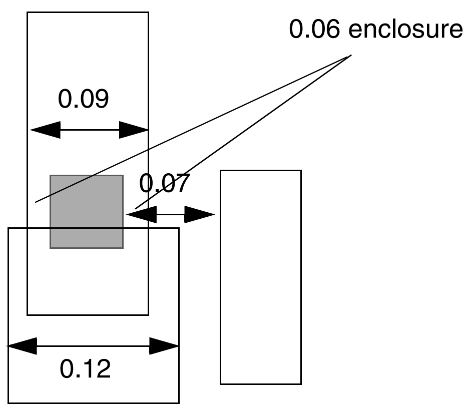
Figure 2-40 Illustration of the Enclosure Rule With PARALLEL and WITHIN



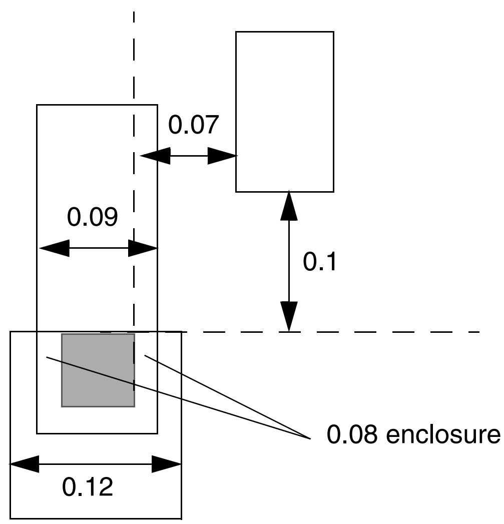
a) Violation, the parallel and within conditions are fulfilled, 0.09 enclosure on all four sides is needed and fails.



b) OK, there are neighbors on both sides such that 0.09 enclosure rule does not apply, and 0.05, 0 enclosure is fulfilled.



c) Violation, the cut is not completely inside the wide wire of 0.12.



d) OK, the cut is completely inside the wide wire of 0.12, which fulfills the corresponding 0.06 0.00 enclosure

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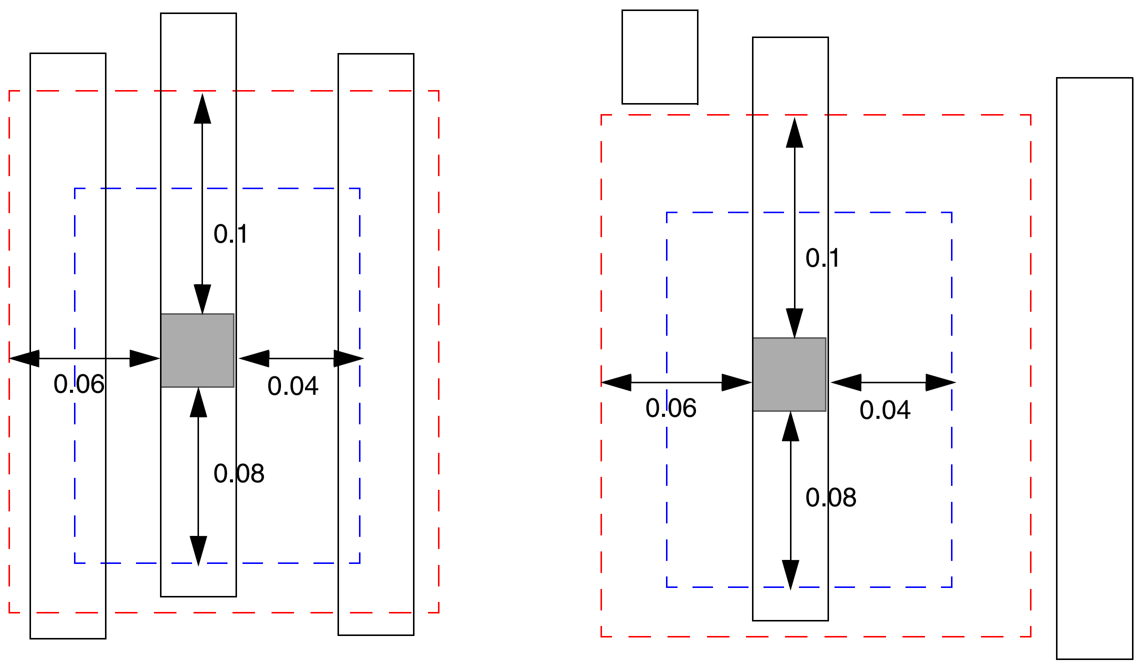
- Figure 2-41 illustrates the following enclosure rules:

```
PROPERTY LEF58_ENCLOSURE
     "ENCLOSURE CUTCLASS VSINGLECUT ABOVE 0.020 0.000 ;
      ENCLOSURE CUTCLASS VSINGLECUT ABOVE 0.015 0.015
      PARALLEL -0.08 -0.1 WITHIN 0.04  0.06 ; " ;
```

Figure 2-41 Illustration of the Enclosure Rule With PARALLEL and WITHIN



a) OK, having neighbors on both sides of the smaller search window is fine.

b) OK, having no neighbor in either sides of the larger search window is fine.

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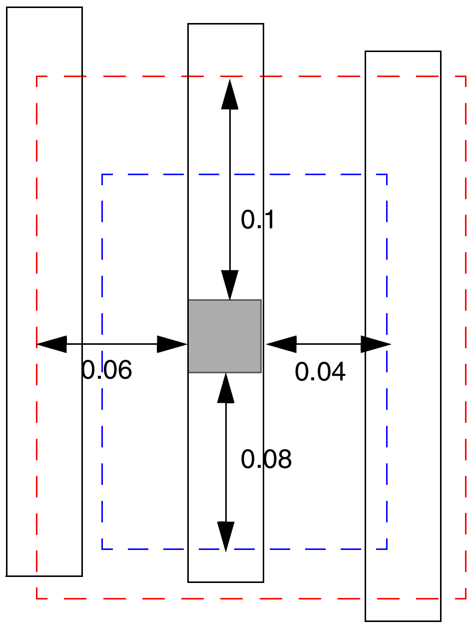
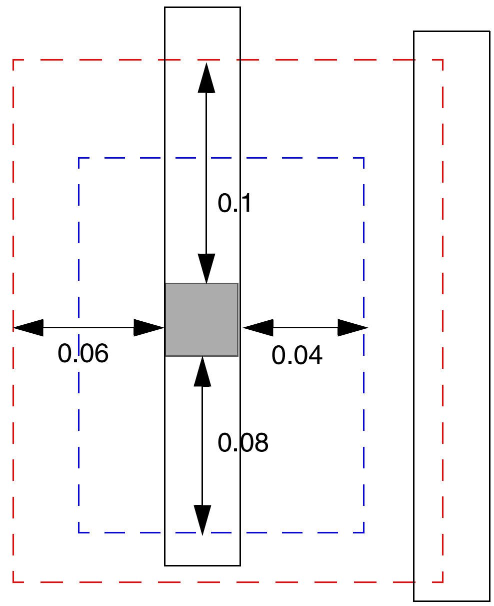
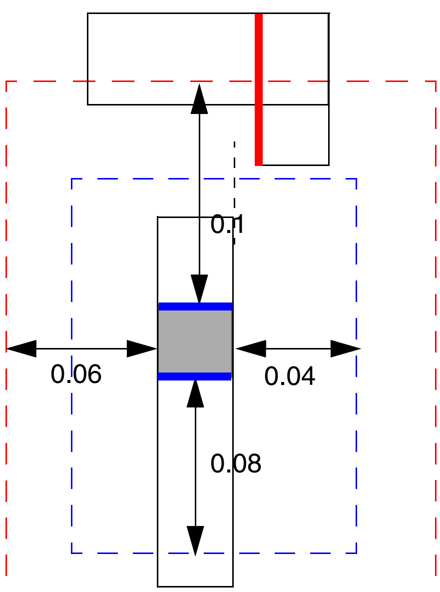


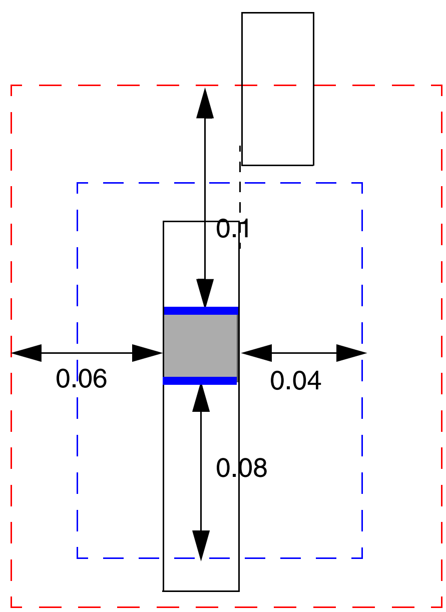
c) Violation, having a neighbor on right side alone of the smaller search window



d) Violation, having a neighbor between the smaller and large search windows.



e) Violation, the vertical red edge of the neighbor wire does not have PRL with the orthogonal blue cut edges; the horizontal edge of the neighbor wire is irrelevant



f) OK, the right neighbor wire has PRL=0 (>= 0)

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- Figure 2-42 illustrates the following enclosure rules:

```
PROPERTY LEF58_ENCLOSURE
  "ENCLOSURE CUTCLASS VSINGLECUT ABOVE 0.020 0.000 ;
   ENCLOSURE CUTCLASS VSINGLECUT ABOVE 0.015 0.015
      PARALLEL -0.08 -0.1 WITHIN 0.04 0.06
      BELOWENCLOSURE 0.05 ; " ;
```


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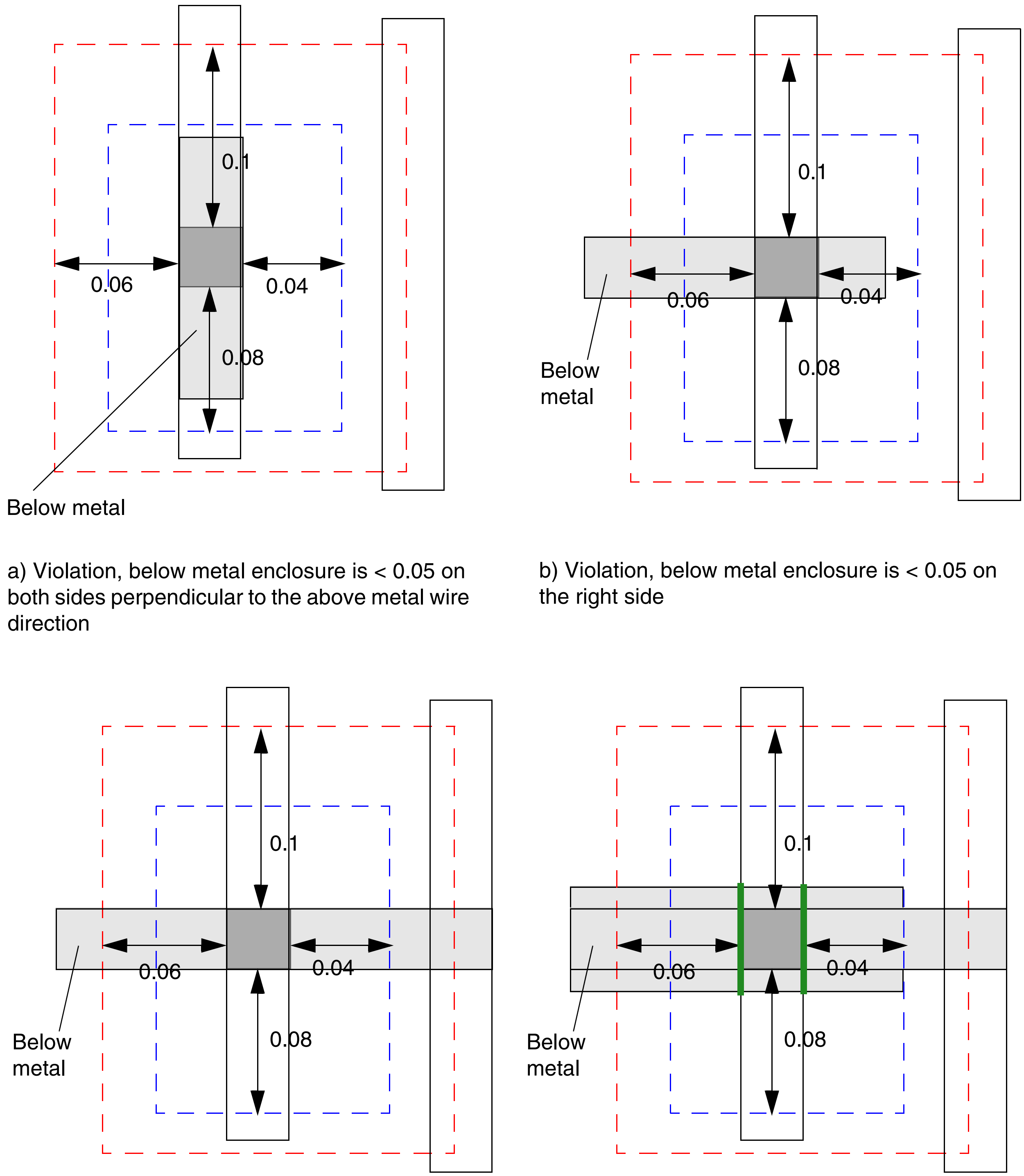
Figure 2-42 Illustration of the Enclosure Rule With PARALLEL and WITHIN



a) Violation, below metal enclosure is < 0.05 on both sides perpendicular to the above metal wire direction

b) Violation, below metal enclosure is < 0.05 on the right side

c) OK, below metal enclosure is >= 0.05 on both sides

d) Violation, the entire green wire edges require 0.05 enclosure and the enclosure on the right one is < 0.05

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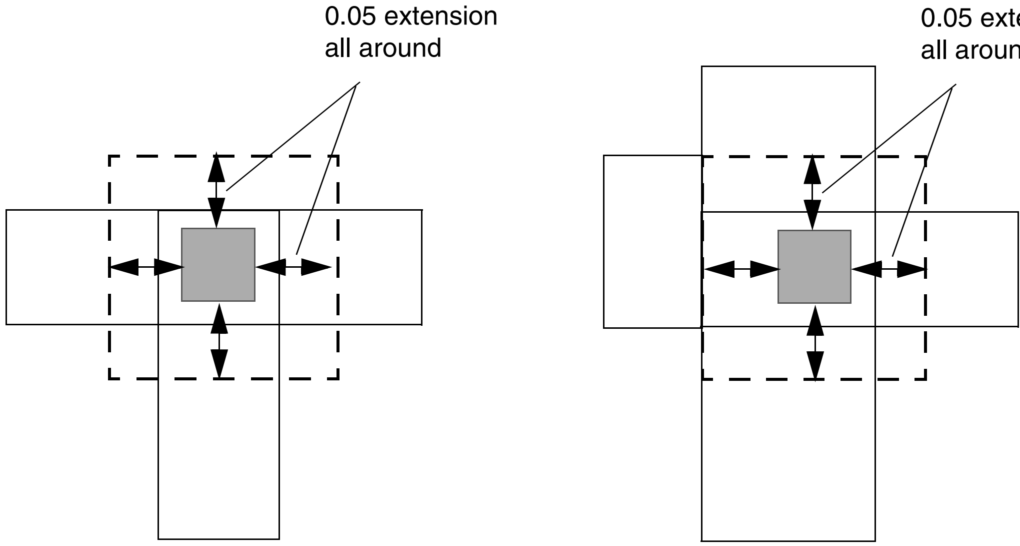
Figure 2-43 Illustration of the Enclosure Rule With CONCAVECORNERS

```
PROPERTY LEF58_ENCLOSURE
    "ENCLOSURE 0.05 0.05 CONCAVECORNERS 2 ; " ;
```



a) OK, only find two concave corners in the dotted search window

b) Violation, find 3 concave corners (> 2), top left corner does not count when that corner collides with a corner of the search window while the bottom left corner counts for aligning to a side edge of the search window

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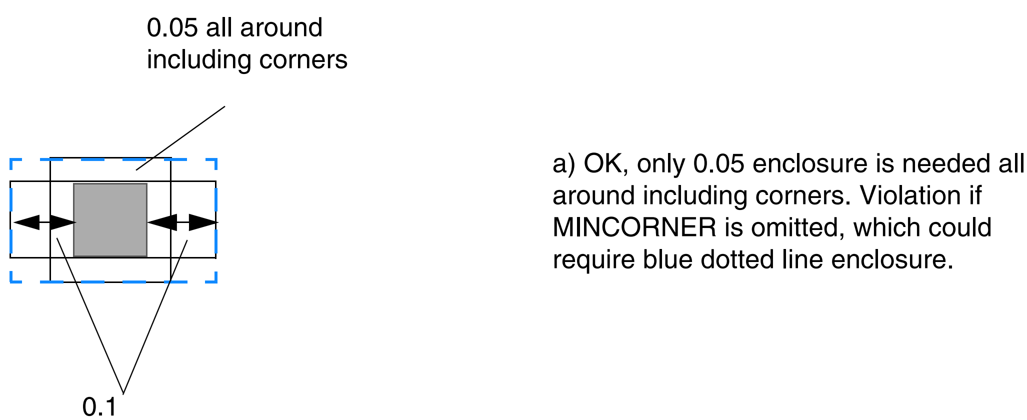
- The following enclosure rule illustrates `MINCORNER`:

```
ENCLOSURE 0.00 0.04 ;
PROPERTY LEF58_ENCLOSURE "
     ENCLOSURE MINCORNER 0.05 0.1 ; " ;
```

Figure 2-44 Illustration of the Enclosure Rule With MINCORNER



- The following diagram illustrates use of `ALLSIDES`:

Figure 2-45 Illustration of the Enclosure Rule With ALLSIDES in BELOWENCLOSURE

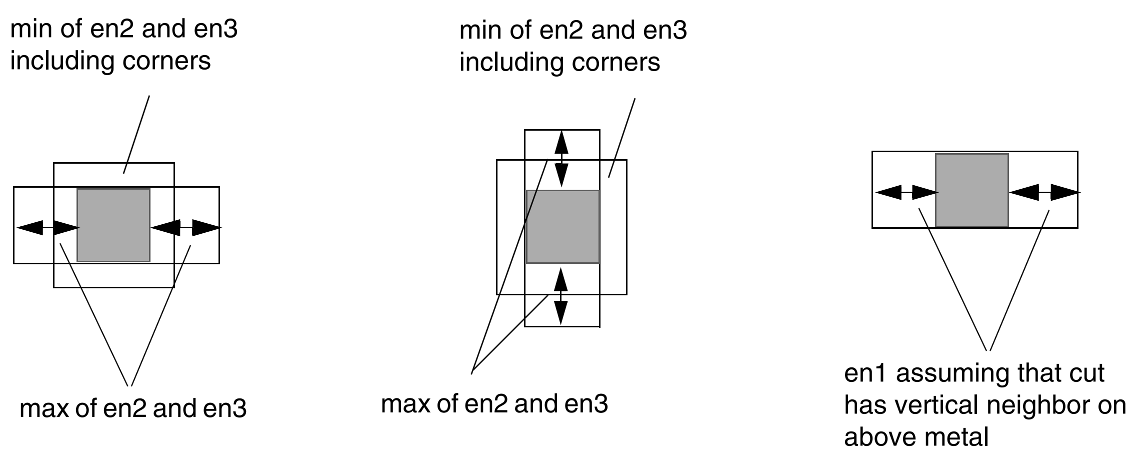


Illustration of minimum enclosure on below metal layer to exempt the rule
with BELOWENCLOSURE en1 ALLSIDES en2 en3

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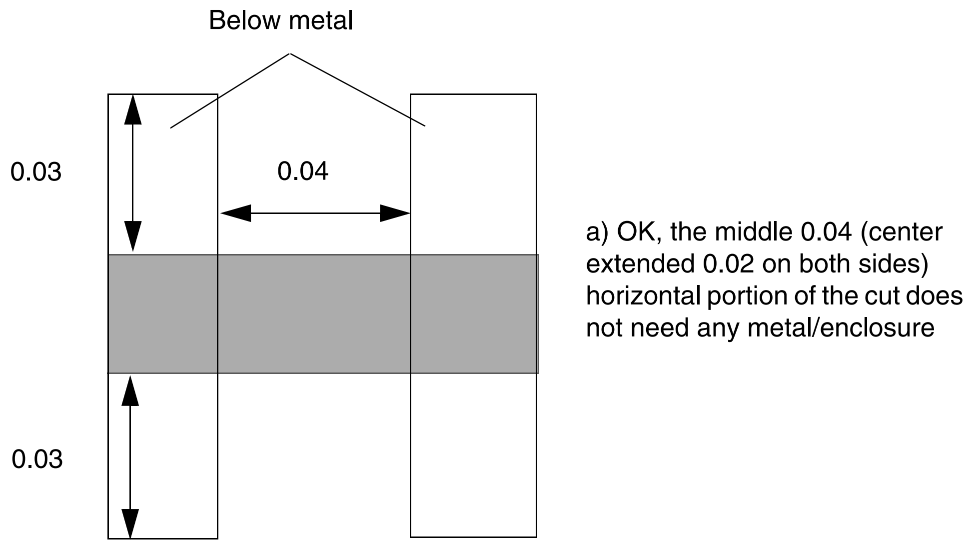
- The following diagram illustrates the enclosure rule with `HOLLOW` given below:

```
ENCLOSURE 0.00 0.04 ;
PROPERTY LEF58_ENCLOSURE "
     ENCLOSURE BELOW 0 0.03 HOLLOW HORIZONTAL 0.02 ; " ;
```

Figure 2-46 Illustration of the Enclosure Rule With HOLLOW



- The following diagram illustrates an enclosure rule with `OTHERWITHINWIDTH`:

Figure 2-47 Illustration of the Enclosure Rule with OTHERWITHINWIDTH

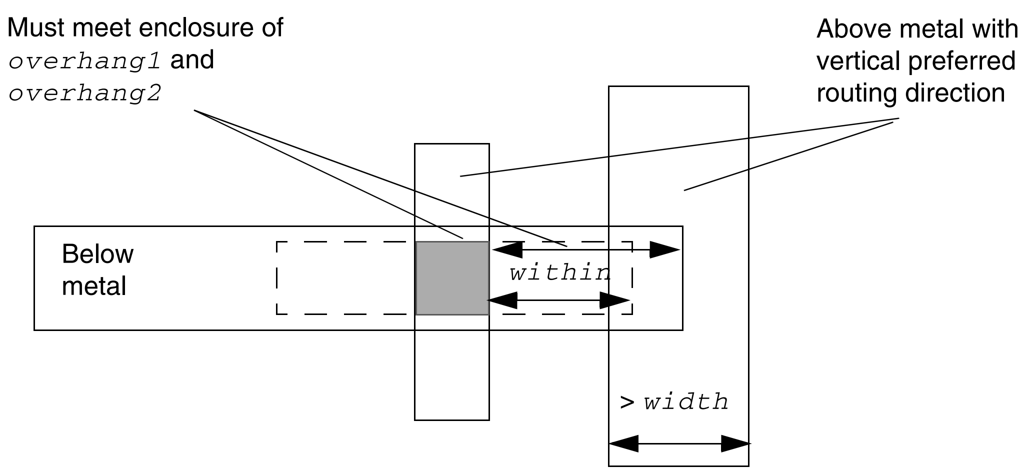


Illustration of `ENCLOSURE BELOW` *`overhang1 overhang2`*
`OTHERWITHINWIDTH` *`width`* `WITHIN` *`within`*

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- The following diagram illustrates an enclosure rule with `OTHERSIDE`:

Figure 2-48 Illustration of Enclosure Rule with OTHERSIDE

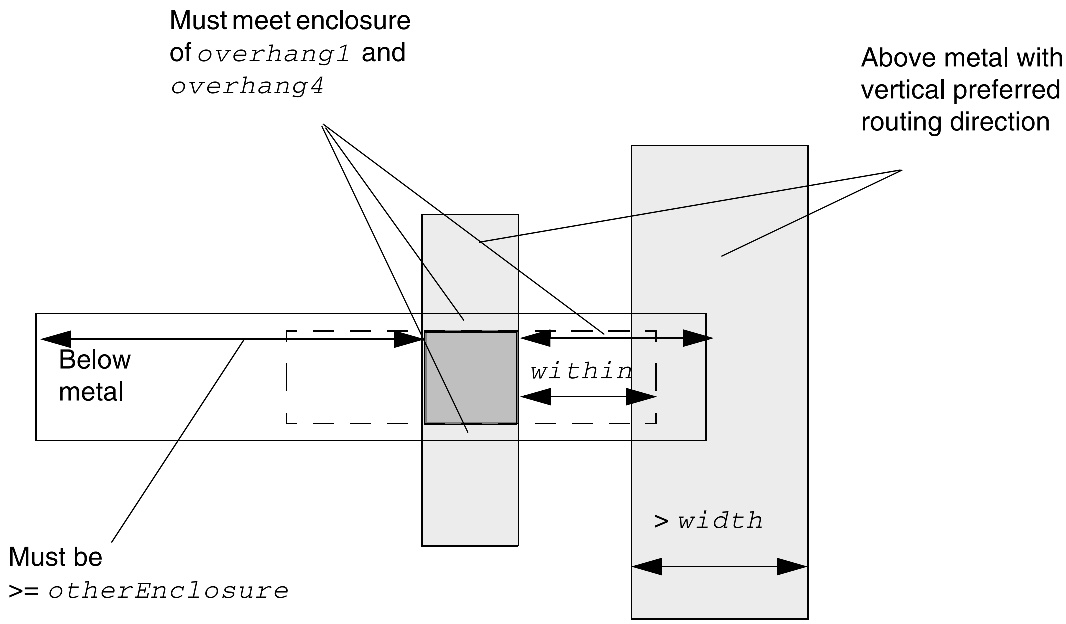

Illustration of `ENCLOSURE BELOW` *overhang1 overhang2*
`OTHERWITHINWIDTH` *width* `WITHIN` *within*
`OTHERSIDE` *otherEnclosure*

Enclosure Edge Rule

You can create a specific cut-edge enclosure rule that does not fit the normal enclosure rule semantics by using the following property definition:

```
PROPERTY LEF58_ENCLOSUREEDGE
     "ENCLOSUREEDGE [CUTCLASS className] [ABOVE | BELOW] overhang
          {OPPOSITE
               {[EXCEPTEOL eolWidth] [NOCONCAVECORNER within]
                    [CUTTOBELOWSPACING spacing
                       [ABOVEMETAL extension]]
               | WRONGDIRECTION
               }
          |[INCLUDECORNER]
           {WIDTH [BOTHWIRE] minWidth [maxWidth]
           | SPANLENGTH minSpanLength [maxSpanLength]}
                PARALLEL parLength
                {WITHIN parWithin | WITHIN minWithin maxWithin}
```


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```
            [EXCEPTEXTRACUT [cutWithin]]
            [EXCEPTTWOEDGES [exceptWithin]]
        |CONVEXCORNERS convexLength adjacentLength
            PARALLEL parWithin LENGTH length
        }
    ;]..." ;
```

Where:

All other keywords are the same as the existing LEF cut layer `ENCLOSUREEDGE` syntax.

`ABOVE | BELOW`

If you specify `ABOVE`, the overhang is required on the routing layers above this cut layer. If you specify `BELOW`, the overhang is required on the routing layers below this cut layer. If you specify neither, the rule applies to both adjacent routing layers.

`ABOVEMETAL` *`extension`*

Specifies that the cut to below metal different-net spacing of a cut edge fulfilling the overhang requirement is measured from the projected above metal edge with extension along the cut edge direction. See Figure 2-50 on page 323.
Type: Float, specified in microns

`BOTHWIRE`

Specifies that the rule only applies if the width of the neighbor wire is greater than or equal to *`minWidth`*.

`CONVEXCORNERS` *`convexLength adjacentLength`*
`PARALLEL` *`parWithin`* `LENGTH` *`length`*

Specifies the overhang on an edge with length less than or equal to *`convexLength`* between two convex corners to be *`overhang`*, if the following conditions are met:

The edge has adjacent edge with length less than or equal to *`adjacentLength`* and is also between two convex corners. The cut has neighbor wires with parallel run length greater than 0 and within (less than) *`parWithin`* on both of that edge and another adjacent edge with length greater than or equal to *`length`*.
Type: Float, specified in microns

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`CUTCLASS` *`className`* Defines the enclosure edge rule for a specific cut class (*`className`*). If `CUTCLASS` is defined in cut layer, then `CUTCLASS` must be specified in the `ENCLOSUREEDGE` statement.

Specify individual rules with the `CUTCLASS` keyword for each cut class, if needed.

`CUTTOBELOWSPACING` *`spacing`*

Defines the spacing of a cut edge fulfilling the overhang requirement to a different-net wire on the below metal layer.
Type: Float, specified in microns

`ENCLOSUREEDGE` *`overhang`* `WIDTH` *`minWidth`* `PARALLEL` *`parLength`* `WITHIN` *`parWithin`*

Indicates that any edge from this cut layer that is enclosed by metal that is greater than or equal to *`minWidth`* wide, and the enclosing metal edge is parallel to another metal edge greater than *`parLength`* in length and less than *`parWithin`* distance away, requires *`overhang`* enclosure.
Type: Float, specified in microns (for all values)

`EXCEPTEXTRACUT [`*`cutWithin`*`]`

Indicates that if there is another via cut in the same metal intersection in both the above and below layers, this rule is not checked. If you specify *`cutWithin`*, the other via cut should be less than and equal to *`cutWithin`* distance away in order to ignore this rule.

`EXCEPTTWOEDGES [`*`exceptWithin`*`]`

Specifies that if the enclosing metal edges have parallel metal edges greater than *`parLength`* that are less than *`exceptWithin`*, if specified, or *`parWithin`* away on opposite sides that covers the same PRL between the cut, then the rule does not apply.

`INCLUDECORNER` Defines the enclosure requirement applied to the corners of the cut in Euclidean measurement. This corner check is essential when the cut is slightly outside the projection of the neighbor wire.

`NOCONCAVECORNER` *`within`*

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Specifies that there should be no concave corner on the above metal layer that is a *within* distance away after the extending *overhang* on the orthogonal edges of the edges that fulfill the *overhang* requirement.
Type: Float, specified in microns

OPPOSITE [EXCEPTEOL *eolWidth*]

Specifies that if overhang of a via cut is less than or equal to *overhang* on the above metal layer (ABOVE must be specified when OPPOSITE is used), the entire opposite edge must also have overhang less than or equal to overhang.

The EXCEPTEOL keyword indicates that EOL edges with width less than to *eolWidth* are exempted from the rules.

Type: Float, specified in microns

SPANLENGTH *minSpanLength* [*maxSpanLength*]

Specifies that the enclosure edge rule is triggered by the span length of the wire containing the cut via in the direction perpendicular to the parallel run length is greater than or equal to *minSpanLength*, instead of the width of the wire. An optional *maxSpanLength* has a similar definition as *maxWidth* in WIDTH.
Type: Float, specified in microns

WIDTH *minWidth* [*maxWidth*]

Specifies an optional *maxWidth*, such that if the width of the wire containing the cut via is greater than or equal to *minWidth* and less than *maxWidth*, then all the corresponding enclosure edge rules that meet the parallel and within conditions will be applied to the cut.

If any one of the ENCLOSUREEDGE statements use *maxWidth*, then all the *ENCLOSUREEDGE* statements must have *maxWidth*. If SPANLENGTH is specified then all of them must have *maxSpanLength*, and all of the enclosure edge rules that meet the width condition in WIDTH statement or span length condition in SPANLENGTH construct (parallel run length is greater than or equal to *minSpanLength* and less than *maxSpanLength*) and the parallel and within conditions will be applied to the cut.
Type: Float, specified in microns

WITHIN *minWithin maxWithin*

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Specifies that the rule only applies if parallel neighbor metal edge is greater than or equal to *minWithin* and less than *maxWithin* distance away.
Type: Float, specified in microns

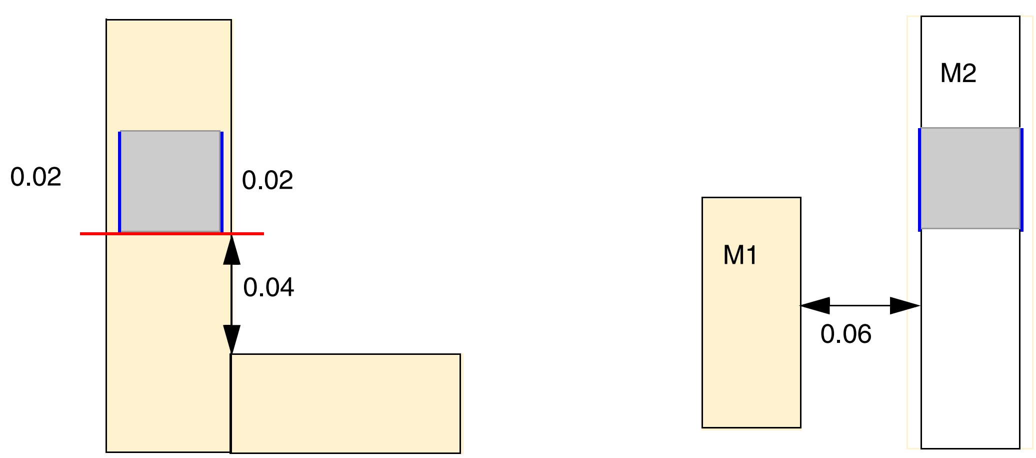
`WRONGDIRECTION` Specifies that overhang in the non-preferred direction on the above metal layer (you must specify the `ABOVE` keyword when when `OPPOSITE` is defined) of a via cut must be exactly equal to the *overhang*.

Enclosure Edge Rule Examples

- Figure 2-49 illustrates enclosure edge rule with `NOCONCAVECORNER` and `CUTTOBELOWSPACING`:

```
PROPERTY LEF58_ENCLOSUREEDGE
    "ENCLOSUREEDGE ABOVE 0.02 OPPOSITE
          NOCONCAVECORNER 0.05
          CUTTOBELOWSPACING  0.07 ; " ;
```

Figure 2-49 Illustration of Enclosure Edge Rules



a) Violation, the blue edges fulfill 0.02 overhang requirement, and by extending 0.02 (in red) on the orthogonal bottom edge, the concave corner is within 0.05 away

b) Violation, the blue edges fulfilling 0.02 overhang requirement need 0.07 spacing to a below metal object

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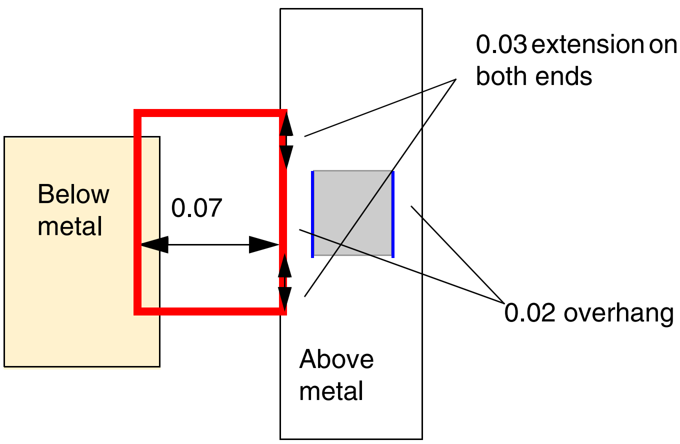
- The following figure illustrates enclosure edge rule with `ABOVEMETAL`:

```
PROPERTY LEF58_ENCLOSUREEDGE
    "ENCLOSUREEDGE ABOVE 0.02 OPPOSITE
          CUTTOBELOWSPACING  0.07
          ABOVEMETAL 0.03 ; " ;
```

Figure 2-50 Illustration of Enclosure Edge Rules



a) Violation, the blue edges fulfilling 0.02 overhang requirement need 0.07 spacing to a below metal object from the projected above metal with 0.03 extension, which means no neighbor object in the red region

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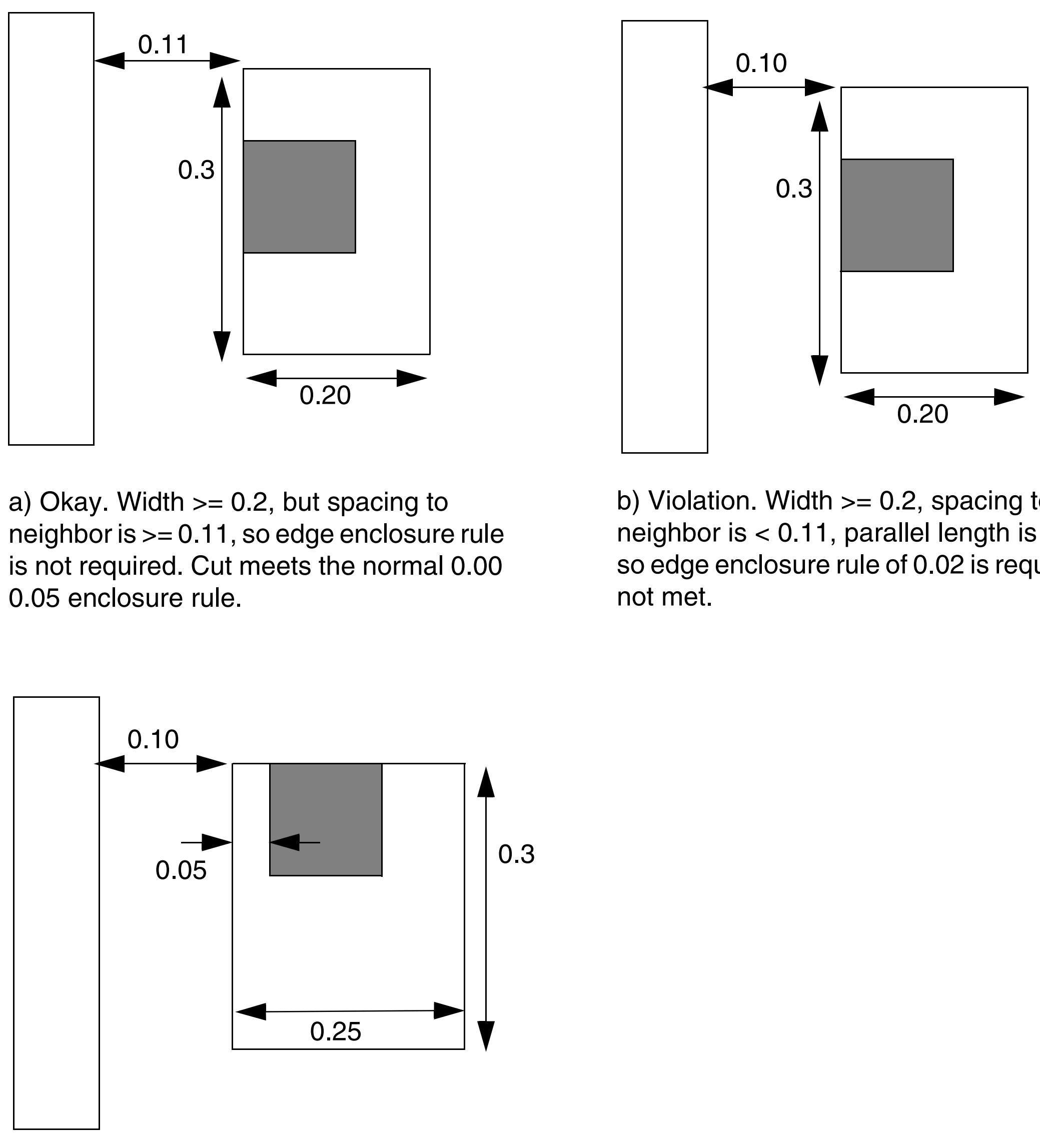
- Figure 2-51 illustrates the following enclosure edge rule:

```
ENCLOSURE 0.0 0.05 ;          #normal enclosure rule
PROPERTY LEF58_ENCLOSUREEDGE
    "ENCLOSUREEDGE 0.02 WIDTH 0.2 PARALLEL 0.25 WITHIN 0.11 ;" ;
```

Figure 2-51 Illustration of Enclosure Edge Rules



a) Okay. Width >= 0.2, but spacing to neighbor is >= 0.11, so edge enclosure rule is not required. Cut meets the normal 0.00 0.05 enclosure rule.

b) Violation. Width >= 0.2, spacing to neighbor is < 0.11, parallel length is > 0.25, so edge enclosure rule of 0.02 is required but not met.

c) Okay. Width >= 0.2, spacing to neighbor is < 0.11, parallel length is > 0.25, so edge enclosure rule of 0.02 is required and met. Cut also meets the normal 0.00 0.05 enclosure rule.

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- Figure 2-52 illustrates the following enclosure edge rule:

```
PROPERTY LEF58_ENCLOSUREEDGE
    "ENCLOSUREEDGE 0.05 WIDTH 0.20
        PARALLEL 0.50 WITHIN 0.08 0.10
        EXCEPTTWOEDGES 0.12 ; " ;
```


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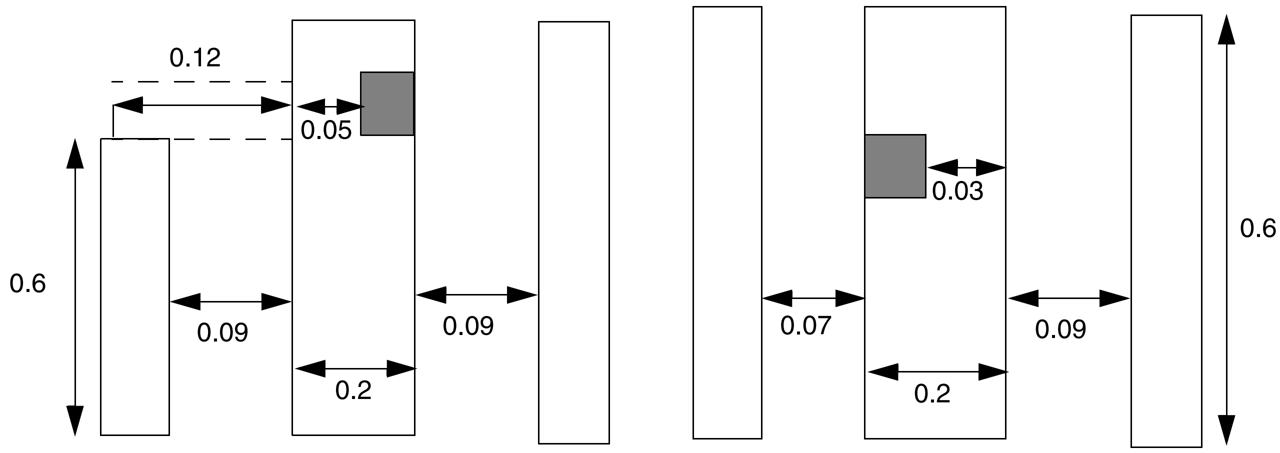
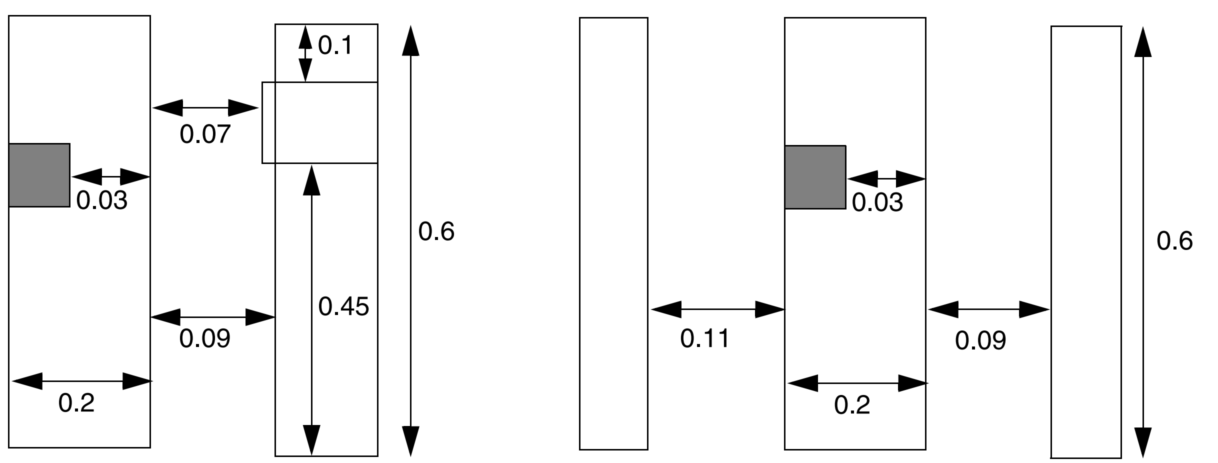
Figure 2-52 Illustration of the Enclosure Edge Rule with EXCEPTTWOEDGES



a) Violation, the right neighbor does not have corresponding left neighbor on the dotted region, and EXCEPTTWOEDGES is not triggered on that region to be a violation

b) OK, the lower bound of WITHIN of 0.08 is irrelevant on EXCEPTTWOEDGES, and the left neighbor is < 0.12 to exempt the rule.



c) OK, the portion with 0.07 (< 0.08) spacing away is ignored, the rest having 0.45 and 0.1 (< 0.50) parallel run length should be treated individually, and the rule is not triggered.

d) OK, there are two neighbor wires < 0.12. Violation if 0.12 is not specified in EXCEPTTWOEDGES.

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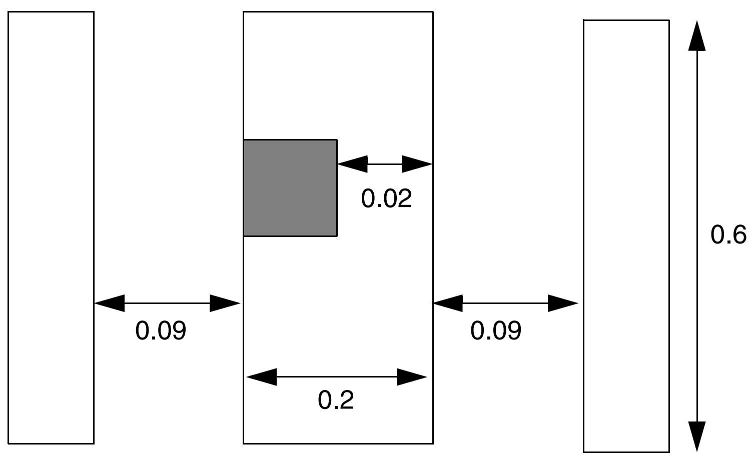
- Figure 2-53 illustrates the following enclosure edge rule:

```
PROPERTY LEF58_ENCLOSUREEDGE
    "ENCLOSUREEDGE 0.05 WIDTH 0.20
     PARALLEL 0.50 WITHIN 0.10 EXCEPTTWOEDGES ;" ;
```

Figure 2-53 Illustration of Enclosure Edge Rules



a) Okay, since it has 2 parallel neighbors,
the rule is ignored.

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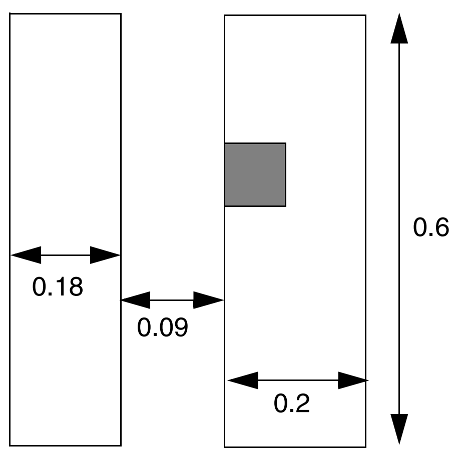
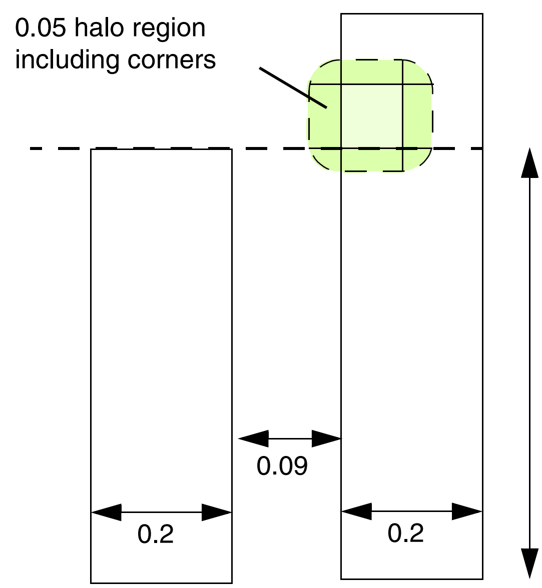
- Figure 2-54 illustrates the following enclosure edge rule:

```
PROPERTY LEF58_ENCLOSUREEDGE
    "ENCLOSUREEDGE 0.05 INCLUDECORNER
         WIDTH BOTHWIRE 0.20
         PARALLEL 0.5 WITHIN 0.10 ; " ;
```

Figure 2-54 Illustration of the Enclosure Edge Rule with BOTHWIRE and INCLUDECORNER



a) OK, the width of the left neighbor wire is 0.18 < 0.20, which will not trigger the rule. Violation, if BOTHWIRE is not specified.



b) Violation, the bottom left corner overlaps with the projection of the neighbor wire. OK, if INCLUDECORNER is not specified.

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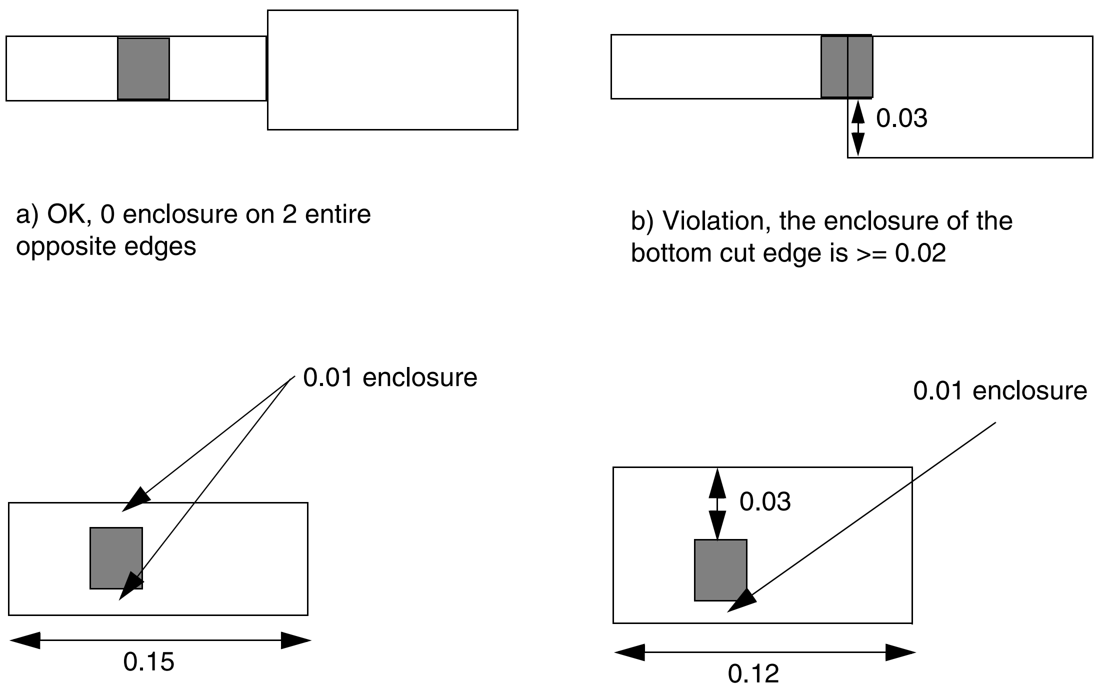
Figure 2-55 Illustration of the Enclosure Edge Rule with OPPOSITE and EXCEPTEOL

```
PROPERTY LEF58_ENCLOSUREEDGE
   "ENCLOSUREEDGE ABOVE 0.02 OPPOSITE
       EXCEPTEOL 0.15 ; " ;
```



a) OK, 0 enclosure on 2 entire opposite edges

b) Violation, the enclosure of the bottom cut edge is >= 0.02

c) OK, 0.01 enclosure on 2 entire opposite edges

d) OK, EOL edges are exempted from the rule

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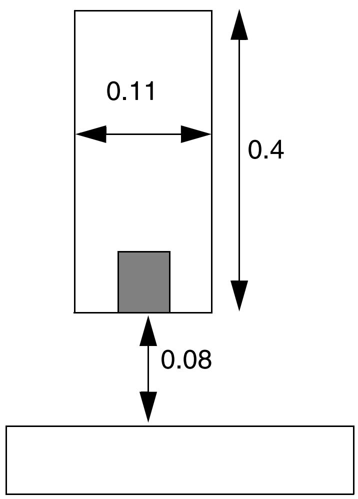
Figure 2-56 Illustration of the Enclosure Edge Rule with SPANLENGTH

```
PROPERTY LEF58_ENCLOSUREEDGE
   "ENCLOSUREEDGE 0.02 SPANLENGTH 0.2
       PARALLEL 0.1 WITHIN 0.09 ; " ;
```



a) Violation, although the width is merely 0.11, the span length in the direction perpendicular to PRL is 0.4 (>= 0.2), with PRL of 0.11 (> 0.1), the rule is triggered.

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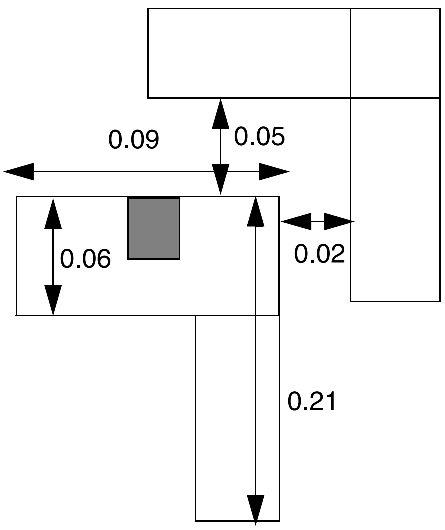
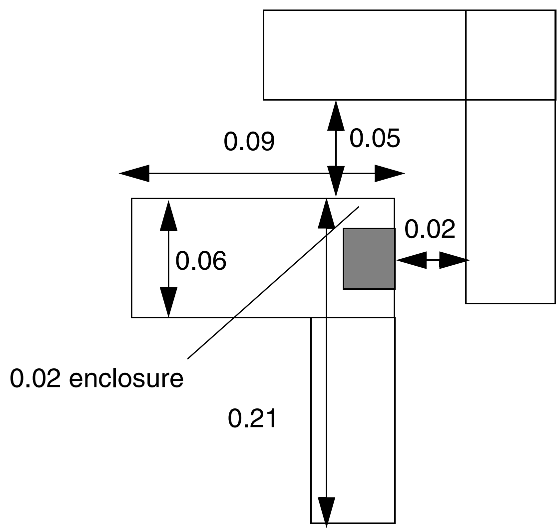
Figure 2-57 Illustration of the Enclosure Edge Rule with CONVEXCORNERS

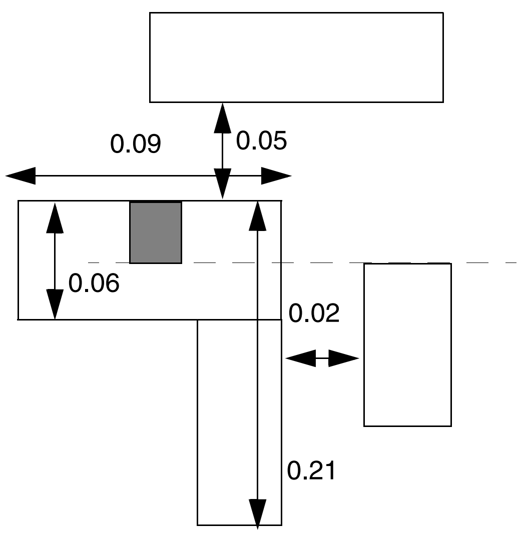
```
PROPERTY LEF58_ENCLOSUREEDGE
   "ENCLOSUREEDGE 0.02 CONVEXCRONERS 0.1 0.08
       PARALLEL 0.06 LENGTH 0.2 ; " ;
```



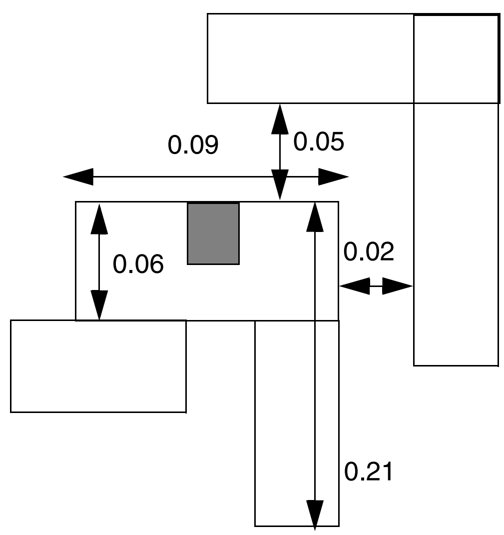
a) Violation, all of the conditions are met, but enclosure is 0 on the top edge



b) OK, enclosure requirement is only on top edge, the enclosure on the left edge is irrelevant



c) OK, no parallel run length with the cut on the right wire



d) OK, the 0.06 adjacent edge is not between 2 convex corners

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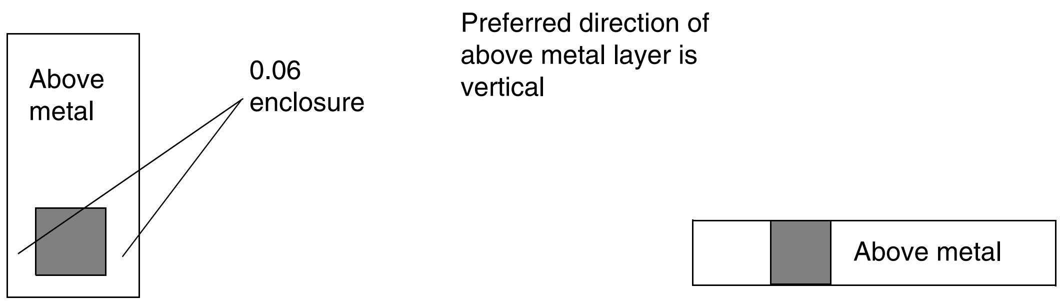
Figure 2-58 Illustration of the Enclosure Edge Rule with WRONGDIRECTION

```
PROPERTY LEF58_ENCLOSUREEDGE
   "ENCLOSUREEDGE ABOVE 0.05 OPPOSITE
       WRONGDIRECTION ; " ;
```



a) Violation, the enclosure in the non-preferred direction (horizontal) must be exactly 0.05. Larger than that is still a violation.

b) Violation, the enclosure requirement in the non-preferred direction also applies to non-preferred wires.

Enclosure Table Rule

You can create an enclosure table rule to specify the enclosure requirement based on the wire width in a table format by using the following property definition:

```
PROPERTY LEF58_ENCLOSURETABLE
     "ENCLOSURETABLE [CUTCLASS className] [MASK maskNum]
     [OTHERMASK otherMaskNum]
               [USEMAXWIDTH | USEMINWIDTH | WITHINFIRSTWIDTH]
          [DEFAULT {[ABOVE | BELOW]
              overhang1 overhang2 overhang3 overhang4}...]
          {WIDTH width {[ABOVE | BELOW]
              overhang1 overhang2 overhang3 overhang4 [MINSUM]
              [MUSTUSE]
              [MAXLENGTH length | LENGTH length]
              [LAYER trimLayer OVERLAP {1|2}
                    [TRIMLENGTHOUTERMETAL trimLength]
              | OTHERWITHINWIDTH width [LAYER layerName]
                    WITHIN within [EXCEPTOVERLAP]
                    [OTHERSIDE otherEnclosure]]
              | LINEENDGAPONLY
              | ROWBOUNDARY within OTHERSIDE otherEnclosure
                    [EXCEPTINSIDECUT]]
              [EXACTZERO]
              [INCLUDEABUTTED]}...
```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```
| OTHERWIDTH otherWidth [PARALLEL parLength WITHIN parWithin]
     {{ABOVE| BELOW}
   overhang1 overhang2 overhang3 overhang4 [MINSUM]}...}...
; " ;
```

Where:

All the other keywords are the same as the existing LEF cut layer `ENCLOSURETABLE` syntax.

`CUTCLASS` *`className`*

Defines enclosure rules for a specific cut class *`className`*. If `CUTCLASS` is defined in cut layer, then `CUTCLASS` must be specified in the `ENCLOSURETABLE` statement.

`DEFAULT {[ABOVE| BELOW]` *`overhang1 overhang2 overhang3 overhang4`*`}...`

Specifies *`overhang1`* and *`overhang2`* to be overhangs of a cut on two opposite sides and *`overhang3`* and *`overhang4`* on the other two opposite sides in a wire of any width. For a rectangular cut, the first two overhang values, *`overhang1`* and *`overhang2`*, would be applied to the end or short cut edges while the last two overhang values, *`overhang3`* and *`overhang4`*, would be applied to the side or long cut edges.

If you specify `BELOW`, the overhang is required on the routing layers below this cut layer. If you specify `ABOVE`, the overhang is required on the routing layers above this cut layer. If you specify neither, the rule applies to both adjacent routing layers.
Type: Float, specified in microns

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`EXACTZERO`
: Specifies that if any of the given overhang values – *overhang1*, *overhang2*, *overhang3*, or *overhang4* – is equal to `0`, the enclosures must be exactly equal to `0` on the given sides and remain as greater than or equal to the given value on the other non-zero overhangs.

For example:

```
PROPERTY LEF58_ENCLOSURETABLE "
ENCLOSURETABLE
   WIDTH 0.0 0.0 0.0 0.030 0.030 EXACTZERO
        0.015 0.015 0.015 0.015 ; " ;
```

means that enclosures must be exactly equal to 0 on two opposite sides and greater than or equal to 0.030 on the other two opposite sides or greater than equal to 0.015 on all four sides to be legal.

`EXCEPTINSIDECUT`
: Specifies that this `ROWBOUNDARY` enclosure rule does not apply if the via cut is completely covered by the window formed by extending the row boundary by *within* on both sides. This includes the cases where the cut edge touches or aligns with the window edge.

`EXCEPTOVERLAP`
: Specifies that this `OTHERWITHINWIDTH` enclosure rule does not apply if the via cut is fully covered with the metal shape on the other layer or *layerName* in `LAYER`, if specified.

`INCLUDEABUTTED`
: Specifies that if a cut is abutted to a wide wire with width greater than *width*, the overhang values corresponding to the wide wire should still be applied on the cut.

`LAYER` *trimLayer* `OVERLAP {1|2}`

Specifies a special enclosure requirement on rectangular pins depending on which pin end overlaps with or touches shapes in *trimLayer*, which must be a layer with `TYPE TRIMMETAL`. If 1 is given in `OVERLAP`, the corresponding enclosure rule could be applied to the pin end that overlaps with the shapes in *trimLayer*. If 2 is given in `OVERLAP`, the corresponding enclosure rule could be applied to both ends of a pin only if both the ends overlap with the shapes in *trimLayer*.

`LENGTH length`
: Specifies that the overhang requirements apply only if the span length of the wire containing the cut is greater than or equal to *length*.

Type: Float, specified in microns

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`LINEENDGAPONLY`
: Specifies that the rule applies only if a wire containing a cut on which the enclosure rule is applied fulfills the `LINEENDGAP` condition on the above or below metal layer.

`MASK` *`maskNum`*
: Specifies the mask on which the metal of this enclosure table rule will be applied. With `ABOVE`/`BELOW`, the mask is referred on the above/below metal layers. Without `ABOVE` or `BELOW`, the mask is referred on both above and below metal layers. The *`maskNum`* must be a positive integer, and most applications only support values of 1, 2, or 3. If `MASK` is used on one mask, separate tables should be defined for other masks.

  Type: Integer

`MAXLENGTH` *`length`*
: Specifies the overhang requirements only applies if the span length of the wire containing the cut is less than or equal to the *`length`*.

  Type: Float, specified in microns

`MINSUM`
: Indicates that it is legal for sum of overhang values on two opposite sides greater than or equal to the sum of the specified overhangs, and the smaller overhang value is greater than or equal to the smaller specified overhangs. This keyword should only be defined for asymmetrical overhang values.

`MUSTUSE`
: Specifies that if the subsequent condition is met, this overhang requirement must be applied. This means that if the conditions in other statements with the same `WIDTH` value are also met, they are to be ignored and only the statement with `MUSTUSE` must be checked. For any width value, `MUSTUSE` must be defined with an optional construct. If multiple `MUSTUSE` statements are defined, they must be defined with the same optional construct.

  Type: Float, specified in microns

`OTHERMASK` *`otherMaskNum`*
: Specifies the mask on which the metal on the other layer in `OTHERWIDTH` will be applied. *`otherMaskNum`* must be a positive integer, and most applications only support the values of 1, 2, or 3.

  Type: Integer

`OTHERSIDE` *`otherEnclosure`*

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Defines the enclosure on the opposite side of a cut edge, which faces the wire fulfilling the `OTHERWITHINWIDTH` condition, as *otherEnclosure*. This means that *otherEnclosure* is only checked for that opposite cut edge, while the remaining three edges must fulfill the smallest three values among *overhang1*, *overhang2*, *overhang3*, and *overhang4*. See Figure 2-61 on page 342.

Type: Float, specified in microns

{OTHERWIDTH *otherWidth* {{ABOVE| BELOW}
overhang1 overhang2 overhang3 overhang4 [MINSUM]}...}...

Specifies that the overhang requirements on the above/below metal layer corresponding to the specified `ABOVE`/`BELOW` keyword will depend on the width of the other (below/above) metal layer. The meaning of overhang and `MINSUM` is the same as in `WIDTH`. See Figure 2-70 on page 350.

Type: Float, specified in microns

OTHERWITHINWIDTH *width* [LAYER *layerName*] WITHIN *within*

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Specifies that the enclosure rule applies only if there is a neighboring wire within *within* of the cut and completely covering a cut edge in the non-preferred routing layer direction having width greater than *width* on the other layer having the RECTONLY construct on which the enclosure applies. This means that ABOVE or BELOW must be specified. If LAYER is specified, *layerName* must be a previously defined metal layer and the neighboring wire search is preformed on the given layer. *layerName* should not be the intermediate below metal layer, which could be specified by ABOVE OTHERWITHINWIDTH without using LAYER.

Type: Float, specified in microns

For example:

```
PROPERTY LEF58_ENCLOSURETABLE "
    ENCLOSURETABLE CUTCLASS VA
       WIDTH 0.00000 BELOW 0.090 0.090 0.000 0.000
       WIDTH 0.00000 BELOW 0.120 0.120 0.000 0.000
          MUSTUSE LENGTH 0.5
       WIDTH 0.00000 BELOW 0.060 0.060 0.000 0.000
          OTHERWITHINWIDTH 0.039 WITHIN 0.008
       ... ; " ;
```

means that if the length on a below routing metal containing a VA cut is greater than or equal to 0.5, "0.120 0.120 0.000 0.000" must be applied even if the OTHERWITHINWIDTH condition is also fulfilled. If the length is less than 0.5 and the OTHERWITHINWIDTH condition is met, "0.060 0.060 0.000 0.000" is applied. If the length is less than 0.5 and the OTHERWITHINWIDTH condition is not met, "0.090 0.090 0.000 0.000" is applied.

PARALLEL *parLength* WITHIN *parWithin*

Specifies that the enclosure rule applies only if there is a metal neighbor at least on one side of the cut within (less than) *parWithin* having common parallel run length greater than or equal to *parLength* on the layer on which the enclosure is applied. If *parLength* is a negative value, the two edges will be checked in WITHIN style. The metal containing the cut is not considered a neighbor. In addition, only the metal that has negative PRL with the cut is considered. See Figure 2-69 on page 349

Type: Float, specified in microns

ROWBOUNDARY *within* OTHERSIDE *otherEnclosure*

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Specifies that the enclosure rule applies only on the cut that is within *within* of any standard cell row boundary. The opposite side of the cut edge facing the boundary must have enclosure greater than or equal to *otherEnclosure*. See Figure 2-71 on page 350.

Type: Float, specified in microns

`TRIMLENGTHOUTERMETAL` *trimLength*

Specifies that the enclosure with a shape on the layer *trimLayer* overlapping with or touching a wire end applies only if the *trimLayer* shape has a length greater than or equal to *trimLength*, and only the outermost wires overlapping or touching the *trimLayer* shape are subjected to this rule.

Type: Float, specified in microns

`USEMAXWIDTH` Specifies that the maximum wire width with which a cut overlaps is used to look up the proper width rows for the overhang requirements. It is similar to the width interpretation in `ENCLOSURE`.

`USEMINWIDTH` Specifies that the minimum wire width with which a cut overlaps or touches is used to look up the proper width rows for the overhang requirements.

{WIDTH *width* {[ABOVE| BELOW]
overhang1 overhang2 overhang3 overhang4}...}...

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Specifies the overhang requirements for a cut in a wire having a certain width. The width values must be the same or in increasing order (from top to bottom) in the table. If any of the overhang values on a routing layer are asymmetrical, that is, *overhang1* is not equal to *overhang2* or *overhang3* is not equal to *overhang4*, then the minimum wire width that the cut touches or overlaps with is determined. The last row with wire width greater than *width* and any one of the corresponding specified overhangs or overhangs defined in DEFAULT must be fulfilled.

In addition, the minimum overhang value on all the four sides, including corners, is applied. The other overhang values per cut edges, without corner consideration, are applied. If all of the overhang values are symmetrical, the maximum wire width that the cut overlaps with and the traditional rectangle enclosure by using one overhang value on the two opposite sides and another overhang value on the other two opposite sides must be used.

The definition of overhangs ABOVE and BELOW have the same meaning as those in DEFAULT.

For a rectangular cut, the first two overhang values, *overhang1* and *overhang2*, would be applied to the end or short cut edges while the last two overhang values, *overhang3* and *overhang4*, would be applied to the side or long cut edges.

Type: Float, specified in microns

WITHINFIRSTWIDTH

Specifies that the minimum wire width that a cut overlaps with should be used to look up the enclosure requirement except for the case in which one of the overlap wire width is applicable to the width on the first row. In that case, if a cut overlaps with the wire widths applicable to multiple rows, the minimum wire width other than the widths for the first row would be used. In other words, the enclosure on the first row would be used only if a cut overlaps solely with the wire widths applicable to the first row width.

Enclosure Table Rule Examples

- The following diagram illustrates the enclosure table rule given below:

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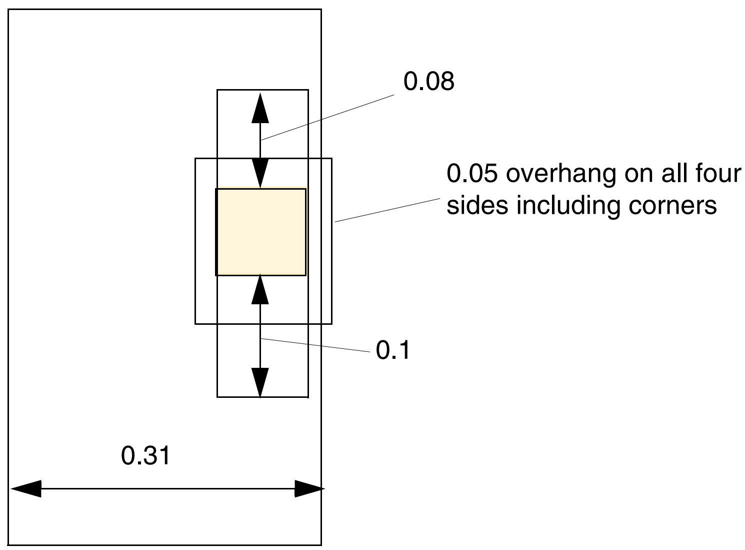
```
PROPERTY LEF58_ENCLOSURETABLE
      "ENCLOSURETABLE
           WIDTH 0.3 0.05 0.05 0.08 0.1; " ;
```

Figure 2-59 Illustration of Enclosure Table Rule



a) OK, 0.05 all around the cut, and 0.08 and 0.1 based on the cut edges

- The following enclosure table rule indicates that:
  - For a cut in a wire of any width, 0.01 µm on two opposite sides and 0.02 µm on the other two opposite sides can be an overhang choice.
  - For a via cut in a wire with width greater than 0.1 µm and less than or equal to 0.2 µm, 0.015 µm overhang on four sides can be applied.
  - For a via cut in a wire with width greater than 0.2 µm and less than or equal to 0.3 µm, 0.015 µm and 0.016 µm overhangs on two opposite sides, and 0.017 µm and 0.018 µm overhangs on the other two opposite sides, or 0.025 µm and 0.03 µm or 0.026 µm and 0.029 µm or 0.027 µm and 0.28 µm on two opposite sides, 0.00 µm on the other two opposite sides can be applied.
  - For a via cut in a wire with width greater than 0.3 µm, 0.019 µm overhang on four sides can be applied.

```
PROPERTY LEF58_ENCLOSURETABLE
"ENCLOSURETABLE
```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```
DEFAULT 0.01 0.01 0.02 0.02
WIDTH 0.1 0.015 0.015 0.015 0.015
WIDTH 0.2 0.015 0.016 0.017 0.018
         0.025 0.03 0.00 0.00 MINSUM
WIDTH 0.3 0.019 0.019 0.019 0.019 ; " ;
```

Figure 2-60 Illustration of Enclosure Table Rule with ABOVE OTHERWITHINWIDTH

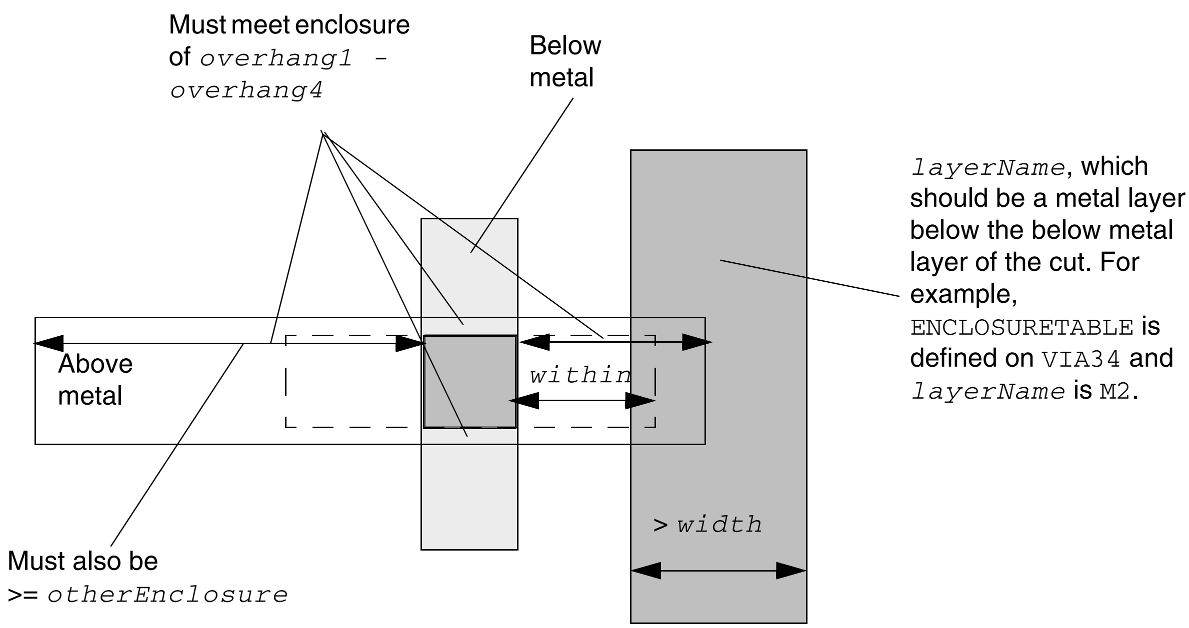

Illustration of `ENCLOSURETABLE ... ABOVE...`
`OTHERWITHINWIDTH` *`width`* `LAYER` *`layerName`*
`WITHIN` *`within`*

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Figure 2-61 Illustration of Enclosure Table Rule with BELOW OTHERWITHINWIDTH

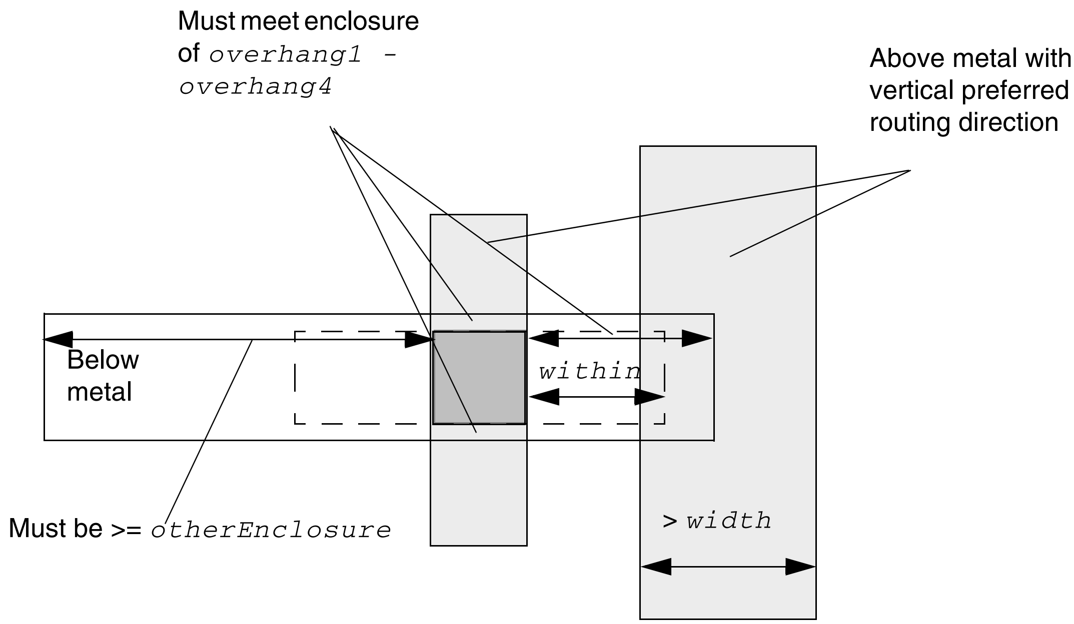

Illustration of `ENCLOSURETABLE ... BELOW ...`
`OTHERWITHINWIDTH` *`width`* `WITHIN` *`within`*
`OTHERSIDE` *`otherEnclosure`*

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Figure 2-62 Illustration of Enclosure Table Rule with EXCEPTOVERLAP in OTHERWITHINWIDTH

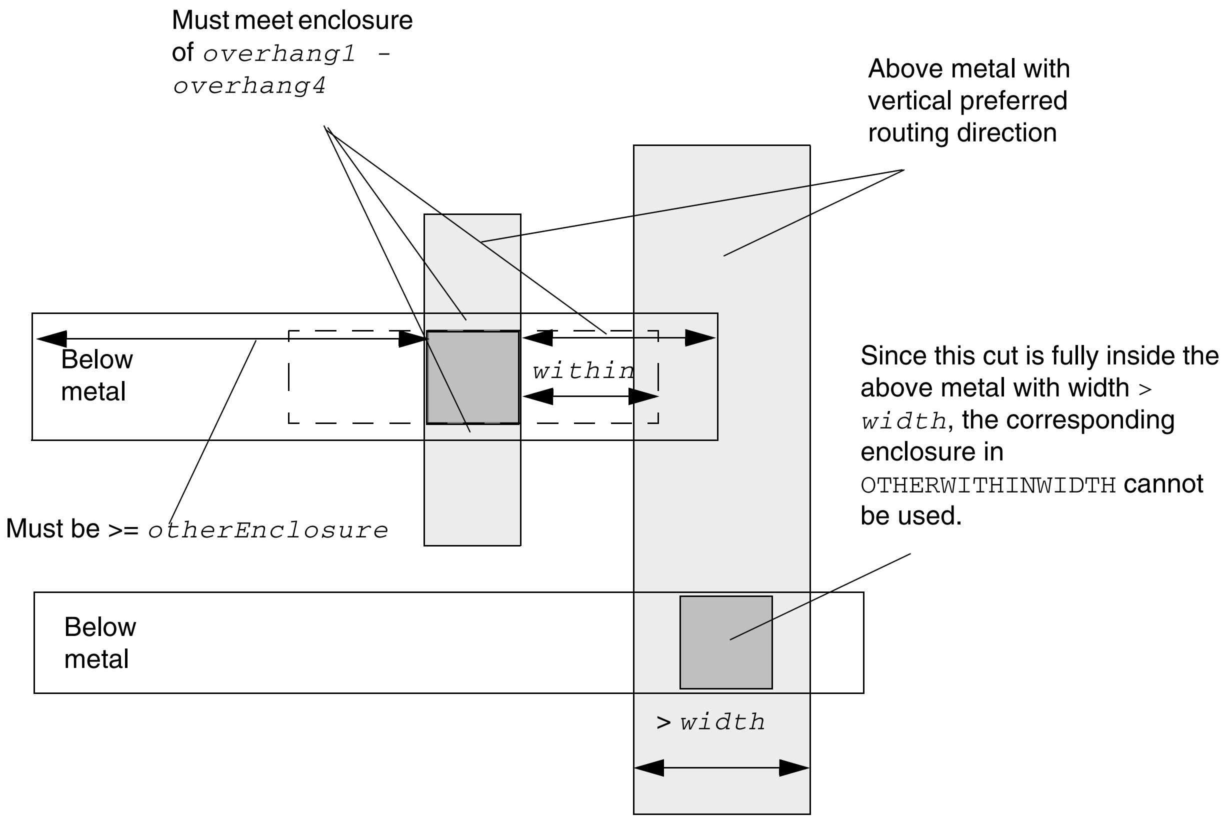

Illustration of ENCLOSURETABLE ... BELOW ...
OTHERWITHINWIDTH *width* WITHIN *within*
OTHERSIDE *otherEnclosure*

- The following diagram illustrates the enclosure table rule with LAYER given below:

```
PROPERTY LEF58_ENCLOSURETABLE
"ENCLOSURETABLE
     WIDTH 0.0 BELOW 0.02 0.02 0.0 0.0
     WIDTH 0.0 BELOW 0.01 0.01 0.0 0.0
          LAYER TRIM1 OVERLAP 1 ; " ;
```


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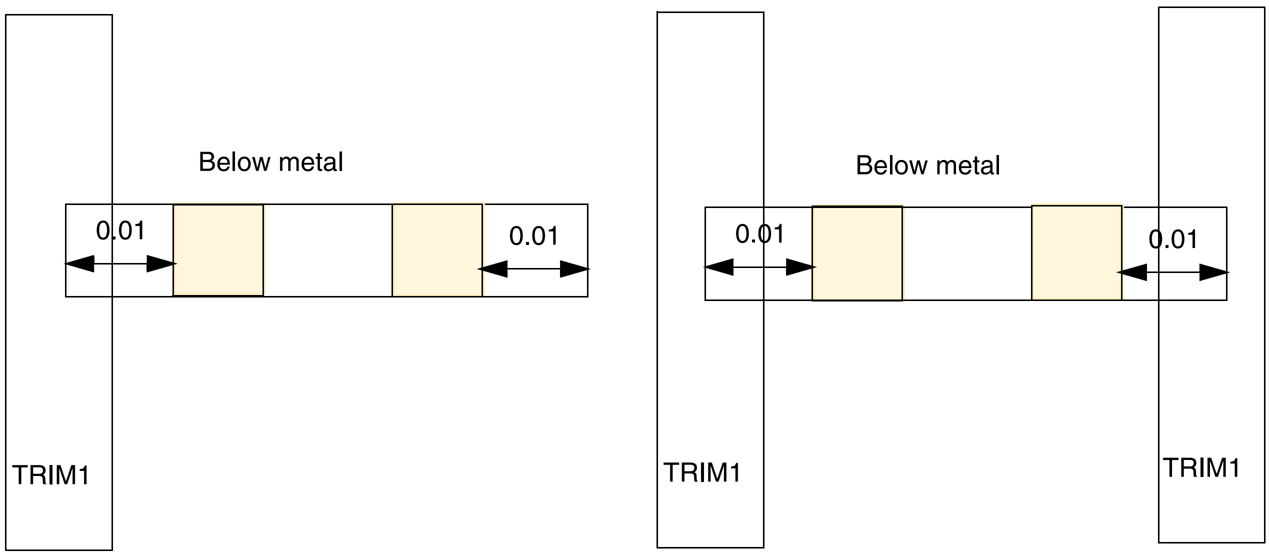
Figure 2-63 Illustration of Enclosure Table Rule with LAYER



a) Violation, the left cut overlaps with TRIM1 shape, and 0.01 enclosure is sufficient, but the right cut needs 0.02 & fails.

b) OK, both ends overlap with TRIM1 shape, 0.01 enclosure could be used on both of them.

- The following diagram illustrates the enclosure table rule with `LAYER` given below:

```
PROPERTY LEF58_ENCLOSURETABLE
"ENCLOSURETABLE
     WIDTH 0.0 BELOW 0.02 0.02 0.0 0.0
     WIDTH 0.0 BELOW 0.01 0.01 0.0 0.0
          LAYER TRIM1 OVERLAP 2 ; " ;
```


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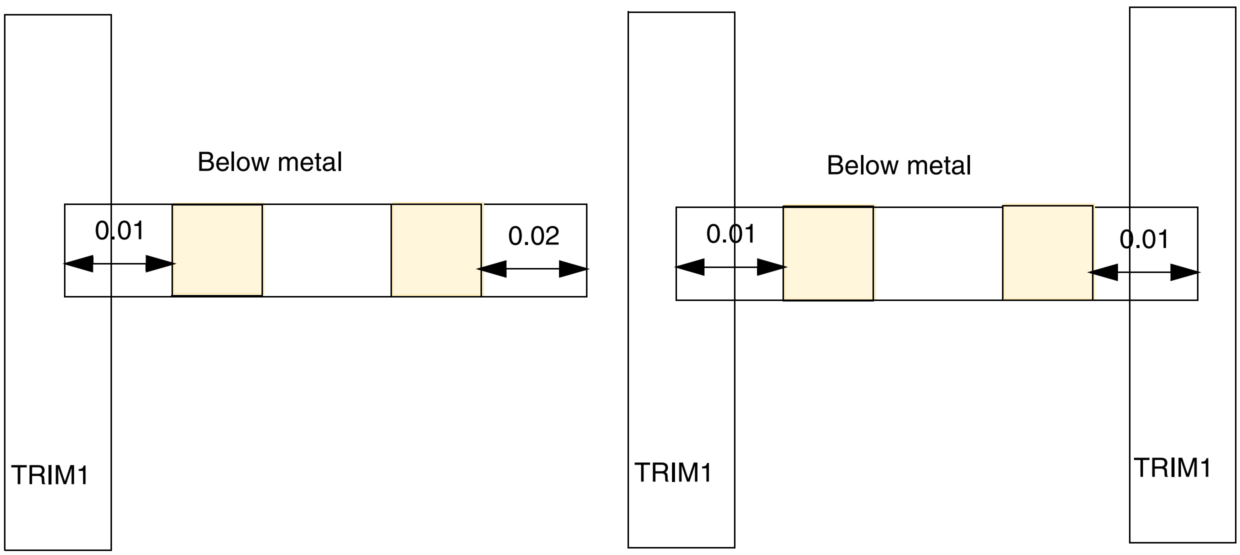
Figure 2-64 Illustration of Enclosure Table Rules with LAYER



a) Violation, only left end overlaps with TRIM1, 0.02 enclosure must be used on both ends. If OVERLAP 1 is given, it would be fine.

b) OK, both ends overlap with TRIM1 shape, 0.01 enclosure could be used on both of them

- The following diagram illustrates an enclosure table rule with `LINEENDGAPONLY`:

```
PROPERTY LEF58_ENCLOSURETABLE
  " ENCLOSURETABLE
      WIDTH 0.1 BELOW 0.02 0.02 0.10 0.10 LINEENDGAPONLY
      WIDTH 0.1 BELOW 0.05 0.05 0.10 0.10 ; " ;
```

On below metal layer,

```
PROPERTY LEF58_LINEENDGAP "
    LINEENDGAP 0.03 WIDTH 0.12 MASK 1 ; " ;
```


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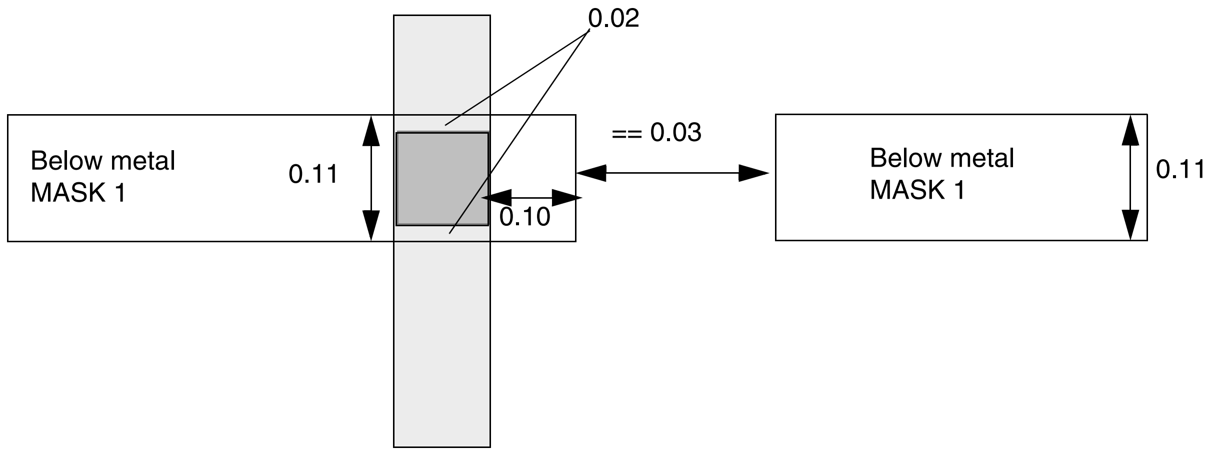
Figure 2-65 Illustration of Enclosure Table Rule with LINEENDGAPONLY

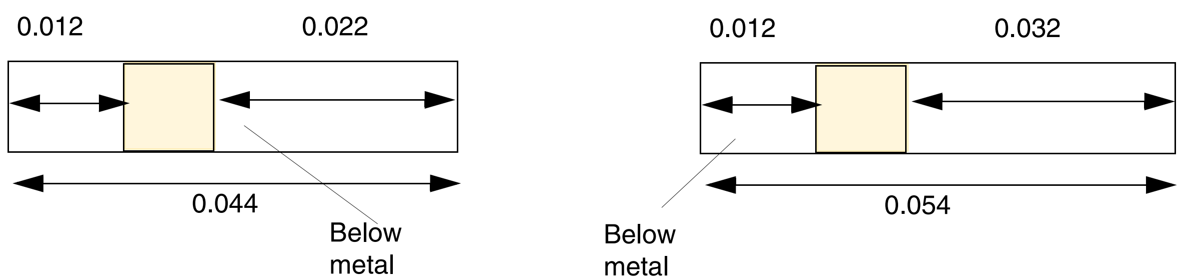


a) OK, the `0.11` wire fulfills the `LINEENDGAP` condition on the below metal layer such that the first statement could be applied

- The following diagram illustrates the enclosure table rule given below with `MAXLENGTH`:

```
PROPERTY LEF58_ENCLOSURETABLE
"ENCLOSURETABLE
    WIDTH 0.0 BELOW 0.0 0.0 0.015 0.015
        BELOW 0.0 0.0 0.012 0.012 MAXLENGTH 0.05 ; " ;
```

Figure 2-66 Illustration of Enclosure Table Rule with MAXLENGTH



a) OK, the span length of 0.044 (<= 0.05), and enclosure of 0.012 is sufficient

b) Violation, the span length of 0.054 requires 0.015 enclosure

- The following diagram illustrates an enclosure table rule with `OTHERMASK`:

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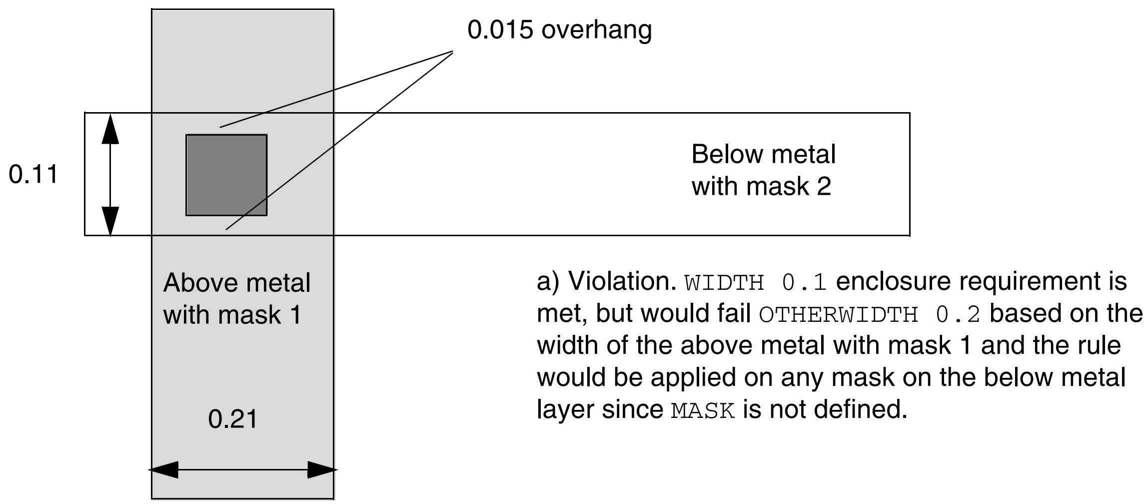
```
PROPERTY LEF58_ENCLOSURETABLE
"ENCLOSURETABLE OTHERMASK 1
    WIDTH 0.1 BELOW 0.015 0.015 0.015 0.015
    OTHERWIDTH 0.2 BELOW 0.019 0.019 0.019 0.019 ; " ;
```

Figure 2-67 Illustration of Enclosure Table Rules with OTHERMASK



- The following diagram illustrates an enclosure table rule with `OTHERWIDTH`:

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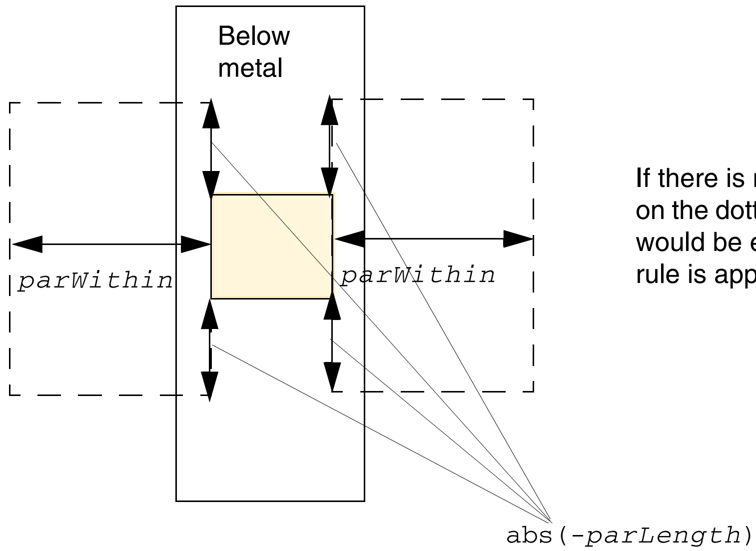
Figure 2-68 Illustration of Enclosure Table Rules with OTHERWIDTH



If there is no neighbor on below metal on the dotted search windows, the rule would be exempted. Otherwise, the rule is applied.

Illustrations of `OTHERWIDTH ...`
`PARALLEL -parLength WITHIN parWithin`
`BELOW ...`

- The following diagram illustrates an enclosure table rule with `OTHERWIDTH`:

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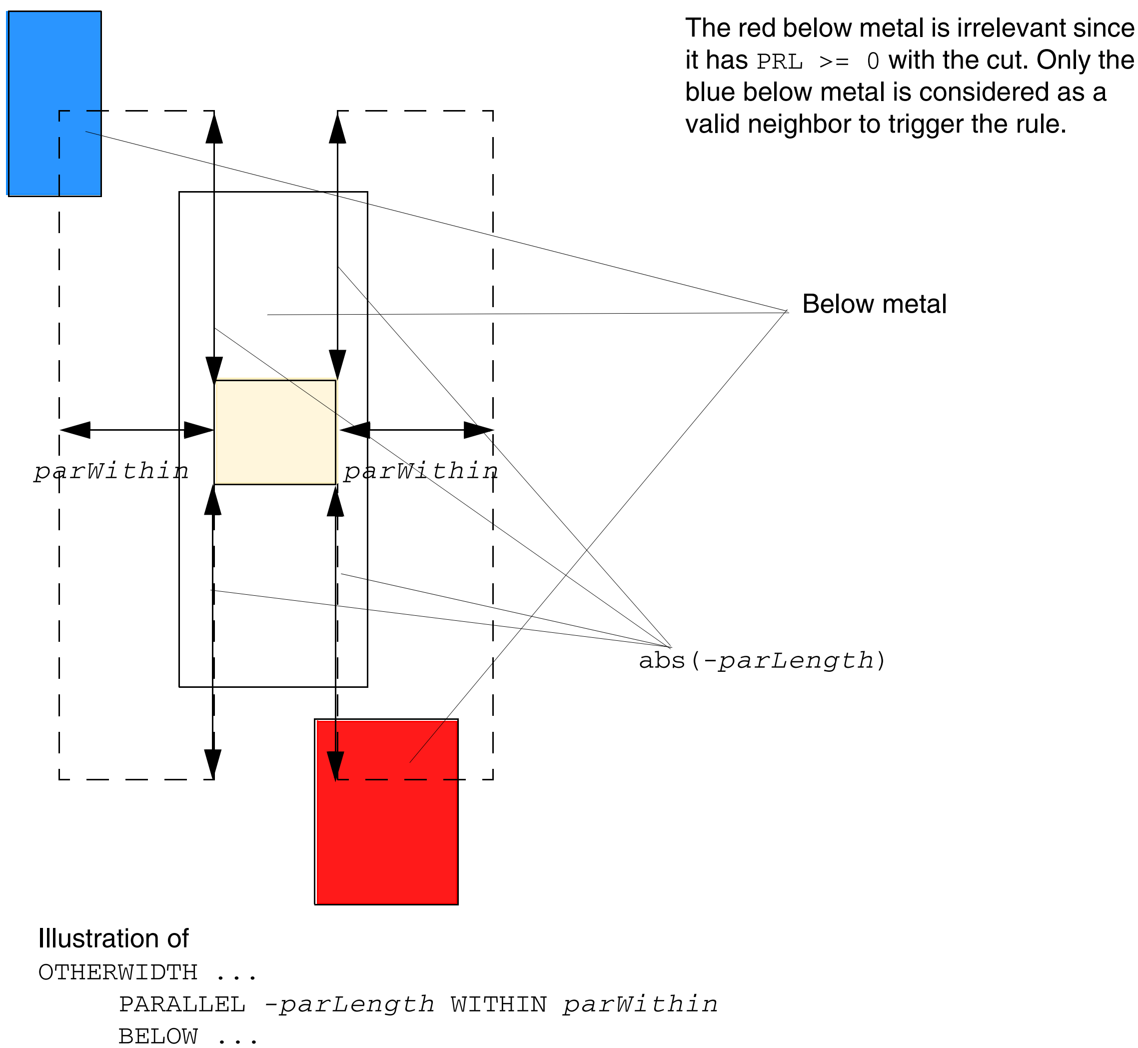
Figure 2-69 Illustration of Enclosure Table Rule with OTHERWIDTH



- The following rule indicates that for a via cut in a wire with width > 0.1 µm, 0.015 overhang on four sides can be applied, for a via cut in a wire with width on the above metal layer > 0.2 µm, 0.019 overhang on the below metal layer on four sides can be applied.

```
PROPERTY LEF58_ENCLOSURETABLE
"ENCLOSURETABLE
    WIDTH 0.1 0.015 0.015 0.015 0.015
    OTHERWIDTH 0.2 BELOW 0.019 0.019 0.019 0.019 ; " ;
```


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Figure 2-70 Illustration of Enclosure Table Rules with OTHERWIDTH

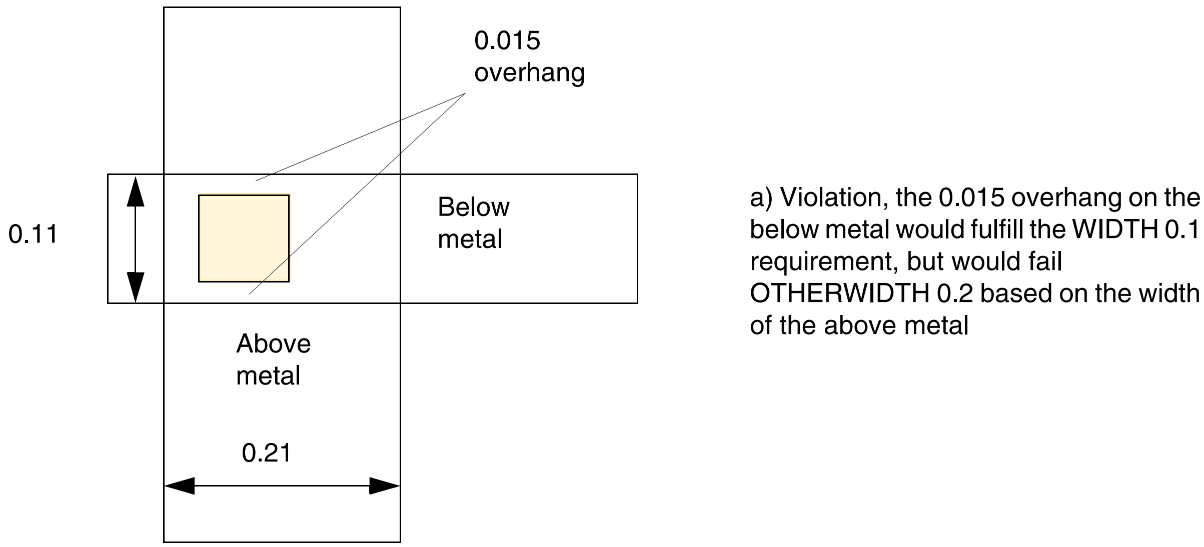

Figure 2-71 Illustration of Enclosure Table Rule with ROWBOUNDARY

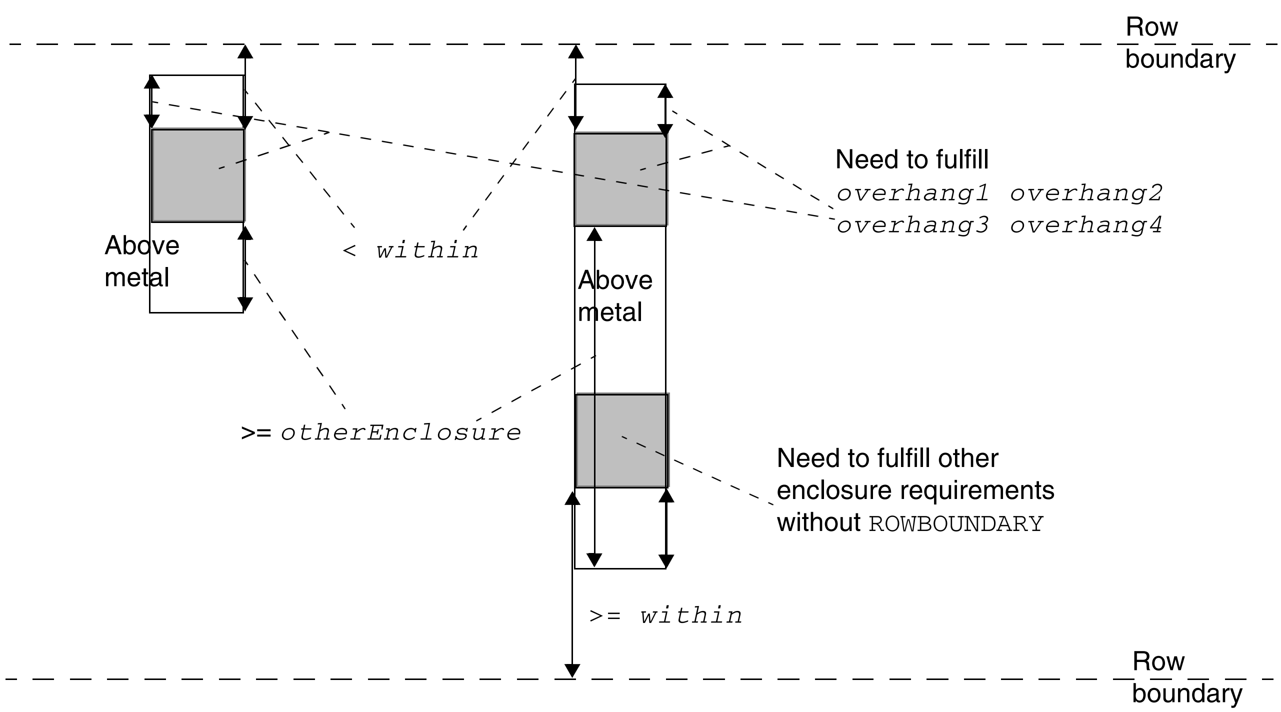

Illustration of `ENCLOSURETABLE ... ABOVE...`
`overhang1 overhang2 overhang3 overhang4`
`ROWBOUNDARY` *`within`* `OTHERSIDE` *`otherEnclosure`*

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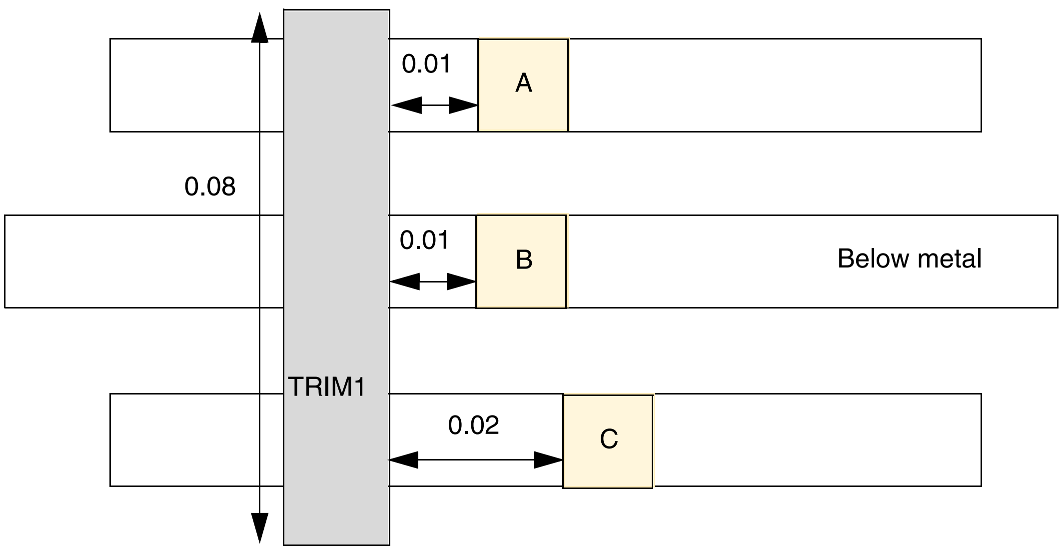
- The following diagram depicts an enclosure table rule with `TRIMLENGTHOUTERMETAL`:

```
PROPERTY LEF58_ENCLOSURETABLE
"ENCLOSURETABLE
    WIDTH 0.0 BELOW 0.01 0.01 0.0 0.0
    WIDTH 0.0 BELOW 0.02 0.02 0.0 0.0
        LAYER TRIM1 OVERLAP 1
        TRIMLENGTHOUTERMETAL 0.08 ; " ;
```

Figure 2-72 Illustration of Enclosure Table Rules with TRIMLENGTHOUTERMETAL



a) Violation, via A fails the second `LAYER` enclosure statement. Via B is not subjected to the second statement and passes the first enclosure statement. Via C passes the second enclosure statement.

- The following diagram illustrates an enclosure table rule with `WITHINFIRSTWIDTH`:

```
PROPERTY LEF58_ENCLOSURETABLE
    "ENCLOSURETABLE WITHINFIRSTWIDTH
          WIDTH 0.000 0.020 0.015 0.000 0.000
          WIDTH 0.024 0.020 0.015 0.001 0.001; " ;
```


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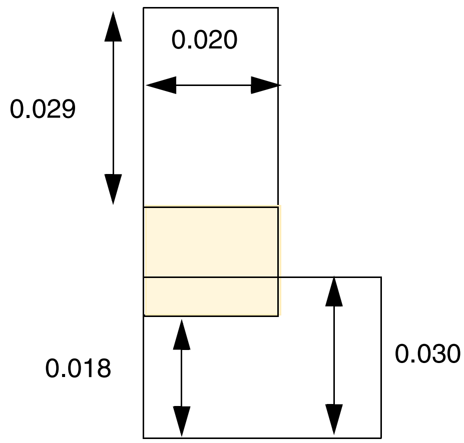
Figure 2-73 Illustration of Enclosure Table Rule with WITHINFIRSTWIDTH



a) Violation, the enclosure of the first row applies only if the cut is completely within the wire portion with width of `0.020`. OK if `WITHINFIRSTWIDTH` is omitted.

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Enclosure Width Rule

You can create a specific cut-width enclosure rule that does not fit the normal enclosure rule semantics by using the following property definition:

```
PROPERTY LEF58_ENCLOSUREWIDTH
    "ENCLOSUREWIDTH {VIAOVERLAPONLY | USEMINWIDTH}
        ; " ;
```

Where:

`ENCLOSUREWIDTH VIAOVERLAPONLY`

Indicates a method to check the enclosure width requirements. If a metal shape of a cut via protrudes on a certain side of the wide wire, the enclosure is not checked on that side (or the enclosure should be checked on the sides overlapped with the wide wire). In addition, the enclosure based on the width of the protrusion must also be checked against the cut.

When `ENCLOSUREWIDTH VIAOVERLAPONLY` is defined in a cut layer, for a given width, if only one `ENCLOSURE` statement with `WIDTH` construct having the same overhang values is found, the special edge based enclosure checking is applied. Otherwise, the traditional enclosure method is applied.

Cuts within a wide wire not only need to fulfill the corresponding `ENCLOSURE WIDTH` statements, but also need to fulfill one of the `ENCLOSURE` statements without `WIDTH`.

`USEMINWIDTH`

Specifies that the minimum width of the wires that overlap with a cut should be used to look up for the corresponding width-based enclosure requirements. This affects only `ENCLOSURE` statements and has no impact on `ENCLOSURETABLE`. When `INCLUDEABUTTED` is specified on an `ENCLOSURE` statement, it has precedence over `USEMINWIDTH`. This means that a cut overlapping or touching a wire with width greater than or equal to *`minWidth`* needs to follow only the specified enclosure requirements.

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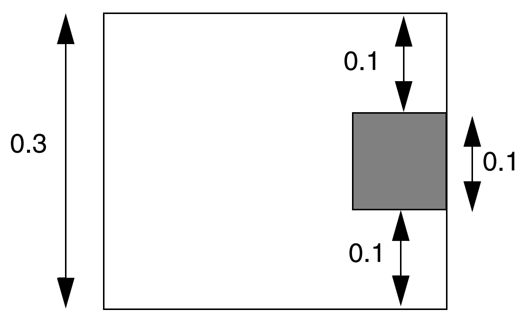
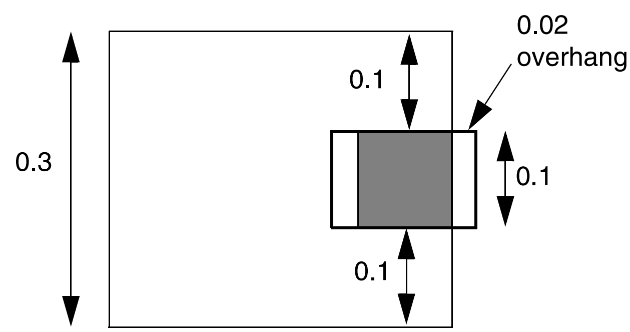
Enclosure Width Rule Examples

Figure 2-74 Illustration of Enclosure Width Rules

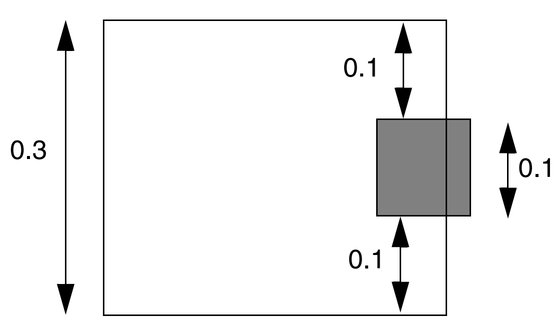
```
PROPERTY LEF58_ENCLOSUREWIDTH
   "ENCLOSUREWIDTH VIAOVERLAPONLY ;" ;
ENCLOSURE 0.0 0.02 ;
ENCLOSURE 0.02 0.02 WIDTH 0.15 ;
ENCLOSURE 0.03 0.03 WIDTH 0.3 ;
```



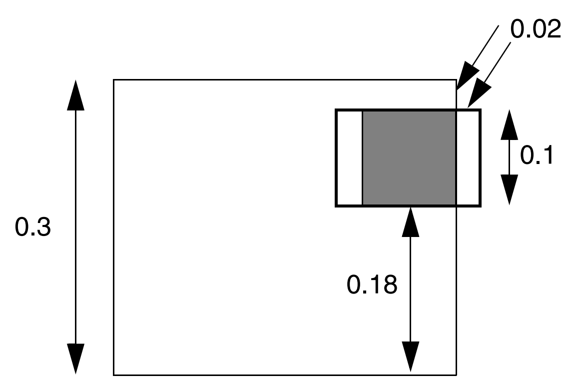
a) Violation, the right edge of the cut does not have 0.03 overhang



b) OK, the 0.02 protrusion would exempt the enclosure checking on the right edge, and the other three sides have enough overhangs



c) Violation, the protrusion exemption is met, but the basic enclosure of 0, 0.02 is not met



d) Violation, the right edge is exempted by the protrusion, but the top edge still fails the 0.03 overhang

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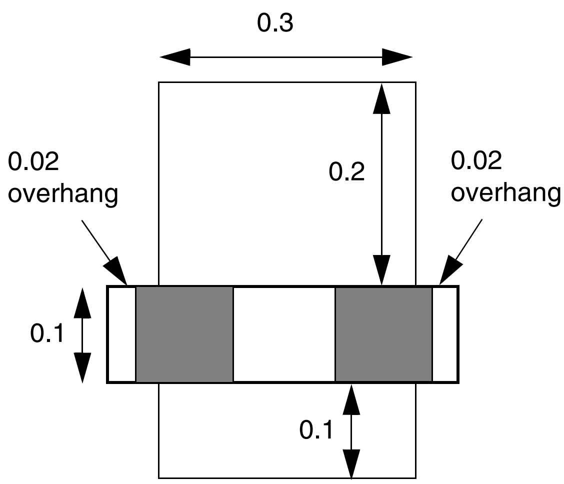
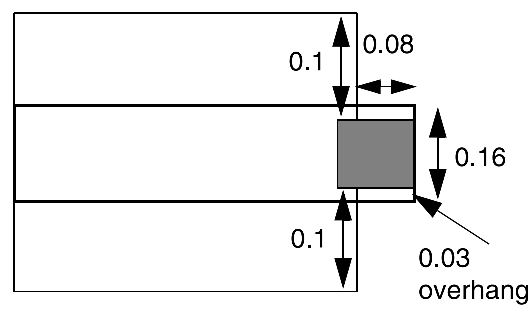


e) OK, the protrusion on left and right edges could exempt the corresponding checking on those edges of the cuts, and the rest of the edges have enough overhangs. In addition, the basic enclosure of 0, 0.02 is met



f) Violation, the width of the protrusion is considered as the width of the max fractured rectangle of 0.16, and the corresponding enclosure 0.02, 0.02 is not met.

Enclosure Joint Rule

You can create enclosure joint rule to specify the overhang between a non-redundant cut to two consecutive joints in either above or both above and below routing layer.

You can create a enclosure joint rule by using the following property definition:

```
PROPERTY LEF58_ENCLOSURETOJOINT
     "ENCLOSURETOJOINT [CUTCLASS className] [ABOVE | BELOW]
             toOneJointOverhang [toBothJointOverhang]
             [EOLMINLENGTH minLength]
             JOINTWIDTH jointWidth JOINTLENGTH spanLength
             ; " ;
```

Where:

`ABOVE | BELOW`

Specifies that the rule only applies if the via cut is in two consecutive joints on the above or below routing layer.

`CUTCLASS` *className*

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Defines the enclosure to joint rule for a specific cut class *className*. If CUTCLASS is defined in cut layer, then CUTCLASS must be specified in ENCLOSURETOJOINT statement.

For each cut class, one individual rule with the CUTCLASS keyword should be specified, if needed.

ENCLOSURETOJOINT *toOneJointOverhang toBothJointOverhang* JOINTWIDTH *jointWidth* JOINTLENGTH *spanLength*

Specifies the overhang between a non-redundant cut to two consecutive joints in either above or both above and below routing layer, with span greater than *spanLength* and not a EOL edge with length equal to the wire width, which is less than *jointWidth*, to be either *toOneJointOverhang* to one of the joints or *toBothJointOverhang* to both of the joints.

For redundant cuts sharing the same metal on the above and below routing layer, one of the cuts fulfilling the rule will be sufficient.
Type: Float, specified in microns

EOLMINLENGTH *minLength*

Indicates that the joint must not be a EOL edge with length greater than or equal to the specified *minLength* along both sides and the length of the EOL edge is no longer necessarily equal to the wire width.
Type: Float, specified in microns

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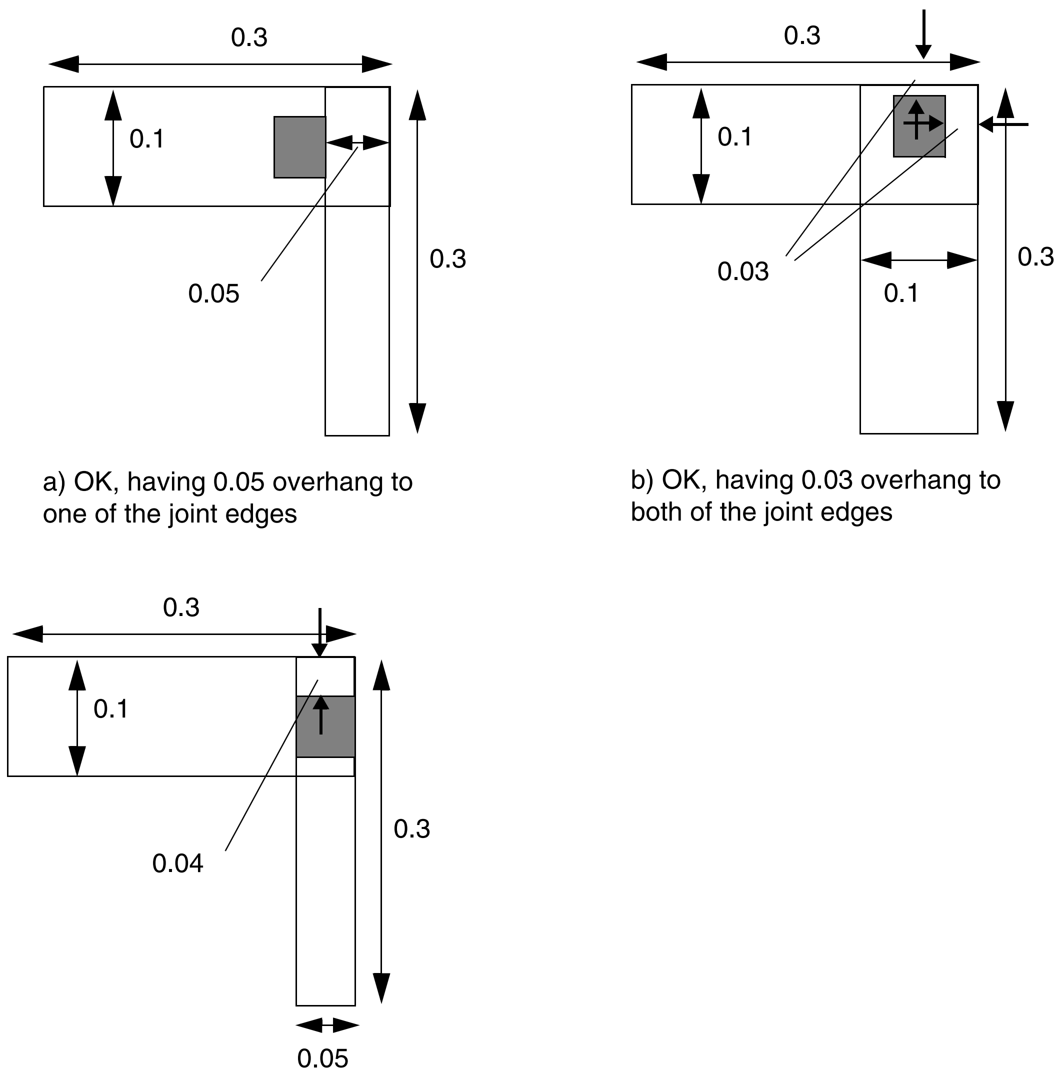
Enclosure Joint Rule Examples

Figure 2-75 Illustration of Enclosure Joint Rule

```
PROPERTY LEF58_ENCLOSURETOJOINT
   "ENCLOSURETOJOINT 0.05 0.03
       JOINTWIDTH 0.15 JOINTLENGTH 0.1  ;" ;
```



a) OK, having 0.05 overhang to one of the joint edges

b) OK, having 0.03 overhang to both of the joint edges

c) Violation, having 0.04 and 0.0 overhang to the joints will not meet either of the requirements

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EOL Enclosure Rule

You can create EOL enclosure rule to specify the enclosure requirement on a EOL edge with width less than the specified EOL width.

You can create a EOL enclosure rule by using the following property definition:

```
PROPERTY LEF58_EOLENCLOSURE
     "EOLENCLOSURE eolWidth [MINEOLWIDTH minEolWidth]
          [HORIZONTAL | VERTICAL]
          [INCLUDEEDGEALIGNED]
          [EQUALRECTWIDTH] [CUTCLASS className] [ABOVE | BELOW]
          {{LONGEDGEONLY | SHORTEDGEONLY} overhang
          | overhang
              [exactOverhang
              | PARALLELEDGE parSpace EXTENSION backwardExt forwardExt
                     [MINLENGTH minLength]
              | MINLENGTH minLength
              | ALLSIDES
              ]
          }
          ; " ;
```

Where:

All other keywords are the same as the existing LEF cut layer EOLENCLOSURE syntax.

ABOVE | BELOW	Specifies that the EOL enclosure rule only applies on the above or below routing layer.
ALLSIDES	Specifies that the specified enclosure, overhang, will be applied on all four sides of the cut.
CUTCLASS *className*	
	Defines the EOL enclosure rule for a specific cut class *className*. If CUTCLASS is defined in cut layer, then CUTCLASS must be specified in the EOLENCLOSURE statement.
	For each cut class, one individual rule with CUTCLASS keyword must be specified, if required.
EOLENCLOSURE *eolWidth*	

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Specifies the enclosure requirement on a EOL edge with width less than *eolWidth*. No overhang requirement is specified for a cut on a non-EOL edge in this statement, which is a supplement ENCLOSURE requirement on a EOL edge. The ENCLOSURE requirement on a non-EOL edge is still governed by other ENLCOSURE statements.
Type: Float, specified in microns

EQUALRECTWIDTH

Specifies that the EOL enclosure rule only applies to EOL edges with length equal to the wire width.

HORIZONTAL | VERTICAL

Specifies that the rule applies only if the EOL edge is in the given direction.

INCLUDEEDGEALIGNED

Specifies that the rule also applies when the cut edge is aligned to the EOL edge. For example, if the bottom cut edge is aligned to the top EOL edge in a rectilinear shape.

LONGEDGEONLY | SHORTEDGEONLY *overhang*

Specifies the ENCLOSURE EOL requirement of *overhang* is only applied to the side/long edges in LONGEDGEONLY or end/short edges in SHORTEDGEONLY of the rectangular cut, which means that a rectangular class name must be defined in CUTCLASS.

MINEOLWIDTH *minEolWidth*

Specifies that the rule only applies when *minEolWidth* is less than or equal to EOL width or length, which should be less than *eolWidth* and the EOL edge must touch a wire with width greater than or equal to *minEolWidth*.
Type: Float, specified in microns

MINLENGTH *minLength*

Indicates that the EOL enclosure only applies if EOL edge have length greater than or equal to the *minLength* along both sides. In other words, if the EOL length is less than the *minLength* along any one side, the edge is not a EOL edge to trigger the rule.
Type: Float, specified in microns

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`overhang [exactOverhang]`

Specifies that the enclosure EOL requirement is either exactly equal to *`exactOverhang`* or greater than or equal to *`overhang`*.
Type: Float, specified in microns

`PARALLELEDGE` *`parSpace`* `EXTENSION` *`backwardExt forwardExt`*
[`MINLENGTH` *`minLength`*]

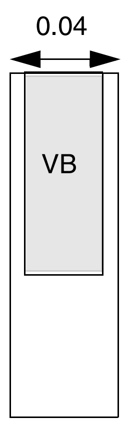
Indicates that the EOLenclosure rule only applies if there is a parallel edge on one side that is less than the *`parSpace`* subtracting the width of the EOL edge away and by extending *`backwardExt`* going backward and *`forwardExt`* going forward in the direction orthogonal to the EOL edge.

The `MINLENGTH` keyword indicates that if the EOL length is less than *`minLength`* along the side, then any parallel edge on that side is ignored, and the rule may not apply.
Type: Float, specified in microns

EOL Enclosure Rule Examples

Figure 2-76 Illustration of EOL Enclosure Rule with MINEOLWIDTH

```
PROPERTY LEF58_EOLENCLOSURE
  "EOLENCLOSURE 0.1 MINEOLWIDTH 0.05
     CUTCLASS VB SHORTEDGEONLY 0.02 ; " ;
```



b) OK, the rule does not apply on EOL width of 0.04 (< 0.05).

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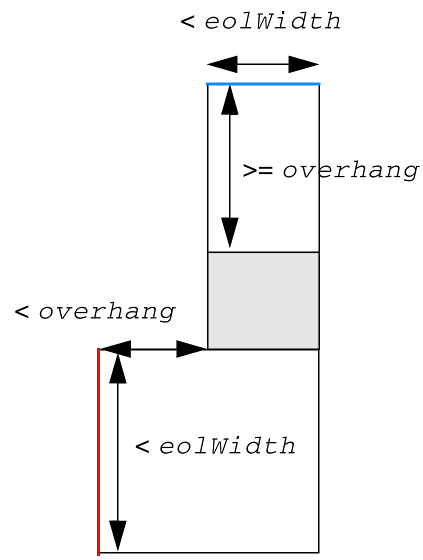
Figure 2-77 Illustration of EOL Enclosure Rule with INCLUDEEDGEALIGNED



It is a violation. The rule is OK based on the top blue EOL edge. However, it is a violation based on the left red EOL edge because the top EOL edge is aligned to the bottom cut edge. It is OK if `INCLUDEEDGEALIGNED` is omitted.

Illustration of
`EOLENCLOSURE` *`eolWidth`* `INCLUDEEDGEALIGNED`
`overhang`

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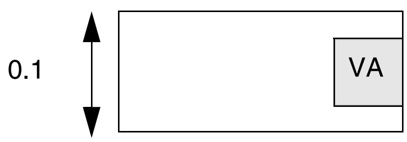
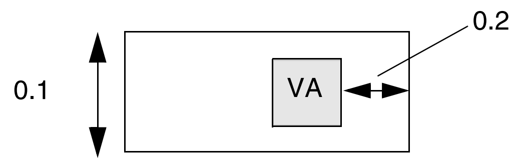
Figure 2-78 Illustration of EOL Enclosure Rule with EQUALRECTWIDTH

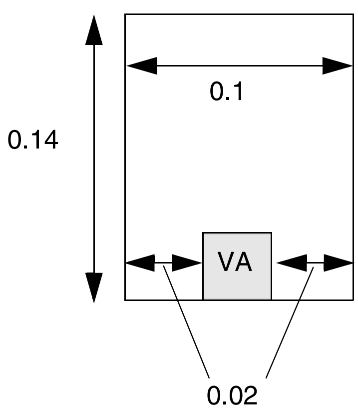
```
PROPERTY LEF58_EOLENCLOSURE
   "EOLENCLOSURE 0.15 EQUALRECTWIDTH
       CUTCLASS VA 0.05 0.0 ; " ;
```



a) OK, zero overhang is allowed



b) Violation, overhang should be either zero or >= 0.05



c) OK, the vertical edges are not EOL edges with length equal to wire width, and the rule does not apply on them. The bottom edge has zero overhang, which is allowed. It is a violation if EQUALRECTWIDTH is not specified.

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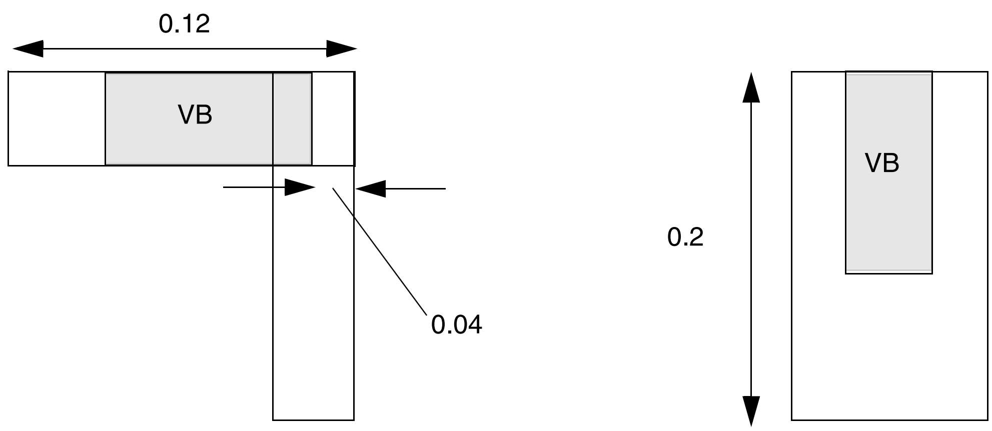
Figure 2-79 Illustration of EOL Enclosure Rule with LONGEDGEONLY

```
PROPERTY LEF58_EOLENCLOSURE
  "EOLENCLOSURE 0.15
     CUTCLASS VB
     LONGEDGEONLY 0.04 ; " ;
```



a) Violation, 0.04 overhang must be on the side/long edge when the cut is on a EOL edge

b) OK, the side/long edge of the cut is not on a EOL edge

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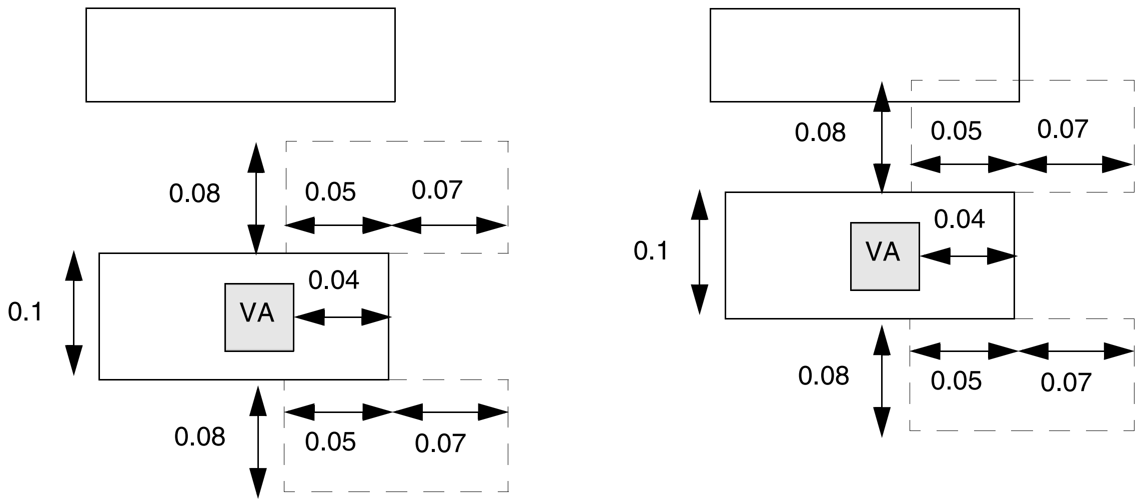
Figure 2-80 Illustration of EOL Enclosure Rule with PARALLELEDGE and EXTENSION

```
PROPERTY LEF58_EOLENCLOSURE
   "EOLENCLOSURE 0.15
       CUTCLASS VA 0.05 PARALLELEDGE 0.18
       EXTENSION 0.05 0.07 ; " ;
```



a) OK, the dotted line neighbor searching window is formed by extending 0.08 (0.18 - width of wire of 0.1) along the EOL edge, 0.05 & 0.07 backward & forward perpendicular to the EOL edge, and it could not find a neighbor, which would not trigger the EOL enclosure rule

b) Violation, having a neighbor would trigger the rule, and 0.04 (<= 0.05) enclosure is not enough

EOL Spacing Rule

You can create EOL spacing rule to specify the cut spacing on certain edges of a cut on a EOL edge above routing layer with width less than the specified EOL width.

You can create a EOL spacing rule by using the following property definition:

```
PROPERTY LEF58_EOLSPACING
     "EOLSPACING cutSpacing1 cutSpacing2
         [CUTCLASS className1 [{TO className2 cutSpacing1 cutSpacing2}...]]
         ENDWIDTH eolWidth PRL prl
         ENCLOSURE smallerOverhang equalOverhang
         EXTENSION sideExt backwardExt SPANLENGTH spanLength
         ; " ;
```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Where:

[CUTCLASS *className1* [{TO *className2 cutSpacing1 cutSpacing2*}...]]

Defines the EOL cut spacing rule for a specific cut class *className* to any other cuts, including those belonging to a different cut class. If TO is defined, the corresponding cut spacings will be applied between *className1* and *className2*. If CUTCLASS is defined in cut layer, then CUTCLASS must be specified in the EOLSPACING statement.

For each cut class, at the most, one individual rule with CUTCLASS keyword should be specified, if required.

ENCLOSURE *smallerOverhang equalOverhang*

Specifies that the EOL cut spacing rule only applies if overhang of the cut on one of the edges is less than *smallerOverhang* and the overhang on one of the orthogonal edges exactly equal to *equalOverhang* on above routing layer.
Type: Float, specified in microns

EOLSPACING *cutSpacing1 cutSpacing2* ENDWIDTH *eolWidth* PRL *prl*

Specifies the cut spacing on certain edges of a cut on a EOL edge on above routing layer with width less than *eolWidth*, if additional conditions (described later) are met. If the cut has a common parallel run length less than *prl* to a neighbor cut, *cutSpacing1* is applied. Otherwise, *cutSpacing2* is applied in a maximum projection style (when the cuts have common parallel run length greater than or equal to *prl*). If *prl* is negative, it means that a neighbor cut within abs(prl) distance, including exactly equal to abs(prl), beyond the edge of a cut will need to be *cutSpacing2* distance from it.
Type: Float, specified in microns

EXTENSION *sideExt backwardExt* SPANLENGTH *spanLength*

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Specifies that the EOL cut spacing rule only applies if certain neighbor wire condition is met. If the wire containing the cut has a span length greater than or equal to *spanLength* (having a EOL edge is not a necessary condition), and a neighbor wire with edge parallel to *backwardExt* direction must overlap with a search window by extending *sideExt* along the edge with *smallerOverhang* and *backwardExt* along the orthogonal edge that the cut has overhang exactly equal to *equalOverhang*. Otherwise (the wire containing the cut has a span length less than *spanLength*), a similar search window on the opposite side must not overlap with any neighbor wires with edge parallel to *backwardExt* direction as an additional condition. Then, the cut spacing requirement is applied to the opposite edges of the edge having enclosure less than *smallerOverhang* and the orthogonal edge that the cut has overhang exactly equal to *equalOverhang* with a neighbor.
Type: Float, specified in microns

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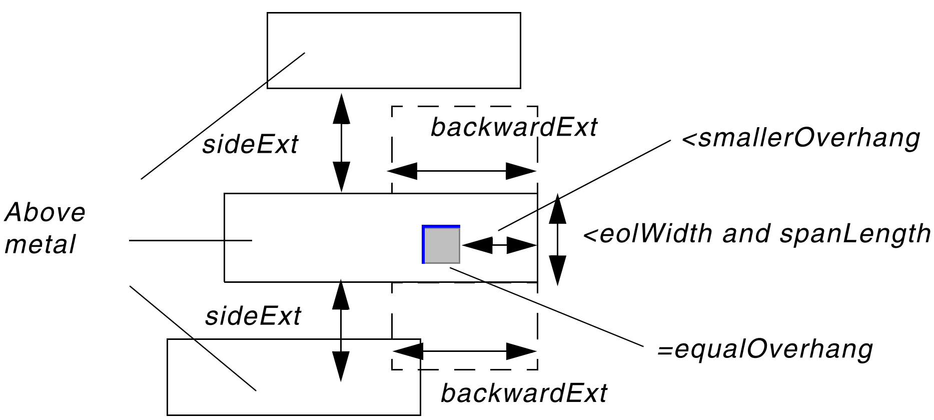
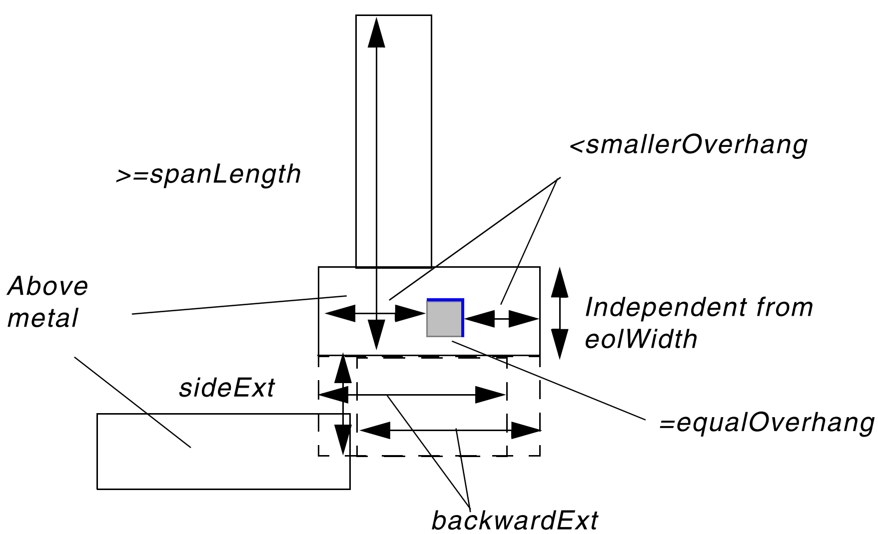
EOL Spacing Rule Examples

Figure 2-81 Illustration of EOLSPACING Rule

Illustration of `EOLSPACING` … `ENDWIDTH` *`eolWidth`* …
`ENCLOSURE` *`smallerOverhang equalOverhang`*
`EXTENSION` *`sideExt backwardExt`* `SPANLENGTH` *`spanLength`*



The blue edges are subject to the cut spacing rule



There are two EOL edges and two search window at the bottom. Since there is a bottom neighbor wire, the blue edges are subject to the cut spacing rule. Particularly, the left edge is opposite to a EOL edge that fulfills the neighboring condition.

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Forbidden Spacing Rule

You can create a forbidden spacing rule to specify forbidden spacing between two cuts.

You can create a forbidden spacing rule by using the following property definition:

```
PROPERTY LEF58_FORBIDDENSPACING
     "FORBIDDENSPACING CUTCLASS className minSpacing maxSpacing
          [SAMEMASK]
          [SHORTEDGEONLY | LONGEDGEONLY | HORIZONTAL | VERTICAL]
          [PRL {prl TO prlClassName}...]
          ; " ;
```

Where:

`FORBIDDENSPACING CUTCLASS` *`className minSpacing maxSpacing`*

Specifies that spacing between two cuts on a given layer belonging to the same *`className`* cut class cannot be in the range *`minSpacing`* <= *`spacing`* <= *`maxSpacing`*.
Type: Float, specified in microns

`HORIZONTAL | VERTICAL`

Specifies that forbidden spacing applies on only the given direction on the cut edge.

`LONGEDGEONLY` Specifies that the forbidden spacing rule only applies to the side/long edge of cuts with a common parallel run length greater than 0. This means that the *`className`* must be a rectangular cut class.

`SAMEMASK` Specifies that the forbidden spacing rule only applies to same-mask cuts.

`SHORTEDGEONLY` Specifies that the forbidden spacing rule only applies to the end/short edge of *`className`* cuts with a common parallel run length greater than 0. This means that *`className`* must be a rectangular cut class.

`PRL {`*`prl`* `TO` *`prlClassName`*`}...`

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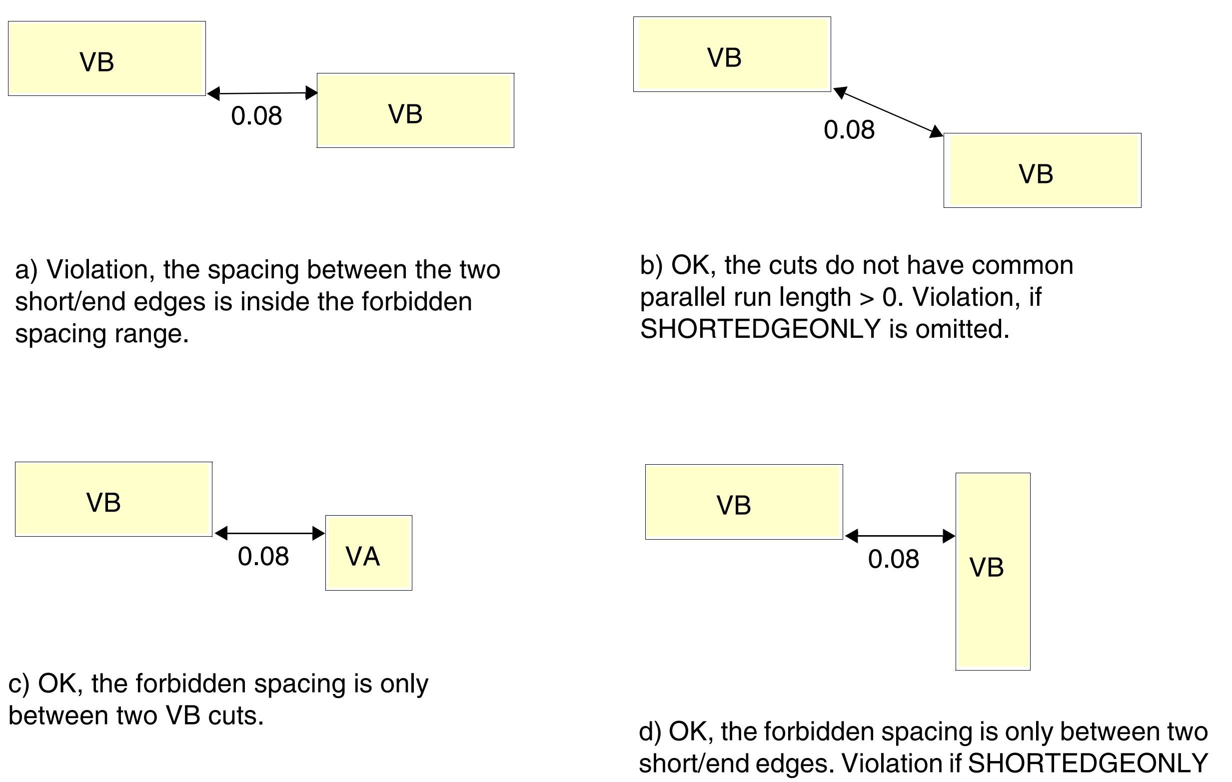
Specifies that forbidden spacing is between a *`className`* cut to *`prlClassName`* cut with common parallel run length greater than *`prl`*. If *`prl`* is negative, it is interpreted in the `WITHIN` style. For example, if `SHORTEDGEONLY` is also defined, it is similar to extending the end or short edge by abs(*`prl`*). If some of the cut classes are not defined in *`prlClassName`*, this means that there is no constraint between *`className`* to those undefined cut classes. See Figure 2-83 on page 370.
Type: Float, specified in microns

Forbidden Spacing Rule Examples

- The following forbidden spacing table rule illustrates forbidden spacing between two VB cuts:

```
PROPERTY LEF58_FORBIDDENSPACING
   "FORBIDDENSPACING CUTCLASS VB 0.07 0.09
     SHORTEDGEONLY ; " ;
```

Figure 2-82 Illustration of Forbidden Spacing Rule



a) Violation, the spacing between the two short/end edges is inside the forbidden spacing range.

b) OK, the cuts do not have common parallel run length > 0. Violation, if SHORTEDGEONLY is omitted.

c) OK, the forbidden spacing is only between two VB cuts.

d) OK, the forbidden spacing is only between two short/end edges. Violation if SHORTEDGEONLY is omitted.

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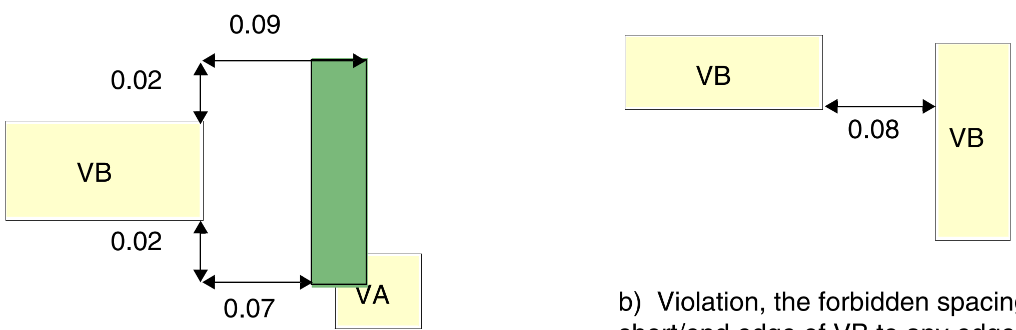
Figure 2-83 Illustration of Forbidden Spacing Rule

```
PROPERTY LEF58_FORBIDDENSPACING
     "FORBIDDENSPACING CUTCLASS VB 0.07 0.09
          SHORTEDGEONLY PRL -0.02 TO VA -0.03 TO VB ; " ;
```



a) Violation, the forbidden spacing is between a short/end edge of VB to VA within abs(-0.02), and the green region shows an forbidden area for any VA

b) Violation, the forbidden spacing is between a short/end edge of VB to any edges of VB. OK if PRL is omitted.

Keep-out Zone Rule

The keep-out zone rule can be used to demarcate a cut spacing zone in which cuts of other classes cannot overlap.

You can create a keep-out zone rule by using the following property definition:

```
PROPERTY LEF58_KEEPOUTZONE
     "KEEPOUTZONE CUTCLASS className1 [TO className2]
          [SAMEMASK] [SAMEMETAL | DIFFMETAL]
          [EXCEPTEXACTALIGNED [SIDE | END] spacing]
          {EXTENSION sideExtension forwardExtension |
          ENDEXTENSION endSideExtension endForwardExtension
          SIDEEXTENSION sideSideExtension sideForwardExtension |
          HORIZONTALEXTENSION horzSideExtension horzForwardExtension
          VERTICALEXTENSION vertSideExtension vertForwardExtension}
          SPIRALEXTENSION extension [CORNERONLY]
          ;" ;
```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Where:

`CORNERONLY`
: Specifies that the `SPIRALEXTENSION` keep-out zone applies only on the corners that are outside the zones of `EXTENSION`, `ENDEXTENSION` and `SIDEEXTENSION`, or `HORIZONTALEXTENSION` and `VERTICALEXTENSION`.

`DIFFMETAL`
: Specifies that the keep-out zone rule applies only for different-metal cuts.

`EXCEPTEXACTALIGNED [SIDE | END]` *`spacing`*

Specifies that if cuts are exactly aligned, they just need to be *`spacing`* apart. These cuts are not subjected to the keepout zone checking. For rectangular cuts, `SIDE` and `END` would mean that side/long edges and end/short edges are exactly aligned correspondingly. If neither of them is specified for rectangular cuts, exactly aligned cuts on any edge would only be spacing apart.

`HORIZONTALEXTENSION` *`horzSideExtension horzForwardExtension`*
`VERTICALEXTENSION` *`vertSideExtension vertForwardExtension`*

Specifies the extensions of the keep-out zone on the neighbor cuts based on the horizontal or vertical cut edge. This is particularly useful in case of a square cut.

Type: Float, specified in microns

`KEEPOUTZONE CUTCLASS` *`className1`* `[TO` *`className2`*`]`
`{EXTENSION` *`sideExtension forwardExtension`* `|`
`ENDEXTENSION` *`endSideExtension endForwardExtension`*
`SIDEEXTENSION` *`sideSideExtension sideForwardExtension`*`}`
`SPIRALEXTENSION` *`extension`*

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Specifies a keep-out zone for cut class *className1* to cut class *className2*, if specified, or any cut classes. ENDEXTENSION and SIDEEXTENSION must be used only on rectangular cuts.
EXTENSION forms a keep-out zone by extending *forwardExtension* on both sides on any cut edges and *sideExtension* in the orthogonal direction toward outside the cut.
SPIRALEXTENSION forms another keep-out zone by extending *extension* on all cut edges in Euclidean measurement. It is a violation if a *className2* cut, if specified, or any cut is inside any of the two keep-out zones. Similarly, ENDEXTENSION and SIDEEXTENSION specify the extensions on the end or short edges and the side or long cut edges of *className1*, respectively.
Type: Float, specified in microns

Note that this keep-out zone rule is an alternative cut spacing definition of CUTCLASS SPACINGTABLE. If any defined cut class does not appear in CUTCLASS SPACINGTABLE, then one or more KEEPOUTZONE statements must be specified for that cut class to define its cut spacing to the other cut classes. For example, suppose there are three cut classes: VA, VB, and VC. If VA is not defined in CUTCLASS SPACINGTABLE, then VA must be defined as *className1* in a KEEPOUTZONE statement. There can either be one KEEPOUTZONE statement without a TO cut class (so that it is applied to both VB and VC) or there can be two separate KEEPOUTZONE statements, one with TO VB and another with TO VC.

SAMEMASK — Specifies the keep-out zone rule applies only to same-mask cuts.

SAMEMETAL — Specifies that the keep-out zone rule applies only for same-metal cuts.

Keep-out Zone Rule Examples

- The following example illustrates the keep-out zone rule:

```
KEEPOUTZONE CUTCLASS VB TO VA
   ENDEXTENSION endSideExtension endForwardExtension
   SIDEEXTENSION sideSideExtension sideForwardExtension
   SPIRIALEXTENSION extension
```


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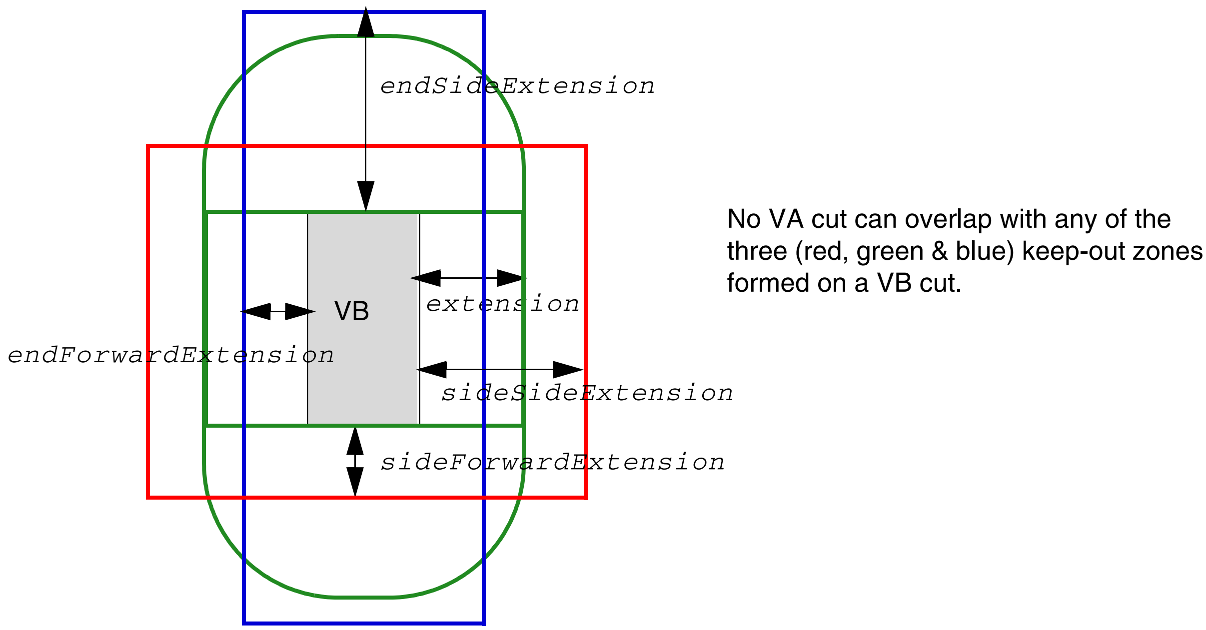
Figure 2-84 Illustration of the Keep-out Zone Rule



■ The following example illustrates the keep-out zone rule with `CORNERONLY`:

```
KEEPOUTZONE CUTCLASS VB TO VA
   ENDEXTENSION endSideExtension endForwardExtension
   SIDEEXTENSION sideSideExtension sideForwardExtension
   SPIRIALEXTENSION extension CORNERONLY
```


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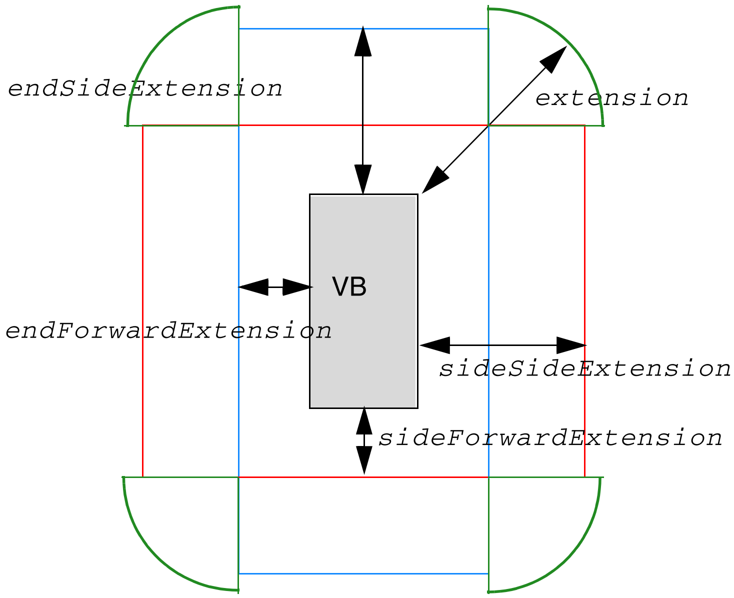
Figure 2-85 Illustration of the Keep-out Zone Rule with CORNERONLY



No VA cut can overlap with any of the three (red, green and blue) keep-out zones formed on a VB cut. The green keep-out zone exists only in the corners that are not covered by the red and blue keep-out zones.

- The following example illustrates the keep-out zone rule with `HORIZONTALEXTENSION` and `VERTICALEXTENSION`:

```
KEEPOUTZONE CUTCLASS VA TO VA
   HORIZONTALEXTENSION horzSideExtension horzForwardExtension
   VERTICALEXTENSION vertSideExtension vertForwardExtension
   SPIRIALEXTENSION extension
```


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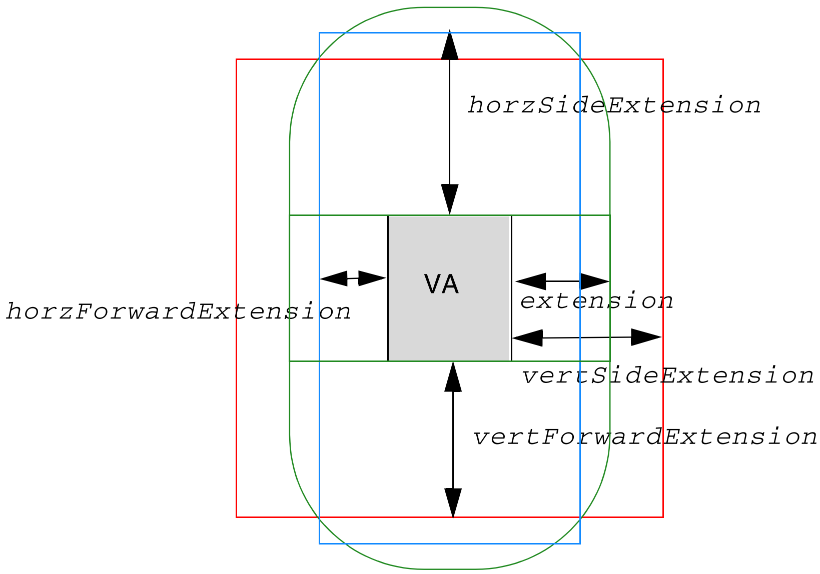
Figure 2-86 Illustration of the Keep-out Zone Rule with HORIZONTALEXTENSION and VERTICALEXTENSION



No VA cut can overlap with any of the three (red, green and blue) keep-out zones formed on a VA cut.

Manufacturing Grid Rule

You can create a manufacturing grid rule to specify the manufacturing grid on a given layer to override the library manufacturing grid value.

You can create a manufacturing grid rule by using the following property definition:

```
PROPERTY LEF58_MANUFACTURINGGRID
     "MANUFACTURINGGRID value
     ; " ;
```

Where:

`MANUFACTURINGGRID` *`value`*

Specifies the manufacturing grid on the given layer that overrides the library manufacturing grid value.
Type: Float, specified in microns

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Maximum Spacing Rule

You can create a maximum spacing rule to specify the maximum spacing between any two cuts.

You can create a maximum spacing rule by using the following property definition:

```
PROPERTY LEF58_MAXSPACING
     "MAXSPACING spacing [CUTCLASS className]
     ;" ;
```

Where:

`CUTCLASS` *className*

Specifies that the maximum spacing between cuts of the given *className* to any cuts. If `CUTCLASS` is defined in cut layer, then `CUTCLASS` must be specified in the `MAXSPACING` statement.

You can specify individual rules with the `CUTCLASS` keyword for each cut class, if needed.

`MAXSPACING` *spacing*

Specifies the maximum spacing between any two cuts.
Type: Float, specified in microns

Minimum Enclosure Rule

You can create a minimum enclosure rule to specify the minimum enclosure requirements for cuts on the given cut class. These requirements are typically applied on cell objects.

You can create a maximum spacing rule by using the following property definition:

```
PROPERTY LEF58_MINENCLOSURE
     "MINENCLOSURE CUTCLASS className [ABOVE | BELOW]
        overhang1 overhang2
        ; " ;
```

Where:

`MINENCLOSURE CUTCLASS` *className* `[ABOVE | BELOW]`
overhang1 overhang2

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Specifies that the cuts on the given cut class of *className* should have minimum enclosures of *overhang1* on two opposite sides and *overhang2* on the other two opposite sides on the above and below metal routing layers. If ABOVE or BELOW is specified, the enclosure applies only on the given above or below metal routing layer. These enclosure requirements are not used for routing. They specify the minimum enclosures typically used on some cell objects.
Type: Float, specified in microns

No Cut Class Rule

You can create a no cut class rule to specify that a given cut class cannot be used on a layer.

You can create a no cut class rule by using the following property definition:

```
PROPERTY LEF58_NOCUTCLASS
     "NOCUTCLASS className [EXCEPTMULTICUTS]
     ; " ;
```

Where:

EXCEPTMULTICUTS

Specifies that only multi-cut via of the given cut class with at least one same metal on a layer could still be used on the cut layer.

NOCUTCLASS *className*

Specifies that the given cut class cannot be used on this layer. A typical usage is to define the same set of cut classes on a base layer and the REGION layer corresponding to that base layer and then to specify one of these cut classes as NOCUTCLASS in the base layer and another cut class as NOCUTCLASS on the REGION layer.
Type: String

No Metal Spacing Rule

You can create a no metal spacing rule by using the following property definition:

```
PROPERTY LEF58_NOMETALSPACING
     "NOMETALSPACING spacing BOTHSPACING bothSpacing [PRL prl]
           [SAMEMASK] [CUTCLASS className]
           NEIGHBORMETAL metalPrl metalWithin
```


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```
   noMetalPrl {noMetalWithin | noMetalLowWithin noMetalHighWithin}
[SAMESIDE]
[ABOVE | BELOW]
[EXCEPTLINEENDSPACING [INCLUDELESSTHAN] lineEndSpacing]
;" ;
```

Where:

`ABOVE | BELOW` Specifies that the metal search windows should be formed only on the given above or below metal layer.

`CUTCLASS` *`className`* Specifies that the cut spacing rule applies only to the cuts of the given cut class. If `CUTCLASS` is defined in a cut layer but `CUTCLASS` is not specified in the `NOMETALSPACING` statement, the rule would be applied on all of the cuts belonging to any cut class.

You can specify individual rules with the `CUTCLASS` keyword for each cut class, if required.

`EXCEPTLINEENDSPACING [INCLUDELESSTHAN]` *`lineEndSpacing`*

Specifies that the rule does not apply if the metal neighbors just outside the metal search window formed by *`noMetalPrl`* and *`noMetalWithin`* have a spacing exactly equal to *`lineEndSpacing`*. If `INCLUDELESSTHAN` is specified, the checking condition is changed to less than or equal to *`lineEndSpacing`*.

This syntax works on the metal search window formed by *`metalPrl`* and *`metalWithin`* as well when `SAMESIDE` is also specified. This means that the rule would be applied if the metal neighbors just outside the metal search window formed by *`metalPrl`* and *`metalWithin`* have a spacing exactly equal to *`lineEndSpacing`*. In other words, when the wire gap is exactly equal to *`lineEndSpacing`*, it is as if the gap were filled with metal. If `INCLUDELESSTHAN` is specified, the checking condition is changed to less than or equal to *`lineEndSpacing`*.

Type: Float, specified in microns

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`NOMETALSPACING` *`spacing`* `BOTHSPACING` *`bothSpacing`* `[PRL` *`prl`*`]`
`NEIGHBORMETAL` *`metalPrl metalWithin`*
`noMetalPrl` `{`*`noMetalWithin`* `|` *`noMetalLowWithin`*
`noMetalHighWithin``}`
`[SAMESIDE]`

Specifies that the spacing of a cut edge that fulfills specific metal neighbor conditions, described later, to any cut should be greater than or equal to *spacing*. The spacing between two edges of two cuts that fulfill those specific metal neighbor conditions is greater than or equal to *bothSpacing* measured in the orthogonal direction of the cut edges. If PRL is defined, the cuts must have PRL greater than *prl* to trigger *bothSpacing*. Form metal search windows on two opposite cut edges, or on the same edge if SAMESIDE is defined, in the preferred direction on the above or below metal layers by *metalPrl* and *metalWithin* as one window and either *noMetalPrl* and *noMetalWithin* or *noMetalLowWithin noMetalHighWithin* as another window. For negative values of *metalPrl* and *noMetalPrl*, the first metal search window must overlap a metal neighbor and the second metal search window should not overlap with a metal neighbor. For positive values of *metalPrl* and *noMetalPrl*, the first metal search window must have a metal neighbor with PRL greater than *metalPrl* with the cut edge and the second metal search window must have no metal neighbor with PRL greater than *noMetalPrl* with the cut edge. The cut edge without a metal neighbor is subjected to this rule.

Type: Float, specified in microns

`SAMEMASK` Specifies that the cut spacing rule applies only to same-mask cuts.

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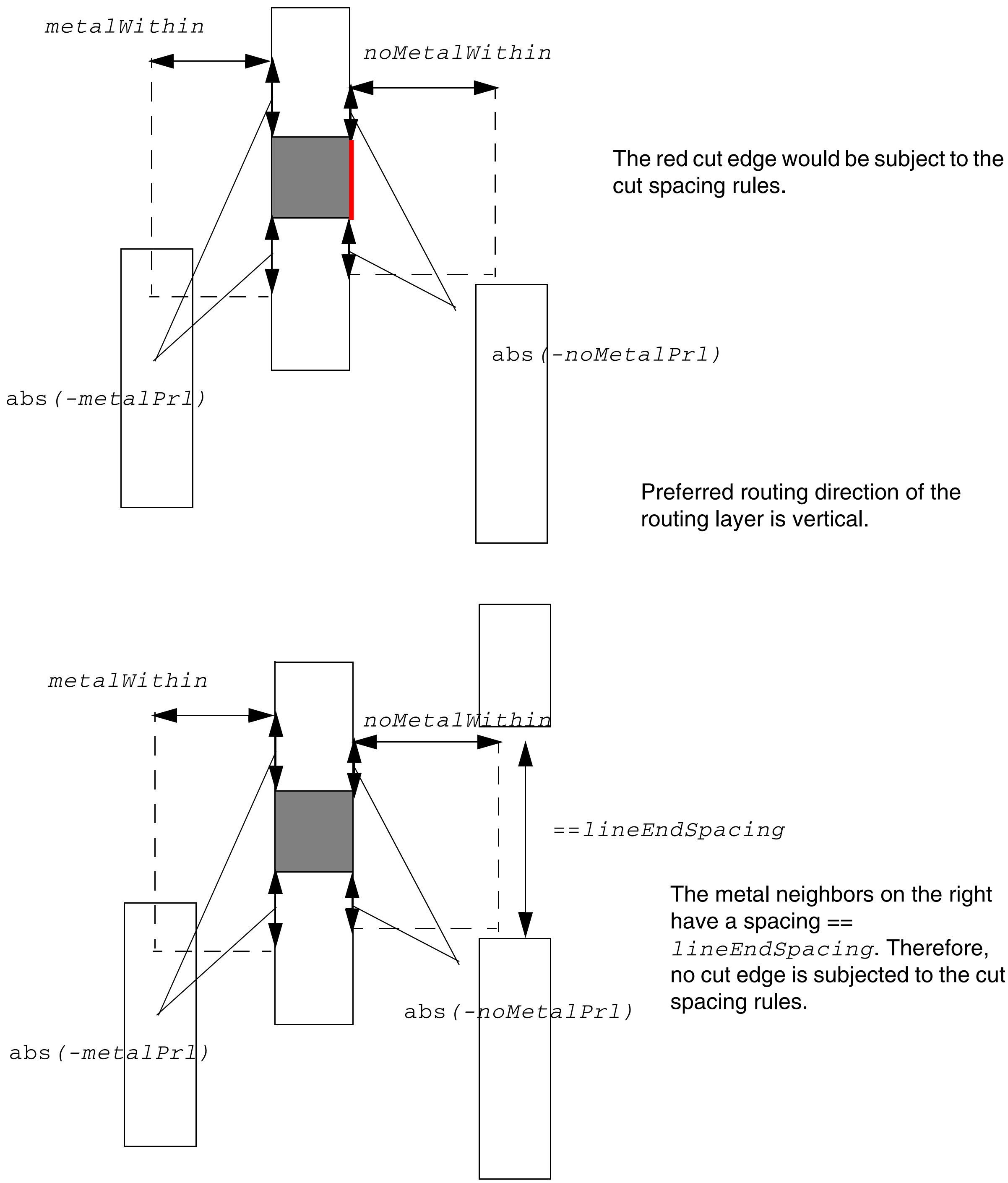
Figure 2-87 Illustration of No Metal Spacing Rule with NEIGHBORMETAL



Illustrations of `NEIGHBORMETAL`
-metalPrl metalWithin -noMetalPrl noMetalWithin
`EXCEPTLINEENDSPACING` *lineEndSpacing*

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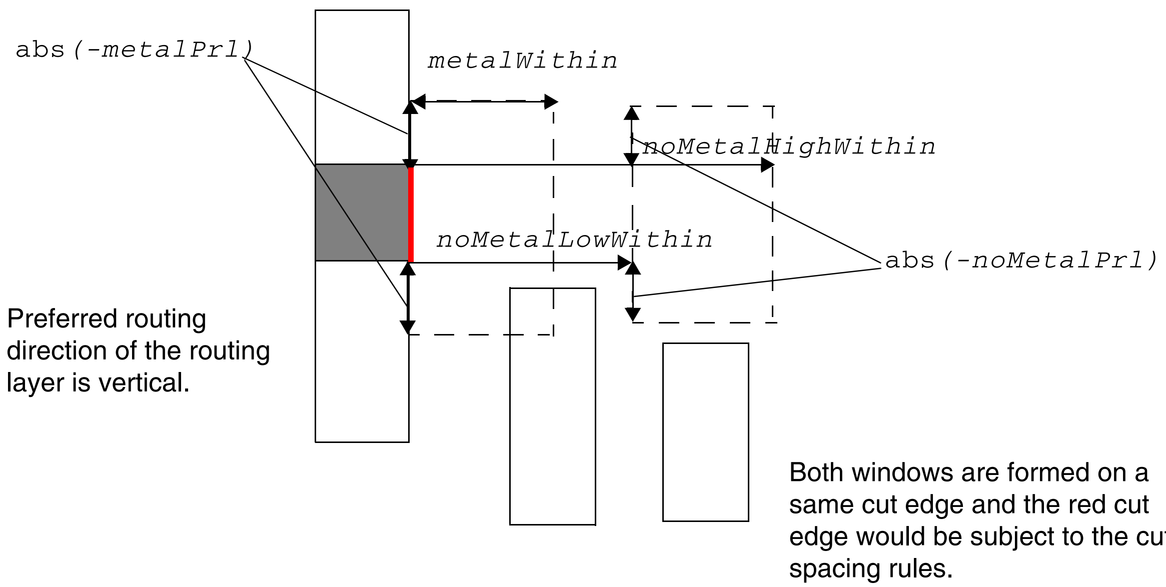
Figure 2-88 Illustration of No Metal Spacing Rule with SAMESIDE in NEIGHBORMETAL



Illustrations of `NEIGHBORMETAL`
-metalPrl metalWithin -noMetalPrl
noMetalLowWithin noMetalHighWithin `SAMESIDE`

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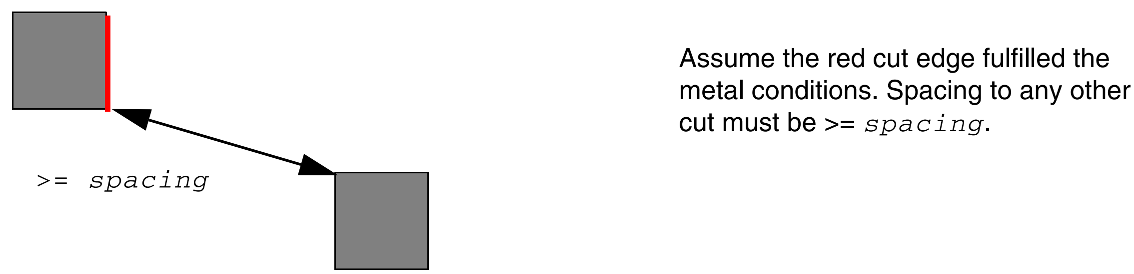
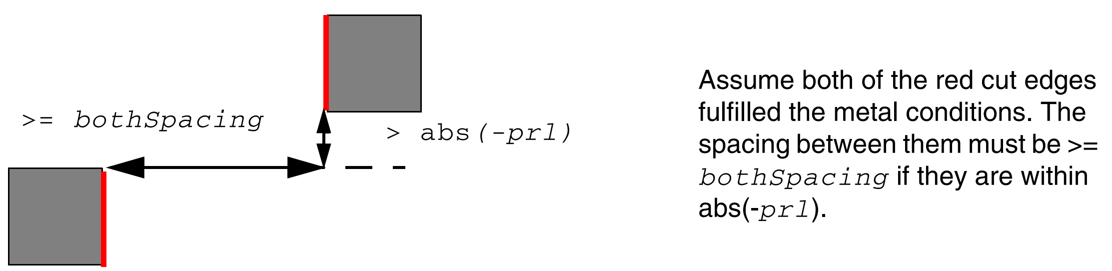
Figure 2-89 Illustration of No Metal Spacing Rule with BOTHSPACING





Illustrations of `NOMETALSPACING` *`spacing`*
`BOTHSPACING` *`bothSpacing`* `PRL` *`-prl`*

Spacing with Same Metal Rule

You can use spacing with same metal rules to:

- Require extra space between cuts on different nets that overlap orthogonally, in order to avoid stress migration between the cuts. This rule does not apply if the cuts have the same metal above or below.
- Require extra spacing between different cut layers unless they are connected by a single metal shape.

You can create a spacing with same metal rule by using the following property definition:

```
PROPERTY LEF58_SPACING
     "SPACING cutSpacing
         [SAMEMASK
         |MAXXY
         |[CENTERTOCENTER]
            [SAMENET | SAMEMETAL | SAMEVIA]
```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```
[ LAYER secondLayerName
    [STACK
    | ORTHOGONALSPACING orthogonalSpacing]
    | CUTCLASS className
          [SHORTEDGEONLY [PRL prl]
          | LONGEDGEONLY
          | CONCAVECORNER
               [WIDTH width ENCLOSURE enclosure
                   EDGELENGTH edgeLength
               | PARALLEL parLength WITHIN parWithin
                   ENCLOSURE enclosure
               | EDGELENGTH edgeLength
                   ENCLOSURE edgeEnclosure adjEnclosure]
               | WRONGDIRECTIONONLY xSize ySize
                   [LAYER adjacentCutLayer]]
          | EXTENSION extension
          | NONEOLCONVEXCORNER eolWidth
               [MINLENGTH minLength]
          | ABOVEWIDTH width [ENCLOSURE enclosure]
          |WRONGDIRECTION
               [EXCEPTRECTANGLE | OVERLAPONLY]
          |DIFFCONCAVECORNER ]]]
 | ADJACENTCUTS {2 | 3 | 4}
    [EXACTALIGNED exactAlignedCut]
    [TWOCUTS twoCuts [TWOCUTSSPACING twoCutsSpacing]
          [SAMECUT]
    | EACHCUTS numCuts [EXCEPTEVENNUMCUT]]
    WITHIN {cutWithin | cutWithin1 cutWithin2}
    [EXCEPTSAMEPGNET]
    [EXCEPTALLWITHIN exceptAllWithin]
    [ENCLOSURE [ABOVE|BELOW] enclosure
          [ANYCUT] [EXCEPTSAMEMETAL]
          [METALMASK metalMaskNum]
          [METALWIDTH maxWidth]
          [EXCEPTLINEENDGAP]]
    [CUTCLASS className [TO {ALL | VIAGROUP groupName
         [CUTCLASSLIST {classNameList}...]}]
     |VIAGROUP groupName TO CUTCLASS {className|ALL}]
    [NOPRL [HORIZONTAL|VERTICAL] | SIDEPARALLELOVERLAP
    |PRL prl [HORIZONTAL|VERTICAL] [SAMESIDE]]
    [SAMEMASK]
| PARALLELOVERLAP [EXCEPTSAMENET | EXCEPTSAMEMETAL
       |EXCEPTSAMEMETALOVERLAP | EXCEPTSAMEVIA]
| PARALLELWITHIN within [EXCEPTSAMENET]
       [CUTCLASS className
          [LONGEDGEONLY
          | ENCLOSURE enclosure {ABOVE | BELOW}
              PARALLEL parLength WITHIN parWithin]]
| SAMEMETALSHAREDEDGE parwithin [ABOVE][CUTCLASS className]
       [EXCEPTTWOEDGES][EXCEPTSAMEVIA numCut]
| AREA cutArea] ;..." ;
```


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Where:

All other keywords are the same as the existing LEF cut layer `SPACING` syntax.

`ABOVEWIDTH` *`width`* `[ENCLOSURE` *`enclosure`*`]`

Specifies the cut to metal spacing applies only on the above metal layer wire with width greater than or equal to the *`width`*. This means that the second layer must be a metal layer. In addition, the spacing applies only to different-net objects.

When you define the `ENCLOSURE` keyword, the spacing applies only if the enclosure on layer *`secondLayerName`* of the cut is less than the specified *`enclosure`*, including the enclosure at the corners.
Type: Float, specified in microns

`ADJACENTCUTS … SAMEMASK`

Specifies that the adjacent cut rule only applies to same mask cuts. The presence of different mask cuts is irrelevant for this adjacent cut rule checking.

`ADJACENTCUTS… WITHIN {`*`cutWithin`* `|` *`cutWithin1 cutWithin2`*`}`

Specifies that the adjacent cut rule only counts neighbor cut greater than or equal to *`cutWithin1`* and less than *`cutWithin2`* when two within values are specified. See Figure 2-110 on page 414.

`ADJACENTCUTS {2 | 3 | 4} [EXACTALIGNED` *`exactAlignedCut`*`] WITHIN` *`cutWithin`* `[SIDEPARALLELOVERLAP]`

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The EXACTALIGNED keyword specifies that the adjacent cut spacing rule applies when the cut has at least *exactAlignedCut* via cuts that are perfectly aligned horizontally and/or vertically and are less than *cutWithin* distance from each other. Otherwise, the adjacent cut spacing rule applies when the number of adjacent cuts is equal to or greater than the specified number of cuts after the ADJACENTCUTS keyword, which must be smaller than *exactAlignedCut*, and at least one adjacent cut is not exactly aligned.

The SIDEPARALLELOVERLAP keyword indicates that the adjacent cut spacing rule applies only when there is a parallel edge overlap greater than 0 side by side between two rectangular cut vias.

Note: Do not use the SIDEPARALLELOVERLAP for square cut classes.

ANYCUT

Specifies that the rule applies if any cut, checking cut or adjacent neighbor cuts, has enclosure less than *enclosure* on any one side in the preferred routing direction. See Figure 2-117 on page 420.

CUTCLASS *className* [TO {ALL | VIAGROUP *groupName*
[CUTCLASSLIST {*classNameList*}...]}]

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Defines the adjacent cut spacing rule for a specific cut class (*className*). If CUTCLASS is defined on a cut layer but this CUTCLASS keyword is not defined, the adjacent cut rule is applied to any cut belonging to any cut class.

You can specify individual rules with the CUTCLASS keyword for each cut class, if needed.

The TO ALL keyword specifies that the adjacent cut rule applies between cuts of *className* to cuts of any cut classes.

When both TO ALL and TWOCUTS are specified, the two cuts having multiple neighbor cuts to violate TWOCUTS must still belong to *className* to be a violation.

The TO VIAGROUP keyword specifies that the adjacent cut rule applies between cuts of *className* to the cuts belonging to *groupName*. Cuts belonging to the same via group should be counted as one neighbor cut. Only the cuts belonging to different via groups should be counted against the adjacent cut rule.

CUTCLASSLIST {*classNameList*}... specifies that the neighbor cuts must be either cuts belonging to the via group *groupName* or non-via group cuts belonging to the given *classNameList*.

See Figure 2-124 on page 427.

CUTCLASS *className* [LONGEDGEONLY]

Specifies that the parallel within cut spacing only applies to cuts belonging to the same *className*. If *className* is a rectangular cut class, then you can specify the LONGEDGEONLY keyword to limit the cut spacing between two long/side edges having a neighbor wire on the above metal between the cuts.

CUTCLASS *className* [SHORTEDGEONLY [PRL *prl*] | CONCAVECORNER]

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Defines the inter-layer cut spacing of *className* to a metal in *secondLayerName*, which must be a routing layer. If *className* is a rectangular cut class, the SHORTEDGEONLY statement will limit the spacing from an end/short edge of the cut to a metal when there is common parallel run length greater than 0 or greater than *prl* if PRL is defined between them. This cut to metal spacing is ignored for same-net. See Figure 2-94 on page 400.

The CONCAVECORNER keyword specifies the spacing of the cut to the concave corner of a wire that contains the cut. See Figure 2-96 on page 402.

DIFFCONCAVECORNER — Specifies a cut to a different-metal concave corner, where the spacing is measured in Euclidean style. Cuts must face the concave corner to be violations. See Figure 2-107 on page 412.

EACHCUTS *numCuts* [EXCEPTEVENNUMCUT]

Specifies that the adjacent cut rule applies only if the following conditions are met. A cut has greater than or equal to the specified number of cuts, defined with the ADAJCENTCUTS keyword, of neighbors within *cutWithin*. A special group of cuts is formed between the triggering cuts and its neighbors, such that each of the cuts in the group has greater than or equal to *numCuts* cuts of the group within *cutWithin*. Then, count the number of neighboring cuts in that group. If the count is greater than or equal to the specified number of cuts defined with the ADAJCENTCUTS keyword, it is a violation. If EXCEPTEVENNUMCUT is defined, and if the count is even, it is not a violation. In other words, it is only a violation if the count is odd with EXCEPTEVENNUMCUT. If there are two or more cuts having greater than *numCuts* neighbors, EXCEPTEVENNUMCUT cannot be applied. *numCuts* should be less than or equal to the specified number of cuts defined with the ADAJCENTCUTS keyword. In addition, when EACHCUTS is defined, *cutSpacing* and *cutWithin* should have same value and *cutSpacing* is not used.

Type: Integer

EDGELENGTH *edgeLength* ENCLOSURE *edgeEnclosure adjEnclosure*

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Specifies that an edge of a concave corner of a wire containing a cut has length less than *edgeLength*, the other end of that edge is a convex corner and the cut has parallel run length greater than or equal to 0 to that edge, the spacing between that edge and the facing cut edge must be greater than or equal to *cutSpacing*. In addition, if that spacing/enclosure is less than *edgeEnclosure* (and greater than or equal to *cutSpacing*, which must be less than *edgeEnclosure*), the enclosure of the adjacent two opposite sides must be greater than or equal to *adjEnclosure*. Otherwise, it is a violation.
Type: Float, specified in microns

ENCLOSURE [ABOVE | BELOW] *enclosure*

Specifies that the adjacent cut rule applies only if the checking cut has enclosure less than *enclosure* on any one side in the preferred routing direction on the above and below metal layers. If ABOVE or BELOW is specified, the enclosure requirement would be applied only on the specified above or below metal layer. See Figure 2-115 on page 418 for an example.

Type: Float, specified in microns

ENCLOSURE *enclosure* {ABOVE | BELOW}
PARALLEL *parLength* WITHIN *parWithin*

Specifies that the cut spacing only applies if the enclosure on either above or below metal layer (depending on whether ABOVE or BELOW is defined) is less than *enclosure* on a side having a neighbor wire within less than *parWithin* and common parallel run length to the cut greater than or equal to *parLength* on one and only one side. The cut spacing is only applied to the side without a neighbor wire and on the facing edge of the neighbor cut. If *parLength* is a negative value, the two edges would be checked in a WITHIN style. In addition, the rule is applied between a cut of *className* to cuts of any cut classes. See Figure 2-130 on page 433.
Type: Float, specified in microns

EXCEPTALLWITHIN *exceptAllWithin*

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Specifies the exception that if a cut has all neighbors within *cutWithin* or range of *cutWithin1* and *cutWithin2*, which are also within *exceptAllWithin*, it is allowed. *exceptAllWithin* should be less than or equal to *cutSpacing*. See Figure 2-116 on page 419.
Type: Float, specified in microns

EXCEPTLINEENDGAP
: Specifies that the adjacent cut rule does not apply if a wire containing a cut on which the enclosure condition is applied fulfills the LINEENDGAP condition on the above or below metal layer.

EXCEPTRECTANGLE
: Specifies that the cut to metal rule does not apply on a rectangular metal shapes.

EXCEPTSAMEMETAL
: Specifies that multiple neighbor cuts sharing the same metal on the above or below metal layer, if ABOVE or BELOW is specified, would count only as one individual neighbor cut, and each of the same-metal cuts is considered separately for the adjacent rule as if the remaining same-metal cuts did not exist.

EXCEPTSAMENET
: Indicates that the parallel within rule does not apply to same-net cuts.

EXCEPTSAMEVIA
: Indicates that the parallel overlap rule does not apply to cuts that share the same metal shapes both above and below the metal layers that cover the overlap area between the cuts.

EXTENSION *extension*
: Specifies that the *extension* should be extended on the cut edges that do not fulfill the overhang defined in ENCLOSUREEDGE OPPOSITE before *cutSpacing* is applied between the extended edges to the metal in *secondLayerName*.
Type: Float, specified in microns

LONGEDGEONLY
: Specifies that the cut-to-metal spacing is applied only on the side or long cut edges with PRL greater than 0. The specified *className* must be a rectangular cut class.

MASKOVERLAP
: Specifies the cut to metal, containing the cut, spacing on the overlap area of two different masks. This means that the second layer must be a metal layer.

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`MAXXY`

Indicates that the *`cutSpacing`* value is used as the largest x or y distance for spacing between objects. This keyword can be applied only when `EUCLIDEAN` is specified in `CLEARANCEMEASURE`. See Figure 2-90 on page 396.

`METALMASK` *`metalMaskNum`*

Specifies that the adjacent cut rule applies only if the wire containing the cut on which the enclosure condition is applied belongs to *`metalMaskNum`*.

Type: Integer

`METALWIDTH` *`maxWidth`*

Specifies that the adjacent cut rule applies only if a wire containing a cut on which the enclosure condition is applied has a width less than or equal to *`maxWidth`*. See Figure 2-118 on page 421.

Type: Float, specified in microns

`NONEOLCONVEXCORNER` *`eolWidth`* `[MINLENGTH` *`minLength`*`]`

Specifies the spacing of a cut to a convex corner that does not touch a EOL edge with width less than *`eolWidth`* of a metal shape containing the cut in form of a triangle formed by the smaller edge length or *`cutSpacing`*. In other words, the triangle is a keep-out region that the cut could not overlap.

The *`MINLENGTH`* keyword indicates that the EOL edge must have length greater than or equal to the *`minLength`* along both sides. In other words, if the EOL length is less than *`minLength`* along any one side, then the edge is not a EOL edge.
Type: Float, specified in microns

`NOPRL [HORIZONTAL|VERTICAL]`

The `NOPRL` keyword specifies that the adjacent cut rule applies only among cuts with no common parallel run length. If `HORIZONTAL` or `VERTICAL` is specified, the cuts with horizontal or vertical common parallel run length, respectively, would not be counted as adjacent cut neighbors. The `EXACTALIGNED` keyword cannot be specified along with `NOPRL`.

`ORTHOGONALSPACING` *`orthogonalSpacing`*

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Specifies the different-net spacing between objects on the cut layer and objects on the *secondLayerName*, which must be a routing layer. The rule does not apply to same-net objects.

The spacing check is done by using the following method:

The difference between *cutSpacing* and *orthogonalSpacing* is calculated, and half of that difference is shrunk on two sides of the cut and grown on the other two sides. Then, a Euclidean spacing of the average of *cutSpacing* and *orthogonalSpacing* is applied to the newly formed rectangle geometry. The shrink and grow operations are done on the alternated sides to form another checking region. A violation occurs if neighbor wires on the *secondLayerName* are found in both the regions. See Figure 2-92 on page 398 and Figure 2-93 on page 399.

Type: Float, specified in microns

Note: You cannot use ORTHOGONALSPACING along with CENTERTOCENTER, SAMENET, SAMEMETAL, or SAMEVIA.

OVERLAPONLY

Specifies the spacing of a cut to the overlap area between a preferred direction wire and a non-preferred direction wire. The cut must be inside this rectilinear wire. See Figure 2-106 on page 411.

```
PARALLELOVERLAP [EXCEPTSAMENET | EXCEPTSAMEMETAL
    |EXCEPTSAMEMETALOVERLAP | EXCEPTSAMEVIA]
```


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Indicates that the cuts that have a parallel edge overlap greater than 0 require *cutSpacing* distance between them. Only one PARALLELOVERLAP spacing value is allowed per cut layer.

EXCEPTSAMENET indicates that the parallel within rule does not apply to same-net cuts.

EXCEPTSAMEMETAL indicates that the parallel overlap rule does not apply to cuts that share the same metal shapes above or below metal layers of the cuts.

EXCEPTSAMEMETALOVERLAP indicates that the parallel overlap rule does not apply to cuts that share the same metal shapes above or below metal layers that cover the projected overlap area between the cuts. This is the default.

EXCEPTSAMEVIA indicates that the parallel overlap rule does not apply to cuts that share the same metal shapes both above and below the metal layers that cover the overlap area between the cuts.

Note: You should not use PARALLELOVERLAP along with SAMENET, SAMEMETAL, or SAMEVIA.

PARALLEL *parLength* WITHIN *parWithin* ENCLOSURE *enclosure*

Specifies that the cut to concave corner spacing only applies for a wire with default wire width containing the cut and all the conditions as described below hold true. The cut has zero enclosure on two opposite sides, and the enclosure is less than *enclosure* on one of the other two opposite sides on the specified *secondLayerName*. There are neighbor wires on both metal layers of the cut on opposite sides. The spacing to the below metal must be exactly equal to *parWithin* - 0.001 (due to WITHIN nature, the given value is offset by 0.001) and the above metal neighbor is within *parWithin* of the cut edge with zero enclosure having parallel run length greater than *parLength*. If *parLength* is a negative value, the cut edges would be extended by abs(*parLength*) along both the sides to search for the neighbors. See Figure 2-98 on page 404.
There is a corner case on the zero enclosure requirement. If the cut edge having a neighbor not on layer *secondLayerName* touches or collides with the concave corner, the rule will be applicable.
Type: Float, specified in microns

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`PARALLELWITHIN` *`within`*

Specifies that if the edge of a neighbor cut is within *within* distance beyond the edge of another cut, *cutSpacing* is required between them. If a cut layer has more than one cut class, the parallel within rule could be applied among all cut classes, including via cuts belonging to different cut classes

Note: You should not use `PARALLELWITHIN` along with `SAMENET`, `SAMEMETAL`, or `SAMEVIA`. In addition, you can either specify `PARALLELOVERLAP` or `PARALLELWITHIN`.

`PRL` *`prl`* `[HORIZONTAL|VERTICAL] [SAMESIDE]`

Specifies a parallel run length (PRL) condition for both *cutSpacing* in `SPACING` and *cutWithin** in `WITHIN`. If cuts have PRL greater than *prl* in one direction and fulfill the `WITHIN` condition in the other direction, the neighbor cut would be counted against the rule. The applied *cutSpacing* is also measured in the other direction, same as the direction in `WITHIN`. If `HORIZONTAL` or `VERTICAL` is specified, PRL is measured in the given direction only. In addition, perfectly aligned neighbor cuts in both directions or in the given `HORIZONTAL` or `VERTICAL` direction are excluded. If `SAMESIDE` is specified, `ENCLOSURE` must also be specified and the neighbor cuts must be on the same side of the checking cut fulfilling enclosure condition. `CENTERTOCENTER` should not be specified along with `PRL`. See Figure 2-122 on page 425 and Figure 2-123 on page 426.

Type: Float, specified in microns

`SAMECUT`

Specifies that the adjacent cut rule only applies between two cuts, each having the same cut neighbors, besides potential neighbor of each other. See Figure 2-111 on page 415.

`SAMEMASK`

Indicates that the minimum spacing rule only applies between objects on the same mask.

`SAMEMETAL`

Indicates that the *cutSpacing* value only applies to cuts that are overlapped with the same metal shape. The `SAMEMETAL` *cutSpacing* value should be smaller than the normal `SPACING` *cutSpacing* value that applies to different-net cuts.

See Figure 2-91 on page 397 for an example of the difference between `SAMEMETAL` and `SAMENET`.

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`SAMEMETALSHAREDEDGE` *`parWithin`*
`[ABOVE][EXCEPTTWOEDGES][EXCEPTSAMEVIA` *`numCut`*`]`

Specifies the spacing greater than and equal to *`cutSpacing`* between two via cuts that have a common parallel run length greater than 0, have common above and /or below metal shapes covering the entire length of common projection between them and have neighbor wire(s) within *`parWithin`* distance from them on the same edge.

`ABOVE` specifies that the rule only applies if the via cut must have a common metal routing layer. This means that having a common below metal routing layer of the cuts is irrelevant.

`EXCEPTTWOEDGES` specifies that if the cuts have two neighbor wires within *`parWithin`* distance on the opposite sides, the spacing rule is ignored.

`EXCEPTSAMEVIA` *`numCut`* specifies that if there are greater than and equal to *`numCut`* of cuts having common same metal on both the above and below layers, the spacing rule is ignored.
Type: Integer

`SAMEVIA` Indicates that the *`cutSpacing`* value only applies to cuts that share the same metal shapes above and below that cover the overlap area between the cuts. The `SAMEVIA` *`cutSpacing`* value should be smaller than the normal `SPACING` *`cutSpacing`* value that applies to different-metal cuts.

`TWOCUTS` *`twoCuts`* Specifies that the cut spacing between two cuts, each having greater than or equal to *`twoCuts`* neighbor cuts that are within (or less than) a *`cutWithin`* distance away, is the *`cutSpacing`*. The value of *`twoCuts`* must be smaller than the specified number of cuts defined with the `ADJACENTCUTS` keyword. See Figure 2-109 on page 414 for an example.
Type: Integer

`TWOCUTSSPACING` *`twoCutsSpacing`*

Specifies that the adjacent cut rule applies only if all of the cuts fulfilling the `TWOCUTS` condition have a spacing less than *`twoCutsSpacing`*. See Figure 2-110 on page 414 for an example.

`VIAGROUP` *`groupName`* `TO CUTCLASS {`*`className`*`|ALL}`

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Specifies an adjacent cut rule applying between the cuts belonging to the same via group of *groupName* as triggering cuts to the cuts of *className* or any cuts in ALL as neighbors. Triggering cuts belonging to the same via group are merged in a maximum bounding box before searching for neighbor cuts. Neighbor cuts should be considered with reference to the closest cut of the via group. See Figure 2-125 on page 428.

WIDTH *width* ENCLOSURE *enclosure* EDGELENGTH *edgeLength*

Specifies that the cut to concave corner spacing applies if the width of an edge forming the concave corner is less than or equal to the *width*, the length of the adjacent edge is greater than or equal to the *edgeLength*, and forms a search window for cuts by extending *cutSpacing* on the concave corner along both sides of the edge with width less than or equal to the *width*. Any cut edge parallel to the metal edge with width less than or equal to the *width* inside that search window must have enclosure greater than or equal to *enclosure*. Otherwise, it will be a violation.
Type: Float, specified in microns

WRONGDIRECTION

Specifies the spacing for a cut to a non-preferred direction metal, when the cut is contained in a preferred direction same-metal wire. This indicates that the second layer must be a metal layer.

WRONGDIRECTIONONLY *xSize ySize* [LAYER *adjacentCutLayer*]

Specifies that only cuts that are fully enclosed within a search window of *xSize* x *ySize* size, formed from the concave corner towards the non-preferred routing wires (wrong direction) on both sides can trigger the rule. Fully enclosed includes edge alignment, such as the right cut edge touching the right edge of the window. If LAYER is specified, only the cuts overlapping or edge-aligned with any cuts on an adjacent cut layer *adjacentCutLayer* in a stacked via case can trigger the rule. See Figure 2-99 on page 405.

Type: Float, specified in microns

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Spacing Rule Examples

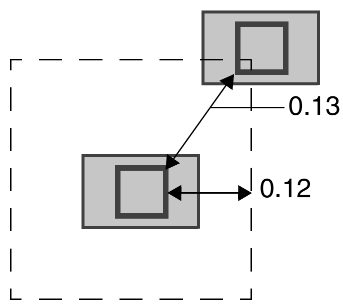
- The following spacing rule indicates that the minimum spacing of objects on the same mask in the routing layer is 0.15 µm while the minimum spacing of objects on different masks is 0.10 µm:

```
SPACING 0.10 ;
PROPERTY LEF58_SPACING
  "SPACING 0.15 SAMEMASK ; " ;
```

- The following spacing rule indicates that cuts should have x or y distance of maximum 0.12 µm between them:

```
PROPERTY LEF58_SPACING
  "SPACING 0.12 MAXXY ;" ;
```

Figure 2-90 Illustration of Spacing Rule with MAXXY



a) Violation, a neighbor cut edge is within 0.02 of a cut edge, 0.1 spacing is required between them.

- The following figure illustrates the `SAMEMETAL` rules for the following examples.

  If the *via3* layer has the following spacing rules:

```
SPACING 1.5 ;              #via3 to via3 spacing
SPACING 1.5 LAYER via2 ;   #via3 to via2 spacing
```

  Then both `a)` and `b)` are violations.

- If the *via3* layer has the following spacing rules, then `a)` is a violation, but `b)` is allowed:

```
SPACING 1.5 ;
SPACING 1.5 LAYER via2 ;
PROPERTY LEF58_SPACING
    "SPACING 0 SAMEMETAL LAYER via2 STACK ;" ;
```


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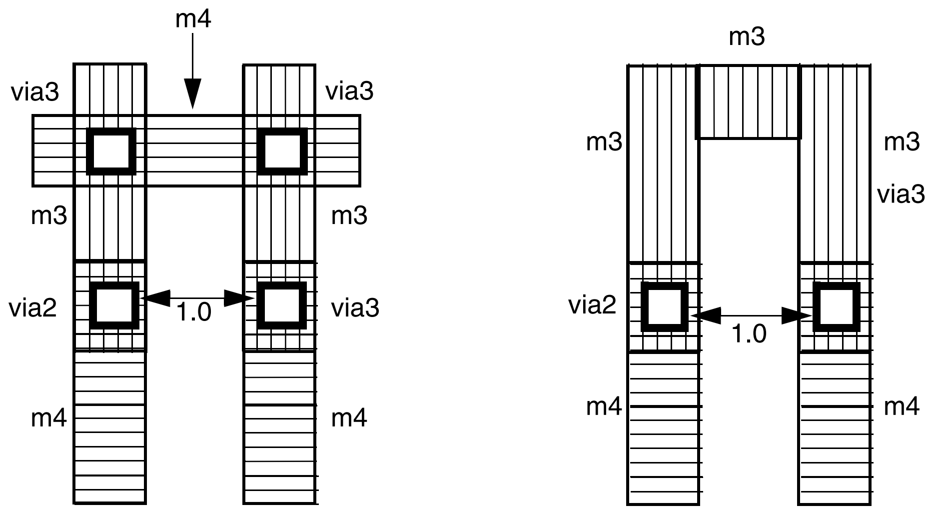
Figure 2-91 Illustration of SAMEMETAL Rules



a) `SAMENET` rules apply, but not `SAMEMETAL`

b) `SAMEMETAL` and `SAMENET` rules apply.

- The following spacing rule indicates that cuts that share the same metal shapes on both above and below metal layers should be 0.12 µm apart:

```
PROPERTY LEF58_SPACING
  "SPACING 0.12 SAMEVIA ;" ;
```

- The following spacing rule shows orthogonal spacing:

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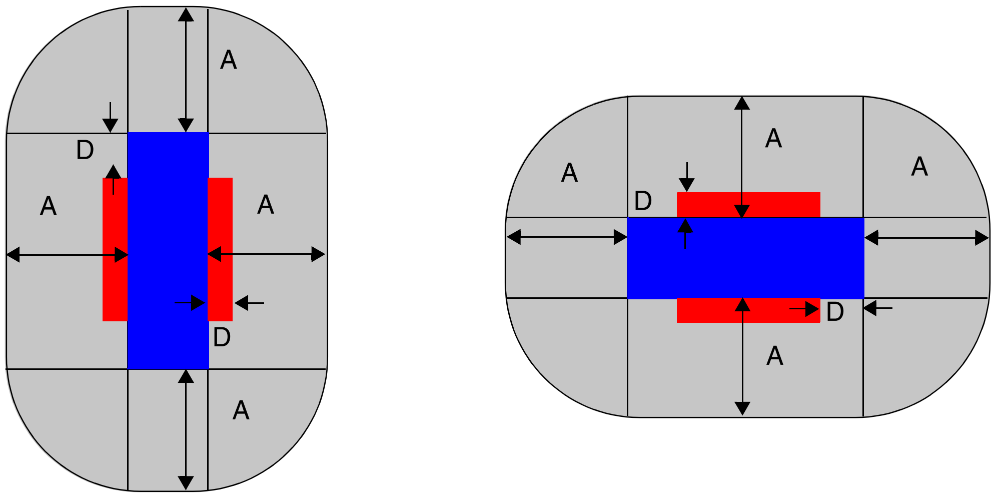
Figure 2-92 Definition of Spacing Rule with ORTHOGONALSPACING



Start with a square red cut, shrink by D, `abs(` *`cutSpacing`* – *`orthogonalSpacing`*)/2 on two sides of the cut and grow by D on the other two sides to result the blue rectangle. Then, spacing of A, (*`cutSpacing`* + *`orthogonalSpacing`*) / 2, is applied in a Euclidean fashion. Vioation, when wires on layer *`secondLayerName`* are found in both the gray regions.

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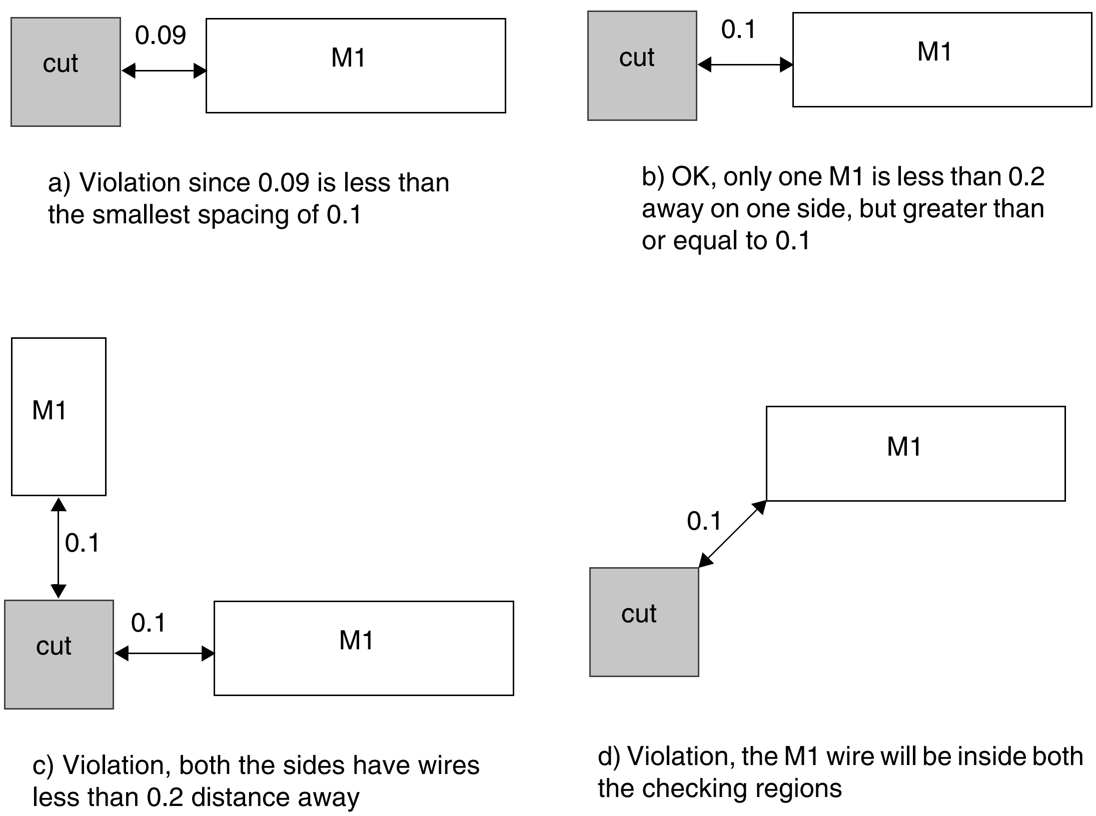
Figure 2-93 Illustration of Spacing Rule with ORTHOGONALSPACING

```
PROPERTY LEF58_SPACING
"SPACING 0.1 LAYER M1 ORTHOGONALSPACING 0.2 ;" ;
```



- The following spacing rule specifies that a `VA` via can have at most one neighboring `VA` via within `0.30` µm center-to-center distance away:

  ```
  PROPERTY LEF58_SPACING
    "SPACING 0.30 CENTERTOCENTER ADJACENTCUTS 2 WITHIN 0.30 CUTCLASS VA ;" ;
  ```

- The following spacing rule specifies that `0.1` µm spacing is required between two cuts if one cut is within `0.02` µm beyond the edge of the other cut:

  ```
  PROPERTY LEF58_SPACING
    "SPACING 0.09 ;" ;
    "SPACING 0.1 PARALLELWITHIN 0.02 ;" ;
  ```

- The following spacing rule indicates that same-net is exempted for the cut to metal rule:

  ```
  PROPERTY LEF58_SPACING
  ```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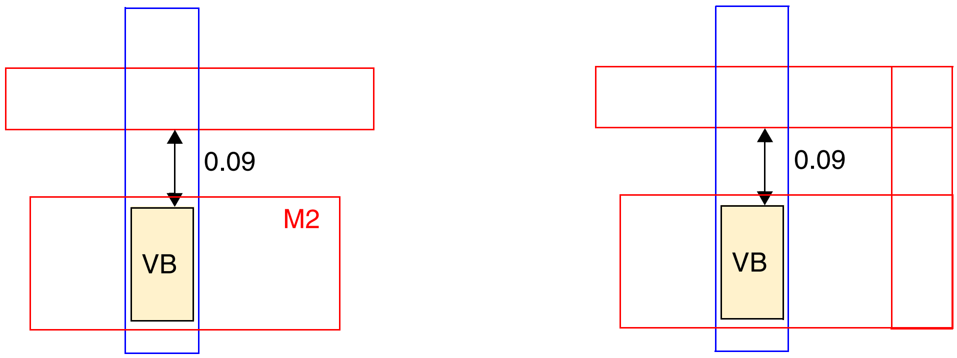
```
"SPACING 0.1 LAYER M2
    CUTCLASS VB SHORTEDGEONLY ;" ;
```

Figure 2-94 Illustration of CUTCLASS with SHORTEDGEONLY



a) Violation, the spacing between the short/ end edge of a VXBAR to an edge of the above metal wire is 0.09 (< 0.1)

b) OK, same-net is exempted for the cut to metal rule

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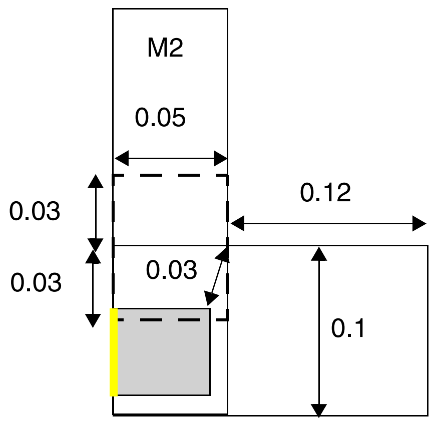
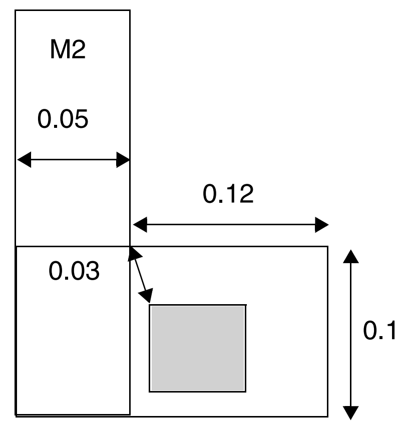
Figure 2-95 Illustration of CONCAVECORNER with WIDTH

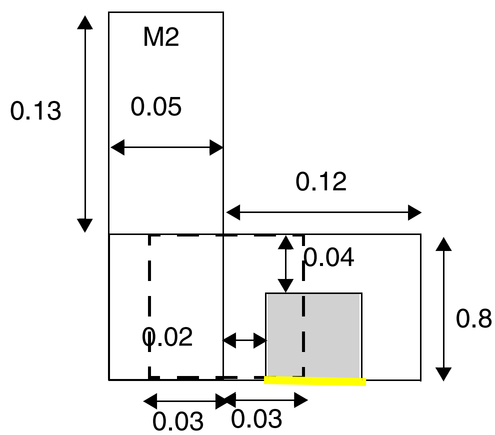
```
PROPERTY LEF58_SPACING
     "SPACING 0.03 LAYER M2 CUTCLASS VA
          CONCAVECORNER
          WIDTH 0.08 ENCLOSURE 0.02 EDGELENGTH 0.05  ; " ;
```



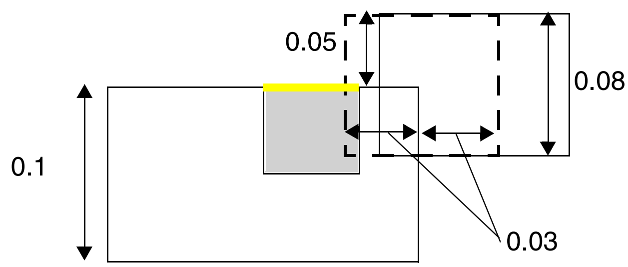
a) Violation, the vertical wire fulfills the width condition, the search window finds a cut, and the yellow cut edge does not have enclosure of 0.02.



b) OK, the cut is on the other side of the edge fulfilling the width condition.



c) Violation, both the wires fulfill the width and length conditions, and the yellow cut edge does not have enclosure of 0.02.



d) Violation, the search will be extended independent of whether there is a jog or not, and the yellow cut edge does not have enclosure of 0.02.

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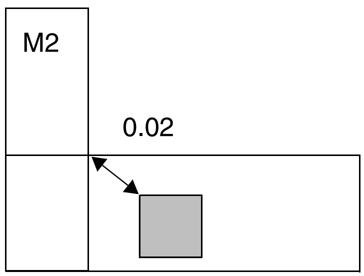
Figure 2-96 Illustration of CUTCLASS with CONCAVECORNER

```
PROPERTY LEF58_SPACING
     "SPACING 0.03 LAYER M2 CUTCLASS VA
        CONCAVECORNER ; " ;
```



a) Violation, the cut needs to be 0.03
away from the concave corner

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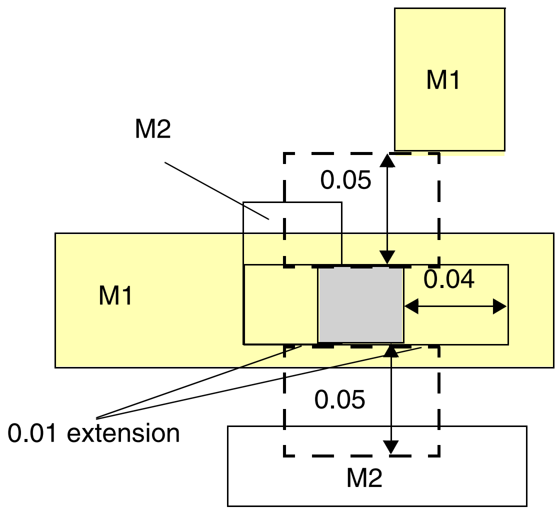
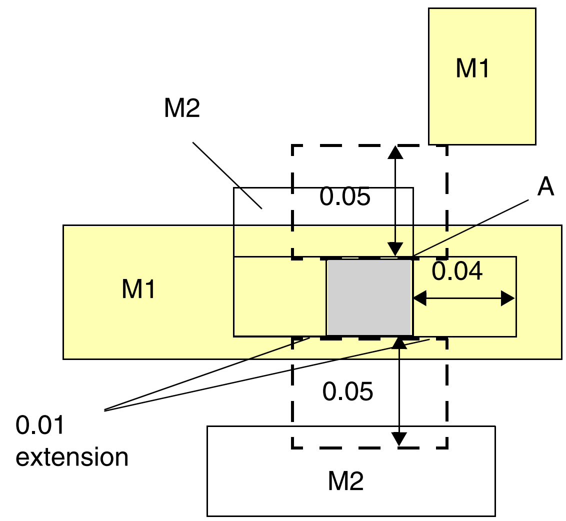
Figure 2-97 Illustration of CONCAVECORNER with WITHIN in PARALLEL

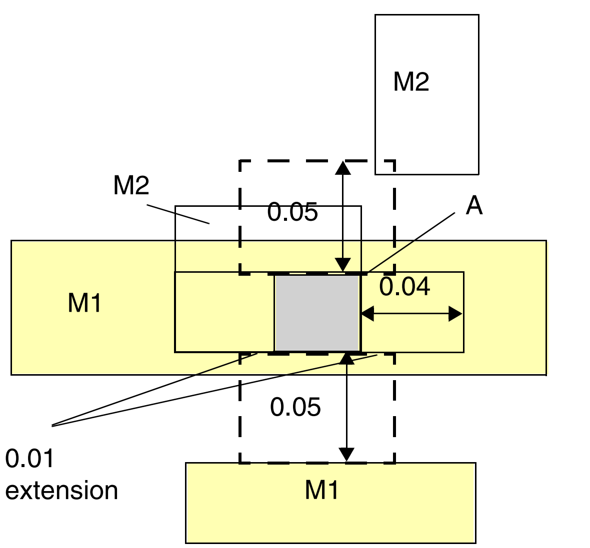
```
PROPERTY LEF58_SPACING
     "SPACING 0.04 LAYER M2 CUTCLASS VA
          CONCAVECORNER
          PARALLEL -0.01 WITHIN 0.051 ENCLOSURE 0.06  ; " ;
```



a) Violation, as long as part of two opposite cut edges having zero enclosure, it is still a violation.



b) Violation, cut corner A having a neighbor on M1 collides with the concave corner is still a violation



c) OK, cut corner A having a neighbor on M2 collides with the concave corner would not trigger the rule

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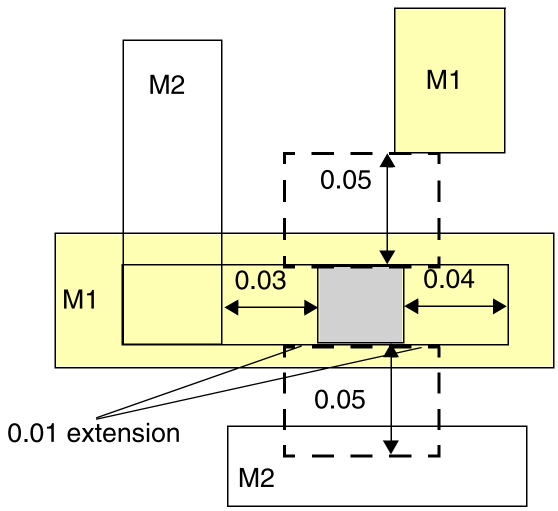
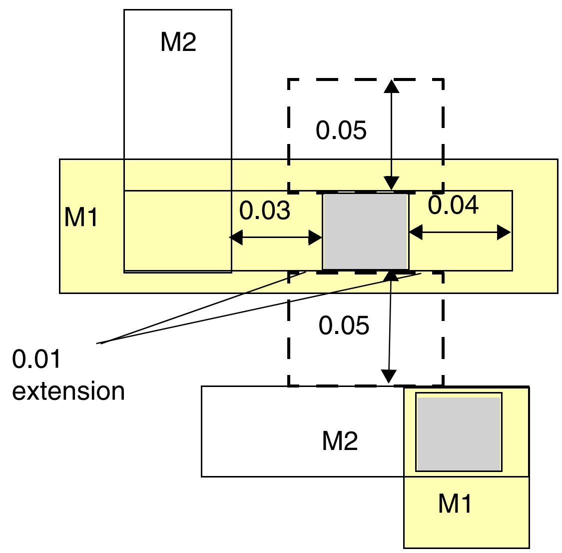
Figure 2-98 Illustration of CONCAVECORNER with WITHIN in PARALLEL

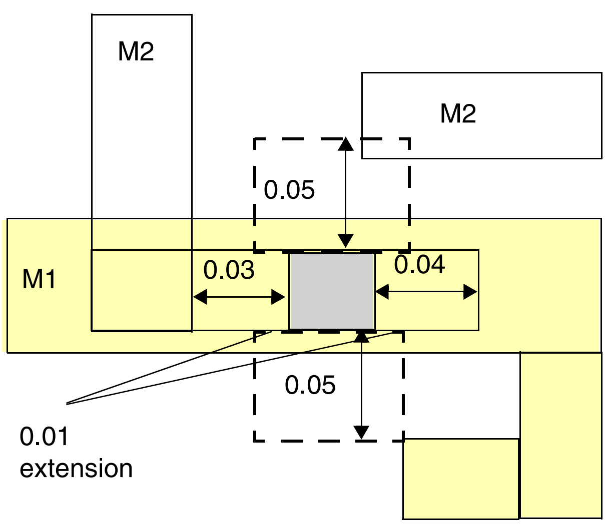
```
PROPERTY LEF58_SPACING
     "SPACING 0.04 LAYER M2 CUTCLASS VA
          CONCAVECORNER
          PARALLEL -0.01 WITHIN 0.051 ENCLOSURE 0.06  ; " ;
```



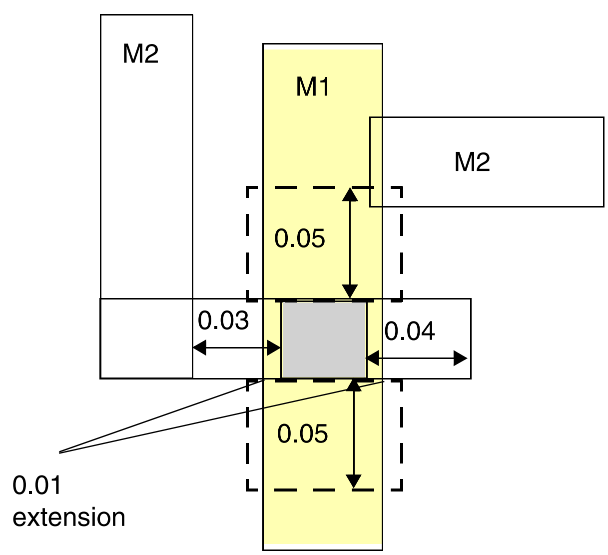
a) Violation, all of the conditions are met, particularly the M1 neighbor is exactly 0.05 apart.



b) OK, neighbor wires are not on opposite side



c) Violation, the same-metal/same-net wire could be a valid neighbor.



d) OK, the wire directly connected to the cut is not a valid neighbor

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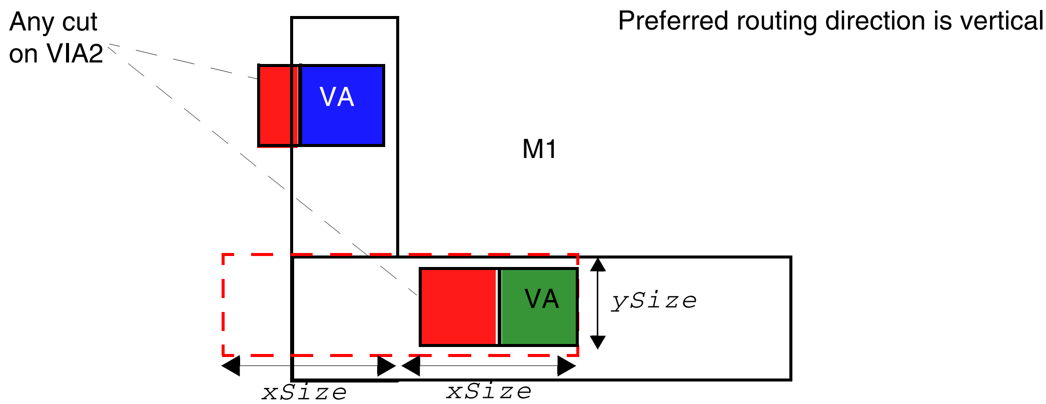
Figure 2-99 Illustration of CONCAVECORNER with WRONGDIRECTIONONLY



Only the green via fully enclosed in the red search window formed from the concave corner with size of *`xSize`* and *`ySize`* toward the non-preferred wrong-way wire on both sides and having an edge aligned (or an overlapped) adjacent cut on VIA2 is subjected to the rule.

Illustration of `SPACING` *`cutSpacing`* `LAYER M1 CUTCLASS VA`
`CONCAVECORNER`
`WRONGDIRECTIONONLY` *`xSize ySize`* `LAYER VIA2`

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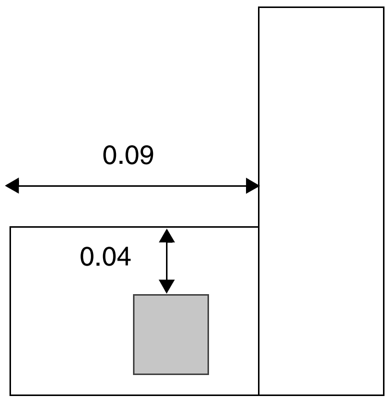
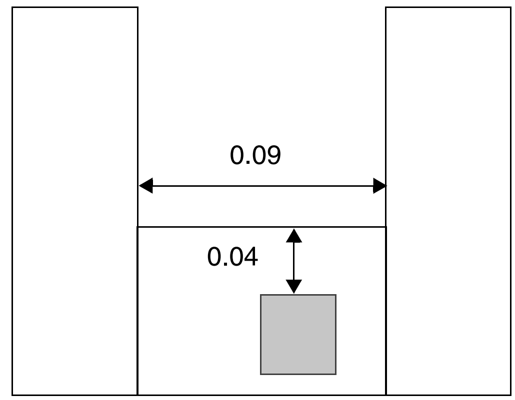
Figure 2-100 Illustration of EDGELENGTH with ENCLOSURE

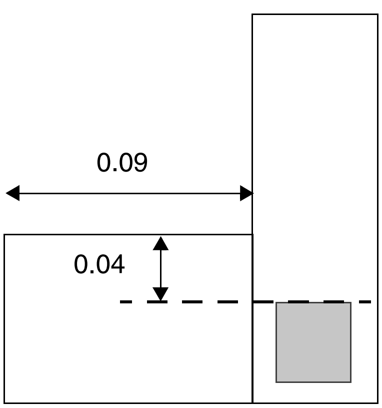
```
PROPERTY LEF58_SPACING  "
      SPACING 0.05 LAYER M2 CUTCLASS VA CONCAVECORNER
          EDGELENGTH 0.1 ENCLOSURE 0.06 0.08 ; " ;
```



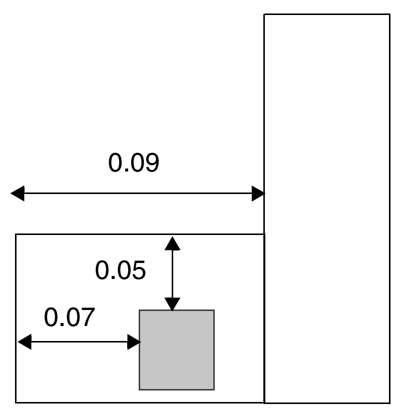
a) Violation, the spacing of the cut to an edge connected to a concave corner with length of 0.09 (< 0.1) is 0.04 (< 0.05).



b) OK, the 0.09 edge does not contain to a convex corner on the other end.



c) OK, the cut does not have PRL >= 0 to the 0.09 edge.



d) Violation, the enclosure of the top cut edge is 0.05 (>= 0.05, but < 0.06), the left adjacent cut edge has enclosure of 0.07 (< 0.08).

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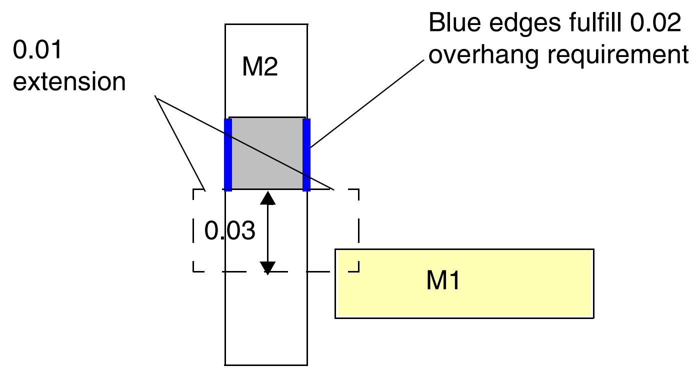
Figure 2-101 Illustration of CUTCLASS with Extension

```
PROPERTY LEF58_ENCLOSUREEDGE
     "ENCLOSUREEDGE ABOVE 0.02 OPPOSITE ; " ;
PROPERTY LEF58_SPACING
     "SPACING 0.03 LAYER M1 CUTCLASS VA
          EXTENSION 0.01 ; " ;
```



a) Violation, apply extension of 0.01 on the non-blue edges, and find M1 metal within 0.03 search window

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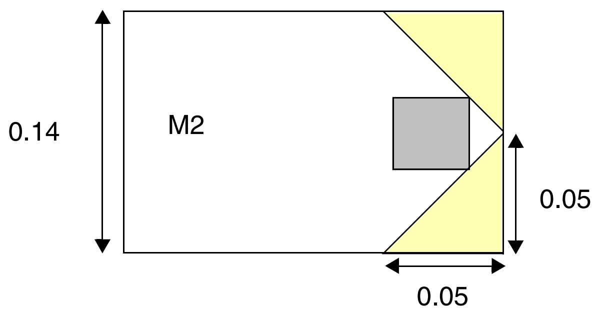
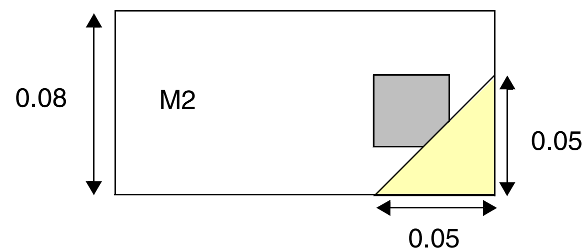
Figure 2-102 Illustration of CUTCLASS with NONEOLCONVEXCORNER

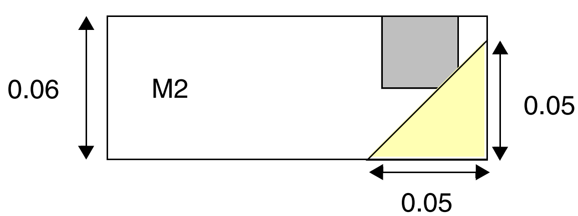
```
PROPERTY LEF58_SPACING
     "SPACING 0.05 LAYER M2 CUTCLASS VA
          NONEOLCONVEXCORNER 0.07 ; " ;
```



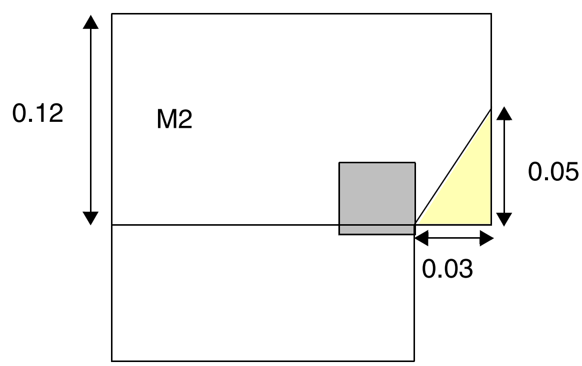
a) OK, the cut is outside of the 0.05 triangle



b) Violation, the cut overlaps with the triangle keepout.



c) OK, the edge is a EOL.



d) OK, the bottom edge of the triangle becomes edge length when it is < 0.05, and touching the keepout is fine

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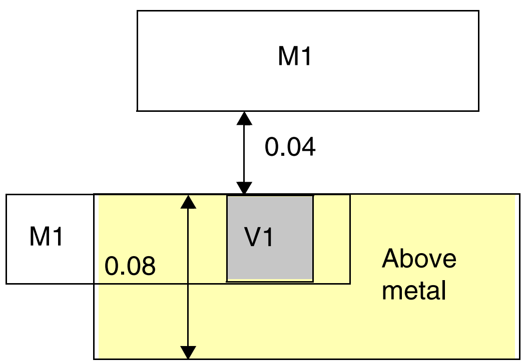
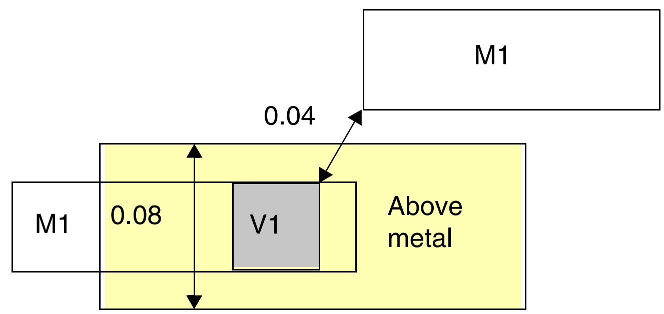
Figure 2-103 Illustration of CUTCLASS with ABOVEWIDTH and ENCLOSURE

On layer V1,

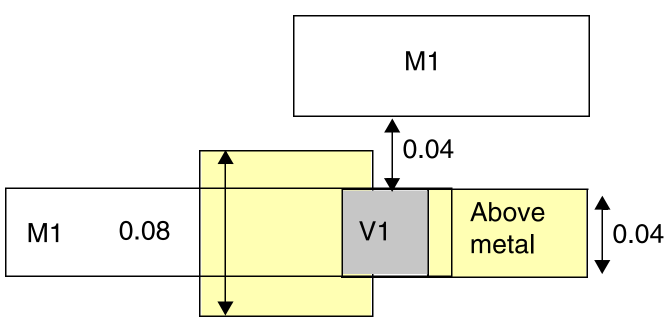
```
PROPERTY LEF58_SPACING
     "SPACING 0.05 LAYER M1 CUTCLASS VA
          ABOVEWIDTH 0.07 ENCLOSURE 0.01 ; " ;
```



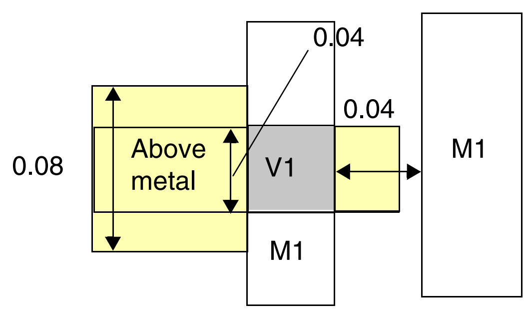
a) Violation, cut to metal spacing is 0.04 (<= 0.05).



b) Violation, cut to metal spacing is measured in Euclidean style.



c) Violation, as long as part of the cut is inside a wide wire, the entire cut edge is subjected for the spacing check.



d) Violation, the left edge of the cut touches a wire with width of 0.08 (>= 0.07).

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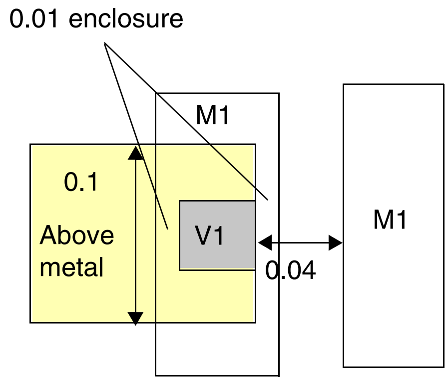
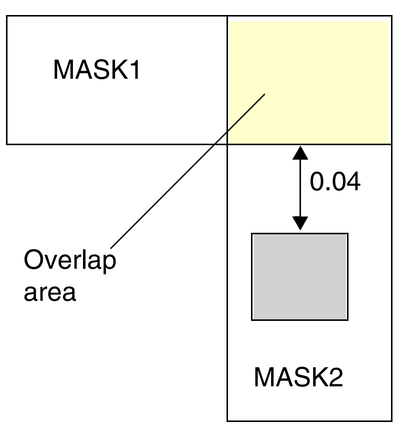


e) OK, both set of opposite cut sides have enclosure >= 0.01

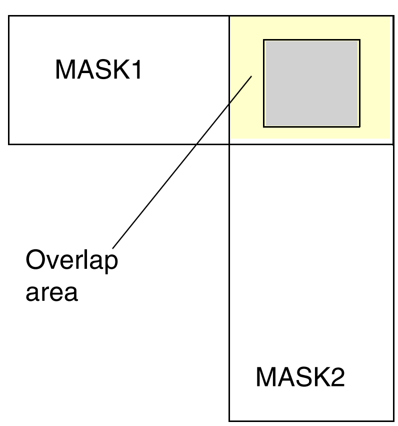
Figure 2-104 Illustration of CUTCLASS with MASKOVERLAP

On layer V1,

```
PROPERTY LEF58_SPACING
     "SPACING 0.05 LAYER M2 CUTCLASS VA
          MASKOVERLAP ; " ;
```



a) Violation, the cut needs 0.05 spacing to the overlap area of the 2 masks.



b) Violation, the cut cannot be inside the overlap area.

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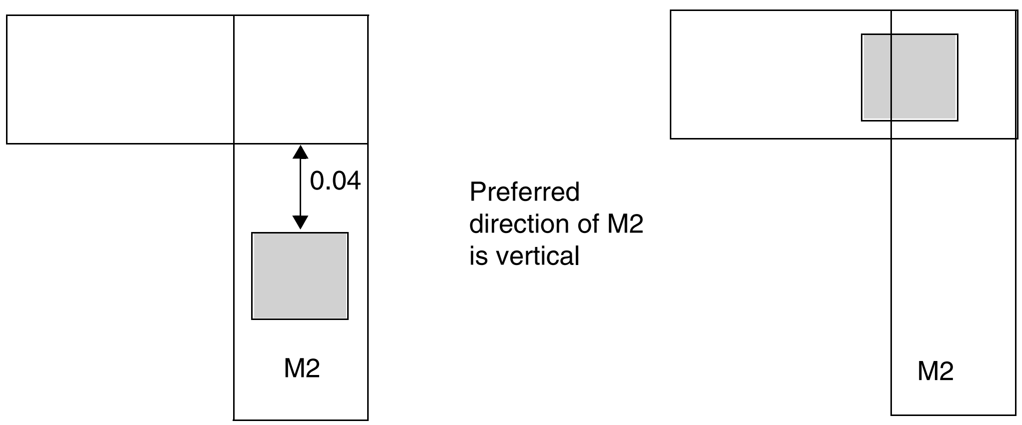
Figure 2-105 Illustration of CUTCLASS with WRONGDIRECTION

```
PROPERTY LEF58_SPACING
     "SPACING 0.05 LAYER M2 CUTCLASS VA
          WRONGDIRECTION ; " ;
```



a) Violation, the cut needs 0.05 spacing to non-preferred wires.

b) Violation, the cut could not be inside or touch non-preferred wires.

Figure 2-106 Illustration of CUTCLASS with WRONGDIRECTION OVERLAPONLY

Illustration of `SPACING` *`spacing`* `LAYER` *`secondLayerName`*
`CUTCLASS` *`className`*
`WRONGDIRECTION OVERLAPONLY`

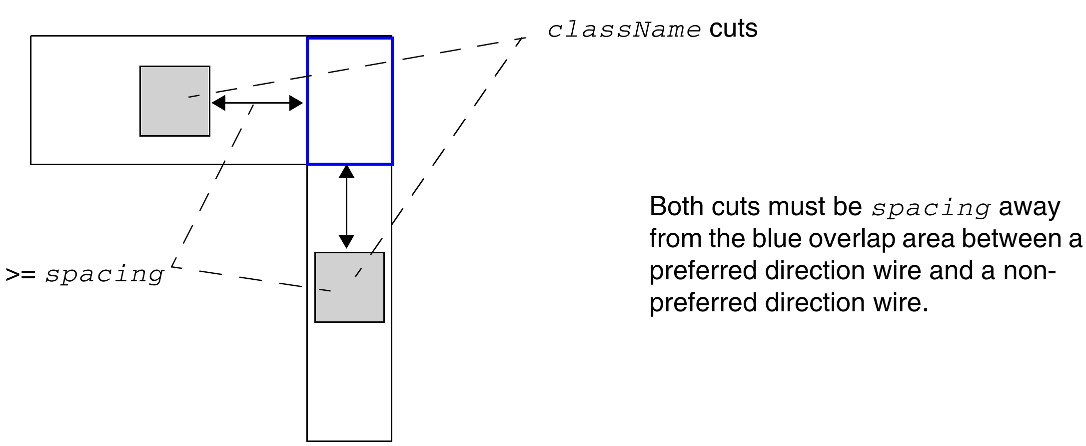


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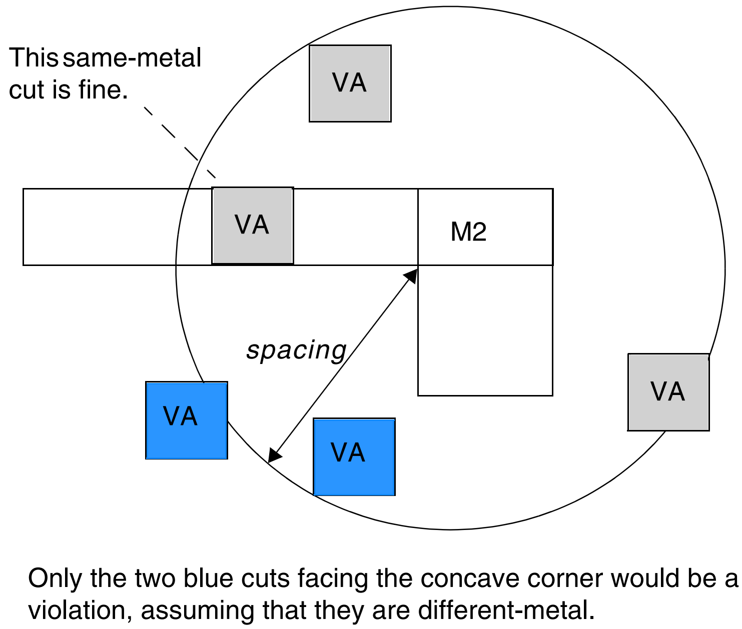
Figure 2-107 Illustration of CUTCLASS with DIFFCONCAVECORNER

Illustration of `SPACING` *`spacing`* `LAYER M2 CUTCLASS VA DIFFCONCAVECORNER`



- The following spacing rule indicates that the center-to-center spacing between cuts must be at least 0.2 if a cut has at least three exactly aligned neighbor cuts, horizontally and/or vertically, less than or equal to 0.2 µm away, or a cut has at least two neighbor cuts less than 0.2 µm away and one of those neighbor cuts is not exactly aligned:

```
PROPERTY LEF58_SPACING
  "SPACING 0.2 CENTERTOCENTER ADJACENTCUTS 2 EXACTALIGNED 3 WITHIN 0.2 ;" ;
```


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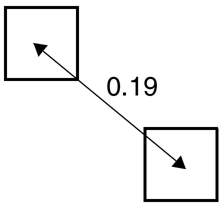
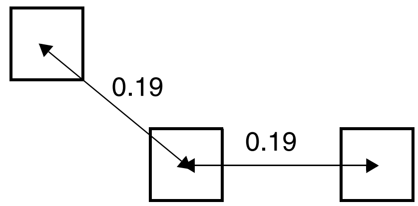
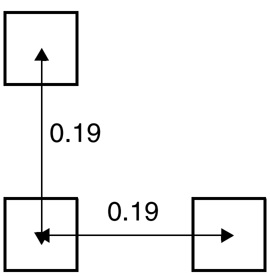
Figure 2-108 Illustration of the ADJACENTCUTS with EXACTALIGNED



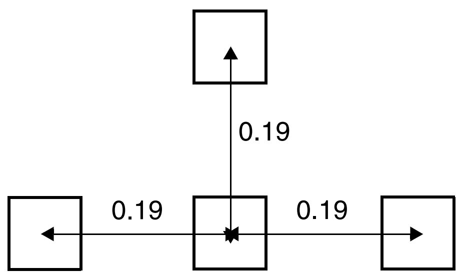
a) Okay. Only 1 neighbor cut within 0.2



b) Violation. When there is 1 non-perfectly aligned neighbor cut (on the left), any other neighbor cut, perfectly aligned or not (on the right) would trigger the rule spacing of 0.2



c) Okay. Two perfectly aligned neighbor cuts in 'L' or a straight line are fine



d) Violation.Three perfectly aligned neighbor cuts would trigger the rule spacing of 0.2

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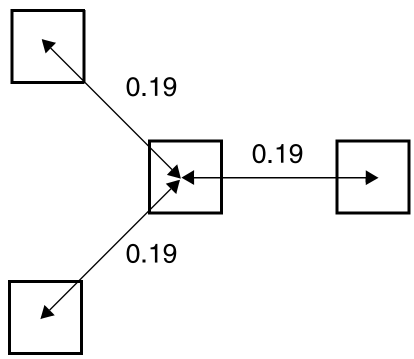
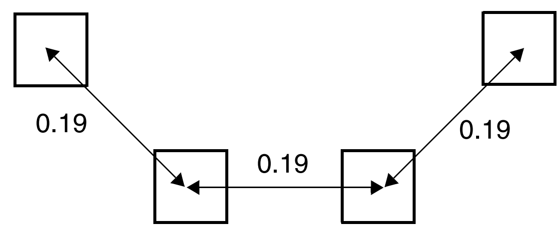
Figure 2-109 Illustration of the ADJACENTCUTS with TWOCUTS

```
PROPERTY LEF58_SPACING
   "SPACING 0.2 CENTERTOCENTER ADJACENTCUTS 3 ;
       TWOCUTS 2 WITHIN 0.2 ;" ;
```



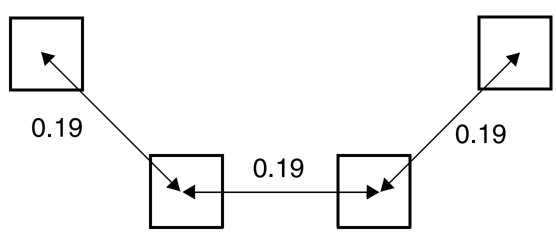
a) Violation, the middle cut has 3 neighbor cuts within 0.2



b) Violation, each of the middle 2 cuts have 2 neighbor cuts within 0.2, and the spacing between them is < 0.2. OK if TWOCUTS is not used.

Figure 2-110 Illustration of the ADJACENTCUTS with TWOCUTSSPACING

```
PROPERTY LEF58_SPACING
   "SPACING 0.2 CENTERTOCENTER ADJACENTCUTS 3
       TWOCUTS 2 TWOCUTSSPACING 0.18 WITHIN 0.2 ;" ;
```



a) OK, the middle two cuts fulfilling the `TWOCUTS` condition do not have a spacing < `0.18`. Violation if `TWOCUTSPACING` is omitted.

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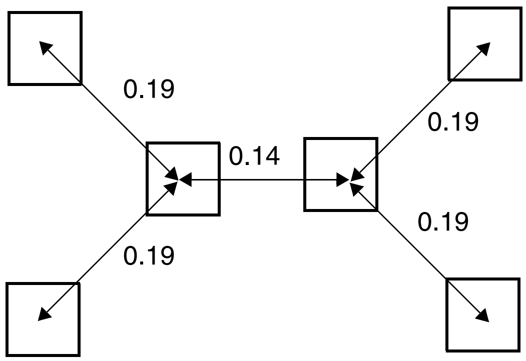
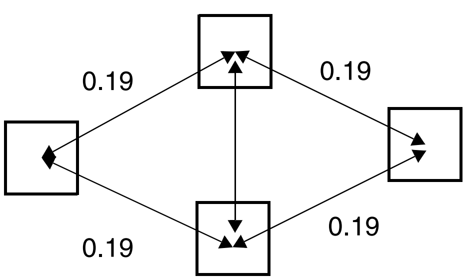
Figure 2-111 Illustration of ADJACENTCUTS with SAMECUT

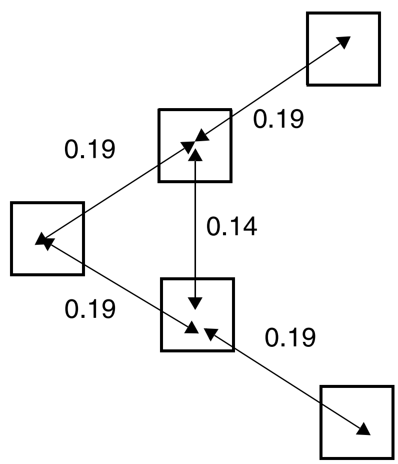
```
PROPERTY LEF58_SPACING
   "SPACING 0.2 CENTERTOCENTER ADJACENTCUTS 4
       TWOCUTS 3 SAMECUT WITHIN 0.2 ;" ;
```



a) OK, the middle two cuts do not have common neighbors. Violation if SAMECUT is omitted



b) Violation, the middle cuts have common neighbor besides neighboring to each other



c) OK, all of the neighbors must be common.

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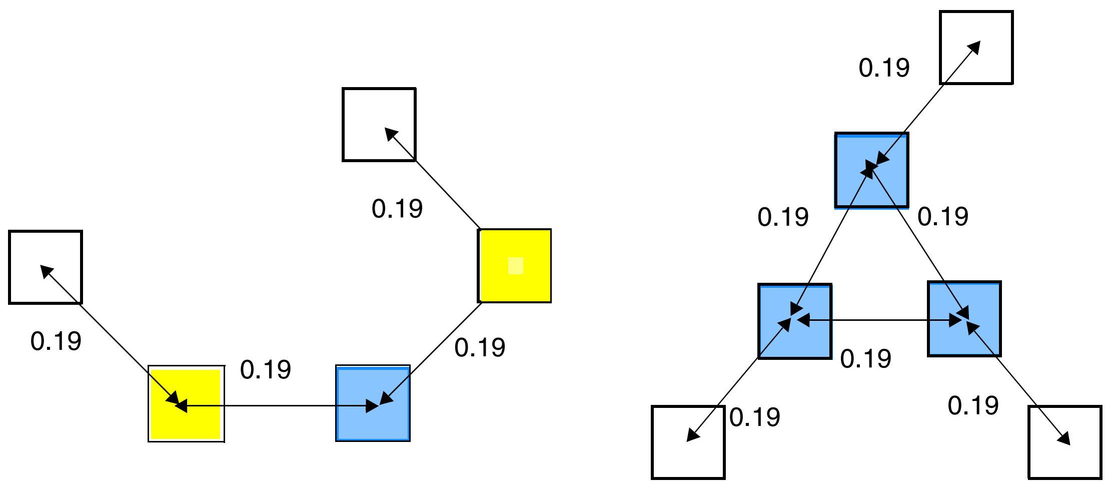
Figure 2-112 Illustration of the ADJACENTCUTS with EACHCUTS

```
PROPERTY LEF58_SPACING
   "SPACING 0.2 CENTERTOCENTER
       ADJACENTCUTS 2 EACHCUTS 2 WITHIN 0.2 ;" ;
```



a) OK, although the blue cut has two yellow neighbor cuts and each of them also has two neighboring cuts, each of the yellow neighbor cuts only has one neighboring cut of the triggering blue cut (or its yellow neighboring cuts).

b) Violation, the blue cuts form a special group. Although each of them has three neighbors, after picking one of them as the triggering cut, each of the two blue neighbors has two neighboring cuts of the group and other white cut neighbors are irrelevant.

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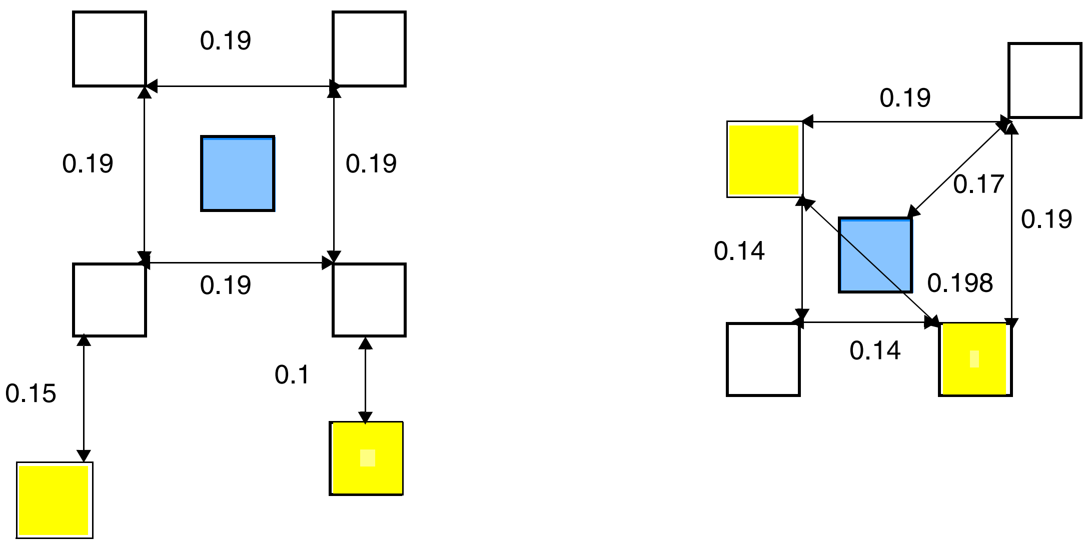
Figure 2-113 Illustration of the ADJACENTCUTS with EXCEPTEVENNUMCUT

```
PROPERTY LEF58_SPACING
   "SPACING 0.2 ADJACENTCUTS 3 EACHCUTS 3
       EXCEPTEVENNUMCUT WITHIN 0.2 ;" ;
```



a) OK, the blue cut has four neighbors (>=`3` in `ADJACENTCUTS`) and each of the neighbors has three cuts of the group formed by the triggering cut and its neighbor(>=`3` in `EACHCUTS`). As there are four neighbors meeting the condition and `EXCEPTEVENNUMCUT` is defined, it is not a violation. It is a violation if `EXCEPTEVENNUMCUT` is omitted. In addition, the presence of the yellow cuts are irrelevant to this rule.

b) Violation, the blue cut has four neighbors and each of them has >=`3` cuts of the group formed by the triggering cut and its neighbor. As there are three cuts (one blue and two yellow) having four (>`3`) neighbors, it is a violation even if `EXCEPTEVENNUMCUT` is defined.

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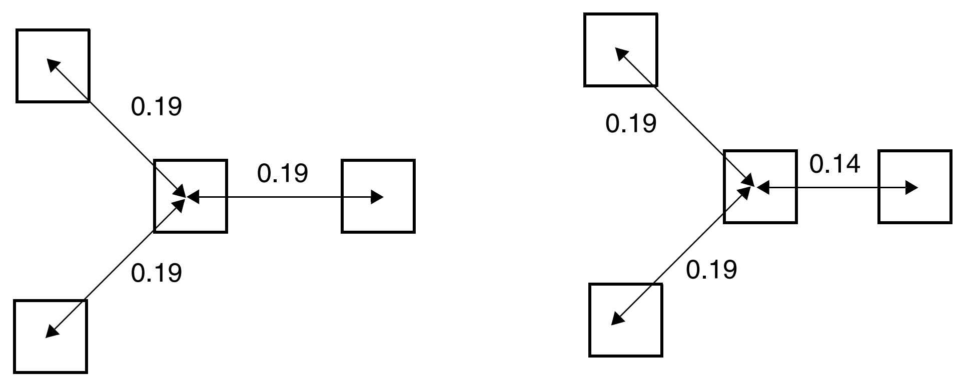
Figure 2-114 Illustration of the ADJACENTCUTS with WITHIN

```
PROPERTY LEF58_SPACING
   "SPACING 0.2 CENTERTOCENTER ADJACENTCUTS 3 ;
       WITHIN 0.15 0.2 ;" ;
```



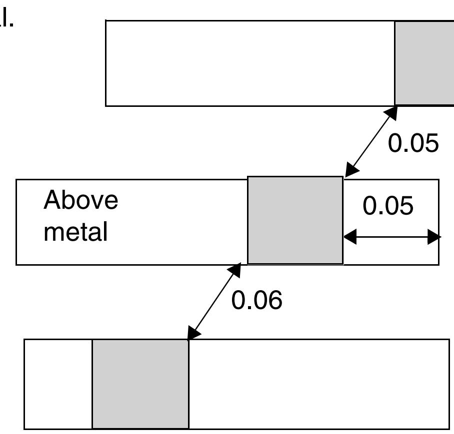
a) Violation, the middle cut has 3 neighbor cuts >= 0.15 and < 0.2

b) OK, one of the neighbor cut is outside the within range

Figure 2-115 Illustration of the ADJACENTCUTS with ENCLOSURE

```
PROPERTY LEF58_SPACING
   "SPACING 0.07 ADJACENTCUTS 2 WITHIN 0.07
       ENCLOSURE ABOVE 0.05 ;" ;
```

Preferred routing direction on above metal is horizontal.



a) OK, the middle has enclosure of `0.05` in the preferred routing direction on the above metal layer. It fails if `ENCLOSURE` is omitted.

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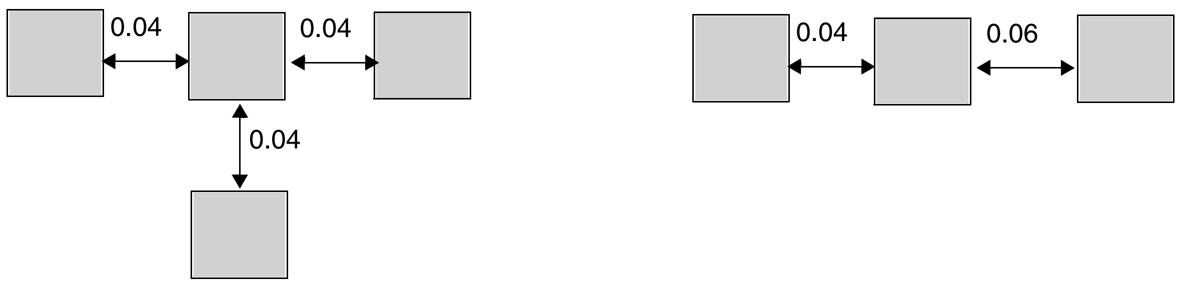
Figure 2-116 Illustration of ADJACENTCUTS with EXCEPTALLWITHIN

Example of

```
PROPERTY LEF58_SPACING  "
     SPACING 0.06 ADJACENTCUTS 2 WITHIN 0.07
         EXCEPTALLWITHIN 0.05 ; " ;
```



a) OK, it does not matter how many neighbors the middle cut has; as long as they are all within 0.05, it is allowed

b) Violation, the middle cut has two neighbor within 0.07, and the left cut has spacing of 0.04 (< 0.06)

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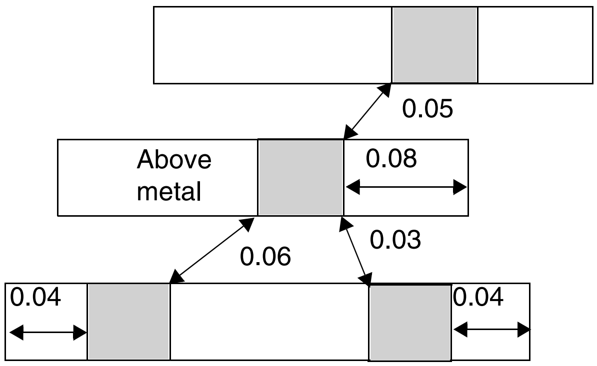
Figure 2-117 Illustration of ADJACENTCUTS with ANYCUT and EXCEPTSAMEMETAL

Preferred routing direction on above metal is horizontal.



a) Violation, any cuts having an enclosure < 0.05 is a violation. Although `EXCEPTSAMEMETAL` would count the below two cuts as one and consider each individual cut separately, the presence of a top neighbor cut would still make this pattern a violation. Without the top neighbor cut, it is OK.

Example of

```
PROPERTY LEF58_SPACING  "
     SPACING 0.07 ADJACENTCUTS 2 WITHIN 0.07
         ENCLOSURE ABOVE 0.05
         ANYCUT EXCEPTSAMEMETAL ; " ;
```


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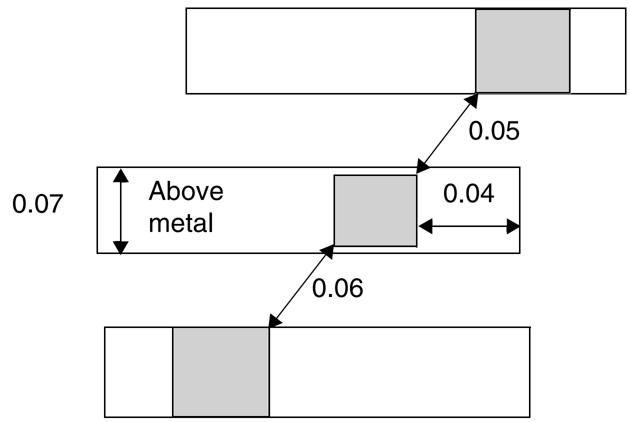
Figure 2-118 Illustration of the ADJACENTCUTS with METALWIDTH in ENCLOSURE

Example of

```
PROPERTY LEF58_SPACING
   "SPACING 0.07 ADJACENTCUTS 2 WITHIN 0.07
       ENCLOSURE ABOVE 0.05 METALWIDTH 0.06;" ;
```

Preferred routing direction on above
metal is horizontal.



a) OK, the width of the middle wire is `0.07` (> `0.06`).
Violation if `METALWIDTH` is omitted.

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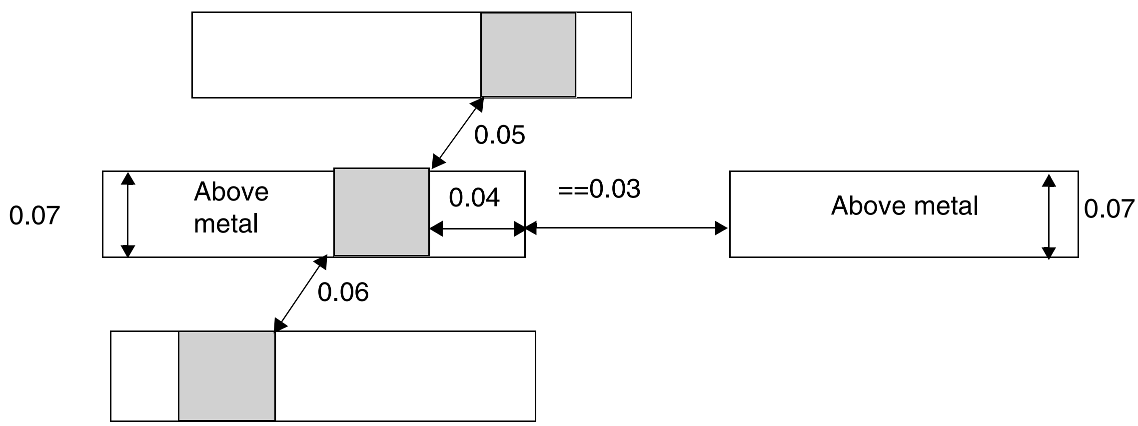
Figure 2-119 Illustration of the ADJACENTCUTS with EXCEPTLINEENDGAP

```
PROPERTY LEF58_SPACING
   "SPACING 0.07 ADJACENTCUTS 2 WITHIN 0.07
       ENCLOSURE ABOVE 0.05 EXCEPTLINEENDGAP;" ;
```

On the above metal layer,

```
PROPERTY LEF58_LINEENDGAP
   "LINEENDGAP 0.03 WIDTH 0.12;" ;
```

Preferred routing direction on
above metal is horizontal.



a) OK, although the middle cut has an enclosure < `0.05`, the wire fulfills the `LINEENDGAP` condition on the above metal layer such that the adjacent cut rule does not apply.

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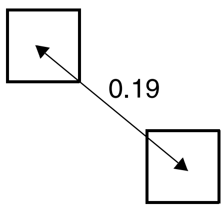
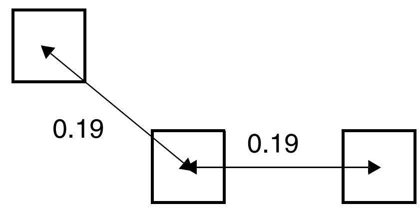
Figure 2-120 Example of ADJACENTCUTS with NOPRL

Example of

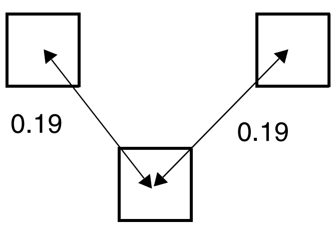
```
PROPERTY LEF58_SPACING
   "SPACING 0.2 CENTERTOCENTER
       ADJACENTCUTS 2 WITHIN 0.2 NOPRL VERTICAL ;" ;
```



a) Okay. Only 1 neighboring cut within 0.2



b) Okay. The right cut has a parallel run length > `0` with the middle cut vertically, and is not counted as a neighbor.



c) Violation, has two neighbor cuts without parallel run length.

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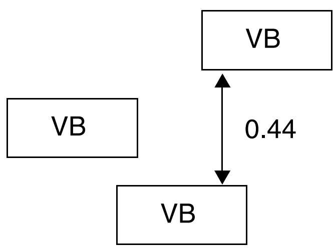
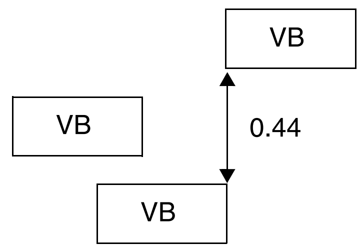
- The following spacing rule indicates that a VB via can at most have one neighbor VB via within 0.45 µm distance if they have a side to side parallel edge overlap greater than 0:

```
PROPERTY LEF58_SPACING
  "SPACING 0.45 ADJACENTCUTS 2 WITHIN 0.45 CUTCLASS VB SIDEPARALLELOVERLAP ;" ;
```

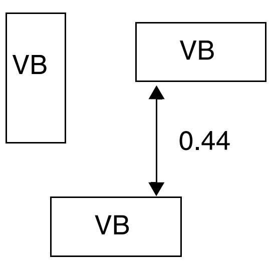
Figure 2-121 Illustration of ADJACENTCUTS with SIDEPARALLELOVERLAP



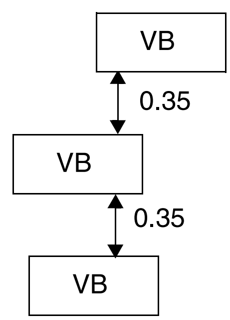
a) Violation, 2 VB < 0.2 with side by side parallel edge overlap > 0



b) OK, the right VB does not have side by side parallel edge overlap > 0



c) OK, the left VB only has a side to end parallel edge overlap > 0



d) Violation, the side by side parallel edge overlap > 0 does not need to be on the same edge

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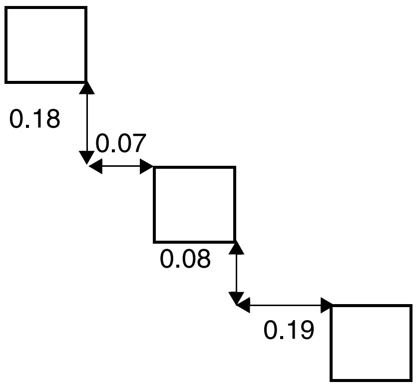
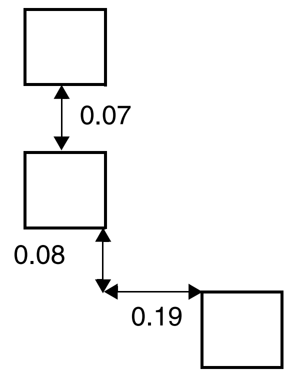
Figure 2-122 Illustration of ADJACENTCUTS with PRL



a) OK, the neighbor cut must be within `0.1` vertically and `0.2` horizontally. Only the bottom cut is qualified for this condition. Violation if `VERTICAL` is omitted since both the top and bottom cuts are within `0.1` in one direction `0.2` in the other direction. The top cut with vertical spacing of `0.18` (< `0.19`) would be a violation since that cut is within `0.1` horizontally.



b) OK, the top perfectly aligned cut is excluded for this rule. There is only one legal bottom cut to be OK.

Example of

```
PROPERTY LEF58_SPACING "
     SPACING 0.19 ADJACENTCUTS 2 WITHIN 0.2
     PRL -0.1 VERTICAL ; " ;
```


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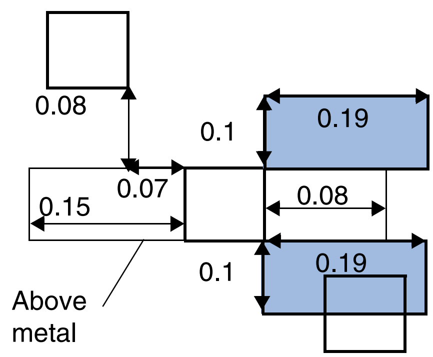
Figure 2-123 Illustration of ADJACENTCUTS with ENCLOSURE and PRL



a) OK, the neighbor cuts must overlap with the blue windows due to `SAMESIDE`, which means the neighbor cuts must be on the same side of the checking cut fulfilling the enclosure condition.

Example of

```
PROPERTY LEF58_SPACING "
     SPACING 0.19 ADJACENTCUTS 2 WITHIN 0.2
     ENCLOSURE ABOVE 0.1
     PRL -0.1 VERTICAL SAMESIDE ; " ;
```


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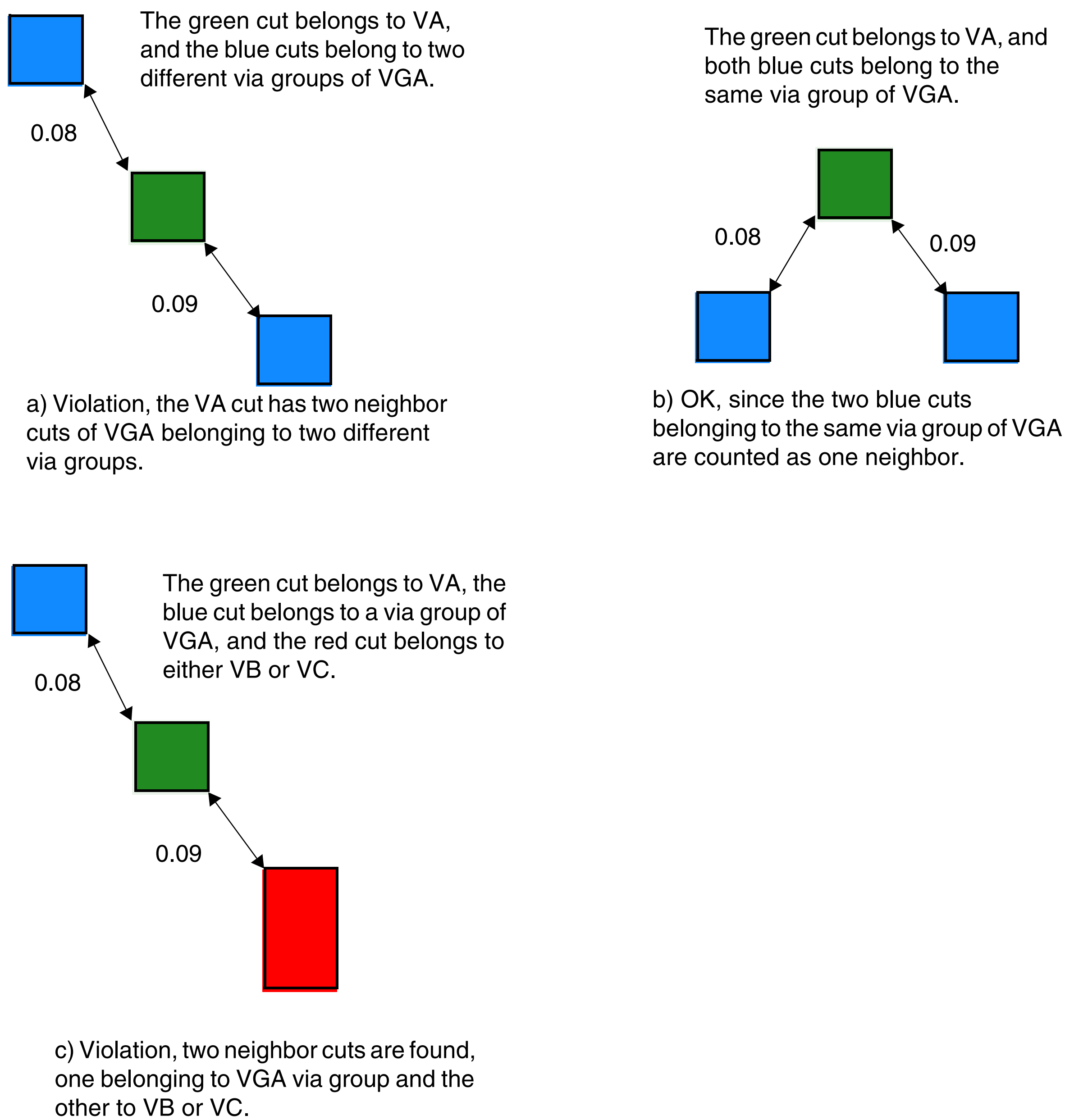
Figure 2-124 Illustration of ADJACENTCUTS with VIAGROUP



Example of

```
PROPERTY LEF58_SPACING "
     SPACING 0.1 ADJACENTCUTS 2 WITHIN 0.1
     CUTCLASS VA TO VIAGROUP VGA
     CUTCLASSLIST VB VC ; ";
```


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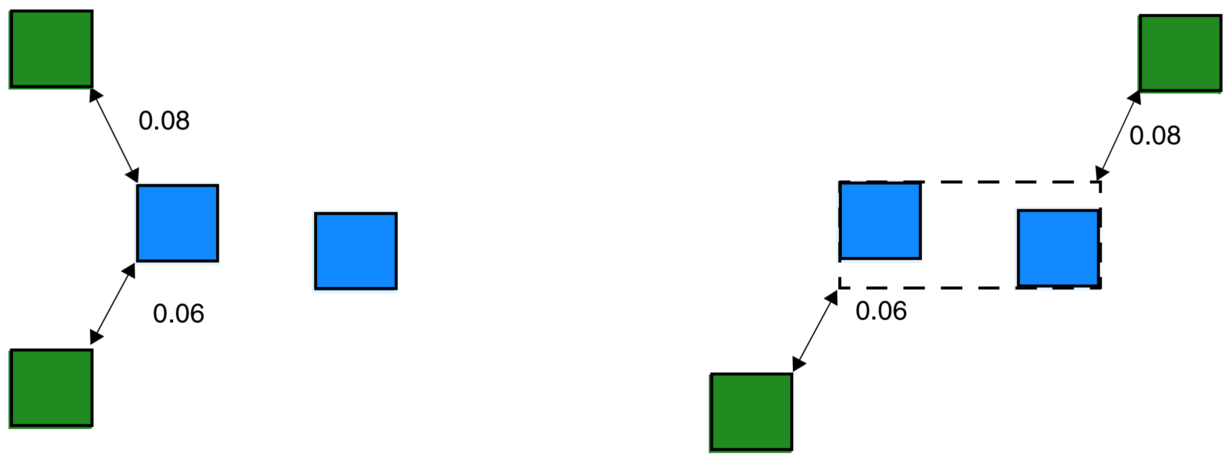
Figure 2-125 Illustration of ADJACENTCUTS with VIAGROUP

The green cuts belong to VA, and the blue cuts belong to the same via group of VGA.



a) Violation, the VGA cut has two neighbor cuts of VA.

b) Violation, all neighbor cuts of the cuts belonging to the same via group of VGA are counted, which is similar to merging the via group cuts within the dotted line.

Example of

```
PROPERTY LEF58_SPACING "
     SPACING 0.1 ADJACENTCUTS 2 WITHIN 0.1
     VIAGROUP VGA TO CUTCLASS VA ; ";
```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- The following spacing rule indicates that cuts that do not share the same metal shapes both above and below metal layers and have a parallel edge overlap greater than 0 must have 0.15 µm distance between them:

```
PROPERTY LEF58_SPACING
  "SPACING 0.12 SAMEVIA ;" ;
  "SPACING 0.13 ;" ;
  "SPACING 0.15 PARALLELOVERLAP EXCEPTSAMEVIA ;" ;
```

Figure 2-126 Illustration of PARALLELOVERLAP Rule with EXCEPTSAMEVIA

a) Violation, the cuts only share metal shape on one metal layer, and 0.15 parallel overlap spacing is needed. Okay, if EXCEPTSAMEVIA is not used.

b) OK, the cuts share metal shapes on both metal layers, and 0.12 SAMEVIA spacing is needed and met

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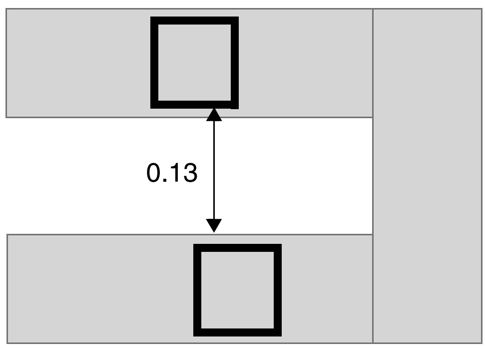
Figure 2-127 Illustration of Spacing Rule with EXCEPTSAMEMETAL

```
SPACING 0.13 ;
PROPERTY LEF58_SPACING
"SPACING 0.15 PARALLELOVERLAP EXCEPTSAMEMETAL ;" ;
```



a) OK, but with the default of EXCEPTSAMEMETALOVERLAP, it will be a violation since the same-metal must cover the projected overlap area of the cuts.

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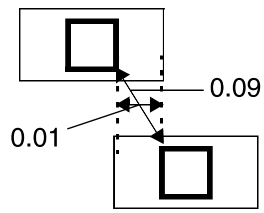
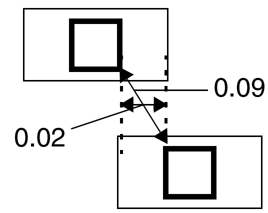
Figure 2-128 Illustration of PARALLELWITHIN Rules

Examples of

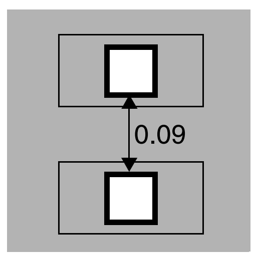
```
PROPERTY LEF58_SPACING   "
     SPACING 0.09 ;
     SPACING 0.1 PARALLELWITHIN 0.02 ; " ;
```



a) Violation, a neighbor cut edge is within 0.02 of a cut edge, 0.1 spacing is required among them.



b) Okay, the neighbor cut is not within 0.02 beyond the other cut edge.



c) Violation, the 0.1 parallel within spacing applies to same-metal as well

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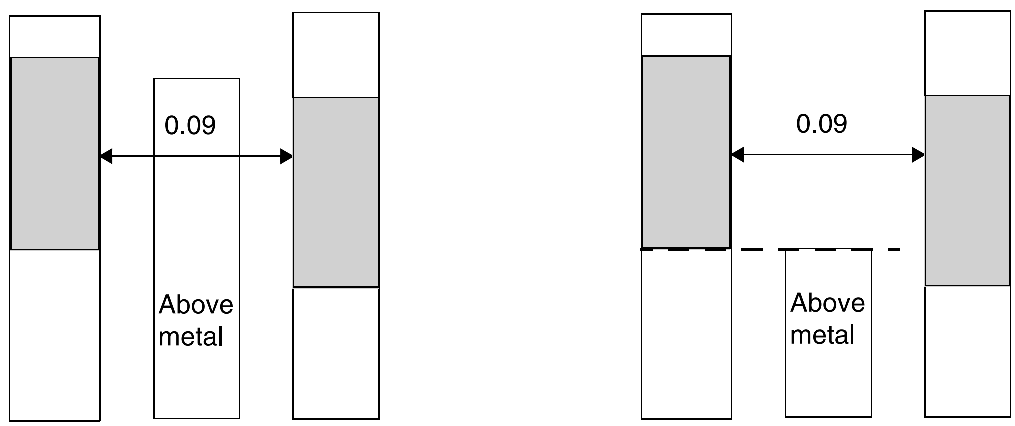
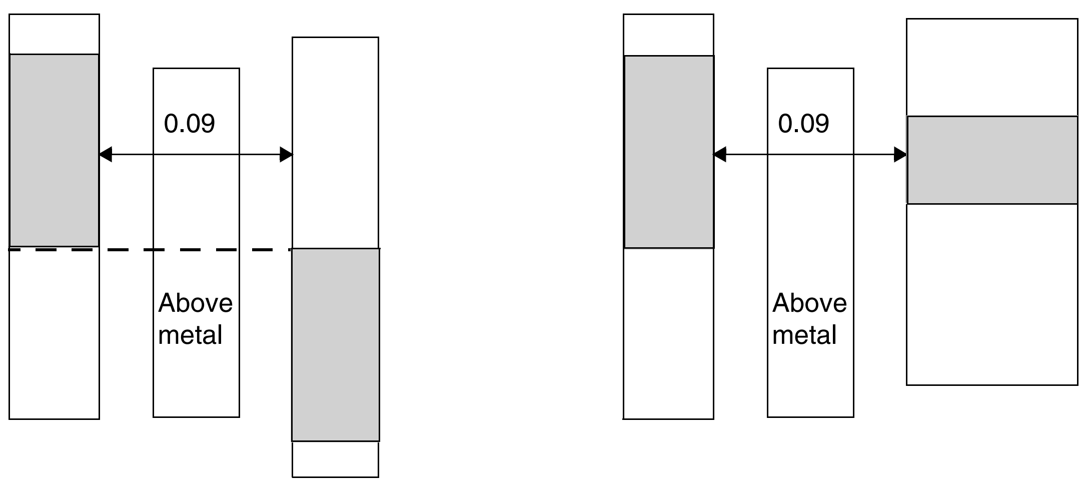
Figure 2-129 Illustration of PARALLELWITHIN with LONGEDGEONLY

```
PROPERTY LEF58_SPACING
     "SPACING 0.1 PARALLELWITHIN 0.0 CUTCLASS VB
          LONGEDGEONLY  ; " ;
```



a) Violation, the long/side cut edges has a middle wire with spacing 0.09 (< 0.1)

b) OK, no middle neighbor wire in the common projected run length of the cuts



c) OK, no common projected run length between the cuts

d) OK, the spacing only applies to two long/side edges. Violation if LONGEDGEONLY is omitted.

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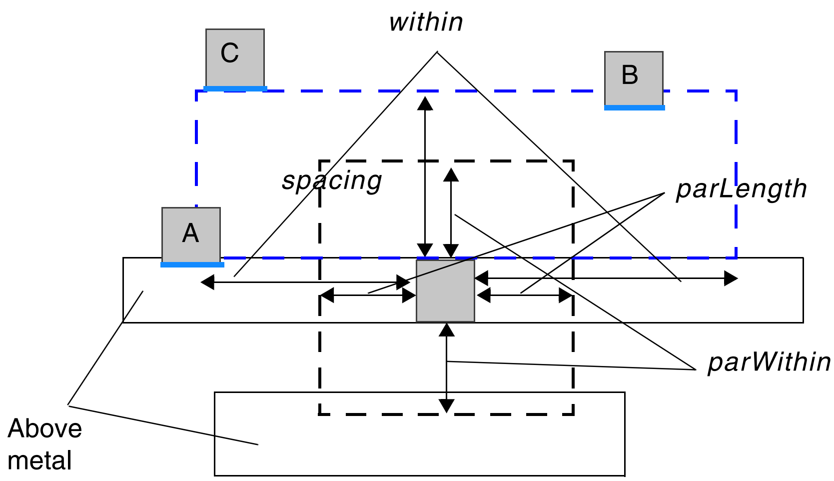
Figure 2-130 Illustration of PARALLELWITHIN with ENCLOSURE ABOVE



The bottom cut edge having a neighbor wire has enclosure less than `enclosure` on the above metal layer. Hence, the facing blue edge of any neighbor cuts must be outside of the blue dotted window. Hence, via A is ok since its blue edge is outside, but via B will be a violation. The via C is ok since the blue edge touches the window.

Illustration of"`SPACING` *`spacing`* `PARALLELWITHIN` *`within`*
`CUTCLASS` *`cutClass`*
`ENCLOSURE` *`enclosure`* `ABOVE`
`PARALLEL -`*`parLength`* `WITHIN` *`parWithin`*

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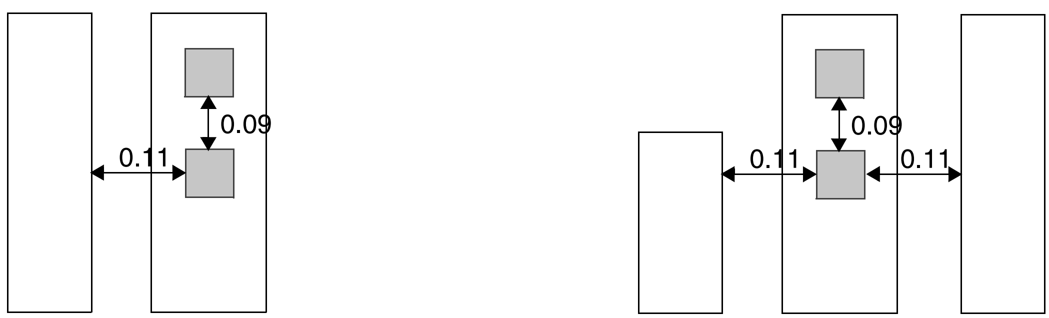
- The following spacing rule indicates that two via cuts having common parallel run length greater than 0, having common above and /or below metal shapes covering the entire length of common projection between them and having one neighbor wire within 0.12 µm distance of them on the same edge must be, at least 0.1 µm spacing apart:

```
PROPERTY LEF58_SPACING
  "SPACING 0.1 SAMEMETALSHAREDEDGE 0.12 EXCEPTTWOEDGES ;" ;
```

Figure 2-131 Illustration of SAMEMETALSHAREDEDGE rule



a) Violation, there is a neighbor within 0.12, and 0.1 spacing needed between the cuts

b) Okay, at least 1 cut having 2 neighbors within 0.12 on opposite sides will exempt the rule

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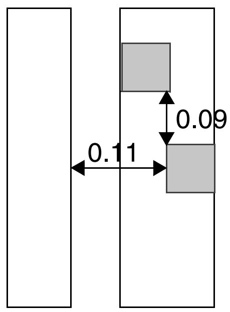
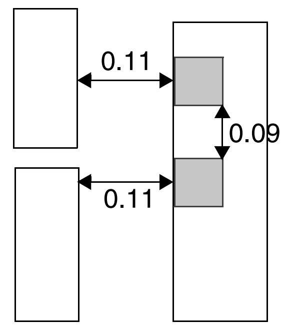
Figure 2-132 Illustration of SAMEMETALSHAREDEDGE Rule

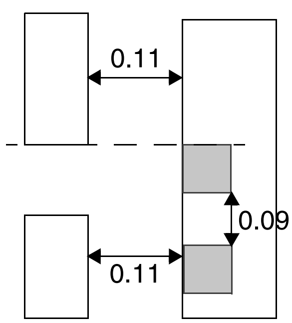
```
PROPERTY LEF58_SPACING "SPACING 0.1 SAMEMETALSHAREDEDGE 0.12 ;" ;
```



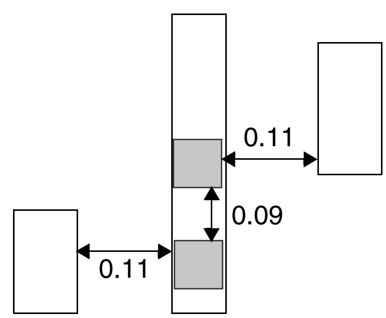
a) Okay, the cuts do not have common parallel run length greater than 0



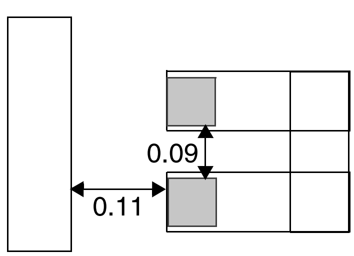
b) Violation, the neighbors could be different wires



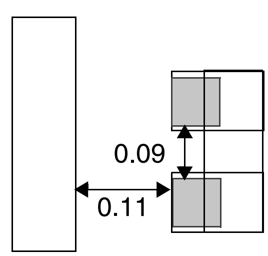
c) Okay, the top cut does not have a neighbor



d) Violation, parallel opposite neighbor edges, one neighbor per cuts, triggers the rule



e) Okay, the cuts do not have common metal shapes



f) Violation, the common metal shapes covering the entire length of the projection between the cuts

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Spacing Table Rule

The spacing table rule can be used to define cut spacing between different cut classes.

You can create a spacing table rule by using the following property definition:

```
[PROPERTY LEF58_SPACINGTABLE
       "SPACINGTABLE
          [ORTHOGONAL
             {WITHIN cutWithin SPACING orthoSpacing} ... ;
          |[DEFAULT defaultCutSpacing]
             [SAMEMASK]
             [SAMENET | SAMEMETAL | SAMEVIA]
             [LAYER secondLayerName
                 [NOSTACK]
                 [NONZEROENCLOSURE
                 | PRLFORALIGNEDCUT
                   {{className1 | ALL} TO {className2 | ALL} }...
                 | EXCEPTENCLOSURE exceptEnclosure [ABOVE|BELOW]]]
             [CENTERTOCENTER
                 {{className1 | ALL}| TO {className2 | ALL}}...]
             [CENTERANDEDGE [NOPRL]
                 {{className1 | ALL}| TO {className2 | ALL}}...]
             [PRLSPACING spacing PRL prl]
             [PRL {prl|USEDEFAULT} [ HORIZONTAL| VERTICAL] [MAXXY]
                 [{{className1 | ALL} TO {className2 | ALL} ccPrl
                        [ HORIZONTAL| VERTICAL] }...] ]
             [PRLTWOSIDES
                  {prl1 prl2 prl3 prl4 [WITHIN within]
                         className1 TO className2 spacing}...]
             [ENDEXTENSION extension [{TO className classExtension}...]
             [SIDEEXTENSION {TO className classExtension}...] ]
             [EXACTALIGNEDSPACING [HORIZONTAL | VERTICAL]
                 {{className | ALL} exactAlignedSpacing}...]
             |EXACTALIGNEDWIREWIDTHSPACING
                   {ABOVE|BELOW}
                   [HORIZONTAL| VERTICAL]
                   {{className|ALL} width exactAlignedSpacing}...]...
             [NONOPPOSITEENCLOSURESPACING
                 {className nonOppositeEnclosureSpacing}...]
             [OPPOSITEENCLOSURERESIZESPACING
                 {className resize1 resize2
                      oppositeEnclosureResizeSpacing}...
                 [EDGEALIGNED [HORIZONTAL|VERTICAL]]]
              CUTCLASS { {className1 | ALL} [SIDE | END]}...
                 {{className2 | ALL} [SIDE | END] {- | cutSpacing1}
                 {- | cutSpacing2}...}...;
          ]
           ; " ;]
```


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Where:

All other keywords are the same as the existing LEF cut layer `SPACINGTABLE` syntax.

`CENTERTOCENTER`
`{{`*`className1`*` | ALL} TO {`*`className2`*` | ALL}}...`

Computes the *`cutSpacing1`* and *`cutSpacing2`* distance from cut-center to cut-center, instead of cut-edge to cut-edge (the default behavior), for the given list of class name pairs. The *`className1`* is one of the cut classes in the first row of the table and *`className2`* is one of the cut classes in the first column of the table. The `ALL` keyword applies to all the vias on a cut layer that has only one cut class without an explicit `CUTCLASS` definition. The keyword should be specified only if one of the layers does not have a cut class.

`CENTERANDEDGE [NOPRL]`
`{{`*`className1`*` | ALL} TO {`*`className2`*` | ALL}}...`

Indicates that center-to-center measurement applies to the larger of the *`cutSpacing1`* and *`cutSpacing2`*, which must be different values, while edge-to-edge measurement applies to the smaller of the two cut spacings.

The `NOPRL` keyword indicates that both center to center and edge to edge spacing must be met, where the required center to center spacing is the maximum of *`cutSpacing1`* and *`cutSpacing2`*, and the required edge to edge spacing is the minimum of *`cutSpacing1`* and *`cutSpacing2`*.

Note: The `CENTERTOCENTER` and `CENTERANDEDGE` keywords can be defined simultaneously, but must be on different cut classes.

`{{`*`className1`*` | ALL} TO {`*`className2`*` | ALL}` *`ccPrl`*
`[HORIZONTAL | VERTICAL]}...`

Specifies a specific *`ccPrl`* value between the given cut classes. If `HORIZONTAL` or `VERTICAL` is specified, the common parallel run length is measured horizontally or vertically, respectively. Any cut class combinations that is not specified would still follow *`prl`* defined in `PRL`.
Type: Float, specified in microns

`CUTCLASS { {`*`className1`*` | ALL} [SIDE | END|]}`

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Specifies a list of cut classes, which may not necessarily be a complete list of all cut classes in the cut layer. In this case, the cut spacing requirements are not needed for any missing cut classes. The `ALL` keyword applies to all vias on a cut layer which has only one cut class without an explicit `CUTCLASS` definition.

If an intercut layer is specified (using the `LAYER` keyword), the cut classes in *`className1`* are defined for the layer for which the spacing table is defined. When intercut layer spacing is needed between a cut layer with multiple cut classes and a cut layer without a cut class, this cut class spacing table should be specified, and the `"SPACING ... LAYER ... ;"` statement cannot be used.

If the cut class has a rectangular cut shape, the `SIDE` and `END` keywords can be used to specify cut spacing on a certain edge - side/long or end/short (see Figure 2-133 on page 446). The diagram indicates the regions that the other cut via should be fully contained on certain edges to be applied.

`{{`*`className2`*` | ALL} [SIDE | END] {- | `*`cutSpacing1`*`} {- |`
`cutSpacing2``}`

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Indicates that *cutSpacing1* and *cutSpacing2* is applied between *className2* and *className1* in the first row of the table.

There are two sets of cut spacing values for each table entry. The first *cutSpacing1* value applies if there is no parallel edge overlap between the via cuts. The second *cutSpacing2* value applies if there is a parallel edge overlap greater than 0. If CENTERTOCENTER keyword is used for two cut classes, then *cutSpacing1* and *cutSpacing2* should be identical. If - is specified, the *defaultCutSpacing* value (specified with the DEFAULT keyword) is used.

If interlayer cut spacing is specified with the LAYER keyword, the cut class in *className2* is defined for the *secondLayerName* (specified with the LAYER keyword). If LAYER is not specified, then the table must be a NxN symmetrical table with spacing values in any (i, j) and (j, i) locations in the table being the same.

The ALL, SIDE, and END keywords are the same as described earlier.
Type: Float, specified in microns

DEFAULT *defaultCutSpacing*

Indicates the default cut spacing between cut classes.
Type: Float, specified in microns

Note: If a table entry contains -, the *defaultCutSpacing* value applies. In this case the DEFAULT keyword must be specified.

An undefined cut class will have the spacing of DEFAULT *defaultCutSpacing*, and the spacing values in any (i, j) and (j, i) location in the table are the same.

EDGEALIGNED [HORIZONTAL|VERTICAL]

Specifies that the opposite enclosure resize spacing is also applied if the edges of the expanded windows are aligned either horizontally or vertically or only in the specified direction (HORIZONTAL or VERTICAL). See Figure 2-136 on page 449.

ENDEXTENSION *extension* [{TO *className classExtension*}...]

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Defines extension values applied to the end/short edge of all rectangular cuts, before *cutSpacing1* and *cutSpacing2*, are measured to the cuts with extensions. If the TO keyword is specified, then *classExtension* is applied to the end/short edge of all rectangular cuts to measure the cut spacing to cuts in *className*. Otherwise, *extension* is applied to the end/short edge of all rectangular cuts to measure the cut spacing to cuts for any non-specified cut classes.
Type: Float, specified in microns

```
EXACTALIGNEDSPACING [HORIZONTAL | VERTICAL]
    {{className | ALL} exactAlignedSpacing}
```

Specifies the spacing between two *className* cuts that are perfectly aligned, either horizontally or vertically, must be greater than or equal to *exactAlignedSpacing*. If HORIZONTAL or VERTICAL is also specified, the spacing applies only to the corresponding direction. In addition, at most two such statements can be defined, one with HORIZONTAL and another with VERTICAL.

ALL means that the exactly aligned cuts of any cut class must have spacing greater than or equal to *exactAlignedSpacing*. The length of the aligned cut edges must be identical to be considered as exactly aligned.

Type: Float, specified in microns

Note: The definition of EXACTALIGNEDSPACING remains edge-to-edge even when CENTERTOCENTER keyword is specified.

```
EXACTALIGNEDWIREWIDTHSPACING {ABOVE|BELOW}
    [HORIZONTAL| VERTICAL]
    {{className|ALL} width exactAlignedSpacing}...
```


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Specifies that the spacing between two *className* cuts that are perfectly aligned, either horizontally or vertically, must be greater than or equal to *exactAlignedSpacing* if both cuts are within a wire with width greater than or equal to *width* on either the metal layer above or below, as indicated by ABOVE or BELOW. If HORIZONTAL or VERTICAL is also specified, the spacing applies only in the corresponding direction. A maximum of two such statements may be defined, one for HORIZONTAL and one for VERTICAL.

ALL means that all exactly aligned cuts, regardless of cut class, must have spacing greater than or equal to *exactAlignedSpacing*. For cuts to be considered exactly aligned, the length of their aligned cut edges must be identical.

Type: Float, specified in microns

EXCEPTENCLOSURE *exceptEnclosure* [ABOVE|BELOW]

Specifies that the inter-layer cut spacing between any cuts in the current layer to any cuts in layer *secondLayerName* applies only for the cuts in the current layer with enclosure on the above metal layer on all four sides greater than *exceptEnclosure* and parallel run length greater than 0 with the cuts in layer *secondLayerName*. In other words, if the above metal layer of the cuts in the current layer is less than or equal to *exceptEnclosure* on any cut edges, the inter-layer cut spacing is not applied. If BELOW is specified, the enclosure condition should be checked against the below metal layer. In addition, *cutSpacing1* should be zero while *cutSpacing2* is the required cut spacing between cuts for the two specified cut classes.

Type: Float, specified in microns

LAYER *secondLayerName*

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Defines the inter-cut-layer spacing between a *className1* via in the first row of the table on the layer that this spacing table is being defined to another *className2* via in the first column on *secondLayerName* cut layer. This cut spacing is ignored for same-net. This second layer must be a previously defined cut layer immediately below the layer that this spacing table is being defined, that is, "one layer look ahead" is not supported.

If an inter-cut-layer spacing table is defined for same-net cuts using the SAMENET keyword, the cuts on two different layers can always be stacked if they are exactly aligned (that is, the centers of the cuts are aligned) for same sized cuts. For different sized cuts, it is legal if the smaller cut is completely covered by the bigger cut. Otherwise, the cuts must have *cutSpacing* between them.

Note: When *secondLayerName* is not defined in a same-layer cut spacing table, then the cut classes in the columns must have the same order as the rows.

NONOPPOSITEENCLOSURESPACING
{*className nonOppositeEnclosureSpacing*}...

Specifies that the center-to-center spacing between a cut of *className* that does not fulfill the overhang defined in ENCLOSUREEDGE OPPOSITE to any cut of *className* is *nonOppositeEnclosureSpacing* independent of parallel run length.
Type: Float, specified in microns

NONZEROENCLOSURE

Specifies that the inter-layer cut spacing between cuts in the current layer to cuts in the specified *secondLayerName* only applies for cut edges with enclosure on the above metal layer of the cuts in the current layer greater than 0 and has parallel run length greater than 0 with the cuts in *secondLayerName* layer. In other words, the *cutSpacing1* should be 0, while *cutSpacing2* is the required cut spacing between cuts for the specified two cut classes.

NOSTACK

Specifies that stacked vias between any cuts in the current layer to any cuts in layer *secondLayerName* are not allowed. This construct must be specified with either SAMENET or SAMEMETAL.

OPPOSITEENCLOSURERESIZESPACING {*className resize1 resize2 oppositeEnclosureResizeSpacing*}...

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Specifies that the center-to-center spacing between two cuts of *className* that fulfill the overhang defined in ENCLOSUREEDGE OPPOSITE is *oppositeEnclosureResizeSpacing* independent of parallel run length if the corners of two windows barely touch by expanding the two opposite cut edges fulfilling the overhang defined in ENCLOSUREEDGE OPPOSITE by *resize1* and the other two opposite cut edges by *resize2*. See Figure 2-135 on page 448.
Type: Float, specified in microns

PRL {*prl*|USEDEFAULT} [HORIZONTAL| VERTICAL] [MAXXY]

Defines the condition when to apply *cutSpacing1* and *cutSpacing2* between cuts. When cuts have a common parallel run length greater than *prl*, *cutSpacing2* is used between the cuts. Otherwise, *cutSpacing1* is used. If *prl* is negative, it indicates that a neighbor cut within abs(*prl*) distance beyond the edge of a cut will need to be *cutSpacing2* distance away from it. This is a global definition that applies to cuts in all cut classes.

USEDEFAULT must be specified only along with a specific *ccPrl* value between the given cut classes in {*className1* | ALL} TO {*className2* | ALL} *ccPrl*. It means that the default PRL condition would be applied on non-specified cut class combinations.

Type: Float, specified in microns

HORIZONTAL specifies that *cutSpacing2* is used only when the common parallel run length measured horizontally is greater than *prl*. Similarly, VERTICAL specifies that *cutSpacing2* is used only when the common parallel run length measured vertically is greater than *prl*. Otherwise, *cutSpacing1* is used.

The MAXXY keyword specifies that the spacing is measured as the maximum projection style when common parallel run length is larger than the given *prl*, particularly for a negative *prl* value. Otherwise, the spacing is measured from edge-to-edge.

PRLFORALIGNEDCUT {{*className1* | ALL} TO {*className2* | ALL} } ...

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Specifies the second *cutSpacing2* value only applies to cut edges of *className2* on *secondLayerName* that fulfill the overhang defined in ENCLOSUREEDGE OPPOSITE to the cut of *className1* when parallel edge overlap is greater than 0 or *prl*, if specified. In other words, even when parallel edge overlap condition is satisfied, the *cutSpacing1* value could be used instead if the cut edges do not entirely align to the above metal. The spacing is measured from the projection on the metal edge, and negative *prl*, if specified, is treated as an extension of the metal edge.

PRLSPACING *spacing* PRL *prl*

Specifies that two cuts with any cut classes less than or equal to *prl* in both directions must have spacing greater than or equal to *spacing*. Typically, *prl* is a negative value. The table entry values would be applied on cases with PRL greater than *prl*.

Type: Float, specified in microns

PRLTWOSIDES {*prl1 prl2 prl3 prl4* [WITHIN *within*] *className1* TO *className2 spacing*}...

Specifies that the center-to-center spacing between cuts of *className1* and *className2* that are greater than or equal to *prl1* and less than or equal to *prl2* on one side and greater than or equal to *prl3* and less than or equal to *prl4* on the other side to be *spacing*. If WITHIN is specified, the cuts must be within *within*, and *spacing* is measured as corner to corner.If the cuts do not fulfill these PRL and WITHIN conditions, it would follow the specified cut class spacing as usual.

Type: Float, specified in microns

SAMEMASK

Specifies that the cut spacing applies between cuts on the same mask. When a cut layer has more than one mask, you can specify up to two cut class SPACINGTABLE rules, one with SAMENET or SAMEMETAL, and one with neither of them for the same cut layer spacing for same-mask cuts (with SAMEMASK on all of those tables). Then you can specify (without the SAMEMASK keyword) up to another two tables for different-mask cuts.

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SAMENET

Indicates that *cutSpacing1* and *cutSpacing2* values only apply to same-net cuts. The SAMENET cut spacing values should be smaller than the normal SPACINGTABLE *cutSpacing* values that apply to different-net cuts.

SAMEMETAL

Indicates that *cutSpacing1* and *cutSpacing2* values only apply to cuts that are overlapped with the same metal shape. The SAMEMETAL cut spacing values should be smaller than the normal SPACINGTABLE *cutSpacing* values that apply to different-metal cuts.

SAMEVIA

Indicates that *cutSpacing1* and *cutSpacing2* values only apply to cuts that are overlapped with the same metal shape on both the above and below metal. The SAMEVIA cut spacing values should be smaller than the normal SPACINGTABLE *cutSpacing* values that apply to different-metal cuts.

SIDEEXTENSION {TO *className classExtension*}

Defines extension value, *classExtension*, applied to the side/long edge of all rectangular cuts before *cutSpacing1* and *cutSpacing2* are measured to the cuts in *className* with extensions. When side extension is applied between certain cut classes, the end extension is not applied. Hence, you cannot apply the same *className* after the TO statement.
Type: Float, specified in microns

You cannot mix with any other cut layer spacing statements, except for ADJACENTCUTS, PARALLELWITHIN, and SAMEMETALSHAREDEDGE.

Multiple spacing among different cut classes is possible. For cut layer shapes in PIN or OBS statement that belong to a macro of class CORE, the cut size of the via should match one of the sizes of the cut classes so that the tools can determine the proper spacing for the cut. If not, specific spacing should be defined for the cut in the macro definition using the SPACING keyword that is part of the layer geometry specification in MACRO. If the SPACING keyword is not specified, minimum spacing in the cut layer is used. If the cut size of the abstracted cuts does not match one of the cut class sizes, a single spacing value is applied to all four sides of the cut. This may cause DRC violations.

As with any OBS shapes, a cut layer OBS shape is always considered to belong to a net that is different from any pin, even if it is overlapping with a pin geometry on the adjacent metal layer. In this case, different-net cut-to-cut spacing is used to compute the cut-to-cut distance between the OBS and any cut that is connected to the corresponding pin.

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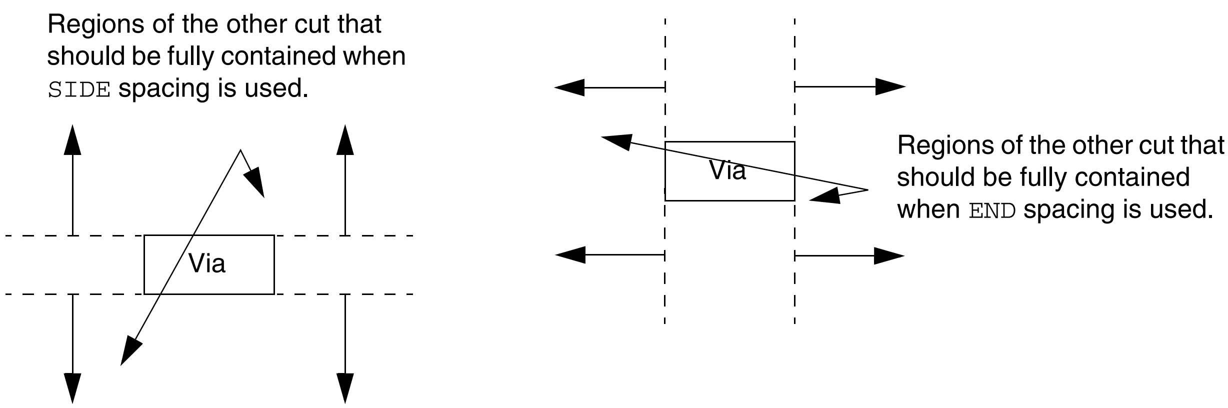
Note: It is recommended not to define a large cut layer OBS shape abstracting cut shapes, even in a macro for a non-standard cell. If defined, minimum cut spacing is applied to prevent blocking via access of nearby pins. This may, however, cause DRC violations. The cut layer blockage shapes (defined using the `BLOCKAGES` keyword) will use minimum cut spacing around them, similar to OBS.

Spacing Table Rule Examples

- The following illustration shows the regions that the cut via should be overlapped with when `SIDE` or `END` keywords are used.

Figure 2-133 Illustration of Spacing Table Rule With Side and End



- The following spacing table rules specify the spacing requirements between `VA` via and end edge of `VB` via, and end edge of `VB` via and `VA` via:

Cut-to-cut spacing between two center-to-center `VA` vias	0.20 µm
Cut-to-cut spacing between two center-to-center `VC` vias	0.50 µm
Edge-to-edge spacing between the end edge of `VB` and `VA` or `VC` via with parallel edge overlap greater than 0	0.30 µm
Edge-to-edge spacing between the end edge of `VB` via and an edge of another `VB` via with a parallel edge overlap greater than 0	0.40 µm
Edge-to-edge spacing for the rest of the combinations	0.15 µm

The rules translate into the following `SPACINGTABLE` property definition:

```
PROPERTY LEF58_SPACINGTABLE
  DEFAULT 0.15 CENTERTOCENTER VA TO VA VC TO VC
```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```
CUTCLASS    VA          VB SIDE      VB END    VC
VA          0.20  0.20  -            -  0.30   -     -
VB SIDE     -     -     -    -       -  0.40   -     -
VB END      -     0.30  -    0.40    -  0.40   -     0.30
VC          -     -     -    -       -  0.30   0.50  0.50
```

- The following spacing table rule indicates that center-to-center spacing between two square cut `VA` vias must be greater than or equal to `0.20` µm:

```
PROPERTY LEF58_SPACINGTABLE CENTERTOCENTER VA TO VA
  CUTCLASS  VA
  VA        0.20  0.20
```

Figure 2-134 Illustration of Spacing Table Rule With CenterToCenter

a) Violation. Center-to-center spacing < 0.20.

b) Okay. Center-to-center spacing >= 0.20.

- The following spacing table rule illustrates opposite enclosure resize spacing:

```
PROPERTY LEF58_ENCLOSUREEDGE
    "ENCLOSUREEDGE ABOVE 0.02 OPPOSITE ; " ;
PROPERTY LEF58_SPACINGTABLE
      "SPACINGTABLE CENTERTOCENTER VA TO VA
           OPPOSITEENCLOSURERESIZESPACING VA 0.05 0.03 0.07
           CUTCLASS VA …
           VA  0.1 0.1 … ; " ;
```


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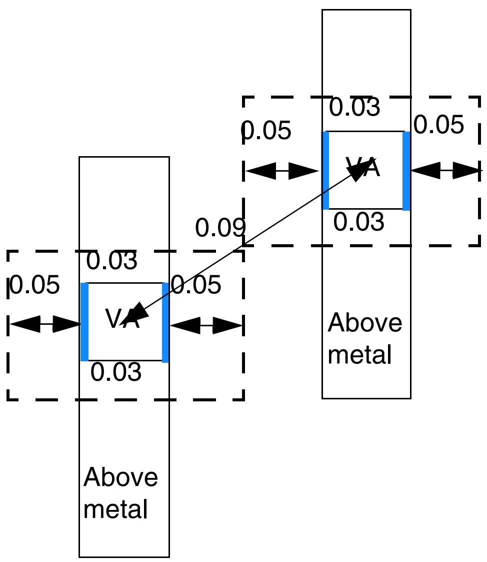
Figure 2-135 Illustration of Spacing Table Rule With Opposite Enclosure Resize Spacing



a) OK, the blue edges fulfill `ENCLOSUREEDGE OPPOSITE`, the dotted windows after expanding `0.05` on the blue edges and `0.03` on the other two opposite edges touch at the corners, and cut spacing of `0.09` (>= `0.07`) is good.

- The following spacing table rule illustrates an aligned edge condition in opposite enclosure resize spacing:

```
PROPERTY LEF58_ENCLOSUREEDGE
    "ENCLOSUREEDGE ABOVE 0.02 OPPOSITE ; " ;
PROPERTY LEF58_SPACINGTABLE
      "SPACINGTABLE CENTERTOCENTER VA TO VA
           OPPOSITEENCLOSURERESIZESPACING VA 0.05 0.03 0.07
           EDGEALIGNED HORIZONTAL
           CUTCLASS VA ...
           VA  0.1 0.1 ... ; " ;
```


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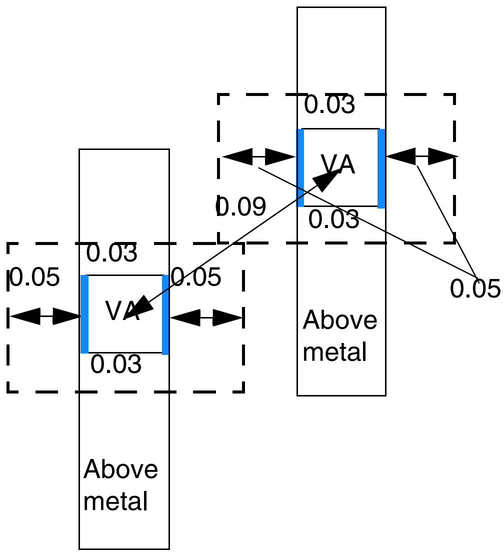
Figure 2-136 Illustration of Spacing Table Rule with an Aligned Edge Condition in Opposite Enclosure Resize Spacing



a) OK, the blue edges fulfill `ENCLOSUREEDGE OPPOSITE`, the dotted windows after expanding `0.05` on the blue edges and `0.03` on the other two opposite edges have horizontally aligned edges, and cut spacing of `0.09` (>= `0.07`) is good.

- The following spacing table rule indicates that a default spacing of `0.15` µm is required between via `VC` and end edge of via `VB` when parallel edge overlap is less than or equal to 0:

```
PROPERTY LEF58_SPACINGTABLE DEFAULT 0.15
CUTCLASS     VB SIDE     VB END      VC
VB SIDE      -   -       -   0.40    -     -
VB END       -   0.40    -   0.40    -     0.30
VC           -   -       -   0.30    0.50  0.50
```


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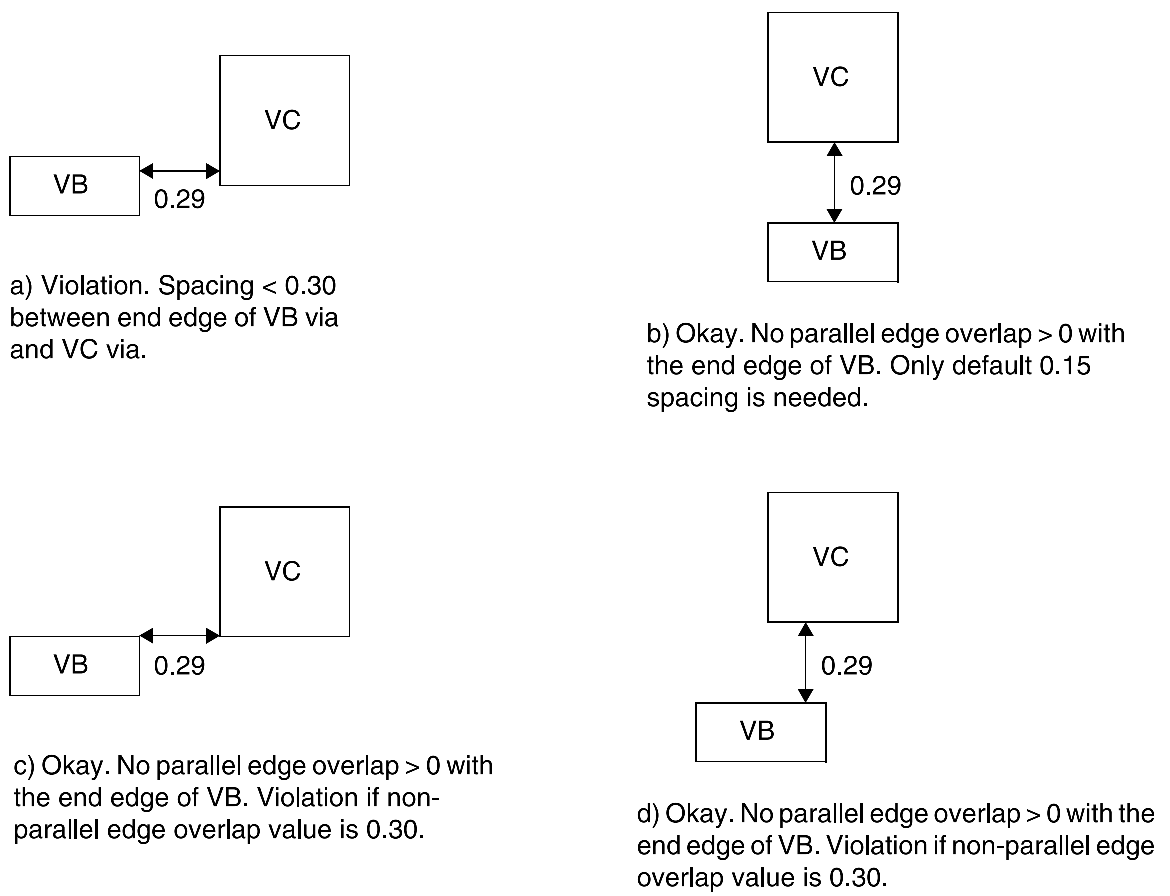
Figure 2-137 Illustration of Spacing Table Rule With Default



- The following spacing table rule indicates the spacing requirements between two `VB` vias when default spacing of `0.15` µm is specified:

```
PROPERTY LEF58_SPACINGTABLE DEFAULT 0.15
CUTCLASS    VB SIDE    VB END
VB SIDE     -   0.40   -   -
VB END      -   -      -   -
```


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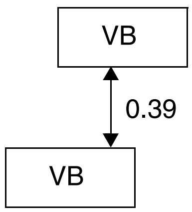
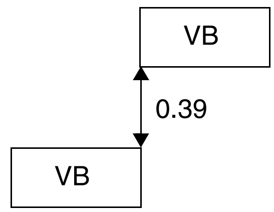
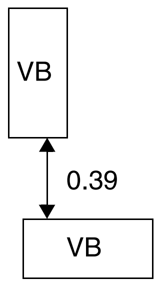
Figure 2-138 Illustration of Spacing Table Rule With Default



a) Violation. Spacing < 0.40 between two VB vias on the side edges.



b) Okay. No parallel edge overlap > 0 between two VB vias. Violation if non-parallel edge overlap value is 0.40.



c) Okay. Spacing (default spacing) >= 0.15 between side edge of VB and end edge of VB.

- The following spacing table rules specify the intercut layer spacing of different metals between vias:

Intercut layer center-to-center spacing between two `VA` vias, when one of the vias is on the current cut layer and the other is on `V1` cut layer	0.10 µm
Intercut layer spacing between the side edge of `VB` to `VA` via, when one of the vias is on the current cut layer and the other is on `V1` cut layer	0.20 µm
Intercut layer spacing between the side edge of two `VB` vias, when one of the vias is on the current cut layer and the other is on `V1` cut layer.	0.30 µm

Note: All the intercut layer spacings are excluded for same metal cuts.

The rules translate into the following `SPACINGTABLE` property definition:

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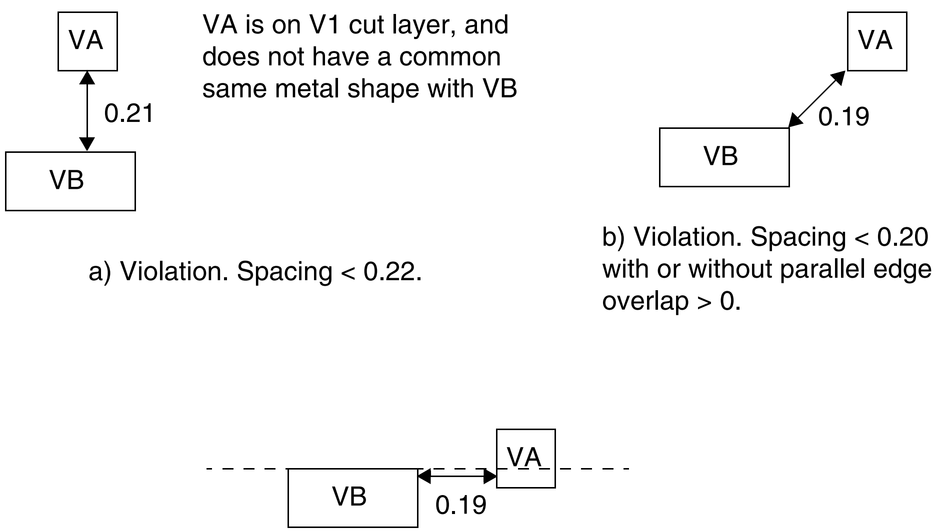
```
PROPERTY LEF58_SPACINGTABLE LAYER V1 CENTERTOCENTER VA TO VA
CUTCLASS     VA            VB SIDE
VA           0.10  0.10    0.20  0.20
VB SIDE      0.20  0.20    0.30  0.30
```

- The following spacing table rule indicates that via `VA` should not overlap with the projection line from the side edge of via `VB`:

```
PROPERTY LEF58_SPACINGTABLE DEFAULT 0.0 LAYER V1
CUTCLASS     VA            VB SIDE
VA           -     -       0.20  0.22
VB SIDE      0.20  0.22    -     -
```

Figure 2-139 Illustration of Spacing Table Rule With Layer



a) Violation. Spacing < 0.22.

b) Violation. Spacing < 0.20 with or without parallel edge overlap > 0.

c) Okay. VA is not fully above the projection line from the SIDE edge

- The following spacing table rule specifies that the intercut layer spacing between `VA` via on the current cut layer to any center-to-center vias on `C1` cut layer, when the cuts do not share a common same metal shape, must be `0.15` µm:

```
PROPERTY LEF58_SPACINGTABLE LAYER C1 CENTERTOCENTER VA TO ALL
CUTCLASS     VA
ALL          0.15  0.15
```

- The following illustration defines `PRL` with `MAXXY`:

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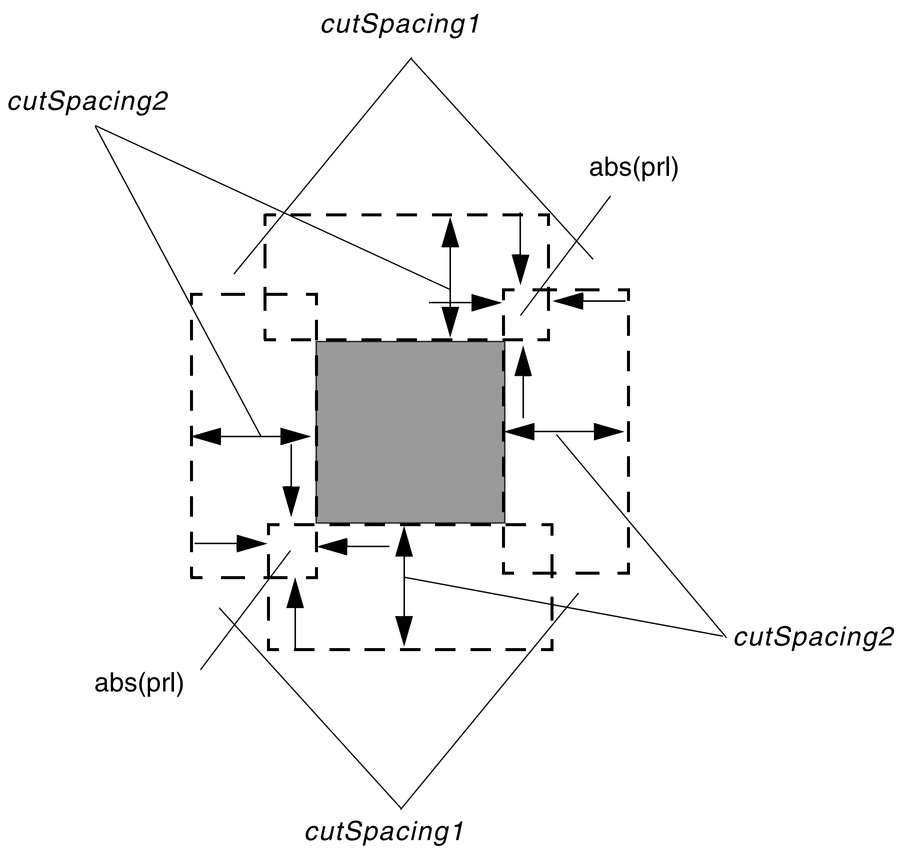
Figure 2-140 Definition of Spacing Table Rule With PRL

Definition of `PRL` *`prl`* [`MAXXY`] on a square cut when the prl value is negative.



The four dotted line regions with *cutSpacing2* from the edge of a cut and abs(*prl*) extended beyond the edges are forbidden from having a neighbor cut if `MAXXY` is specified. Otherwise, *cutSpacing2* is measured as edge-to-edge style in those regions. If a neighbor cut is in the four corners, *cutSpacing1* should be observed.

- The following spacing table rule illustrates the use of `PRLTWOSIDES`:

```
PROPERTY LEF58_SPACINGTABLE "
  SPACINGTABLE CENTERTOCENTER VA TO VA
     PRLTWOSIDES -0.07 -0.05 -0.1 -0.08 VA TO VA 0.15
     CUTCLASS VA ...
     VA     0.2 0.2 ... ; " ;
```


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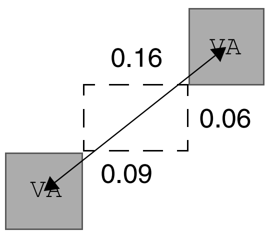
Figure 2-141 Definition of Spacing Table Rule With PRL

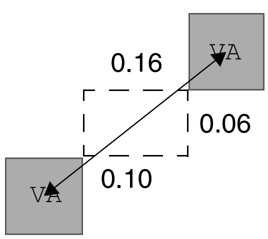


a) OK, the PRL conditions of PRLTWOSIDES is fulfilled, 0.16 (>= 0.15) is OK

- The following spacing table rule illustrates the use of `PRLTWOSIDES` with `WITHIN`:

```
PROPERTY LEF58_SPACINGTABLE "
  SPACINGTABLE CENTERTOCENTER VA TO VA
     PRLTWOSIDES -0.07 -0.05 -0.1 -0.08 WITHIN 0.11
         VA TO VA 0.09
     CUTCLASS VA ...
     VA     0.2 0.2 ... ; " ;
```

Figure 2-142 Definition of Spacing Table Rule With WITHIN in PRLTWOSIDES



a) Violation, corner to corner spacing is 0.117 (> 0.11). The `WITHIN` condition is not fulfilled, 0.2 center to center is needed & failed.

- The following spacing table rule indicates that 0.20 µm is the cut to cut spacing between two VA vias measuring center-to-center when the cuts have parallel edge overlap greater than 0 while 0.10 µm is the cut to cut spacing between two VA vias measuring edge-to-edge when the cuts have no parallel edge overlap:

```
PROPERTY LEF58_SPACINGTABLE
  "SPACINGTABLE
     CENTERANDEDGE VA TO VA
     CUTCLASS    VA
     VA         0.10  0.20 ; " ;
```

- The following spacing table rule indicates that the cut to cut spacing between two VA vias is the maximum of 0.20 µm measuring center-to-center and 0.10 µm measuring edge-to-edge:

```
PROPERTY LEF58_SPACINGTABLE
```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```
  "SPACINGTABLE
     CENTERANDEDGE NOPRL VA TO VA
     CUTCLASS    VA
     VA         0.10  0.20 ; " ;
```

- The following spacing table rule indicates that two perfectly aligned `VA` cuts must have spacing greater than or equal to 0.1 µm, two cuts with parallel edge overlap greater than 0 must have spacing greater than or equal to 0.12 µm, and two cuts without parallel edge overlap must have spacing greater than or equal to 0.11 µm:

```
PROPERTY LEF58_SPACINGTABLE
  "EXACTALIGNEDSPACING VA 0.1
   CUTCLASS  VA ...
   VA  0.11  0.12 ...;  " ;
```

- The following spacing table rule indicates that 0.15 µm extension is applied to the end/short edge of rectangular VB cuts before 0.10 µm spacing is required to square VC cuts, -0.05 µm extension (pulled in by 0.05 µm) is applied to the end/short edge of rectangular VB cuts before 0.10 µm spacing is required to square VA cuts, and 0.03 µm extension is applied to the side/long edge of rectangular VB cuts before 0.10 µm spacing is required between two VB cuts.

```
PROPERTY LEF58_SPACINGTABLE
  "SPACINGTABLE
     DEFAULT 0.10
     ENDEXTENSION -0.05 TO VC 0.15 SIDEEXTENSION TO VB 0.03
     CUTCLASS     VA    VB END    VB SIDE   VC
     VA         -    - -    -    -    -    -    -
     VB END     -    - -    -    -    -    -    -
     VB SIDE    -    - -    -    -    -    -    -
     VC         -    - -    -    -    -    -    -  ; " ;
```

- The following example shows spacing table rule with `ENDEXTENSION`:

```
PROPERTY LEF58_SPACINGTABLE
  "SPACINGTABLE
     DEFAULT 0.10
     ENDEXTENSION 0.05 TO VC 0.15
     CUTCLASS     VA    VB END    VB SIDE   VC
     VA         -    - -    -    -    -    -    -
     VB END     -    - -    -    -    -    -    -
     VA SIDE    -    - -    -    -    -    -    -
     VC         -    - -    -    -    -    -    -  ; " ;
```


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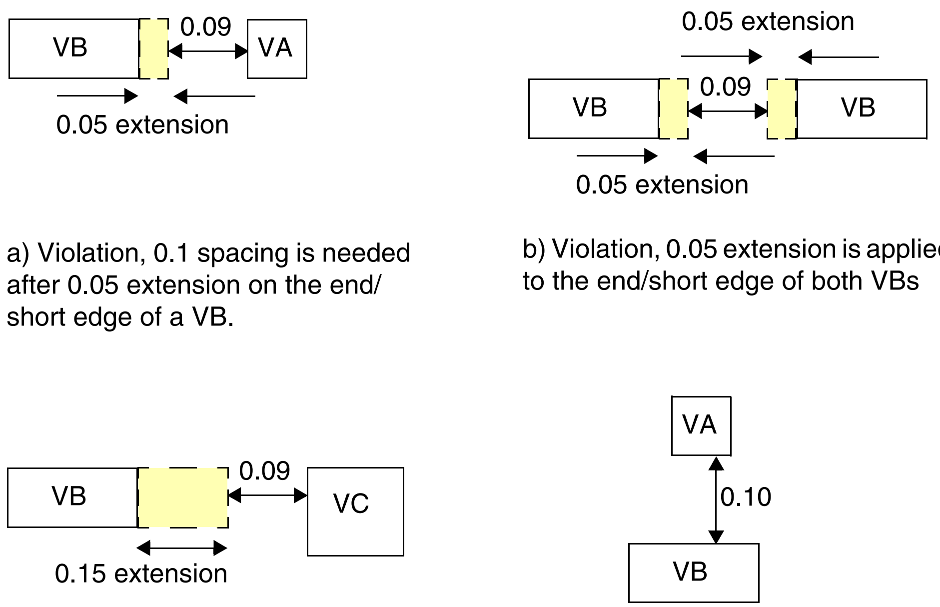
Figure 2-143 Illustration of Spacing Table Rule with ENDEXTENSION



a) Violation, 0.1 spacing is needed after 0.05 extension on the end/ short edge of a VB.

b) Violation, 0.05 extension is applied to the end/short edge of both VBs

c) Violation, 0.15 extension is applied when the neighbor cut belongs to VC

d) OK, no extension is applied to the side/long edge of a VB

- The following spacing table rule illustrates use of `NONZEROENCLOSURE` on layer `V2`:

```
PROPERTY LEF58_SPACINGTABLE
  "SPACINGTABLE
     LAYER V1
           NONZEROENCLOSURE
     CUTCLASS     V2....
     V1           0.000   0.05...; " ;
```


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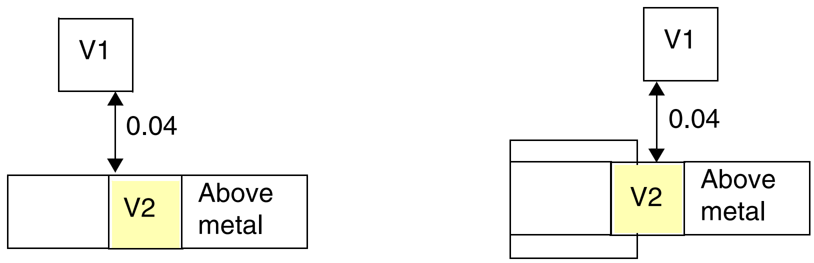
Figure 2-144 Illustration of Spacing Table Rule with NONZEROENCLOSURE



a) OK, the top edge of V2 has zero enclosure, and the rule is not triggered.

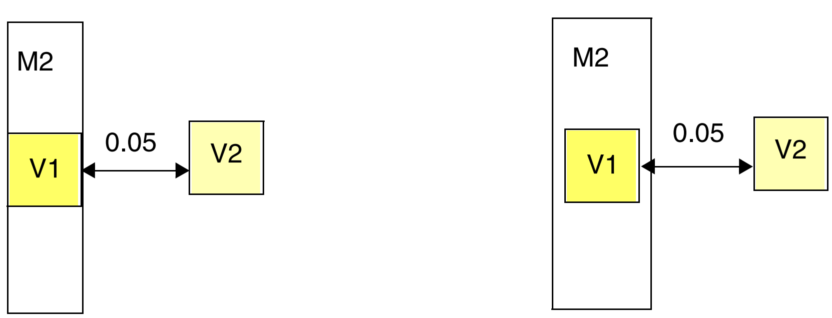
b) Violation, as long as part of the top edge of V2 has non-zero enclosure, and that edge has a neighbor cut with parallel run length > 0

- The following example shows parallel run length for aligned cuts:

  On layer V2,

```
PROPERTY LEF58_SPACINGTABLE
    "SPACINGTABLE
      LAYER V1
         PRLFORALIGNEDCUT V2 TO V1
      CUTCLASS          V2 ...
      V1                0.06    0.05 ... ; " ;
```

Figure 2-145 Example of Spacing Table Rule with PRLFORALIGNEDCUT



a) OK, the right edge of V1 aligns with above metal of M2, 0.05 could be applied

b) Violation, the right edge does not align with above metal of M2, 0.06 spacing is used and fails

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- The following example shows parallel run length for aligned cuts:

  On layer V2,

```
PROPERTY LEF58_ENCLOSUREEDGE "
    ENCLOSUREEDGE ABOVE 0.02 OPPOSITE ; " ;
PROPERTY LEF58_SPACINGTABLE
    "SPACINGTABLE
      LAYER V1
         PRLFORALIGNEDCUT V2 TO V1
         PRL -0.04
      CUTCLASS          V2 ...
      V1                0.09    0.05 ... ; " ;
```

Figure 2-146 Example of Spacing Table Rule with PRLFORALIGNEDCUT

a) OK, the right edge of V1 fulfills `ENCLOSUREEDGE OPPOSITE`, and V2 is barely outside the dotted search window on the projected metal edge by extending 0.04 of prl vertically and 0.05 of spacing horizontally.

b) Violation, the right edge does not does not fulfill `ENCLOSUREEDGE OPPOSITE`, 0.09 spacing is used and fails

Spacing Table with Center Spacing Rule

You can create a center spacing rule to specify the intercut layer spacing among cuts in center-to-center style.

You can use the following property definition:

```
PROPERTY LEF58_SPACINGTABLE
       "SPACINGTABLE CENTERSPACING
            LAYER secondLayerName
            [OVERLAPMETALSPACING metalLayerName WIDTH width
                {className1 TO className2 spacing}...]
```


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```
        CUTCLASS {{className1 | ALL} }...
            {{className2 | ALL} {cutSpacing}...}...
    ; " ;
```

Where:

`LAYER` *`secondLayerName`*

The *`secondLayerName`* variable defines the intercut layer spacing between a *`className1`* via in the first row of the table on the layer that this spacing table is being defined to another *`className2`* via in the first column on the *`secondLayerName`* cut layer. This cut spacing is ignored for same-net. This second layer must be a previously defined cut layer immediately below the layer that this spacing table is being defined, that is., "one layer look ahead" is not supported.

`CUTCLASS {{`*`className1`*` | ALL}}...`

Specifies a list of cut classes, which may not necessarily be a complete list of all cut classes in the cut layer. Cut spacing requirements are not needed for any missing cut classes.

The `ALL` keyword applies to all vias on a cut layer which has only one cut class without an explicit `CUTCLASS` definition.

`{{`*`className2`*` | ALL} {`*`cutSpacing`*`}...}...`

The *`className2`* variable indicates that the center-to-center *`cutSpacing`* is applied between itself on *`secondLayerName`* to the first row *`className1`*.

The `ALL` keyword has the same meaning as defined previously.

Type: Float, specified in microns

`[OVERLAPMETALSPACING` *`metalLayerName`* `WIDTH` *`width`*
    `{`*`className1`* `TO` *`className2 spacing`*`}...]`

Specifies that when a *`className1`* cut overlaps with a wire with width greater than or equal to *`width`* on *`metalLayerName`*, center-to-center *`spacing`* would be applied between that cut and a *`className2`* cut. *`metalLayerName`* must be a metal layer.

Type: Float, specified in microns

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For each cut layer, there can be at the most one such inter-layer `CUTCLASS CENTERSPACING SPACINGTABLE`. If such a spacing table is specified, then any other inter-layer spacing table cannot be defined.

Center Spacing Rule Examples

- The following example indicates that 0.1 µm is the center-to-center spacing between two `VA` via cuts, 0.20 µm is the center-to-center spacing between two `VB` via cuts, and 0.15 µm is the center-to-center spacing between a `VA` via cut and a `VB` via cut:

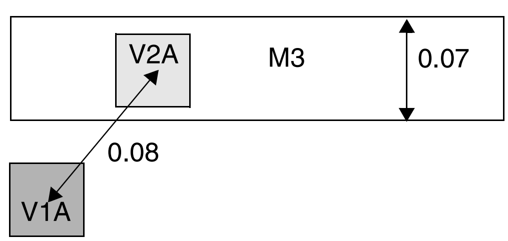
```
PROPERTY LEF58_SPACINGTABLE
  "SPACINGTABLE CENTERSPACING
    LAYER V1
        CUTCLASS     VA        VB
        VA           0.10      0.15
        VB           0.15      0.20 ; " ;
```

  One of the via is on the current layer with the spacing table definition while the other via is on layer `V1`.

- The following example is an illustration of spacing table with a center spacing rule defined on V2:

```
PROPERTY LEF58_SPACINGTABLE "
SPACINGTABLE CENTERSPACING
    LAYER V1 OVERLAPMETALSPACING M3 WIDTH 0.06
    V2A TO V1A 0.07
    CUTCLASS V2A ..
    V1A     0.09  ... ; " ;
```

Figure 2-147 Illustration of Spacing Table with Center Spacing Rule



a) OK because `V2A` overlaps with an `M3` wire with width of `0.07` (>= `0.06`), `0.07` center-to-center spacing is required and is met. Violation if `M3` metal width is changed to `0.05`.

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One Array Rule

You can create a one array rule to define a 1xN array via by using the following property definition:

```
PROPERTY LEF58_ONEDARRAY
      "ONEDARRAY CUTCLASS className CUTSPACING cutSpacing
            ARRAYCUTS arrayCuts SPACING spacing
            LAYER secondLayerName SPACING interLayerSpacing
            ENCLOSURE overhang1 overhang2
            ; " ;
```

Where:

`ARRAYCUTS` *`arrayCuts`*

Specifies the N, which is greater than or equal to *`arrayCuts`* in the 1xN array via.
Type: Integer

`ENCLOSURE` *`overhang1 overhang2`*

Specifies that the via cuts in this 1XN array must have *`overhang1`* on two opposite sides and *`overhang2`* on the other two opposite sides on the above metal layer.
Type: Float, specified in microns

`LAYER` *`secondLayerName`* `SPACING` *`interLayerSpacing`*

Specifies the edge-to-edge cut spacing to be *`interLayerSpacing`* between this 1xN array via to any other different-net via cuts of any cut classes in layer *`secondLayerName`*.
Note: Forward look-up reference is allowed for `LAYER` *`secondLayerName`*.
Type: Float, specified in microns

`ONEDARRAY CUTCLASS` *`className`* `CUTSPACING` *`cutSpacing`*

Defines a 1xN array via of *`className`*, which cuts, with edge-to-edge spacing of *`cutSpacing`*, must be perfectly aligned in the preferred direction of the routing layer below this cut layer.
Type: Float, specified in microns

`SPACING` *`spacing`*

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Specifies the edge-to-edge cut spacing of this 1xN array via to any other via cuts of any cut classes or another 1xN array via to be *spacing* when the cuts have common parallel run length are greater than 0. The 1xN array via will be merged together as one rectangular shape before the spacing check is applied. that will override other cut spacing rules (for common parallel run length greater than 0).
Type: Float, specified in microns

Opposite Overlap Cut Spacing Rule

You can create an opposite overlap cut spacing rule to specify the inter-layer cut spacing between a cut on the given layer to two cuts on another layer.

You can use the following property definition:

```
PROPERTY LEF58_OPPOSITEOVERLAPCUTSPACING
     "OPPOSITEOVERLAPCUTSPACING spacing PRL prl
             LAYER secondLayerName
            ; " ;
```

Where:

`OPPOSITEOVERLAPCUTSPACING` *spacing* `PRL` *prl* `LAYER` *secondLayerName*

Specifies the inter-layer cut spacing between a cut of any cut class on the given layer to two cuts of any cut class on *secondLayerName* to be *spacing* when both the cuts on *secondLayerName* have a parallel run length greater than or equal to *prl* to the cut on the given layer and both are on opposite sides of that cut. It is a violation only if both of the inter-layer neighboring cuts are at less than *spacing*. *secondLayerName* could be an adjacent cut layer with forward look-up reference (above cut layer). This rule does not apply when all three cuts are on the same net.
Type: Float, specified in microns

Opposite Overlap Cut Spacing Example

- The following example is an illustration of an opposite overlap cut spacing rule defined on layer `VIA2` as follows:

```
PROPERTY LEF58_OPPOSITEOVERLAPCUTSPACING "
```


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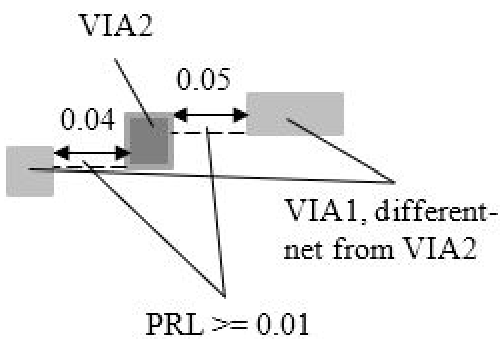
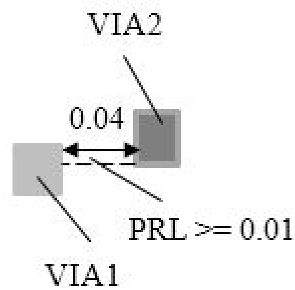
```
OPPOSITEOVERLAPCUTSPACING 0.06 PRL 0.01
LAYER VIA1 ; " ;
```

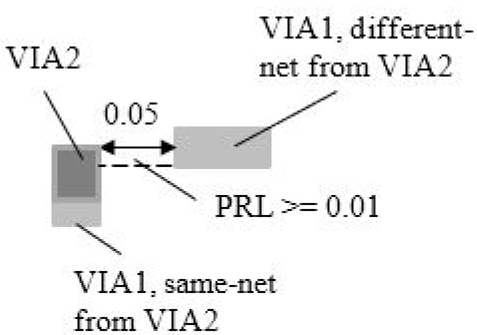
Figure 2-148 Illustration of Opposite Overlap Cut Spacing Rule



a) Violation. A `VIA2` cut has two `VIA1` cuts on two opposite sides with `PRL >= 0.01` on each of them and spacing of `< 0.06`.



b) OK. Only one `VIA1` cut would not trigger the rule.



c) Violation. An overlapping same-net `VIA1` would always be a violation when there is another different-net `VIA2` cut fulfilling the PRL condition.

Orthogonal Spacing Rule

You can create an orthogonal spacing rule to define additional spacing constraints for two exactly aligned cuts.

You can use the following property definition:

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```
PROPERTY LEF58_ORTHOGONALSPACING
       "ORTHOGONALSPACING cutSpacing
           [CUTCLASS className [TO ALL]| CUTCLASSLIST {classNameList}...]
           {ABOVE | BELOW}
           PARALLEL parLength WITHIN parWithin WIDTH width
           [METALPRL metalPrl METALWITHIN metalWithin]
           [HORIZONTAL | VERTICAL]
           ; " ;
```

Where:

`CUTCLASSLIST {`*`classNameList`*`}...`

Specifies that the rule applies only on cuts belonging the list of the given cut classes.

`HORIZONTAL | VERTICAL`

Specifies that *`cutSpacing`* applies only in the given direction.

`METALPRL `*`metalPrl`*` METALWITHIN `*`metalWithin`*

Specifies that the search window for metal neighbor wire is formed by *`metalPrl`* and *`metalWithin`* on the cut having a neighbor cut. If the wire containing the neighboring cut touches or overlaps with the metal search window, it is a violation.
Type: Float, specified in microns

`ORTHOGONALSPACING `*`cutSpacing`*
`[CUTCLASS `*`className`*`] {ABOVE | BELOW}`
`PARALLEL `*`parLength`*` WITHIN `*`parWithin`*` WIDTH `*`width`*

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The orthogonal spacing rule specifies that if there are two exactly aligned cuts of cut class *className*, if specified, or of any cut classes with spacing less than *cutSpacing*, and there is another cut of cut class *className*, if specified, or of any cut classes within *parWithin* having parallel run length greater than *parLength* in the orthogonal direction, it is a violation to have a wire with width less than or equal to *width* on the above or below metal layer in ABOVE or BELOW overlapping with, or even touching, a search window formed by extended *parLength* on both sides on the cut edge having a neighbor to the neighbor cut edge. Metal containing the cuts is excluded.

If zero is specified as *cutSpacing*, this rule will be applicable for two cuts instead of three. There is no need to have an exactly aligned cut. Just a neighboring cut fulfilling *parWithin* and *parLength* conditions would trigger the rule. See Figure 2-150 on page 466.

Type: Float, specified in microns

TO ALL — Specifies that the rule is triggered if at least one cut belongs to the given cut class, *className*; the other cuts could belong to any cut class.

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Figure 2-149 Illustration of Orthogonal Spacing Rule

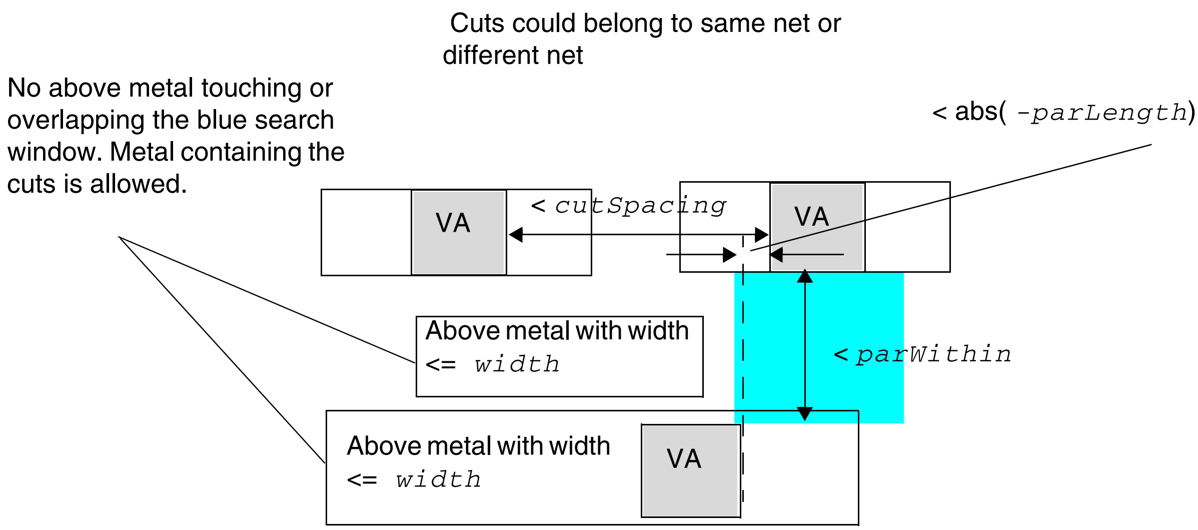

Illustration of `ORTHOGONALSPACING` *`cutSpacing`*
`CUTCLASS VA ABOVE`
`PARALLEL` *`-parLength`* `WITHIN` *`parWithin`*
`WIDTH` *`width`*

Figure 2-150 Illustration of Orthogonal Spacing Rule with Zero Cut Spacing

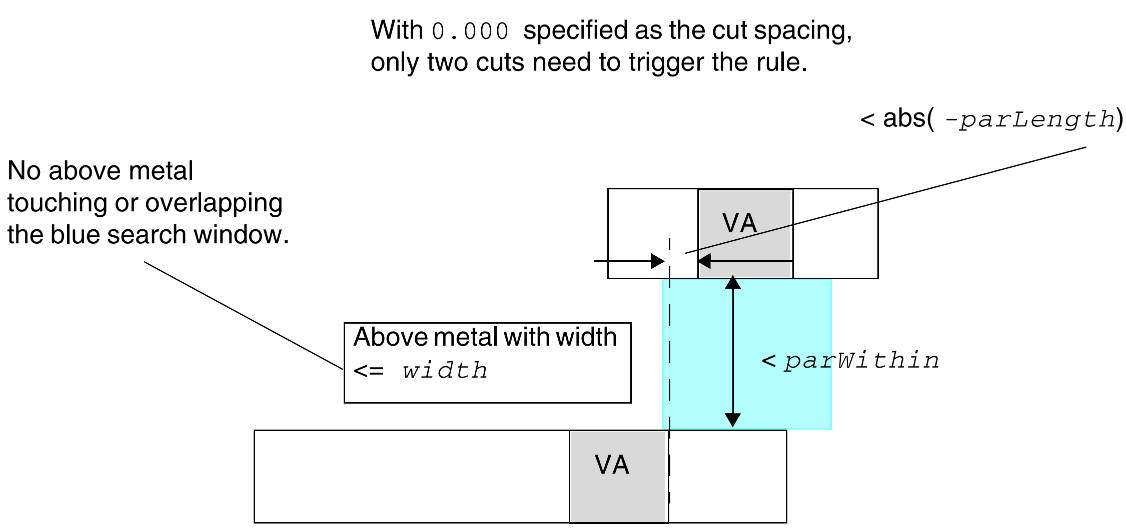

Illustration of `ORTHOGONALSPACING 0.000`
`CUTCLASS VA ABOVE`
`PARALLEL` *`-parLength`* `WITHIN` *`parWithin`*
`WIDTH` *`width`*

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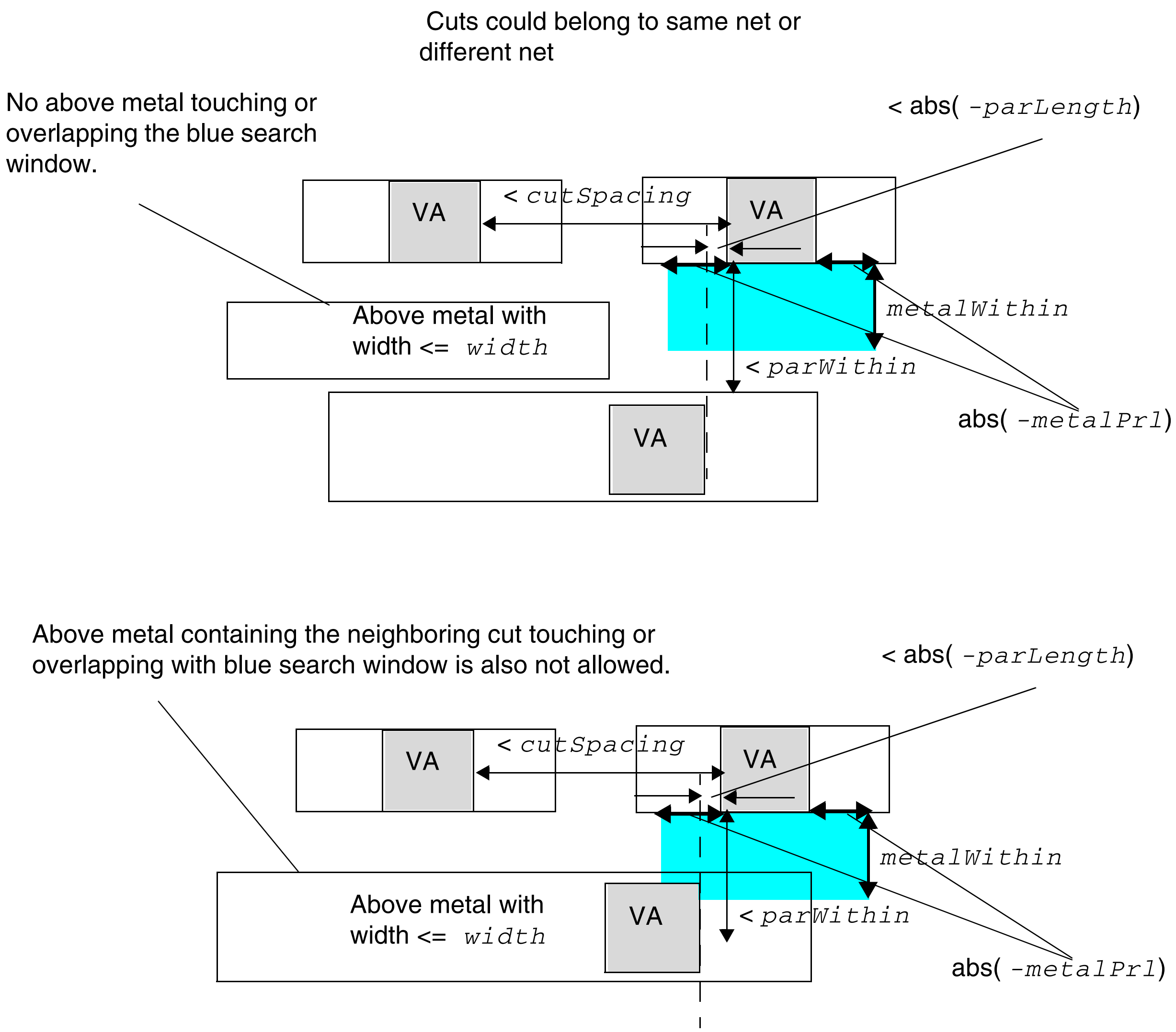
Figure 2-151 Illustrations of the Orthogonal Spacing Rule with METALPRL and METALWITHIN



Illustrations of `ORTHOGONALSPACING` *`cutSpacing`*
`CUTCLASS VA ABOVE`
`PARALLEL` *`-parLength`* `WITHIN` *`parWithin`*
`WIDTH` *`width`*
`METALPRL` *`metalPrl`* `METALWITHIN` *`metalWithin`*

PRL Two Sides Spacing Rule

You can create a PRL two sides spacing rule to define additional spacing constraints for cut spacing.

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You can use the following property definition:

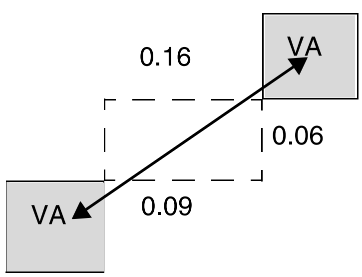
```
PROPERTY LEF58_PRLTWOSIDESSPACING
       "PRLTWOSIDESSPACING prlSpacing nonPrlSpacing
           CUTCLASS className1 TO className2
           PRL prl1 prl2 prl3 prl4
           ; " ;
```

The PRL two sides spacing rule specifies that the center-to-center spacing between a cut of `className1` and another cut of `className2` that are greater than or equal to `prl1` and less than or equal to `prl2` on one side, and greater than or equal to `prl3` and less than or equal to `prl4` on the other side to be `prlSpacing`. Otherwise, the center-to-center spacing of `nonPrlSpacing` should be applied on them. This spacing is checked in addition to the spacing defined in `CUTCLASS SPACINGTABLE`. This means that it is a violation if cut spacing fails in `PRLTWOSIDESSPACING` or in the spacing table.

Figure 2-152 Illustration of the PRL Two Sides Spacing Rule

Example of

```
PROPERTY LEF58_PRLTWOSIDESSPACING  "
        PRLTWOSIDESSPACING 0.15 0.2 CUTCLASS VA TO VA
                 PRLTWOSIDES -0.07 -0.05 -0.1 -0.08 ; " ;
```



a) OK, the PRL conditions of `PRLTWOSIDES` is fulfilled, 0.16 (>= 0.15) is OK

Region Rule

You can create a region rule to define area-based rules on a cut layer.

You can use the following property definition:

```
PROPERTY LEF58_REGION
       "REGION regionLayerName BASEDLAYER cutLayerName
       ; " ;
```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Where:

`REGION` *`regionLayerName`* `BASEDLAYER` *`cutLayerName`*

Specifies a set of region- or area-based rules on a cut layer *`cutLayerName`*. *`regionLayerName`* is a masterslice layer with type `REGION`. Users can create a blockage on that masterslice layer on a design to indicate the areas of the region. In addition, `MACRO` may also have `OBS` on that masterslice layer to indicate that the areas of its instance should be subjected to these region-based rules. All the rules specified on this cut layer with `REGION` are applied to the corresponding areas of *`regionLayerName`* on *`cutLayerName`*. The areas outside *`cutLayerName`* honor the set of rules on the given cut layer without type `REGION`. If a rule check crosses the boundaries of a region, it is subjected to the rules for both inside and outside the region, with the tighter rule values of the two sets dominating.

The set of rules defined in the layer with type `REGION` should be very small as compared to the layer without type `REGION`. The rules defined in this layer with type `REGION` should be a replacement of the corresponding rule (having the same set of the constructs of a LEF statement or a complete replacement on a table, like `SPACINGTABLE`) on the layer without type `REGION`. All the other rules defined on the layer without type `REGION` are also applicable on the layer with type `REGION`.

Region Rule Example

The following example means that a cut overlapping region `R1` should have spacing of `0.7` and a cut spacing check outside region `R1` should have spacing of `0.5`. Adjacent cut and any other rules on layer `V1` (without `REGION`) would also be applied on region `R1`:

```
LAYER R1
    TYPE MASTERSLICE ;
    PROPERTY LEF58_TYPE "TYPE REGION ; " ;
END R1
...
LAYER V1
    TYPE CUT ;
    ...
    SPACING 0.5 ;
    SPACING 0.8 ADJACENTCUTS 2 WITHIN 0.8 ;
```


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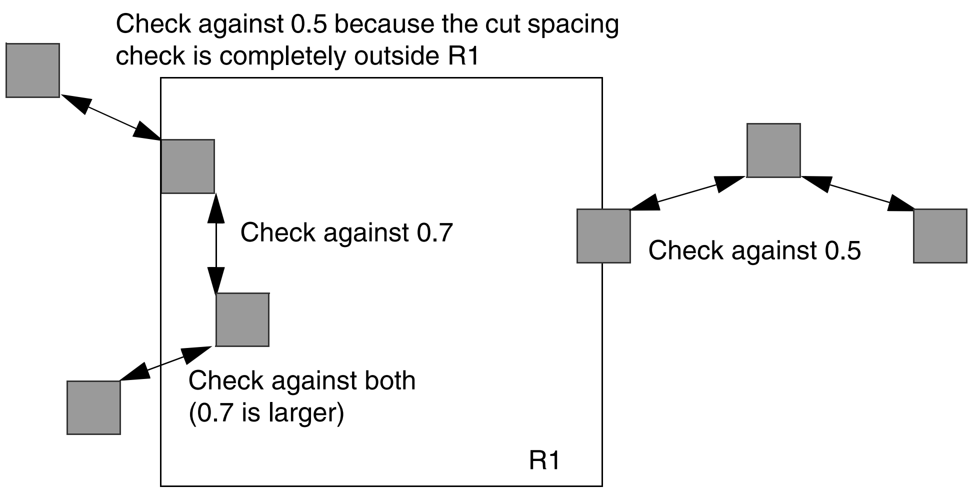
```
    ...
END V1

LAYER V1R1
    TYPE CUT ;
    PROPERTY LEF58_REGION "REGION R1 BASEDLAYER V1 ; " ;
    ...
    SPACING 0.7 ;
    ...
END V1R1
```

Figure 2-153 Illustration of the Region Rule



Same-Metal Aligned Cuts Rule

You can create a same-metal aligned cuts rule to define the number of center-aligned cuts that share the same metal on a cut layer.

You can use the following property definition:

```
PROPERTY LEF58_SAMEMETALALIGNEDCUTS
       "SAMEMETALALIGNEDCUTS numCuts
           CUTCLASS {className | ALL}
           | {CUTCLASSLIST {classNameList}...}...}
           [ABOVE | BELOW] WIDTH width SPACING spacing
           [METALMASK metalMaskNum]
           [EXCEPTENCLOSURE exceptEnclosure {ABOVE|BELOW}]
           [PARALLEL parLength WITHIN parWithin
           | METALWITHIN metalWithin [WITHCUT]
           | CUTWITHIN cutWithin
```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```
| NOVIAOVERLAPMETAL within]
[CENTERTOCENTER | EDGETOEDGE]
; " ;
```

Where:

`ABOVE | BELOW` Specifies that the rule applies only if the cuts share the same metal on the given above or below metal layer.

`CENTERTOCENTER | EDGETOEDGE`

Specifies that cut spacing is measured as center-to-center or edge-to-edge if `CENTERTOCENTER` or `EDGETOEDGE` is specified. This would overwrite any built-in cut spacing style.

`CUTCLASS {`*`className`*` | ALL}`

Specifies that the rule applies only if the cuts belong to the given cut class of *`className`* or any cut class if `ALL` is specified.

`{CUTCLASSLIST {`*`classNameList`*`}...}...`

Specifies that the rule applies only if there is at least one cut belonging to a cut class in each set of the given cut class list in *`classNameList`* and all of the cuts must belong to a cut class in all of the given cut classes.

For example,

```
PROPERTY LEF58_ SAMEMETALALIGNEDCUTS "
    SAMEMETALALIGNEDCUTS 3 CUTCLASSLIST VA VB
    CUTCLASSLIST VA VC ....″ ;
```

means that the rule applies only with a cut of `VA` or `VB`, another cut of `VA` or `VC` and at least two more cuts of `VA`, `VB`, and `VC`. For example, four `VA` cuts, one `VA` cut and three `VB` cuts, or two `VB` cuts and two `VC` cuts could trigger the rule.

`CUTWITHIN `*`cutWithin`*

Specifies that the rule applies only if the current layer has a neighboring cut on the adjacent cut layer that has a parallel run length (PRL) greater than `0` with a window formed by the merging of the violating cuts and is within *`cutWithin`* from the wire containing the aligned cuts. The neighboring wire containing the neighboring cut does not need to cover all of the aligned cuts. In addition, spacing is measured in the edge-to-edge style.

Type: Float, specified in microns

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`EXCEPTENCLOSURE` *`exceptEnclosure`* `{ABOVE|BELOW}`

Specifies that the rule does not apply if all same-metal aligned cuts have enclosure greater than or equal to *`exceptEnclosure`* on all four sides, either in the `ABOVE` or `BELOW` metal layer.

Type: Float, specified in microns

`METALMASK` *`metalMaskNum`*

Specifies that the rule applies only if the metal containing the aligned cuts belongs to the mask *`metalMaskNum`*.

`METALWITHIN` *`metalWithin`* `[WITHCUT]`

Specifies that the rule applies only if the layer with all the shared same-metal cuts has neighboring wires within *`metalWithin`*, and their combined PRL touches or covers more than *`numCuts`* of cuts. The PRL of the neighboring wires on the opposite sides of the aligned cuts could be combined if their projection touches or overlaps. In addition, the width of this neighboring wire must also be less than or equal to *`width`*, and *`spacing`* is measured in the edge-to-edge style.

`WITHCUT` means that the rule applies only if a neighboring wire contains a cut on the adjacent cut layer and that cut needs to touch or overlap with a window formed by the merging of the violating cut.

Type: Float, specified in microns

`NOVIAOVERLAPMETAL` *`within`*

Specifies that the rule applies only if there is a metal on the other layer overlapping with the metal covering the aligned cuts by extending *`within`* on the other metal wire connected to one of the end cuts of the aligned cuts and there is no via cut in the overlap metal. In addition, *`spacing`* is measured in the edge-to-edge style.

Type: Float, specified in microns

`PARALLEL` *`parLength`* `WITHIN` *`parWithin`*

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Specifies that the rule applies only if the layer with shared same-metal cuts has a neighboring wire within *parWithin* and having a parallel run length greater than *parLength* to the cuts. If *parLength* is negative, it is equivalent to extending the bounding box of the aligned cuts by abs(*parLength*) and any neighboring wire overlapping with this extended bounding box would trigger the rule.

In addition, the width of this neighboring wire must also be less than or equal to *width* and *spacing* is measured in the edge-to-edge style.

Type: Float, specified in microns

SAMEMETALALIGNEDCUTS *numCuts* WIDTH *width* SPACING *spacing*

Specifies the number of center-aligned cuts to be less than or equal to *numCuts* if the cuts share the same metal on the above or below metal layer with width less than or equal to *width* and center-to-center spacing less than *spacing*.

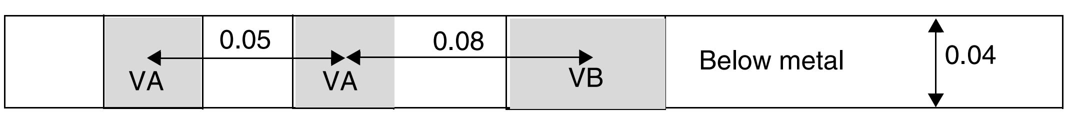
Type: Float, specified in microns

Same-Metal Aligned Cuts Rule Examples

- The following example is an illustration of the same-metal aligned cuts rule:

```
PROPERTY LEF58_SAMEMETALALIGNEDCUTS "
    SAMEMETALALIGNEDCUTS 2 CUTCLASS ALL
        BELOW WIDTH 0.04 SPACING 0.1 ; " ;
```

Figure 2-154 Illustration of the Same-Metal Aligned Cuts Rule



a) Violation, the number of center-aligned cuts of any cut class sharing the same metal on the below metal layer with center-to-center spacing < 0.1 is 3 (> 2).

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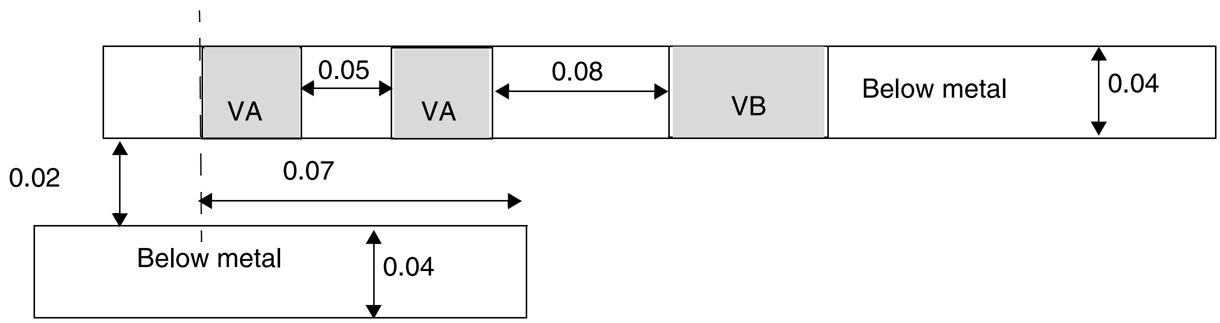
- The following example is an illustration of the same-metal aligned cuts rule with a metal neighbor:

```
PROPERTY LEF58_SAMEMETALALIGNEDCUTS "
    SAMEMETALALIGNEDCUTS 2 CUTCLASS ALL
        BELOW WIDTH 0.04 SPACING 0.1
             PARALLEL 0.05 WITHIN 0.03; " ;
```

Figure 2-155 Illustration of the Same-Metal Aligned Cuts Rule with PARALLEL and WITHIN



a) Violation. the number of center-aligned cuts of any cut class sharing the same metal on the below metal layer with edge-to-edge spacing `< 0.1` is `3` (`> 2`), and there is a below metal layer within `0.03` and having a parallel run length of `0.07` (`> 0.05`) to the cuts.

- The following example is an illustration of the same-metal aligned cuts rule with a metal neighbor and `METALWITHIN` specified:

```
PROPERTY LEF58_SAMEMETALALIGNEDCUTS "
    SAMEMETALALIGNEDCUTS 2 CUTCLASS VA
        BELOW WIDTH 0.04 SPACING 0.1 METALWITHIN 0.03; " ;
```


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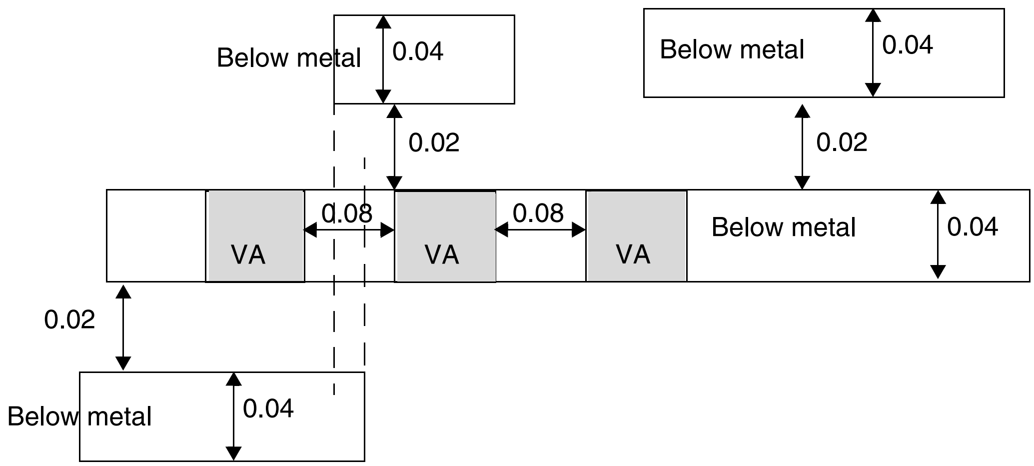
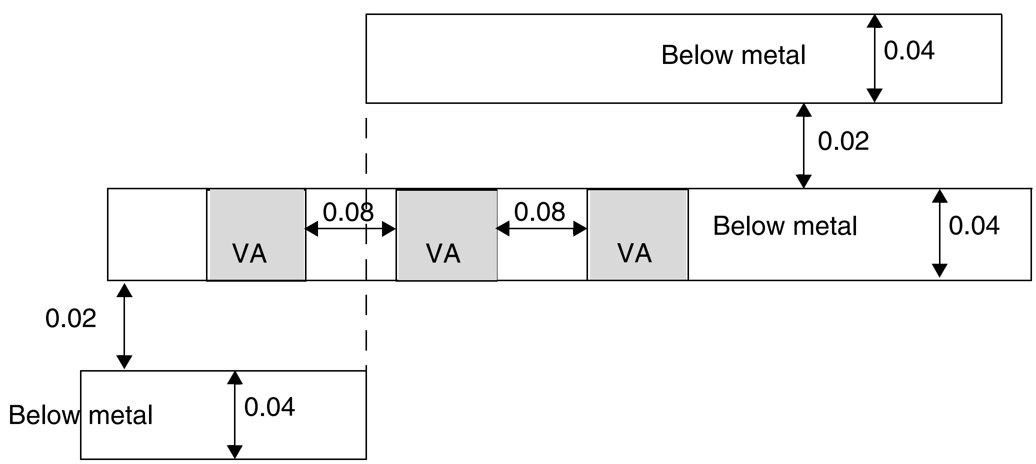
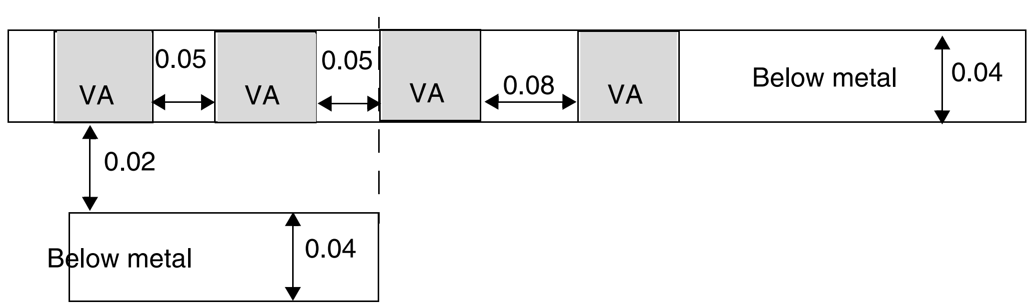
Figure 2-156 Illustration of the Same-Metal Aligned Cuts Rule with METALWITHIN



a) OK, the left two wires combined to overlap with two cuts and the right wire overlaps with only one cut.



b) Violation, the two neighboring wires combined to overlap with three (>2) aligned cuts.



c) Violation, as long as there is a neighboring wire touching or overlapping with three (> 2) aligned cuts, the rule is triggered. Having yet another aligned fourth cut is irrelevant.

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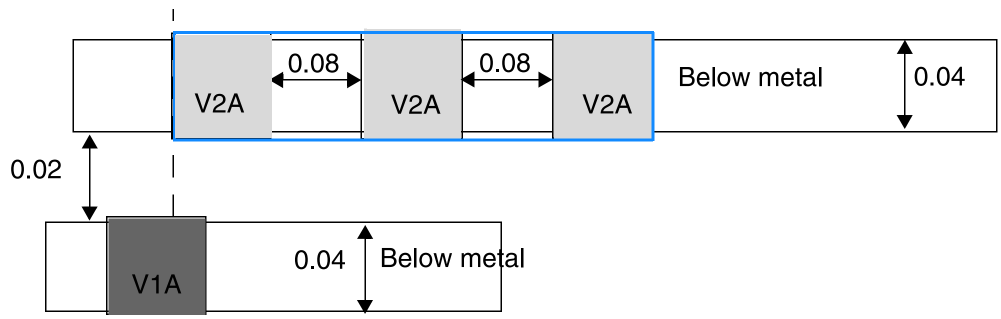
- The following example is an illustration of the same-metal aligned cuts rule with `CUTWITHIN` specified:

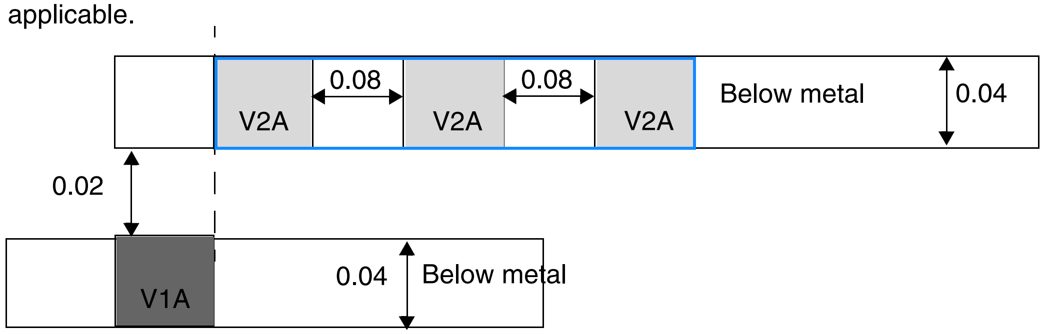
  On V2,

```
PROPERTY LEF58_SAMEMETALALIGNEDCUTS "
    SAMEMETALALIGNEDCUTS 2 CUTCLASS V2A
        BELOW WIDTH 0.04 SPACING 0.1 CUTWITHIN 0.03; " ;
```

Figure 2-157 Illustration of the Same-Metal Aligned Cuts Rule with CUTWITHIN



a) Violation, because the adjacent layer cut, `V1A`, has PRL > 0 with the blue window formed by merging the violating cuts of `V2A` and that neighbor cut of `V1A` is `0.02` (<= `0.03`) from the wire containing the aligned cuts of `V2A`. Note that the wire containing the neighbor cut need not cover all of the aligned cuts for the rule to be applicable.



b) OK, no adjacent layer cut is found having PRL > 0 with the blue window since `V1A` is just outside of that window.

- The following example is an illustration of the same-metal aligned cuts rule with `NOVIAOVERLAPMETAL` specified:

```
PROPERTY LEF58_SAMEMETALALIGNEDCUTS "
    SAMEMETALALIGNEDCUTS 2 CUTCLASS V2A
        BELOW WIDTH 0.04 SPACING 0.1
            NOVIAOVERLAPMETAL 0.06; " ;
```


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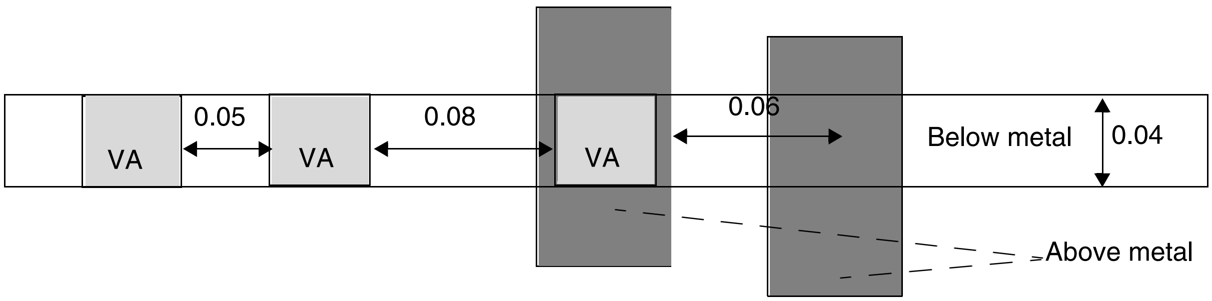
Figure 2-158 Illustration of the Same-Metal Aligned Cuts Rule with NOVIAOVERLAPMETAL



a) Violation because there is an above metal within `0.06` on the above metal wire connected to the right cut and that metal overlaps with the below metal without a via cut.

Via Cluster Rule

You can create a via cluster rule to define a list of critiques to form a via cluster for via cuts belonging to a cut class, and the spacing requirements on them.

You can use the following property definition:

```
PROPERTY LEF58_VIACLUSTER
     "VIACLUSTER CUTCLASS className
           CUTS perpendicularNumCut
           [DIAGONAL diagonalNumCut]
               [NOPRLSPACING diagonalSpacing bendSpacing WITHIN within
               EXTENSION sideExtension edgeExtension]]
           WITHIN minWithin maxWithin
           ; " ;
```

Where:

`CUTS perpendicularNumCut [DIAGONAL diagonalNumCut]`

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Specifies that the maximum consecutive number of via cuts of a cluster perfectly aligned in a direction perpendicular to the cut edges that fulfill the overhang defined in `ENCLOSUREEDGE OPPOSITE` to be *`perpendicularNumCut`* or diagonally aligned without bend/notch to be *`diagonalNumCut`*, if defined. In other words, there is no limit to the consecutive number of via cuts perfectly aligned in a direction parallel to the cut edges that fulfill the overhang defined using `ENCLOSUREEDGE OPPOSITE` or diagonally if `DIAGONAL` keyword is not defined.
Type: Integer

`NOPRLSPACING` *`diagonalSpacing bendSpacing`* `WITHIN` *`within`*
`EXTENSION` *`sideExtension edgeExtension`*

Specifies the cluster criteria when *`diagonalNumCut`* is set to `0`, which implies that there is no limit to the number of cuts that can be inserted diagonally and bend/notch is allowed. If the cuts are inserted diagonally without bend/notch, at the most two neighbor cuts can be found within *`within`* having spacing greater than or equal to *`diagonalSpacing`* in center-to-center measurements (on both *`within`* and *`diagonalSpacing`*). Otherwise (with bend/notch), at the most two non-perfectly aligned neighbor cuts can be found within *`within`* having spacing greater than or equal to the *`bendSpacing`* in center-to-center measurements (on both *`within`* and *`bendSpacing`*).

In addition to forming a via cluster, one neighbor cut inserted diagonally without any bend/notch or two neighbor cuts having a bend/notch must be found touching a search window formed by extending *`sideExtension`* along both sides on the cut edges, that fulfill the overhang defined in `ENCLOSUREEDGE OPPOSITE` rule, and with the *`edgeExtension`* perpendicular to those edges.
Type: Float, specified in microns

`VIACLUSTER CUTCLASS` *`className`*

Defines a list of critiques to form a via cluster for via cuts belonging to cut class *`className`*, and the spacing requirements on them.

`WITHIN` *`minWithin maxWithin`*

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Specifies that the via cuts must be larger than or equal to *`minWithin`* from each other in square corner (maximum x and y) style. For diagonal cluster, the via cuts must be *`minWithin`* apart in both x and y direction in corner to corner style. To form a via cluster, the via cuts must be within *`maxWithin`* in square corner style.

In other words, the cut to cut spacing among via cuts in the same cluster will be *`minWithin`* while a via cut of a cluster to a via cut of another cluster or not belonging to a cluster will be subject to the regular cut spacing rules.
Type: Float, specified in microns

Figure 2-159 Illustration of the Via Cluster Rule with WITHIN

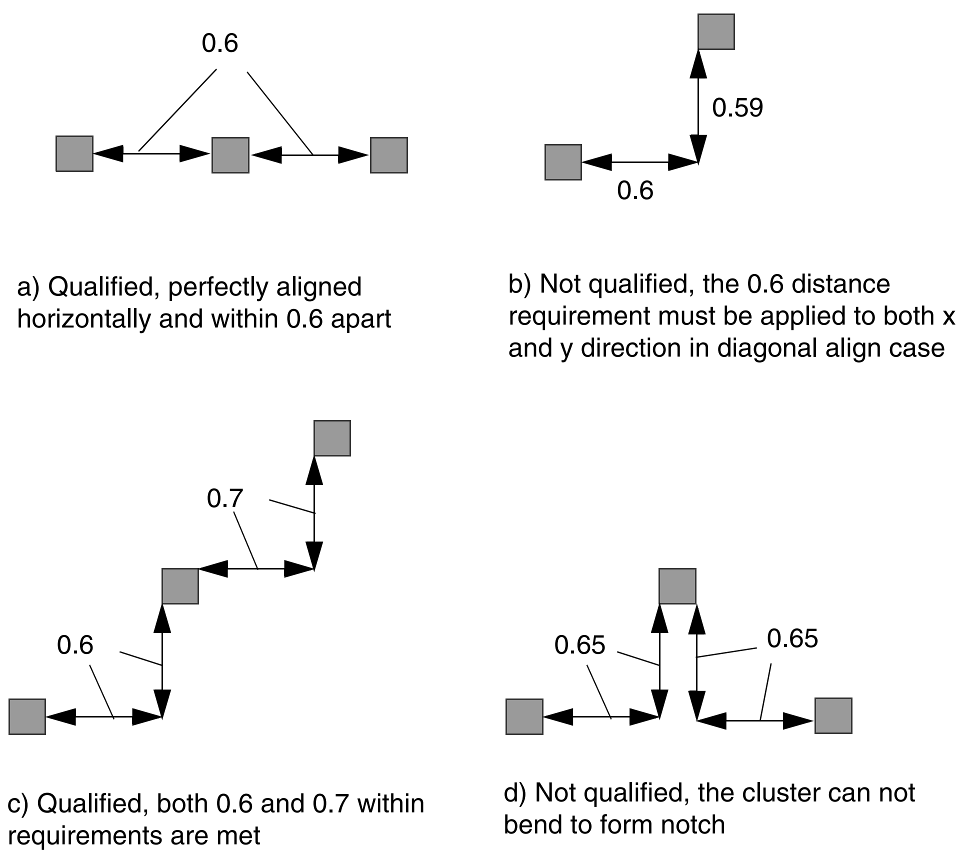

Illustration of `CUTS 3 DIAGONAL 3 WITHIN 0.6 0.7`

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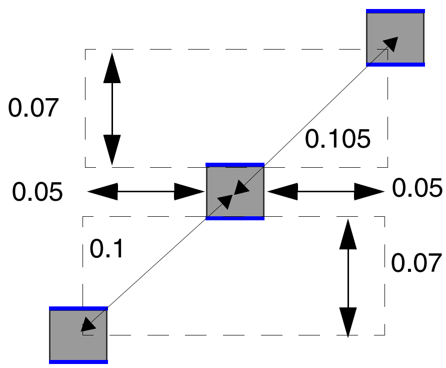
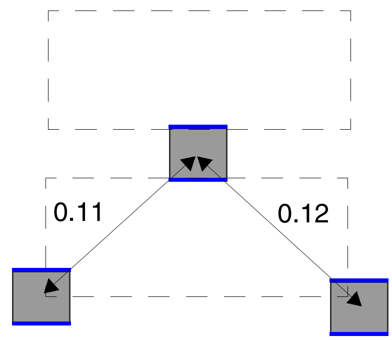
Figure 2-160 Illustration of the Via Cluster Rule with NOPRLSPACING

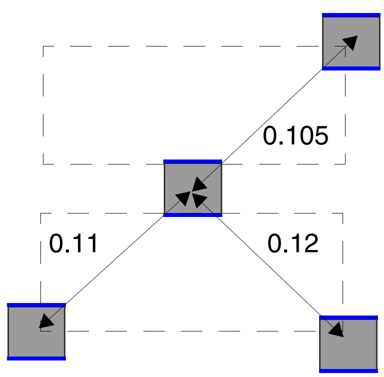
The blue edges fulfill the `ENCLOSUREEDGE OPPOSITE` rule



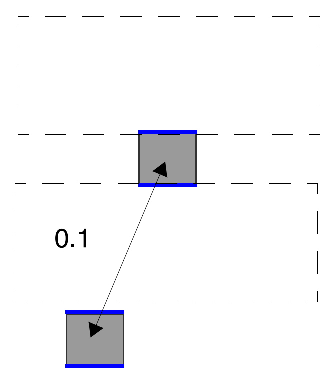
a) Qualified, 2 neighbor cuts diagonally with center-to-center spacing > 0.1



b) Qualified, 2 neighbor cuts in bend/ notch style



c) Not qualified, too many neighbor cuts within 0.13 distance



d) Not qualified, for non perfectly aligned cuts with parallel run length > 0, the neighbor cut would be outside the search window

Illustration of
```
CUTS 3 DIAGONAL 0
    NOPRLSPACING 0.1 0.11 WITHIN 0.13
    EXTENSION 0.05 0.07
```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Via Group Rule

You can create a via group rule by using the following property definition:

```
PROPERTY LEF58_VIAGROUP
       "VIAGROUP groupName CUTS numCut CUTCLASS {{className}... | ALL}
            [MIXEDCUTCLASSES] [SAMEMASK]
            {{[PRLTWOSIDES prl1 prl2 prl3 prl4] SPACING minSpacing maxSpacing
              | PRL prl PRLSPACING prlSpacing [HORIZONTAL|VERTICAL]}
                 [ALLOWBEND [NOHORIZONTAL|NOVERTICAL]]
            | ALIGNEDSPACING {HORIZONTAL | VERTICAL}
                 minSpacing maxSpacing
            | LGROUP HORIZONTALSPACING horizontalSpacing PRL verticalPrl
                 VERTICALSPACING verticalSpacing PRL horizontalPrl}
            ; " ;]
```

Where:

`ALIGNEDSPACING {HORIZONTAL | VERTICAL}`
    `minSpacing maxSpacing`

Specifies that the cuts that are greater than *`minSpacing`* and less than *`maxSpacing`* in the given direction of `HORIZONTAL` or `VERTICAL` with `PRL` greater than `0` in the orthogonal direction will form a via group.

`ALLOWBEND [NOHORIZONTAL|NOVERTICAL]`

Specifies that cuts could be formed in a bend/notch pattern. This construct is only supported when *`numCut`* is `3`.

`NOHORIZONTAL/NOVERTICAL` means that a horizontal/vertical bend with two cuts having `PRL` greater than `0` horizontally/vertically would be a violation by itself. Then, a vertical/horizontal bend could be a legal via group.

`LGROUP HORIZONTALSPACING` *`horizontalSpacing`* `PRL` *`verticalPrl`*
    `VERTICALSPACING` *`verticalSpacing`* `PRL` *`horizontalPrl`*

Specifies that a via group must be formed with three cuts in an 'L' shape. A cut must have a neighbor cut within *`horizontalSpacing`* horizontally with PRL greater than *`verticalPrl`* and another neighbor cut within *`verticalSpacing`* vertically with PRL greater than *`horizontalPrl`*. *`numCut`* must be `3` when `LGROUP` is defined. See Figure 2-164 on page 485.

Type: Float, specified in microns

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`MIXEDCUTCLASSES`
: Specifies that when `CUTCLASS ALL` or a list of cut classes are provided, the cuts belonging to the same cut class would not form a via group. In other words, only mixed cuts belonging to at least two different cut classes could form a via group. Currently, this construct can work only with *`numCut`* equal to `2`.

`PRL` *`prl`* `PRLSPACING` *`prlSpacing`* `[HORIZONTAL|VERTICAL]`

: Specifies that cuts having PRL greater than *`prl`* and spacing less than *`prlSpacing`* will form a via group. If `HORIZONTAL` or `VERTICAL` is specified, *`prlSpacing`* would be measured in the given direction and *`prl`* would be measured in the orthogonal direction.

  Type: Float, specified in microns

`SAMEMASK`
: Specifies that the via group is only formed among same-mask cuts.

```
VIAGROUP groupName CUTS numCut CUTCLASS {{className}... | ALL}
     [PRLTWOSIDES prl1 prl2 prl3 prl4]
     SPACING minSpacing maxSpacing
```

: Specifies the maximum number of cuts within a via group of cuts belonging to any of the given cut classes in *`className`* or any cut classes in `ALL` to be *`numCut`*. Tools can support only up to four defined cut classes. The via group has the name of *`groupName`*. Any two neighboring cuts of a via group must have PRL greater than *`prl1`* and less than *`prl2`* on one side and PRL greater than *`prl3`* and less than *`prl4`* on the other side if `PRLTWOSIDES` is defined and the cuts must have spacing greater than *`minSpacing`* and less than *`maxSpacing`*. In addition, all the cuts must be diagonal without any bend or notch.

  Type: Float, specified in microns

Via Group Rule Examples

- The following example is an illustration of the via group rule:

```
PROPERTY LEF58_VIAGROUP "
    VIAGROUP groupName CUTS 3 CUTCLASS VA
        PRLTWOSIDES prl1 prl2 prl3 prl4
        SPACING minSpacing maxSpacing " ;
```


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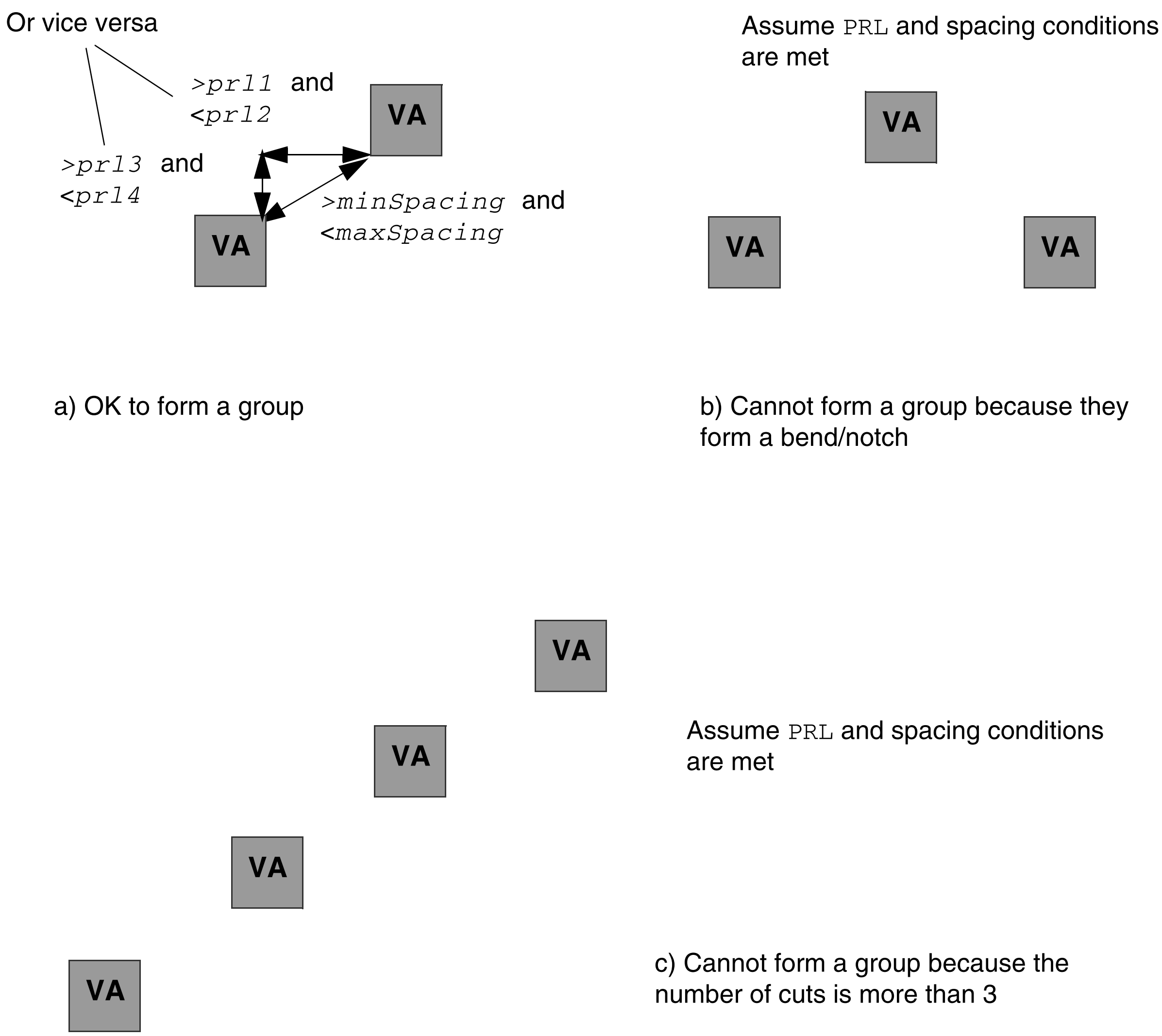
Figure 2-161 Illustration of the Via Group Rule



■ The following diagram illustrates a via group rule with `ALIGNEDSPACING`:

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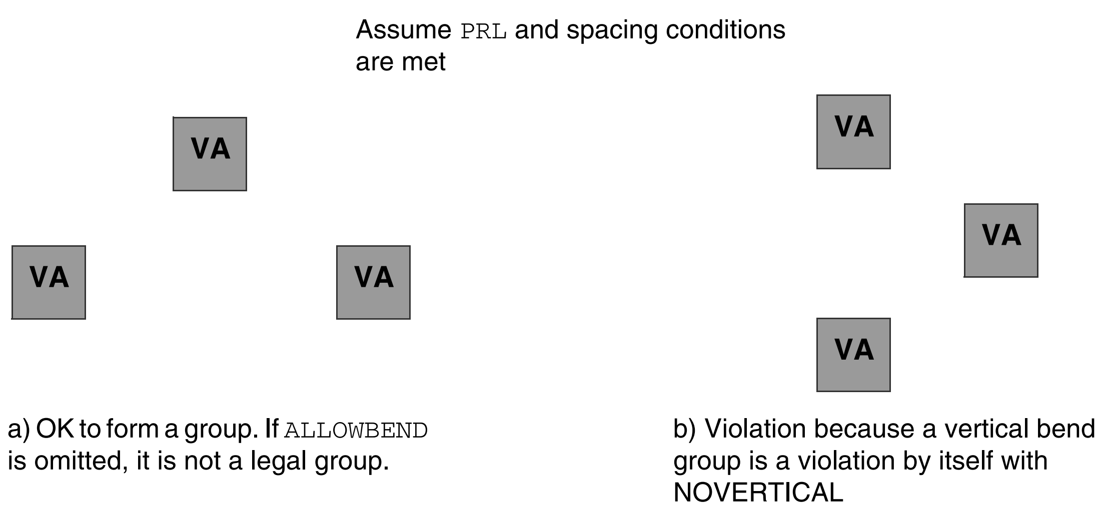
Figure 2-162 Illustration of the Via Group Rule with ALIGNEDSPACING

Illustration of:
`VIAGROUP` *`groupName`* `CUTS 2 CUTCLASS VA`
`ALIGNEDSPACING HORIZONTAL` *`minSpacing maxSpacing`*

- The following example is an illustration of the via group rule with `ALLOWBEND`:

```
PROPERTY LEF58_VIAGROUP "
    VIAGROUP groupName CUTS 3 CUTCLASS VA
        PRLTWOSIDES prl1 prl2 prl3 prl4
        SPACING minSpacing maxSpacing
        ALLOWBEND NOVERTICAL" ;
```

Figure 2-163 Illustration of the Via Group Rule with ALLOWBEND



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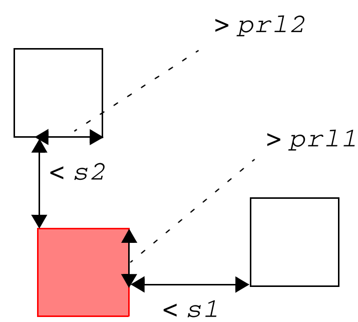
- The following example is an illustration of the via group rule with `LGROUP`:

```
PROPERTY LEF58_VIAGROUP "
    VIAGROUP groupName CUTS 3 CUTCLASS ALL
        LGROUP HORIZONTALSPACING s1 PRL prl1
        VERTICALSPACING S2 PRL prl2";
```

Figure 2-164 Illustration of the Via Group Rule with LGROUP



The 3 cuts in an 'L' shape can form a via group. The red cut must have a neighboring cut on the left or right side and another neighboring cut on the top or bottom side that fulfill the corresponding spacing and PRL criteria.

Via Group Cluster Rule

You can create a via group cluster rule by using the following property definition:

```
PROPERTY LEF58_VIAGROUPCLUSTER
       "VIAGROUPCLUSTER groupName CUTS numCut
           CUTCLASS {className | ALL}
           TWOCUTS verticalSpacing horziontalPrl
           TWOCUTSCLUSTER horziontalSpacing verticalPrl
           ; " ;
```

Where:

```
VIAGROUPCLUSTER groupName CUTS numCut
   CUTCLASS {className | ALL}
   TWOCUTS verticalSpacing horziontalPrl
   TWOCUTSCLUSTER horziontalSpacing verticalPrl
```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Specifies a via group with exactly *numCut* cuts, where *numCut* must be an even number. They are in sets of two cuts within *verticalSpacing* vertically with PRL greater than *horziontalPrl*. Each of the two-cut sets must be within *horziontalSpacing* horizontally with PRL greater than *verticalPrl* based on the maximum bounding box of the two cuts. All cuts must belong to *className*, if specified. Otherwise, any cuts can form the via group when ALL is specified.

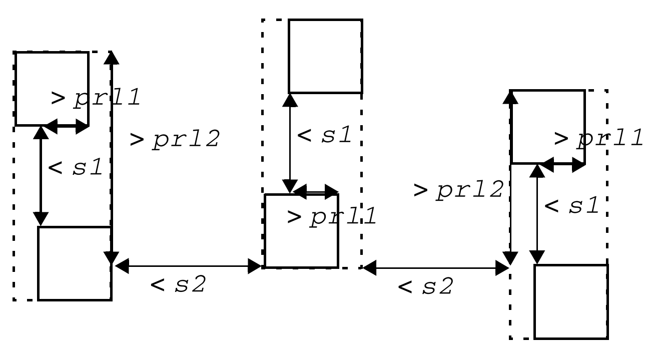
Type: Float, specified in microns

Via Group Cluster Rule Example

- The following example is an illustration of the via group cluster rule:

```
PROPERTY LEF58_VIAGROUPCLUSTER "
    VIAGROUPCLUSTER groupName CUTS 6
        CUTCLASS ALL TWOCUTS s1 prl1
        TWOCUTSCLUSTER S2 prl2 " ;
```

Figure 2-165 Illustration of the Via Group Cluster Rule



These 6 cuts can form a via group. They are in 3 sets of 2 cuts. Each of the 2-cut sets has cuts within *s1* vertically with PRL > *prl1*, and the max boundary boxes of the 2-cut sets are within *s2* horizontally with PRL > *prl2*.

Via Group Override Spacing Rule

You can create a via group override spacing rule by using the following property definition:

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```
PROPERTY LEF58_VIAGROUPOVERRIDESPACING
       "VIAGROUPOVERRIDESPACING
           ; " ;]
```

Where:

`VIAGROUPOVERRIDESPACING`

Specifies that all of the same-layer cut spacing rules, including adjacent cut rules, are not checked for the cuts that could form a via group based on the `VIAGROUP` construct.

Via Group Spacing Rule

You can create a via group spacing rule by using the following property definition:

```
PROPERTY LEF58_VIAGROUPSPACING
       "VIAGROUPSPACING spacing [MAXSPACING]
           [SAMEMASK | DIFFMASK]
            [{HORIZONTAL| VERTICAL} [PRL prl] [TWOSIDES]]
           {VIAGROUP groupName | VIAGROUPCLUSTER groupName}
           CUTS numCut
           [LAYER cutLayerName] CUTCLASS {className | All}
           [TOVIAGROUP]
           [MERGEDCUTS | NOMERGEDCUTS]
           [PARALLEL parLength WITHIN parWithin
                OTHERCUTCLASS otherClassName]
           ; " ;]
```

Where:

`DIFFMASK` Specifies that the spacing applies only to different-mask cuts.

`HORIZONTAL| VERTICAL [PRL` *prl*`]`

Specifies that the spacing applies only on the given direction, either horizontally or vertically, with cuts having a parallel run length (PRL) greater than `0` or greater than *prl* if `PRL` is explicitly specified.

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`LAYER` *`cutLayerName`* Specifies an inter-layer cut spacing between a via group on the current layer to a cut of *`className`* on an adjacent cut layer of *`cutLayerName`*.
Note: Forward look-up reference is allowed for `LAYER` *`cutLayerName`*.

Type: String

`MAXSPACING` Designates the spacing rule as a maximum spacing rule. This means that there must be another neighbor cut at less than *`spacing`* distance from any cut within the via group.

`MERGEDCUTS | NOMERGEDCUTS`

`MERGEDCUTS` specifies that a bounding box is formed by the cuts on a group before cut spacing is applied on the bounding box. `NOMERGEDCUTS` should be applied only along with `TOVIAGROUP` to disable the built-in cut merging behavior.

See Table 2-1 on page 489 for the impact of using `MERGEDCUTS`, `NOMERGEDCUTS`, and `TOVIAGROUP` in various combinations.

`PARALLEL` *`parLength`* `WITHIN` *`parWithin`*
`OTHERCUTCLASS` *`otherClassName`*

Specifies the spacing rule applies only if any cuts belonging to via group *`groupName`* is a neighbor cut of a top or bottom edge of a cut with cut class *`otherClassName`* within *`parWithin`* having a parallel run length greater than *`parLength`*.
Type: Float, specified in microns

`SAMEMASK` Specifies that the spacing applies only on same-mask cuts.

`TOVIAGROUP` Specifies that the spacing applies only between cuts belonging to two different *`groupName`* via groups. In addition, cuts must be merged before spacing is applied.

`TWOSIDES` Specifies that the spacing applies only if there are two neighboring cuts within *`spacing`* on the opposite sides in the given direction.

`VIAGROUPSPACING` *`spacing`* `VIAGROUP` *`groupName`* `CUTS` *`numCut`*
`CUTCLASS {`*`className`* `| ALL}`

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Specifies the spacing of cuts belonging to the via group *groupName* with number of cuts *numCut* to cuts belonging to the cut class *className* or any cut classes in ALL to be *spacing*.

Type: Float, specified in microns

VIAGROUPCLUSTER *groupName*

Specifies that the spacing rule is applied on a via group *groupName* defined in VIAGROUPCLUSTER.

Table 2-1 Impact of Using MERGEDCUTS, NOMERGEDCUTS, and TOVIAGROUP

	MERGEDCUTS	NOMERGEDCUTS	**Without either of them**
With TOVIAGROUP	Irrelevant because cut merging is default in TOVIAGROUP	No cut merging	Cut merging
Without TOVIAGROUP	Cut merging	Irrelevant because cuts are not merged by default	No cut merging

Via Group Spacing Rule Examples

- The following diagram illustrates the via group spacing rules defined below:

```
VIAGROUP VGA CUTS 3 CUTCLASS VA
   PRLTWOSIDES prl1 prl2 prl3 prl4
   SPACING minSpacing maxSpacing
VIAGROUPSPACING spacing1
   VIAGROUP VGA CUTS 3 CUTCLASS VA
VIAGROUPSPACING spacing2
   VIAGROUP VGA CUTS 2 CUTCLASS VB
```


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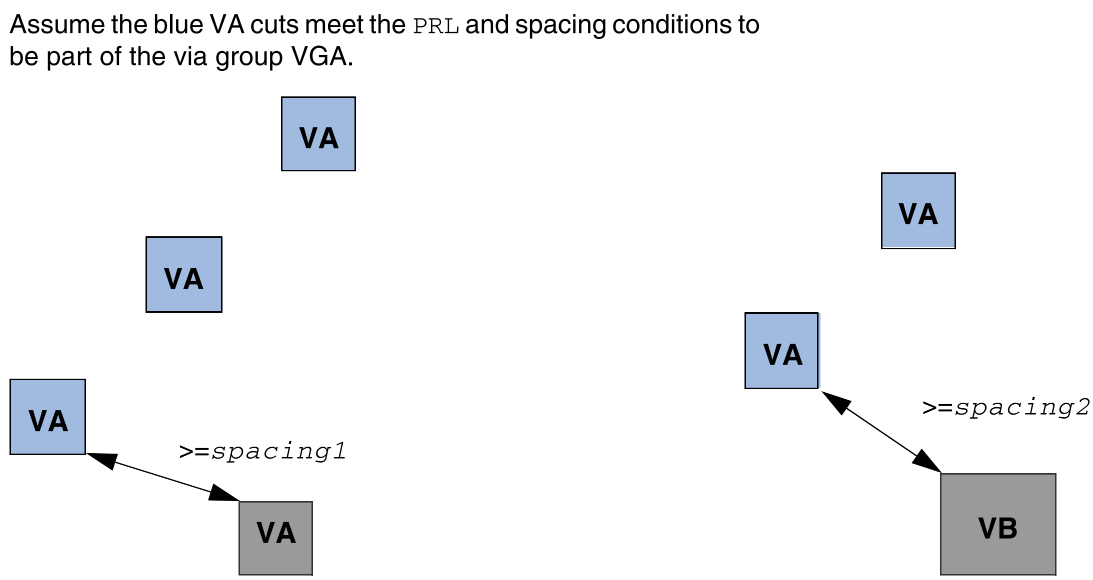
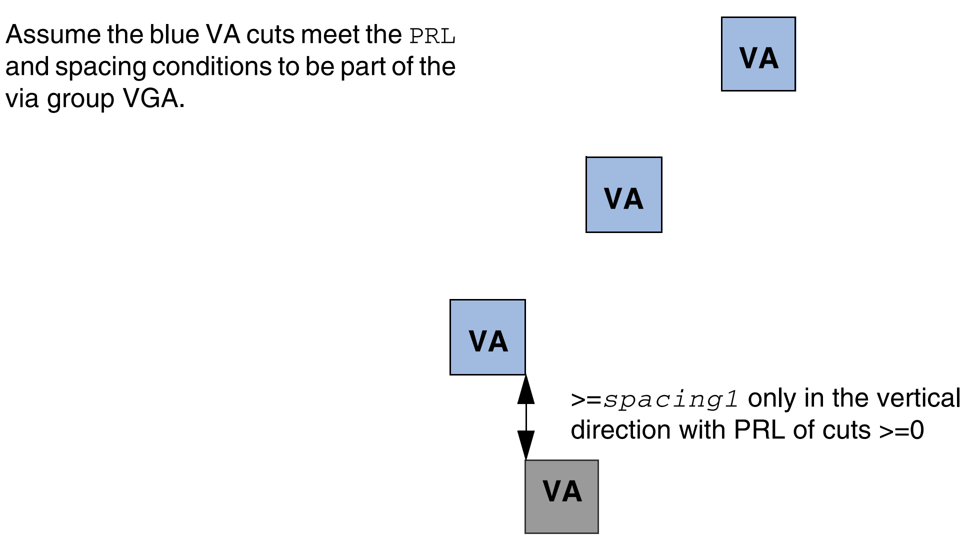
Figure 2-166 Illustration of the Via Group Spacing Rule



- The following diagram illustrates the via group spacing rule with `VERTICAL` defined below:

```
VIAGROUP VGA CUTS 3 CUTCLASS VA
   PRLTWOSIDES prl1 prl2 prl3 prl4
   SPACING minSpacing maxSpacing
VIAGROUPSPACING spacing1 VERTICAL
   VIAGROUP VGA CUTS 3 CUTCLASS VA
```

Figure 2-167 Illustration of the Via Group Spacing Rule with VERTICAL



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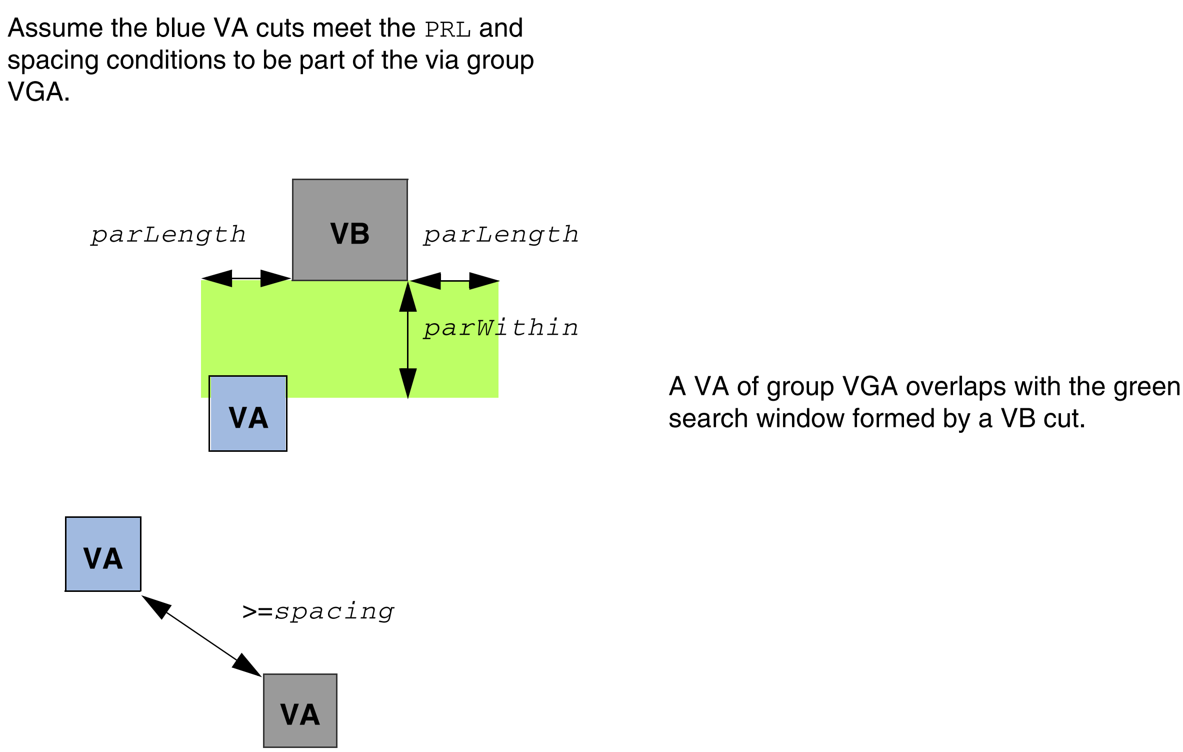
- This example illustrates the following via group spacing rule:

```
VIAGROUP VGA CUTS 3 CUTCLASS VA
   PRLTWOSIDES prl1 prl2 prl3 prl4
   SPACING minSpacing maxSpacing
VIAGROUPSPACING spacing
   VIAGROUP VGA CUTS 2 CUTCLASS VA
   PARALLEL parLength WITHIN parWithin
   OTHERCUTCLASS VB
```

Figure 2-168 Illustration of the Via Group Spacing Rule with OTHERCUTCLASS



- The following diagram illustrates the via group spacing rule with `TOVIAGROUP` defined below:

```
VIAGROUP VGA CUTS 2 CUTCLASS VA
   ALIGNEDSPACING VERTICAL
   SPACING minSpacing maxSpacing
VIAGROUPSPACING spacing HORIZONTAL PRL prl
   VIAGROUP VGA CUTS 2 CUTCLASS VA TOVIAGROUP
```


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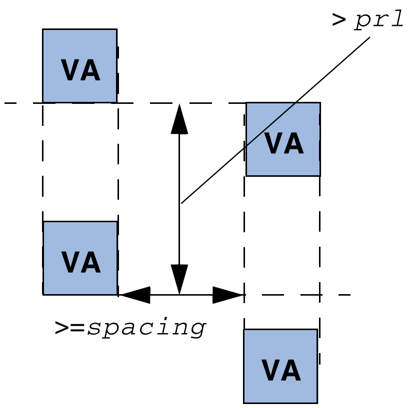
Figure 2-169 Illustration of the Via Group Spacing Rule with TOVIAGROUP

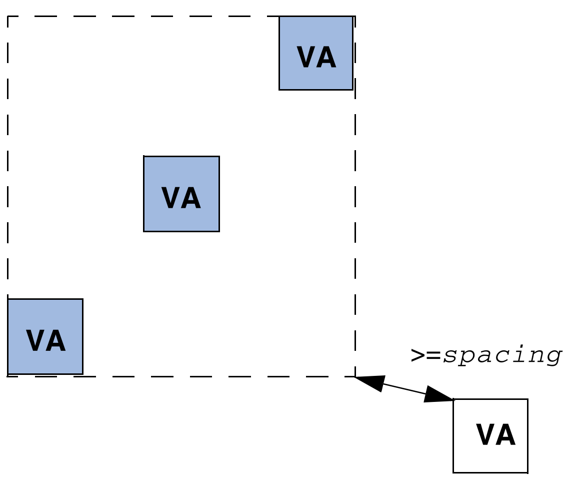


Assume the blue VA cuts meet the spacing conditions to be part of the via group VGA.

- The following diagram illustrates the via group spacing rule with `MERGEDCUTS` defined below:

```
VIAGROUPSPACING spacing
   VIAGROUP VGA CUTS 3 CUTCLASS VA MERGEDCUTS
```

Figure 2-170 Illustration of the Via Group Spacing Rule with MERGEDCUTS



Assume the blue VA cuts meet the spacing conditions to be part of the via group VGA. Via group spacing is applied between a bounding box formed by the cuts on the group against a neighboring cut.

- The following diagram illustrates the via group spacing rule with `TWOSIDES` defined below:

```
VIAGROUPSPACING spacing
```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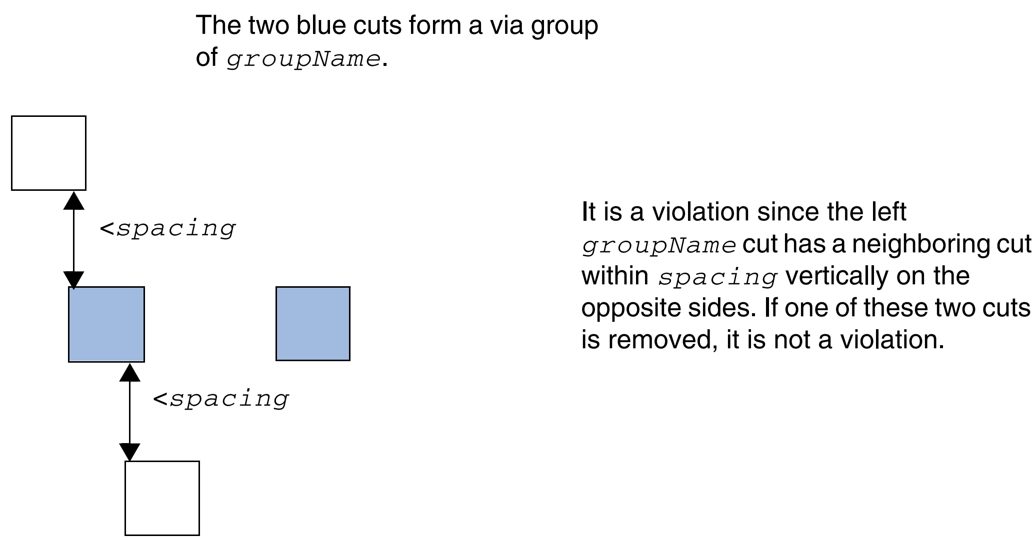
```
VERTICAL TWOSIDES VIAGROUP groupName CUTS 2
CUTCLASS ALL
```

Figure 2-171 Illustration of the Via Group Spacing Rule with TWOSIDES



Voltage Spacing Rule

Voltage spacing rules can be used to define spacing among vias objects on the same cut layer in different voltages.

You can create a voltage spacing rule by using the following property definition:

```
PROPERTY LEF58_VOLTAGESPACING
     "VOLTAGESPACING {voltage spacing} ...
          ; " ;
```

Where:

```
VOLTAGESPACING {voltage spacing} ...
```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Defines the spacing among the vias in the same cut layer for different voltages. Both the voltage and spacing values should be increasing monotonically. The spacing value is required for any net with a worst-case voltage between its neighbors, including a negative minimum voltage that is greater than or equal to the voltage specified in the voltage-spacing definition table. For example, if you have `1.5`v and `2.0`v rules, the spacing between a net with a maximum voltage of `1.3`v and a minimum voltage of `0.0`v and a net with a maximum voltage of `1.4`v and a minimum voltage of `-0.5`v should follow the `1.5`v spacing rule because the worst-case voltage is `max (1.3v - (-0.5v) = 1.8v, 1.4v - 0.0v = 1.4v) = 1.8v`.

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3

LEF Syntax - Layer (Implant)

This chapter contains information about the following topics:

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Layer (Implant)

```
LAYER layerName
      TYPE IMPLANT ;
      [MASK maskNum ;]
      [WIDTH minWidth ;]
      [SPACING minSpacing [LAYER layerName2]   ;] ...
      [PROPERTY propName propVal ;] ...
      [PROPERTY LEF58_AREA
         "AREA minArea
         ; " ;]
      [PROPERTY LEF58_COREEDGELENGTH
         "COREEDGELENGTH minLength
              [EXCEPTADJACENTLENGTH
                 {exactEdgeLength adjLength [EXACTADJACENTLENGTH]}...]
              [WITHINLAYER {layerName}...
              |EXCEPTLAYER {exceptLayerName}...]
         ; " ;]
      [PROPERTY LEF58_CORNERSPACING
         "CORNERSPACING spacing [ALIGNEDONLY]
              [CHECKIMPLANTGROUPONLY]
         ; " ;]
      [PROPERTY LEF58_FORBIDDENWIDTH
         "FORBIDDENWIDTH width
         ; " ;]
      [PROPERTY LEF58_MAXSPACING
         "MAXSPACING maxSpacing WIDTH width LENGTH length
         ; " ;]
      [PROPERTY LEF58_MINENCLOSEDAREA
         "MINENCLOSEDAREA area
         ; " ;]
      [PROPERTY LEF58_MINHEIGHT
         "MINHEIGHT minHeight WIDTH maxWidth OVERLAP maxOverlap
         ; " ;]
      [PROPERTY LEF58_MINSTEP
         "MINSTEP minStepLength MINADJACENTLENGTH minAdjLength
         ; " ;]
      [PROPERTY LEF58_SPACING
         "SPACING minSpacing [LAYER layerName2]
              [HORIZONTAL|VERTICAL PRL prl ][EXCEPTABUTTED]
                    [EXCEPTCORNERTOUCH] [LENGTH length]
                    [INTERSECTLAYERS layerNameList...]
         ; " ;]
      [PROPERTY LEF58_WIDTH
         "WIDTH minWidth [LAYER {layerName2 | ANY} [NODIFFMERGE]]
              [CELLHEIGHT height LAYER layerName3]
              [WITHINLAYER layerName
              |EXCEPTLAYER exceptLayerName]
              [ZEROPRL [MAXWIDTH maxWidth]
                    [NOTALIGNEDROWWIDTH width LENGTH length]]
```


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```
            [EXCEPTCORNERTOUCH][LENGTH length]
            [CHECKIMPLANTGROUP groupName]
            ; " ;]
END layerName
```

Defines implant layers in the design. Each layer is defined by assigning it a name and simple spacing and width rules. These spacing and width rules only affect the legal cell placements. These rules interact with the library methodology, detailed placement, and filler cell support. You must define implant layers separately, with their own layer statements.

`LAYER` *`layerName`*

Specifies the name for the layer. This name is used in later references to the layer.

`LAYER` *`layerName2`*

Specifies the name of another implant layer that requires extra spacing that is greater than or equal to *`minSpacing`* from this implant layer.

`MASK` *`maskNum`*

Specifies how many masks for double- or triple-patterning will be used for this layer. The *`maskNum`* variable must be an integer greater than or equal to 2. Most applications only support values of 2 or 3.

`PROPERTY` *`propName propVal`*

Specifies a numerical or string value for a layer property defined in the `PROPERTYDEFINITIONS` statement. The *`propName`* you specify must match the *`propName`* listed in the `PROPERTYDEFINITIONS` statement.

`SPACING` *`minSpacing`*

Specifies the minimum spacing for the layer. This value affects the legal cell placement.
Type: Float, specified in microns

`TYPE IMPLANT`

Identifies the layer as an implant layer.

`WIDTH` *`minWidth`*

Specifies the minimum width for this layer. This value affects the legal cell placement.
Type: Float, specified in microns

Example 3-1 Implant Layer

Typically, you define high-drive cells on one implant layer and low-drive cells on another implant layer. The following example defines high-drive cells on *implant1* and low-drive cells on *implant2*. Both implant layers cover the entire cell. The placer and filler cell creation

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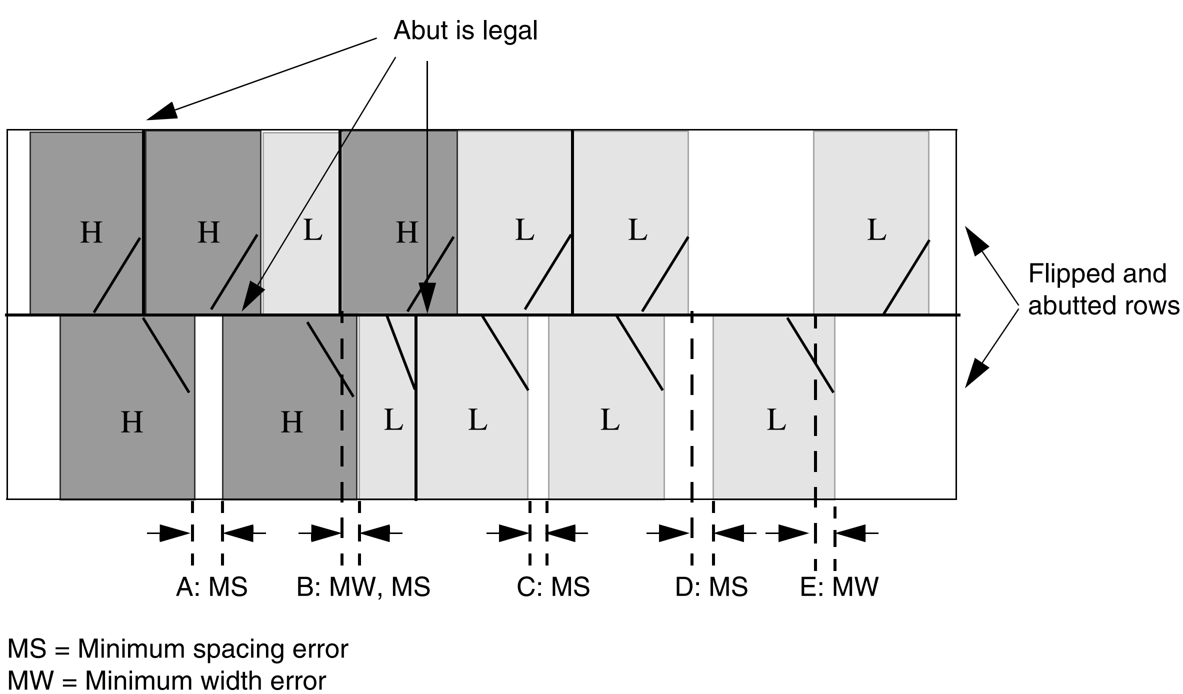
attempt to legalize the cell overlaps in abutting rows to ensure that the minimum width and spacing values are met.

```
LAYER implant1          #high-drive implant layer
    TYPE IMPLANT ;
    WIDTH 0.50 ;        #implant rectangles must be >=0.50 microns wide
    SPACING 0.50 ;      #implant rectangles must be >=0.50 microns apart
END implant1

LAYER implant2          #low-drive implant layer
       TYPE IMPLANT ;
       WIDTH 0.50 ;     #implant rectangles must be >=0.50 microns wide
       SPACING 0.50 ;   #implant rectangles must be >=0.50 microns apart
END implant2
```

Assume that the high-drive cells and low-drive cells are completely covered by their respective implant layers. Because there is no spacing between *implant1* and *implant2* specified, you might see a placement like that illustrated in Figure 3-1 on page 498.

Figure 3-1



Note that you can correct A, C, D, and E by putting in filler cells with the appropriate implant type. However, B cannot be corrected by a filler cell—either the placer must avoid it, or you

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must allow the filler cell or post-process command to move cells or modify the implant layer to correct the error.

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Defining Implant Layer Properties to Create 32/28 nm and Smaller Nodes Rules

You can include implant layer properties in your LEF file to create DRC rules that currently are not supported by existing LEF syntax. The properties are specified inside the `LAYER IMPLANT` statements, where they can be seen with other rules.

Before you can reference them, properties must be defined at the beginning of the LEF file in the `PROPERTYDEFINITIONS` statement, immediately before the first `LAYER` statement.

- Properties belong to the `LAYER` object and have a type of `STRING`.
- The property names used for these rules all start with `LEF58_`.

All properties use the following syntax within the LEF `PROPERTYDEFINITIONS` statement:

```
PROPERTYDEFINITIONS
     LAYER propName STRING ["stringValue"] ;
END PROPERTYDEFINITIONS
```

The property definitions for the implant layer properties are as follows:

```
PROPERTYDEFINITIONS
    LAYER LEF58_AREA STRING ;
    LAYER LEF58_COREEDGELENGTH STRING ;
    LAYER LEF58_CORNERSPACING STRING ;
    LAYER LEF58_MAXSPACING STRING ;
    LAYER LEF58_MINENCLOSEDAREA STRING ;
    LAYER LEF58_MINSTEP STRING ;
    LAYER LEF58_SPACING STRING ;
    LAYER LEF58_WIDTH STRING ;
END PROPERTYDEFINITIONS
```

Area Rule

An area rule can be used to specify the minimum area on the implant layer.

You can use the following property definition:

```
PROPERTY LEF58_AREA
     "AREA minArea
     ; " ;
```


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Where:

AREA minArea Specifies the minimum area on the implant layer.
Type: Float, specified in microns.

Core Edge Length Rule

A core edge length rule can be used to indicate that the edge length of an implant shape of a `CORE` (standard) cell must be greater than or equal to the specified minimum length.

You can create a core edge length rule by using the following property definition:

```
PROPERTY LEF58_COREEDGELENGTH
      "COREEDGELENGTH minLength
          [EXCEPTADJACENTLENGTH
             {exactEdgeLength adjLength [EXACTADJACENTLENGTH]}...]
            [WITHINLAYER {layerName}...
            |EXCEPTLAYER {exceptLayerName}...]
       ; " ;
```

Where:

`COREEDGELENGTH` *minLength*

Specifies that the edge length of the implant shape of a `CORE` (standard) cell, in the direction of the cell row, must be greater than or equal to the minLength.
Type: Float, specified in microns

`EXCEPTADJACENTLENGTH`
`{`*exactEdgeLength adjLength* `[EXACTADJACENTLENGTH]}...]`

Specifies that the edge length is exempted if it's length is exactly equal to *exactEdgeLength*, with an adjacent length less than or equal to *adjLength*, or exactly equal to *adjLength* if `EXACTADJACENTLENGTH` is defined.
Type: Float, specified in microns

`EXCEPTLAYER {`*exceptLayerName*`}...`

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Specifies that the implant shapes overlapping with the shapes in *exceptLayerName*, which should be a masterslice layer with ABUTFILLER type property, should not be merged with other implant shapes that are not overlapping with the shapes in *exceptLayerName*, on which the core edge length rule is checked. Typically, only standard cells contain OBS shapes in *exceptLayerName*.

Type: String

WITHINLAYER {*layerName*}...

Specifies that the implant shapes overlapping with the shapes in *layerName*, which should be a masterslice layer with the ABUTFILLER type property, should be checked against *minLength* by themselves. In other words, those implant shapes should not be merged with other implant shapes that are not overlapping with the shapes in *layerName*. If multiple *layerName* layers are defined, only the implant shapes overlapping with the shapes on all those layers are checked individually. The original implant shapes overlapping with the shapes in *layerName* should be kept without being cut off by shapes in *layerName*. Typically, only standard cells contain OBS shapes in *layerName*.

Type: String

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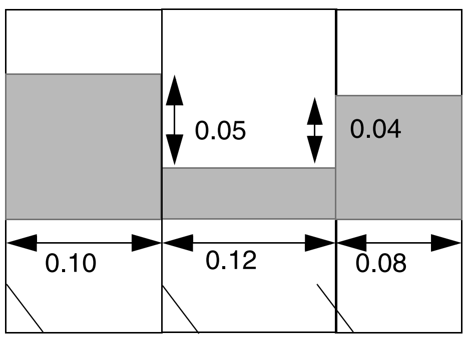
Core Edge Length Rule Examples

Figure 3-2 Example of Core Edge Length Rule

```
PROPERTY LEF58_COREEDGELENGTH "
     COREEDGELENGTH 0.12 EXCEPTADJACENTLENGTH
      0.08 0.04 EXACTADJACENTLENGTH 0.10 0.06 ; " ;
```



a) OK, the left edge has length of 0.10 with an adjacent length of 0.05 (<= 0.06), the middle edge has length of 0.12 (>= 0.12) and the right edge has length of 0.08 with an adjacent length of 0.04, which the left & right edges fulfill the exemption

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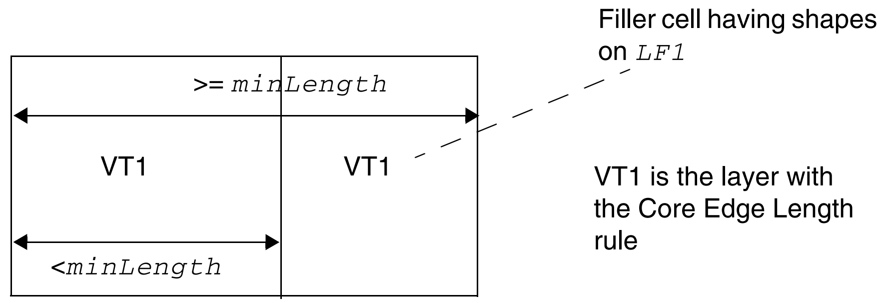
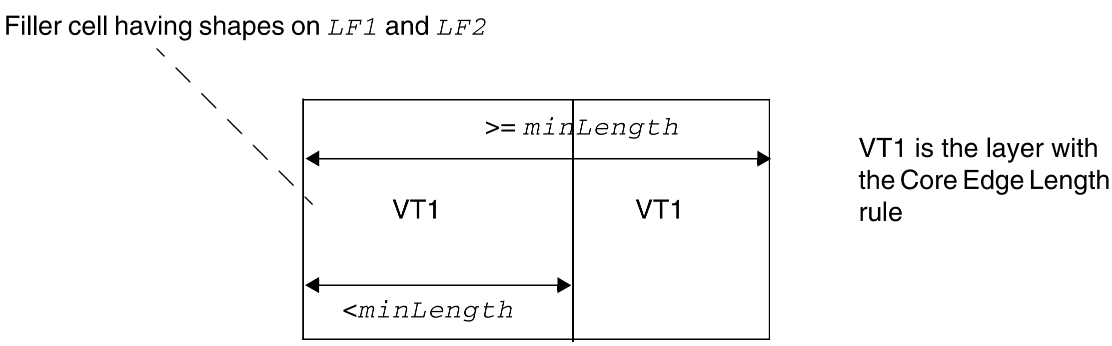
Figure 3-3 Example of the Core Edge Length Rule with EXCEPTLAYER



Violation, the right filler cell is excluded and the core edge length check is applied on the left cell alone. OK if `EXCEPTLAYER` is omitted.

Illustration of `COREEDGELENGTH` *`minLength`*
`EXCEPTLAYER` *`LF1 LF2`*

Figure 3-4 Example of the Core Edge Length Rule with WITHINLAYER



Violation, the left filler cell should be checked alone. OK if `WITHINLAYER` is omitted.

Illustration of `COREEDGELENGTH` *`minLength`*
`WITHINLAYER` *`LF1 LF2`*

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Corner Spacing Rule

A corner spacing rule can be used to specify the spacing between two corner touch points.

You can create a corner spacing rule by using the following property definition:

```
PROPERTY LEF58_CORNERSPACING
        "CORNERSPACING spacing [ALIGNEDONLY]
            [CHECKIMPLANTGROUPONLY]
        ; " ;
```

Where:

`CORNERSPACING` *`spacing`*

Specifies the spacing between two corner touch points, where each of the touch points is formed by two implant shapes.

Type: Float, specified in microns.

`ALIGNEDONLY`

Specifies the rule applies only when the two corner touch points are vertically aligned.

`CHECKIMPLANTGROUPONLY`

Specifies that this corner spacing rule does not apply to shapes of this implant layer, but applies only to the merged shapes of any implant group with `BASEDLAYER` pointing to this implant layer.

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Corner Spacing Rule Examples

Figure 3-5 Illustration of Corner Spacing Rule

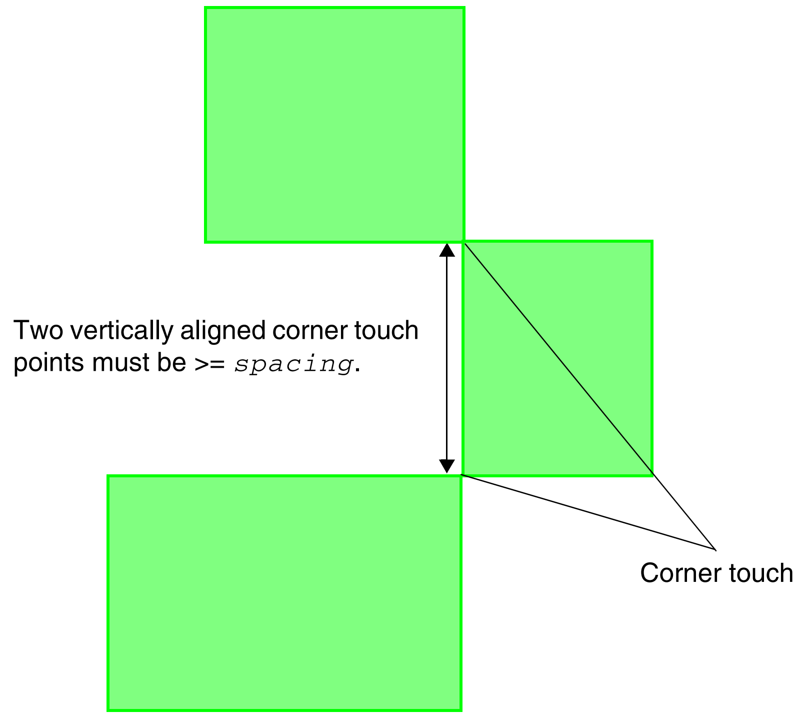

Illustration of `CORNERSPACING` *`spacing`* `ALIGNEDONLY`

- The following example illustrates a corner spacing rule with `CHECKIMPLANTGROUPONLY`:

```
PROPERTYDEFINITIONS
LIBRARY LEF58_IMPLANTGROUP STRING "
      IMPLANTGROUP IPG1 LAYER IP1 IP2 BASEDLAYER IP1 ;";
END PROPERTYDEFINITIONS
```

On layer `IP1`:

```
PROPERTY LEF58_CORNERSPACING "
      CORNERSPACING 0.15 ALIGNEDONLY
          CHECKIMPLANTGROUPONLY ; " ;
```


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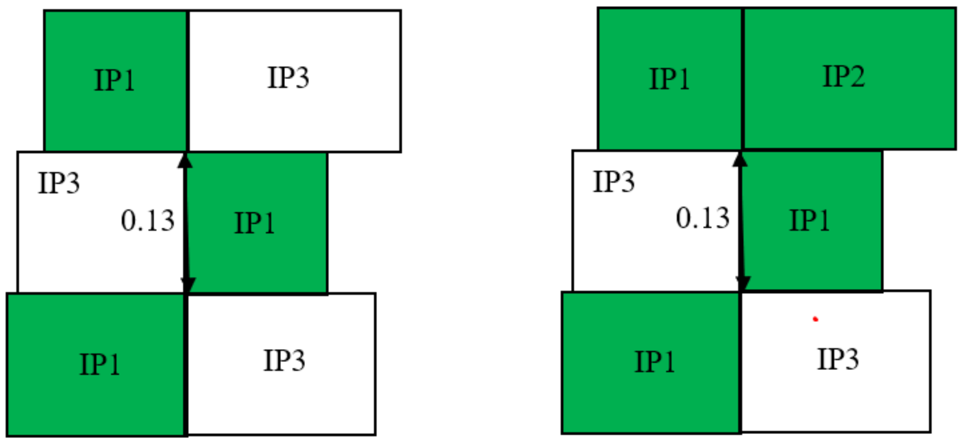
Figure 3-6 Example of the Corner Spacing Rule with CHECKIMPLANTGROUPONLY

Green indicates `IPG1`



a) Violation, the `IP1` shapes do not need to check this corner spacing rule, but the merged shapes of `IPG1`, which are `IP1` shapes only in this case, fail.

b) OK, the `IPG1` merged shapes do not form a corner in the top row and the `IP1` shapes alone do not subject to this rule. Violation if `CHECKIMPLANTGROUPONLY` is omitted.

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Forbidden Width Rule

You can create a forbidden width rule by using the following property definition:

```
PROPERTY LEF_CDN_FORBIDDENWIDTH
        "FORBIDDENWIDTH width
        ; " ;
```

Where:

`FORBIDDENWIDTH` *`width`*

Specifies that if the implant layer shape abuts two different implant layer shapes and all three shapes have a width exactly equal to *`width`*, it is a violation if both neighboring shapes are also each abutted to a different implant layer shape on their opposite sides. All same-layer implant shapes must be merged before considering this rule.

Type: Float, specified in microns

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Maximum Spacing Rule

You can create a maximum spacing rule by using the following property definition:

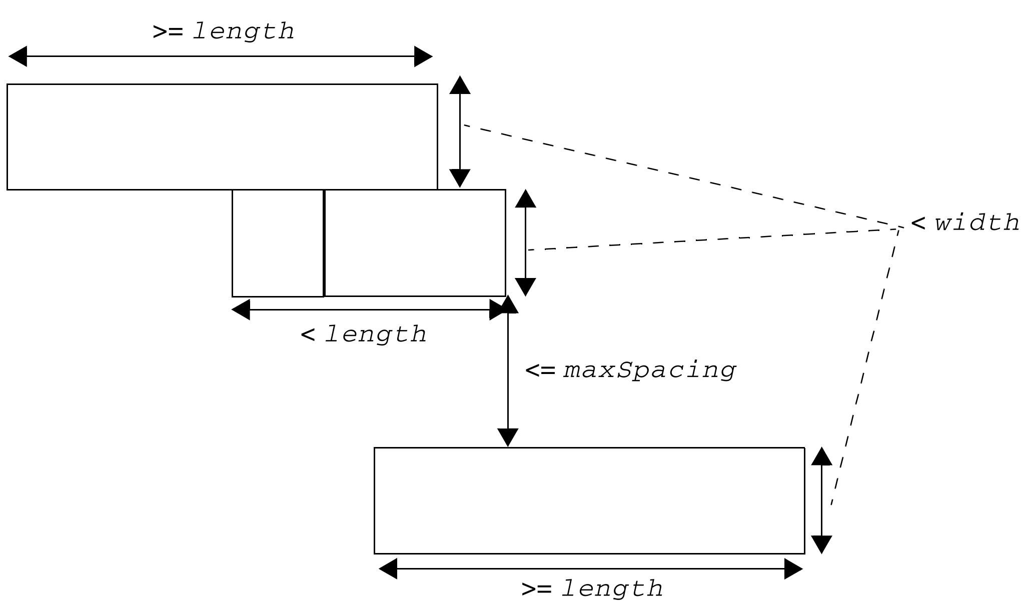
```
PROPERTY LEF58_MAXSPACING
      "MAXSPACING maxSpacing WIDTH width LENGTH length
       ; " ;
```

Where:

`MAXSPACING` *`maxSpacing`* `WIDTH` *`width`* `LENGTH` *`length`*

Specifies that an implant shape with width less than or equal to *`width`* vertically and length less than *`length`* horizontally in horizontal cell rows must be fully covered horizontally by other implant shapes that are vertically on two opposite sides less than or equal to *`maxSpacing`*. Abutting implant shapes should be merged if on the same row but not merged if on adjacent rows. *Type*: Float, specified in microns square

Figure 3-7 Illustration of the Max Spacing Rule



It is OK. The top and bottom implant shapes have length >= *`length`*, which would not trigger the rule. The middle shape triggers the rule. It has neighbor shapes on both the top and bottom opposite sides <= *`maxSpacing`*. Those neighbor shapes together can fully cover the entire horizontal length of the middle shape to not be a violation.

Illustration of `MAXSPACING` *`maxSpacing`* `WIDTH` *`width`* `LENGTH` *`length`*

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Min Enclosed Area Rule

A min enclosed area rule can be used to specify the minimum area that is enclosed by shapes on the current implant layer.

You can create a min enclosed area rule by using the following property definition:

```
PROPERTY LEF58_MINENCLOSEDAREA
      "MINENCLOSEDAREA area
       ; " ;
```

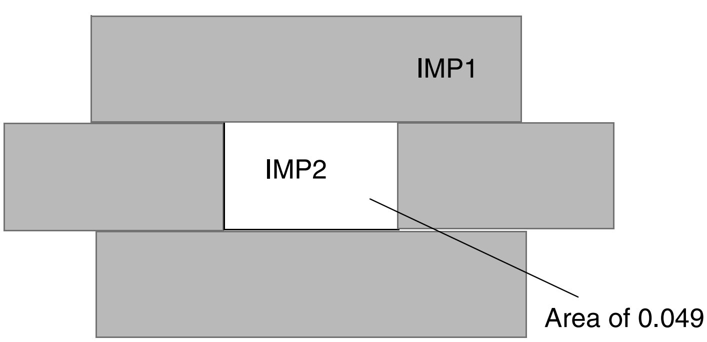
Where:

`MINENCLOSEDAREA` *`area`*

Specifies the minimum area that is enclosed by shapes on the current implant layer.
Type: Float, specified in microns square

Figure 3-8 Example of the Min Enclosed Area Rule

```
LAYER IMP1
     TYPE IMPLANT
     PROPERTY LEF58_MINENCLOSEDAREA "
           MINENCLOSEDAREA 0.5  ; " ;
END IMP1
```



a) Violation, the area doughnut hole of IMP1 < 0.5

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Minimum Height Rule

You can create a minimum height rule by using the following property definition:

```
PROPERTY LEF58_MINHEIGHT
        "MINHEIGHT minHeight WIDTH maxWidth OVERLAP maxOverlap
        ; " ;]
```

Where:

`MINHEIGHT` *`minHeight`* `WIDTH` *`maxWidth`* `OVERLAP` *`maxOverlap`*

Specifies a minimum height rule for three implant layer shapes arranged in a staircase or notch pattern across three consecutive vertical cell rows. If the middle shape has a width less than or equal to *`maxWidth`*, and its horizontal overlap with both the top and bottom shapes does not exceed *`maxOverlap`*, then at least one of the three shapes must have a vertical height greater than or equal to *`minHeight`*. If all three shapes have height less than *`minHeight`*, it is a violation.
Type: Float, specified in microns.

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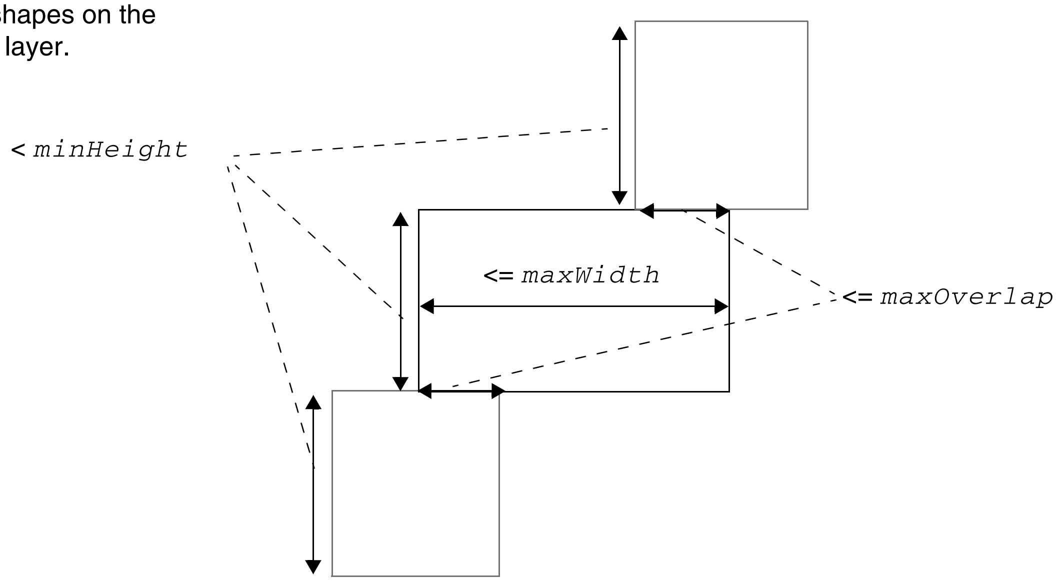
Figure 3-9 Illustration of the Minimum Height Rule

Illustration of
`MINHEIGHT` *`minHeight`*
`WIDTH` *`maxWidth`* `OVERLAP` *`maxOverlap`*

Three shapes on the implant layer.



The middle shape has a width <= *`maxWidth`* horizontally and an overlap with both the top and bottom shapes with a length <= *`maxOverlap`*, one of the three shapes must have a height >= *`minHeight`* vertically. As they are all < *`minHeight`*, it is a violation.

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Min Step Rule

You can create a min step rule by using the following property definition:

```
PROPERTY LEF58_MINSTEP
        "MINSTEP minStepLength MINADJACENTLENGTH minAdjLength
        ; " ;
```

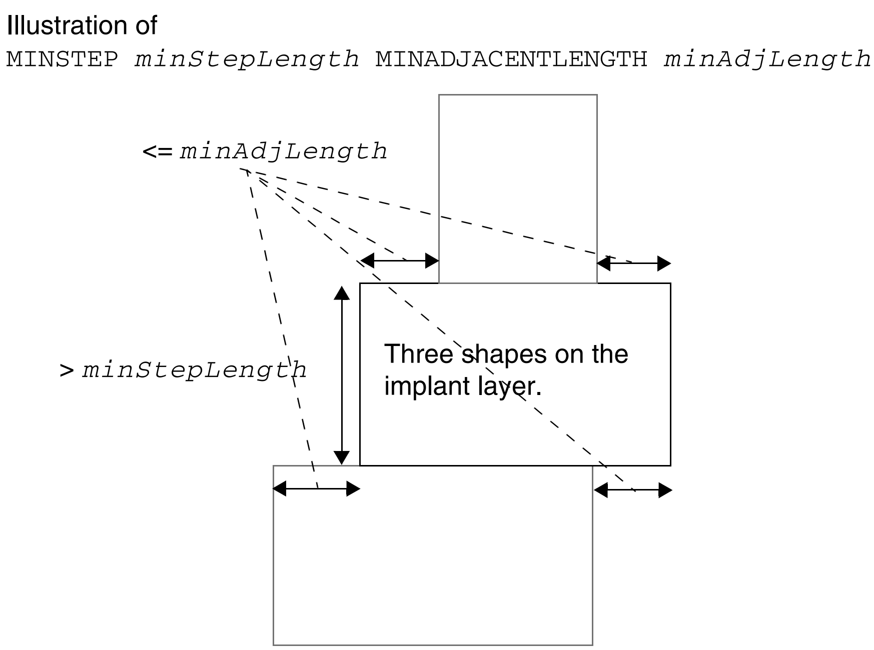
Where:

`MINSTEP` *`minStepLength`* `MINADJACENTLENGTH` *`minAdjLength`*

Specifies a min step length rule vertically on a pattern with three implant layer shapes forming a staircase on three consecutive vertical cell rows. The middle vertical cell length of the staircase must be greater than *minStepLength* if the lengths of both the adjacent horizontal edges are less than or equal to *minAdjLength*. The staircase pattern means that consecutive edges must form a concave corner and a convex corner.

Type: Float, specified in microns

Figure 3-10 Illustration of the Min Step Rule



Based on the left three consecutive edges, the middle vertical length edge must be greater than *minStepLength* since they form a staircase with two adjacent horizontal edge lengths less than or equal to *minAdjLength*. The right three consecutive edges are irrelevant since they do not form a staircase.

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Spacing Rule

A spacing rule can be used to specify the spacing requirements for the current implant layer. You can create a spacing rule by using the following property definition:

```
PROPERTY LEF58_SPACING
        "SPACING minSpacing [LAYER layerName2]
            [HORIZONTAL|VERTICAL PRL prl ][EXCEPTABUTTED]
                 [EXCEPTCORNERTOUCH] [LENGTH length]
                 [INTERSECTLAYERS layerNameList...]
            ; " ;
```

Where:

`EXCEPTABUTTED` Specifies that the implant spacing rule is exempted if the two implant shapes are abutted. `PRL` in the orthogonal direction of the spacing (that is, vertical `PRL` in `HORIZONTAL` or horizontal `PRL` in `VERTICAL`) must be greater than `0` to trigger the exemption. In other words, the corner or point touch in the `VERTICAL` case for horizontal cell rows is not exempted.

`EXCEPTCORNERTOUCH`

Specifies that the implant spacing rule is exempted if the two implant shapes are corner or point touch in the `VERTICAL` case for horizontal cell rows.

`HORIZONTAL|VERTICAL PRL` *`prl`*

Specifies that the implant spacing rule applies only in the given direction. By default, the spacing is applied horizontally for implant shapes on the same row. In the `VERTICAL` case, the rule applies only if the horizontal projected `PRL` is greater than *`prl`*. If *`prl`* is negative, the rule is checked for two implant shapes within abs(*`prl`*).

Type: Float, specified in microns

`INTERSECTLAYERS` *`layerNameList...`*

Specifies that the spacing rule applies only if the spacing gap is covered by shapes in any implant layers in *`layerNameList`*. For `VERTICAL`, this construct should be applied only when a negative value is given in `PRL`. It is a violation only if the spacing gap is covered by shapes in any implant layers in *`layerNameList`* on both sides on two adjacent rows.

Type: String

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`LAYER` *`layerName2`*

Specifies the name of another implant layer that requires extra spacing that is greater than or equal to *`minSpacing`* from this implant layer.

`LENGTH` *`length`*

Specifies that the spacing rule applies only to implant shapes with length less than *`length`*, which is the vertical height for horizontal cell rows. Implant shapes are not merged across different cell rows for length or height determination. In addition, an implant shape with length less than *`length`* does not need to honor any other implant spacing rule without `LENGTH`.

Type: Float, specified in microns

`SPACING` *`minSpacing`*

Specifies the minimum spacing for the layer. This value affects the legal cell placement.

Type: Float, specified in microns

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Spacing Rule Examples

Figure 3-11 Illustration of the Spacing Rule

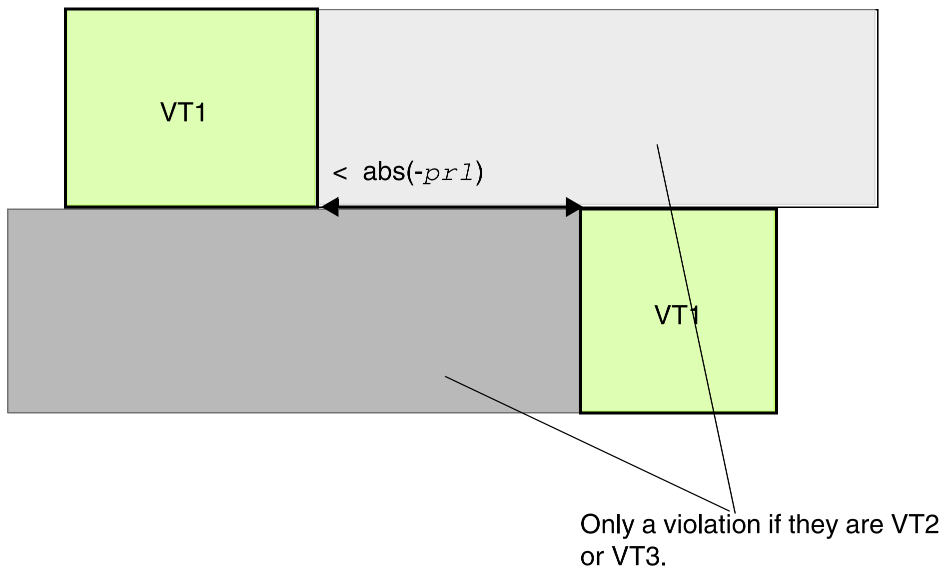

Illustration of
`SPACING 0.001 VERTICAL PRL` *`-prl`*
`INTERSECTLAYERS` *`VT2 VT3`*
on VT1 layer

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Figure 3-12 Illustration of the Spacing Rule

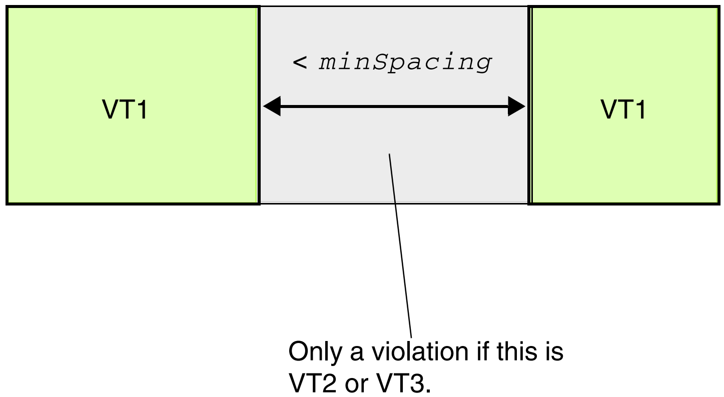

Illustration of
`SPACING` *`minSpacing`* `HORIZONTAL`
`INTERSECTLAYERS` *`VT2 VT3`*
on VT1 layer

Width Rule

A width rule can be used to specify the width requirements for the current implant layer. You can create a spacing rule by using the following property definition:

```
[PROPERTY LEF58_WIDTH
        "WIDTH minWidth [LAYER {layerName2 | ANY}[NODIFFMERGE]]
            [CELLHEIGHT height LAYER layerName3]
            [WITHINLAYER layerName
            |EXCEPTLAYER exceptLayerName]
            [ZEROPRL [MAXWIDTH maxWidth]
                 [NOTALIGNEDROWWIDTH width LENGTH length]]
            [EXCEPTCORNERTOUCH][LENGTH length]
            [CHECKIMPLANTGROUP groupName]
            ]; " ;
```

Where:

`CELLHEIGHT` *`height`* `LAYER` *`layerName3`*

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Specifies that the implant shapes in standard cells with height exactly equal to *height* should not be merged with the same implant shapes in horizontally abutted cells with shapes on *layerName3*. *layerName3* is typically a masterslice layer. When *height* is specified as 0, it means that cells with any height should not be merged with the same implant shapes in horizontally abutted cells with shapes on *layerName3*.

Type: String

CHECKIMPLANTGROUP *groupName*

Specifies that it is a violation only if the given implant group, *groupName*, to which the implant layer belongs also violates the width requirement at the same location.
Type: String

EXCEPTCORNERTOUCH

Specifies that the width requirement does not apply if corners touch.

EXCEPTLAYER *exceptLayerName*

Specifies that the implant shapes overlapping with the shapes in *exceptLayerName*, which should be a masterslice layer with the ABUTFILLER type property, in the same cell should not be merged with other implant shapes that are not overlapping with the shapes in *exceptLayerName*, on which the width rule is checked. Typically, only standard cells contain OBS shapes in *exceptLayerName*.

Type: String

LAYER {*layerName2* | ANY}

Specifies the name of another implant layer that requires the width check after its shapes merge with the shapes on the current implant layer. If ANY is specified, the width check is performed one by one between the current implant layer and any other implant layers.
Type: String

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`LENGTH` *`length`* Specifies that the width rule applies only to implant shapes with length less than *`length`*, which is the vertical height for horizontal cell rows. Implant shapes should be merged across different cell rows for length or height determination. In addition, an implant shape with length less than *`length`* does not need to honor any other implant width rule without `LENGTH`.
Type: Float, specified in microns

`MAXWIDTH` *`maxWidth`*

Specifies that the adjacent row width checking applies only if the width of any shapes on a row is less than or equal to *`maxWidth`*.
Type: Float, specified in microns

`NODIFFMERGE` Specifies that shapes belonging to different implant layers defined in `LAYER` must not be merged and only the shapes from the same implant layer should be merged prior to applying the width rule check.

`NOTALIGNEDROWWIDTH` *`width`* `LENGTH` *`length`*

Specifies that the rule for width in the direction perpendicular to the row is checked only if one of the implant shapes with a width less than or equal to *`width`* is located within a cell whose length is less than *`length`*, which is the vertical height for horizontal cell rows. This rule is checked only on the edges that are not aligned with the row boundary.
Type: Float, specified in microns

Specifies that the adjacent row width checking applies only if the width of any shapes on a row is less than or equal to *`maxWidth`*.
Type: Float, specified in microns

`WIDTH` *`minWidth`*

Specifies the minimum width for this layer. This value affects the legal cell placement.
Type: Float, specified in microns

`WITHINLAYER` *`layerName`*

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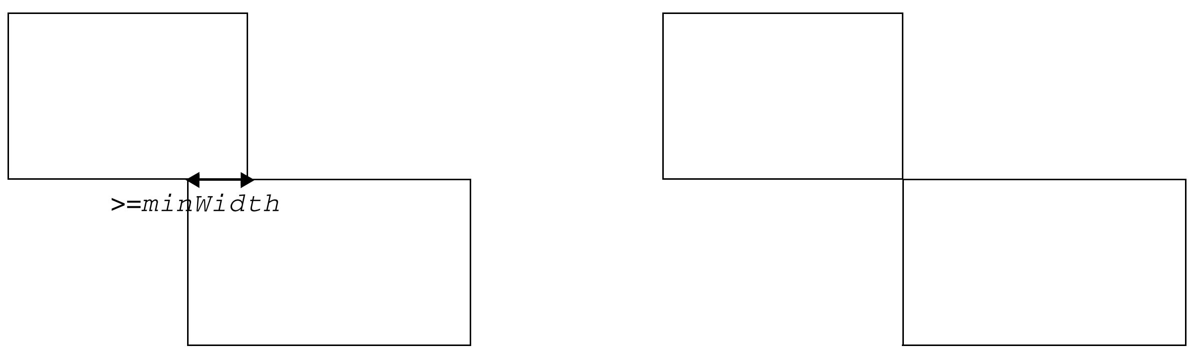
Specifies that the implant shapes overlapping with the shapes in *layerName*, which should be a masterslice layer, should be checked against *minWidth* by themselves. In other words, those implant shapes should not be merged with other implant shapes that are not overlapping with the shapes in *layerName*. Typically, only standard cells contain OBS shapes in *layerName*.

Type: String

`ZEROPRL` Specifies that the width requirement applies only for two implant shapes on adjacent rows with zero `PRL` in the direction perpendicular to the row direction.

Width Rule Examples

Figure 3-13 Example of the Width Rule with EXCEPTCORNERTOUCH



Horizontal width must be >= *minWidth*

Corner touching is allowed. Violation if `EXCEPTCORNERTOUCH` is omitted.

Illustration of `WIDTH` *minWidth* `ZEROPRL EXCEPTCORNERTOUCH`

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Figure 3-14 Example of the Width Rule with EXCEPTLAYER

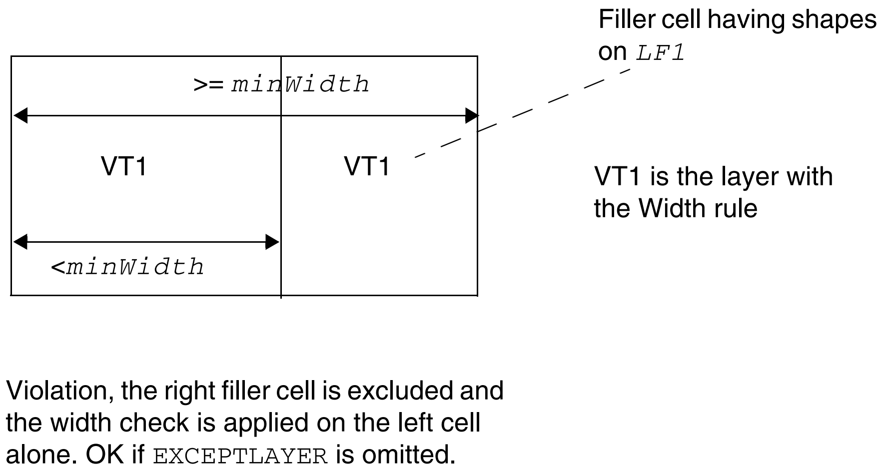

Illustration of `WIDTH` *`minWidth`* `EXCEPTLAYER` *`LF1`*

- The following example illustrates a Width rule with `CHECKIMPLANTGROUP`:

```
PROPERTYDEFINITIONS
LIBRARY LEF58_IMPLANTGROUP STRING "
      IMPLANTGROUP IPG1 LAYER IP1 IP2 IP3 BASEDLAYER IP1 ;
      IMPLANTGROUP IPG2 LAYER IP1 IP4 BASEDLAYER IP1 ; " ;
END PROPERTYDEFINITIONS
```

On layer `IP1`:

```
PROPERTY LEF58_WIDTH  "
      WIDTH 0.15 LAYER IP2 ZEROPRL
          CHECKIMPLANTGROUP IPG1 ; " ;
```


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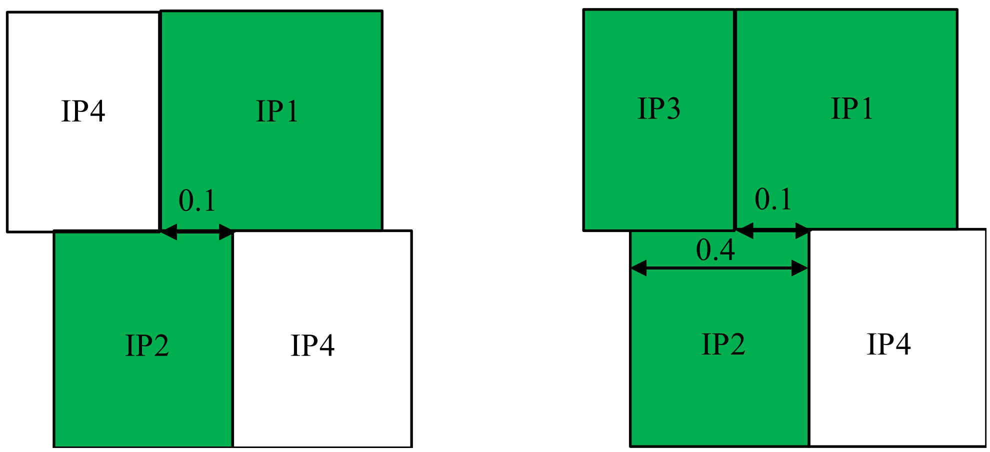
Figure 3-15 Example of the Width Rule with CHECKIMPLANTGROUP

Green indicates `IPG1`



a) Violation. `IP1` and `IP2` have vertical overlap (width) of `0.1` (< `0.15`) and so does `IPG1`. Overlap with `IP4`, which belongs to another group of `IP1`, is irrelevant.

b) OK. `IPG1` has vertical overlap (width) of `0.4`. Violation if `CHECKIMPLANTGROUP` is omitted.

- The following example illustrates a Width rule with `MAXWIDTH`:

  On layer `IP1`:

```
PROPERTY LEF58_WIDTH  "
      WIDTH 0.15 ZEROPRL MAXWIDTH 0.4 ;
      WIDTH 0.15 LAYER IP2 ZEROPRL MAXWIDTH 0.4 ; " ;
```


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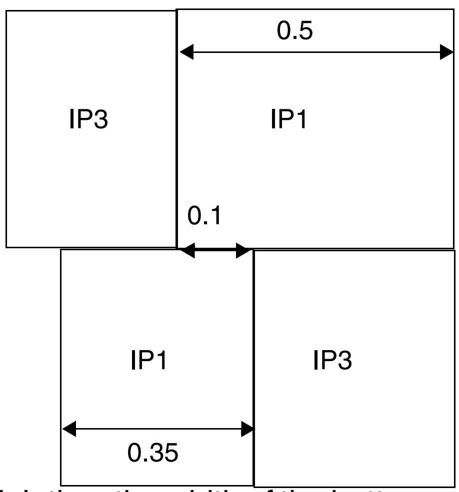
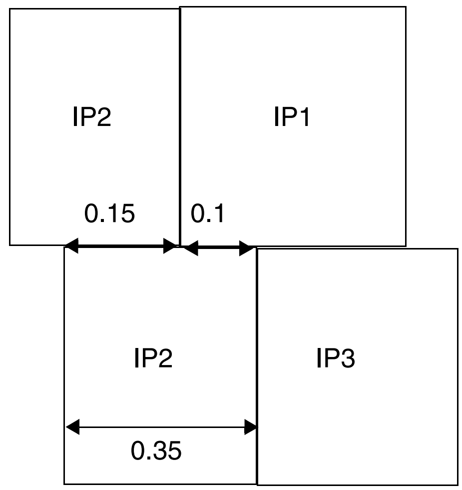
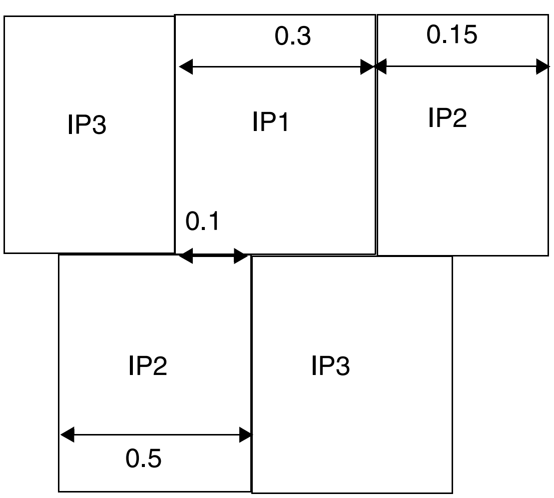
Figure 3-16 Example of the Width Rule with MAXWIDTH



a) Violation, the width of the bottom `IP1` shape is `0.35`, which could trigger the rule even though the width of the top `IP1` shape is `0.5` (> `0.4`). If the width of the bottom IP1 shape is changed to > `0.4`, it would not be a violation.



b) OK, the `IP1` and `IP2` shapes would be merged on the top row, the cross row width is `0.15 + 0.1 = 0.25` (>= `0.15`) to pass the rule.



c) OK, the `P1` and `IP2` shapes would be merged on the top row and the combined width of `0.3 + 0.15 = 0.45` (> `0.4`) does not trigger the rule.

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4

LEF Syntax - Layer (Masterslice)

This chapter contains information about the following topics:

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Layer (Masterslice)

```
LAYER layerName
     TYPE {MASTERSLICE} ;
     [MASK maskNum ;]
     [PROPERTY propName propVal ;] ...
     [PROPERTY LEF58_TYPE
        "TYPE [NWELL | PWELL | ABOVEDIEEDGE | BELOWDIEEDGE | DIFFUSION | TRIMPOLY
              | TRIMMETAL [VIRTUAL] | REGION | MEOL | WELLDISTANCE [BOUNDARYTAP]
              | CPODE | RCBLOCKAGE metalLayer | TRIMDIFFUSION | ABUTFILLER
              | ABUTLOGIC | CUTPGRAIL metalLayer | PROBE | TS | MASKSPLIT];" ;
     [PROPERTY LEF58_ABUTFILLER
        "ABUTFILLER distance LAYER abutFillerLayerName
          [EXCEPTCELL otherAbutlogicLayerName
             EXACTSPACING spacing PRL {prl}...]
          [SAMEIMPLANTLAYER]
        ; " ;]
     [PROPERTY LEF58_ADJACENTTRIMS
        "ADJACENTTRIMS trimNum SPACING spacing [WITHIN within]
          [SAMEMASK | MASK maskNum]
          ; " ;]
     [PROPERTY LEF58_AREA
        "AREA minArea
          ; " ;]
     [PROPERTY LEF58_BOUNDARYSPACING
        "BOUNDARYSPACING spacing1 spacing2 LAYER layerName
          ; " ;]
     [PROPERTY LEF58_BOUNDINGBOX
        "BOUNDINGBOX layerName
          ; " ;]
     [PROPERTY LEF58_COREEDGELENGTH
        "COREEDGELENGTH minLength [MAXLENGTH maxLength]
          [WIDTH width]
          [CONCAVECORNER | CONVEXCORNER | MIXEDCORNER]
          [EXTENDTOALIGN width spacing [EXTENDONLY | ALIGNONLY]]
          [FILLERONLY]
          [EXCEPTLAYER {exceptLayerName}...]
          ; " ;]
     [PROPERTY LEF58_CPODEGROUP
        "CPODEGROUP num PARALLEL parLength WITHIN within
          [INCLUDESIDESHIFT commonNum]
          ; " ;]
     [PROPERTY LEF58_CPODEGROUPSPACING
        "CPODEGROUPSPACING spacing CPODES num
          ; " ;]
     [PROPERTY LEF58_DIFFIMPLANTMUSTHAVE
        "DIFFIMPLANTMUSTHAVE
          ; " ;]
     [PROPERTY LEF58_DIFFUSIONJOGMUSTHAVE
        "DIFFUSIONJOGMUSTHAVE
```


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```
      ; " ;]
[PROPERTY LEF58_DIFFUSIONJOGMUSTHAVE
   "DIFFUSIONJOGMUSTHAVE
      ; " ;]
[PROPERTY LEF58_ENCLOSURE
   "ENCLOSURE overhang LAYER secondLayerName;..." ;]
[PROPERTY LEF58_FORBIDDENARRAY
   FORBIDDENARRAY numRows numCols XSPACING xLow xHigh
      YSPACING yLow yHigh
      [SAMEMASK]
      ; " ;]
[PROPERTY LEF58_MAXLENGTH
  "MAXLENGTH length [HORIZONTAL|VERTICAL]
  ; " ;]
[PROPERTY LEF58_MAXLENGTH
  "MAXLENGTH maxLength
      [HORIZONTAL|VERTICAL] [GAPMERGED gap]
      [ORTHOGONALLENGTH minLength]
      [EXCEPTWIDTH width MAXLENGTH widthMaxLength]
      ; " ;]
[PROPERTY LEF58_MAXLENGTH
  "MAXLENGTH maxLength {HORIZONTAL|VERTICAL}
      [GAPMERGED gap][EXCEPTWIDTH width]
      ; " ;]
[PROPERTY LEF58_MAXLENGTH
  "MAXLENGTH maxLength {HORIZONTAL|VERTICAL}
      [GAPMERGED gap][EXCEPTWIDTH width]
  ; " ;]
[PROPERTY LEF58_MAXLENGTH
  "MAXLENGTH maxLength [WIDTH width]
  ; " ;]
[PROPERTY LEF58_MINENCLOSEDAREA
  "MINENCLOSEDAREA area
  ; " ;]
[PROPERTY LEF58_MINLENGTH
  "MINLENGTH minLength {HORIZONTAL|VERTICAL}
      [EXACTWIDTH width]
  ; " ;]
[PROPERTY LEF58_MINSTEPTABLE
  "MINSTEPTABLE {yLength xLength}...
  ; " ;]
[PROPERTY LEF58_NOBOUNDARYJOG
  "NOBOUNDARYJOG
  ; " ;]
[PROPERTY LEF58_PROBEAREA
  "PROBEAREA width length LAYER layerName
  ; " ;]
[PROPERTY LEF58_RECTONLY
  "RECTONLY
  ; " ;]
[PROPERTY LEF58_REGION
```


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```
  "REGION regionLayerName BASEDLAYER trimLayerName
  ; " ;]
[PROPERTY LEF58_ROWBASED
  "ROWBASED
  ; " ;]
[PROPERTY LEF58_SPACING
   "SPACING minSpacing
   ; " ;]
[PROPERTY LEF58_SPACING
   "SPACING spacing LENGTH length
   ; " ;]
[PROPERTY LEF58_SPACING
   "SPACING spacing PRL prl
   ; " ;]
[PROPERTY LEF58_SPACING
   "SPACING spacing LAYER layerName
   ; " ;]
[PROPERTY LEF58_SPACING
   "SPACING spacing [MERGEDTRIM [length]]
   [PRLSPACING prlSpacing1 prlSpacing2]
   [ENDTOEND endToEndSpacing [PRL prl]
        [EXACTALIGNED exactAlignedSpacing [MIDDLEWIREONLY]
            [EXCEPTDIFFMASKALIGNED]]
        [EDGEALIGNED edgeAlignedSpacing ]
        [EXCEPTMIDMETALWIDTH width]]
   [SAMEMASK] [MASK maskNum [VIRTUAL]]
   ; " ;...]
[PROPERTY LEF58_SPACING
   "SPACING spacing
   ; " ;]
[PROPERTY LEF58_SPACING
   "SPACING spacing [SKIPPROTRUDEDFILLER]
   ; " ;]
[PROPERTY LEF58_SPACING
   "SPACING minSpacing [HORIZONTAL|VERTICAL]
   [SAMENET | LAYER secondLayerName]
   [LENGTH length1 length2]
   ;..." ;]
[PROPERTY LEF58_SPACING
   "SPACING minSpacing [HORIZONTAL|VERTICAL]
   [SAMENET | LAYER secondLayerName]
   [LENGTH length1 length2]
   [INCLUDEENDCAPREGULAR]
   ;..." ;]
[PROPERTY LEF58_TRIMMEDMETAL
   "TRIMMEDMETAL metalLayer [MASK maskNum [INCLUDEVIRTUAL]
     [VIRTUAL virtualMaskNum]]
     ]; " ;
[PROPERTY LEF58_TRIMSHAPE
   "TRIMSHAPE [METALMASK metalMaskNum] EXTENSIONMODEL
   {ADJACENTTRACK | MIDTRACK | EXTENSION extension [METALEDGE]}
```


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```
        EXACTWIDTH exactWidth [EXACTVIRTUALWIDTH exactVirtualWidth]
        [MAXOVERLAPWIDTH maxOverlapWidth]
        [GAPMERGED trimGapLength]
        [MAXLENGTH maxLength
             [ROWBOUNDARYMAXLENGTH rowBoundaryMaxLength]
             [GAPMERGEDFORLENGTH trimGapLength]
             [MASK maskNum]]
        [EXCEPTSPACING spacing]
        [EXCEPTWIDTH width [EXACTWIDTH]
             [EXCEPTVIRTUALWIDTH virtualWidth]]
        [USEMETALMASK]
        ; " ;]
     [PROPERTY LEF58_WIDTH
        "WIDTH minWidth [WITHINLAYER {layerName}...]
        ;" ;]
     [PROPERTY LEF58_WIDTH
        "WIDTH minWidth
        ;" ;]
     [PROPERTY LEF58_WIDTH
        "WIDTH minWidth [ZEROPRL]
        ;" ;]
     [PROPERTY LEF58_WIDTHLENGTHRATIO
        "WIDTHLENGTHRATIO ratio WIDTH width
        ;" ;]
     [PROPERTY LEF58_YMAXLENGTH
        "YMAXLENGTH maxLength
        ;" ;]

END layerName
```

Defines masterslice (non-routing) or overlap layers in the design. Masterslice layers are typically polysilicon layers and are only needed if the cell `MACROs` have pins on the polysilicon layer.

Each layer is defined by assigning it a name and design rules. You must define masterslice or overlap layers separately, with their own layer statements.

You must define layers in process order from bottom to top. For example:

```
poly        masterslice
cut01       cut
metal1      routing
cut12       cut
metal2      routing
cut23       cut
metal3      routing
```

`LAYER` *layerName*

Specifies the name for the layer. This name is used in later references to the layer.

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`TYPE`

Specifies the purpose of the layer.

`MASTERSLICE`	Layer is fixed in the base array. If pins appear in the masterslice layers, you must define vias to permit the routers to connect those pins and the first routing layer. Wires are not allowed on masterslice layers. Routing tools can use only one masterslice layer. If a masterslice layer is defined, exactly one cut layer must be defined between the masterslice layer and the adjacent routing layers.

`MASK` *`maskNum`*

Specifies how many masks for double- or triple-patterning will be used for this layer. The *`maskNum`* variable must be an integer greater than or equal to 2. Most applications only support values of 2 or 3.

`PROPERTY` *`propName propVal`*

Specifies a numerical or string value for a layer property defined in the `PROPERTYDEFINITIONS` statement. The *`propName`* you specify must match the propName listed in the `PROPERTYDEFINITIONS` statement.

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Defining Masterslice Layer Properties to Create 32/28 nm and Smaller Nodes Rules

You can include masterslice layer properties in your LEF file to create 32/28nm and smaller nodes rules that currently are not supported by existing LEF syntax. The properties are specified inside the `LAYER MASTERSLICE` statements, where they can be seen with other rules.

Before you can reference them, properties must be defined at the beginning of the LEF file in the `PROPERTYDEFINITIONS` statement, immediately before the first `LAYER` statement.

- Properties belong to the `LAYER` object and have a type of `STRING`.
- The property names used for these rules all start with `LEF58_`.

All properties use the following syntax within the LEF `PROPERTYDEFINITIONS` statement:

```
PROPERTYDEFINITIONS
     LAYER propName STRING ["stringValue"] ;

END PROPERTYDEFINITIONS
```

The property definitions for the masterslice layer properties are as follows:

```
PROPERTYDEFINITIONS
    LAYER LEF58_ABUTFILLER STRING ;
    LAYER LEF58_ADJACENTTRIMS STRING ;
    LAYER LEF58_AREA STRING ;
    LAYER LEF58_BOUNDARYSPACING STRING ;
    LAYER LEF58_BOUNDINGBOX STRING ;
    LAYER LEF58_COREEDGELENGTH STRING ;
    LAYER LEF58_CPODEGROUP STRING ;
    LAYER LEF58_CPODEGROUPSPACING STRING ;
    LAYER LEF58_DIFFIMPLANTMUSTHAVE STRING ;
    LAYER LEF58_DIFFUSIONJOGMUSTHAVE STRING ;
    LAYER LEF58_ENCLOSURE STRING ;
    LAYER LEF58_FORBIDDENARRAY STRING ;
    LAYER LEF58_MAXLENGTH STRING ;
    LAYER LEF58_MINENCLOSEDAREA STRING ;
    LAYER LEF58_MINLENGTH STRING ;
    LAYER LEF58_MINSTEPTABLE STRING ;
    LAYER LEF58_NOBOUNDARYJOG STRING ;
    LAYER LEF58_PROBEAREA STRING ;
    LAYER LEF58_RECTONLY STRING ;
    LAYER LEF58_REGION STRING ;
```


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```
    LAYER LEF58_ROWBASED STRING ;
    LAYER LEF58_SPACING STRING ;
    LAYER LEF58_TAPDISTANCE STRING ;
    LAYER LEF58_TRIMMEDMETAL STRING ;
    LAYER LEF58_TRIMSHAPE STRING ;
    LAYER LEF58_TYPE STRING ;
    LAYER LEF58_WIDTH STRING ;
    LAYER LEF58_WIDTHLENGTHRATIO STRING ;
    LAYER LEF58_YMAXLENGTH STRING ;
END PROPERTYDEFINITIONS
```

Type Rule

A type rule can be used to further classify a masterslice layer.

You can create a type rule by using the following property definition:

```
TYPE MASTERSLICE;
     PROPERTY LEF58_TYPE
       "TYPE [NWELL | PWELL | ABOVEDIEEDGE | BELOWDIEEDGE | DIFFUSION | TRIMPOLY
              | TRIMMETAL [VIRTUAL] | REGION | MEOL | WELLDISTANCE [BOUNDARYTAP]
              | CPODE | RCBLOCKAGE metalLayer | TRIMDIFFUSION | ABUTFILLER
              | ABUTLOGIC | CUTPGRAIL metalLayer | PROBE | TS | MASKSPLIT]
              ;" ;
```

Where:

`ABOVEDIEEDGE | BELOWDIEEDGE`

Specifies that the masterslice layer is a special one for a hard macro to define an OBS on the layer such that the die boundary of the above or below die do not overlap.

The following property statement can be defined on layers with type `ABOVEDIEEDGE` or `BELOWDIEEDGE`:

```
PROPERTY LEF58_SPACING
     "SPACING minSpacing
     ; " ;
```


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`ABUTFILLER` Specifies a special layer for abutment consideration. The standard cells with shapes on this layer should be filler cells with `CLASS CORE SPACER`.

The following property could be applied on this `ABUTFILLER` layer with type `ABUTFILLER`.

```
PROPERTY LEF58_MAXLENGTH
     "MAXLENGTH length [HORIZONTAL|VERTICAL]
     ; " ;
```

where

`MAXLENGTH` *`length`* `[HORIZONTAL|VERTICAL]`

Specifies that the maximum length of any shape on this layer must be less than or equal to *`length`*. If `HORIZONTAL` or `VERTICAL` is specified, the maximum length is checked only in the given direction.

Type: Float, specified in microns

`ABUTLOGIC` Specifies a special layer for abutment consideration. The standard cells with shapes on this layer should be logic cells. This layer should be defined after layer with type `ABUTFILLER`.

The following property could be applied on this `ABUTLOGIC` layer with type `ABUTLOGIC`.

```
PROPERTY LEF58_ABUTFILLER
     "ABUTFILLER distance LAYER abutFillerLayerName
     ; " ;

PROPERTY LEF58_WIDTH
     "WIDTH minWidth
     ;" ;
```


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`CPODE`

Specifies that this masterslice layer is a special layer related to poly and diffusion.

The following properties could be defined on a `CPODE` masterslice layer:

```
PROPERTY LEF58_BOUNDARYSPACING
     "BOUNDARYSPACING spacing1 spacing2 LAYER layerName
        ; " ;

PROPERTY LEF58_MAXLENGTH
     "MAXLENGTH maxLength
        [[HORIZONTAL|VERTICAL] [GAPMERGED gap]
        [ORTHOGONALLENGTH minLength]]
        ; " ;

PROPERTY LEF58_CPODEGROUP
     "CPODEGROUP num PARALLEL parLength WITHIN within
        [INCLUDESIDESHIFT commonNum]
        ; " ;

PROPERTY LEF58_CPODEGROUPSPACING
     "CPODEGROUPSPACING spacing CPODES num
        ; " ;

PROPERTY LEF58_DIFFIMPLANTMUSTHAVE
        "DIFFIMPLANTMUSTHAVE
          ; " ;

PROPERTY LEF58_DIFFUSIONJOGMUSTHAVE
     "DIFFUSIONJOGMUSTHAVE
          ; " ;

PROPERTY LEF58_SPACING
     "SPACING spacing LENGTH length
        ; " ;

PROPERTY LEF58_SPACING
     "SPACING spacing PRL prl
     ; " ;
```

`CUTPGRAIL` *`metalLayer`*

Specifies that the shapes on this layer should cut the overlapping PG rails in *`metalLayer`*. These layer shapes should exist only in some special standard cells as `MACRO OBS`.

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`DIFFUSION`

Defines a diffusion layer.

The following property statements can be defined on this layer:

```
PROPERTY LEF58_COREEDGELENGTH
     "COREEDGELENGTH minLength [MAXLENGTH maxLength]
          [WIDTH width]
          [CONCAVECORNER | CONVEXCORNER | MIXEDCORNER]
          [EXTENDTOALIGN width spacing [EXTENDONLY |
             ALIGNONLY]]
          [FILLERONLY]
          [EXCEPTLAYER {exceptLayerName}...]
          ; " ;

PROPERTY LEF58_MAXLENGTH
     "MAXLENGTH maxLength [HORIZONTAL|VERTICAL]
           [GAPMERGED gap][EXCEPTWIDTH width]
           ; " ;

PROPERTY LEF58_MINENCLOSEDAREA
     "MINENCLOSEDAREA area
     ; " ;

PROPERTY LEF58_SPACING
     "SPACING minSpacing
     ; " ;

PROPERTY LEF58_WIDTHLENGTHRATIO
      "WIDTHLENGTHRATIO ratio WIDTH width; " ;
```

`MASKSPLIT`

Specifies a mask split layer to include the mask split region created in the design on this layer. This region should zigzag around blocks, and specific standard cells with shapes on this layer must be placed under this region.

The following properties can be applied on this `MASKSPLIT` layer:

```
PROPERTY LEF58_BOUNDINGBOX
     "BOUNDINGBOX layerName
     ; " ;

PROPERTY LEF58_MINSTEPTABLE
     "MINSTEPTABLE {yLength xLength}...
     ; " ;

PROPERTY LEF58_YMAXLENGTH
     "YMAXLENGTH maxLength
     ; " ;
```


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`MEOL`

Specifies that the masterslice layer is a middle-end-of-line (MEOL) layer.

The following property statement can be defined on this layer:

```
PROPERTY LEF58_MAXLENGTH
     "MAXLENGTH maxLength
       ; " ;
```

where

`MAXLENGTH maxLength`

Specifies that the maximum length of any shape on an `MEOL` layer must be less than or equal to *maxLength*. Typically, some standard cells would have shapes on an `MEOL` layer. If placement abuts too many such cells, it could violate this rule. Placement needs to avoid such violations by honoring the rule.

`NWELL`

Indicates that the layer is an `NWELL` layer.

The following property statements can be defined on this layer:

```
PROPERTY LEF58_ENCLOSURE
     "ENCLOSURE overhang LAYER secondLayerName;..." ;
```

```
PROPERTY LEF58_SPACING
     "SPACING minSpacing [HORIZONTAL|VERTICAL]
        [SAMENET | LAYER secondLayerName]
        [LENGTH length1 length2]
        ;..." ;
```

```
PROPERTY LEF58_WIDTH
     "WIDTH minWidth"WIDTH minWidth [WITHINLAYER
     {layerName}...]
     ;" ;
```

`PROBE`

Specifies a special layer for test probing purposes. At most, one such layer can be defined.

The following properties could be applied on this `PROBE` layer:

```
PROPERTY LEF58_PROBEAREA
     "PROBEAREA width length LAYER layerName
     ; " ;
```

```
PROPERTY LEF58_SPACING
     "SPACING spacing LAYER layerName
     ; " ;
```


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`PWELL`
: Indicates that the layer is a `PWELL` layer.

The following property statements can be defined on this layer:

```
PROPERTY LEF58_ENCLOSURE
     "ENCLOSURE overhang LAYER secondLayerName;..." ;
```

```
PROPERTY LEF58_SPACING
     "SPACING minSpacing [SAMENET | LAYER
               secondLayerName] ;..." ;
```

```
PROPERTY LEF58_WIDTH
     "WIDTH minWidth;" ;
```

`RCBLOCKAGE`
: Specifies that if a shape on this masterslice layer overlaps with a shape on the specified metal layer, *`metalLayer`*, the metal line-end that is covered by the shape in the masterslice layer would not be extended for RC calculation.

`REGION`
: Defines a special masterslice layer that is used to define the areas of a region on which a set of rules defined in the metal, cut, and/or trim metal layers with the `REGION` property would be applied.

The following property statements can be defined on this layer:

```
PROPERTY LEF58_ROWBASED
     "ROWBASED
     ;";
```

where

`ROWBASED`

Specifies that certain cell rows would be in this `REGION` layer, and the corresponding rules should be applied on them. This is typically used in mixed/hybrid row designs in which certain cell height row areas may be added in this `REGION` layer. The actual cell rows would be identified by the site name with a separate command.

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`TRIMDIFFUSION` Specifies a trim layer for a diffusion layer.

The following properties could be applied on this `TRIMDIFFUSION` layer.

```
PROPERTY LEF58_DIFFUSIONJOGMUSTHAVE
     "DIFFUSIONJOGMUSTHAVE
     ; " ;

PROPERTY LEF58_MAXLENGTH
     "MAXLENGTH maxLength {HORIZONTAL|VERTICAL}
            [GAPMERGED gap][EXCEPTWIDTH width]
     ; " ;

PROPERTY LEF58_MINLENGTH
     "MINLENGTH minLength {HORIZONTAL|VERTICAL}
            [EXACTWIDTH width]
     ; " ;

PROPERTY LEF58_SPACING
     "SPACING spacing
     ; " ;

PROPERTY LEF58_RECTONLY
     "RECTONLY
     ; " ;
```

where

`RECTONLY`

Specifies that shapes must be rectangular on this layer.

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`TRIMMETAL`

Defines a trim metal layer. This layer type is only used along with metal layers manufactured with self-aligned double patterning (SADP) technology. The `TRIMMETAL` layer has the shapes for the SADP mask used to “trim” or “cut” or “block” the self-aligned metal lines created during the first mask step of SADP processing. These shapes could be predefined in macros/cells or added at the line-end of a wire during routing. There are additional rules in cut and metal layers to define constraints to those shapes on a `TRIMMETAL` layer.

The `TRIMMETAL` layer can have the following property to indicate which metal layer or which mask of the metal layer can be trimmed by the `TRIMMETAL` shapes. The metal layer should have the `MASK` construct with value of `2` or larger to indicate that it is SADP (or DPT) layer. The `TRIMMETAL` layer could also have the `MASK` construct to indicate number of masks on that layer.

```
PROPERTY LEF58_TRIMMEDMETAL
     "TRIMMEDMETAL metalLayer [MASK maskNum [INCLUDEVIRTUAL]
       [VIRTUAL virtualMaskNum]]
       ; " ;
```


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In addition, the following property statements can be defined on this layer:

```
PROPERTY LEF58_ADJACENTTRIMS
     "ADJACENTTRIMS trimNum SPACING spacing [WITHIN within]
     [SAMEMASK | MASK maskNum]
       ; " ;

PROPERTY LEF58_AREA
     "AREA minArea
       ; " ;

PROPERTY LEF58_FORBIDDENARRAY
     "FORBIDDENARRAY numRows numCols XSPACING xLow xHigh
     YSPACING yLow yHigh
     [SAMEMASK]
     ; " ;

PROPERTY LEF58_MAXLENGTH
     "MAXLENGTH maxLength [WIDTH width]
     ; … " ;

PROPERTY LEF58_SPACING
     "SPACING minSpacing
       ; " ;

PROPERTY LEF58_SPACING
     "SPACING spacing LAYER cutLayer [CUTCLASS className]
         [MASK maskNum] [TRIMSPACING trimSpacing [DIFFMASK]]
         ; " ;...

PROPERTY LEF58_SPACING
     "SPACING spacing [MERGEDTRIM] [length]]
         [PRLSPACING prlSpacing1 prlSpacing2]
         [ENDTOEND endToEndSpacing [PRL prl]
          [EXACTALIGNED exactAlignedSpacing [MIDDLEWIREONLY]
              [EXCEPTDIFFMASKALIGNED]]
          [EDGEALIGNED edgeAlignedSpacing]
          [EXCEPTMIDMETALWIDTH width]]
         [SAMEMASK] [MASK maskNum [VIRTUAL]]
         ; " ;...
```


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```
PROPERTY LEF58_TRIMSHAPE
     "TRIMSHAPE [METALMASK metalMaskNum] EXTENSIONMODEL
        {ADJACENTTRACK | MIDTRACK | EXTENSION extension
             [METALEDGE]}
        EXACTWIDTH exactWidth
        [EXACTVIRTUALWIDTH exactVirtualWidth]
        [MAXOVERLAPWIDTH maxOverlapWidth]
        [GAPMERGED trimGapLength]
        [MAXLENGTH maxLength
             [ROWBOUNDARYMAXLENGTH rowBoundaryMaxLength]
             [GAPMERGEDFORLENGTH trimGapLength]
             [MASK maskNum]]
        [EXCEPTSPACING spacing]
        [EXCEPTWIDTH width [EXACTWIDTH]
             [EXCEPTVIRTUALWIDTH virtualWidth]]
         [USEMETALMASK]
         ; " ;
```

Typically, shapes on a trim metal layer could be defined in macros/cells only. There are additional rules in cut and metal layers to define constraints for the shapes on a `TRIMMETAL` layer.

`TRIMPOLY` Defines a TPO layer for self-aligned double patterning (SADP) technology. Only cells would have shapes on that layer. The following property statements define the constraints that placement needs to honor:

```
PROPERTY LEF58_AREA
     "AREA minArea
       ; " ;
```

```
PROPERTY LEF58_SPACING
     "SPACING spacing [SKIPPROTRUDEDFILLER]
       ; " ;
```

```
PROPERTY LEF58_COREEDGELENGTH
     "COREEDGELENGTH minLength
       ; " ;
```

```
PROPERTY LEF58_NOBOUNDARYJOG
     "NOBOUNDARYJOG
       ; " ;
```

```
PROPERTY LEF58_WIDTH
     "WIDTH minWidth  [ZEROPRL]
     ; " ;
```

`TS` Specifies a through-silicon layer between the back-side and front-side layers.

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`VIRTUAL`

Specifies that it is a virtual trim layer. No trims would be displayed or output. It is used to achieve a fully filled track requirement with line-end gaps equal to the trim width.

This means that:

(1) `EXCEPTSPACING` should not be defined in `TRIMSHAPE`.

(2) Trim spacing could still be defined and would be translated to the spacing requirement between two line-end gaps.

This virtual trim layer could be defined to trim one of the metal masks virtually while a real trim layer could be defined to trim the other metal mask. Two virtual trim layers could also be defined to trim different metal masks with different critiques virtually.

`WELLDISTANCE [BOUNDARYTAP]`

Specifies that this masterslice layer is a special layer for defining the well tap distance. When a well tap cell or core boundary overlaps with the layer, its neighboring well tap cell and the well tap cell overlapping the layer should use the defined tap distance on this layer. `BOUNDARYTAP` indicates a region that requires a special way for inserting the well tap.

The following property can be defined on this layer:

```
PROPERTY LEF58_TAPDISTANCE
     "TAPDISTANCE {distance | { HALFROW rowNum
             [LEFT|RIGHT] distance}...}
          TAPTYPE typeName
          ; " ;
```

Type Rule Examples

- The following example indicates that macro `A` defines a region of (0,0) to (100, 100) with respect to the placement of that macro, such that the boundary of the above die does not overlap:

```
LAYER TOPDIE
    TYPE MASTERSLICE
    PROPERTY LEF58_TYPE "TYPE ABOVEDIEEDGE ;" ;
END TOPDIE

MACRO A
...
   OBS
```


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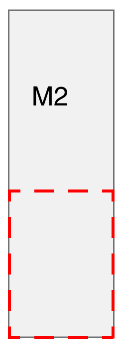
```
        LAYER TOPDIE
        RECT 0.000 0.000 100.000 100.000 ;
     END
  ...
  END A
```

- The following example depicts a masterslice layer that defines an `RCBLOCKAGE` shape:

```
TYPE MASTERSLICE
PROPERTY LEF58_TYPE "
    TYPE RCBLOCKAGE M2 ; " ;
```

Figure 4-1 Illustration of RCBLOCKAGE Type Rule



Abut Filler Rule for Abut Logic Layers

You can create an abut filler rule for abut logic layer of type `ABUTLOGIC` by using the following property definition:

```
PROPERTY LEF58_ABUTFILLER
     "ABUTFILLER distance LAYER abutFillerLayerName
          [EXCEPTCELL otherAbutlogicLayerName
             EXACTSPACING spacing PRL {prl}...]
          [SAMEIMPLANTLAYER]
     ; " ;
```

Where:

`ABUTFILLER` *`distance`* `LAYER` *`abutFillerLayerName`*

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Specifies that standard cells with shapes on this layer must be abutted on both sides with filler cells with shapes in *abutFillerLayerName*, which must be a previously defined layer with type ABUTFILLER, on the same row. The standard cells with shapes on this layer, or any other ABUTLOGIC layers, could also be abutted together. Then, the leftmost and rightmost edges should fulfill this filler abutment condition. In addition, the filler cells with shapes in *abutFillerLayerName* must be inserted within *distance* of the standard cells with shapes in this layer. Note that the shapes in this layer and in *abutFillerLayerName* could overlap when the cells are abutted.

Type: Float, specified in microns

EXCEPTCELL *otherAbutlogicLayerName* EXACTSPACING *spacing* PRL {*prl*}...

Specifies an exemption when a standard cell with shapes on this layer has a neighboring cell with shapes on *otherAbutlogicLayerName*, which has a horizontal spacing exactly equal to *spacing* and a vertical PRL exactly equal to one of the given *prl* values. Here, *otherAbutlogicLayerName* is another masterslice layer with the property type of ABUTLOGIC. Then, a certain portion of this cell with shapes on this layer does not need to fulfill the abutted filler condition.

Type: Float, specified in microns

SAMEIMPLANTLAYER

Specifies that the abutted filler cells with shapes in *abutFillerLayerName* must have identical implant layer shapes to the standard cells with shapes on this layer.

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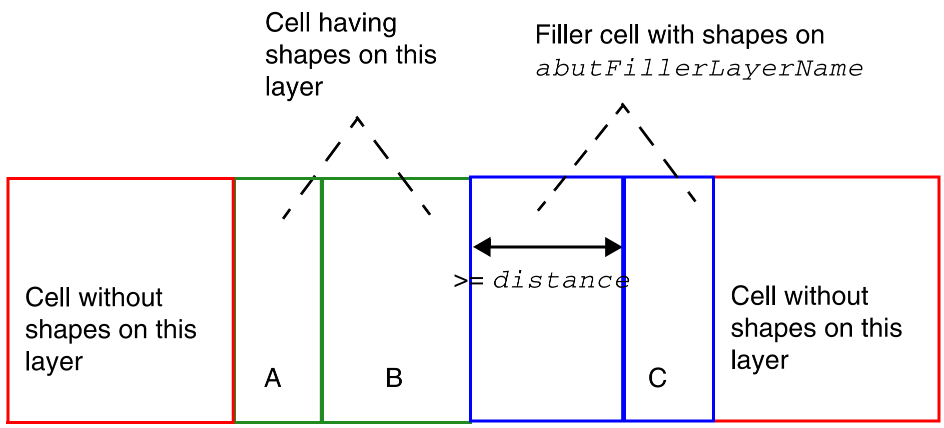
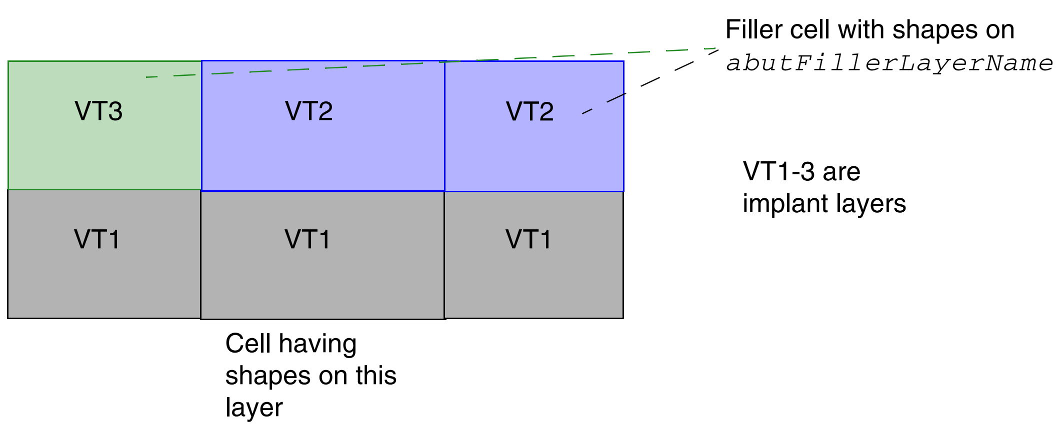
Figure 4-2 Illustration of Abut Filler Rule



There are two problems. The left edge of cell A is not abutted with the filler cell with shapes on `abutFillerLayerName`. Cell C is not within *distance* of cell B or any cells with shapes on this layer.

Illustration of `ABUTFILLER` *`distance`* `LAYER` *`abutFillerLayerName`*

Figure 4-3 Illustration of Abut Filler Rule with SAMEIMPLANTLAYER



Violation, implant layer shapes do not match on the left.
OK if `SAMEIMPLANTLAYER` is omitted.

Illustration of `ABUTFILLER` *`distance`* `LAYER` *`abutFillerLayerName`*
`SAMEIMPLANTLAYER ;`

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Adjacent Trim Rule for Trim Metal Layers

You can create an adjacent trim rule for trim metal layers by using the following property definition:

```
PROPERTY LEF58_ADJACENTTRIMS
     "ADJACENTTRIMS trimNum SPACING spacing [WITHIN within]
       [SAMEMASK | MASK maskNum]
       ; " ;
```

Where:

`ADJACENTTRIMS` *`trimNum`* `SPACING` *`spacing`*

Specifies that it is a violation if two adjacent trim tracks have greater than or equal to *`trimNum`* trims and each of the trims has a spacing less than *`spacing`* to another trim on the adjacent trim track. This means that the trims would be in a zig-zag pattern on two adjacent trim tracks. Adjacent trim tracks mean the adjacent metal tracks, which the trim would cut. For example, if the trims cut only `MASK 2` metal wires, adjacent trim tracks would be adjacent `MASK 2` metal tracks. If the trims cut all the metal wires, adjacent trim tracks would be adjacent metal tracks of any masks.

Type: Float, specified in microns

`MASK` *`maskNum`* Specifies that the trim rule is applied only to trims with the specified mask. *`maskNum`* must be a positive integer, and most applications support only the values `1`, `2`, or `3`.

Type: Integer

`SAMEMASK` Specifies that the rule is applied only on same-mask trims.

`WITHIN` *`within`* Specifies that only adjacent track trims with vertical spacing between 0 and *`within`* are considered for this rule. *`spacing`* is measured horizontally.

Type: Float, specified in microns

Adjacent Trim Rule Examples

- The following example illustrates the adjacent trim rule for trim metal shapes:

```
LAYER TRIM1
    TYPE MASTERSLICE ;
    MASK 2 ;
```


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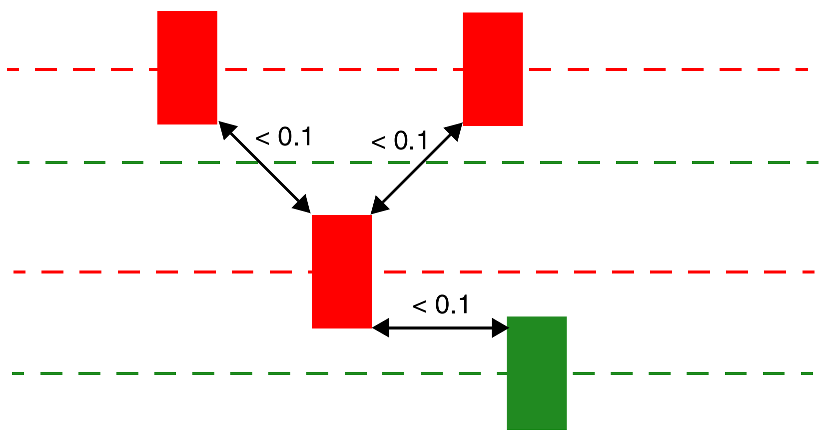
```
        PROPERTY LEF58_TYPE "TYPE TRIMMETAL ; " ;
        ....
        PROPERTY LEF58_TRIMSHAPE "TRIMSHAPE ... USEMETALMASK ; " ;
        PROPERTY LEF58_ ADJACENTTRIMS "
            ADJACENTTRIMS 3 SPACING 0.1 SAMEMASK ; " ;
    END TRIM1
```

Figure 4-4 Example of Adjacent Trim Rule



a) Violation, with `SAMEMASK`, the adjacent trim tracks would be adjacent same-mask metal tracks because `USEMETALTRACK` is used in `TRIMSHAPE`. Without `SAMEMASK`, the adjacent trim tracks would be adjacent metal tracks (alternated red and green tracks). As there is only one red trim and one green trim on the adjacent tracks, it is not a violation.

- The following example illustrates the adjacent trim rule for trim metal shapes with `WITHIN`:

```
    LAYER TRIM1
        TYPE MASTERSLICE ;
        MASK 2 ;
        PROPERTY LEF58_TYPE "TYPE TRIMMETAL ; " ;
        ....
        PROPERTY LEF58_TRIMSHAPE "TRIMSHAPE ... USEMETALMASK ; " ;
        PROPERTY LEF58_ADJACENTTRIMS "
            ADJACENTTRIMS 3 SPACING 0.1 WITHIN 0.05 SAMEMASK ; " ;
    END TRIM1
```


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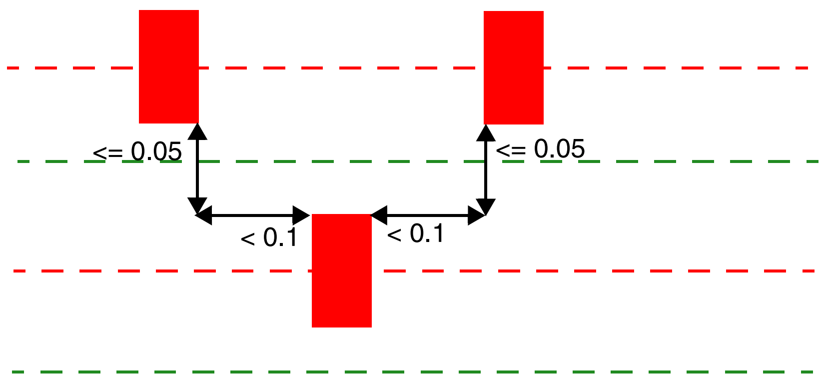
Figure 4-5 Example of Adjacent Trim Rule with WITHIN



a) Violation, with `WITHIN 0.05`, the adjacent track trims must be `<= 0.05` vertically and `< 0.1` horizontally to be recognized as neighbors. As there are 3 such trims in total, it is a violation.

Area Rule for a Trim Metal Layer

You can create an area rule for the masterslice `TRIMMETAL` layer by using the following property definition:

```
PROPERTY LEF58_AREA
        "AREA minArea
        ; " ;
```

Where:

`AREA` *`minArea`*	Specifies the minimum area on the masterslice `TRIMMETAL` layer. *Type:* Float, specified in microns

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Boundary Spacing Rule for a CPODE Layer

You can create a boundary spacing rule for the masterslice `CPODE` layer by using the following property definition:

```
PROPERTY LEF58_BOUNDARYSPACING
        "BOUNDARYSPACING spacing1 spacing2 LAYER layerName
        ; " ;
```

Where:

`BOUNDARYSPACING` *`spacing1 spacing2`* `LAYER` *`layerName`*

Specifies that if two CPODE shapes overlap or touch the two opposite sides of the boundaries of an area in *`layerName`* and both CPODE shapes extend vertically to the cell containing the area, any additional CPODE shape overlapping the area between them must be more than *`spacing1`* from one of the boundary CPODE shape or more than *`spacing2`* from the other boundary CPODE shape. A value of `0` for either *`spacing1`* or *`spacing2`* means that particular spacing is not checked. If the neighbor CPODE shape is within the other non-zero value spacing from any boundary CPODE shape, it is a violation. *`layerName`* must be a masterslice layer. The shapes on that layer must be defined in cells, slightly outside of the cell bounding box.

Type: Float, specified in microns

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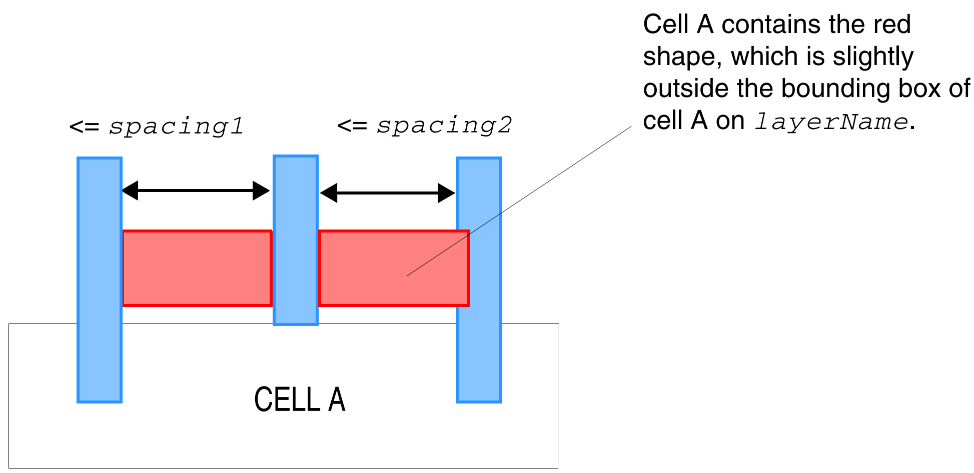
Figure 4-6 Illustration of Boundary Spacing Rule



It is a violation because the left and right CPODE shapes overlapping/touching the boundaries of the red shape on *layerName* and there is a middle neighbor CPODE shape, which is `<=` *spacing1* to one of the boundary CPODE shapes and `<=` *spacing2* to the other boundary CPODE shape.

If *spacing2* is `0`, it is still a violation because the actual left spacing is `<=` *spacing1* (or the right spacing is `<=` *spacing1*).

Illustration of `BOUNDARYSPACING` *spacing1 spacing2*
`LAYER` *layerName*

Bounding Box Rule for a Mask Split Layer

You can create a bounding box rule for the masterslice `MASKSPLIT` layer by using the following property definition:

```
PROPERTY LEF58_BOUNDINGBOX
        "BOUNDINGBOX layerName
        ; " ;
```

Where:

`BOUNDINGBOX` *layerName*

Specifies a rectangular bounding box on layer *layerName*, which should be a masterslice layer, to enclose the mask split region on this layer.

Type: String

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Core Edge Length Rule with Diffusion

A core edge length rule can be used to indicate that the edge length of a diffusion shape of a standard cell must be greater than or equal to the specified minimum length.

You can create a core edge length rule by using the following property definition:

```
PROPERTY LEF58_COREEDGELENGTH
     "COREEDGELENGTH minLength [MAXLENGTH maxLength]
          [WIDTH width]
          [CONCAVECORNER | CONVEXCORNER | MIXEDCORNER]
          [EXTENDTOALIGN width spacing [EXTENDONLY | ALIGNONLY]]
          [FILLERONLY]
          [EXCEPTLAYER {exceptLayerName}...]
          ; " ;
```

Where:

`[CONCAVECORNER | CONVEXCORNER | MIXEDCORNER]`

Specifies that the edge length is checked if the edge is between 2 concave corners in `CONCAVECORNER`, or if the edge is between two convex corners in `CONVEXCORNER`, or if the edge is between a concave and a convex corners in `MIXEDCORNER`. See Example of Core Edge Length Rule with CONVEXCORNER and MIXEDCORNER on page 554

`COREEDGELENGTH` *`minLength`*

Specifies that the edge length of the diffusion shape of a CORE (standard) cell, in the direction of the cell row, must be greater than or equal to the *`minLength`*.
Type: Float, specified in microns

`EXCEPTLAYER {`*`exceptLayerName`*`}...`

Specifies that the diffusion shapes overlapping with the shapes in *`exceptLayerName`*, which should be a masterslice layer with the `ABUTFILLER` type property, should not be merged with other diffusion shapes that are not overlapping with the shapes in *`exceptLayerName`*, on which the core edge length rule is checked. Typically, only standard cells contain OBS shapes in *`exceptLayerName`*.

Type: String

`EXTENDTOALIGN` *`width spacing`* `[EXTENDONLY | ALIGNONLY]`

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Specifies that if the diffusion shapes on the top and bottom edges within a cell row have width exactly equal to *`width`*, the shapes need to be extended and aligned. In addition, any gap on the same edge that is less than or equal to *`spacing`* should be filled and merged for length consideration. When `MAXLENGTH` is also defined, this alignment and extension operation is only considered for the maximum length check. In other words, *`minLength`* remains to be checked against the original diffusion shapes.
`EXTENDONLY` or `ALIGNONLY` specifies that only the extension or alignment operation is performed, respectively.
Type: Float, specified in microns

`FILLERONLY` Specifies a larger minimum core edge length value, which would be considered only as a soft rule in filler insertion in placement. The defined *`minLength`* should be larger than another `COREEDGELENGTH` without `FILLERONLY`.

Type: Float, specified in microns

`MAXLENGTH` *`maxLength`*

Specifies that the edge length of the diffusion shape of a `CORE` (standard) cell, in the direction of the cell row, must be less than or equal to *`maxLength`*.

Type: Float, specified in microns

`WIDTH` *`width`* Specifies that the rule applies only on the diffusion shapes with width exactly equal to *`width`*. Rules without `WIDTH` would be applied to all diffusion shapes, independent of their width.
Type: Float, specified in microns

Core Edge Length Rule Examples

- The following example illustrates the core edge length rule:

```
PROPERTY LEF58_COREEDGELENGTH
   "COREEDGELENGTH 0.1 ;
```


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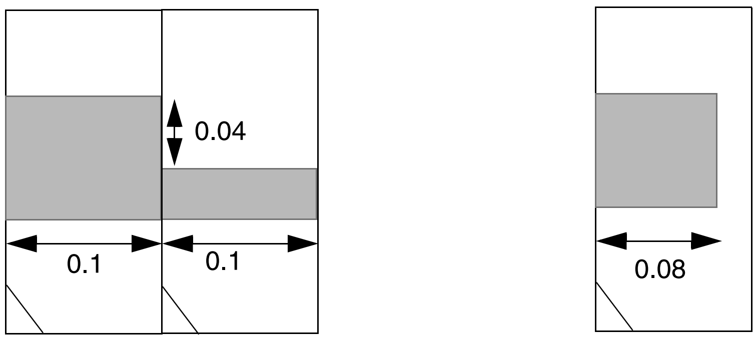
Figure 4-7 Example of Core Edge Length Rule



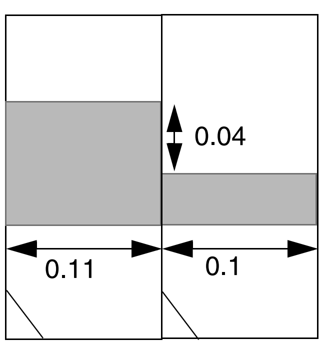
a) OK, the edge length is only checked in the direction of the cell row, which is horizontal.

b) Violation, any horizontal (direction of the cell row) length must be checked.

- The following example illustrates the core edge length rule with `CONVEXCORNER` and `MIXEDCORNER`:

```
PROPERTY LEF58_COREEDGELENGTH
     "COREEDGELENGTH 0.12 CONVEXCORNER ;
      COREEDGELENGTH 0.1 MIXEDCORNER ; " ;
```

Figure 4-8 Example of Core Edge Length Rule with CONVEXCORNER and MIXEDCORNER



a) Violation, the 0.1 edge fulfills the `MIXEDCORNER` requirement (>= 0.1), but the 0.11 edge fails the `CONVEXCORNER` requirement

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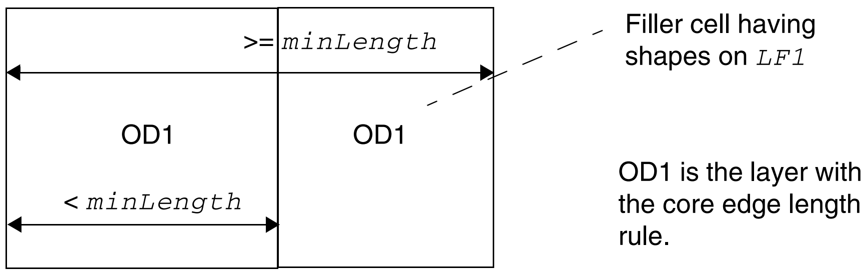
- The following example illustrates the core edge length rule with `EXCEPTLAYER`:

Figure 4-9 Example of Core Edge Length Rule with `EXCEPTLAYER`



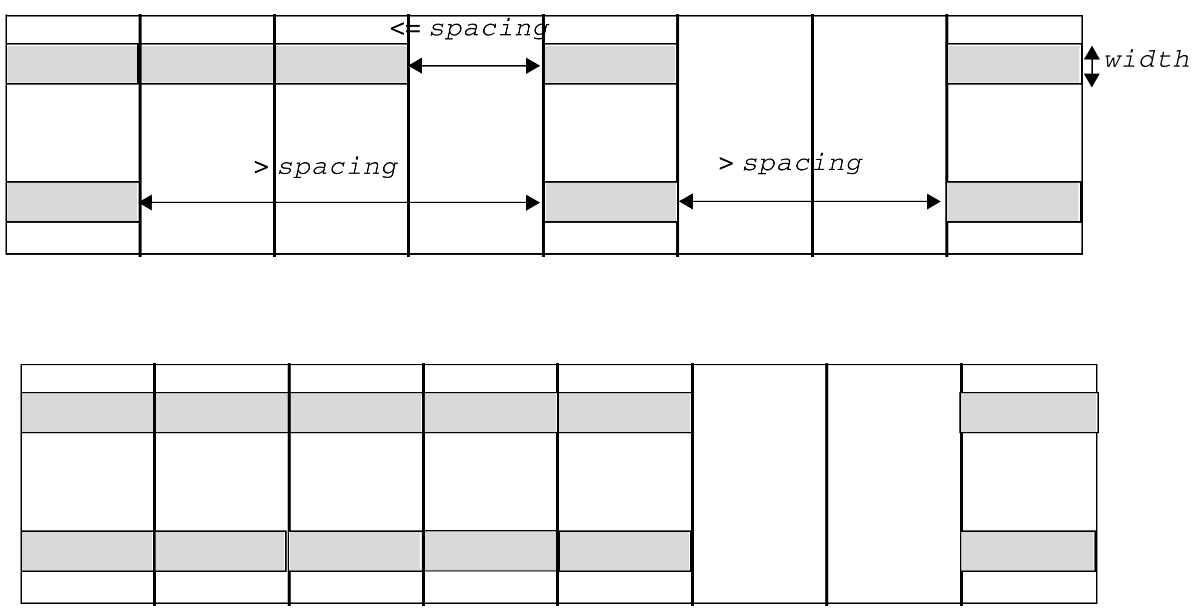
Violation, the right filler cell should be excluded and the core edge length checking should be applied on the left cell alone. OK if `EXCEPTLAYER` is omitted.

Illustration of `COREEDGELENGTH` *`minLength`*
`EXCEPTLAYER` *`LF1 LF2`*

- The following example illustrates the core edge length rule with `EXTENDTOALIGN`:

Figure 4-10 Example of Core Edge Length Rule with EXTENDTOALIGN

Original diffusion shapes



After extension and alignment for length calculation

Illustration of `COREEDGELENGTH ... EXTENDTOALIGN` *`width spacing`*

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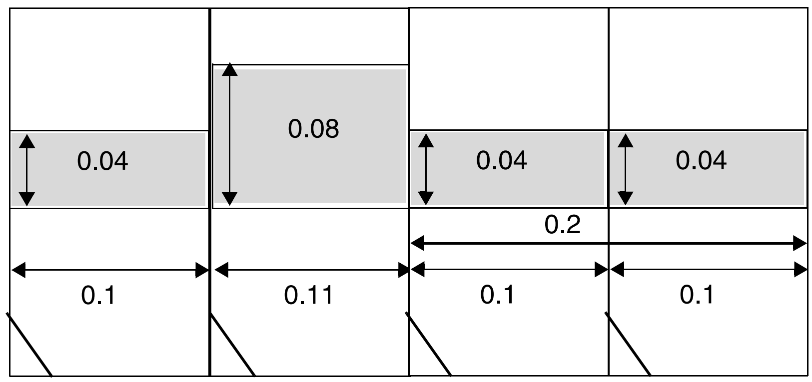
- The following example illustrates the core edge length rule with `WIDTH`:

```
PROPERTY LEF58_COREEDGELENGTH "
      COREEDGELENGTH 0.12 WIDTH 0.04 ;  " ;
```

Figure 4-11 Example of Core Edge Length Rule with WIDTH



a) Violation. All the shapes with width of `0.04` should be considered separately. The `0.11` length is irrelevant since the width is not `0.04`. The right sum length of `0.2` passes the rule. However, the left `0.1` length would be a violation.

CPODE Group Rule for CPODE Layers

You can create a CPODE group rule for `CPODE` layers by using the following property definition:

```
PROPERTY LEF58_CPODEGROUP
     "CPODEGROUP num PARALLEL parLength WITHIN within
         [INCLUDESIDESHIFT commonNum]
         ; " ;
```

Where:

`CPODEGROUP` *`num`* `PARALLEL` *`parLength`* `WITHIN` *`within`*

Specifies that at most *`num`* `CPODE` shapes with PRL greater than or equal to *`parLength`* and within *`within`* will form a `CPODE` group. If zero is given in *`num`*, it means that two or more `CPODE` shapes will form a group.

Type: Float, specified in microns

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`INCLUDESIDESHIFT` *`commonNum`*

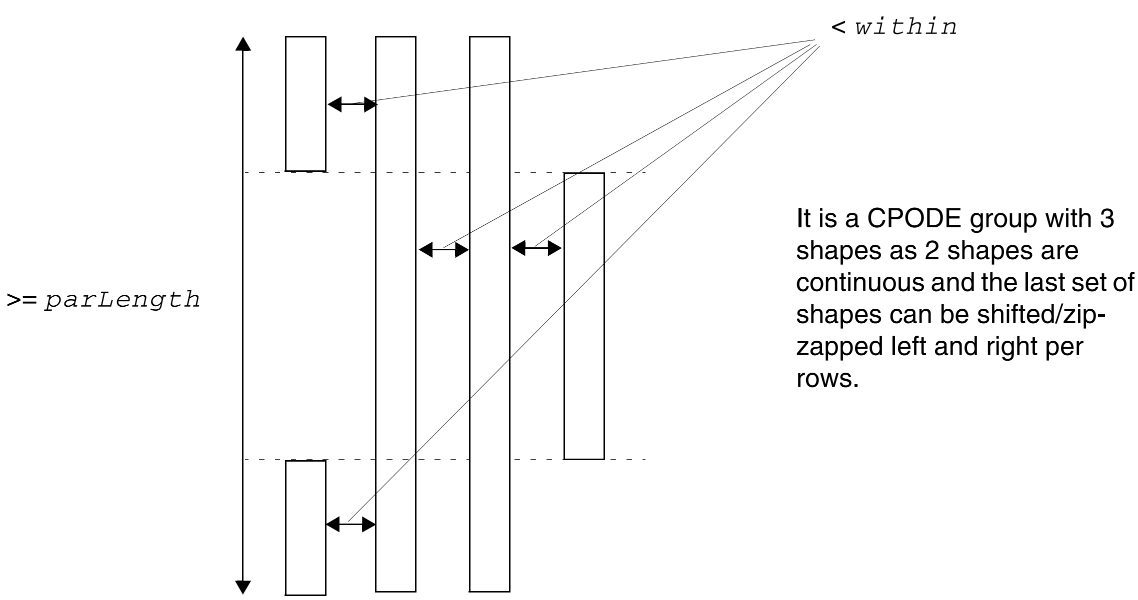
Specifies that *`commonNum`* CPODE shapes must be continuous, while the remaining neighbor shapes may be shifted or arranged side by side in rows. *`commonNum`* must be less than *`num`*. If *`num`* has a special value of zero, *`commonNum`* must be at least `1`.

CPODE Group Rule Example

- The following example illustrates a `CPODEGROUP` rule:

```
PROPERTY LEF58_CPODEGROUP
"CPODEGROUP 3 PARALLEL parLength
   WITHIN within INCLUDESIDESHIFT 2
   ; " ;
```

Figure 4-12 Example of CPODE Group Rule with INCLUDESIDESHIFT



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CPODE Group Spacing Rule for CPODE Layers

You can create a CPODE group spacing rule for `CPODE` layers by using the following property definition:

```
PROPERTY LEF58_CPODEGROUPSPACING
        "CPODEGROUPSPACING spacing CPODES num
          ; " ;
```

Where:

`CPODEGROUPSPACING` *`spacing`* `CPODES` *`num`*

Specifies that the spacing between two `CPODE` groups with at least one of them having greater than or equal to *`num`* shapes is *`spacing`*.

Type: Float, specified in microns

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Different Implant Must Have Rule for CPODE Layers

You can create a different implant must have rule for a `CPODE` layer by using the following property definition:

```
PROPERTY LEF58_DIFFIMPLANTMUSTHAVE
        "DIFFIMPLANTMUSTHAVE
          ; " ;
```

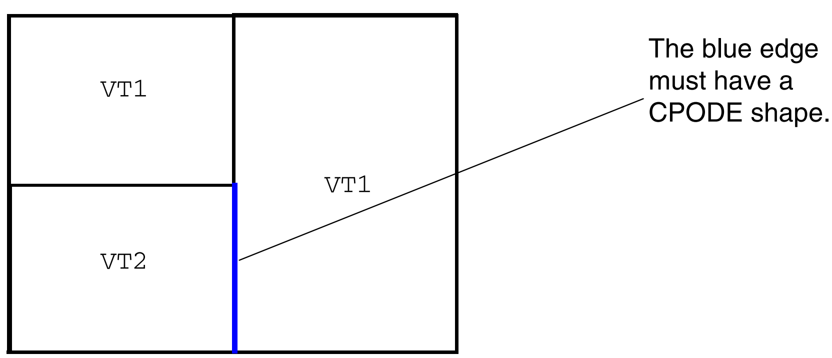
Where:

`DIFFIMPLANTMUSTHAVE`

Specifies that a `CPODE` shape must exist on the boundary between any two different implant shapes.

Different Implant Must Have Rule Example

Figure 4-13 Illustration of Different Implant Must Have Rule



A cell can have mixed implant shapes, like the left one. All different implant shape boundaries must be considered. The bottom implant shape transition edge must have a CPODE shape.

Illustrations of `DIFFIMPLANTMUSTHAVE`

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Diffusion Jog Must Have Rule for CPODE Layers

You can create a diffusion jog must have rule for a `CPODE` layer by using the following property definition:

```
PROPERTY LEF58_DIFFUSIONJOGMUSTHAVE
        "DIFFUSIONJOGMUSTHAVE
          ; " ;
```

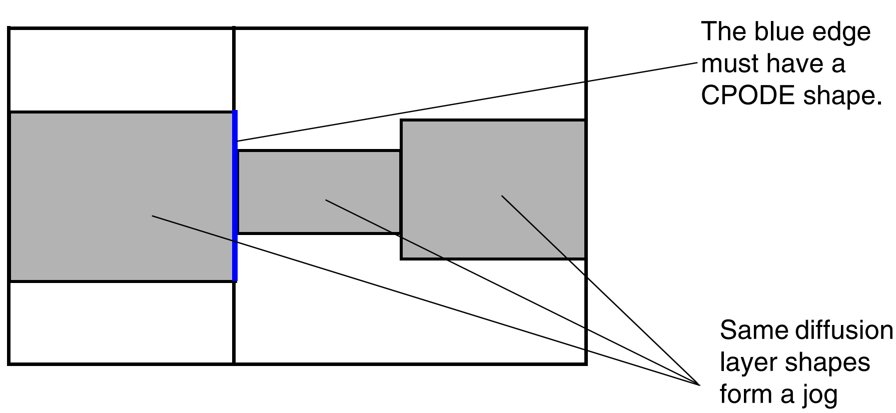
Where:

`DIFFUSIONJOGMUSTHAVE`

Specifies that a `CPODE` shape must exist on the boundary between any jog formed between two cells by shapes on the same diffusion layer. If the jog happens inside a cell, it need not be covered by a `CPODE` shape.

Diffusion Jog Must Have Rule Example

Figure 4-14 Illustration of Diffusion Jog Must Have Rule



When same diffusion layer shapes form a jog on different cells at the cell boundary, a CPODE shape must cover the jog. The right jog inside a cell does not necessarily need to be covered.

Illustration of `DIFFUSIONJOGMUSTHAVE`

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Diffusion Jog Must Have Rule for TRIMDIFFUSION Layers

You can create a diffusion jog must have rule for a `TRIMDIFFUSION` layer by using the following property definition:

```
PROPERTY LEF58_DIFFUSIONJOGMUSTHAVE
        "DIFFUSIONJOGMUSTHAVE
          ; " ;
```

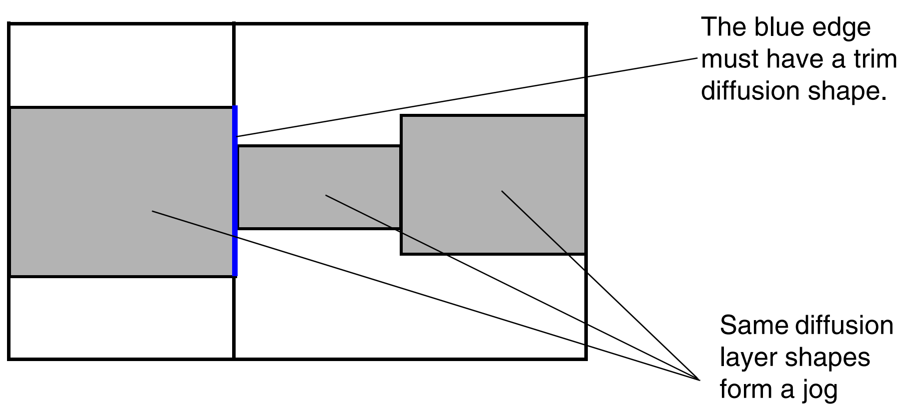
Where:

`DIFFUSIONJOGMUSTHAVE`

Specifies that a trim diffusion shape must cover a jog formed by shapes on the same diffusion layer if the jog occurs between two cells. If the jog is within a cell, it need not be covered by a trim diffusion shape.

Diffusion Jog Must Have Rule Example

Figure 4-15 Illustration of Diffusion Jog Must Have Rule



When same diffusion layer shapes form a jog on different cells at the cell boundary, a trim diffusion shape must cover the jog. The right jog inside a cell does not necessarily need to be covered.

Illustration of `DIFFUSIONJOGMUSTHAVE`

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Enclosure Rule for Well Layers

Enclosure rules can be used to specify that objects on the current well layer must be enclosed by at least overhang amount of objects in the specified second well layer on all the four sides.

You can create an enclosure rule by using the following property definition:

```
PROPERTY LEF58_ENCLOSURE
     "ENCLOSURE overhang LAYER secondLayerName;..." ;
```

Where:

`ENCLOSURE` *`overhang`* `LAYER` *`secondLayerName`*

Specifies that objects on the current well layer must be enclosed by at least *`overhang`* amount of objects in the *`secondLayerName`* well layer on all the four sides. The *`secondLayerName`* can be any well layer, including forward reference of more than one layer.

This syntax is only allowed on a well masterslice layer with property statement of `LEF58_TYPE "TYPE NWELL ; "` or `LEF58_TYPE "TYPE PWELL ; "`. This check is optional and is not performed if the enclosing layer does not exist around the well layer.
Type: Float, specified in microns

Enclosure Rule Example

- The following example indicates that an enclosure of 0.1 µm is required for objects in `PWELLA` layer by objects in `NWELLA` layer:

```
LAYER NWELLA
  TYPE MASTERSLICE
  PROPERTY LEF58_TYPE "TYPE NWELL ;" ;
END NWELLA
LAYER PWELLA
  TYPE MASTERSLICE
  PROPERTY LEF58_TYPE "TYPE PWELL ;" ;
  PROPERTY LEF58_ENCLOSURE "ENCLOSURE 0.1 LAYER NWELLA;" ;
END PWELLA
```


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Forbidden Array Rule for Trim Metal Layers

You can create a forbidden array rule for trim metal layers by using the following property definition:

```
PROPERTY LEF58_FORBIDDENARRAY
     "FORBIDDENARRAY numRows numCols XSPACING xLow xHigh
     YSPACING yLow yHigh
     [SAMEMASK]
     ; " ;
```

Where:

`FORBIDDENARRAY` *`numRows numCols`* `XSPACING` *`xLow xHigh`*
`YSPACING` *`yLow yHigh`*

Specifies that the given trim array of *`numRows`* `x` *`numCols`* is not allowed if all the trims are perfectly aligned in both directions. Each of the two nearest trims are greater than or equal to *`xLow`* and less than or equal to *`xHigh`* horizontally and greater than or equal to *`yLow`* and less than or equal to *`yHigh`* vertically. This means that it is not a violation if any extra trim is present inside the trim array. In addition, there is a built-in exemption. If an extra perfectly aligned trim exists on the next adjacent track or the next same-mask adjacent track in `SAMEMASK` to any of trims in the array, it is not a violation. However, one or more additional perfectly aligned trim to that extra trim would be a violation.

Type: Float, specified in microns

`SAMEMASK` Specifies that the rule applies only on the same-mask trims.

Forbidden Array Rule Examples

- The following example illustrates the forbidden array rule below for trim metal shapes:

```
PROPERTY LEF58_FORBIDDENARRAY "
    FORBIDDENARRAY 3 2 XSPACING 0.09 0.11
       YSPACING 0.05 0.07 ; " ;
```


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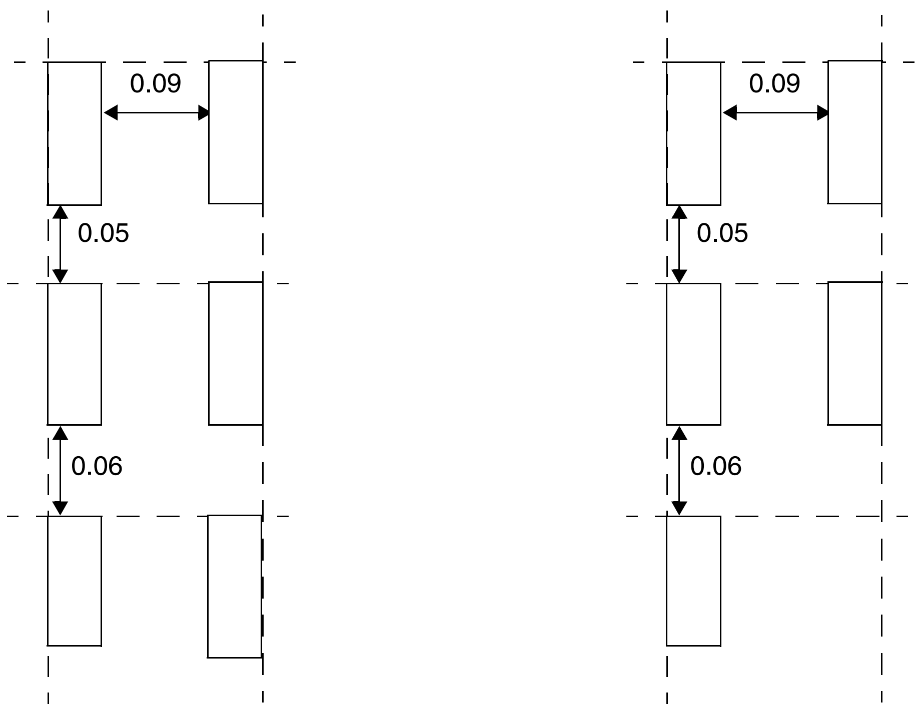
Figure 4-16 Example of Forbidden Array Rule



a) Violation, this is a 3 x 2 trim array with horizontal spacing of `0.09` (>= `0.09` and <= `0.11`) and vertical spacing of `0.05` and `0.06` (>= `0.05` and <= `0.07`).

b) OK, this is an incomplete 3 x 2 trim array.

- The following example illustrates the forbidden array rule below for trim metal shapes:

```
PROPERTY LEF58_FORBIDDENARRAY "
    FORBIDDENARRAY 2 2 XSPACING 0.15 0.20
       YSPACING 0.05 0.07 ; " ;
```


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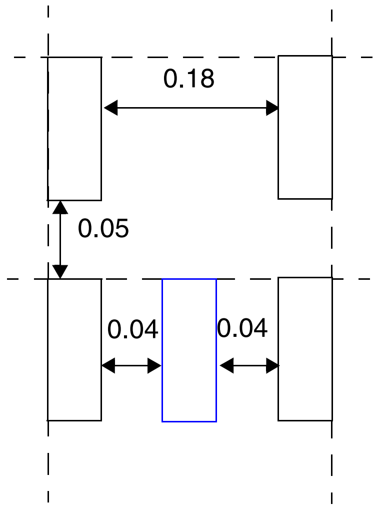
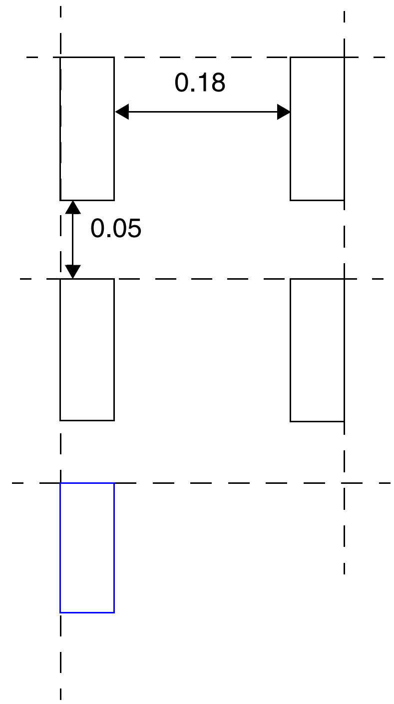
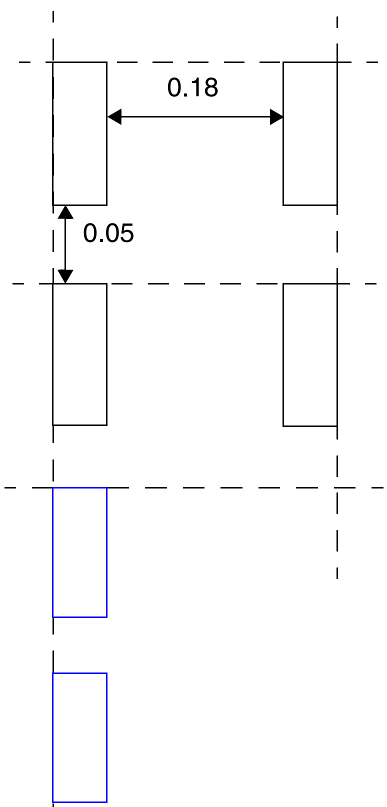
Figure 4-17 Example of Forbidden Array Rule



a) OK, the extra blue trim inside the 2x2 trim array would become OK.



b) OK, the extra perfectly aligned blue trim on the next adjacent track to any one of the trims in the array would be an exemption and would make it OK.



c) Violation, two extra perfectly aligned blue trims would not trigger the exemption and the original 2x2 trim array would be a violation.

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Max Length Rule for CPODE Layers

You can create a max length rule for `CPODE` layers by using the following property definition:

```
PROPERTY LEF58_MAXLENGTH
     "MAXLENGTH maxLength
        [HORIZONTAL|VERTICAL] [GAPMERGED gap]
        [ORTHOGONALLENGTH minLength]
        [EXCEPTWIDTH width MAXLENGTH widthMaxLength]
        ; " ;
```

Where:

`HORIZONTAL|VERTICAL`

Specifies that the maximum length rule is checked only in the given direction.

`EXCEPTWIDTH` *`width`* `MAXLENGTH` *`widthMaxLength`*

Specifies that if all of the filler cells forming the `CPODE` shape have width less than or equal to *`width`*, the maximum length must be less than or equal to *`widthMaxLength`* instead of *`maxLength`*.

Type: Float, specified in microns

`GAPMERGED` *`gap`*

Specifies that if two shapes are within *`gap`* in the given direction, they should be merged for this maximum length check.

`MAXLENGTH` *`maxLength`*

Specifies that the maximum length of any shape on this layer must be less than or equal to *`maxLength`*.

Type: Float, specified in microns

`ORTHOGONALLENGTH` *`minLength`*

Specifies that the rule applies only if the orthogonal length is greater than *`minLength`*. This construct must be specified along with `HORIZONTAL` or `VERTICAL`.

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Max Length Rule for Diffusion Layers

You can create a max length rule for a `DIFFUSION` layer by using the following property definition:

```
PROPERTY LEF58_MAXLENGTH
    "MAXLENGTH maxLength {HORIZONTAL|VERTICAL}
         [GAPMERGED gap][EXCEPTWIDTH width]
      ; " ;
```

Where:

`EXCEPTWIDTH` *`width`*

Specifies that the maximum length rule applies only if the width on the orthogonal direction of the given direction of `HORIZONTAL` or `VERTICAL` is exactly equal to *`width`*.

Type: Float, specified in microns

`GAPMERGED` *`gap`*

Specifies that if two shapes are less than or equal to *`gap`* on the orthogonal direction of the given direction of `HORIZONTAL` or `VERTICAL`, they should be merged for this maximum length check.

Type: Float, specified in microns

`MAXLENGTH` *`maxLength`* `{HORIZONTAL|VERTICAL}`

Specifies that the maximum length of any shape on this layer must be less than or equal to *`maxLength`* in the given direction of `HORIZONTAL` or `VERTICAL`.

Type: Float, specified in microns

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Max Length Rule for Trim Diffusion Layers

You can create a max length rule for a TRIMDIFFUSION layer by using the following property definition:

```
PROPERTY LEF58_MAXLENGTH
     "MAXLENGTH maxLength {HORIZONTAL|VERTICAL}
            [GAPMERGED gap][EXCEPTWIDTH width]
         ; " ;
```

Where:

EXCEPTWIDTH *width*

Specifies that the maximum length rule applies only if the width on the orthogonal direction of the given direction of HORIZONTAL or VERTICAL is exactly equal to *width*.

Type: Float, specified in microns

GAPMERGED *gap*

Specifies that if two shapes are less than or equal to *gap* on the orthogonal direction of the given direction of HORIZONTAL or VERTICAL, they should be merged for this maximum length check.

Type: Float, specified in microns

MAXLENGTH *maxLength* {HORIZONTAL|VERTICAL}

Specifies the maximum length when shapes are abutted horizontally or vertically to be less than or equal to *maxLength* when HORIZONTAL or VERTICAL is specified.

Type: Float, specified in microns

Max Length Rule for Trim Metal Layers

You can create a max length rule for trim metal layers by using the following property definition:

```
PROPERTY LEF58_MAXLENGTH
     "MAXLENGTH maxLength [WIDTH width]
     ; … " ;
```


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Where:

`MAXLENGTH` *`maxLength`*

Specifies that the maximum length of any shape on this trim metal layer must be less than or equal to *`maxLength`*. This rule is honored only during placement by considering the cell trim shape abutment.

Type: Float, specified in microns

`WIDTH` *`width`*

Specifies that the maximum length rule applies only if the width of a shape is less than *`width`*.

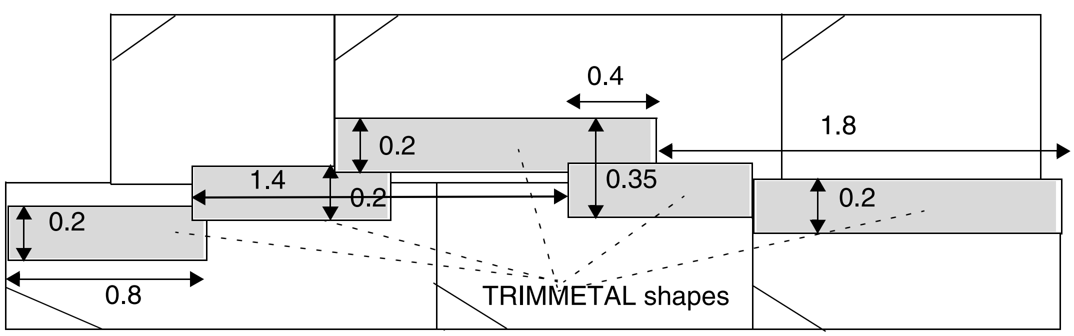
Type: Float, specified in microns

Max Length Rule Example

- The following example illustrates the max length rule for trim metal shapes:

```
PROPERTY LEF58_ MAXLENGTH "
     MAXLENGTH 2.0 WIDTH 0.3 ; " ;
```

Figure 4-18 Example of Max Length Rule



a) OK, `1.4` and `1.8` length `TRIMMETAL` shapes are not continuous and not summed due to the middle shape with width `0.35` (>= `0.3`) and they individually would not violate the length rule of `2.0`. The zig zag patterns of `0.8` and `1.4` length are also not summed up because there is no straight line going through all of the shapes.

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Min Enclosed Area Rule

You can create a minimum enclosed area rule on a diffusion layer by using the following property definition:

```
PROPERTY LEF58_MINENCLOSEDAREA
       "MINENCLOSEDAREA area
       ; " ;
```

Where:

`MINENCLOSEDAREA` *`area`*

Specifies the minimum area size of the hole to be greater than or equal to *`area`*.

Type: Float, specified in microns

Min Length Rule

You can create a minimum length rule for a trim diffusion layer by using the following property definition:

```
PROPERTY LEF58_MINLENGTH
       "MINLENGTH minLength {HORIZONTAL|VERTICAL}
           [EXACTWIDTH width]
       ; " ;
```

Where:

`EXACTWIDTH` *`width`*

Specifies that the minimum length rule applies only if the width on the orthogonal direction of the given direction of `HORIZONTAL` or `VERTICAL` is exactly equal to *`width`*.

Type: Float, specified in microns

`MINLENGTH` *`minLength`* `{HORIZONTAL|VERTICAL}`

Specifies that the minimum length of any shape on this layer must be greater than or equal to *`minLength`* in the given direction of `HORIZONTAL` or `VERTICAL`.

Type: Float, specified in microns

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Min Step Rule for a Mask Split Layer

You can create a min step rule for a Mask Split layer using the following property definition:

```
PROPERTY LEF58_MINSTEPTABLE
       "MINSTEPTABLE {yLength xLength}...
       ; " ;
```

Where:

`MINSTEPTABLE {yLength xLength}...`

Sets *xLength* as the minimum horizontal length of the region created in the design level on this layer on both sides adjacent to a vertical length greater than the maximum possible *yLength*. Both *yLength* and *xLength* must increase monotonically.

Type: Float, specified in microns

No Boundary Jog Rule

You can create a no boundary jog rule for a TRIMPOLY layer using the following property definition:

```
PROPERTY LEF58_NOBOUNDARYJOG
       "NOBOUNDARYJOG
       ; " ;
```

Where:

`NOBOUNDARYJOG` Specifies that the shapes on abutted cells cannot form a jog on the left or right boundaries of a cell.

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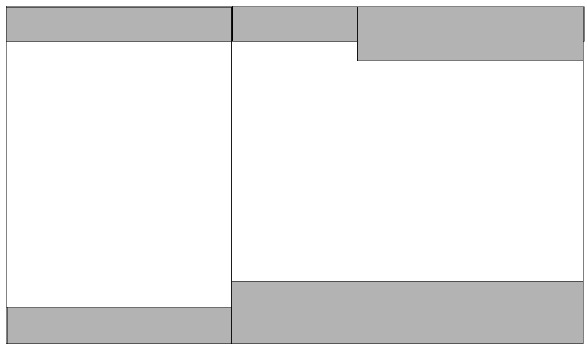
No Boundary Jog Rule Example

Figure 4-19 Illustration of No Boundary Jog Rule



The top jog inside a cell is fine. The bottom jog is a violation as the jog happens on the cell boundary.

Illustrations of `NOBOUNDARYJOG`

Probe Area Rule

You can create a probe area rule to define a probe area on a probe layer.

You can use the following property definition:

```
PROPERTY LEF58_PROBEAREA
       "PROBEAREA width length LAYER layerName
       ; " ;
```

Where:

`PROBEAREA` *`width length`* `LAYER` *`layerName`*

Specifies a probe area with dimension of *`width`* x *`length`*, which should be created on some special pins on *`layerName`*, which should be a routing layer. This construct must be defined on a `PROBE` layer.

Type: Float, specified in microns

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Region Rule

You can create a region rule to define area-based rules on a trim layer.

You can use the following property definition:

```
PROPERTY LEF58_REGION
       "REGION regionLayerName BASEDLAYER trimLayerName
       ; " ;
```

Where:

`REGION` *`regionLayerName`* `BASEDLAYER` *`trimLayerName`*

Specifies a set of region- or area-based rules on a trim layer *`trimLayerName`*. *`regionLayerName`* is a masterslice layer with type `REGION`. Users can create a blockage on that masterslice layer on a design to indicate the areas of the region. In addition, `MACRO` may also have `OBS` on that masterslice layer to indicate that the areas of its instance should be subjected to these region-based rules. All the rules specified on this trim layer with `REGION` are applied to the corresponding areas of *`regionLayerName`* on *`trimLayerName`*. The areas outside *`trimLayerName`* honor the set of rules on the given trim layer without type `REGION`. If a rule check crosses the boundaries of a region, it is subjected to the rules for both inside and outside the region, with the tighter rule values of the two sets dominating.

The set of rules defined in the layer with type `REGION` should be very small as compared to the layer without type `REGION`. The rules defined in this layer with type `REGION` should be a replacement of the corresponding rule (having the same set of the constructs of a LEF statement or a complete replacement on a table, like `SPACINGTABLE`) on the layer without type `REGION`. All the other rules defined on the layer without type `REGION` are also applicable on the layer with type `REGION`.

Type: String

Region Rule Example

The following example means that a trim metal completely inside region `R1` should have spacing of `0.4.` Otherwise, spacing of `0.6` should be applied. Any other rules on layer `TM1` (without `REGION`) would also be applied on region `R1`:

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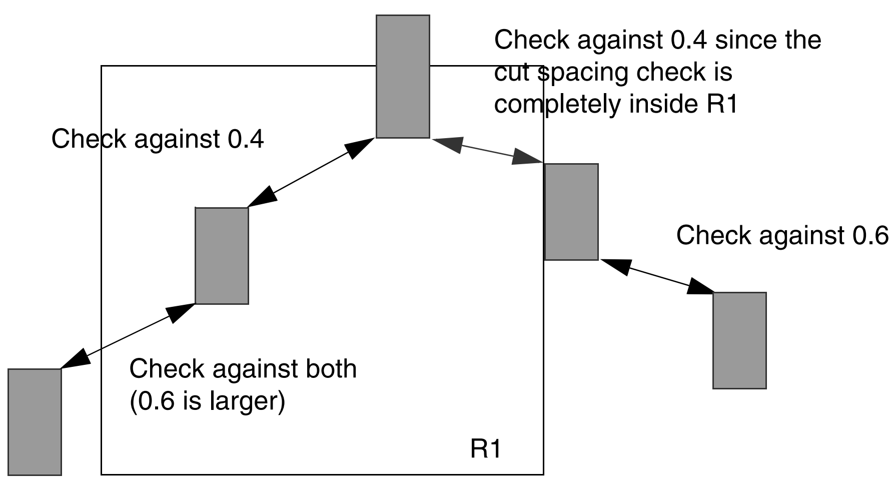
```
LAYER R1
    TYPE MASTERSLICE ;
    PROPERTY LEF58_TYPE “TYPE REGION ; ” ;
END R1
...
LAYER TM1
    TYPE MASTERSLICE ;
    PROPERTY LEF58_TYPE "TYPE TRIMMETAL ; " ;
    ...
    PROPERTY LEF58_SPACING "SPACING 0.6 ;" ;
    ...
END TM1

LAYER TM1R1
    TYPE MASTERSLICE ;
    PROPERTY LEF58_REGION "REGION R1 BASEDLAYER TM1 ; " ;
    ...
    PROPERTY LEF58_SPACING "SPACING 0.4 ;" ;
    ...
END TM1R1
```

Figure 4-20 Illustration of Region Rule



Spacing Rule for Above Die Edge, Below Die Edge, and Diffusion Layers

The following spacing rule can be defined on a masterslice layer for which `ABOVEDIEEDGE`, `BELOWDIEEDGE`, or `DIFFUSION` type rules have been defined:

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```
PROPERTY LEF58_SPACING
     "SPACING minSpacing
     ; " ;
```

Where:

`SPACING minSpacing`

Specifies a spacing between an OBS on the masterslice layer with type `ABOVEDIEEDGE`, `BELOWDIEEDGE`, or `DIFFUSION` to the die boundary of the above or below die to be *`minSpacing`*.
Type: Float, specified in microns

Spacing Rule for Abutting CPODE Shapes

A spacing rule can be defined for abutting `CPODE` shapes as follows:

```
PROPERTY LEF58_SPACING
     "SPACING spacing LENGTH length
     ; " ;
```

Where:

`SPACING spacing LENGTH length`

Specifies that if a CPODE shape has a vertical length less than *`length`*, form a window by extending *`spacing`* horizontally on both sides. If neighbor CPODE shapes abut with both the top and bottom window edges, it is a violation.

Type: Float, specified in microns

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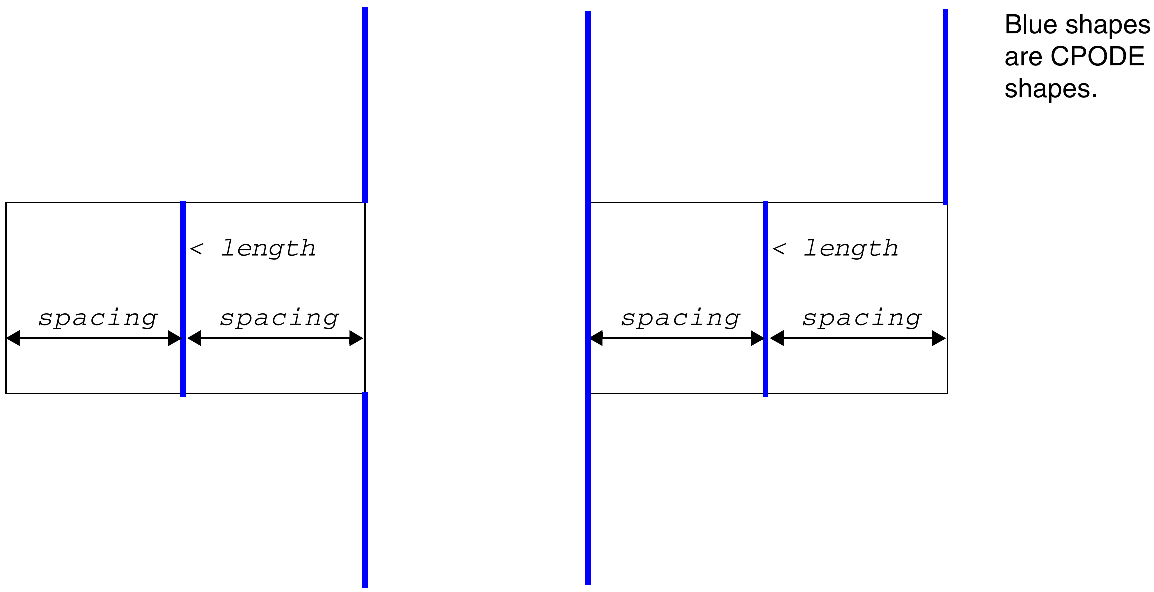
Example for Spacing Rule for Abutting CPODE Shapes

Figure 4-21 Illustration of Spacing Rule for Abutting CPODE Shapes



a) Violation, the neighbors on the same right side abut to both the top and bottom edges of the search window.

b) OK, the through CPODE shape on the left is irrelevant. The right shape abuts only to the top edge of the search window so it is OK.

Illustrations of `SPACING` *`spacing`* `LENGTH` *`length`*

Spacing Rule for Objects with PRL on a CPODE Layer

A spacing rule can be defined for objects on a `CPODE` layer as follows:

```
PROPERTY LEF58_SPACING
     "SPACING spacing PRL prl
     ; " ;
```

Where:

`SPACING` *`spacing`* `PRL` *`prl`*

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Specifies that the spacing between two objects with parallel run length greater than *prl* must be greater than or equal to *spacing*.

Type: Float, specified in microns

Spacing Rule for a Probe Layer

A spacing rule can be defined for objects on a PROBE layer as follows:

```
PROPERTY LEF58_SPACING
     "SPACING spacing LAYER layerName
     ; " ;
```

Where:

SPACING *spacing* LAYER *layerName*

Specifies an inter-layer spacing, *spacing*, between the probe areas to the shapes on the given layer, *layerName*, which must be either a routing or a cut layer.

Type: Float, specified in microns

Spacing Rule for a Trim Metal Layer

A TRIMMETAL spacing rule can be used to specify the spacing among trim metal shapes.

```
PROPERTY LEF58_SPACING
        "SPACING spacing [MERGEDTRIM [length]]
            [PRLSPACING prlSpacing1 prlSpacing2]
            [ENDTOEND endToEndSpacing [PRL prl]
               [EXACTALIGNED exactAlignedSpacing [MIDDLEWIREONLY]
                  [EXCEPTDIFFMASKALIGNED]]
               [EDGEALIGNED edgeAlignedSpacing]
               [EXCEPTMIDMETALWIDTH width]]
            [SAMEMASK] [MASK maskNum [VIRTUAL]]
            ; " ;...
```

Where:

ENDTOEND *endToEndSpacing* [PRL *prl*]

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Specifies the end-to-end spacing with parallel run length greater than *prl*, if specified, or 0 in the preferred routing direction on the metal layer that is trimmed.
Type: Float, specified in microns

EDGEALIGNED *edgeAlignedSpacing*

Specifies that the end-to-end spacing is *edgeAlignedSpacing* if the opposite long/side edges of the trim metal shapes are exactly aligned. That is the left/top edge of a trim metal shape is exactly aligned to the right/bottom edge of another trim metal. Otherwise, *endToEndSpacing* is applied.
Type: Float, specified in microns

EXACTALIGNED *exactAlignedSpacing*

Specifies that the end-to-end spacing is *exactAlignedSpacing* if the short/end side of the trim metal shapes are exactly aligned. Otherwise, *endToEndSpacing* is applied in Euclidean measurement independent of parallel run length if PRL is not defined. If PRL is also defined, *endToEndSpacing* is applied as MAXXY style in the orthogonal direction of the parallel run length.
Type: Float, specified in microns

EXCEPTDIFFMASKALIGNED

Specifies that there is no spacing requirement on two same-mask, exactly aligned trims if there is another exactly aligned, different-mask trim between them. In other words, it is legal to have perfectly aligned, consecutive same-mask and different-mask trims. This construct must be specified with SAMEMASK.

EXCEPTMIDMETALWIDTH *width*

Specifies that all the end-to-end spacing rules would be exempted if there is a metal of any mask with width greater than or equal to width on a layer that the trim metal trims.
Type: Float, specified in microns

MASK *maskNum*

Specifies that the trim spacing rule is applied only to a trim with the given mask to another trim with any mask. *maskNum* must be a positive integer, and most applications support only the values 1, 2, or 3.

Type: Integer

MERGEDTRIM [*length*]

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Specifies that the trim spacing rule applies only between a merged trim and another trim, which may be merged or non-merged. If *length* is defined, the rule applies only on merged trims with length greater than or equal to *length*.

MIDDLEWIREONLY Specifies that the spacing applies only if a metal shape on the layer on which the trim is defined overlaps with the gap or area between the perfectly aligned trims.

SAMEMASK Specifies that the spacing values are applied to same-mask objects. It is allowed to have just one spacing statement with SAMEMASK, which means that there is no constraint on different-mask shapes.
Type: Float, specified in microns

SPACING *spacing* [PRLSPACING *prlSpacing1 prlSpacing2*]

Specifies the minimum spacing to be *spacing*. If PRLSPACING is specified, *prlSpacing1* is applied with parallel run length equal to 0, *prlSpacing2* with parallel run length greater than 0 and *spacing* with parallel run length less than 0.
Type: Float, specified in microns

VIRTUAL Specifies the spacing on virtual *maskNum* trims. There is no spacing between real and virtual *maskNum* trims. This construct must be used only if INCLUDEVIRTUAL is specified within TRIMMEDMETAL.

Examples of Spacing Rule for a Trim Metal Layer

- The following example illustrates the spacing rule for TRIMMETAL layer:

```
# No MASK

PROPERTY LEF58_TRIMSHAPE "
   TRIMSHAPE EXTENSIONMODEL MIDTRACK
      EXACTWIDTH 0.2 MAXLENGTH 0.6 ; " ;

PROPERTY LEF58_SPACING "
   SPACING 0.5 PRLSPACING 0.3 0.4 ENDTOEND 0.8 ; " ;
```


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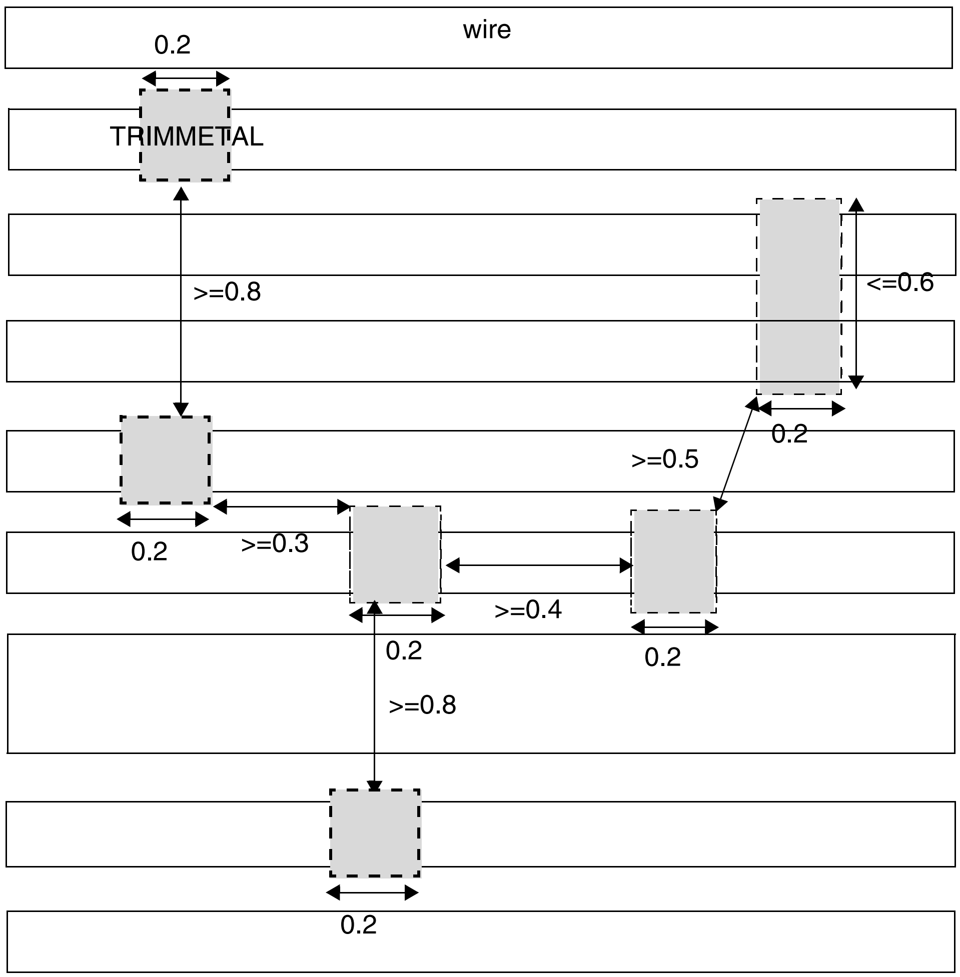
Figure 4-22 Illustration of Spacing Rule for a Trim Metal Layer



TRIMMETAL shapes (in gray color) must have width of 0.2 & max length of 0.6. Spacing with `PRL` > 0 must be >= 0.4, `PRL` = 0 must be >= 0.3, `PRL` < 0 must be >= 0.5 and end to end with `PRL` > 0 must be >= 0.8.

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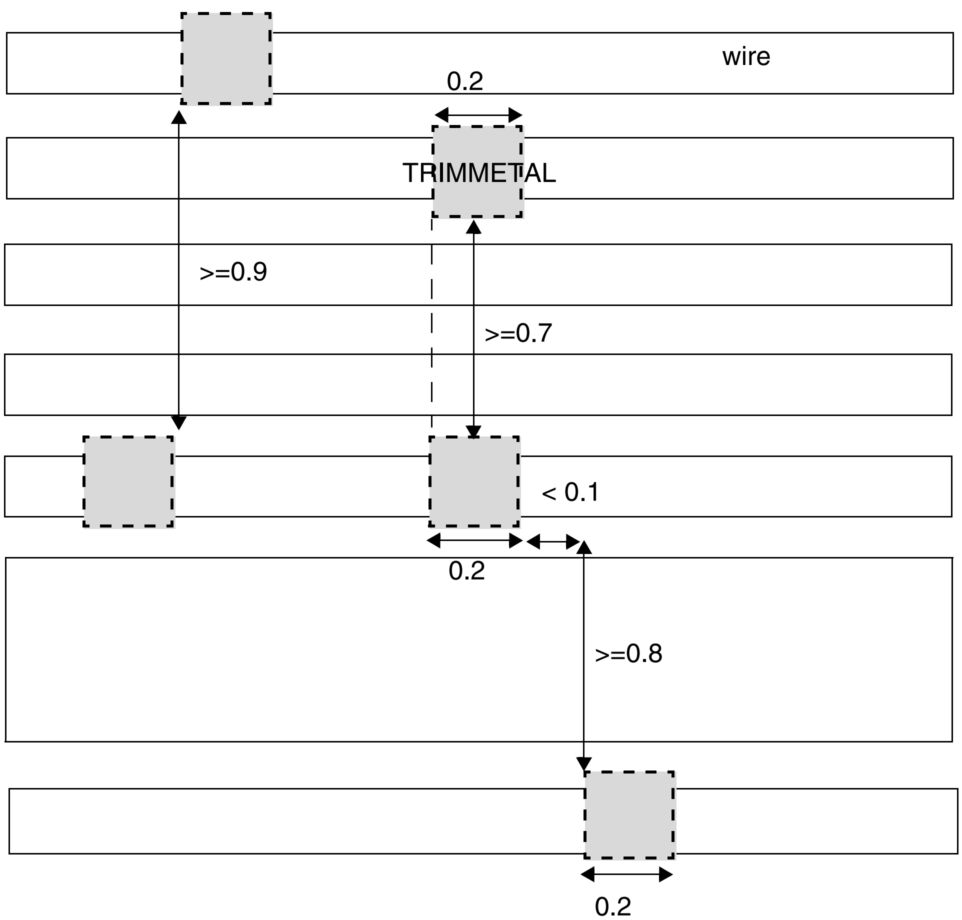
- The following example illustrates the spacing rule for `TRIMMETAL` layer with `EXACTALIGNED` and `EDGEALIGNED`:

```
# No MASK
PROPERTY LEF58_TRIMSHAPE "
     TRIMSHAPE EXTENSIONMODEL MIDTRACK
          EXACTWIDTH 0.2 ; " ;
PROPERTY LEF58_SPACING "
     SPACING 0.5 ENDTOEND 0.8 PRL -0.1 EXACTALIGNED 0.7
          EDGEALIGNED 0.9 ; " ;
```

Figure 4-23 Illustration of Spacing Rule for a Trim Metal Layer



- The following example illustrates the spacing rule for `TRIMMETAL` layer with `EXCEPTDIFFMASKALIGNED`:

```
PROPERTY LEF58_SPACING "
     SPACING 0.05 ENDTOEND 0.1 EXACTALIGNED 0.15
```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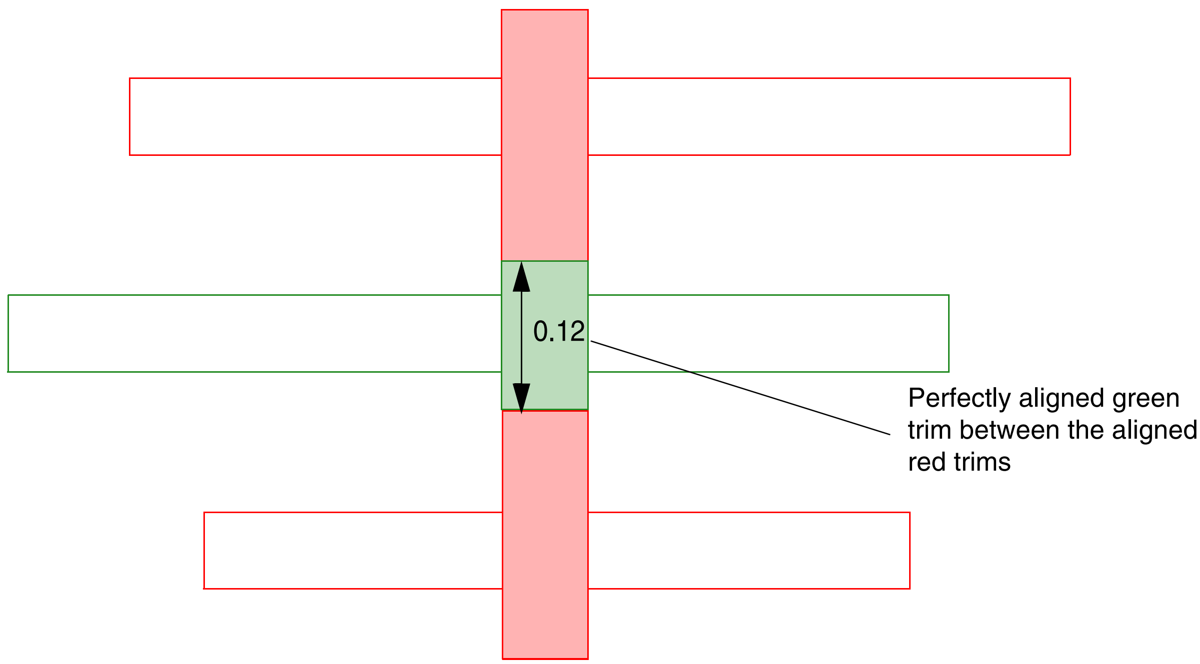
```
EXCEPTDIFFMASKALIGNED SAMEMASK ; " ;
```

Figure 4-24 Illustration of Spacing Rule for a Trim Metal Layer with EXCEPTDIFFMASKALIGNED



OK, the perfectly aligned red trims need to honor 0.15 spacing, but with the presence of another perfectly aligned green trim between them, this spacing rule is exempted. Violation if `EXCEPTDIFFMASKALIGNED` is omitted.

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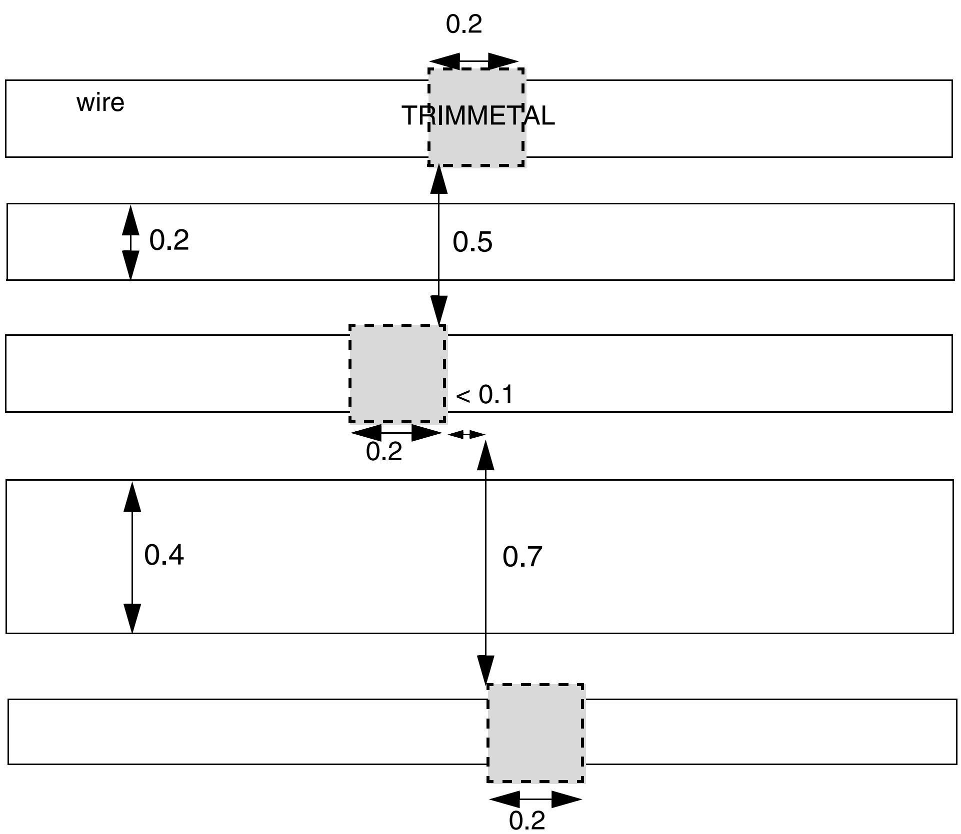
- The following example illustrates the spacing rule for `TRIMMETAL` layer with `EXCEPTMIDMETALWIDTH`:

```
# No MASK
PROPERTY LEF58_TRIMSHAPE "
     TRIMSHAPE EXTENSIONMODEL MIDTRACK
          EXACTWIDTH 0.2 ; " ;
PROPERTY LEF58_SPACING "
     SPACING 0.5 ENDTOEND 0.8 PRL -0.1
          EXCEPTMIDMETALWIDTH 0.4 ; " ;
```

Figure 4-25 Illustration of Spacing Rule for a Trim Metal Layer with EXCEPTMIDMETALWIDTH



The top 0.5 spacing is a violation, but the bottom 0.7 spacing is OK by `EXCEPTMIDMETALWIDTH` exemption.

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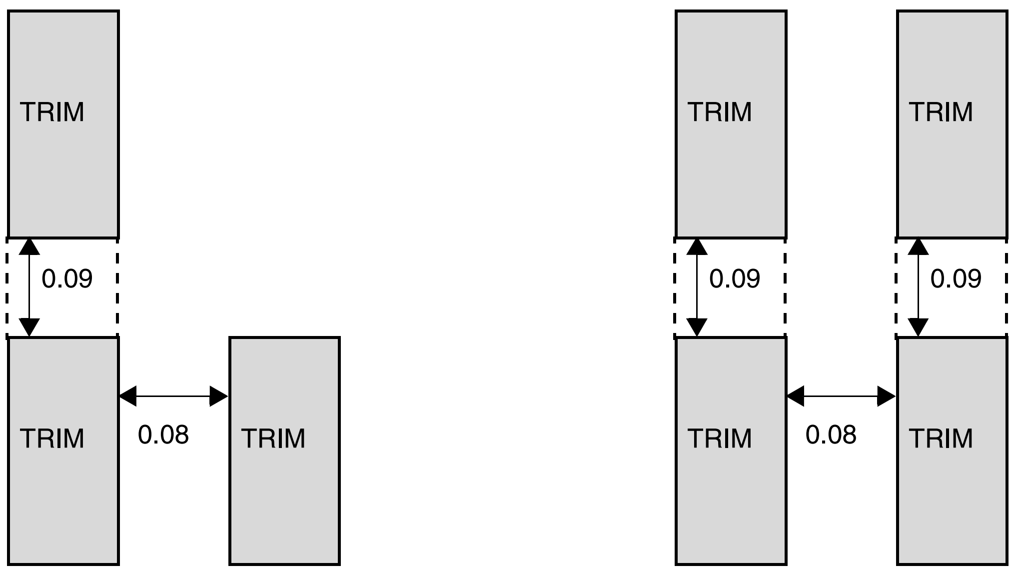
- The following example illustrates the spacing rule for `TRIMMETAL` layer with `MERGEDTRIM`:

```
PROPERTY LEF58_TRIMSHAPE "
    TRIMSHAPE EXTENSIONMODEL ...
        GAPMERGED 0.1 ; " ;
PROPERTY LEF58_SPACING "
        SPACING 0.05
        SPACING 0.09 MERGEDTRIM ; " ;
```

Figure 4-26 Illustration of Spacing Rule for a Trim Metal Layer with MERGEDTRIM



a) Violation, the spacing between a left merged trim to a right non-merged trim still needs to be >= `0.09`.

b) Violation, the spacing between two merged trims is `0.08` (< `0.09` in the second spacing)

- The following example illustrates the spacing rule for `TRIMMETAL` layer with `MERGEDTRIM [`*`length`*`]` and `MIDDLEWIREONLY`:

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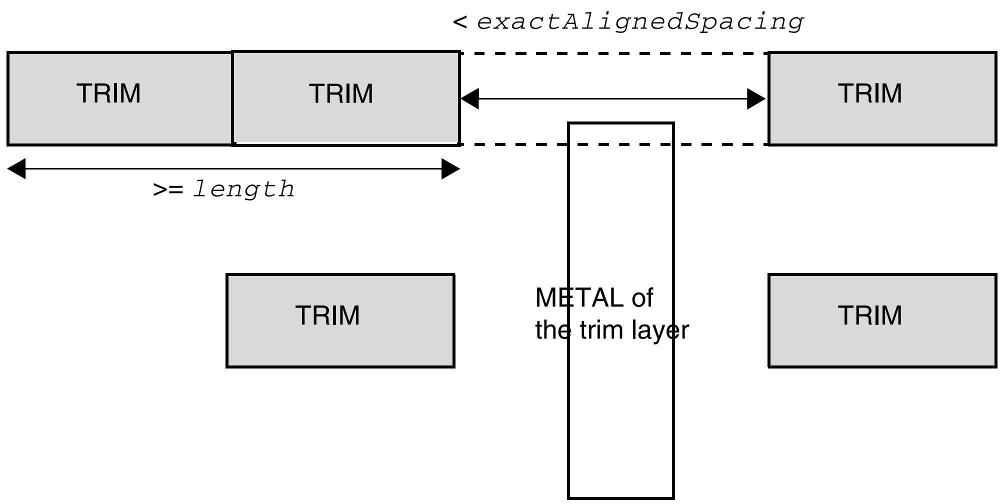
Figure 4-27 Illustration of Spacing Rule with MERGEDTRIM and MIDDLEWIREONLY



a) Violation, the bottom trims are OK since there are no merged trims. The top trims are not okay since the length of the merged trims is >= *length* and there is a metal shape overlapping with the dotted area between the perfectly aligned trims.

Illustration of

```
PROPERTY LEF58_SPACING "
      SPACING spacing ;
      SPACING spacing MERGEDTRIM length
          ENDTOEND endToEndSpacing
          EXACTALIGNED exactAlignedSpacing MIDDLEWIREONLY ; " ;
```

Spacing Rule for a Trim Diffusion Layer

A spacing rule can be defined on a trim diffusion layer as follows:

```
PROPERTY LEF58_SPACING
     "SPACING spacing
     ; " ;
```

Where:

`SPACING` *`spacing`*

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Specifies the spacing between two non-abutted/overlapping shapes to be greater than or equal to *spacing*. This spacing is applied horizontally, vertically, and in Euclidean style. Two corner shapes are not allowed.

Type: Float, specified in microns

Spacing Rule for a Trim Poly Layer

A spacing rule can be defined on a `TRIMPOLY` layer as follows:

```
PROPERTY LEF58_SPACING
     "SPACING spacing [SKIPPROTRUDEDFILLER]
     ; " ;
```

Where:

`SKIPPROTRUDEDFILLER`

Specifies that the spacing rule does not apply when one shape is a padding filler for any protruded cells.

`SPACING` *spacing*

Specifies the spacing between two shapes to be greater than or equal to *spacing*.

Type: Float, specified in microns

Spacing Rule for NWell Layers

A spacing rule can be used to specify the minimum spacing allowed between objects on the same NWell layer or objects on different NWell layers.

```
PROPERTY LEF58_SPACING
     "SPACING minSpacing [HORIZONTAL|VERTICAL]
        [SAMENET | LAYER secondLayerName]
        [LENGTH length1 length2]
        ;..." ;
```

Where:

`HORIZONTAL|VERTICAL`

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Specifies that spacing is applied in the given direction.

`LENGTH` *`length1 length2`*

Specifies that the spacing rule applies only between two shapes where one shape has a length greater than *`length1`* and the other has a length greater than *`length2`*.

Type: Float, specified in microns

`minSpacing` Specifies the default minimum spacing in a well layer.
Type: Float, specified in microns

`SAMENET` Indicates that the minimum spacing only applies to same-net cuts. The `SAMENET` *`minSpacing`* value should be smaller than the normal cut spacing value that applies to different-net cuts.

`LAYER` *`secondLayerName`*

Applies the spacing rule between objects on this well layer and objects on another previously defined well layer (*`secondLayerName`*) for specifying an inter-well layer spacing.

Spacing Rule for PWell Layers

A spacing rule can be used to specify the minimum spacing allowed between objects on the same PWell layer or objects on different PWell layers.

```
PROPERTY LEF58_SPACING
     "SPACING minSpacing [HORIZONTAL|VERTICAL]
        [SAMENET | LAYER secondLayerName]
        [LENGTH length1 length2]
        [INCLUDEENDCAPREGULAR]
        ;..." ;
```

Where:

`HORIZONTAL|VERTICAL`

Specifies that spacing is applied in the given direction.

`INCLUDEENDCAPREGULAR`

Specifies that a macro with `CLASS ENDCAP REGULAR` is assumed to have a PWELL shape with length greater than both *`length1`* and *`length2`* individually.

`LENGTH` *`length1 length2`*

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Specifies that the spacing rule applies only between two shapes where one shape has a length greater than *length1* and the other has a length greater than *length2*.

Type: Float, specified in microns

minSpacing Specifies the default minimum spacing in a well layer.
Type: Float, specified in microns

SAMENET Indicates that the minimum spacing only applies to same-net cuts. The SAMENET *minSpacing* value should be smaller than the normal cut spacing value that applies to different-net cuts.

LAYER *secondLayerName*

Applies the spacing rule between objects on this well layer and objects on another previously defined well layer (*secondLayerName*) for specifying an inter-well layer spacing.

Trim Spacing Rule for a Trim Metal Layer

A trim spacing rule can be used to specify the spacing between any trims on the current layer to any cuts on the cut layer.

```
PROPERTY LEF58_SPACING
     "SPACING spacing LAYER cutLayer [CUTCLASS className]
        [MASK maskNum] [TRIMSPACING trimSpacing [DIFFMASK]]
        ; " ;...
```

Where:

CUTCLASS *className*

Specifies that the trim-to-cut spacing rule applies only to cuts of the given cut class. If CUTCLASS is defined in the cut layer, CUTCLASS must be specified in this statement.

You can specify individual rules with the CUTCLASS keyword for each cut class, if required.

MASK *maskNum* Specifies the cut spacing rule applies only on trims of the given mask.

SPACING *spacing* LAYER *cutLayer*

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Specifies a spacing between any trims on the current layer to any cuts on *cutLayer*, which must be an adjacent cut layer to the metal layer that this trim metal layer trims.

Type: Float, specified in microns

TRIMSPACING *trimSpacing* [DIFFMASK]

Specifies that the rule applies only on a trim with a neighbor trim within *trimSpacing*. Only cuts on a wire abutted to that neighbor trim would be checked against the rule. DIFFMASK means that the neighbor trim must have different mask from the triggering trim.

Type: Float, specified in microns

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Figure 4-28 Illustration of a Trim Spacing Rule for a Trim Metal Layer

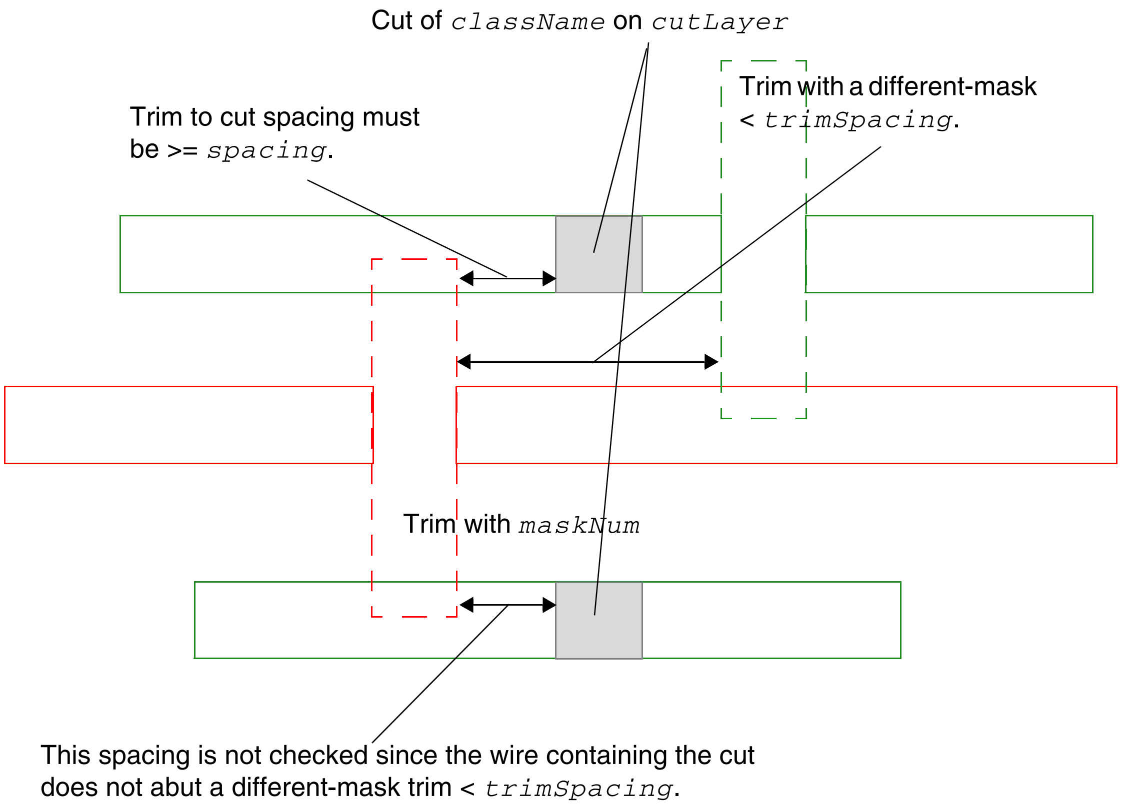

Illustration of
`SPACING` *`spacing`* `LAYER` *`cutLayer`* `CUTCLASS` *`className`*
`MASK` *`maskNum`* `TRIMSPACING` *`trimSpacing`* `DIFFMASK`

Tap Distance Rule for a Well Distance Layer

You can create a tap distance rule to specify distance between well tap cells.

```
PROPERTY LEF58_TAPDISTANCE
"TAPDISTANCE {distance | {HALFROW rowNum [LEFT|RIGHT] distance}...}
     TAPTYPE typeName
     ; " ;
```

Where:

```
TAPDISTANCE {distance|{HALFROW rowNum [LEFT|RIGHT] distance}...}
     TAPTYPE typeName
```


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Specifies that the effective distance of a well tap cell with type *typeName* to be *distance* on both the left and right sides of the cell, which the cell could cover. The distance is measured from the center of the cell. For example, the center-to-center distance between two well tap cells of the same type must be less than or equal to 2 * *distance*.

Type: Float, specified in microns

HALFROW specifies that different distances could be defined on a half row basis. *rowNum* specifies the half row, with respect to the cell origin, on which the distance is defined.

LEFT | RIGHT specifies that different distances could be defined on the left or right sides on a half cell row basis. Left and right is referred with respect to the cell origin.

Trimmed Metal Rule for a Trim Metal Layer

Trimmed metal rules can be used to specify the metal layer that the shapes on the TRIMMETAL layer tries to trim.

You can create a trimmed metal rule by using the following property definition:

```
PROPERTY LEF58_TRIMMEDMETAL
        "TRIMMEDMETAL metalLayer [MASK maskNum [INCLUDEVIRTUAL]
        [VIRTUAL virtualMaskNum]]
        ; " ;
```

Where:

INCLUDEVIRTUAL Specifies that *maskNum* of *metalLayer* has two types of trims. Wires with specific widths are cut by real trims, while wires with alternate widths are cut by virtual trims. The TRIMSHAPE rule is used to specify which wire widths correspond to which trim type.

TRIMMEDMETAL *metalLayer* [MASK *maskNum*]

Specifies the metal layer *metalLayer* that the shapes on the TRIMMETAL layer tries to trim. If *maskNum* is given, only *maskNum* on *metalLayer* is trimmed.

Type: Integer

VIRTUAL *virtualMaskNum*

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Specifies the mask of a virtual trim to be *virtualMaskNum*. The other mask, *maskNum*, is a real trim. This means that this layer must be a two-mask trim layer. One mask, 1 or 2, is a real trim while the other mask, 2 or 1, is a virtual trim.

Type: Integer

Example of Trimmed Metal Rule

- The following is an example of a double patterned layer TM1 used to trim both masks of M1. As both TM1 and M1 are double-patterned, and the TRIMMEDMETAL property does not specify a mask, it implies that MASK 1 of TM1 trims MASK 1 of M1, and MASK 2 of TM1 trims MASK 2 of M1.

```
LAYER TM1
    TYPE MASTERSLICE ;
    MASK 2 ;
    PROPERTY LEF58_TYPE "TYPE TRIMMETAL ; " ;
    PROPERTY LEF58_TRIMMEDMETAL "TRIMMEDMETAL M1 ; " ;
...
END TM1

...
LAYER M1
   TYPE ROUTING ;
   MASK 2 ;
...
END M1
```

- The following example is of a single patterned TRIMMETAL TM2 layer, which trims only one mask of the double-patterned M2 layer. This is indicated by the MASK 1 portion of TM2's TRIMMEDMETAL property:

```
LAYER TM2
    TYPE MASTERSLICE ;
    PROPERTY LEF58_TYPE "TYPE TRIMMETAL ; " ;
    PROPERTY LEF58_TRIMMEDMETAL "TRIMMEDMETAL M2 MASK 1 ; " ;
...
END TM2


LAYER M2
   TYPE ROUTING ;
    MASK 2 ;
...
END M2
```

Trim Shape Rule for a Trim Metal Layer

Trim shape rules can be used to specify that shapes on this layer should be formed based on the line-end wires on the metal layer that is trimmed.

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You can create a trim shape rule by using the following property definition:

```
PROPERTY LEF58_TRIMSHAPE
     "TRIMSHAPE [METALMASK metalMaskNum] EXTENSIONMODEL
        {ADJACENTTRACK | MIDTRACK | EXTENSION extension [METALEDGE]}
        EXACTWIDTH exactWidth [EXACTVIRTUALWIDTH exactVirtualWidth]
        [MAXOVERLAPWIDTH maxOverlapWidth]
        [GAPMERGED trimGapLength]
        [MAXLENGTH maxLength
             [ROWBOUNDARYMAXLENGTH rowBoundaryMaxLength]
             [GAPMERGEDFORLENGTH trimGapLength]
             [MASK maskNum]]
        [EXCEPTSPACING spacing]
        [EXCEPTWIDTH width [EXACTWIDTH]
             [EXCEPTVIRTUALWIDTH virtualWidth]]
        [USEMETALMASK]
        ; " ;
```

Where:

EXACTVIRTUALWIDTH *exactVirtualWidth*

Specifies that the virtual trims on the specified metal mask, *metalMaskNum*, must be exactly equal to *exactVirtualWidth*. This construct must be used only if METALMASK is defined and INCLUDEVIRTUAL is specified within TRIMMEDMETAL. *exactWidth* is still used as the trim width of real trims.

Type: Float, specified in microns

EXACTWIDTH

Specifies that only wires with width exactly equal to *width* are cut by trims.

EXCEPTVIRTUALWIDTH *virtualWidth*

Specifies that if *virtualWidth* exceeds *width*, wires with width less than or equal to *width* are cut by real trims, while wires with width greater than *width* less than or equal to *virtualWidth* are cut by virtual trims. Conversely, if width is greater than *virtualWidth*, wires with width less than or equal to *virtualWidth* are cut by virtual trims, and wires with width greater than *virtualWidth* and less than or equal to *width* are cut by real trims. This construct must be defined if METALMASK is defined and INCLUDEVIRTUAL is specified within TRIMMEDMETAL.

Type: Float, specified in microns

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`GAPMERGED` *`trimGapLength`*

Specifies that perfectly aligned trims with spacing/gap length less than or equal to *`trimGapLength`* should be merged. Only same-mask trims can be merged. No trim spacing check would be applied among the merged trims. Merged trims would still check spacing against other trims as usual.

Type: Float, specified in microns

`GAPMERGEDFORLENGTH` *`trimGapLength`*

Specifies that perfectly aligned trims with spacing/gap length less than or equal to *`trimGapLength`* should be merged only for the `MAXLENGTH` check. No trim spacing check would be applied among the merged trims. Merged trims would still check spacing against other trims as usual.

Type: Float, specified in microns

`MASK` *`maskNum`*

Specifies that the maximum length rule applies on the given mask. *`maskNum`* must be a positive integer, and most applications support only the values 1, 2, or 3.

Type: Integer

`MAXOVERLAPWIDTH` *`maxOverlapWidth`*

Specifies a special case that two individual trims with width of *`exactWidth`* may overlap, provided their combined width does not exceed *`maxOverlapWidth`*. This construct must be defined along with `TRIMMETALTRACK` with `ONTRACK`. The corresponding metal shapes must snap to the nearest on-grid location. If needed, two overlap on-grid trims should be inserted.

Type: Float, specified in microns

`METALEDGE`

Specifies that extension is extended from the metal edges that the trim metal trims.

`METALMASK` *`metalMaskNum`*

Specifies the trim shape definition when the trim is added on the metal wires belonging to the given mask of *`metalMaskNum`*.

Type: Integer

`ROWBOUNDARYMAXLENGTH` *`rowBoundaryMaxLength`*

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Specifies that perfectly aligned trims overlapping with any cell row boundaries must have length less than or equal to *rowBoundaryMaxLength*. Trims not overlapping with the cell row boundaries must have length less than or equal to *maxLength*. This means that only horizontal trims could specify this construct.

Type: Float, specified in microns

```
TRIMSHAPE EXTENSIONMODEL
   {ADJACENTTRACK | MIDTRACK | EXTENSION extension}
   EXACTWIDTH exactWidth [MAXLENGTH maxLength]
   [EXCEPTSPACING spacing] [EXCEPTWIDTH width]
```


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Specifies that shapes on this layer should be formed based on the line-end wires on the metal layer (TRIMMEDEMETAL) that is trimmed. Typically, this trim metal technology would impose a strict 1D on grid routing methodology on TRIMMEDMETAL. Tracks in preferred routing direction of TRIMMEDMETAL could be non-uniform, but must be homogeneous across the entire design. In addition, wires with a same width are always used on any track.

The shapes must be a rectangle with width measured in the preferred routing layer on TRIMMEDMETAL exactly equal to *exactWidth* and length measured in the orthogonal direction less than or equal to *maxLength*, if specified. Shapes are always formed at the line-end of wires on TRIMMEDMETAL. The 'width' dimension is always sandwiched between two line-end wires along the preferred direction on TRIMMEDMETAL.

With ADJACENTTRACK specified in EXTENSIONMODEL, the 'height' would extend to the center line of the adjacent tracks on the orthogonal direction.

With MIDTRACK specified in EXTENSIONMODEL, the 'height' would extend by half of the required spacing of the wires. The shapes could be merged if they are perfectly aligned.

With EXTENSION *extension* specified in EXTENSIONMODEL, the 'height' of a trim metal would extend on both sides from the center by *extension*.

When the line-end to line-end spacing is greater than or equal to *spacing*, if specified, the shape is not formed. In addition, *spacing* must be greater than *exactWidth*.

When the width of line-end is greater than *width*, if specified, the shape is not formed.

USEMETALMASK — Specifies that the mask of a shape on this layer must follow the mask of a wire on TRIMMEDMETAL. This means that *maskNum* on this layer must match the mask number on the TRIMMEDMETAL.

Trim Shape Rule Examples

- The following example is an illustration of TRIMSHAPE rule with MIDTRACK extension model:

```
MASK 2 ;
```


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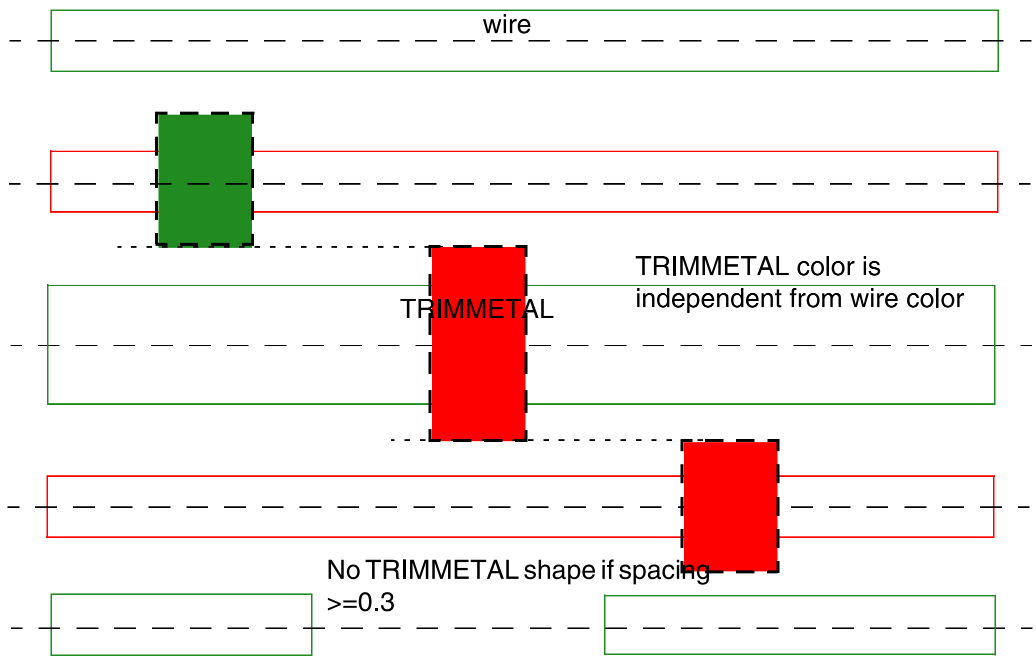
```
PROPERTY LEF58_TRIMSHAPE "
   TRIMSHAPE  EXTENSIONMODEL MIDTRACK
      EXACTWIDTH 0.2 EXCEPTSPACING 0.3 ; " ;
```

Figure 4-29 Illustration of Trim Shape Rule with MIDTRACK Extension Model



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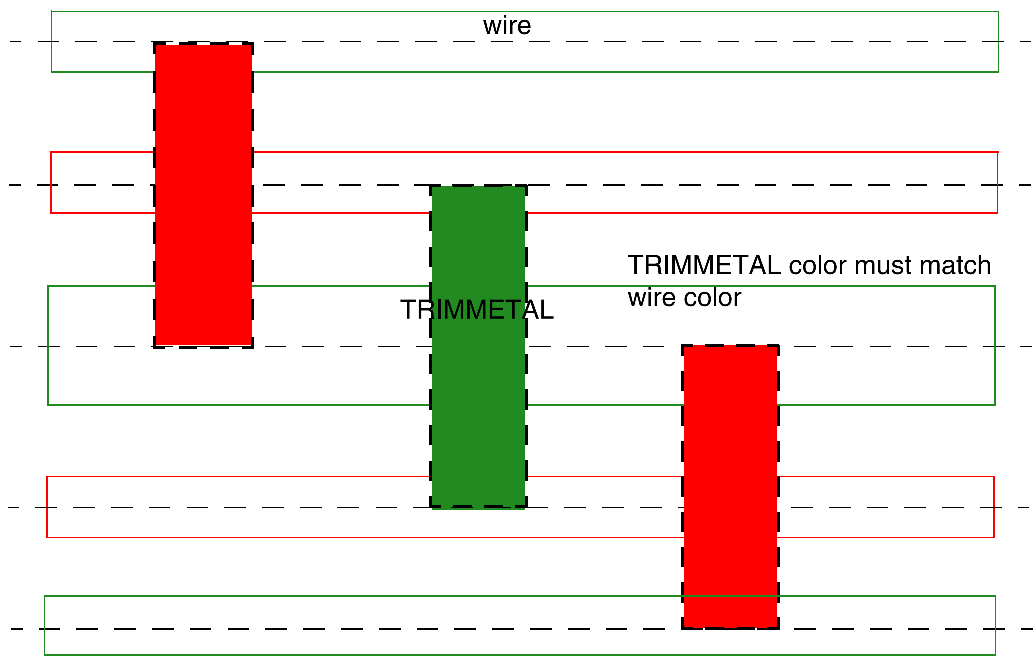
- The following example is an illustration of `TRIMSHAPE` rule with `ADJACENTTRACK` extension model:

```
MASK 2 ;
PROPERTY LEF58_TRIMSHAPE "
     TRIMSHAPE EXTENSIONMODEL ADJACENTTRACK
          EXACTWIDTH 0.2 USEMETALMASK ; " ;
```

Figure 4-30 Illustration of Trim Shape Rule with ADJACENTTRACK Extension Model



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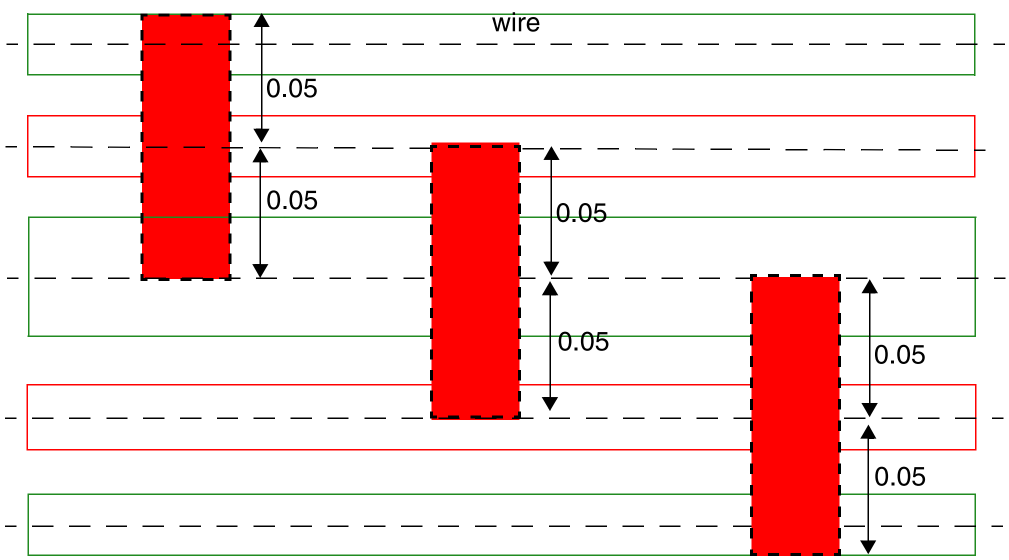
- The following example is an illustration of `TRIMSHAPE` rule with `EXTENSION` extension model:

```
PROPERTY LEF58_TRIMSHAPE "
     TRIMSHAPE EXTENSIONMODEL EXTENSION 0.05
          EXACTWIDTH 0.2 ; " ;
```

Figure 4-31 Illustration of Trim Shape Rule with EXTENSION Extension Model



Trim metal is always extended by `0.05` on both ends.

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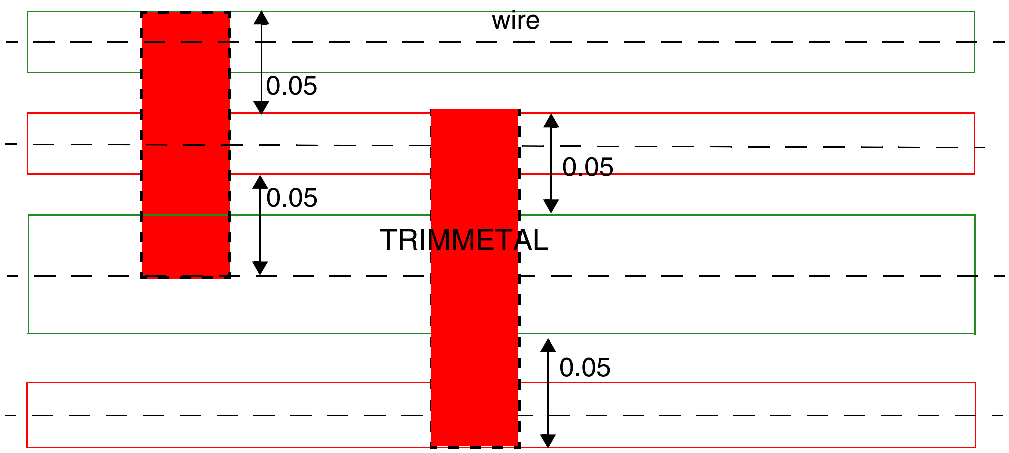
- The following example is an illustration of `TRIMSHAPE` rule with extension from metal edges:

```
PROPERTY LEF58_TRIMSHAPE "
     TRIMSHAPE EXTENSIONMODEL EXTENSION 0.05
          METALEDGE EXACTWIDTH 0.2 ; " ;
```

Figure 4-32 Example of Trim Shape Rule with Extension from Metal Edges



Trim metal is always extended by `0.05` on metal edges.

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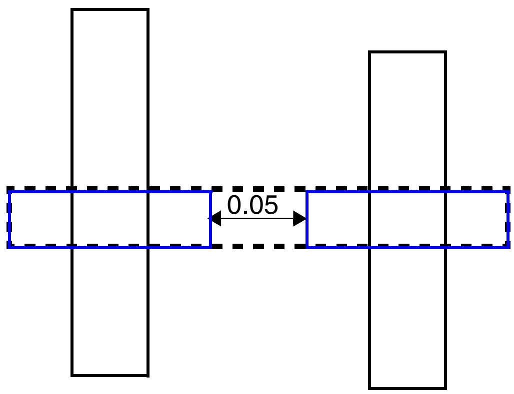
- The following example is an illustration of `TRIMSHAPE` rule with `GAPMERGED`:

```
PROPERTY LEF58_TRIMSHAPE "
     TRIMSHAPE EXTENSIONMODEL ...
          GAPMERGED 0.05 ; " ;
```

Figure 4-33 Example of Trim Shape Rule with GAPMERGED



The two perfectly aligned blue trims would be merged as the dotted line.

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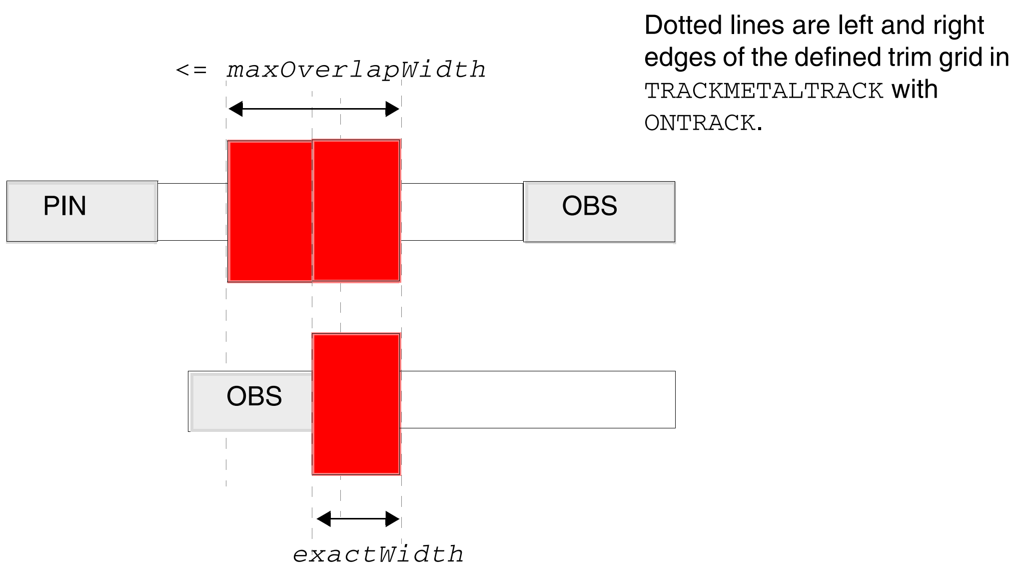
- The following example is an illustration of `TRIMSHAPE` rule with `MAXOVERLAPWIDTH`:

Figure 4-34 Illustration of Trim Shape Rule with MAXOVERLAPWIDTH



In the top case, the pin must be extended to the left edge of the nearest trim grid and the OBS must be extended to the right edge of the nearest trim grid. Two overlap trims must be inserted with width `<=` *`maxOverlapWidth`*. As the OBS is aligned to the trim grid in the bottom case already, only one trim can be inserted.

Illustrations of `TRIMSHAPE ... EXACTWIDTH` *`exactWidth`*
`MAXOVERLAPWIDTH` *`maxOverlapWidth`*

Width Rule for an ABUTLOGIC Layer

A width rule can be used to specify a minimum width for cell shapes on an `ABUTLOGIC` layer.

You can create a width rule by using the following property definition:

```
PROPERTY LEF58_WIDTH
     "WIDTH minWidth
     ;" ;
```

Where:

`WIDTH` *`minWidth`*

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Specifies that the cell shapes on this layer must have a width of at least *minWidth*. Abutted cell shapes are merged before checking against this width condition.

Type: Float, specified in microns

Width Rule for an NWell Layer

A width rule can be used to specify a default width for an NWELL layer.

You can create a width rule by using the following property definition:

```
PROPERTY LEF58_WIDTH
     "WIDTH minWidth [WITHINLAYER {layerName}...]
     ;" ;
```

Where:

WIDTH *minWidth*

Specifies the default width for the NWELL layer.

WITHINLAYER {*layerName*}...

Specifies that the NWELL shapes overlapping with the shapes in on any given layer *layerName*, which should be a masterslice layer, should be checked against *minWidth* by themselves. In other words, those NWELL shapes should not be merged with other NWELL shapes that are not overlapping with the shapes in *layerName*. Typically, only standard cells contain OBS shapes in *layerName*.

Type: String

Width Rule for a PWell Layer

A width rule can be used to specify a default width for a PWELL layer.

You can create a width rule by using the following property definition:

```
PROPERTY LEF58_WIDTH
     "WIDTH minWidth;" ;
```


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Where:

`WIDTH` *`minWidth`*

Specifies the default width for the `PWELL` layer.

Width Rule for a TRIMPOLY Layer

A width rule can be used to specify a minimum width for a `TRIMPOLY` layer.

You can create a width rule by using the following property definition:

```
PROPERTY LEF58_WIDTH
     "WIDTH minWidth [ZEROPRL]
     ;" ;
```

Where:

`WIDTH` *`minWidth`*

Specifies the minimum width for the `TRIMPOLY` layer.

`ZEROPRL` Specifies that the width requirement applies only for two `TRIMPOLY` shapes on adjacent rows with zero parallel run length in the direction perpendicular to the row direction.

Width Length Ratio Rule for a Diffusion Layer

A width length ratio rule can be used to specify a length ratio between diffusion shapes with a given width on a masterslice layer.

You can create a width length ratio rule by using the following property definition:

```
PROPERTY LEF58_WIDTHLENGTHRATIO
     "WIDTHLENGTHRATIO ratio WIDTH width; " ;
```

Where:

`WIDTHLENGTHRATIO` *`ratio`* `WIDTH` *`width`*

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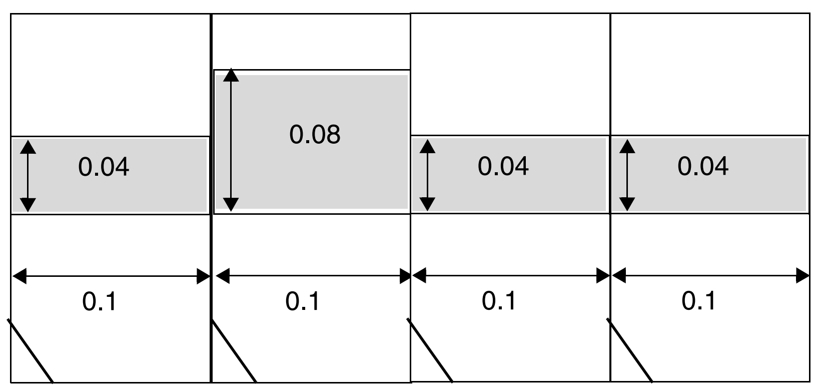
Specifies that the length ratio between the diffusion shapes with width exactly equal to *`width`* and all the diffusion shapes should be less than or equal to *`ratio`*, which is represented as a value between `0.0` and `100.0` in percentage. In other words, dividing the length of the diffusion shapes with width exactly equal to *`width`* by the length of all diffusion shapes should be less than or equal to *`ratio`*.
Type: Float, between `0.0` and `100.0`

Width Length Rule Example

- The following example illustrates the width length ratio rule:

```
PROPERTY LEF58_WIDTHLENGTHRATIO "
      WIDTHLENGTHRATIO 0.3 WIDTH 0.08 ;  " ;
```

Figure 4-35 Example of Width Length Ratio Rule



a) OK. The length of diffusion shapes with width of `0.08` is `0.1` and total length is `0.4`, which makes the length ratio of `0.1/0.4 = 0.25`, which is <= 0.3.

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YMax Length Rule for a Mask Split Layer

You can create a Y max length rule for a Mask Split layer using the following property definition:

```
PROPERTY LEF58_YMAXLENGTH
     "YMAXLENGTH maxLength
     ;" ;
```

Where:

`YMAXLENGTH` *`maxLength`*

Specifies the maximum vertical length for the region created in the design level on this layer as less than or equal to *`maxLength`*.

Type: Float, specified in microns

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5

LEF Syntax - Layer (Overlap)

This chapter contains information about the following topics:

Layer (Overlap)

```
LAYER layerName
     TYPE {OVERLAP} ;
     [PROPERTY propName propVal ;] ...
END layerName
```

Defines overlap layers in the design. The overlap layer should normally be named `OVERLAP`. It can be used in `MACRO` definitions to form rectilinear-shaped cells and blocks (that is, an “L”-shaped block).

Each layer is defined by assigning it a name and design rules. You must define an overlap layer separately, with its own layer statements.

You must define layers in process order from bottom to top.

`LAYER` *`layerName`*

Specifies the name for the layer. This name is used in later references to the layer.

`PROPERTY` *`propName propVal`*

Specifies a numerical or string value for a layer property defined in the `PROPERTYDEFINITIONS` statement. The *`propName`* you specify must match the propName listed in the `PROPERTYDEFINITIONS` statement.

`TYPE`

Specifies the purpose of the layer.

`OVERLAP`	Layer used for overlap checking for rectilinear blocks. Obstruction descriptions in the macro obstruction statements refer to the overlap layer.

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6

LEF Syntax - Layer (Routing)

This chapter contains information about the following topics:

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Layer (Routing)

```
LAYER layerName
     TYPE ROUTING ;
     [MASK maskNum ;]
     DIRECTION {HORIZONTAL | VERTICAL | DIAG45 | DIAG135} ;
     PITCH {distance | xDistance yDistance} ;
     [DIAGPITCH {distance | diag45Distance diag135Distance} ;]
     WIDTH defaultWidth ;
     [OFFSET {distance | xDistance yDistance} ;]
     [DIAGWIDTH diagWidth ;]
     [DIAGSPACING diagSpacing ;]
     [DIAGMINEDGELENGTH diagLength ;]
     [AREA minArea ;]
     [MINSIZE minWidth minLength [minWidth2 minLength2] ... ;]
     [[SPACING minSpacing
         [ RANGE minWidth maxWidth
            [ USELENGTHTHRESHOLD
            | INFLUENCE value [RANGE stubMinWidth stubMaxWidth]
            | RANGE minWidth maxWidth]
         | LENGTHTHRESHOLD maxLength [RANGE minWidth maxWidth]
         | ENDOFLINE eolWidth WITHIN eolWithin
              [PARALLELEDGE parSpace WITHIN parWithin [TWOEDGES]]
         | SAMENET [PGONLY]
         | NOTCHLENGTH minNotchLength
         | ENDOFNOTCHWIDTH endOfNotchWidth NOTCHSPACING minNotchSpacing
               NOTCHLENGTH minNotchLength
         ]
       ;] ...
     [SPACINGTABLE
          {PARALLELRUNLENGTH {length} ...
             {WIDTH width {spacing} ...} ... ;
             [SPACINGTABLE INFLUENCE
               {WIDTH width WITHIN distance
                SPACING spacing} ... ;]
          |TWOWIDTHS {WIDTH width [PRL runLength]
               {spacing} ...} ... ;
          }
       ;" ;]
     [WIREEXTENSION value ; ]
     [MINIMUMCUT numCuts WIDTH width [WITHIN cutDistance]
        [FROMABOVE | FROMBELOW]
        [LENGTH length WITHIN distance] ;] ...
     [MAXWIDTH width ;]
     [MINWIDTH width ;]
     [MINSTEP minStepLength
        [ [INSIDECORNER | OUTSIDECORNER | STEP] [LENGTHSUM maxLength]
        | MAXEDGES maxEdges] ;]
     [MINENCLOSEDAREA area [WIDTH width] ;] ...
     [PROTRUSIONWIDTH width1 LENGTH length WIDTH width2 ;]
```


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```
[RESISTANCE RPERSQ value ;]
[CAPACITANCE CPERSQDIST value ;]
[HEIGHT distance ;]
[THICKNESS distance ;]
[SHRINKAGE distance ;]
[CAPMULTIPLIER value ;]
[EDGECAPACITANCE value ;]
[MINIMUMDENSITY minDensity ;]
[MAXIMUMDENSITY maxDensity ;]
[DENSITYCHECKWINDOW windowLength windowWidth ;]
[DENSITYCHECKSTEP stepValue ;]
[FILLACTIVESPACING spacing ;]
[ANTENNAMODEL {OXIDE1 | OXIDE2 | OXIDE3 | OXIDE4 | OXIDE5 | ...
   | OXIDE32}
   ;] ...
[ANTENNAAREARATIO value ;] ...
[ANTENNADIFFAREARATIO {value | PWL ( ( d1 r1 ) ( d2 r2 ) ...) } ;] ...
[ANTENNACUMAREARATIO value ;] ...
[ANTENNACUMDIFFAREARATIO {value | PWL ( ( d1 r1 ) ( d2 r2 ) ...) } ;] ...
[ANTENNAAREAFACTOR value [DIFFUSEONLY] ;] ...
[ANTENNASIDEAREARATIO value ;] ...
[ANTENNADIFFSIDEAREARATIO {value | PWL ( ( d1 r1 ) ( d2 r2 ) ...) } ;] ...
[ANTENNACUMSIDEAREARATIO value ;] ...
[ANTENNACUMDIFFSIDEAREARATIO {value | PWL ( ( d1 r1 ) ( d2 r2 ) ...) } ;] ...
[ANTENNASIDEAREAFACTOR value [DIFFUSEONLY] ;] ...
[ANTENNACUMROUTINGPLUSCUT ;]
[ANTENNAGATEPLUSDIFF plusDiffFactor ;]
[ANTENNAAREAMINUSDIFF minusDiffFactor ;]
[ANTENNAAREADIFFREDUCEPWL ( ( diffArea1 diffMetalFactor1 )
    ( diffArea2 diffMetalFactor2 ) ...) ;]
[PROPERTY propName propVal ;] ...
[ACCURRENTDENSITY {PEAK | AVERAGE | RMS}
   { value
   | FREQUENCY freq_1 freq_2 ... ;
       [WIDTH width_1 width_2 ... ;]
       TABLEENTRIES
         v_freq_1_width_1 v_freq_1_width_2 ...
         v_freq_2_width_1 v_freq_2_width_2 ...
         ...
   } ;]
[DCCURRENTDENSITY AVERAGE
   { value
   | WIDTH width_1 width_2 ... ;
       TABLEENTRIES value_1 value_2 ...
   } ;]
[PROPERTY LEF58_ACCURRENTDENSITY
  "ACCURRENTDENSITY {PEAK | AVERAGE | RMS}
     [TEMPPWL temp_1 multi_1 temp_2 multi_2  ...]
     [HOURSPWL hours_1 multi_1 hours_2 multi_2  ...]
     ; " ;]
     ; " ;]
```


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```
[PROPERTY LEF58_ANTENNACUMAREARATIO
  "ANTENNACUMAREARATIO {OXIDE1 | OXIDE2 | OXIDE3 | OXIDE4 | OXIDE5 | ...
   | OXIDE32} value
   [GATEEXPONENT index] [GATEAREA area]
   [DIFFONLY]
   ;" ;]
[PROPERTY LEF58_ANTENNADIFFGATEPWL
  "ANTENNADIFFGATEPWL {OXIDE1 | OXIDE2 | OXIDE3 | OXIDE4 | OXIDE5 | ...
   | OXIDE32}
   ((gateArea1 effectiveGateArea1) (gateArea2 effectiveGateArea2) ... )
   ;" ;]
[PROPERTY LEF58_ANTENNADIFFONLYAREARATIO
  "ANTENNADIFFONLYAREARATIO {OXIDE1 | OXIDE2 | OXIDE3 | OXIDE4
     | OXIDE5 | ... | OXIDE32}  value
   ;..." ;]
   ;..." ;]
[PROPERTY LEF58_ANTENNADIFFPROTECTORAREARATIO
  "ANTENNADIFFPROTECTORAREARATIO {OXIDE1 | OXIDE2 | OXIDE3
     | OXIDE4 | OXIDE5 | ...| OXIDE32}  value
   ;..." ;]
[PROPERTY LEF58_ANTENNAGATEPWL
  "ANTENNAGATEPWL {OXIDE1 | OXIDE2 | OXIDE3 | OXIDE4 | OXIDE5 | ...
   | OXIDE32}
   ((gateArea1 effectiveGateArea1) (gateArea2 effectiveGateArea2)... )
   ;" ;]
[PROPERTY LEF58_ANTENNAGATEPLUSDIFF
  "ANTENNAGATEPLUSDIFF {OXIDE1 | OXIDE2 | OXIDE3 | OXIDE4 | OXIDE5 | ...
    | OXIDE32}
    {plusDiffFactor |
    PWL ((diffArea1 plusDiffProtect1) (diffArea1 plusDiffProtect2)...)}
     ;" ;]
[PROPERTY LEF58_AREA
  "AREA minArea
     [MASK maskNum]
     [EXCEPTMINWIDTH minWidth
     | [EXCEPTEDGELENGTH {minEdgeLength maxEdgeLength | edgeLength}]
           [EXCEPTMINSIZE {minWidth minLength}...]
           [EXCEPTSTEP length1 length2]
     | RECTWIDTH rectWidth
     | EXCEPTRECTANGLE
     | LAYER trimLayer OVERLAP {1 | 2}
     ]
  ;..." ;]
[PROPERTY LEF58_BACKSIDE
  "BACKSIDE ;" ;]
[PROPERTY LEF58_BLOCKEOLBLOCKAGE
  "BLOCKEOLBLOCKAGE size OFFSET offset
     ;" ;]
[PROPERTY LEF58_BOUNDARYBLOCKAGE
  "BOUNDARYBLOCKAGE size
     ;" ;]
```


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```
[PROPERTY LEF58_BOUNDARYEOLBLOCKAGE
  "BOUNDARYEOLBLOCKAGE size OFFSET offset
     [PARALLEL parLength WITHIN parWithin SPACING spacing]
     ;" ;]
[PROPERTY LEF58_COREEOLBLOCKAGE
  "COREEOLBLOCKAGE outwardSize inwardSize
     [SIDEEXTENSION sideExtension]
     ;" ;]
[PROPERTY LEF58_BUMPSPACING
  "BUMPSPACING spacing WIDTH bumpWidth
     ;" ;]
[PROPERTY LEF58_CORNEREOLKEEPOUT
  "CORNEREOLKEEPOUT WIDTH eolWidth EOLSPACING eolSpacing
     { SPACING spacing1 spacing2 WITHIN within1 within2
     | EXTENSION backwardExt sideExt forwardExt }
     ;" ;]
[PROPERTY LEF58_CORNERFILLSPACING
  "CORNERFILLSPACING spacing EDGELENGTH length1 length2
     ADJACENTEOL eolWidth
     ; " ;]
[PROPERTY LEF58_CORNERSPACING
  "CORNERSPACING
     {CONVEXCORNER [SAMEMASK]
          [CORNERONLY within
          | CORNERTOCORNER [ORTHOGONAL]]
          [EXCEPTEOL eolWidth [EXCEPTJOGLENGTH length
              [EDGELENGTH] [INCLUDELSHAPE]]
          | EOLONLY eolWidth]
     |CONCAVECORNER
          [MINLENGTH minLength]
          [EXCEPTNOTCH [notchLength]]
          [TOCONVEXCORNER]}
     [EXCEPTSAMENET | EXCEPTSAMEMETAL]
     {WIDTH width SPACING spacing
     | WIDTH width SPACING horizontalSpacing verticalSpacing}...
  ; " ;]
[PROPERTY LEF58_ENCLOSURESPACING
  "ENCLOSURESPACING
     [CUTCLASS cutClass [LONGEDGEONLY] ]
     [FROMABOVE | FROMBELOW]
     {ENCLOSURE enclosure SPACING spacing}...
  ; ... " ;]
[PROPERTY LEF58_ENCLOSUREWITHEOL
  "ENCLOSUREWITHEOL eolWidth
     [MASK maskNum] [CUTWITHIN cutWithin]
     [EXCEPTNOCUTINOTHERMETALLENGTH
          otherCutDistance otherLength]
     [CUTCLASS className] [FROMABOVE | FROMBELOW]
     {ENCLOSURE enclosure EXCEPTLINEENDSPACING spacing
     | {ENCLOSURESPACING spacing enclosure}... }
  ; ... " ;]
```


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```
[PROPERTY LEF58_EOLEXTENSIONSPACING
  "EOLEXTENSIONSPACING spacing [SAMEMASK]
   [PARALLELONLY] [NONEOL]]
   [OTHERWIDTH otherWidth]
     {ENDOFLINE eolWidth EXTENSION extension
         [ENDTOEND endToEndExtension]} ...
   [MINLENGTH minLength [TWOSIDES]]
     ;" ;]
[PROPERTY LEF58_EOLKEEPOUT
  "EOLKEEPOUT {eolWidth | minEolWidth maxEolWidth}
    [RIGHTWAYONLY | WRONGWAYONLY]
     EXTENSION backwardExt sideExt forwardExt
     [EXCEPTWITHIN lowSideExt highSideExt]
     [CLASS className [OTHERENDEOL]]
     [CORNERONLY
         [EXCEPTFROM { BACKEDGE [FORWARDGAP forwardGap]
            [EXACTALIGNED exactForwardExt]
            [EXTRAFORWARDKEEPOUT minForwardExt
                 maxForwardExt]
            | FRONTEDGE }
         [OTHEREOL [minEolWidth maxEolWidth]]
         [MASK maskNum [TWOSIDES] [SAMEMASK]
         | DIFFMASK
         | SAMEMASK]
         [EXCEPTSAMESIDEMETAL backwardExt1 sideExt1 forwardExt1]
         [EXCEPTNOOPPSIDEMETAL backwardExt2 sideExt2 forwardExt2]
         [EXCEPTLINEENDSPACING spacing | EXCEPTLINEENDGAP]
         [EOLWITHINCUT withinCut [FROMABOVE|FROMBELOW]]
         [EXCEPTEOLSPACING eolSpacing]
         ]]
     [EXCEPTSAMEMETAL]
     ; " ;]
[PROPERTY LEF58_EOLTRACK
  "EOLTRACK eolWidth PITCH pitch TRACKSDISTANCE distance
     [EXCEPTEOLSPACING eolSpacing [WITHIN within]]
     ; " ;]
[PROPERTY LEF58_EOLVIAKEEPOUT
  "EOLVIAKEEPOUT eolWidth MASK maskNum
     EXTENSION backwardExt sideExt forwardExt
     OTHERWIDTH otherWidth
     OTHERENDEXTENSION otherEndExtension
     [EXCEPTBACKSIDEALIGNED]
     ; " ;]
[PROPERTY LEF58_FILLTOFILLSPACING
  "FILLTOFILLSPACING spacing ;" ;]
[PROPERTY LEF58_FIVEWIRESEOLSPACING
  "FIVEWIRESEOLSPACING eolSpacing WITHIN eolWithin
      PRL prl
      ENCLOSECUT {BELOW|ABOVE} encloseDist CUTWITHIN cutWithin
      NOMETALEOLEXTENSION eolSideExtension eolForwradExtension
;...";]
```


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```
[PROPERTY LEF58_FIVEWIRESFORBIDDENSPACING
  "FIVEWIRESFORBIDDENSPACING minSpacing maxSpacing
       [EXCEPTWIRES numWire]
       {HORIZONTAL|VERTICAL} PRL prl
;...";]
[PROPERTY LEF58_FORBIDDENSPACING
  "FORBIDDENSPACING minSpacing maxSpacing [minSpacing2 maxSpacing2]
     [WIDTH minWidth WITHIN within PRL prl
     | [SAMEMASK] WIDTH maxWidth PRL prl
         [TWOEDGES within | EXACTSPACINGEDGE exactSpacing
             [IGNOREMIDDLE width spacing]
             [SPANLENGTH spanLength [WITHIN within]]
             [EXCEPTWIREWIDTHSPACING {width spacing}...]]
     | EXACTWIDTH exactWidth PRL prl
         OTHERWIDTH otherWidth EXACTSPACINGEDGE exactSpacing
     | [SAMEMASK | MASK maskNum]
         WIDTHRANGE minWidth maxWidth PRL prl
         OTHERWIDTH otherWidth WITHIN within [OTHERSAMEMASK]]
     ; " ;]
[PROPERTY LEF58_FORBIDDENSPANLENGTH
  "FORBIDDENSPANLENGTH spanLength
     EXCEPTJOGLENGTH orthSpanLength [INCLUDELSHAPE]
     ; " ;]
[PROPERTY LEF58_GAP
  "GAP EXACTWIDTH exactWidth MAXLENGTH maxLength
     SPACING {{0 | 1 | 2} spacing}...[ENDTOEND endToEndSpacing]
     ; " ;]
[PROPERTY LEF_CDN_JOGCORNERKEEPOUTZONE
  "JOGCORNERKEEPOUTZONE innerEdgeExtension innerOrthoExtension
     OUTER outerEdgeExtension outerOrthoExtension
     BACK backEdgeExtension backOrthoExtension
     ; " ;]
[PROPERTY LEF58_JOGKEEPOUTZONE
  "JOGKEEPOUTZONE horzExtension vertExtension
     INNER innerEdgeExtension innerOrthoExtension
     OUTER outerEdgeExtension outerOrthoExtension
     BACK backEdgeExtension backOrthoExtension
     FRONT frontEdgeExtension frontOrthoExtension
     ; " ;]
[PROPERTY LEF58_JOGKEEPOUTZONETABLE
  "JOGKEEPOUTZONETABLE
     {SPANLENGTH spanLength rightWayExtension wrongWayExtension}...
     ; " ;]
[PROPERTY LEF58_JOINTCORNERSPACING
  "JOINTCORNERSPACING spacing [SAMEMASK]
     JOINTWIDTH jointWidth [MINLENGTH minLength]
     JOINTLENGTH spanLength [EDGELENGTH edgeLength]
     ; " ;]
[PROPERTY LEF58_JOINTFORBIDDENSPACING
  "JOINTFORBIDDENSPACING minSpacing maxSpacing
     WIDTH width WRONGDIRWIDTH wrongDirWidth
```


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```
     OTHERWIDTH otherWidth [PRL prl]
     ; " ;]
[PROPERTY LEF58_LINEENDALIGNEDWITHCUT
  "LINEENDALIGNEDWITHCUT alignedDistance ENCLOSURE enclosure
     [CUTCLASS className]
     {FROMABOVE | FROMBELOW} [CUTONANYWIRES]
     ; " ;]
[PROPERTY LEF58_LINEENDALIGNEDWITHCUTSPACING
  "LINEENDALIGNEDWITHCUTSPACING minAlignedDistance
     maxAlignedDistance SPACING minSpacing maxSpacing
     PRL prl [CUTCLASS className] {FROMABOVE | FROMBELOW}
     ; " ;]
[PROPERTY LEF58_LINEENDGAP
  "LINEENDGAP gap WIDTH width [MASK maskNum]
  ; " ;]
[PROPERTY LEF58_LITHOMACROHALO
  "LITHOMACROHALO horizontalHalo verticalHalo
     {WIRESPACING wireSpacing | NOSAMEMETALJOG}
  ; " ;]
[PROPERTY LEF58_MANUFACTURINGGRID
  "MANUFACTURINGGRID value
      ; " ;]
[PROPERTY LEF58_MAXDIAGLENGTH
  "MAXDIAGLENGTH diagonalLength manhattanLength
      ; " ;]
[PROPERTY LEF58_MAXLENGTHWITHCUT
  "MAXLENGTHWITHCUT length WIDTH width
     WITHIN cutDistance {FROMABOVE | FROMBELOW}
     OTHERWIDTH otherWidth EXCEPTOTHERLENGTH otherLength
     [LAYER secondLayerName SECONDWIDTH secondWidth
        EXCEPTSECONDLENGTH secondLength]
     [PGONLY]
     ; " ;]
[PROPERTY LEF58_MAXOVERLAPLENGTH
  "MAXOVERLAPLENGTH length LAYER secondLayerName
      ; " ;]
[PROPERTY LEF58_MAXWIDTH
  "MAXWIDTH width [WRONGDIRECTION]
      ; " ;]
[PROPERTY LEF_CDN_MINAREAWITHCUT
  "MINAREAWITHCUT minArea
     CUTCLASSLIST {className}... {FROMABOVE | FROMBELOW}
      ; " ;]
[PROPERTY LEF58_MINIMUMCUT
  "MINIMUMCUT {numCuts | {CUTCLASS className numCuts
     [DIFFCUTCLASS]} ... }
     WIDTH width [WITHIN cutDistance [MAXXY]]
     [FROMABOVE | FROMBELOW]
     [EXCEPTOTHERWIDTH otherWidth]
     [LENGTH length WITHIN distance
     |AREA area [WITHIN distance]
```


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```
          |SAMEMETALOVERLAP
          | FULLYENCLOSED
          | OVERLAPONLY]
          ; " ;]
     [PROPERTY LEF58_MINLENGTHPARALLEL
       "MINLENGTHPARALLEL minLength MASK maskNum WIDTH width
          {FROMABOVE|FROMBELOW} CUTCLASS className
          EXCEPTEOLSPACING eolSpacing EXCEPTWITHIN within
          ; " ;]
     [PROPERTY LEF58_MINLENGTHWITHCUT
       "MINLENGTHWITHCUT minLength WIDTH minWidth maxWidth
          CUTCLASSLIST {className}... {FROMABOVE | FROMBELOW}
          ; " ;]
     [PROPERTY LEF58_MINSIZE
       "MINSIZE [RECTONLY] minWidth minLength [minWidth minLength]...
          ;..." ;]
     [PROPERTY LEF58_MINSTEP
       "MINSTEP minStepLength
          [[INSIDECORNER | OUTSIDECORNER | STEP] [LENGTHSUM maxLength]
          |MAXEDGES maxEdges
            [HORIZONTAL|VERTICAL]
            [EXCEPTRECTANGLE]
            [MASK maskNum]
            [ MINADJACENTLENGTH minAdjLength
               [CONVEXCORNER [EXCEPTWITHIN exceptWithin]
               |CONCAVECORNER
               |THREECONCAVECORNERS [CENTERWIDTH width]
               | minAdjLength2]
           | MINBETWEENLENGTH minBetweenLength
               [EXCEPTSAMECORNERS]
           | NOADJACENTEOL eolWidth
               [EXCEPTADJACENTLENGTH minAdjLength
               |MINADJACENTLENGTH minAdjLength]
               [CONCAVECORNERS]
           | NOBETWEENEOL eolWidth
           | CONCAVEANDCONVEX
           | ADJACENTLENGTHRANGE minLength maxLength]]
          ] ; " ;]
     [PROPERTY LEF58_OFFSET
       "OFFSET [FROMDIEBOUNDARY] {distance | xDistance yDistance}
          ;..." ;]
     [PROPERTY LEF58_OPPOSITEEOLSPACING
       "OPPOSITEEOLSPACING [SAMEMASK] WIDTH width
          ENDWIDTH eolWidth [MINLENGTH minLength]
          [JOINTWIDTH jointWidth] JOINTLENGTH spanLength
              {[JOINTTOEDGEEND jointToEdgeEndLength]
              [JOINTEXTENSION jointExtension]
              [JOINTCORNERONLY]
          [SIDELENGTH length [TOSIDE toSideLength]]
          [SIDEEXTENSION sideExtension [TOSIDE toSideExtension] ]
          [SIDEEDGELENGTH sideEdgeLength]
```


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```
      [PRL prl]
      {[EXCEPTEDGELENGTH edgeLength [PRL maxPRL]]}...
      ENDTOEND endSpacing endSpacing [PRL individualPrl]
      ENDTOJOINT endSpacing jointSpacing [PRL individualPrl]
      JOINTTOEND jointSpacing endSpacing [PRL individualPrl]
      JOINTTOJOINT jointSpacing jointSpacing [PRL individualPrl]
      [SIDETOEND sideSpacing endSpacing [PRL individualPrl]
      SIDETOJOINT sideSpacing jointSpacing] [PRL individualPrl]
      [JOINTTOSIDE jointSpacing sideSpacing] [PRL individualPrl]
      [SIDETOSIDE sideSpacing sideSpacing] [PRL individualPrl]
   ;" ;]
 [PROPERTY LEF58_PINCONNECTBLOCKAGE
   "PINCONNECTBLOCKAGE size ENCLOSEDLENGTH enclosedLength
   ; " ;]
 [PROPERTY LEF58_PITCH
   "PITCH {distance | xDistance yDistance} [FIRSTLASTPITCH firstLastPitch]
   ; " ;]
 [PROPERTY LEF58_PROTRUSIONWIDTH
   "PROTRUSIONWIDTH width1
      {LENGTH length WIDTH width2
      | {WIDTH width2
         {MINSIZE {minWidth minLength | minLength
            CUTCLASS className {FROMABOVE | FROMBELOW}}
         | MINLENGTH minLength
            [EXCEPTCUT cutDistance [FROMABOVE | FROMBELOW]]}...}
   ; " ;]
 [PROPERTY LEF58_RECTONLY
   "RECTONLY
       [EXCEPTNONCOREPINS
          [MACROCLASS {RING | BLOCK | PAD | BUMP}...]]
       [EXCEPTRIGHTWAYWIDTH maxWidth
       | RIGHTWAYWIDTHONLY exactWidth]
       [EXCEPTVIAEDGEALIGNED | EXCEPTVIA]
       [PARTIALTRACKS]
       ;" ;]
 [PROPERTY LEF58_REGION
   "REGION regionLayerName BASEDLAYER routingLayerName
   ; " ;]
 [PROPERTY LEF58_RIGHTWAYONGRIDONLY
   "RIGHTWAYONGRIDONLY [CHECKMASK]
       [EXCEPTWIDTH exceptMinWidth]
       [EXCEPTVIA]
       ; " ;]
 [PROPERTY LEF58_SIXWIRESSPACING
   "SIXWIRESSPACING spacing {SAMESIDE | OPPOSITESIDE}
       EOLSPACING eolSpacing
       ; " ;]
 [PROPERTY LEF58_SPACING
   "SPACING eolSpace EOLPERPENDICULAR eolWidth perWidth ;" ;]
 [PROPERTY LEF58_SPACING
   "SPACING minSpacing AREA maxArea ;" ;]
```


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```
[PROPERTY LEF58_SPACING
  "SPACING minSpacing LAYER trimLayer [TRIMMASK trimMaskNum]
       [MASK maskNum] EXCEPTOVERLAP overlapLength
        [TRIMENDSPACING trimEndSpacing [PRL prl]]
   ;" ;]
[PROPERTY LEF58_SPACING
  "SPACING minSpacing SAMEMASK ;" ;]
[PROPERTY LEF58_SPACING
  "SPACING minSpacing [WRONGDIRECTION [NONEOL eolWidth]
       [PRL prl] ]
  ;" ;]
[PROPERTY LEF58_SPACING
  "SPACING minSpacing NOTCHLENGTH minNotchLength
     [EXCEPTWITHIN lowExcludeSpacing highExcludeSpacing]
     [WITHIN within SPANLENGTH sideOfNotchSpanLength
     |{WIDTH | CONCAVEENDS} sideOfNotchWidth
     |NOTCHWIDTH notchwidth]
  ;" ;]
[PROPERTY LEF58_SPACING
  "SPACING minSpacing NOTCHSPAN span NOTCHSPACING notchSpacing
    EXCEPTNOTCHLENGTH notchLength
  ; " ;
[PROPERTY LEF58_SPACING
  "SPACING eolSpace
    ENDOFLINE eolWidth
    {[EXACTWIDTH]
    [WRONGDIRSPACING wrongDirSpace]
    {[OPPOSITEWIDTH oppositeWidth]
     WITHIN eolWithin [wrongDirWithin]
     [SAMEMASK]
     [EXCEPTEXACTWIDTH exactWidth otherWidth]
     [FILLCONCAVECORNER fillTriangle
     | FILLCONCAVECORNERONLY fillTriangle]
     [WITHCUT [CUTCLASS cutClass] [ABOVE] withCutSpace
        [ENCLOSUREEND enclosureEndWidth
           [WITHIN enclosureEndWithin]]]
     [ENDPRLSPACING endPrlSpace PRL endPrl]
     [ENDTOEND endToEndSpace [oneCutSpace twoCutSpace]
        [EXTENSION extension [wrongDirExtension]]
        [OTHERENDWIDTH otherEndWidth]]
     [MAXLENGTH maxLength | MINLENGTH minLength [TWOSIDES]]
     [EQUALRECTWIDTH]
     [PARALLELEDGE [SUBTRACTEOLWIDTH] parSpace
        WITHIN parWithin [PRL prl]
        [MINLENGTH minLength] [TWOEDGES]
        [SAMEMETAL] [NONEOLCORNERONLY]
        [PARALLELSAMEMASK]]
     [ENCLOSECUT [BELOW | ABOVE] encloseDist
         CUTSPACING cutToMetalSpace [ALLCUTS]]
    | TOCONCAVECORNER [MINLENGTH minLength]
        [MINADJACENTLENGTH
```


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```
            {minAdjLength | minAdjLength1 minAdjLength2}]
      | TONOTCHLENGTH notchLength}
    ;..." ;]
  [PROPERTY LEF58_SPACING
    "SPACING {spacing | minSpacing maxSpacing }
       CONVEXCORNERS EXTENSION sideExt orthogonalExt
       [ SAMESIDE extension
       | SINGLE edgeForwardExt cornerForwardExt cornerBackwardExt
             SPANLENGTH spanLength OPPOSITEWIDTH oppWidth
             OPPOSITEEXTENSION oppSideExt
                 oppForwardExt1 oppForwardExt2
       | MIDDLEWIRES {0 | 1 | 2} [INCLUDEPARTIALOVERALP]
             [BACKWARDWIRE sideExt backwardExt]
             [REVERSE
                 [EXCEPTWRONGDIRWIRE edgeExt wrongDirEdgeExt]
                 [EXCEPTCONVEXCORNER edgeBackwardExt
                     edgeForwardExt wrongDirEdgeExt]]]
       ; " ;]
  [PROPERTY LEF58_SPACINGTABLE
    "SPACINGTABLE
       PARALLELRUNLENGTH [DIAGONAL] [WRONGDIRECTION]
       [SAMEMASK]
       [EXCEPTEOL eolWidth]
       {length} ...
       {WIDTH width
            [EXCEPTWITHIN lowExcludeSpacing highExcludeSpacing]
              {spacing}...} ... ;
       [SPACINGTABLE
            INFLUENCE {WIDTH width WITHIN distance SPACING spacing} ... ;]
    ;" ;]
  [PROPERTY LEF58_SPACINGTABLE
    "SPACINGTABLE JOGTOJOGSPACING jogToJogSpacing
        JOGWIDTH jogWidth SHORTJOGSPACING shortJogSpacing
         {WIDTH width PARALLEL parLength WITHIN parWithin
              [EXCEPTWITHIN lowExcludeSpacing highExcludeSpacing]
              LONGJOGSPACING longJogSpacing
              [SHORTJOGSPACING widthShortJogSpacing]}
    ];" ;
  [PROPERTY LEF58_SPACINGTABLE
    "SPACINGTABLE
       DIRECTIONALSPANLENGTH [WRONGDIRECTION]
       [SAMEMASK]
       [EXCEPTJOGLENGTH length [EDGELENGTH] [INCLUDELSHAPE]]
       [EXCEPTEOL eolWidth]
       [EXACTSPANLENGTHSPACING spanLength1 TO spanLength2
           [PRL prl] {exactSpacing}...]...
       PRL {prl} ...
       {SPANLENGTH spanLength
           [EXACTSPACING exactSpacing
           | SPACINGTOMINSPAN spacingToMinSpan]
           [EXACTSELFSPACING exactSelfSpacing]
```


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```
         [NOEXCEPTEOL|EOLSPACING eolspacing] {spacing}...}...
  ]; " ;
[PROPERTY LEF58_SPACINGTABLE
  "SPACINGTABLE
     TWOWIDTHS [WRONGDIRECTION]
       [SAMEMASK | MASK maskNum]
       [EXCEPTEOL eolWidth]
       [FILLCONCAVECORNER fillTriangle]
       {WIDTH width [PRL runLength]{spacing} ...} ...
  ]; " ;
[PROPERTY LEF58_SPANLENGTHENCLOSURESPACING
  "SPANLENGTHENCLOSURESPACING spacing
       SPANLENGTH spanLength MINLENGTH minLength
       {FROMABOVE|FROMBELOW} ENCLOSURE enclosure
       PRL prl
       ; " ;]
[PROPERTY LEF58_SPANLENGTHTABLE
  "SPANLENGTHTABLE {spanLength}... [MASK maskNum][WRONGDIRECTION]
     [ORTHOGONAL length]
     [EXCEPTOTHERSPAN otherSpanlength
     | {OTHERSPAN otherSpanLength MINSPANLENGTH minSpanLength
           [BOTHVIAS minBothViasSpanLength]}...]
      [JOGSPANONLY {jogSpanLength}... [LASTINCLUDEGREATERTHAN]
           [INCLUDELSHAPE]]
     ; " ;]
[PROPERTY LEF58_TWOWIRESFORBIDDENSPACING
  "TWOWIRESFORBIDDENSPACING [SAMEMASK | MASK maskNum]
       {minSpacing maxSpacing}...
       [HORIZONTAL|VERTICAL]
       [IGNOREMIDDLE]
       { MINSPANLENGTH minSpanLength [EXACTSPANLENGTH]
         MAXSPANLENGTH maxSpanLength [EXACTSPANLENGTH]
        |FIRSTSPANLENGTH spanLength1 spanLength2
            [RECTONLY | EXCEPTRECTONLY]
            [OTHERSPANLENGTH otherSpanLength1
               [EXACTEQUAL | LESSTHAN]
            |OTHEREDGELENGTH otherEdgeLength1
               [EXCEPTSPANLENGTH minSpanLength1 maxSpanLength1]]
         SECONDSPANLENGTH spanLength3 spanLength4}
            [RECTONLY | EXCEPTRECTONLY]
            [OTHERSPANLENGTH otherSpanLength2]
               [EXACTEQUAL | LESSTHAN]
            |OTHEREDGELENGTH otherEdgeLength1
               [EXCEPTSPANLENGTH minSpanLength1 maxSpanLength1]]
        |OFFGRIDONLY minWidth1 maxWidth1 [minWidth2 maxWidth2]
        |ORTHOGONALNONEOL eolWidth
            SECONDNONEOL minWidth maxWidth }
       PRL prl [UNION spanLength]
       ; " ;]
[PROPERTY LEF58_TYPE
  "TYPE {POLYROUTING | MIMCAP | HIGHR
```


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```
        | TSVMETAL | PADMETAL | STACKEDMIMCAP }
        ;" ;]
  [PROPERTY LEF58_VOLTAGESPACING
    "VOLTAGESPACING [TOCUT [ABOVE | BELOW]]
        {voltage spacing} ...
        ;..." ;]
  [PROPERTY LEF58_WIDTHTABLE
    "WIDTHTABLE {width}...[WRONGDIRECTION] [ORTHOGONAL]
      ;" ;]
  [PROPERTY LEF58_WIDTH
    "WIDTH minWidth [WRONGDIRECTION]
      ;" ;]
  [PROPERTY LEF58_WRONGDIREOLKEEPOUT
    "WRONGDIREOLKEEPOUT edgeExt wrongDirEdgeExt ENDOFLINE eolWidth
        EDGELENGTH minLength maxLength
        ;" ;]
END layerName
```

Defines routing layers in the design. Each layer is defined by assigning it a name and design rules. You must define routing layers separately, with their own layer statements.

You must define layers in process order from bottom to top. For example:

```
poly       masterslice
cut01      cut
metal1     routing
cut12      cut
metal2     routing
cut23      cut
metal3     routing
```

ACCURRENTDENSITY

Specifies how much AC current a wire on this layer of a certain width can handle at a certain frequency in units of milliamps per micron (mA/µm).

Note: The true meaning of current density would have units of milliamps per square micron (mA/µm^2); however, the thickness of the metal layer is implicitly included, so the units in this table are milliamps per micron, where only the wire width varies.

The ACCURRENTDENSITY syntax is defined as follows:

```
{PEAK | AVERAGE | RMS}
{ value
| FREQUENCY freq_1 freq_2 ... ;
    [WIDTH width_1 width_2 ...  ; ]
    TABLEENTRIES
     v_freq_1_width_1 v_freq_1_width_2 ...
     v_freq_2_width_1 v_freq_2_width_2 ...
```


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```
    ...
} ;
```

`PEAK`	Specifies the peak current limit of the layer.
`AVERAGE`	Specifies the average current limit of the layer.
`RMS`	Specifies the root mean square current limit of the layer.
`value`	Specifies a maximum current for the layer in mA/µm. *Type:* Float
`FREQUENCY`	Specifies frequency values, in megahertz. You can specify more than one frequency. If you specify multiple frequency values, the values must be specified in ascending order. If you specify only one frequency value, there is no frequency dependency, and the table entries are assumed to apply to all frequencies. *Type:* Float
`WIDTH`	Specifies wire width values, in microns. You can specify more than one wire width. If you specify multiple width values, the values must be specified in ascending order. If you specify only one width value, there is no width dependency, and the table entries are assumed to apply to all widths. *Type:* Float
`TABLEENTRIES`	Defines the maximum current for each of the frequency and width pairs specified in the `FREQUENCY` and `WIDTH` statements, in mA/µm. The pairings define each width for the first frequency in the `FREQUENCY` statement, then the widths for the second frequency, and so on. The final value for a given wire width and frequency is computed from a linear interpolation of the table values. the widths are not adjusted for any process shrinkage, so the should be correct for the “drawn width”. *Type:* Float

Example 6-1 AC Current Density Statements

Most LEF files do not include `PEAK` or `AVERAGE` limits. The `PEAK` limits are not a practical problem for digital signal routing. The `AVERAGE` limits are only needed for DC limits and not AC currents.

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Most technologies do not have frequency dependency for RMS limits, but the LEF syntax requires a frequency value, so in practice the frequency value is a single value of 1, as shown in the example below. In this case the RMS limit does not vary with the frequency.

The following examples define AC current density tables:

The RMS current density at 0.7 µm is 9.0 + (7.5 - 9.0) x (0.8 - 0.7) / (0.8 - 0.4) = 8.625 mA/µm at frequency 300Mhz. Therefore, a 0.7 µm wide wire can carry 8.625 x 0.7 = 6.035 mA of RMS current.

The RMS current density at 0.7 µm is 7.5 + (6.8 - 7.5) x (0.8 - 0.7) / (0.8 - 0.4) = 7.325 mA/µm at frequency 600Mhz. Therefore, a 0.7 µm wide wire can carry 7.325 x 0.7 = 5.1275 mA of RMS current.

```
LAYER met1
    ...
    ACCURRENTDENSITY PEAK        #peak AC current limit for met1
    FREQUENCY 100 400 ;          #2 freq values in MHz
    WIDTH
    0.4 0.8 1.6 5.0 10.0 ;       #5 width values in microns
    TABLEENTRIES
    9.0 7.5 6.5 5.4 4.7          #mA/um for 5 widths and freq_1 (when the frequency
                                 #is 100 Mhz)
    7.5 6.8 6.0 4.8 4.0 ;        #mA/um for 5 widths and freq_2 (when the frequency
                                 #is 400 Mhz)
END met1 ;
```

The PEAK current density at 0.7 µm for 100 Mhz is 9.0 + (7.5 - 9.0) x (0.8 - 0.7) / (0.8 - 0.4) = 8.625 mA/µm, and at 0.7 µm for 400 Mhax is 7.5 + (6.8 - 7.5) x (0.8 - 0.7) / (0.8 - 0.4) = 7.325 mA/mm. Then interpolating between the frequencies at 300Mhz gives 8.625 + (7.325 - 8.625) x (400 - 300) / (400 - 100) = 8.192 mA/µm.

The RMS current density at 0.4 µm is 7.5 mA/µm. Therefore, a 0.4 µm wide wire can carry 7.5 x .4 = 3.0 µm of RMS current.

```
LAYER cut12
    ...
    ACCURRENTDENSITY PEAK       #peak AC current limit for one cut
    FREQUENCY 10 200 ;          #2 freq values in MHz
      CUTAREA 0.16 0.32 ;       #2 cut areas in um squared
    TABLEENTRIES
    0.5 0.4                     #mA/um squared for 2 cut areas at freq_1 (10 Mhz)
    0.4 0.35 ;                  #mA/um squared for 2 cut areas at freq_2 (200 Mhz)
    ACCURRENTDENSITY AVERAGE    #average AC current limit for via cut12
    10.0 ;                      #mA/um squared for any cut area at any frequency
```


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```
    ACCURRENTDENSITY RMS        #RMS AC current limit for via cut12
    FREQUENCY 1 ;               #1 freq (required by syntax; not really used)
      CUTAREA 0.16 1.6 ;        #2 cut areas in um squared
    TABLEENTRIES
    10.0 9.0 ;                  #mA/um squared for 2 cut areas at any frequency
    ....
END cut12 ;
```

ANTENNAAREADIFFREDUCEPWL ((*diffArea1 diffMetalFactor1*) (*diffArea2 diffMetalFactor2*) ...)

Indicates that the metal area is multiplied by a *diffMetalReduceFactor* that is computed from a piece-wise linear interpolation based on the diff_area attached to the metal. (See Example 4 in Appendix C, "Calculating and Fixing Process Antenna Violations.") This means that the ratio is calculated as:

ratio = (*metalFactor* x metal_area x *diffMetalReduceFactor*) / gate_area

The *diffArea* values are floats, specified in microns squared. The *diffArea* values should start with `0` and monotonically increase in value to the maximum size *diffArea* allowed. The *diffMetalFactor* values are floats with no units. The *diffMetalFactor* values are normally between `0.0` and `1.0`. If no rule is defined, the *diffMetalReduceFactor* value in the PAR(m_i) equation defaults to `1.0`.

For more information on the PAR(m_i) equation and process antenna models, see Appendix C, "Calculating and Fixing Process Antenna Violations."

ANTENNAAREAFACTOR *value* [DIFFUSEONLY]

Specifies the multiply factor for the antenna metal area calculation. `DIFFUSEONLY` specifies that the current antenna factor should only be used when the corresponding layer is connected to the diffusion.
Default: `1.0`
Type: Float

For more information on process antenna calculation, see Appendix C, "Calculating and Fixing Process Antenna Violations."

Note: If you specify a value that is greater than 1.0, the computed areas will be larger, and violations will occur more frequently.

ANTENNAAREAMINUSDIFF *minusDiffFactor*

Indicates that the antenna ratio metal area should subtract the diffusion area connected to it. This means that the ratio is calculated as:

ratio = (*metalFactor* x metal_area - *minusDiffFactor* x diff_area) /gate_area

If the resulting value is less than 0, it should be truncated to 0. For example, if a *metal2* shape has a final ratio that is less than 0 because it connects to a diffusion shape, then the cumulative check for *metal3* (or *via2*) connected to the *metal2* shape adds in a cumulative value of 0 from the *metal2* layer. (See Example 1 in Appendix C, "Calculating

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and Fixing Process Antenna Violations.”)
Type: Float
Default: `0.0`

For more information on process antenna models, see Calculating a PAR, in Appendix C, “Calculating and Fixing Process Antenna Violations.”

`ANTENNAAREARATIO` *`value`*

Specifies the maximum legal antenna ratio, using the area of the metal wire that is not connected to the diffusion diode. For more information on process antenna calculation, see Appendix C, “Calculating and Fixing Process Antenna Violations.”
Type: Float

`ANTENNACUMAREARATIO` *`value`*

Specifies the cumulative antenna ratio, using the area of the wire that is not connected to the diffusion diode. For more information on process antenna calculation, see Appendix C, “Calculating and Fixing Process Antenna Violations.”
Type: Float

`ANTENNACUMDIFFAREARATIO {`*`value`*` | PWL ( ( `*`d1 r1`*` ) ( `*`d2 r2`*` )...)}`

Specifies the cumulative antenna ratio, using the area of the metal wire that is connected to the diffusion diode. You can supply and explicit ratio *`value`* or specify piece-wise linear format (`PWL`), in which case the cumulative ratio value is calculated using linear interpolation of the diffusion area and ratio input values. The diffusion input values must be specified in ascending order.
Type: Float

For more information on process antenna calculation, see Appendix C, “Calculating and Fixing Process Antenna Violations.”

`ANTENNACUMDIFFSIDEAREARATIO {`*`value`*` | PWL ( ( `*`d1 r1`*` ) ( `*`d2 r2`*` )...)}`

Specifies the cumulative antenna ratio, using the side wall area of the metal wire that is connected to the diffusion diode. You can supply and explicit ratio *`value`* or specify piece-wise linear format (`PWL`), in which case the cumulative ratio value is calculated using linear interpolation of the diffusion area and ratio input values. The diffusion input values must be specified in ascending order.
Type: Float

For more information on process antenna calculation, see Appendix C, “Calculating and Fixing Process Antenna Violations.”

`ANTENNACUMROUTINGPLUSCUT`

Indicates that the cumulative ratio rules (`ANTENNACUMAREARATIO` and `ANTENNACUMDIFFAREARATIO`) accumulate with the previous cut layer instead of the

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previous metal layer. Use this to combine metal and cut area ratios into one cumulative ratio rule.

Note: This rule does not affect `ANTENNACUMSIDEAREARATIO` and `ANTENNACUMDIFFSIDEAREA` models.

For more information on process antenna models, see Calculating a CAR, in Appendix C, "Calculating and Fixing Process Antenna Violations."

`ANTENNACUMSIDEAREARATIO` *`value`*

Specifies the cumulative antenna ratio, using the side wall area of the metal wire that is not connected to the diffusion diode. For more information on process antenna calculation, see Appendix C, "Calculating and Fixing Process Antenna Violations."

`ANTENNADIFFAREARATIO {`*`value`*` | PWL ( ( `*`d1 r1`*` ) ( `*`d2 r2`*` )...)}`

Specifies the antenna ratio, using the area of the metal wire that is connected to the diffusion diode. You can supply and explicit ratio *`value`* or specify piece-wise linear format (`PWL`), in which case the ratio value is calculated using linear interpolation of the diffusion area and ratio input values. The diffusion input values must be specified in ascending order.
Type: Float

For more information on process antenna calculation, see Appendix C, "Calculating and Fixing Process Antenna Violations."

`ANTENNADIFFSIDEAREARATIO {`*`value`*` | PWL ( ( `*`d1 r1`*` ) ( `*`d2 r2`*` )...)}`

Specifies the antenna ratio, using the side wall area of the metal wire that is connected to the diffusion diode. You can supply and explicit ratio *`value`* or specify piece-wise linear format (`PWL`), in which case the ratio value is calculated using linear interpolation of the diffusion area and ratio input values. The diffusion input values must be specified in ascending order.
Type: Float

For more information on process antenna calculation, see Appendix C, "Calculating and Fixing Process Antenna Violations."

`ANTENNAGATEPLUSDIFF` *`plusDiffFactor`*

Indicates that the antenna ratio gate area includes the diffusion area multiplied by *`plusDiffFactor`*. This means that the ratio is calculated as:

ratio = (*metalFactor* x metal_area) / (gate_area + *plusDiffFactor* x diff_area)

The ratio rules without "`DIFF`" (the `ANTENNAAREARATIO`, `ANTENNACUMAREARATIO`, `ANTENNASIDEAREARATIO`, and `ANTENNACUMSIDEAREARATIO` statements), are unnecessary for this layer if the `ANTENNAGATEPLUSDIFF` rule is specified because a zero diffusion area already is accounted for by the `ANTENNADIFF*RATIO` statements. (See Example 3 in Routing Layer Process Antenna Model Examples in Appendix C, "Calculating and Fixing Process Antenna Violations.")

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Type: Float
Default: `0.0`

For more information on process antenna models, see Calculating a PAR, in Appendix C, "Calculating and Fixing Process Antenna Violations."

`ANTENNAMODEL {OXIDE1 | OXIDE2 | OXIDE3 | OXIDE4 | OXIDE5 | ... | OXIDE32}`

Specifies the oxide model for the layer. If you specify an `ANTENNAMODEL` statement, that value affects all `ANTENNA*` statements for the layer that follow it until you specify another `ANTENNAMODEL` statement.
Default: `OXIDE1`, for a new `LAYER` statement

Because LEF is sometimes used incrementally, if an `ANTENNA` statement occurs twice for the same oxide model, the last value specified is used. For any given `ANTENNA` keyword, only one value or PWL table is stored for each oxide metal on a given layer.

Example 6-2 Antenna Model Statement

The following example defines antenna information for oxide models on layer *metal1*.

```
LAYER metal1
    ANTENNAMODEL OXIDE1 ;          #OXIDE1 not required, but good practice
    ANTENNACUMAREARATIO 5000 ;     #OXIDE1 values
    ANTENNACUMDIFFAREARATIO 8000 ;
    ANTENNAMODEL OXIDE2 ;          #OXIDE2 model starts here
    ANTENNACUMAREARATIO 500 ;      #OXIDE2 values
    ANTENNACUMDIFFAREARATIO 800 ;
    ANTENNAMODEL OXIDE3 ;
    ANTENNACUMAREARATIO 300 ;
    ANTENNACUMDIFFAREARATIO 600 ;
    ...
END metal1
```

`ANTENNASIDEAREAFACTOR` *value* `[DIFFUSEONLY]`

Specifies the multiply factor for the antenna metal side wall area calculation. `DIFFUSEONLY` specifies that the current antenna factor should only be used when the corresponding layer is connected to the diffusion.
Default: `1.0`
Type: Float

For more information on process antenna calculation, see Appendix C, "Calculating and Fixing Process Antenna Violations."

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`ANTENNASIDEAREARATIO` *`value`*

Specifies the antenna ratio, using the side wall area of the metal wire that is not connected to the diffusion diode. For more information on process antenna calculation, see Appendix C, "Calculating and Fixing Process Antenna Violations."
Type: Float

`AREA` *`minArea`*

Specifies the minimum metal area required for polygons on the layer. All polygons must have an area that is greater than or equal to *`minArea`*, if no `MINSIZE` rule exists. If a `MINSIZE` rule exists, all polygons must meet either the `MINSIZE` or the `AREA` rule. For an example using these rules, see Example 6-7 on page 641.
Type: Float, specified in microns squared

`CAPACITANCE CPERSQDIST` *`value`*

Specifies the capacitance for each square unit, in picofarads per square micron. This is used to model wire-to-ground capacitance.

`CAPMULTIPLIER` *`value`*

Specifies the multiplier for interconnect capacitance to account for increases in capacitance caused by nearby wires.
Default: `1`
Type: Integer

`DCCURRENTDENSITY`

Specifies how much DC current a wire on this layer of a given width can handle in units of milliamps per micron (mA/µm).

The true meaning of current density would have units of milliamps per square micron (mA/μm^2); however, the thickness of the metal layer is implicitly included, so the units in this table are milliamps per micron, where only the wire width varies.

The `DCCURRENTDENSITY` syntax is defined as follows:

```
AVERAGE
{ value
| WIDTH width_1 width_2 ... ;
    TABLEENTRIES value_1 value_2 ...
} ;
```

`AVERAGE`	Specifies the average current limit of the layer.
`value`	Specifies the current limit for the layer, in mA/µm.

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`WIDTH` Specifies wire width values, in microns. You can specify more than one wire width. If you specify multiple width values, the values must be specified in ascending order.
Type: Float

`TABLEENTRIES` Specifies the value of current density for each specified width, in mA/µm.

The final value for a given wire width is computed from a linear interpolation of the table values. The widths are not adjusted for any process shrinkage, so they should be correct for the “drawn width”.
Type: Float

Example 6-3 DC Current Density Statements

The following examples define DC current density tables:

```
LAYER met1
    ...
    DCCURRENTDENSITY AVERAGE    #avg. DC current limit for met1
    50.0 ;                      #mA/um for any width
```

(or)

```
    DCCURRENTDENSITY AVERAGE    #avg. DC current limit for met1
    WIDTH
      0.4 0.8 1.6 5.0 20.0 ;      #5 width values in microns
    TABLEENTRIES
    7.5 6.8 6.0 4.8 4.0 ;       #mA/um for 5 widths
    ...
END met1 ;
```

The `AVERAGE` current density at 0.4 µm is 7.5 mA/µm. Therefore, a 0.4 µm wide wire can carry 7.5 x .4 = 3.0 mA of `AVERAGE` DC current.

```
LAYER cut12
    ...
    DCCURRENTDENSITY AVERAGE    #avg. DC current limit for via cut12
    10.0 ;                      #mA/um squared for any cut area
```

(or)

```
    DCCURRENTDENSITY AVERAGE    #avg. DC current limit for via cut12
      CUTAREA 0.16 0.32 ;        #2 cut areas in µm2
    TABLEENTRIES
    10.0 9.0 ;                  #mA/um squared for 2 cut areas
```


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```
    ...
END cut12 ;
```

DENSITYCHECKSTEP *stepValue*

Specifies the stepping distance for metal density checks, in distance units.
Type: Float

DENSITYCHECKWINDOW *windowLength windowWidth*

Specifies the dimensions of the check window, in distance units.
Type: Float

DIAGMINEDGELENGTH *diagLength*

Specifies the minimum length for a diagonal edge. Any 45-degree diagonal edge must have a length that is greater than or equal to *diagLength*.
Type: Float, specified in microns

DIAGPITCH {*distance* | *diag45Distance diag135Distance*}

Specifies the 45-degree routing pitch for the layer. Pitch is used by the router to get the best routing density.
Default: None
Type: Float, specified in microns

distance — Specifies one pitch value that is used for both the 45-degree angle and 135-degree angle directions.

diag45Distance diag135Distance

Specifies the 45-degree angle pitch (the center-to-center space between 45-degree angle routes) and the 135-degree angle pitch.

DIAGSPACING *diagSpacing*

Specifies the minimum spacing allowed for a 45-degree angle shape.
Default: None
Type: Float, specified in microns

DIAGWIDTH *diagWidth*

Specifies the minimum width allowed for a 45-degree angle shape.
Default: None
Type: Float, specified in microns

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`DIRECTION {HORIZONTAL | VERTICAL | DIAG45 | DIAG135}`

Specifies the preferred routing direction. Automatic routing tools attempt to route in the preferred direction on a layer. A typical case is to route horizontally on layers *metal1* and *metal3*, and vertically on layer *metal2*.

`HORIZONTAL`	Routing parallel to the x axis is preferred.
`VERTICAL`	Routing parallel to the y axis is preferred.
`DIAG45`	Routing along a 45-degree angle is preferred.
`DIAG135`	Routing along a 135-degree angle is preferred.

Note: Angles are measured counterclockwise from the positive x axis.

`EDGECAPACITANCE` *`value`*

Specifies a floating-point value of peripheral capacitance, in picofarads per micron. The place-and-route tool uses this value in two situations:

- Estimating capacitance before routing
- Calculating segment capacitance after routing

For the second calculation, the tool uses *`value`* only if you set layer thickness, or layer height, to `0`. In this situation, the peripheral capacitance is used in the following formula:

segment capacitance = (*layer capacitance per square* x *segment width* x *segment length*) + (*peripheral capacitance* x 2 (*segment width* + *segment length*))

`FILLACTIVESPACING` *`spacing`*

Specifies the spacing between metal fills and active geometries.
Type: Float

`HEIGHT` *`distance`*

Specifies the distance from the top of the ground plane to the bottom of the interconnect.
Type: Float

`LAYER` *`layerName`*

Specifies the name for the layer. This name is used in later references to the layer.

`MASK` *`maskNum`*

Specifies how many masks for double- or triple-patterning will be used for this layer. The *`maskNum`* variable must be an integer greater than or equal to 2. Most applications only support values of 2 or 3.

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`MAXIMUMDENSITY` *`maxDensity`*

Specifies the maximum metal density allowed for the layer, as a percentage. The *`minDensity`* and *`maxDensity`* values represent the metal density range within which all areas of the design must fall. The metal density must be greater than or equal to *`minDensity`* and less than or equal to *`maxDensity`*.
Type: Float
Value: Between `0.0` and `100.0`

Example 6-4 Minimum and Maximum Density

The following example specifies a metal density range in which the minimum metal density must be greater than or equal to 20 percent and the maximum metal density must be less than or equal to 70 percent.

```
MINIMUMDENSITY 20.0 ;
MAXIMUMDENSITY 70.0 ;
```

`MAXWIDTH` *`width`*

Specifies the maximum wire width, in microns, allowed on the layer. Maximum wire width is defined as the smaller value of the width and height of the maximum enclosed rectangle. For example, `MAXWIDTH 10.0` specifies that the width of every wire on the layer must be less than or equal to 10.0 µm.
Type: Float

`MINENCLOSEDAREA` *`area`* `[WIDTH` *`width`*`]`

Specifies the minimum area size limit for an empty area that is enclosed by metal (that is, a donut hole formed by the metal).

`area`	Specifies the minimum area size of the hole, in microns squared. *Type:* Float
`width`	Applies the minimum area size limit only when a hole is created from a wire that has a width that is greater than *`width`*, in microns. If any of the wires that surround the donut hole are larger than this value, the rule applies. *Type:* Float

Example 6-5 Min Enclosed Area Statement

The following `MINENCLOSEDAREA` example specifies that a hole area must be greater than or equal to 0.40 μm^2.

```
LAYER m1
    ...
```


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```
    MINENCLOSEDAREA 0.40 ;
```

The following `MINENCLOSEDAREA` example specifies that a hole area must be greater than or equal to 0.30 μm^2. However, if any of the wires enclosing the hole have a width that is greater than 0.15 μm, then the hole area must be greater than or equal to 0.40 μm^2. If any of the wires enclosing the hole are larger than 0.50 μm, then the hole area must be greater than or equal to 0.80 μm^2.

```
LAYER m1
    ...
    MINENCLOSEDAREA 0.30 ;
    MINENCLOSEDAREA 0.40 WIDTH 0.15 ;
    MINENCLOSEDAREA 0.80 WIDTH 0.50 ;
```

`MINIMUMCUT`

Specifies the number of cuts a via must have when it is on a wide wire or pin whose width is greater than *`width`*. The `MINIMUMCUT` rule applies to all vias touching this particular metal layer. You can specify more than one `MINIMUMCUT` rule per layer. (See Example 6-6 on page 638.)

The `MINIMUMCUT` syntax is defined as follows:

```
[MINIMUMCUT numCuts WIDTH width
     [WITHIN cutDistance]
  [FROMABOVE | FROMBELOW]
  [LENGTH length WITHIN distance]
;] ...
```

`numCuts` — Specifies the number of cuts a via must have when it is on a wire or pin whose width is greater than *`width`*.
Type: Integer

`WIDTH` *`width`* — Specifies the width of the wire or pin, in microns.
Type: Float

`WITHIN` *`cutDistance`*

Indicates that *`numCuts`* via cuts must be less than *`cutDistance`* from each other in order to be counted together to meet the minimum cut rule. (See Figure 6-2 on page 640.)

`FROMABOVE | FROMBELOW`

Indicates whether the rule applies only to connections from above this layer or from below.
Default: The rule applies to connections from above and below.

`LENGTH` *`length`* `WITHIN` *`distance`*

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Indicates that the rule applies for thin wires directly connected to wide wires, if the wide wire has a width that is greater than *width* and a length that is greater than *length*, and the vias on the thin wire are less than *distance* from the wide wire. (See Figure 6-1 on page 639). The *length* value must be greater than or equal to the *width* value.

If LENGTH and WITHIN are present, this rule only checks the thin wire connected to a wide wire, and does not check the wide wire itself. A separate MINIMUMCUT *x* WIDTH *y* ; statement without LENGTH and WITHIN is required for any wide wire minimum cut rule.
Type: Float, specified in microns

Example 6-6 Minimum Cut Rules

The following MINIMUMCUT definitions show different ways to specify a MINIMUMCUT rule.

- **Minimum Cut Rule Example 1**

  The following syntax specifies that two via cuts are required for *metal4* wires that are greater than 0.5 µm when connecting from *metal3* or *metal5*.

  ```
  LAYER metal4
      MINIMUMCUT 2 WIDTH 0.5 ;
  ```

- **Minimum Cut Rule Example 2**

  The following syntax specifies that four via cuts are required for *metal4* wires that are greater than 0.7 µm, when connecting from *metal3*.

  ```
  LAYER metal4
      MINIMUMCUT 4 WIDTH 0.7 FROMBELOW ;
  ```

- **Minimum Cut Rule Example 3**

  The following syntax specifies that four via cuts are required for *metal4* wires that are greater than 1.0 µm, when connecting from *metal5*.

  ```
  LAYER metal4
      MINIMUMCUT 4 WIDTH 1.0 FROMABOVE ;
  ```

- **Minimum Cut Rule Example 4**

  The following syntax specifies that two via cuts are required for *metal4* wires that are greater than 1.1 µm wide and greater than 20.0 µm long, and the via cut is less than 5.0 µm from the wide wire. Figure 6-1 on page 639 illustrates this example.

  ```
  LAYER metal4
  ```


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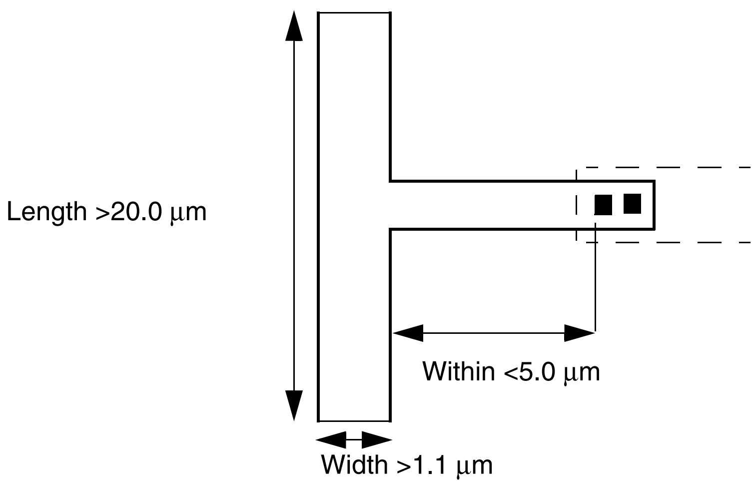
```
MINIMUMCUT 2 WIDTH 1.1 LENGTH 20.0 WITHIN 5.0 ;
```

Figure 6-1 Minimum Cut Rule



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■ **Minimum Cut Rule Example 5**

The following syntax specifies that two via cuts are required for *metal4* wires that are greater than 1.0 µm wide. The via cuts must be less than 0.3 µm from each other in order to meet the minimum cut rule. Figure 6-2 on page 640 illustrates this example.

```
MINIMUMCUT 2 WIDTH 1.0 WITHIN 0.3 ;
```

Figure 6-2 Minimum Cut Within Rule

a) Violation. The wire width is > 1.0 µm, therefore 2 cuts are needed. However, the 2 cuts are >= 0.3 µm apart, therefore they cannot be counted together.

b) Okay. The wire width is > 1.0 µm, therefore 2 cuts are needed. The 2 cuts are < 0.3 µm apart, therefore they are counted together and meet the rule.

`MINIMUMDENSITY` *`minDensity`*

Specifies the minimum metal density allowed for the layer, as a percentage. The *`minDensity`* and *`maxDensity`* values represent the metal density range within which all areas of the design must fall. The metal density must be greater than or equal to *`minDensity`* and less than or equal to *`maxDensity`*. For an example of this statement, see Example 6-4 on page 636.
Type: Float
Value: Between `0.0` and `100.0`

`MINSIZE` *`minWidth minLength`* `[`*`minWidth2 minLength2`*`]`

Specifies the minimum width and length of a rectangle that must be able to fit somewhere within each polygon on this layer (see Figure 6-3 on page 642). All polygons must meet this `MINSIZE` rule, if no `AREA` rule is specified. If an `AREA` rule is specified, all polygons must meet either the `MINSIZE` or the `AREA` rule.

You can specify multiple rectangles by specifying a list of *`minWidth2`* and *`minLength2`* values. If more than one rectangle is specified, the `MINSIZE` rule is satisfied if any of the rectangles can fit within the polygon.
Type: Float, specified in microns, for all values

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Example 6-7 Minimum Size and Area Rules

Assume the following minimum size and area rules:

```
LAYER metal1
    TYPE ROUTING ;
    AREA 0.07 ;           #0.20 um x 0.35 um = 0.07 um2
    MINSIZE 0.14 0.30 ;   #0.14 um x 0.30 um = 0.042 um2
    ....
```

Figure 6-3 on page 642 illustrates how these rules behave when one or both of the rules are present in the `LAYER` statement:

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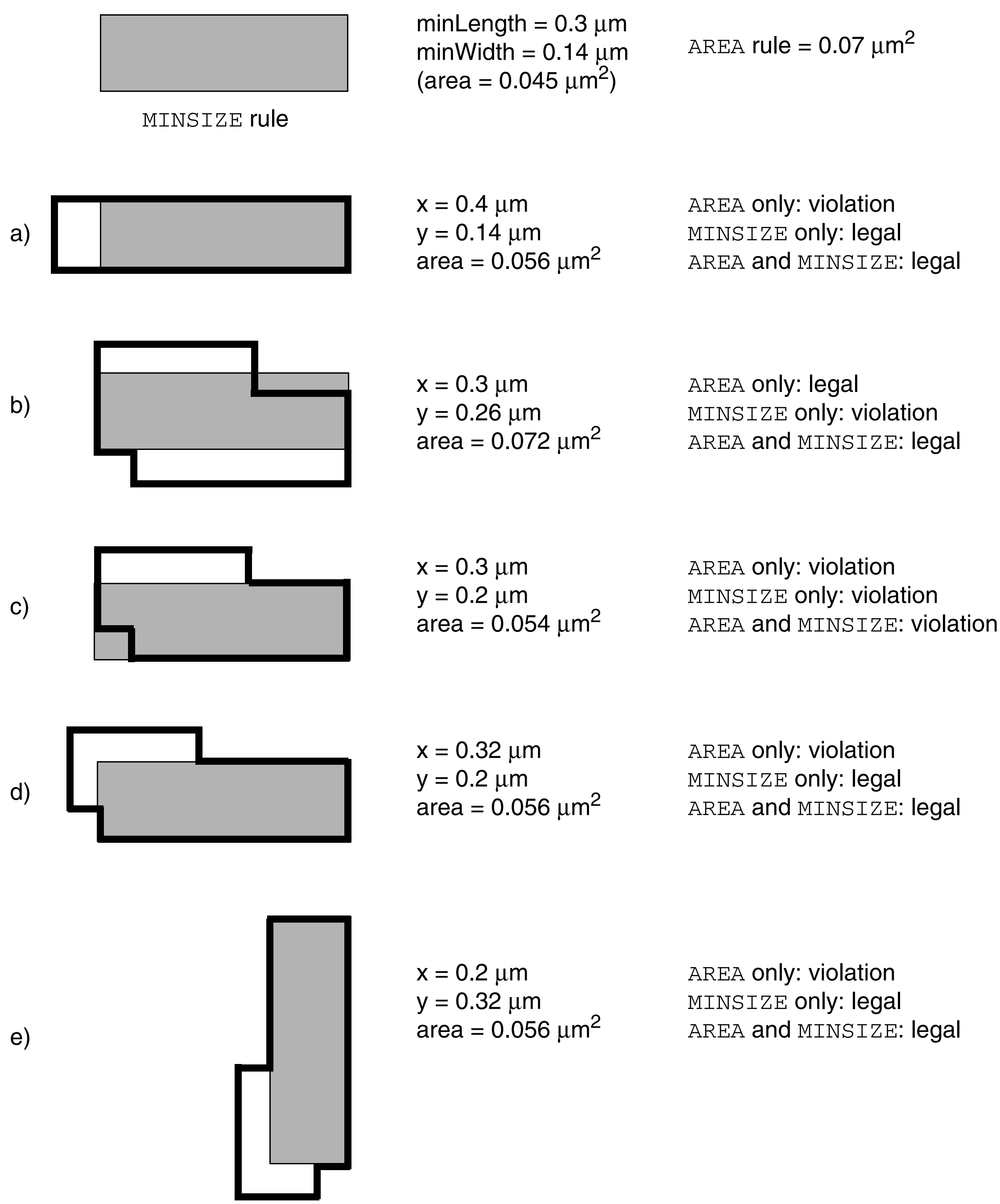
Figure 6-3 Minimum Size and Area Rules



The following statement defines a `MINSIZE` rule that specifies that every polygon must have a minimum area of 0.07 µm^2, or that a rectangle of 0.14 x 0.30 µm must be able to fit within the polygon, or that a rectangle of 0.16 x 0.26 µm must be able to fit within the polygon:

```
LAYER metal1
```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```
    TYPE ROUTING ;
    AREA 0.07 ;                       #0.20 x 0.35 um = 0.07 um2
    MINSIZE 0.14 0.30 0.16 0.26 ;    #0.14 x 0.30 um = 0.042 um2
                                      #0.16 x 0.26 um = 0.0416 um2
    ...
END metal1
```

MINSTEP

Specifies the minimum step size, or shortest edge length, for a shape. The MINSTEP rule ensures that Optical Pattern Correction (OPC) can be performed during mask creation for the shape.

Note: A single layer should only have one type of MINSTEP rule. It should include either INSIDECORNER, OUTSIDECORNER, or STEP statements (with an optional LENGTHSUM value), or one LENGTHSUM statement, or one MAXEDGES statement.

For an illustration of the MINSTEP rules, see Figure 6-4 on page 647. For an example, see Example 6-8 on page 645.

The syntax for MINSTEP is as follows:

```
[MINSTEP minStepLength
  [ [INSIDECORNER | OUTSIDECORNER | STEP]
       [LENGTHSUM maxLength]
  | [MAXEDGES maxEdges] ;]
```

minStepLength Specifies the minimum step size, or shortest edge length, for a shape. The edge of a shape must be greater than or equal to this value, or a violation occurs.
Type: Float, specified in microns

INSIDECORNER Indicates that a violation occurs if two or more consecutive edges of an inside corner are less than *minStepLength*.

If LENGTHSUM is also defined, a violation only occurs if the total length of all consecutive edges (that are less than *minStepLength*) is greater than *maxLength*.

Shape b in Figure 6-4 on page 647 shows an inside corner. It is considered an inside corner because the two edges >= *minStepLength* (shown with thick lines) that abut the consecutive short edges < *minStepLength* (shown with dashed lines) form an inside corner (or concave shape).
Default: OUTSIDECORNER

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`OUTSIDECORNER` Indicates that a violation occurs if two or more consecutive edges of an outside corner are less than *minStepLength*.

If `LENGTHSUM` is also defined, a violation only occurs if the total length of all consecutive edges (that are less than *minStepLength*) is greater than *maxLength*.

Shape `a` in Figure 6-4 on page 647 shows an outside corner. It is considered an outside corner because the two edges >= *minStepLength* (shown with thick lines) that abut the consecutive short edges < *minStepLength* (shown with dashed lines) form an outside corner (or convex shape).

Note: This is the default rule, if `INSIDECORNER`, `OUTSIDECORNER`, or `STEP` is not specified.

`STEP` Indicates that a violation occurs if one or more consecutive edges of a step are less than *minStepLength*.

If `LENGTHSUM` is also defined, a violation only occurs if the total length of all consecutive edges (that are less than *minStepLength*) is greater than *maxLength*.

Shape f in Figure 6-4 on page 647 shows a step. It is considered a step because the two edges >= *minStepLength* (shown with thick lines) that abut the consecutive short edges < *minStepLength* (shown with dashed lines) form a step instead of a corner.
Default: `OUTSIDECORNER`

`LENGTHSUM` *maxLength*

Specifies the maximum total length of consecutive short edges (edges that are less than *minStepLength*) that OPC can correct without causing new DRC violations.

If the total length of the edges is greater than *maxLength*, a violation occurs. No violation occurs if the total length is less than or equal to *maxLength*.

`MAXEDGES` *maxEdges*

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Specifies that up to *maxEdges* consecutive edges that are less than *minStepLength* in length are allowed, but more than *maxEdges* in a row is a violation. Typically, most tools only allow a *maxEdges* value of `0`, `1`, or `2`. A *maxEdges* value of `0` means that no edge can be less than *minStepLength*.
Type: Integer

Note: The *maxEdges* value of 1 will check the cases covered by `OUTSIDECORNER` and `INSIDECORNER`. However, there is no relationship between `MAXEDGES` and `STEP`.

Example 6-8 Minimum Step Rules

- The following table shows the results of the specified `MINSTEP` rules using the shapes in Figure 6-4 on page 647. For these rules, assume *minStepLength* equals 0.05 µm, and that each dashed edge is 0.04 µm in length.

MINSTEP Rule	Result
`MINSTEP 0.05 ;`	`OUTSIDECORNER` is the default behavior. Therefore, shapes `a` and `d` are violations because their consecutive edges are less than 0.05 µm. Shapes `b`, `c`, `e`, and `f` are not outside corner checks.
`MINSTEP 0.04 ;`	`OUTSIDECORNER` is the default behavior. Therefore, shapes `a` and `d` are checked and are legal because their consecutive edges are greater than or equal to 0.04 µm.
`MINSTEP 0.05 LENGTHSUM 0.08 ;`	Shape `a` is legal because its consecutive edges are less than 0.05 µm, and the total length of the edges is less than or equal to 0.08 µm. Shape `d` is a violation because even though its consecutive edges are less than 0.05 µm, the total length of the edges is greater than 0.08 µm.
`MINSTEP 0.05 LENGTHSUM 0.16 ;`	Shapes `a` and `d` are legal because the total length of their consecutive edges is less than or equal to 0.16 µm.

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MINSTEP Rule	Result
`MINSTEP 0.05 INSIDECORNER ;`	Shapes `b` and `e` are violations because their consecutive edges are less than 0.05 µm. Shapes `a`, `c`, `d`, and `f` are not inside corner checks.
`MINSTEP 0.05 INSIDECORNER LENGTHSUM 0.15 ;`	Shape `b` is legal because its consecutive edges are less than 0.05 µm, and the total length of the edges is less than or equal to 0.15 µm. Shape `e` is a violation because even though its consecutive edges are less than 0.05 µm, the total length of the edges is greater than 0.15 µm.
`MINSTEP 0.05 STEP ;`	Shapes `c` and `f` are violations because their consecutive edges are less than 0.05 µm. Shapes `a`, `b`, `d`, and `e` are not step checks.
`MINSTEP 0.05 STEP LENGTHSUM 0.08 ;`	Shape `c` is legal because its consecutive edges are less than 0.05 µm, and the total length of the edges is less than or equal to 0.08 µm. Shape `f` is a violation because even though its consecutive edges are less than 0.05 µm, the total length of the edges is greater than 0.08 µm.
`MINSTEP 0.04 STEP ;`	Shapes `c` and `f` are legal because their consecutive edges are greater than or equal to 0.04 µm.

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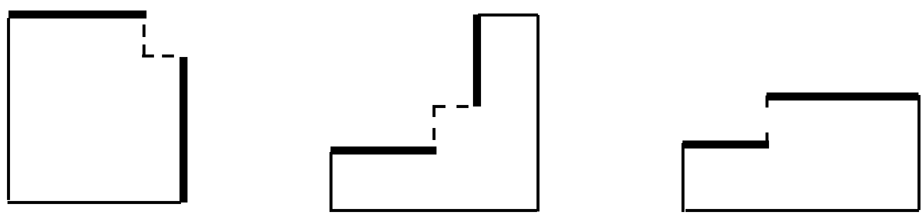
Figure 6-4

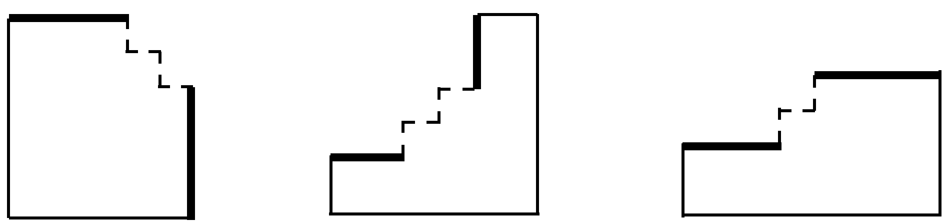
Note: All dashed edges are 0.04 µm in length.



a) `OUTSIDECORNER` 0.08 µm `LENGTHSUM`

b) `INSIDECORNER` 0.08 µm `LENGTHSUM`

c) `STEP` 0.04 µm



d) `OUTSIDECORNER` 0.16 µm `LENGTHSUM`

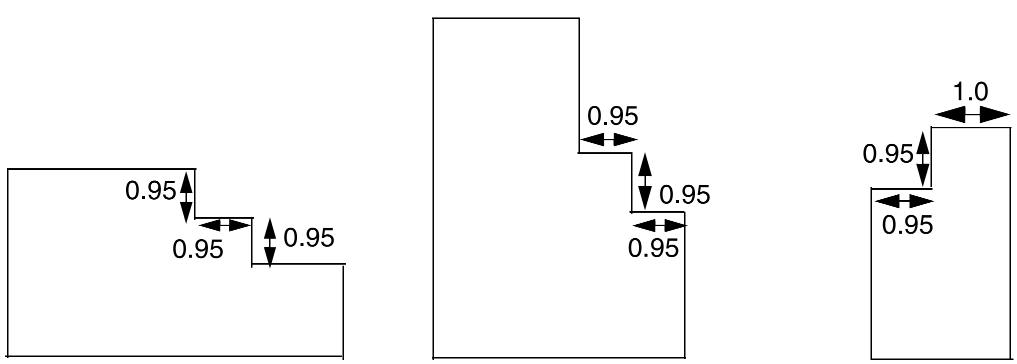
e) `INSIDECORNER` 0.16 µm `LENGTHSUM`

f) `STEP` 0.12 µm `LENGTHSUM`

- Figure 6-5 on page 647 shows the results of the following `MINSTEP MAXEDGES` rule:

```
MINSTEP 1.0 MAXEDGES 2 ;
```

Figure 6-5



a) Violation; there is more than two edges in a row that are < 1.0 µm in length.

b) Violation; there is more than two edges in a row that are < 1.0 µm in length.

c) No violation; there are only two edges in a row that ares < 1.0 µm in length.

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`MINWIDTH` *`width`*

Specifies the minimum legal object width on the routing layer. For example, `MINWIDTH 0.15` specifies that the width of every object must be greater than or equal to 0.15 µm. This value is used for verification purposes, and does not affect the routing width. The `WIDTH` statement defines the default routing width on the layer.
Default: The value of the `WIDTH` statement
Type: Float, specified in microns

`OFFSET {`*`distance`*` | `*`xDistance yDistance`*`}`

Specifies the offset for the routing grid from the design origin for the layer. This value is used to align routing tracks with standard cell boundaries, which helps routers get good on-grid access to the cell pin shapes. For best routing results, most standard cells have a 1/2 pitch offset between the `MACRO SIZE` boundary and the center of cell pins that should be aligned with the routing grid. Normally, it is best to not set the `OFFSET` value, so the software can analyze the library to determine the best offset values to use, but in some cases it is necessary to force a specific offset.

Generally, it is best for all of the horizontal layers to have the same offset and all of the vertical layers to have the same offset, so that routing grids on different layers align with each other. Higher layers can have a larger pitch, but for best results, they should still align with a lower layer routing grid every few tracks to make stacked-vias more efficient.

Default: The software is allowed to determine its own offset values for preferred and non-preferred routing tracks.
Type: Float, specified in microns

`distance` Specifies the offset value that is used for the preferred direction routing tracks.

`xDistance yDistance`

Specifies the x offset for vertical routing tracks, and the y offset for horizontal routing tracks.

`PITCH {`*`distance`*` | `*`xDistance yDistance`*`}`

Specifies the required routing pitch for the layer. Pitch is used to generate the routing grid (the DEF `TRACKS`). For more information, see “Routing Pitch” on page 969.
Type: Float, specified in microns

`distance` Specifies one pitch value that is used for both the x and y pitch.

`xDistance yDistance`

Specifies the x pitch (the space between each vertical routing track), and the y pitch (the space between each horizontal routing track).

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`PROPERTY` *`propName propVal`*

Specifies a numerical or string value for a layer property defined in the `PROPERTYDEFINITIONS` statement. The *`propName`* you specify must match the *`propName`* listed in the `PROPERTYDEFINITIONS` statement.

`PROTRUSIONWIDTH` *`width1`* `LENGTH` *`length`* `WIDTH` *`width2`*

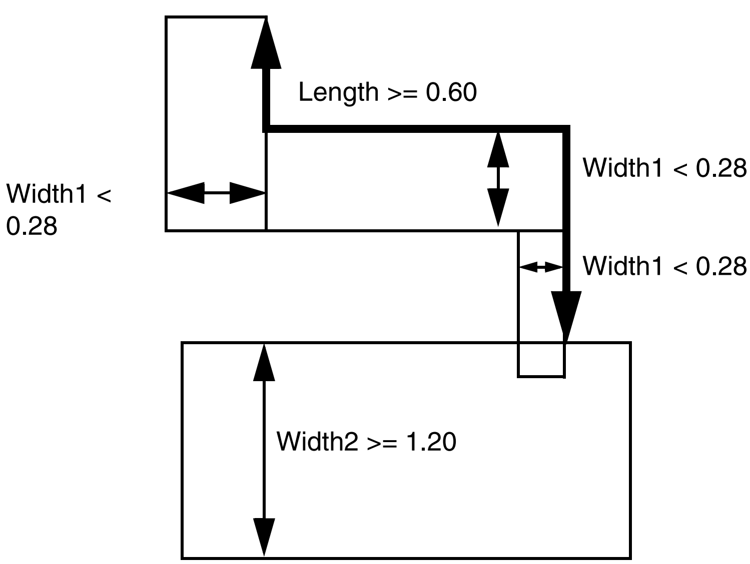
Specifies that the width of a protrusion must be greater than or equal to *`width1`* if it is shorter than *`length`*, and it connects to a wire that has a width greater than or equal to *`width2`* (see Figure 6-6 on page 649). Length is determined by the shortest possible path among all of the protrusion wires with width smaller *`width1`*, and is measured by the shortest outside edges of the wires.
Type: Float, specified in microns

Example 6-9 Protrusion

The following example specifies that a protrusion must have a width that is greater than or equal to 0.28 µm, if the length of the protrusion is less than 0.60 µm and the wire it connects to has a width that is greater than or equal to 1.20 µm.

```
LAYER m1
    ...
    PROTRUSIONWIDTH 0.28 LENGTH 0.60 WIDTH 1.20 ;
    ...
```

Figure 6-6



a) Okay; length >= 0.60 among all protrusion wires.

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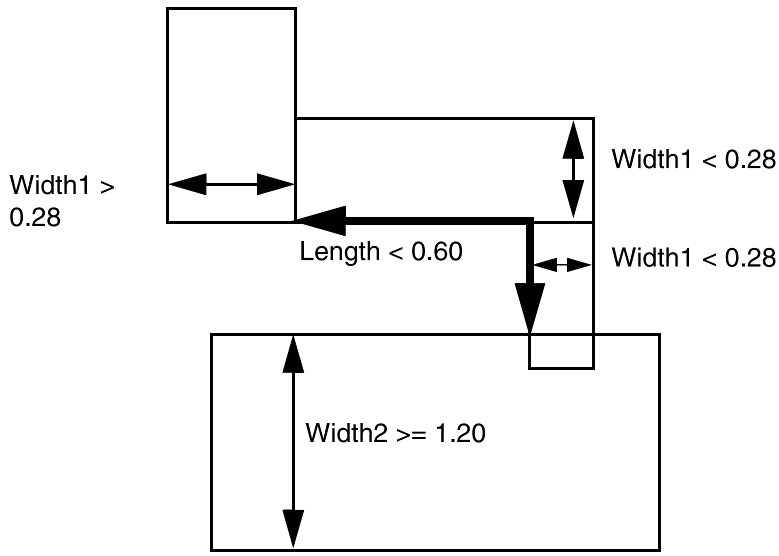
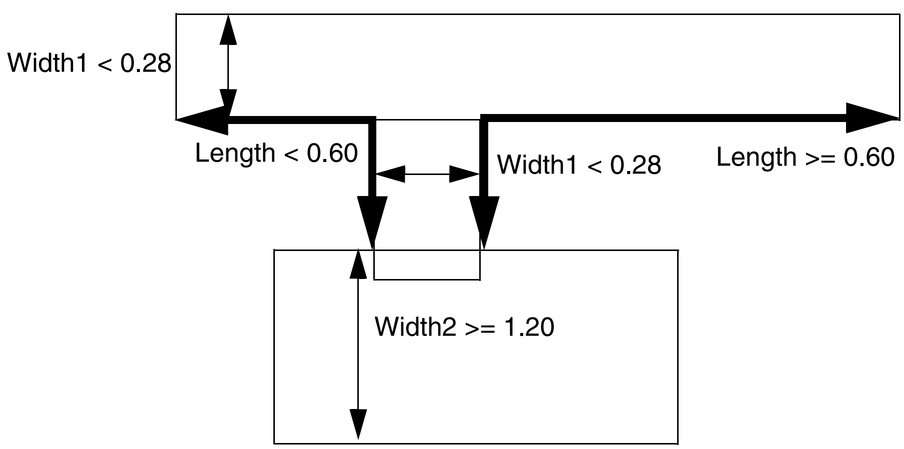


b) Violation; length < 0.60 is measured among all protrusion wires with width < 0.28.



c) Violation; length < 0.60 for shortest possible path.

If the given value of `LENGTH` in `PROTRUSIONWIDTH` is zero, then the length of the protrusion wire is irrelevant. In this case, the width of the protrusion wire should always be checked independent of the length of the wire. The following example illustrates this rule:

```
PROTRUSIONWIDTH 0.05 LENGTH 0 WIDTH 0.11 ; " ;
```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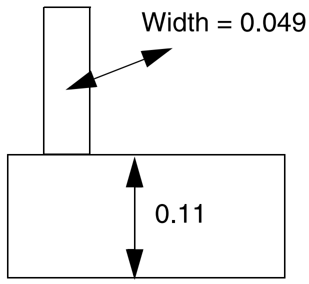
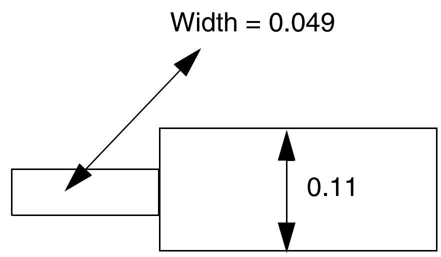


a) Violation, the width of the protrusion wire must be >= 0.05, and its length is irrelevant.



b) Violation, it does not matter which sides the stub wire protruded from.

`RESISTANCE RPERSQ` *`value`*

Specifies the resistance for a square of wire, in ohms per square. The resistance of a wire can be defined as:

RPERSQU x *wire length/wire width*

`SHRINKAGE` *`distance`*

Specifies the value to account for shrinkage of interconnect wiring due to the etching process. Actual wire widths are determined by subtracting this constant value.
Type: Float

`SPACING`

Specifies the spacing rules to use for wiring on the layer. You can specify more than one spacing rule for a layer. See "Using Spacing Rules" on page 657.

The syntax for describing spacing rules is defined as follows:

```
[SPACING minSpacing
  [ RANGE minWidth maxWidth
      [ USELENGTHTHRESHOLD
      | INFLUENCE influenceLength
          [RANGE stubMinWidth stubMaxWidth]
      | RANGE minWidth maxWidth]
  | LENGTHTHRESHOLD maxLength
      [RANGE minWidth maxWidth]
  | ENDOFLINE eolWidth WITHIN eolWithin
      [PARALLELEDGE parSpace WITHIN parWithin
          [TWOEDGES]]
  | SAMENET [PGONLY]
  | NOTCHLENGTH minNotchLength
  | ENDOFNOTCHWIDTH endOfNotchWidth
      NOTCHSPACING minNotchSpacing
        NOTCHLENGTH minNotchLength
```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```
  ]
;] ...
```

`SPACING` *`minSpacing`*

Specifies the default minimum spacing, in microns, allowed between two geometries on different nets.
Type: Float

`RANGE` *`minWidth maxWidth`*

Indicates that the minimum spacing rule applies to objects on the layer with widths in the indicated `RANGE` (that is, widths that are greater than or equal to *`minWidth`* and less than or equal to *`maxWidth`*). If you do not specify a range, the rule applies to all objects.
Type: Float

Note: If you specify multiple `RANGE` rules, the range values should not overlap.

`USELENGTHTHRESHOLD`

Indicates that the threshold spacing rule should be used if the other object meets the previous `LENGTHTHRESHOLD` value.

Note: `USELENGTHTHRESHOLD` and `LENGTHTHRESHOLD` are being made obsolete in the next LEF/DEF release. Use `SPACINGTABLE PARALLELRUNLENGTH` instead.

`INFLUENCE` *`influenceLength`*
`[RANGE` *`stubMinWidth stubMaxWidth`*`]`

Indicates that any length of the stub wire that is less than or equal to *`influenceLength`* from the wide wire inherits the wide wire spacing.
Type: Float

The influence rule applies to stub wires on the layer with widths in the indicated `RANGE` (that is, widths that are greater than or equal to *`stubMinWidth`* and less than or equal to *`stubMaxWidth`*). If you do not specify a range, the rule applies to all stub wires.
Type: Float

Note: Specifying the `INFLUENCE` keyword denotes that the statement only checks the influence rule, and does *not* check normal spacing. You must also specify a separate `SPACING` statement for normal spacing checks.

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`RANGE` *minWidth maxWidth*

Specifies an optional second width range. The spacing rule applies if the widths of both objects fall in the ranges defined (each object in a different range). For an object's width to fall in a range, it must be greater than or equal to *minWidth* and less than or equal to *maxWidth*.
Type: Float

Note: If you specify multiple `RANGE` rules, the range values should not overlap.

`LENGTHTHRESHOLD` *maxLength*
`[RANGE` *minWidth maxWidth*`]`

Specifies the maximum parallel run length or projected length with an adjacent metal object for this spacing value. The *minSpacing* value should be less than or equal to the "default" *minSpacing* value when no `LENGTHTHRESHOLD` is specified for this range of widths. For an example, see "Using Spacing Rules" on page 657.

The threshold spacing rule applies to objects with widths in the indicated `RANGE` (that is, widths that are greater than or equal to *minWidth* and less than or equal to *maxWidth*). If you do not specify a range, the rule applies to all objects.
Type: Float

Note: If you specify multiple `RANGE` rules, the range values should not overlap.

Note: `USELENGTHTHRESHOLD` and `LENGTHTHRESHOLD` are being made obsolete in the next LEF/DEF release. Use `SPACINGTABLE PARALLELRUNLENGTH` instead.

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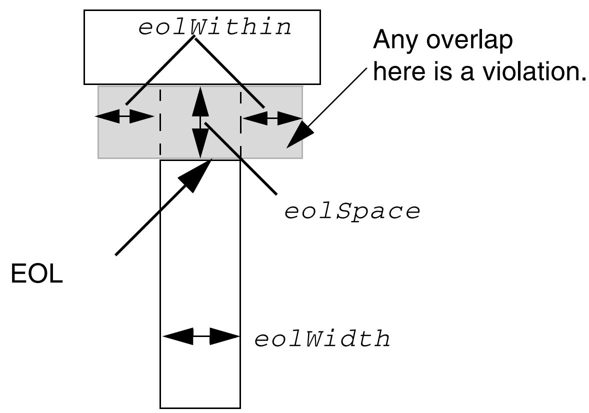
`ENDOFLINE` *`eolWidth`* `WITHIN` *`eolWithin`*

Indicates that an edge that is shorter than *`eolWidth`*, noted as end-of-line (EOL from now on) edge requires spacing greater than or equal to *`eolSpace`* beyond the EOL anywhere within (that is, less than) *`eolWithin`* distance (see Figure 6-7 on page 654).

Typically, *`eolSpace`* is slightly larger than the minimum allowed spacing on the layer. The *`eolWithin`* value must be less than the minimum allowed spacing.

Figure 6-7



a) EOL width < *`eolWidth`* requires *`eolSpace`* beyond EOL to either side by < *`eolWithin`* distance.

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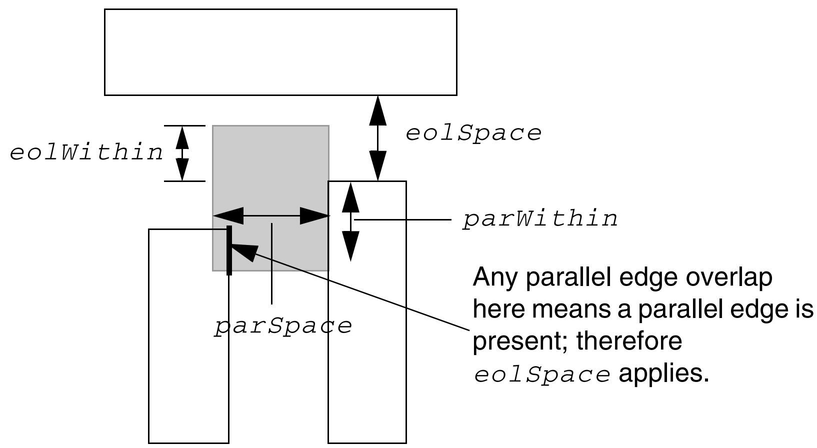
```
PARALLELEDGE parSpace WITHIN parWithin
    [TWOEDGES]
```

Indicates the EOL rule applies only if there is a parallel edge that is less than *parSpace* away, and is also less than *parWithin* from the EOL and *eolWithin* beyond the EOL (see Figure 6-8 on page 655).

Figure 6-8



b) EOL space rule with PARALLELEDGE is triggered only if there is a parallel edge that overlaps inside the illustrated shaded box.

If TWOEDGES is specified, the EOL rule applies only if there are two parallel edges that meet the PARALLELEDGE *parSpace*, *eolWithin*, and *parWithin* parameters (see Figure 6-9 on page 656).

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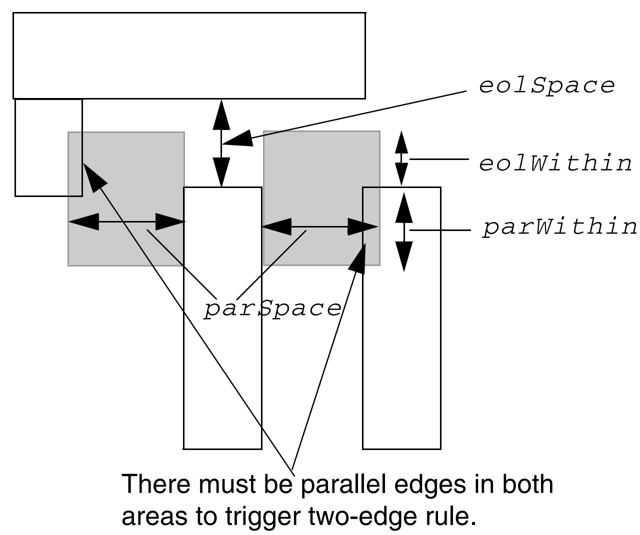
Figure 6-9



c) EOL rule with `TWOEDGES` is triggered only if both sides have parallel edge overlaps inside the illustrated shaded boxes.

`SAMENET [PGONLY]`

Indicates that the *`minSpacing`* value only applies to same-net metal. If `PGONLY` also is specified, the *`minSpacing`* value only applies to same-net metal that is a power or ground net.

This rule typically is used when a technology has wider spacing for wider width wires; however, it still allows minimum spacing for same-net wires, even if they are wide. (See Example 6-11 on page 664.)

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`NOTCHLENGTH` *`minNotchLength`*

Indicates that any notch with a notch length less than *`minNotchLength`* must have notch spacing greater than or equal to *`minSpacing`*. (See illustration `a` in Figure 6-16 on page 665.)

The value you specify for *`minSpacing`* should be only slightly larger than the normal minimum spacing rule (typically, between 1x and 1.5x minimum spacing).
Type: Float, specified in microns

If the value of the specified notch length is zero, then the length of the notch is irrevelant. In other words, the spacing of a notch should always be checked independent of its length.

Note: You can specify only one `NOTCHLENGTH` rule per layer.

`ENDOFNOTCHWIDTH` *`endOfNotchWidth`*
  `NOTCHSPACING` *`minNotchSpacing`*
    `NOTCHLENGTH` *`minNotchLength`*

Indicates that the notch metal at the bottom end of a U-shaped notch requires spacing that is greater than or equal to *`minSpacing`*, if the notch has a width that is less than *`endOfNotchWidth`*, notch spacing that is less than or equal to *`minNotchSpacing`*, and notch length that is greater than or equal to *`minNotchLength`*. The spacing is required for the extent of the notch.

The values you specify for *`notchSpacing`* and *`minSpacing`* should be only slightly larger than the normal minimum spacing rule (typically between 1x and 1.5x minimum spacing). The value you specify for *`endOfNotchWidth`* should be only slightly larger than the minimum width rule (typically, between 1x and 1.5x minimum width).
Type: Float, specified in microns (for all values)

Note: You can specify only one `ENDOFNOTCHWIDTH` rule per layer.

Using Spacing Rules

Spacing rules apply to pin-to-wire, obstruction-to-wire, via-to-wire, and wire-to-wire spacing. These requirements specify the default minimum spacing allowed between two geometries on different nets.

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When defined with a `RANGE` argument, a spacing value applies to all objects with widths within a specified range. That is, the rule applies to objects whose widths are greater than or equal to the specified minimum width and less than or equal to the specified maximum width.

Note: If you specify multiple `RANGE` arguments, the `RANGE` values should not overlap.

In the following example, the default minimum allowed spacing between two adjacent objects is 0.3 µm. However, for objects between 0.5 and 1.0 µm in width, the spacing is 0.4 µm. For objects between 1.01 and 2.0 µm in width, the spacing is 0.5 µm.

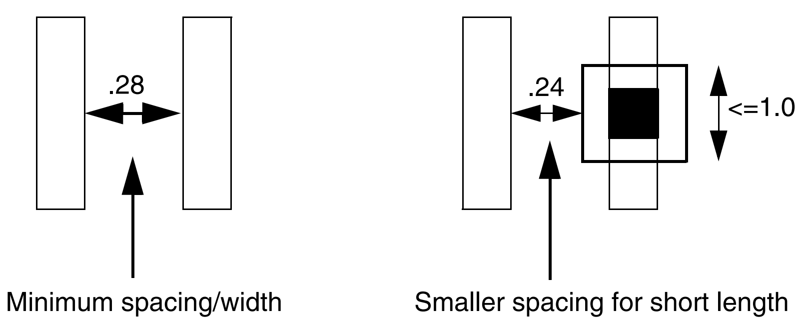
```
SPACING 0.3 ;
SPACING 0.4 RANGE 0.5  1.0 ;
SPACING 0.5 RANGE 1.01 2.0 ; #The RANGE begins at 1.01 and not 1.0 because
                             #RANGE values should not overlap.
```

Threshold spacing is a function of both the wire width and the length of the neighboring object. It is typically used when vias are wider than the wire to allow tighter wire-to-wire spacing, even when the vias are present.

In the following example, a slightly tighter spacing of .24 µm is needed if the other object is less than or equal to 1.0 µm in length (see Figure 6-10 on page 658).

```
SPACING 0.28 ;
SPACING 0.24 LENGTHTHRESHOLD 1.0 ;
```

Figure 6-10



The `USELENGTHTHRESHOLD` argument specifies that the threshold spacing rule should be applied if the other object meets the previous `LENGTHTHRESHOLD` value.

In the following example, a larger spacing of 0.32 µm is needed for wire widths between 1.5 and 9.99 µm. However, if the other object is less than or equal to 1.0 µm in length, the smaller .0.28 µm spacing is applied (see Figure 6-11 on page 659).

```
SPACING 0.28 ;                       #Default minimum spacing is >=0.28 um.
SPACING 0.28 LENGTHTHRESHOLD 1.0 ;   #For short parallel lengths of <= 1.0 um,
```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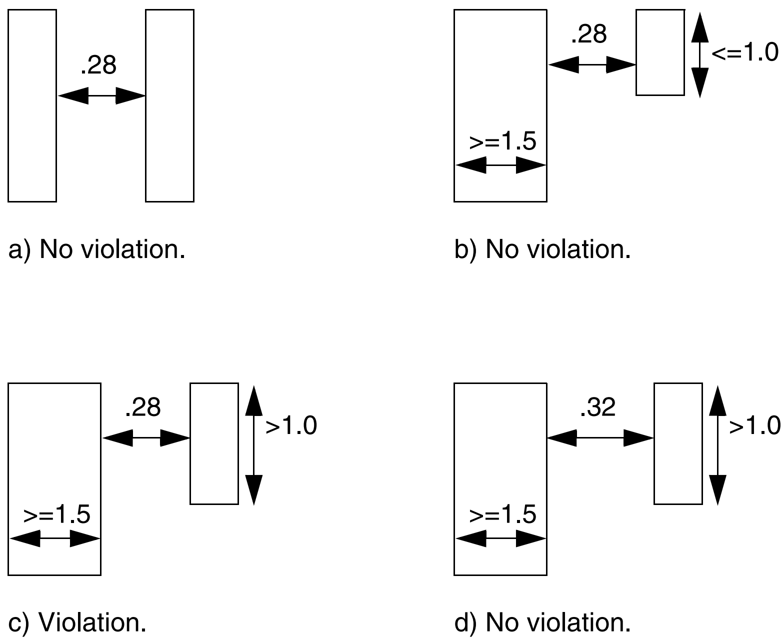
```
                                    #0.28 spacing is allowed.
SPACING 0.32 RANGE 1.5 9.99 USELENGTHTHRESHOLD ;
                                    #Wide wires with 1.5 <= width <=9.99 need
                                    #0.32 spacing unless the parallel run
                                    #length is <= 1.0 from the previous rule.
```

Figure 6-11



Influence spacing rules are used to support the inheritance of wide wire spacing by nets connected to the wide wires. For example, a larger spacing is needed for stub wires attached to large objects like pre-routed power wires. A piece of metal connecting to a wider wire will inherit spacing rules for a user-defined distance from the wider wire.

In Figure 6-12 on page 660, a minimum space of N is required between two metal lines when at least one metal line has a width that is >= Y. This spacing must be maintained for any small piece of metal (<Y) that is connected to the wide metal within X range of the wide metal. Outside of this range, normal spacing rules (Z) apply.

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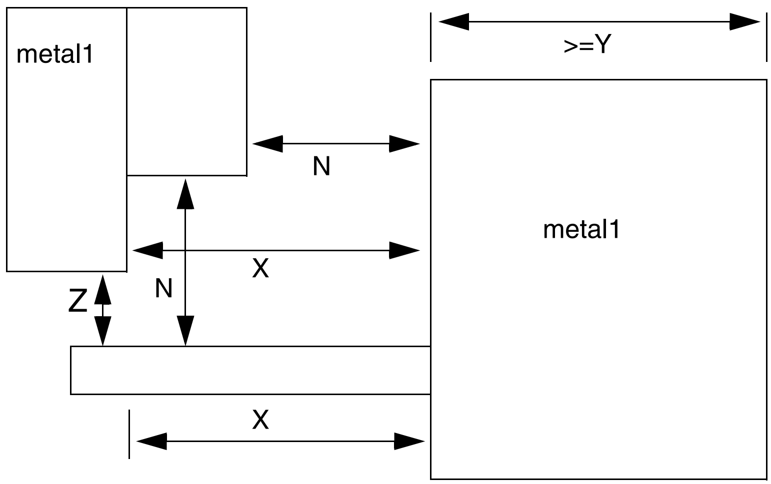
Figure 6-12



In the following example, the 0.5 µm spacing applies for the first 1.0 µm of the stub sticking out from the large object. This rule only applies to the stub wire; the previous rule must be included for the wide wire spacing. The `SPACING 0.5 RANGE 2.01 2000.0` statement is required to get extra spacing for the wide-wire itself.

```
SPACING 0.5 RANGE 2.01 2000.0 ;
SPACING 0.28 ;              #Minimum spacing is >= 0.28 um.
SPACING 0.5 RANGE 2.01 2000.0 ;     #wide-wire >= 2.01 um wide requires 0.5um spacing
SPACING 0.5 RANGE 2.01 2000.0 INFLUENCE 1.000 ;
                            #Stub wires <= 1.0 um from wide wires >= 2.01
                            #require 0.5 um spacing.
```

Some processes only need the `INFLUENCE` rule for certain widths of the stub wire. In the following example, the 0.5 µm spacing is required only for stub wires between 0.5 and 1.0 µm in width.

```
SPACING 0.28 ;         #Minimum spacing is >= 0.28 um.
SPACING 0.5 RANGE 2.01 2000.0 ;     #wide-wire >= 2.01 um wide requires 0.5um spacing
SPACING 0.5 RANGE 2.01 2000.0 INFLUENCE 1.00 RANGE 0.5 1.0 ;
                       #Stub wires with 0.5 <= width <= 1.0, and <= 1.0 um from
                       #wide wide wires >= 2.01 require 0.5 um spacing.
```

Example 6-10 EOL Spacing Rules

- If you include the following routing layer rules in your LEF file:

  ```
  SPACING 1.0 ;             #minimum spacing is 1.0 µm
  SPACING 1.2 ENDOFLINE 1.3 WITHIN 0.6 ;
  ```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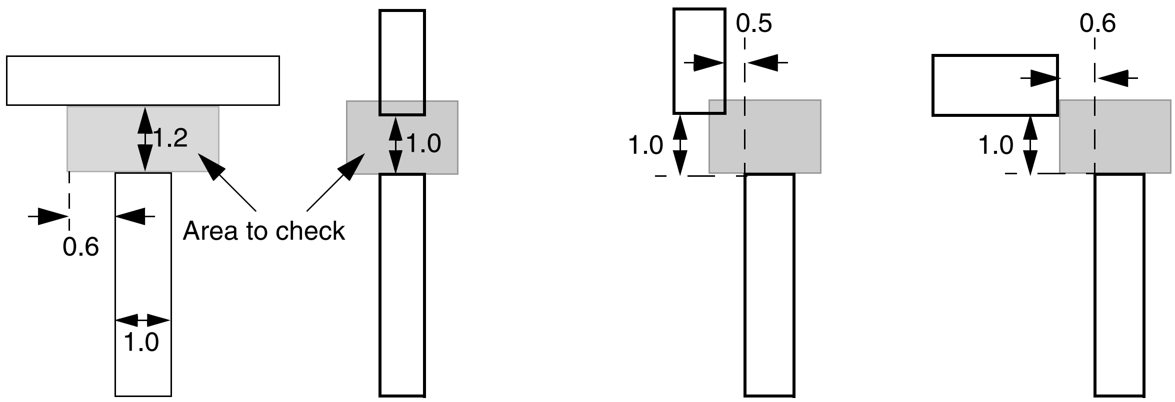
Any EOL that is less than 1.3 µm wide requires spacing that is greater than or equal to 1.2 µm beyond the EOL, within 0.6 µm to either side. Figure 6-13 on page 661 includes examples of legal spacing for, and violations of, this rule.

Figure 6-13



a) No violation. Has 1.2 µm spacing.

b) Violation; has only 1.0 µm spacing.

c) Violation; has <1.2 µm spacing within 0.6 µm away.

d) No violation. Has >= 1.2 µm spacing within (<) 6 µm away.

■ If you include the following routing layer rules in your LEF file:

```
SPACING 1.0 ;             #minimum spacing is 1.0 µm
SPACING 1.2 ENDOFLINE 1.3 WITHIN 0.6 PARALLELEDGE 1.1 WITHIN 0.5 ;
```

Any line that is less than 1.3 µm wide, with a parallel edge that is less than 1.1 µm away, and is within 0.5 µm of the EOL, requires spacing greater than or equal to 1.2 µm beyond the EOL, within 0.6 µm to either side of the EOL. Figure 6-14 on page 662 includes examples of legal spacing for, and violations of, this rule.

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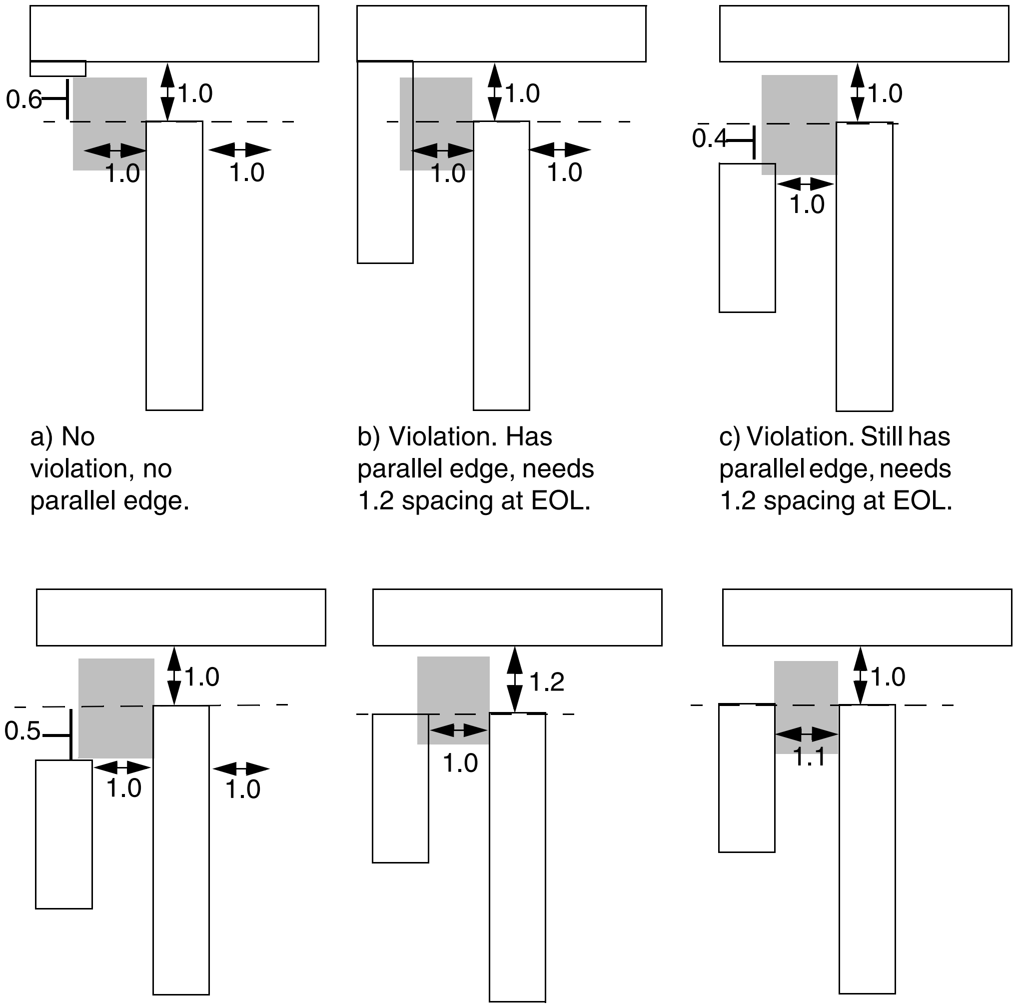
Figure 6-14



a) No violation, no parallel edge.

b) Violation. Has parallel edge, needs 1.2 spacing at EOL.

c) Violation. Still has parallel edge, needs 1.2 spacing at EOL.

d) No violation, no parallel edge.

e) No violation. Has parallel edge, but has 1.2 spacing at EOL.

f) No violation, no parallel edge (left edge is >= 1.1 away).

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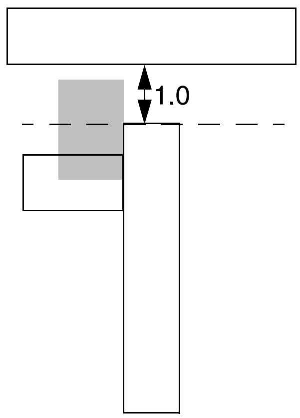

g) No violation. Has overlap on the left side, but no parallel edge.

- The following routing layer rule creates an EOL spacing rule for two parallel edges:

```
SPACING 1.0 ;            #minimum spacing is 1.0 µm
SPACING 1.2 ENDOFLINE 1.3 WITHIN 0.6 PARALLELEDGE 1.1 WITHIN 0.5 TWOEDGES ;
```


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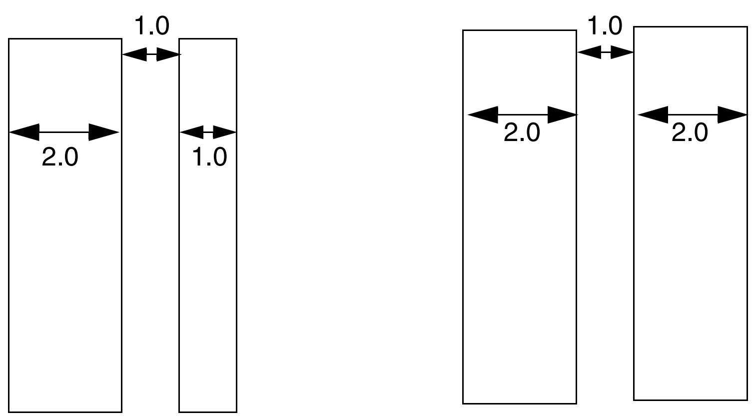
Example 6-11 Same Net Spacing Rule

If you include the following routing layer rules in your LEF file, same-net power or ground nets can use 1.0 µm spacing, even if they are 2 µm to 5 µm wide, as shown in Figure 6-15 on page 664:

```
LAYER M1
TYPE ROUTING ;
SPACING 1.0 ;                #min spacing is 1.0
SPACING 1.5 RANGE 2.0 5.0 ;  #need 1.5 spacing for 2 to 5 µm wide wires
SPACING 1.0 SAMENET PGONLY ;
```

Figure 6-15



a) Okay if both wires are the same net and are either a power or ground net.

b) Okay if both wires are the same net and are either a power or ground net.

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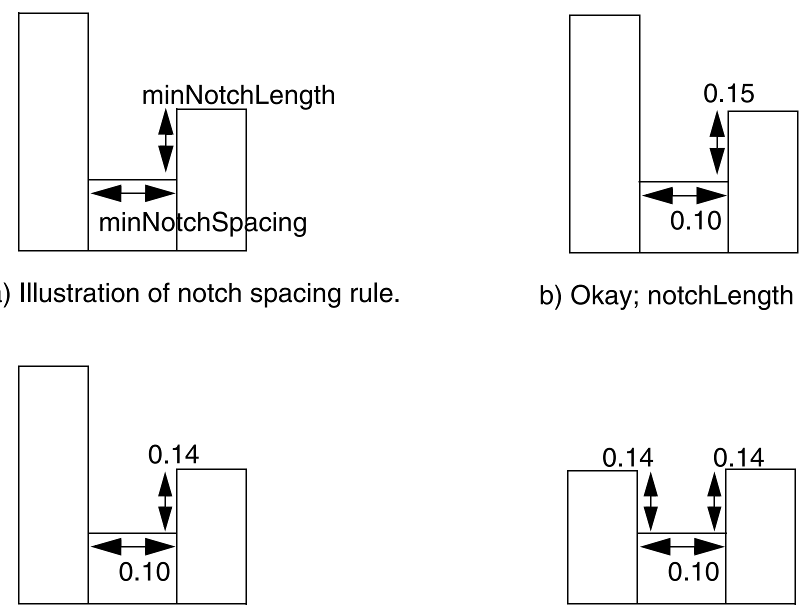
Example 6-12 Notch Length Spacing Rule

The figure below illustrates the following routing layer rules:

```
SPACING 0.10 ;
SPACING 0.12 NOTCHLENGTH 0.15 ;
```

Figure 6-16 Notch Length Rule Definitions



a) Illustration of notch spacing rule.

b) Okay; notchLength is not < 0.15.

c) Violation

d) Violation

Example 6-13 End Of Notch Width Spacing Rule

If you include the following routing layer rules in your LEF file, the notch metal at the bottom end of a U-shaped notch must have spacing that is greater than or equal to 0.14 µm, if the notch metal has a width that is less than 0.15 µm, notch spacing that is less than or equal to 0.16 µm, and notch length that is greater than or equal to 0.08 µm. See Figure 6-17 on page 666 for different layout examples for these rules.

```
SPACING 0.10 ;       #default spacing
SPACING 0.14 ENDOFNOTCHWIDTH 0.15 NOTCHSPACING 0.16 NOTCHLENGTH 0.08 ;
```


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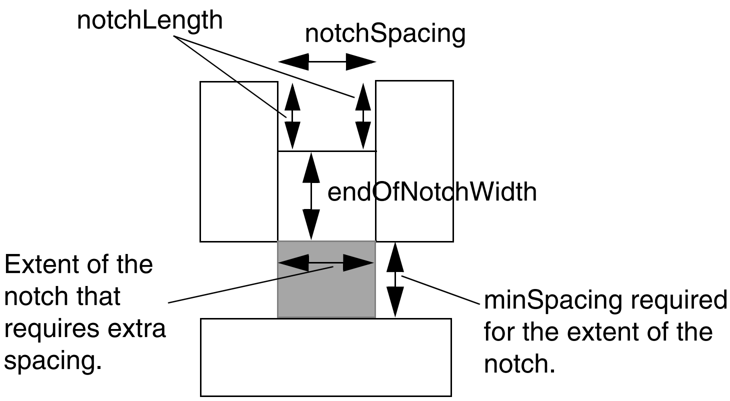
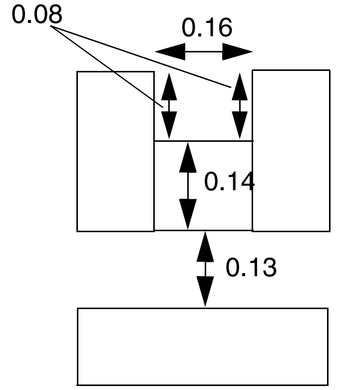
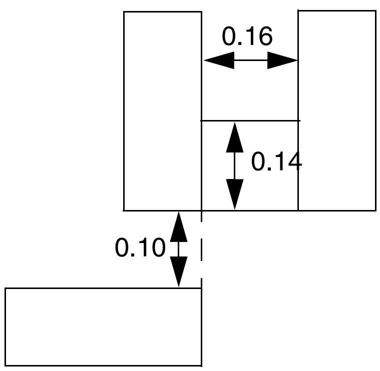
Figure 6-17 End Of Notch Width Rule Definitions



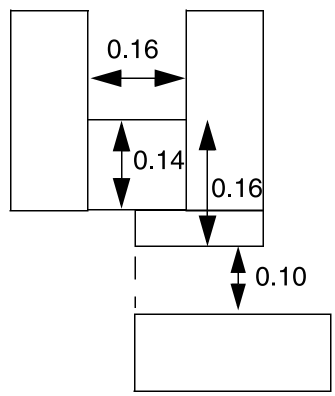
a) ENDOFNOTCHWIDTH rule definitions. The gray box shows where extra space is required.



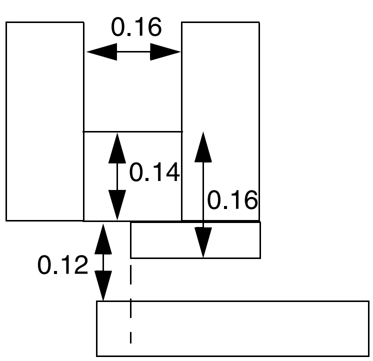
b) Violation; at least 0.14 µm spacing required.



c) Okay. No overlap with notch above; therefore only default spacing of 0.10 required.



d) Okay. Notch metal width is >= 0.15 for the overlap; therefore extra space is not required.



e) Violation. Notch metal width is < 0.15 for part of the overlap; therefore 0.14 spacing is required.

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`SPACINGTABLE`

Specifies the spacing tables to use for wiring on the layer. You can specify only one parallel run length and one influence spacing table for a layer. For information on and examples of using spacing tables, see "Using Spacing Tables" on page 668.

The syntax for describing spacing tables is defined as follows:

```
[SPACINGTABLE
   {PARALLELRUNLENGTH {length} ...
     {WIDTH width {spacing} ...}... ;
     [SPACINGTABLE INFLUENCE
       {WIDTH width WITHIN distance
        SPACING spacing} ... ;]
   | TWOWIDTHS {WIDTH width [PRL runLength]
       {spacing} ...} ... ;
   }
;]
```

`PARALLELRUNLENGTH {length} ...`
`{WIDTH width {spacing} ...}`

Specifies the maximum parallel run length between two objects, in microns. If the maximum width of the two objects is greater than *width*, and the parallel run length is greater than *length*, then the spacing between the objects must be greater than or equal to *spacing*. The first spacing value is the minimum spacing for a given width, even if the PRL value is not met.

You must specify *length*, *width*, and *spacing* values in increasing order.
Type: Float, specified in microns (for all values)

`TWOWIDTHS {WIDTH width [PRL runLength] {spacing} ...}`

Creates a table in which the spacing between two objects depends on the widths of both objects (instead of just the widest width). Optionally, it also can depend on the parallel run length between the two objects (PRL). For more information, see "Two-Width Spacing Tables."

The first width value should be 0 without an accompanied run length definition.

The PRL values in SPACINGTABLE TWOWIDTHS statement can be negative, which should be interpreted in the same way as in SPACINGTABLE PARLLELRUNLENGTH rules.
Type: Float, specified in microns (for all values)

`INFLUENCE {WIDTH width WITHIN distance SPACING spacing ...}`

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Creates a table that enforces wide wire spacing rules between nearby perpendicular wires. If an object has a width that is greater than *width*, and is located less than *distance* from two perpendicular wires, then the spacing between the perpendicular wires must be greater than or equal to *spacing*.

You must specify *width* values in increasing order.
Type: Float, specified in microns (for all values)

Note: You can only specify an `INFLUENCE` table if you specify a `PARALLELRUNLENGTH` table first.

Specifying SPACING Statements with SPACINGTABLE

You can specify some of the `SPACING` statements with the `SPACINGTABLE` statements. For example, the following `SPACING` statements can be specified with `SPACINGTABLE`:

```
SPACING x SAMENET ___ ;
SPACING x ENDOFLINE ___ ;
SPACING x NOTCHLENGTH ___ ;
SPACING x ENDOFNOTCHWIDTH ___ ;
```

These `SPACING` checks are orthogonal to the `SPACINGTABLE` checks, except `SAMENET` spacing will override `SPACINGTABLE` for same-net objects.

However, you cannot specify some `SPACING` statements (as given below) with `SPACINGTABLE` as these would generate semantic errors.

```
SPACING x ;
SPACING x RANGE ___ ;
SPACING x LENGTHTHRESHOLD ___ ;
```

Using Spacing Tables

Some processes have complex width and length threshold rules. Instead of creating multiple `SPACING` rules with different `LENGTHTHRESHOLD` and `RANGE` statements, you can define the information in a spacing table.

For example, for Figure 6-18 on page 669, a typical 90nm DRC manual might have the following rules described:

Minimum spacing	0.15 µm spacing
Either width>0.25 µm and parallel length>0.50 µm	0.20 µm spacing

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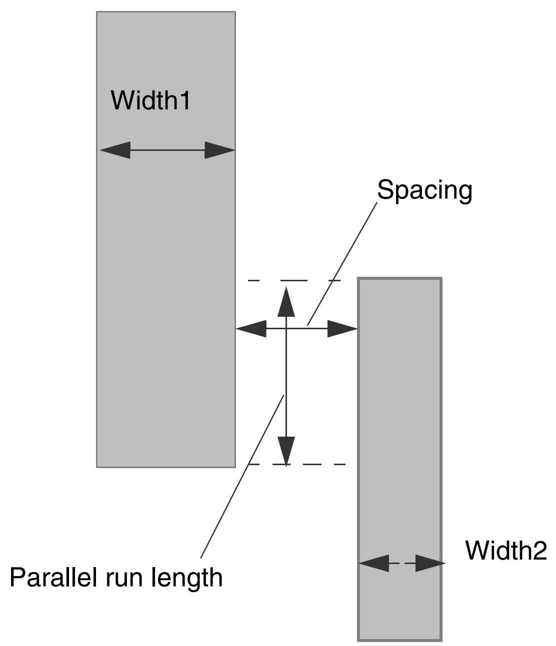
Either width>1.50 µm and parallel length>0.50 µm	0.50 µm spacing
Either width>3.00 µm and parallel length>3.00 µm	1.00 µm spacing
Either width>5.00 µm and parallel length>5.00 µm	2.00 µm spacing

Figure 6-18



These rules translate into the following `SPACINGTABLE PARALLELRUNLENGTH` statement:

```
LAYER metal1
    ...
    SPACINGTABLE
     PARALLELRUNLENGTH 0.00 0.50 3.00 5.00        #lengths must be increasing
       WIDTH 0.00      0.15 0.15 0.15 0.15        #max width>0.00
       WIDTH 0.25      0.15 0.20 0.20 0.20        #max width>0.25
       WIDTH 1.50      0.15 0.50 0.50 0.50        #max width>1.50
       WIDTH 3.00      0.15 0.50 1.00 1.00        #max width>3.00
       WIDTH 5.00      0.15 0.50 1.00 2.00 ;      #max width>5.00
    ...
END metal1
```

Using the `SPACINGTABLE PARALLELRUNLENGTH` statement, the rules can be described in the following way:

1. Find the maximum width of the two objects.

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2. Find the lowest table row where the maximum width is greater than the table-row width value. The first row is used even if the maximum width is less than and equal to the table-row width.

3. Find the right-most table column where the parallel run length is greater than the table PRL value. The first column spacing value is used even if the object's parallel run length is less than and equal to the table PRL value. The spacing value listed where the row and column intersect is the required spacing for that maximum width and parallel run length.

By definition, the width is the smaller dimension of the object (that is, the width of each object must be less than or equal to its length).

Influence Spacing Tables

Processes often require a second spacing table to enforce the wide wire spacing rules between nearby perpendicular wires, even if the wires are narrow. Figure 6-19 on page 671 illustrates this situation. Use the following `SPACINGTABLE INFLUENCE` syntax to describe this table:

```
SPACINGTABLE INFLUENCE
     {WIDTH width WITHIN distance SPACING spacing} ... ;
```

If a wire has a width that is greater than *`width`*, and the distance between it and two other wires is less than *`distance`*, the other wires must be separated by spacing that is greater than or equal to *`spacing`*. Typically, the *`distance`* and *`spacing`* values are the same. Note that the distance halo extends horizontally, but not into the corners.

By definition, the width is the smaller dimension of the object (that is, the width is less than or equal to the length of the large wire).

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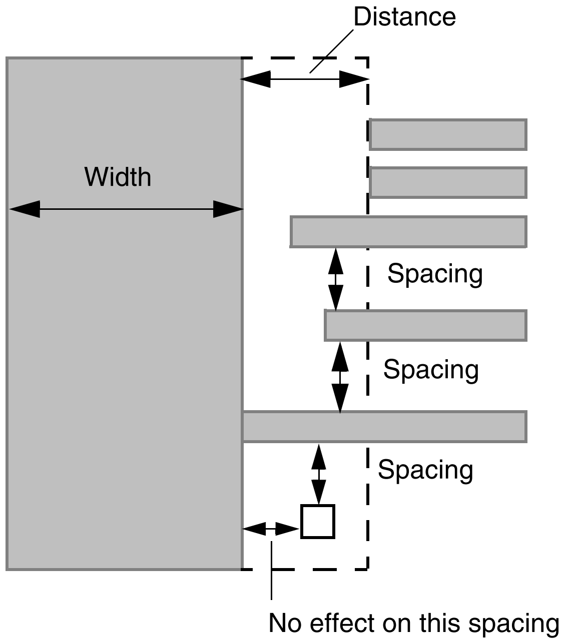
Figure 6-19



The wide wire rules often match the larger width and spacing values in the `SPACINGTABLE PARALLELRUNLENGTH` values. The previously described rules translate into the following `SPACINGTABLE INFLUENCE` statement:

```
LAYER metal1
    ...
    SPACINGTABLE INFLUENCE
      WIDTH 1.50 WITHIN 0.50 SPACING 0.50 #w>1.50, dist<0.50, needs sp>=0.50
      WIDTH 3.00 WITHIN 1.00 SPACING 1.00 #widths must be increasing
      WIDTH 5.00 WITHIN 2.00 SPACING 2.00 ;
    ...
END metal1
```

Two-Width Spacing Tables

You can create a table that enforces spacing rules that depends on the width of both objects instead of just the widest width, and optionally depends on the parallel run length between the two objects. You can use this table to replace existing `SPACING...RANGE...RANGE` rules to make it easier to read, and to include parallel run length effects in one common table. Use the following `SPACINGTABLE TWOWIDTHS` syntax to describe this table:

```
SPACINGTABLE
     TWOWIDTHS {WIDTH width [PRL runLength] {spacing} ... } ... ;
```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To find the required spacing, a 2-dimensional table is used that implicitly has the same widths (and optional parallel run lengths) for the row and column headings. There must be exactly as many *spacing* values in each WIDTH row as there are WIDTH rows. The *width* and *runLength* values must be the same or increasing from top to bottom in the table. The *spacing* values must be the same or increasing from left to right, and from top to bottom in the table.

Given two objects with *width1*, *width2*, and a parallel overlap of *runLength*, you find the spacing using the following method:

1. Find the last row where both *width1* is greater than the table row width, and *runLength* is greater than the table row run length. If no table row run length exists, the *runLength* value is not checked for that row (only that *width1* is greater than table row width is checked).
2. Find the right-most column where both *width2* is greater than table column width and *runLength* is greater than table column run length. If no table column run length exists, the *runLength* value is not checked for that column (only that *width2* is greater than table column width is checked).
3. The intersection of the matching row and column gives the required spacing.

For example, assume a DRC manual has the following rules described:

Minimum spacing	0.15 µm spacing
Either width>0.25 µm and parallel length>0.0 µm	0.20 µm spacing
Both width>0.25 µm and parallel length>0.0 µm	0.25 µm spacing
Either width>1.50 µm and parallel length>1.50 µm	0.50 µm spacing
Both width>1.50 µm and parallel length>1.50 µm	0.60 µm spacing
Either width>3.00 µm and parallel length>3.00 µm	1.00 µm spacing
Both width>3.00 µm and parallel length>3.00 µm	1.20 µm spacing

The rules translate into the following SPACINGTABLE:

```
SPACINGTABLE  TWOWIDTHS
#             width=     0.00  0.25  1.50  3.0
#               prl=     none  0.00  1.50  3.0
#                        ----------------------
WIDTH 0.00               0.15  0.20  0.50  1.00
WIDTH 0.25 PRL 0.0       0.20  0.25  0.50  1.00
WIDTH 1.50 PRL 1.50      0.50  0.50  0.60  1.00
WIDTH 3.00 PRL 3.00      1.00  1.00  1.00  1.20 ;
```


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Note that both width *and* parallel run length (if specified) must be exceeded to index into the row and column. Therefore, in this example:

If width1 = 0.25, width2 = 0.25, and prl = 0.0, then spacing = 0.15.

If width1 = 0.25, width2 = 0.26, and prl = 0.0, then spacing = 0.15.

If width1 = 0.25, width2 = 0.26, and prl = 0.1, then spacing = 0.20.

If width1 = 0.26, width2 = 0.26, and prl = 0.1, then spacing = 0.25.

`THICKNESS` *`distance`*

Specifies the thickness of the interconnect.
Type: Float

`TYPE ROUTING`

Identifies the layer as a routable layer.

`WIDTH` *`defaultWidth`*

Specifies the default routing width to use for all regular wiring on the layer.
Type: Float

`WIREEXTENSION` *`value`*

Specifies the distance by which wires are extended at vias. You must specify a value that is more than half of the routing width.
Default: Wires are extended half of the routing width
Type: Float

Note: The `WIREEXTENSION` statement only extends wires and not vias. For 65nm and below, `WIREEXTENSION` is no longer recommended because it may generate some advance rule violations if wires and vias have different widths.

The following figure shows `WIREEXTENSION` with same and different wire and via widths:

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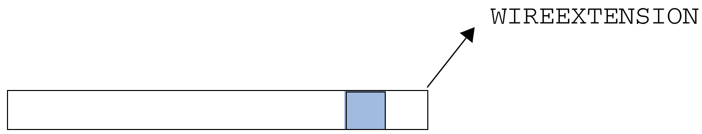
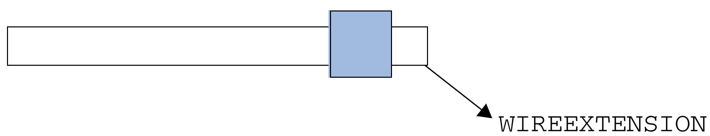
Figure 6-20 Illustration of WIREEXTENSION



a) Wire and via with same width



b) Wire and via with different widths

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Defining Routing Layer Properties to Create 32/28 nm and Smaller Nodes Rules

You can include routing layer properties in your LEF file to create 32/28 nm and smaller nodes rules that currently are not supported by existing LEF syntax. The properties are specified inside the `LAYER ROUTING` statements, where they can be seen with other rules.

Before you can reference them, properties must be defined at the beginning of the LEF file in the `PROPERTYDEFINITIONS` statement, immediately before the first `LAYER` statement.

- Properties belong to the `LAYER` object and have a type of `STRING`.
- The property names used for these rules all start with `LEF58_`.

All properties use the following syntax within the LEF `PROPERTYDEFINITIONS` statement:

```
PROPERTYDEFINITIONS
     LAYER propName STRING ["stringValue"] ;

END PROPERTYDEFINITIONS
```

The property definitions for the routing layer properties are as follows:

```
PROPERTYDEFINITIONS
   LAYER LEF58_ACCURRENTDENSITY STRING ;
   LAYER LEF58_ANTENNACUMAREARATIO STRING ;
   LAYER LEF58_ANTENNADIFFGATEPWL STRING ;
   LAYER LEF58_ANTENNADIFFONLYAREARATIO STRING ;
   LAYER LEF58_ANTENNADIFFPROTECTORAREARATIO STRING ;
   LAYER LEF58_ANTENNAGATEPWL STRING ;
   LAYER LEF58_ANTENNAGATEPLUSDIFF STRING ;
   LAYER LEF58_AREA STRING ;
   LAYER LEF58_BACKSIDE STRING ;
   LAYER LEF58_BOUNDARYBLOCKAGE STRING ;
   LAYER LEF58_BOUNDARYEOLBLOCKAGE STRING ;
   LAYER LEF58_BUMPSPACING STRING ;
   LAYER LEF58_COREEOLBLOCKAGE STRING ;
   LAYER LEF58_CORNEREOLKEEPOUT STRING ;
   LAYER LEF58_CORNERFILLSPACING STRING ;
   LAYER LEF58_CORNERSPACING STRING ;
   LAYER LEF58_ENCLOSURESPACING STRING ;
   LAYER LEF58_ENCLOSUREWITHEOL STRING ;
   LAYER LEF58_EOLKEEPOUT STRING ;
   LAYER LEF58_EOLTRACK STRING ;
   LAYER LEF58_EOLVIAKEEPOUT STRING ;
```


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```
LAYER LEF58_EOLEXTENSIONSPACING STRING ;
LAYER LEF58_FILLTOFILLSPACING STRING ;
LAYER LEF58_FIVEWIRESEOLSPACING STRING ;
LAYER LEF58_FIVEWIRESFORBIDDENSPACING STRING ;
LAYER LEF58_FORBIDDENSPACING STRING ;
LAYER LEF58_FORBIDDENSPANLENGTH STRING ;
LAYER LEF58_GAP STRING ;
LAYER LEF58_JOGCORNERKEEPOUTZONE STRING ;
LAYER LEF58_JOGKEEPOUTZONE STRING ;
LAYER LEF58_JOGKEEPOUTZONETABLE STRING ;
LAYER LEF58_JOINTCORNERSPACING STRING;
LAYER LEF58_JOINTFORBIDDENSPACING STRING;
LAYER LEF58_LINEENDALIGNEDWITHCUT STRING;
LAYER LEF58_LINEENDALIGNEDWITHCUTSPACING STRING ;
LAYER LEF58_LINEENDGAP STRING ;
LAYER LEF58_LITHOMACROHALO STRING ;
LAYER LEF58_MANUFACTURINGGRID STRING ;
LAYER LEF58_MAXDIAGLENGTH STRING ;
LAYER LEF58_MAXLENGTHWITHCUT STRING ;
LAYER LEF58_MAXOVERLAPLENGTH STRING ;
LAYER LEF58_MAXWIDTH STRING ;
LAYER LEF58_MINAREAWITHCUT STRING ;
LAYER LEF58_MINIMUMCUT STRING ;
LAYER LEF58_MINLENGTHPARALLEL STRING ;
LAYER LEF58_MINLENGTHWITHCUT STRING ;
LAYER LEF58_MINSIZE STRING ;
LAYER LEF58_MINSTEP STRING ;
LAYER LEF58_OFFSET STRING ;
LAYER LEF58_OPPOSITEEOLSPACING STRING ;
LAYER LEF58_PINCONNECTBLOCKAGE STRING ;
LAYER LEF58_PITCH STRING ;
LAYER LEF58_PROTRUSIONWIDTH STRING;
LAYER LEF58_RECTONLY STRING;
LAYER LEF58_REGION STRING ;
LAYER LEF58_RIGHTWAYONGRIDONLY STRING ;
LAYER LEF58_SIXWIRESSPACING STRING ;
LAYER LEF58_SPACINGTABLE STRING ;
LAYER LEF58_SPANLENGTHENCLOSURESPACING STRING ;
LAYER LEF58_SPANLENGTHTABLE STRING ;
LAYER LEF58_SPACING STRING ;
LAYER LEF58_TWOWIRESFORBIDDENSPACING STRING ;
```


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```
   LAYER LEF58_TYPE STRING ;
   LAYER LEF58_VOLTAGESPACING STRING ;
   LAYER LEF58_WIDTHTABLE STRING ;
   LAYER LEF58_WIDTH STRING ;
   LAYER LEF58_WRNGDIREOLKEEPOUT STRING ;
END PROPERTYDEFINITIONS
```

Type Rule

A type rule can be used to further classify a routing layer.

You can create a type rule by using the following property definition:

```
TYPE ROUTING;
     PROPERTY LEF58_TYPE
       "TYPE {POLYROUTING | MIMCAP | HIGHR
              | TSVMETAL | PADMETAL | STACKEDMIMCAP }
              ;" ;
```

Where:

`HIGHR`	Specifies that the layer is a high resistance routing layer that is not used as a regular routing layer.
`MIMCAP`	Indicates that the layer is a mimcap layer. A mimcap layer is a metal layer that is not to be used as a routing layer.
`PADMETAL`	Specifies a pad metal layer that is used to connect the topmost routing layer or the bottom-most `BACKSIDE` routing layer to outside the die. The pad metal layer can either be on the top or at the `BACKSIDE` in the layer stack. This layer is not used for regular routing.
`POLYROUTING`	Indicates that the polysilicon layer should be considered as a routing layer. Polysilicon layers provide extra routing resources for designs with limited metal routing layers.
`STACKEDMIMCAP`	Specifies a special type of mimcap layer to which a cut shape above a `STACKEDMIMCAP` shape connects but, instead of stopping, continues to the next regular routing layer. The cut can pass through multiple `STACKEDMIMCAP` layers before it reaches the next regular routing layer. A `MASTERSLICE TRIMMETAL` layer shape can be used to cut holes in a `STACKEDMIMCAP` shape to allow the cut shape to pass through the shape without being connected to it. The `STACKEDMIMCAP` layer is not used for regular routing.

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`TSVMETAL` Specifies an intermediate layer on which a special Through Silicon Via (TSV) crossing multiple layers would first stop. A separate via is used to connect the `TSVMETAL` layer to the destined regular routing layer. This layer is not used for regular routing.

The `POLYROUTING` and `MIMCAP` keywords are mutually exclusive; you cannot specify them together.

AC Current Density Rule

The AC density rule can be used to specify how much AC current a wire on a layer of a certain width can handle at a certain frequency in units of milliamps per micron (mA/mm).

You can define a AC current density rule by using the following property definition:

```
PROPERTY LEF58_ACCURRENTDENSITY
       "ACCURRENTDENSITY {PEAK | AVERAGE | RMS}
          [TEMPPWL temp_1 multi_1 temp_2 multi_2  ...]
          [HOURSPWL hours_1 multi_1 hours_2 multi_2  ...]
          ; " ;
```

All other keywords are the same as the existing LEF routing layer `ACCURRENTDENSITY` syntax.

Note that the property statement is a compliment to the original syntax, which must be specified first.

Where:

`HOURSPWL` *`hours_1 multi_1 hours_2 multi_2 ...`*

Specifies a multiplier table based on the hours of operation. When the operation hour is greater than or equal to a certain hour in the table, the corresponding multiplier is used to scale the current by multiplying the multiplier with linear interpolation in the current calculation.
Type: Float

`TEMPPWL` *`temp_1 multi_1 temp_2 multi_2 ...`*

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Specifies a multiplier table based on the temperature in Celsius. When the operation temperature is >= certain temperature in the table, the corresponding multiplier is used to scale the current by multiplying the multiplier with linear interpolation in the current calculation.
Type: Float

AC Current Density Rule Examples

The following AC current density rule indicates that the `RMS` table entry values will be defined for 100C and 10000 hours since the corresponding multipliers are 1.0 for those values. At 80C and 15000 hours, the multiplier is 1.5 x 0.8 = 1.2. Hence, the wires can carry 1.2x times the `RMS` table values.

```
PROPERTY LEF58_ACCURRENTDENSITY
       "ACCURRENTDENSITY RMS
          TEMPPWL 50 2.2 80 1.5 100 1.0 120 0.75
          HOURSPWL 5000 1.8 10000 1.0 15000 0.8 ; " ;
```

Antenna Cumulative Area Ratio Rule

The antenna cumulative area ratio rule is used to specify the cumulative antenna ratio, using the area of the metal wire that is not connected to the diffusion diode.

You can create an antenna cumulative area ratio rule by using the following property definition:

```
PROPERTY LEF58_ANTENNACUMAREARATIO
       "ANTENNACUMAREARATIO {OXIDE1 | OXIDE2 | OXIDE3 | OXIDE4 | OXIDE5 | ...
        | OXIDE32} value
        [GATEEXPONENT index] [GATEAREA area]
        [DIFFONLY]
        ;" ;
```

Where:

`ANTENNACUMAREARATIO {OXIDE1 | OXIDE2 | OXIDE3 | OXIDE4 | OXIDE5 | ...| OXIDE32}` *`value`*

Specifies the cumulative antenna ratio, using the area of the metal wire that is not connected to the diffusion diode.

Type: Float

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`DIFFONLY` Specifies the cumulative antenna ratio, using the area of the metal wire connected to a normal diffusion device, which is defined in `ANTENNADIFFAREA` on a macro pin. This check is skipped if the area is connected to the gate.

`GATEAREA` *`area`* Specifies that the rule applies only if the gate areas are less than *`area`*.

Type: Float

`GATEEXPONENT` *`index`*

Specifies an exponential cumulative antenna ratio, using the electrical connected gateArea into the rule ratio formula as *`value * gateArea`*$^{\mathtt{index}}$.

Type: Float

Antenna Diff Gate PWL Rule

The antenna diff gate PWL rule is used to specify that the antenna gate connected to the diffusion diode can be used to define a piece-wise linear (PWL) table on the given antenna model that is indexed by the real gate area, and returns an “effective gate area” interpolated from the table.

You can create an antenna diff gate PWL rule by using the following property definition:

```
PROPERTY LEF58_ANTENNADIFFGATEPWL
     "ANTENNADIFFGATEPWL {OXIDE1 | OXIDE2 | OXIDE3 | OXIDE4 | OXIDE5 | ...
        | OXIDE32}
        ((gateArea1 effectiveGateArea1) (gateArea2 effectiveGateArea2) ... )
        ;" ;
```

Where:

```
ANTENNADIFFGATEPWL {OXIDE1 | OXIDE2 | OXIDE3 | OXIDE4 |
     OXIDE5 | ... | OXIDE32}
     ((gateArea1 effectiveGateArea1) (gateArea2
       effectiveGateArea2)... )
```


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Specifies that the antenna gate connected to the diffusion diode can be used to define a piece-wise linear (PWL) table on the given antenna model that is indexed by the real gate area, and returns an “effective gate area” interpolated from the table.
Type: Float, specified in microns

effectiveGateArea indicates the effective gate area interpolated from the PWL table. If this table it not defined, the real gate area is used.

gateArea indicates the real gate area.

Note that if ANTENNADIFFGATEPWL is defined, the ANTENNAGATEPWL rule is applied only to the antenna gate that is not connected to the diffusion diode. If ANTENNADIFFGATEPWL is not defined, the ANTENNAGATEPWL rule is applied to all the antenna gates.

Antenna Diff Only Area Ratio Rule

The antenna diff only area rule is used to specify a diffusion-area-based antenna ratio.

You can create an antenna diff only area rule by using the following property definition:

```
PROPERTY LEF58_ANTENNADIFFONLYAREARATIO
       “ANTENNADIFFONLYAREARATIO {OXIDE1 | OXIDE2 | OXIDE3 | OXIDE4
          | OXIDE5 | ... | OXIDE32}  value
        ;..." ;
```

Where:

ANTENNADIFFONLYAREARATIO {OXIDE1 | OXIDE2 | OXIDE3 | OXIDE4 | OXIDE5 | ... | OXIDE32} *value*

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Specifies a diffusion-area-based antenna ratio, using the area of the metal wire connected to a diffusion device, which is defined in `ANTENNADIFFAREA` on a macro pin. This ratio is associated with the given `OXIDE` model.

Type: Float, specified in microns

The partial area ratio (PAR) is calculated for diffusion and gate area separately as:

$PAR(N_{i,j}, D_k)$ = Drawn area $(N_{i,j})$ / Total area of diffusion area below $N_{i,j}$

$PAR(N_{i,j}, G_k)$ = Drawn area $(N_{i,j})$ / Total area of gate area below $N_{i,j}$

The following two rule checks are preformed individually:

- The PAR for node $N_{i,j}$ with respect to diffusion D_k by dividing the drawn area of the node by the area of diffusions that electrically connected to it must be less than or equal to the given value in `ANTENNADIFFONLYAREARATIO`.
- The PAR for node $Ni_{,j}$ with respect to gate G_k by dividing the drawn area of the node by the area of gates that electrically connected to it must be less than or equal to the given value in `ANTENNAAREARATIO`.

Antenna Diff Protector Area Ratio Rule

The antenna diff protector area rule is used to specify a protector diffusion-area-based antenna ratio.

You can create an antenna diff protector area rule by using the following property definition:

```
PROPERTY LEF58_ANTENNADIFFPROTECTORAREARATIO
       "ANTENNADIFFPROTECTORAREARATIO {OXIDE1 | OXIDE2 | OXIDE3
          | OXIDE4 | OXIDE5 | ... | OXIDE32}  value
        ;..." ;
```

Where:

```
ANTENNADIFFPROTECTORAREARATIO {OXIDE1 | OXIDE2 | OXIDE3
      | OXIDE4 | OXIDE5 | ... | OXIDE32} value
```


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Specifies a protector diffusion-area-based antenna ratio, using the area of the metal wire connected to a protector diffusion device, which is defined in ANTENNADIFFAREA with PROTECTOR on a macro pin. This ratio is associated with the given OXIDE model.

Type: Float, specified in microns

The metal connected to a pin with a protector diffusion device would only be checked against this given ratio. This includes the case when the metal also connects to gates or non-protector diffusions. Other ANTENNA* rules would be ignored.

The partial area ratio (PAR) is calculated for protector diffusion as:

$PAR(N_{i,j}, D_k)$ = Drawn area $(N_{i,j})$ / Total area of protector diffusion area below $N_{i,j}$

The PAR for node $N_{i,j}$ with respect to protector diffusion D_k by dividing the drawn area of the node by the area of protector diffusions that electrically connected to it must be less than or equal to the given value in ANTENNADIFFPROTECTORAREARATIO.

Antenna Gate PWL Rule

The antenna gate PWL rule can be used to define a PWL (piece-wise linear) table on the given antenna model that is indexed by the real gate area, and returns an “effective gate-area” interpolated from the table.

You can define antenna gate PWL rule by using the following property definition:

```
PROPERTY LEF58_ANTENNAGATEPWL
       "ANTENNAGATEPWL {OXIDE1 | OXIDE2 | OXIDE3 | OXIDE4 | OXIDE5 | ...
        | OXIDE32}
        ((gateArea1 effectiveGateArea1) (gateArea2 effectiveGateArea2) ... )
        ;" ;
```

Where:

ANTENNAGATEPWL	Defines the PWL table that is indexed by the real gate area, and returns an "effective gate-area" interpolated from the table.

(*gateArea1 effectiveGateArea1*) (*gateArea2 effectiveGateArea2*)

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The *effectiveGateArea* is used rather than the real gate area in the PAR equation. If the PWL table is not defined, then the real gate area is used.

Type: Float, specified in units of square microns.

Antenna Gate Plus Diffusion Rule

The antenna gate plus diffusion rule can be used for representing protection provided by the diffusion area that is added to the gate-area value in the PAR equation on the given antenna model, which can be considered as “additional effective gate-area".

You can define the antenna gate plus diffusion rule by using the following property definition:

```
PROPERTY LEF58_ANTENNAGATEPLUSDIFF
       "ANTENNAGATEPLUSDIFF {OXIDE1 | OXIDE2 | OXIDE3 | OXIDE4 | OXIDE5 | ...
         | OXIDE32}
         {plusDiffFactor |
         PWL ((diffArea1 plusDiffProtect1) (diffArea1 plusDiffProtect2) ...)}
         ;" ;
```

All other keywords are the same as the existing LEF routing layer ANTENNAGATEPLUSDIFF syntax.

Where:

ANTENNAGATEPLUSDIFF

Indicates the antenna gate area plus diffusion area.

plusDiffFactor | *PWL* ((*diffArea1 plusDiffProtect1*) (*diffArea1 plusDiffProtect2*) ...)

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Indicates protection provided by the diffusion area that is added to the gate area value in the PAR equation, which can be considered as “additional effective gate-area". The PAR equation is as follows:

PAR = {(*metalFactor* * metal) * *diffReducePWL*(diff) - (*minusDiffFactor* * diff) } / {*gatePWL*(gate) + *plusDiffProtect*(diff) }

If *plusDiffFactor* value is specified, then:

plusDiffProtect = *plusDiffFactor* * diffusion area

If the PWL table is defined, then:

plusDiffProtect = interpolated PWL value indexed by diffusion area.

If PAR is greater than the *metalGateRatio* (from ANTENNAAREARATIO), and no diffusion is connected then there will be a violation.

If PAR is greater than *metalGateDiffRatio* (from ANTENNADIFFAREARATIO), then there will be a violation. This is checked even if the diffusion area is equal to 0 (that is, no diffusion is connected).

The *diffArea* and *plusDiffProtect* values are floats in units of square microns.

Antenna Gate Plus Diffusion Rule Examples

The following rule shows antenna gate plus diffusion rule for metal1:

```
metal_area <= 100 * (funcGate(gate_area) + funcDiff(diff_area))
funcGate = 1 for gate_area < 1, and equals 2 for gate_area >= 1
funcDiff = 10 for diff_area < 2, and equals 50 for diff_area  >= 2.
```

Then

```
100 >= metal_area / (funcGate(gate) + funcDiff(diff))
```

This fits the PAR formula above using *gatePWL*(gate) and *plusDiffProtect*(diff) values.

Then the LEF file will have the following:

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```
#Max-ratio = 100 for all diff-area values because the diff-protect value is
#accounted for inside the PAR equation. Note, there is no need for ANTENNAAREARATIO
# which is only checked for metal with no diff connected, while ANTENNADIFFAREARATIO
#is checked for diff and no diff connected.
ANTENNAMODEL OXIDE1 ;
ANTENNADIFFAREARATIO  ( 0 100 ) ( 10000.0 100 ) ;

#This models funcGate(gate_area) above: 0 to 0.99 is 1, above 1.0 is 2
PROPERTY LEF58_ANTENNAGATEPWL OXIDE1
"ANTENNAGATEPWL ( ( 0  1 ) ( 0.99 1 ) ( 1.0 2 ) ( 10000.0 2 ) ) ;" ;

#This models funcDiff(diff_area) above: 0 to 1.99 is 10, above 1.0 is 50
PROPERTY LEF58_ANTENNAGATEPLUSDIFF OXIDE1
"ANTENNAGATEPLUSDIFF
 ( ( 0  10 ) ( 1.99 10 ) ( 2.0 50 ) ( 10000.0 50 ) ) ;" ;
```

Area Rule

In some cases, it is necessary to allow a smaller minimum area for "simple rectangles" and simple polygon shapes, but require a larger minimum area for complex polygons. This is done with multiple `AREA` statements.

You can define a minimum area rule that requires a larger area

1. When all the edges of the polygon are short

   or

2. If a minimum-sized rectangle cannot fit inside the polygon.

You can also combine the two into one statement, in which case the larger area is required if all the edges are short and a minimum-sized rectangle cannot fit inside the polygon.

You can create a minimum area rule by using the following property definition:

```
PROPERTY LEF58_AREA
      "AREA minArea
     [MASK maskNum]
     [EXCEPTMINWIDTH minWidth]
     | [EXCEPTEDGELENGTH {minEdgeLength maxEdgeLength | edgeLength}]
          [EXCEPTMINSIZE {minWidth minLength}...]
          [EXCEPTSTEP length1 length2]
     | RECTWIDTH rectWidth
     | EXCEPTRECTANGLE
```


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```
| LAYER trimLayer OVERLAP {1 | 2}
]
;..." ;
```

Where:

All the other keywords are the same as the existing LEF routing layer `AREA` syntax.

`EXCEPTMINWIDTH` *`minWidth`*

Specifies that the rule does not apply if the width of a wire is greater than or equal to the *`minWidth`*.
Type: Float, specified in microns

`EXCEPTEDGELENGTH` *`minEdgeLength maxEdgeLength`* | *`edgeLength`*

The *`edgeLength`* specifies that the minimum area rule applies for a given polygon, except if at least one edge-length is greater than or equal to *`edgeLength`*.

If both *`minEdgeLength`* and *`maxEdgeLength`* are specified instead, the minimum area rule does not apply if the entire edge length is less than *`minEdgeLength`* or at least one edge length is greater than or equal to the *`maxEdgeLength`*. In other words, the minimum area rule only applies if at least one edge length is greater than or equal to the *`minEdgeLength`* and the edge length is less than the *`maxEdgeLength`*.

Type: Float, specified in microns

`EXCEPTMINSIZE {`*`minWidth minLength`*`}...`

Indicates the *`minArea`* rule applies for a given polygon except if a minimum-sized rectangle of dimensions *`minWdith minLength`* can fit inside the polygon.

Multiple width/length pairs are allowed. The minimum area rule does not apply, if multiple width/length pairs are specified, and if any one of the given min-sized rectangles fit inside a given polygon.
Type: Float, specified in microns (for both values)

`EXCEPTRECTANGLE`

Specifies that the minimum area rule only applies to non-rectangular objects. It is illegal to define `AREA EXCEPTRECTANGLE` by itself, and it should be complemented by a separate `AREA` statement without any exception clauses.

`EXCEPTSTEP` *`length1 length2`*

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Specifies that the minimum area rule does not apply for a polygon having an edge with length greater than or equal to *length1* with an adjacent edge with length less than *length2*.
Type: Float, specified in microns

`LAYER` *trimLayer* `OVERLAP {1 | 2}`

Specifies that the area rule applies only when:

- One line-end of a wire overlaps or touches with shapes in *trimLayer*, which must be a layer with `TYPE TRIMMETAL`, if `OVERLAP` is specified as `1`.
- Both line-ends of a wire overlap or touch with shapes in *trimLayer*, if `OVERLAP` is specified as `2`.

`MASK` *maskNum*

Specifies which mask the area rule is applied on. The *maskNum* must be a positive integer, and most applications only support values of 1, 2, or 3. If any one of the `AREA` statements uses `MASK`, all of the `AREA` statements must use `MASK`, and each mask must have at least one corresponding `AREA` statement.
Type: Integer

`RECTWIDTH` *rectWidth*

Specifies the minimum area of a rectangle with width less than or equal to *rectWidth* to be greater than or equal to *minArea*. For a given rectangle, find the smallest *rectWidth* that is greater than or equal to the width of the rectangle. If such an `AREA` statement can be found, the corresponding area rule should only be checked for the rectangle. If no such statement can be found, the rest of the `AREA` statements will apply to the rectangle.
Type: Float, specified in microns

Area Rule Examples

- If you have the following `AREA` definition in your routing layer statement:

  ```
  AREA 0.4 ;
  ```

  All polygons on the layer must have a minimum area of 0.4 μm^2.

- If the following minimum area rule exists:

  ```
  PROPERTY LEF58_AREA "AREA 0.6 EXCEPTEDGELENGTH 0.5 ;" ;
  ```


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All polygons on the layer must have a minimum area of 0.6 μm^2, except if a polygon has at least one edge that is greater than or equal to 0.5 µm.

- If the following minimum area rule exists:

```
PROPERTY LEF58_AREA
  "AREA 0.6 EXCEPTEDGELENGTH 0.5 EXCEPTMINSIZE 0.1 0.5 ;" ;
```

Any polygon on the layer must have a minimum area of 0.6 μm^2 when neither of the following conditions hold:

  - The polygon has at least one edge that is greater than or equal to 0.5 µm
  - A rectangle of size 0.1 µm by 0.5 µm can fit inside the polygon

- If the following minimum area rule exists:

```
AREA 0.1 ;
PROPERTY LEF58_AREA
   "AREA 0.15 EXCEPTMINWIDTH 0.05 ;" ;
```

All polygons with any width less than 0.05 µm must have a minimum area of 0.15 μm^2. Otherwise, a minimum area of 0.1 μm^2 is needed.

- The following minimum area rule indicates that any rectangles with width less than or equal to 0.05 µm must have an area greater than or equal to 0.08 μm^2, any rectangles with width greater than 0.05 µm and less than or equal to 0.1 µm must have an area greater than or equal to 0.12 μm^2 while any other objects must have an area greater than or equal to 0.15 μm^2.

```
AREA 0.15 ;
PROPERTY LEF58_AREA
   "AREA 0.08 RECTWIDTH 0.05 ;
    AREA 0.12 RECTWIDTH 0.1 ; " ;
```


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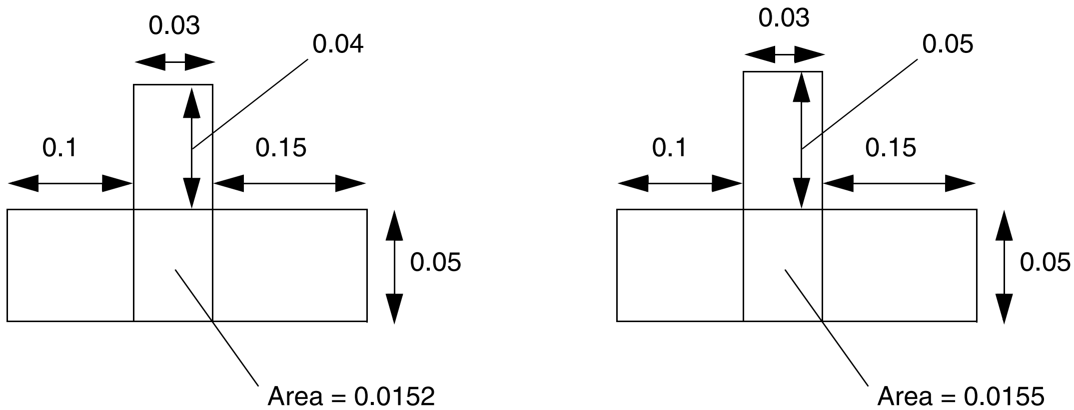
Figure 6-21 Illustration of Area Rule with EXCEPTSTEP

```
AREA 0.015 ;
PROPERTY LEF58_AREA
   "AREA 0.017 EXCEPTSTEP 0.12 0.05 ; " ;
```



a) OK, the edge of 0.15 (>= 0.12) has an adjacent edge of 0.04 (< 0.05), and only 0.015 area should be followed and is met

b) Violation, there is no edge >= 0.12 having an adjacent edge < 0.05, 0.017 area should be followed and is failed.

- The following minimum area rule indicates that any rectangles must have an area greater than or equal to 0.08 μm^2, while any other polygons must have an area greater than or equal to 0.10 μm^2:

```
AREA 0.08 ;
PROPERTY LEF58_AREA
   "AREA 0.10 EXCEPTRECTANGLE ;" ;
```

Figure 6-22 Area Rule with EXCEPTRECTANGLE

a) OK, a rectangle only requires minimum area of 0.08 and is met

b) Violation, this is not a rectangle, and minimum area must be >= 0.10

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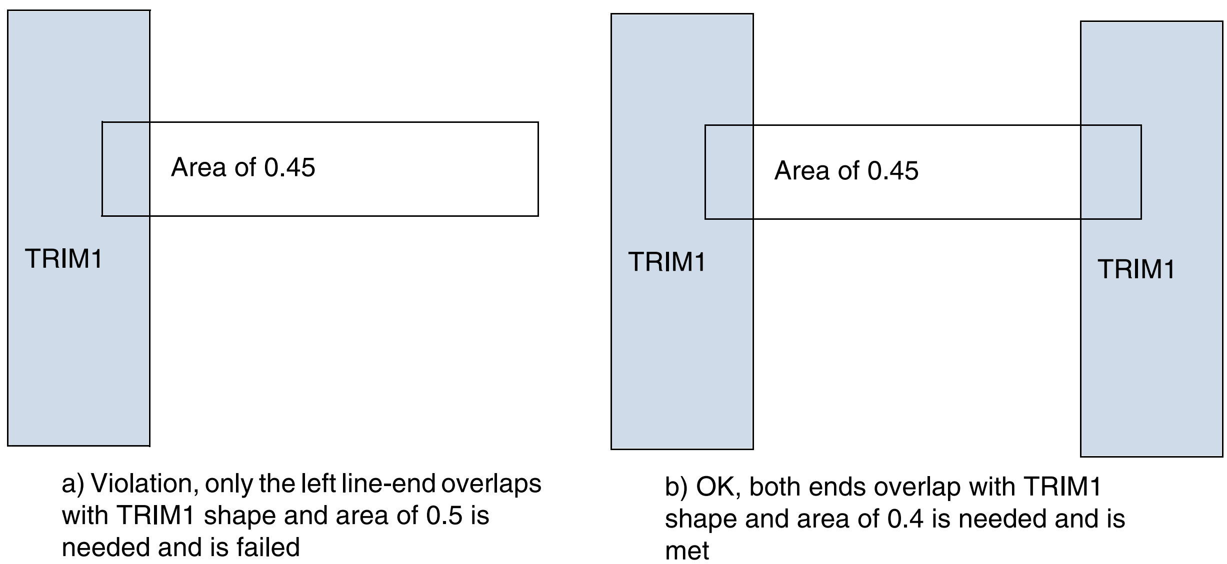
- The following example illustrates minimum area rule for trim layer with overlap:

```
AREA 0.06 ;
PROPERTY LEF58_AREA "
   AREA 0.5 LAYER TRIM1 OVERLAP 1
   AREA 0.4 LAYER TRIM1 OVERLAP 2 ;" ;
```

Figure 6-23 Area Rule for Trim Layer with Overlap



Backside Rule

A backside rule can be used to specify that the routing layer is used on the underside of the die.

You can create a backside rule by using the following property definition:

```
PROPERTY LEF58_BACKSIDE
     "BACKSIDE ;" ;
```

Where:

`BACKSIDE`	Indicates that the routing layer is a backside routing layer.

The `POLYROUTING`, `MIMCAP` (`PROPERTY LEF58_TYPE)`, and `BACKSIDE` keywords are mutually exclusive; you cannot specify them together.

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Block EOL Blockage Rule

A block EOL blockage rule can be used to specify line-end blockages formed on all blocks.

You can create a block EOL blockage rule by using the following property definition:

```
PROPERTY LEF58_BLOCKEOLBLOCKAGE
     "BLOCKEOLBLOCKAGE size OFFSET offset
     ;" ;
```

Where:

`BLOCKEOLBLOCKAGE` *`size`* `OFFSET` *`offset`*

Specifies line-end blockages formed on all blocks, MACRO with `CLASS` BLOCK, in the orthogonal direction of the preferred routing direction of the layer in the following method. The block OBS shapes with `SPACING` override on this layer are truncated by the block boundaries if the OBS shapes extend outside the block. The line-end blockages are formed by going toward the core by *`offset`* from the truncated OBS shapes and having the size of *`size`*. The line-end blockages prevent the line-end of a wire from ending inside these blockages.

Type: Float, specified in microns

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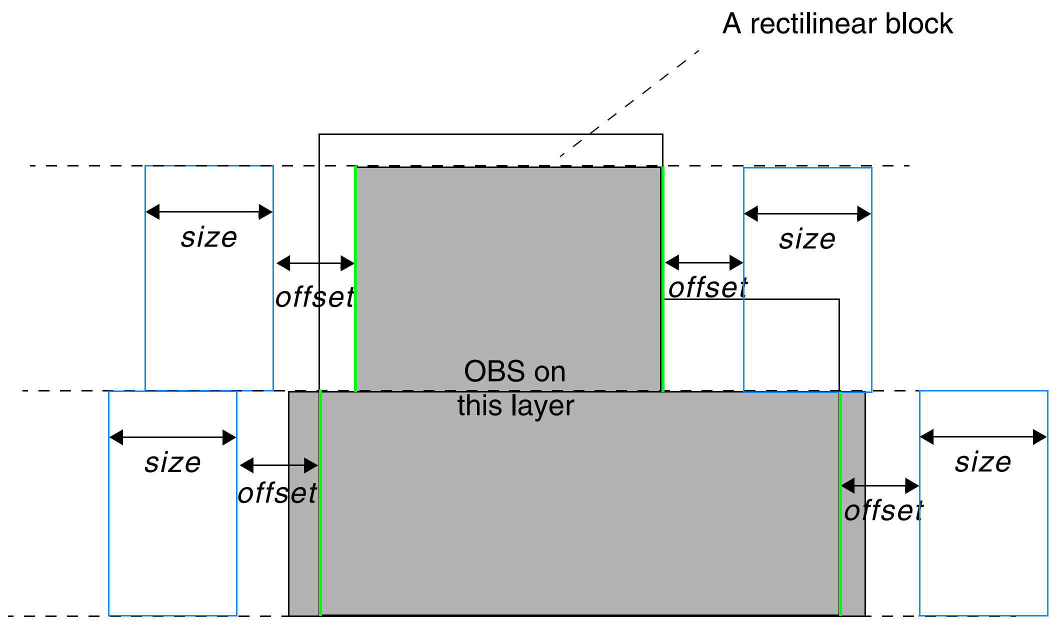
Figure 6-24 Illustration of the Block EOL Blockage Rule



The OBS with the `SPACING` override on this layer is truncated by the block boundaries. The green edges are used to form the line-end blockages, which have the same height as the OBS and go toward the core by *`offset`* and have the size of *`size`*.

Illustration of `BLOCKEOLBLOCKAGE` *`size`* `OFFSET` *`offset`*

Boundary Blockage Rule

A boundary blockage rule is used to specify line-end blockages. You can create a boundary blockage rule by using the following property definition:

```
PROPERTY LEF58_BOUNDARYBLOCKAGE
       "BOUNDARYBLOCKAGE size
          ;" ;
```

Where:

`BOUNDARYBLOCKAGE` *`size`*

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Specifies the routing blockages formed from the boundary edges in the preferred routing direction of the layer of a design. The blockage is formed from the boundary edges by going `size` inward to the core. These blockages prevent any wires overlapping with it.

Type: Float, specified in microns

Figure 6-25 Illustration of the Boundary Blockage Rule

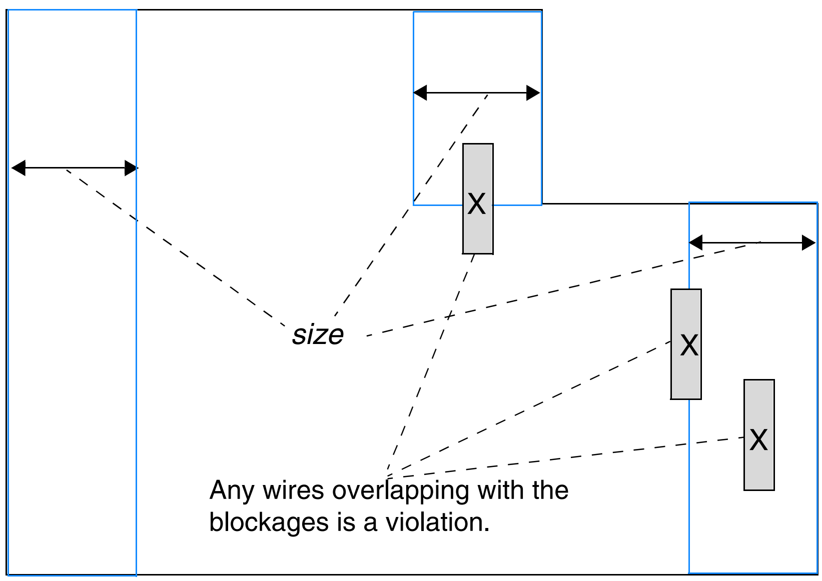

Illustration of `BOUNDARYBLOCKAGE` *size*

Boundary EOL Blockage Rule

A Boundary EOL blockage rule is used to specify line-end blockages. You can create a boundary EOL blockage rule by using the following property definition:

```
PROPERTY LEF58_BOUNDARYEOLBLOCKAGE
        "BOUNDARYEOLBLOCKAGE size OFFSET offset
           [PARALLEL parLength WITHIN parWithin SPACING spacing]
           ;" ;
```


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Where:

`BOUNDARYEOLBLOCKAGE` *`size`* `OFFSET` *`offset`*

Specifies line-end blockages formed near the boundary edges in the orthogonal direction of the preferred routing direction of the layer of a design. The blockage is formed by going inward from the boundary edge to the core by *`offset`* and having the size of *`size`* going further inward to the core. These blockages prevent the line-end of a wire from ending inside the blockages if it does not cross the edge closer to the boundary.

Type: Float, specified in microns

`PARALLEL` *`parLength`* `WITHIN` *`parWithin`* `SPACING` *`spacing`*

Specifies exemptions when an EOL in the blockage has certain neighboring wires. A search window is formed by extending *`parLength`* on EOL on both sides and *`parWithin`* in the beyond orthogonal direction. If there is a neighboring wire covering the entire length of the EOL, such as a wire on the same track, the violation is exempted. If there are neighboring wires on both sides, the violation is exempted if the spacing between the two neighboring wires is less than *`spacing`*. The neighboring wire(s) should be at least partially outside or touch the edge of the boundary EOL blockage on the side toward the design boundary edge that forms the blockage.

Type: Float, specified in microns

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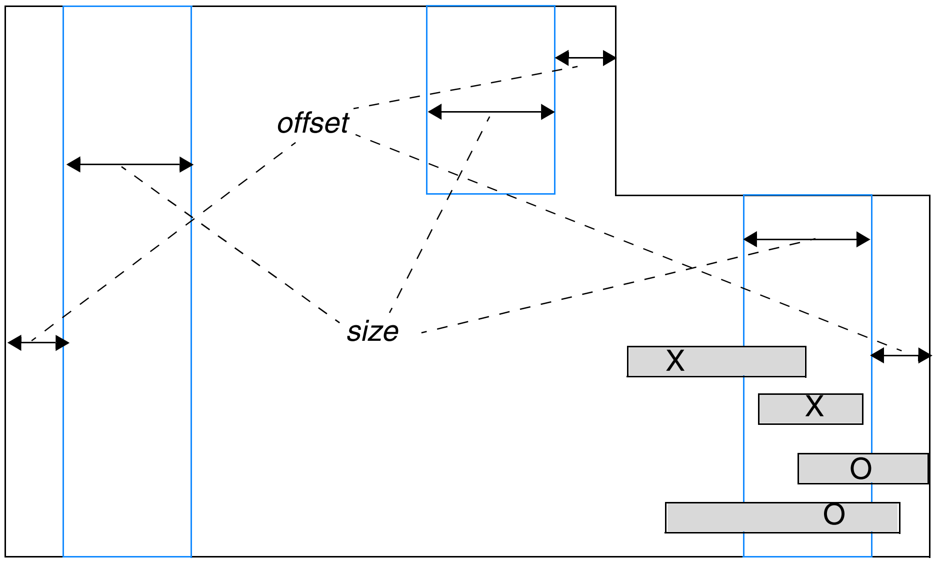
Figure 6-26 Illustration of the Boundary EOL Blockage Rule



Preferred routing direction is horizontal

The top wire crossing the left edge of the blue line-end blockage on the right boundary would be a violation. The second wire that is completely enclosed by the blockage is also a violation. The third wire crossing the right edge of the blockage and the fourth wire going through the blockage are both legal.

Illustration of `BOUNDARYEOLBLOCKAGE` *`size`* `OFFSET` *`offset`*

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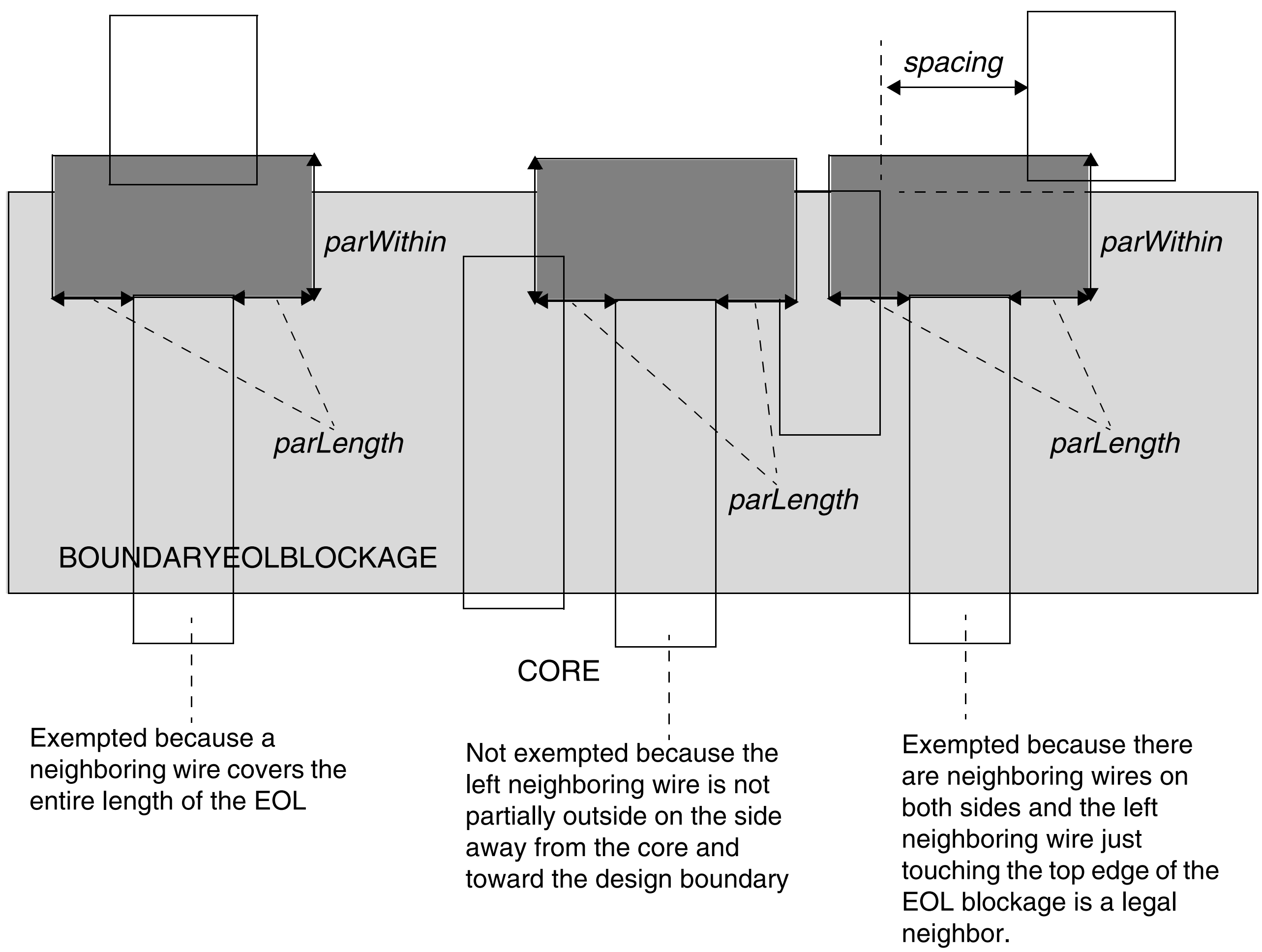
Figure 6-27 Illustration of the Boundary EOL Blockage Rule with PARALLEL, WITHIN, and SPACING



Illustrations of `BOUNDARYEOLBLOCKAGE` *`size`* `OFFSET` *`offset`* `PARALLEL` *`parLength`* `WITHIN` *`parWithin`* `SPACING` *`spacing`*

Bump Spacing Rule

A bump spacing rule is used to specify the spacing between a bump with width larger than the specified value and a wire.

You can create a bump spacing rule by using the following property definition:

```
PROPERTY LEF58_BUMPSPACING
       "BUMPSPACING spacing WIDTH bumpWidth
          ;" ;
```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Where:

`BUMPSPACING` *`spacing`* `WIDTH` *`bumpWidth`*

Specifies a bump to wire spacing if the width of the bump is greater than *`bumpWidth`*. The width of the wire is irrelevant.

Type: Float, specified in microns

Core EOL Blockage Rule

A Core EOL blockage rule is used to specify line-end blockages formed from the core edges. You can create a core EOL blockage rule by using the following property definition:

```
PROPERTY LEF58_COREEOLBLOCKAGE
       "COREEOLBLOCKAGE outwardSize inwardSize
          [SIDEEXTENSION sideExtension]
          ;" ;
```

Where:

`COREEOLBLOCKAGE` *`outwardSize inwardSize`*
`SIDEEXTENSION` *`sideExtension`*

Defines the line-end blockages formed from the core edges in the orthogonal direction of the preferred routing direction of the layer of a design. The blockage is formed by going outward from the core by *`outwardSize`* and inward by *`inwardSize`*. If `SIDEEXTENSION` is defined, the blockage is extended by *`sizeExtension`* along the direction of the core edge on both sides. These blockages prevent any line-end of a wire end inside it. If a wire goes through the blockage and just touches the far end edge, it is still a violation. If a wire just touches an edge without extending inside the blockage, it is fine.
Type: Float, specified in microns

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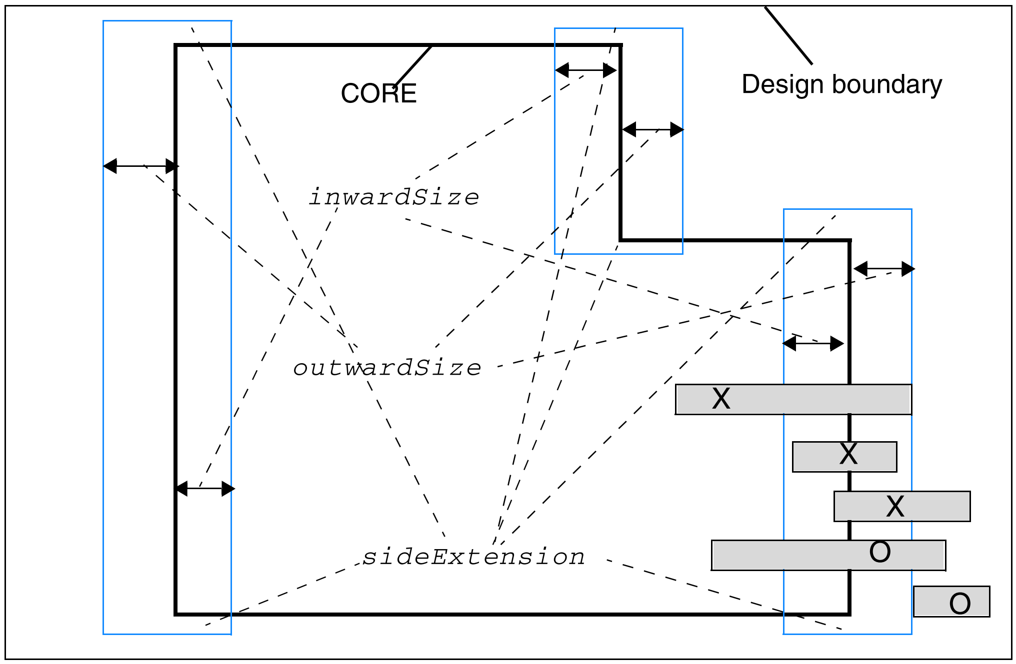
Figure 6-28 Illustration of the Core EOL Blockage Rule



Preferred routing direction is horizontal

The first wire that goes through the blockage and just touches the edge is still a violation. The next two wires have at least one line-end inside the blockage and are, therefore, violations. The fourth wire going through the blockage is legal. The bottommost wire that just touches the blockage edge is also fine.

Illustration of

```
COREEOLBLOCKAGE outwardSize inwardSize
   SIDEEXTENSION sideExtension
```

Corner EOL Keep-out Rule

A corner EOL keep-out rule specifies a keep-out region for an EOL edge. You can create a corner EOL keep-out rule by using the following property definition:

```
PROPERTY LEF58_CORNEREOLKEEPOUT
      "CORNEREOLKEEPOUT WIDTH eolWidth EOLSPACING eolSpacing
         { SPACING spacing1 spacing2 WITHIN within1 within2
         | EXTENSION backwardExt sideExt forwardExt }
         ;" ;
```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Where:

`CORNEREOLKEEPOUT WIDTH` *`eolWidth`* `EOLSPACING` *`eolSpacing`*
`SPACING` *`spacing1 spacing2`* `WITHIN` *`within1 within2`*

Specifies that if the distance between two corner-to-corner end-of-lines (EOLs) in the direction of the EOL edge is greater than or equal to *`within1`* and less than or equal to *`within2`* and the distance between them in the direction perpendicular to the EOL edge is greater than or equal to *`spacing1`* and less than or equal to *`spacing2`*, then a neighbor wire is not allowed in the search window formed by their facing corners.

Both of the EOLs must have width less than *`eolWidth`* and the spacing to a neighbor wire with `PRL` greater than `0` to the EOL must be greater than *`eolSpacing`*.

`spacing1` and *`spacing2`* could be negative, which means that if the two end-of-lines are within abs(*`spacing1`*), the rule would be triggered.

Type: Float, specified in microns

`CORNEREOLKEEPOUT WIDTH` *`eolWidth`* `EOLSPACING` *`eolSpacing`*
`EXTENSION` *`backwardExt`* `sideExt` *`forwardExt`*

Specifies a keep-out region for an EOL edge with width less than *`eolWidth`* and spacing greater than *`eolSpacing`* with `PRL` greater than `0` to a neighbor wire. The keep-out region is formed by extending *`backwardExt`* going backward, *`sideExt`* on the side, and *`forwardExt`* going forward from the corners of EOL. Any corners falling within the keep-out region will be a violation.

Type: Float, specified in microns

Corner EOL Keep-out Rule Examples

The following diagrams illustrate the corner EOL keep-out rule:

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Figure 6-29 Illustration of the Corner EOL Keep-out Rule with SPACING

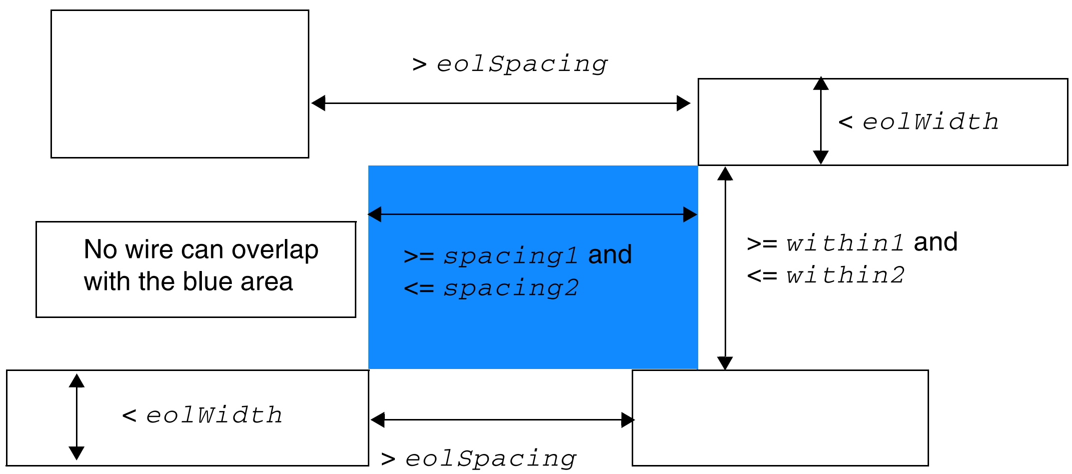

Illustration of `CORNEREOLKEEPOUT WIDTH` *`eolWidth`*
`EOLSPACING` *`eolSpacing`* `SPACING` *`spacing1`* `spacing2`
`WITHIN` *`within1 within2`*

Figure 6-30 Illustration of the Corner EOL Keep-out Rule with EXTENSION

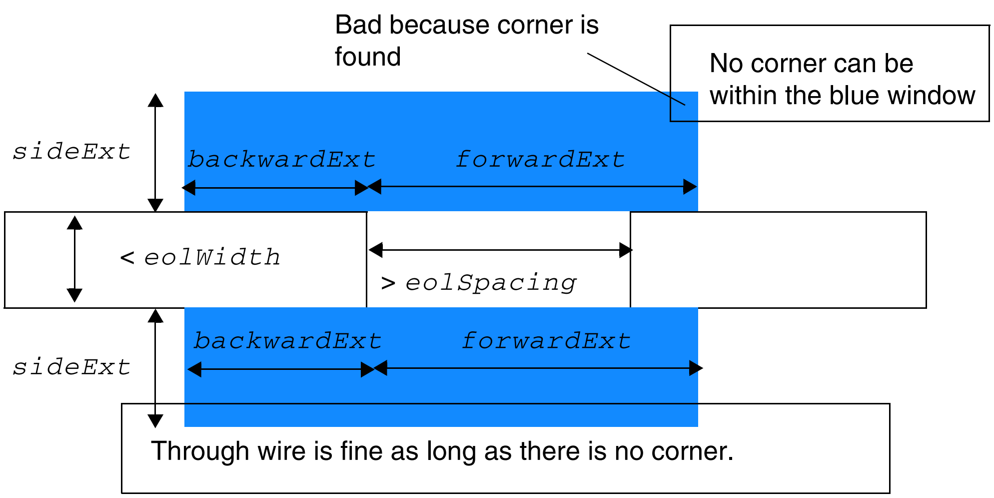

Illustration of `CORNEREOLKEEPOUT WIDTH` *`eolWidth`*
`EOLSPACING` *`eolSpacing`*
`EXTENSION` *`backwardExt sideExt forwardExt`*

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Corner Fill Spacing Rule

A corner fill spacing rule can be used to define spacing of missing corners.

You can create a corner fill spacing rule by using the following property definition:

```
PROPERTY LEF58_CORNERFILLSPACING
     "CORNERFILLSPACING spacing EDGELENGTH length1 length2
        ADJACENTEOL eolWidth ;" ;
```

Where:

`CORNERFILLSPACING` *`spacing`* `EDGELENGTH` *`length1 length2`*
`ADJACENTEOL` *`eolWidth`*

Specifies the spacing of a missing corner after the corner is filled to be *`spacing`*, if the following conditions are met:

The missing concave corner is composed of two edges with length less than *`length1`* and *`length2`* respectively. The edge with length less than *`length2`* has an adjacent EOL edge with width less than *`eolWidth`*.
Type: Float, specified in microns

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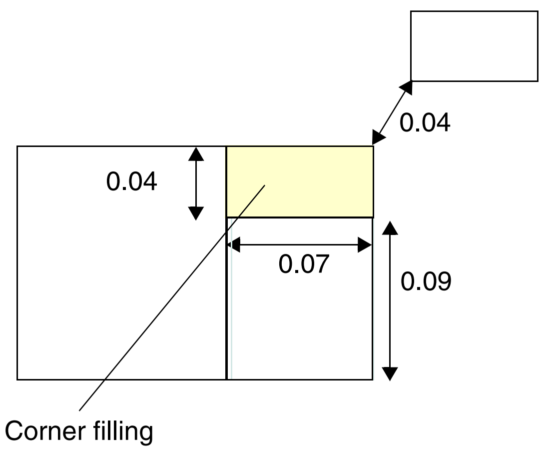
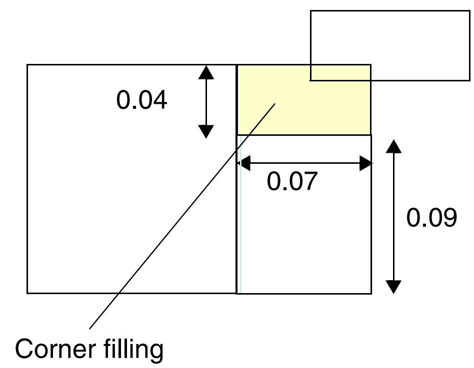
Figure 6-31 Illustration of Corner Fill Spacing Rule

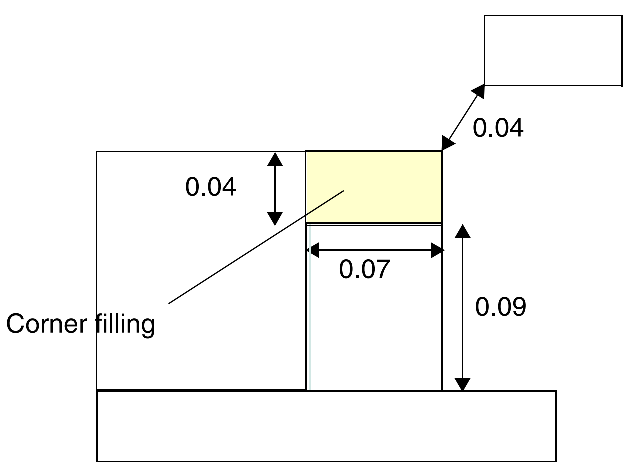
```
PROPERTY LEF58_CORNERFILLSPACING
  "CORNERFILLSPACING 0.05 EDGELENGTH 0.06 0.08
      ADJACENTEOL 0.10 ; " ;
```



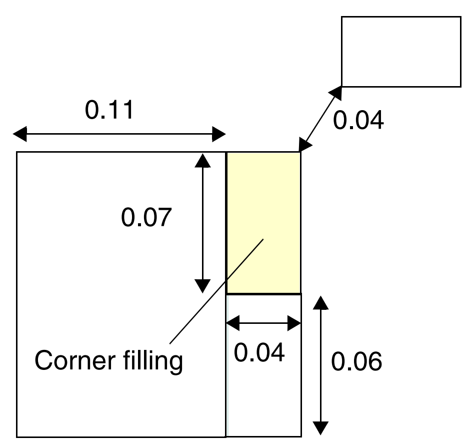
a) Violation, all of the conditions are met, the spacing after the corner filling is 0.04 (< 0.05)



b) Violation, neighbor wire overlaps with filling area is also bad



c) OK, the 0.09 adjacent edge is not a EOL edge



d) OK, 0.07 cannot be the first edge length, being the second edge length, its adjacent edge length is more than 0.10 (0.11>0.10).

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Corner Spacing Rule

You can create corner spacing rule to define spacing between convex/concave corners and any edges, depending on whether a convex or concave corner is defined.

You can create a corner spacing rule by using the following property definition:

```
PROPERTY LEF58_CORNERSPACING
     "CORNERSPACING
        {CONVEXCORNER [SAMEMASK]
            [CORNERONLY within
            | CORNERTOCORNER [ORTHOGONAL]]
            [EXCEPTEOL eolWidth [EXCEPTJOGLENGTH length
                 [EDGELENGTH] [INCLUDELSHAPE]]
            | EOLONLY eolWidth]
        |CONCAVECORNER
            [MINLENGTH minLength]
            [EXCEPTNOTCH [notchLength]]
            [TOCONVEXCORNER]}
        [EXCEPTSAMENET | EXCEPTSAMEMETAL]
        {WIDTH width SPACING spacing
        | WIDTH width SPACING horizontalSpacing verticalSpacing}...
        ; " ;
```

Where:

`CORNERONLY` *`within`*

Specifies that the corner spacing rule only applies to a neighbor corner, which is within spacing in the direction parallel to the specified direction in DIRECTION on a Manhattan routing layer and *`within`* in the perpendicular direction.

`CORNERSPACING {CONVEXCORNER | CONCAVECORNER}`
`{WIDTH width SPACING` *`spacing`*`}...`

Specifies the spacing, which is measured in `MAXXY` style, between a convex or concave corner and any edges depending on whether `CONVEXCORNER` or `CONCAVECORNER` keyword is defined. For convex corner cases, parallel run length to the neighbor wire must be less than or equal to 0 to trigger the rule. If the width of a wire containing the corner is greater than *`width`*, then the corresponding *`spacing`* is applied.

Type: Float, specified in microns

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`CORNERTOCORNER` Specifies that the rule applies only between two facing convex corners and the spacing is measured in Euclidean. In addition, the rule applies only when the PRL of the wires is strictly less than zero.

`EOLONLY` *`eolWidth`* Specifies that the rule applies only between two line ends with width less than *`eolWidth`*.

Type: Float, specified in microns

`EXCEPTEOL` *`eolWidth`* Specifies that the corner spacing rule does not apply to a corner connected to a EOL edge with width less than the *`eolWidth`*.
Type: Float, specified in microns

`EXCEPTJOGLENGTH` *`length`* `[EDGELENGTH] [INCLUDELSHAPE]`

Specifies that the minimum default (right-way) spacing will be applied between a wrong way jog with minimum wrong way width and length less than *`length`* sandwiched by two right way wires with projected `PRL` less than 0 in wrong way direction in a 'Z' shape and a right way wire.

If `INCLUDELSHAPE` is specified, this minimum default spacing will also be applied between a wrong way jog with minimum wrong way width and length less than *`length`* connected to a right way wire in a 'L' shape and a right way wire.

If `EDGELENGTH` is specified, *`length`* is measured against the length of the edges in the wrong way direction of a wrong way jog in a ‘Z’ shape instead of the entire span length of the jog. If `INCLUDELSHAPE` is also specified to include an ‘L’ shape exemption, the longest edge length would be the same as the span length of the jog, and this `EDGELENGTH` construct would have no impact on it.
Type: Float, specified in microns

`EXCEPTNOTCH [`*`notchLength`*`]`

Specifies that the rule does not apply to a notch (U shape) that is a neighbor of the concave corner. *`notchLength`* specifies that the notch length must be less than the given value to be exempted from the rule. See Figure 6-37 on page 712.

`EXCEPTSAMENET | EXCEPTSAMEMETAL`

Specifies that the rule does not apply to same-net or same-metal objects of the corner.

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MINLENGTH *minLength*

Specifies that the corner spacing rule applies only if both edges of the concave corner have length >= *minLength*.
Type: Float, specified in microns

ORTHOGONAL
Specifies that the rule applies only on the corners of two orthogonal direction wires. This means that the directions of the width of the wires of the corners should be opposite for the rule to apply.

SAMEMASK
Specifies that the corner spacing rule only applies to objects on the same mask.

TOCONVEXCORNER
Specifies that this corner spacing rule applies only between a concave corner and a convex corner. In addition, the spacing is measured in Euclidean.

WIDTH *width* SPACING *horizontalSpacing verticalSpacing*

Specifies that the rule would require *horizontalSpacing* horizontally and *verticalSpacing* vertically from the concave corner in MAXXY style to avoid a neighbor wire. If EXCEPTNOTCH without a given *notchLength* is also defined, the notch length must be less than the minimum of *horizontalSpacing* and *verticalSpacing* to be exempted from the rule.
Type: Float, specified in microns

Corner Spacing Rule Examples

- The following example illustrates the corner spacing rule:

```
PROPERTY LEF58_CORNERSPACING
    "CORNERSPACING CONVEXCORNER EXCEPTEOL 0.090
       WIDTH 0.000 SPACING 0.110 ... ;
    CORNERSPACING CONCAVECORNER EXCEPTNOTCH
       WIDTH 0.000 SPACING 0.150 ; " ;
```


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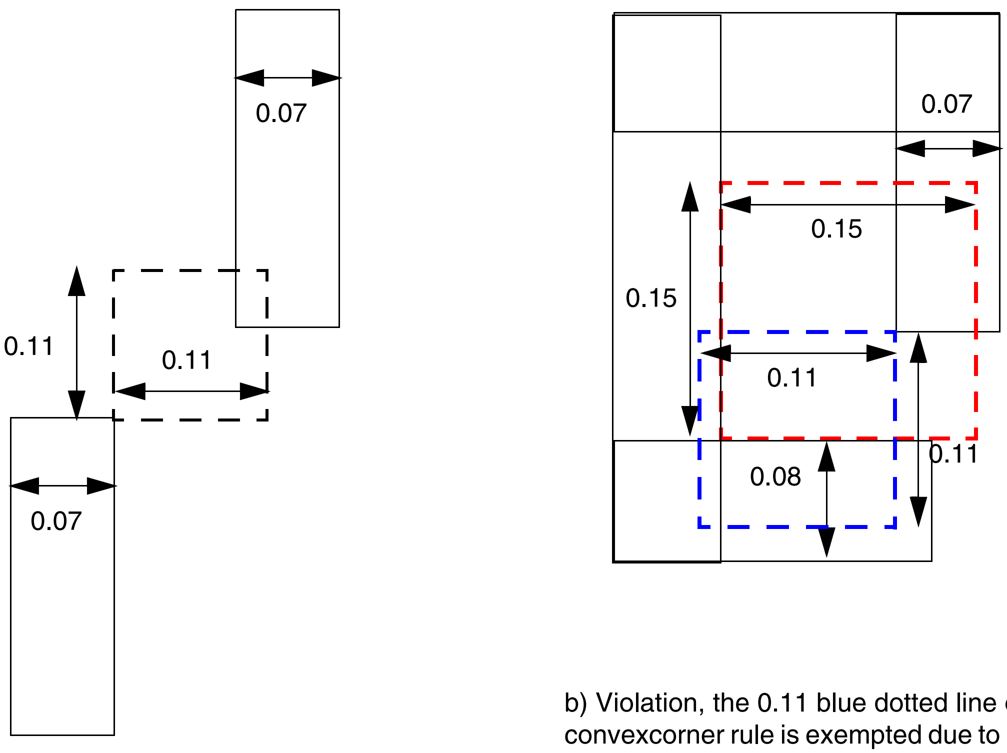
Figure 6-32 Illustration of Corner Spacing Rule



a) OK, end of line is exempted for convexcorner rule

b) Violation, the 0.11 blue dotted line of convexcorner rule is exempted due to PRL > 0, but the 0.15 red dotted line of concavecorner rule still applies and fails

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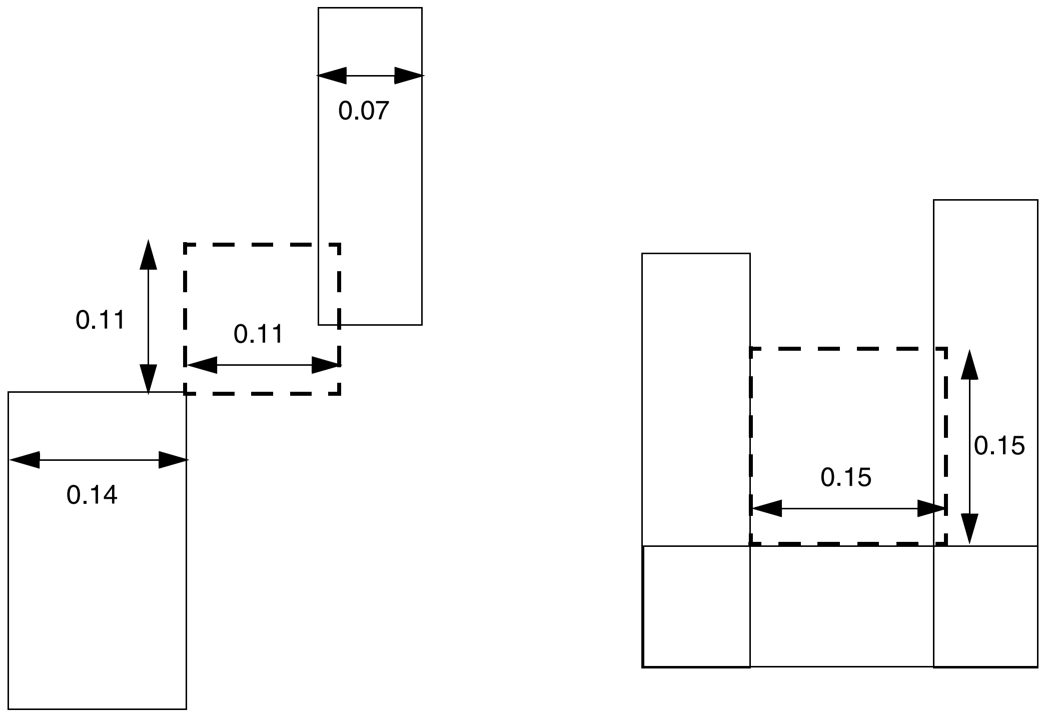
Figure 6-33 Corner Spacing Rule



c) Violation, one EOL edge would not exempt the rule.

d) OK, notch is exempted for concave corner

- The following example illustrates corner spacing rule with `CORNERONLY`:

```
DIRECTION VERTICAL ;
PROPERTY LEF58_CORNERSPACING "
     CORNERSPACING CONVEXCORNER CORNERONLY 0.050
        WIDTH 0.000 SPACING 0.100  ; " ;
```


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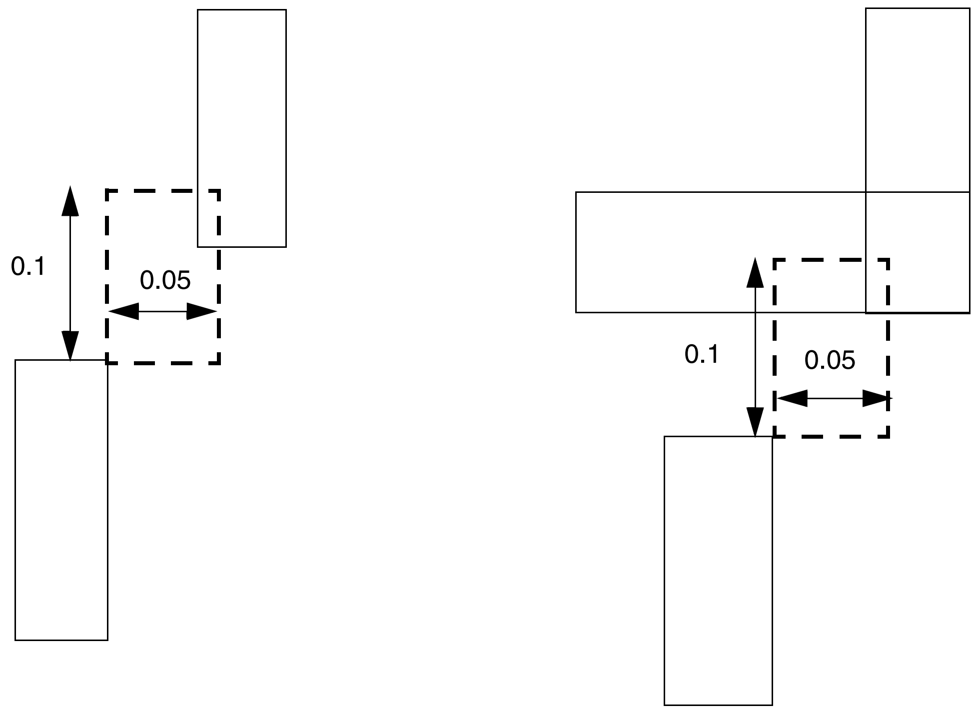
Figure 6-34 Corner Spacing Rule with CORNERONLY



a) Violation, find a neighbor corner within the search window.

b) OK, no corner found.

- The following example illustrates the corner spacing rule:

```
DIRECTION VERTICAL ;
PROPERTY LEF58_WIDTHTABLE "
    WIDTHTABLE 0.050 ... ;
    WIDTHTABLE 0.090 ... WRONGDIRECTION ; " ;
PROPERTY LEF58_CORNERSPACING "
    CORNERSPACING CONVEXCORNER EXCEPTEOL 0.090
       EXCEPTJOGLENGTH 0.170 INCLUDELSHAPE
       WIDTH 0.000 SPACING 0.110 ... ;
    CORNERSPACING CONCAVECORNER MINLENGTH 0.05
       WIDTH 0.000 SPACING 0.150 ; " ;
```


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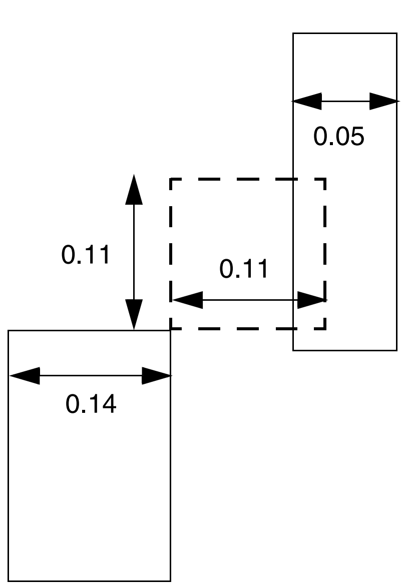
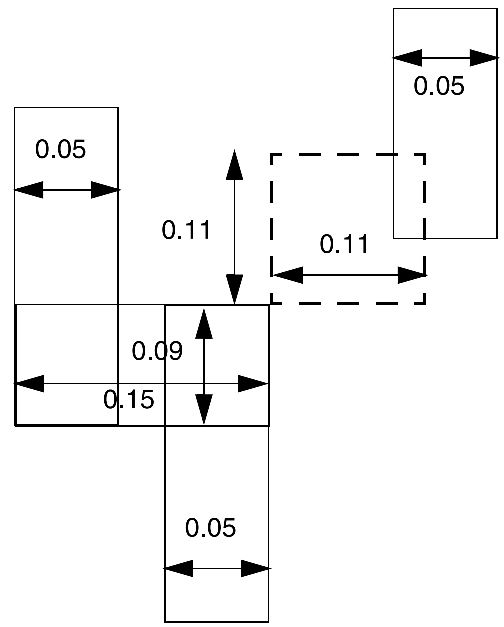
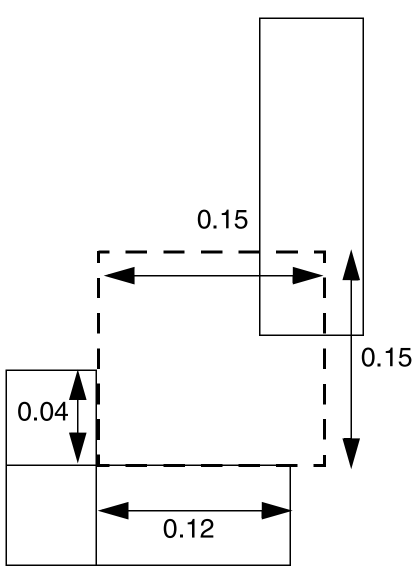
Figure 6-35 Illustration of Corner Spacing Rule



a) OK, the neighbor having PRL > 0 would not trigger the rule



b) OK, the rule is exempted between this 'Z' jog and a EOL edge.



c) OK, the vertical edge of the concave corner has length of 0.04 (< 0.05)

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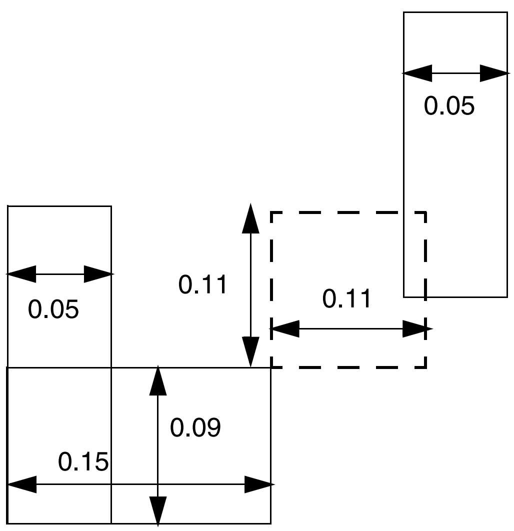
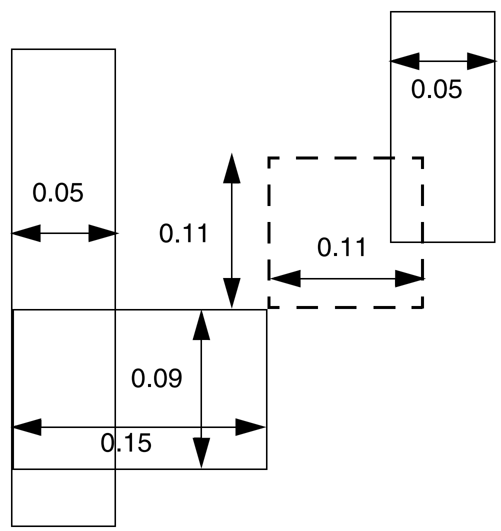
■ The following example illustrates the corner spacing rule:

```
DIRECTION VERTICAL ;
PROPERTY LEF58_WIDTHTABLE "
   WIDTHTABLE 0.050 ... ;
   WIDTHTABLE 0.090 ... WRONGDIRECTION ; " ;
PROPERTY LEF58_CORNERSPACING "
   CORNERSPACING CONVEXCORNER EXCEPTEOL 0.090
      EXCEPTJOGLENGTH 0.170
      WIDTH 0.000 SPACING 0.110 ... ; " ;
```

Figure 6-36 Illustration of Corner Spacing Rule



a) OK, the rule is exempted between this 'L' jog and a EOL edge



b) Violation, the rule does not exempt between this 'T' jog and a EOL edge

■ The following example illustrates the corner spacing rule:

```
PROPERTY LEF58_CORNERSPACING "
   CORNERSPACING CONCAVECORNER EXCEPTNOTCH 0.08
      WIDTH 0.000 SPACING 0.100 0.120 ; " ;
```


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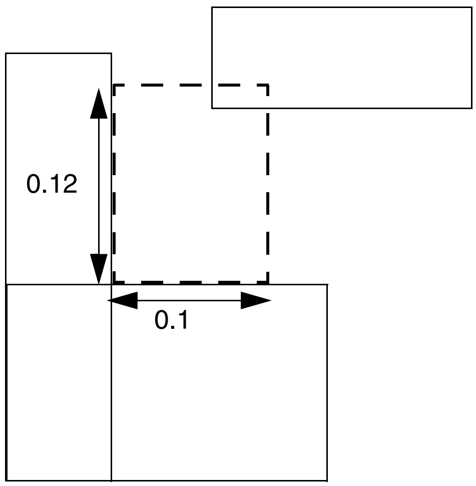
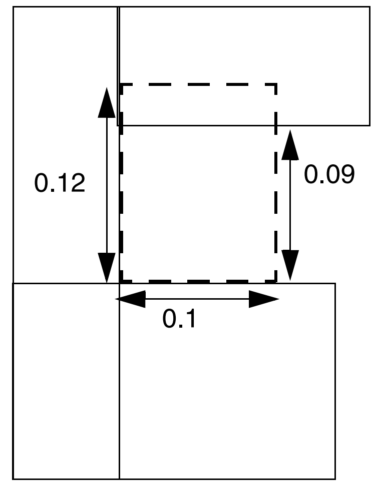
Figure 6-37 Illustration of Corner Spacing Rule with EXCEPTNOTCH in CONCAVECORNER



a) Violation, no neighbor could be inside the dotted line search window



b) Violation, notch length of `0.09` (>=`0.08`). OK if notch length of `0.08` is omitted because `0.09` is less than minimum of (0.1, 0.12).

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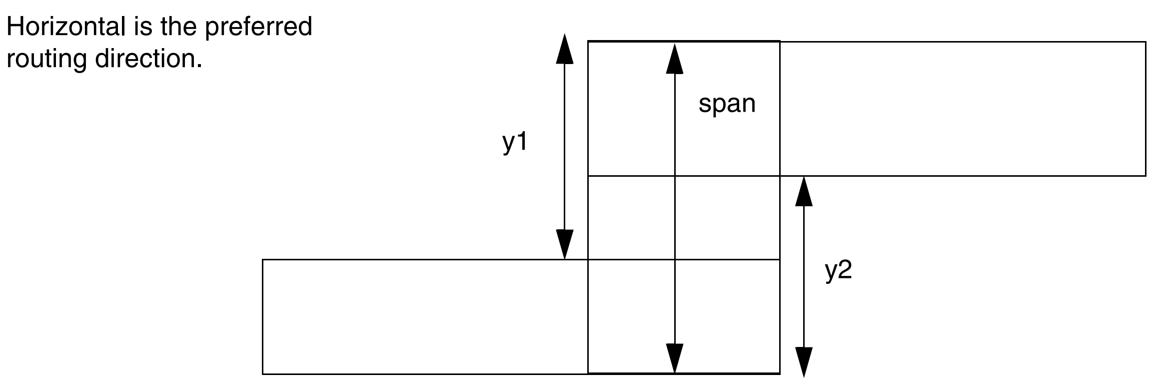
Figure 6-38 Illustration of CORNERSPACING with EDGELENGTH in EXCEPTJOGLENGTH



Both y1 and y2 must be < *`length`* to be qualified as a 'Z' shape for `EXCEPTJOGLENGTH` exemption. If `EDGELENGTH` is not specified, span must be < *`length`* to be qualified.

Illustration of `EXCEPTJOGLENGTH` *`length`* `EDGELENGTH`

- The following example illustrates the corner spacing rule with `CORNERTOCORNER`:

```
DIRECTION VERTICAL
PROPERTY LEF58_CORNERSPACING
   "CORNERSPACING CONVEXCORNER CORNERTOCORNER
      WIDTH 0.000 SPACING 0.100 ; " ;
```


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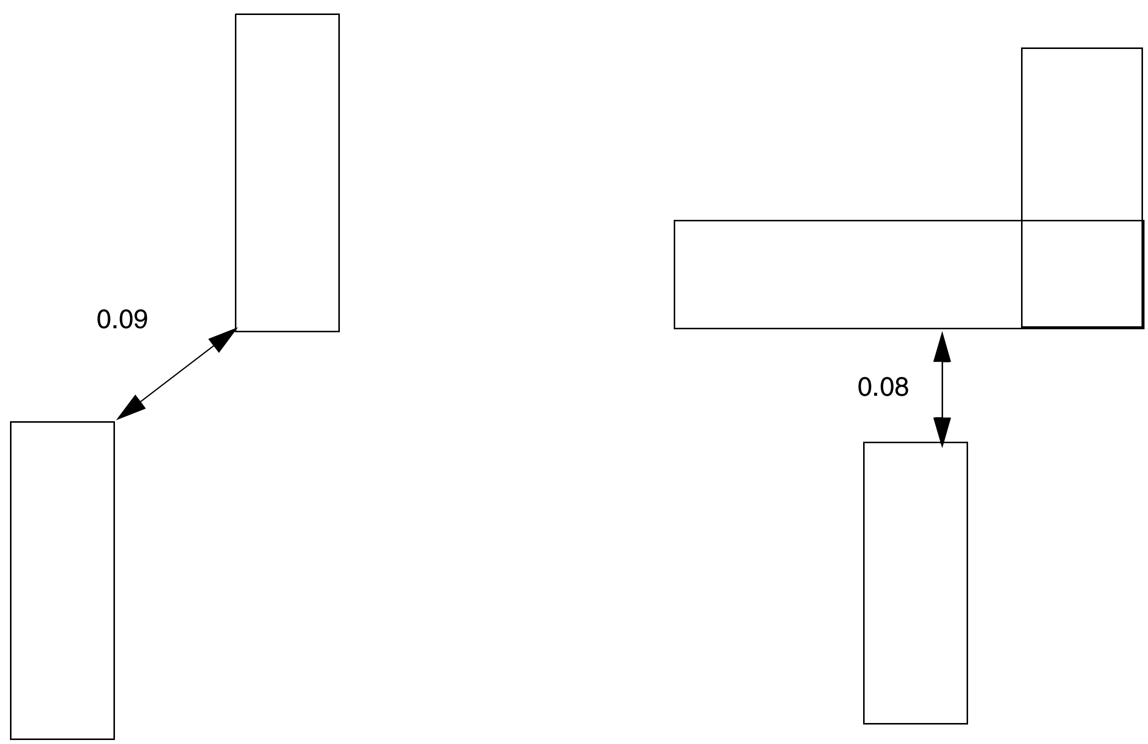
Figure 6-39 Illustration of Corner Spacing Rule with CORNERTOCORNER



a) Violation, spacing between the two convex corners is `0.09` (< `0.1`)

b) OK, no corner found

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Figure 6-40 Illustration of Corner Spacing Rule with CORNERTOCORNER ORTHOGONAL

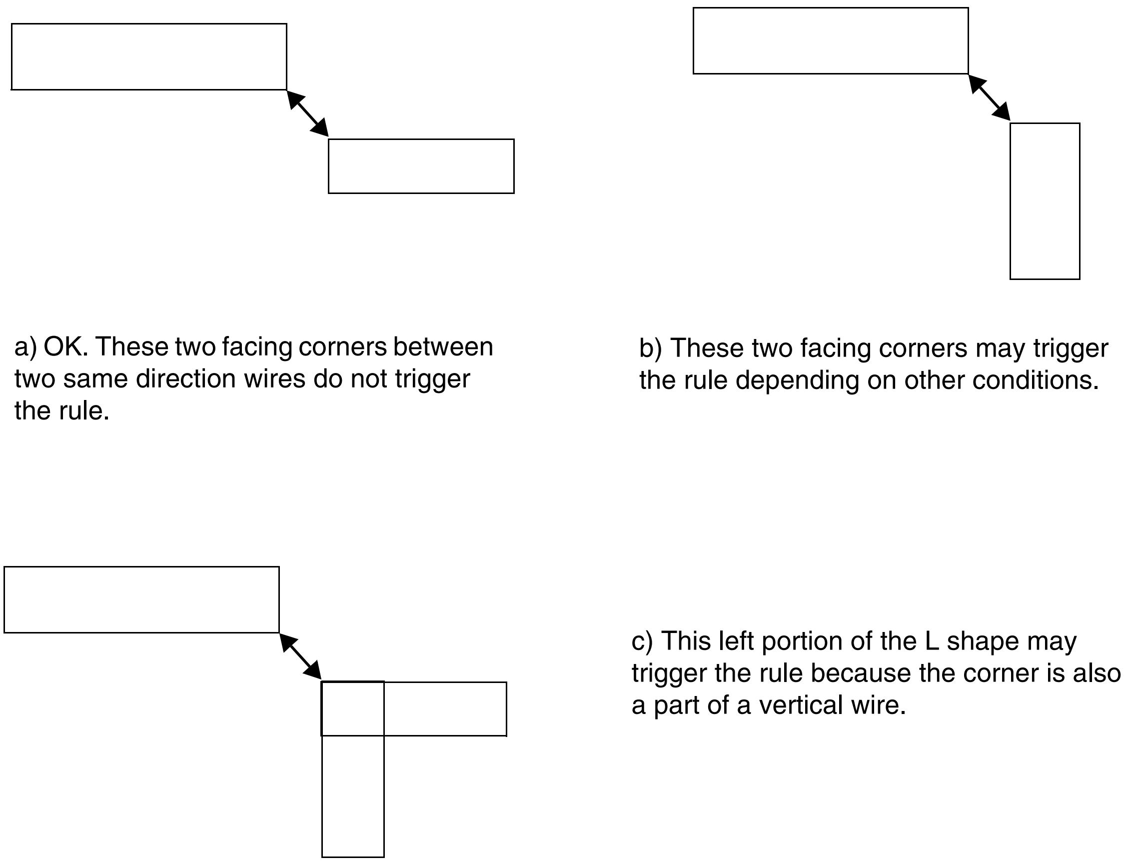

Illustration of `CORNERSPACING CONVEXCORNER CORNERTOCORNER ORTHOGONAL …`

Enclosure Spacing Rule

You can use the enclosure spacing rule to specify the spacing on an edge with enclosure less than the specified enclosure.

You can define an enclosure spacing rule by using the following property definition:

```
PROPERTY LEF58_ENCLOSURESPACING
     "ENCLOSURESPACING
          [CUTCLASS cutClass [LONGEDGEONLY] ]
```


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```
        [FROMABOVE | FROMBELOW]
        {ENCLOSURE enclosure SPACING spacing}...
    ; ... " ;
```

Where:

`CUTCLASS` *`cutClass`*

Specifies that the spacing applies only to an edge with a cut of the given *`cutClass`*. If `CUTCLASS` is defined in cut layer, then `CUTCLASS` must be specified in the `ENCLOSURESPACING` statement.

You can specify individual rules with the `CUTCLASS` keyword for each cut class, if required.

`ENCLOSURESPACING` {`ENCLOSURE` *`enclosure`* `SPACING` *`spacing`*}

Specifies the spacing on an edge with enclosure less than *`enclosure`* to be *`spacing`*. You can determine the portion of a cut having enclosure less than *`enclosure`*, extending the cut corner by *`enclosure`* in Euclidean style to intersect with the metal edge. The extended edge is checked against spacing. *Type*: Float, specified in microns

`FROMABOVE` | `FROMBELOW`

Specifies that the spacing applies only to a via cut from above or below the specified layer.

`LONGEDGEONLY` Specifies that the spacing applies only to a metal edge containing a side/long edge of a rectangular cut with enclosure less than the *`enclosure`*.

Enclosure Spacing Rule Example

The following rule indicates enclosure spacing:

```
PROPERTY LEF58_ENCLOSURESPACING  "
  ENCLOSURESPACING CUTCLASS VB LONGEDGEONLY
      ENCLOSURE 0.03 SPACING 0.1 ; " ;
```


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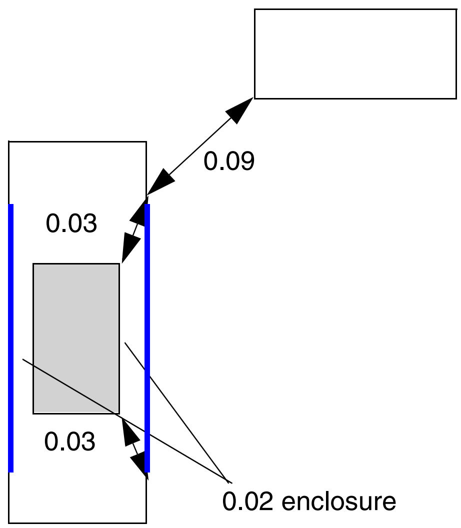
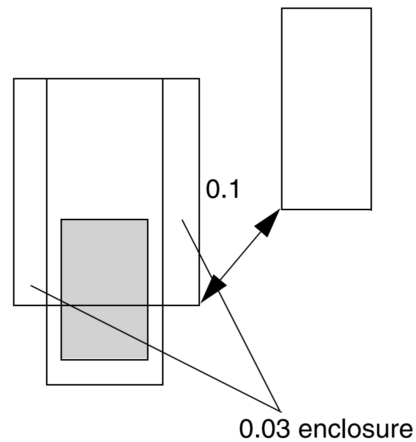
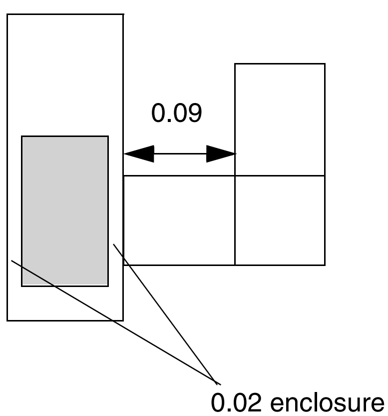
Figure 6-41 Illustration of Enclosure Spacing Rule



a) Violation, the cut corners need to extend by 0.03 to form the blue edges, where spacing of 0.1 is applied



b) OK, the metal spacing only applies to the portion of cut with enclosure < 0.03



c) Violation, the spacing applies to same-metal jog

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Enclosure with EOL Rule

You can use the enclosure with EOL rule to specify the conditions for the enclosure for a cut on the above or below cut layer with respect to an end-of-line edge.

You can define an enclosure with EOL rule by using the following property definition:

```
PROPERTY LEF58_ENCLOSUREWITHEOL
       "ENCLOSUREWITHEOL eolWidth
          [MASK maskNum] [CUTWITHIN cutWithin]
          [EXCEPTNOCUTINOTHERMETALLENGTH
               otherCutDistance otherLength]
          [CUTCLASS className] [FROMABOVE | FROMBELOW]
          {ENCLOSURE enclosure EXCEPTLINEENDSPACING spacing
          | {ENCLOSURESPACING spacing enclosure}... }
       ; ... " ;
```

Where:

`CUTCLASS` *className*

Specifies that the rule applies only to an end-of-line edge with a cut of the given cut class. If `CUTCLASS` is defined in the cut layer, `CUTCLASS` must be specified in the `ENCLOSUREWITHEOL` statement.

You can specify individual rules with the `CUTCLASS` keyword for each cut class, if required.

`CUTWITHIN` *cutWithin*

Specifies that the rule applies only if there is another same-metal cut with *cutWithin* on the rule-defined layer.

Type: Float, specified in microns

`{ENCLOSURESPACING` *spacing enclosure*`}...`

Specifies that if the EOL spacing is greater than *spacing*, the corresponding enclosure would be applied to the cut on the EOL on the above and/or below cut layer depending on whether `FROMABOVE` or `FROMBELOW` is specified.

Type: Float, specified in microns

`ENCLOSUREWITHEOL` *eolWidth*
`ENCLOSURE` *enclosure* `EXCEPTLINEENDSPACING` *spacing*

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Specifies that a cut on either the above or below cut layer must have enclosure greater than or equal to *enclosure* to an end-of-line edge with width less than *eolWidth* unless the end-of-line spacing is exactly equal to *spacing*.
Type: Float, specified in microns

EXCEPTNOCUTINOTHERMETALLENGTH *otherCutDistance otherLength*

Specifies that the rule is exempted if the neighboring wire has length greater than or equal to *otherLength* and there is no cut on the neighboring wire belonging to the given cut class *className* if defined or any cut class on the given layer in FROMABOVE or FROMBELOW, if defined, or on any above or below cut layer having a distance to the triggering cut less than *otherCutDistance*. That distance would be equivalent to enclosure of the two cuts plus the EOL spacing of the two wires.

Type: Float, specified in microns

FROMABOVE | FROMBELOW

Specifies that the spacing applies only to a via cut from above or below the specified layer.

MASK *maskNum*

Specifies that the rule is applied only to metal with the given mask. *maskNum* must be a positive integer, and most applications only support values of 1, 2, or 3.

Type: Integer

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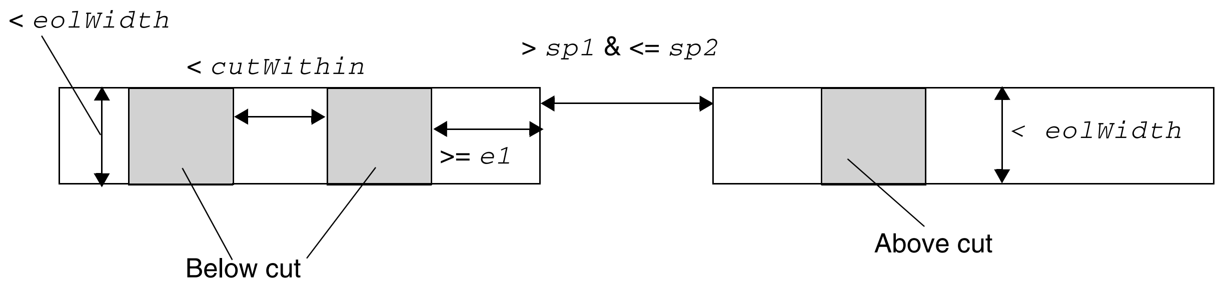
Enclosure with EOL Rule Examples

Figure 6-42 Illustration of Enclosure with EOL Rule with CUTWITHIN



As EOL spacing is > `sp1` & <= `sp2`, `e1` enclosure is applied on the below cut. The enclosure is not applied on the above cut because it does not have a neighbor cut within `cutWithin`.

Illustration of

```
ENCLOSUREWITHEOL eolWidth
   CUTWITHIN cutWithin
   ENCLOSURESPACING sp1 e1
   ENCLOSURESPACING sp2 e2
```


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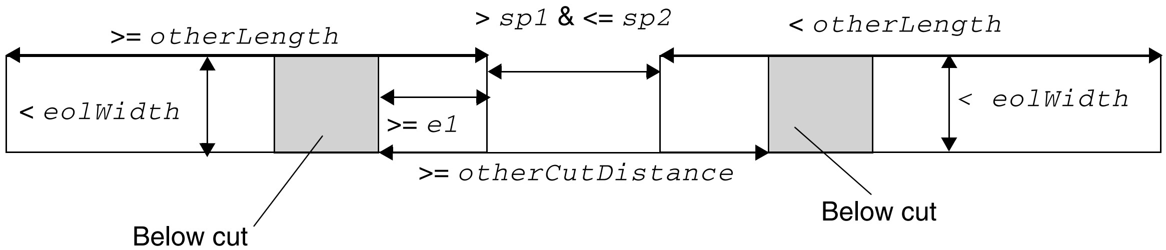
Figure 6-43 Illustration of Enclosure with EOL Rule with EXCEPTNOCUTINOTHERMETALLENGTH



As EOL spacing is > `sp1` & <= `sp2`, `e1` enclosure is applied on the left cut because the length of the right wire is < `otherLength` and does not trigger the `EXCEPTNOCUTINOTHERMETALLENGTH` exemption. The right cut is exempted because both the length and cut distance conditions are met in `EXCEPTNOCUTINOTHERMETALLENGTH`.

Illustration of

```
ENCLOSUREWITHEOL eolWidth
    EXCEPTNOCUTINOTHERMETALLENGTH
        otherCutDistance otherLength
    ENCLOSURESPACING sp1 e1
    ENCLOSURESPACING sp2 e2
```


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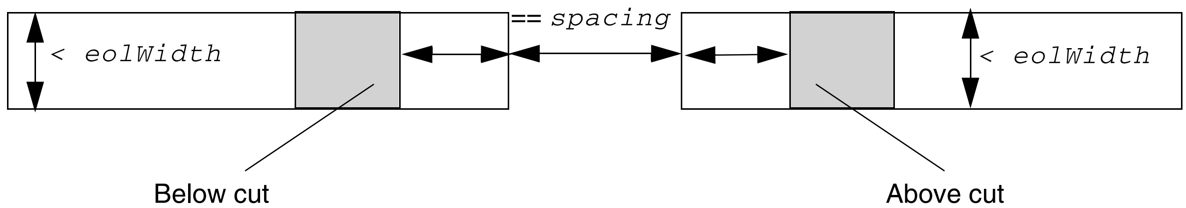
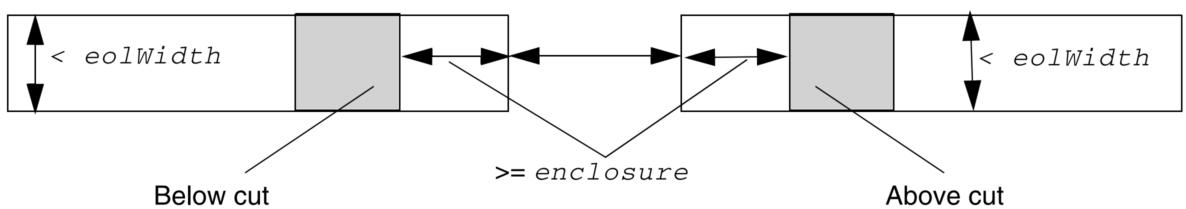
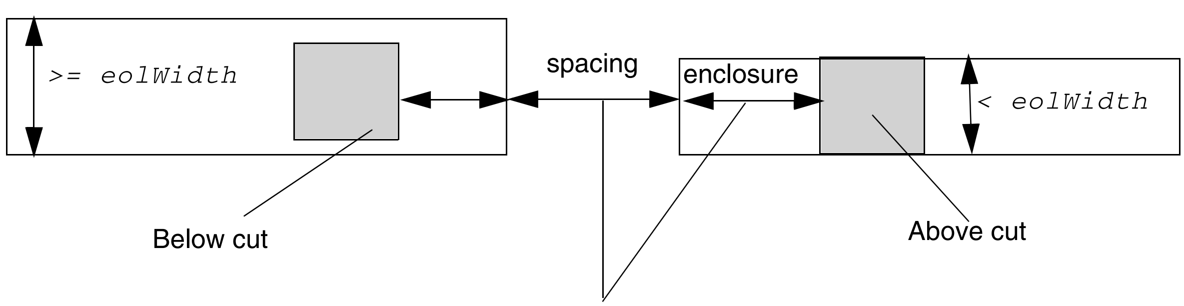
Figure 6-44 Illustrations of Enclosure with EOL Rule with EXCEPTLINEENDSPACING



Since EOL spacing is exactly equal to `spacing`, the enclosure only needs to fulfill basic enclosure and does not needs to be checked against `enclosure`.



Since enclosure is >= `enclosure` on both cuts, the end-of-line spacing only needs to fulfill basic EOL spacing and does not needs to be checked against `spacing`.



The left cut does not matter since it is not against an EOL. The right cut should have enclosure >= `enclosure` or the right EOL has spacing == `spacing`.

Illustrations of
`ENCLOSUREWITHEOL` *`eolWidth`*
`ENCLOSURE` *`enclosure`*
`EXCEPTLINEENDSPACING` *`spacing`*

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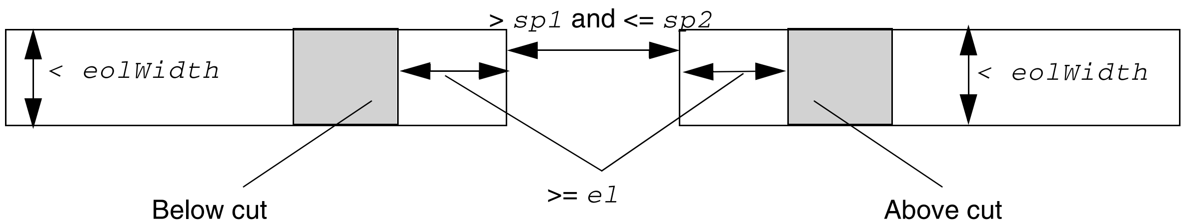
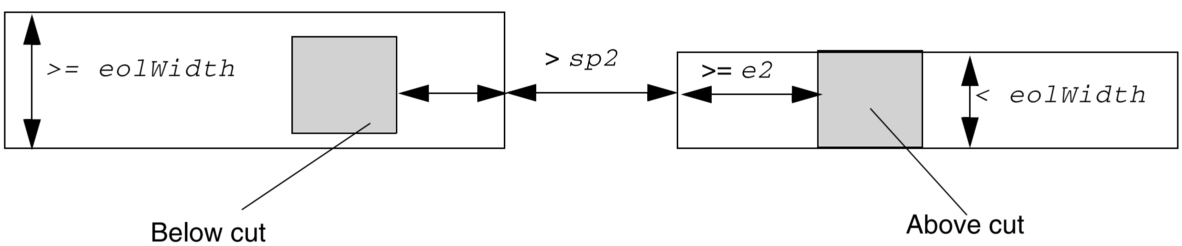
Figure 6-45 Illustrations of Enclosure with EOL Rule with ENCLOSURESPACING



As EOL spacing is greater than *sp1* and less than or equal to *sp2*, *e1* enclosure is applied on both below and above cuts.



The left cut does not matter because it is not against an EOL.
The right cut should have enclosure >= *e2*.

Illustrations of

```
ENCLOSUREWITHEOL eolWidth
    ENCLOSURESPACING sp1 e1
    ENCLOSURESPACING sp2 e2
```

EOL Extension Spacing Rule

EOL extension spacing rule can be used to indicate that for a given width of an end-of-line certain extension should be applied to the EOL edge before checking for edge-to-edge spacing to any neighbor wires.

You can define a EOL extension spacing rule by using the following property definition:

```
PROPERTY LEF58_EOLEXTENSIONSPACING
       "EOLEXTENSIONSPACING spacing [SAMEMASK]
         [PARALLELONLY] [NONEOL]]
```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```
        [OTHERWIDTH otherWidth]
        {ENDOFLINE eolWidth EXTENSION extension
           [ENDTOEND endToEndExtension]} ...
        [MINLENGTH minLength [TWOSIDES]]
      ;" ;
```

Where:

EOLEXTENSIONSPACING *spacing* {ENDOFLINE *eolWidth* EXTENSION *extension* [ENDTOEND *endToEndExtension*]} ...

Specifies that for a given width of an end-of-line, find the last row where the width is less than *eolWidth*, the corresponding extension should be applied to the EOL edge before checking for edge-to-edge spacing to any neighbor wires. The *eolWidth* values should be increased for each subsequent ENDOFLINE statement, and at the most three statements can be specified and supported.
Type: Float, specified in microns

ENDTOEND *endToEndExtension* specifies that *endToEndExtension* is applied if the neighbor wire is also an end-of-line, and extension is applied for an end-to-side situation.
Type: Float, specified in microns

MINLENGTH *minLength* [TWOSIDES]

Indicates that the rule does not apply if the end-of-line length is less than *minLength* along both sides.
Type: Float, specified in microns

TWOSIDES means that the rule only applies when the end-of-line length is greater than or equal to *minLength* along both the sides. In other words, the rule does not apply if the end-of-line length is less than *minLength* along any one side.

OTHERWIDTH *otherWidth*

Specifies that the rule applies only if the width of the neighbor wire is less than *otherWidth*. See Figure 6-47 on page 727.

PARALLELONLY [NONEOL]]

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Specifies that the rule only applies if a facing neighbor parallel edge to the EOL edge is found. If `NONEOL` is specified, the neighbor parallel edge cannot be a EOL. See Figure 6-48 on page 727.

`SAMEMASK` Specifies that the EOL extension spacing only applies to same-mask objects.

Note: Multiple `EOLEXTENSIONSPACING` statements with `SAMEMASK` can be defined.

EOL Extension Spacing Rule Examples

- The following EOL extension spacing rule indicates that an end-of-line edge with width less than 0.11 µm should have 0.14 µm extension, and an end-of-line edge with width less than 0.15 µm and greater than or equal to 0.11 µm should have 0.13 µm extension in end-to-end situation or 0.12 µm extension in end-to-side situation. Then 0.1 µm edge to edge spacing must be enforced with proper extension on the end-of-line edge(s).

```
PROPERTY LEF58_EOLEXTENSIONSPACING
    "EOLEXTENSIONSPACING 0.1 ENDOFLINE 0.11 EXTENSION 0.14
     ENDOFLINE 0.15 EXTENSION 0.12 ENDTOEND 0.13 ;" ;
```


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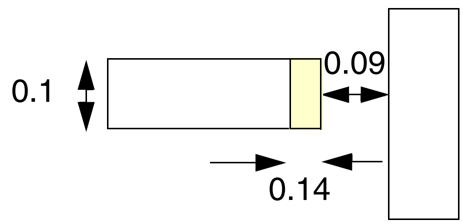
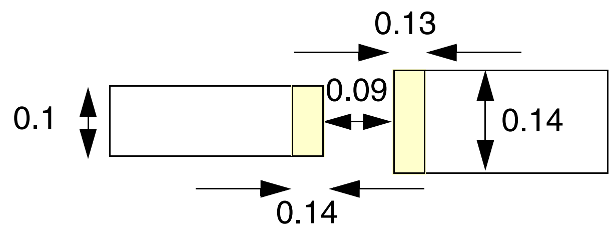
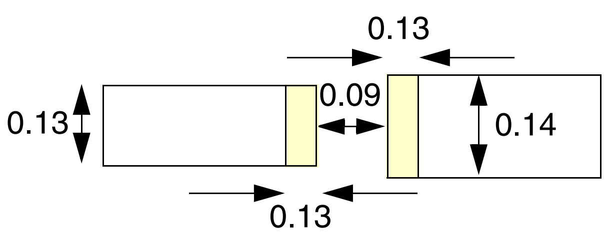
Figure 6-46 EOL Extension Spacing Rule Example



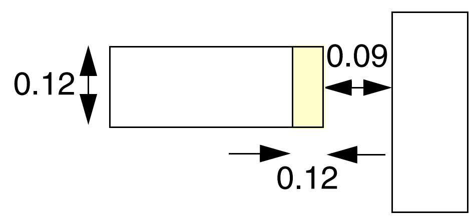
a) Violation, 0.14 extension is applied for width < 0.11, and 0.1 spacing is required after the extension



b) Violation, for end-to-end situation, 0.14 extension is applied for width < 0.11 on the left, 0.13 extension is applied for width >= 0.11, and 0.1 spacing is required



c) Violation, for end-to-end situation, 0.13 extension is applied for both wires with width >= 0.11, and 0.1 spacing is required



d) Violation, 0.12 extension is applied for width >= 0.11 for end-to-side, and 0.1 spacing is required

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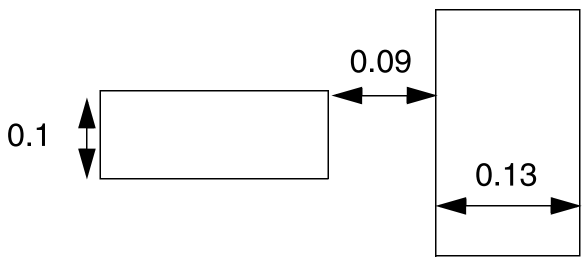
Figure 6-47 EOL Extension Spacing Rule with OTHERWIDTH



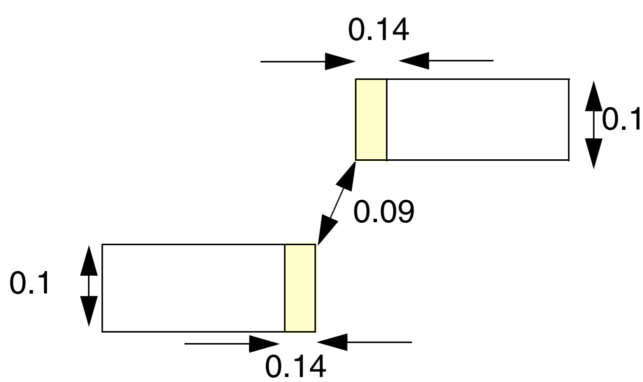
a) OK, the neighbor wire has width of 0.13 (>= 0.12). Violation if `OTHERWIDTH` is omitted.

Example of `PROPERTY LEF58_EOLEXTENSIONSPACING "`

```
    EOLEXTENSIONSPACING 0.1
        OTHERWIDTH 0.12
        ENDOFLINE 0.11 EXTENSION 0.00 ; " ;
```

Figure 6-48 EOL Extension Spacing Rule with NONEOL

```
PROPERTY LEF58_EOLEXTENSIONSPACING "
    EOLEXTENSIONSPACING 0.1
        PARALLELONLY NONEOL
        ENDOFLINE 0.11 EXTENSION 0.14 ; " ;
```



a) OK, the neighbor edge must not be a EOL.
Violation if NONEOL is omitted.

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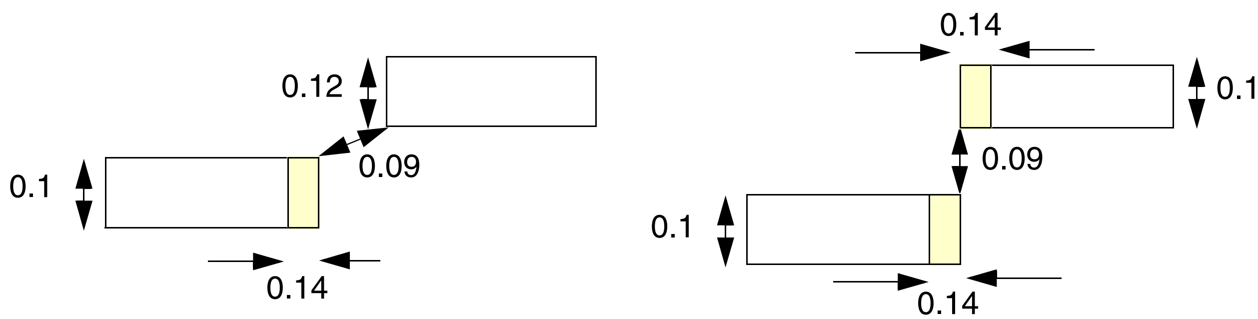
- The following EOL extension spacing rule indicates that if a facing neighbor parallel edge to the EOL edge is found, then without `PARALLELONLY` it will be a violation:

```
PROPERTY LEF58_EOLEXTENSIONSPACING
   "EOLEXTENSIONSPACING 0.1  PARALLELONLY
       ENDOFLINE 0.11 EXTENSION 0.14 ; " ;
```

Figure 6-49 EOL Extension Spacing Rule Example



a) Violation, after extending 0.14, the EOL edge to a parallel (vertical) neighbor edge spacing is 0.09 (< 0.1)

b) OK, there is no facing parallel (vertical) neighbor edge. Violation if PARALLELONLY is omitted.

EOL Keep-out Rule

EOL keep-out rule can be used to define a keep-out region for an end-of-line edge.

You can create an EOL keep-out rule by using the following property definition:

```
PROPERTY LEF58_EOLKEEPOUT
     "EOLKEEPOUT {eolWidth | minEolWidth maxEolWidth}
         [RIGHTWAYONLY | WRONGWAYONLY]
         EXTENSION backwardExt sideExt forwardExt
         [EXCEPTWITHIN lowSideExt highSideExt]
         [CLASS className [OTHERENDEOL]]
         [CORNERONLY
             [EXCEPTFROM { BACKEDGE [FORWARDGAP forwardGap]
                 [EXACTALIGNED exactForwardExt]
                 [EXTRAFORWARDKEEPOUT minForwardExt
                      maxForwardExt]
                 | FRONTEDGE }
              [OTHEREOL [minEolWidth maxEolWidth]]
              [MASK maskNum [TWOSIDES] [SAMEMASK]
              | DIFFMASK
              | SAMEMASK]
              [EXCEPTSAMESIDEMETAL backwardExt1 sideExt1 forwardExt1]
```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```
        [EXCEPTNOOPPSIDEMETAL backwardExt2 sideExt2 forwardExt2]
        [EXCEPTLINEENDSPACING spacing | EXCEPTLINEENDGAP]
        [EOLWITHINCUT withinCut [FROMABOVE|FROMBELOW]]
       [EXCEPTEOLSPACING eolSpacing]
       ]]
   [EXCEPTSAMEMETAL]
  ; " ;
```

Where:

`CLASS` *`className`*

Assigns a class name to the rule. It is a violation if all the rules belonging to the same class fail. An entire set of failed class rules will be a violation. If any one of the rules have `CLASS`, then `CLASS` must be specified for all of the rules. Typically, `CLASS` is used along with `EXCEPTFROM` - one being `BACKEDGE` and another being `FRONTEDGE`. It is a violation if both of them fail on the same side of the wire.

`CORNERONLY`

Specifies that it is a violation if there is a corner falling within the keep-out, and a pass-through edge is allowed.

`DIFFMASK`

Specifies that the rule applies only to different-mask objects.

`EOLKEEPOUT` *`eolWidth`* `EXTENSION` *`backwardExt sideExt forwardExt`*

Defines a keep-out region for an end-of-line edge with length less than the *`eolWidth`* by extending *`backwardExt`* going backward, *`sideExt`* on the side, and *`forwardExt`* going forward. Any objects falling within the region will be a violation.

Type: Float, specified in microns

`EOLWITHINCUT` *`withinCut`* `[FROMABOVE|FROMBELOW]`

Specifies that the rule applies only if there is a cut either on the below or above cut layer that is within *`withinCut`* from the end of line (EOL).If `FROMABOVE|FROMBELOW` is specified, the cut must be from the above or below cut layer. If `OTHEREOL` is also specified, a separate cut must also exist on the neighboring EOL wire to trigger the rule.

Type: Float, specified in microns

`EXACTALIGNED` *`exactForwardExt`*

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Specifies that *exactForwardExt* applies only between two perfectly aligned EOL edges with the same width and the original `EXTENSION` values do not need to be checked. See Figure 6-58 on page 741.

Type: Float, specified in microns

`EXCEPTFROM {BACKEDGE | FRONTEDGE}`

Specifies that the rule only applies if the neighbor wire containing the corner is not coming from the front edge of the keep-out window in `EXCEPTFROM FRONTEDGE` or not coming from the back edge of the search window in `EXCEPTFROM BACKEDGE`.

`EXCEPTEOLSPACING` *eolSpacing*

Specifies that the rule does not apply if there is no neighboring wire less than or equal to *eolSpacing* and parallel run length (PRL) greater than `0` to the EOL.

Type: Float, specified in microns

`EXCEPTLINEENDGAP`

Specifies that the rule does not apply if the `LINEENDGAP` condition on this layer is met. This exemption would be applied on a triggering EOL or as a neighboring wire meeting the `LINEENDGAP` condition.

`EXCEPTLINEENDSPACING` *spacing*

Specifies that the rule does not apply if there is another wire of the neighboring wire with spacing exactly equal to *spacing* with PRL greater than `0` to the wire edge of the neighbor triggering the rule.

Type: Float, specified in microns

`EXCEPTNOOPPSIDEMETAL` *backwardExt2 sideExt2 forwardExt2*

Specifies that the rule does not apply if there is no wire on the opposite side of the neighboring wire in a window formed by extensions of *backwardExt2*, *sideExt2*, and *forwardExt2* on the end-of-line (EOL) edge.

Type: Float, specified in microns

`EXCEPTSAMESIDEMETAL` *backwardExt1 sideExt1 forwardExt1*

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Specifies that the rule does not apply if there is another wire on the same side of the neighboring wire in a window formed by extensions of *backwardExt1*, *sideExt1*, and *forwardExt1* on the end-of-line (EOL) edge.

Type: Float, specified in microns

`EXCEPTWITHIN` *lowSideExt highSideExt*

Specifies that the rule is not checked on a neighbor object overlapping with a region greater than or equal to *lowSideExt* and less than or equal to *highSideExt* away from the sides of the EOL edge, which covers the entire range of *backwardExt* and *forwardExt*.

Type: Float, specified in microns

`EXCEPTSAMEMETAL`

Specifies that the keep-out region does not apply to same-metal objects.

`EXTRAFORWARDKEEPOUT` *minForwardExt maxForwardExt*

Specifies an additional keep-out window on an EOL edge by extending *sideExt* on the side, and between *minForwardExt* and *maxForwardExt* going forward.

Type: Float, specified in microns

`FORWARDGAP` *forwardGap*

Specifies if there is a face-to-face object within the search window of *backwardExt sideExt forwardGap*, the objects must be extended such that the face-to-face gap must become *forwardExt*. See Figure 6-52 on page 735.

Note: `FORWARDGAP` is no longer supported by tools.

Type: Float, specified in microns

`MASK` *maskNum*

Specifies which mask the triggering wire of this rule belongs to, *maskNum* must be a positive integer, and most applications only support values of 1, 2, or 3. See Figure 6-59 on page 742.
Type: Integer

minEolWidth maxEolWidth

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Specifies that the rule applies only if the EOL width is greater than or equal to *minEolWidth* and less than *maxEolWidth*.

Type: Float, specified in microns

OTHERENDEOL

Specifies that if the object found in the extension window is also an EOL, it is only a violation if that EOL also fails all of the EOLKEEPOUT statements in the same CLASS. OTHERENDEOL must be specified along with EXCEPTFROM BACKEDGE since it would be a symmetrical situation meaning either one of the two EOLs as a trigger edge would find the other EOL, and it makes sense for both EOLs to fail other EOLKEEPOUT statements to be a violation.

OTHEREOL [*minEolWidth maxEolWidth*]

Specifies that the rule applies only if the neighboring wire is also an EOL wire. If two EOL width values are defined, the neighboring EOL must be greater than or equal to *minEolWidth* and less than *maxEolWidth*.

RIGHTWAYONLY | WRONGWAYONLY

Specifies that the rule applies only if the EOL edge is in the preferred routing direction in RIGHTWAYONLY or in the non-preferred routing direction in WRONGWAYONLY. Tools can only support WRONGWAYONLY at present.
See Figure 6-67 on page 749.

SAMEMASK

Specifies that the rule applies only to same-mask objects.

TWOSIDES

Specifies that the rule only applies if it has 2 different-mask wires on both sides.

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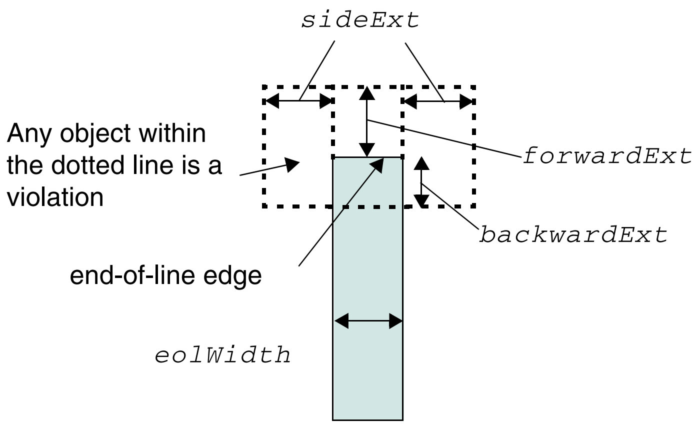
Figure 6-50 Definition of EOL Keep Out Rule



Definition of `EOLKEEPOUT` *`eolWidth`* `EXTENSION` *`backwardExt sideExt forwardExt`*

EOL Keep Out Rule Examples

- The following example illustrates `EOLKEEPOUT` with `EXTENSION:`

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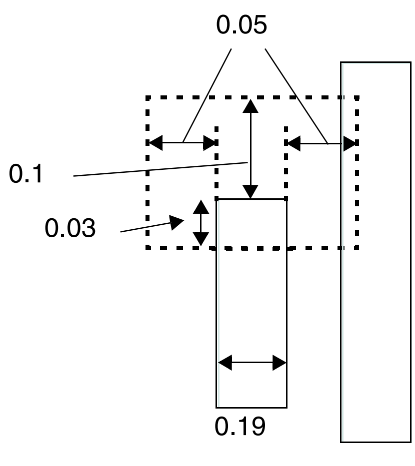
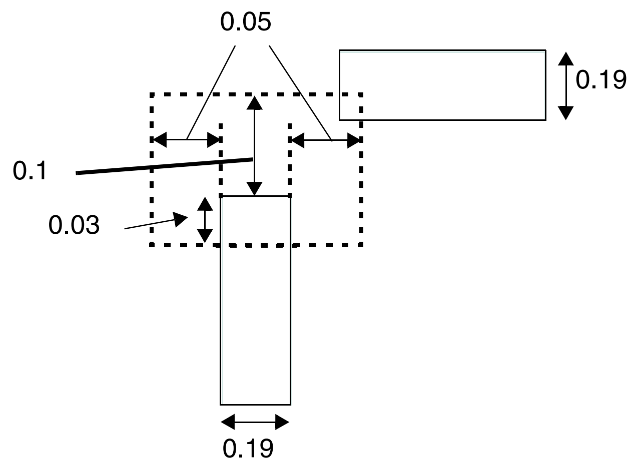
Figure 6-51 Illustration of EOL Keep Out Rule with EXTENSION

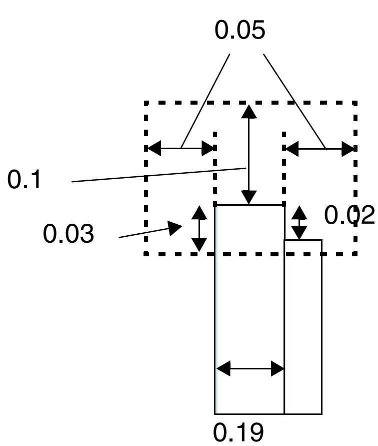
```
PROPERTY LEF58_EOLKEEPOUT
  "EOLKEEPOUT 0.2 EXTENSION 0.03 0.05 0.1 ; " ;
```



a) Violation, side neighbor is also a violation. With CORNERONLY, it is OK since there is no corner neighbor.



b) Violation, any neighbor in the dotted region is a violation.



c) Violation, any neighbors, including same-metal, would be a violation. With EXCEPTSAMEMETAL, it would be OK.

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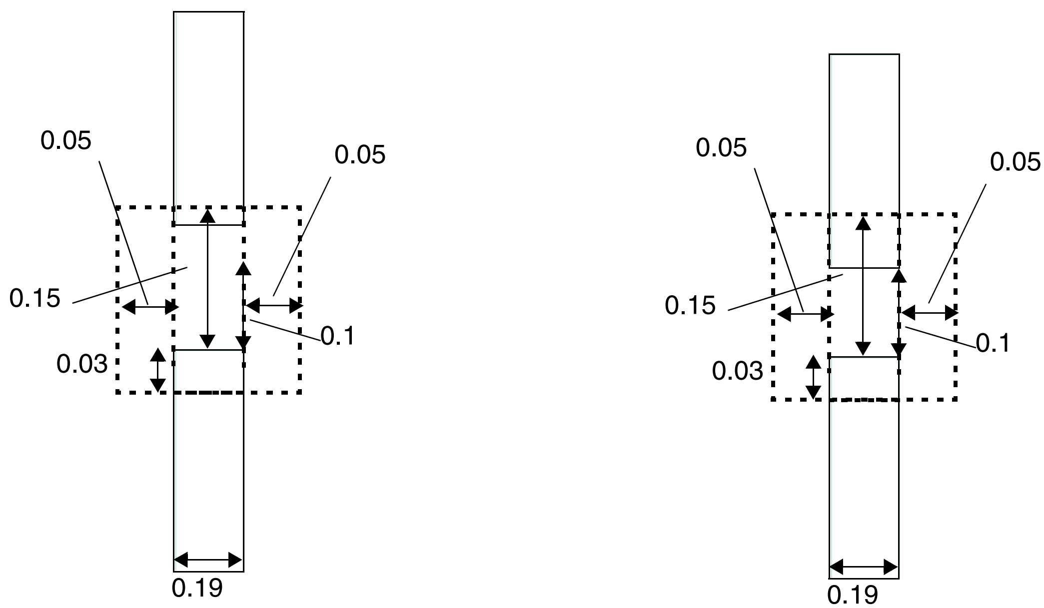
Figure 6-52 Illustration of EOL Keep Out Rule with FORWARDGAP

```
PROPERTY LEF58_EOLKEEPOUT "
    EOLKEEPOUT 0.2 EXTENSION 0.03 0.05 0.1
        CORNERONLY EXCEPTFROM BACKEDGE
        FORWARDGAP 0.15  ; " ;
```



a) Violation, if there is a face-to-face object within the dotted search window, the spacing between the object must be exactly equal to 0.1

b) OK, the spacing is exactly 0.1

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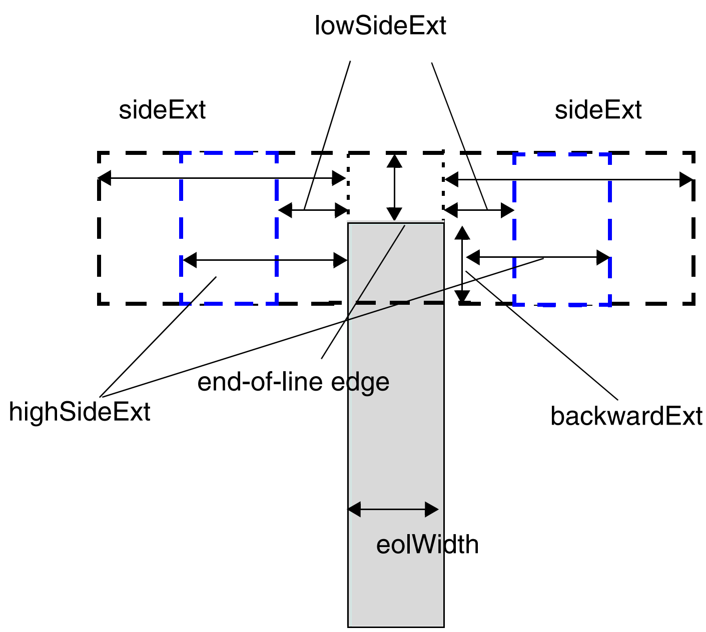
Figure 6-53 Illustration of EOL Keep Out Rule with eolWidth and EXCEPTWITHIN

Illustration of `EOLKEEPOUT` *`eolWidth`*
`EXTENSION` *`backwardExt sideExt forwardExt`*
`EXCEPTWITHIN` *`lowSideExt highSideExt`*



Any objects overlapping with the blue dotted regions is exempted from the rule

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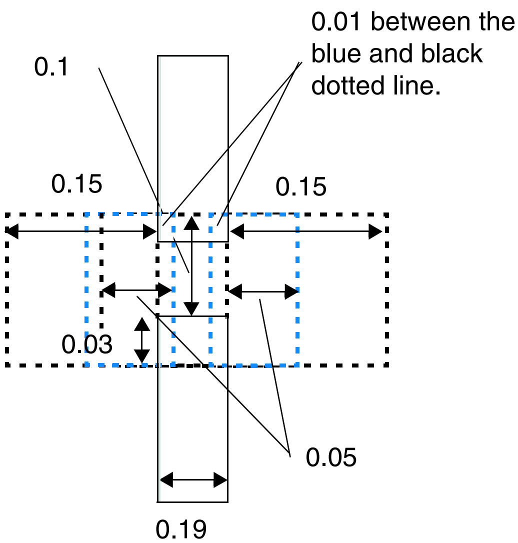
Figure 6-54 Illustration of EOL Keep Out Rule with EXCEPTWITHIN

```
PROPERTY LEF58_EOLKEEPOUT
  "EOLKEEPOUT 0.2 EXTENSION 0.03 0.15 0.1 ; " ;
     EXCEPTWITHIN -0.01 0.05 ;";
```



a) OK, having negative EXCEPTWITHIN value would exclude neighbor on the same track

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Figure 6-55 Illustration of EOL Keep Out Rule with EXTRAFORWARDKEEPOUT

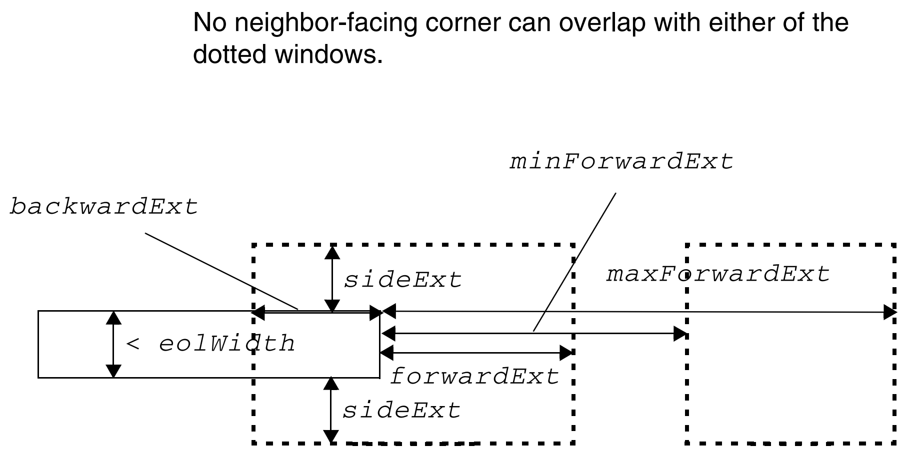

Illustration of:

```
EOLKEEPOUT eolWidth
  EXTENSION backwardExt sideExt forwardExt
  CORNERONLY EXCEPTFROM BACKEDGE
      EXTRAFORWARDKEEPOUT minForwardExt maxForwardExt
```


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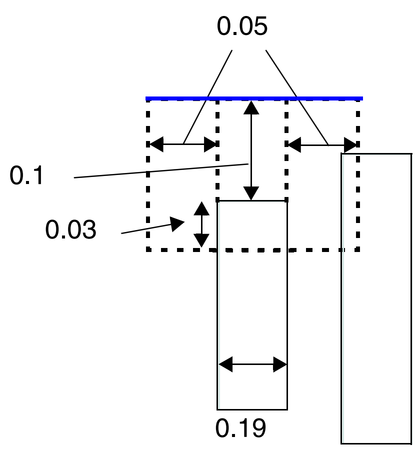
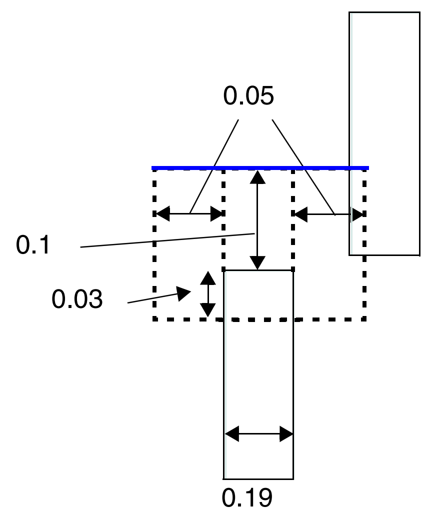
Figure 6-56 Illustration of EOL Keep Out Rule with EXCEPTFROM and FRONTEDGE

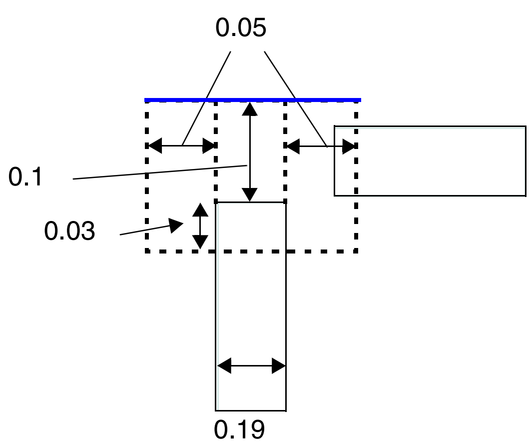
```
PROPERTY LEF58_EOLKEEPOUT
  "EOLKEEPOUT 0.2 EXTENSION 0.03 0.05 0.1
      CORNERONLY EXCEPTFROM FRONTEDGE ; " ;
```



a) Violation, neighbor is not coming from the front edge in blue



b) OK, neighbor wire is coming from the front edge. It would be a violation without EXCEPTFROM or with EXCEPTFROM BACKEDGE.



c) Violation, neighbor wire coming from the side edges is also bad

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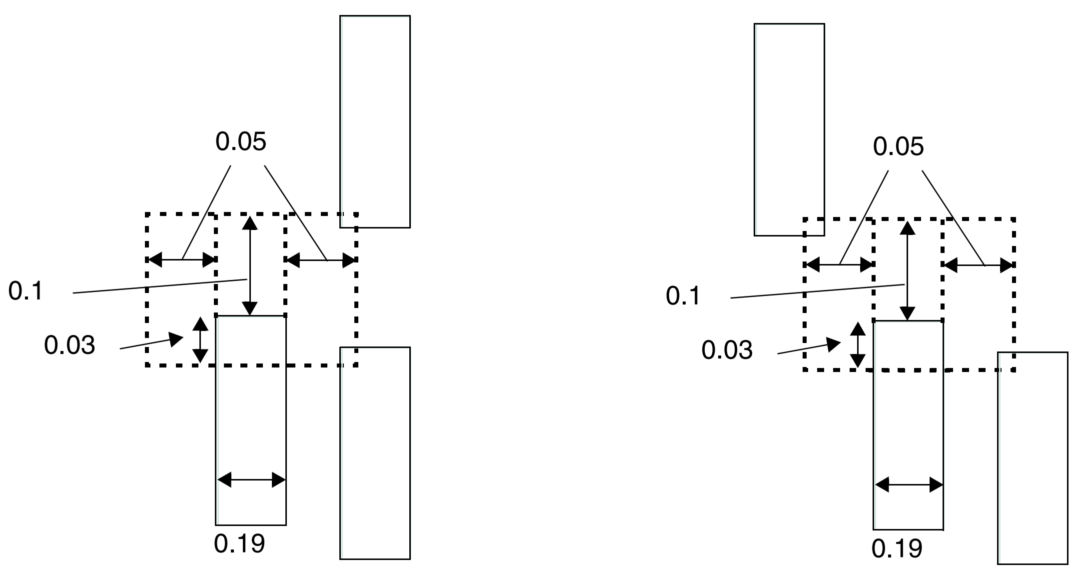
- The following example indicates that it is a violation if all first three rules fail or both the last two rules fail：

```
PROPERTY LEF58_EOLKEEPOUT "
    EOLKEEPOUT 0.06 EXTENSION 0.00 0.08 0.12  CLASS A  CORNERONLY
       EXCEPTFROM FRONTEDGE ;
    EOLKEEPOUT 0.06 EXTENSION 0.04 0.08 0.00  CLASS A CORNERONLY
       EXCEPTFROM BACKEDGE ;
    EOLKEEPOUT 0.06 EXTENSION 0.04 0.06 0.12  CLASS A CORNERONLY ;
    EOLKEEPOUT 0.09 EXTENSION 0.04 0.08 0.12  CLASS B CORNERONLY
       EXCEPTFROM FRONTEDGE ;
    EOLKEEPOUT 0.09 EXTENSION 0.04 0.00 0.12  CLASS B CORNERONLY
       EXCEPTFROM BACKEDGE ; " ;
```

- The following example indicates that the rule fails if both the statements with neighbors are on the same right side:

```
PROPERTY LEF58_EOLKEEPOUT
    "EOLKEEPOUT 0.2 EXTENSION 0.03 0.05 0.0 CLASS A
        CORNERONLY EXCEPTFROM FRONTEDGE ;
    EOLKEEPOUT 0.2 EXTENSION 0.0 0.05 0.1 CLASS A
        CORNERONLY EXCEPTFROM BACKEDGE ; " ;
```

Figure 6-57 Illustration of EOL Keep Out Rule with FRONTEDGE and BACKEDGE



a) Violation, both of the statements fail with neighbors on the same right side

b) OK, failing neighbors are on opposite side of the wire

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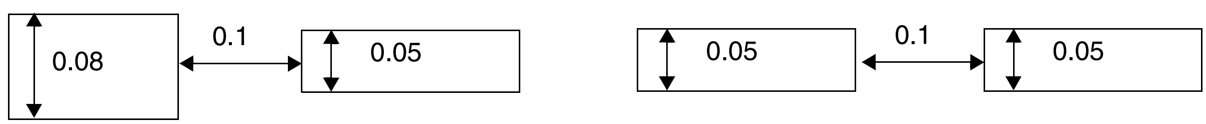
- The following example indicates how the rule is applied for two perfectly aligned EOL edges with the same width:

```
PROPERTY LEF58_EOLKEEPOUT
   "EOLKEEPOUT 0.09 EXTENSION 0 0.01 0.11
      CORNERONLY EXCEPTFROM BACKEDGE
      EXACTALIGNED 0.1; " ;
```

Figure 6-58 Illustration of EOL Keep Out Rule with Exactly Aligned EOL Edges



a) Violation, the two EOL edges do not have the same width and 0.11 spacing is not met.

b) OK, spacing of 0.1 is met for these perfectly aligned EOL edges with the same width.

- The following example indicates that the rule fails if the mask 1 wire has two different-mask (mask 2) wires on both sides:

```
PROPERTY LEF58_EOLKEEPOUT
   "EOLKEEPOUT 0.2 EXTENSION 0.03 0.05 0.1
      CORNERONLY EXCEPTFROM BACKEDGE
      MASK 1 TWOSIDES ; " ;
```


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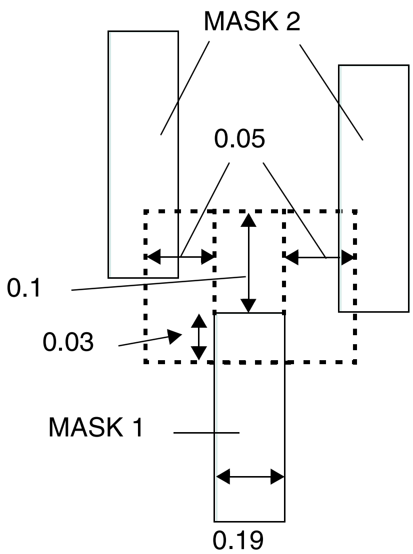
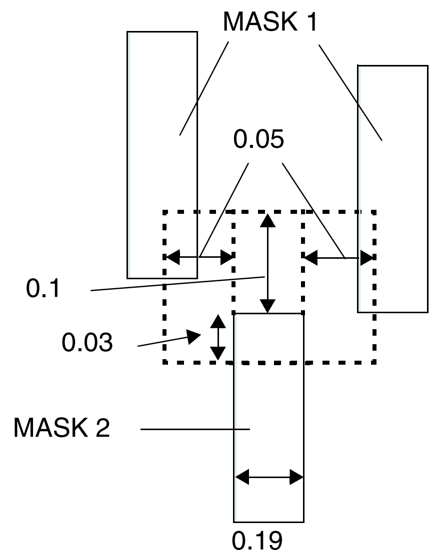
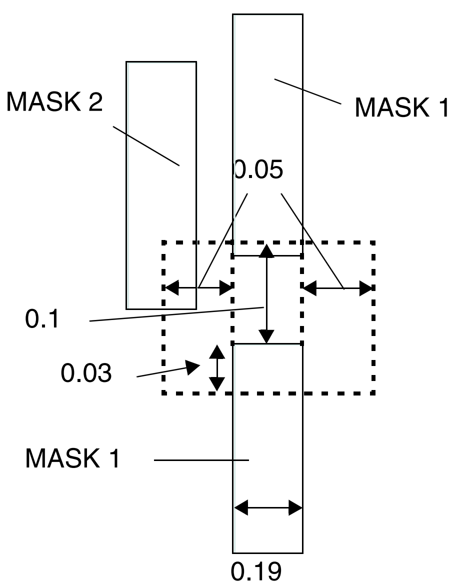
Figure 6-59 Illustration of EOL Keep Out Rule with Mask and Two Sides



a) Violation, the mask 1 wire has two different-mask (mask 2) wires on both sides.



b) OK, the triggering wire must be a mask 1 wire.



c) OK, only one mask 2 neighbor

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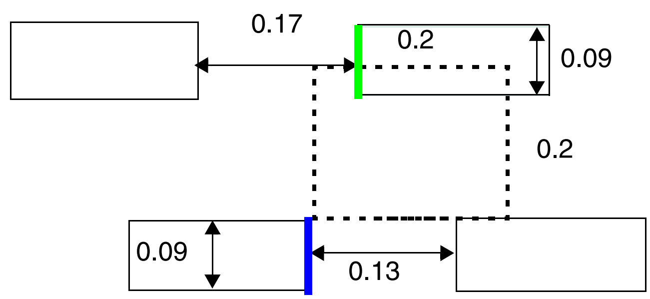
Figure 6-60 Illustration of EOL Keep Out Rule with OTHERENDEOL

```
DIRECTION HORIZONTAL ;
PROPERTY LEF58_EOLKEEPOUT "
            EOLKEEPOUT 0.1 EXTENSION 0 0.2 0.2
EXCEPTWITHIN -0.01 0 CLASS A OTHERENDEOL CORNERONLY
EXCEPTFROM BACKEDGE ;
            EOLKEEPOUT 0.1 EXTENSION 0 0 0.15 CLASS A ; " ;
```



a) OK, when checking the blue EOL, it would fall both CLASS A statements. However, OTHERENDEOL is specified, the neighbor green EOL would pass the second statement, and overall, it would not be a violation.

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Figure 6-61 Illustration of EOL Keep Out Rule with EXCEPTSAMESIDEMETAL

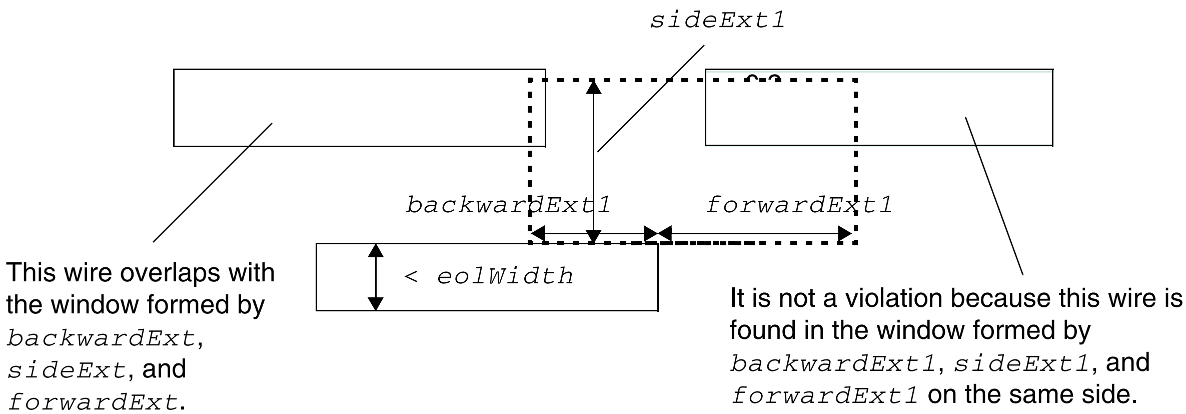

Illustration of:
```
EOLKEEPOUT eolWidth
  EXTENSION backwardExt sideExt forwardExt
  CORNERONLY EXCEPTFROM FRONTEDGE
      EXCEPTSAMESIDEMETAL backwardExt1 sideExt1 forwardExt1
```


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Figure 6-62 Illustration of EOL Keep Out Rule with EXCEPTNOOPPSIDEMETAL

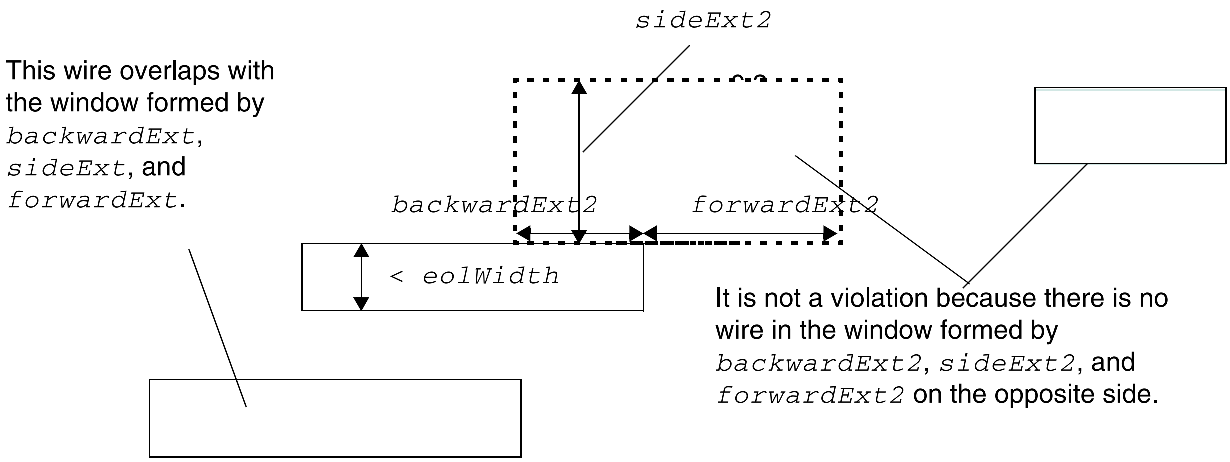

Illustration of:

```
EOLKEEPOUT eolWidth
  EXTENSION backwardExt sideExt forwardExt
  CORNERONLY EXCEPTFROM FRONTEDGE
      EXCEPTNOOPPSIDEMETAL backwardExt2 sideExt2 forwardExt2
```


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Figure 6-63 Illustration of EOL Keep Out Rule with EXCEPTLINEENDSPACING

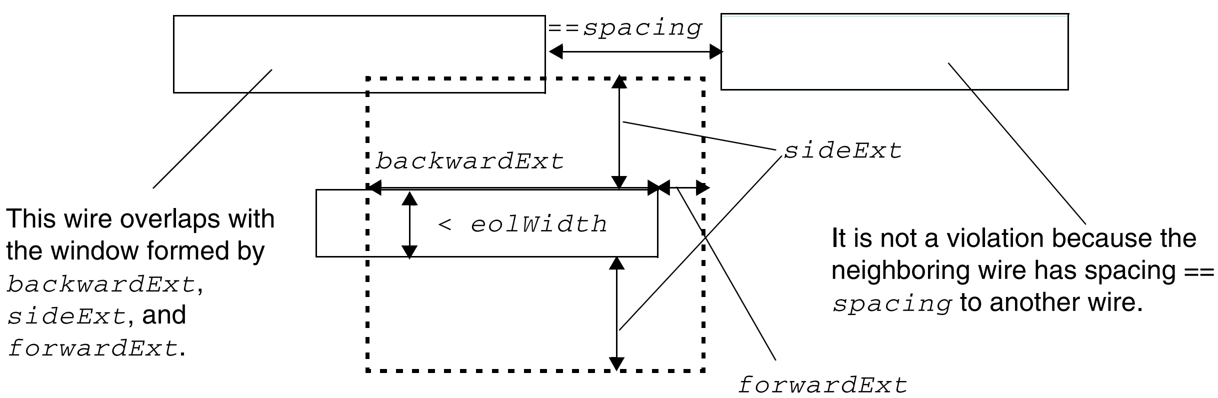

Illustration of:

```
EOLKEEPOUT eolWidth
  EXTENSION backwardExt sideExt forwardExt
  CORNERONLY EXCEPTFROM FRONTEDGE
  EXCEPTLINEENDSPACING spacing
```

Figure 6-64 Illustration of EOL Keep Out Rule with EOLWITHINCUT

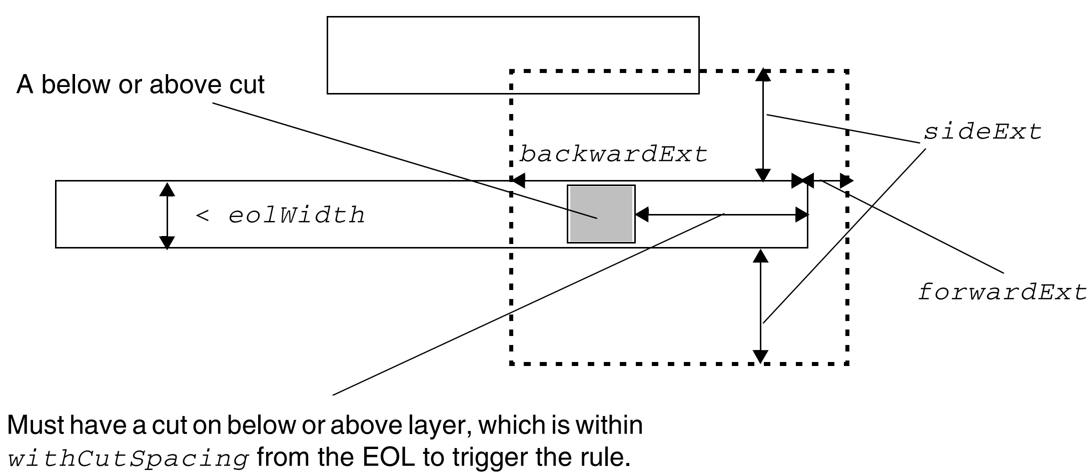

Illustration of:

```
EOLKEEPOUT eolWidth
  EXTENSION backwardExt sideExt forwardExt
  CORNERONLY EXCEPTFROM FRONTEDGE
  EOLWITHINCUT withinCut
```


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Figure 6-65 Illustration of EOL Keep Out Rule with EXCEPTLINEENDGAP

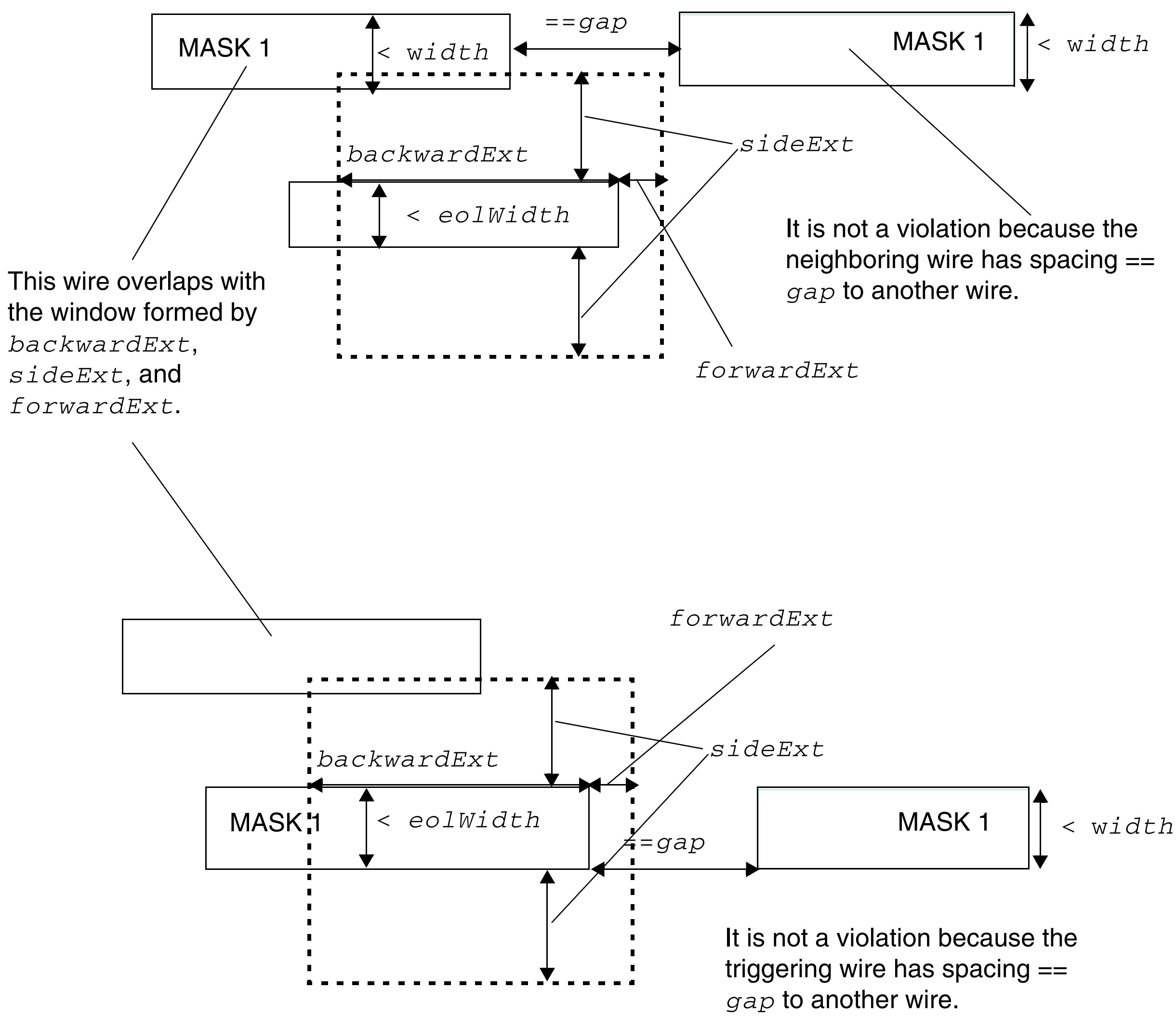

Illustration of:

```
LINEENDGAP gap WIDTH width MASK 1

EOLKEEPOUT eolWidth
  EXTENSION backwardExt sideExt forwardExt
  CORNERONLY EXCEPTFROM FRONTEDGE
  EXCEPTLINEENDGAP
```


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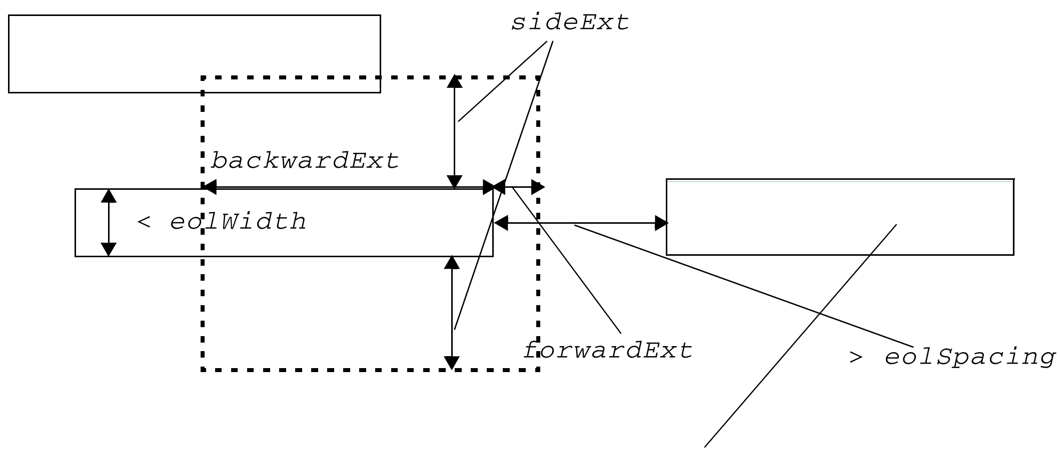
Figure 6-66 Illustration of EOL Keep Out Rule with EXCEPTEOLSPACING



This is not a violation since the nearest neighbor wire with `PRL > 0` is > *eolSpacing*.

Illustration of:

```
EOLKEEPOUT eolWidth
  EXTENSION backwardExt sideExt forwardExt
  CORNERONLY EXCEPTFROM FRONTEDGE
  EXCEPTEOLSPACING eolSpacing
```


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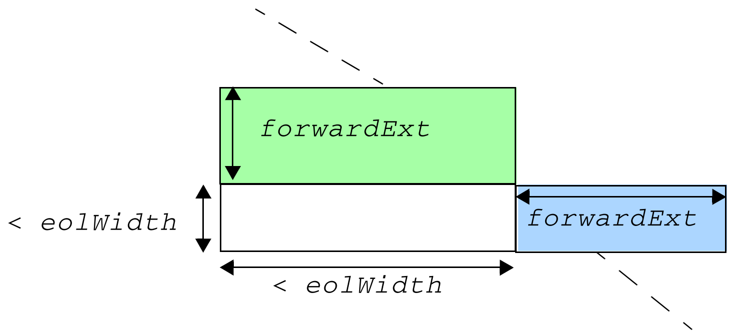
Figure 6-67 Illustration of EOL Keep Out Rule with RIGHTWAYONLY | WRONGWAYONLY

If `RIGHTWAYONLY` is specified, the green EOLKEEPOUT window would be formed on both the top and bottom EOL edges.



If `WRONGWAYONLY` is specified, the blue EOLKEEPOUT window would be formed on both the left and right EOL edges.

Illustration of `DIRECTION HORIZONTAL` and `EOLKEEPOUT` *`eolWidth`*
`RIGHTWAYONLY | WRONGWAYONLY`
`EXTENSION 0 0` *`forwardExt`*

EOL Track Rule

An EOL track rule can be used to define alignment of EOL with certain tracks.

You can create an EOL track rule by using the following property definition:

```
PROPERTY LEF58_EOLTRACK
       "EOLTRACK eolWidth PITCH pitch TRACKSDISTANCE distance
          [EXCEPTEOLSPACING eolSpacing [WITHIN within]]
          ; " ;
```

Where:

`EOLTRACK` *`eolWidth`* `PITCH` *`pitch`* `TRACKSDISTANCE` *`distance`*

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Specifies that any EOL with width/length less than *eolWidth* must align to certain tracks, which are *pitch* apart. There are two sets of tracks, which are apart by *distance*; the bottom EOL of a vertical wire or the left EOL of a horizontal wire aligns to one set of tracks while the top EOL of a vertical wire or the right EOL of a horizontal wire aligns to the other set of tracks. As the locations of the EOL tracks may vary depending on the design, the first track location would be specified by a separate command option as an offset to a reference point, like the core or design boundary.

Type: Float, specified in microns

`EXCEPTEOLSPACING` *eolSpacing* `WITHIN` *within*

Specifies that EOL with spacing less than or equal to *eolSpacing* to a neighboring wire with PRL greater than `0` does not need to align to the given tracks. If `WITHIN` is specified, a window would be formed by extending *within* along the EOL on both sides and *eolSpacing* in the orthogonal direction. Any neighboring wire overlaps with the window means that the EOL does not need to align to the given tracks.

Type: Float, specified in microns

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EOL Track Rule Examples

Figure 6-68 Illustration of EOL Track Rule

O would be specified by a separate command option.

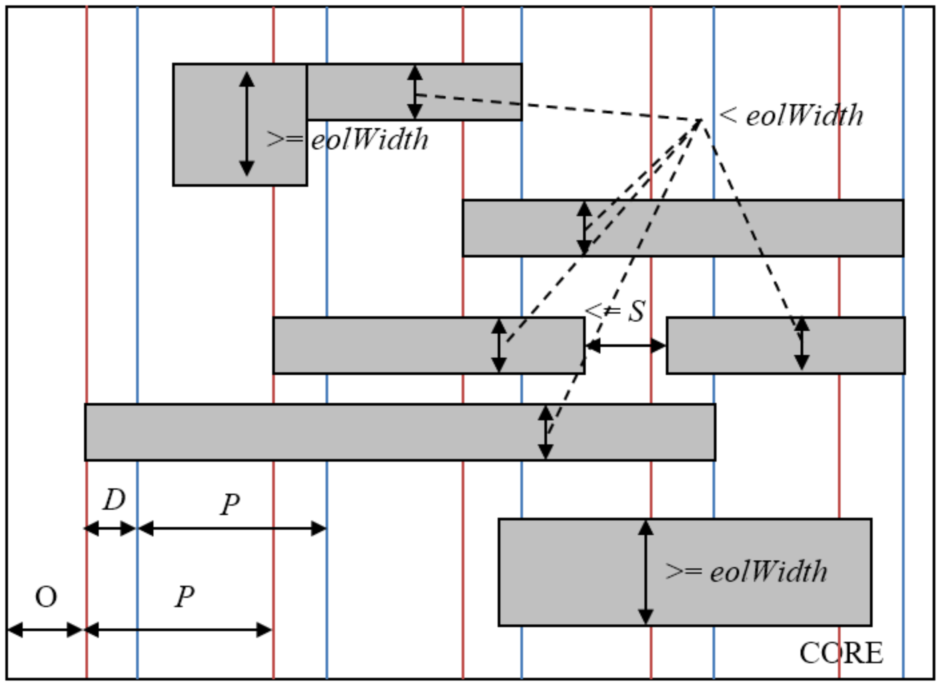

Illustration of:

```
DIRECTION HORIZONTAL ;
PROPERTY LEF58_EOLTRACK "
     EOLTRACK eolWidth  PITCH P TRACKSDISTANCE D
         EXCEPTEOLSPACING S ; " ;
```


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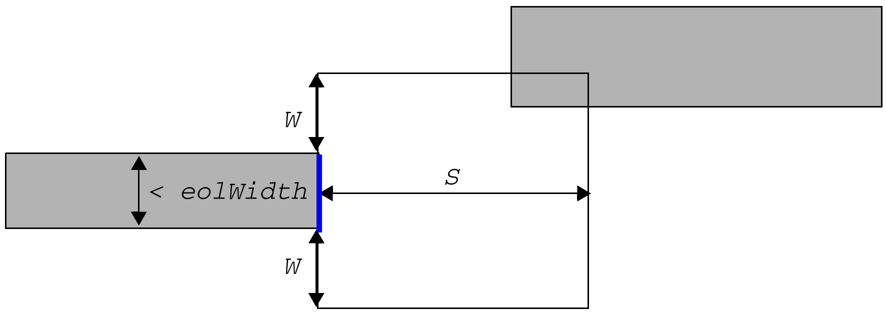
Figure 6-69 Illustration of EOL Track Rule with WITHIN in EXCEPTEOLSPACING



The blue edge does not need to align to the given tracks as there is a neighboring wire in the window formed in the EOL edge.

Illustration of:

```
DIRECTION HORIZONTAL ;
PROPERTY LEF58_EOLTRACK "
     EOLTRACK eolWidth PITCH P TRACKSDISTANCE D
         EXCEPTEOLSPACING S WITHIN W ; " ;
```

EOL Via Keep-out Rule

An EOL via keep-out rule can be used to define keep-out regions for end-of-line edges.

You can create an EOL via keep-out rule by using the following property definition:

```
PROPERTY LEF58_EOLVIAKEEPOUT
       "EOLVIAKEEPOUT eolWidth MASK maskNum
          EXTENSION backwardExt sideExt forwardExt
          OTHERWIDTH otherWidth
          OTHERENDEXTENSION otherEndExtension
          [EXCEPTBACKSIDEALIGNED]
          ; " ;]
```

Where:

```
EOLVIAKEEPOUT eolWidth MASK maskNum
    EXTENSION backwardExt sideExt forwardExt
    OTHERWIDTH otherWidth
    OTHERENDEXTENSION otherEndExtension
```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Defines a keep-out region for an end-of-line edge with length less than *eolWidth* belonging to mask *maskNum* by extending *backwardExt* going backward, *sideExt* on the side and *forwardExt* going forward. Any above or below via cut belonging to a wire with width less than or equal to *otherWidth* and with mask other than *maskNum* touching or overlapping with the keep-out region would be a violation if there is a same-mask wire fulfilling the OTHERENDEXTENSION condition. OTHERENDEXTENSION means that a same-mask neighbor wire with width less than *eolWidth* is extended by *otherEndExtension* at both line ends and the extended wire overlaps with the keep-out region on the same side as the neighboring via. This same-mask neighbor must not have PRL greater than or equal to 0 with the EOL edge.
Type: Float, specified in microns

EXCEPTBACKSIDEALIGNED

Specifies that the rule does not apply if the same-mask neighbor wire without extension is aligned to the EOL coming from the back edge of the keep-out window.

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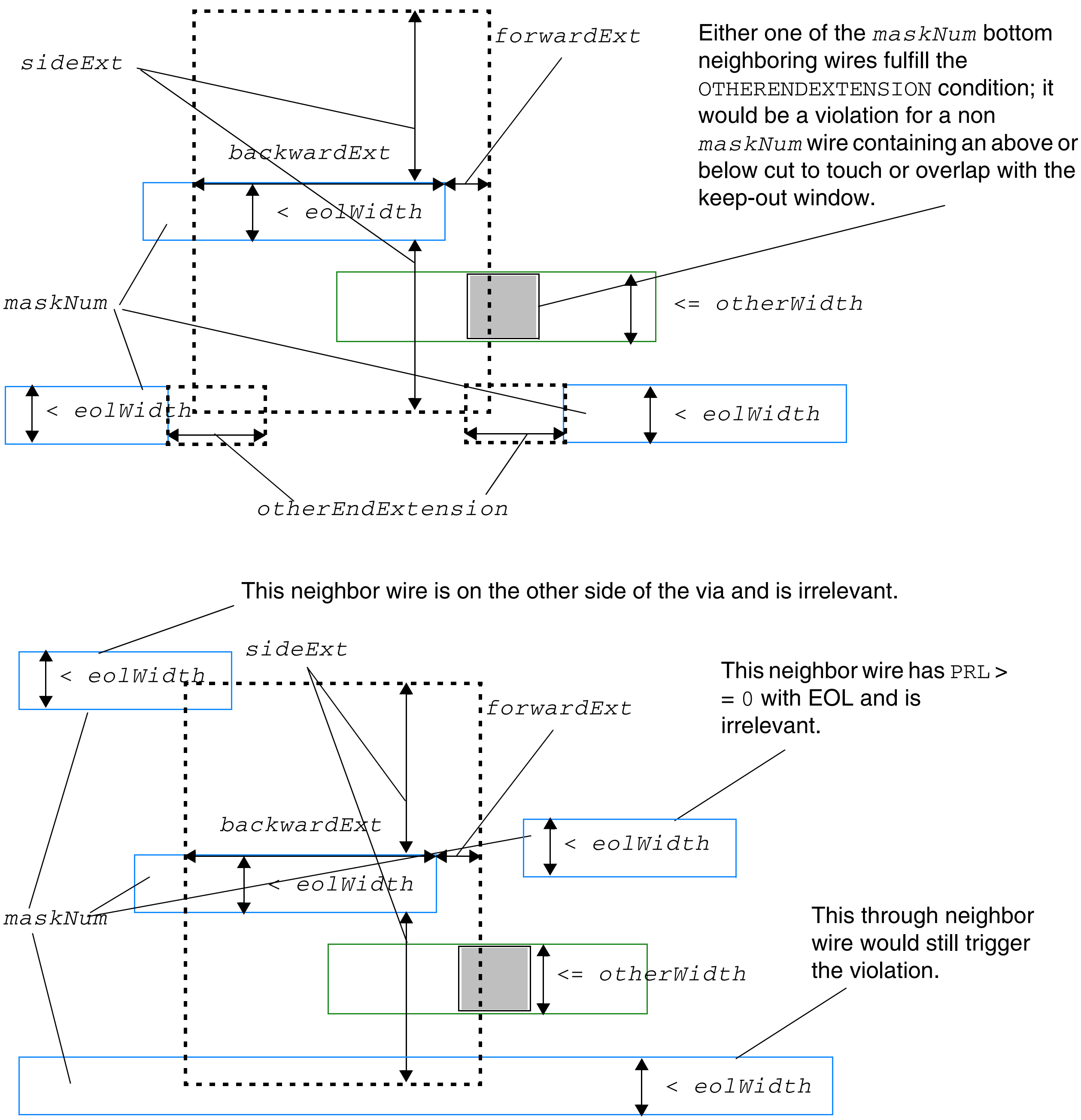
EOL Via Keep-out Rule Examples

Figure 6-70 Illustrations of EOL Via Keep-out Rule



Illustrations of:

EOLVIAKEEPOUT *eolWidth* MASK *maskNum*
  EXTENSION *backwardExt sideExt forwardExt*
  OTHERWIDTH *otherWidth*
  OTHERENDEXTENSION *otherEndExtension*

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Figure 6-71 Illustration of EOL Via Keep-out Rule with EXCEPTBACKSIDEALIGNED

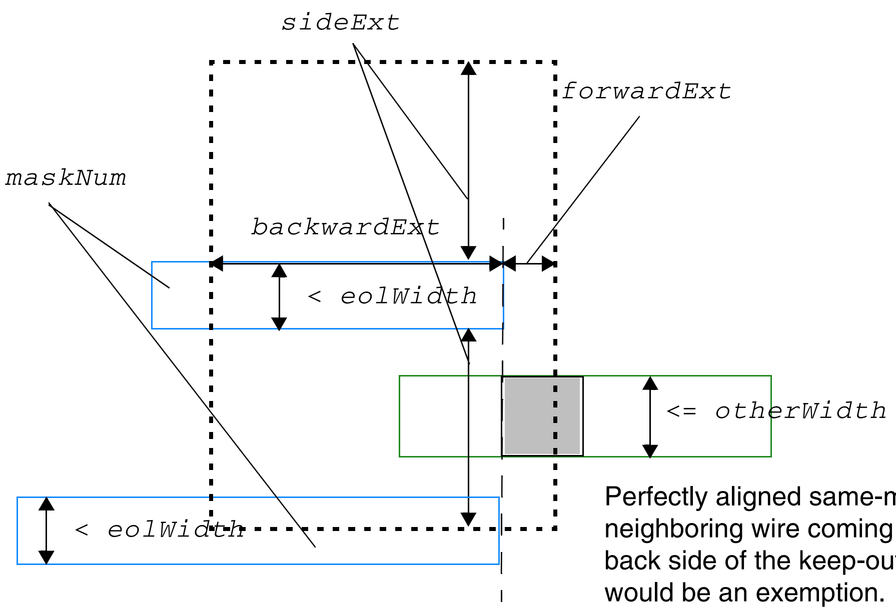

Illustration of:

```
EOLVIAKEEPOUT eolWidth MASK maskNum
     EXTENSION backwardExt sideExt forwardExt
     OTHERWIDTH otherWidth
     OTHERENDEXTENSION otherEndExtension
     EXCEPTBACKSIDEALIGNED
```

Fill to Fill Spacing Rule

A fill to fill spacing rule can be used to define spacing between metal fills.

You can create a fill to fill spacing rule by using the following property definition:

```
PROPERTY LEF58_FILLTOFILLSPACING
     "FILLTOFILLSPACING spacing ;" ;
```

Where:

`FILLTOFILLSPACING`	Specifies the spacing between metal fills. *Type:* Float, specified in microns

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Five Wires EOL Spacing Rule

The five wires EOL spacing rule is used to define spacing constraints for five consecutive wires meeting certain conditions.

You can create a five wire EOL spacing rule by using the following property definition:

```
PROPERTY LEF58_FIVEWIRESEOLSPACING
     "FIVEWIRESEOLSPACING eolSpacing WITHIN eolWithin
         PRL prl
        ENCLOSECUT {BELOW|ABOVE} encloseDist CUTWITHIN cutWithin
         NOMETALEOLEXTENSION eolSideExtension eolForwradExtension
     ;...";
```

Where:

`FIVEWIRESEOLSPACING` *`eolSpacing`* `WITHIN` *`eolWithin`*
`PRL` *`prl`*
`ENCLOSECUT {BELOW|ABOVE}` *`encloseDist`* `CUTWITHIN` *`cutWithin`*
`NOMETALEOLEXTENSION` *`eolSideExtension eolForwradExtension`*

Specifies that if there are five consecutive minimum default width wires that are minimum spacing apart and have a parallel run length greater than *`prl`*, it is a violation to have a neighbor wire overlapping with the region formed by the line end of the middle wire by extending *`eolWithin`* along the EOL edge and *`eolSpacing`* along the orthogonal direction. In addition, the following conditions must also be met to be a violation:

- There is a cut on below or above cut layer in case `BELOW` or `ABOVE` is specified on the middle wire within *`cutWithin`* of parallel wires and having enclosure less than *`encloseDist`* from the line end.
- There is no wire completely covering or touching the entire range along the orthogonal direction of the EOL of one of the two regions by extending *`eolSideExtension`* along the EOL corners and *`eolForwradExtension`* along the orthogonal direction.

Type: Float, specified in microns

Five Wires EOL Spacing Rule Example

- The following example is an illustration of `FIVEWIRESEOLSPACING`:

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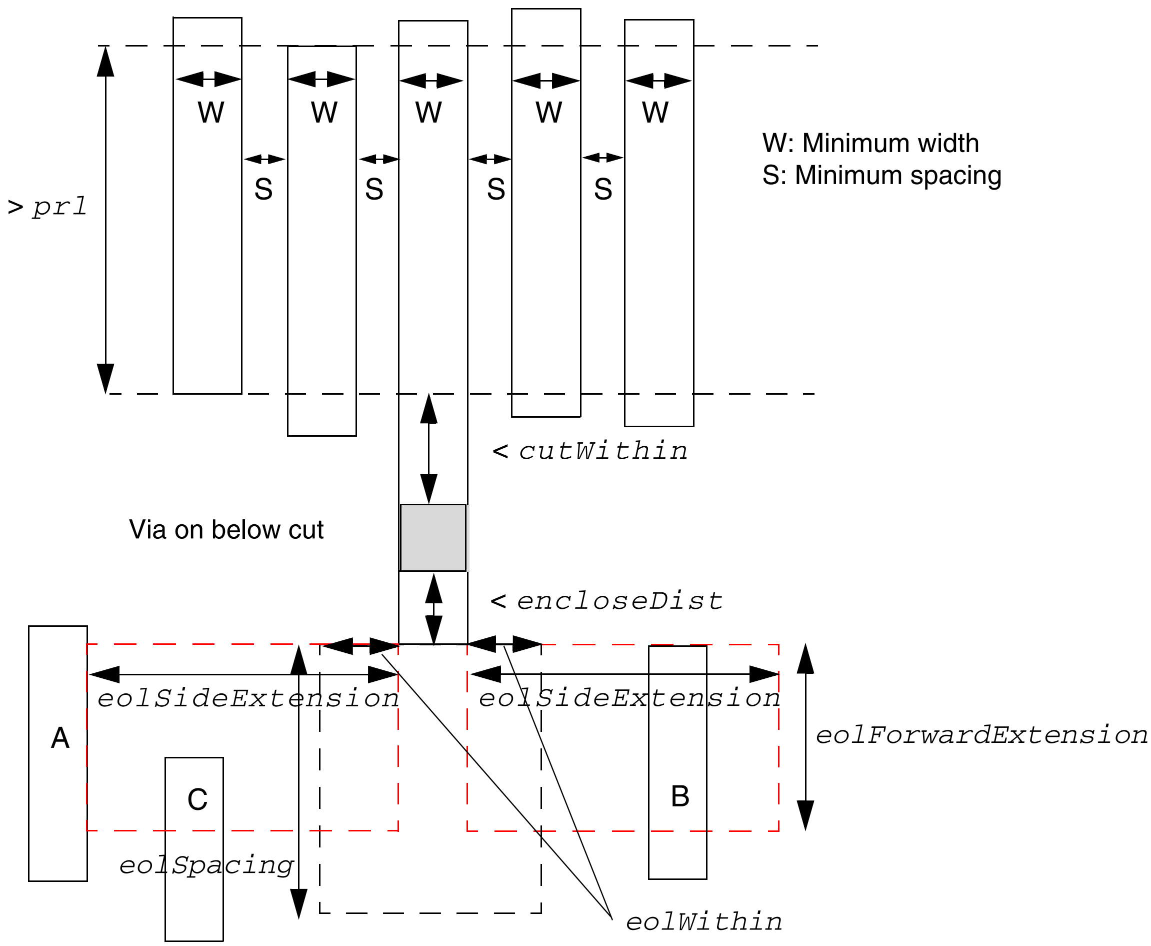
```
FIVEWIRESEOLSPACING eolSpacing WITHIN eolWithin
     PRL prl
     ENCLOSECUT BELOW encloseDist CUTWITHIN cutWithin
     NOMETALEOLEXTENSION eolSideExtension eolForwardExtension
```

Figure 6-72 Illustration of FIVEWIRESEOLSPACING



It is a violation if there is no metal wire covering the entire y range of the red regions, like A or B (C is irrelevant) and a metal wire overlaps with the black region.

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Five Wires Forbidden Spacing Rule

The five wires forbidden spacing rule is used to define spacing constraints for five consecutive wires meeting certain conditions.

You can create a five wire forbidden spacing rule by using the following property definition:

```
PROPERTY LEF58_FIVEWIRESFORBIDDENSPACING
     "FIVEWIRESFORBIDDENSPACING minSpacing maxSpacing
           [EXCEPTWIRES numWire]
           {HORIZONTAL|VERTICAL} PRL prl
           ;...";
```

Where:

```
FIVEWIRESFORBIDDENSPACING minSpacing maxSpacing
     [EXCEPTWIRES numWire]
     {HORIZONTAL|VERTICAL} PRL prl
```

Specifies that if there are five consecutive minimum default width wires that are minimum spacing apart in the given direction of `HORIZONTAL` or `VERTICAL` having parallel run length (`PRL`) greater than *`prl`*, it is a violation if there are neighboring wires on both sides with spacing greater than or equal to *`minSpacing`* and less than or equal to *`maxSpacing`* in the given direction having `PRL` greater than *`prl`*.

If `EXCEPTWIRES` is specified, the maximum number of consecutive wires allowed is *`numWire`*. This rule is applied if there are between five and *`numWire`* consecutive wires.

Type: Float, specified in microns

Five Wires Forbidden Spacing Rule Example

- The following example is an illustration of `FIVEWIRESFORBIDDENSPACING`:

```
FIVEWIRESFORBIDDENSPACING minSpacing maxSpacing
    HORIZONTAL PRL prl
```


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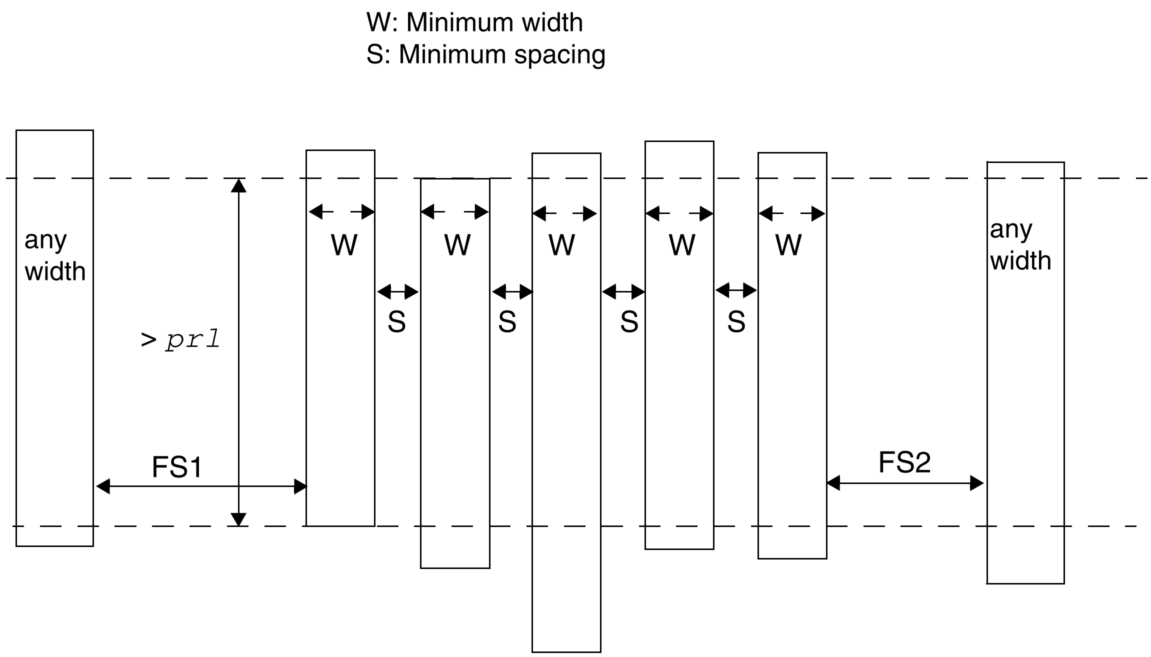
Figure 6-73 Illustration of FIVEWIRESFORBIDDENSPACING



It is a violation if both FS1 and FS2 >= *minSpacing* and <= *maxSpacing*. In other words, if only FS1 or FS2 is in the given spacing range, it is not a violation.

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Forbidden Spacing Rule

A forbidden spacing rule can be used to define spacing between two wires, such that under certain conditions a violation may occur if there is a different-metal polygon wire between the wires.

You can create a forbidden spacing rule by using the following property definition:

```
PROPERTY LEF58_FORBIDDENSPACING
     "FORBIDDENSPACING minSpacing maxSpacing [minSpacing2 maxSpacing2]
        [WIDTH minWidth WITHIN within PRL prl
        | [SAMEMASK] WIDTH maxWidth PRL prl
            [TWOEDGES within | EXACTSPACINGEDGE exactSpacing
                [IGNOREMIDDLE width spacing]
                [SPANLENGTH spanLength [WITHIN within]]
                [EXCEPTWIREWIDTHSPACING {width spacing}...]]
        | EXACTWIDTH exactWidth PRL prl
            OTHERWIDTH otherWidth EXACTSPACINGEDGE exactSpacing
        | [SAMEMASK | MASK maskNum]
            WIDTHRANGE minWidth maxWidth PRL prl
            OTHERWIDTH otherWidth WITHIN within [OTHERSAMEMASK]]
       ; " ;
```

Where:

`EXACTSPACINGEDGE` *`exactSpacing`*

Specifies forbidden spacing only applies if the wire width, less than the *`maxWidth`*, has a neighbor with exact spacing of *`exactSpacing`* on one side, and the forbidden spacing range will be applied on the opposite side. The forbidden spacing measurement will become the traditional edge-to-edge between wires from right/left or top/bottom of a wire to left/right or bottom/top of another wire. In addition, this rule will apply to right way wires only while the forbidden spacing range is measured in the direction perpendicular to the specified direction in `DIRECTION` on a Manhattan routing layer.

Type: Float, specified in microns

`EXACTWIDTH` *`exactWidth`* `PRL` *`prl`* `OTHERWIDTH` *`otherWidth`*
`EXACTSPACINGEDGE` *`exactSpacing`*

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Specifies that the spacing between a wire with width exactly equal to *exactWidth* and another wire with width greater than or equal to *otherWidth* must not be greater than or equal to *minSpacing* and less than or equal to *maxSpacing* if the first wire has another wire on the other side with the exact spacing of *exactSpacing* and the three wires have parallel run length greater than *prl*. See Figure 6-83 on page 774.

Type: Float, specified in microns

`EXCEPTWIREWIDTHSPACING {`*`width spacing`*`}...`

Specifies that neighbor wires, with at least one of them not in the forbidden spacing range, having width exactly equal to *width* with `PRL` greater than `0` and the spacing less than *spacing* would be merged as if the gap between the wires were filled. Then, the merged wires may be able to shield another wire, which is within the forbidden spacing range, such that it is not a violation.

Type: Float, specified in microns

`FORBIDDENSPACING` *`minSpacing maxSpacing`* `WIDTH` *`maxWidth`* `PRL` *`prl`*

Specifies that if the spacing between the right/left or top/bottom edge of a wire with width less than the *maxWidth* to the right/ left or top/bottom of another wire is greater than or equal to the *minSpacing* and less than or equal to the *maxSpacing* with parallel run length greater than *prl*, then it will be a violation if and only if there is a different-metal polygon wire between these two wires.

Type: Float, specified in microns

`IGNOREMIDDLE` *`width spacing`*

Specifies that a wire between the triggering wire and the forbidden spacing range wire would be ignored if that wire's width is not the same as *width* or its spacing to the triggering wire is not the same as *spacing*. See Figure 6-80 on page 771.

Type: Float, specified in microns

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`MASK` *maskNum*

Specifies that the forbidden spacing rule applies only to the objects belonging to the given mask. In other words, forbidden spacing is checked against the objects in the given mask, and the mask of other neighboring objects is irrelevant. *maskNum* must be a positive integer, and most applications support only the values 1, 2, or 3.

Type: Integer

minSpacing2 maxSpacing2

Specifies an additional second set of forbidden spacing range. This second set can be defined only if the `TWOEDGES` or `EXACTSPACINGEDGE` keyword is also specified.

Type: Float, specified in microns

`OTHERSAMEMASK`

Specifies that the forbidden spacing rule applies only if the two outer wires have the same mask. The mask of the middle wire is irrelevant. See Figure 6-85 on page 776.

`TWOEDGES` *within*

Indicates that the forbidden spacing rule only applies if the wire width is less than the *maxWidth* that has neighbors on both the sides within *within*. The forbidden spacing measurement will become the traditional edge-to-edge between wires from right/left or top/bottom of a wire to left/right or bottom/top of another wire.

Type: Float, specified in microns

`SAMEMASK`

Specifies that the forbidden spacing rule applies only to objects on the same mask. The objects are the ones that the forbidden spacing is checked against, and the colors of other neighbor objects are irrelevant.

```
[SAMEMASK]
    WIDTHRANGE minWidth maxWidth PRL prl
    OTHERWIDTH otherWidth WITHIN within
```


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Specifies that the spacing between a wire with width greater than *minWidth* and less than *maxWidth*, which is sandwiched by two wires within *within* with width greater than *otherWidth*, and any of two wires must not be greater than or equal to *minSpacing* and less than or equal to *maxSpacing* if the three wires have parallel run length greater than *prl*.

Type: Float, specified in microns

SAMEMASK specifies that the forbidden spacing rule applies only to objects on the same mask. The objects are the ones that the forbidden spacing is checked against, and the mask of the other neighbor object is irrelevant.

SPANLENGTH *spanLength*

Specifies that the forbidden spacing range rule only applies if the span length of the neighbor wire is less than the *spanLength*.

Type: Float, specified in microns

WIDTH *minWidth* WITHIN *within* PRL *prl*

Specifies that it is a violation if two wires are apart by a distance that is greater than or equal to the *minSpacing* and less than or equal to the *maxSpacing* and are within *within* distance from a wire with width greater than or equal to the *minWidth* and has parallel run length greater than *prl*.

Type: Float, specified in microns

WITHIN *within*

Specifies that the forbidden spacing range rule only applies if the neighbor wire has another neighbor within (less than) a *within* distance on either side. See Figure 6-79 on page 770.

Type: Float, specified in microns

Forbidden Spacing Rule Examples

- The following example shows forbidden spacing rule with WIDTH and PRL:

```
PROPERTY LEF58_FORBIDDENSPACING
    "FORBIDDENSPACING 0.2 0.25 WIDTH 0.05 PRL 0.15 ; " ;
```


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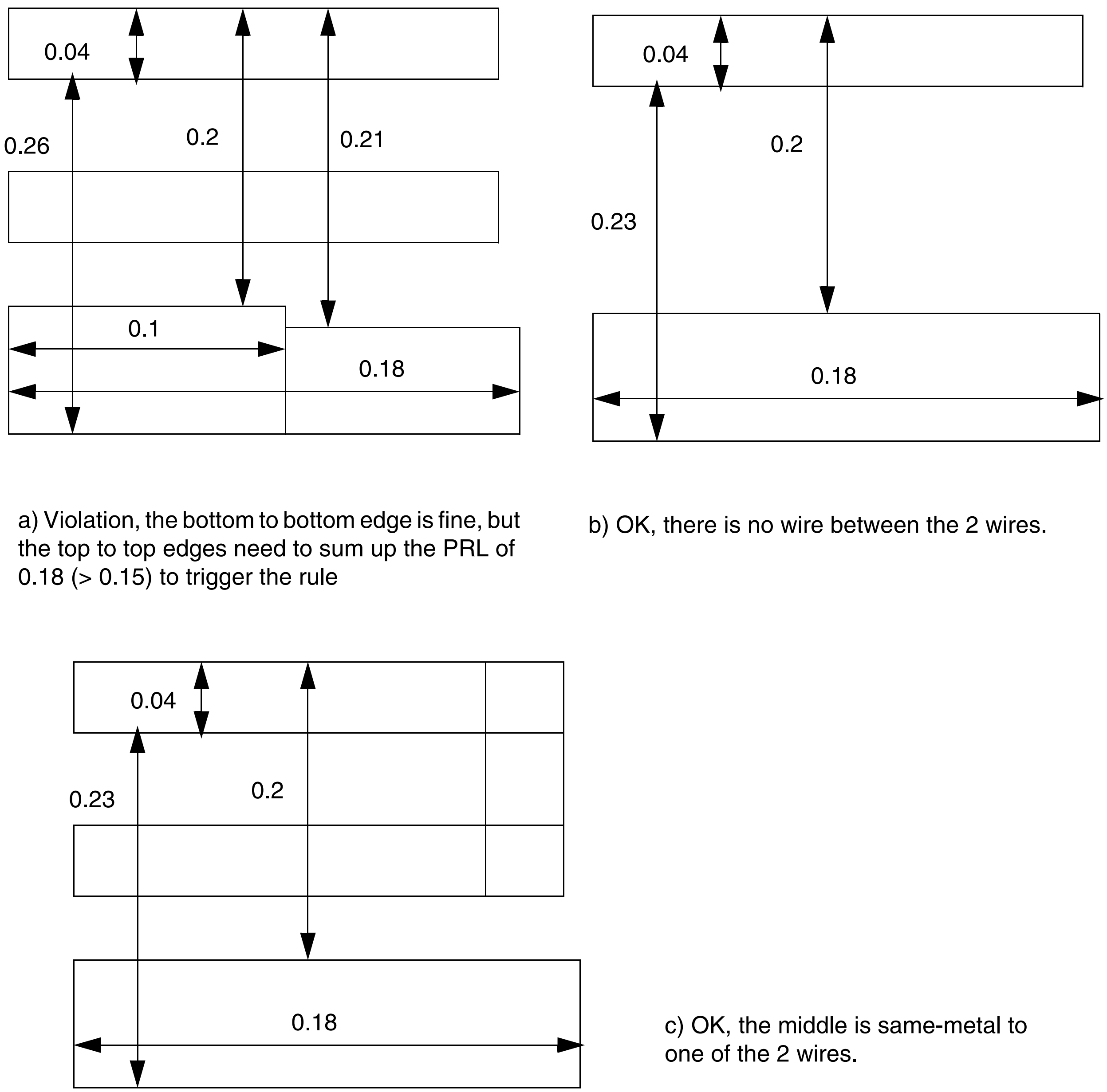
Figure 6-74 Illustration of FORBIDDENSPACING with WIDTH and PRL



a) Violation, the bottom to bottom edge is fine, but the top to top edges need to sum up the PRL of 0.18 (> 0.15) to trigger the rule

b) OK, there is no wire between the 2 wires.

c) OK, the middle is same-metal to one of the 2 wires.

- The following example shows forbidden spacing rule with `WIDTH` and `WITHIN`:

```
PROPERTY LEF58_FORBIDDENSPACING
    "FORBIDDENSPACING 0.07 0.09
        WIDTH 0.2 WITHIN 0.4 PRL 0.1 ; " ;
```


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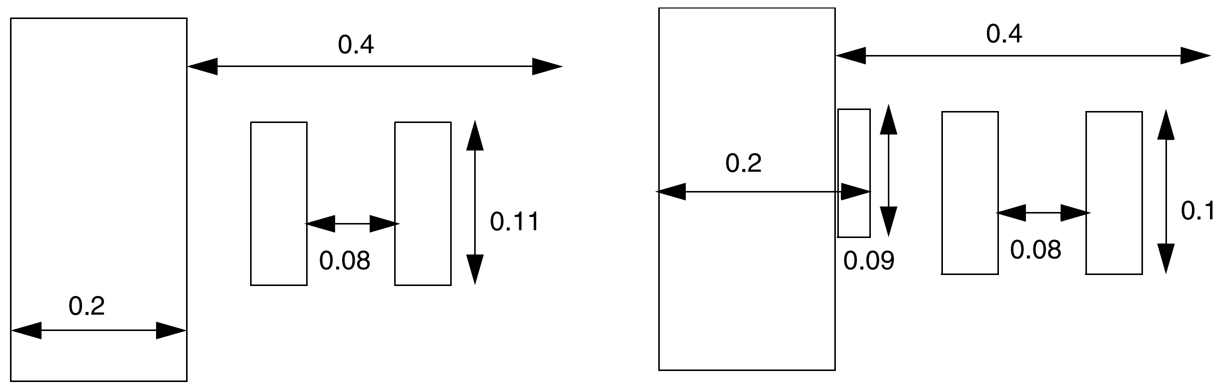
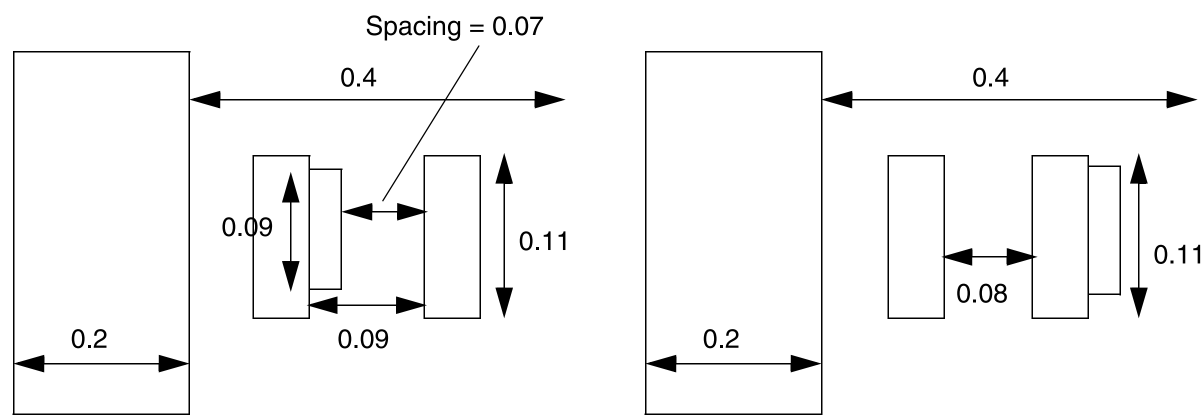
Figure 6-75 Illustration of FORBIDDENSPACING with WIDTH and WITHIN



a) Violation, the 2 neighbor wires have a spacing of 0.08, which is inside the forbidden spacing zone

b) OK, after shrink and grow of half width of 0.2 on the wide wire, the final geometry has width less than 0.2, and the rule is not triggered.



c) Violation, since 0.07 spacing in the jog and 0.09 spacing are in the forbidden range, PRL is the sum of them, 0.11 (> 0.1).

d) Violation, a jog on the other side does not exempt the rule.

- The following example shows forbidden spacing rule with `PRL`:

```
PROPERTY LEF58_FORBIDDENSPACING
    "FORBIDDENSPACING 0.07 0.09
        WIDTH 0.2 WITHIN 0.4 PRL 0.1 ; " ;
```


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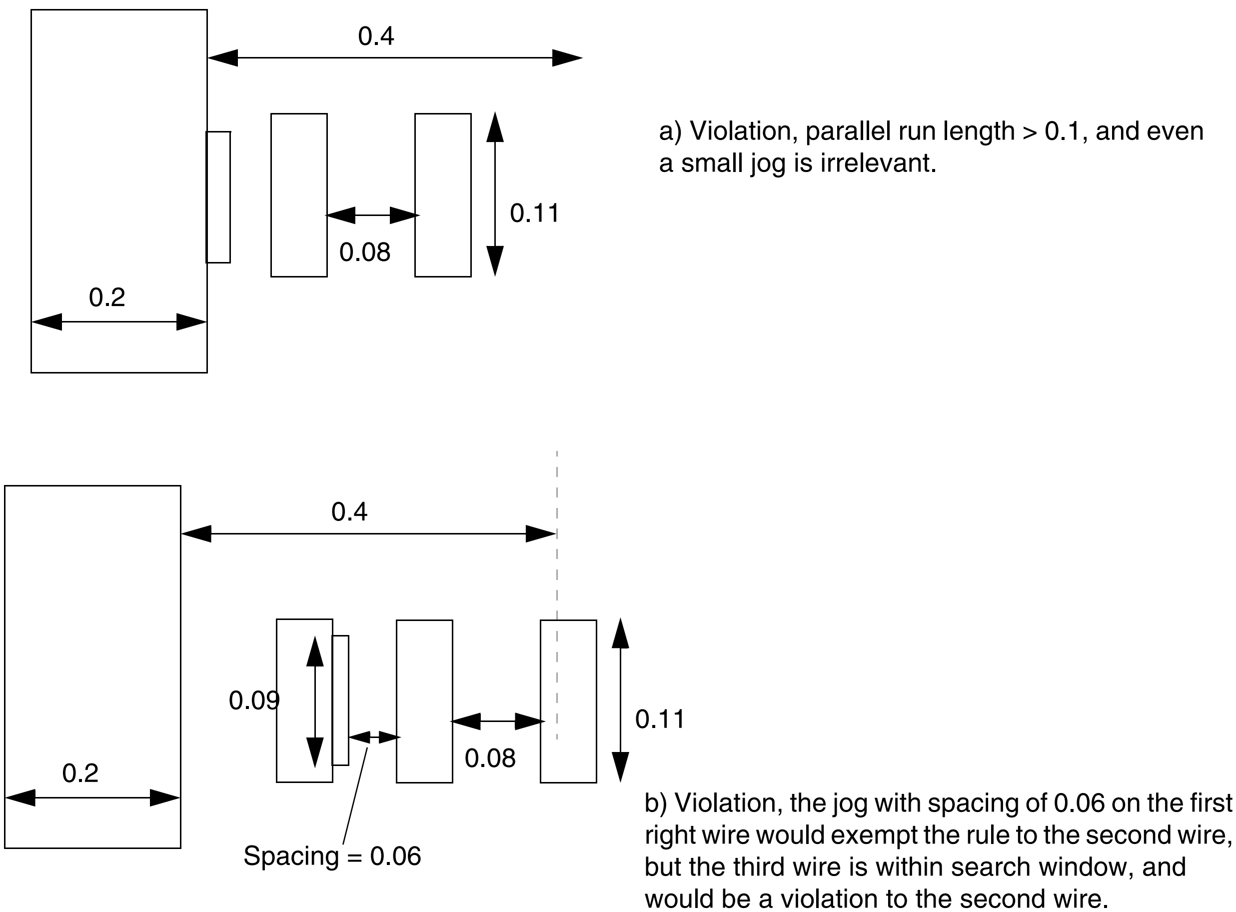
Figure 6-76 Illustration of FORBIDDENSPACING with PRL



- The following example shows forbidden spacing rule with `TWOEDGES`:

```
PROPERTY LEF58_FORBIDDENSPACING
   "FORBIDDENSPACING 0.2 0.25 0.3 0.33 WIDTH 0.05 PRL 0.15
      TWOEDGES 0.06 ; " ;
```


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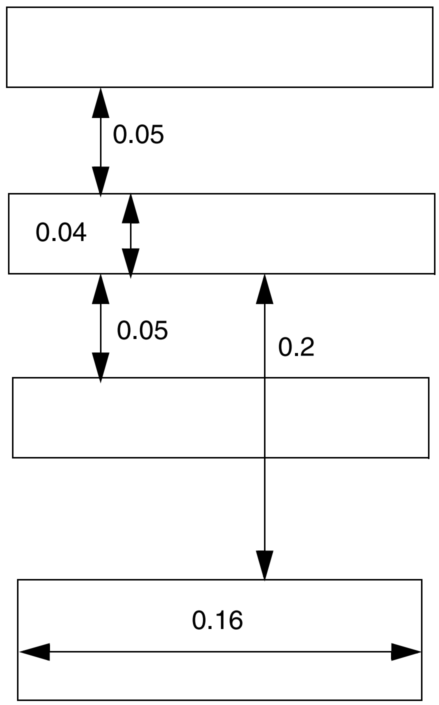
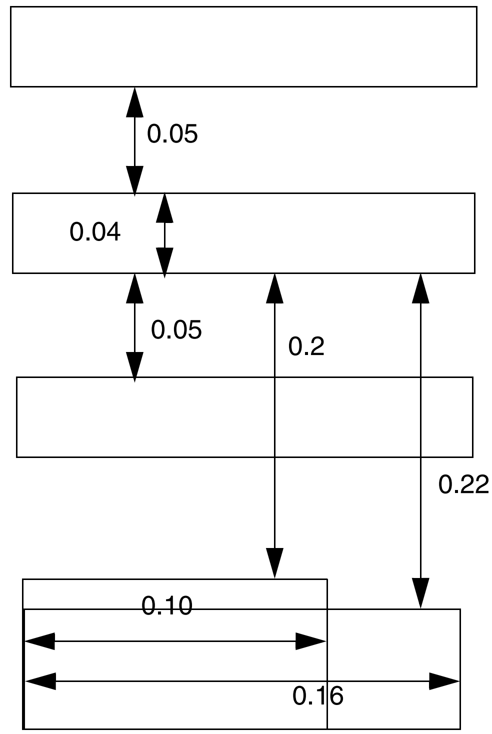
Figure 6-77 Illustration of FORBIDDENSPACING with TWOEDGES



a) Violation, all of the conditions are met, and yet the spacing is 0.2 (>= 0.2 & <= 0.25)



b) Violation, the PRL is considered as 0.16 when both edges are in forbidden spacing range.

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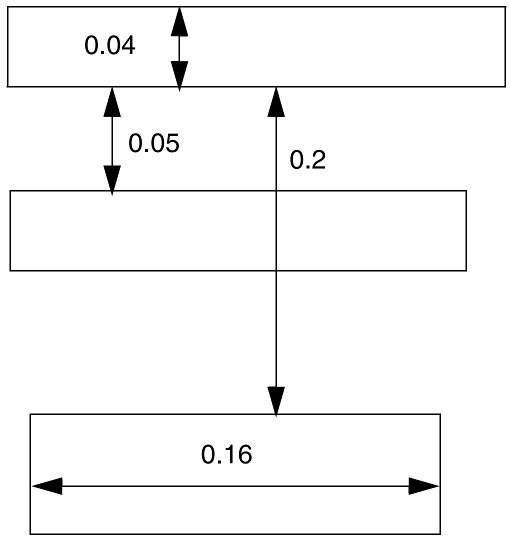
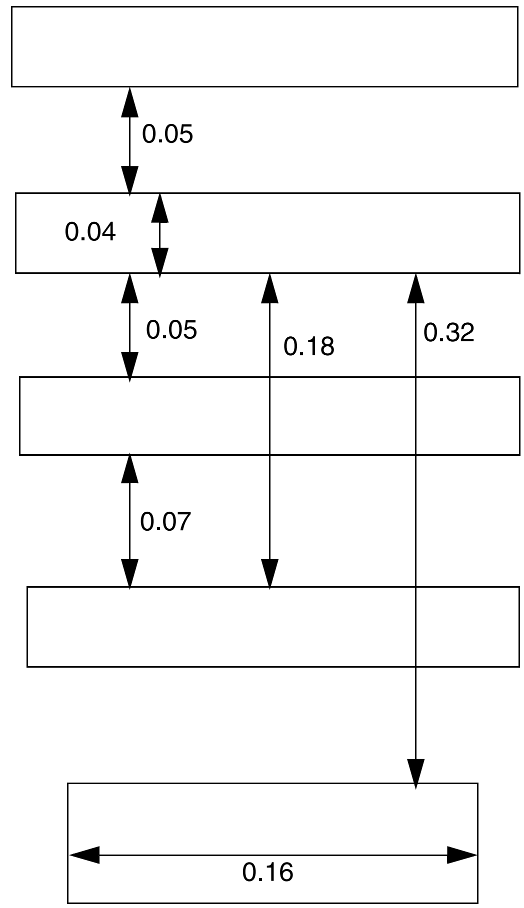


c) OK, the top wire only has 1 neighbor.



d) Violation, the bottom most wire should be checked independently

- The following example shows forbidden spacing rule with `EXACTSPACINGEDGE`:

```
DIRECTION HORIZONTAL ;
PROPERTY LEF58_FORBIDDENSPACING
    "FORBIDDENSPACING 0.2 0.25 WIDTH 0.05 PRL 0.04
       EXACTSPACINGEDGE 0.06 SPANLENGTH 0.09 ; " ;
```


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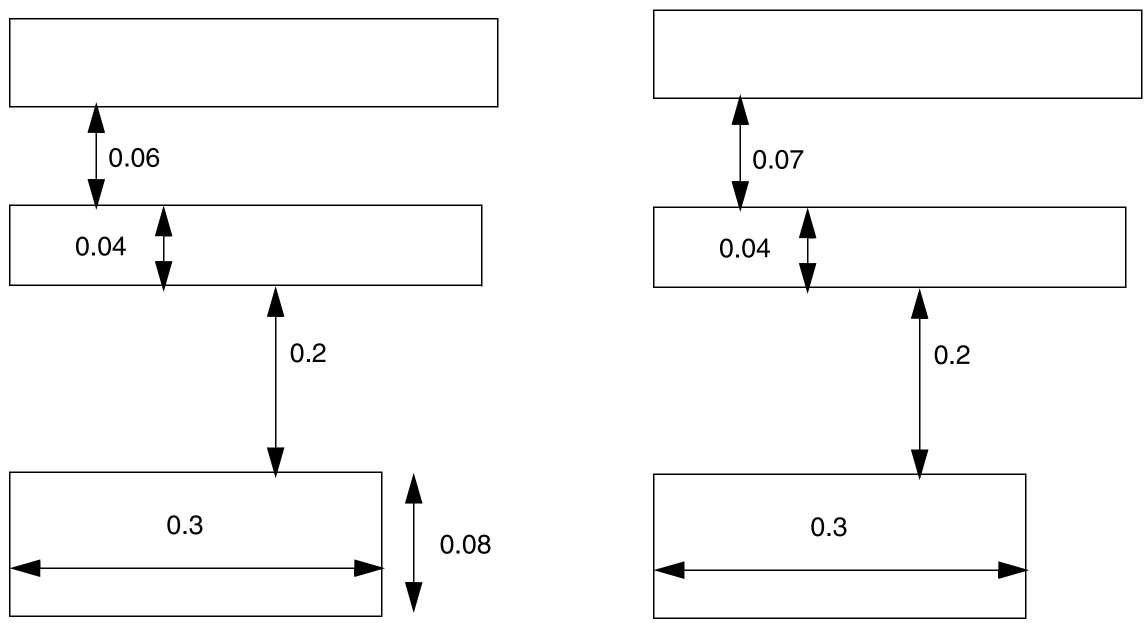
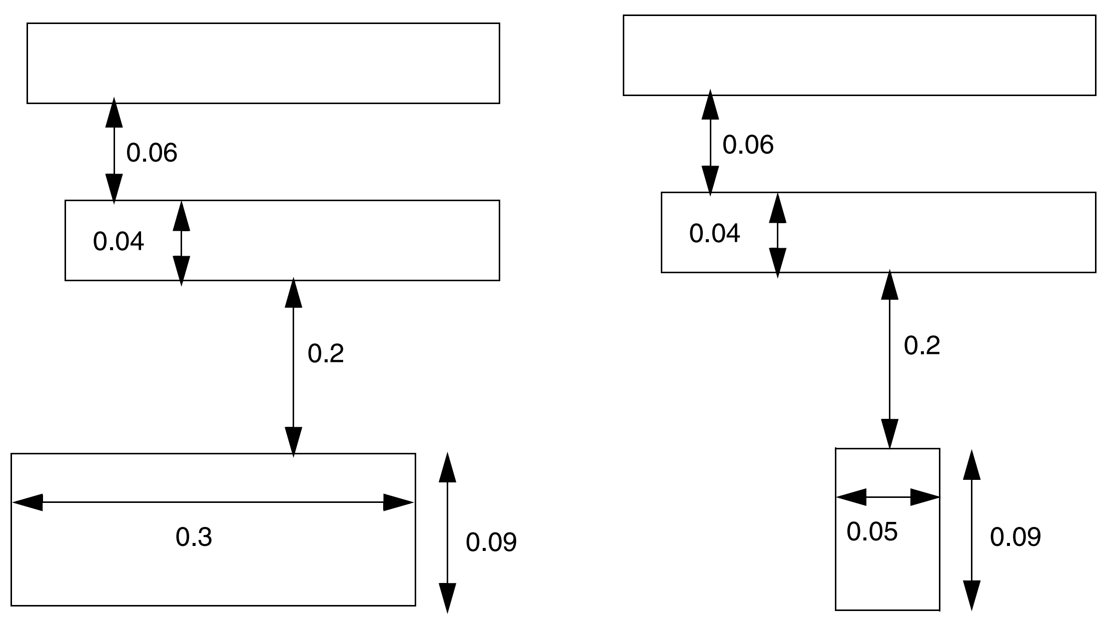
Figure 6-78 Illustration of FORBIDDENSPACING with EXACTSPACINGEDGE



a) Violation, exact spacing of 0.06 neighbor on top & spacing to bottom neighbor is 0.2 (>= 0.2 & <= 0.25)

b) OK, top neighbor is not exactly 0.06 spacing away.



c) OK, the span condition is not met

d) OK, the span, not width, of 0.09 is checked

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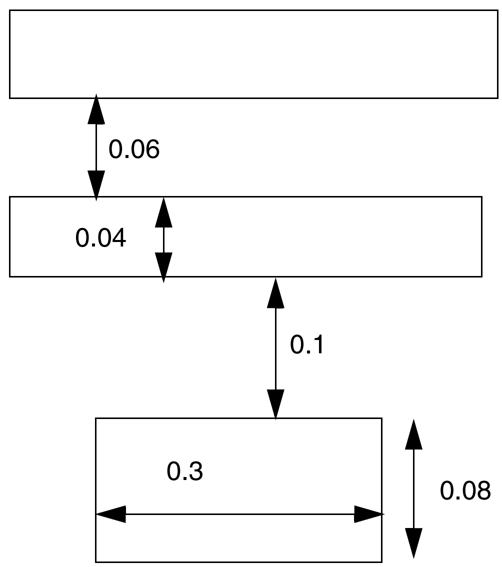
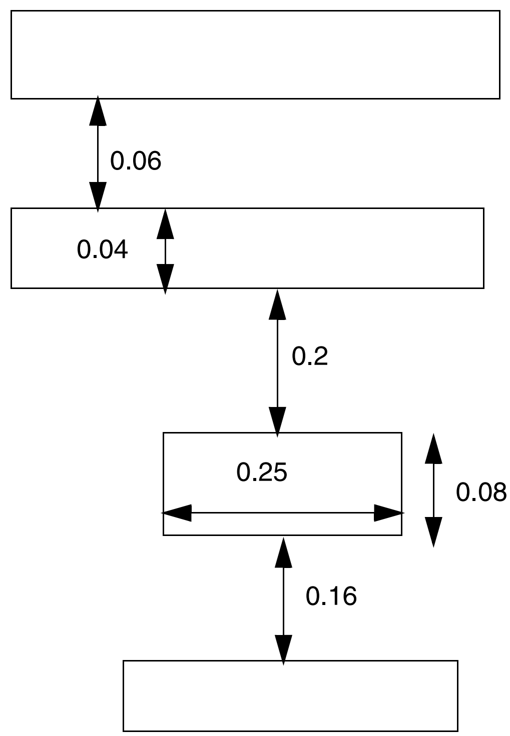
- The following example shows forbidden spacing rule with `WITHIN` in `EXACTSPACINGEDGE SPANLENGTH`:

```
DIRECTION HORIZONTAL ;
PROPERTY LEF58_FORBIDDENSPACING
    "FORBIDDENSPACING 0.08 0.12 0.2 0.25 WIDTH 0.05 PRL 0.04
       EXACTSPACINGEDGE 0.06 SPANLENGTH 0.09
         WITHIN 0.15 ; " ;
```

Figure 6-79 Illustration of FORBIDDENSPACING with EXACTSPACINGEDGE WITHIN



a) Violation, bottom neighbor is within the first forbidden range (>= 0.08 & <= 0.12), and the 0.04 width is within 0.15 of the 0.08 wire.



b) OK, the 0.08 wire does not have a neighbor within 0.15. It would be a violation if WITHIN is omitted.

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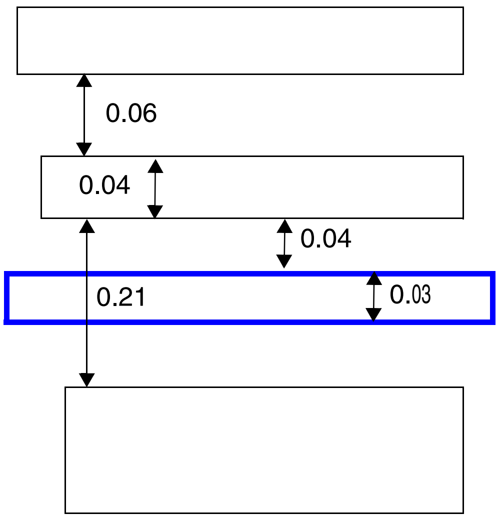
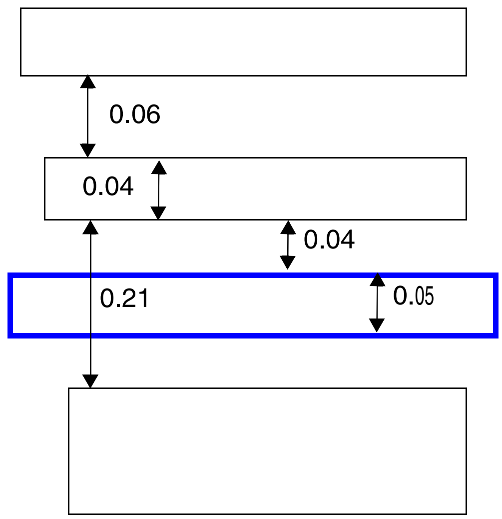
- The following example shows forbidden spacing rule with `IGNOREMIDDLE`:

```
DIRECTION HORIZONTAL ;
PROPERTY LEF58_FORBIDDENSPACING "
    FORBIDDENSPACING 0.2 0.25 WIDTH 0.05 PRL 0.00
       EXACTSPACINGEDGE 0.06
       IGNOREMIDDLE 0.03 0.04 ; " ;
```

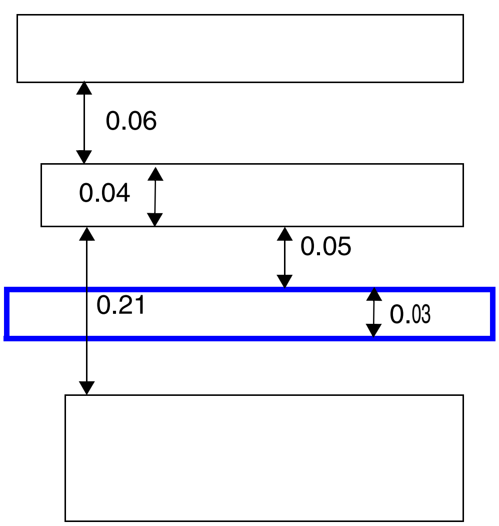
Figure 6-80 Illustration of FORBIDDENSPACING with IGNOREMIDDLE



a) OK, the blue wire has a width exactly equal to `0.03` and is exactly `0.04` from the triggering wire. It would not be ignored and would shield the violation.



b) Violation, the blue wire has width of `0.05`, which is not exactly equal to `0.03`. It would be ignored as if it did not exist.



c) Violation, the blue wire has spacing of 0.05 from the triggering wire, which is not exactly equal to 0.04. It would be ignored as if it did not exist.

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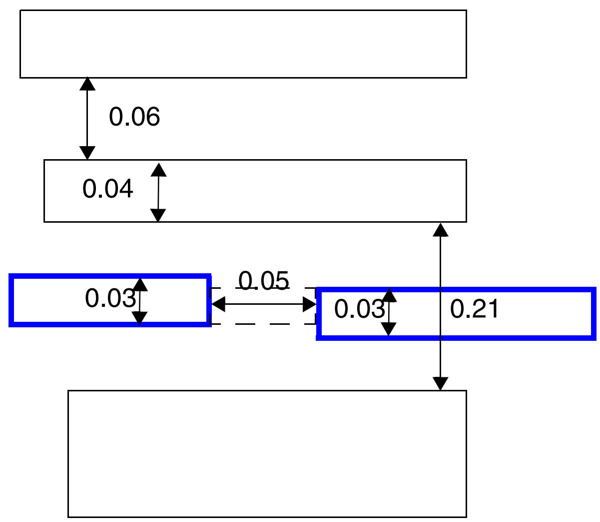
- The following example shows forbidden spacing rule with `EXCEPTWIREWIDTHSPACING`:

```
DIRECTION HORIZONTAL ;
PROPERTY LEF58_FORBIDDENSPACING "
    FORBIDDENSPACING 0.2 0.25 WIDTH 0.05 PRL 0.00
       EXACTSPACINGEDGE 0.06
       EXCEPTWIREWIDTHSPACING 0.03 0.07 ; " ;
```

Figure 6-81 Illustration of FORBIDDENSPACING with EXCEPTWIREWIDTHSPACING



a) OK, the width of the blue wires is exactly equal to `0.03` and the spacing is `0.05` (< `0.07`). The gap in the dotted line would be filled to merge the blue wires such that they would shield the wide bottom wire within the forbidden spacing range to be OK.

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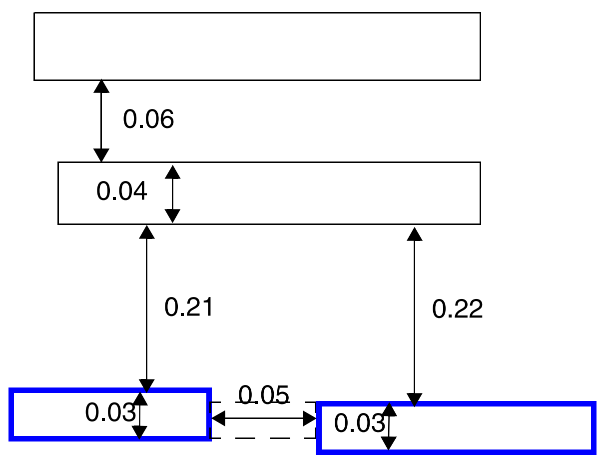
- The following example shows forbidden spacing rule with `EXCEPTWIREWIDTHSPACING`:

```
DIRECTION HORIZONTAL ;
PROPERTY LEF_CDN_FORBIDDENSPACING "
    FORBIDDENSPACING 0.2 0.25 WIDTH 0.05 PRL 0.00
       EXACTSPACINGEDGE 0.06 SPANLENGTH 0.03
       EXCEPTWIREWIDTHSPACING 0.03 0.07 ; " ;
```

Figure 6-82 Illustration of FORBIDDENSPACING with EXCEPTWIREWIDTHSPACING



This is a violation. The two blue wires are in the forbidden spacing range, but they do not fulfill the `SPANLENGTH` condition to be violations. However, the width of the blue wires is exactly equal to `0.03` and the spacing is `0.05` (< `0.07`), which would trigger the `EXCEPTWIREWIDTHSPACING` condition. The gap of the dotted line having a span length less than `0.03` is also checked against the forbidden spacing rule to be a violation.

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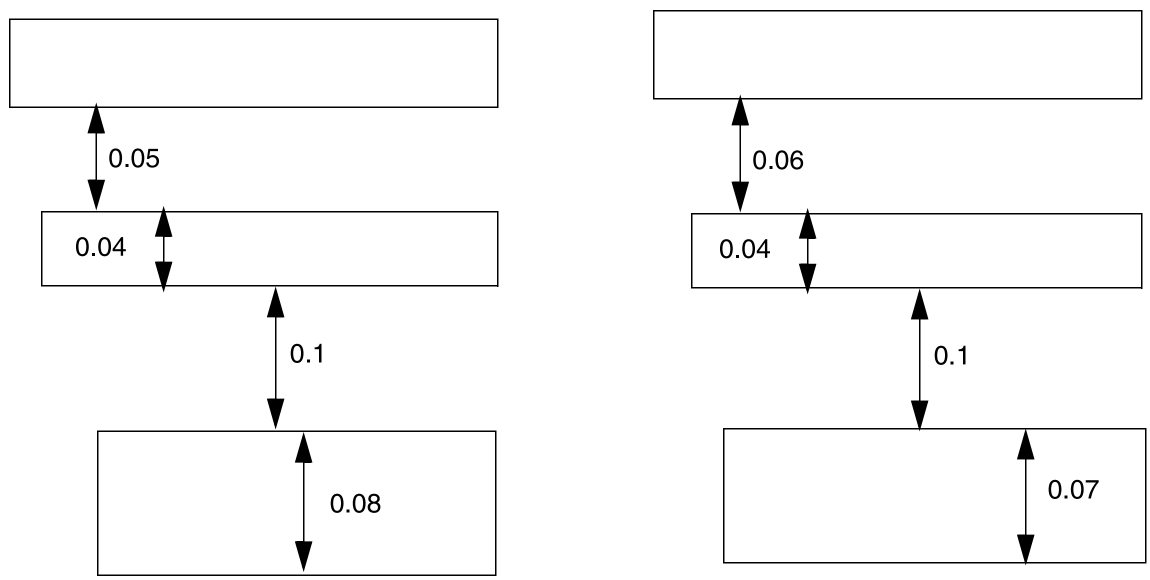
- The following example shows forbidden spacing rule with `EXACTWIDTH`:

```
DIRECTION HORIZONTAL ;
PROPERTY LEF58_FORBIDDENSPACING "
    FORBIDDENSPACING 0.08 0.12 EXACTWIDTH 0.04 PRL 0
       OTHERWIDTH 0.08 EXACTSPACINGEDGE 0.06 ; " ;
```

Figure 6-83 Illustration of FORBIDDENSPACING with EXACTWIDTH



a) OK, the middle 0.04 wire does not have an exact spacing of 0.05 on the other (top) side

b) OK, the bottom neighbor has width of 0.07 (< 0.08)

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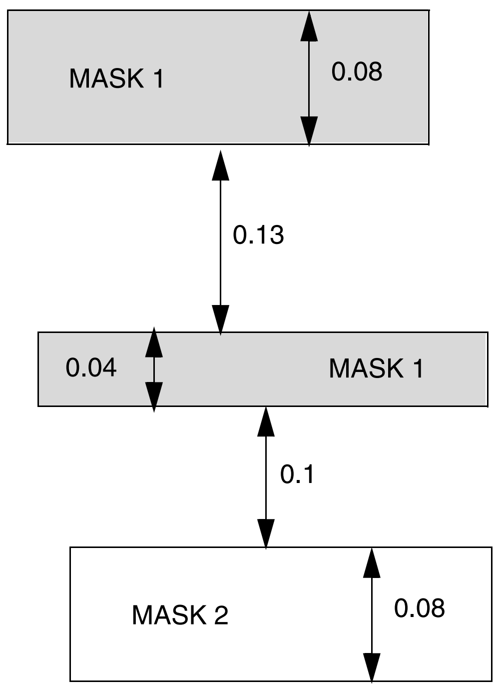
- The following example shows forbidden spacing rule with `WIDTHRANGE`:

```
DIRECTION HORIZONTAL ;
PROPERTY LEF58_FORBIDDENSPACING "
    FORBIDDENSPACING 0.08 0.12 SAMEMASK
       WIDTHRANGE 0.00 0.05 PRL 0.00
       OTHERWIDTH 0.07 WITHIN 0.35 ; " ;
```

Figure 6-84 Illustration of FORBIDDENSPACING with WIDTHRANGE



a) OK, the spacing between the bottom two wires are within the forbidden spacing range of `0.08` and `0.12`, but they are different-mask; the top two same-mask wires have spacing outside the forbidden spacing range. Violation if `SAMEMASK` is omitted.

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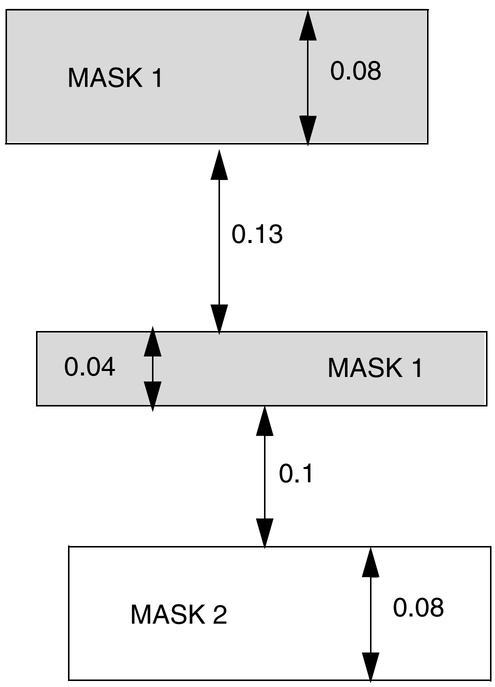
- The following example shows forbidden spacing rule with `WIDTHRANGE` and `OTHERSAMEMASK`:

```
DIRECTION HORIZONTAL ;
PROPERTY LEF58_FORBIDDENSPACING "
    FORBIDDENSPACING 0.08 0.12
       WIDTHRANGE 0.00 0.05 PRL 0.00
       OTHERWIDTH 0.07 WITHIN 0.35 OTHERSAMEMASK ; " ;
```

Figure 6-85 Illustration of FORBIDDENSPACING with OTHERSAMEMASK



a) OK, the bottom two wires could potentially violate the rule, but the two outer wires must be same-mask to trigger the rule. Violation if `OTHERSAMEMASK` is omitted.

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Forbidden Span Length Rule

A forbidden span length rule can be used to specify the conditions under which a wire of the specified span length in the non-preferred routing direction is allowed or forbidden.

You can create a forbidden span length rule by using the following property definition:

```
PROPERTY LEF58_FORBIDDENSPANLENGTH
       "FORBIDDENSPANLENGTH spanLength
          EXCEPTJOGLENGTH orthSpanLength [INCLUDELSHAPE]
       ; " ;
```

Where:

`FORBIDDENSPANLENGTH` *`spanLength`* `EXCEPTJOGLENGTH` *`orthSpanLength`*

Specifies that a wire with span length of *`spanLength`* in the non-preferred routing direction is forbidden except when it is part of a wrong-way Z-shaped jog with span length of *`orthSpanLength`* in the orthogonal/preferred routing direction.

Type: Float, specified in microns

`INCLUDELSHAPE` Specifies that a wire with span length of *`spanLength`* in the non-preferred routing direction is also allowed when it is part of an L-shaped jog with span length of *`orthSpanLength`* in the orthogonal/preferred routing direction.

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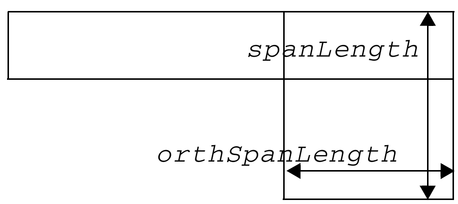
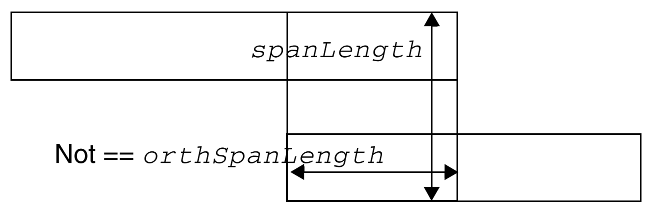
Forbidden Span Length Rule Example

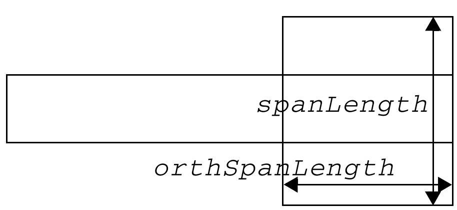
Figure 6-86 Illustrations of Forbidden Span Length Rule



a) OK, because the vertical span length of `spanLength` is part of an ‘L’ shape with horizontal span length of `orthSpanLength`. It is a violation if `INCLUDELSHAPE` is omitted.



b) It is a violation since the vertical span length of `spanLength` is part of a ‘Z’ shape with a horizontal span length not equal to *orthSpanLength.*



c) It is a violation since the vertical span length of `spanLength` is not part of a ‘Z’ or ‘L’ shape although the horizontal span length is equal to `orthSpanLength`.

Illustrations of:
`FORBIDDENSPANLENGTH` *`spanLength`*
`EXCEPTJOGLENGTH` *`orthSpanLength`* `INCLUDELSHAPE`

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Gap Rule

A gap rule can be used to specify rectangular gaps in a design with certain width and length.

You can create a gap rule by using the following property definition:

```
PROPERTY LEF58_GAP
     "GAP EXACTWIDTH exactWidth MAXLENGTH maxLength
         SPACING {{0 | 1 | 2} spacing}...[ENDTOEND endToEndSpacing]
         ; " ;
```

Where:

`ENDTOEND` *`endToEndSpacing`*

Specifies that the end-to-end spacing of two gaps with parallel run length greater than 0 in non-preferred direction of the routing layer must be greater than or equal to *`endToEndSpacing`*.
Type: Float, specified in microns

`GAP EXACTWIDTH` *`exactWidth`* `MAXLENGTH` *`maxLength`*

Specifies the empty area without wires must be a rectangle with width exactly equal to *`exactWidth`* and length less than or equal to *`maxLength`*. The 'width' is measured along the preferred direction of the routing layer. The gap between two parallel wires with the corresponding required spacing is not formed. The gaps are formed/ aligned to line end of wires and are extended half of the required spacing of the wires in the perpendicular direction.
Type: Float, specified in microns

`SPACING {{0 | 1 | 2}` *`spacing`*`}...`

Specifies the spacing among those gaps based on whether they are on the same track (0), adjacent track (1), or 2 tracks apart (2), where track is defined along the preferred direction of the routing layer. At the most three spacing values with different track number can be defined.
Type: Float, specified in microns

Gap Rule Example

- The following rule illustrates the Gap rule with `ENDTOEND`:

```
DIRECTION HORIZONTAL ;
PROPERTY LEF58_GAP "
    GAP EXACTWIDTH 0.2 MAXLENGTH 0.6
```


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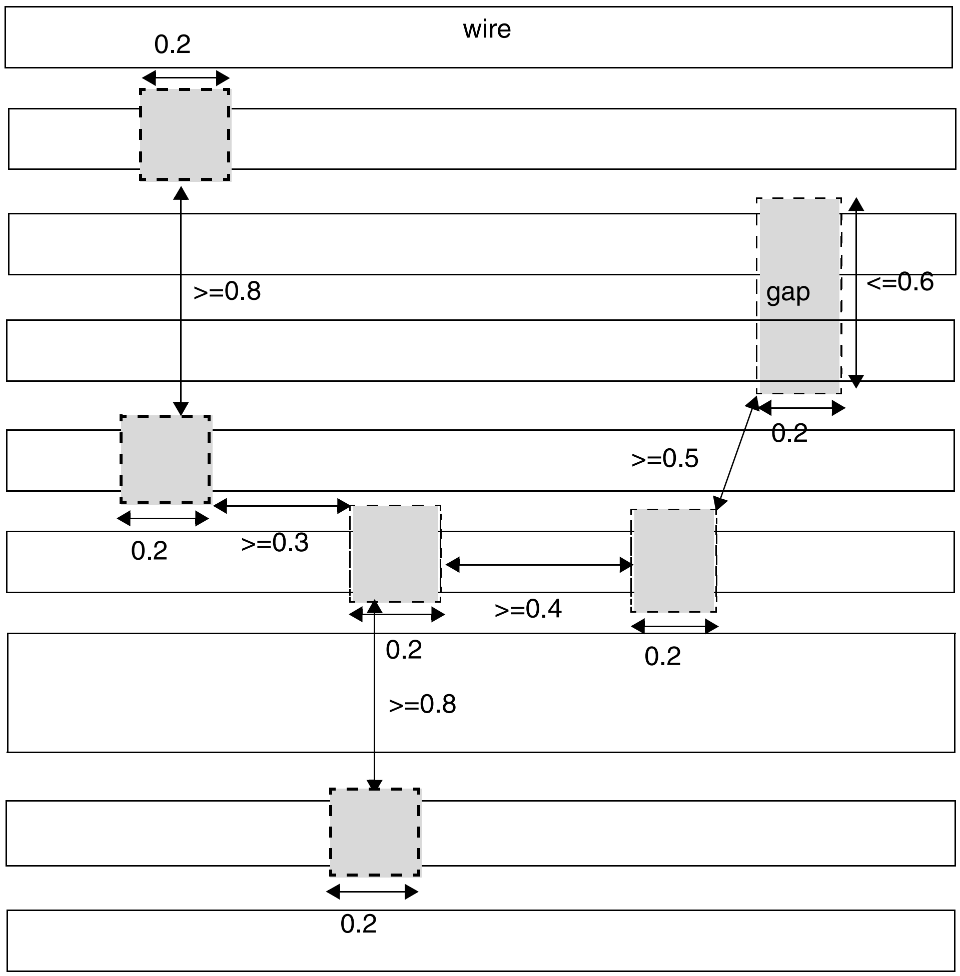
```
SPACING 0 0.4 1 0.3 2 0.5 ENDTOEND 0.8; " ;
```

Figure 6-87 Illustration of Gap Rule with ENDTOEND



Gaps (in gray color) must have width of 0.2 & max length of 0.6. Spacing on the same track must be >= 0.4, on adjacent track must be >= 0.3, two track apart must be >= 0.5 and end to end with PRL > 0 must be >= 0.8.

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Jog Corner Keep-Out Zone Rule

A jog corner keep-out zone rule can be used to specify two keep-out zones on a wrong-way jog connected to a right-way wire in an "L" shape if there is a neighboring wire in a window formed along the wrong-way wire edge on the concave corner of the jog.

You can create a jog corner keep-out zone rule by using the following property definition:

```
PROPERTY LEF_CDN_JOGCORNERKEEPOUTZONE
       "JOGCORNERKEEPOUTZONE innerEdgeExtension innerOrthoExtension
          OUTER outerEdgeExtension outerOrthoExtension
          BACK backEdgeExtension backOrthoExtension
          ; " ;
```

Where:

```
JOGCORNERKEEPOUTZONE innerEdgeExtension innerOrthoExtension
     OUTER outerEdgeExtension outerOrthoExtension
     BACK backEdgeExtension backOrthoExtension
```


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Specifies two keep-out zones on a wrong-way jog connected to a right-way wire in an 'L' shape if there is a neighboring wire in a window formed along the wrong-way wire edge on the concave corner of the jog. The keep-out zones do not allow any neighboring wire to overlap with them.

Type: Float, specified in microns

Using the right orientation of the L-shaped jog as an illustration, form a window on the top edge of the wrong-way jog by extending *innerEdgeExtension* along the edge or horizontally and *innerOrthoExtension* along the orthogonal direction or vertically. If a neighboring wire overlaps with that window:

- Form a keep-out zone on the bottom edge of the wrong-way jog by extending *outerEdgeExtension* along the edge or horizontally and *outerOrthoExtension* along the orthogonal direction or vertically.
- Form a keep-out zone on the left edge of the wrong-way jog by extending *backEdgeExtension* along the edge or vertically and *backOrthoExtension* along the orthogonal direction or horizontally.

If the orientation of the L shape is rotated or flipped, the neighbor search window and all keep-out zones are rotated or flipped accordingly.

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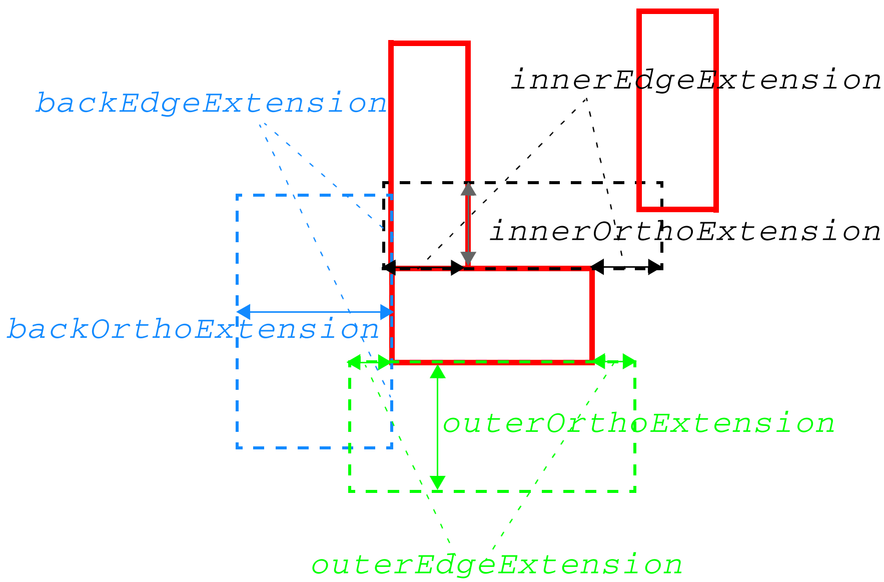
Jog Corner Keep-Out Zone Rule Example

Figure 6-88 Illustration of Jog Corner Keep-Out Zone Rule



As a neighbor wire is overlapping with the black window, the blue and green keep-out zones are formed to prevent any neighbor wire.

Illustration of `JOGSCORNERKEEPOUTZONE` *`innerEdgeExtension innerOrthoExtension`*
`OUTER` *`outerEdgeExtension outerOrthoExtension`*
`BACK` `backEdgeExtension backOrthoExtension ;`

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Jog Keep-Out Zone Rule

A jog keep-out zone rule can be used to specify keep-out zones for a wrong-way jog connected to a right-way wire in an "L" shape.

You can create a jog keep-out zone rule by using the following property definition:

```
PROPERTY LEF58_JOGKEEPOUTZONE
       "JOGKEEPOUTZONE horzExtension vertExtension
          INNER innerEdgeExtension innerOrthoExtension
          OUTER outerEdgeExtension outerOrthoExtension
          BACK backEdgeExtension backOrthoExtension
          FRONT frontEdgeExtension frontOrthoExtension
          ; " ;
```

Where:

```
JOGKEEPOUTZONE horzExtension vertExtension
    INNER innerEdgeExtension innerOrthoExtension
    OUTER outerEdgeExtension outerOrthoExtension
    BACK backEdgeExtension backOrthoExtension
    FRONT frontEdgeExtension frontOrthoExtension
```


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Specifies five keep-out zones on a wrong-way jog connected to a right way-wire in an "L" shape. The keep-out zones do not allow any neighboring wire to overlap with them.

Type: Float, specified in microns

Using the right orientation of the L-shaped jog as an illustration:

- Form a keep-out zone on the top edge of the wrong way jog by extending *innerEdgeExtension* along the edge or horizontally and *innerOrthoExtension* along the orthogonal direction or vertically.
- Form a keep-out zone on the bottom edge of the wrong way jog by extending *outerEdgeExtension* along the edge or horizontally and *outerOrthoExtension* along the orthogonal direction or vertically.
- Form a keep-out zone on the left edge of the wrong way jog by extending *backEdgeExtension* along the edge or vertically and *backOrthoExtension* along the orthogonal direction or horizontally.
- Form a keep-out zone on the right edge of the wrong way jog by extending *frontEdgeExtension* along the edge or vertically and *frontOrthoExtension* along the orthogonal direction or horizontally.
- Form a last keep-out zone on the bottom-right convex corner by extending *horzExtension* horizontally and *vertExtension* vertically.
- If the orientation of the L shape is rotated or flipped, all keep-out zones are rotated or flipped accordingly.

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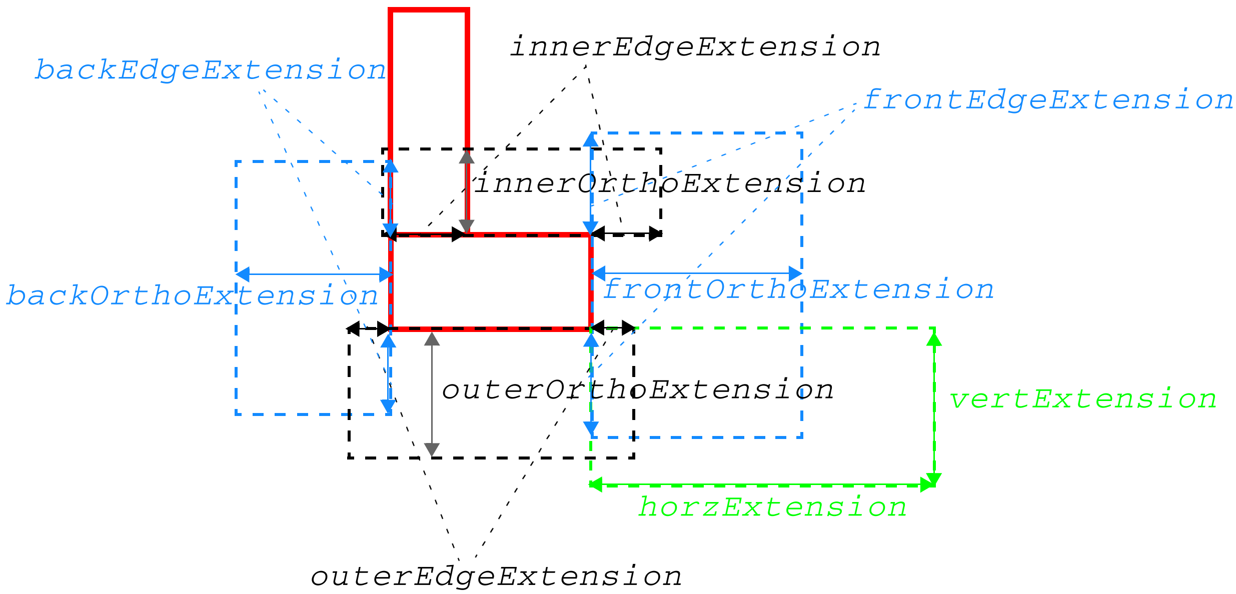
Jog Keep-Out Zone Rule Example

Figure 6-89 Illustration of Jog Keep-Out Zone Rule



The five keep-out zones are formed on the L-shaped jog.

Illustration of `JOGKEEPOUTZONE` *`horzExtension vertExtension`*
```
    INNER innerEdgeExtension innerOrthoExtension
    OUTER outerEdgeExtension outerOrthoExtension
    BACK backEdgeExtension backOrthoExtension
    FRONT frontEdgeExtension frontOrthoExtension ;
```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Jog Keep-Out Zone Table Rule

A jog keep-out zone table rule can be used to specify a keep-out zone on a concave corner of a wrong-way jog connected to a right-way wire in an "L" shape.

You can create a jog keep-out zone table rule by using the following property definition:

```
PROPERTY LEF58_JOGKEEPOUTZONETABLE
       "JOGKEEPOUTZONETABLE
          {SPANLENGTH spanLength rightWayExtension wrongWayExtension}...
          ; " ;
```

Where:

`JOGKEEPOUTZONETABLE`
`{SPANLENGTH` *`spanLength rightWayExtension`*
`wrongWayExtension``}...`

Specifies a keep-out zone on the concave corner of a wrong way jog connected to a right way wire in an 'L' shape. The keep-out zone does not allow any neighboring wire to overlap with it. Find the largest *spanLength* from which the actual span length of the jog in the non-preferred routing direction is larger. The keep-out zone is formed on the concave corner by extending the corresponding *rightWayExtension* along the right way wire edge and *wrongWayExtension* along the wrong way wire edge.

Type: Float, specified in microns

Jog Keep-Out Zone Table Rule Example

- The following rule illustrates the Jog Keep-Out Zone Table rule:

```
DIRECTION VERTICAL ;
PROPERTY LEF58_JOGKEEPOUTZONETABLE "
    JOGKEEPOUTZONETABLE
    SPANLENGTH sp1 rwe1 wwe1
    SPANLENGTH sp2 rwe2 wwe2
    SPANLENGTH sp3 rwe3 wwe3 ; " ;
```


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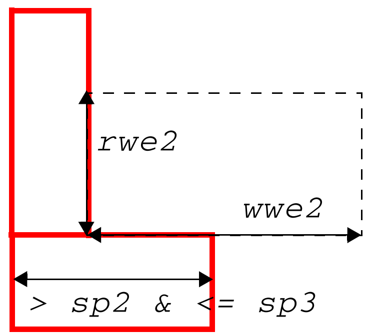
Figure 6-90 Illustration of Jog Keep-Out Zone Table Rule



The span length of the jog in the non-preferred routing direction is > *sp2* and <= *sp3*. Therefore, the corresponding extensions, *rwe2* and *wwe2*, are used to form the keep-out zone, which prevents any neighbor wire.

Example of `JOGKEEPOUTZONE` *`horzExtension vertExtension`*

```
    INNER innerEdgeExtension innerOrthoExtension
    OUTER outerEdgeExtension outerOrthoExtension
    BACK backEdgeExtension backOrthoExtension
    FRONT frontEdgeExtension frontOrthoExtension ;
```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Joint Corner Spacing Rule

A joint corner spacing rule can be used to define the spacing between two facing joints of joint corners.

You can create a joint corner spacing rule by using the following property definition:

```
PROPERTY LEF58_JOINTCORNERSPACING
     "JOINTCORNERSPACING spacing [SAMEMASK]
        JOINTWIDTH jointWidth [MINLENGTH minLength]
        JOINTLENGTH spanLength [EDGELENGTH edgeLength]
     ; " ;
```

Where:

All other keywords are the same as the existing LEF routing layer joint corner spacing rule syntax.

`EDGELENGTH` *`edgeLength`*

Specifies that the joint corner spacing only applies if the edge length of both the joints is less than or equal to the specified *`edgeLength`*.
Type: Float, specified in microns

`JOINTCORNERSPACING` *`spacing`*
`JOINTWIDTH` *`jointWidth`* `[MINLENGTH` *`minLength`*`]`
`JOINTLENGTH` *`spanLength`*

Specifies the spacing between two facing joints of joint corners with parallel run length less than or equal to zero to be *`spacing`*. A joint corner is a convex corner consisting of two consecutive joints with span greater than *`spanLength`* and not a EOL edge with length less than the *`jointWidth`* and having a length greater than or equal to the *`minLength`* along both the sides, if the `MINLENGTH` keyword is specified.
Type: Float, specified in microns

`SAMEMASK` Specifies that the corner spacing rule only applies to objects on the same mask.

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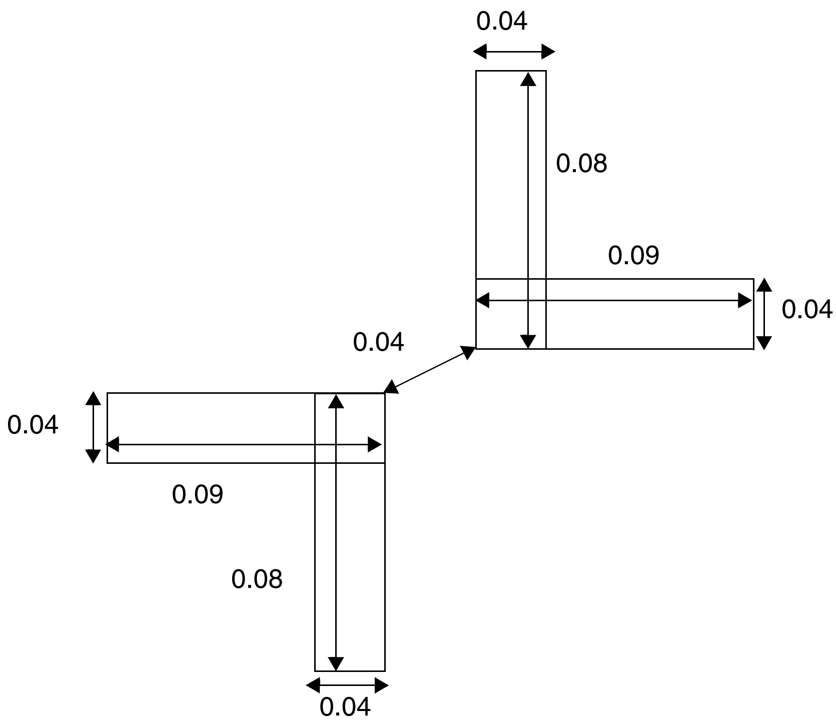
Joint Corner Spacing Rule Example

Figure 6-91 Illustration of Joint Corner Spacing Rule

```
PROPERTY LEF58_JOINTCORNERSPACING
  "JOINTCORNERSPACING 0.05
      JOINTWIDTH 0.07 JOINTLENGTH 0.06
      EDGELENGTH 0.1 ; " ;
```



a) Violation, the spacing is applied between two facing joints with no common parallel run length

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Joint Forbidden Spacing Rule

A joint forbidden spacing rule can be used to define forbidden spacing between a joint and a neighboring wire.

You can create a joint forbidden spacing rule by using the following property definition:

```
PROPERTY LEF58_JOINTFORBIDDENSPACING
     "JOINTFORBIDDENSPACING minSpacing maxSpacing
     WIDTH width WRONGDIRWIDTH wrongDirWidth
     OTHERWIDTH otherWidth [PRL prl]
     ; " ;
```

Where:

```
JOINTFORBIDDENSPACING minSpacing maxSpacing
     WIDTH width WRONGDIRWIDTH wrongDirWidth
     OTHERWIDTH otherWidth [PRL prl]
```

Specifies a forbidden spacing between a joint and a neighboring wire in the orthogonal direction of the preferred routing direction of the layer to be greater than or equal to *minSpacing* and less than or equal to *maxSpacing*. A joint is a jog in the L or T shape, which consists of a preferred routing wire with width less than or equal to *width* and a wrong way non-preferred routing wire with width less than or equal to *wrongDirWidth*. In addition, the neighboring wire must have a width less than or equal to *otherWidth*. The rule is applied when the parallel run length (PRL) of the two wires is greater than `0` by default or greater than *prl* if `PRL` is defined. The forbidden spacing is applied on both the opposite sides of the joint in the non-preferred routing direction.

Type: Float, specified in microns

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Figure 6-92 Illustration of Joint Forbidden Spacing Rule

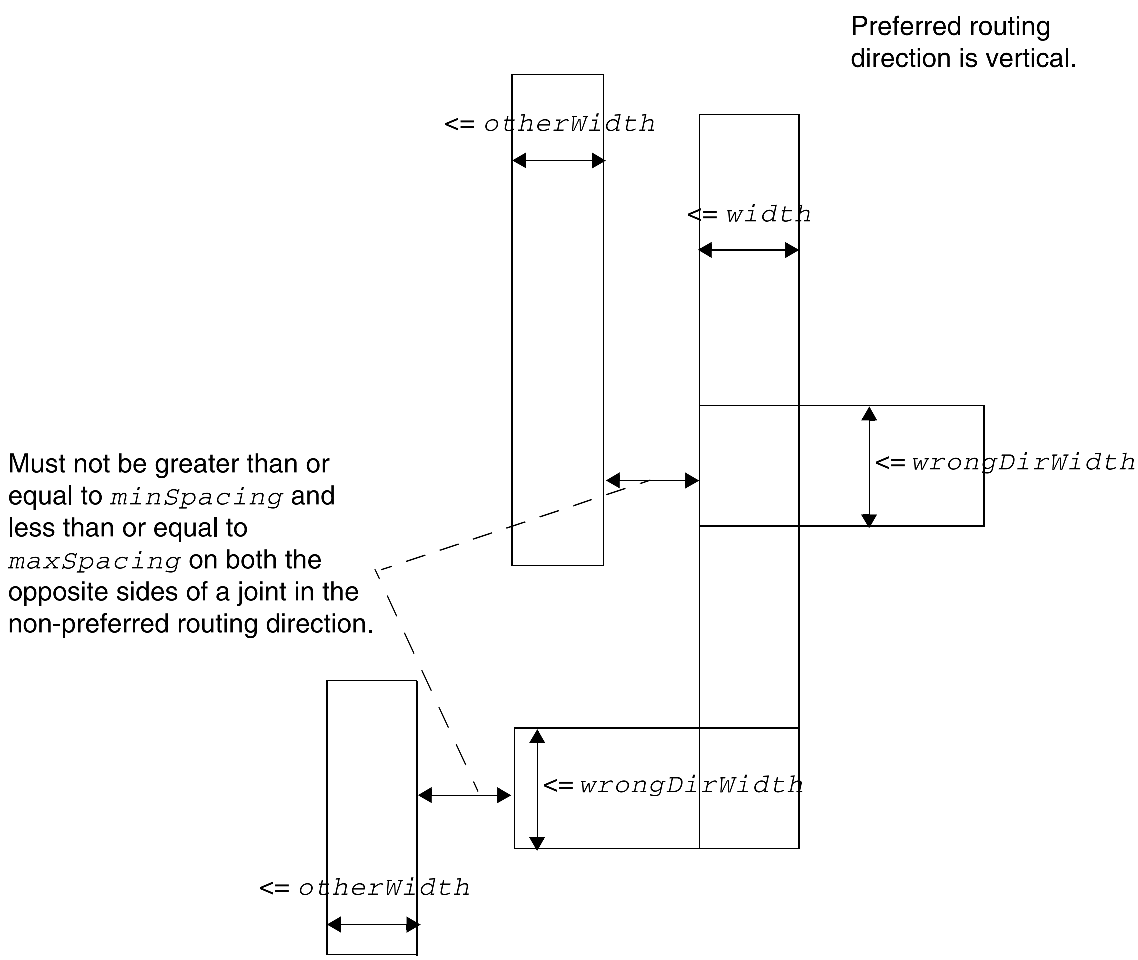

Illustration of `JOINTFORBIDDENSPACING` *`minSpacing maxSpacing`*
`WIDTH` *`width`* `WRONGDIRWIDTH` *`wrongDirWidth`*
`OTHERWIDTH` *`otherWidth`*

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Line End Aligned With Cut Rule

You can create a line end aligned with cut rule by using the following property definition:

```
PROPERTY LEF58_LINEENDALIGNEDWITHCUT
       "LINEENDALIGNEDWITHCUT alignedDistance ENCLOSURE enclosure
          [CUTCLASS className]
          {FROMABOVE | FROMBELOW} [CUTONANYWIRES]
          ; " ;
```

Where:

`LINEENDALIGNEDWITHCUT` *`alignedDistance`* `ENCLOSURE` *`enclosure`*
`{FROMABOVE | FROMBELOW}`

Specifies that three minimum default width wires with minimum spacing is a violation if the line end of the middle wire is exactly *alignedDistance* projected distance from the line ends of the two wires on the opposite sides and the middle wire contains a cut with enclosure less than or equal to *enclosure* on the above or below cut layer if `FROMABOVE` or `FROMBELOW` is specified. A positive *alignedDistance* value means that the neighbor line ends go beyond the middle line end, and a negative *alignedDistance* value means that the middle line end extends beyond the neighbor line ends. Zero *alignedDistance* means that all the three line ends are exactly aligned.

Type: Float, specified in microns

`CUTCLASS` *`className`*
Specifies that the cut must belong to the given cut class to trigger the rule. If `CUTCLASS` is not defined, any cuts belonging to any cut class could trigger the rule.

`CUTONANYWIRES`
Specifies that the rule is applied if the cut with enclosure less than or equal to *enclosure* is on any of the three wires.

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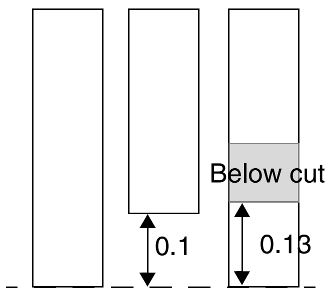
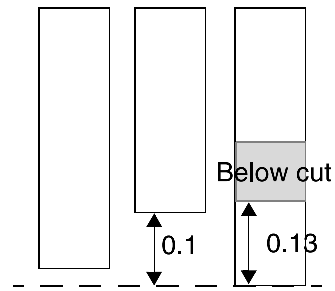
Figure 6-93 Line End Aligned With Cut Rule Example



a) Violation, the cut could be on any wire to trigger the rule. OK if `CUTONANYWIRES` is omitted because the middle wire does not have a cut.



b) OK, the left wire is not exactly 0.1 beyond the middle wire.

Example of

```
PROPERTY LEF58_LINEENDALIGNEDWITHCUT
  "LINEENDALIGNEDWITHCUT 0.1
      ENCLOSURE 0.15 FROMBELOW
      CUTONANYWIRES ; " ;
```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Line End Aligned With Cut Spacing Rule

You can create a line end aligned with cut spacing rule by using the following property definition:

```
PROPERTY LEF58_LINEENDALIGNEDWITHCUTSPACING
       "LINEENDALIGNEDWITHCUTSPACING minAlignedDistance
          maxAlignedDistance SPACING minSpacing maxSpacing
          PRL prl [CUTCLASS className] {FROMABOVE | FROMBELOW}
          ; " ;
```

Where:

`LINEENDALIGNEDWITHCUTSPACING` *`minAlignedDistance maxAlignedDistance`* `SPACING` *`minSpacing maxSpacing`* `PRL` *`prl`* `{FROMABOVE │ FROMBELOW}`	Specifies that three minimum default width wires with minimum spacing is a violation if the line-end of the middle wire goes beyond the neighboring line ends by the exact same distance, which is greater than or equal to *`minAlignedDistance`* and less than or equal to *`maxAlignedDistance`*, the middle line-end has a spacing greater than or equal to *`minSpacing`* and less than or equal to *`maxSpacing`* to a neighbor wire with PRL greater than *`prl`* and the middle wire contains a cut overlapping with the stick out portion on the above or below cut layer when `FROMABOVE` or `FROMBELOW` is specified. *Type*: Float, specified in microns
`CUTCLASS` *`className`*	Specifies that the cut must belong to the given cut class to trigger the rule. If `CUTCLASS` is not defined, any cuts belonging to any cut class could trigger the rule.

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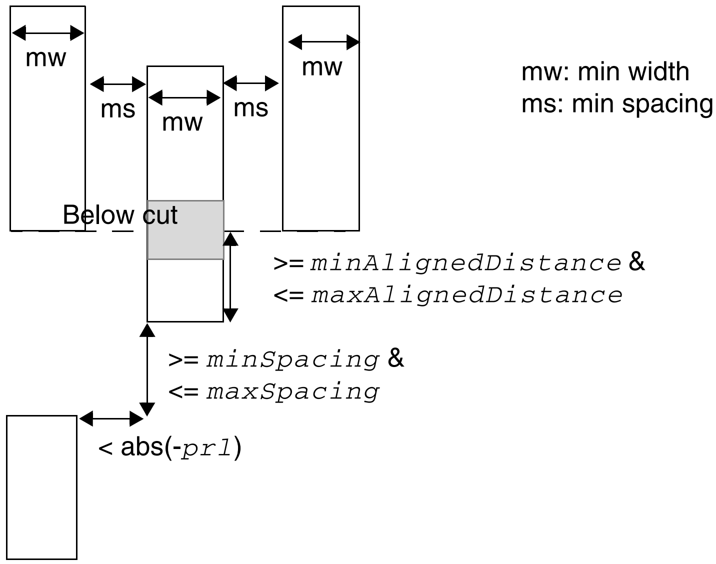
Figure 6-94 Line End Aligned With Cut Spacing Rule Example



Violation, the middle wire sticks out the same amount, which fulfills the given distance conditions, from the left and right wires. It has a neighbor wire at the bottom, which fulfills the given spacing and PRL values, and there is a below cut overlapping with the stick out portion. All three wires have min width and are min spacing apart.

Illustration of:

```
LINEENDALIGNEDWITHCUTSPACING
    minAlignedDistance maxAlignedDistance
    SPACING minSpacing maxSpacing
    PRL -prl FROMBELOW ; " ;
```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Line End Gap Rule

You can create a line end gap rule by using the following property definition:

```
PROPERTY LEF58_LINEENDGAP
       "LINEENDGAP gap WIDTH width [MASK maskNum]
       ; " ;
```

Where:

`LINEENDGAP` *`gap`* `WIDTH` *`width`* `[MASK` *`maskNum`*`]`

Specifies that two perfectly aligned wires having width less than *width* and having exactly the spacing of *gap* could be an exemption or a triggering condition on some other rules. If `MASK` is defined, both of the wires must belong to *maskNum*. By itself, it is not a rule and is just a line-end spacing condition. Some other rules would have a Boolean `EXCEPTLINEENDGAP` or `LINENEDGAPONLY` to refer back to this condition.

Type: Float, specified in microns

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Litho Macro Halo Rule

A litho macro halo rule is used to define a routing halo that allows only planar connections to its pins.

You can create a litho macro halo rule by using the following property definition:

```
PROPERTY LEF58_LITHOMACROHALO
     "LITHOMACROHALO horizontalHalo verticalHalo
          {WIRESPACING wireSpacing | NOSAMEMETALJOG}
        ; " ;
```

Where:

`LITHOMACROHALO` *`horizontalHalo verticalHalo`* `WIRESPACING` *`wireSpacing`*

Specifies halo values for litho purpose on a block macro. A routing halo with *`horizontalHalo`* and *`verticalHalo`* offset horizontally and vertically from the bounding box of the macro could be formed. Inside this routing halo, only planar connections to macro pins are allowed. Any other wires or above/below via insertion is also disallowed. In addition, the wires connected to the macro pins must have spacing greater than or equal to *`wireSpacing`*. A separate command specifies which block macro should create such a routing halo on them.

Type: Float, specified in microns

`NOSAMEMETALJOG` Specifies that the halo does not allow same-layer same-metal jogs. In other words, only rectangular wires are allowed. Layer change is still not allowed inside the halo, but rectangular through wires are allowed.

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Manufacturing Grid Rule

Manufacturing grid rules can be used to specify the manufacturing grid on a given layer.

You can create a manufacturing grid rule by using the following property definition:

```
PROPERTY LEF58_MANUFACTURINGGRID
     "MANUFACTURINGGRID value
        ; " ;
```

Where:

`MANUFACTURINGGRID` *`value`*

Specifies the manufacturing grid on the given layer that will override the library manufacturing grid value.
Type: Float, specified in microns

Maximum Diagonal Length Rule

The maximum length rule can be used to specify the maximum length of an edge on a routing layer allowing diagonal routing.

You can create a maximum diagonal length rule by using the following property definition:

```
PROPERTY LEF58_MAXDIAGLENGTH
       "MAXDIAGLENGTH diagonalLength manhattanLength
           ; " ;
```

Where:

`MAXDIAGLENGTH` *`diagonalLength manhattanLength`*

Specifies that any diagonal edge length must be less than or equal to *`diagonalLength`* and any Manhattan edge length must be less than or equal to *`manhattanLength`* in an RDL routing layer. This construct can be defined only on a routing layer that allows diagonal routing.
Type: Float, specified in microns

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Maximum Length with Cut Rule

A maximum length with cut rule can be used to specify the maximum length of a wire that has a single cut without a neighboring cut within the specified cut distance.

You can create a maximum length with cut rule by using the following property definition:

```
PROPERTY LEF58_MAXLENGTHWITHCUT
     "MAXLENGTHWITHCUT length WIDTH width
         WITHIN cutDistance {FROMABOVE | FROMBELOW}
         OTHERWIDTH otherWidth EXCEPTOTHERLENGTH otherLength
         [LAYER secondLayerName SECONDWIDTH secondWidth
             EXCEPTSECONDLENGTH secondLength]
          [PGONLY]
     ; " ;
```

Where:

`MAXLENGTHWITHCUT` *`length`* `WIDTH` *`width`*
`WITHIN` *`cutDistance`* `{FROMABOVE | FROMBELOW}`
`OTHERWIDTH` *`otherWidth`* `EXCEPTOTHERLENGTH` *`otherLength`*

Specifies that if a wire with width less than *`width`* and length greater than *`length`* has a cut from the cut layer above or below if `FROMABOVE` or `FROMBELOW` is defined, and that cut is connected to another metal layer wire with width less than *`otherWidth`*, there must be another such cut within *`cutDistance`*. In other words, if a wire has a single cut without a neighboring cut within *`cutDistance`*, the maximum length of the wire must be less than or equal to *`length`*. If the other metal layer wire has length greater than *`otherLength`*, this rule is exempted. If *`otherLength`* is set as `0`, this condition cannot be used for the exemption.

Type: Float, specified in microns

`LAYER` *`secondLayerName`* `SECONDWIDTH` *`secondWidth`*
`EXCEPTSECONDLENGTH` *`secondLength`*

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	Specifies a routing layer to be *`secondLayerName`*, which is a layer below or above the bottom or top metal layer of the cut when `FROMBELOW` or `FROMABOVE` is specified. If the other layer wire is connected to a wire with width less than *`secondWidth`* and length greater than *`secondLength`* on that *`secondLayerName`*, this rule is exempted. The rule is also exempted if the other layer wire is not connected to a wire in *`secondLayerName`*. *Type*: Float, specified in microns
`PGONLY`	Specifies that this maximum length with cut rule applies only on PG nets.

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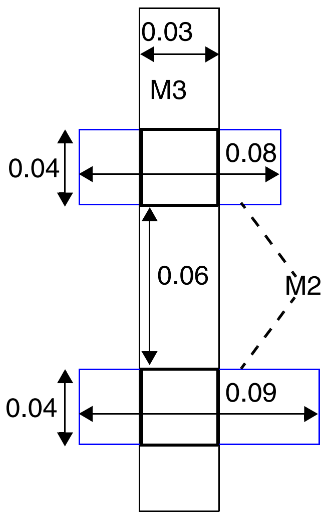
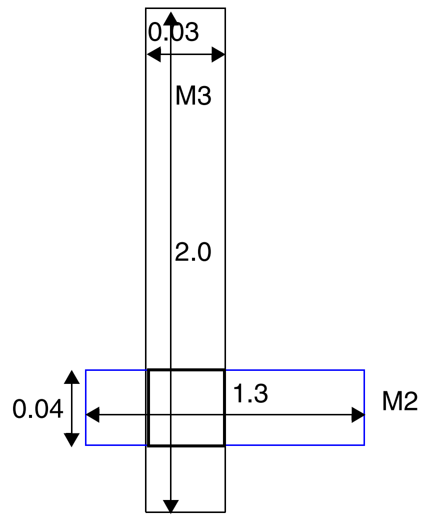
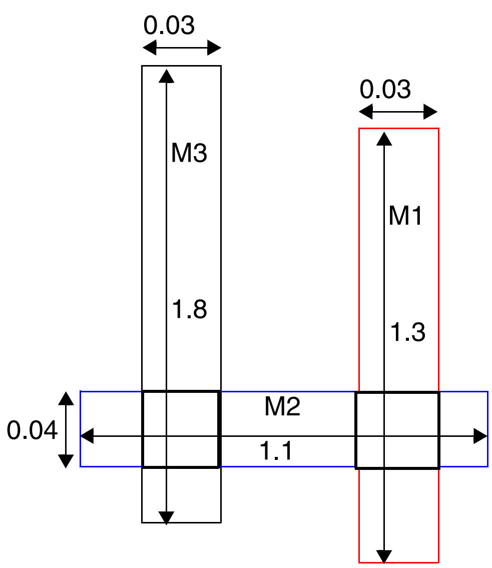
Figure 6-95 Maximum Length With Cut Rule Example



a) OK, there are two cuts within 1.0.



b) OK, the below metal layer wire has a length of 1.3 (> 1.2) so the rule is exempted even if the M2 wire is connected to M1.



c) OK, although the below metal layer wire has length of 1.1 (<= 1.2) to not exempt the rule, the second below metal layer (M1) wire has a length of 1.3 (> 1.1) to exempt the rule.

Example of
On M3

```
PROPERTY LEF58_MAXLENGTHWITHCUT "
  MAXLENGTHWITHCUT 0.5 WIDTH 0.05
     WITHIN 1.0 FROMBELOW
     OTHERWIDTH 0.06 EXCEPTOTHERLENGTH 1.2
     LAYER M1 SECONDWIDTH 0.04 EXCEPTSECONDLENGTH 1.1 ; " ;
```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Maximum Overlap Length Rule

You can create a maximum overlap length rule between two consecutive routing layers, with one of them being a RDL layer, by using the following property definition:

```
PROPERTY LEF58_MAXOVERLAPLENGTH
       "MAXOVERLAPLENGTH length LAYER secondLayerName
           ; " ;
```

Where:

`MAXOVERLAPLENGTH` *`length`* `LAYER` *`secondLayerName`*

Specifies that the overlap between a shape on this layer, which must be an RDL routing layer, and a shape on *`secondLayerName`*, which must be an adjacent below metal layer, must have a length less than or equal to *`length`*.
Type: Float, specified in microns

Figure 6-96 Illustration of Maximum Overlap Length Rule

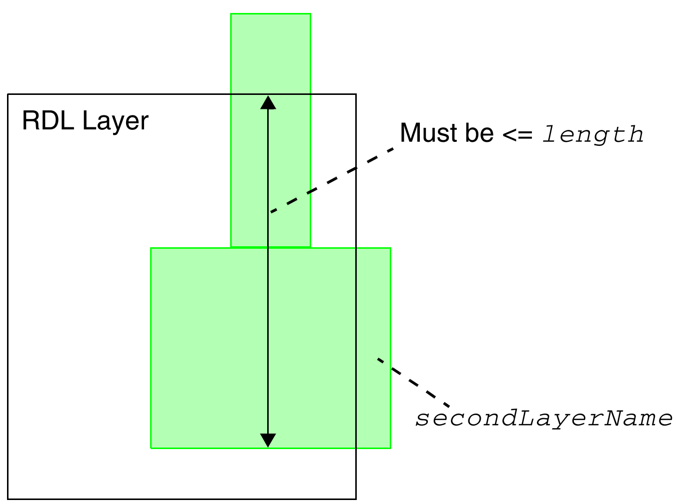

Illustration of
`MAXOVERLAPLENGTH` *`length`* `LAYER` *`secondLayerName`*

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Minimum Area With Cut Rule

A minimum area with cut rule allows you to specify the minimum area of a wire with a cut of the specified class.

You can define a minimum area with cut rule by using the following property definition:

```
PROPERTY LEF_CDN_MINAREAWITHCUT
       "MINAREAWITHCUT minArea
          CUTCLASSLIST {className}... {FROMABOVE | FROMBELOW}
          ; " ;
```

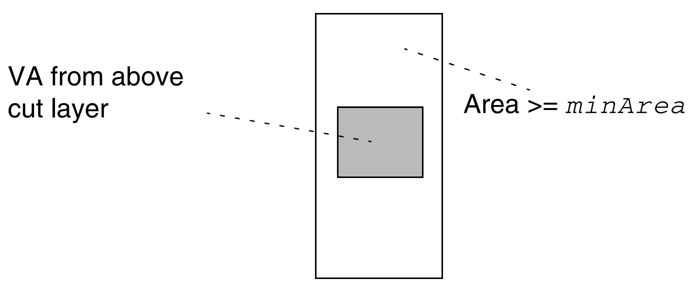
Where:

`MINAREAWITHCUT` *`minArea`*
    `CUTCLASSLIST {`*`className`*`}... {FROMABOVE | FROMBELOW}`

Specifies that a wire containing a cut belonging to *`className`* on the above or below cut layer when `FROMABOVE` or `FROMBELOW` is specified must have an area greater than or equal to *`minArea`*.

Type: Float, specified in microns

Figure 6-97 Illustration of Minimum Area With Cut Rule



The area must be >= *`minArea`* since the wire contains a VA cut from the above cut layer. If the wire does not contain a VA cut, the rule does not apply.

Illustration of `MINAREAWITHCUT` *`minArea`*
`CUTCLASSLIST VA FROMABOVE`

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Maximum Width Rule

A maximum width rule can be used to specify the maximum wire width on a given layer.

You can create a maximum width rule by using the following property definition:

```
PROPERTY LEF58_MAXWIDTH
     "MAXWIDTH width [WRONGDIRECTION]
        ; " ;
```

Where:

All other keywords are the same as the existing LEF routing layer `MAXWIDTH` syntax.

`WRONGDIRECTION`	Specifies the maximum wire width with direction perpendicular to the specified direction in `DIRECTION` on a Manhattan routing layer.

Minimum Cut Rule

Minimum cut rules exist for thin wires connected to a wide wire or pin.

You can define a minimum cut rule by using the following property definition:

```
PROPERTY LEF58_MINIMUMCUT
     "MINIMUMCUT {numCuts | {CUTCLASS className numCuts
         [DIFFCUTCLASS]}... }
         WIDTH width [WITHIN cutDistance [MAXXY]]
         [FROMABOVE | FROMBELOW]
         [EXCEPTOTHERWIDTH otherWidth]
         [LENGTH length WITHIN distance
         |AREA area [WITHIN distance]
         |SAMEMETALOVERLAP
         | FULLYENCLOSED
         | OVERLAPONLY]
         ; " ;
```

All other keywords are the same as the existing LEF routing layer `MINIMUMCUT` syntax.

Where:

`AREA` *`area`* `[WITHIN` *`distance`*`]`

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Applies the minimum cut rule when the wide object has a width that is greater than *width*, and an area that is greater than *area*.

The area of a polygon is determined by a process in which the polygon is shrunk by an amount equal to *width*/2, then grown by an amount equal to *width*/2. The resulting polygon at the connection location is used for area comparison. If the connection is made at a thin wire (width less than and equal to *width*) that connects to the wide object, the biggest remaining neighbor should be checked individually with their own areas. If it is followed by a `WITHIN` *distance* syntax, the within distance is measured from the edges of the remaining neighbors.

If *width* is less than width of the default routing wire is used along with the `AREA` keyword, the minimum cut requirement on the routing vias can vary depending on the area of the routing wire on the layer. This should be used cautiously. A small area can result in longer routing run times, and more DRC violations.

`WITHIN` indicates that the rule applies for thin wires directly connected to a wide object, if the cuts on the thin wire are less than *distance* from the wide object.

If `AREA` and `WITHIN` are defined, this rule only checks the thin wire connected to a wide wire; it does not check the wide wire itself. A separate `MINIMUMCUT` *numCuts* `WIDTH` *width* statement without `AREA` and `WITHIN` is required for any wide wire minimum cut rule.
Type: Float, specified in microns

Note: You can specify either `AREA WITHIN` or `LENGTH WITHIN` in a routing layer.

If `WITHIN` *cutDistance* is absent, only cuts belonging to the same via are considered as multiple cuts.

`CUTCLASS` *className numCuts*

Defines the minimum cut rule for a specific cut class *className* in the cut layers either above or below the current routing layer to be *numCuts*. Multiple `CUTCLASS` keywords can be defined to specify different *numCuts* for each of them. If `CUTCLASS` is defined in cut layer, then `CUTCLASS` must be specified in the `MINIMUMCUT` statement.

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`DIFFCUTCLASS`
: Specifies that if a wide wire has one cut belonging to *`className`*, then it must have at least *`numCuts`* `- 1` cuts not belonging to *`className`*.

`EXCEPTOTHERWIDTH` *`otherWidth`*
: Specifies that the minimum cut rule is exempted if the width on the other metal of all of the cuts is greater than or equal to *`otherWidth`*.

  Type: Float, specified in microns

`FULLYENCLOSED`
: Specifies that all the cuts must be completely inside a wide wire with width greater than *`minWidth`* to be counted against the minimum cut requirement of *`numCuts`*. See Figure 6-101 on page 811.

`MAXXY`
: Specifies that the cut neighbor search distance in `WITHIN` is measured in `MAXXY` style.

`OVERLAPONLY`
: Specifies that only the cuts overlapping a wide wire with width greater than *`width`* are to be counted against the minimum cut requirement of *`numCuts`*. See Figure 6-103 on page 812.

`SAMEMETALOVERLAP`
: Specifies that if the common metal on both the top and bottom layers containing the cut overlaps with the wide wire having a width greater than *`minWidth`*, then the rule still applies even if the cut itself does not overlap with the wide wire. See Figure 6-100 on page 810.

Minimum Cut Rule Examples

- The following minimum cut rule indicates that if an object has width greater than 0.1 µm, either a VA via with minimum three cuts or a VB via with minimum two cuts are needed on that object:

```
PROPERTY LEF58_MINIMUMCUT
    "MINIMUMCUT CUTCLASS VA 3 CUTCLASS VB 2 WIDTH 0.1 ; " ;
```

- The following minimum cut rule indicates that vias with `2` cuts (placed at a distance less than `0.05` µm apart) are required, if the wire has a width greater than `0.09` µm, an area greater than `2.0` µm^2, and *`width`* is less than the default routing width of `0.10` µm.

```
PROPERTY LEF58_MINIMUMCUT
    "MINIMUMCUT 2 WIDTH 0.09 WITHIN 0.05 AREA 2.0 ;" ;
```


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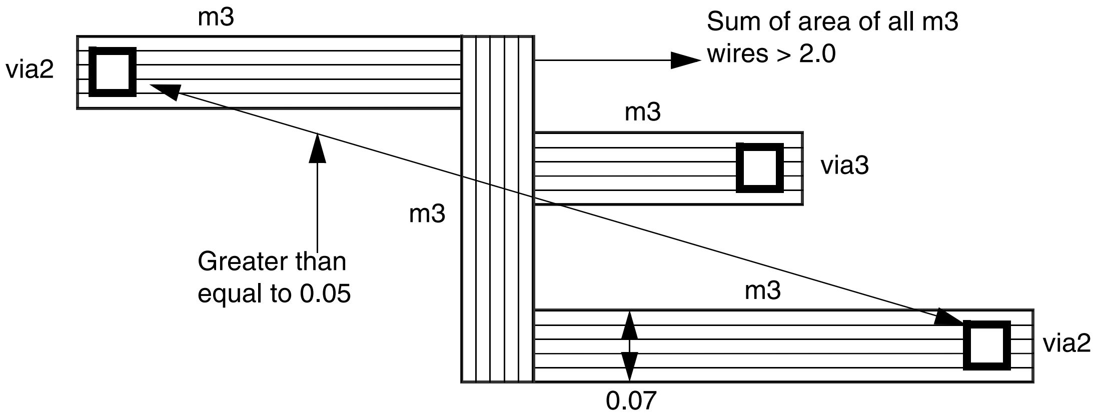
Figure 6-98 Minimum Cut Rule Example



Violation; all three vias should have two cuts. If `WITHIN 0.05` is not specified, it is still a violation as two cuts are needed for all end points by definition.

- The following minimum cut rules indicate that `3` via cuts are required, if a thin wire is connected to a wide wire with a width greater than `0.24` µm and an area greater than `1.6` µm^2, and the distance between the vias and the wide wire is less than `3.0` µm.

```
PROPERTY LEF58_MINIMUMCUT
    "MINIMUMCUT 2 WIDTH 0.24 ;
    "MINIMUMCUT 3 WIDTH 0.40 ;
    "MINIMUMCUT 2 WIDTH 0.24 AREA 0.3 WITHIN 10.0 ;
    "MINIMUMCUT 3 WIDTH 0.24 AREA 1.6 WITHIN 3.0 ;" ;
```


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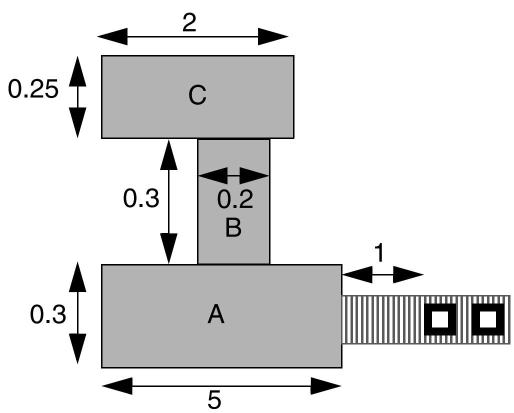
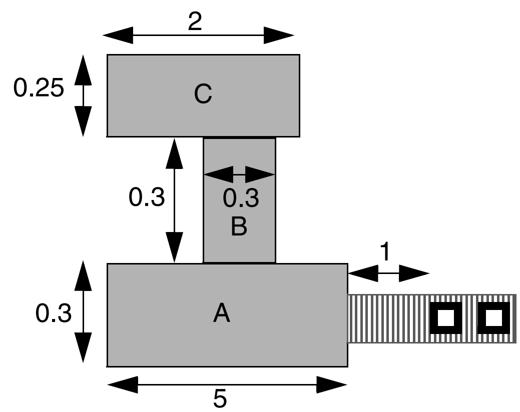
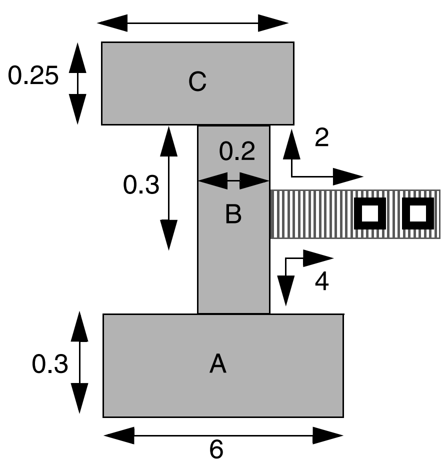
Figure 6-99 Minimum Cut Rule Example



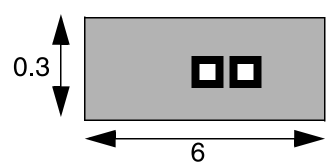
a) Okay; width of B < 0.24, only area of A, 1.5, would require two cuts.



b) Violation; sum of all three areas (A, B, and C) is 2.09, which would require three cuts.



c) Okay; the area of C only needs 2 cuts, and the area of A needs 3 cuts, but the cut is farther than the `WITHIN` distance of 3, and it also fulfills the 2 cuts requirement. Same situation occurs if a 2 cut via is directly connected to B instead of a wire.



d) Okay; `MINIMUMCUT 2 WIDTH 0.24` rule applies. The `WITHIN` rules do not apply when direct via is dropped.

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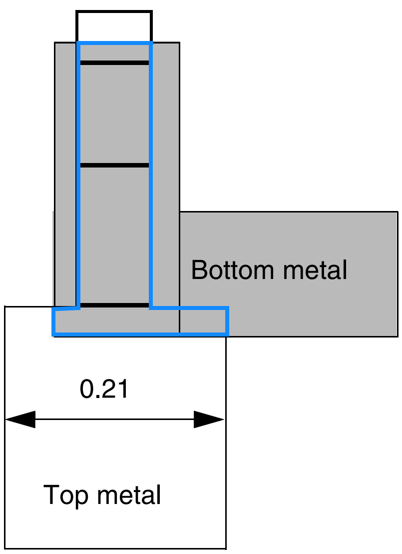
- The following minimum cut rule indicates that if an object has width greater than 0.1 µm, either a VA via with minimum three cuts or a VB via with minimum two cuts are required on that object.

```
PROPERTY LEF58_MINIMUMCUT
    "MINIMUMCUT CUTCLASS VA 3 CUTCLASS VB 2 WIDTH 0.1 ; " ;
```

- The following minimum cut rule indicates that if a common metal, between top and bottom layers, overlaps with a wide wire having a width of 0.2 µm, the rule still applies even if the cut itself does not overlap with the wide wire:

```
PROPERTY LEF58_MINIMUMCUT
    "MINMUMCUT 2 WIDTH 0.2 SAMEMETALOVERLAP ; " ;
```

Figure 6-100 Illustration of Minimum Cut Rule with SAMEMETALOVERLAP



a) Violation, the common metal on both top and bottom metal outlined in blue overlaps with the wide wire although the cut itself does not overlap with it.

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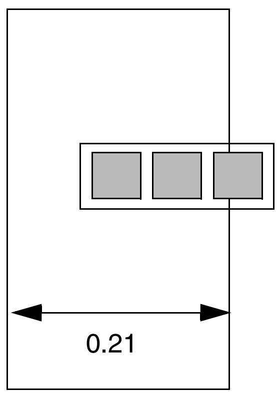
- The following example is an illustration of the minimum cut rule with the `FULLYENCLOSED` keyword:

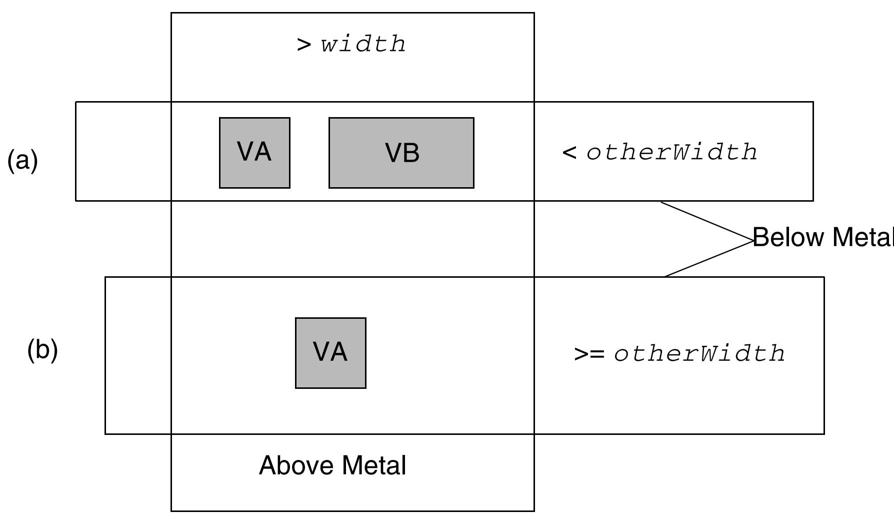
```
PROPERTY LEF58_MINMUMCUT "
MINMUMCUT 3 WIDTH 0.2 FULLYENCLOSED ; " ;
```

Figure 6-101 Illustration of Minimum Cut Rule with FULLYENCLOSED



a) Violation, only two cuts are completely inside the wide wire of 0.21. It is OK if `FULLYENCLOSED` is omitted.

Figure 6-102 Illustration of Minimum Cut Rule with DIFFCUTCLASS and EXCEPTOTHERWIDTH



Case (a) is fine because a cut of a different cut class, VB, is present next to a VA cut. Case (b) is fine because the width on the other layer is >= *otherWidth*. It would be a violation if `EXCEPTOTHERWIDTH` is omitted.

Illustration of `MINIMUMCUT 2 CUTCLASS VA DIFFCUTCLASS`
`WIDTH` *width* `FROMBELOW`
`EXCEPTOTHERWIDTH` *otherWidth*

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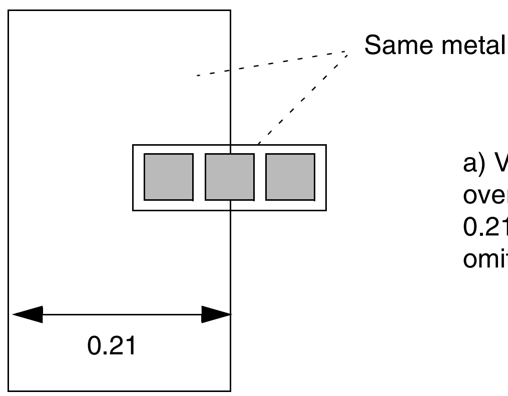
- The following example is an illustration of the minimum cut rule with the `OVERLAPONLY` keyword:

```
PROPERTY LEF58_MINMUMCUT "
     MINMUMCUT 3 WIDTH 0.2 OVERLAPONLY ; " ;
```

Figure 6-103 Illustration of Minimum Cut Rule with OVERLAPONLY



a) Violation, only two cuts overlap with the wide wire of 0.21. It is OK if `OVERLAPONLY` is omitted.

Minimum Length Parallel Rule

You can define a minimum length parallel rule by using the following property definition:

```
PROPERTY LEF58_MINLENGTHPARALLEL
       "MINLENGTHPARALLEL minLength MASK maskNum WIDTH width
          {FROMABOVE|FROMBELOW} CUTCLASS className
          EXCEPTEOLSPACING eolSpacing EXCEPTWITHIN within
          ; " ;
```

Where:

```
MINLENGTHPARALLEL minLength MASK maskNum WIDTH width
     {FROMABOVE|FROMBELOW} CUTCLASS className
     EXCEPTEOLSPACING eolSpacing EXCEPTWITHIN within
```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Specifies that a wire with mask of *maskNum* and width less than *width* containing a cut with *className* either from above or below the cut layer in `FROMABOVE` or `FROMBELOW`, respectively, must have a length greater than or equal to *minLength*. If the wire is completely covered by neighboring wires within *within* on both sides, this minimum length rule is exempted. If *within* is large enough to include multiple wires on one side, the combined projection to the triggering wire should be considered. If the neighboring wire has spacing to another wire less than or equal to *eolSpacing* with `PRL` greater than `0`, the gap would be virtually filled for the neighbor coverage purpose.

Type: Float, specified in microns

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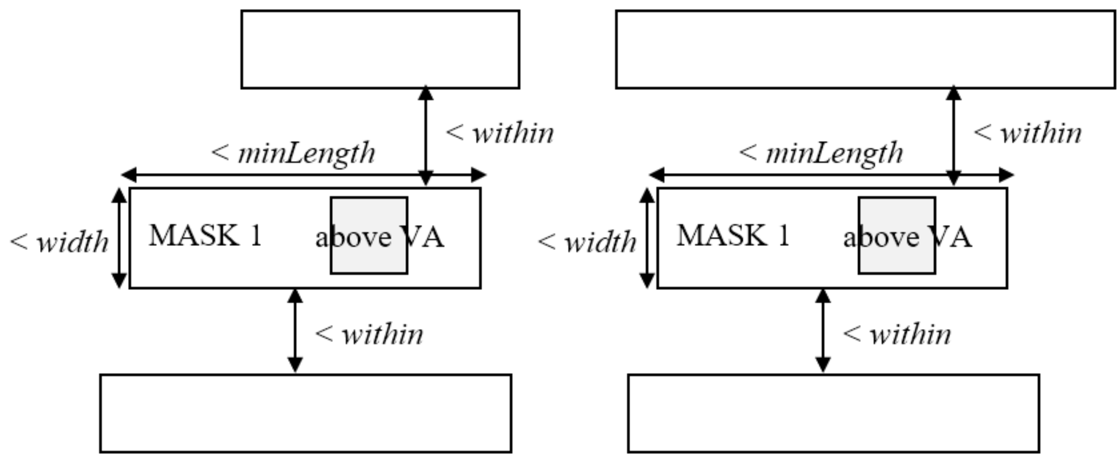
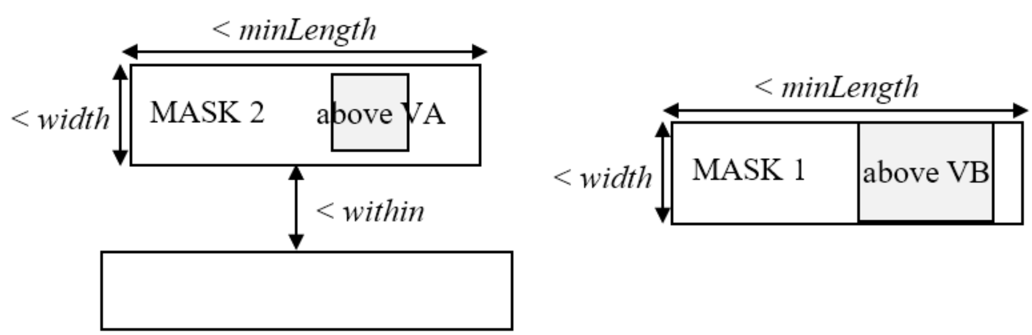
Minimum Length Parallel Rule Examples

Figure 6-104 Illustrations of Minimum Length Parallel Rule



a) Violation, the top neighboring wire does not completely cover the middle wire.

b) OK, there are two neighboring wires within `within` on both sides.



c) OK, the wire is not `Mask 1`.

d) OK, the above cut is not `VA`.

Illustrations of `MINLENGTHPARALLEL` *`minLength`* `MASK 1`
`WIDTH` *`width`* `FROMABOVE CUTCLASS VA`
`EXCEPTEOLSPACING` *`eolSpacing`* `EXCEPTWITHIN` *`within`*

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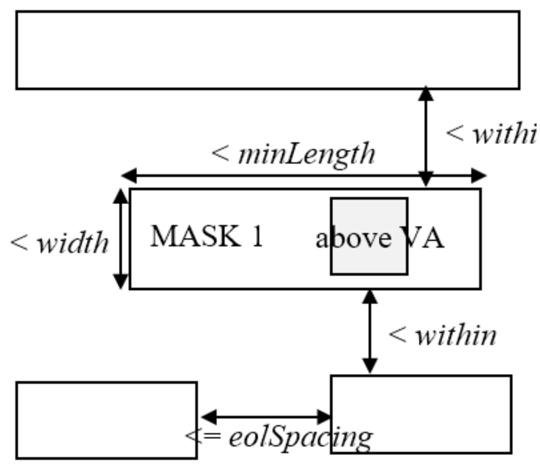
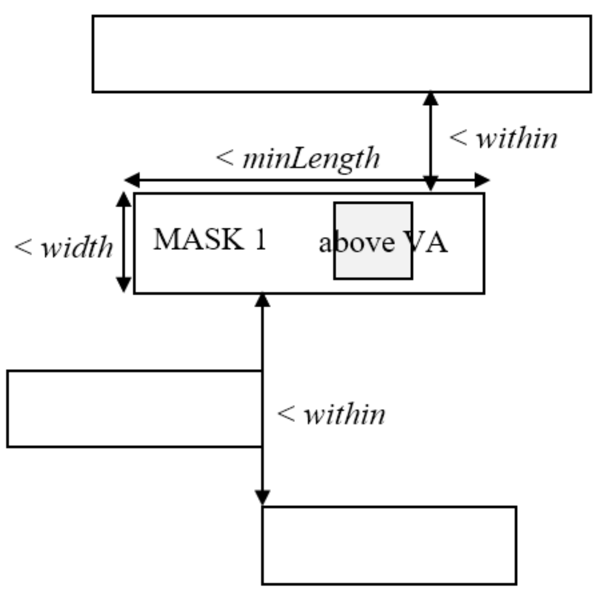
Figure 6-105 Illustrations of Minimum Length Parallel Rule



a) OK, the two bottom wires have spacing <= *`eolSpacing`*, the gap could be considered to be filled to cover the trigger wire on the bottom completely.



b) OK, the two bottom wires are within *`within`*, they are combined to cover the trigger wire on the bottom completely.

Illustrations of `MINLENGTHPARALLEL` *`minLength`* `MASK 1`
`WIDTH` *`width`* `FROMABOVE CUTCLASS VA`
`EXCEPTEOLSPACING` *`eolSpacing`* `EXCEPTWITHIN` *`within`*

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Minimum Length With Cut Rule

A minimum length with cut rule allows you to specify the minimum length of a wire within the specified width range with a cut of the specified class.

You can define a minimum length with cut rule by using the following property definition:

```
PROPERTY LEF_CDN_MINLENGTHWITHCUT
       "MINLENGTHWITHCUT minLength WIDTH minWidth maxWidth
          CUTCLASSLIST {className}... {FROMABOVE | FROMBELOW}
          ; " ;
```

Where:

`MINLENGTHWITHCUT` *`minLength`* `WIDTH` *`minWidth maxWidth`*
`CUTCLASSLIST {`*`className`*`}... {FROMABOVE | FROMBELOW}`

Specifies a wire with width greater than or equal to *`minWidth`* and less than *`maxWidth`* containing a cut belonging to *`className`* on the above or below cut layer when `FROMABOVE` or `FROMBELOW` is specified must have a length greater than or equal to *`minLength`*.

Type: Float, specified in microns

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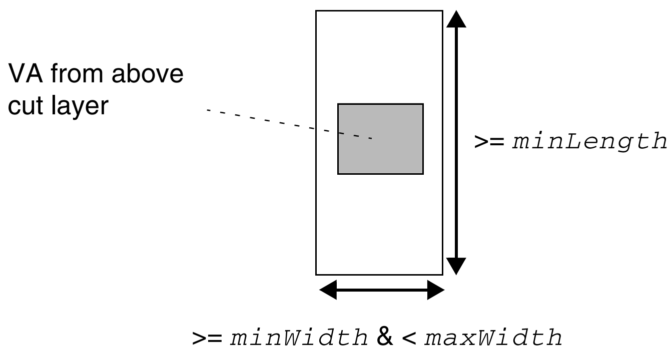
Figure 6-106 Illustration of Minimum Length With Cut Rule



The vertical length must be >= *minLength* since the wire contains a VA cut from the above cut layer and the width of the wire is >= *minWidth* and < *maxWidth*. If the wire does not contain a VA cut or if the width condition is not met, the rule does not apply.

Illustration of `MINLENGTHWITHCUT` *minLength*
`WIDTH` *minWidth maxWidth*
`CUTCLASSLIST VA FROMABOVE`

Minimum Size Rule

Minimum size rules allow you to specify the minimum width and length of a rectangle that must be able to fit somewhere within each polygon on a layer.

You can define a minimum size rule by using the following property definition:

```
PROPERTY LEF58_MINSIZE
     "MINSIZE [RECTONLY] minWidth minLength [minWidth minLength]...
         ] ;..." ;
```

Where:

All the other keywords are the same as the existing LEF routing layer `MINSIZE` syntax.

`RECTONLY`	Specifies that the minimum size rule applies only to rectangular wires.

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Minimum Step Rule

Minimum step rules allow you to require a minimum adjacent edge length following edges that are less than the specified minimum step length. You can define at the most two `MINSTEP` statements.

You can define a minimum step rule by using the following property definition:

```
PROPERTY LEF58_MINSTEP
     "MINSTEP minStepLength
         [[INSIDECORNER | OUTSIDECORNER | STEP] [LENGTHSUM maxLength]
         |MAXEDGES maxEdges
           [HORIZONTAL|VERTICAL]
           [EXCEPTRECTANGLE]
           [MASK maskNum]
           [ MINADJACENTLENGTH minAdjLength
                [CONVEXCORNER [EXCEPTWITHIN exceptWithin]
                |CONCAVECORNER
                |THREECONCAVECORNERS [CENTERWIDTH width]
                |minAdjLength2]
           | MINBETWEENLENGTH minBetweenLength
                [EXCEPTSAMECORNERS]
           | NOADJACENTEOL eolWidth
                [EXCEPTADJACENTLENGTH minAdjLength
                |MINADJACENTLENGTH minAdjLength]
                [CONCAVECORNERS]
           | NOBETWEENEOL eolWidth
           | CONCAVEANDCONVEX
           | ADJACENTLENGTHRANGE minLength maxLength]]
          ] ; " ;
```

Where:

All the other keywords are the same as the existing LEF routing layer `MINSTEP` syntax.

`ADJACENTLENGTHRANGE` *`minLength maxLength`*

Specifies that the rule applies only if the length of the edge adjacent to the min-step edges that are less than *`minStepLength`* is greater than or equal to *`minLength`* and less than *`maxLength`*.

Type: Float, specified in microns

`CONCAVEANDCONVEX` Specifies that the rule applies only if all of the involved edges are between a concave corner and a convex corner.

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`CONCAVECORNER`
: Specifies if a concave corner is between two convex corners, and if one of the length of the edges to form the concave corner is less than *`minAdjLength`*, then the other length must be greater than or equal to the *`minStepLength`*.

  This has similar definition as `CONVEXCORNER` where the corner conditions are reversed.

`CONCAVECORNERS`
: Specifies that the adjacent EOL minimum step rules only apply if both of the neighbor edges of the EOL have concave corners at the other end.

`EXCEPTADJACENTLENGTH` *`minAdjLength`*
: Indicates that the adjacent EOL minimum step rule does not apply if the other neighbor edge of the minstep edge has length greater than or equal to *`minAdjLength`*.
  Type: Float, specified in microns

`EXCEPTRECTANGLE`
: Specifies that the rule does not apply on any rectangular shapes.

`EXCEPTSAMECORNERS`
: Indicates that a *`minBetweenLength`* length is *not* required for an edge that has the same type of 90-degree corner at each end (that is, both corners are convex, or both are concave). See Figure 6-109 on page 823 for an illustration of `EXCEPTSAMECORNERS` in a `MINSTEP` rule.

`EXCEPTWITHIN` *`exceptWithin`*
: Specifies that if there is a neighbor object, same-net or different-net, is within *`exceptWithin`* measured in Euclidean distance from the convex corner or the two adjacent edges, the minstep rule does not apply.
  Type: Float, specified in microns

`HORIZONTAL|VERTICAL`
: Specifies that the rule applies only if the min-step edge is in the given direction. This directional construct must work only with *`maxEdges`* as `1`.

`MASK` *`maskNum`*
: Specifies the mask on which the rule is applied. *`maskNum`* must be a positive integer, and most applications only support values of `1`, `2`, or `3`.

  Type: Integer

`MINADJACENTLENGTH` *`minAdjLength`* `[CONVEXCORNER |` *`minAdjLength2`*`]`

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Indicates that the edges adjacent to min-step edges that are less than *minStepLength* must be greater than or equal to *minAdjLength* in length in order to be allowed; otherwise, it is considered a violation. If 0 is specified in *minAdjLength*, which only works along with CONCAVECORNER or CONVEXCORNER, it has a special meaning that the adjacent edge length condition is ignored and both the edges of the rule must be greater than or equal to *minStepLength*. In addition, the concave or convex corner does not need to be sandwiched between two convex or concave corners.
If *minAdjLength2* is specified, then one adjacent edge must be greater than or equal to *minAdjLength* and the other adjacent edge must be greater than or equal to *minAdjLength2*. See Minimum Step Rule Examples on page 821.
Type: Float, specified in microns

The CONVEXCORNER keyword indicates that if a convex corner is between two concave corners, and if one of the length of the edges to form the convex corner is less than *minAdjLength*, then the other length must be greater than or equal to *minStepLength*.

MINADJACENTLENGTH *minAdjLength*

Specifies that the adjacent EOL minimum step rule applies only if the other neighbor edge forming a concave corner with the minimum step edge has length greater than *minAdjLength*. See Figure 6-113 on page 825.
Type: Float, specified in microns

MINBETWEENLENGTH *minBetweenLength*

Indicates that one of the edges between min-step edges that are less than *minStepLength* must be greater than or equal to *minBetweenLength* in length in order to be allowed; otherwise, it is considered a violation.
Type: Float, specified in microns

NOADJACENTEOL *eolWidth*

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Indicates that the adjacent edges to min-step edges less than *minStepLength* must not be EOL edge with length less than *eolWidth* in order to be allowed; otherwise, it is considered a violation. In addition, `MAXEDGES 1` should be defined along with `NOADJACENTEOL`.
Type: Float, specified in microns

`NOBETWEENEOL` *eolWidth*

Indicates that the edge between two minstep edges less than *minStepLength* must not be a EOL edge with length less than *eolWidth*. In addition, `MAXEDGES 1` should be defined along with `NOBETWEENEOL`.
Type: Float, specified in microns

`THREECONCAVECORNERS [CENTERWIDTH` *width*`]`

Specifies that if a polygon has consecutive edges having three concave corners with two convex corners between them in a stair case, it is a violation if one of the edge touching the middle concave corner has length less than *minStepLength* and the other edge has length less than *minAdjLength*. If `CENTERWIDTH` is specified, it is only a violation if the width of both of the edges of the middle concave corner is less than or equal to *width*. See Figure 6-118 on page 829.
Type: Float, specified in microns

Minimum Step Rule Examples

- The following minimum step rule indicates that a 0.9 µm EOL edge must not be between two minstep edges:

```
PROPERTY LEF58_MINSTEP
  "MINSTEP 0.5 MAXEDGES 1 NOBETWEENEOL 1.0 ;" ;
```


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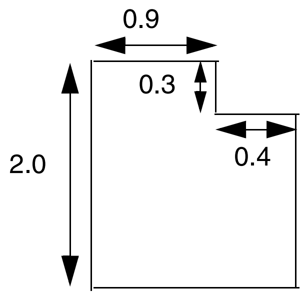
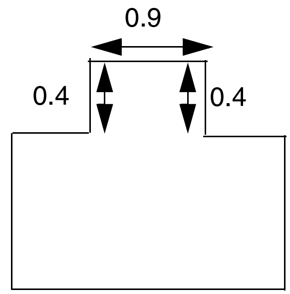
Figure 6-107 MinStep Rule with NOBETWEENEOL



a) OK, there is only one 0.3 minstep next to a 0.9 EOL, and having two consecutive minstep edges without a EOL between them does not trigger the rule

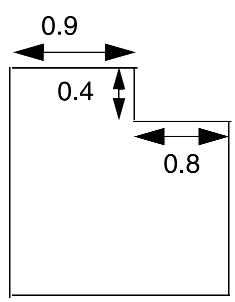


b) Violation, a 0.9 EOL is in between two minstep edges

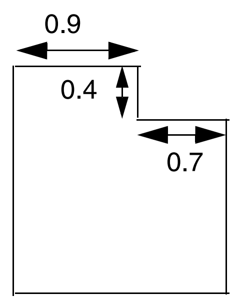
■ The following minimum step rule indicates that the rule applies if the non-EOL adjacent edge has a length that is greater than or equal to 0.8 µm:

```
PROPERTY LEF58_MINSTEP
  "MINSTEP 0.5 MAXEDGES 1
      NOADJACENTEOL 1.0
      EXCEPTADJACENTLENGTH 0.8 ;" ;
```

Figure 6-108 MinStep Rule with EXCEPTADJACENTLENGTH



a) OK, the non-EOL adjacent edge has length of 0.8 (>= 0.8)



b) Violation, both conditions on the neighbor edges fulfill

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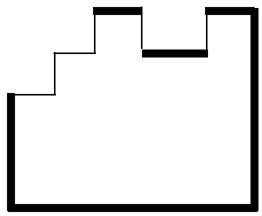
Figure 6-109 Illustration of ExceptSameCorners Definition in the MinStep Rule



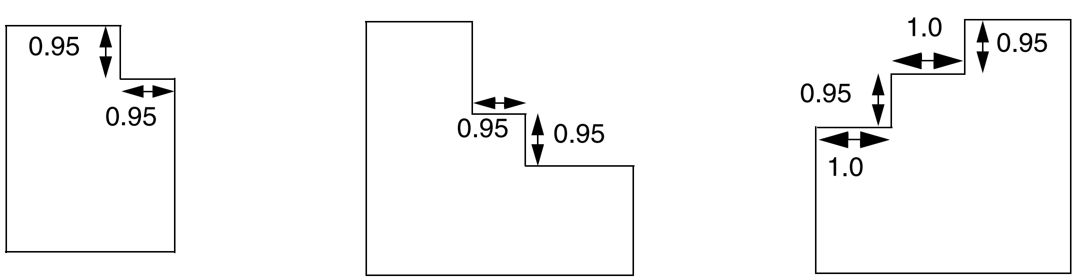
The EXCEPTSAMECORNERS definition of same-corners edges refers to any edge that ends in two convex or two concave 90-degree corners. All of the thick edges in the object above have same-corner edges, and would not be checked by this rule.. “Stair-step” edges are shown in thin lines.

- The following minimum step rule indicates that there can only be one edge less than 1.0 µm in a row. The adjacent edges must be greater than or equal to 1.0 µm.

```
PROPERTY LEF58_MINSTEP
    "MINSTEP 1.0 MAXEDGES 1 ;" ;
```

Figure 6-110 MinStep Rule with MAXEDGES

```
PROPERTY LEF58_MINSTEP “MINSTEP 1.0 MAXEDGES 1 ;” ;
```



a) Violation, more than 1 edges in a row that are < 1.0 in length.

b) Violation, more than 1 edges in a row that are < 1.0 in length.

c) OK, only 1 edge in a row that is < 1.0 in length.

- The following minimum step rule indicates that an edge less than 0.6 µm must have adjacent edges greater than or equal to 1.0 µm.

```
PROPERTY LEF58_MINSTEP
"MINSTEP 0.6 MAXEDGES 1 MINADJACENTLENGTH 1.0 ;" ;
```


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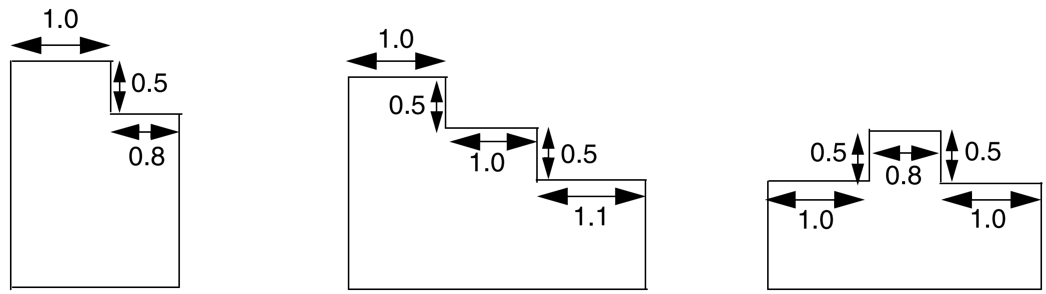
Figure 6-111 MinStep Rule with Adjacent Edge



a) Violation, an edge < 0.6 long has adjacent edges < 1.0.

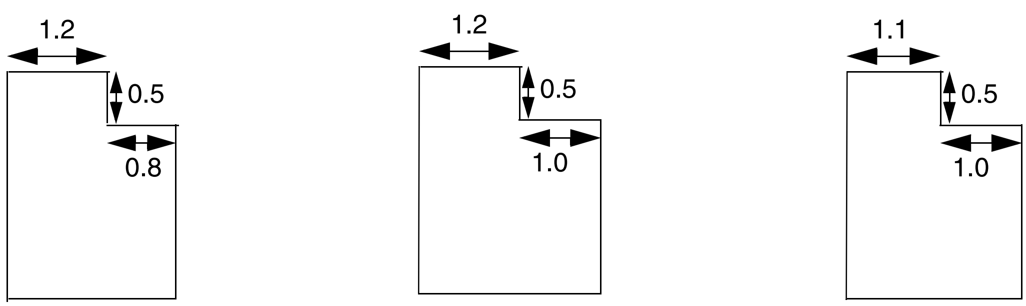
b) Okay, all edges < 0.6 long have >=1.0 adjacent edges.

c) Violation, the edges < 0.5 have an adjacent edge < 1.0.

- The following minimum step rule indicates that no edge less than 0.2 µm, and any edge less than 0.6 µm must have one adjacent edge greater than or equal to 1.0 µm, and the other adjacent edge greater than or equal to 1.2 µm.

```
MINSTEP 0.2 MAXEDGES 0 ;
PROPERTY LEF58_MINSTEP
"MINSTEP 0.6 MAXEDGES 1 MINADJACENTLENGTH 1.0 1.2 ;" ;
```

Figure 6-112 MinStep Rule with MINADJACENTLENGTH



a) Violation: an edge < 0.6 long has adjacent edge < 1.0.

b) OK, an edge < 0.6 long has adjacent edge >= 1.0, and >= 1.2.

c) Violation, an edge < 0.6 long has adjacent edge >= 1.0, but only 1.1 for the other edge (>= 1.2 required).

- The following rule illustrates that the minimum step rule does not apply if the non-EDO adjacent edge has length greater than the specifed minimum adjacent length:

```
PROPERTY LEF58_MINSTEP "
    MINSTEP 1.5 MAXEDGES 1
```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```
NOADJACENTEOL 1.0
MINADJACENTLENGTH 0.7 ; " ;
```

Figure 6-113 MinStep Rule with MINADJACENTLENGTH in NOADJACENTEOL

a) Violation, the non-EOL adjacent edge has length of 0.8 (> 0.7)

b) OK, the vertical min step edge does not form a concave corner with its neighbor

- The following minimum step rule illustrates `CONVEXCORNER`:

```
PROPERTY LEF58_MINSTEP
"MINSTEP 0.6 MAXEDGES 1 MINADJACENTLENGTH 1.0 CONVEXCORNER ;" ;
```


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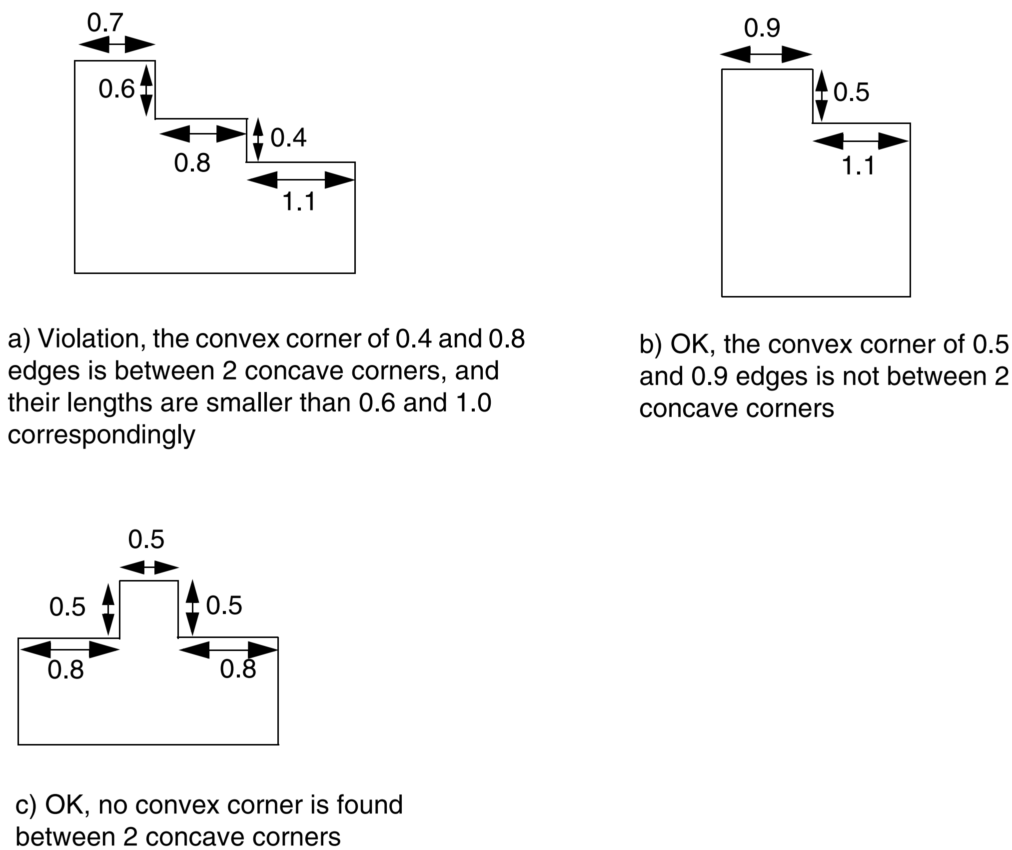
Figure 6-114 MinStep Rule with CONVEXCORNER



- The following figure defines `EXCEPTWITHIN` *`exceptWithin`* in `CONVEXCORNER`:

Figure 6-115 EXCEPTWITHIN exceptWithin in CONVEXCORNER in the MinStep Rule

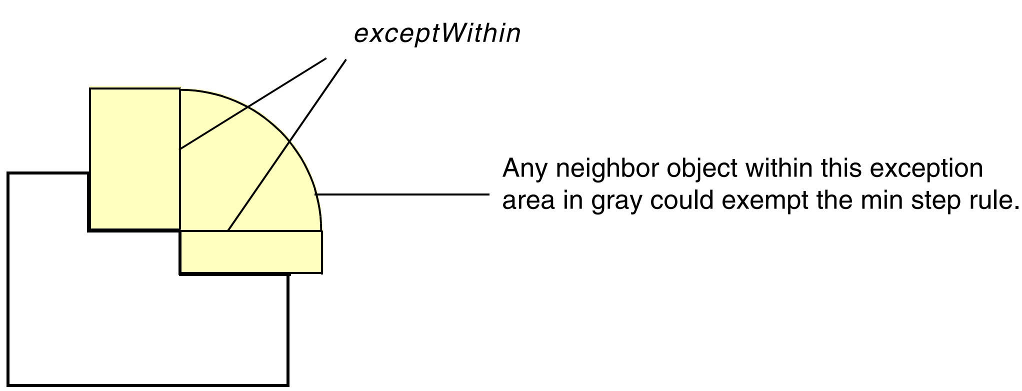


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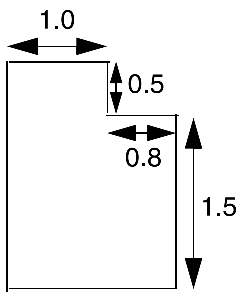
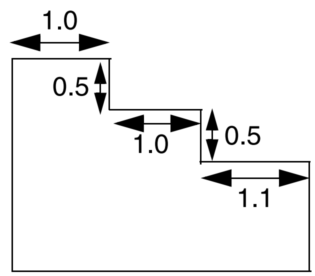
Figure 6-116 MinStep Rule with EXCEPTSAMECORNERS

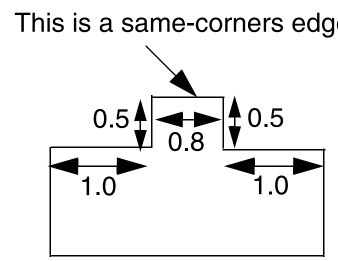
```
PROPERTY LEF58_MINSTEP "MINSTEP 0.6 MAXEDGES 1 MINBETWEENLENGTH 1.0 ;" ;
```



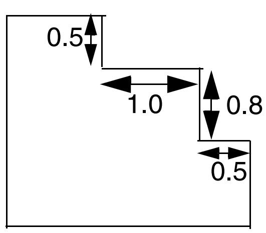
a) Okay, an edge < 0.6 long has a >1.0 edge between it and any other < 0.6 edge.



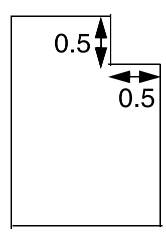
b) Okay, the edges < 0.6 µm long have a >=1.0 µm edge between them.



c) Violation, the edges < 0.6 do not have an edge >= 1.0 between them.
d) Okay if the rule includes `EXCEPTSAMECORNERS`, because then 1.0 length is not required for the same corners edge.



e) Okay, there is an edge >= 1.0 between the 0.5 edges.



f) Violation, an edge < 0.6 long has no edge >= 1.0 before another edge < 0.6.

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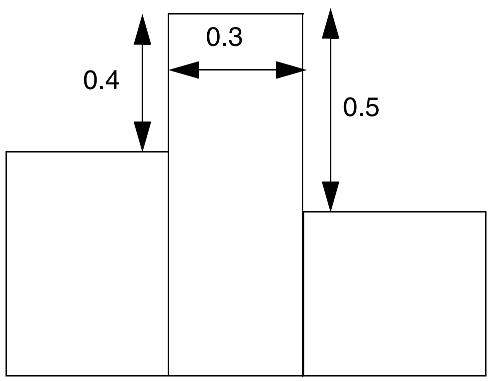
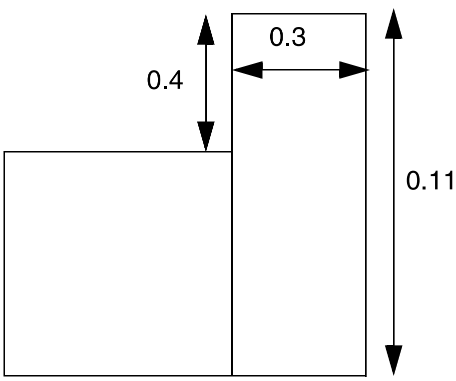
Figure 6-117 MinStep Rule with CONCAVECORNERS

```
PROPERTY LEF58_MINSTEP
    "MINSTEP 0.5 MAXEDGES 1
        NOADJACENTEOL 1.0 CONCAVECORNERS ; " ;
```



a) Violation, the 0.4 min step is next to 0.3 EOL while both 0.4 and 0.5 edges have concave corners on the other end



b) OK, the 0.11 edge does not have a concave corner on the other end. Violation if CONCAVECORNERS is omitted.

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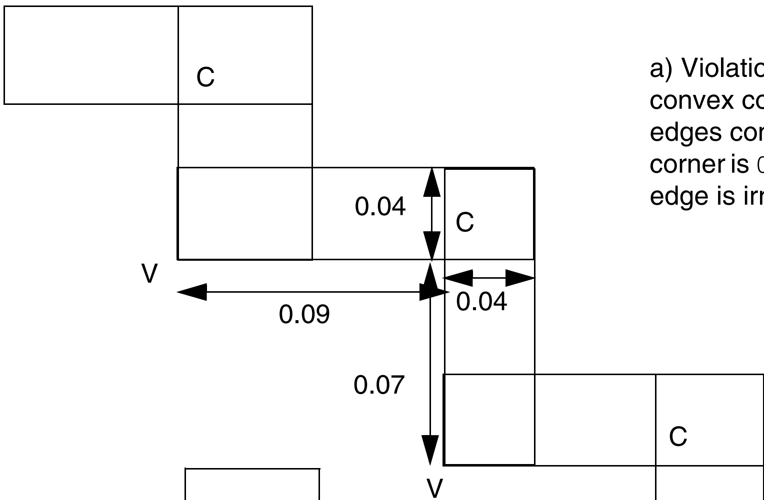
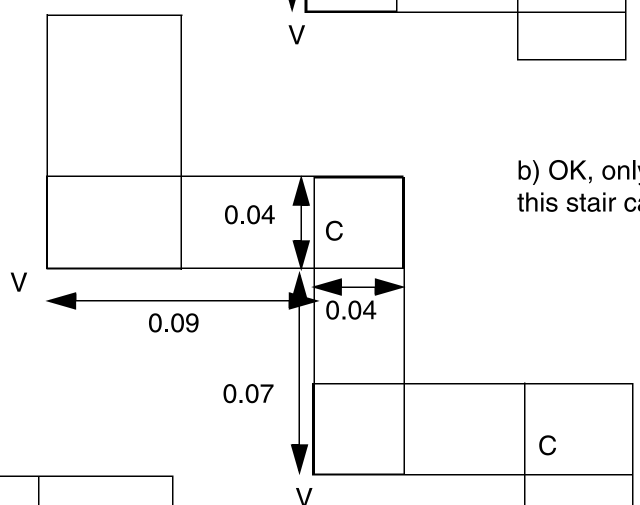
Figure 6-118 MinStep Rule with THREECONCAVECORNERS

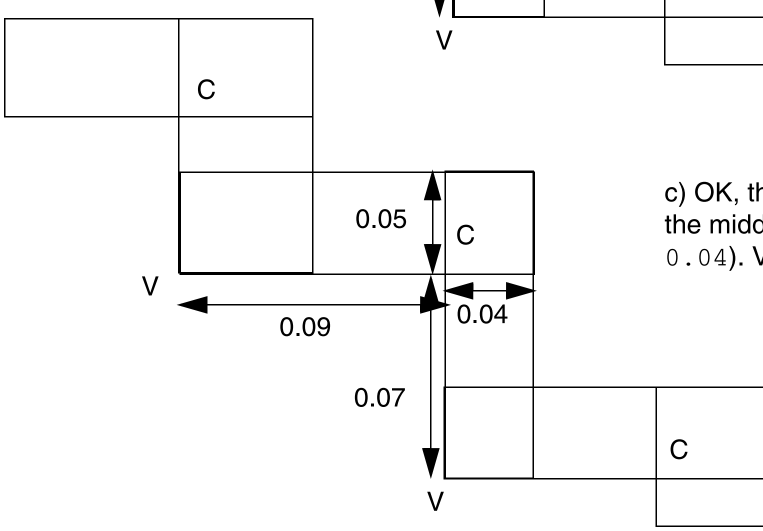
```
PROPERTY LEF58_MINSTEP "
    MINSTEP 0.1 MAXEDGES 1 MINADJACENTLENGTH 0.08
        THREECONCAVECORNERS CENTERWIDTH 0.04; " ;
```



a) Violation, three concave corners (`C`) with two convex corners (`V`) are found, and width of the edges connected to the middle center concave corner is `0.04` (<= `0.04`) and the width of the other edge is irrelevant.



b) OK, only two concave corners are present on this stair case pattern.



c) OK, the width of one of the edges connected to the middle center concave corner is `0.05` (> `0.04`). Violation if `CENTERWIDTH` is omitted.

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Offset Rule

An offset rule allows you to specify the offset for the routing grid from the design origin for the layer.

You can define an offset rule by using the following property definition:

```
PROPERTY LEF58_OFFSET
     "OFFSET [FROMDIEBOUNDARY] {distance | xDistance yDistance}
     ;..." ;
```

Where:

All the other keywords are the same as the existing LEF routing layer `OFFSET` syntax.

`FROMDIEBOUNDARY`	Specifies the offset from the die instead of the core boundary.

Opposite EOL Spacing Rule

An opposite EOL spacing rule can be used to define spacing on a wire with two neighbor wires on the opposite edges.

You can create an opposite EOL spacing rule by using the following property definition:

```
PROPERTY LEF58_OPPOSITEEOLSPACING
     "OPPOSITEEOLSPACING [SAMEMASK] WIDTH width
          ENDWIDTH eolWidth [MINLENGTH minLength]
          [JOINTWIDTH jointWidth] JOINTLENGTH spanLength
              [JOINTTOEDGEEND jointToEdgeEndLength]
              [JOINTEXTENSION jointExtension]
              [JOINTCORNERONLY]
          [SIDELENGTH length[TOSIDE toSideLength]]
          [SIDEEXTENSION sideExtension [TOSIDE toSideExtension]]
          [SIDEEDGELENGTH sideEdgeLength]
          [PRL prl]
          {[EXCEPTEDGELENGTH edgeLength [PRL maxPRL]]}...
          ENDTOEND endSpacing endSpacing [PRL individualPrl]
          ENDTOJOINT endSpacing jointSpacing [PRL individualPrl]
          JOINTTOEND jointSpacing endSpacing [PRL individualPrl]
          JOINTTOJOINT jointSpacing jointSpacing [PRL individualPrl]
          [SIDETOEND sideSpacing endSpacing [PRL individualPrl]
          SIDETOJOINT sideSpacing jointSpacing] [PRL individualPrl]
          [JOINTTOSIDE jointSpacing sideSpacing] [PRL individualPrl]
          [SIDETOSIDE sideSpacing sideSpacing] [PRL individualPrl]
            ;" ;
```


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Where:

`OPPOSITEEOLSPACING`
: Defines the spacing requirements on a wire with two neighbor wires on opposite edges that have a projected parallel run length greater than 0. The neighbor wires are classified either as a EOL or a T or L joint.

`SAMEMASK`
: Specifies that the rule applies only if all three objects to trigger the rule belong to the same mask.

`WIDTH` *`width`*
: Specifies that the rule applies only if the width of the middle wire is less than *`width`*.
Type: Float, specified in microns

`ENDWIDTH` *`eolWidth`* `[MINLENGTH` *`minLength`*`]`

: Specifies that the rule applies only if the neighbor end-of-line has width less than *`eolWidth`*.

: `MINLENGTH` indicates that the edge is only an end when the end-of-line length is greater than or equal to *`minLength`* along both the sides. In other words, if end-of-line length is less than *`minLength`* along any one side, it is not an end, but may be a joint.
Type: Float, specified in microns

`JOINTEXTENSION` *`jointExtension`*

: Specifies the extension on both sides of a joint to be *`jointExtension`*. The extension should be treated as if it is a part of the joint.
Type: Float, specified in microns

`[JOINTWIDTH` *`jointWidth`*`] JOINTLENGTH` *`spanLength`*
`[JOINTTOEDGEEND` *`jointToEdgeEndLength`*`]`

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Specifies that the neighbor wire end edge is a joint if its width is less than *jointWidth*, if specified, or less than *eolWidth*, its span is greater than *spanLength*, and it is not a EOL edge.

If JOINTTOEDGEEND is specified, then at least one of the distances from the end points of the joint to the ends of the edge, that contain the joint, must be less than or equal to *jointToEdgeEndLength*. A T or L configuration (see Figure 6-119 on page 835) is a typical joint. However, joints are not restricted to such configurations only. A joint can be any edge that fulfills the above definition.
Type: Float, specified in microns

JOINTLENGTH *spanLength*

Specifies that the neighbor wire end edge is a joint if it has width less than *eolWidth* and span greater than *spanLength*, and it is not an EOL, that is, it is either a T or L joint pattern.
Type: Float, specified in microns

{EXCEPTEDGELENGTH *edgeLength* [PRL *maxPRL*]}...

Specifies that the rule does not apply if both the end or joint neighbor edges have a length greater than and equal to *edgeLength* and projected parallel run length is less than and equal to *maxPRL*, if PRL is also specified. At the most, two such statements can be specified and supported.
Type: Float, specified in microns

ENDTOEND *endSpacing endSpacing*

ENDTOJOINT *endSpacing jointSpacing*

JOINTTOEND *jointSpacing endSpacing*

JOINTTOJOINT *jointSpacing jointSpacing*

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Specifies the spacing between the neighbor edges to the middle wire. There are four groups of two spacings. The keywords define the category of the neighbors, either as an end or a joint. For example, in the case of `ENDTOJOINT`, the first spacing, *`endSpacing`*, specifies the minimum spacing between the end neighbor edge to the middle wire, and the second spacing, *`jointSpacing`*, specifies the minimum spacing between the joint neighbor edge to the middle wire. To satisfy the rule, for both end/joint neighbors, either both the neighbor spacings must be greater than and equal to the minimum of the specified spacings, or at least one neighbor spacing must be greater than and equal to the maximum of the specified spacings in `ENDTOEND` or `JOINTTOJOINT`. For end and joint neighbors, both `ENDTOJOINT` and `JOINTOTEND` statements must be fulfilled individually. To fulfill one statement either joint spacing greater than or equal to *`jointSpacing`*, or end spacing greater than or equal to *`endSpacing`*, must be true.
Type: Float, specified in microns (for all values)

`JOINTCORNERONLY`
Specifies that the joint must form a joint corner, which is a convex corner consisting of two consecutive joints.

`JOINTTOSIDE` *`jointSpacing sideSpacing`*

Specifies a spacing requirement similar to `SIDETOJOINT`, but having joint spacing to be the first spacing.
Type: Float, specified in microns

`PRL` *`prl`*
Specifies that the rule only applies if the projected parallel run length of a wire and the two neighbor wires is greater than *`prl`*. If prl is negative, it is similar to extending the end, joint, or side edges by abs(prl) in a `WITHIN` style.
Type: Float, specified in microns

`PRL` *`individualPrl`*
Specifies that the parallel run length on the given `END`, `JOINT`, and `SIDE` combination must be greater than the specified *`individualPrl`* value to trigger the rule. If it is not specified on certain combinations, those combinations would be checked against the global *`prl`*.
Type: Float, specified in microns

`SIDEEDGELENGTH` *`sideEdgeLength`*

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Specifies that a side fulfilling other conditions must also be an edge with length greater than the *sideEdgeLength*.
Type: Float, specified in microns

SIDEEXTENSION *sideExtension* [TOSIDE *toSideExtension*]

Specifies the extension of a side of EOL edge to be *toSideExtension* only in the direction going beyond the EOL edge in case of *SIDETOSIDE*, and this extension definition is only applied to one of the SIDE only while the other SIDE would equivalent to having zero extension. The extension should be treated as if it is part of the side. The *sideExtension* is still used for SIDE to END or JOINT cases.
Type: Float, specified in microns

SIDELENGTH *length* [TOSIDE *toSideLength*]

Specifies the length on the sides of a EOL edge to be *toSideLength* in case of SIDETOSIDE, and this length definition is only applied to one of the SIDE only while the other SIDE will be equivalent to having zero length. The *length* is still used for SIDE to END or JOINT cases.
Type: Float, specified in microns

SIDETOEND *sideSpacing endSpacing*
SIDETOJOINT *sideSpacing jointSpacing*

Specifies the spacing between a side of a EOL edge defined in SIDELENGTH, which must be specified along with these constructs, to a wire which has a neighbor end or joint on the opposite side with a certain spacing. The spacings have similar definitions of *jointSpacing* and *endSpacing*.
Type: Float, specified in microns

SIDETOSIDE *sideSpacing sideSpacing*

Specifies a spacing requirement between two sides of a EOL edge with a middle wire. In addition, any defined PRL value is ignored when considering SIDETOSIDE spacing.
Type: Float, specified in microns

Opposite EOL Spacing Examples

- Figure 6-119 on page 835 illustrates END and JOINT with ENDWIDTH 0.1, MINLENGTH 0.05, JOINTLENGTH 0.15, and JOINTTOEDGEEND 0.08:

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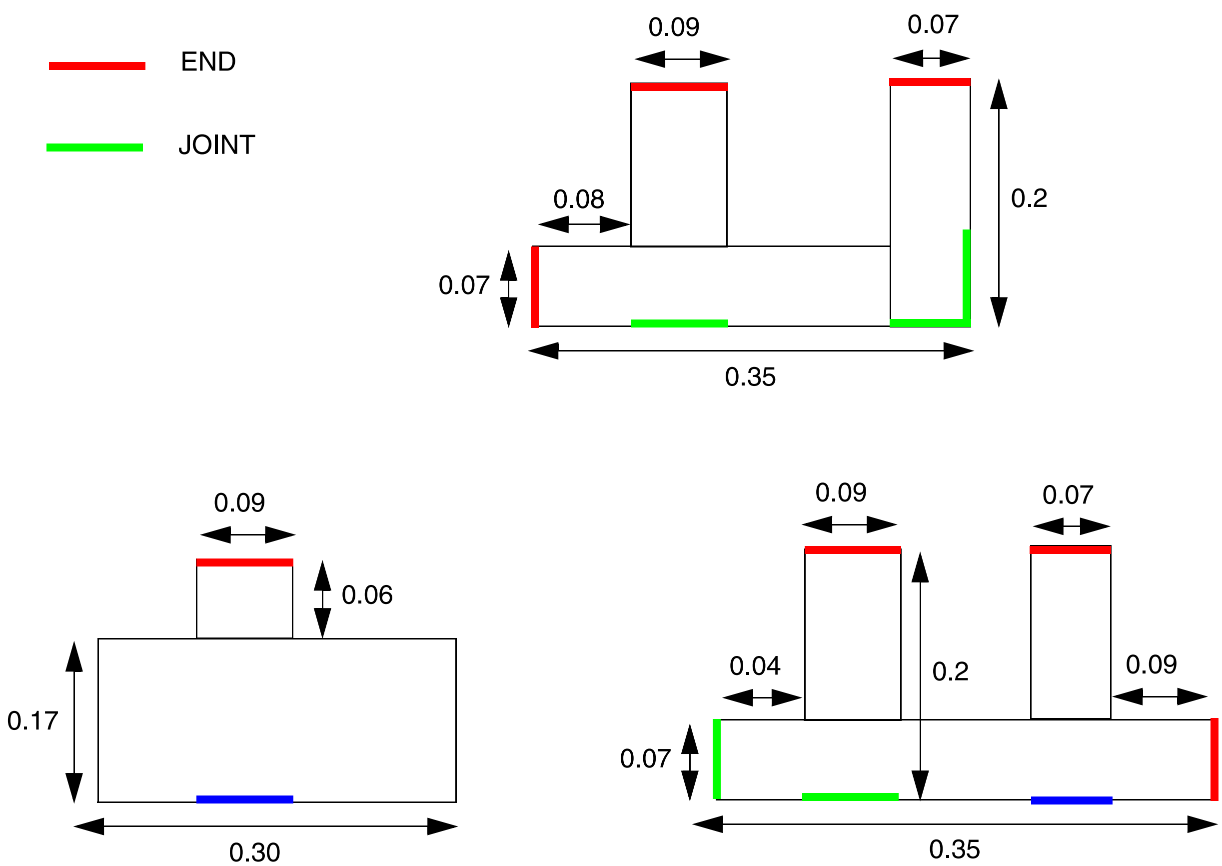
Figure 6-119 Illustration of END and JOINT



Since the whole bottom edge has span length > 0.15, and its length/width of 0.3 >= 0.1, it is not a joint. The blue segment is also not a joint since it is merely part of the bottom edge with span length > 0.15.

The blue segment is not a joint since both lengths to the end of the edge > 0.08. The vertical left segment is not an end since one of the end-of-line length is 0.04 < 0.05, but it is a joint.

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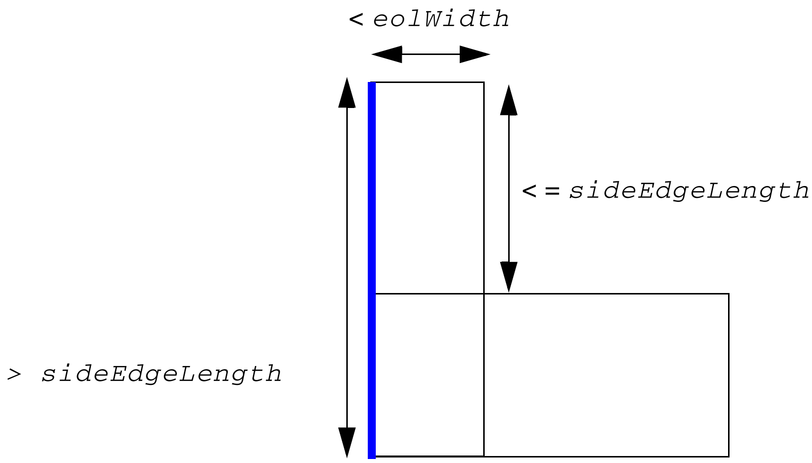
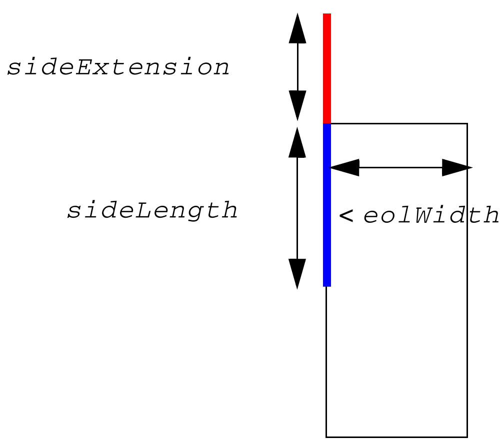
Figure 6-120 Illustration of SIDEEDGELENGTH



Only the blue edge satisfies the SIDEEDGELENGTH condition to be a SIDE.

Figure 6-121 Illustration of SIDELENGTH and SIDEEXTENSION



Blue edge represents the SIDELENGTH while red edge represents the SIDEEXTENSION.

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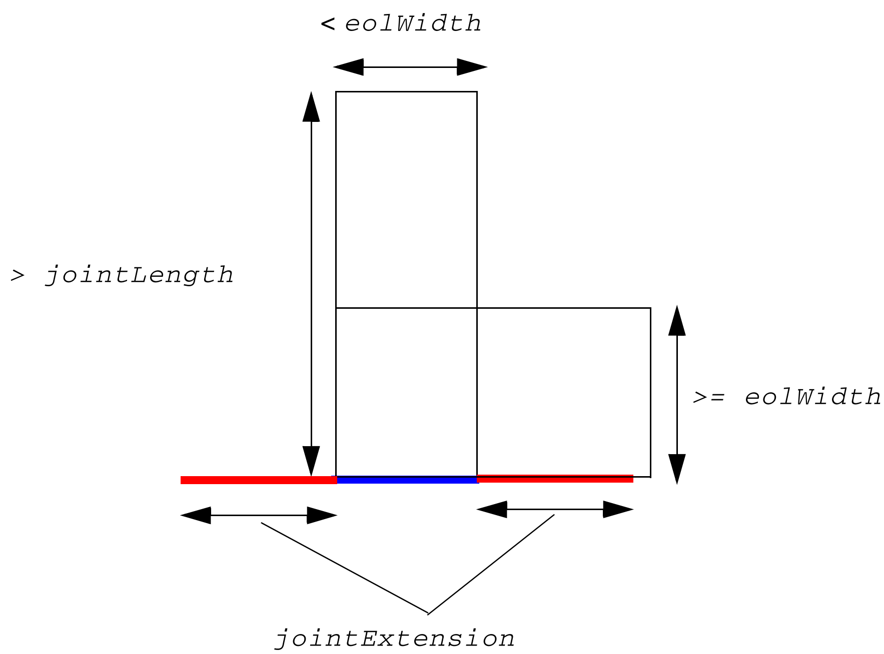
Figure 6-122 Illustration of JOINTEXTENSION



Blue edge is a JOINT while the red edges represent the JOINTEXTENSION.

- The following example illustrates the Opposite EOL Spacing rule given below:

```
PROPERTY LEF58_OPPOSITEEOLSPACING
    WIDTH 0.08
    ENDWIDTH 0.1
    JOINTLENGTH 0.15
    EXCEPTEDGELENGTH 0.08 PRL 0.03
    ENDTOEND 0.11 0.13
    ENDTOJOINT 0.12 0.10
    JOINTTOEND 0.12 0.10
    JOINTTOJOINT 0.12 0.16;
```


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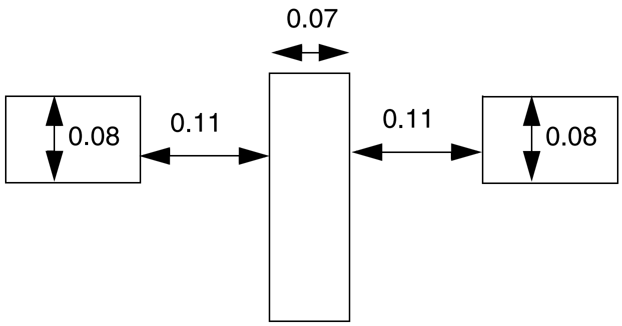
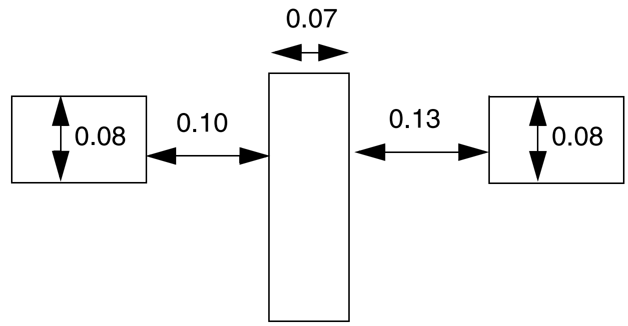
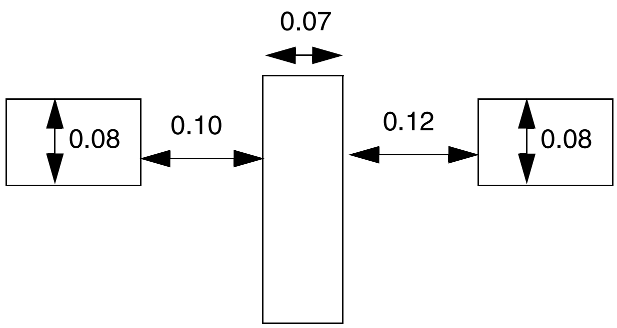
Figure 6-123 Illustration of OPPOSITEEOLSPACING



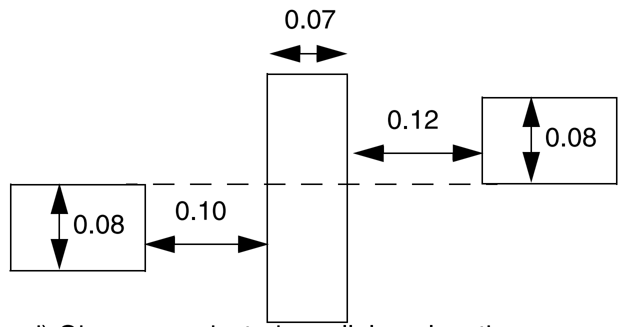
a) Okay, both of the neighbor spacings >= 0.11 (minimum of the spacings)



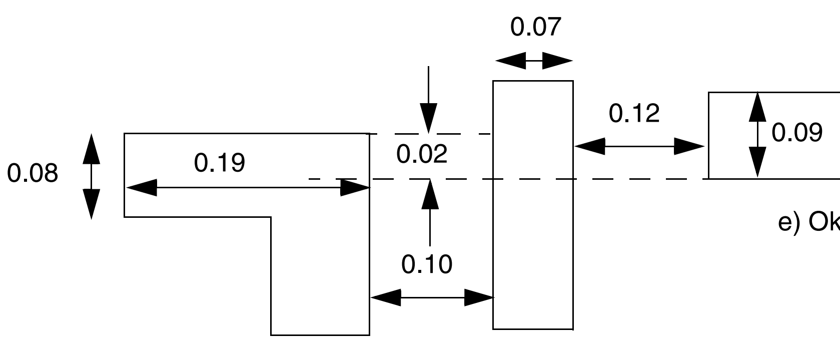
b) Okay, right spacing >= 0.13 (maximum of the spacings)



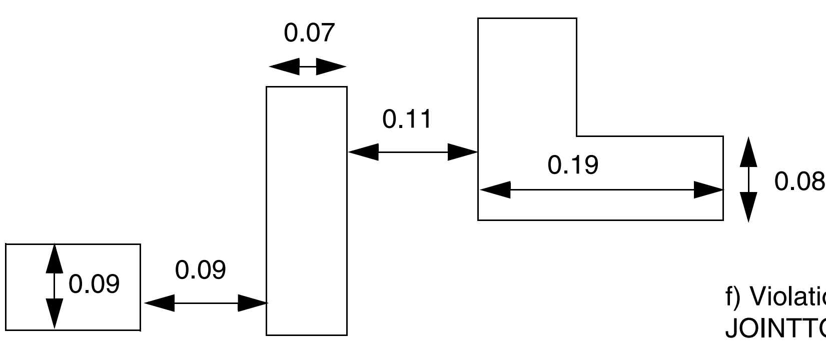
c) Violation, left spacing < 0.11 to trigger the rule



d) Okay, no projected parallel run length



e) Okay, the exception conditions are met



f) Violation, ENDTOJOINT is fine, but JOINTTOEND is not

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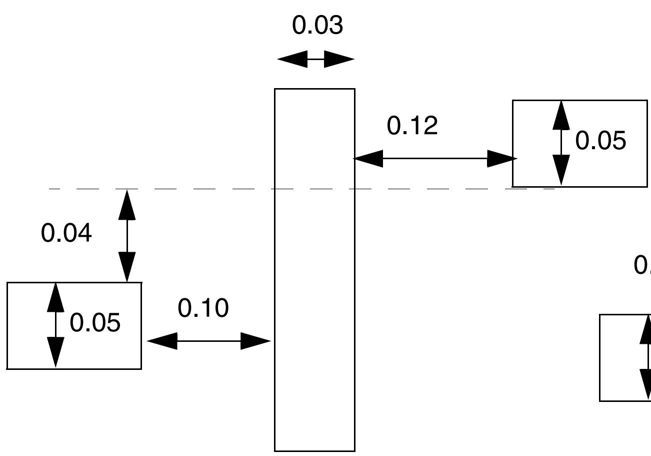
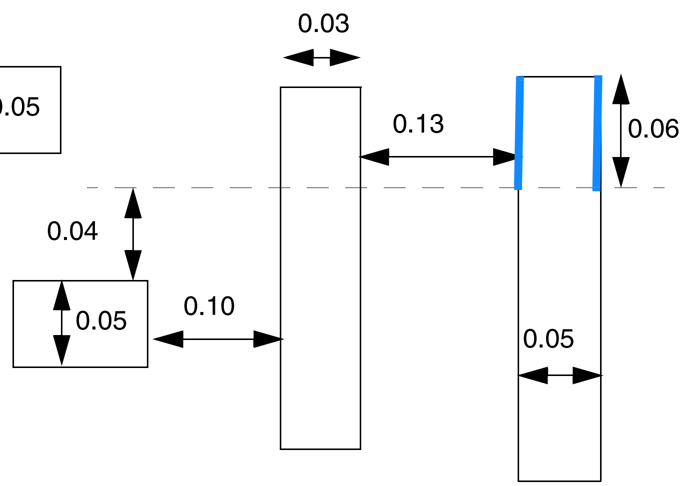
Figure 6-124 Illustration of OPPOSITEEOLSPACING

```
PROPERTY LEF58_OPPOSITEEOLSPACING
   "OPPOSITEEOLSPACING
      WIDTH 0.08
      ENDWIDTH 0.1
      JOINTLENGTH 0.15
      SIDELENGTH 0.06
      PRL -0.05
      ENDOTEND 0.11 0.13
      ENDTOJOINT 0.12 0.10
      JOINTTOEND 0.12 0.10
      JOINTTOJOINT 0.12 0.16
      SIDETOEND 0.14 0.15
      SIDETOJOINT 0.14 0.15 ;";
```



a) Violation, the neighbor wires are within 0.05 (PRL -0.05) of each other



b) Violation, the blue edges are SIDE of a EOL edge, with 0.13 (< 0.14) spacing to the middle wire, and the opposite end wire has 0.10 (< 0.15) spacing

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Figure 6-125 Illustration of SIDELENGTH in OPPOSITEEOLSPACING

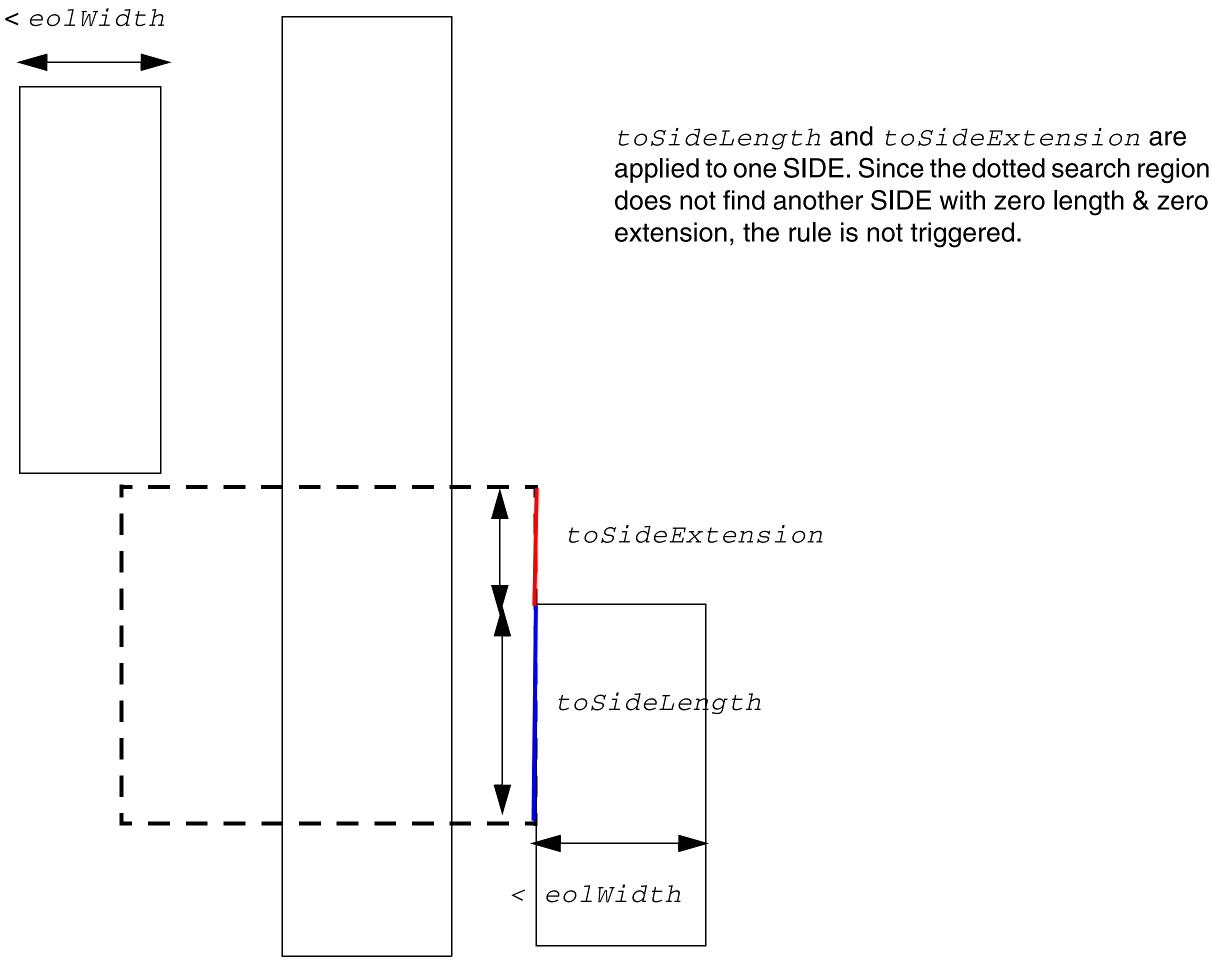

Illustration of `SIDELENGTH` *`sideLength`* `TOSIDE` *`toSideLength`* `SIDEEXTENSION` *`sideExtension`* `TOSIDE` *`tosideExtension`*

Pin Connect Blockage Rule

You can create a pin connect blockage rule to define a blockage area when a wire is connected to pins on a redistribution (RDL) layer.

You can create a pin connect blockage rule by using the following property definition:

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```
PROPERTY LEF58_PINCONNECTBLOCKAGE
     "PINCONNECTBLOCKAGE size ENCLOSEDLENGTH enclosedLength
     ; " ;
```

Where:

`PINCONNECTBLOCKAGE` *`size`* `ENCLOSEDLENGTH` *`enclosedLength`*

Specifies a blockage area when a wire is connected to any pin on the RDL layer. The blockage is formed by expanding along the given *`size`* value on the connected corners. Other corners or through wires, such as 45-degree wires, cannot overlap with the blockage. However, another corner connection formed between another wire connected to the pin is allowed. Any enclosed empty area (that is, a dough-nut hole formed by the metal) would have a length along the pin edge that is greater than or equal to *`enclosedLength`* if it overlaps with the blockage area.
Type: Float, specified in microns

Pin Connect Blockage Rule Example

- The following example illustrates a pin connect blockage rule:

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Figure 6-126 Example of a Pin Connect Blockage Rule

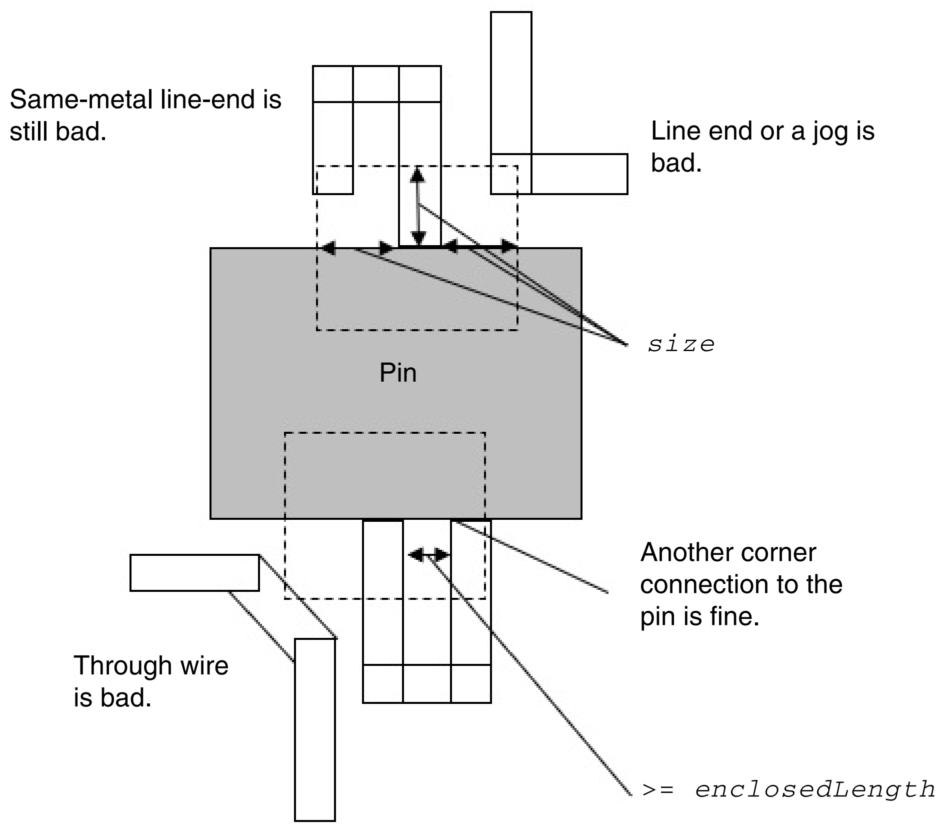

Illustration of: `PINCONNECTBLOCKAGE` *`size`*
`ENCLOSEDLENGTH` *`enclsoedLength`*

Pitch Rule

You can create a pitch rule to define non-uniform pitch or track on layers, whose preferred direction must the same as the direction of the standard cell rows.

You can create a pitch rule by using the following property definition:

```
PROPERTY LEF58_PITCH
     "PITCH {distance | xDistance yDistance } [FIRSTLASTPITCH firstLastPitch]
     ; " ;
```

All other keywords are the same as the existing LEF routing layer `PITCH` syntax.

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Where:

`FIRSTLASTPITCH` *`firstLastPitch`*

Specifies a non-uniform pitch/track on this layer, whose preferred direction must be in the same direction as that of the standard cell rows (normally horizontal). The routing tracks are repeated for every standard-cell row (see example below). The *`firstLastPitch`* defines the distance of the first and last tracks from the standard-cell row boundary and pitch value on this layer defines the distance of the intermediate tracks. In other words, subtracting the cell row height by 2 * *`firstLastPitch`* should be divisible by pitch value on this layer. The pitch value of the property should be identical with the pitch value in the native/regular `PITCH` statement. If not, the pitch value of the property will be honored.
Type: Floats, specified in microns

Pitch Rule Examples

- The following example shows routing tracks:

```
PROPERTY LEF58_PITCH STRING "
    PITCH 0.1 FIRSTLASTPITCH 0.14 ; ";
```


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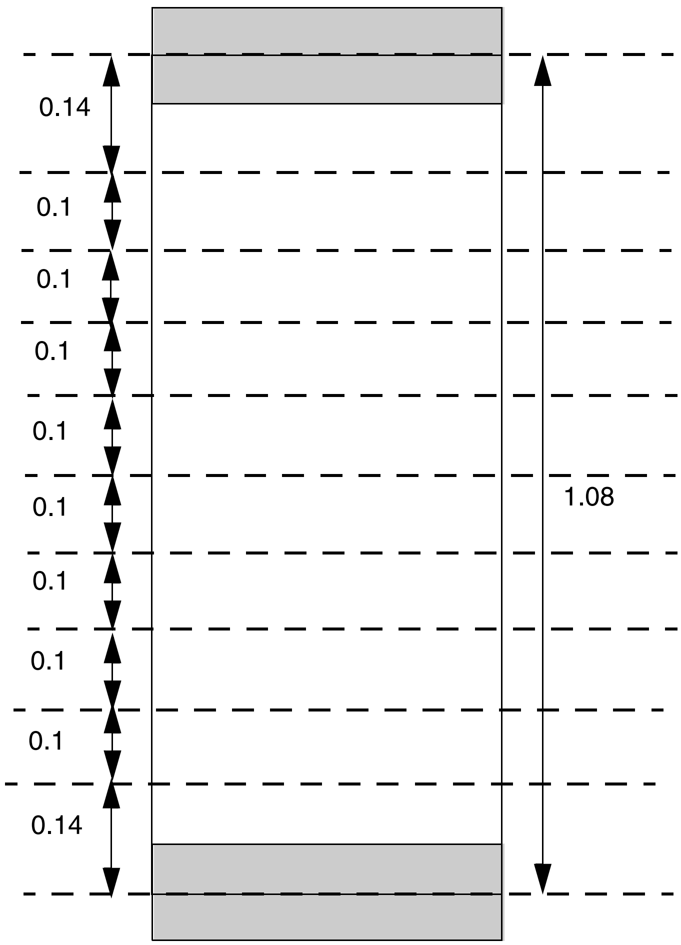
Figure 6-127 Example of Pitch Rule



In this example, the cell height is 1.08. The dotted line represents the routing tracks. The first and last tracks are 0.14 from the cell row boundary while the intermediate tracks are 0.1 apart making 2 * 0.14 + 8 * 0.1 = 1.08. The routing track definition would be repeated based on cell rows.

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Protrusion Width Rule

You can use the protrusion width rule for a protrusion wire, connected to a wire, that has width greater than or equal to the specified width, or whose length is greater than or equal to the specified length.

You can define a protrusion width rule by using the following property definition:

```
PROPERTY LEF58_PROTRUSIONWIDTH
     "PROTRUSIONWIDTH width1
          {LENGTH length WIDTH width2
          |{WIDTH width2
              {MINSIZE {minWidth minLength | minLength
                  CUTCLASS className {FROMABOVE | FROMBELOW}}
              | MINLENGTH minLength
                  [EXCEPTCUT cutDistance[FROMABOVE | FROMBELOW]]}...}
          ; " ;
```

Where:

All other keywords are same as the existing LEF routing layer `PROTRUSIONWIDTH` syntax.

`EXCEPTCUT` *`cutDistance`* `[FROMABOVE | FROMBELOW]`

Specifies that if there is a cut of any cut classes in above, below, or either cut layer and if `FROMABOVE`, `FROMBELOW` or none is specified less than or equal to the *`cutDistance`* from wire with width greater than or equal to the *`width2`*, then the rule does not apply.
Type: Float, specified in microns

`MINSIZE` *`minWidth minLength`*

Specifies that it is a violation if the area of protruded wire with width greater than or equal to *`width1`* outside of the wide wire of width greater than or equal to *`width2`* does not enclose a size of *`minWidth minLength`*. The rule does not apply if there is no such area. See Figure 6-130 on page 848.
Type: Float, specified in microns

`MINSIZE` *`minLength`* `CUTCLASS` *`className`* `{FROMABOVE | FROMBELOW}`

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Specifies that it is a violation if the area outside the wide wire of width greater than or equal to *width2* does not enclose a size of *width1 minLength*, and the protruded area contains one and only one cut with cut class of *className*, either on the above or below cut layer determined by ABOVE/BELOW. It is not a violation if multiple cuts share common metal on both the above and below layers, irrespective of whether they are in the protruded or wide wire. See Figure 6-131 on page 849.

Type: Float, specified in microns

{WIDTH *width2* MINLENGTH *minLength*}...

Specifies that when a protrusion wire is connected to a wire with width greater than or equal to *width2*, then it must have length greater than or equal to the *minLength* and width greater than or equal to *width1*.
Type: Float, specified in microns

Protrusion Width Rule Examples

- The following protrusion width rule indicates that the width of the protrusion wire should be greater than or equal to the specified width of 0.05, and its length should be greater than or equal to the specified minimum lengths of 0.12 and 0.14 when it is connected to wide wire with width of 0.11 and 0.15.

```
PROPERTY LEF58_PROTRUSIONWIDTH
  "PROTRUSIONWIDTH 0.05
    WIDTH 0.11 MINLENGTH 0.12
    WIDTH 0.15 MINLENGTH 0.14 ; " ;
```


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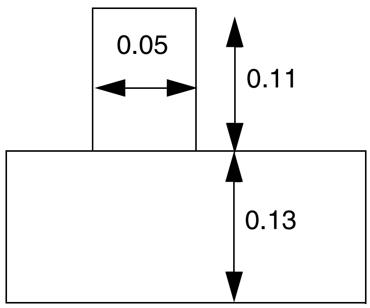
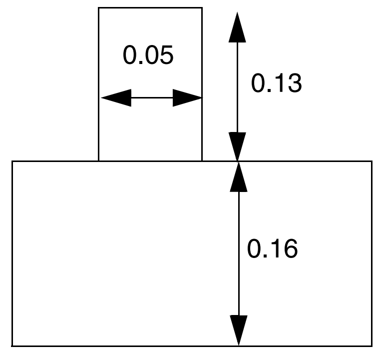
Figure 6-128 Illustration of Protrusion Width Rule



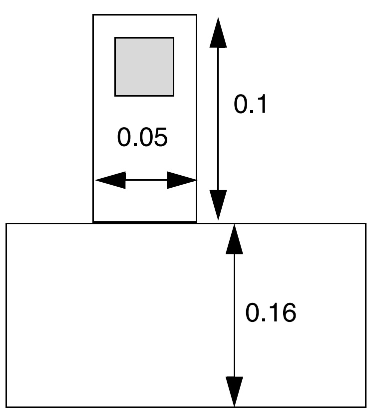
a) Violation, the width of the protrusion wire is fine, but it is too short.



b) Violation, the length requirement varies by the width of the wide wire

Figure 6-129 Illustration of Protrusion Width Rule with EXCEPTCUT

```
PROPERTY LEF58_PROTRUSIONWIDTH "
        PROTRUSIONWIDTH 0.05
              WIDTH 0.11 MINLENGTH 0.12 EXCEPTCUT 0.1 ; " ;
```



a) OK, cut exemption is kicked in.

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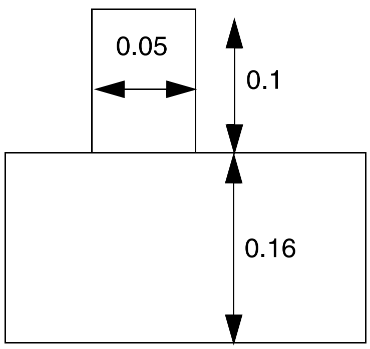
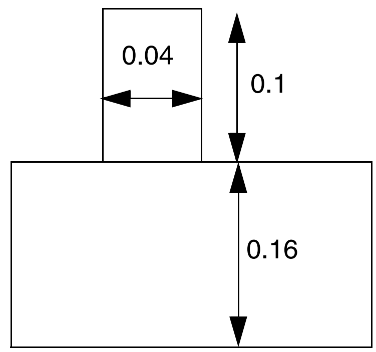
Figure 6-130 Illustration of Protrusion Width Rule with MINSIZE

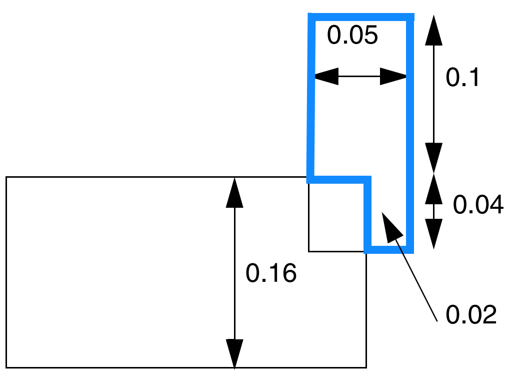
```
PROPERTY LEF58_PROTRUSIONWIDTH
        "PROTRUSIONWIDTH 0.05
             WIDTH 0.11 MINSIZE 0.02 0.12 ; " ;
```



a) Violation, the protruded length is < 0.12



b) OK, there is no protruded area with width >= 0.05.



c) OK, the blue area is the protruded area with width >= 0.05, and it fits size of 0.02 0.12

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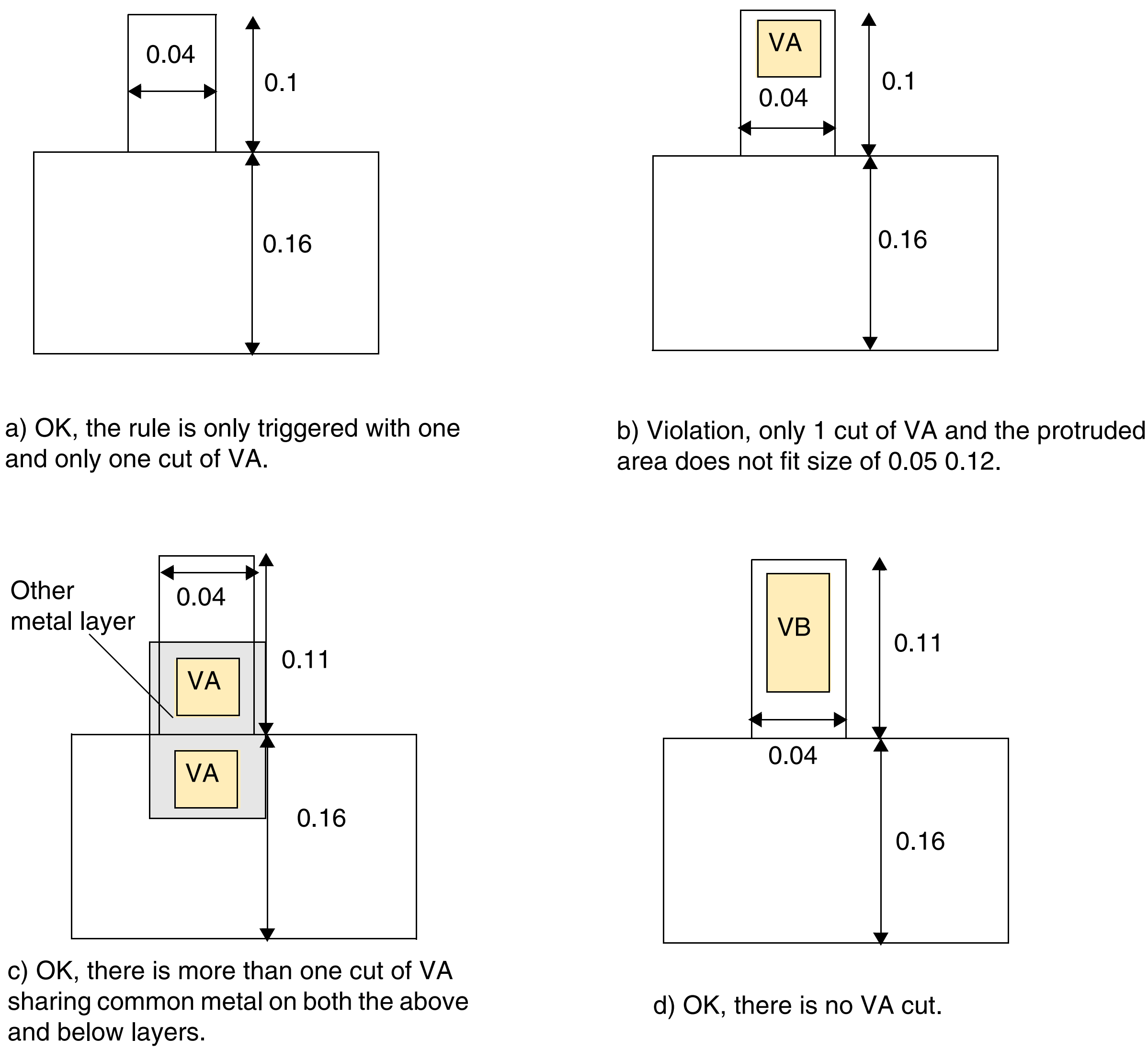
Figure 6-131 Illustration of Protrusion Width Rule with WIDTH and MINSIZE

```
PROPERTY LEF58_PROTRUSIONWIDTH
        "PROTRUSIONWIDTH 0.05
             WIDTH 0.11 MINSIZE  0.12
             CUTCLASS VA FROMABOVE  ; " ;
```



■ The following protrusion width rule indicates that the rule will fail when the given value of `LENGTH` is 0, and if the protruded wire touches the wide wire, and the aligned portion has a width less than the specified first width:

```
PROPERTY LEF58_PROTRUSIONWIDTH
     "PROTRUSIONWIDTH 0.05 LENGTH 0 WIDTH 0.11 ; " ;
```


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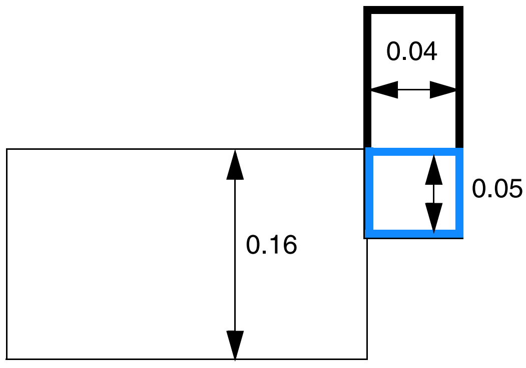
Figure 6-132 Illustration of Protrusion Width Rule with LENGTH and WIDTH



a) OK, the blue touching portion has width >= 0.05.

b) Violation, this is not a touching case.

Rectangle Only Rule

Rectangle only rules can be used to indicate that a layer allows rectangular objects only.

You can create a rectangle only rule by using the following property definition:

```
PROPERTY LEF58_RECTONLY
     "RECTONLY
           [EXCEPTNONCOREPINS
              [MACROCLASS {RING | BLOCK | PAD | BUMP}...]]
           [EXCEPTRIGHTWAYWIDTH maxWidth
           | RIGHTWAYWIDTHONLY exactWidth]
          [EXCEPTVIAEDGEALIGNED | EXCEPTVIA]
          [PARTIALTRACKS]
          ; " ;
```

Where:

`EXCEPTNONCOREPINS`	Specifies that the rectangular object check requirement is exempted for a connection to the pins that are not in a standard cell (with `CLASS CORE`). In other words, the connection of a default-width wire to a wide I/O pad or block pin is allowed. However, I/O pins are not exempted, and they are still subjected to the rectangular object check.

`EXCEPTRIGHTWAYWIDTH` *`maxWidth`*

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Specifies that a rectilinear shape is exempted from the rectangular object check requirement if all of the right way width, which is in the non-preferred direction on the layer, is less than or equal to *maxWidth*.

Type: Float, specified in microns

`EXCEPTVIA`

Specifies that a rectilinear shape is exempted from the rectangular object check requirement if it is caused by the metal of a via. If the via is removed, the remaining shape must still be a rectangle.

`EXCEPTVIAEDGEALIGNED`

Specifies that a rectilinear shape is exempted from the rectangular object check requirement if it is caused by the metal of a via. If the via is removed, the remaining shape must still be a rectangle. In addition, one of the metal edges of the via must be aligned to the wire metal edge.

`MACROCLASS {RING | BLOCK | PAD | BUMP}...`

Specifies that the `EXCEPTNONCOREPINS` exemption applies only on pins on macros with the specified class types. In addition, wires with any width can connect to those pins in a rectilinear style.

Type: String

`PARTIALTRACKS`

Specifies that the `RECTONLY` behavior is required only on certain tracks on the layer. A separate command would define which subset of the tracks would allow only rectangular shapes. If that command is not run properly, no tracks would have the rectangular-shape-only tag and `RECTONLY` would not be checked on the layer. Any object that overlaps or touches a track with `RECTONLY` must be a rectangle.

`RECTONLY`

Specifies that the layer can only allow rectangular objects. In other words, rectilinear shapes with jogs or concave corners are illegal.

Type: Float, specified in microns

`RIGHTWAYWIDTHONLY` *exactWidth*

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Specifies that this rectangular check applies only for the right-way width, which is in the non-preferred direction on the layer, when it is exactly equal to *exactWidth*. Specifically, if any portion of a non-rectangle shape has a right-way width exactly equal to *exactWidth*, it is a violation.

Type: Float, specified in microns

Rectangle Only Rule Example

Figure 6-133 Illustration of Rectangle Only Rule

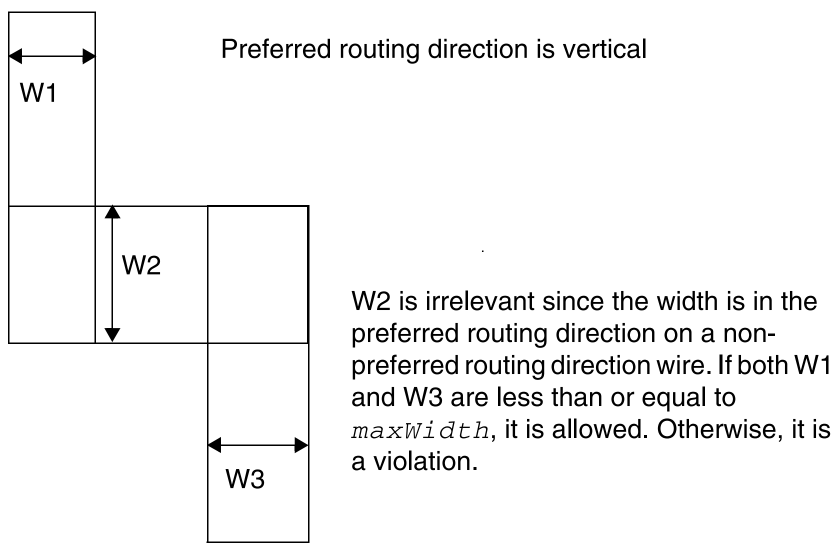

Illustration of `RECTONLY EXCEPTRIGHTWAYWIDTH` *maxWidth*

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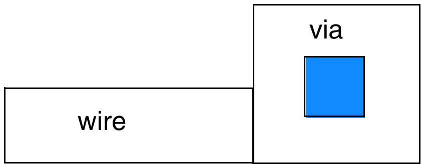
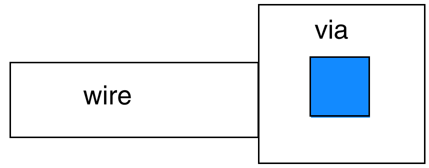
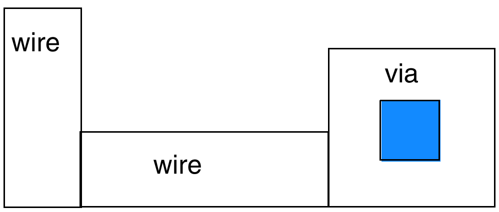
Figure 6-134 Illustration of Rectangle Only Rule with EXCEPTVIAEDGEALIGNED



a) OK, because the bottom metal edge of the via is aligned with the wire metal edge. If the via is removed, the remaining wire is still a rectangle.



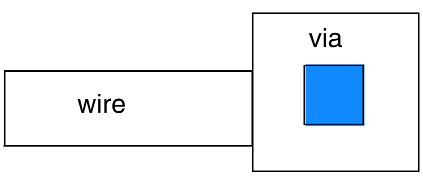
b) It is a violation because there is no metal edge alignment between the wire and the via.



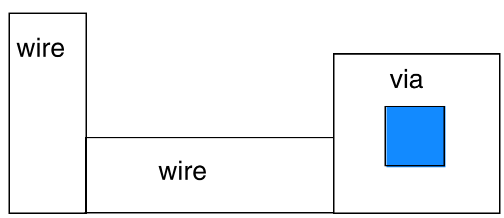
c) It is a violation because the left two wires form a rectilinear shape, which is not allowed.

Illustration of `RECTONLY EXCEPTVIAEDGEALIGNED`

Figure 6-135 Illustration of Rectangle Only Rule with EXCEPTVIA



a) It is OK because if the via is removed, the remaining wire is still a rectangle.



b) It is a violation because the left two wires form a rectilinear shape, which is not allowed.

Illustration of `RECTONLY EXCEPTVIA`

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Region Rule

You can create a region rule to define area-based rules on a routing layer.

You can use the following property definition:

```
PROPERTY LEF58_REGION
       "REGION regionLayerName BASEDLAYER routingLayerName
       ; " ;
```

Where:

`REGION` *`regionLayerName`* `BASEDLAYER` *`routingLayerName`*

Specifies a set of region- or area-based rules on a routing layer *routingLayerName*. *regionLayerName* is a masterslice layer with type `REGION`. Users can create a blockage on that masterslice layer on a design to indicate the areas of the region. In addition, `MACRO` may also have `OBS` on that masterslice layer to indicate that the areas of its instance should be subjected to these region-based rules. All the rules specified on this routing layer with `REGION` are applied to the corresponding areas of *regionLayerName* on *routingLayerName*. The areas outside *routingLayerName* honor the set of rules on the given routing layer without type `REGION`. If a rule check crosses the boundaries of a region, it is subjected to the rules for both inside and outside the region, with the tighter rule values of the two sets dominating.

The set of rules defined in the layer with type `REGION` should be very small as compared to the layer without type `REGION`. The rules defined in this layer with type `REGION` should be a replacement of the corresponding rule (having the same set of the constructs of a LEF statement or a complete replacement on a table like `SPACINGTABLE`) on the layer without type `REGION`. All the other rules defined on the layer without type `REGION` are also applicable on the layer with type `REGION`.

Region Rule Example

The following example means that a rectangular object overlapping region `R1` should have spacing of `0.7` and a rectangular object outside region `R1` should have spacing of `0.5`. Non-rectangular object in region `R1` would still follow `0.8` and any other rules on layer `M1` (without `REGION`) would also be applied on region `R1`:

```
LAYER R1
```


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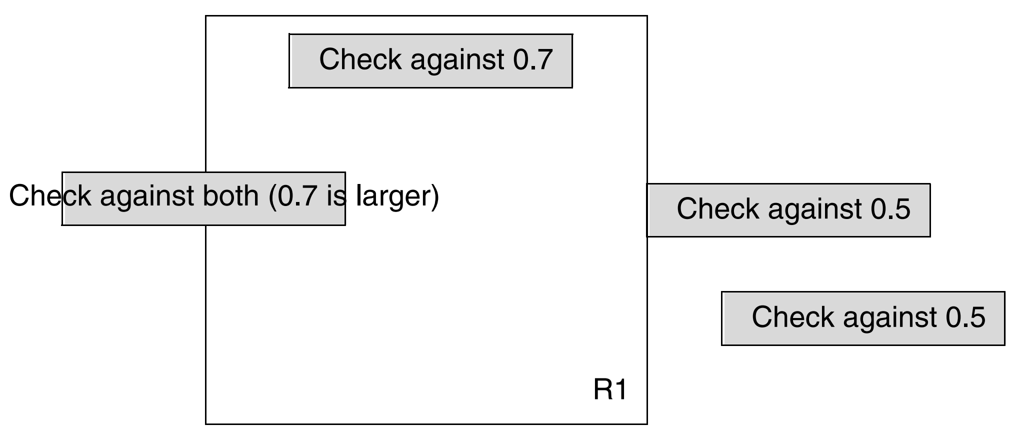
```
    TYPE MASTERSLICE ;
    PROPERTY LEF58_TYPE "TYPE REGION ; " ;
END R1
...
LAYER M1
    TYPE ROUTING ;
    ...
    AREA 0.5 ;
    PROPERTY LEF58_AREA "AREA 0.8 EXCEPTRECTANGLE ; " ;
    ...
END M1

LAYER M1R1
    TYPE ROUTING ;
    PROPERTY LEF58_REGION "REGION R1 BASEDLAYER M1 ; " ;
    ...
    AREA 0.7 ;
    ...
END M1R1
```

Figure 6-136 Illustration of Region Rule



Right Way on Grid Only Rule

The right way on grid only rule can be used to specify that the wires on the preferred routing direction of the layer must be on grid/track.

You can create a right way on grid only rule by using the following property definition:

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```
PROPERTY LEF58_RIGHTWAYONGRIDONLY
     "RIGHTWAYONGRIDONLY [CHECKMASK]
         [EXCEPTWIDTH exceptMinWidth]
         [EXCEPTVIA]
         ; " ;
```

Where:

`CHECKMASK`
: Specifies that the mask/color of the wires on the preferred routing direction should be checked against the track color. For default-width wires, the wire and track color should be the same. If wide wires could block fewer tracks based on spacing by taking the reverse track color, the wire color should be the reverse of the track color.

`EXCEPTVIA`
: Specifies that the on-grid wire checking does not apply on the metal shape of a via.

`EXCEPTWIDTH` *`exceptMinWidth`*
: Specifies that the on-grid wire checking does not apply if the width of the wire is greater than *`exceptMinWidth`*.
Type: Float, specified in microns

`RIGHTWAYONGRIDONLY`
: Specifies that the wires on the preferred routing direction must be put on a grid/track.

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Six Wires Spacing Rule

A six wires spacing rule can be used to specify a spacing rule on a six-wire pattern.

You can create a six wires spacing rule by using the following property definition:

```
PROPERTY LEF58_SIXWIRESSPACING
       "SIXWIRESSPACING spacing {SAMESIDE | OPPOSITESIDE}
           EOLSPACING eolSpacing
          ; " ;
```

Where:

`SIXWIRESSPACING` *`spacing`* `{SAMESIDE | OPPOSITESIDE}`
`EOLSPACING` *`eolSpacing`*

Specifies a spacing rule on a six-wire pattern. All of the six wires should be right-way wires with the default wire width. There are two sets of two perfectly aligned wires at the middle and two other outermost wires on the two opposite sides of the two sets of two wires. The EOL spacing between the two perfectly aligned wires must be less than or equal to *`eolSpacing`* on both the sets. The combined gap of these two sets of line-ends must be fully covered by the two outermost wires. The side-to-side spacing between any two wires in the non-preferred routing direction must be the default minimum spacing.

For `SAMESIDE`, if the spacing/gap of the two line-ends of the two wires on the same side, one each from the two sets of wires, in the preferred routing direction is less than *`spacing`*, it is a violation.

For `OPPOSITESIDE`, if the overlap of the two line-ends of the two wires in the opposite side, one each from the two sets of wires, in the preferred routing direction is greater than or equal to `0` and less than *`spacing`*, it is a violation.

Type: Float, specified in microns

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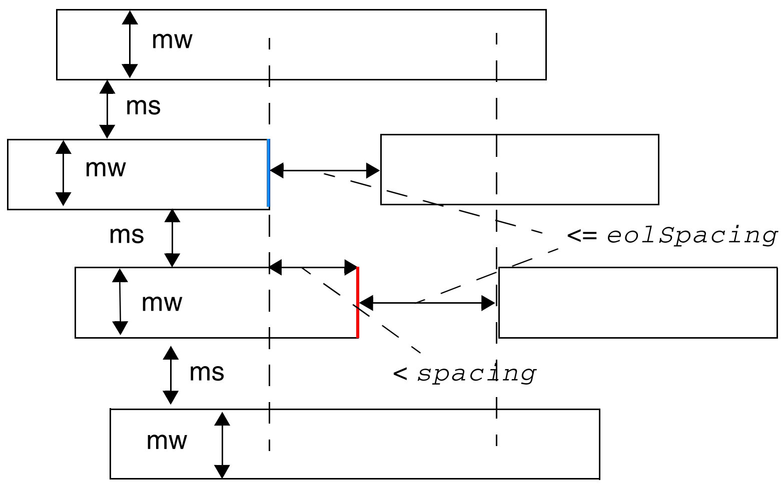
Six Wires Spacing Rule Examples

Figure 6-137 Six Wires Rule with SAMESIDE



Preferred routing direction is horizontal

mw: default width
ms: default minimum spacing

It is a violation. There are six default width wires with default minimum spacing between any two wires vertically. Two sets of two wires are perfectly aligned with spacing <= `eolSpacing`. The combined gap of the two sets of line-ends are covered by the two outermost wires showed in the dotted lines. The spacing/gap between the red and blue edges of the two wires on the same side is < `spacing`.

Illustration of

```
SIXWIRESSPACING spacing SAMESIDE
     EOLSPACING eolSpacing
```


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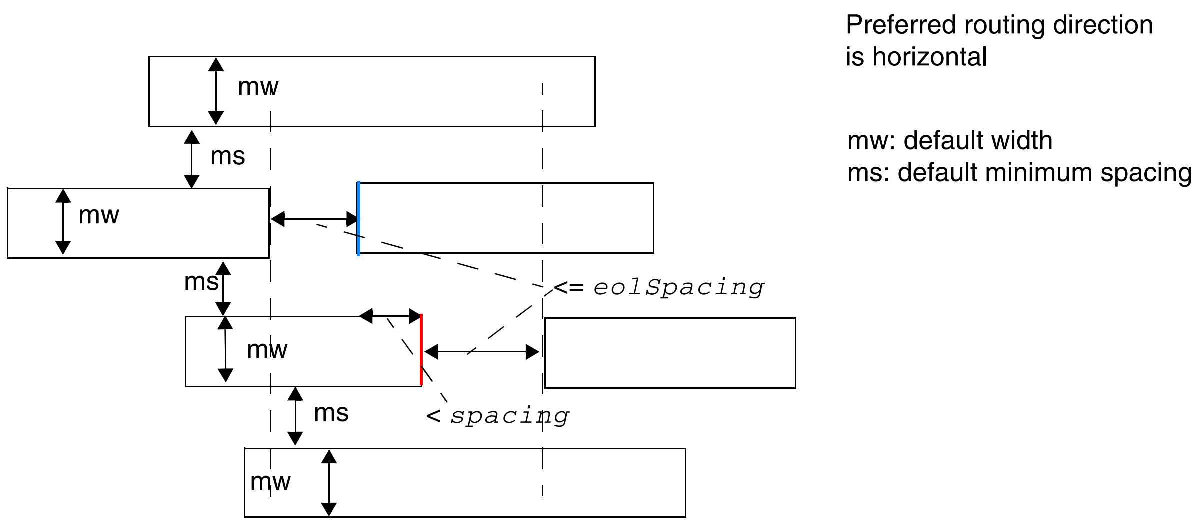
Figure 6-138 Six Wires Rule with OPPOSITESIDE



It is a violation. There are six default width wires with default minimum spacing between any two wires vertically. Two sets of two wires are perfectly aligned with spacing <= `eolSpacing`. The combined gap of the two sets of line-ends are covered by the two outermost wires showed in the dotted lines. The overlap between the red and blue edges of the two wires on the opposite sides is < `spacing`.

Illustration of
```
SIXWIRESSPACING spacing OPPOSITESIDE
     EOLSPACING eolSpacing
```

Span Length Enclosure Spacing Rule

A span length enclosure spacing rule can be used to specify spacing on an edge with a certain span length and edge length having a cut with a certain extension on that edge to any wires with given `PRL`.

You can create a span length enclosure spacing rule by using the following property definition:

```
PROPERTY LEF58_SPANLENGTHENCLOSURESPACING
     "SPANLENGTHENCLOSURESPACING spacing
          SPANLENGTH spanLength MINLENGTH minLength
          {FROMABOVE|FROMBELOW} ENCLOSURE enclosure
          PRL prl
          ; " ;
```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Where:

```
SPANLENGTHENCLOSURESPACING spacing
   SPANLENGTH spanLength MINLENGTH minLength
   {FROMABOVE|FROMBELOW} ENCLOSURE enclosure
   PRL prl
```

Specifies the spacing of an edge with span length greater than or equal to *spanLength* to any parallel edge of a wire with a common parallel run length greater than *prl*, which must be zero or a positive value if:

- Both edges have length greater than *minLength*, where the edge length is the entire edge, which may be longer than the portion having span length greater than or equal to *spanLength*.

And

- The edge with span length greater than or equal to *spanLength* has a cut in the range of the common parallel run length with the neighbor edge from above or below this layer, depending on whether FROMABOVE or FROMBELOW is specified, with enclosure less than *enclosure* from that edge.

Type: Float, specified in microns

Span Length Enclosure Spacing Rule Example

- The following example is an illustration of SPANLENGTHENCLOSURESPACING:

```
SPANLENGTHENCLOSURESPACING spacing
   SPANLENGTH spanLength MINLENGTH minLength
   FROMABOVE ENCLOSURE enclosure
   PRL prl
```


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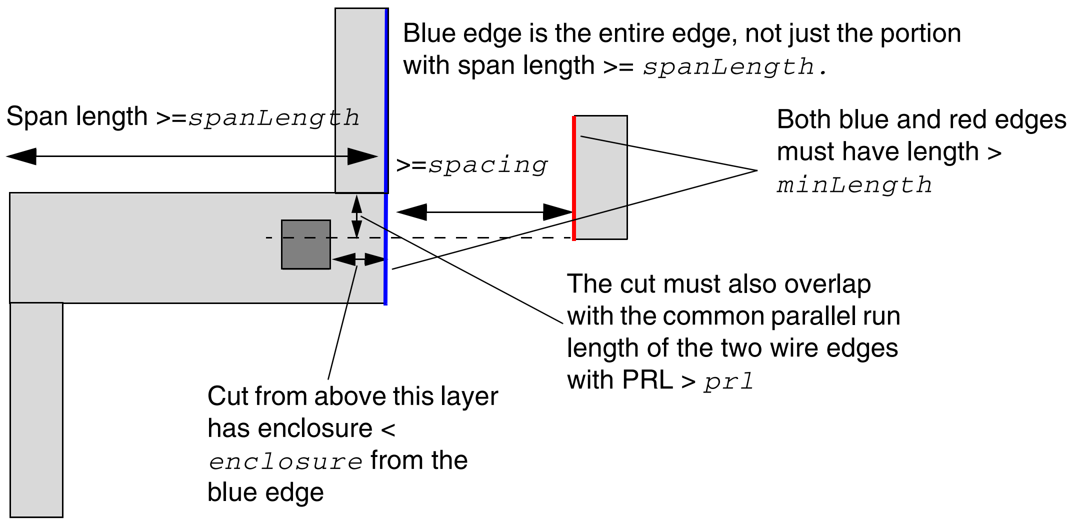
Figure 6-139 Illustration of SPANLENGTHENCLOSURESPACING



Span Length Table Rule

A span length table rule can be used to specify all the allowable legal span lengths on the routing layer.

You can create a span length table rule by using the following property definition:

```
PROPERTY LEF58_SPANLENGTHTABLE
     "SPANLENGTHTABLE {spanLength} ...  [MASK maskNum][WRONGDIRECTION]
          [ORTHOGONAL length]
          [EXCEPTOTHERSPAN otherSpanlength
          | {OTHERSPAN otherSpanLength MINSPANLENGTH minSpanLength
                [BOTHVIAS minBothViasSpanLength]}...]
           [JOGSPANONLY {jogSpanLength}... [LASTINCLUDEGREATERTHAN]
                [INCLUDELSHAPE]]
       ; " ;
```

Where:

BOTHVIAS *minBothViasSpanLength*

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Specifies that if a wire fulfilling the `OTHERSPAN` condition contains both the above and below layer vias or is floating, the span length can be greater than or equal to *`minBothViasSpanLength`*. As `MINSPANLENGTH` already allows a span length greater than or equal to *`minSpanLength`*, checking for the existence of both above and below layer vias can be done only for a span length greater than or equal to *`minBothViasSpanLength`* and less than *`minSpanLength`*. This means that *`minBothViasSpanLength`* must be less than *`minSpanLength`*.

Type: Float, specified in microns

`EXCEPTOTHERSPAN` *`otherSpanlength`*

Indicates that the span length rule only applies if the span length on the perpendicular direction is greater than the specified *`otherSpanlength`*. This construct must be defined on a layer with RECTONLY. Otherwise, the span length in the orthogonal direction could be ambiguous in a polygon shape. See Figure 6-141 on page 865.

Type: Float, specified in microns

`INCLUDELSHAPE` Specifies that the given values in *`jogSpanLength`* must also be applied on a wrong way jog connected to a right way wire in an 'L' shape.

`JOGSPANONLY {`*`jogSpanLength`*`}...`

Specifies that the span length of a wrong way jog sandwiched between two right way wires in a 'Z' shape must be equal to one of the given values in *`jogSpanLength`*. The original table span length values in *`spanLength`* cannot be applied.

`JOGSPANONLY` must be defined in both the span length tables – with and without `WRONGDIRECTION`. *`jogSpanLength`* in a table with `WRONGDIRECTION` defines the span length in the preferred-routing direction. *`jogSpanLength`* in a table without `WRONGDIRECTION` defines the span length in the non preferred-routing direction. See Figure 6-142 on page 866.

Type: Float, specified in microns

`LASTINCLUDEGREATERTHAN`

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Specifies that the last value given in *jogSpanLength* is interpreted as 'greater than or equal to'.

`MASK` *maskNum*

Specifies that the span length rule applies only on the given mask of the current layer. *maskNum* must be a positive integer, and most applications support only the values 1, 2, or 3.

`ORTHOGONAL` *length*

Specifies that the length between two inside facing corners of a rectilinear object must be greater than or equal to the *length*.

Type: Float, specified in microns

`{OTHERSPAN` *otherSpanLength* `MINSPANLENGTH` *minSpanLength*`}...`

Specifies that if the span length on the orthogonal direction of the span length being checked is less than or equal to *otherSpanlength*, the span length being checked needs to be only greater than or equal to the *minSpanLength*. In addition, if the span length is smaller than *minSpanLength* but is exactly equal to one of the values in *spanLength*, it is also legal. If multiple OTHERSPAN and MINSPANLENGTH are specified, the first *otherSpanLength* that the current span length is less than or equal to would be applied. This construct must be defined on a layer with RECTONLY. In a polygon shape, the span length in the orthogonal direction could be ambiguous. See Figure 6-143 on page 866.

Type: Float, specified in microns

`SPANLENGTHTABLE {`*spanLength*`)}...`

Specifies all of the allowable legal span lengths on the routing layer. All of the given span lengths are exact values, except for the last one, which is greater than equal to the value. All of the possible span lengths of an object in both directions should be checked against the given span length values.

If MASK is not specified, at most two SPANLENGTHTABLE spacing tables can be defined - one with WRONGDIRECTION and the other without.

If MASK is specified, each mask for the layer can have separate SPANLENGTHTABLE spacing tables, one with and one without WRONGDIRECTION.

See Figure 6-140 on page 865.

Type: Float, specified in microns

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`WRONGDIRECTION`

Specifies all of the allowable legal span lengths in the direction parallel to the specified direction in `DIRECTION` on a Manhattan routing layer.

Note that using `WRONGDIRECTION` changes the interpretation of any wrong-way routing widths in the `DEF NETS` section. If `WRONGDIRECTION` is specified, then any wrong-way routing in the `DEF NETS` section will use the `WRONGDIRECTION` width for that layer unless the net or route has a `NONDEFAULTRULE` with a `WIDTH` greater than the `WRONGDIRECTION` width. But, the implicit default route-extension is still half of the preferred direction width.

Some older tools may not understand this behavior. Normally, they will still read/write and round-trip the DEF routing properly, but will not understand that the width is slightly larger for wrong-way routes. If these tools check wrong-way width, then the DRC rules may flag false violations. RC extraction with the wrong width will also flag errors, although wrong-way routes are generally short and the width difference is small, so the RC error is normally negligible.

Span Length Table Rule Examples

- The following example is an illustration of `SPANLENGTHTABLE {`*`spanLength`*`}...` `ORTHOGONAL` *`length`* and `SPANLENGTHTABLE {`*`spanLength`*`}... WRONGDIRECTION` with preferred direction being horizontal:

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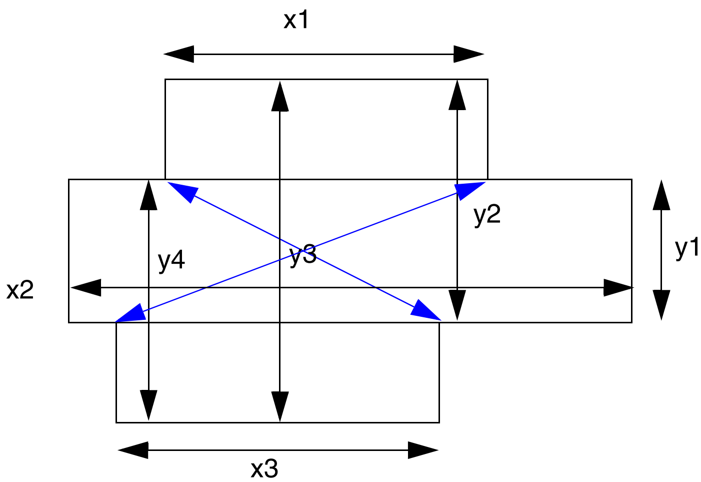
Figure 6-140 Illustration of SPANLENGTHTABLE

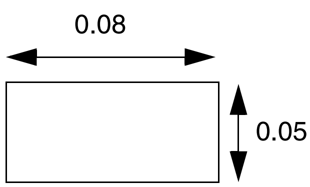


y1, *y2*, *y3*, and *y4* should be checked against the given span lengths in `SPANLENGTHTABLE` without `WRONGDIRECTION` while *x1*, *x2*, and *x3* should be checked against the span lengths in `SPANLENGTHTABLE` with `WRONGDIRECTION`. In addition, the length of the blue arrows should be checked against length.

- The following example is an illustration of `SPANLENGTHTABLE` with `EXCEPTOTHERSPAN`:

```
DIRECTION HORIZONTAL ;
PROPERTY LEF58_RECTONLY "
    RECTONLY ; " ;
PROPERTY LEF58_SPANLENGTHTABLE "
    SPANLENGTHTABLE 0.05 0.06 0.07 0.080 0.090 0.10 ;
    SPANLENGTHTABLE 0.07 0.10 WRONGDIRECTION
        EXCEPTOTHERSPAN 0.05 ; " ;
```

Figure 6-141 Illustration of SPANLENGTHTABLE with EXCEPTOTHERSPAN



a) OK, span length of 0.05 fulfills the first span length table, and span length of 0.08 is not checked because `EXCEPTOTHERSPAN` condition is not fulfilled. Violation if `EXCEPTOTHERSPAN` is omitted.

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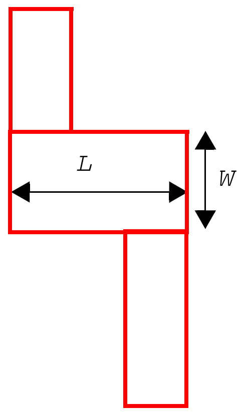
- The following example is an illustration of `SPANLENGTHTABLE` with `JOGSPANONLY`:

Figure 6-142 Illustration of SPANLENGTHTABLE with JOGSPANONLY



As only one value is specified in `JOGSPANONLY` in each table, the span length of the 'Z' shape jog must be equal to the given values. That is *L* in the non preferred-routing direction and *W* in the preferred-routing direction.

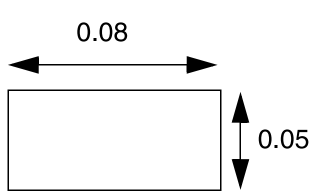
Illustration of:

```
SPANLENGTHTABLE ... JOGSPANONLY L ;
SPANLENGTHTABLE...WRONGDIRECTION
    JOGSPANONLY W ;
```

- The following example is an illustration of `SPANLENGTHTABLE` with `OTHERSPAN`:

```
DIRECTION HORIZONTAL ;
PROPERTY LEF58_RECTONLY "
    RECTONLY ; " ;
PROPERTY LEF58_SPANLENGTHTABLE "
    SPANLENGTHTABLE 0.04 0.05 0.06 0.07 0.080 0.090 0.10 ;
    SPANLENGTHTABLE 0.07 0.10 WRONGDIRECTION
        OTHERSPAN 0.04 MINSPANLENGTH 0.09
        OTHERSPAN 0.06 MINSPANLENGTH 0.08 ; " ;
```

Figure 6-143 Illustration of SPANLENGTHTABLE with OTHERSPAN



a) OK, span length of 0.05 fulfills the first span length table, and span length of 0.08 is checked against `OTHERSPAN` of 0.06, which would pass the corresponding requirement of >= 0.08.

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EOL Spacing Rule

An EOL spacing rule ensures that Optical Proximity Correction (OPC) can be performed without interference between the OPC shapes added at the EOLs.

You can create an EOL spacing rule by using the following property definition:

```
PROPERTY LEF58_SPACING
       "SPACING eolSpace
          ENDOFLINE eolWidth
          [{EXACTWIDTH]
          [WRONGDIRSPACING wrongDirSpace]
          {[OPPOSITEWIDTH oppositeWidth]
           WITHIN eolWithin [wrongDirWithin]
           [SAMEMASK]
           [EXCEPTEXACTWIDTH exactWidth otherWidth]
           [FILLCONCAVECORNER fillTriangle
           | FILLCONCAVECORNERONLY fillTriangle]
           [WITHCUT [CUTCLASS cutClass] [ABOVE] withCutSpace
              [ENCLOSUREEND enclosureEndWidth
                 [WITHIN enclosureEndWithin]]]
           [ENDPRLSPACING endPrlSpace PRL endPrl]
           [ENDTOEND endToEndSpace [oneCutSpace twoCutSpace]
              [EXTENSION extension [wrongDirExtension]]
              [OTHERENDWIDTH otherEndWidth]]
           [MAXLENGTH maxLength | MINLENGTH minLength [TWOSIDES]]
           [EQUALRECTWIDTH]
           [PARALLELEDGE [SUBTRACTEOLWIDTH] parSpace
               WITHIN parWithin [PRL prl]
               [MINLENGTH minLength] [TWOEDGES]
               [SAMEMETAL][NONEOLCORNERONLY]
               [PARALLELSAMEMASK]]
           [ENCLOSECUT [BELOW | ABOVE] encloseDist
              CUTSPACING cutToMetalSpace [ALLCUTS]]
          | TOCONCAVECORNER [MINLENGTH minLength]
              [MINADJACENTLENGTH
                 {minAdjLength | minAdjLength1 minAdjLength2}]
          | TONOTCHLENGTH notchLength
      }
       ;..." ;
```

Where:

The keywords `SPACING` (including `ENDOFLINE` and `WITHIN`), `PARALLELEDGE`, and `TWOEDGES` are the same as the existing LEF syntax.

`ENCLOSECUT [BELOW | ABOVE]` *`encloseDist`* `CUTSPACING` *`cutToMetalSpace`* `[ALLCUTS]`

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Indicates that the rule only applies if there is a cut below or above this metal that is less than *encloseDist* from the end-of-line edge and the cut-edge to metal-edge space beyond the EOL edge less than *cutToMetalSpace*. If there is more than one cut connecting the same metal shapes above and below, only one cut needs to meet this rule, unless ALLCUTS is also specified which checks the rule against all of the cuts. (See Figure 6-152 on page 881, and Figure 6-154 on page 883.).

If you specify BELOW, the *encloseDist* and *cutToMetalSpace* is checked for the cut layer below this routing layer. If you specify ABOVE, they are checked for the cut layer above this routing layer. If you specify neither, the rule applies to both adjacent cut layers.
Type: Float, specified in microns (for both values)

ENCLOSUREEND *enclosureEndWidth* [WITHIN *enclosureEndWithin*]

Specifies that the EOL spacing rule applies to a EOL edge touching a via cut with length equal to the wire width that is less than *enclosureEndWidth*.

The WITHIN keyword specifies the within search distance for a EOL edge touching a via cut to be *enclosureEndWithin* instead of *eolWithin*.
Type: Float, specified in microns

ENDPRLSPACING *endPrlSpace* PRL *endPrl*

Specifies that *endPrlSpace* is applied between the end-of-line (EOL) edge to any neighbor wire with PRL greater than 0 but less than or equal to *endPrl*. Otherwise, eolSpace is applied to the end-to-line situation.

Type: Float, specified in microns

ENDTOEND *endToEndSpace* [*oneCutSpace twoCutSpace*]

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Specifies the two EOL spacings. For end-to-end situation when there is a parallel run length greater than 0 between the two EOL edges with *eolWithin* extension on the checking EOL edge, *endToEndSpace* is applied. Otherwise, *eolSpace* is applied to end-to-line situation.

An end-to-end situation with one of the EOL edge touching a via cut (and the other does not touching a via cut) must have spacing greater than or equal to *oneCutSpace*. If both EOL edges touching a via cut, the spacing must be greater than or equal to *twoCutSpace*. These additional spacing values must be specified along with WITHCUT.
Type: Float, specified in microns

EQUALRECTWIDTH

Indicates that if the length of the EOL edge is larger than the wire width, the rule does not apply. If there are multiple EOL statements with the EQUALRECTWIDTH keyword for a given layer, they must all have the EQUALRECTWIDTH keyword.

EXACTWIDTH

Specifies that this end-of-line spacing rule only applies if the edge width/length is exactly equal to *eolWidth*. This construct would typically make sense only if WIDTHTABLE or SPANLENGTHTABLE is also defined on the layer, and the end-of-line spacing rule will only be triggered for one of the discrete widths.

EXTENSION *extension* [*wrongDirExtension*]

Specifies the extension on two EOL edges facing each other as *extension*. The *wrongDirExtension* variable specifies the extension on two EOL edges parallel to the specified direction in DIRECTION on a Manhattan routing layer. The end to end spacing is applied as if the extensions are part of the EOL edges. When only the extensions have projected parallel run length, but the EOL edges do not, the end to end spacing still applies.
Type: Float, specified in microns

EXCEPTEXACTWIDTH *exactWidth otherWidth*

Specifies that the end-of-line spacing rule does not apply if the EOL edge has width/length exactly equal to *exactWidth* and the width/length of the other neighbor edge is less than or equal to the *otherWidth*. See Figure 6-160 on page 888.
Type: Float, specified in microns

FILLCONCAVECORNER *fillTriangle*

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Specifies that the EOL next to any concave corner should be filled by a triangle along the edges, using the value of *`fillTriangle`* to complete the corner triangle before spacing is checked. If either edge of the concave corner is shorter than *`fillTriangle`*, that entire edge is used to form the triangle. See Figure 6-159 on page 888.
Type: Float, specified in microns

`FILLCONCAVECORNERONLY` *`fillTriangle`*

Specifies that the end-of-line (EOL) next to a concave corner should be filled with a triangle along the edges, using the value of *`fillTriangle`* to complete the corner triangle before spacing is checked. This rule applies only between the EOL and the concave corner. If either edge of the concave corner is shorter than *`fillTriangle`*, that entire edge is used to form the triangle.

Type: Float, specified in microns

`MAXLENGTH` *`maxLength`*

Indicates that if the EOL is more than *`maxLength`* along both sides, the rule does not apply. (See Figure 6-153 on page 882.)
Type: Float, specified in microns

`MINADJACENTLENGTH {`*`minAdjLength`* `|` *`minAdjLength1 minAdjLength2`*`}`

Specifies that the EOL to concave corner spacing rule applies only if both of the edge lengths forming the concave corner are greater than *`minAdjLength`*, or one of the edge length is greater than *`minAdjLength1`* and the other edge length is greater than *`minAdjLength2`*. See Figure 6-157 on page 886.
Type: Float, specified in microns

`MINLENGTH` *`minLength`*

Indicates that if the EOL length is less than *`minLength`* along both sides, the rule does not apply.
Type: Float, specified in microns

`MINLENGTH` *`minLength`* `[TWOEDGES]`

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Indicates that if the EOL length is less than *minLength* along the side, then any parallel-edge on that side is ignored, and the rule may not apply. If TWOEDGES is specified, the EOL rule applies only if there are two parallel edges on each side of the EOL edge that meet the PARALLELEDGE.
Type: Float, specified in microns

NONEOLCORNERONLY

Specifies that the parallel edge neighbor wire must contain a corner that does not belong to another EOL edge.

OPPOSITEWIDTH *oppositeWidth*

Indicates that the rule applies only if a wire beyond the end of the line edge has a perpendicular span to the EOL edge less than *oppositeWidth*.
Type: Float for all parameters, specified in microns

OTHERENDWIDTH *otherEndWidth*

Indicates that the rule only applies if the width of the other wire is less than the *otherEndWidth*.

PARALLELEDGE [SUBTRACTEOLWIDTH] *parSpace* WITHIN *parWithin*

Indicates that the EOL rule applies only if there is a parallel-edge less than *parSpace* away that is also less than *parWithin* from the end of the wire.

PARALLELSAMEMASK

Specifies that the end-of-line spacing rule only applies if there is a side neighbor having the same-mask of the wire with the EOL edge found in the side search window. Having a different-mask side neighbor in the search is irrelevant.

PRL *prl*

Indicates that the end-of-line (EOL) spacing rule applies only if the side neighbor and the neighbor beyond the EOL edge have parallel run length greater than the specified *prl* after extending by the WITHIN values. When there are two nearby EOL edges on opposite or orthagonal sides, their search window for the side neighbor could be merged when each of the individual EOL edges has a neighbor beyond the EOL within *eolSpace*, and the PRL is determined by the merged window.
Type: Float, specified in microns

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`SAMEMASK`
: Indicates that the EOL spacing rule only applies between objects on the same mask.

Note: The `SAMEMASK` keyword can only be specified with `MINLENGTH`, `OPPOSITEWIDTH`, and `ENDTOEND`, or `PARALLELEGDE` keywords.

`SAMEMETAL`
: Specifies that the EOL spacing rule applies only if the neighbors on the side(s) of and beyond the EOL edge are connected as same-metal. In addition, if a side neighbor wire covering the entire length of *parWithin*, only if it is non-zero, then it will be exempted as a valid neighbor.

`SUBTRACTEOLWIDTH`
: Indicates that the *parSpace* value should be subtracted by the width of the EOL edge to define the distance required to search for a parallel neighbor edge.

`TOCONCAVECORNER`
: Specifies that the spacing between a EOL edge with width less than *eolWidth* and a concave corner of neighbor wires must be greater than or equal to *spacing*.
Type: Float, specified in microns

`TONOTCHLENGTH` *notchLength*
: Specifies the end-of-line spacing to a notch with length less than *notchLength*. See Figure 6-161 on page 889.
Type: Float, specified in microns

`TWOSIDES`
: Indicates that the rule applies only when the EOL length is greater than and equal to *minLength* along both sides. In other words, if the EOL length is less than *minLength* along any one side, the rule does not apply.

`WITHCUT [CUTCLASS` *cutClass*`] [ABOVE]` *withCutSpace*
: Specifies the EOL spacing, *withCutSpace*, only applies if a EOL edge touches a via cut in either below or above cut layer.
Type: Float, specified in microns

The `CUTLCASS` keyword defines the EOL spacing rule for certain *cutClass* only. If `CUTCLASS` is defined in the routing layer, then `CUTCLASS` must be specified in the `SPACING` statement.

The `ABOVE` keyword defines the EOL spacing rule only applies if a EOL edge touches a via cut in above cut layer.

`WITHIN` *eolWithin* `[`*wrongDirWithin*`]`

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The `wrongDirWithin` variable specifies the within distance if a EOL edge is parallel to the specified direction in `DIRECTION` on a Manhattan routing layer, while `eolWithin` is the within distance of a EOL edge perpendicular to that direction.
Type: Float, specified in microns

`WRONGDIRSPACING` *`wrongDirSpace`*

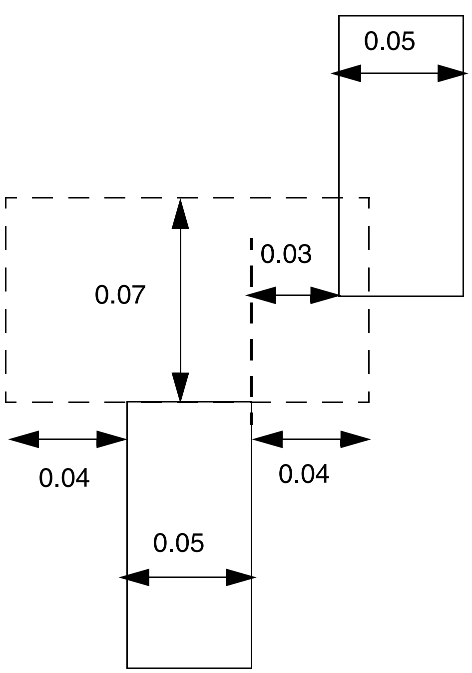
Specifies an optional end-of-line spacing for a EOL edge parallel to the specified direction in `DIRECTION` on a Manhattan routing layer to be `wrongDirSpace`.
Type: Float, specified in microns

EOL Spacing Rule Examples

The following example illustrates EOL spacing with exact width:

```
PROPERTY LEF58_WIDTHTABLE
    "WIDTHTABLE 0.05 0.10 … ; " ;
SPACING 0.09 ENDOFLINE 0.051 WITHIN 0.02 ;
PROPERTY LEF58_SPACING  "
    SPACING 0.07 ENDOFLINE 0.10 EXACTWIDTH WITHIN 0.04 ; " ;
```

Figure 6-144 Example of Spacing Rule with EXACTWIDTH



a) OK, the 0.051 EOL width statement does not apply since the neighbor wire is outside within of 0.02, and the 0.1 EOL width statement also does not apply since the edge width/length of 0.05 is not exactly equal to 0.10.

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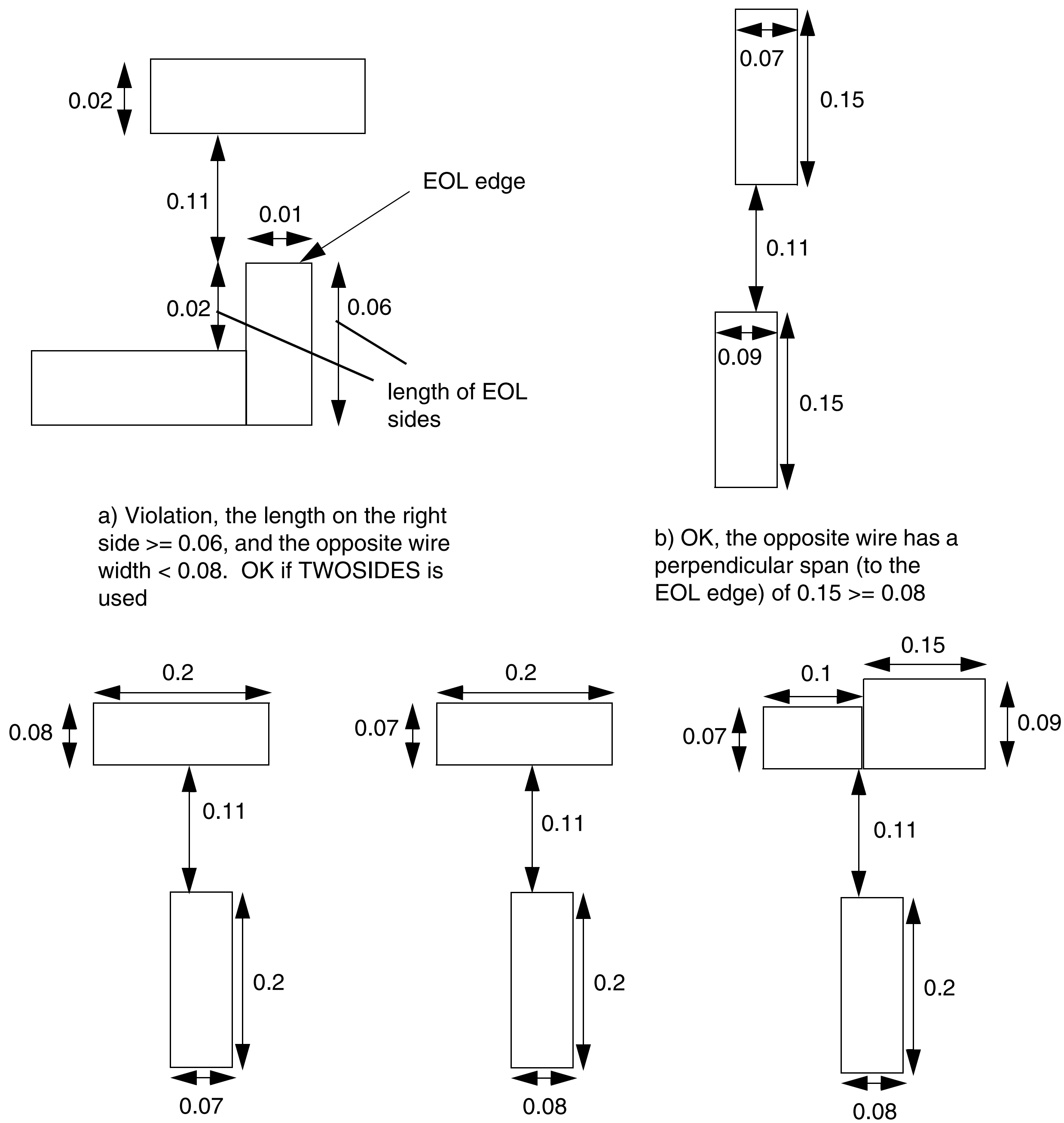
Figure 6-145 EOL Spacing Rule Illustrations

```
PROPERTY LEF58_SPACING "SPACING 0.12 ENDOFLINE 0.1 OPPOSITEWIDTH 0.08
WITHIN 0.05 MINLENGTH 0.06 ;" ;
```



a) Violation, the length on the right side >= 0.06, and the opposite wire width < 0.08. OK if TWOSIDES is used

b) OK, the opposite wire has a perpendicular span (to the EOL edge) of 0.15 >= 0.08

c) OK, opposite wire width >= 0.08, and the rule does not apply

d) Violation, swapping the wire width of the wires becomes a violation

e) Violation, if part of the opposite wire width < 0.08 (the left segment), it is a violation

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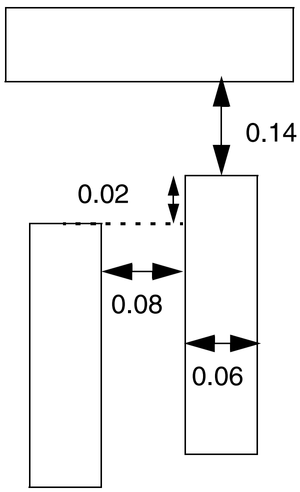
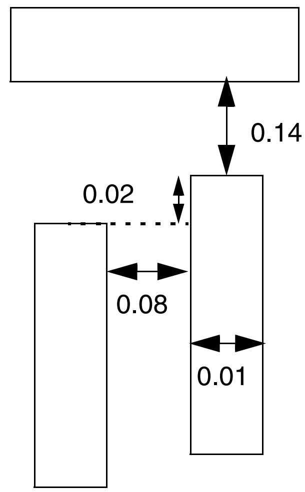
Figure 6-146 EOL Spacing Rule Illustrations

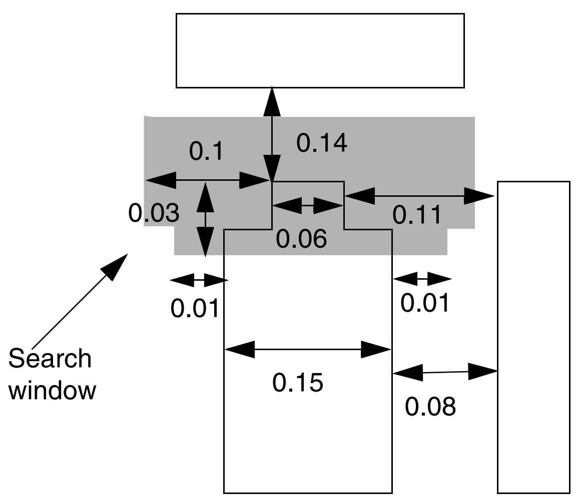
```
PROPERTY LEF58_SPACING "SPACING 0.15 ENDOFLINE 0.1 WITHIN 0.05
PARALLELEDGE SUBTRACTEOLWIDTH 0.16 WITHIN 0.03 ;" ;
```



a) Violation, the top wire needs to be 0.15 away to avoid the EOL violation



b) OK, the combined EOL wire width & the gap of the parallel edge is 0.17 (>= 0.16), and the EOL rule does not apply



d) Okay, the parallel edge is searched by extending the outline of the wire dynamically based on the wire width. Hence, 0.1 (0.16 - 0.06) on top, and 0.01 (0.16 - 0.15) on bottom.

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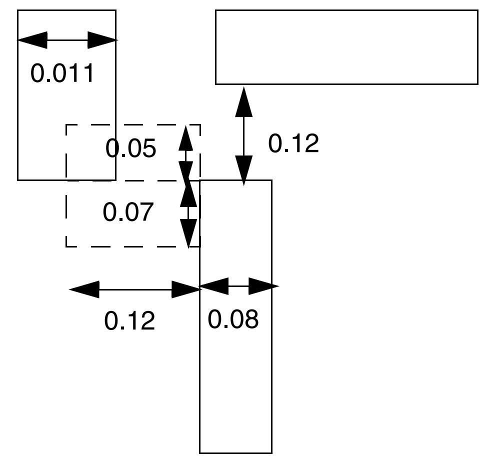
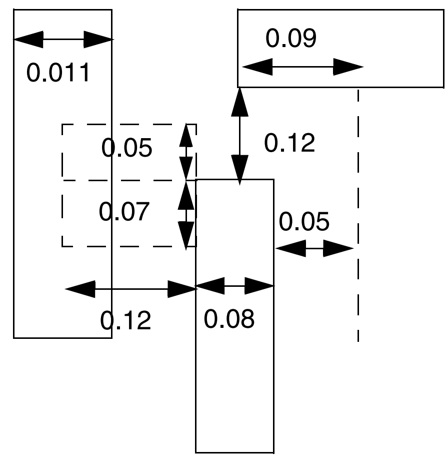
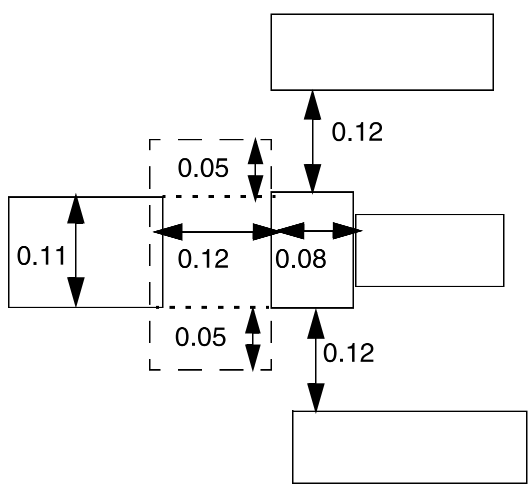
Figure 6-147 EOL Spacing Rule Illustrations



a) OK, the left neighbor only has PRL of 0.05 (<= 0.1). Violation if PRL is omitted.



b) OK, the top neighbor beyond the EOL edge has PRL of 0.09 (<= 0.1). Violation if PRL is omitted.



c) Violation, if the side neighbor search window is considered for the top and bottom 0.08 EOL individually, PRL would be 0.07 to not trigger the rule. However, the correct search window should be merged between these two EOL edges as shown in the dotted window, and PRL is 0.11.

```
PROPERTY LEF58_SPACING "
      SPACING 0.13 ENDOFLINE 0.1 WITHIN 0.05
           PARALLELEDGE 0.12 WITHIN 0.07
           PRL 0.1 ;" ;
```

Figure 6-148 on page 877 illustrates the following EOL spacing rule:

```
PROPERTY LEF58_SPACING
   "SPACING 0.10 ENDOFLINE 0.15 WITHIN 0.05 ENDTOEND 0.12
      MINLENGTH 0.11 TWOSIDES ;" ;
```


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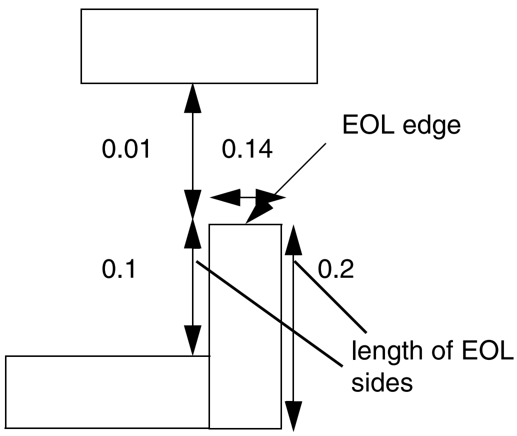
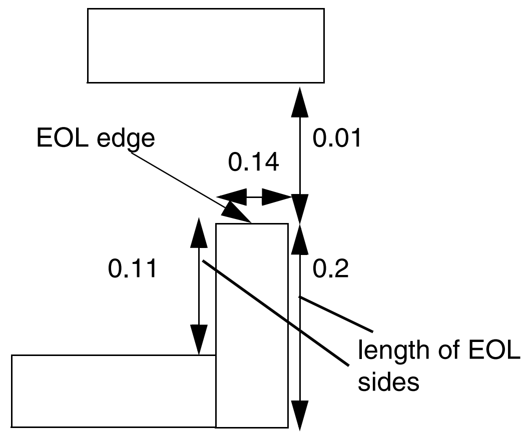
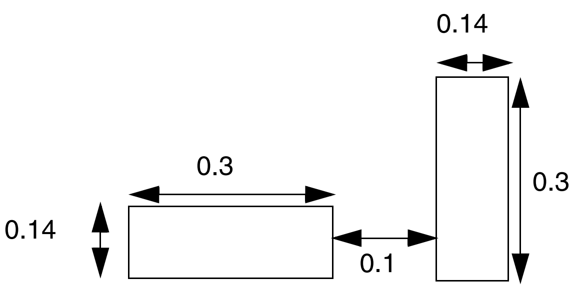
Figure 6-148 EOL Spacing Rule Illustrations



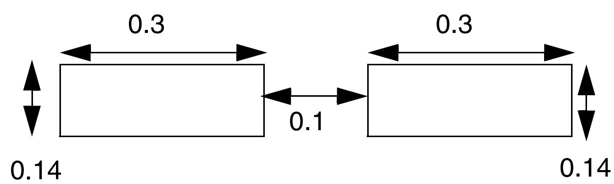
a) OK, the length of the left EOL side < 0.11, and the rule does not apply. Violation without TWOSIDES



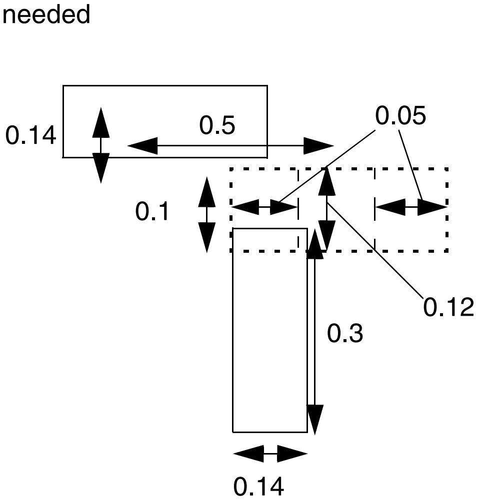
b) Violation, the length of both of the sides >= 0.11. and EOL spacing < 0.1



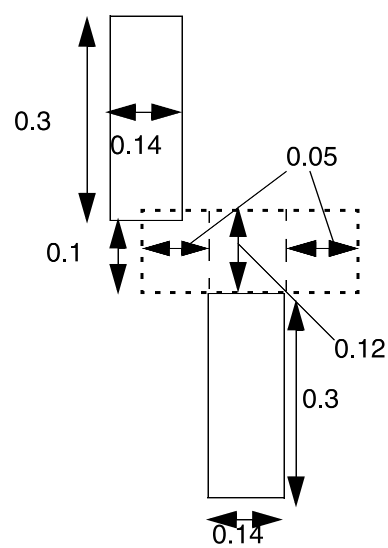
c) OK, the right edge is not a EOL edge, and 0.1 is needed



d) Violation, this is an end-to-end situation, and EOL spacing < 0.12



e) OK, it fulfills the end-to-line EOL spacing of 0.1, and the 2 EOL edges do not have common parallel run length such that end-to-end EOL spacing is ignored.



f) Violation, it does not satisfy end-to-end EOL spacing

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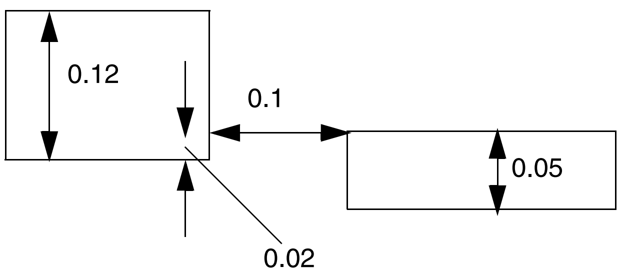
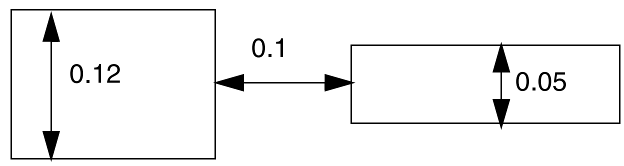
Figure 6-149 EOL Spacing Rule Illustrations with ENDPRLSPACING

```
PROPERTY LEF58_SPACING
   "SPACING 0.1 ENDOFLINE 0.06 WITHIN 0
       ENDPRLSPACING 0.12 PRL 0.03 ;" ;
```



a) Violation, `ENDPRLSPACING` of `0.12` fails.



b) OK, `PRL` of the `EOL` is 0.05 (> 0.03), only `EOL` spacing of `0.1` is required and is met.

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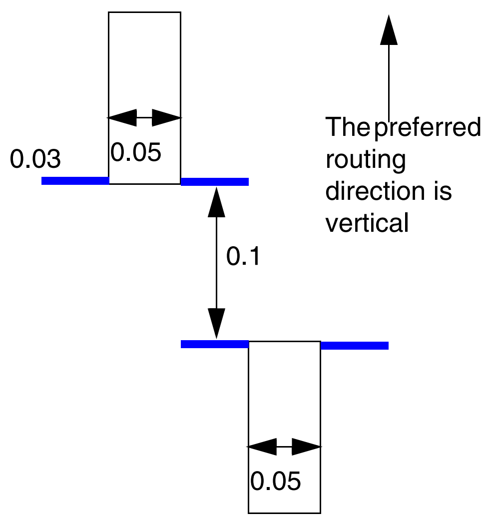
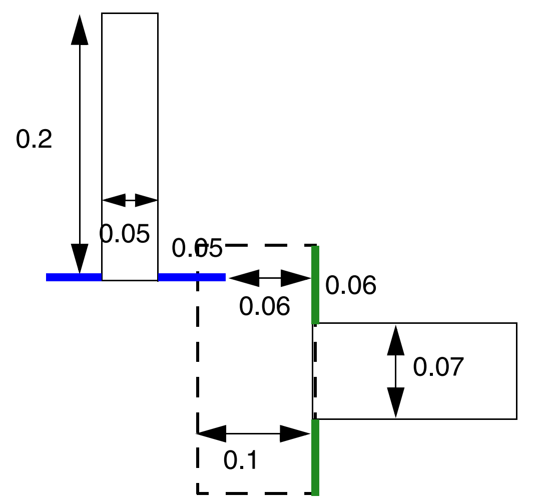
Figure 6-150 EOL Spacing Rule Illustrations

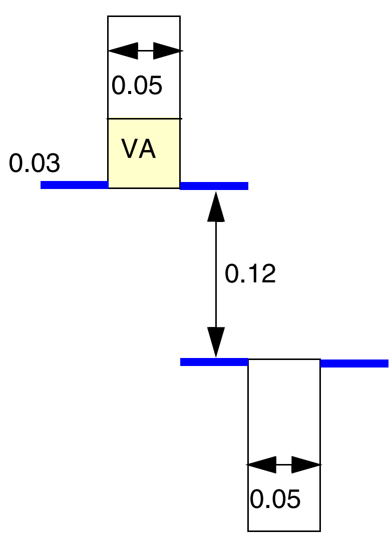
```
DIRECTION VERTICAL ;
PROPERTY LEF58_SPACING
   "SPACING 0.1 ENDOFLINE 0.08 WITHIN 0.05 0.06
      WITHCUT CUTCLASS VA 0.12 ENCLOSUREEND 0.14
      ENDTOEND 0.11 0.13 0.15
      EXTENSION 0.03 0.08 ; " ;
```



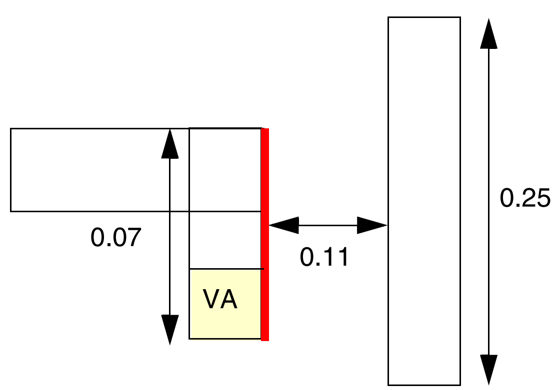
a) Violation, as long as the extensions have common projected run length, 0.11 spacing is needed



b) OK, the blue 0.05 is merely a within distance for searching the neighbor, 0.11 (>= 0.1) is the spacing between the vertical edge of to the green 0.06 within extension.



c) Violation, with touching one cut in one of the EOL edges, 0.13 spacing is needed



d) OK, the red edge is not an enclosure end edge, and 0.12 spacing does not apply. It is a EOL edge, and 0.1 spacing is met.

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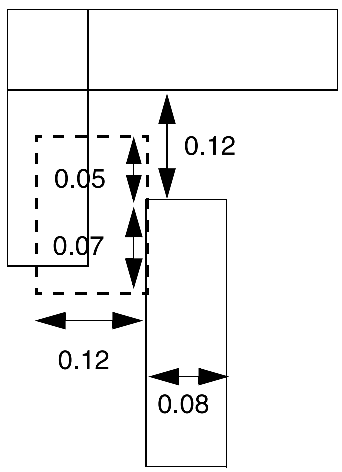
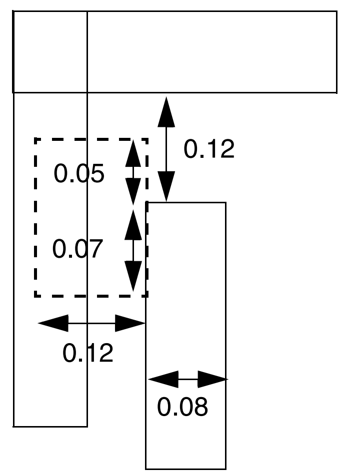
Figure 6-151 EOL Spacing Rule Illustrations

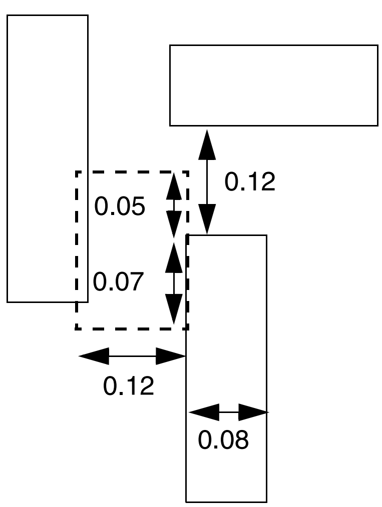
```
PROPERTY LEF58_SPACING
   "SPACING 0.13 ENDOFLINE 0.1 WITHIN 0.05
       PARALLELEDGE 0.12 WITHIN 0.07
       SAMEMETAL ; " ;
```



a) Violation, neighbor on the side is found and EOL spacing is 0.12 (<= 0.13)



b) OK, the neighbor covers the entire length of the searching window, the rule does not apply



c) OK, the neighbors do not belong to same-metal. Violation, if SAMEMETAL is not specified.

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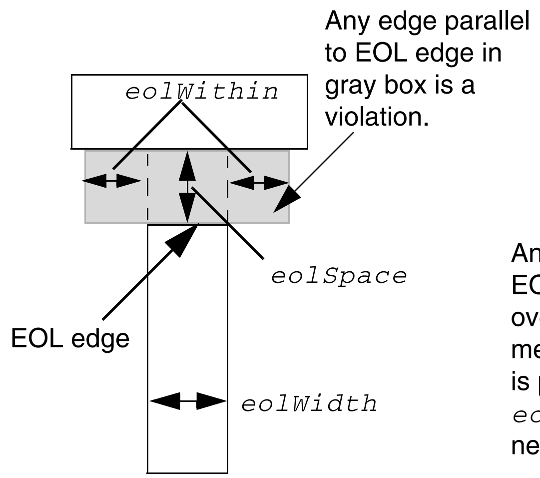
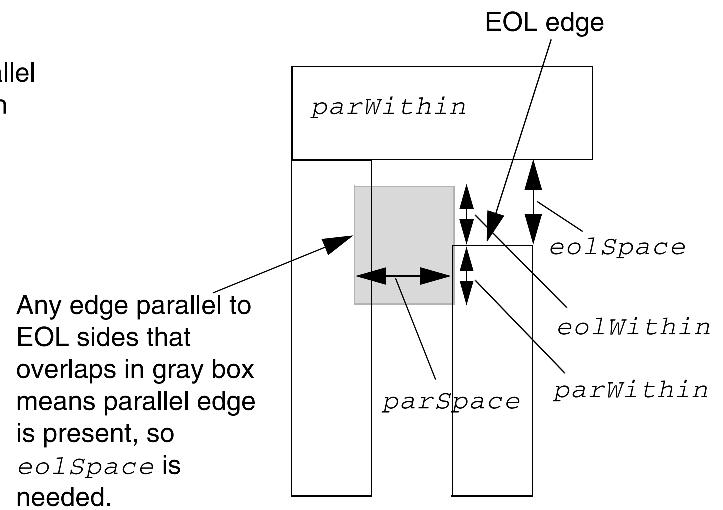
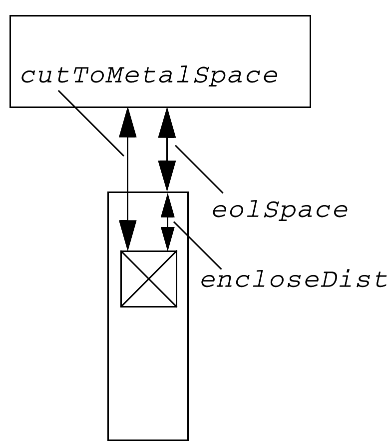
Figure 6-152 EOL Spacing Rule Illustrations



a) Basic EOL rule: EOL width < *eolWidth* requires >= *eolSpace* beyond EOL edge to opposite edges to either side by < *eolWithin* distance (gray box).



b) EOL with `PARALLELEDGE` is triggered only if an EOL side also has a parallel edge < *parSpace* and < *eolWithin* above, or < *parWithin* below, EOL edge (gray box). If *minLength* is given, and the EOL side is < *minLength*, the parallel edge does not matter, and no extra space is needed.



c) EOL with `ENCLOSECUT` rule is triggered if EOL edge has a cut edge that has < *encloseDist* enclosure, and the cut edge is < *cutToMetalSpace* from the metal beyond the EOL edge.

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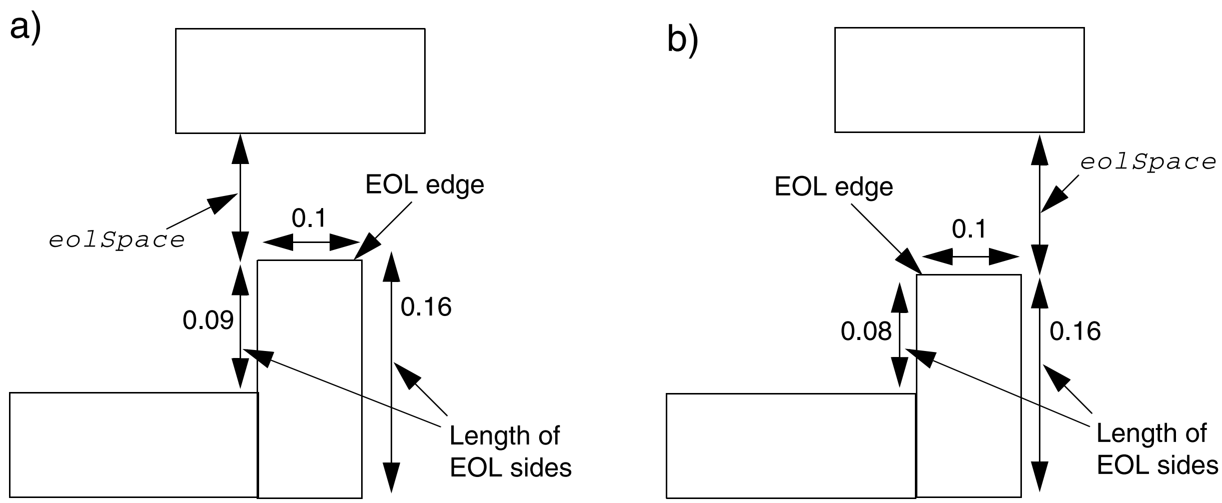
Figure 6-153 Examples of EOL Spacing Rule with MaxLength and MinLength




```
PROPERTY LEF58_SPACING
   "SPACING 0.12 ENDOFLINE 0.11 WITHIN 0.05 MAXLENGTH 0.08 ;" ;
```

a) Both sides have a maximum length > 0.08, so do not check *eolSpace*.
b) One side has a maximum length <= 0.08, so check *eolSpace* >= 0.12.

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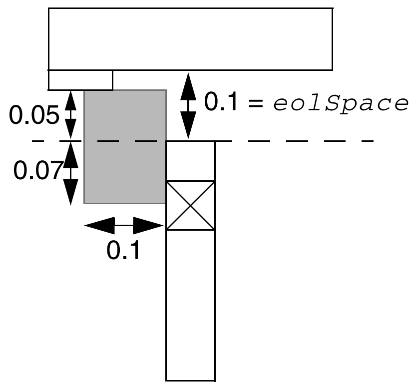
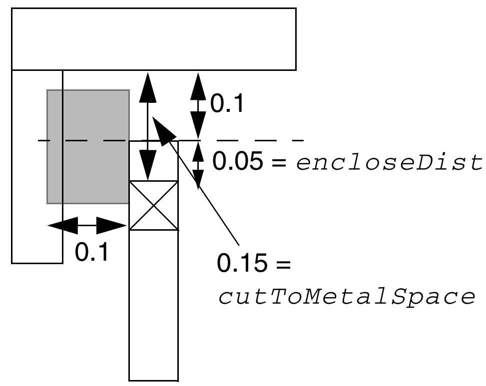
Figure 6-154 Example of More Complex EOL Spacing Rule

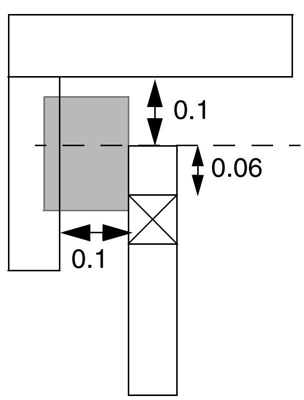
```
PROPERTY LEF58_SPACING
   "SPACING 0.15 ENDOFLINE 0.10 WITHIN 0.05 PARALLELEDGE 0.10
    WITHIN 0.07 MINLENGTH 0.10 ENCLOSECUT BELOW 0.06 CUTSPACING 0.16 ;" ;
```



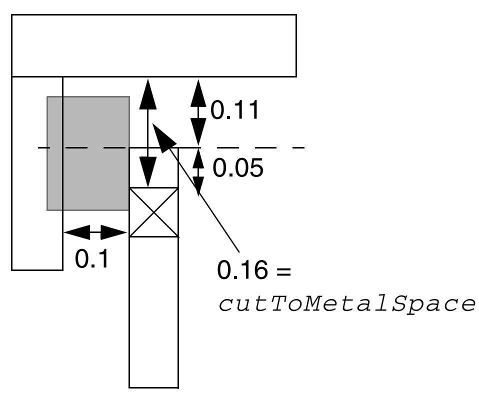
a) No violation. No parallel edge inside gray box (`parSpace` < 0.10, `eolWithin` < 0.05, and `parWithin` < 0.07), so do not check `eolSpace`.



b) Violation. Meets min length, has parallel edge, cut enclosure `encloseDist` < 0.06, `cutToMetalSpace` < 0.16, so needs 0.15 `eolSpace`.



c) No violation. Meets min length, has parallel edge, but cut enclosure `encloseDist` >= 0.06, so do not check `eolSpace`.



d) No violation. Meets min length, has parallel edge, cut enclosure `encloseDist` < 0.06, but `cutToMetalSpace` >= 0.16, so do not check `eolSpace`.

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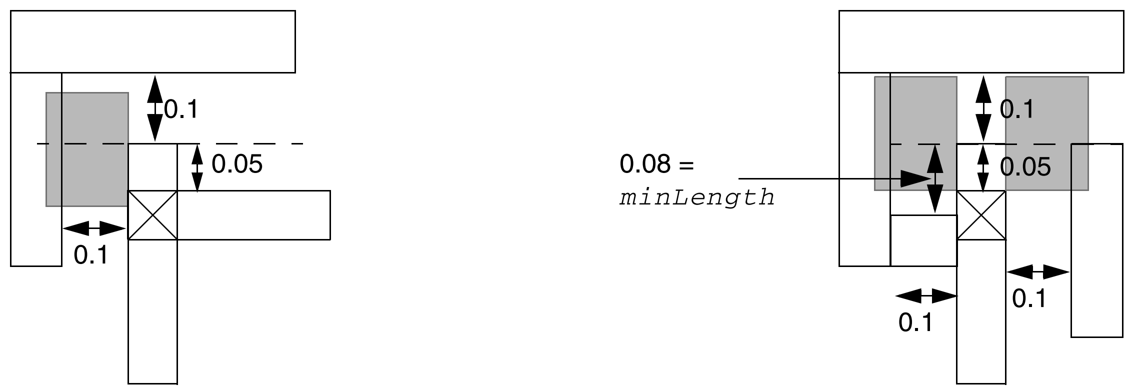
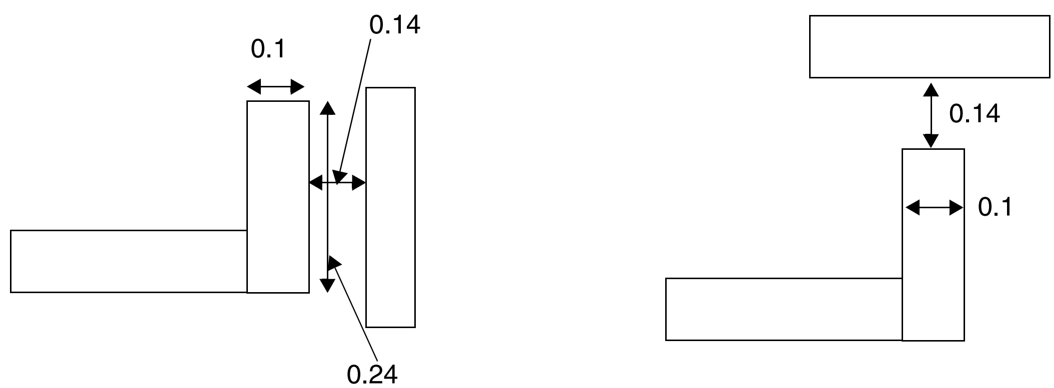


e) No violation. Does not meet *minLength* >= 0.06, so do not check *eolSpace*.

f) Violation. The *minLength* = 0.08 means no left parallel edge, but the right side has a parallel edge, so needs 0.15 *eolSpace* or *cutToMetalSpace* >= 0.16.

Figure 6-155 Example of EOL Spacing Rule with EQUALRECTWIDTH



a) No violation. The 0.24 edge is not a valid EOL edge with the `EQUALRECTWIDTH` keyword since its length is not the same as the wire width of 0.1.

b) Violation. The 0.1 edge is a valid EOL edge, even with the `EQUALRECTWIDTH` keyword, and EOL spacing is < 0.15.

```
PROPERTY LEF58_SPACING
   "SPACING 0.15 ENDOFLINE 0.25 WITHIN 0.10 EQUALRECTWIDTH;
```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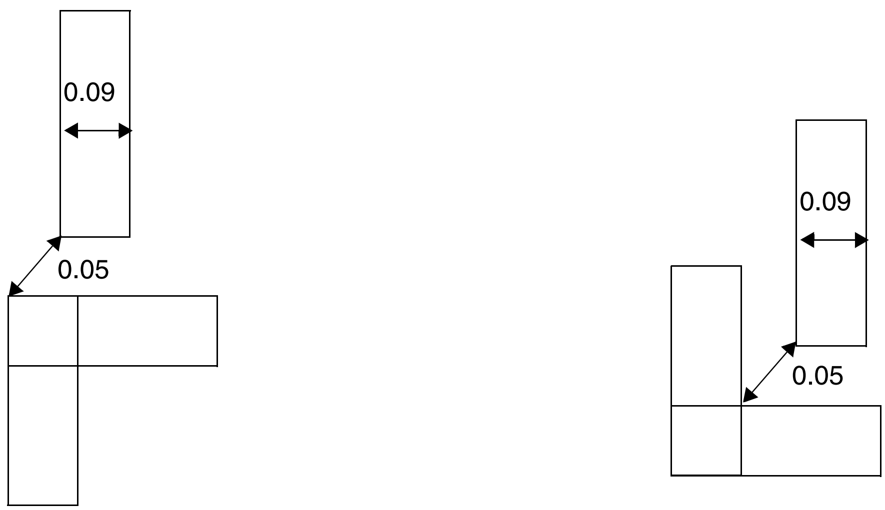
- The following example indicates that the 0.15 µm EOL spacing is applied to an EOL edge less than 0.10 µm to objects on the same mask in the routing layer. This statement does not apply to objects on different masks, and a separate EOL rule without `SAMEMASK` can be defined to an EOL spacing for different mask objects.

```
PROPERTY LEF58_SPACING
     "SPACING 0.15 ENDOFLINE 0.10 WITHIN 0.0 SAMEMASK ; " ;
```

Figure 6-156 Illustration of Spacing Rule with TOCONCAVECORNER

```
PROPERTY LEF58_SPACING
    "SPACING 0.06 ENDOFLINE 0.1 TOCONCAVECORNER ; " ;
```



a) OK, the spacing applies to concave corner only

b) Violation, the spacing to the concave corner is 0.05 (<= 0.06)

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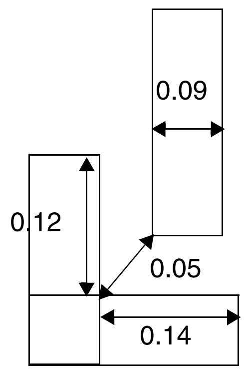
Figure 6-157 Illustration of Spacing Rule with MINADJACENTLENGTH

```
PROPERTY LEF58_SPACING
    "SPACING 0.06 ENDOFLINE 0.1
        TOCONCAVECORNER
        MINADJACENTLENGTH 0.12 ; " ;
```



a) OK, the vertical edge forming the concave corner is 0.12 (<= 0.12). It is a violation if MINADJACENTLENGTH is omitted.

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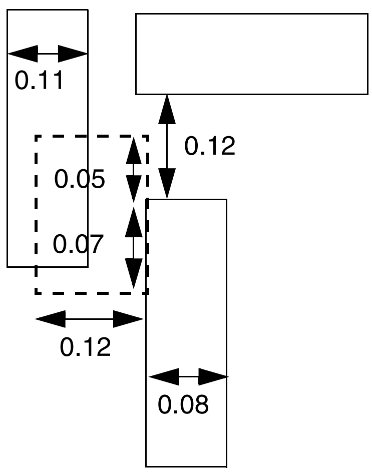
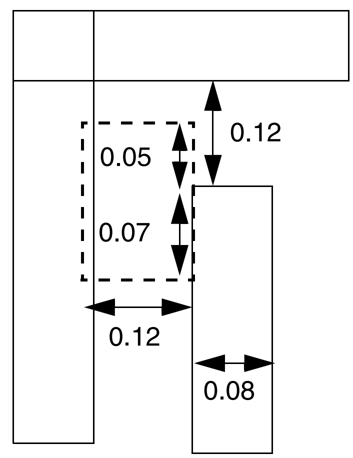
Figure 6-158 EOL Spacing Rule with NONEOLCORNERONLY

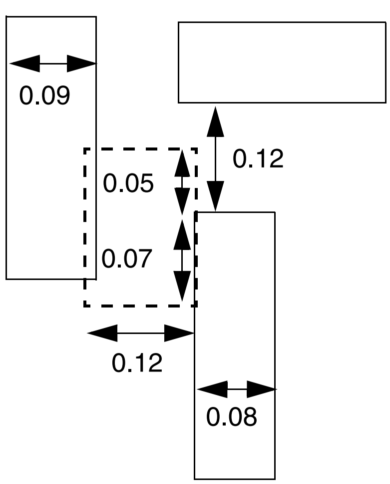
```
PROPERTY LEF58_SPACING
   "SPACING 0.13 ENDOFLINE 0.1 WITHIN 0.05
       PARALLELEDGE 0.12 WITHIN 0.07
       NONEOLCORNERONLY ; " ;
```



a) Violation, a neighbor corner is found and EOL spacing is 0.12 (<= 0.13).



b) OK, the neighbor does not have a corner in the search window



c) OK, the neighbor corner belonging to another EOL edge

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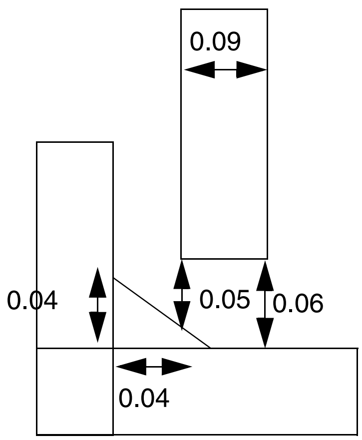
Figure 6-159 EOL Spacing Rule with FILLCONCAVECORNER

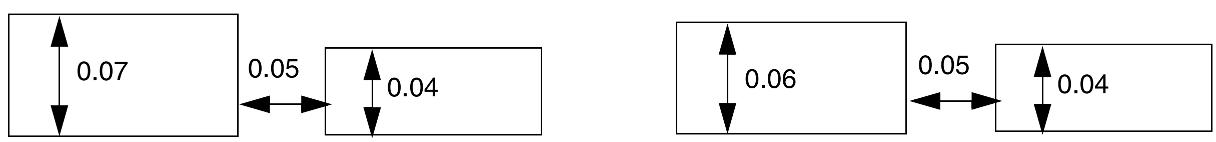
```
PROPERTY LEF58_SPACING  "
      SPACING 0.06 ENDOFLINE 0.1 WITHIN 0
         FILLCONCAVECORNER 0.04 ; " ;
```



a) Violation, spacing is only 0.05 to the triangular fill with length of 0.04 along the edges of the concave corner

Figure 6-160 EOL Spacing Rule with EXCEPTEXACTWIDTH

```
SPACING 0.05 ;
PROPERTY LEF58_SPACING  "
   SPACING 0.06 ENDOFLINE 0.1 WITHIN 0
     EXCEPTEXACTWIDTH 0.07 0.05 ; " ;
```



a) OK, the width of the left EOL edge is exactly equal to 0.07 while the right edge has width less than 0.05 to exempt the end-of-line spacing rule

b) Violation, there is no EOL edge with width exactly equal to 0.07

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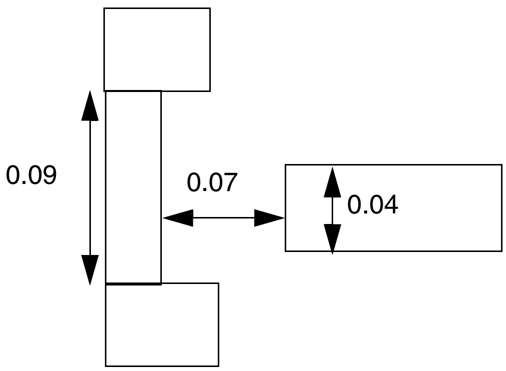
Figure 6-161 EOL Spacing Rule with TONOTCHLENGTH

```
PROPERTY LEF58_SPACING  "
   SPACING 0.08 ENDOFLINE 0.05
     TONOTCHLENGTH 0.1 ; " ;
```



a) Violation, the EOL edge is 0.07 away from
a notch with length of 0.09 (< 0.1)

Convex Corners Spacing Rule

A convex corners spacing rule can be used to define spacing between two convex corners of two parallel wires

You can create convex corners spacing rule by using the following property definition:

```
PROPERTY LEF58_SPACING
     "SPACING {spacing | minSpacing maxSpacing }
          CONVEXCORNERS EXTENSION sideExt orthogonalExt
          [ SAMESIDE extension
          | SINGLE edgeForwardExt cornerForwardExt cornerBackwardExt
                SPANLENGTH spanLength OPPOSITEWIDTH oppWidth
                OPPOSITEEXTENSION oppSideExt
                    oppForwardExt1 oppForwardExt2
          | MIDDLEWIRES {0 | 1 | 2} [INCLUDEPARTIALOVERALP]
                [BACKWARDWIRE sideExt backwardExt]
                [REVERSE
                    [EXCEPTWRONGDIRWIRE edgeExt wrongDirEdgeExt]
                    [EXCEPTCONVEXCORNER edgeBackwardExt
                        edgeForwardExt wrongDirEdgeExt]]]
         ; " ;
```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Where:

`BACKWARDWIRE` *`sideExt backwardExt`*

Specifies that a window is formed from the opposite corners of the two facing convex corners of the rule by extending *`sideExt`* along the edge and going backward *`backwardExt`* in the orthogonal direction. The rule applies only if there is a neighboring wire overlapping with at least one of the windows. In other words, if there is no wire overlap with either of the windows, the rule is exempted.

Type: Float, specified in microns

`EXCEPTCONVEXCORNER` *`edgeBackwardExt edgeForwardExt wrongDirEdgeExt`*

Specifies that if a convex corner of a default minimum-width wire overlaps with the window formed by extending *`edgeBackwardExt`* backward and *`edgeForwardExt`* forward in the preferred routing direction and *`wrongDirEdgeExt`* in the non-preferred routing direction from a facing neighbor convex corner, it would not be a violation. However, the facing neighbor convex corner must not touch an edge with the default minimum-width end-of-line, with two convex corners.

Type: Float, specified in microns

`EXCEPTWRONGDIRWIRE` *`edgeExt wrongDirEdgeExt`*

Specifies that a wrong-direction wire with a width equal to the minimum width/span length in the preferred routing direction would form a window by extending *`edgeExt`* in the preferred routing direction and *`wrongDirEdgeExt`* in the non-preferred routing direction based on the wrong direction wire. Any convex corners of a default minimum-width wire overlapping with the window would not be a violation.

Type: Float, specified in microns

`INCLUDEPARTIALOVERALP`

Specifies that if the neighboring wires merely touch or overlap with the window formed by the two facing convex corners, it is a violation.

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`MIDDLEWIRES {0 | 1 | 2}`

Specifies the number of wires that should be present between the two corners to trigger the rule. By default, two minimum-width neighboring wires with minimum spacing fully across the window formed by the corners must be present to trigger the rule. `0` means that the two corners are projected minimum spacing apart without any middle wires. In addition, the two facing convex corners must belong to wires with default minimum width. The metal search windows would also be extended from the opposite convex corners of the two facing convex corners by *`sideExt`* inwardly.

`REVERSE`

Specifies that the rule has reverse meaning in `MIDDLEWIRES`. Passing in `MIDDLEWIRES` would mean violation in `REVERSE` and vice versa. The rule is still checked on a convex corner belonging to a default minimum-width wire. Violations can only happen on that corner, or the edge containing the corner. For example, if a convex corner cannot find a facing convex corner in the given spacing, it would be a violation. For `MIDDELEWIRES 1`, if there is no minimum-width neighboring wire with minimum spacing fully across the window formed by the corners, it is a violation. If any one of the search windows formed on the convex corners by the given values in `EXTENSION` overlaps with a neighbor, it is a violation.

In addition, if multiple statements are defined with `REVERSE`, it is only a violation if all of the statements with `REVERSE` fail. If there are multiple statements with some having `REVERSE` and some without, only the statements with `REVERSE` would be considered together. The statements without `REVERSE` would be checked separately. In addition, the statements with `REVERSE` would not be checked in routing and the DRC checker. They are used by another command, which would use the violation markers for some other special purposes.

`SPACING {`*`spacing`*` | `*`minSpacing maxSpacing`*` }`
`    CONVEXCORNERS EXTENSION `*`sideExt orthogonalExt`*

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Specifies that the spacing between two convex corners of two parallel wires in the opposite sides facing each other must be greater than or equal to *spacing* in the direction of the parallel wires if the window formed by the corners has exactly two minimum-width neighboring wires fully across the window. Also, all of the wires, including the ones containing the corners, are exactly minimum spacing apart. If two spacing values are given, the spacing between the two convex corners must be less than or equal to *minSpacing* or greater than or equal to *maxSpacing*. If *minSpacing* is negative, the two convex corners must go past each other by greater than or equal to abs(*minSpacing*).

In addition, both the search windows by extending sideways away from the two middle neighboring wires by *sideExt* and forward along the parallel wires by *orthogonalExt* from the corners do not contain any neighboring wires.

Type: Float, specified in microns

SAMESIDE *extension*

Specifies that the parallel wires are on the same side instead of facing each other, and the window formed by the corners should be extended by *extension* along the wire direction. In addition, if there is a neighbor wire touching the edge of the search windows from the convex corners that faces the two middle parallel wires, it is considered a valid neighbor. If a neighbor wire touches any of the other edges of the search windows, the wire is not counted as neighbor.
Type: Float, specified in microns

SINGLE *edgeForwardExt cornerForwardExt cornerBackwardExt*
SPANLENGTH *spanLength* OPPOSITEWIDTH *oppWidth*
OPPOSITEEXTENSION *oppSideExt oppForwardExt1 oppForwardExt2*

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Specifies a forbidden zone of a single convex corner to any wires to be a *spacing* distance beyond a virtual edge from the corner and extending minimum spacing in the direction parallel to the edge that contains the corner, if all of the conditions described below are met.

The wire containing the corner must have width exactly equal to minimum width and span length greater than or equal to *spanLength*. There is no wire beyond the edge containing the corner by *edgeForwardExt*. A search window by extending side way away from the corner by *sideExt* and backward by *orthogonalExt* does not contain any neighbor wires. There is a neighbor wire with width less than or equal to the *oppWidth* fully abutted to a search window formed from the opposite corner by extending minimum spacing on the side and *cornerForwardExt* and *cornerBackwardExt* in forward and backward direction. A search window by extending sideways from the opposite side of that neighbor wire by *oppSideExt* and projected forward from the corner from *oppForwardExt1* to *oppForwardExt2* must overlap or touch a wire.
Type: Float, specified in microns

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Figure 6-162 Illustration of Spacing Rule with CONVEXCORNERS

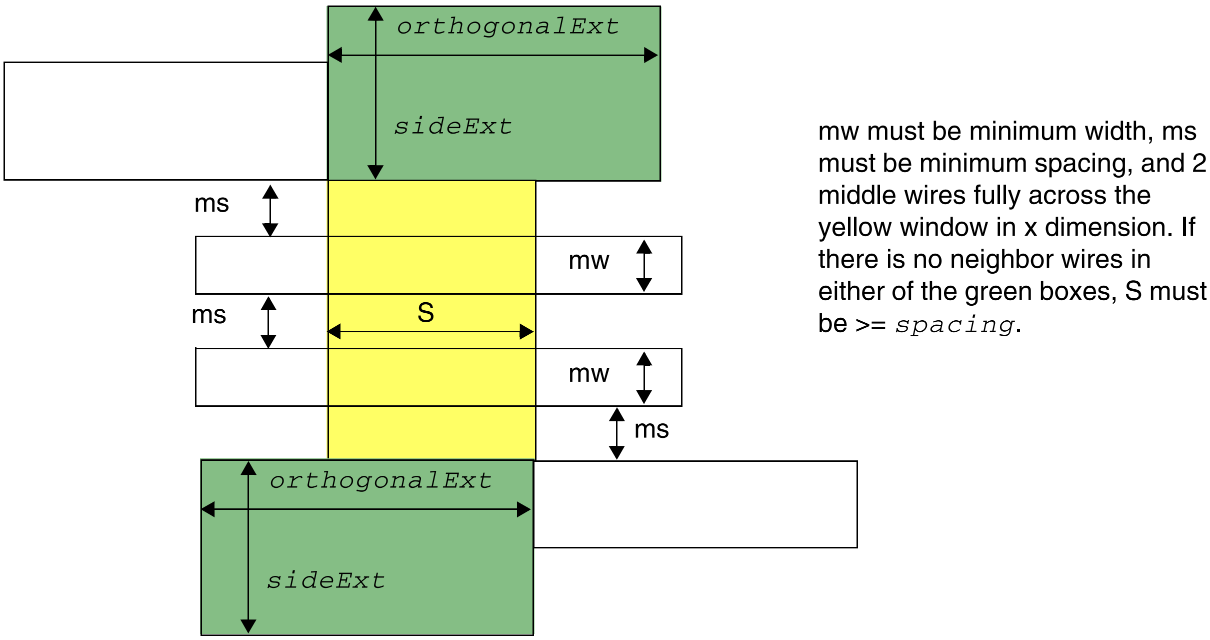

Illustration of `SPACING` *`spacing`* `CONVEXCORNERS EXTENSION` *`sideExt orthogonalExt`*

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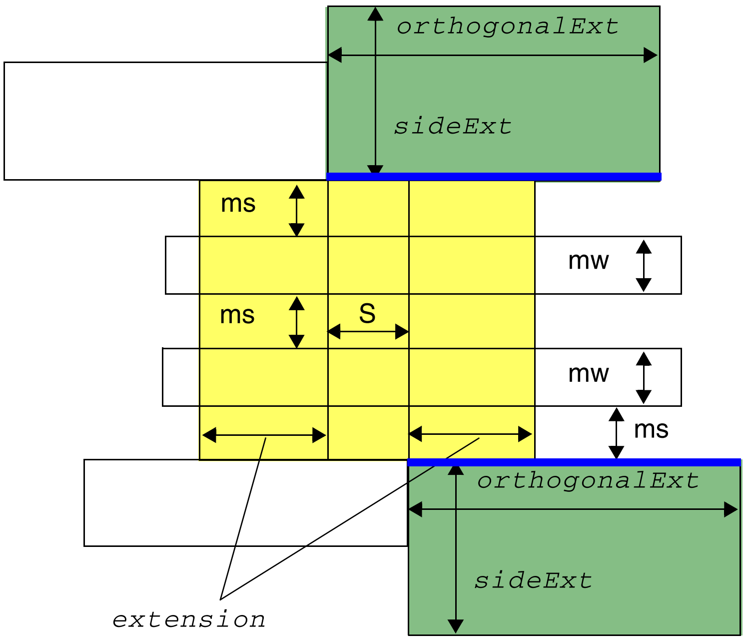
Figure 6-163 Illustration of Spacing Rule with CONVEXCORNERS



mw must be minimum width, ms must be minimum spacing, and 2 middle wires fully across the yellow windows in x dimension. If there is no neighbor wires in either of the green boxes, S must be >= spacing. Objects that barely touch the blue edges are considered as neighbor.

Illustration of `SPACING` *`spacing`* `CONVEXCORNERS EXTENSION` *`sideExt orthogonalExt`* `SAMESIDE` *`extension`*

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Figure 6-164 Illustration of Spacing Rule with CONVEXCORNERS

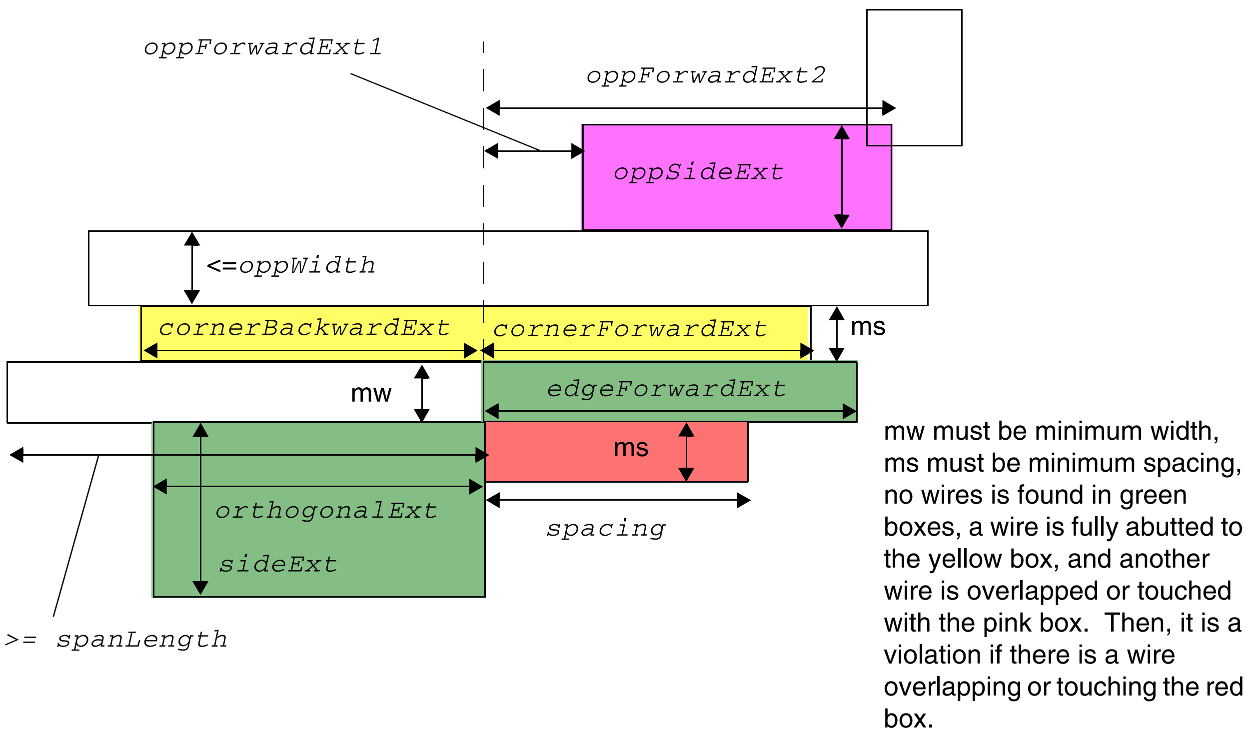

Illustration of `SPACING` *`spacing`* `CONVEXCORNERS`
`EXTENSION` *`sideExt orthogonalExt`*
`SINGLE` *`edgeForwardExt cornerForwardExt`*
`cornerBackwardExt`
`SPANLENGTH` *`spanLength`*
`OPPOSITEWIDTH` *`oppWidth`*
`OPPOSITEEXTENSION` *`oppSideExt oppForwardExt1`*
`oppForwardExt2`

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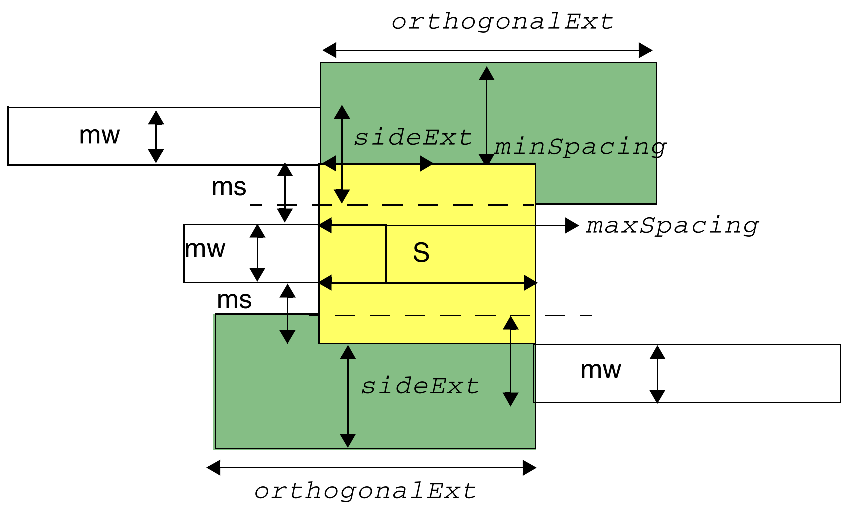
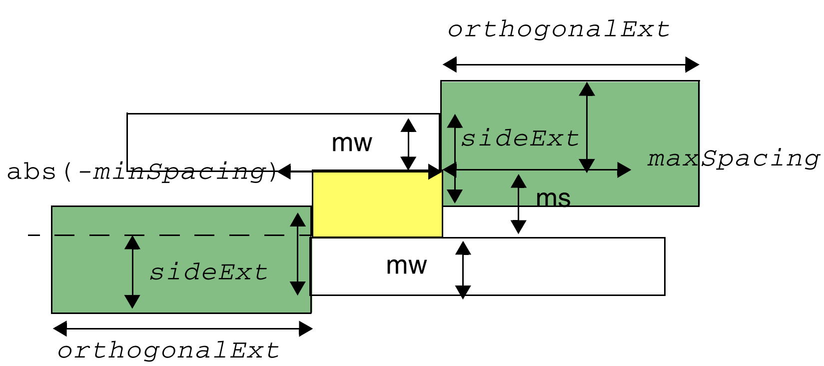
Figure 6-165 Illustrations of Spacing Rule with MIDDLEWIRES in CONVEXCORNERS



One middle wire touches or overlaps the yellow window in the x dimension and `S` is greater than *`minSpacing`* and less than *`maxSpacing`*. It is a violation.

Illustration of `SPACING` *`minSpacing maxSpacing`*
`CONVEXCORNERS EXTENSION` *`sideExt orthogonalExt`*
`MIDDLEWIRES 1 INCLUDEPARTIALOVERLAP`



The two convex corners are within `abs(-`*`minSpacing`*`)` to be a violation.

Illustration of `SPACING` -*`minSpacing maxSpacing`*
`CONVEXCORNERS EXTENSION` *`sideExt orthogonalExt`*
`MIDDLEWIRES 0`

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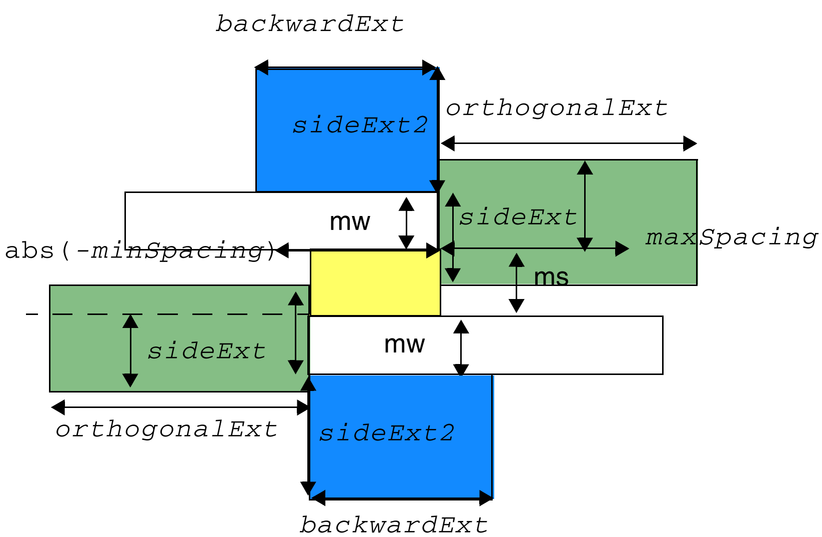
Figure 6-166 Illustration of Spacing Rule with BACKWARDWIRE in CONVEXCORNERS



OK because no wire is overlapping with either of the blue windows.

Illustration of `SPACING` -*minSpacing maxSpacing*
`CONVEXCORNERS EXTENSION` *sideExt orthogonalExt*
`MIDDLEWIRES 0 BACKWARDWIRE` *sideExt2 backwardExt*

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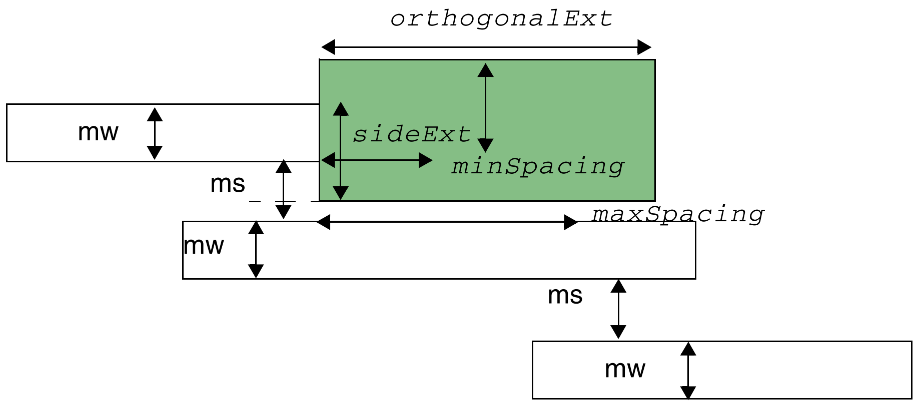
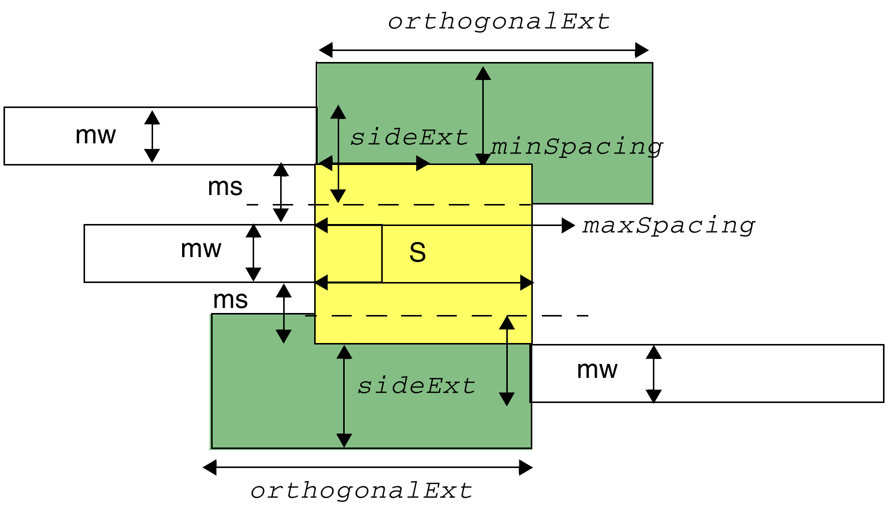
Figure 6-167 Illustrations of Spacing Rule with REVERSE in CONVEXCORNERS

It is a violation since there is no facing convex corner in the given spacing.





It is a violation since the middle wire does not fully cover the yellow window.

It is a violation since the bottom green search window has a neighbor.

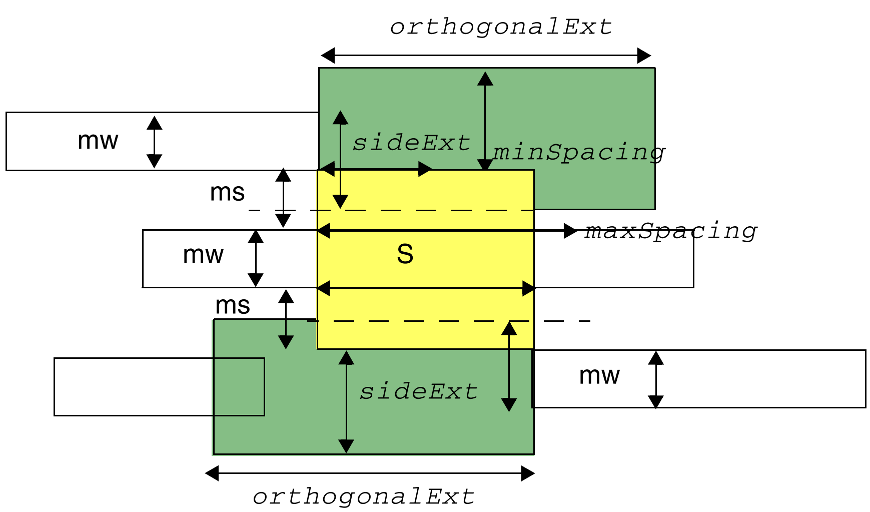


Illustration of `SPACING` *`minSpacing maxSpacing`*
`CONVEXCORNERS EXTENSION` *`sideExt orthogonalExt`*
`MIDDLEWIRES 1 REVERSE`

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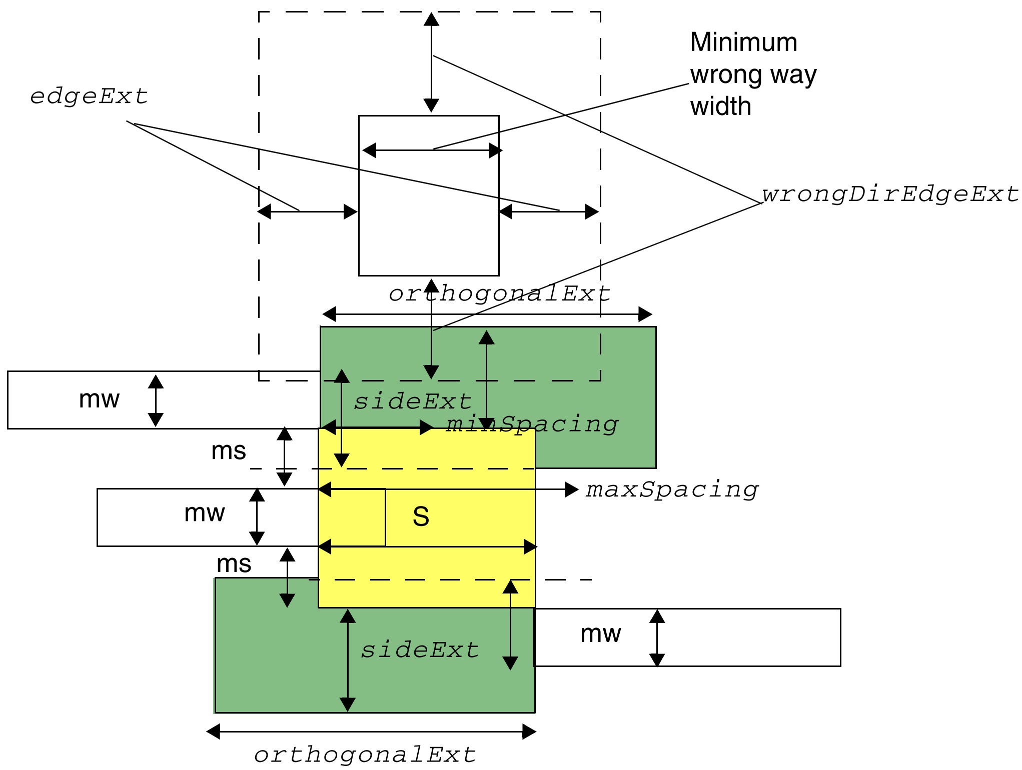
Figure 6-168 Illustration of Spacing Rule with EXCEPTWRONGDIRWIRE in CONVEXCORNERS



It is OK since the top mw wire overlaps with a window formed by a wrong way wire in `EXCEPTWRONGDIRWIRE`.

Illustration of `SPACING` *`minSpacing maxSpacing`*
`CONVEXCORNERS EXTENSION` *`sideExt orthogonalExt`*
`MIDDLEWIRES 1 REVERSE`
`EXCEPTWRONGDIRWIRE` *`edgeExt wrongDirEdgeExt`*

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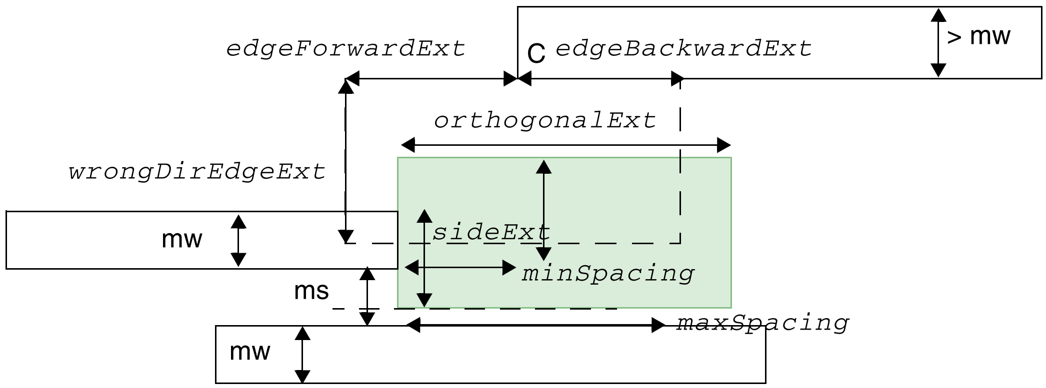
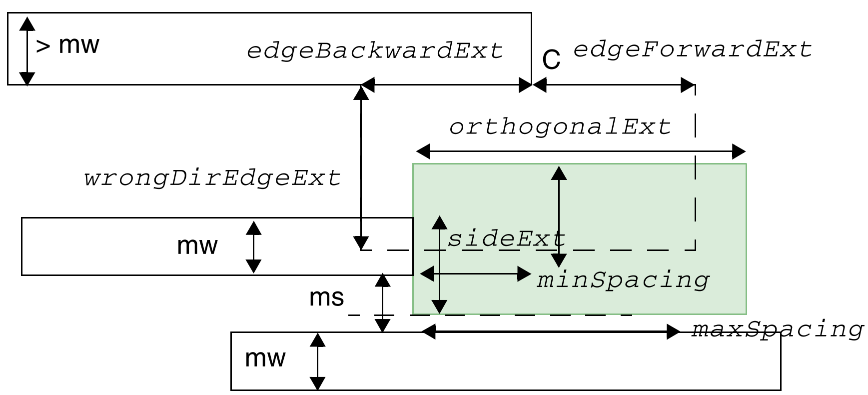
Figure 6-169 Illustration of Spacing Rule with EXCEPTCONVEXCORNER in CONVEXCORNERS



It is OK since the top mw wire overlaps with a window formed at a convex corner `C` in `EXCEPTCONVEXCORNER`.



Violation because corner `C` is not a facing convex corner.

Illustrations of `SPACING` *`minSpacing maxSpacing`*
`CONVEXCORNERS EXTENSION` *`sideExt orthogonalExt`*
`MIDDLEWIRES 0 REVERSE`
`EXCEPTCONVEXCORNER` *`edgeBackwardExt`*
`edgeForwardExt wrongDirEdgeExt`

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Perpendicular EOL Spacing Rule

Perpendicular EOL spacing rules can be specified to allow a different spacing rule for perpendicular edges versus opposite edges.

You can create a perpendicular EOL spacing rule by using the following property definition:

```
PROPERTY LEF58_SPACING
     "SPACING eolSpace EOLPERPENDICULAR eolWidth perWidth ;" ;
```

Where:

`SPACING` *`eolSpace`* `EOLPERPENDICULAR` *`eolWidth perWidth`*

Indicates that an EOL edge with a width that is less than *`eolWidth`* requires spacing greater than or equal to *`eolSpace`* beyond the EOL, to a perpendicular edge with a width that is less than *`perWidth`*. Typically, *`eolSpace`* is slightly larger than the minimum allowed spacing on this layer.
Type: Float, specified in microns (for all values)

Perpendicular EOL Spacing Rule Examples

Figure 6-170 Illustration of Perpendicular EOL Spacing Parameters

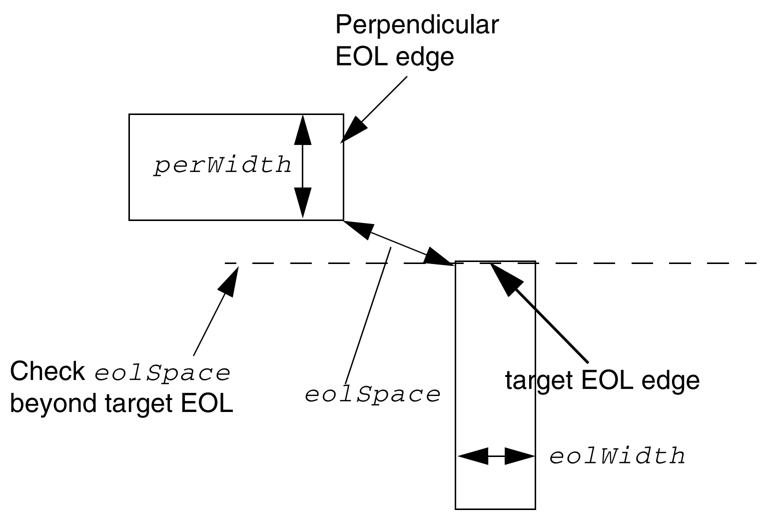

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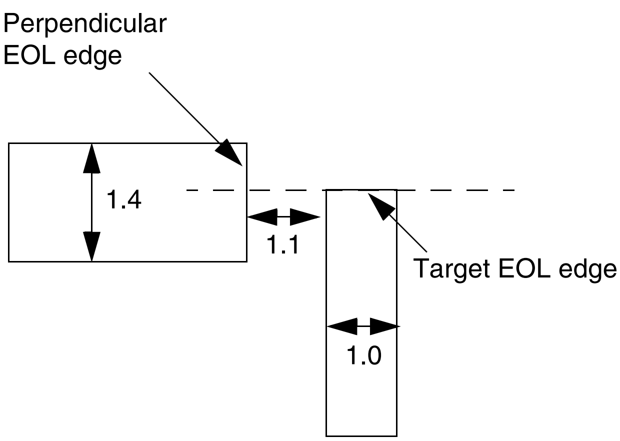
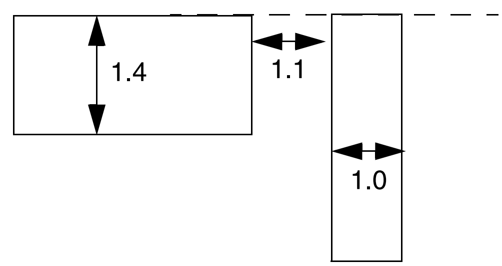
Figure 6-171 Example of Perpendicular EOL Spacing Rule

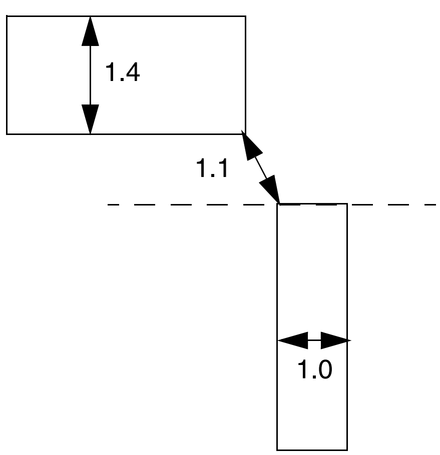
```
PROPERTY LEF58_SPACING "SPACING 1.2 EOLPERPENDICULAR 1.1 1.5 ;" ;
```



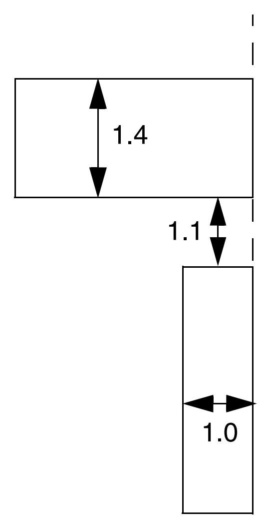
a) Violation. EOL is < 1.1 wide, and has a perpendicular edge < 1.5 wide that is < 1.2 distance from EOL edge.



b) No violation. Perpendicular edge is not "seen" by EOL edge, so it is not checked.



c) Violation. EOL is < 1.1 wide, and has a perpendicular edge < 1.5 wide that is < 1.2 distance from EOL edge.



d) No violation. Perpendicular edge is not "seen" by EOL edge, so it is not checked.

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Spacing Area Rule

Area spacing rules can be specified between objects having area less than the maximum area to any other objects in the layer.

You can create an area spacing rule by using the following property definition:

```
PROPERTY LEF58_SPACING
     "SPACING minSpacing AREA maxArea ;" ;
```

Where:

`AREA maxArea`

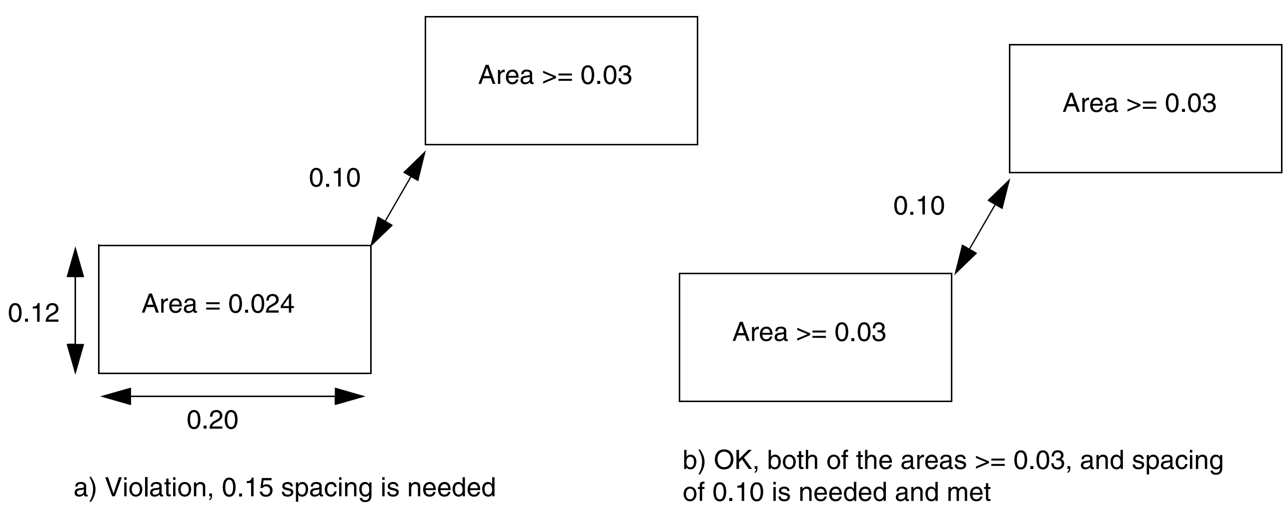
Specifies the minimum spacing rule only applies between an object with area less than *maxArea* to any other objects in the layer.
Type: Float, specified in microns squared

Spacing Area Rule Examples

The following spacing rule indicates that the minimum spacing of an object with area less than 0.03 µm² is 0.15 µm, while the minimum spacing of another object in the layer is 0.10 µm:

```
SPACING 0.10 ;
PROPERTY LEF58_SPACING
   "SPACING 0.15 AREA 0.03 ;" ;
```

Figure 6-172 Illustration of Spacing Area Rule



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Spacing Layer Rule

Spacing layer rules can be specified for shapes in a TRIMMETAL layer.

You can create a spacing layer rule by using the following property definition:

```
PROPERTY LEF58_SPACING
     "SPACING minSpacing LAYER trimLayer [TRIMMASK trimMaskNum]
     [MASK maskNum] EXCEPTOVERLAP overlapLength
     [TRIMENDSPACING trimEndSpacing [PRL prl]]
     ;" ;
```

Where:

`PRL prl`

Specifies that the trim end edge-to-metal spacing applies when PRL between them is greater than *prl*. If *prl* is negative, it is measured in the WITHIN style. The *trimEndSpacing* would be measured in the MAXXY style in the orthogonal direction.

Type: Float, specified in microns

`SPACING minSpacing LAYER trimLayer EXCEPTOVERLAP overlapLength`

Specifies that the spacing to shapes in *trimLayer*, which must be a layer with TYPE TRIMMETAL, should be greater than or equal to *minSpacing*. However, there is an exemption. When a shape in the current layer overlaps with a *trimLayer* shape by a length that is greater than or equal to *overlapLength* as measured in the preferred routing direction in the current layer, the spacing check against that *trimLayer* shape is skipped. If *overlapLength* is a positive value, which is larger than the trim width, the metal needs to go beyond the '*overlapLength* value – trim width value' for the rule to be exempted. If *overlapLength* is a negative value and if the two shapes are within the abs(*overlapLength*) value, the spacing rule is not applied. This is particularly meaningful when the trim metal shape and the metal shape are in a corner-to-corner case.
Type: Float, specified in microns

`MASK maskNum`

Specifies that spacing to shapes in *trimLayer*, applies only on the given mask of the current layer. *maskNum* must be a positive integer, and most applications only support values of 1, 2 or 3.

`TRIMENDSPACING trimEndSpacing`

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Specifies that the metal spacing to the end edge, which is parallel to the preferred routing direction of the metal routing layer, of shapes in *trimLayer* should be greater than or equal to *trimEndSpacing*.
Type: Float, specified in microns

TRIMMASK *trimMaskNum*

Specifies that the spacing for shapes in *trimLayer* applies only on the given mask of the *trimLayer* layer. *trimMaskNum* must be a positive integer, and most applications only support values of 1, 2, or 3.

Spacing Layer Examples

- The following spacing rule indicates that the spacing to shapes in TRIMMETAL layer TRIM1 should be greater than or equal to 0.1 µm, and the spacing to the end edges of shapes in the TRIMMETAL layer TRIM1 should be greater than or equal to 0.12 µm. When a shape in the current layer overlaps with a TRIM1 shape by a length greater than or equal to 0.05 in the horizontal direction in the current layer, the spacing check against that TRIM1 shape is skipped:

```
DIRECTION HORIZONTAL ;
PROPERTY LEF58_SPACING "
   SPACING 0.1 LAYER TRIM1 EXCEPTOVERLAP 0.05
        TRIMENDSPACING 0.12;" ;
```


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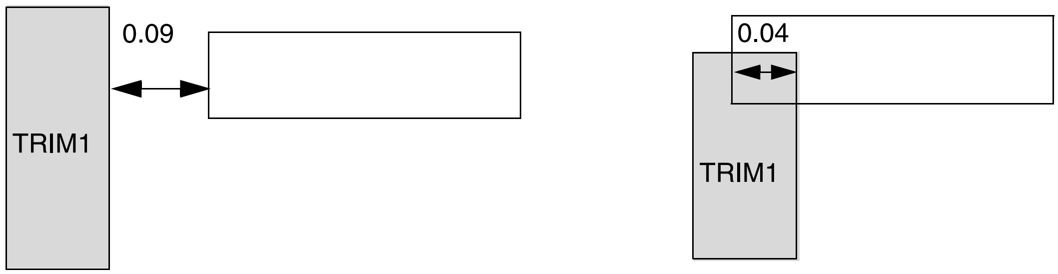
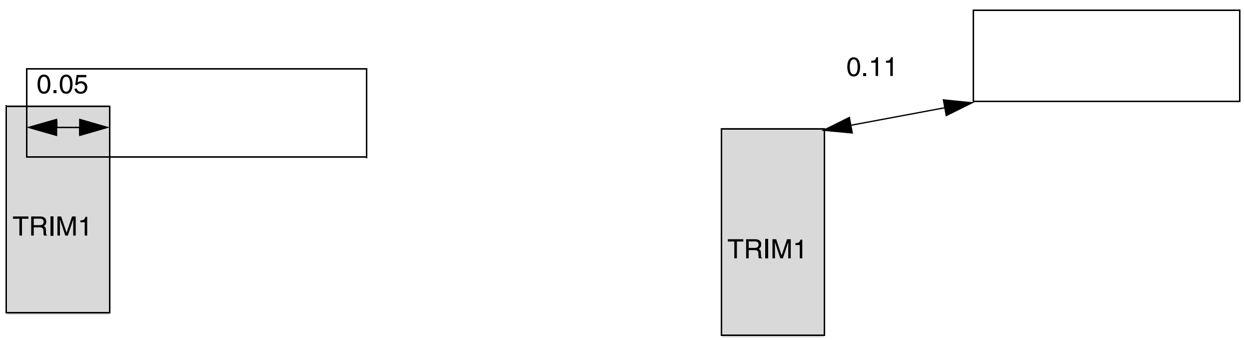
Figure 6-173 Illustration of Spacing Layer Rule



a) Violation, spacing to TRIM1 needs to be >= 0.1

b) Violation, horizontal (preferred routing direction) overlap must be > = 0.05 to be exempted



c) OK, overlap exemption condition is met

d) Violation, 0.12 spacing is needed to the end edge of the TRIM1 shape and is not met

- The following example illustrates a spacing layer rule with negative overlap length:

```
DIRECTION HORIZONTAL ;
PROPERTY LEF58_SPACING “
   SPACING 0.1 LAYER TRIM1 EXCEPTOVERLAP -0.05 ; ” ;
```


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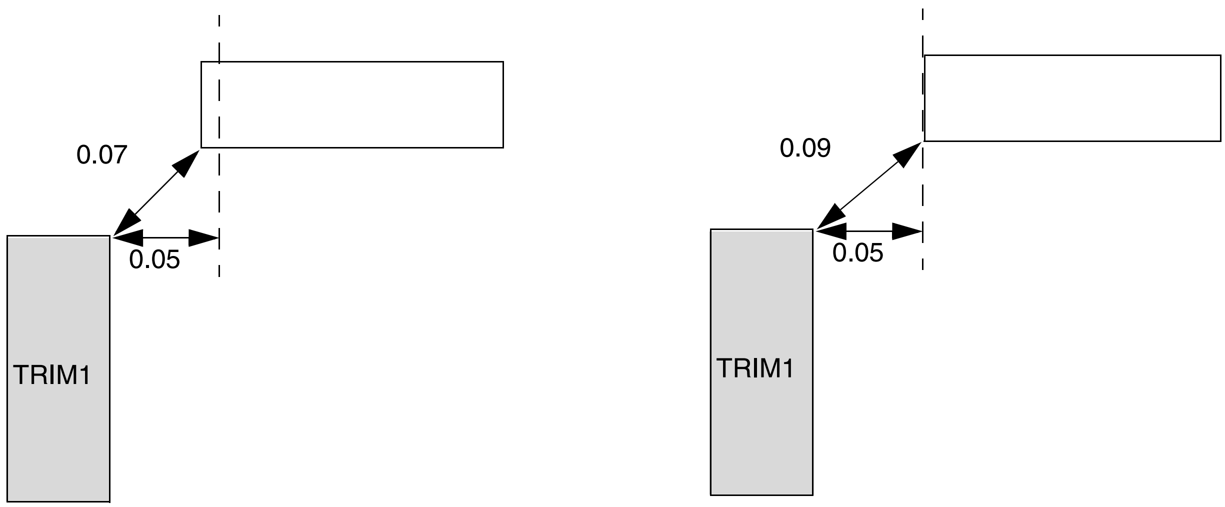
Figure 6-174 Illustration of Spacing Layer Rule with Negative Overlap Length



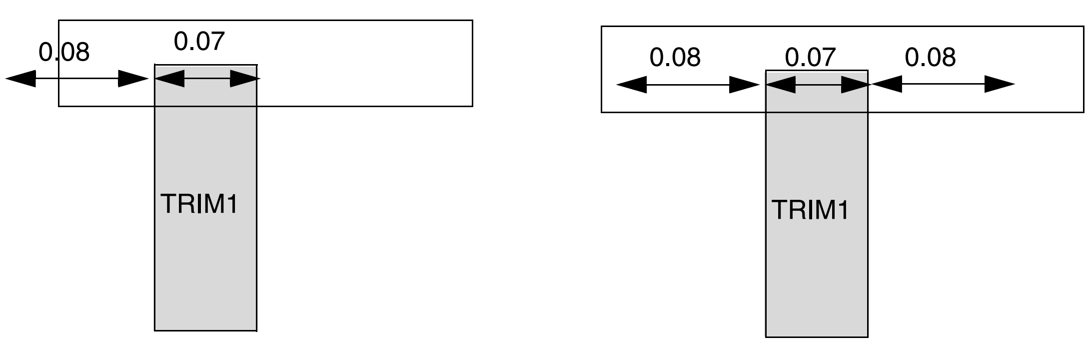
a) OK, the metal shape is within `0.05`, so the rule is not applicable

b) Violation, the metal shape is just outside `0.05`, and spacing of `0.09` (< `0.1`) fails

- The following example illustrates a spacing layer rule with positive overlap length:

```
DIRECTION HORIZONTAL ;
PROPERTY LEF58_SPACING "
   SPACING 0.1 LAYER TRIM1 EXCEPTOVERLAP 0.15 ; " ;
```

Figure 6-175 Illustration of Spacing Layer Rule with Positive Overlap Length



a) Violation, the metal does not go beyond 0.08 (0.15 – 0.07) to be exempted from the rule.

b) OK, the metal goes beyond `0.08` to fulfill the `EXCEPTOVERLAP` condition and be exempted from the rule.

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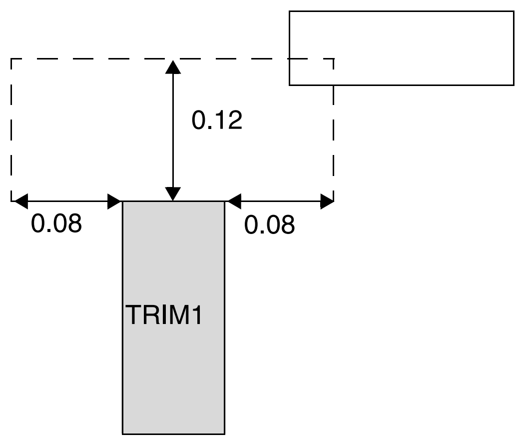
- The following example illustrates a spacing layer rule with negative PRL:

```
DIRECTION HORIZONTAL ;
PROPERTY LEF58_SPACING "
   SPACING 0.1 LAYER TRIM1 EXCEPTOVERLAP -0.05
       TRIMENDSPACING 0.12 PRL -0.08 ; " ;
```

Figure 6-176 Illustration of Spacing Layer Rule with Negative PRL



a) Violation, a wire overlaps with the dotted window formed by `abs(-0.08)` of PRL and `0.12` of trim end spacing.

Same Mask Spacing Rule

Same mask spacing rule exists for double patterning technology, when two masks with different spacing requirements among them are applied on one routing layer. This adds a same mask spacing.

You can create same mask spacing rule by using the following property definition:

```
PROPERTY LEF58_SPACING
     "SPACING minSpacing SAMEMASK ;" ;
```

Where:

`SAMEMASK`	Specifies the minimum spacing rule only applies between objects on the same mask.

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Same Mask Spacing Rule Examples

- The following same mask spacing rule indicates that the minimum spacing of objects on the same mask in the routing layer is 0.15 µm while the minimum spacing of objects on different masks is 0.10 µm.

```
SPACING 0.10 ;
PROPERTY LEF58_SPACING
    "SPACING 0.15 SAMEMASK ; " ;
```

Wrong Direction Spacing Rule

Wrong direction spacing rule exists for minimum spacing, which is larger than the right direction spacing, between the long or side edges of two wires with direction perpendicular to the specified direction in `DIRECTION` on a Manhattan routing layer.

You can create wrong direction spacing rule by using the following property definition:

```
PROPERTY LEF58_SPACING
     "SPACING minSpacing [WRONGDIRECTION [NONEOL eolWidth]
            [PRL prl]]
            ;" ;
```

Where:

All other keywords are the same as the existing LEF routing layer `SPACING` syntax.

`NONEOL` *`eolWidth`*	Specifies the minimum spacing between two edges having parallel run length greater than 0 with direction perpendicular to the specified direction in `DIRECTION` on a Manhattan routing layer. Those edges are not end-of-line edges with length less than the *`eolWidth`*. *Type*: Float, specified in microns
`PRL` *`prl`*	Specifies that the wrong direction spacing is applied only if the two edges have common parallel run length greater than *`prl`* in the non-preferred routing direction. If *`prl`* is a negative value, the two edges will be checked in `WITHIN` style. *Type*: Float, specified in microns
`WRONGDIRECTION`	Specifies the minimum spacing, which is larger than the right direction spacing, between two edges having parallel run length greater than 0 with direction perpendicular to the specified direction in `DIRECTION` on a Manhattan routing layer.

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Examples

- The following example illustrates wrong direction spacing rule:

```
DIRECTION VERTICAL ;
SPACING 0.10 ;
PROPERTY LEF58_SPACING
    "SPACING 0.12 WRONGDIRECTION ; " ;
```


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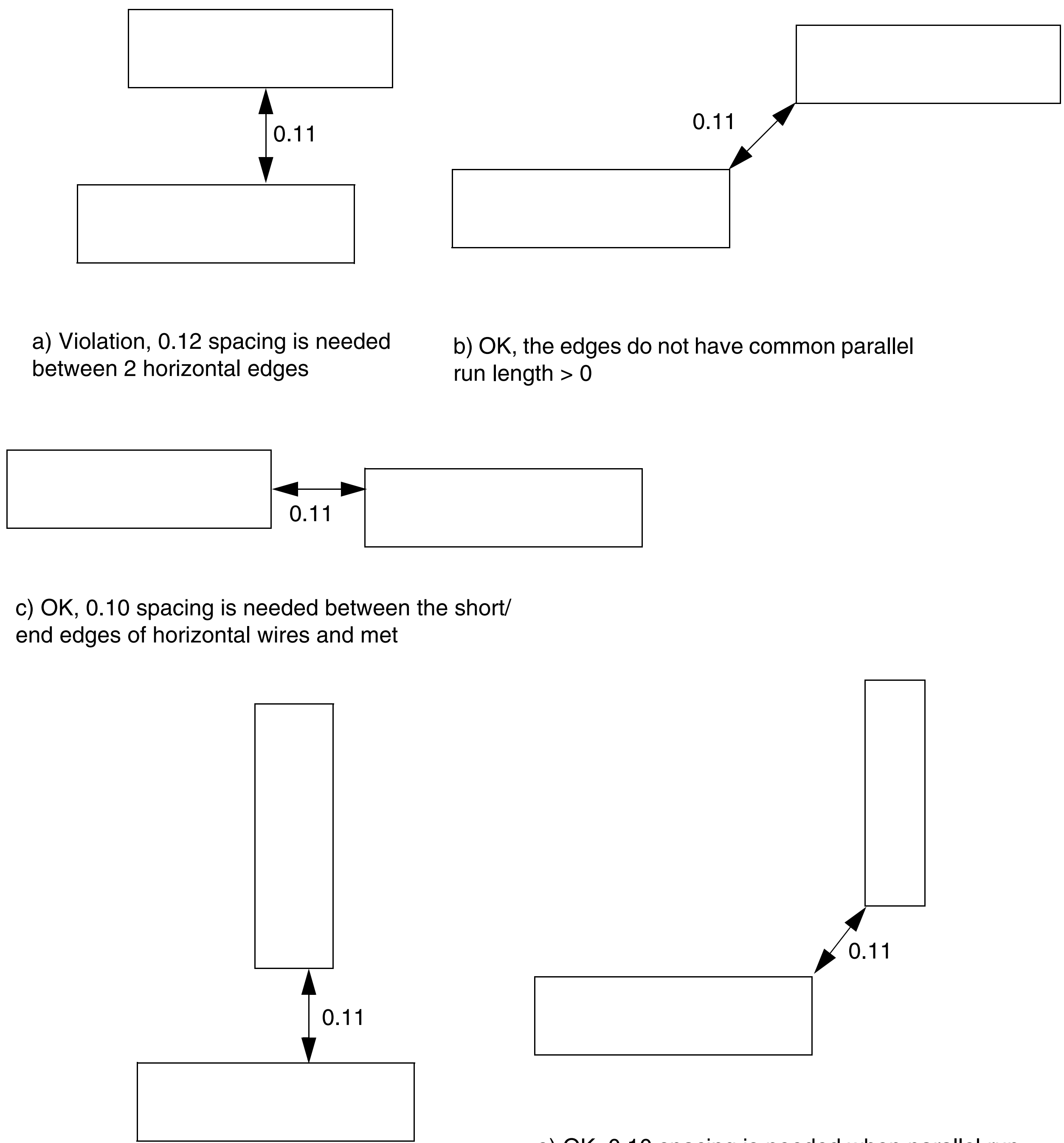
Figure 6-177 Illustration of Spacing with WRONGDIRECTION



a) Violation, 0.12 spacing is needed between 2 horizontal edges

b) OK, the edges do not have common parallel run length > 0

c) OK, 0.10 spacing is needed between the short/ end edges of horizontal wires and met

d) Violation, 0.12 spacing is needed between 2 horizontal edges independent of the wire direction

e) OK, 0.10 spacing is needed when parallel run length <= 0

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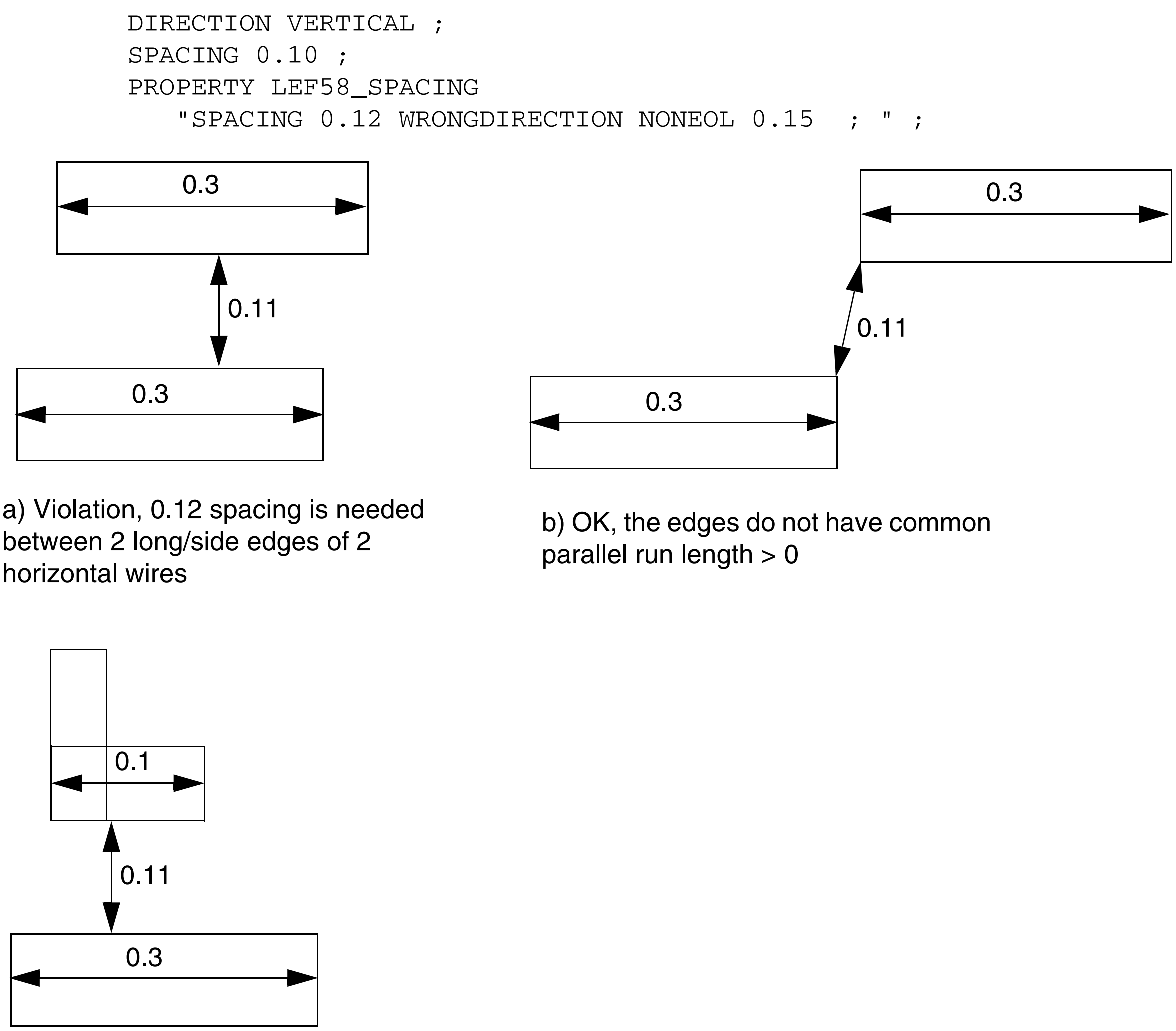
- The following wrong direction spacing rule indicates that the long/side edges between two horizontal wires must have a spacing of 0.12 μm if they are not EOL edges with length less than 0.15 μm and they have common parallel run length greater than 0. Otherwise, 0.10 μm is the minimum spacing.

```
DIRECTION VERTICAL ;
SPACING 0.10 ;
PROPERTY LEF58_SPACING
    "SPACING 0.12 WRONGDIRECTION NONEOL 0.15 ; " ;
```

Figure 6-178 Illustration of WRONGDIRECTION with NONEOL



a) Violation, 0.12 spacing is needed between 2 long/side edges of 2 horizontal wires

b) OK, the edges do not have common parallel run length > 0

c) OK, the top horizontal edge is an EOL edge, only 0.10 spacing is needed and met

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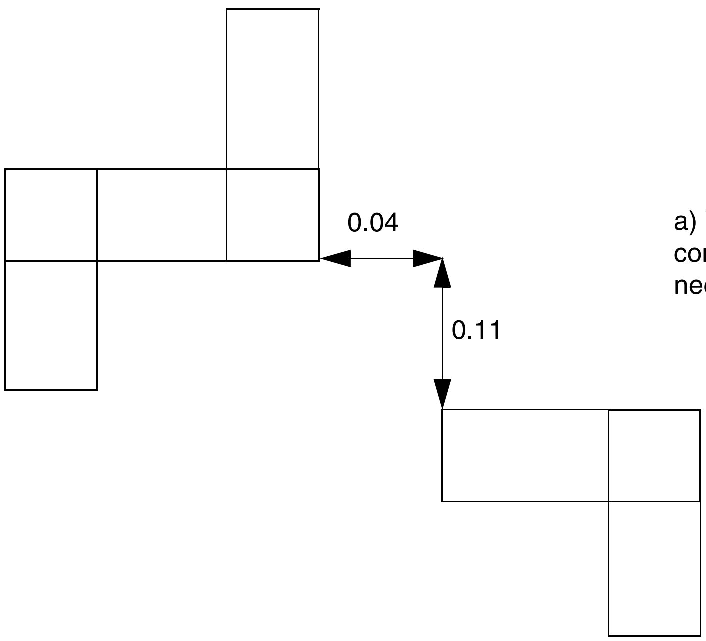
Figure 6-179 Illustration of WRONGDIRECTION with PRL

```
DIRECTION VERTICAL ;
SPACING 0.10 ;
PROPERTY LEF58_SPACING
   "SPACING 0.12 WRONGDIRECTION
        PRL -0.05 ; " ;
```



a) Violation, the parallel run length condition is met, 0.12 spacing is needed

Notch Length Spacing Rule

Notch length spacing rule exists for any notch with a notch length less than the minimum notch length with notch spacing of less than or equal to the mininmum spacing.

You can create notch length spacing rule by using the following property definition:

```
PROPERTY LEF58_SPACING
     "SPACING minSpacing NOTCHLENGTH minNotchLength
     [EXCEPTWITHIN lowExcludeSpacing highExcludeSpacing]
         [WITHIN within SPANLENGTH sideOfNotchSpanLength
         | {WIDTH | CONCAVEENDS} sideOfNotchWidth
         | NOTCHWIDTH notchWidth]
          ;" ;
```

Where:

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All other keywords are the same as the existing LEF routing layer `SPACING` syntax.

`CONCAVEENDS` *`sideOfNotchWidth`*

Specifies the minimum notch length spacing only applies if the width of both sides of the notch must be less than or equal to *`sideOfNotchWidth`*. In addition, one of the side edges with length less than *`minNotchLength`* must be between two concave corners at the ends, and the length of the opposite edge must be greater than or equal to *`minNotchLength`*.
Type: Float, specified in microns

`EXCEPTWITHIN` *`lowExcludeSpacing highExcludeSpacing`*

Specifies that it is legal to have notch spacing greater than or equal to *`lowExcludeSpacing`* but less than *`highExcludeSpacing`*.
Type: Float, specified in microns

`NOTCHWIDTH` *`notchWidth`*

Specifies that the notch length rule applies only if the width of the notch is less than or equal to *`notchWidth`*.
Type: Float, specified in microns

`WIDTH` *`sideOfNotchWidth`*

Specifies that the minimum notch length spacing only applies if the width of at least one side of the notch is greater than or equal to the specified *`sideOfNotchWidth`*.

`WITHIN` *`within`* `SPANLENGTH` *`sideOfNotchSpanLength`*

Specifies that if one of the side edges of the notch is within *`within`* distance and any portion of that edge has span length less than the *`sideOfNotchSpanLength`* and that portion has length greater than or equal to *`minNotchLength`* (this has a reverse definition of less than the *`minNotchLength`* without `WITHIN`), the notch spacing must be greater than or equal to the *`minSpacing`*. Otherwise, it would be a violation.
Type: Float, specified in microns

Notch Length Spacing Rule Examples

- The following spacing rule example shows notch length with `CONCAVEENDS`:

```
SPACING 0.10 ;
```


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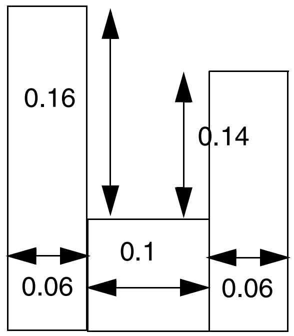
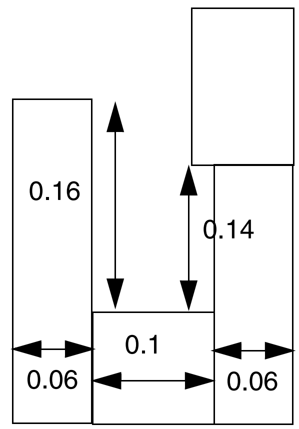
```
PROPERTY LEF58_SPACING
    "SPACING 0.12 NOTCHLENGTH 0.15 CONCAVEENDS 0.08  ; " ;
```

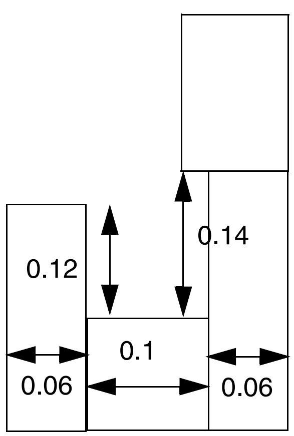
Figure 6-180 Illustration of Spacing Rule with NOTCHLENGTH and CONCAVEENDS



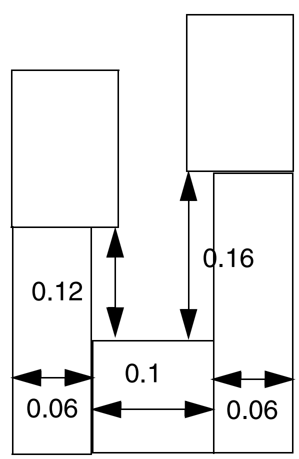
a) OK, the notch length of 0.14 is not between 2 concave corners. Without CONCAVEENDS, it is a violation.



b) Violation, both of the notch length conditions are met



c) OK, the length of the side edge without 2 concave corners is < 0.15



d) OK, this situation with concave corners on both of the side edges is fine

- The following spacing rule example shows notch length with `EXCEPTWITHIN` and `NOTCHWIDTH`:

```
PROPERTY LEF58_SPACING "
```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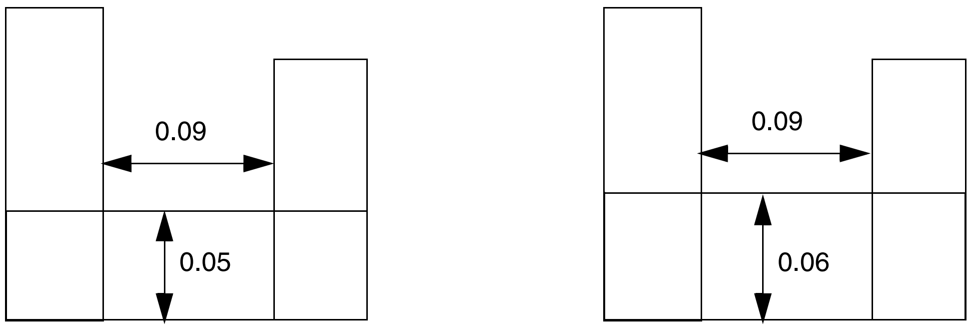
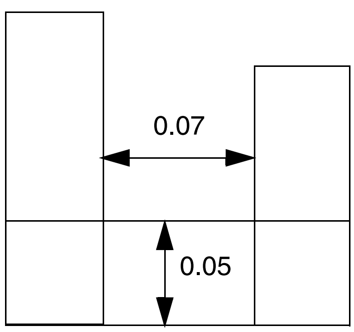
```
SPACING 0.1 NOTCHLENGTH 0
EXCEPTWITHIN 0.06 0.08 NOTCHWIDTH 0.05 ; " ;
```

Figure 6-181 Illustration of Spacing Rule with EXCEPTWITHIN and NOTCHWIDTH



a) Violation, notch width of 0.05 (<= 0.05) and notch spacing of 0.09 (< 0.1 and is not within the except within range)

b) OK, the notch width of 0.06 (> 0.05) would not trigger



c) OK, notch spacing of 0.07 is within the except within range (>= 0.06 and < 0.08)

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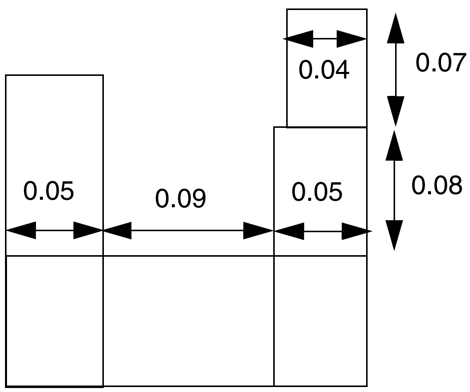
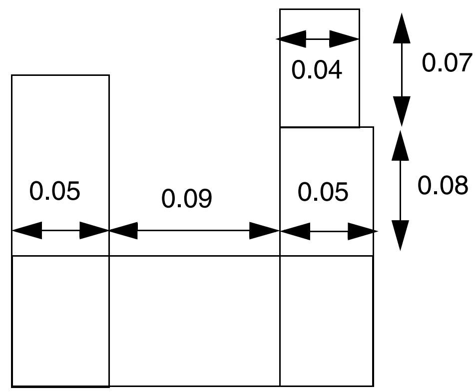
- The following spacing rule example shows notch length with `WITHIN` and `SPANLENGTH`:

```
SPACING 0.09 ;
PROPERTY LEF58_SPACING
   "SPACING 0.1 NOTCHLENGTH 0.06 WITHIN 0.2
      SPANLENGTH 0.05 ; " ;
```

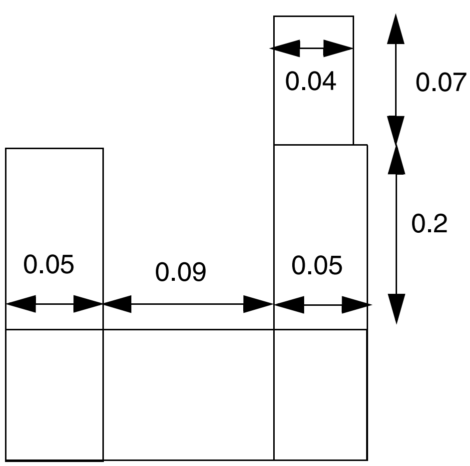
Figure 6-182 Illustration of Spacing Rule with WITHIN and SPANLENGTH



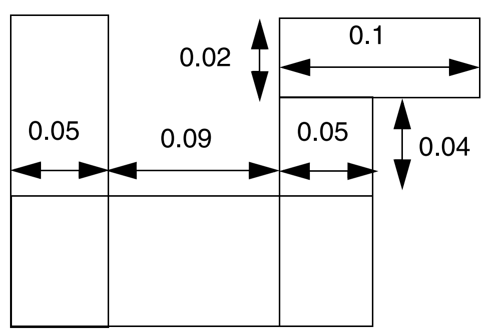
a) OK, only the bottom portion of right edge with width of 0.05 forming the notch is considered, and the top portion of 0.04 width/ span length is ignored.



b) Violation, the top portion of the right side notch is within 0.2 and has span length of 0.04 (< 0.05) with length of 0.07 (>= 0.06)



c) OK, the top portion of the right side notch is outside the length range > 0.2



d) OK, the bottom portion of the right side notch has length of 0.04 (< 0.06), and the top portion has span length >= 0.05 (width of 0.02 is irrelevant)

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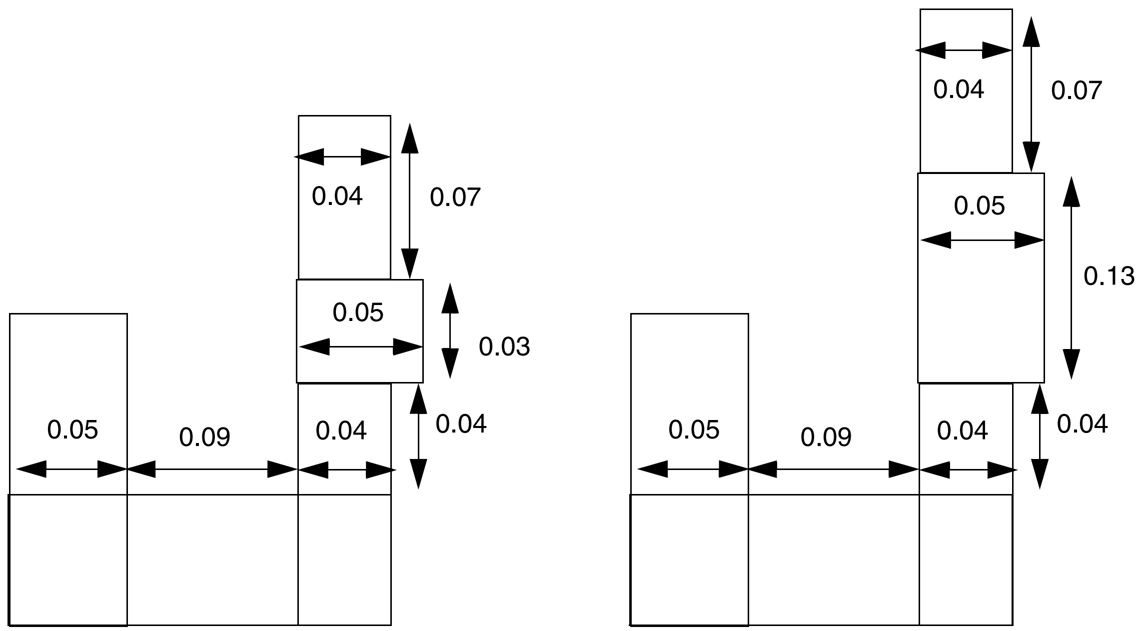
Figure 6-183 Illustration of Spacing Rule with WITHIN and SPANLENGTH

```
PROPERTY LEF58_SPACING
    "SPACING 0.1 NOTCHLENGTH 0.06 WITHIN 0.2
        SPANLENGTH 0.05 ; " ;
```



a) Violation, the top portion of the right side notch is within 0.2 and has span length of 0.04 (< 0.05) with length of 0.07 (>= 0.06)

b) OK, the top portion of the right side notch only has length of 0.03 (< 0.06) within 0.2

Notch Span Spacing Rule

Notch span spacing rule exists for any notch with span less than the specified span on both sides.

You can create notch span spacing rule by using the following property definition:

```
PROPERTY LEF58_SPACING
     "SPACING minSpacing NOTCHSPAN span NOTCHSPACING notchSpacing
     EXCEPTNOTCHLENGTH notchLength
     ; " ;
```

Where:

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All other keywords are the same as the existing LEF routing layer `SPACING` syntax.

`SPACING` *`minSpacing`* `NOTCHSPAN` *`span`* `NOTCHSPACING` *`notchSpacing`* `EXCEPTNOTCHLENGTH` *`notchLength`*

Specifies that any notch with span less than *`span`* on both sides and notch spacing less than *`notchSpacing`*, the portion of the notch length edge having common parallel run length greater than 0 on the opposite side must have a spacing greater than or equal to *`minSpacing`* to any neighbor wire with common parallel run length greater than 0. In addition, multiple consecutive U notches fulfilling the span and notch spacing conditions will be a violation, irrespective of how far the neighbor wires are. The rule does not apply if the minimum notch length is exactly equal to *`notchLength`* and the outer edges of the notches have neighbor wires that are greater than or equal to *`minSpacing`* away.
Type: Float, specified in microns

Examples

- The following example shows spacing rule with notch span:

```
SPACING 0.08 ;
PROPERTY LEF58_SPACING
    "SPACING 0.1 NOTCHSPAN 0.15 NOTCHSPACING 0.12
       EXCEPTNOTCHLENGTH 0.13 ; " ;
```


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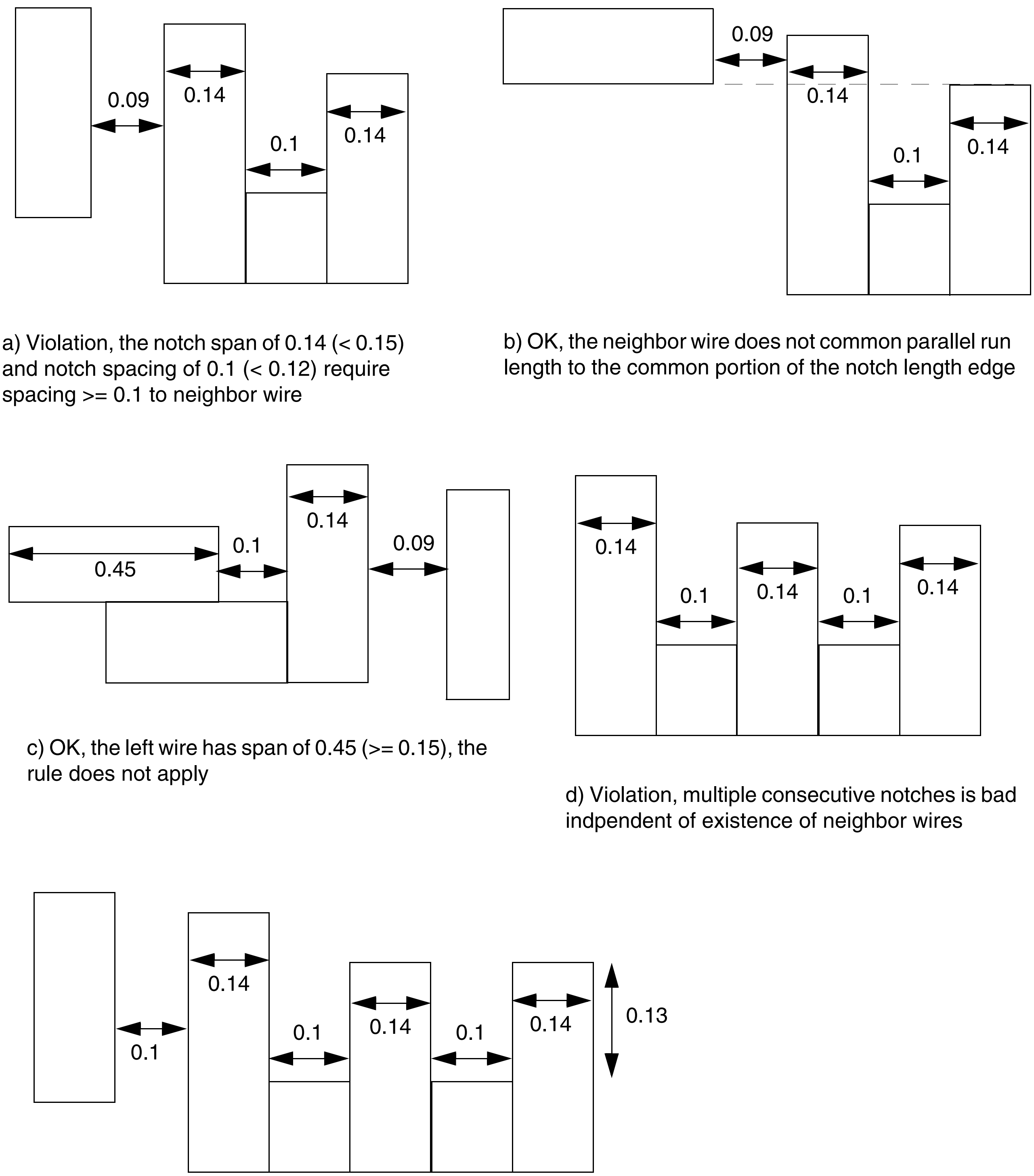
Figure 6-184 Illustration of Spacing Rule with NOTCHSPAN



a) Violation, the notch span of 0.14 (< 0.15) and notch spacing of 0.1 (< 0.12) require spacing >= 0.1 to neighbor wire

b) OK, the neighbor wire does not common parallel run length to the common portion of the notch length edge

c) OK, the left wire has span of 0.45 (>= 0.15), the rule does not apply

d) Violation, multiple consecutive notches is bad indpendent of existence of neighbor wires

e) OK, the notch length exception is fulfilled

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Jog to Jog Spacing Table Rule

Jog to Jog spacing table rules can be used to define jog or protrusion spacing requirements.

You can define a jog to jog spacing table rule by using the following property definition:

```
PROPERTY LEF58_SPACINGTABLE
       "SPACINGTABLE JOGTOJOGSPACING jogToJogSpacing
          JOGWIDTH jogWidth SHORTJOGSPACING shortJogSpacing
            {WIDTH width PARALLEL parLength WITHIN parWithin
                 [EXCEPTWITHIN lowExcludeSpacing highExcludeSpacing]
                 LONGJOGSPACING longJogSpacing
                 [SHORTJOGSPACING widthShortJogSpacing]} ...
       ; " ;
```

Where:

EXCEPTWITHIN *lowExcludeSpacing highExcludeSpacing*

Specifies that any wire that is greater than or equal to *lowExcludeSpacing* and less than *highExcludeSpacing* away from the wide wire should be ignored as if it does not exist.
Type: Float, specified in microns

SPACINGTABLE JOGTOJOGSPACING *jogToJogSpacing*
JOGWIDTH *jogWidth* SHORTJOGSPACING *shortJogSpacing*
{WIDTH *width* PARALLEL *parLength* WITHIN *parWithin*
LONGJOGSPACING *longJogSpacing* } ...

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Specifies jog or protrusion spacing requirements as described below:

Find the last row that a wire with width greater than *width* has neighbor wires with a parallel run length greater than *parLength* that is less than *parWithin* distance away. If there is a jog or protrusion wire/edge in the wide wire or a neighbor wire, with continuous common projected length less than or equal to *jogWidth*, spacing to the opposite metal must be greater than or equal to *shortJogSpacing*. If the continous common projected length is greater than *jogWidth*, then spacing to the opposite metal must be greater than or equal to *longJogSpacing*. If there are two or more jog or protrusion wires with continuous common projected length less than or equal to *jogWidth* and spacing on the jog or protrusion wires is less than *longJogSpacing*, (projected) spacing in the wire direction of the wide wire between any two jog or protrusion wires must be greater than or equal to *jogToJogSpacing*. All of the values in the table must be the same or increasing, and *longJogSpacing* must be larger than *shortJogSpacing*.
Type: Float, specified in microns

SHORTJOGSPACING *widthShortJogSpacing*

Specifies individual *widthShortJogSpacing* for the given width, which would override and has the same meaning as the global *shortJogSpacing*.
Type: Float, specified in microns

Jog to Jog Spacing Table Rule Examples

- Figure 6-185 on page 924 illustrates the following jog to jog spacing table rules:

```
PROPERTY LEF58_SPACINGTABLE "
   "SPACINGTABLE JOGTOJOGSPACING 0.7
       JOGWIDTH 0.2 SHORTJOGSPACING 0.08
       WIDTH 0.4 PARALLEL 0.2 WITHIN 0.5
          LONGJOGSPACING 0.13
       WIDTH 0.8 PARALLEL 0.3 WITHIN 0.5
          LONGJOGSPACING 0.15 ; " ;
```


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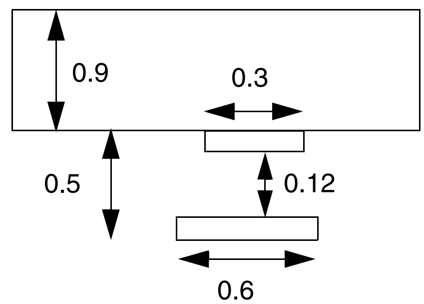
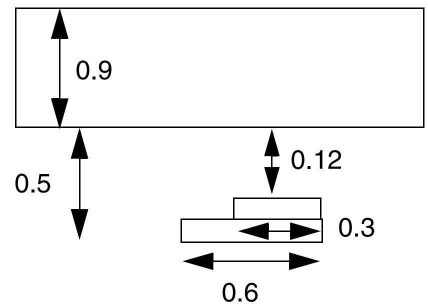
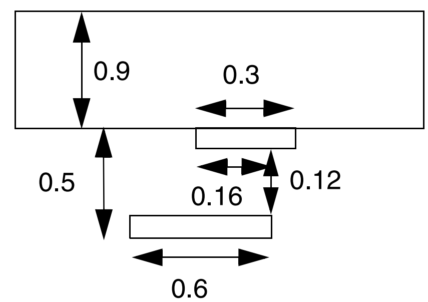
Figure 6-185 Illustration of Spacing Table with JOGTOJOGSPACING



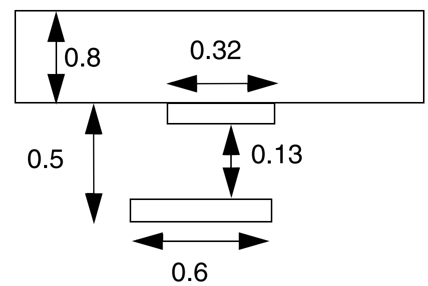
a) Violation, all the triggering conditions are met, and the spacing is 0.12 (< 0.15)



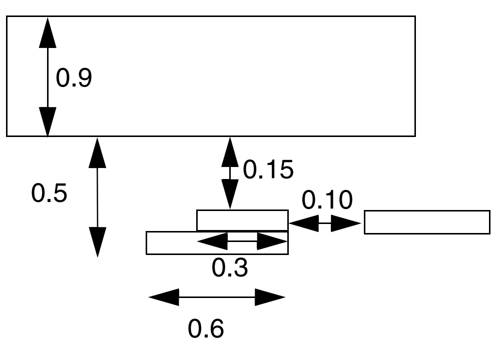
b) Violation, the jog/protrusion can be in the neighbor wire



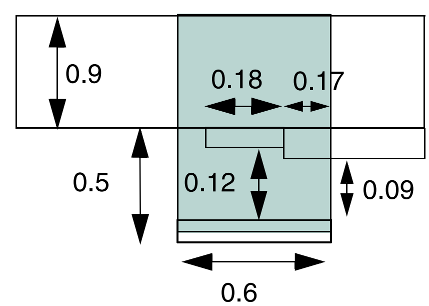
c) OK, the run length of the jog/protrusion is 0.16, spacing of 0.12 >= 0.08



d) OK, the wide wire without the jog/protrusion must have width > 0.8 to trigger the jog spacing of 0.15 while the jog spacing of 0.13 based on width > 0.4 is met.



e) OK, the 0.15 jog spacing is applied only to the opposite metal, but not to the right side neighbor 0.10 away



f) Violation, the continuous run length with spacing < 0.15 is 0.35 (0.18 + 0.17), which would require spacing of 0.15

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- Figure 6-186 on page 926 illustrates the following jog to jog spacing table rules:

```
PROPERTY LEF58_SPACINGTABLE "
   "SPACINGTABLE JOGTOJOGSPACING 0.7
      JOGWIDTH 0.2 SHORTJOGSPACING 0.08
      WIDTH 0.8 PARALLEL 0.3 WITHIN 0.5
         LONGJOGSPACING 0.15 ; " ;
```


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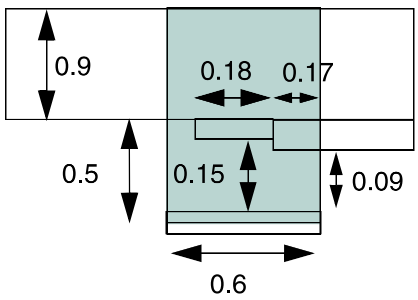
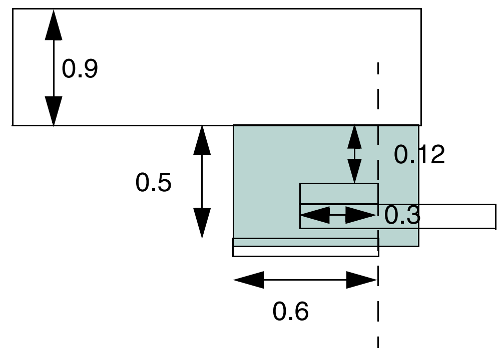
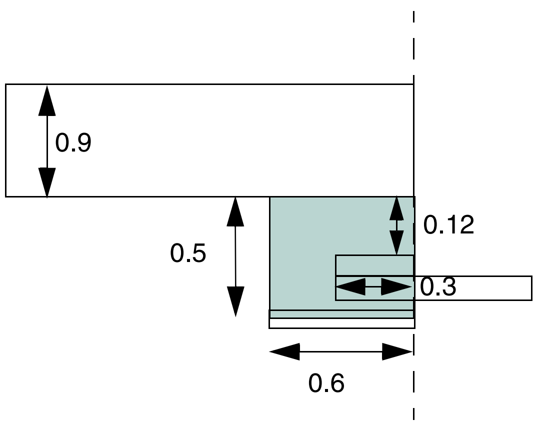
Figure 6-186 Illustration of Spacing Table with JOGTOJOGSPACING



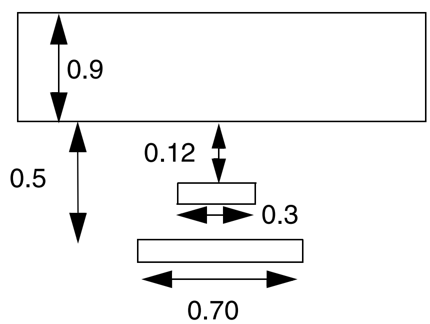
a) OK, only 0.17 segment is within 0.15, spacing of 0.08 is required and met



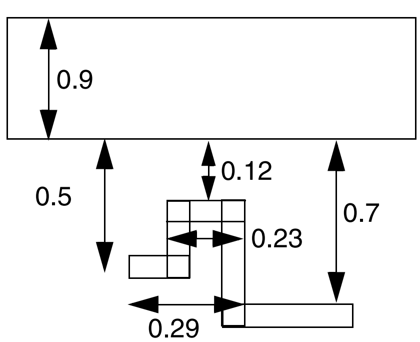
b) Violation, all neighbor wires in the green region are considered



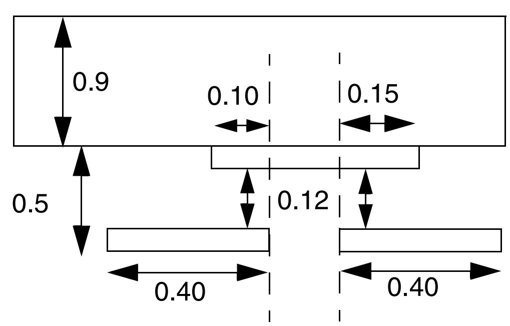
c) Violation, any neighbor wires with run length > 0.2 must have spacing >= 0.15



d) Violation, the neighbor wire does not need to be a jog



e) OK, the parallel run length is 0.29 within the 0.32 away, the rule is not checked



f) OK, the common projected run length of the jog/ protrusion is 0.10 and 0.15 individually, 0.08 spacing is needed and met

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- Figure 6-187 on page 928 illustrates the following jog to jog spacing table rules:

```
PROPERTY LEF58_SPACINGTABLE "
   "SPACINGTABLE JOGTOJOGSPACING 0.7
     JOGWIDTH 0.2 SHORTJOGSPACING 0.08
     WIDTH 0.8 PARALLEL 0.3 WITHIN 0.5
       LONGJOGSPACING 0.15 ; " ;
```


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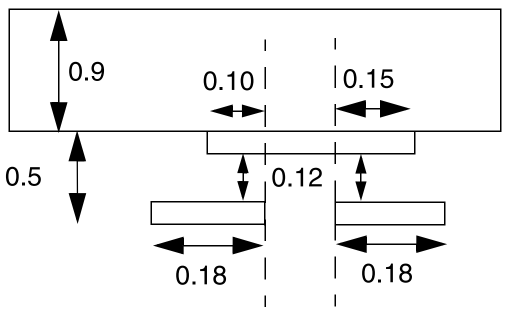
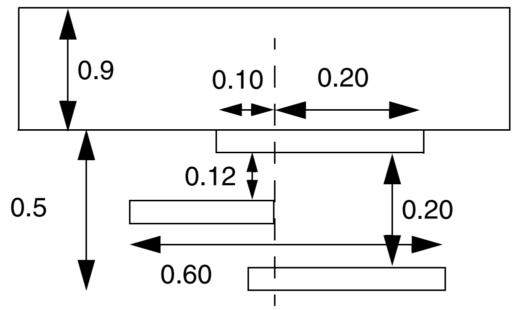
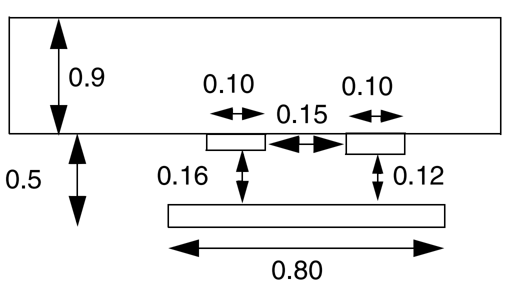
Figure 6-187 Illustration of Spacing Table with JOGTOJOGSPACING



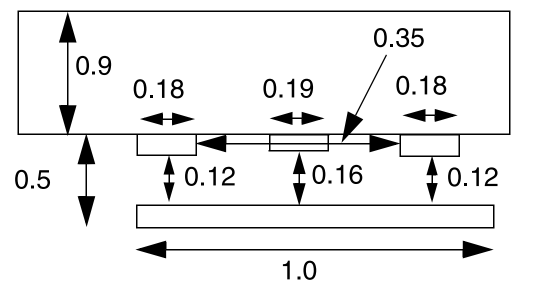
a) OK, parallel run length of 0.18 individually would not trigger the rule



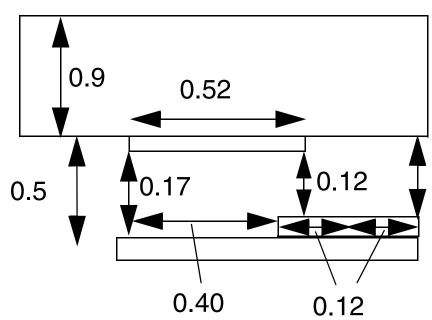
b) OK, only the 0.10 segment is within 0.15, spacing of 0.08 is required and met



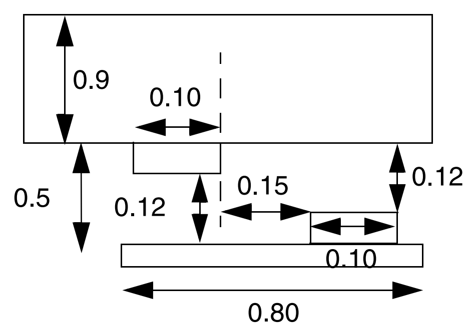
c) OK, the left jog/protrusion having a jog spacing of 0.16 (>= 0.15) is not a valid jog/ protrusion for considering spacing among jog/protrusion wires



d) Violation, having a 'good' 0.19 jog/protrusion at the middle is irrelevant, the distance of 0.35 between the left and right jog/protrusion wires would still violate >= 0.7 jog to jog spacing.



e) OK, the top 0.52 jog/protrusion is broken into two segments for the rule, and 0.40 segment has spacing of 0.17 (>= 0.15), and the 0.12 segment has spacing of 0.12 (>= 0.08). The bottom 0.24 jog/protrusion is broken into two segments of 0.12 with spacing of 0.17 and 0.12 (>= 0.08)



f) Violation, jog/protrusion wires on opposite wires still need to follow the 0.7 spacing

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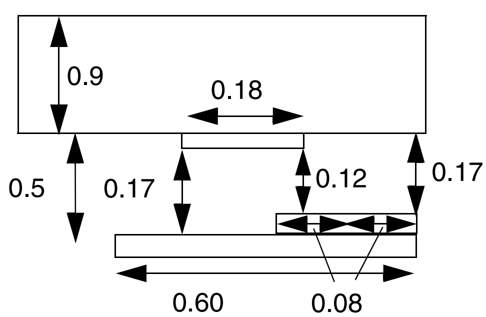
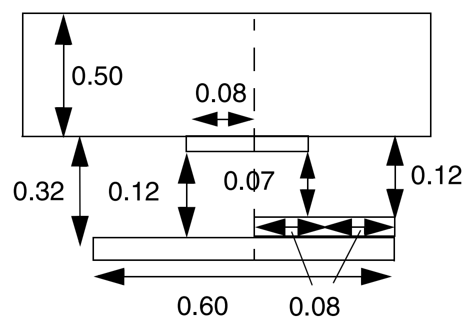
- Figure 6-188 on page 929 illustrates the following jog to jog spacing table rules with `JOGTOJOGSPACING 0.3`:

```
PROPERTY LEF58_SPACINGTABLE "
   "SPACINGTABLE JOGTOJOGSPACING 0.7
      JOGWIDTH 0.2 SHORTJOGSPACING 0.08
      WIDTH 0.8 PARALLEL 0.3 WITHIN 0.5
        LONGJOGSPACING 0.15 ; " ;
```

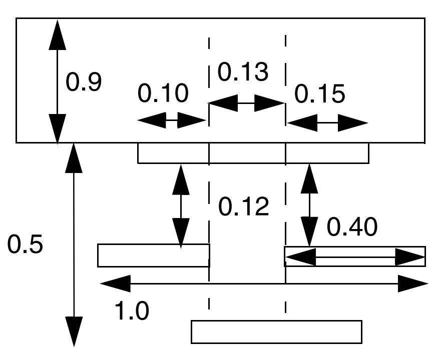
Figure 6-188 Illustration of Spacing Table with JOGTOJOGSPACING



a) OK, the jogs/protrusions are broken into two segments each. Eventually, there is only one common segment with length 0.08 and spacing < 0.15, jog to jog spacing of 0.7 is not checked.



b) Violation, the continuous run length with spacing < 0.13 is 0.24 (three segments of 0.08), which would require spacing of 0.13



c) Violation, 0.10 and 0.15 segments are considered individually, and would fail the jog to jog spacing of 0.7.

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- The following example indicates that the short jog spacing is 0.08 µm for width greater than 0.4 µm, 0.09 µm for width greater than 0.8 µm. Any neighbor wire greater than or equal to 0.04 µm and less than 0.06 µm away is ignored (for width greater than 0.8 µm only).

```
PROPERTY LEF58_SPACINGTABLE "
    "SPACINGTABLE JOGTOJOGSPACING 0.7
        JOGWIDTH 0.2 SHORTJOGSPACING 0.08
        WIDTH 0.4 PARALLEL 0.2 WITHIN 0.5
           LONGJOGSPACING 0.13 SHORTJOGSPACING 0.08
        WIDTH 0.8 PARALLEL 0.3 WITHIN 0.5
           EXCEPTWITHIN 0.04 0.06
           LONGJOGSPACING 0.15 SHORTJOGSPACING 0.09 ; " ;
```

Direction Span Length Spacing Table Rule

You can create a direction span length spacing table rule to create a table in which the spacing between two objects depends on the maximum span length of the two objects and their parallel run length.

You can define a direction span length spacing table using the following property definition:

```
PROPERTY LEF58_SPACINGTABLE
     "SPACINGTABLE
         DIRECTIONALSPANLENGTH [WRONGDIRECTION]
         [SAMEMASK]
         [EXCEPTJOGLENGTH length [EDGELENGTH] [INCLUDELSHAPE]]
         [EXCEPTEOL eolWidth]
         [EXACTSPANLENGTHSPACING spanLength1 TO spanLength2
             [PRL prl] {exactSpacing}...]...
         PRL {prl}...
         {SPANLENGTH spanLength
             [EXACTSPACING exactSpacing
             | SPACINGTOMINSPAN spacingToMinSpan]
             [EXACTSELFSPACING exactSelfSpacing]
             [NOEXCEPTEOL|EOLSPACING eolspacing] {spacing}...}...
     ; " ;
```

Where:

```
DIRECTIONALSPANLENGTH PRL {prl}...
{ SPANLENGTH spanLength {spacing}...}...
```


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Creates a table in which spacing between two objects depends on the span length of the objects and their parallel run length. There must be exactly as many spacing values in each `SPANLENGTH` row as there are `PRL` columns. All of the values, *`prl`*, *`spanLength`* and *`spacing`* must be monotonically increasing. In other words, the *`prl`* values must be increasing from left to right, the spanLength values must be increasing from top to bottom. If *`prl`* value is negative, the spacing is measured in a `MAXXY` style. In other words, no neighbor can be within the search window with dimension of abs(prl) and spacing. The spacing values must be the same or increasing from left to right, but could have any value relationship, from top to bottom in the table. To find the required spacing, locate the last row where the individual span length of the two objects is, greater than the table row span length, and the right-most column where the parallel run length of the two objects is, greater than the table column *`prl`*, the maximum of the corresponding spacing values is the required spacing. When parallel run length of the two objects is less than or equal to the first PRL value, then these rules do not apply, but some other rules will. In addition, spacing value can be zero, which can typically be defined in negative `PRL`, indicating that there is no constraint on that object, and spacing requirement of the other object is not checked. See Figure 6-191 on page 937.
Type: Float, specified in microns

`EOLSPACING` *`eolspacing`*

Specifies a spacing value that should be applied when the `EXCEPTEOL` condition is met.

Type: Float, specified in microns

`EXACTSELFSPACING` *`exactSelfSpacing`*

Specifies that two wires with span length that belong to the same row could have exact spacing of *`exactSelfSpacing`*. This should not be used on the first row since it would have the same meaning of `EXACTSPACING`.

`EXACTSPACING` *`exactSpacing`*

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Specifies that if the spacing of a wire with span length greater than specified span length of a row with EXACTSPACING and less than or equal to the specified span length of the next row is exactly *exactSpacing*, and it (or its) neighbor wire has a span length between the specified span length of the first two rows, then it is not a violation. The *exactSpacing* should be smaller than all of the spacing values on that row, except when the spacing value is zero for a certain PRL, which means that there is no constraint between wires belonging to that row to any other wires with parallel run length less than or equal to the specified value.
Type: Float, specified in microns

EXACTSPANLENGTHSPACING *spanLength1* TO *spanLength2*
  [PRL *prl*] {*exactSpacing*}...

Specifies a list of exact spacing values in {*exactSpacing*}..., which are also allowed between a wire with exact span length of *spanLength1* and another wire with exact span length of *spanLength2* with parallel run length greater than *prl* or zero if PRL is not specified. This is in addition to spacing value of *spacing* in the table. In addition, this construct should not be used along with either EXACTSPACING or EXACTSELFSPACING in individual rows.
Type: Float, specified in microns

EXCEPTEOL *eolWidth*

Specifies that the directional span length spacing does not apply to EOL edges with width less than the *eolWidth*. Those EOL edges should follow the SPACING ENDOFLINE rules.
Type: Float, specified in microns

EXCEPTJOGLENGTH *length* [EDGELENGTH] [INCLUDELSHAPE]

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Specifies that the minimum default (right-way) spacing would be applied between a wrong way jog with minimum wrong way width and length less than *length* sandwiched by two right way wires with projected PRL less than or equal to 0 in the wrong way direction in a 'Z' shape and a right way wire or two 'Z' shapes that can fulfill this exemption condition. See Figure 6-189 on page 935.

If INCLUDELSHAPE is specified without EDGELENGTH, this minimum default spacing will also be applied between a wrong way jog with minimum wrong way width and length less than *length* connected to a right way wire in a 'L' shape and a right way wire or two 'L' shapes that can fulfill this exemption condition or an 'L' shape and a 'Z' shape that can fulfill this exemption condition.

If EDGELENGTH is specified, *length* is measured against the length of the edges in the wrong way direction of a wrong way jog in a ‘Z’ shape instead of the entire span length of the jog. If INCLUDELSHAPE is also specified to include an ‘L’ shape exemption, the longest edge length would be the same as the span length of the jog, and the EDGELENGTH construct would have no impact on it.

Type: Float, specified in microns

NOEXCEPTEOL — Specifies that EXCEPTEOL does not apply to the rows that have NOEXCEPTEOL keyword.

SAMEMASK — Specifies that the directional span length spacing rule only applies to objects on the same mask.

SPACINGTOMINSPAN *spacingToMinSpan*

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Specifies that if the spacing of a wire with span length greater than specified span length of a row with `SPACINGTOMINSPAN` and less than or equal to specified span length of the next row is greater than or equal to *`spacingToMinSpan`*, and it (or it's) neighbor wire has a span length between the specified span length of the first two rows, then it is not a violation, and spacing requirement of the other object is not checked. When the spacing value is zero for a certain PRL, this indicates that there is no constraint between wires belonging to that row to any other wires with parallel run length less than or equal to the PRL value of the first non-zero spacing value. In other words, *`spacingToMinSpan`* will not override the zero spacing value in this case. See Figure 6-192 on page 938.
Type: Float, specified in microns

`WRONGDIRECTION`

Specifies that the spacing is applied to wrong direction edges. In other words, the spacing is `applied` to the direction parallel to the specified direction in DIRECTION on a Manhattan routing layer. You must define two tables - one with `WRONGDIRECTION` and the other without `WRONGDIRECTION`. The spacing of the one without `WRONGDIRECTION` is applied to the direction perpendicular to the specified direction in `DIRECTION`.

Direction Span Length Spacing Table Rule Examples

- The following example illustrates the following spacing table rules with direction span length:

```
DIRECTION VERTICAL ;
PROPERTY LEF58_WIDTHTABLE
     "WIDTHTABLE 0.050 ... ;
     WIDTHTABLE 0.090 ... WRONGDIRECTION ; " ;
PROPERTY LEF58_SPACINGTABLE "
  SPACINGTABLE
    DIRECTIONALSPANLENGTH
        EXCEPTJOGLENGTH 0.170
        PRL                          0.000 ...
        SPANLENGTH 0.000             0.050 ...
        SPANLENGTH 0.140             0.100 ... ;
```


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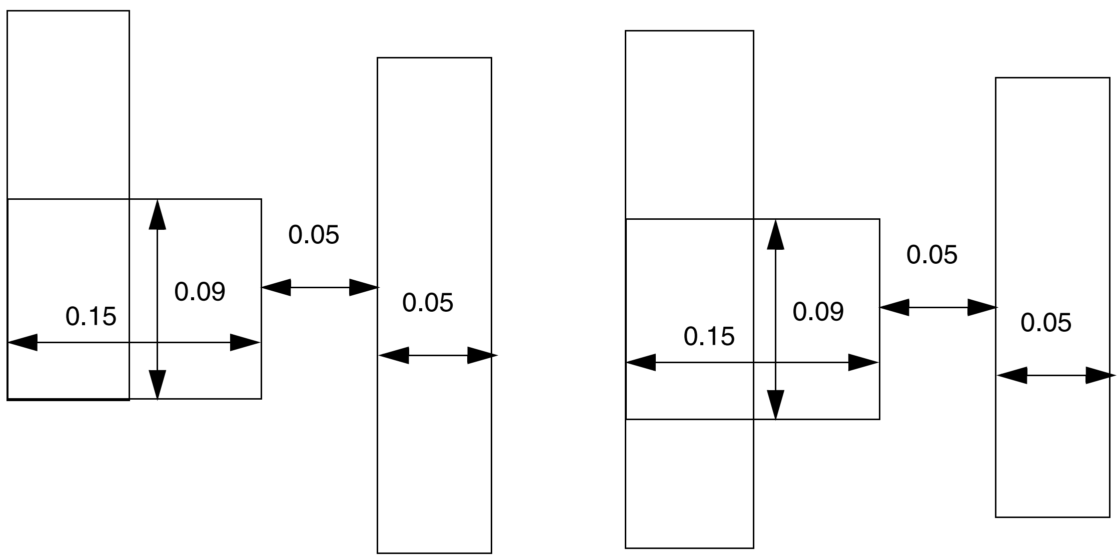
Figure 6-189 Illustration of Spacing Table Rule with DIRECTIONALSPANLENGTH



a) OK, EXCEPTJOGLENGTH applies to L shape configuration

b) Violation, EXCEPTJOGLENGTH does not apply to T shape configuration.

- The following example illustrates use of `EDGELENGTH`:

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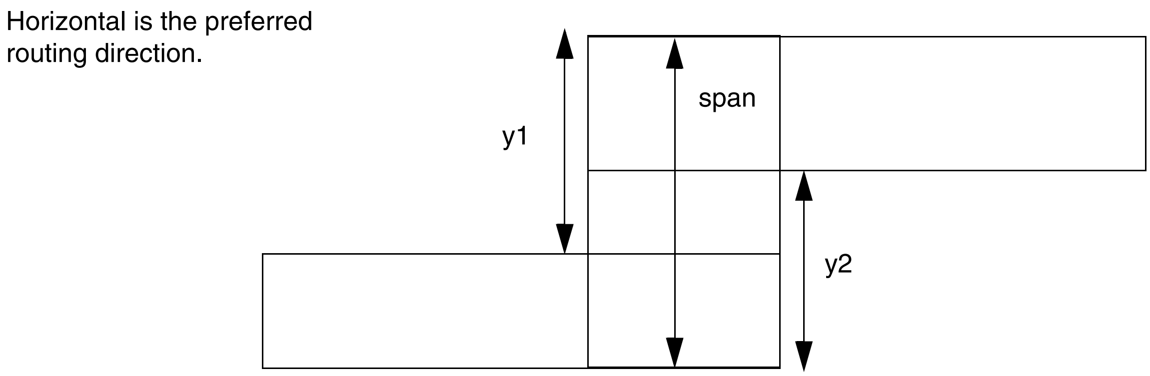
Figure 6-190 Illustration of DIRECTIONALSPANLENGTH with EDGELENGTH in EXCEPTJOGLENGTH



Both y1 and y2 must be < *length* to be qualified as a 'Z' shape for EXCEPTJOGLENGTH exemption. If EDGELENGTH is not specified, span must be < *length* to be qualified.

Illustration of EXCEPTJOGLENGTH *length* EDGELENGTH

- The following example illustrates direction spanlength with negative parallel run length:

```
DIRECTION HORIZONTAL ;
PROPERTY LEF58_SPACINGTABLE "
    SPACINGTABLE
       DIRECTIONALSPANLENGTH
          PRL                     -0.050 ...
          SPANLENGTH    0.000      0.030 ... ; " ;
```


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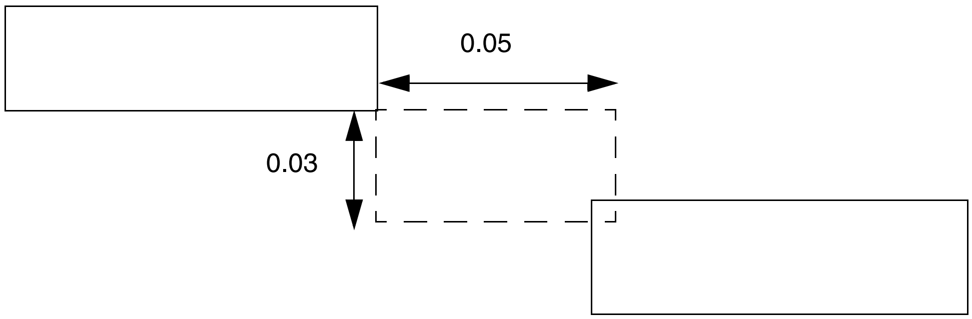
Figure 6-191 Illustration of DIRECTIONALSPANLENGTH with PRL



a) Violation, when PRL is negative, a search window of 0.03 x 0.05 is formed for neighbors (0.03 being the direction perpendicular to the preferred routing direction in the table without WRONGDIRECTION)

- The following example illustrates direction span length with spacing to minimum span:

```
DIRECTION VERTICAL ;
PROPERTY LEF58_SPACINGTABLE "
  SPACINGTABLE
    DIRECTIONALSPANLENGTH
    PRL                                      0.000 ...
    SPANLENGTH 0.000  EXACTSPACING 0.050      0.070 ...
    SPANLENGTH 0.090  SPACINGTOMINSPAN 0.060  0.080 ...
    SPANLENGTH 0.110                          0.090 ...   ; " ;
```


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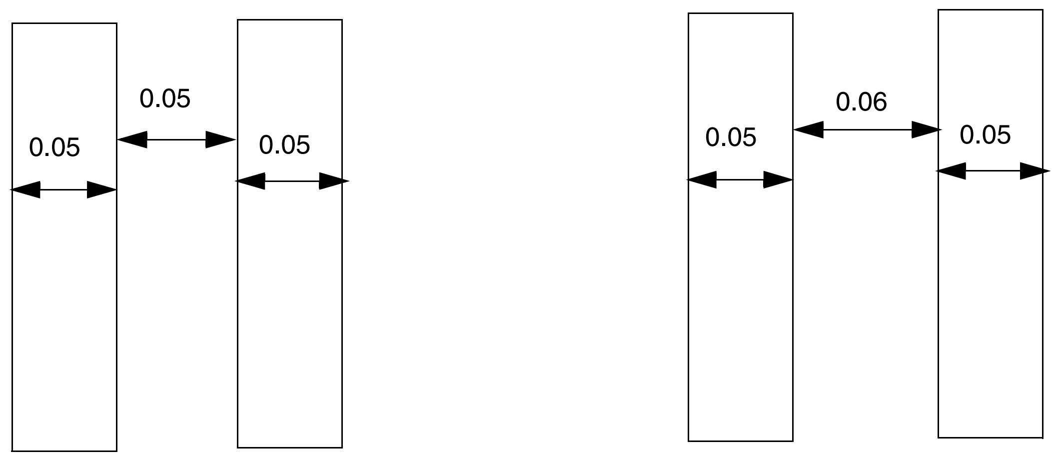
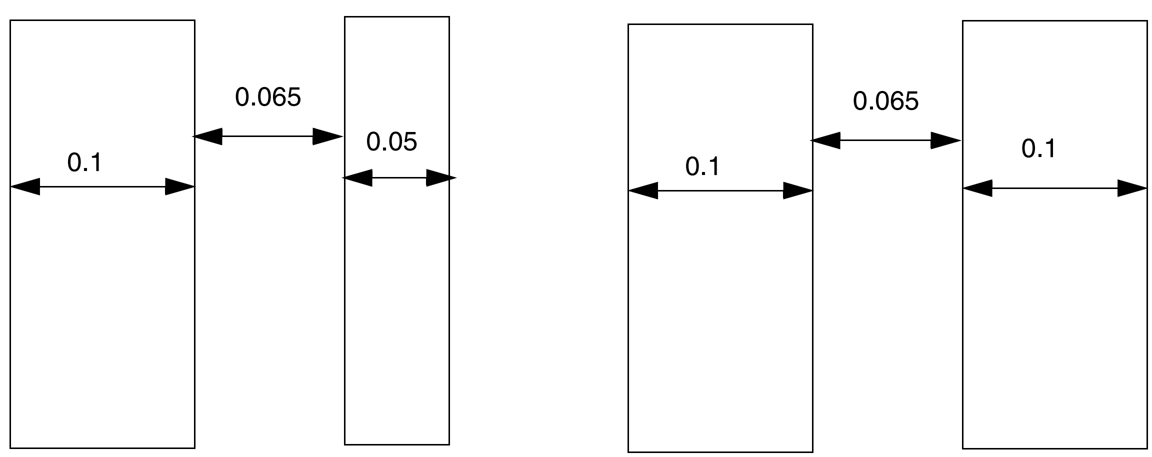
Figure 6-192 Illustration of Spacing Table Rule with DIRECTIONALSPANLENGTH



a) OK, exact spacing of 0.05 is fine.

b) Violation, if the spacing is not exactly 0.05, it needs to be >= 0.07



c) OK, spacing of span length of 0.1 to a span length of 0.05 between the first 2 rows (> 0 & <= 0.09) must be >= 0.06, which is fulfilled

d) Violation, spacing of span length of 0.1 of 2 wires must be >= 0.08

Two-Widths Spacing Table Rule

You can create a two-widths spacing table rule to create a table in which spacing between two objects depends on the widths of both the objects.

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You can define a two-widths spacing table rule by using the following property definition:

```
PROPERTY LEF58_SPACINGTABLE
     "SPACINGTABLE
          TWOWIDTHS [WRONGDIRECTION]
              [SAMEMASK | MASK maskNum]
              [EXCEPTEOL eolWidth]
              [FILLCONCAVECORNER fillTriangle]
              {WIDTH width [PRL runLength] {spacing} ...} ...
          ; " ;
```

Where:

All other keywords are the same as the existing LEF routing layer `SPACINGTABLE` syntax.

`EXCEPTEOL` *`eolWidth`*
: Specifies that the two widths table does not apply to EOL edges with width less than *`eolWidth`*. Those EOL edges should follow the `SPACING ENDOFLINE` rules instead.
Type: Float, specified in microns

`FILLCONCAVECORNER` *`fillTriangle`*
: Specifies that any concave corners should be filled by a triangle with value of *fillTriangle* along the edges to form the corner before minimum spacing of the table is checked. This fill corner operation does not apply to any other spacing values. See Figure 6-193 on page 940.
Type: Float, specified in microns

`MASK` *`maskNum`*
: Specifies that the spacing rule applies only between two wires with the given mask. In addition, the spacing in the table entry is always applied between two wires of the corresponding width values even if there is another wire with any mask between them. *`maskNum`* must be a positive integer, and most applications only support the values 1, 2, or 3.
Type: Integer

`SAMEMASK`
: Specifies that spacing applies only to the objects that belong to the same mask. The same mask objects recognize the first spacing table, but not the second one. The spacing in the second table is incorporated in the first one.
Type: Float, specified in microns

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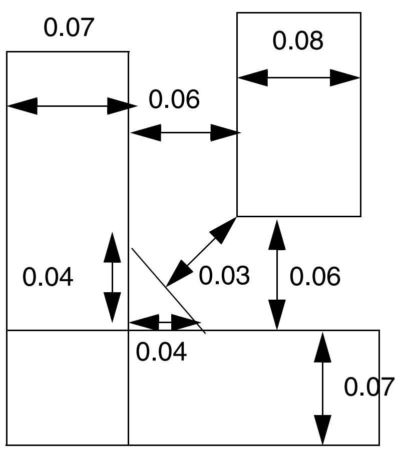
`WRONGDIRECTION`

Specifies the spacings of an edge with direction perpendicular to the specified direction in `DIRECTION` on a Manhattan routing layer to any other edges on that layer. This has a similar definition in `SPACINGTABLE PARALLELRUNLENGTH` spacing table.
Type: Float, specified in microns

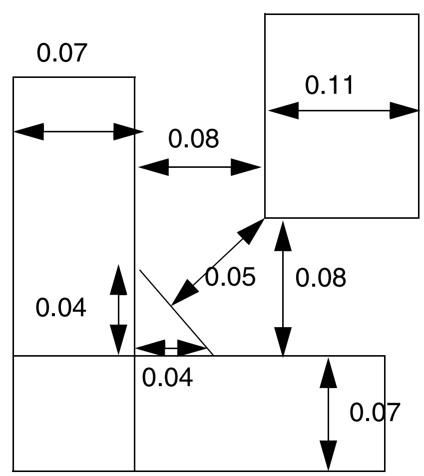
Two-Widths Spacing Table Rule Examples

Figure 6-193 Two-Widths Spacing Table Rule with FILLCONCAVECORNER

```
PROPERTY LEF58_SPACINGTABLE
     "TWOWIDTHS FILLCONCAVECORNER 0.04
     WIDTH    0.06   0.08
     WIDTH    0.08   0.09 ; " ;
```



a) Violation, spacing is only 0.03 to the triangular fill with length of 0.04 along the edges of the concave corner.



b) OK, the corner fill does not apply to non-minimum spacing check

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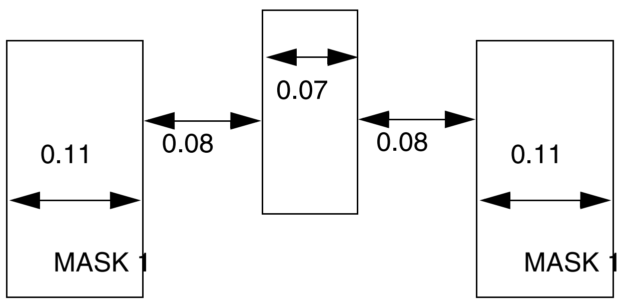
Figure 6-194 Two-Widths Spacing Table Rule with MASK

```
PROPERTY LEF58_SPACINGTABLE
     "TWOWIDTHS MASK 1
     WIDTH 0      0.04   0.06
     WIDTH 0.1    0.06   0.30 ; " ;
```



a) Violation between two 0.11 wires; having a middle wire with any mask is irrelevant.

Parallel Span Length Spacing Table Rule

Parallel span length spacing table rules can be used to define supplement spacing constraints based on span length of the objects.

You can define a parallel span length spacing table rule by using the following property definition:

```
PROPERTY LEF58_SPACINGTABLE
       "SPACINGTABLE
          PARALLELRUNLENGTH [DIAGONAL] [WRONGDIRECTION]
          [SAMEMASK]
          [EXCEPTEOL eolWidth]
          {length}...
          {WIDTH width
                [EXCEPTWITHIN lowExcludeSpacing highExcludeSpacing]
               {spacing}...}... ;
          [SPACINGTABLE
                 INFLUENCE {WIDTH width WITHIN distance SPACING spacing} ... ;]
       ; " ;
```

Where:

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All other keywords are the same as the existing LEF routing layer `SPACINGTABLE` syntax.

`DIAGONAL`
: Specifies that the defined spacing is applied between a diagonal 45-degree wire to any wire in an RDL routing layer.

`EXCEPTEOL` *`eolWidth`*
: Specifies that the parallel run length table does not apply to EOL edges with width less than *`eolWidth`*. Those EOL edges should follow the `SPACING ENDOFLINE` rules instead.
Type: Float, specified in microns

`EXCEPTWITHIN` *`lowExcludeSpacing highExcludeSpacing`*

Specifies that any wire that is at a distance greater than or equal to the *`lowExcludeSpacing`* and less than *`highExcludeSpacing`* away from the wide wire should be ignored, as if it does not exist. The conditions *`lowExcludeSpacing`* and *`highExcludeSpacing`* only apply to spacing if both of them are less than the specified *`spacing`*. In other words, spacing between the two wires must not be greater than 0 to less than *`lowExcludeSpacing`* and greater than or equal to *`highExcludeSpacing`* to less than *`spacing`*.
Type: Float, specified in microns

Note that at the most one such `SAMEMASK` spacing table and another parallel run length spacing table without the `SAMEMASK` keyword can be defined together.

The *`length`* can be negative that will act as extending the edges by abs(*`length`*) in a `WITHIN` style and enforcing *`spacing`* in `MAXXY` style.

`PARALLELSPANLENGTH PRL` *`runLength`* `{SPANLENGTH` *`spanLength`* `{`*`spacing`*`}}`

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Creates a table in which spacing between two objects depends on the span length of both the objects that have a parallel run length greater than *runLength*. To find the required spacing, a NxN two-dimensional table is used that implicitly has the same span lengths for row and column headings. There must be exactly as many spacing values in each SPANLENGTH row as the number of SPANLENGTH rows. The *spanLength* values must increase from top to bottom in the table. The spacing values must be the same, or increase from left to right and from top to bottom across the table. Consider two objects with *spanLength1* and *spanLength2*. You need to find the last row where *spanLength1* is greater than the table row *spanLength*, and the right-most column where *spanLength2* is greater than the table column *spanLength*. The intersection of the matching row and column provides the required spacing.

The parallel run length is measured as a sum of lengths between objects. Turning corners may break up the parallel run length thus resulting in inaccurate calculations. These are supplement spacing constraints, in addition to the regular spacing based on wire width.
Type: Float, specified in microns (for all values)

SAMEMASK

Specifies that the spacings only apply to objects belonging to the same mask.
Type: Float, specified in microns

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`WRONGDIRECTION`

Specifies the spacings of an edge with direction perpendicular to the specified direction in `DIRECTION` on a Manhattan routing layer to any other edges on that layer.
Type: Float, specified in microns

Note that at the most one such spacing table and another parallel run length spacing table without the `WRONGDIRECTION` keyword can be defined together. In addition, `SPACING` with `WRONGDIRECTION` keyword is also allowed on that layer.

With the existence of `PARALLELRUNLENGTH WRONGDIRECTION`, there is a built-in condition for `PARALLELRUNLENGTH` for both with or without `WRONGDIRECTION` spacing tables. The edge length of an edge having width greater than *`width`* must be greater than the default wrong way minimum width (if not defined, use right way minimum width), in order for the spacing in the second and beyond columns to be applied. If edge length condition is not met, the spacing in the first column will be applied independent of parallel run length. This behavior will also work for a combination of a `PARALLELRUNLENGTH WRONGDIRECTION` and `TWOWIDTHS` spacing tables. See Figure 6-199 on page 950.

Spacing Table Rule Examples

- Figure 6-195 on page 945 illustrates the following spacing table rules with `SPANLENGTH 0`, `0.10`, and `0.20`:

```
PROPERTY LEF58_SPACINGTABLE
   "SPACINGTABLE
    PARALLELSPANLENGTH PRL 0.15
    SPANLENGTH    0       0.10    0.15    0.20
    SPANLENGTH    0.10    0.15    0.17    0.23
    SPANLENGTH    0.20    0.20    0.23    0.25 ; " ;
```


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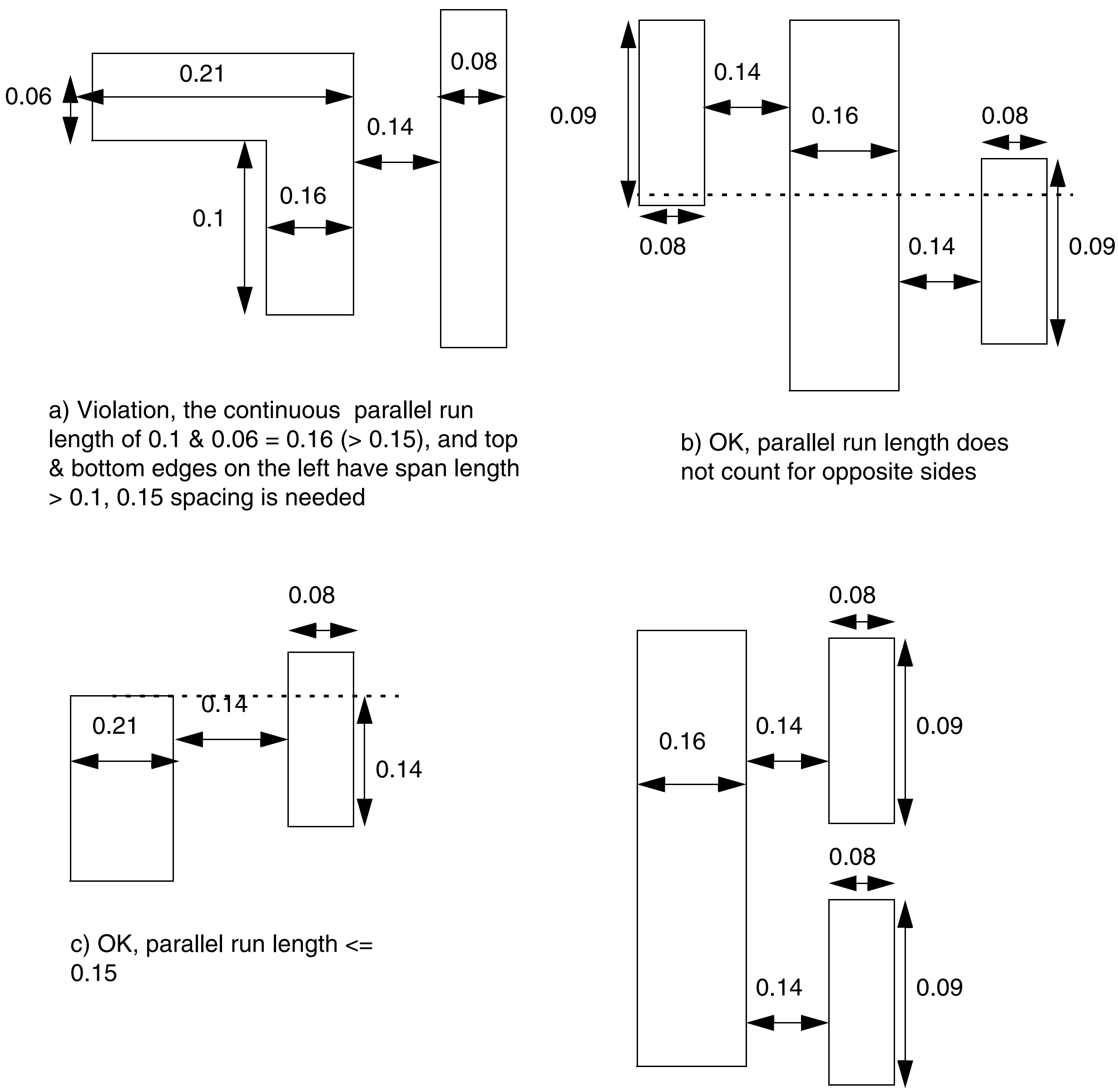
Figure 6-195 Spacing Table Rule Example



a) Violation, the continuous parallel run length of 0.1 & 0.06 = 0.16 (> 0.15), and top & bottom edges on the left have span length > 0.1, 0.15 spacing is needed

b) OK, parallel run length does not count for opposite sides

c) OK, parallel run length <= 0.15

d) OK, parallel run length is not continuous, and each of the individual ones of 0.09 <= 0.15 would not trigger the rule

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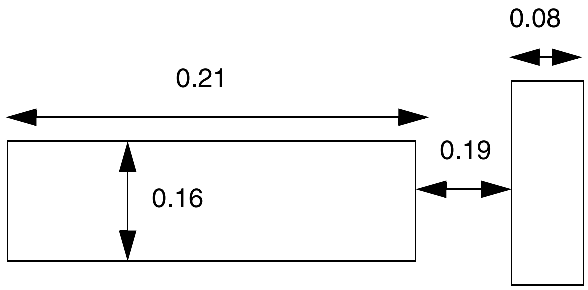

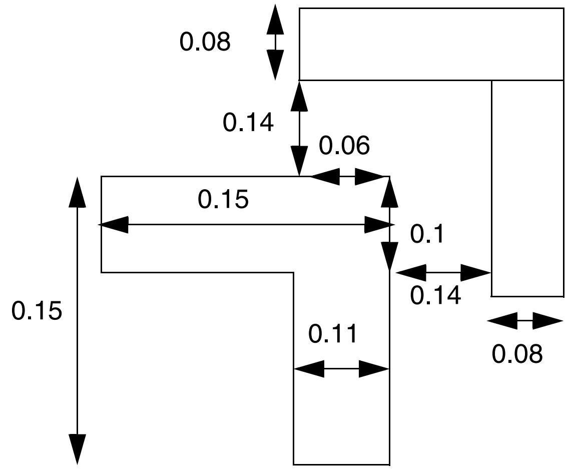

e) Violation, it can happen in EOL when the conditions are fulfilled. With span length of 0.21, 0.2 spacing is needed

f) OK, the top & side parallel run length is calculated separately, and each of them <= 0.15

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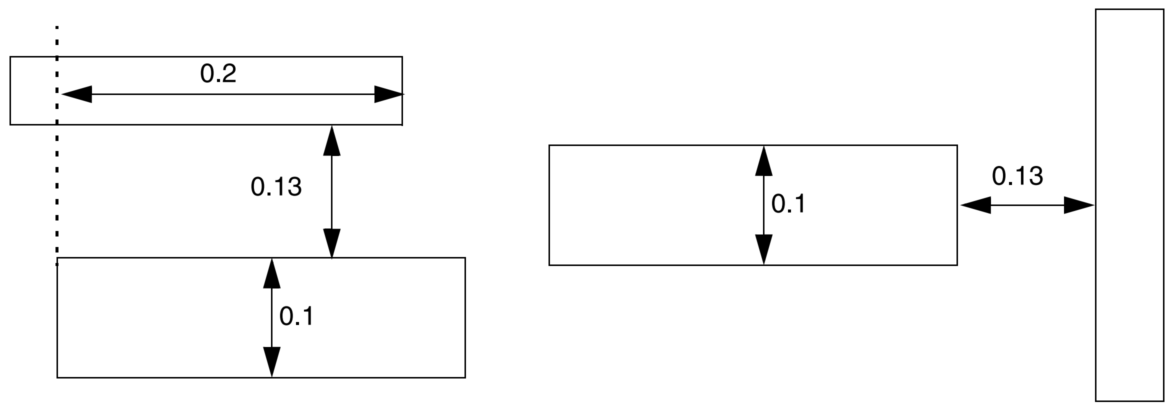
Figure 6-196 Illustration of Spacing Table Rule with WRONGDIRECTION

```
DIRECTION VERTICAL ;
PROPERTY LEF58_SPACINGTABLE
    "SPACINGTABLE
        PARALLELRUNLENGTH WRONGDIRECTION 0.0   0.08
           WIDTH 0.0                     0.05  0.05
           WIDTH 0.09                    0.05  0.14 ; " ;
```



a) Violation, 0.14 spacing is needed

b) Violation, 0.14 spacing is also needed on the side

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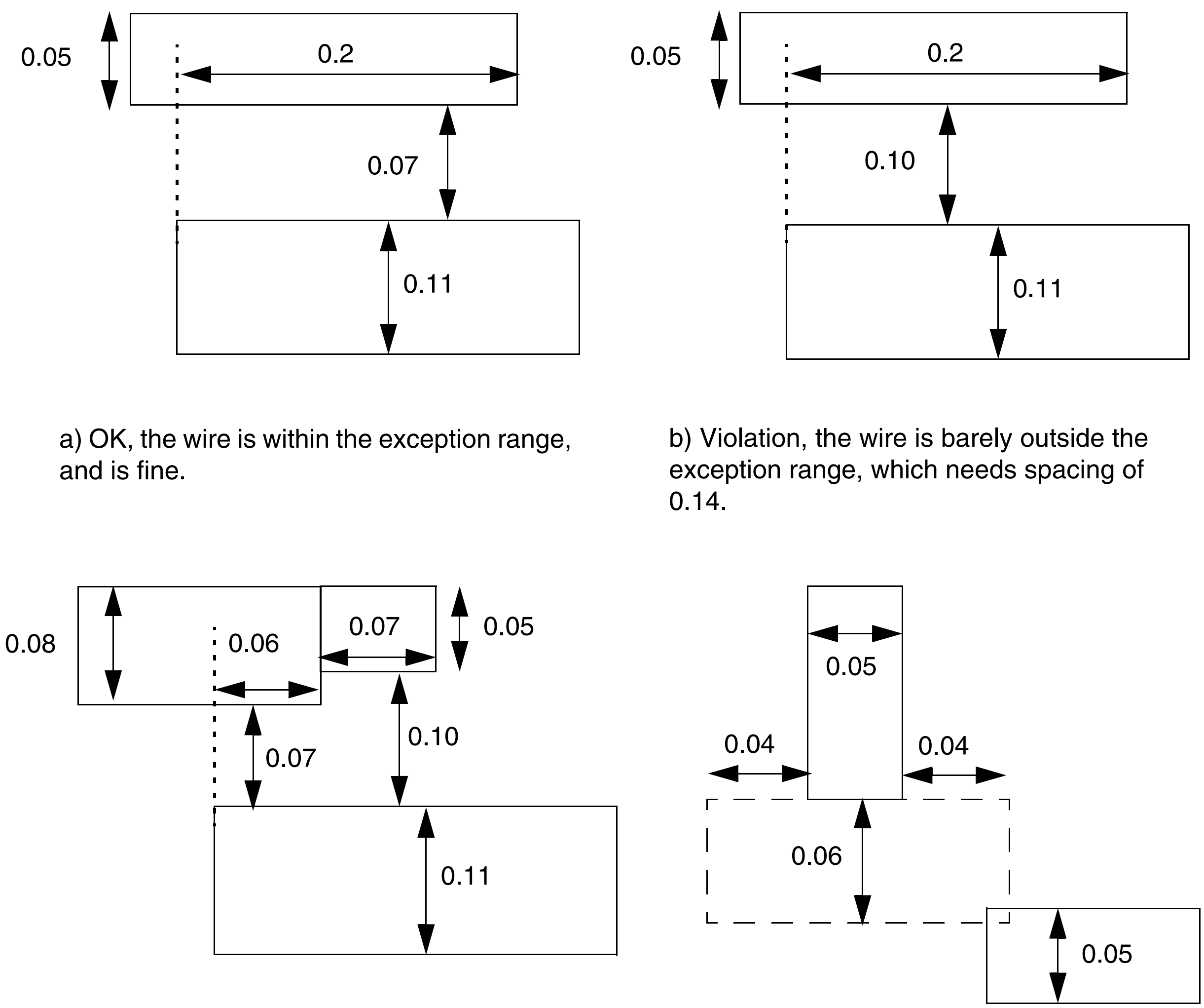
Figure 6-197 Illustration of Spacing Table Rule with EXCEPTWITHIN

```
PROPERTY LEF58_SPACINGTABLE
  "SPACINGTABLE
      PARALLELRUNLENGTH                       -0.041  -0.04  0.08
         WIDTH 0.0                             0.05    0.06  0.06
         WIDTH 0.1 EXCEPTWITHIN 0.07 0.10      0.05    0.06  0.14 ; " ;
```



a) OK, the wire is within the exception range, and is fine.

b) Violation, the wire is barely outside the exception range, which needs spacing of 0.14.

c) OK, PRL of the portion of wire outside the exception range is 0.07, which would not trigger the rule

d) Violation, the neighbor wire is within 0.04, and although the corner to corner spacing would be > 0.06, the spacing is checked in MAXXY style, which would fail.

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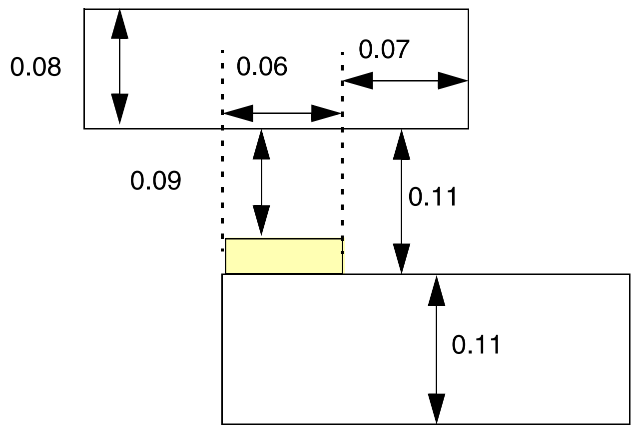
Figure 6-198 Illustration of Spacing Table Rule with EXCEPTWITHIN



e) Violation, PRL measurement should be 0.06 + 0.07 = 0.13 (> 0.08) because after the shrink & grow operation to determine the wide wire width, the yellow portion of the jog will be gone, and the spacing between the 2 objects will be 0.11 for the entire 0.13 length, which is a violaton (< 0.14).

- The following example illustrates `PARALLELRUNLENGTH` with `WRONGDIRECTION:`

```
DIRECTION VERTICAL ;
PROPERTY LEF58_WIDTH
    "WIDTH 0.12 WRONGDIRECTION ;
SPACINGTABLE
      PARALLELRUNLENGTH      0.00      0.07
         WIDTH 0.0           0.05      0.05
         WIDTH 0.1           0.05      0.08 ; " ;
PROPERTY LEF58_SPACINGTABLE
    "SPACINGTABLE
      PARALLELRUNLENGTH WRONGDIRECTION ... ; " ;
```


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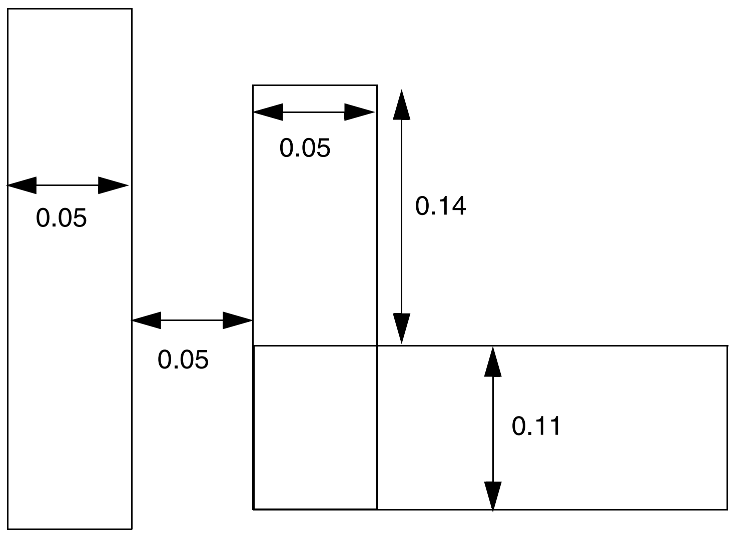
Figure 6-199 Illustration of PARALLELRUNLENGTH with WRONGDIRECTION



a) OK, the entire left edge of the right wire is 0.25, but edge length with width > 0.1 is only 0.11 (<= 0.12, default wrong direction minimum width). Hence, 0.05 spacing is needed and is met.

Two Wires Forbidden Spacing Rule

Two wires forbidden spacing rules can be used to define a list of forbidden spacing ranges between two wires.

You can create a two wires forbidden spacing rule by using the following property definition:

```
PROPERTY LEF58_TWOWIRESFORBIDDENSPACING
     "TWOWIRESFORBIDDENSPACING [SAMEMASK | MASK maskNum]
        {minSpacing maxSpacing}...
        [HORIZONTAL|VERTICAL]
        [IGNOREMIDDLE]
        { MINSPANLENGTH minSpanLength [EXACTSPANLENGTH]
          MAXSPANLENGTH maxSpanLength [EXACTSPANLENGTH]
            |FIRSTSPANLENGTH spanLength1 spanLength2
                [RECTONLY | EXCEPTRECTONLY]
                [OTHERSPANLENGTH otherSpanLength1
                   [EXACTEQUAL | LESSTHAN]
                |OTHEREDGELENGTH otherEdgeLength1
                   [EXCEPTSPANLENGTH minSpanLength1 maxSpanLength1]]
             SECONDSPANLENGTH spanLength3 spanLength4}
                [RECTONLY | EXCEPTRECTONLY]
```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```
              [OTHERSPANLENGTH otherSpanLength2]
                 [EXACTEQUAL | LESSTHAN]
              |OTHEREDGELENGTH otherEdgeLength1
                 [EXCEPTSPANLENGTH minSpanLength1 maxSpanLength1]]
           |OFFGRIDONLY minWidth1 maxWidth1 [minWidth2 maxWidth2]
           |ORTHOGONALNONEOL eolWidth
              SECONDNONEOL minWidth maxWidth }
          PRL prl [UNION spanLength]
       ; " ;
```

Where:

`EXCEPTSPANLENGTH` *`minSpanLength maxSpanLength`*

Specifies that the edge length should not be summed up if the span length of the portion that does not fulfill the span length condition has a span length greater than or equal to *`minSpanLength`* and less than or equal to *`maxSpanLength`*.

Type: Float, specified in microns

`FIRSTSPANLENGTH` *`spanLength1 spanLength2`*
`SECONDSPANLENGTH` *`spanLength3 spanLength4`*

Specifies that the span length of one wire must be greater than or equal to *`spanLength1`* and less than or equal to *`spanLength2`* and the other wire must be greater than or equal to *`spanLength3`* and less than or equal to *`spanLength4`* to trigger the rule.

Type: Float, specified in microns

`HORIZONTAL|VERTICAL` Specifies that the check applies only on the given direction on the spacing.

`IGNOREMIDDLE` Specifies that any wires between the two wires that trigger the rule do not have any impact. It is as if the middle wires did not exist.

`MASK` *`maskNum`* Specifies that the forbidden spacing applies only on the given mask. *`maskNum`* must be a positive integer, and most applications support only the values 1, 2, or 3.

Type: Integer

`OFFGRIDONLY` *`minWidth1 maxWidth1`* `[`*`minWidth2 maxWidth2`*`]`

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Specifies that the rule applies only if at least one of the wires in a certain direction is an off-grid wire with width greater than or equal to *minWidth1* and less than or equal to *maxWidth1* and the other wire has width greater than or equal to *minWidth2* and less than or equal to *maxWidth2*, if *minWidth2* and *maxWidth2* are specified. HORIZONTAL|VERTICAL must be defined with this construct, and it should be the non-preferred routing direction of the layer. The off-grid rule would be checked only when the width of both wires is in the specified direction. For example, on a vertical preferred routing layer, HORIZONTAL should be defined along with OFFGRIDONLY. Only vertical wires with a horizontal width would be checked for this off-grid rule.

Type: Float, specified in microns

OTHEREDGELENGTH *otherEdgeLength*

Specifies that the rule applies only if the entire edge length in the orthogonal direction is greater than *otherEdgeLength*. The entire edge includes the portion that does not fulfill the span length condition as long as some other portion fulfills the span length condition in a rectilinear shape. OTHEREDGELENGTH should be defined along with either HORIZONTAL or VERTICAL.

Type: Float, specified in microns

ORTHOGONALNONEOL *eolWidth* SECONDNONEOL *minWidth maxWidth*

Specifies one or more set of forbidden spacing on two non-EOL edges in the orthogonal direction of the spacing. A non-EOL edge is not EOL with width less than *eolWidth*. One of the two non-EOL edges must have width greater than or equal to *minWidth* and less than or equal to *maxWidth* on the entire edge. This construct should typically be used along with HORIZONTAL or VERTICAL.

Type: Float, specified in microns

OTHERSPANLENGTH *otherSpanLength* [EXACTEQUAL | LESSTHAN]

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Specifies that the rule applies only if the span length in the orthogonal direction is greater than *otherSpanLength*. If EXACTEQUAL is specified, the span length in the orthogonal direction must be exactly equal to *otherSpanLength* to trigger the rule. If LESSTHAN is specified, the span length in the orthogonal direction must be less than *otherSpanLength* to trigger the rule. OTHERSPANLENGTH should be defined along with either HORIZONTAL or VERTICAL.

Type: Float, specified in microns

RECTONLY | EXCEPTRECTONLY

Specifies that the rule applies only on rectangular wire in RECTONLY or does not apply on rectangular wire in EXCEPTRECTONLY.

SAMEMASK

Specifies that the forbidden spacing rule applies only to two wires on the same mask. This rule is applied independent of whether there is a neighbor wire with a different mask between them.

TWOWIRESFORBIDDENSPACING {*minSpacing maxSpacing*}...
MINSPANLENGTH *minSpanLength* [EXACTSPANLENGTH]
MAXSPANLENGTH *maxSpanLength* [EXACTSPANLENGTH]
PRL *prl*

Specifies that one or more set of forbidden spacing between two wires to be greater than or equal to *minSpacing* and less than or equal to *maxSpacing*. One of the wires must have a span length greater than or equal to *minSpanLength* or exactly equal to the given span length if EXACTSPANLENGTH is defined. The other wire must have a span length less than or equal to *maxSpanLength* or exactly equal to the given span length if EXACTSPANLENGTH is defined. The two wires must also have a parallel run length greater than *prl*.

Type: Float, specified in microns

UNION *spanLength*

Specifies that the PRL is the union sum of the edges with span length greater than or equal to *spanLength* against the entire neighbor edges. UNION should be defined along with FIRSTSPANLENGTH and SECONDSPANLENGTH.

Type: Float, specified in microns

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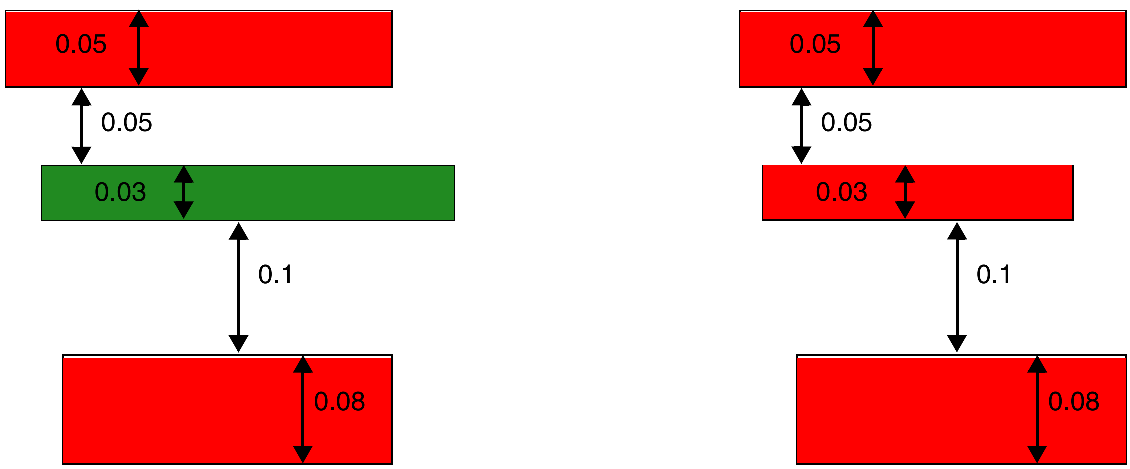
Two Wires Forbidden Spacing Examples

- In the following example, the specified forbidden spacing rules are applied only in the horizontal direction:

```
DIRECTION HORIZONTAL ;
PROPERTY LEF58_TWOWIRESFORBIDDENSPACING "
       TWOWIRESFORBIDDENSPACING SAMEMASK 0.16 0.20
         MINSPANLENGTH 0.05 EXACTSPANLENGTH
         MAXSPANLENGTH 0.08 EXACTSPANLENGTH  PRL 0 ; " ;
```

Figure 6-200 Illustration of TWOWIRESFORBIDDENSPACING



a) Violation, the spacing between the top and bottom wires is 0.18, which is within the forbidden spacing of >= 0.16 and <= 0.20. It is a violation with or without the middle 0.03 different-mask wire. If the middle wire was red, it would not be a violation.

b) Violation, the middle 0.03 same-mask wire does not fully cover the top and bottom wires. It is a violation based on the parallel run length on the right.

- Consider the following forbidden spacing rule with `EXCEPTSPANLENGTH`:

```
PROPERTY LEF58_TWOWIRESFORBIDDENSPACING "
       TWOWIRESFORBIDDENSPACING 0.16 0.20 VERTICAL
         FIRSTSPANLENGTH 0.04 0.50
         SECONDSPANLENGTH 0.12 0.15 OTHEREDGELENGTH 0.09
         EXCEPTSPANLENGTH 0.05 0.05 PRL 0.14 ; " ;
```


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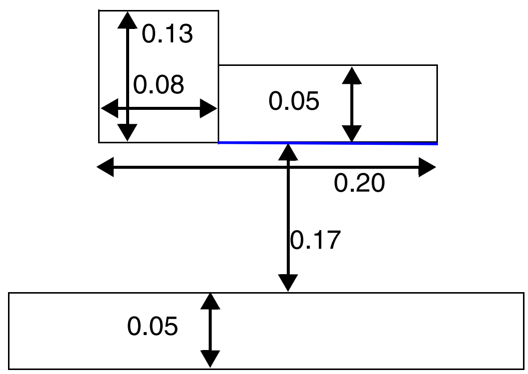
Figure 6-201 Illustration of TWOWIRESFORBIDDENSPACING with EXCEPTSPANLENGTH



a) OK, the blue portion has span length of 0.05, which is in the range of `EXCEPTSPANLENGTH` and the corresponding edge length is not summed up to check against `OTHEREDGELENGTH`. Violation if `EXCEPTSPANLENGTH` is omitted.

- Consider the following forbidden spacing rule:

```
DIRECTION HORIZONTAL ;
PROPERTY LEF58_TWOWIRESFORBIDDENSPACING "
       TWOWIRESFORBIDDENSPACING 0.16 0.20 IGNOREMIDDLE
         MINSPANLENGTH 0.05 EXACTSPANLENGTH
         MAXSPANLENGTH 0.08 EXACTSPANLENGTH  PRL 0.1 ; " ;
```


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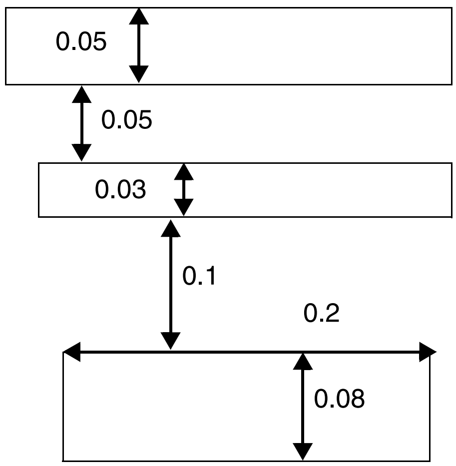
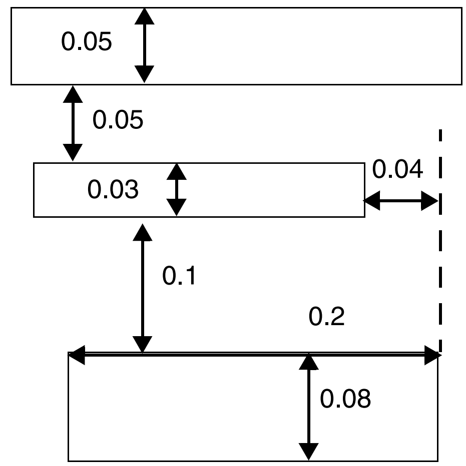
Figure 6-202 Illustration of TWOWIRESFORBIDDENSPACING with IGNOREMIDDLE



a) Violation, the middle wire does not impact the rule as if it did not exist.



b) Violation, the middle wire does not impact the rule and does not change the PRL between the two wires

- Consider the following forbidden spacing rule with `ORTHOGONALNONEOL`:

```
PROPERTY LEF_CDN_TWOWIRESFORBIDDENSPACING "
       TWOWIRESFORBIDDENSPACING 0.16 0.20 VERTICAL
          ORTHOGONALNONEOL 0.14
          SECONDNONEOL 0.07 0.07 PRL 0.000 ; " ;
```


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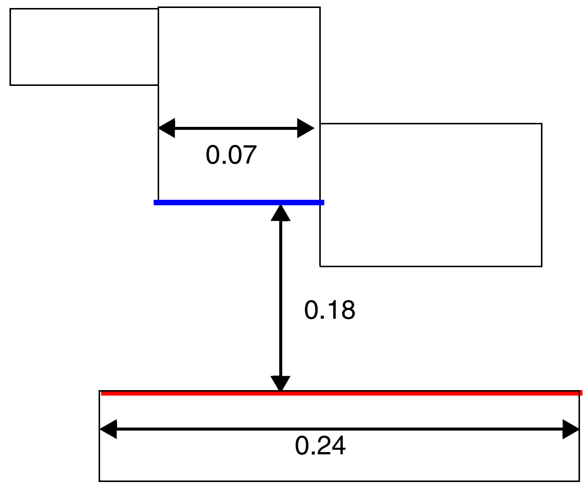
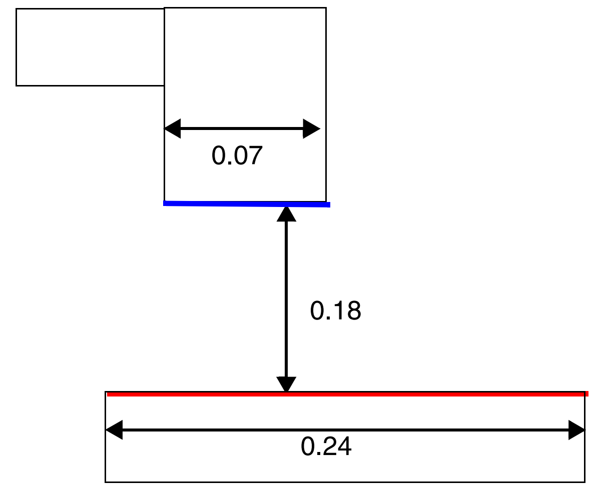
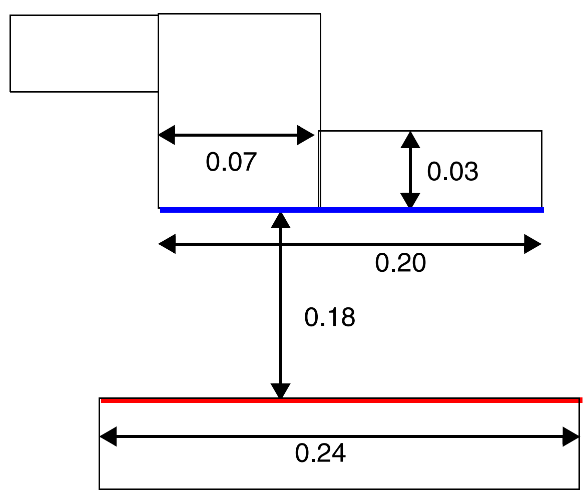
Figure 6-203 Illustrations of TWOWIRESFORBIDDENSPACING with ORTHOGONALNONEOL



a) Violation, the blue non-EOL edge has a width of `0.07` and the spacing to the red non-EOL is `0.18` (>= `0.16` and <= `0.20`).



b) OK, the blue edge is an EOL.



c) OK, only the left portion of the blue non-EOL has width of `0.07`.

- Consider the following forbidden spacing rule with `OTHEREDGELENGTH`:

```
PROPERTY LEF_CDN_TWOWIRESFORBIDDENSPACING "
       TWOWIRESFORBIDDENSPACING 0.16 0.20 VERTICAL
          FIRSTSPANLENGTH 0.04 0.05
          SECONDSPANLENGTH 0.12 0.15 OTHEREDGELENGTH 0.09
          PRL 0.14 UNION 0.12 ; " ;
```


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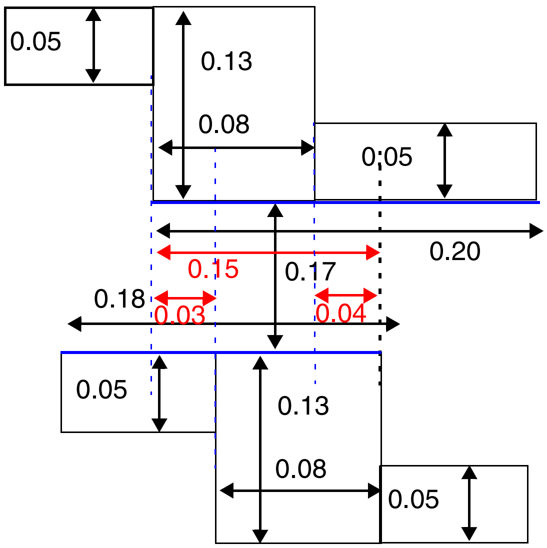
Figure 6-204 Illustration of TWOWIRESFORBIDDENSPACING with OTHEREDGELENGTH



Violation, `OTHEREDGELENGTH` is the entire blue edges of length `0.18` and `0.20` (> `0.09`) although the horizontal span length is just `0.08` with the vertical span length of `0.13` fulfilling the `SECONDSPANLENGTH` condition and `UNION` means that `PRL` is the red value of `0.15`, not only `0.08`, not only `0.03` or `0.04` individually.

Figure 6-205 Illustration of TWOWIRESFORBIDDENSPACING with OTHERSPANLENGTH

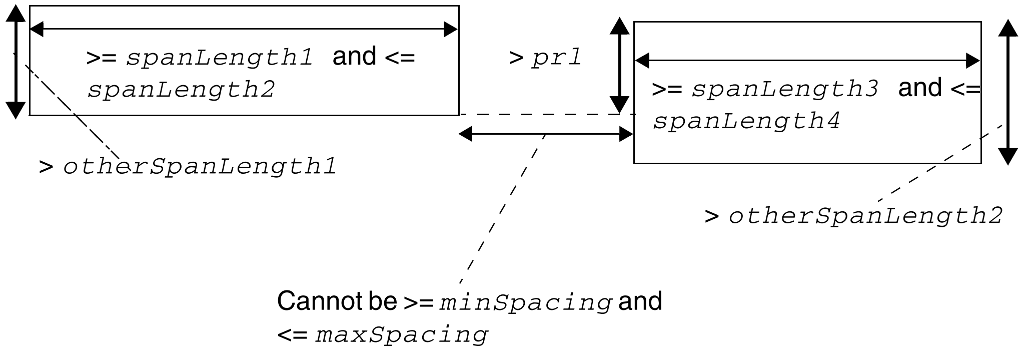


Illustration of `TWOWIRESFORBIDDENSPACING` *`minSpacing maxSpacing`* `HORIZONTAL`

```
FIRSTSPANLENGTH spanLength1 spanLength2
   OTHERSPANLENGTH otherSpanLength1
SECONDSPANLENGTH spanLength3 spanLength4
   OTHERSPANLENGTH otherSpanLength2
PRL prl
```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Voltage Spacing Rule

Voltage spacing rules can be used to define different-net spacing between objects on the metal layer and objects on the metal/cut layer above or below it according to different voltages.

You can create a voltage spacing rule by using the following property definition:

```
PROPERTY LEF58_VOLTAGESPACING
     "VOLTAGESPACING [TOCUT [ABOVE | BELOW]]
        {voltage spacing} ...
          ;..." ;
```

Where:

`VOLTAGESPACING {voltage spacing} ...`

Defines different-net spacing among wires in the same routing layer in different voltages. Both the voltage and spacing values should be monotonically increasing. The spacing value is required for any net with a worst-case voltage between its neighbors, including negative minimum voltage greater than or equal to the voltage in the table. For example, if you have 1.5v and 2.0v rules, then the spacing between a net with a max 1.3v and a min 0.0v and a net with a max 1.4v and a min -0.5v should follow the 1.5v spacing since the worst-case voltage is max (1.3v - (-0.5v) = 1.8v, 1.4v - 0.0v = 1.4v) = 1.8v.

`TOCUT [ABOVE | BELOW]`

Defines the spacing for wires in different voltages in routing layer to all cut vias in adjacent above and below cut layers. If `ABOVE` or `BELOW` is specified, the rule only applies to cuts in the above or below cut layer correspondingly.

Type: Float, *voltage* in volts and *spacing* in microns

Voltage Spacing Rule Examples

- The following example indicates that 0.1 µm is the wire to wire spacing for voltage greater than or equal to 2.0V and less than 3.0V, 0.15 µm is the wire to wire spacing for voltage greater than or equal to 3.0V and less than 4.0V, and 0.2 µm is the wire to wire spacing for voltage greater than or equal to 4.0V:

```
VOLTAGESPACING
2.0      0.1
```


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```
3.0      0.15
4.0      0.2 ;
```

- The following example indicates the same as above, except that the spacing is between wires in metal layer to all cut vias in the adjacent above and below cut layers:

```
VOLTAGESPACING TOCUT
2.0      0.1
3.0      0.15
4.0      0.2 ;
```

Width Table Rule

You can use width table rules to define all the allowable legal widths on the routing layer.

You can define a width table rule by using the following property definition:

```
PROPERTY LEF58_WIDTHTABLE
     "WIDTHTABLE {width}... [WRONGDIRECTION] [ORTHOGONAL]
        ;" ;
```

Where:

`ORTHOGONAL`	Specifies that one of the right or wrong direction width between two inside corners of a rectilinear object must be greater than or equal to the first width value in the `WIDTHTABLE` in the corresponding direction, if `WIDTHTABLE WRONGDIRECTION` is specified. Otherwise, the width between the corners must be greater than or equal to the first width value in the `WIDTHTABLE` (for both directions) either vertically or horizontally. *Type:* Float, specified in microns

`WIDTHTABLE {`*`width`*`}...`

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Defines all the allowable legal widths on the routing layer. All the given widths are exact width values, except for the last one, which is equal to or greater than the value.

The `WIDTH` syntax should be used to define the default routing width on the layer, which should match one of the values in the `WIDTHTABLE` statement. In case that the last value of `WIDTHTABLE` denotes the exact width also, the `MAXWIDTH` statement with the last value can be used to represent it.

A polygon will be fractured into rectangles, and only the shorter dimension of the rectangles will be checked against the allowable legal widths.

At the most, two `WIDTHTABLE` spacing tables can be defined, one with `WRONGDIRECTION` and the other without it.

Type: Float, specified in microns.

`WRONGDIRECTION`

Specifies that the allowable legal widths are for wires with direction perpendicular to the specified direction in `DIRECTION` on a Manhattan routing layer.

Note that using `WRONGDIRECTION` changes the interpretation of any wrong-way routing widths in the `DEF NETS` section. If `WRONGDIRECTION` is specified, then any wrong-way routing in the `DEF NETS` section will use the `WRONGDIRECTION` width for that layer unless the net or route has a `NONDEFAULTRULE` with a `WIDTH` greater than the `WRONGDIRECTION` width. But, the implicit default route-extension is still half of the preferred direction width.

Some older tools may not understand this behavior. Normally, they will still read/write and round-trip the DEF routing properly, but will not understand that the width is slightly larger for wrong-way routes. If these tools check wrong-way width, then the DRC rules may flag false violations. RC extraction with the wrong width will also flag errors, although wrong-way routes are generally short and the width difference is small, so the RC error is normally negligible.

- The following width table rule indicates that width must be 0.10 µm, 0.15 µm, 0.20 µm, 0.25 µm, 0.30 µm, and greater than or equal to 0.40 µm.

```
MAXWIDTH 0.40 ;       # To define legal width =, instead of >=, 0.40 µm.
PROPERTY LEF58_WIDTHTABLE
```


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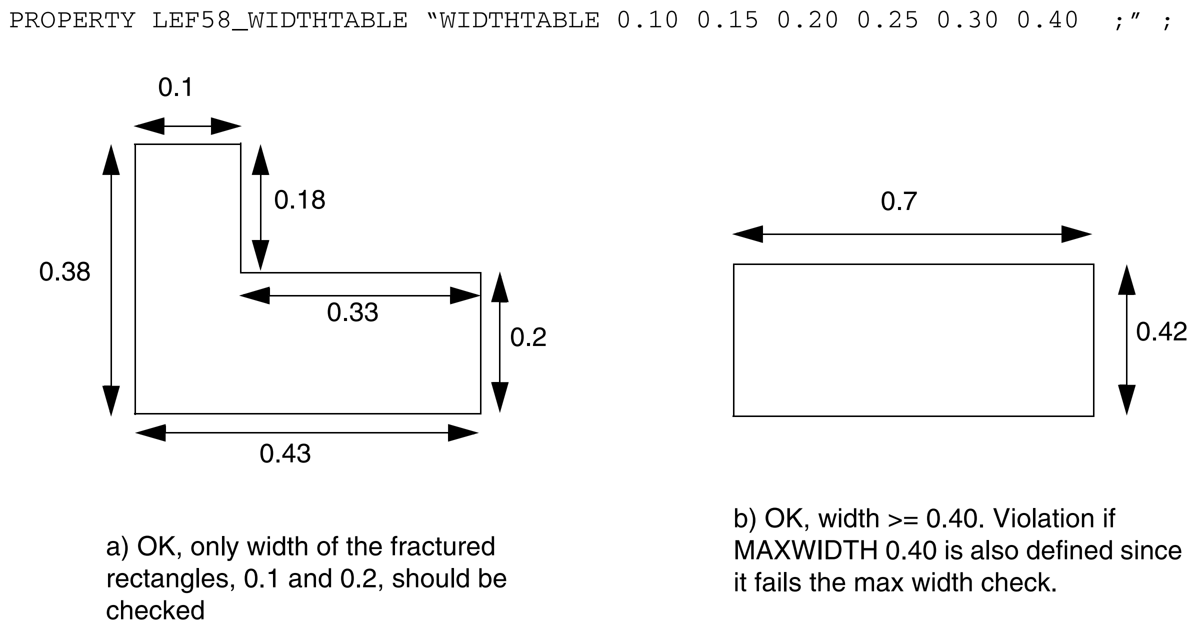
```
    "WIDTHTABLE 0.10 0.15 0.20 0.25 0.30 0.40  ;" ;
```

Figure 6-206 Illustration of WIDTHTABLE Rule



- The following width table rule indicates that width must be 0.05 µm, 0.1 µm, or greater than or equal to 0.15 µm for horizontal wires.

```
DIRECTION VERTICAL ;
PROPERTY LEF58_WIDTHTABLE
    "WIDTHTABLE 0.05 0.1 0.15 WRONGDIRECTION ; " ;
```


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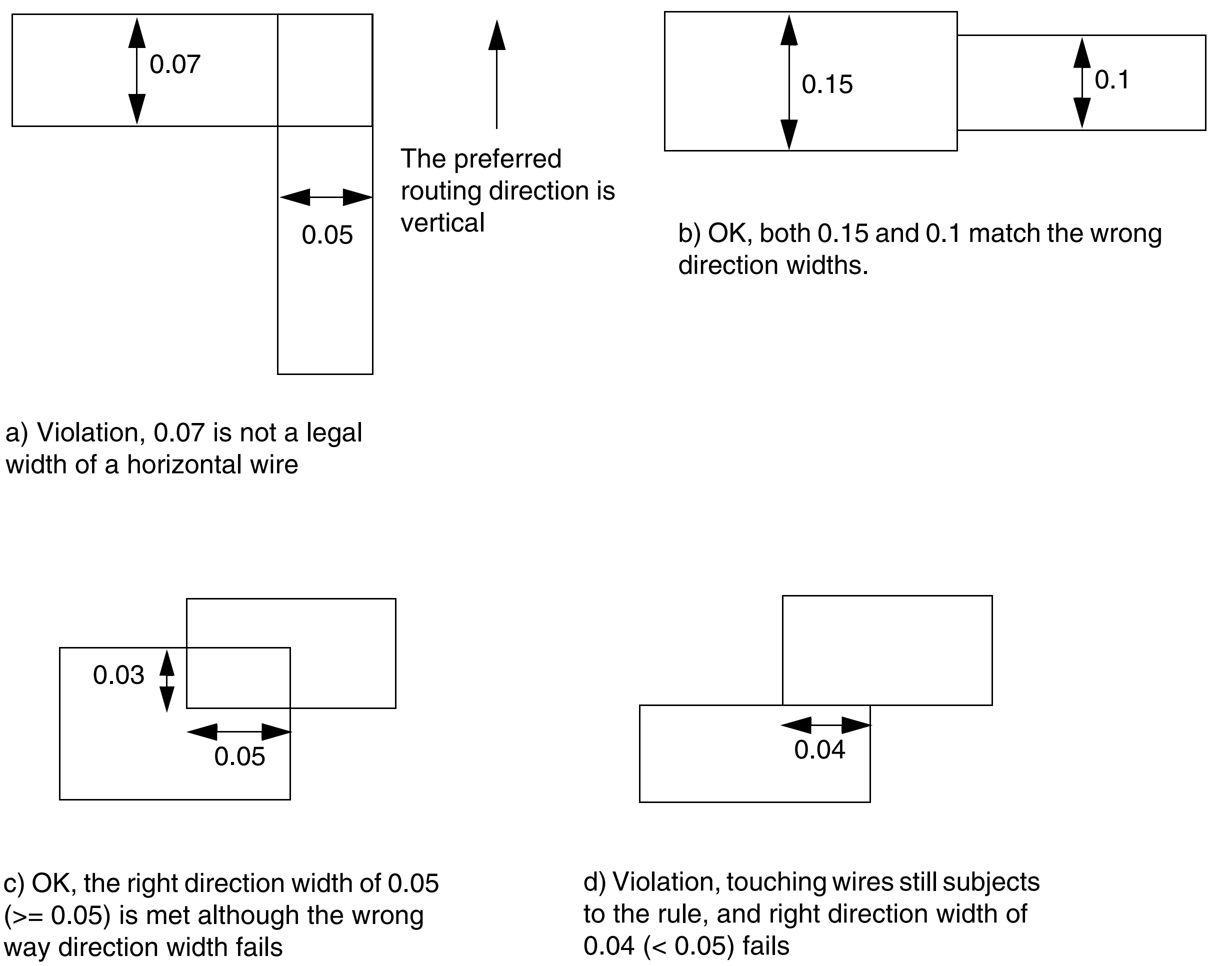
Figure 6-207 Illustration of Width Table Rule with ORTHOGONAL and WRONGDIRECTION

```
DIRECTION VERTICAL ;
PROPERTY LEF58_WIDTHTABLE
   "WIDTHTABLE 0.05 0.07 0.1 0.12 0.15 ORTHOGONAL ;
    WIDTHTABLE 0.1 0.15 WRONGDIRECTION ; " ;
```



Width Rule

Width rules can be used to define the default routing width to use for all regular wiring on the layer.

You can define a width rule by using the following property definition:

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```
PROPERTY LEF58_WIDTH
     "WIDTH minWidth [WRONGDIRECTION]
        ;" ;
```

Where:

WIDTH is the same as the existing routing layer WIDTH syntax.

WRONGDIRECTION	Specifies the default routing width to use for all regular wiring with direction perpendicular to the specified direction in DIRECTION on a Manhattan routing layer. Note that using WRONGDIRECTION changes the interpretation of any wrong-way routing widths in the DEF NETS section. If WRONGDIRECTION is specified, then any wrong-way routing in the DEF NETS section will use the WRONGDIRECTION width for that layer unless the net or route has a NONDEFAULTRULE with a WIDTH greater than the WRONGDIRECTION width. But, the implicit default route-extension is still half of the preferred direction width. Some older tools may not understand this behavior. Normally, they will still read/write and round-trip the DEF routing properly, but will not understand that the width is slightly larger for wrong-way routes. If these tools check wrong-way width, then the DRC rules may flag false violations. RC extraction with the wrong width will also flag errors, although wrong-way routes are generally short and the width difference is small, so the RC error is normally negligible.

Width Rule Examples

- The following width rule indicates that the default routing width of a vertical route is 0.1 µm, while the default routing width of a horizontal route is 0.14 µm:

  ```
  DIRECTION VERTICAL;
  WIDTH 0.1;
  PROPERTY LEF58_WIDTH
      "WIDTH 0.14 WRONGDIRECTION ;" ;
  ```

  In the DEF, a vertical default route in the NETS section will have a width of 0.10 µm with extension of 0.05 µm, while a horizontal route will have a width of 0.14 µm with extension of 0.05 µm.

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In more advanced nodes, wider widths are required for non-preferred direction. The following example shows route with wrong-way segment:

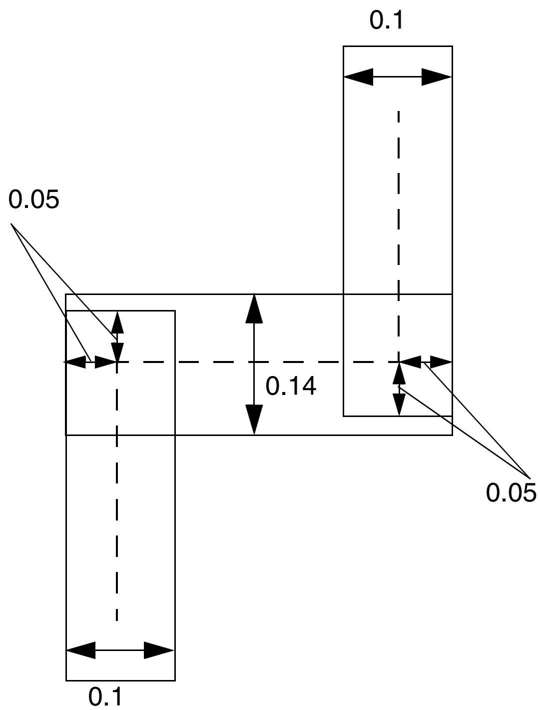

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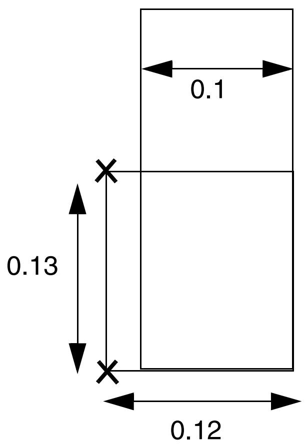
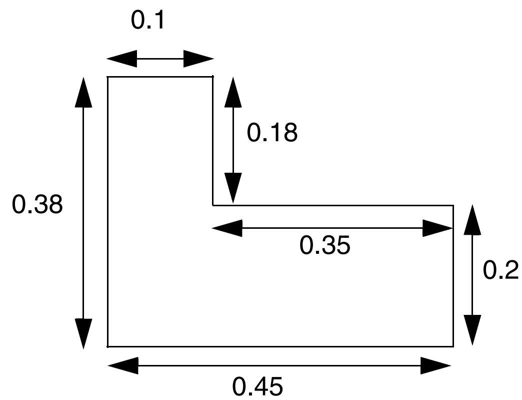
Figure 6-208 Illustration of WIDTH Rule with WRONGDIRECTION

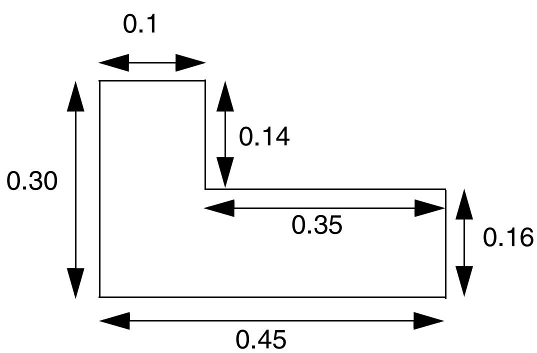
```
DIRECTION VERTICAL;
WIDTH 0.1;
PROPERTY LEF58_WIDTH "WIDTH 0.2 WRONGDIRECTION  ;" ;
```



a) Violation. The 0.13 distance between the two convex corners marked with X is less than the wrong direction width of 0.2. It does not matter that 0.13 is larger or smaller than the 0.12 value in the other direction.



b) OK, the 0.18 vertical segment does not count as the wrong direction width. The wrong direction width along that edge is actually 0.2 + 0.18 = 0.38.



c) Violation, the 0.16 vertical segment would fail the wrong direction width requirement of 0.2.

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Wrong Direction EOL Keep Out Rule

You can define a wrong direction EOL keep out rule by using the following property definition:

```
PROPERTY LEF58_WRONGDIREOLKEEPOUT
       "WRONGDIREOLKEEPOUT edgeExt wrongDirEdgeExt ENDOFLINE eolWidth
            EDGELENGTH minLength maxLength
            ;" ;
```

Where:

`WRONGDIREOLKEEPOUT` *`edgeExt wrongDirEdgeExt`* `ENDOFLINE` *`eolWidth`*
`EDGELENGTH` *`minLength maxLength`*

Specifies that a wrong-direction wire with a width equal to the minimum width/span length in the preferred routing direction would form a window by extending *`edgeExt`* in the preferred routing direction and *`wrongDirEdgeExt`* in the non-preferred routing direction based on the wrong direction wire. The wrong-direction wire must have an edge in the non-preferred direction greater than *`minLength`* and less than *`maxLength`*. In addition, the wrong-direction wire must be part of a rectilinear shape in the form of Z, like a same-metal jog, or L, like a thick via in the orthogonal direction of a wire. Then, the window could not overlap with any EOL with width/length less than *`eolWidth`*.

Type: Float, specified in microns

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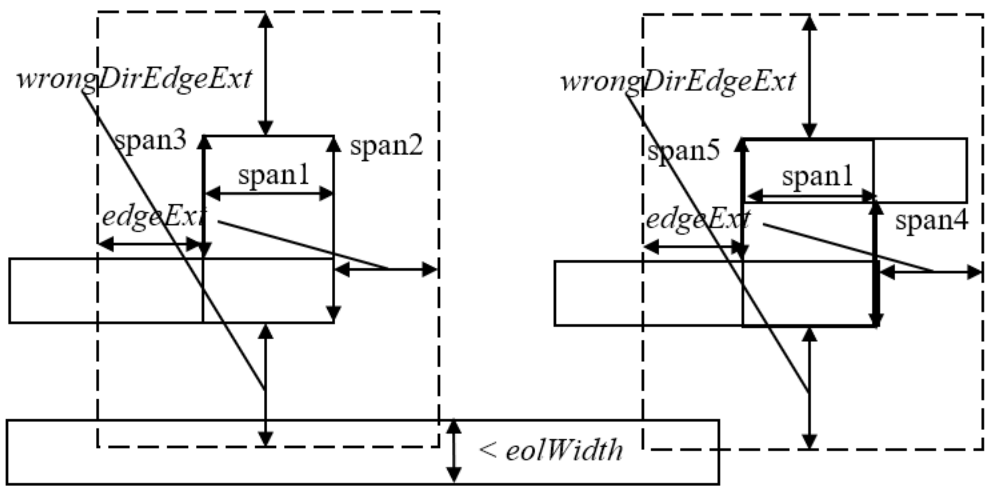
Wrong Direction EOL Keep Out Rule Example

Figure 6-209 Illustration of WRONGDIRECTIONEOLKEEPOUT Rule



Both `span2` to `span5` are greater than *`minLength`* and less than *`maxLength`*. The left pattern is fine since the neighboring wire at the bottom is a through wire without an EOL. The right pattern has two violations: the bottom wire has an EOL overlapping with the dotted window, and the right EOL of the Z shape is also a violation.

Illustration of

```
DIRECTION HORIZONTAL ;
PROPERTY LEF58_SPANLENGTHTABLE "
        SPANLENGTH ...
        SPANLENGTH span1 ... WRONGDIRECTION ; " ;
PROPERTY LEF58_WRONGDIREOLKEEPOUT "
WRONGDIREOLKEEPOUT edgeExt wrongDirEdgeExt
        ENDOFLINE eolWidth
        EDGELENGTH minLength maxLength ; " ;
```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Routing Pitch

The `PITCH` statements define the detail routing grid generated when you initialize a floorplan. The pitch for a given routing layer defines the distance between routing tracks in the preferred direction for that layer. The complete routing grid is the union of the tracks generated for each routing layer.

The spacing of the grid should be no less than line-to-via spacing in both the horizontal and vertical directions. Grid spacing less than line-to-via spacing can result in routing problems and can decrease the utilization results.

The grid should normally allow for diagonal vias. Via spacing on all layers included in the via definition in LEF determines whether or not diagonal vias can be used. The router is capable of avoiding violations between diagonal vias. If you allow diagonal vias, less time is needed for routing and the layout creates a smaller design.

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7

ALIAS Statements

This chapter contains information about the following topics.

ALIAS Statements

You can use alias statements in LEF and DEF files to define commands or parameters associated with the library or design. An alias statement can appear anywhere in a LEF or DEF file as follows:

```
&ALIAS  &&aliasName  =  aliasDefinition   &ENDALIAS
```

`&ALIAS` and `&ENDALIAS` are both reserved keywords and are not case sensitive. An alias statement has the following requirements:

- `&ALIAS` must be the first token in the line in which it appears.
- *aliasName* is string name and must appear on the same line as *&ALIAS.* It is case sensitive based on the value of `NAMESCASENSITIVE` in the LEF input, or the value of `Input.Lef.Names.Case.Sensitive`.
- *aliasName* cannot contain any of the following special characters: #, space, tab, or control characters.
- `&ENDALIAS` must be the last token in the line in which it appears.
- Multiple commands can appear in the alias definition, separated by semicolons. However, the last command must not be terminated by a semicolon.

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ALIAS Definition

The alias name (*aliasName*) is an identifier for the associated alias definition (*aliasDefinition*). The data reader stores the alias definition in the database. If the associated alias name already exists in the database, a warning is issued and the existing definition is replaced.

Alias definitions are text strings with the following properties:

- *aliasDefinition* is any text excluding "`&ENDALIAS`".
- All EOL, space, and tab characters are preserved.
- *aliasDefinition* text can expand to multiple lines.

ALIAS Examples

The following examples include legal and illegal alias statements:

- The following statement is legal.

  ```
  &ALIAS &&MAC = SROUTE ADDCELL AREA &&CORE &ENDALIAS
  ```

- The following statement is illegal because `MAC` does not start with "`&&`".

  ```
  &ALIAS MAC = SROUTE AREA &&CORE &ENDALIAS
  ```

- The following statement is illegal because `&ALIAS` is not the first token in this line.

  ```
  ( 100 200 ) &ALIAS &&MAC = SROUTE AREA &&CORE &ENDALIAS
  ```

- The following statement is legal. It contains multiple commands; the last command is not terminated by a semicolon.

  ```
  $ALIAS $$ = INPUT LEF myfile.txt;
  VERIFY LIBRARY
  ENDALIAS
  ```

The following examples show legal and illegal alias names:

- "`Engineer_change`" is a legal alias name.

  ```
  &&Engineer_change
  ```

- "`&Version&History&&`" is a legal alias name.

  ```
  &&&Version&History&&
  ```

- "`design history`" is an illegal alias name. It contains a space character and is considered as two tokens: an *aliasName* token "`&&design`," and a non-*aliasName* token "`history`".

  ```
  &&design history
  ```


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- "`someName#IO-pin-Num`" is an illegal alias name. It contains a "#" character and is translated as one *aliasName* token "`&&someName`". The "#" is considered a comment character.

  `&&someName#IO-pin-Num`

ALIAS Expansion

Alias expansion is the reverse operation of alias definition. The following is the syntax for alias expansion.

`&&`*`aliasName`*

where *aliasName* is any name previously defined by an alias statement. If an *aliasName* does not exist in the database, no substitution occurs.

You use aliases as string expansion parameters for LEF or DEF files. An alias can substitute for any token of a LEF or DEF file.

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8

DEF Syntax

This chapter contains information about the following topics:

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About Design Exchange Format Files

A Design Exchange Format (DEF) file contains the design-specific information of a circuit and is a representation of the design at any point during the layout process. The DEF file is an ASCII representation using the syntax conventions described in "Typographic and Syntax Conventions" on page 9.

DEF conveys logical design data to, and physical design data from, place-and-route tools. Logical design data can include internal connectivity (represented by a netlist), grouping information, and physical constraints. Physical data includes placement locations and orientations, routing geometry data, and logical design changes for back-annotation. Place-and-route tools also can read physical design data, for example, to perform ECO changes.

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For standard-cell-based/ASIC flow tools, floorplanning is part of the design flow. You typically use the various floorplanning commands to create a floorplan interactively. This data then becomes part of the physical data output for the design using the `ROWS`, `TRACKS`, `GCELLGRID`, and `DIEAREA` statements. You also can manually enter this data into DEF to create the floorplan.

It is legal for a DEF file to contain only floorplanning information, such as `ROWS`. In many cases, the DEF netlist information is in a separate format, such as Verilog, or in a separate DEF file. It is also common to have a DEF file that only contains a `COMPONENTS` section to pass placement information.

General Rules

Note the following information about creating DEF files:

- Identifiers like net names and cell names are limited to 2,048 characters.
- DEF statements end with a semicolon (`;`). You *must* leave a space before the semicolon.
- Each section can be specified only once. Sections end with `END` *`SECTION`*.
- You must define all objects before you reference them except for the `+ ORIGINAL` argument in the `NETS` section.

Character Information

LEF and DEF identifiers can contain any printable ASCII character, except space, tab, or new-line characters. This means the following characters are allowed along with all alpha-numeric characters:

```
! " # $ % & ' ( ) * + , / : ; < = > ? @ [ \ ] ^ _ ` { | } ~
```

A LEF or DEF property string value is contained inside a pair of quotes, like this "*<string>*". The *<string>* value can contain any printable ASCII character as shown above, including space, tab, or new-line characters.

Some characters have reserved meanings unless escaped with \.

`[ ]`	Default special characters for bus bits inside a net or pin name unless overridden by `BUSBITCHARS`
`/`	Default special character for hierarchy inside a net or component name unless overridden by `DIVIDERCHAR`

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`#`	The comment character. If preceded by a space, tab, or new-line, everything after # until the next new-line is treated as a comment.
`*`	Matches any sequence of characters for `SPECIALNETS` or `GROUPS` component identifiers.
`%`	Matches any single character for `SPECIALNETS` or `GROUPS` component identifiers.
`"`	The start and end character of a property string value. It has no special meaning for an identifier.
`\`	The escape character

You can use the backslash (`\`) as an escape character before each of the special characters shown above. When the backslash precedes a character that has a special meaning in LEF or DEF, the special meaning of the character is ignored.

Name Escaping Semantics for Identifiers

Here are some examples depicting the use of the escape character (`\`) in identifiers:

- A DEF file with `BUSBITCHARS "[]"` and net or pin name:

`A[0]`	is the 0th member of bus A
`A<0>`	is a scalar named A<0>
`A\[0\]`	is a scalar named A[0]

- A DEF file with `DIVIDERCHAR "/"` and net or component names like:

`A/B/C`	is a 3-level hierarchical name, where C is inside B and B is inside A
`A/B\/C`	is a 2-level hierarchical name where B/C is inside A
`A\/B`	is a flat name A/B

- The `"` character has no special meaning for an identifier. So an identifier would not need a `\` before a `"`. An identifier like:

  ```
  name_with_"_in_it
  ```

  or even

  ```
  "_name_starts_with_qoute
  ```


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is legal without a \.

- An identifier that starts with # would be treated as a comment unless the \ is present like this:

  ```
  \#_name_with_hash_for_first_char
  ```

  Note, the # needs to be preceded by white-space to be treated as a comment char, so it has no special meaning inside an identifier. So an identifier like

  ```
  name_with_#_in_it
  ```

  is legal without a \.

- Pattern matching characters * or % inside SPECIALNETS or GROUPS component identifiers can be disabled like this:

  ```
      GROUPS  1 ;
           - myGroup i1/i2/\* ...
  ```

  or

  ```
     SPECIALNETS 1 ;
           - VDD ( i1/i2/\*  VDD ) ...
  ```

  These will match the exact name i1/i2/* and not match i1/i2/i3 or other components starting with i1/i2/.

  Note, the * and % have no special meaning in other identifiers, so no \ is needed for them.

- A real \ char in an identifier needs to be escaped like this:

  ```
  name_with_\\_in_it ....
  ```

  The first \ escapes the second \, so the real name is just name_with__in_it.

Escaping Semantics for Quoted Property Strings

Properties may have string type values, placed within double quotes ("). However, if you need to use a double quote as a part of the string value itself, you would need to precede it with the escape character (\) to avoid breaking the property syntax. The escape sequence '\"' is converted to '"' during parsing.

The example below depicts the use of the escape character in a quoted property string:

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```
PROPERTY stringQuotedProp "string with \" quote and single
backslash \and double backslash \\" ;
```

The actual value of the property in the database will be:

```
string with " quote and single backslash and double backslash\
```

Here:

- The first `\` escapes the `"` so it does not end the property string.
- The next `\` has no effect on the subsequent `a` character.
- The first `\` in `\\` escapes the second `\` character. This means that the second `\` in `\\` is treated as a real `\` character, and it does not escape the final `"` character, which ends the property string.

Note that the other special characters like `[ ] / # * %` have no special meaning inside a property string and do not need to be escaped.

LEF/DEF to LEF/DEF Equivalence

In DEF syntax, `\` is only used to escape characters that have a special meaning if they are not escaped.

Consider the following LEF/DEF header specification:

- `LEFDEF/[]` is equivalent to LEF or DEF with `DIVIDERCHAR "/"` and `BUSBITCHARS "[]"`
- `LEFDEF|<>` is equivalent to LEF or DEF with `DIVIDERCHAR "|"` and `BUSBITCHARS "<>"`

In the following examples, `<>` are not special characters for `LEFDEF/[]` files and `[]` are not special characters for `LEFDEF|<>` files. Observe how the header settings (listed above) affect the semantic meaning of the names:

- `A<0>` with `LEFDEF/[]` is not equivalent to `A<0>` with `LEFDEF|<>`
- `A<0>` with `LEFDEF/[]` is equivalent to `A\<0\>` with `LEFDEF|<>`
- `A[0]` with `LEFDEF/[]` is equivalent to `A<0>` with `LEFDEF|<>`

Verilog and DEF Equivalence

For Verilog and DEF equivalence, consider the following DEF header specification:

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- `DEF/[]` is equivalent to DEF with `DIVIDERCHAR "/"` and `BUSBITCHARS "[]"`
- `DEF|<>` is equivalent to DEF with `DIVIDERCHAR "|"` and `BUSBITCHARS "<>"`

In the following examples (showing net names), `<>` are not special characters for `DEF/[]` files and `[]` are not special characters for `DEF|<>` files:

- `A<0>` in `DEF/[]` is equivalent to `\A<0>` in Verilog

  `A<0>` in `DEF|<>` is equivalent to `A[0]` in Verilog (bit 0 of bus `A`)
- `A[0]` in `DEF/[]` is equivalent to `A[0]` in Verilog (bit 0 of bus `A`)

  `A[0]` in `DEF|<>` is equivalent to `\A[0]` in Verilog
- `A\<0\>` in `DEF/[]` is equivalent to `\A<0>` in Verilog

  `A\<0\>` in `DEF|<>` is equivalent to `\A<0>` in Verilog
- `A\[0\]` in `DEF/[]` is equivalent to `\A[0]` in Verilog

  `A\[0\]` in `DEF|<>` is equivalent to `\A[0]` in Verilog *

The following example shows instance path names for Verilog and DEF equivalence:

- `A/B` in `DEF/[]` represents instance path `A.B` (instance `A` in the top module, with instance `B` inside the module referenced by instance `A`) in Verilog.
- `A\/B` in `DEF/[]` represents instance `\A/B` in Verilog.
- `A\/B/C` in `DEF/[]` represents `\A/B .C` in Verilog (escaped instance `\A/B` in the top module, with instance `C` inside the module referenced by instance `\A/B`).
- The net and instance path `A\/B/C/D[0]` in `DEF/[]` will represent `\A/B .C.D[0]` in Verilog (escaped instance `\A/B` in the top module, with instance `C` inside the module referenced by instance `\A/B`, and bus `D` in that module with bit 0 being specified).

Comparison of DEF and Verilog Escaping Semantics

The DEF escape `\` applies only to the next character and prevents the character from having a special meaning.

The Verilog escape `\` affects the complete "token" and is terminated by a trailing white space (" ", Tab, Enter, etc.).

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Order of DEF Statements

Standard DEF files can contain the following statements and sections. You can define the statements and sections in any order; however, data must be defined before it is used. For example, you must specify the `UNITS` statement before any statements that use values dependent on `UNITS` values, and `VIAS` statements must be defined before statements that use via names. If you specify statements and sections in the following order, all data is defined before being used.

```
[ VERSION statement ]
[ DIVIDERCHAR statement ]
[ BUSBITCHARS statement ]
DESIGN statement
[ TECHNOLOGY statement ]
[ UNITS statement ]
[ HISTORY statement ] ...
[ PROPERTYDEFINITIONS section ]
[ DIEAREA statement ]
[ ROWS statement ] ...
[ TRACKS statement ] ...
[ GCELLGRID statement ] ...
[ VIAS statement ]
[ STYLES statement ]
[ NONDEFAULTRULES statement ]
[ REGIONS statement ]
[ COMPONENTMASKSHIFT statement ]
[ COMPONENTS section ]
[ PINS section ]
[ PINPROPERTIES section ]
[ BLOCKAGES section ]
[ SLOTS section ]
[ FILLS section ]
[ SPECIALNETS section ]
[ NETS section ]
[ SCANCHAINS section ]
[ GROUPS section ]
[ BEGINEXT section ] ...
END DESIGN statement
```

DEF Statement Definitions

The following definitions describe the syntax arguments for the statements and sections that make up a DEF file. The statements and sections are listed in alphabetical order, *not* in the order they must appear in a DEF file. For the correct order, see Order of DEF Statements on page 982.

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Blockages

```
[BLOCKAGES numBlockages ;
     [- LAYER layerName
          [ + COMPONENT compName | + SLOTS | + FILLS
            | + PUSHDOWN | + EXCEPTPGNET]
          [ + SPACING minSpacing | + DESIGNRULEWIDTH effectiveWidth]
          [ + MASK maskNum]
               {RECT pt pt | POLYGON pt pt pt ...} ...
     ;] ...
     [- PLACEMENT
          [ + SOFT | + PARTIAL maxDensity]
          [ + PUSHDOWN]
          [ + COMPONENT compName]
            {RECT pt pt} ...
     ;] ...
END BLOCKAGES]
```

Defines placement and routing blockages in the design. You can define simple blockages (blockages specified for an area), or blockages that are associated with specific instances (components). Only placed instances can have instance-specific blockages. If you move the instance, its blockage moves with it.

COMPONENT *compName*

Specifies a component with which to associate a blockage. Specify with LAYER *layerName* to create a blockage on *layerName* associated with a component. Specify with PLACEMENT to create a placement blockage associated with a component.

DESIGNRULEWIDTH *effectiveWidth*

Specifies that the blockage has a width of *effectiveWidth* for the purposes of spacing calculations. If you specify DESIGNRULEWIDTH, you cannot specify SPACING. As a lot of spacing rules in advanced nodes no longer just rely on wire width, DESIGNRULEWIDTH is not allowed for 20nm and below nodes.
Type: DEF database units

EXCEPTPGNET

Indicates that the blockage only blocks signal net routing, and does not block power or ground net routing.

This can be used above noise sensitive blocks, to prevent signal routing on specific layers above the block, but allow power routing connections.

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`FILLS`
: Creates a blockage on the specified layer where metal fill shapes cannot be placed.

`LAYER` *`layerName`*
: Normally only cut or routing layers have blockages, but it is legal to create a blockage on any layer.

  Note: Cut-layer blockages will prevent vias from being placed in that area.

`MASK` *`maskNum`*
: Specifies which mask for double or triple patterning lithography to use for the specified shapes. The *`maskNum`* variable must be a positive integer. Most applications support values of only 1, 2, or 3. Shapes without any defined mask have no mask set (are uncolored).

  For example,

```
- LAYER metal1 + PUSHDOWN + MASK 1
    RECT ( -300 -310 ) ( 320 330 )   #rectangle on mask 1
    RECT ( -150 -160 ) ( 170 180 );  #rectangle on mask 1
```

`numBlockages`
: Specifies the number of blockages in the design specified in the `BLOCKAGES` section.

`PARTIAL` *`maxDensity`*
: Indicates that the initial placement should not use more than *`maxDensity`* percentage of the blockage area for standard cells. Later placement of clock tree buffers, or buffers added during timing optimization ignore this blockage. The *`maxDensity`* value is calculated as:

  standard cell area in blockage area/blockage area <= *maxDensity*

  This can be used to reduce the density in a locally congested area, and preserve it for buffer insertion.
  Type: Float
  Value: Between `0.0` and `100.0`

`PLACEMENT`
: Creates a placement blockage. You can create a simple placement blockage, or a placement blockage attached to a specific component.

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`POLYGON` *`pt pt pt`*	Specifies a sequence of at least three points to generate a polygon geometry. Most fab DRC rules require each polygon edge to be parallel to the x or y axis, or at a 45-degree angle. However, odd-angles are allowed sometimes on a few layers, such as bump layers. Each `POLYGON` statement defines a polygon generated by connecting each successive point, and then by connecting the first and last points. The *`pt`* syntax corresponds to a coordinate pair, such as *`x y`*. Specify an asterisk (`*`) to repeat the same value as the previous *`x`* or *`y`* value from the last point.
`PUSHDOWN`	Specifies that the blockage was pushed down into the block from the top level of the design.
`RECT` *`pt pt`*	Specifies the coordinates of the blockage geometry. The coordinates you specify are absolute. If you associate a blockage with a component, the coordinates are not relative to the component's origin.
`SOFT`	Indicates that the initial placement should not use the area, but later phases, such as timing optimization or clock tree synthesis, can use the blockage area. This can be used to preserve certain areas (such as small channels between blocks) for buffer insertion after the initial placement.
`SLOTS`	Creates a blockage on the specified layer where slots cannot be placed.
`SPACING` *`minSpacing`*	Specifies the minimum spacing allowed between this particular `blockage` and any other shape. The *`minSpacing`* value overrides all other normal LAYER-based spacing rules, including wide-wire spacing rules, end-of-line rules, parallel run-length rules, and so on. A `blockage` with `SPACING` is not "seen" by any other DRC check, except the simple check for *`minSpacing`* to any other routing shape on the same layer. The *`minSpacing`* value cannot be larger than the maximum spacing defined in the `SPACING` or `SPACINGTABLE` for that layer. Tools may change larger values to the maximum spacing value with a warning. *Type:* Integer, specified in DEF database units

Example 8-1 Blockages Statements

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- The following `BLOCKAGES` section defines eight blockages in the following order: two *metal2* routing blockages, a pushed down routing blockage, a routing blockage attached to component `|i4`, a floating placement blockage, a pushed down placement blockage, a placement blockage attached to component `|i3`, and a fill blockage.

```
BLOCKAGES 7 ;
    - LAYER metal2
         RECT ( -300 -310 ) ( 320 330 )
         RECT ( -150 -160 ) ( 170 180 ) ;
    - LAYER metal1 + PUSHDOWN
         RECT ( -150 -160 ) ( 170 180 ) ;
    - LAYER metal1 + COMPONENT |i4
         RECT ( -150 -160 ) ( 170 180 ) ;
    - PLACEMENT
         RECT ( -150 -160 ) ( 170 180 ) ;
    - PLACEMENT + PUSHDOWN
         RECT ( -150 -160 ) ( 170 180 ) ;
    - PLACEMENT + COMPONENT |i3
         RECT ( -150 -160 ) ( 170 180 ) ;
    - LAYER metal1 + FILLS
         RECT ( -160 -170 ) ( 180 190 ) ;
END BLOCKAGES
```

- The following `BLOCKAGES` section defines two blockages. One requires minimum spacing of 1000 database units for its rectangle and polygon. The other requires that its rectangle's width be treated as 1000 database units for DRC checking.

```
BLOCKAGES 2 ;
    - LAYER metal1
        + SPACING 1000     #RECT and POLYGON require at least 1000 dbu spacing
        RECT ( -300 -310 ) ( 320 300 )
        POLYGON ( 0 0 ) ( * 100 ) ( 100 * ) ( 200 200 ) ( 200 0 ) ; #Has 45-degree
                                                                   #edge
    - LAYER metal1
        + DESIGNRULEWIDTH 1000 #Treat the RECT as 1000 dbu wide for DRC checking
        RECT ( -150 -160 ) ( 170 180 ) ;
END BLOCKAGES
```

Bus Bit Characters

BUSBITCHARS "*delimiterPair*" ;

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Specifies the pair of characters used to specify bus bits when DEF names are mapped to or from other databases. The characters must be enclosed in double quotation marks. For example:

```
BUSBITCHARS "()" ;
```

If one of the bus bit characters appears in a DEF name as a regular character, you must use a backslash (\) before the character to prevent the DEF reader from interpreting the character as a bus bit delimiter.

If you do not specify the `BUSBITCHARS` statement in your DEF file, the default value is `"[]"`.

Component Mask Shift

`[COMPONENTMASKSHIFT` *`layer1`* `[`*`layer2`* `...] ;]`

Defines which layers of a component may potentially be shifted from the original mask colors in the LEF. This can be useful to shift all the layers of a specific component in order to align the masks with other component or router mask settings to increase routing density. This definition allows a specific component to compactly describe the mask shifting for that component.

All the listed layers must have a LEF MASK statement to indicate that the specified layer is either a two or three mask layer. The order of the layers must be increasing from the highest layer down to the lowest layer in the LEF layer order.

Example 8-2 Component Mask Shift

The following example indicates that any given component can shift the mask on layers `M3`, `M2`, `VIA1`, or `M1`:

```
COMPONENTMASKSHIFT M3 M2 VIA1 M1 ;
```

This layer list is used to interpret the `+ MASKSHIFT` *`shiftLayerMasks`* value for a specific component as shown in the example given below:

```
-  i1/i2 AND2
    + MASKSHIFT 1102  #M3 shifted by 1, M2 by 1, VIA1 by 0, M1 by 2
    ...
```

For details on components, see the Components section.

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Components

```
COMPONENTS numComps ;
     [- compName modelName
         [+ EEQMASTER macroName]
         [+ SOURCE {NETLIST | DIST | USER | TIMING}]
         [+ {FIXED pt orient | COVER pt orient | PLACED pt orient
                | UNPLACED} ]
         [+ MASKSHIFT shiftLayerMasks]
         [+ HALO [SOFT] left bottom right top]
         [+ ROUTEHALO haloDist minLayer maxLayer]
         [+ WEIGHT weight]
         [+ REGION regionName]
         [+ PROPERTY {propName propVal} ...]...
     ;] ...
     [PROPERTY DEF_CDN_ROUTEHALO
        "ROUTEHALO left bottom right top minLayer maxLayer
        ;" ;]
END COMPONENTS
```

Defines design components, their location, and associated attributes.

`compName modelName`	Specifies the component name in the design, which is an instance of *modelName*, the name of a model defined in the library. A *modelName* must be specified with each *compName*.
`COVER pt orient`	Specifies that the component has a location and is a part of a cover macro. A `COVER` component cannot be moved by automatic tools or interactive commands. You must specify the component's location and its orientation.
`EEQMASTER macroName`	
	Specifies that the component being defined should be electrically equivalent to the previously defined *macroName*.
`FIXED pt orient`	Specifies that the component has a location and cannot be moved by automatic tools, but can be moved using interactive commands. You must specify the component's location and orientation.

`HALO [SOFT] left bottom right top`

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Specifies a placement blockage around the component. The halo extends from the LEF macro's left edge(s) by *left*, from the bottom edge(s) by *bottom*, from the right edge(s) by *right*, and from the top edge(s) by *top*. The LEF macro edges are either defined by the rectangle formed by the MACRO SIZE statement, or, if OVERLAP obstructions exist (OBS shapes on a layer with TYPE OVERLAP), the polygon formed by merging the OVERLAP shapes.

If SOFT is specified, the placement halo is honored only during initial placement; later phases, such as timing optimization or clock tree synthesis, can use the halo area. This can be used to preserve certain areas (such as small channels between blocks) for buffer insertion.
Type: Integer, specified in DEF database units

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`MASKSHIFT` *shiftLayerMasks*

Specifies shifting the cell-master masks used in double or triple patterning for specific layers of an instance of the cell-master. This is mostly used for standard cells where the placer or router may shift one or more layer mask assignments for better density.

The *shiftLayerMasks* variable is a hex-encoded digit, with one digit per multi-mask layer:

...*<thirdLayerShift><secondLayerShift><bottomLayerShift>*

The *bottomLayerShift* value is the mask-shift for the bottom-most multi-mask layer defined in the `COMPONENTMASKSHIFT` statement. The *secondMaskShift*, *thirdMaskShift*, and so on, are the shift values for each layer in order above the bottom-most multi-mask layer. The missing digits indicate that no shift is needed so 002 and 2 have the same meaning.

For 2-mask layers, the *LayerShift* value must be 0 or 1 and indicates:

0 - No mask-shift

1 - Shift the mask colors by 1 (mask 1->2, and 2->1)

For 3-mask layers, the *LayerShift* value can be 0, 1, 2, 3, 4, or 5 that indicates:

0 - No mask shift

1 - Shift by 1 (1->2, 2->3, 3->1)

2 - Shift by 2 (1->3, 2->1, 3->2)

3 - Mask 1 is fixed, swap 2 and 3

4 - Mask 2 is fixed, swap 1 and 3

5 - Mask 3 is fixed, swap 1 and 2

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The purpose of 3, 4, 5 is for standard cells that have a fixed power-rail mask color, but the pins between the power-rails can still be shifted. Suppose you had a standard cell with mask 1 power rails, and three signal pins on mask 1, 2, and 3. A *`LayerShift`* of 3, will keep the mask 1 power rails and signal pin fixed on mask 1, while mask 2 and mask 3 signal pin shapes will swap mask colors.

Note: The `MASKSHIFT` definition does not depend on the LEF constructs of `FIXEDMASK` or `LAYERMASKSHIFT`. However, the LEF and DEF definitions should be consistent. For example, if `FIXEDMASK` is defined in LEF and `LAYERMASKSHIFT` is not, none of the components should have `MASKSHIFT` because the instance color should always be the same as the cell-master. If `LAYERMASKSHIFT` is defined in LEF, non-zero `MASKSHIFT` values for a component should be specified only on the layers that are allowed to have mask shift in `LAYERMASKSHIFT`.

See Example 8-3 on page 991.

Example 8-3 Mask Shift Layers for Components

The following example shows a LEF section that has a three-mask layer defined for `M1`, and two-mask layer defined for layers `VIA1` and `M2`:

```
COMPONENTMASKSHIFT M2 VIA1 M1 ;
COMPONENTS 100 ;
-  i1/i2 AND2
    + MASKSHIFT 2          #M1 layer masks are shifted by 2, no shift for others
    ...
- i1/i3 OR2
    + MASKSHIFT 103        ##M1 layer has shift 3, VIA1 0, M2 1
    ...
```

If an application shifts the layers M1, VIA1, and M2, then the above example indicates that the instance of AND2 cell shifts the M1 layer masks by 2. Since M1 is a three-mask layer, this shows that the cell-master M1 layer mask 1 shifts to 3, mask 2 shifts to 1, and mask 3 shifts to 2. The other layer masks are not shifted. The instance of OR2 cell shifts the M1 layer masks by 3. For a 3-mask layer this means keep mask 1 fixed, mask 2 shifts to 3, and mask 3 shifts to 2). The VIA1 layer does not shift and the M2 layer masks are shifted by 1 (for a two-mask layer this means that mask 1 shifts to 2, and 2 shifts to 1).

Example 8-4 Component Halo

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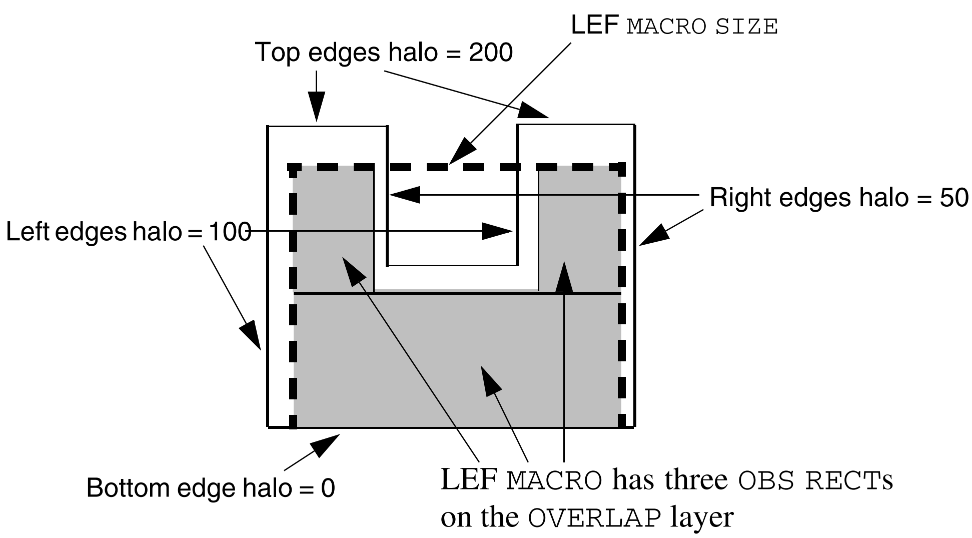
The following statement creates a placement blockage for a “U-shaped” LEF macro, as illustrated in Figure 8-1 on page 992:

```
- i1/i2
    + PLACED ( 0 0 ) N
    + HALO 100 0 50 200 ;
```

Figure 8-1 Component Halo



numComps

Specifies the number of components defined in the COMPONENTS section.

PLACED *pt orient*

Specifies that the component has a location, but can be moved using automatic layout tools. You must specify the component’s location and orientation.

PROPERTY *propName propVal*

Specifies a numerical or string value for a component property defined in the PROPERTYDEFINITIONS statement. The *propName* you specify must match the *propName* listed in the PROPERTYDEFINITIONS statement.

REGION *regionName*

Specifies a region in which the component must lie. *regionName* specifies a region already defined in the REGIONS section. If the region is smaller than the bounding rectangle of the component itself, the DEF reader issues an error message and ignores the argument. If the region does not contain a legal location for the component, the component remains unplaced after the placement step.

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`ROUTEHALO` *`haloDist minLayer maxLayer`*

Specifies that signal routing in the "halo area" around the block boundary should be perpendicular to the block edge in order to reach the block pins. The halo area is the area within *`haloDist`* of the block boundary (see the Figure below). A routing-halo is intended to be used to minimize cross coupling between routing at the current level of the design, and routing inside the block. It has no effect on power routing. Note that this also means it is allowed to route in the "halo corners" because routing in the "halo corner" is not adjacent to the block boundary, and will not cause any significant cross-coupling with routing inside the block.

The routing halo exists for the routing layers between *`minLayer`* and *`maxLayer`*. The layer you specify for *`minLayer`* must be a lower routing layer than *`maxLayer`*.

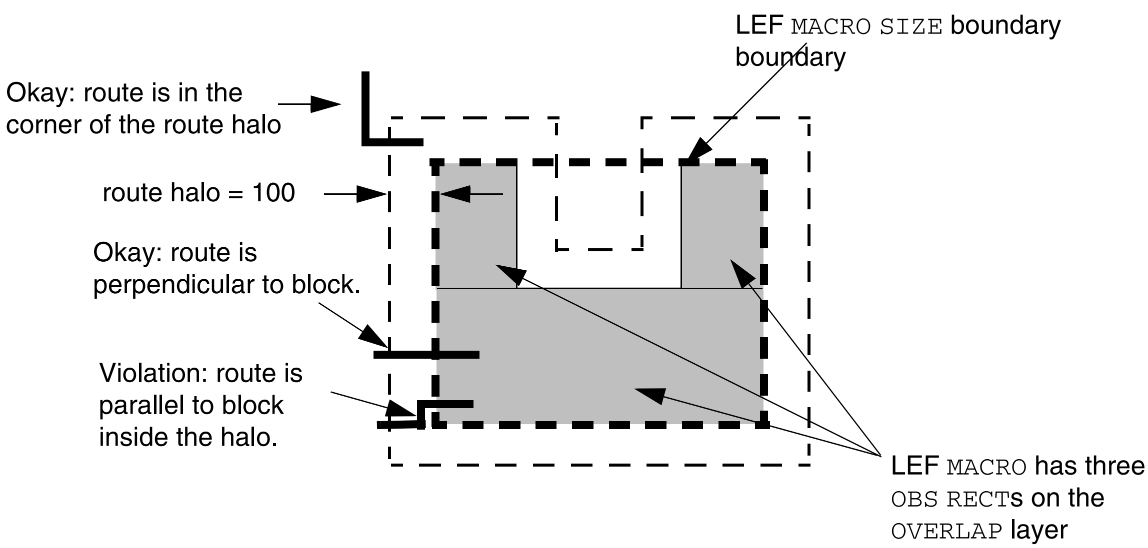
Type: Integer, specified in DEF database units (*`haloDist`*); string that matches a LEF routing layer name (*`minLayer`* and *`maxLayer`*)

Example 8-5 Route Halo Example

For a U-shaped macro, the following component description results in the halo shown in Figure 8-2 on page 993.

```
- i1/i2
    + PLACED ( 0 0 ) N
    + ROUTEHALO 100 metal1 metal3 ;
```

Figure 8-2 Route Halo



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`SOURCE {NETLIST | DIST | USER | TIMING}`

Specifies the source of the component.
Value: Specify one of the following:

`DIST`	Component is a physical component (that is, it only connects to power or ground nets), such as filler cells, well-taps, and decoupling caps.
`NETLIST`	Component is specified in the original netlist. This is the default value, and is normally not written out in the DEF file.
`TIMING`	Component is a logical rather than physical change to the netlist, and is typically used as a buffer for a clock-tree, or to improve timing on long nets.
`USER`	Component is generated by the user for some user-defined reason.

`UNPLACED`

Specifies that the component does not have a location.

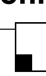
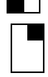
`WEIGHT` *`weight`*

Specifies the weight of the component, which determines whether or not automatic placement attempts to keep the component near the specified location. *`weight`* is only meaningful when the component is placed. All non-zero weights have the same effect during automatic placement.
Default: `0`

Specifying Orientation

If a component has a location, you must specify its location and orientation. A component can have any of the following orientations: N, S, W, E, FN, FS, FW, or FE.

Orientation terminology can differ between tools. The following table maps the orientation terminology used in LEF and DEF files to the OpenAccess database format.

LEF/DEF	OpenAccess	Definition
N (North)	R0	
S (South)	R180	

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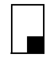
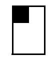
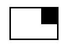
LEF/DEF	OpenAccess	Definition
W (West)	R90	
E (East)	R270	
FN (Flipped North)	MY	
FS (Flipped South)	MX	
FW (Flipped West)	MX90	
FE (Flipped East)	MY90	

Components are always placed such that the lower left corner of the cell is the origin (0,0) after any orientation. When a component flips about the y axis, it flips about the component center. When a component rotates, the lower left corner of the bounding box of the component's sites remains at the same placement location.

Defining Component Properties to Create 32/28 nm and Smaller Nodes Rules

You can include component properties in your DEF file to create 32/28 nm and smaller nodes rules that currently are not supported by existing DEF syntax. The properties are specified inside the `COMPONENT` statements where they can be seen with other rules.

Before you can reference them, properties must be defined at the beginning of the DEF file in the `PROPERTYDEFINITIONS` statement, immediately before the first `COMPONENT` statement.

- Properties belong to the `COMPONENT` object and have a type of `STRING`.
- The property names used for these rules all start with `DEF_CDN_`.

All properties use the following syntax within the DEF `PROPERTYDEFINITIONS` statement:

```
PROPERTYDEFINITIONS
     COMPONENT propName STRING ["stringValue"] ;

END PROPERTYDEFINITIONS
```

The property definitions for the component properties are as follows:

```
PROPERTYDEFINITIONS
     COMPONENT DEF_CDN_ROUTEHALO STRING ;
END PROPERTYDEFINITIONS
```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Route Halo Rule

The route halo rule for a component can be defined using the following property definition:

```
PROPERTY DEF_CDN_ROUTEHALO
        "ROUTEHALO left bottom right top minLayer maxLayer
        ; " ;
```

Where:

`ROUTEHALO` *`left bottom right top minLayer maxLayer`*

Specifies different halo distances for each side by extending *`left`* along the block's left edge, *`bottom`* at the block's bottom edge, *`right`* on the right edge, and *`top`* at the top edge. The other definitions are the same as the native `ROUTEHALO` syntax.

Design

```
DESIGN designName ;
```

Specifies a name for the design. The DEF reader reports a warning if this name is different from that in the database. In case of a conflict, the just specified name overrides the old name.

Die Area

```
[DIEAREA pt pt [pt] ... ;]
```

If two points are defined, specifies two corners of the bounding rectangle for the design. If more than two points are defined, specifies the points of a polygon that forms the die area. The edges of the polygon must be parallel to the x or y axis (45-degree shapes are not allowed), and the last point is connected to the first point. All points are integers, specified as DEF database units.

Geometric shapes (such as blockages, pins, and special net routing) can be outside of the die area, to allow proper modeling of pushed down routing from top-level designs into sub blocks. However, routing tracks should still be inside the die area.

Example 8-6 Die Area Statements

The following statements show various ways to define the die area.

```
DIEAREA ( 0 0 ) ( 100 100 ) ;                         #Rectangle from 0,0 to 100,100
```


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```
DIEAREA ( 0 0 ) ( 0 100 ) ( 100 100 ) ( 100 0 ) ;    #Same rectangle as a polygon
DIEAREA ( 0 0 ) ( 0 100 ) ( 50 100 ) ( 50 50 ) ( 100 50 ) ( 100 0 ) ; #L-shaped polygon
```

Divider Character

DIVIDERCHAR "*character*" ;

Specifies the character used to express hierarchy when DEF names are mapped to or from other databases. The character must be enclosed in double quotation marks. For example:

```
DIVIDERCHAR "/" ;
```

If the divider character appears in a DEF name as a regular character, you must use a backslash (\) before the character to prevent the DEF reader from interpreting the character as a hierarchy delimiter.

If you do not specify the `DIVIDERCHAR` statement in your LEF file, the default value is `"/"`.

Extensions

```
[BEGINEXT "tag"
     extensionText
ENDEXT]
```

Adds customized syntax to the DEF file that can be ignored by tools that do not use that syntax. You can also use extensions to add new syntax not yet supported by your version of LEF/DEF, if you are using version 5.1 or later. Add extensions as separate sections.

extensionText

Defines the contents of the extension.

"tag"

Identifies the extension block. You must enclose *tag* in quotes.

Example 8-7 Extension Statement

```
BEGINEXT "1VSI Signature 1.0"
    CREATOR "company name"
    DATE "timestamp"
    REVISION "revision number"
ENDEXT
```


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Fills

```
[FILLS numFills ;
     [- LAYER layerName [+ MASK maskNum] [+ OPC]
         {RECT pt pt | POLYGON pt pt pt ...} ... ;] ...
     [- VIA viaName [+ MASK viaMaskNum] [+ OPC] pt ... ;] ...
END FILLS]
```

Defines the rectangular shapes that represent metal fills in the design. Each fill is defined as an individual rectangle.

LAYER *layerName*	Specifies the layer on which to create the fill.
MASK *maskNum*	Specifies which mask for double or triple patterning lithography to use for the given rectangles or polygons. The *maskNum* variable must be a positive integer. Most applications support values of 1, 2, or 3 only. Shapes without any defined mask have no mask setting (are uncolored).

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`MASK` *`viaMaskNum`*	Specifies which mask for double or triple patterning lithography to be applied to via shapes on each layer. The *`viaMaskNum`* variable is a hex-encoded three digit value of the form: <*`topMaskNum`*><*`cutMaskNum`*><*`bottomMaskNum`*> For example, MASK 113 means that the top metal and cut layer *`maskNum`* is 1, and the bottom metal layer *`maskNum`* is 3. A value of 0 means the shape on that layer has no mask assignment (is uncolored), so 013 means the top layer is uncolored. If either the first or second digit is missing, they are assumed to be 0, so 013 and 13 mean the same thing. Most applications support *`maskNums`* of 0, 1, 2, or 3 for double or triple patterning. The *`topMaskNum`* and *`bottomMaskNum`* variables specify which mask the corresponding metal shape belongs to. The via-master metal mask values have no effect. For the cut layer, the *`cutMaskNum`* defines the mask for the bottom-most, and then the left-most cut. For multi-cut vias, the via-instance cut masks are derived from the via-master cut mask values. The via-master must have a mask defined for all the cut shapes and every via-master cut mask is "shifted" (from 1 to 2, 2 to1 for two mask layers, and from 1 to 2, 2 to 3, and 3 to 1 for three mask layers) so the lower-left cut matches the *`cutMaskNum`* value. Similarly, for the metal layer, the *`topMaskNum`*/ *`bottomMaskNum`* would define the mask for the bottom-most, then leftmost metal shape. For multiple disjoint metal shapes, the via-instance metal masks are derived from the via-master metal mask values. The via-master must have a mask defined for all of the metal shapes, and every via-master metal mask is “shifted” (1->2, 2->1 for two mask layers, 1->2, 2->3, 3->1 for three mask layers) so the lower-left cut matches the *`topMaskNum`*/*`bottomMaskNum`* value. See Example 8-9 on page 1002.
`numFills`	Specifies the number of `LAYER` statements in the `FILLS` statement, *not* the number of rectangles.
`OPC`	Indicates that the `FILL` shapes require OPC correction during mask generation.

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`POLYGON` *pt pt pt*	Specifies a sequence of at least three points to generate a polygon geometry. Most fab DRC rules require each polygon edge to be parallel to the x or y axis, or at a 45-degree angle. However, odd-angles are allowed sometimes on a few layers, such as bump layers. Each `POLYGON` statement defines a polygon generated by connecting each successive point, and then by connecting the first and last points. The *pt* syntax corresponds to a coordinate pair, such as *x y*. Specify an asterisk (*) to repeat the same value as the previous *x* or *y* value from the last point.
`RECT` *pt pt*	Specifies the lower left and upper right corner coordinates of the fill geometry.
`VIA` *viaName pt*	Places the via named *viaName* at the specified (x y) location (*pt*). *viaName* must be a previously defined via in the DEF `VIAS` or LEF `VIA` section. *Type:* (*pt*) Integers, specified in DEF database units

Example 8-8 Fills Statements

- The following `FILLS` statement defines fill geometries for layers *metal1* and *metal2*:

```
FILLS 2 ;
    - LAYER metal1
        RECT ( 1000 2000 ) ( 1500 4000 )
        RECT ( 2000 2000 ) ( 2500 4000 )
        RECT ( 3000 2000 ) ( 3500 4000 ) ;
    - LAYER metal2
        RECT ( 1000 2000 ) ( 1500 4000 )
        RECT ( 1000 4500 ) ( 1500 6500 )
        RECT ( 1000 7000 ) ( 1500 9000 )
        RECT ( 1000 9500 ) ( 1500 11500 ) ;
END FILLS
```

- The following `FILLS` statement defines two rectangles and one polygon fill geometries:

```
FILLS 1 ;
    -LAYER metal1
       RECT ( 100 200 ) ( 150 400 )
       POLYGON ( 100 100 ) ( 200 200 ) ( 300 200 ) ( 300 100 )
       RECT ( 300 200 ) ( 350 400 ) ;
END FILLS
```


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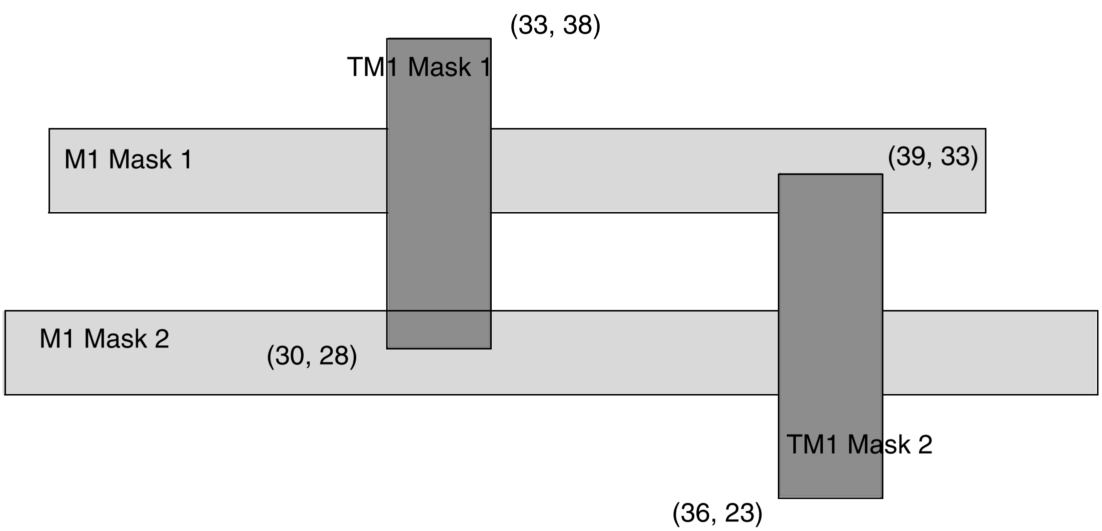
- Shapes on the `TRIMMETAL` layers are written out in the `FILLS` section in DEF in the following format:

  ```
  - LAYER TM1 + MASK x RECT (x x) (x x) ;
  ```

  The following `FILLS` statement defines two rectangular shapes on the `TRIMMETAL` layer `TM1`, which is defined in a property statement in the masterslice layer section in LEF:

  ```
  FILLS 2 ;
  - LAYER TM1 + MASK 1 RECT (30 28) (33 38)
  - LAYER TM1 + MASK 2 RECT (36 23) (39 33)
  ...
  END FILLS
  ```

Figure 8-3 Trim Metal Layer Shapes in the FILLS Section



- The following `FILLS` statement defines two rectangles and two via fill geometries for layer *metal1*. The rectangles and one of the via fill shapes require OPC correction.

  ```
  FILLS 3 ;
      -LAYER metal1 + OPC
         RECT ( 0 0 ) ( 100 100 )
         RECT ( 200 200 ) ( 300 300 ) ;
      -VIA via26
         ( 500 500 )
         ( 800 800 ) ;
      -VIA via28 + OPC
         ( 900 900 ) ;
  END FILLS
  ```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Example 8-9 Multi-Mask Patterns for Fills

The following example shows multi-mask patterning for fills:

```
- LAYER M1 + MASK 1 RECT ( 10 10 ) ( 11 11 ) ;  #RECT on MASK 1
- LAYER M2 RECT ( 10 10 ) ( 11 11 ) ;           #RECT is uncolored
- VIA VIA1_1  + MASK 031 ( 10 10 ) ;            #VIA with top-cut-bot mask 031
```

This indicates that the:

- `M1` rectangle shape is on `MASK 1`
- `M2` rectangle shape has no mask set and is uncolored
- `VIA1_1` via will have:
  - no mask set for the top metal shape - it is uncolored (*`topMaskNum`* is 0 in the 031 value). Note that the via-master color of the top metal shape does not matter.
  - `mask 1` for the bottom metal shape (*`bottomMaskNum`* is 1 in the 031 value)
  - for the cut layer, the bottom-most, and then the left-most cut of the via-instance is mask 3. The mask for the other cuts of the via-instance are derived from the via-master by "shifting" the via-master cut masks to match. So if the via-master's bottom- left cut is `MASK 1`, then the via-master cuts on mask 1 become mask 3 for the via-instance, and similarly cuts on 2 become 1, and cuts on 3 become 2. See Figure 8-11 on page 1067.

GCell Grid

```
[GCELLGRID
     {X start DO numColumns+1 STEP space} ...
     {Y start DO numRows+1 STEP space ;} ...]
```

Defines the gcell grid for a standard cell-based design. Each `GCELLGRID` statement specifies a set of vertical (x) and horizontal (y) lines, or tracks, that define the gcell grid.

Typically, the `GCELLGRID` is automatically generated by a particular router, and is not manually created by the designer.

`DO` *`numColumns+1`*

Specifies the number of columns in the grid.

`DO` *`numRows+1`*

Specifies the number of rows in the grid.

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`STEP` *`space`*

Specifies the spacing between tracks.

`X` *`start`*, `Y` *`start`*

Specify the location of the first vertical (`x`) and first horizontal (`y`) track.

GCell Grid Boundary Information

The boundary of the gcell grid is the rectangle formed by the extreme vertical and horizontal lines. The gcell grid partitions the routing portion of the design into rectangles, called gcells. The lower left corner of a gcell is the origin. The x size of a gcell is the distance between the upper and lower bounding vertical lines, and the y size is the distance between the upper and lower bounding horizontal lines.

For example, the grid formed by the following two `GCELLGRID` statements creates gcells that are all the same size (100 x 200 in the following):

```
GCELLGRID X 1000 DO 101 STEP 100 ;
GCELLGRID Y 1000 DO 101 STEP 200 ;
```

A gcell grid in which all gcells are the same size is called a uniform gcell grid. Adding `GCELLGRID` statements can increase the granularity of the grid, and can also result in a nonuniform grid, in which gcells have different sizes.

For example, adding the following two statements to the above grid generates a nonuniform grid:

```
GCELLGRID X 3050 DO 61 STEP 100 ;
GCELLGRID Y 5100 DO 61 STEP 200 ;
```

When a track segment is contained inside a gcell, the track segment belongs to that gcell. If a track segment is aligned on the boundary of a gcell, that segment belongs to the gcell only if it is aligned on the left or bottom edges of the gcell. Track segments aligned on the top or right edges of a gcell belong to the next gcell.

GCell Grid Restrictions

Every track segment must belong to a gcell, so gcell grids have the following restrictions:

- The x coordinate of the last vertical track must be less than, and not equal to, the x coordinate of the last vertical gcell line.
- The y coordinate of the last horizontal track must be less than, and not equal to, the y coordinate of the last horizontal gcell line.

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Gcells grids also have the following restrictions:

- Each GCELLGRID statement must define two lines.
- Every gcell need not contain the vertex of a track grid. But, those that do must be at least as large in both directions as the default wire widths on all layers.

Groups

```
[GROUPS numGroups ;
     [- groupName [compNamePattern ... ]
        [+ REGION regionNam]
        [+ PROPERTY {propName propVal} ...] ...
     ;] ...
END GROUPS]
```

Defines groups in a design.

compNamePattern

Specifies the components that make up the group. Do not assign any component to more than one group. You can specify any of the following:

- A component name, for example C3205
- A list of component names separated by spaces, for example, I01 I02 C3204 C3205
- A pattern for a set of components, for example, IO* and C320*

Note: An empty group with no component names is allowed.

groupName

Specifies the name for a group of components.

numGroups

Specifies the number of groups defined in the GROUPS section.

PROPERTY *propName propVal*

Specifies a numerical or string value for a group property defined in the PROPERTYDEFINITIONS statement. The *propName* you specify must match the *propName* listed in the PROPERTYDEFINITIONS statement.

REGION *regionName*

Specifies a rectangular region in which the group must lie. *regionName* specifies a region previously defined in the REGIONS section. If region restrictions are specified in

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both COMPONENT and GROUP statements for the same component, the component restriction overrides the group restriction.

History

```
[HISTORY anyText ;] ...
```

Lists a historical record about the design. Each line indicates one record. Any text excluding a semicolon (;) can be included in *anyText*. The semicolon terminates the HISTORY statement. Linefeed and Return do not terminate the HISTORY statement. Multiple HISTORY lines can appear in a file.

Nets

```
NETS numNets ;
     [– { netName
            [ ( {compName pinName | PIN pinName} [+ SYNTHESIZED] ) ] ...
        | MUSTJOIN ( compName pinName ) }
        [+ SHIELDNET shieldNetName ] ...
        [+ VPIN vpinName [LAYER layerName] pt pt
            [PLACED pt orient | FIXED pt orient | COVER pt orient] ] ...
        [+ SUBNET subnetName
            [ ( {compName pinName | PIN pinName | VPIN vpinName} ) ] ...
            [NONDEFAULTRULE rulename]
            [regularWiring] ...] ...
        [+ XTALK class]
        [+ NONDEFAULTRULE ruleName]
        [regularWiring] ...
        [+ SOURCE {DIST | NETLIST | TEST | TIMING | USER}]
        [+ FIXEDBUMP]
        [+ FREQUENCY frequency]
        [+ ORIGINAL netName]
        [+ USE {ANALOG | CLOCK | GROUND | POWER | RESET | SCAN | SIGNAL
                  | TIEOFF}]
        [+ PATTERN {BALANCED | STEINER | TRUNK | WIREDLOGIC}]
        [+ ESTCAP wireCapacitance]
        [+ WEIGHT weight]
        [+ PROPERTY {propName propVal} ...] ...
     ;] ...

END NETS
```

Defines netlist connectivity and regular-routes for nets containing regular pins. These routes are normally created by a signal router that can rip-up and repair the routing. The SPECIALNETS statement defines netlist connectivity and routing for special-routes that are created by "special routers" or "manually" and should not be modified by a signal router.

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Special routes are normally used for power-routing, fixed clock-mesh routing, high-speed buses, critical analog routes, or flip-chip routing on the top-metal layer to bumps.

Input arguments for a net can appear in the `NETS` section or the `SPECIALNETS` section. In case of conflicting values, the DEF reader uses the last value encountered. `NETS` and `SPECIALNETS` statements can appear more than once in a DEF file. If a particular net has mixed wiring or pins, specify the special wiring and pins first.

Arguments

`*compName pinName*`

Specifies the name of a regular component pin on a net or a subnet. LEF `MUSTJOIN` pins, if any, are not included; only the master pin (that is, the one without the `MUSTJOIN` statement) is included. If a subnet includes regular pins, the regular pins must be included in the parent net.

`COVER *pt orient*`

Specifies that the pin has a location and is a part of the cover macro. A `COVER` pin cannot be moved by automatic tools or by interactive commands. You must specify the pin's location and orientation.

`ESTCAP *wireCapacitance*`

Specifies the estimated wire capacitance for the net. `ESTCAP` can be loaded with simulation data to generate net constraints for timing-driven layout.

`FIXED *pt orient*`

Specifies that the pin has a location and cannot be moved by automatic tools, but can be moved by interactive commands. You must specify the pin's location and orientation.

`FIXEDBUMP`

Indicates that the bump net cannot be reassigned to a different pin.
It is legal to have a pin without geometry to indicate a logical connection, and to have a net that connects that pin to two other instance pins that have geometry. Area I/Os have a logical pin that is connected to a bump and an input driver cell. The bump and driver cell have pin geometries (and, therefore, should be routed and extracted), but the logical pin is the external pin name without geometry (typically the Verilog pin name for the chip). Because bump nets are usually routed with special routing, they also can be specified in the `SPECIALNETS` statement. If a net name appears in both the `NETS` and `SPECIALNETS` statements, the `FIXEDBUMP` keyword also should appear in both statements. However, the value only exists once within a given application's database for the net name.

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Because DEF is often used incrementally, the last value read in is used. Therefore, in a typical DEF file, if the same net appears in both statements, the `FIXEDBUMP` keyword (or lack of it) in the `NETS` statement is the value that is used, because the `NETS` statement is defined after the `SPECIALNETS` statement.
For an example specifying the `FIXEDBUMP` keyword, see "Fixed Bump" on page 1058.

`FREQUENCY` *`frequency`*

Specifies the frequency of the net, in hertz. The frequency value is used by the router to choose the correct number of via cuts required for a given net, and by validation tools to verify that the AC current density rules are met. For example, a net described with `+ FREQUENCY 100` indicates the net has 100 rising and 100 falling transitions in 1 second.
Type: Float

`LAYER` *`layerName`*

Specifies the layer on which the virtual pin lies.

`MUSTJOIN (`*`compName pinName`*`)`

Specifies that the net is a mustjoin. If a net is designated `MUSTJOIN`, its name is generated by the system. Only one net should connect to any set of mustjoin pins. Mustjoin pins for macros are defined in LEF. The only reason to specify a `MUSTJOIN` net in DEF (identified arbitrarily by one of its pins) is to specify prewiring for the `MUSTJOIN` connection.
Otherwise, nets are generated automatically where needed for mustjoin connections specified in the library. If the input file specifies that a mustjoin pin is connected to a net, the DEF reader connects the set of mustjoin pins to the same net. If the input file does not specify connections to any of the mustjoin pins, the DEF reader creates a local `MUSTJOIN` net.

`netName`

Specifies the name for the net. Each statement in the `NETS` section describes a single net. There are two ways of identifying the net: *`netName`* or `MUSTJOIN`. If the *`netName`* is given, a list of pins to connect to the net also can be specified. Each pin is identified by a component name and pin name pair (*`compName pinName`*) or as an I/O pin (`PIN` *`pinName`*). Parentheses ensure readability of output. The keyword `MUSTJOIN` cannot be used as a *`netName`*.

`NONDEFAULTRULE` *`ruleName`*

By default the width of any route segment in the `NETS` *regularWiring* section is defined by the default width (`LEF WIDTH` statement value for the routing layer).

This keyword specifies a nondefault rule to use instead of the default rule when creating the net and wiring. When specified with `SUBNET`, identifies the nondefault rule to use when creating the subnet and its wiring.

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The width of any route segment is defined by the corresponding NONDEFAULTRULE WIDTH for that layer.

Wrong-way Width Rules

Some technologies required larger widths for wrong-way routing than in the preferred direction. If the wrong-way width is larger than the default or NDR width, then the wrong-way width is used for wrong-way routes on that layer. The implicit routing extension is still half of the default or NDR width, even for wrong-way routes.

The following LEF DRC rules allow a WRONGDIRECTION keyword that defines wrong-way widths that will affect the width of any wrong-way routes in the DEF NETS section:

LEF58_WIDTH

LEF58_WIDTHTABLE

LEF58_SPANLENGTHTABLE

See the "Impact of Wrong-way Width Rules on page 1019" section for examples and more details.

numNets

Specifies the number of nets defined in the NETS section.

ORIGINAL *netName*

Specifies the original net partitioned to create multiple nets, including the net being defined.

PATTERN {BALANCED | STEINER | TRUNK | WIREDLOGIC}

Specifies the routing pattern used for the net.
Default: STEINER
Value: Specify one of the following:

BALANCED	Used to minimize skews in timing delays for clock nets.
STEINER	Used to minimize net length.
TRUNK	Used to minimize delay for global nets.

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`WIREDLOGIC`	Used in ECL designs to connect output and mustjoin pins before routing to the remaining pins.

`PIN` *`pinName`*

Specifies the name of an I/O pin on a net or a subnet.

`PLACED` *`pt orient`*

Specifies that the pin has a location, but can be moved during automatic layout. You must specify the pin's location and orientation.

`PROPERTY` *`propName propVal`*

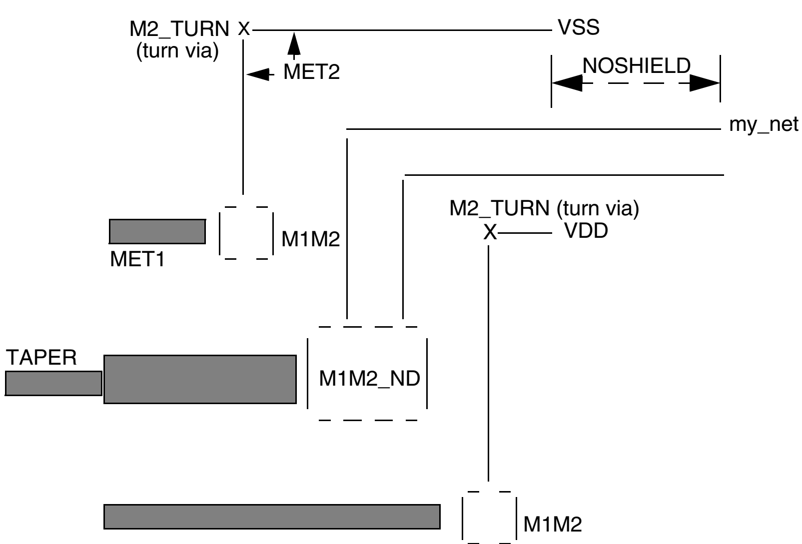
Specifies a numerical or string value for a net property defined in the `PROPERTYDEFINITIONS` statement. The *`propName`* you specify must match the *`propName`* listed in the `PROPERTYDEFINITIONS` statement.

`regularWiring`

Specifies the regular physical wiring for the net or subnet. For regular wiring syntax, see "Regular Wiring Statement" on page 1012.

`SHIELDNET` *`shieldNetName`*

Specifies the name of a special net that shields the regular net being defined. A shield net for a regular net is defined earlier in the DEF file in the `SPECIALNETS` section.



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`SOURCE {DIST | NETLIST | TEST | TIMING | USER}`

Specifies the source of the net. The value of this field is preserved when input to the DEF reader.

Value: Specify one of the following:

`DIST`	Net is the result of adding physical components (that is, components that only connect to power or ground nets), such as filler cells, well-taps, tie-high and tie-low cells, and decoupling caps.
`NETLIST`	Net is defined in the original netlist. This is the default value, and is not normally written out in the DEF file.
`TEST`	Net is part of a scanchain.
`TIMING`	Net represents a logical rather than physical change to netlist, and is used typically as a buffer for a clock-tree, or to improve timing on long nets.
`USER`	Net is user defined.

`SUBNET` *`subnetName`*

Names and defines a subnet of the regular net *`netName`*. A subnet must have at least two pins. The subnet pins can be virtual pins, regular pins, or a combination of virtual and regular pins. A subnet pin cannot be a mustjoin pin.

`SYNTHESIZED`

Used by some tools to indicate that the pin is part of a synthesized scan chain.

`USE {ANALOG | CLOCK | GROUND | POWER | RESET | SCAN | SIGNAL | TIEOFF}`

Specifies how the net is used.

Default: `SIGNAL`

Value: Specify one of the following:

`ANALOG`	Used as an analog signal net.
`CLOCK`	Used as a clock net.
`GROUND`	Used as a ground net.
`POWER`	Used as a power net.
`RESET`	Used as a reset net.
`SCAN`	Used as a scan net.

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`SIGNAL`	Used as a digital signal net.
`TIEOFF`	Used as a tie-high or tie-low net.

`VPIN` *`vpinName pt pt`*

Specifies the name of a virtual pin, and its physical geometry. Virtual pins can be used only in subnets. A `SUBNET` statement refers to virtual pins by the *`vpinName`* specified here. You must define each virtual pin in a `+ VPIN` statement before you can list it in a `SUBNET` statement.

Example 8-10 Virtual Pin

The following example defines a virtual pin:

```
+ VPIN M7K.v2 LAYER MET2 ( -10 -10 ) ( 10 10 ) FIXED ( 10 10 )
    + SUBNET M7K.2 ( VPIN M7K.v2 ) ( /PREG_CTRL/I$73/A I )
      NONDEFAULTRULE rule1
      ROUTED MET2 ( 27060 341440 ) ( 26880 * ) ( * 213280 )
      M1M2 ( 95040 * ) ( * 217600 ) ( 95280 * )
      NEW MET1 ( 1920 124960 ) ( 87840 * )
      COVER MET2 ( 27060 341440 ) ( 26880 * )
```

`WEIGHT` *`weight`*

Specifies the weight of the net. Automatic layout tools attempt to shorten the lengths of nets with high weights. A value of `0` indicates that the net length for that net can be ignored. The default value of `1` specifies that the net should be treated normally. A larger weight specifies that the tool should try harder to minimize the net length of that net. For normal use, timing constraints are generally a better method to use for controlling net length than net weights. For the best results, you should typically limit the maximum weight to 10, and not add weights to more than 3 percent of the nets.
Default: `1`
Type: Integer

`XTALK` *`class`*

Specifies the crosstalk class number for the net. If you specify the default value (`0`), `XTALK` will not be written to the DEF file.
Default: `0`
Type: Integer
Value: `0` to `200`

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Regular Wiring Statement

```
{+ COVER | + FIXED | + ROUTED | + NOSHIELD}
     layerName [TAPER | TAPERRULE ruleName] [STYLE styleNum]
        routingPoints
     [NEW layerName [TAPER | TAPERRULE ruleName] [STYLE styleNum]
        routingPoints
     ] ...
```

Specifies regular wiring for the net.

`COVER`

Specifies that the wiring cannot be moved by either automatic layout or interactive commands. If no wiring is specified for a particular net, the net is unrouted. If you specify `COVER`, you must also specify *`layerName`*.

`FIXED`

Specifies that the wiring cannot be moved by automatic layout, but can be changed by interactive commands. If no wiring is specified for a particular net, the net is unrouted. If you specify `FIXED`, you must also specify *`layerName`*.

`layerName`

Specifies the layer on which the wire lies. You must specify *`layerName`* if you specify `COVER`, `FIXED`, `ROUTED`, or `NEW`. Specified layers must be routable; reference to a cut layer generates an error.

`NEW` *`layerName`*

Indicates a new wire segment (that is, there is no wire segment between the last specified coordinate and the next coordinate), and specifies the name of the layer on which the new wire lies. Noncontinuous paths can be defined in this manner.

`NOSHIELD`

Specifies that the last wide segment of the net is not shielded. If the last segment is not shielded, and is tapered, specify `TAPER` under the `LAYER` argument, instead of `NOSHIELD`.

`ROUTED`

Specifies that the wiring can be moved by the automatic layout tools. If no wiring is specified for a particular net, the net is unrouted. If you specify `ROUTED`, you must also specify *`layerName`*. An example of `DEF NETS` routing is shown in DEF NETS Examples on page 1021.

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routingPoints

Defines the center line coordinates of the route on *layerName*. For information about using routing points, see "Defining Routing Points" on page 1018.

As described above, the width of the routes is defined by the default width (e.g., LEF WIDTH statement on the routing layer) or a NONDEFAULTRULE width for the routing layer. In addition, some technologies require larger widths for wrong-way routes that may increase the width. See the "Impact of Wrong-way Width Rules on page 1019" section for more details.

The *routingPoints* syntax is defined as follows:

```
{ ( x y [extValue] )
    {[MASK maskNum] ( x y [extValue] )
    |[MASK viaMaskNum] viaName [orient]
    |[MASK maskNum] RECT ( deltax1 deltay1       deltax2
deltay2 )
    | VIRTUAL ( x y ) } } ...
```

extValue

Specifies the amount by which the wire is extended past the endpoint of the segment. The extension value must be greater than or equal to 0 (zero).
Default: Half the wire width
Type: Integer, specified in database units

Some tools only allow 0 or the WIREEXTENSION value from the LAYER or NONDEFAULTRULE statement.

MASK *maskNum*

Specifies which mask for double or triple patterning lithography to use for the next wire or RECT. The *maskNum* variable must be a positive integer - most applications support values of 1, 2, or 3 only. Shapes without any defined mask have no mask set (that is, they are uncolored).

MASK *viaMaskNum*

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Specifies which mask for double or triple patterning lithography is to be applied to the next via's shapes on each layer.

The *viaMaskNum* variable is a hex-encoded three-digit value of the form:

<*topMaskNum*><*cutMaskNum*><*bottomMaskNum*>

For example, MASK 113 means that the top metal and cut layer *maskNum* is 1, and the bottom metal layer *maskNum* is 3. A value of 0 means that the shape on the layer has no mask assignment (is uncolored), so 013 means the top layer is uncolored. If either the first or second digit is missing, they are assumed to be 0, so 013 and 13 mean the same thing. Most applications support *maskNums* of 0, 1, 2, or 3 only for double or triple patterning.

The *topMaskNum* and *bottomMaskNum* variables specify the masks to which the corresponding metal shapes belong. In the case of a metal layer with a single shape, the via-master metal mask values have no effect.

For the cut layer, the *cutMaskNum* variable defines the mask for the bottom-most, and then the left-most cut. For multi-cut vias, the via-instance cut masks are derived from the via-master cut mask values. The via-master must have a mask defined for all the cut shapes and every via-master cut mask is "shifted" (from 1 to 2, and 2 to 1 for two mask layers, and from 1 to 2, 2 to 3, and 3 to1 for three mask layers), so the lower-left cut matches the *cutMaskNum* value.

Similarly, for a metal layer with multiple shapes, the *topMaskNum*/*bottomMaskNum* would define the mask for the bottom-most, then leftmost metal shape. For multiple disjoint metal shapes, the via-instance metal masks are derived from the via-master metal mask values. The via-master must have a mask defined for all of the metal shapes, and every via-master metal mask is “shifted” (1->2, 2->1 for two mask layers, 1->2, 2->3, 3->1 for three mask layers) so the lower-left cut matches the *topMaskNum*/*bottomMaskNum* value.

See Example 8-11 on page 1016.

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orient — Specifies the orientation of the *viaName* that precedes it, using the standard DEF orientation values of `N`, `S`, `E`, `W`, `FN`, `FS`, `FE`, and `FW` (See "Specifying Orientation" on page 994).

If you do not specify *orient*, `N` (North) is the default non-rotated value used. All other orientation values refer to the flipping or rotation around the via origin (the `0,0` point in the via shapes). The via origin is still placed at the `(`*x y*`)` value given in the routing statement just before the *viaName*.

Note: Some tools do not support orientation of vias inside their internal data structures; therefore, they are likely to translate vias with an orientation into a different but equivalent via that does not require an orientation.

`RECT` (*deltax1 deltay1 deltax2 deltay2*)

Indicates that a rectangle is created from the previous (*x y*) routing point using the delta values. The `RECT` values leave the current point and layer unchanged.

See Example 8-12 on page 1017.

viaName — Specifies a via to place at the last point. If you specify a via, *layerName* for the next routing coordinates (if any) is implicitly changed to the other routing layer for the via. For example, if the current layer is *metal1*, a *via12* changes the layer to *metal2* for the next routing coordinates.

`VIRTUAL` (*x y*)

Indicates that there is a virtual (non-physical zero-width) connection between the previous point and the new (*x y*) point. An '*' indicates that the *x* or *y* value is to be used from the previous point. The layer remains unchanged.

You can use this keyword to retain the symbolic routing graph.

See Example 8-12 on page 1017.

(*x y*) — Specifies the route coordinates. You cannot specify a route with zero length.

For more information, see "Specifying Coordinates" on page 1019.
Type: Integer, specified in database units

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`STYLE` *`styleNum`*

Specifies a previously defined style from the `STYLES` section in this DEF file. If a style is specified, the wire's shape is defined by the center line coordinates and the style.

`TAPER`

Specifies that the next contiguous wire segment on *`layerName`* is created using the default rule.

`TAPERRULE` *`ruleName`*

Specifies that the next contiguous wire segment on *`layerName`* is created using the specified nondefault rule.

Example 8-11 Multi-mask Patterns for Routing Points

The following example shows a routing statement that specifies three-mask layers `M1` and `VIA1`, and a two-mask layer `M2`:

```
 + ROUTED M1 (10 0 ) MASK 3 (10 20 ) VIA1_1
    NEW M2 ( 10 10 ) (20 10) MASK 1 ( 20 20 ) MASK 031 VIA1_2 ;
```

This indicates that the:

- `M1` wire shape (10 0) to (10 20) belongs to mask 3
- `VIA1_1` via has no preceding `MASK` statement so all the metal and cut shapes have no mask and are uncolored
- first `NEW M2` wire shape (10 10) to (20 10) has no mask set and is uncolored
- second `M2` wire shape (20 10) to (20 20) is on mask 1
- `VIA1_2` via has a MASK 031 (it can be MASK 31 also) so:
  - *`topMaskNum`* is 0 in the 031 value, so no mask is set for the top metal (`M2`) shape
  - *`bottomMaskNum`* is 1 in the 031 value, so mask 1 is used for the bottom metal (`M1`) shape
  - *`cutMaskNum`* is 3 in the 031 value, so the bottom-most, and then the left-most cut of the via-instance is mask 3. The mask for the other cuts of the via-instance are derived from the via-master by "shifting" the via-master's cut masks to match. So if the via-master's bottom-left cut is mask 1, then the via-master cuts on mask 1 become mask 3 for the via-instance, and similarly cuts on 2 shift to 1, and cuts on 3 shift to 2, as shown in Figure 8-4 on page 1017.

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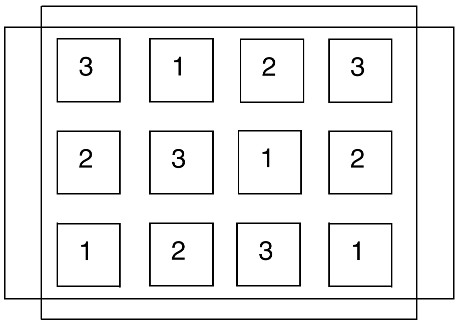
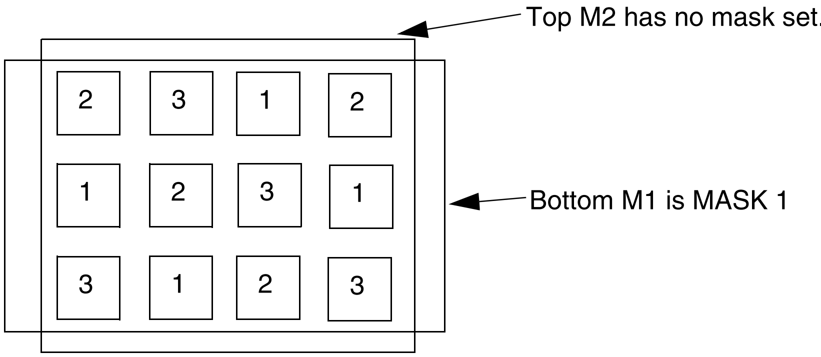
Figure 8-4 Via-master multi-mask patterns



Via-master cut masks for VIA1_2.



Masks for via-instance: ... (20 20) MASK 031 VIA1_2;
Bottom M1 is MASK 1.Top M2 has no mask set. Lower-left cut is MASK 3, and other via-instance cut shape masks are derived from via-master cut-masks as shown above by "shifting" the via-master masks to match: 1->3, 2->1, 3->2.

Example 8-12 Routing Points - Usage of Virtual and Rect

Figure 8-5 on page 1018 shows the results of the following routing statement:

```
+ ROUTED M1 ( 0 0 ) ( 5 0 ) VIRTUAL ( 7 1 ) RECT ( -3 0 -1 2 ) ( 7 7 ) ;
```


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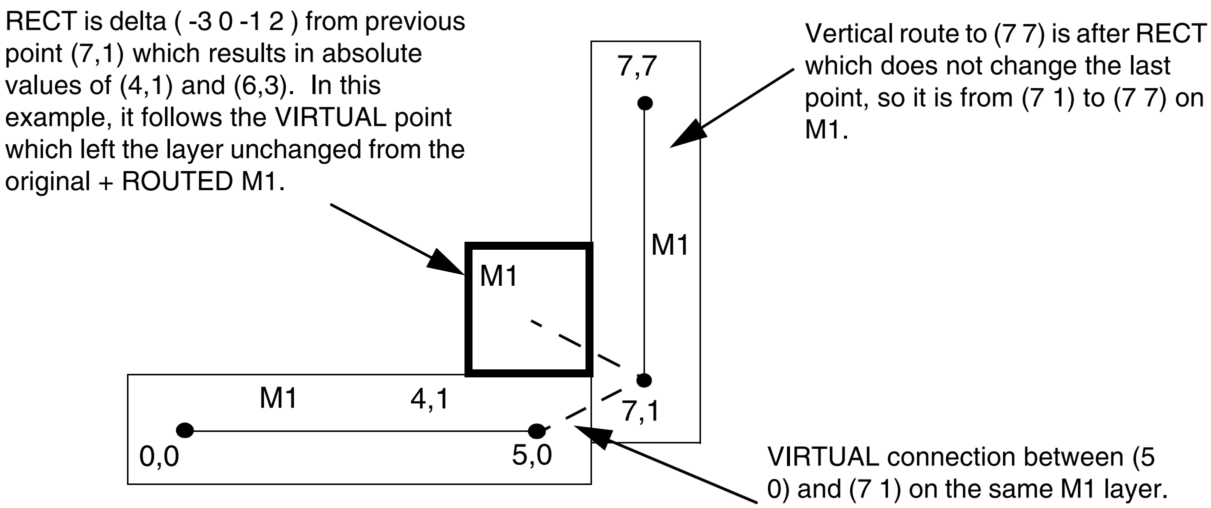
Figure 8-5 Routing Points - Usage of Virtual and Rect



Defining Routing Points

Routing points define the center line coordinates of the route for a specified layer. Routes that are 90 degrees, have a width defined by the routing rule for this wire, and extend from one coordinate (*`x y`*) to the next coordinate.

If either endpoint has an extension value (*`extValue`*), the wire is extended by that amount past the endpoint. Some applications require the extension value to be `0`, half of the wire width, or the same as the routing rule wire extension value. If you do not specify an extension value, the default value of half of the wire width is used.

If a coordinate with an extension value is specified after a via, the wire extension is added to the beginning of the next wire segment after the via (zero-length wires are not allowed).

If the wire segment is a 45-degree edge, and no `STYLE` is specified, the default octagon style is used for the endpoints. The routing rule width must be an even multiple of the manufacturing grid in order to keep all of the coordinates of the resulting outer wire boundary on the manufacturing grid.

If a `STYLE` is defined for 90-degree or 45-degree routes, the routing shape is defined by the center line coordinates and the style. No corrections, such as snapping to manufacturing grid, can be applied, and any extension values are ignored. The DEF file should contain values that are already snapped, if appropriate. The routing rule width indicates the desired user width, and represents the minimum allowed width of the wire that results from the style when the 45-degree edges are properly snapped to the manufacturing grid.

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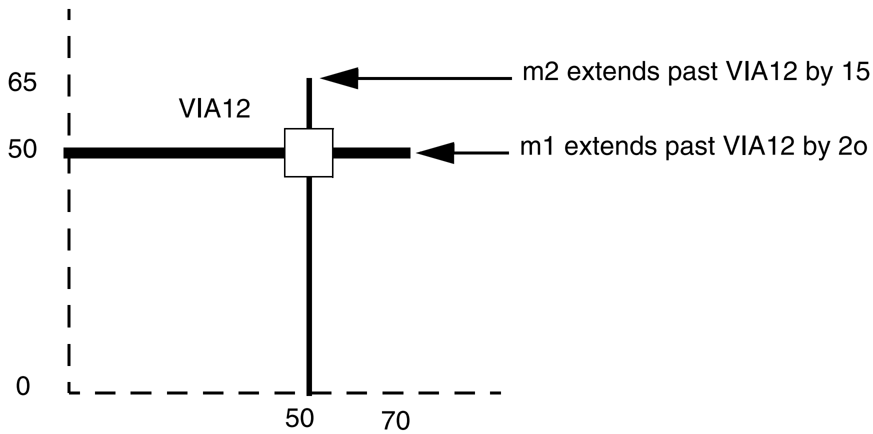
Specifying Coordinates

To maximize compactness of the design files, the coordinates allow for the asterisk (*) convention. Here, (*`x *`*) indicates that the y coordinate last specified in the wiring specification is used; (*`* y`*) indicates that the x coordinate last specified is used. Use (*`* * extValue`*) to specify a wire extension at a via.

ROUTED M1 (0 50) (50 * 20) VIA12 (* * 15) (* 0)



Each coordinate sequence defines a connected orthogonal path through the points. The first coordinate in a sequence must not have an * element.

Because nonorthogonal segments are not allowed, subsequent points in a connected sequence must create orthogonal paths. For example, the following sequence is a valid path:

```
( 100 200 ) ( 200 200 ) ( 200 500 )
```

The following sequence is an equivalent path:

```
( 100 200 ) ( 200 * ) ( * 500 )
```

The following sequence is not valid because it represents a nonorthogonal segment.

```
( 100 200 ) ( 300 500 )
```

Impact of Wrong-way Width Rules

Some technologies require larger widths for wrong-way routing than in the preferred direction. If the wrong-way width is larger than the default or NDR width, then the wrong-way width is used for wrong-way routes on that layer. The implicit routing extension is still half of the default or NDR width, even for wrong-way routes.

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Some older tools may not understand this behavior. Normally, they will still read/write and round-trip the DEF routing properly, but they may not understand that the width is slightly larger for wrong-way routes. If these tools check wrong-way width, then the DRC rules may flag false violations. RC extraction that does not understand the wrong-way width will also be incorrect, although wrong-way routes are generally short and the width difference is small, so the RC error is normally negligible.

The following LEF DRC rules allow a `WRONGDIRECTION` keyword that defines wrong-way widths that will affect the width of any wrong-way routes in the `DEF NETS` section:

`LEF58_WIDTH`	Defines the default routing width to use for all regular wiring on the layer.
`LEF58_WIDTHTABLE`	Defines all the allowable legal widths on the routing layer.
`LEF58_SPANLENGTHTABLE`	Defines all the allowable legal span lengths on the routing layer.

These width rules are mutually exclusive, so only one of the 3 rules is allowed on one routing layer.

The full syntax for `WIDTHTABLE` and `SPANLENGTHTABLE` have an optional `ORTHOGONAL` keyword or `ORTHOGONAL` keyword with a value. The `ORTHOGONAL` keyword and any value after it can be ignored, and has no effect on `DEF NETS` routing interpretation.

The full syntax for these rules is:

```
WIDTH defaultWidth ;
```

Along with:

```
PROPERTY LEF58_WIDTH
   "WIDTH minWidth [WRONGDIRECTION] ;" ;
```

Or

```
PROPERTY LEF58_SPANLENGTHTABLE
"SPANLENGTHTABLE {spanLength} ... [WRONGDIRECTION]
   [ORTHOGONAL length]
; " ;
```

Or

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```
PROPERTY LEF58_WIDTHTABLE
   "WIDTHTABLE {width}...[WRONGDIRECTION] [ORTHOGONAL]
     ];" ;
```

For more information on LEF width rules, see Layer (Routing) in the LEF/DEF Language Reference.

DEF NETS Examples

Example 8-13 Impact of default and nondefault rules on wrong-way segment

- This following example shows how LEF width rules will affect the width of any wrong-way routes in the `DEF NETS` section. An example of each type of rule (`WIDTH`, `WIDTHTABLE`, and `SPANLENGTHTABLE`) is shown below for `METAL2`:

```
LAYER METAL2
   ...
   DIRECTION VERTICAL ;
   #0.6 is the default routing rule width in the vertical direction
   WIDTH 0.06 ;
   #wrong direction (horizontal) metal width must be >= 0.12
   PROPERTY LEF58_WIDTH
      "WIDTH 0.12 WRONGDIRECTION ; " ;
...
END METAL2
```

or

```
LAYER METAL2
   ...
   DIRECTION VERTICAL ;
   #0.06 is the default routing rule width in the vertical direction
   WIDTH 0.06 ;
   #wrong direction (horizontal) metal width must be 0.12, 0.16 or >= 0.20
   PROPERTY LEF58_WIDTHTABLE
      "WIDTHTABLE 0.06 0.08 0.12 0.16 0.20 ;
       WIDTHTABLE 0.12 0.16 0.20 WRONGDIRECTION ; " ;
...
END METAL2
```

or

```
LAYER METAL2
   ...
```


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```
      DIRECTION VERTICAL ;
      #0.06 is the default routing rule width in the vertical direction
      WIDTH 0.06 ;
      #wrong direction (horizontal) metal width must be 0.12, 0.16 or >= 0.20
      PROPERTY LEF58_SPANLENGTHTABLE
         "SPANLENGTHTABLE 0.06 0.08 0.12 0.16 0.20 ;
          SPANLENGTHTABLE 0.12 0.16 0.20 WRONGDIRECTION ; " ;
...
END METAL2
```

For the above rules, any METAL2 vertical routes are in the preferred direction so they will have the normal widths and extensions as given by the default rule width, or the NONDEFAULTRULE width definition. The horizontal routes are in the wrong-direction, so they will use the first WRONGDIRECTION value in the rules above, that is greater than or equal to the preferred-direction width.

If the rule width is larger than the largest wrong-direction value, then the wrong-direction width is the same as the rule width as shown for NDR7 below.

The table below shows examples of different routing rule widths and the corresponding vertical and horizontal route widths and extensions for the WIDTHTABLE and SPANLENGTHTABLE rules shown above. They both have 0.12, 0.16, and 0.20 as the legal WRONGDIRECTION width values.

Rule	**Rule width**	**Vertical Route Width**	**Vertical Route Extension**	**Horizontal Route Width**	**Horizontal Route Extension**
default	0.06	0.06	0.03	0.12	0.03
NDR1	0.08	0.08	0.04	0.12	0.04
NDR2	0.12	0.12	0.06	0.12	0.06
NDR3	0.14	0.14	0.07	0.16	0.07
NDR4	0.16	0.16	0.08	0.16	0.08
NDR5	0.18	0.18	0.09	0.20	0.09
NDR6	0.20	0.20	0.10	0.20	0.10
NDR7	0.30	0.30	0.15	0.30	0.15

For example, the default rule in the table shows that a vertical route in the NETS section will have a width of 0.06 µm with extension of 0.03 µm, while a horizontal route will have

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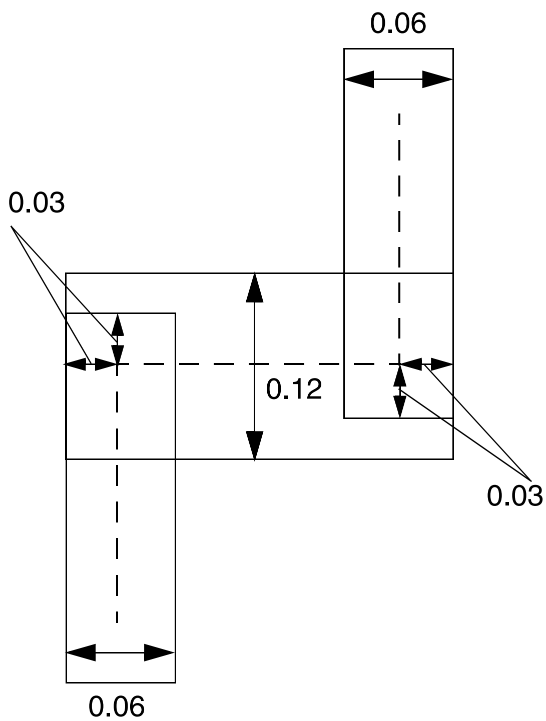
a wrong-way width of 0.12 µm with extension 0.03 µm, as shown in the `DEF NETS` routing example below:

```
- NET1 (...)
   + ROUTED METAL2 ( 1 0 ) ( 1 2 ) ( 3 2 ) ( 3 4 ) ;
```

Figure 8-6 Default rule on wrong way segment



- `NDR1` has a width of 0.08 µm, so it will use the wrong-way width of 0.12, because 0.8 is less than or equal to 0.12 (the first wrong-direction width value). So a vertical route will have a width of 0.08 µm with extension of 0.04 µm on a vertical route, while a horizontal route will have a width of 0.12 µm with extension 0.04 µm:

```
- NET2 (...)
   + NONDEFAULTRULE NDR1
   + ROUTED METAL2 ( 1 0 ) ( 1 3 ) ( 4 3 ) ( 4 5 ) ;
```


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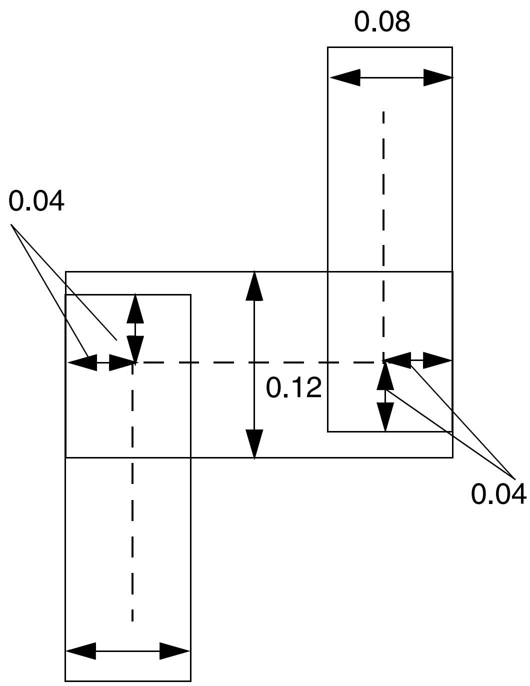
Figure 8-7 Non-Default rule on wrong way segment



- NDR4 in the table has a width of 0.16 µm. Width of 0.16 is larger than the first wrongdirection value of 0.12, but is less than or equal to the second wrong-direction value of 0.16, so it will use the wrong-way width of 0.16. In this case both the vertical and horizontal routes will have a width of 0.16 µm with extension of 0.08 µm.

```
- NET3 (...)
   + NONDEFAULTRULE NDR4
   + ROUTED METAL2 (1 0) (1 3) (4 3) (4 4) ;
```


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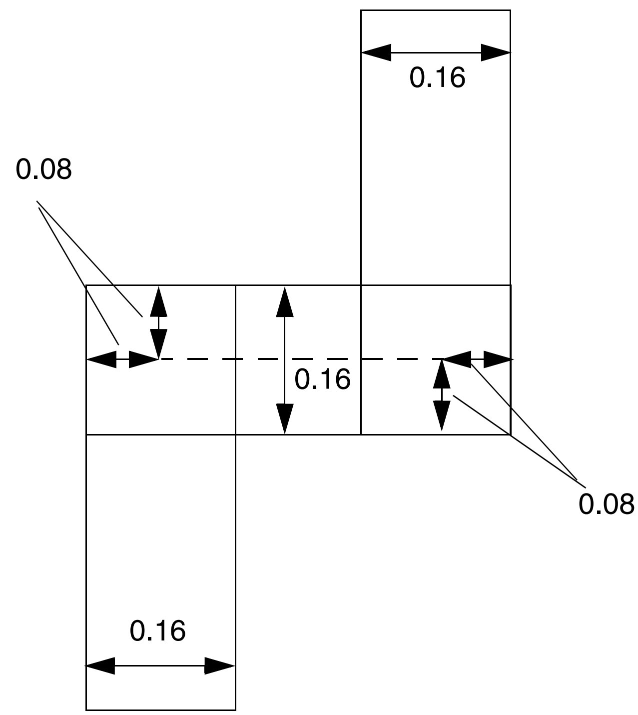
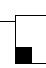
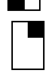
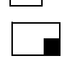
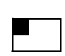
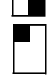
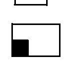
Figure 8-8 Non-Default rule on wrong way segment



Specifying Orientation

If you specify the pin's placement status, you must specify its location and orientation. A pin can have any of the following orientations: N, S, W, E, FN, FS, FW, or FE.

Orientation terminology can differ between tools. The following table maps the orientation terminology used in LEF and DEF files to the OpenAccess database format.

LEF/DEF	OpenAccess	Definition
N (North)	R0	
S (South)	R180	
W (West)	R90	
E (East)	R270	
FN (Flipped North)	MY	
FS (Flipped South)	MX	
FW (Flipped West)	MX90	

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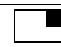
LEF/DEF	OpenAccess	Definition
FE (Flipped East)	MY90	

Example 8-14 Shielded Net

The following example defines a shielded net:

```
NETS 1 ;
    - my_net ( I1 CLK ) ( BUF OUT )
    + SHIELDNET VSS
    + SHIELDNET VDD
      ROUTED
    MET2 ( 14000 341440 ) ( 9600 * ) ( * 282400 )
    M1M2 ( 2400 * )
    + NOSHIELD MET2 ( 14100 341440 ) ( 14000 * )
    + TAPER MET1 ( 2400 282400 ) ( 240 * )
END NETS
```

Nondefault Rules

```
NONDEFAULTRULES numRules ;
     {- ruleName
           [+ HARDSPACING]
           {+ LAYER layerName
                WIDTH minWidth
                [DIAGWIDTH diagWidth]
                [SPACING minSpacing]
                [WIREEXT wireExt]
           } ...
           [+ VIA viaName] ...
           [+ VIARULE viaRuleName] ...
           [+ MINCUTS cutLayerName numCuts] ...
           [+ PROPERTY {propName propVal} ...] ...
           [PROPERTY LEF58_USEVIACUTCLASS
           "USEVIACUTCLASS cutLayerName className
              [ROWCOL numCutRows numCutCols]
              ;... " ;]
     ;} ...
END NONDEFAULTRULES
```

Defines any nondefault rules used in this design that are not specified in the LEF file. This section can also contain the default rule and LEF nondefault rule definitions for reference. These nondefault rule names can be used anywhere in the DEF `NETS` section that requires a nondefault rule name.

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If a nondefault rule name collides with an existing LEF or DEF nondefault rule name that has different parameters, the application should use the DEF definition when reading this DEF file, though it can change the DEF nondefault rule name to make it unique. This is typically done by adding a unique extension, such as `_1` or `_2` to the rule name.

All vias must be previously defined in the LEF `VIA` or DEF `VIAS` sections. Every nondefault rule must specify a width for every layer. If a nondefault rule does not specify a via or via rule for a particular routing-cut-routing layer combination, then there must be a `VIARULE GENERATE DEFAULT` rule that it inherited for that combination.

`DIAGWIDTH` *`diagWidth`*

Specifies the diagonal width for *`layerName`*, when 45-degree routing is used.
Default: `0` (no diagonal routing allowed)
Type: Integer, specified in DEF database units

`HARDSPACING`

Specifies that any spacing values that exceed the LEF `LAYER ROUTING` spacing requirements are “hard” rules instead of “soft” rules. By default, routers treat extra spacing requirements as soft rules that are high cost to violate, but not real spacing violations. However, in certain situations, the extra spacing should be treated as a hard, or real, spacing violation, such as when the route will be modified with a post-process that replaces some of the extra space with metal.

`LAYER` *`layerName`*

Specifies the layer for the various width and spacing values. *`layerName`* must be a routing layer. Each routing layer must have at least a minimum width specified.

`MINCUTS` *`cutLayerName numCuts`*

Specifies the minimum number of cuts allowed for any via using the specified cut layer. All vias (generated or fixed vias) used for this nondefault rule must have at least *`numCuts`* cuts in the via.
Type: (*`numCuts`*) Positive integer

`numRules`

Specifies the number of nondefault rules defined in the `NONDEFAULTRULES` section.

`PROPERTY` *`propName propValue`*

Specifies a property for this nondefault rule. The *`propName`* must be defined as a `NONDEFAULTRULE` property in the `PROPERTYDEFINITIONS` section, and the *`propValue`* must match the type for *`propName`* (that is, integer, real, or string).

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rulename

Specifies the name for this nondefault rule. This name can be used in the NETS section wherever a nondefault rule name is allowed. The reserved name DEFAULT can be used to indicate the default routing rule used in the NETS section.

SPACING *minSpacing*

Specifies the minimum spacing for *layerName*. The LEF LAYER SPACING or SPACINGTABLE definitions always apply; therefore it is only necessary to add a SPACING value if the desired spacing is larger than the LAYER rules already require.
Type: Integer, specified in DEF database units.

VIA *viaName*

Specifies a previously defined LEF or DEF via to use with this rule.

VIARULE *viaRuleName*

Specifies a previously defined LEF VIARULE GENERATE to use with this routing rule. If no via or via rule is specified for a given routing-cut-routing layer combination, then a VIARULE GENERATE DEFAULT via rule must exist for that combination, and it is implicitly inherited.

WIDTH *minWidth*

Specifies the required minimum width allowed for *layerName*.
Type: Integer, specified in DEF database units

WIREEXT *wireExt*

Specifies the distance by which wires are extended at vias. Enter 0 (zero) to specify no extension. Values other than 0 must be greater than or equal to half of the routing width for the layer, as defined in the nondefault rule.
Default: Wires are extended half of the routing width
Type: Integer, specified in DEF database units

Example 8-15 Nondefault Rules

The following NONDEFAULTRULES statement is based on the assumption that there are VIARULE GENERATE DEFAULT rules for each routing-cut-routing combination, and that the default width is 0.3 µm.

```
NONDEFAULTRULES 5 ;
    - doubleSpaceRule #Needs extra space, inherits default via rules
        + LAYER metal1 WIDTH 200 SPACING 1000
        + LAYER metal2 WIDTH 200 SPACING 1000
        + LAYER metal3 WIDTH 200 SPACING 1000 ;
```


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```
    - lowerResistance #Wider wires and double cut vias for lower resistance
                  #and higher current capacity. No special spacing rules,
                  #therefore the normal LEF LAYER specified spacing rules
                  #apply. Inherits the default via rules.
        + LAYER metal1 WIDTH 600 #Metal1 is thinner, therefore a little wider
        + LAYER metal2 WIDTH 500
        + LAYER metal3 WIDTH 500
        + MINCUTS cut12 2 #Requires at least two cuts
        + MINCUTS cut23 2 ;
    - myRule             #Use default width and spacing, change via rules. The
                    #default via rules are not inherited.
        + LAYER metal1 WIDTH 200
        + LAYER metal2 WIDTH 200
        + LAYER metal3 WIDTH 200
        + VIARULE myvia12rule
        + VIARULE myvia23rule ;
    - myCustomRule       #Use new widths, spacing and fixed vias. The default
                    #via rules are not inherited because vias are defined.
        + LAYER metal1 WIDTH 500 SPACING 1000
        + LAYER metal2 WIDTH 500 SPACING 1000
        + LAYER metal3 WIDTH 500 SPACING 1000
        + VIA myvia12_custom1
        + VIA myvia12_custom2
        + VIA myvia23_custom1
        + VIA myvia23_custom2 ;
END NONDEFAULTRULES
```

Use Via Cut Class Rule

You can use the use via cut class only rule to specify the cut class to be used with the routing rule.

You can create a use via cut class rule by using the following property definition:

```
PROPERTY LEF58_USEVIACUTCLASS
     USEVIACUTCLASS cutLayerName className
        [ROWCOL numCutRows numCutCols]
        ;... " ;
```

Where:

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USEVIACUTCLASS *cutLayerName className*
[ROWCOL *numCutRows numCutCols*]

Specifies `className` cuts on `cutLayerName`, which must be a cut layer, to be used with this routing rule. `ROWCOL` specifies the number of cut rows and columns of a multiple-cut via of the cut class `className` to be used with this routing rule. By default, a single cut via or the given cut number defined in `MINCUTS` is used.

Pins

```
[PINS numPins ;
    [ – pinName + NET netName
        [+ SPECIAL]
        [+ DIRECTION {INPUT | OUTPUT | INOUT | FEEDTHRU}]
        [+ NETEXPR "netExprPropName defaultNetName"]
        [+ SUPPLYSENSITIVITY powerPinName]
        [+ GROUNDSENSITIVITY groundPinName]
        [+ USE {SIGNAL | POWER | GROUND | CLOCK | TIEOFF | ANALOG
                  | SCAN | RESET}]
        [+ ANTENNAPINPARTIALMETALAREA value [LAYER layerName]] ...
        [+ ANTENNAPINPARTIALMETALSIDEAREA value [LAYER layerName]] ...
        [+ ANTENNAPINPARTIALCUTAREA value [LAYER layerName]] ...
        [+ ANTENNAPINDIFFAREA value [LAYER layerName]] ...
        [+ ANTENNAMODEL {OXIDE1 | OXIDE2 | OXIDE3 | OXIDE4}] ...
        [+ ANTENNAPINGATEAREA value [LAYER layerName]] ...
        [+ ANTENNAPINMAXAREACAR value LAYER layerName] ...
        [+ ANTENNAPINMAXSIDEAREACAR value LAYER layerName] ...
        [+ ANTENNAPINMAXCUTCAR value LAYER layerName] ...
        [[+ PORT]
         [+ LAYER layerName
             [MASK maskNum]
             [SPACING minSpacing | DESIGNRULEWIDTH effectiveWidth]
               pt pt
         |+ POLYGON layerName
             [MASK maskNum]
             [SPACING minSpacing | DESIGNRULEWIDTH effectiveWidth]
               pt pt pt ...
         |+ VIA viaName
             [MASK viaMaskNum]
                 pt
         ] ...
         [+ COVER pt orient | FIXED pt orient | PLACED pt orient]
        ]...
    ; ] ...
END PINS]
```


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Defines external pins. Each pin definition assigns a pin name for the external pin and associates the pin name with a corresponding internal net name. The pin name and the net name can be the same.

When the design is a chip rather than a block, the `PINS` statement describes logical pins, without placement or physical information.

`ANTENNAMODEL {OXIDE1 | OXIDE2 | OXIDE3 | OXIDE4}`

Specifies the oxide model for the pin. If you specify an `ANTENNAMODEL` statement, that value affects all `ANTENNAGATEAREA` and `ANTENNA*CAR` statements for the pin that follow it until you specify another `ANTENNAMODEL` statement. The `ANTENNAMODEL` statement does not affect `ANTENNAPARTIAL*AREA` and `ANTENNADIFFAREA` statements because they refer to the total metal, cut, or diffusion area connected to the pin, and do not vary with each oxide model.
Default: `OXIDE1`, for a new `PIN` statement

Because DEF is often used incrementally, if an `ANTENNA` statement occurs twice for the same oxide model, the last value specified is used.

Usually, you only need to specify a few `ANTENNA` values; however, for a block with six routing layers, it is possible to have six different `ANTENNAPARTIAL*AREA` values and six different `ANTENNAPINDIFFAREA` values per pin. It is also possible to have six different `ANTENNAPINGATEAREA` and `ANTENNAPINMAX*CAR` values for each oxide model on each pin.

See Example 8-16 on page 1038.

`ANTENNAPINDIFFAREA` *`value`* `[LAYER` *`layerName`*`]`

Specifies the diffusion (diode) area to which the pin is connected on a layer. If you do not specify *`layerName`*, the value applies to all layers. This is not necessary for output pins.
Type: Integer
Value: Area specified in (DEF database units)^2

For more information on process antenna calculation, see Appendix C, "Calculating and Fixing Process Antenna Violations."

`ANTENNAPINGATEAREA` *`value`* `[LAYER` *`layerName`*`]`

Specifies the gate area to which the pin is connected on a layer. If you do not specify *`layerName`*, the value applies to all layers. This is not necessary for input pins.
Type: Integer
Value: Area specified in (DEF database units)^2

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For more information on process antenna calculation, see Appendix C, "Calculating and Fixing Process Antenna Violations."

`ANTENNAPINMAXAREACAR` *value* `LAYER` *layerName*

For hierarchical process antenna effect calculation, specifies the maximum cumulative antenna ratio value, using the metal area at or below the current pin layer, excluding the pin area itself. Use this to calculate the actual cumulative antenna ratio on the pin layer, or the layer above it.
Type: Integer

For more information on process antenna calculation, see Appendix C, "Calculating and Fixing Process Antenna Violations."

`ANTENNAPINMAXCUTCAR` *value* `LAYER` *layerName*

For hierarchical process antenna effect calculation, specifies the maximum cumulative antenna ratio value, using the cut area at or below the current pin layer, excluding the pin area itself. Use this to calculate the actual cumulative antenna ratio for the cuts above the pin layer.
Type: Integer

For more information on process antenna calculation, see Appendix C, "Calculating and Fixing Process Antenna Violations."

`ANTENNAPINMAXSIDEAREACAR` *value* `LAYER` *layerName*

For hierarchical process antenna effect calculation, specifies the maximum cumulative antenna ratio value, using the metal side wall area at or below the current pin layer, excluding the pin area itself. Use this to calculate the actual cumulative antenna ratio on the pin layer, or the layer above it.
Type: Integer

For more information on process antenna calculation, see Appendix C, "Calculating and Fixing Process Antenna Violations."

`ANTENNAPINPARTIALCUTAREA` *value* `[LAYER` *cutLayerName*`]`

Specifies the partial cut area above the current pin layer and inside the macro cell on a layer. If you do not specify *layerName*, the value applies to all layers. For hierarchical designs, only the cut layer above the I/O pin layer is needed for partial antenna ratio calculation.
Type: Integer
Value: Area specified in (DEF database units)^2

For more information on process antenna calculation, see Appendix C, "Calculating and Fixing Process Antenna Violations."

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`ANTENNAPINPARTIALMETALAREA` *`value`* `[LAYER` *`layerName`*`]`

Specifies the partial metal area connected directly to the I/O pin and the inside of the macro cell on a layer. If you do not specify *`layerName`*, the value applies to all layers. For hierarchical designs, only the same metal layer as the I/O pin, or the layer above it, is needed for partial antenna ratio calculation.
Type: Integer
Value: Area specified in (DEF database units)^2

For more information on process antenna calculation, see Appendix C, "Calculating and Fixing Process Antenna Violations."

`ANTENNAPINPARTIALMETALSIDEAREA` *`value`* `[LAYER` *`layerName`*`]`

Specifies the partial metal side wall area connected directly to the I/O pin and the inside of the macro cell on a layer. If you do not specify *`layerName`*, the value applies to all layers. For hierarchical designs, only the same metal layer as the I/O pin, or the layer above it, is needed for partial antenna ratio calculation.
Type: Integer
Value: Area specified in (DEF database units)^2

For more information on process antenna calculation, see Appendix C, "Calculating and Fixing Process Antenna Violations."

`DIRECTION {INPUT | OUTPUT | INOUT | FEEDTHRU}`

Specifies the pin type. Most current tools do not usually use this keyword. Typically, pin directions are defined by timing library data, and not from DEF.
Value: Specify one of the following:

`INPUT`	Pin that accepts signals coming into the cell.
`OUTPUT`	Pin that drives signals out of the cell.
`INOUT`	Pin that can accept signals going either in or out of the cell.
`FEEDTHRU`	Pin that goes completely across the cell.

`GROUNDSENSITIVITY` *`groundPinName`*

Specifies that if this pin is connected to a tie-low connection (such as `1'b0` in Verilog), it should connect to the same net to which *`groundPinName`* is connected.

`groundPinName` must match another pin in this `PINS` section that has a `+ USE GROUND` attribute. The ground pin definition can follow later in this `PINS` section; it does not have to be defined before this pin definition. It is a semantic error to put this attribute on an existing ground pin.

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Note: GROUNDSENSITIVITY is useful only when there is more than one ground net connected to pins in the PINS section. By default, if there is only one net connected to all + USE GROUND pins, the tie-low connections are already implicitly defined (that is, tie-low connections are connected to the same net as any ground pin).

NETEXPR "*netExprPropName defaultNetName*"

Specifies a net expression property name (such as power1 or power2) and a default net name. If *netExprPropName* matches a net expression property higher up in the netlist (for example, in Verilog, VHDL, or OpenAccess), then the property is evaluated, and the software identifies a net to which to connect this pin. If the property does not exist, *defaultNetName* is used for the net name.

netExprPropName must be a simple identifier in order to be compatible with other languages, such as Verilog and CDL. Therefore, it can only contain alphanumeric characters, and the first character cannot be a number. For example, power2 is a legal name, but 2power is not. You cannot use characters such as $ and !. The *defaultName* can be any legal DEF net name.

If more than one pin connects to the same net, only one pin should have a NETEXPR added to it. It is redundant and unnecessary to add NETEXPR to every ground pin connected to one ground net, and it is illegal to have different NETEXPR values for pins connected to the same net.

See Example 8-17 on page 1039.

numPins

Specifies the number of pins defined in the PINS section.

pinName + NET *netName*

Specifies the name for the external pin, and the corresponding internal net (defined in NETS or SPECIALNETS statements).

PORT

Indicates that the following LAYER, POLYGON, and VIA statements are all part of one PORT connection, until another PORT statement occurs. In a PIN definition without any PORT statement, all of the LAYER, POLYGON, and VIA statements are assumed to be part of a single implicit PORT for the PIN.

This commonly occurs for power and ground pins. All of the shapes of one port (rectangles, polygons, and vias) should already be connected with just the port shapes; therefore, the router only needs to connect to one of the shapes for the port. Separate ports should each be connected by routing inside the block (and each DEF PORT should

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map to a single LEF `PORT` in the equivalent LEF abstract for this block).

The syntax for describing `PORT` statements is defined as follows:

```
[[+ PORT]
  [+ LAYER layerName
     [MASK maskNum]
     [SPACING minSpacing | DESIGNRULEWIDTH effectiveWidth]
       pt pt
  |+ POLYGON layerName
     [MASK maskNum]
     [SPACING minSpacing | DESIGNRULEWIDTH effectiveWidth]
       pt pt pt ...
  |+ VIA viaName
     [MASK viaMaskNum]
       pt
  ] ...
  [+ COVER pt orient | FIXED pt orient | PLACED pt orient]
 ] ...
```

`COVER` *`pt orient`*

Specifies the pin's location, orientation, and that it is a part of the cover macro. A `COVER` pin cannot be moved by automatic tools or by interactive commands. If you specify a placement status for a pin, you must also include a `LAYER` statement.

`DESIGNRULEWIDTH` *`effectiveWidth`*

Specifies that this pin has a width of *`effectiveWidth`* for the purpose of spacing calculations. If you specify `DESIGNRULEWIDTH`, you cannot specify `SPACING`. As a lot of spacing rules in advanced nodes no longer just rely on wire width, `DESIGNRULEWIDTH` is not allowed for 20nm and below nodes. (See Example 8-18 on page 1040.)
Type: Integer, specified in DEF database units

`FIXED` *`pt orient`*

Specifies the pin's location, orientation, and that its location cannot be moved by automatic tools, but can be moved by interactive commands. If you specify a placement status for a pin, you must also include a `LAYER` statement

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`LAYER` *`layerName pt pt`*

Specifies the routing layer used for the pin, and the pin geometry on that layer. If you specify a placement status for a pin, you must include a `LAYER` statement.

`MASK` *`maskNum`*

Specifies which mask from double or triple patterning to use for the specified shape. The *`maskNum`* variable must be a positive integer - most applications support values of 1, 2, or 3 only. Shapes without any defined mask do not have a mask set (are uncolored).

`MASK` *`viaMaskNum`*

Specifies which mask for double or triple patterning lithography is to be applied to via shapes on each layer.

The *`viaMaskNum`* variable is a hex-encoded three digit value of the form:
`<topMaskNum><cutMaskNum><bottomMaskNum>`

For example, MASK 113 means the top metal and cut layer *`maskNum`* is 1, and the bottom metal layer *`maskNum`* is 3. A value of 0 indicates that the shape on that layer does not have a mask assignment (is uncolored), so 013 means the top layer is uncolored. If either the first or second digit is missing, they are assumed to be 0, so 013 and 13 mean the same thing. Most applications support *`maskNums`* of 0, 1, 2, or 3 for double or triple patterning.

The *`topMaskNum`* and *`bottomMaskNum`* variables specify the mask to which the corresponding metal shape belongs. The via-master metal mask values have no effect.

For the cut-layer, the *`cutMaskNum`* variable defines the mask for the bottom-most, then the left-most cut in the North (R0) orientation. For multi-cut vias, the via-instance cut masks are derived from the via-master cut mask values. The via-master must have a mask defined for all of the cut shapes and every via-master cut mask is "shifted" (from 1 to 2 and 2 to1 for two mask layers, and from 1 to 2, 2 to 3, and 3 to1 for three mask layers), so the lower-left cut matches the *`cutMaskNum`* value. See Example 8-16 on page 1038.

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Similarly, for the metal layer, the *topMaskNum*/*bottomMaskNum* would define the mask for the bottom-most, then leftmost metal shape. For multiple disjoint metal shapes, the via-instance metal masks are derived from the via-master metal mask values. The via-master must have a mask defined for all of the metal shapes, and every via-master metal mask is "shifted" (1->2, 2->1 for two mask layers, 1->2, 2->3, 3->1 for three mask layers) so the lower-left cut matches the *topMaskNum*/*bottomMaskNum* value.

Shapes without any defined mask, that need to be assigned, can be assigned to an arbitrary choice of mask by applications.

`PLACED` *pt orient*

Specifies the pin's location, orientation, and that it's location is fixed, but can be moved during automatic layout. If you specify a placement status for a pin, you must also include a `LAYER` statement.

`POLYGON` *layerName pt pt pt*

Specifies the layer and a sequence of at least three points to generate a polygon for this pin. The polygon edges must be parallel to the x axis, the y axis, or at a 45-degree angle.

Each `POLYGON` statement defines a polygon generated by connecting each successive point, and then the first and last points. The *pt* syntax corresponds to a coordinate pair, such as *x y*. Specify an asterisk (`*`) to repeat the same value as the previous *x* or *y* value from the last point. (See Example 8-21 on page 1042.)

`SPACING` *minSpacing*

Specifies the minimum spacing allowed between this pin and any other routing shape. This distance must be greater than or equal to *minSpacing*. If you specify `SPACING`, you cannot specify `DESIGNRULEWIDTH`. (See Example 8-18 on page 1040.)
Type: Integer, specified in DEF database units

`VIA` *viaName pt*

Places the via named *viaName* at the specified (x y) location (*pt*). *viaName* must be a previously defined via in the DEF `VIAS` or LEF `VIA` section.
Type: (*pt*) Integers, specified in DEF database units

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`SPECIAL`

Identifies the pin as a special pin. Regular routers do not route to special pins. The special router routes special wiring to special pins.

`SUPPLYSENSITIVITY` *`powerPinName`*

Specifies that if this pin is connected to a tie-high connection (such as `1'b1` in Verilog), it should connect to the same net to which *`powerPinName`* is connected. *`powerPinName`* must match another pin in this `PINS` section that has a `+ USE POWER` attribute. The power pin definition can follow later in this `PINS` section; it does not have to be defined before this pin definition. It is a semantic error to put this attribute on an existing power pin. For an example, see Example 8-17 on page 1039.

Note: `POWERSENSITIVITY` is useful only when there is more than one power net connected to pins in the `PINS` section. By default, if there is only one net connected to all `+ USE POWER` pins, the tie-high connections are already implicitly defined (that is, tie-high connections are connected to the same net as the single power pin).

`USE {ANALOG | CLOCK | GROUND | POWER | RESET | SCAN | SIGNAL | TIEOFF}`

Specifies how the pin is used.
Default: `SIGNAL`
Value: Specify one of the following:

`ANALOG`	Pin is used for analog connectivity.
`CLOCK`	Pin is used for clock net connectivity.
`GROUND`	Pin is used for connectivity to the chip-level ground distribution network.
`POWER`	Pin is used for connectivity to the chip-level power distribution network.
`RESET`	Pin is used as reset pin.
`SCAN`	Pin is used as scan pin.
`SIGNAL`	Pin is used for regular net connectivity.
`TIEOFF`	Pin is used as tie-high or tie-low pin.

Example 8-16 Antenna Model Statement

The following example describes the `OXIDE1` and `OXIDE2` models for pin `clock1`. Note that the `ANTENNAPINPARTIALMETALAREA` and `ANTENNAPINDIFFAREA` values are not affected by the oxide values.

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```
PINS 100 ;
   - clock1 + NET clock1
      ...
      + ANTENNAPINPARTIALMETALAREA 1000 LAYER m1
      + ANTENNAPINDIFFAREA 500 LAYER m1
      ...
      + ANTENNAMODEL OXIDE1                      #not required, but good practice
      + ANTENNAPINGATEAREA 1000
      + ANTENNAMAXAREACAR 300 LAYER m1
      ...
      + ANTENNAMODEL OXIDE2                      #start of OXIDE2 values
      + ANTENNAPINGATEAREA 2000
      + ANTENNAMAXAREACAR 100 LAYER m1
      ...
```

Example 8-17 Net Expression and Supply Sensitivity

The following `PINS` statement defines sensitivity and net expression values for five pins in the design `myDesign`:

```
DESIGN myDesign
...
PINS 4 ;
    - in1 + NET myNet
        ...
        + SUPPLYSENSITIVITY vddpin1 ; #If in1 is connected to 1'b1, use
                                      #net that is connected to vddpin1.
                                      #No GROUNDSENSITIVITY is needed because
                                      #only one ground net is used by PINS.
                                      #Therefore, 1'b0 implicitly means net
                                      #from any +USE GROUND pin.
    - vddpin1 + NET VDD1 + USE POWER
        ...
        + NETEXPR "power1 VDD1" ; #If an expression named power1 is defined in
                                  #the netlist, use it to fine the net.
                                  #Otherwise, use net VDD1.
    - vddpin2 + NET VDD2 + USE POWER
        ...
        + NETEXPR "power2 VDD2" ; #If an expression named power2 is defined in
                                  #the netlist, use it to find the net.
                                  # Otherise, use net VDD2.
    - gndpin1 + NET GND + USE GROUND
        ...
        + NETEXPR "gnd1 GND" ; #If an expression named gnd1 is defined in
                               #the netlist, use it to find net
                               #connection. Otherwise, use net GND.
```


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```
END PINS
```

Example 8-18 Design Rule Width and Spacing Rules

The following statements create spacing rules using the `DESIGNRULEWIDTH` and `SPACING` statements:

```
PINS 3 ;
    - myPin1 + NET myNet1 + DIRECTION INPUT
        + LAYER metal1
              DESIGNRULEWIDTH 1000   #Pin is effectively 1000 dbu wide
              ( -100 0 ) ( 100 200 ) #Pin is 200 x 200 dbu
        + FIXED ( 10000 5000 ) S ;
    - myPin2 + NET myNet2 + DIRECTION INPUT
        + LAYER metal1
              SPACING 500            #Requires >= 500 dbu spacing
              ( -100 0 ) ( 100 200 ) #Pin is 200 x 200 dbu
        + COVER ( 10000 5000 ) S ;
    - myPin3 + NET myNet1            #Pin with two shapes
        + DIRECTION INPUT
        + LAYER metal2 ( 200 200 ) ( 300 300 ) #100 x 100 dbu shape
        + POLYGON metal1 ( 0 0 ) ( 100 100 ) ( 200 100 ) ( 200 0 ) #Has 45-degree edge
        + FIXED ( 10000 5000 ) N ;
END PINS
```

Example 8-19 Multi-Mask Patterns for Pins

The following example shows via master masks:

```
clock + NET clock
    + LAYER M1 MASK 2 ( -25 0 ) ( 25 50 )           #m1 rectangle is on mask 2
    + LAYER M2 ( -10 0 ) ( 10 75 )                  #m2 rectangle, no mask
    + VIA VIA1 MASK 031 ( 0 25 )                    #via1 with mask 031
    ...
```

The `VIA1` via will have:

- no mask set for the top metal shape (*`topMaskNum`* is 0 in the 031 value)
- `MASK 1` for the bottom metal shape (*`botMaskNum`* is 1 in the 031 value)
- the bottom-most, and then the left-most cut of the via-instance is `MASK 3`. The mask for the other cuts of the via-instance are derived from the via-master by "shifting" the via-master cut masks to match. So if the via-master's bottom-left cut is `MASK 1`, then the via-

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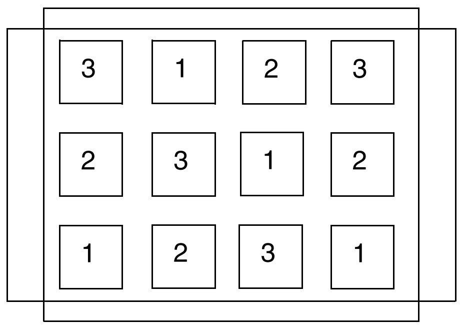
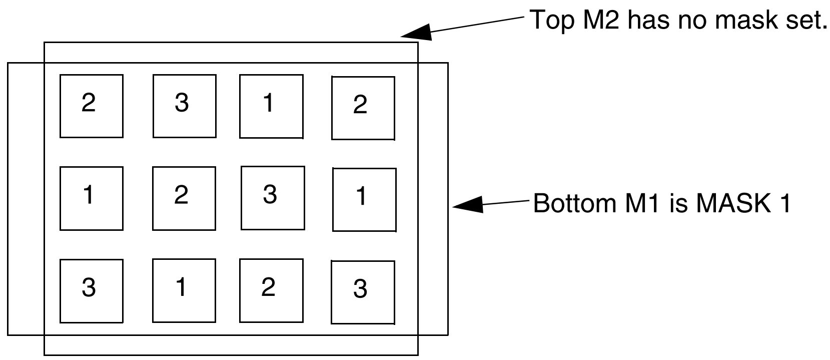
master cuts on `MASK 1` become `MASK 3` for the via-instance. Similarly cuts on 2 shift to 1, and cuts on 3 shift to 2. See Figure 8-9 on page 1041.

Figure 8-9 Multi-Mask Patterns for Pins



Via-master cut masks for VIA1.



Masks for via-instance: + VIA VIA1 MASK 031 (0 25)
Bottom M1 is MASK 1.Top M2 has no mask set. Lower-left cut is MASK 3, and other via-instance cut shape masks are derived from via-master cut-masks as shown above by "shifting" the via-master masks to match: 1->3, 2->1, 3->2.

Example 8-20 Port Example

Assume a block that is 5000 x 5000 database units with a 0,0 origin in the middle of the block. If you have the following pins defined, Figure 8-10 on page 1042 illustrates how pin `BUSA[0]` is created for two different placement locations and orientations:

```
PINS 2 ;
    - BUSA[0] + NEY BUSA[0] + DIRECTION IN{UT + USE SIGNAL
       + LAYER M1 ( -25 0 ) ( 25 50 )    #m1, m2, and via12
       + LAYER M2 ( -10 0 ) ( 10 75 )
```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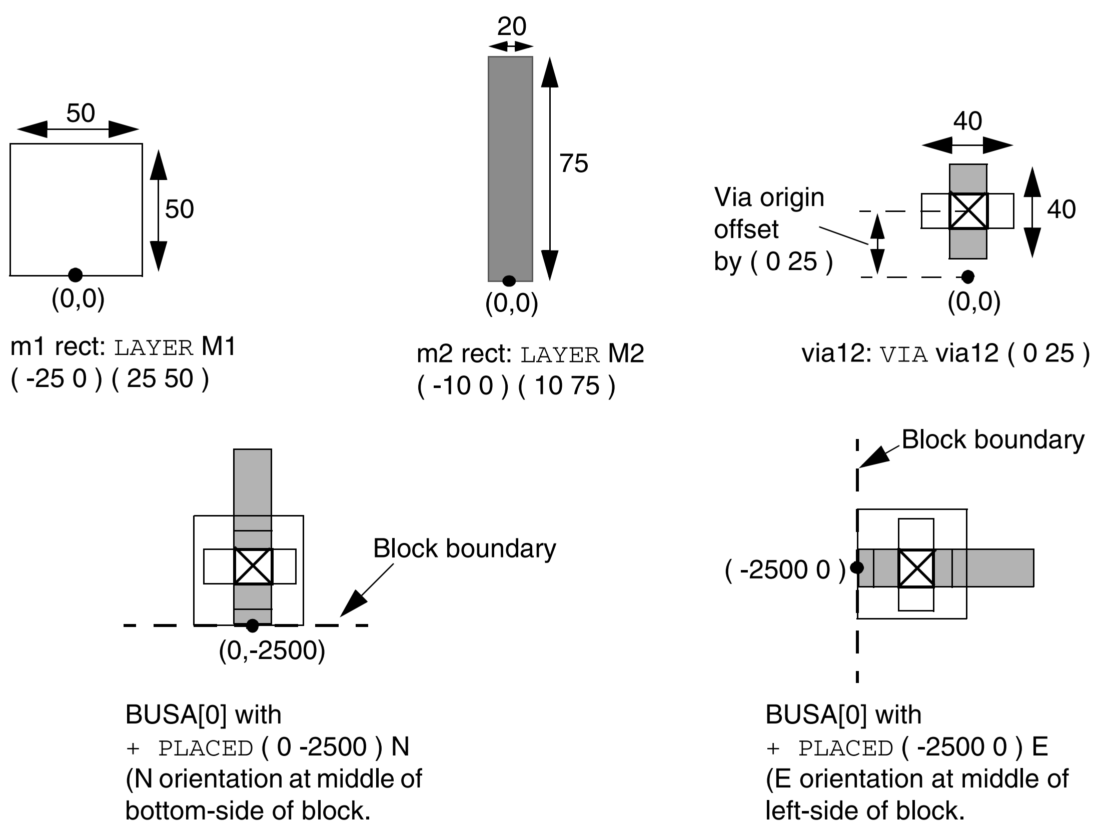
```
       + VIA via12  ( 0 25 )
       + PLACED ( 0 -2500 ) N ;          #middle of bottom side
    - VDD + NET VDD + DIRECTION INOUT + USE POWER + SPECIAL
       + PORT
           + LAYER M2 ( -25 0 ) ( 25 50 )
           + PLACED ( 0 2500 ) S        #middle of top side
        + PORT
           + LAYER M1 (-25 0 ) ( 25 50 )
           + PLACED ( -2500 0 ) E        #middle of left side
        + PORT
           + LAYER M1 ( -25 0 ) ( 25 50 )
           + PLACED ( 2500 0 ) W ;       #middle of right side
END PINS
```

Figure 8-10 Port Illustration



Example 8-21 Port Statement With Polygon

The following `PINS` statement creates a polygon with a 45-degree angle:

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```
PINS 2 ;
    - myPin3 + NET myNet1 + DIRECTION INPUT
        + PORT
          + POLYGON metal1 ( 0 0 ) ( 100 100 ) ( 200 100 ) ( 200 0 ) #45-degree angle
          + FIXED ( 10000 5000 ) N ;
    ...
END PINS
```

Extra Physical PIN(S) for One Logical PIN

In the design of place and route blocks, you sometimes want to add extra physical connection points to existing signal ports (usually to enable the signal to be accessed from two sides of the block). One pin has the same name as the net it is connected to. Any other pins added to the net must use the following naming conventions.

For extra non-bus bit pin names, use the following syntax:

pinname.extra*N*

N is a positive integer, incremented as the physical pins are added
For example:

```
PINS n ;
    - a + NET a .... ;
    - a.extra1 + NET a ... ;
```

For extra bus bit pin names, use the following syntax:

basename.extra*N*[*index*]

basename is simple part of bus bit pin/net name. *N* is a positive integer, incremented as the physical pins are added. [*index*] identifies the specific bit of the bus, if it is a bus bit.
For example:

```
PINS n ;
    - a[0] + net a[0] ... ;
    - a.extra1[0] + net a[0] ... ;
```

Note: The brackets [ ] are the BUSBITCHARS as defined in the DEF BUSBITCHARS statement.

Specifying Orientation

If you specify the pin's placement status, you must specify its location and orientation. A pin can have any of the following orientations: N, S, W, E, FN, FS, FW, or FE.

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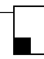
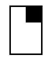
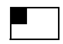
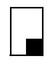
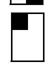
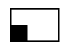
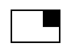
Orientation terminology can differ between tools. The following table maps the orientation terminology used in LEF and DEF files to the OpenAccess database format.

LEF/DEF	OpenAccess	Definition
N (North)	R0	
S (South)	R180	
W (West)	R90	
E (East)	R270	
FN (Flipped North)	MY	
FS (Flipped South)	MX	
FW (Flipped West)	MX90	
FE (Flipped East)	MY90	

Example 8-22 Pin Statements

The following example describes a physical I/O pin.

```
# M1 width = 50, track spacing = 120
# M2 width = 60, track spacing = 140

DIEAREA ( -5000 -5000 ) ( 5000 5000 ) ;
TRACKS Y -4900 DO 72 STEP 140 LAYER M2 M1 ;
TRACKS X -4900 DO 84 STEP 120 LAYER M1 M2 ;
PINS 4 ;
    # Pin on the left side of the block
    - BUSA[0]+ NET BUSA[0] + DIRECTION INPUT
                + LAYER M1 ( -25 0 ) ( 25 165 ) # .5 M1 W + 1 M2 TRACK
                + PLACED ( -5000 2500 ) E ;
    # Pin on the right side of the block
    - BUSA[1] + NET BUSA[1] + DIRECTION INPUT
                + LAYER M1 ( -25 0 ) ( 25 165 ) # .5 M1 W + 1 M2 TRACK
                + PLACED ( 5000 -2500 ) W ;
    # Pin on the bottom side of the block
    - BUSB[0] + NET BUSB[0] + DIRECTION INPUT
                + LAYER M2 ( -30 0 ) ( 30 150 ) # .5 M2 W + 1 M1 TRACK
                + PLACED ( -2100 -5000 ) N ;
    # Pin on the top side of the block
    - BUSB[1] + NET BUSB[1] + DIRECTION INPUT
```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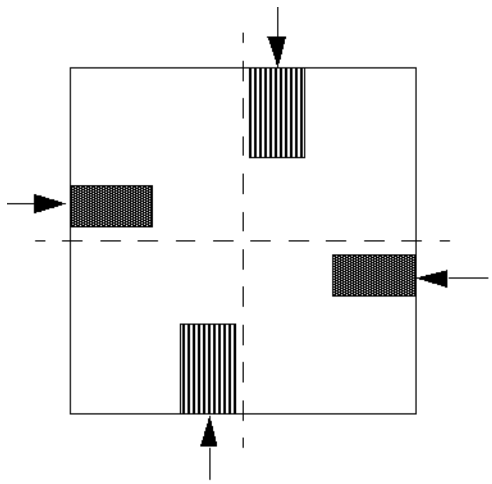
```
            + LAYER M2 ( -30 0 ) ( 30 150 ) # .5 M2 W + 1 M1 TRACK
            + PLACED ( 2100 5000 ) S ;
END PINS
```



The following example shows how a logical I/O pin would appear in the DEF file. The pin is first defined in Verilog for a chip-level design.

```
module chip (OUT, BUSA, BUSB) ;
    input [0:1] BUSA, BUSB;
    output OUT;
....
endmodule
```

The following description for this pin is in the `PINS` section in the DEF file:

```
PINS 5 ;
    - BUSA[0] + NET BUSA[0] + DIRECTION INPUT ;
    - BUSA[1] + NET BUSA[1] + DIRECTION INPUT ;
    - BUSB[0] + NET BUSB[0] + DIRECTION INPUT ;
    - BUSB[1] + NET BUSB[1] + DIRECTION INPUT ;
    - OUT + NET OUT + DIRECTION OUTPUT ;
END PINS
```

Pin Properties

```
[PINPROPERTIES num;
     [- {compName pinName | PIN pinName}
          [+ PROPERTY {propName propVal} ...] ...
     ;] ...
END PINPROPERTIES]
```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Defines pin properties in the design.

compName pinName

Specifies a component pin. Component pins are identified by the component name and pin name.

num

Specifies the number of pins defined in the PINPROPERTIES section.

PIN *pinName*

Specifies an I/O pin.

PROPERTY *propName propVal*

Specifies a numerical or string value for a pin property defined in the PROPERTYDEFINITIONS statement. The *propName* you specify must match the *propName* listed in the PROPERTYDEFINITIONS statement.

Example 8-23 Pin Properties Statement

```
PINPROPERTIES 3 ;
    - CORE/g76 CKA + PROPERTY CLOCK "FALLING" ;
    - comp1 A + PROPERTY CLOCK "EXCLUDED" ;
    - rp/regB clk + PROPERTY CLOCK "INSERTION" ;
END PINPROPERTIES
```

Property Definitions

```
[PROPERTYDEFINITIONS
     [objectType propName propType [RANGE min max]
        [value | stringValue]
     ;] ...
END PROPERTYDEFINITIONS]
```

Lists all properties used in the design. You must define properties in the PROPERTYDEFINITIONS statement before you can refer to them in other sections of the DEF file.

objectType

Specifies the object type being defined. You can define properties for the following object types:

- COMPONENT
- COMPONENTPIN

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- DESIGN
- GROUP
- NET
- NONDEFAULTRULE
- REGION
- ROW
- SPECIALNET

propName

Specifies a unique property name for the object type.

propType

Specifies the property type for the object type. You can specify one of the following property types:

- INTEGER
- REAL
- STRING

RANGE *min max*

Limits real number and integer property values to a specified range.

value | *stringValue*

Assigns a numeric value or a name to a DESIGN object.

Note: Assign values to properties for component pins in the PINPROPERTIES section. Assign values to other properties in the section of the LEF file that describes the object to which the property applies.

Regions

```
[REGIONS numRegions ;
    [- regionName {pt pt} ...
       [+ TYPE {FENCE | GUIDE}]
       [+ PROPERTY {propName propVal} ...] ...
    ;] ...
END REGIONS]
```


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Defines regions in the design. A region is a physical area to which you can assign a component or group.

numRegions

Specifies the number of regions defined in the design.

`PROPERTY` *propName propVal*

Specifies a numerical or string value for a region property defined in the `PROPERTYDEFINITIONS` statement. The *propName* you specify must match the *propName* listed in the `PROPERTYDEFINITIONS` statement.

regionName pt pt

Names and defines a region. You define a region as one or more rectangular areas specified by pairs of coordinate points.

`TYPE {FENCE | GUIDE}`

Specifies the type of region.
Default: All instances assigned to the region are placed inside the region boundaries, and other cells are also placed inside the region.
Value: Specify one of the following:

`FENCE`	All instances assigned to this type of region must be exclusively placed inside the region boundaries. No other instances are allowed inside this region.
`GUIDE`	All instances assigned to this type of region should be placed inside this region; however, it is a preference, not a hard constraint. Other constraints, such as wire length and timing, can override this preference.

Example 8-24 Regions Statement

```
REGIONS 1 ;
    - REGION1 ( 0 0 ) ( 1200 1200 )
        + PROPERTY REGIONORDER 1 ;
```

Rows

```
[ROW rowName siteName origX origY siteOrient
       [DO numX BY numY [STEP stepX stepY]]
       [+ PROPERTY {propName propVal} ...] ... ;] ...
```


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Defines rows in the design.

`DO` *`numX`* `BY` *`numY`*

Specifies a repeating set of sites that create the row. You must specify one of the values as `1`. If you specify `1` for *`numY`*, then the row is horizontal. If you specify `1` for *`numX`*, the row is vertical.
Default: Both *`numX`* and *`numY`* equal `1`, creating a single site at this location (that is, a horizontal row with one site).

`origX origY`

Specifies the location of the first site in the row.
Type: Integer, specified in DEF database units

`PROPERTY` *`propName propVal`*

Specifies a numerical or string value for a row property defined in the `PROPERTYDEFINITIONS` statement. The *`propName`* you specify must match the *`propName`* listed in the `PROPERTYDEFINITIONS` statement.

`rowName`

Specifies the row name for this row.

`siteName`

Specifies the LEF `SITE` to use for the row. A site is a placement location that can be used by LEF macros that match the same site. *`siteName`* can also refer to a site with a row pattern in its definition, in which case, the row pattern indicates a repeating set of sites that are abutted. For more information, see “Site” and “Macro” in “LEF Syntax.”

`siteOrient`

Specifies the orientation of all sites in the row. *`siteOrient`* must be one of `N`, `S`, `E`, `W`, `FN`, `FS`, `FE`, or `FW`. For more information on orientations, see “Specifying Orientation” on page 994.

`STEP` *`stepX stepY`*

Specifies the spacing between sites in horizontal and vertical rows.

Example 8-25 Row Statements

Assume `siteA` is 200 by 900 database units.

```
ROW row_0 siteA 1000 1000 N ; #Horizontal row is one-site wide at 1000, 1000
ROW row_1 siteA 1000 1000 N DO 1 BY 1 ;            #Same as row_0
ROW row_2 siteA 1000 1000 N DO 1 BY 1 STEP 200 0 ; #Same as row_0
ROW row_3 siteA 1000 1000 N DO 10 BY 1 ; #Horizontal row is 10 sites wide,
```


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```
                                          #so row width is 200*10=2000 dbu
ROW row_4 siteA 1000 1000 N DO 10 BY 1 STEP 200 0 ; #Same as row_3
ROW row_5 siteA 1000 1000 N DO 1 BY 10 ; #Vertical row is 10 sites high, so
                                          #total row height is 900*10=9000 dbu
ROW row_6 siteA 1000 1000 N DO 1 BY 10 STEP 0 900 ; #Same as row_5
```

Scan Chains

```
[SCANCHAINS numScanChains ;
     [- chainName
        [+ PARTITION partitionName [MAXBITS maxbits]]
        [+ COMMONSCANPINS [ ( IN pin )] [( OUT pin ) ] ]
         + START {fixedInComp | PIN} [outPin]
        [+ FLOATING
           {floatingComp [ ( IN pin ) ] [ ( OUT pin ) ] [ ( BITS numBits ) ]} ...]
        [+ ORDERED
           {fixedComp [ ( IN pin ) ] [ ( OUT pin ) ] [ ( BITS numBits ) ]} ...
        ] ...
         + STOP {fixedOutComp | PIN} [inPin] ]
     ;] ...
END SCANCHAINS]
```

Defines scan chains in the design. Scan chains are a collection of cells that contain both scan-in and scan-out pins. These pins must be defined in the `PINS` section of the DEF file with `+ USE SCAN`.

chainName

Specifies the name of the scan chain. Each statement in the `SCANCHAINS` section describes a single scan chain.

`COMMONSCANPINS [( IN` *pin* `)] [( OUT` *pin* `)]`

Specifies the scan-in and scan-out pins for each component that does not have a scan-in and scan-out pin specified. You must specify either common scan-in and scan-out pins, or individual scan-in and scan-out pins for each component.

`FLOATING {`*floatingComp* `[( IN` *pin* `)] [( OUT` *pin* `)] [( BITS` *numBits* `)]}`

Specifies the floating list. You can have one or zero floating lists. If you specify a floating list, it must contain at least one component.

floatingComp Specifies the component name.

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`( IN` *`pin`* `)`	Specifies the scan-in pin. If you do not specify a scan-in pin, the router uses the pin you specified for the common scan pins.
`( OUT` *`pin`* `)`	Specifies the scan-out pin. If you do not specify a scan-out pin, the router uses the pin you specified for the common scan pins.
`BITS` *`numBits`*	Specifies the sequential bit length of any chain element. This allows application tools that do not have library access to determine the sequential bit length contribution of any chain element to ensure the `MAXBITS` constraints are not violated for chains in a given partition. You can specify `0` to indicate when elements are non-sequential. *Default:* `1` *Type:* Integer

Note: Scan chain reordering commands can use floating components in any order to synthesize a scan chain. Floating components cannot be shared with other scan chains unless they are in the same `PARTITION`. Each component should only be used once in synthesizing a scan chain.

`MAXBITS` *`maxBits`*

When specified with chains that include the `PARTITION` keyword, sets the maximum bit length (flip-flop bit count) that the chain can grow to in the partition.
Default: `0` (tool-specific defaults apply, which is probably the number of bits in each chain)
Type: Integer
Value: Specify a value that is at least as large as the size of the current chain.

`numScanChains`

Specifies the number of scan chains to synthesize.

`ORDERED {`*`fixedComp`* `[( IN` *`pin`* `)] [( OUT` *`pin`* `)] [( BITS` *`numBits`* `)]}`

Specifies an ordered list. You can specify none or several ordered lists. If you specify an ordered list, you must specify at least two fixed components for each ordered list.

`fixedComp`	Specifies the component name.
`( IN` *`pin`* `)`	Specifies the scan-in pin. If you do not specify a scan-in pin, the router uses the pin you specified for the common scan pins.

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`( OUT pin )` Specifies the scan-out pin. If you do not specify a scan-out pin, the router uses the pin you specified for the common scan pins.

`BITS numBits` Specifies the sequential bit length of any chain element. This allows application tools that do not have library access to determine the sequential bit length contribution of any chain element to ensure the `MAXBITS` constraints are not violated for chains in a given partition. You can specify `0` to indicate when elements are non-sequential.
Default: `1`
Type: Integer

Note: Scan chain reordering commands should synthesize these components in the same order that you specify them in the list. Ordered components cannot be shared with other scan chains unless they are in the same `PARTITION`. Each component should only be used once in synthesizing a scan chain.

`PARTITION partitionName`

Specifies a partition name. This statement allows reordering tools to determine inter-chain compatibility for element swapping (both `FLOATING` elements and `ORDERED` elements). It associates each chain with a partition group, which determines their compatibility for repartitioning by swapping elements between them.

Chains with matching `PARTITION` names constitute a swap-compatible group. You can change the length of chains included in the same partition (up to the `MAXBITS` constraint on the chain), but you cannot eliminate chains or add new ones; the number of chains in the partition is always preserved.

If you do not specify the `PARTITION` keyword, chains are assumed to be in their own single partition, and reordering can be performed only within that chain.

Example 8-26 Partition Scanchain

In the following definition, chain `chain1_clock1` is specified without a `MAXBITS` keyword. The maximum allowed bit length of the chain is assumed to be the sequential length of the longest chain in any `clock1` partition.

```
SCANCHAINS 77 ;
    - chain1_clock1
        + PARTITION clock1
        + START block1/bsr_reg_0 Q
        + FLOATING
```


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```
         block1/pgm_cgm_en_reg_reg ( IN SD ) ( OUT QZ )
         ...
         block1/start_reset_dd_reg ( IN SD ) ( OUT QZ )
      + STOP block1/start_reset_d_reg SD ;
```

In the following definition, chain `chain2_clock2` is specified with a `PARTITION` statement that associates it with `clock2`, and a maximum bit length of `1000`. The third element statement in the `FLOATING` list is a scannable register bank that has a sequential bit length of `4`. The `ORDERED` list element statements have total bit lengths of `1` each because the muxes are specified with a maximum bit length of `0`.

```
   - chain2_clock2
      + PARTITION clock2
        MAXBITS 1000
      + START block1/current_state_reg_0_QZ
      + FLOATING
         block1/port2_phy_addr_reg_0_ ( IN SD ) ( OUT QZ )
         block1/port2_phy_addr_reg_4_ ( IN SD ) ( OUT QZ )
         block1/port3_intfc ( IN SD ) ( OUT MSB ) ( BITS 4 )
              ...
      + ORDERED
         block1/mux1 ( IN A ) ( OUT X ) ( BITS 0 )
         block1/ff1 ( IN SD ) ( OUT Q )
      + ORDERED
         block1/mux2 ( IN A ) ( OUT X ) ( BITS 0 )
         block1/ff2 ( IN SD ) ( OUT Q ) ;
```

In the following definition, chain `chain3_clock2` is also specified with a `PARTITION` statement that associates it with `clock2`. This means it is swap-compatible with `chain2_clock2`. The specified maximum bit length for this chain is `1200`.

```
   - chain3_clock2
      + PARTITION clock2
        MAXBITS 1200
      + START block1/LV_testpoint_0_Q_reg Q
      + FLOATING
         block1/LV_testpoint_0_Q_reg  ( IN SE ) ( OUT Q )
         block1/tm_state_reg_1_ (IN SD ) ( OUT QZ )
         ...
```

In the following definition, chain `chain4_clock3` is specified with a `PARTITION` statement that associates it with `clock3`. The second element statement in the `FLOATING` list is a scannable register bank that has a sequential bit length of `8`, and default pins. The `ORDERED`

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list element statements have total bit lengths of `2` each because the mux is specified with a maximum bit length of `0`.

```
- chain4_clock3
    + PARTITION clock3
    + START block1/prescaler_IO/lfsr_reg1
    + FLOATING
       block1/dp1_timers
       block1/bus8 ( BITS 8 )
       ...
    + ORDERED
       block1/ds1/ff1 ( IN SD ) ( OUT Q )
       block1/ds1/mux1 ( IN B ) ( OUT Y ) ( BITS 0 )
       block1/ds1/ff2 ( IN SD ) ( OUT Q )
       ...
```

`START {`*`fixedInComp`*` | PIN} [`*`outPin`*`]`

Specifies the start point of the scan chain. You must specify this point. The starting point can be either a component, *`fixedInComp`*, or an I/O pin, `PIN`. If you do not specify *`outPin`*, the router uses the pin specified for common scan pins.

`STOP {`*`fixedOutComp`*` | PIN} [`*`inPin`*`]`

Specifies the endpoint of the scan chain. You must specify this point. The stop point can be either a component, *`fixedOutComp`*, or an I/O pin, `PIN`. If you do not specify *`inPin`*, the router uses the pin specified for common scan pins.

Scan Chain Rules

Note the following when defining scan chains.

- Each scan-in/scan-out pin pair of adjacent components in the ordered list cannot have different owning nets.
- No net can connect a scan-out pin of one component to the scan-in pin of a component in a different scan chain.
- For incremental DEF, if you have a `COMPONENTS` section and a `SCANCHAINS` section in the same DEF file, the `COMPONENTS` section must appear before the `SCANCHAINS` section. If the `COMPONENTS` section and `SCANCHAINS` section are in different DEF files, you must read the `COMPONENTS` section or load the database before reading the `SCANCHAINS` section.

Example 8-27 Scan Chain Statements

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```
Nets 100; #Number of nets resulting after scan chain synthesis
    - SCAN-1 ( C1 SO + SYNTHESIZED )
             ( C4 SI + SYNTHESIZED ) + SOURCE TEST ;
    - ...
    - N1 ( C3 SO + SYNTHESIZED )
        ( C11 SI + SYNTHESIZED ) ( AND1 A ) ;
    - ...
END NETS
SCANCHAINS 2; #Specified before scan chain ordering
    - S1
        + COMMONSCANPINS ( IN SI ) ( OUT SO )
        + START SIPAD OUT
        + FLOATING C1 C2 ( IN D ) ( OUT Q ) C3 C4 C5...CN
        + ORDERED A1 ( OUT Q ) A2 ( IN D ) ( OUT Q ) ...
              AM ( N D )
        + ORDERED B1 B2 ... BL
        + STOP SOPAD IN ;
    - S2 ... ;
END SCANCHAINS
SCANCHAINS 2 ; #Specified after scan chain ordering
    - S1
        + START SIPAD OUT
        + FLOATING C1 ( IN SI ) ( OUT SO )
          C2 ( IN D ) ( OUT Q )
          C3 ( IN SI ( OUT SO ) ... CN ( IN SI ) ( OUT SO )
        + ORDERED A1 ( IN SI ) ( OUT Q )
          A2 ( IN D ) ( OUT Q ) ... AM ( IN D ) ( OUT SO )
        + ORDERED B1 ( IN SI ) ( OUT SO )
          B2 ( IN SI ) ( OUT SO ) ...
        + STOP SOPAD IN ;
    - S2 ... ;
END SCANCHAINS
```

Slots

```
[SLOTS numSlots ;
     [- LAYER layerName
         {RECT pt pt | POLYGON pt pt pt ... } ...
     ;] ...
END SLOTS]
```


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Defines the rectangular shapes that form the slotting of the wires in the design. Each slot is defined as an individual rectangle.

`LAYER` *`layerName`*

Specifies the layer on which to create slots.

`numSlots`

Specifies the number of `LAYER` statements in the `SLOTS` statement, *not* the number of rectangles.

`POLYGON` *`pt pt pt`*

Specifies a sequence of at least three points to generate a polygon geometry. Most fab DRC rules require each polygon edge to be parallel to the x or y axis, or at a 45-degree angle. However, odd-angles are allowed sometimes on a few layers, such as bump layers. Each `POLYGON` statement defines a polygon generated by connecting each successive point, and then by connecting the first and last points.
The *`pt`* syntax corresponds to a coordinate pair, such as *`x y`*. Specify an asterisk (`*`) to repeat the same value as the previous *`x`* or *`y`* value from the last point.
Type: DEF database units

`RECT` *`pt pt`*

Specifies the lower left and upper right corner coordinates of the slot geometry.

Example 8-28 Slots Statements

The following statement defines slots for layers `MET1` and `MET2`.

```
SLOTS 2 ;
    - LAYER MET1
        RECT ( 1000 2000 ) ( 1500 4000 )
        RECT ( 2000 2000 ) ( 2500 4000 )
        RECT ( 3000 2000 ) ( 3500 4000 ) ;
    - LAYER MET2
        RECT ( 1000 2000 ) ( 1500 4000 )
        RECT ( 1000 4500 ) ( 1500 6500 )
        RECT ( 1000 7000 ) ( 1500 9000 )
        RECT ( 1000 9500 ) ( 1500 11500 ) ;
END SLOTS
```

The following `SLOTS` statement defines two rectangles and one polygon slot geometries:

```
SLOTS 1 ;
    - LAYER metal1
        RECT ( 100 200 ) ( 150 400 )
```


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```
        POLYGON ( 100 100 ) ( 200 200 ) ( 300 200 ) ( 300 100 )
        RECT ( 300 200 ) ( 350 400 ) ;
END SLOTS
```

Special Nets

```
[SPECIALNETS numNets ;
     [– netName
         [ ( {compName pinName | PIN pinName} [+ SYNTHESIZED] ) ] ...
       [+ VOLTAGE volts]
       [specialWiring] ...
       [+ SOURCE {DIST | NETLIST | TIMING | USER}]
       [+ FIXEDBUMP]
       [+ ORIGINAL netName]
       [+ USE {ANALOG | CLOCK | GROUND | POWER | RESET | SCAN | SIGNAL | TIEOFF}]
       [+ PATTERN {BALANCED | STEINER | TRUNK | WIREDLOGIC}]
       [+ ESTCAP wireCapacitance]
       [+ WEIGHT weight]
       [+ PROPERTY {propName propVal} ...] ...
     ;] ...
END SPECIALNETS]
```

Defines netlist connectivity and special-routes for nets containing special pins. Special-routes are created by "special routers" or "manually", and should not be modified by a signal router. Special routes are normally used for power-routing, fixed clock-mesh routing, high-speed buses, critical analog routes, or flip-chip routing on the top-metal layer to bumps.

Input parameters for a net can appear in the `NETS` section or the `SPECIALNETS` section. In case of conflicting values for an argument, the DEF reader uses the last value encountered for the argument. `NETS` and `SPECIALNETS` statements can appear more than once in a DEF file. If a particular net has mixed wiring or pins, specify the special wiring and pins first.

You can also specify the netlist in the `COMPONENTS` statement. If the netlist is specified in both `NETS` and `COMPONENTS` statements, and if the specifications are not consistent, an error message appears. On output, the writer outputs the netlist in either format, depending on the command arguments of the output command.

compNamePattern pinName

Specifies the name of a special pin on the net and its corresponding component. You can use a *compNamePattern* to specify a set of component names. During evaluation of the pattern match, components that match the pattern but do not have a pin named *pinName* are ignored. The pattern match character is `*` (asterisk). For example, a component name of `abc/def` would be matched by `a*`, `abc/d*`, or `abc/def`.

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`ESTCAP` *`wireCapacitance`*

Specifies the estimated wire capacitance for the net. `ESTCAP` can be loaded with simulation data to generate net constraints for timing-driven layout.

`FIXEDBUMP`

Indicates that the bump net cannot be reassigned to a different pin.

It is legal to have a pin without geometry to indicate a logical connection and to have a net that connects that pin to two other instance pins that have geometry. Area I/Os have a logical pin that is connected to a bump and an input driver cell. The bump and driver cell have pin geometries (and, therefore, should be routed and extracted), but the logical pin is the external pin name without geometry (typically the Verilog pin name for the chip).

Bump nets also can be specified in the `NETS` statement. If a net name appears in both the `NETS` and `SPECIALNETS` statements, the `FIXEDBUMP` keyword also should appear in both statements. However, the value only exists once within a given application's database for the net name.

Because DEF is often used incrementally, the last value read in is used. Therefore, in a typical DEF file, if the same net appears in both statements, the `FIXEDBUMP` keyword (or lack of it) in the `NETS` statement is the value that is used because the `NETS` statement is defined after the `SPECIALNETS` statement.

Example 8-29 Fixed Bump

The following example describes a logical pin that is connected to a bump and an input driver cell. The I/O driver cell and bump cells are specified in the `COMPONENTS` statement. Bump cells are usually placed with + `COVER` placement status so they cannot be moved manually by mistake.

```
COMPONENTS 200
    - driver1 drivercell + PLACED ( 100 100 ) N ;
    ...
    - bumpa1 bumpcell + COVER ( 100 100 ) N ;
    - bumpa2 bumpcell + COVER ( 200 100 ) N ;
```

The pin is assigned in the `PIN` statement.

```
PINS 100
    - n1 + NET n1 + SPECIAL + DIRECTION INPUT ;
    - n2 + NET n2 + SPECIAL + DIRECTION INPUT ;
```

In the `SPECIALNETS` statement, the net `n1` is assigned to `bumpa1` and cannot be reassigned. Note that another net `n2` is assigned to `bumpa2`; however, I/O optimization commands are allowed to reassign `bumpa2` to a different net.

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```
SPECIALNETS 100
    - n1 ( driver1 in ) ( bumpa1 bumpin ) + FIXEDBUMP ;
    - n2 ( driver2 in ) ( bumpa2 bumpin ) ;
```

netName

Specifies the name of the net.

ORIGINAL *netName*

Specifies the original net partitioned to create multiple nets, including the current net.

PATTERN {BALANCED | STEINER | TRUNK | WIREDLOGIC}

Specifies the routing pattern used for the net.
Default: STEINER
Value: Specify one of the following:

BALANCED	Used to minimize skews in timing delays for clock nets.
STEINER	Used to minimize net length.
TRUNK	Used to minimize delay for global nets.
WIREDLOGIC	Used in ECL designs to connect output and mustjoin pins before routing to the remaining pins.

PIN *pinName*

Specifies the name of an I/O pin on a net or a subnet.

PROPERTY *propName propVal*

Specifies a numerical or string value for a net property defined in the PROPERTYDEFINITIONS statement. The *propName* you specify must match the *propName* listed in the PROPERTYDEFINITIONS statement.

specialWiring

Specifies the special wiring for the net. For syntax information, see "Special Wiring Statement" on page 1061.

SOURCE {DIST | NETLIST | TIMING | USER}

Specifies how the net is created. The value of this field is preserved when input to the DEF reader.

DIST	Net is the result of adding physical components (that is, components that only connect to power or ground nets), such as filler cells, well-taps, tie-high and tie-low cells, and decoupling caps.

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`NETLIST`	Net is defined in the original netlist. This is the default value, and is not normally written out in the DEF file.
`TEST`	Net is part of a scanchain.
`TIMING`	Net represents a logical rather than physical change to netlist, and is used typically as a buffer for a clock-tree, or to improve timing on long nets.
`USER`	Net is user defined.

`SYNTHESIZED`

Used by some tools to indicate that the pin is part of a synthesized scan chain.

`USE {ANALOG | CLOCK | GROUND | POWER | RESET | SCAN | SIGNAL | TIEOFF}`

Specifies how the net is used.
Default: `SIGNAL`
Value: Specify one of the following:

`ANALOG`	Used as an analog signal net.
`CLOCK`	Used as a clock net.
`GROUND`	Used as a ground net.
`POWER`	Used as a power net.
`RESET`	Used as a reset net.
`SCAN`	Used as a scan net.
`SIGNAL`	Used as a digital signal net.
`TIEOFF`	Used as a tie-high or tie-low net.

`VOLTAGE` *`volts`*

Specifies the voltage for the net, as an integer in units of .001 volts. For example, `VOLTAGE 1500` in DEF is equal to 1.5 V.

`WEIGHT` *`weight`*

Specifies the weight of the net. Automatic layout tools attempt to shorten the lengths of nets with high weights. Do not specify a net weight larger than 10, or assign weights to more than 3 percent of the nets in a design.

Note: The net constraints method of controlling net length is preferred over using net weights.

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Special Wiring Statement

```
[[+ COVER | + FIXED | + ROUTED | + SHIELD shieldNetName]
     [+ SHAPE shapeType] [+ MASK maskNum]
      + POLYGON layerName pt pt pt ...
     | + RECT layerName pt pt
     | + VIA viaName [orient] pt ...
|{+ COVER | + FIXED | + ROUTED | + SHIELD shieldNetName}
     layerName  routeWidth
          [+ SHAPE
               {RING | PADRING | BLOCKRING | STRIPE | FOLLOWPIN
               | IOWIRE | COREWIRE | BLOCKWIRE | BLOCKAGEWIRE | FILLWIRE
               | FILLWIREOPC | DRCFILL}]
           [+ STYLE styleNum]
           routingPoints
     [NEW layerName routeWidth
           [+ SHAPE
               {RING | PADRING | BLOCKRING | STRIPE | FOLLOWPIN
               | IOWIRE | COREWIRE | BLOCKWIRE | BLOCKAGEWIRE | FILLWIRE
               | FILLWIREOPC | DRCFILL}]
           [+ STYLE styleNum]
           routingPoints
        ] ...
] ...
```

Defines the wiring for both routed and shielded nets.

`COVER`

Specifies that the wiring cannot be moved by either automatic layout or interactive commands. If no wiring is specified for a particular net, the net is unrouted. If you specify `COVER`, you must also specify *`layerName width`*.

`FIXED`

Specifies that the wiring cannot be moved by automatic layout, but can be changed by interactive commands. If no wiring is specified for a particular net, the net is unrouted. If you specify `FIXED`, you must also specify *`layerName width`*.

`layerName routeWidth`

Specifies the width for wires on *`layerName`*. Normally, only routing layers use this syntax, but it is legal for any layer (cut layer shapes or other layers like masterslice layers normally use the `RECT` or `POLYGON` statements). For more information, see "Defining Routing Points" on page 1070.

Vias do not change the route width. When a via is used in special wiring, the previously established *`routeWidth`* is used for the next wire in the new layer. To change the *`routeWidth`*, a new path must be specified using `NEW` *`layerName routeWidth`*.

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Many applications require *routeWidth* to be an even multiple of the manufacturing grid in order to be fabricated, and to keep the center line on the manufacturing grid.
Type: Integer, specified in database units

`NEW` *layerName routewidth*

Indicates a new wire segment (that is, that there is no wire segment between the last specified coordinate and the next coordinate) on *layerName*, and specifies the width for the wire. Noncontinuous paths can be defined in this manner. For more information, see "Defining Routing Points" on page 1070.
Type: Integer, specified in database units

`POLYGON` *layerName pt pt pt*

Specifies a sequence of at least three points to generate a polygon geometry on *layerName*. The polygon edges must be parallel to the x axis, the y axis, or at a 45-degree angle. Each polygon statement defines a polygon generated by connecting each successive point, then connecting the first and last points. The *pt* syntax corresponds to a coordinate pair, such as *x y*. Specify an asterisk (`*`) to repeat the same value as the previous *x* or *y* value from the last point.
Type: (*x y*) Integer, specified in database units

`RECT` *layerName pt pt*

Specifies a rectangle on layer *layerName*. The two points define opposite corners of the rectangle. The *pt* syntax corresponds to a coordinate pair, such as *x y*. You cannot define the same *x* and *y* values for both points (that is, a zero-area rectangle is not legal).
Type: (*x y*) Integer, specified in database units

`ROUTED`

Specifies that the wiring can be moved by automatic layout tools. If no wiring is specified for a particular net, the net is unrouted. If you specify `ROUTED`, you must also specify *layerName width*.

routingPoints

Defines the center line coordinates of the route on *layerName*. For information on using routing points, see "Defining Routing Points" on page 1070. For an example of special wiring with routing points, see Example 8-31 on page 1065.

The *routingPoints* syntax is defined as follows:

```
( x y [extValue])
  { [MASK maskNum] ( x y [extValue])
  | [MASK viaMaskNum] viaName [orient]
```


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```
    [DO numX BY numY STEP stepX stepY]
 } ...
```

DO *numx* BY *numY* STEP *stepX stepY*

Creates an array of power vias of the via specified with *viaName*.

numX and *numY* specify the number of vias to create, in the x and y directions. Do not specify 0 as a value.
Type: Integer

stepX and *stepY* specify the step distance between vias, in the x and y directions, in DEF distance database units.
Type: Integer

For an example of a via array, see Example 8-30 on page 1065.

extValue
Specifies the amount by which the wire is extended past the endpoint of the segment.
Type: Integer, specified in database units
Default: 0

MASK *maskNum*

Specifies which mask for double or triple patterning lithography to use for the next wire. The *maskNum* variable must be a positive integer. Most applications support values of 1, 2, or 3 only. Shapes without any defined mask have no mask set (that is, they are uncolored).

MASK *viaMaskNum*

Specifies which mask for double or triple patterning lithography is to be applied to the next via's shapes on each layer.

The *viaMaskNum* variable is a hex-encoded three digit value of the form:

<*topMaskNum*><*cutMaskNum*><*bottomMaskNum*>

For example, MASK 113 means the top metal and cut layer *maskNum* is 1, and the bottom metal layer *maskNum* is 3. A value of 0 means the shape on that layer has no mask assignment (is uncolored), so 013 means the top layer is uncolored. If either the first or second digit is missing, they are assumed to be 0, so 013 and 13 mean the same thing. Most applications support *maskNum* values of 0, 1, 2, or 3 only for double or triple patterning.

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The *topMaskNum* and *bottomMaskNum* variables specify the masks to which the corresponding metal shapes belong. In the case of a metal layer with a single shape, the via-master metal mask values have no effect.

For the cut-layer, the *cutMaskNum* variable will define the mask for the bottom-most, and then the left-most cut. For multi-cut vias, the via-instance cut masks are derived from the via-master cut mask values. The via-master must have a mask defined for all the cut shapes and every via-master cut mask is "shifted" (from 1 to 2, and 2 to 1 for two mask layers, and from 1 to 2, 2 to 3, and 3 to 1 for three mask layers), so the lower-left cut matches the *cutMaskNum* value.

Similarly, for a metal layer with multiple shapes, the *topMaskNum*/ *bottomMaskNum* would define the mask for the bottom-most, then leftmost metal shape. For multiple disjoint metal shapes, the via-instance metal masks are derived from the via-master metal mask values. The via-master must have a mask defined for all of the metal shapes, and every via-master metal mask is "shifted" (1->2, 2->1 for two mask layers, 1->2, 2->3, 3->1 for three mask layers) so the lower-left cut matches the *topMaskNum*/ *bottomMaskNum* value.

See Example 8-32 on page 1065.

orient

Specifies the orientation of the *viaName* that precedes it, using the standard DEF orientation values of `N`, `S`, `E`, `W`, `FN`, `FS`, `FE`, and `FW` (See "Specifying Orientation" on page 994).

If you do not specify *orient*, `N` (North) is the default non-rotated value used. All other orientation values refer to the flipping or rotation around the via origin (the `0,0` point in the via shapes). The via origin is still placed at the `(`*x y*`)` value given in the routing statement just before the *viaName*.

Note: Some tools do not support orientation of vias inside their internal data structures; therefore, they are likely to translate vias with an orientation into a different but equivalent via that does not require an orientation.

viaName

Specifies a via to place at the last point. If you specify a via, *layerName* for the next routing coordinates (if any) is implicitly changed to the other routing layer for the via. For example, if the current layer is *metal1*, a *via12* changes the layer to *metal2* for the next routing coordinates.

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`( x y )` Specifies the route coordinates. You cannot specify a route with zero length.

For more information, see "Specifying Coordinates" on page 1071.
Type: Integer, specified in database units

Example 8-30 Via Arrays

The following example specifies arrays of via `VIAGEN21_2` on *metal1* and *metal2*.

```
SPECIALNETS 2 ;
    -vdd ( * vdd )
       + ROUTED metal1 150 ( 100 100 ) ( 200 * )
       NEW metal1 0 ( 200 100 ) VIAGEN21_2 DO 10 BY 20 STEP 10000 20000
       NEW metal2 0 (-900 -30 ) VIAGEN21_2 DO 1000 BY 1 STEP 5000 0
       ...
```

As with any other `VIA` statement, the `DO` statement does not change the previous coordinate. Therefore, the following statement creates a *metal1* wire of width 50 from (200 100) to (200 200) along with the via array that starts at (200 100).

```
    NEW metal1 50 ( 200 100 ) VIAGEN21_2 DO 10 BY 20 STEP 1000 2000 ( 200 200 )
```

Example 8-31 Special Wiring With Routing Points

```
SPECIALNETS 1 ;
    - vdd ( * vdd )
        + USE POWER
        + POLYGON metal1 ( 0 0 ) ( 0 100 ) ( 100 100 ) ( 200 200 ) ( 200 0 )
        + POLYGON metal2 ( 100 100 ) ( * 200 ) ( 200 * ) ( 300 300 ) ( 300 100 )
        + RECT metal1 ( 0 0 ) ( 100 200 )
        + ROUTED metal1 100 ( 0 0 50 ) ( 100 0 50 ) via12 ( 100 100 50 )
        + ROUTED metal2 100 + SHAPE RING + STYLE 1 ( 0 0 ) ( 100 100 ) ( 200 100 )
        ;
END SPECIALNETS
```

Example 8-32 Multi-Mask Layers with Special Wiring

The following example shows a routing statement that specifies three-mask layers `M1` and `VIA1`, and a two-mask layer `M2`:

```
+ FIXED + SHAPE RING + MASK 2 + RECT M3 ( 0 0 ) ( 10 10 )
+ ROUTED M1 2000 (10 0 ) MASK 3 (10 20 ) VIA1_1
    NEW M2 1000 ( 10 10 ) (20 10) MASK 1 ( 20 20 ) MASK 031 VIA1_2
+ SHAPE STRIPE + VIA VIA3_3 ( 30 30 ) ( 40 40 )
```


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`;`

This indicates that the:

- `M3` rectangle shape is on mask 2, has `FIXED` route status, and shape `RING`
- `M1` wire shape from (10 0) to (10 20) is on mask 3.
- `VIA1_1` via has no preceding `MASK` statement so all the metal and cut shapes have no mask and are uncolored
- first `NEW M2` wire shape (10 10) to (20 10) has no mask set and is uncolored
- second `M2` wire shape (20 10) to (20 20) is on mask 1
- `VIA1_2` via has a MASK 031 (it can be MASK 31 also) so:
  - *`topMaskNum`* is 0 in the 031 value, so no mask is set for the top metal (`M2`) shape
  - *`bottomMaskNum`* is 1 in the 031 value, so mask 1 is used for the bottom metal (`M1`) shape
  - *`cutMaskNum`* is 3 in the 031 value, so the bottom-most, then left-most cut of the via-instance is mask 3. The mask for the other cuts of the via-instance are derived from the via-master by "shifting" the via-master's cut masks to match. So, if the via-master's bottom-left cut is mask 1, then the via-master cuts on mask 1 become mask 3 for the via-instance, and similarly cuts on 2 shift to 1, and cuts on 3 shift to 2. See Figure 8-11 on page 1067.

The `VIA3_3` has shape `STRIPE`, and the via is at both (30 30) and (40 40). There is no wire segment between (30 30) and (40 40). If the route status is not specified, it is considered as `+ ROUTED`.

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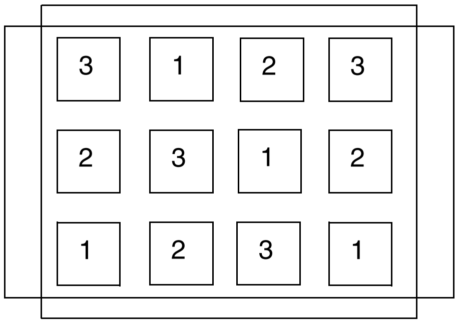
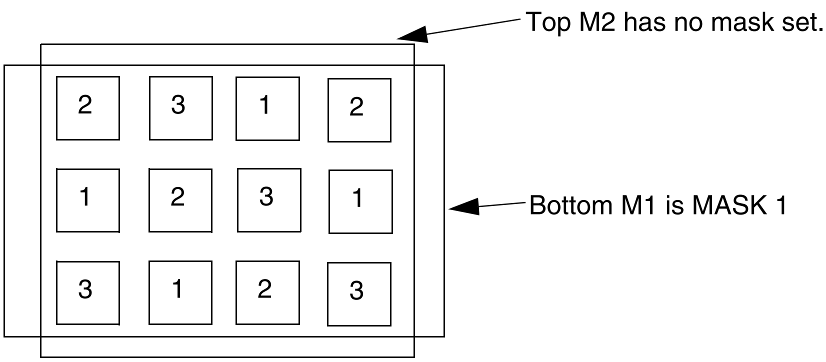
Figure 8-11 Multi-Mask Patterns with Special Wiring



Via-master cut masks for VIA1.



Masks for via-instance: ... (20 20) MASK 031 VIA1_2
Bottom M1 is MASK 1.Top M2 has no mask set. Lower-left cut is MASK 3, and other via-instance cut shape masks are derived from via-master cut-masks as shown above by "shifting" the via-master masks to match: 1->3, 2->1, 3->2.

`SHAPE`

Specifies a wire with special connection requirements because of its shape. This applies to vias as well as wires.
Value: Specify one of the following:

`RING`	Used as ring, target for connection
`PADRING`	Connects padrings
`BLOCKRING`	Connects rings around the blocks
`STRIPE`	Used as stripe
`FOLLOWPIN`	Connects standard cells to power structures.

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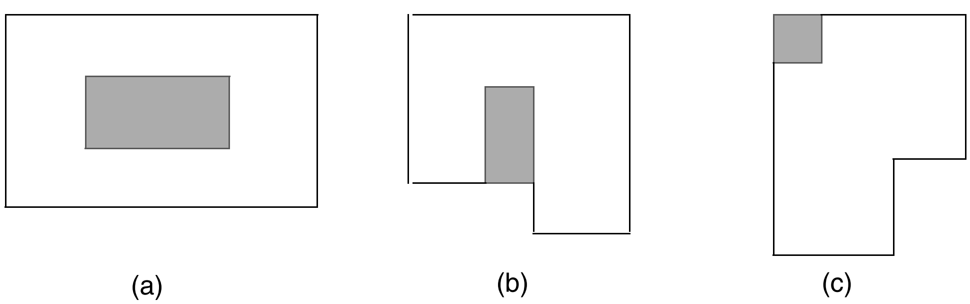
`IOWIRE`	Connects I/O to target
`COREWIRE`	Connects endpoints of followpin to target
`BLOCKWIRE`	Connects block pin to target
`BLOCKAGEWIRE`	Connects blockages
`FILLWIRE`	Represents a fill shape that does not require OPC. It is normally connected to a power or ground net. Floating fill shapes should be in the `FILL` section.
`FILLWIREOPC`	Represents a fill shape that requires OPC. It is normally connected to a power or ground net. Floating fill shapes should be in the `FILL` section.
`DRCFILL`	Used as a fill shape to correct DRC errors, such as `SPACING`, `MINENCLOSEDAREA`, or `MINSTEP` violations on wires and pins of the same net (see Figure 8-12 on page 1068.)

Figure 8-12 Fill Shapes



Examples of fill inside (a) a `MINENCLOSEDAREA` violation, (b) a `SPACING` violation, and (c) a `MINSTEP` violation.

`SHIELD` *`shieldNetName`*

Specifies the name of a regular net to be shielded by the special net being defined.

After describing shielded routing for a net, use `+ ROUTED` to return to the routing of the special net being defined.

`STYLE` *`styleNum`*

Specifies a previously defined style from the `STYLES` section in this DEF file. The style is used with the endpoints of each routing segment to define the routing shape, and applies to all routing segments defined in one *`routingPoints`* statement.

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`VIA` *`viaName`* [*`orient`*] *`pt`* **...**

Specifies the name of the via placed at every point in the list with an optional orientation. For example, the statement
`VIA myVia` (`0 0`) (`1 1`) indicates an instance of `myVia` at 0,0 and at 1,1.

`...`

Example 8-33 Special Nets Statements

Signoff DRC tools may require metal shapes under the trim metal shapes to fill up the gaps between the line-end of wires sandwiched by the trim metal shape to be output in DEF. Those metal shapes would be written out in `_TRIMMETAL_FILLS_RESERVED` with the `DRCFILL` tag in the `SPECIALNETS` section in the following format:

```
- _TRIMMETAL_FILLS_RESERVED
      + ROUTED + SHAPE DRCFILL + MASK x + RECT M1 (x x) (x x)
```

As trim metal shapes need to be aligned and merged, dummy patches are often added even on OBS and unconnected pins. Those patches would be written out in `_SADP_FILLS_RESERVED` with the `DRCFILL` tag in the `SPECIALNETS` section in the following format:

```
- _SADP_FILLS_RESERVED
      + ROUTED + SHAPE DRCFILL + MASK x + RECT M1 (x x) (x x)
```

The following `SPECIALNETS` statement defines two metal shapes under the trim metal shapes and a dummy patch written out with the `DRCFILL` tag:

```
SPECIALNETS 2 ;
- _TRIMMETAL_FILLS_RESERVED
    + ROUTED + SHAPE DRCFILL + MASK 1 + RECT M1 (30 32) (33 34)
    + ROUTED + SHAPE DRCFILL + MASK 2 + RECT M1 (36 27) (39 29)
- _SADP_FILLS_RESERVED
    + ROUTED + SHAPE DRCFILL + MASK 2 + RECT M1 (30 27) (36 29)
END SPECIALNETS
```


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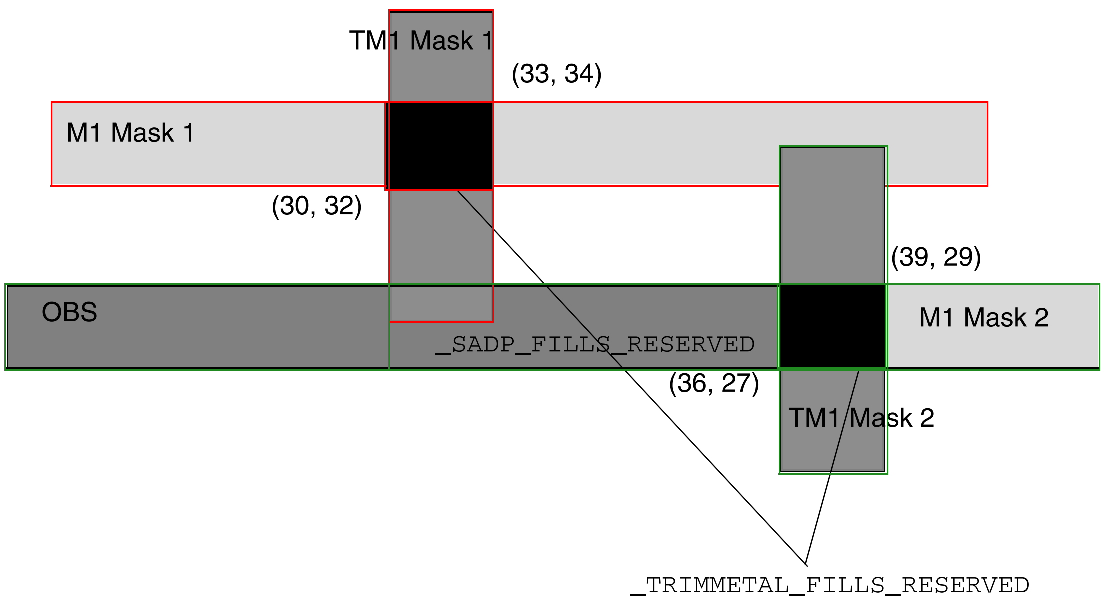
Figure 8-13 Trim Metal and SADP Fills in the SPECIALNETS Section



Defining Routing Points

Routing points define the center line coordinates of a route. If a route has a 90-degree edge, it has a width of *`routeWidth`*, and extends from one coordinate (*`x y`*) to the next coordinate.

If either endpoint has an optional extension value (*`extValue`*), the wire is extended by that amount past the endpoint. If a coordinate with an extension value is specified after a via, the wire extension is added to the beginning of the next wire segment after the via (zero-length wires are not allowed). Some applications convert the extension value to an equivalent route that has the *`x`* and *`y`* points already extended, with no extension value. If no extension value is defined, the wire extension is 0, and the wire is truncated at the endpoint.

The *`routeWidth`* must be an even value to ensure that the corners of the route fall on a legal database coordinate without round off. Because most vendors specify a manufacturing grid, *`routeWidth`* must be an even multiple of the manufacturing grid in order to be fabricated.

If the wire segment is a 45-degree edge, and no `STYLE` is specified, the default octagon style is used for the endpoints. The *`routeWidth`* must be an even multiple of the manufacturing grid in order to keep all of the coordinates of the resulting outer wire boundary on the manufacturing grid.

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If a `STYLE` is defined for 90-degree or 45-degree routes, the routing shape is defined by the center line coordinates and the style. No corrections, such as snapping to manufacturing grid, should be applied, and any extension values are ignored. The DEF file should contain values that are already snapped, if appropriate. The *`routeWidth`* indicates the desired user width, and represents the minimum allowed width of the wire that results from the style when the 45-degree edges are snapped to the manufacturing grid. See Figure 8-15 on page 1075 through Figure 8-24 on page 1086 for examples.

Specifying Coordinates

To maximize compactness of the design files, the coordinates allow for the asterisk (*) convention. For example, (*`x`* `*`) indicates that the y coordinate last specified in the wiring specification is used; (`*` *`y`*) indicates that the x coordinate last specified is used.

Each coordinate sequence defines a connected orthogonal or 45-degree path through the points. The first coordinate in a sequence must not have an * element.

All subsequent points in a connected sequence must create orthogonal or 45-degree paths. For example, the following sequence is a valid path:

```
( 100 200 ) ( 200 200 ) ( 200 500 )
```

The following sequence is an equivalent path:

```
( 100 200 ) ( 200 * ) ( * 500 )
```

The following sequence is not valid because it is not an orthogonal or 45-degree segment.

```
( 100 200 ) ( 300 500 )
```

Special Pins and Wiring

Pins that appear in the `SPECIALNETS` statement are special pins. Regular routers do not route to these pins. The special router routes special wiring to special pins. If you use a component-based format to input the connectivity for the design, special pins to be routed by the special router also must be specified in the `SPECIALNETS` statement, because pins included in the `COMPONENTS` statement are considered regular.

The following example inputs connectivity in a component-based format, specifies `VDD` and `VSS` pins as special pins, and marks `VDD` and `VSS` nets for special routing:

```
COMPONENTS 3 ;
  C1 AND N1 N2 N3 ;
  C2 AND N4 N5 N6 ;
END COMPONENTS
```


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```
SPECIALNETS 2 ;
  VDD ( * VDD ) + WIDTH M1 5 ;
  VSS ( * VSS ) ;
END SPECIALNETS
```

Shielded Routing

If, in a non-routed design, a net has `+ SHIELDNET` attributes, the router adds shielded routing to this net. `+ NOSHIELD` indicates the last wide segment of the net is not shielded. If the last segment is not shielded and is tapered, use the `+ TAPER` keyword instead of `+ NOSHIELD`. For example:

```
+ SHIELDNET VSS      # both sides will be shielded with VSS
+ SHIELDNET VDD      # one side will be shielded with VDD and
+ SHIELDNET VSS     # one side will be shielded with VSS
```

After you add shielded routing to a special net, it has the following syntax:

```
+ SHIELD regularNetName
  MET2 regularWidth ( x y )
```

A shield net specified for a regular net should be defined earlier in the DEF file in the `SPECIALNETS` section. After describing shielded routing for a net, use `+ ROUTED` to return to the routing of the current special net.

For example:

```
SPECIALNETS 2 ;
    - VSS
       + ROUTED MET2 200
    ...
    + SHIELD my_net MET2 100 ( 14100 342440 ) ( 13920 * )
         M2_TURN ( * 263200 ) M1M2 ( 2400 * ) ;
    - VDD
       + ROUTED MET2 200
    ...
    + SHIELD my_net MET2 100 ( 14100 340440 ) ( 8160 * )
         M2_TURN ( * 301600 ) M1M2 ( 2400 * );
END SPECIALNETS
```

Styles

```
[STYLES numStyles ;
     {- STYLE styleNum pt pt ... ;} ...
```


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```
END STYLES]
```

Defines a convex polygon that is used at each of the endpoints of a wire to precisely define the wire's outer boundary. A style polygon consists of two to eight points. Informally, half of the style polygon defines the first endpoint wire boundary, and the other half of the style polygon defines the second endpoint wire boundary. Octagons and squares are the most common styles.

numStyles

Specifies the number of styles specified in the `STYLES` section.

`STYLE` *styleNum pt pt*

Defines a new style. *styleNum* is an integer that is greater than or equal to 0 (zero), and is used to reference the style later in the DEF file. When defining multiple styles, the first *styleNum* must be 0 (zero), and any following *styleNum* should be numbered consecutively so that a table lookup can be used to find them easily.

Style numbers are keys used locally in the DEF file to reference a particular style, but not actual numbers preserved in the application. Each style number must be unique. Style numbers can only be used inside the same DEF file, and are not preserved for use in other DEF files. Because applications are not required to preserve the style number itself, an application that writes out an equivalent DEF file might use different style numbers.
Type: Integer

The *pt* syntax specifies a sequence of at least two points to generate a polygon geometry. The syntax corresponds to a coordinate pair, such as *x y*. Specify an asterisk (`*`) to repeat the same value as the previous *x* (or *y*) value from the last point. The polygon must be convex. The polygon edges must be parallel to the x axis, the y axis, or at a 45-degree angle, and must enclose the point (`0 0`).
Type: Integer, specified in DEF database units

Example 8-34 Styles Statement

The following `STYLES` statement defines the basic octagon shown in Figure 8-14 on page 1074.

```
STYLES 1 ;
    - STYLE 1 ( 30 10 ) ( 10 30 ) ( -10 30 ) ( -30 10 ) ( -30 -10 )
                ( -10 -30 ) ( 10 -30 ) ( 30 -10 ) ;
END STYLES
```


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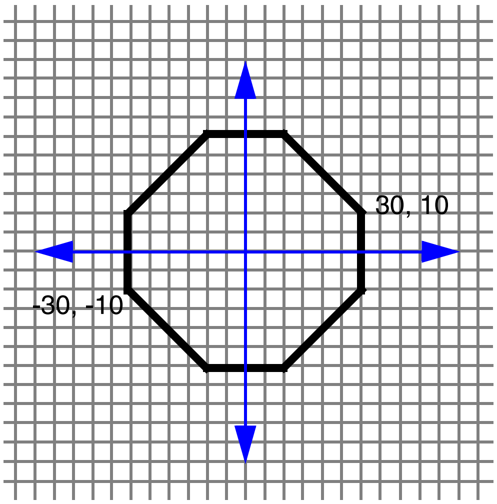
Figure 8-14



Defining Styles

A style is defined as a polygon with points P1 through P*n*. The center line is given as (X0, Y0) to (X1, Y1). Two sets of points are built (P0,1 through P0,*n* and P1,1 through P1,*n*) as follows:

```
P0,i = Pi + (X0, Y0) for 1 <= i <= n
P1,i = Pi + (X1, Y1) for 1 <= i <= n
```

The resulting wire segment shape is a counterclockwise, eight-sided polygon (`S1` through `S8`) that can be computed in the following way:

`S1` = lowest point in (left-most points in (P0,1 through P0,*n* P1,1 through P1,*n*))

`S2` = left-most point in (lowest points in (P0,1 through P0,*n* P1,1 through P1,*n*))

`S3` = right-most point in (lowest points in (P0,1 through P0,*n* P1,1 through P1,*n*))

`S4` = lowest point in (right-most points in (P0,1 through P0,*n* P1,1 through P1,*n*))

`S5` = highest point in (right-most points in (P0,1 through P0,*n* P1,1 through P1,*n*))

`S6` = right-most point in (highest points in (P0,1 through P0,*n* P1,1 through P1,*n*))

`S7` = left-most point in (highest points in (P0,1 through P0,*n* P1,1 through P1,*n*))

`S8` = highest point in (left-most points in (P0,1 through P0,*n* P1,1 through P1,*n*))

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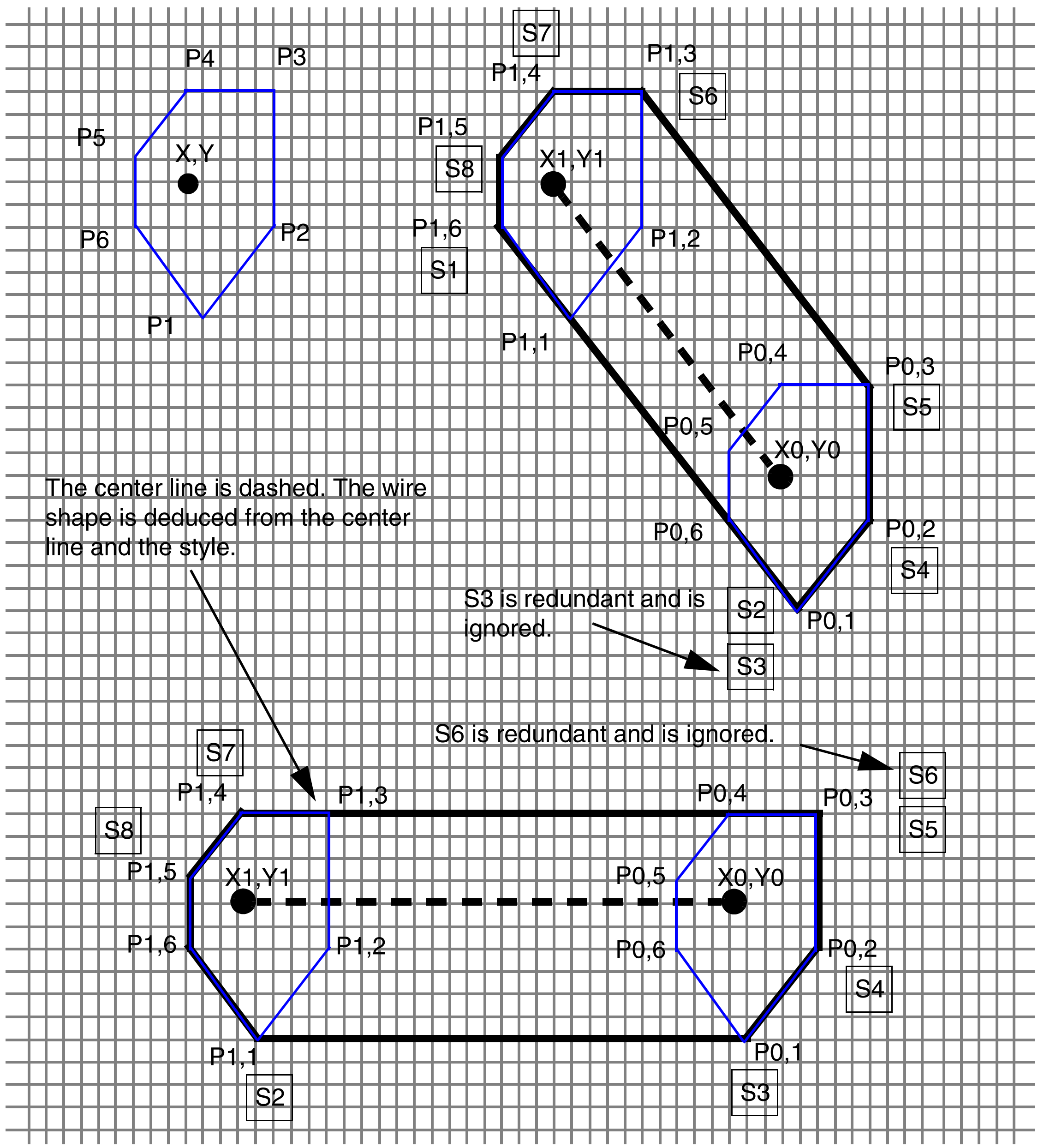
When consecutive points are collinear, only one of them is relevant, and the resulting shape has less than eight sides, as shown in Figure 8-15 on page 1075. A more advanced algorithm can order the points and only have to check a subset of the points, depending on which endpoint was used, and whether the wire was horizontal, vertical, a 45-degree route, or a 135-degree route.

Figure 8-15



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Examples of X Routing with Styles

The following examples illustrate the use of styles for X routing. In two cases, there are examples of `SPECIALNETS` syntax and `NETS` syntax that result in the same geometry.

Example 1

The following statements define an X wire with octagonal ends, as shown in Figure 8-16 on page 1077.

```
STYLES 1 ;
    - STYLE 0 ( 30 10 ) ( 10 30 ) ( -10 30 ) ( -30 10 ) ( -30 -10 ) ( -10 -30 )
         ( 10 -30 ) ( 30 -10 ) ;          #An octagon.
END STYLES

SPECIALNETS 1 ;
    - VSS ...
        + ROUTED metal3 50 + STYLE 0 ( 0 0 ) ( 150 150 ) ( 300 0 ) ( 400 0 ) ;
          #The style applies to all the segments until a NEW statement or ";"
          #at the end of the net.
END SPECIALNETS

NETS 1 ;
    - mySignal ...
        + ROUTED metal3 STYLE 0 ( 0 0 ) ( 150 150 ) ( 300 0 ) ( 400 0 ) ;
        #The style applies to all the segments in the ROUTED statement
END NETS
```


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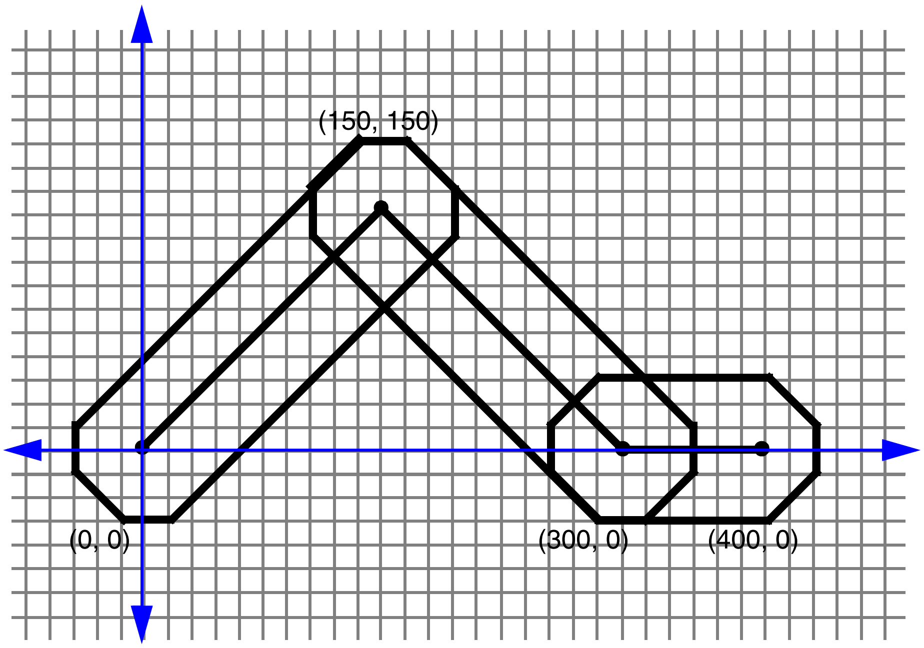
Figure 8-16



Example 2

The following statements define the same X wire with mixed octagonal and manhattan styles, as shown in Figure 8-17 on page 1078.

```
STYLES 2 ;
    - STYLE 0 ( 30 10 ) ( 10 30 ) ( -10 30 ) ( -30 10 ) ( -30 -10 ) ( -10 -30 )
        ( 10 -30 ) ( 30 -10 ) ;                              #An octagon
    - STYLE 1 ( 25 25 ) ( -25 25 ) ( -25 -25 ) ( 25 -25 ) ;  #A square
END STYLES

SPECIALNETS 1 ;
    - POWER (* power)
        + ROUTED metal3 50 + STYLE 0 ( 0 0 ) ( 150 150 )
        NEW metal3 50 + STYLE 1 ( 150 150 ) ( 300 0 ) ( 400 0 ) ;
END SPECIALNETS

NETS 1 ;
    - mySignal ...
        + ROUTED metal3 STYLE 0 ( 0 0 ) ( 150 150 )
        NEW metal3 STYLE 1 ( 150 150 ) ( 300 0 ) ( 400 0 ) ;
END NETS
```


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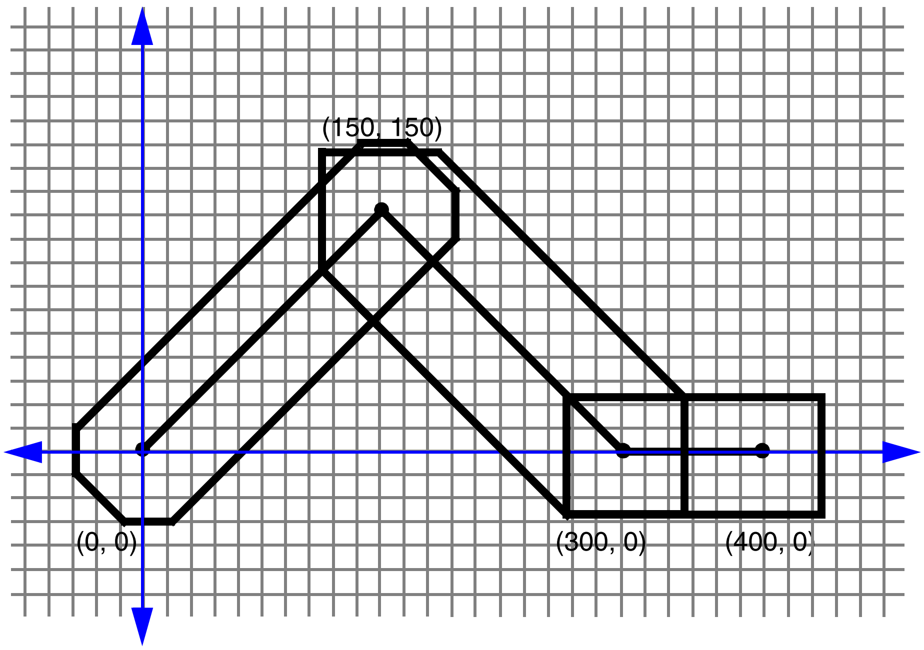
Figure 8-17



Note: The square ends might be necessary for connecting to manhattan wires or pins, or in cases where vias have a manhattan shape even on X routing layers. In practice, the middle wire probably would not use a simple square, such as `style2`; it would use a combination of an octagon and a square for the middle segment style, in order to smooth out the resulting outline at the (150,150) point.

Example 3

The following statements define a manhattan wire with a width of 70, as shown in Figure 8-18 on page 1079.

This example emphasizes that the style overrides the width of 100 units. In this case, the style polygon is a square 70 x 70 units wide, and the vias (`via12`) are 100 x 100 units wide. The application that creates the styles is responsible for meeting any particular width requirements. Normally, the resulting style-computed width is equal to or larger than the wire width given in the routing statement.

```
STYLES 1 ;
    - STYLE 0 ( 35 35 ) ( -35 35 ) ( -35 -35 ) ( 35 -35 ) ;
END STYLES

SPECIALNETS 1 ;
    - POWER ...
        + ROUTED metal1 100 + STYLE 0 ( 0 0 ) via12 ( 600 * ) via12 ;
```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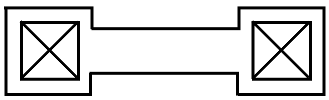
```
END SPECIALNETS
```

Figure 8-18



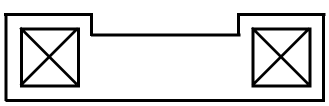
Example 4

The following statements define a similar wire that is offset from the center, as shown in Figure 8-19 on page 1079. Similar to Example 3, the center line in both runs through the middle of the X in the vias.

```
STYLES 1 ;
    - STYLE 0 ( 35 20 ) ( -35 20 ) ( -35 -50 ) ( 35 -50 ) ; #70 x 70 offset square
END STYLES

SPECIALNETS 1 ;
    - POWER ...
        + ROUTED metal1 100 + STYLE 0 ( 0 0 ) via12 ( 600 * ) via12 ;
END SPECIALNETS
```

Figure 8-19



Default Orthogonal Routing Style

In `NETS` for orthogonal segment routing, the half-width extension segment style is used.

In `SPECIALNETS` for orthogonal segments, the default style is a vertical line for horizontal segments and a horizontal line for vertical segments, resulting in a DEF path with zero extensions. The two `SPECIALNETS` examples below have the same routing geometry.

```
SPECIALNETS 1 ;
    - POWER ...
        + ROUTED metal1 20 ( 0 0 ) ( 100 0 ) ;
END SPECIALNETS
```

The following statements define a wire that uses a “2-point line” style, as shown in Figure 8-20 on page 1080.

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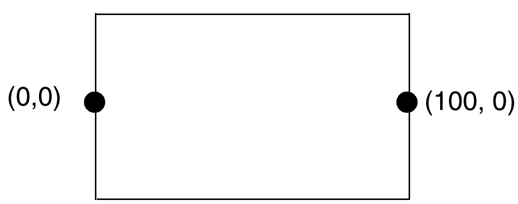
Note: This example shows the simplest style possible, which is a 2-point line. Generally, it would be easier to use a normal route without a style.

```
STYLES 1 ;
    - STYLE 0 ( 0 -10 ) ( 0 10 ) ; #a vertical line
END STYLES

SPECIALNETS 1 ;
    - POWER ...
        + ROUTED metal1 20 + STYLE 0 ( 0 0 ) ( 100 0 ) ;
END SPECIALNETS
```

Figure 8-20



45-Degree Routing Without Styles

As many applications store only the wire endpoints and the width of the wire, DEF includes a specific style default definition. If a style is not explicitly defined, the default style is implicitly included with any 45-degree routing segment. It is computed directly from the wire width and endpoints, at the expense of some loss in flexibility.

The default style is an octagon (shown in Figure 8-21 on page 1081) whose coordinates are computed from the wire width and the manufacturing grid.

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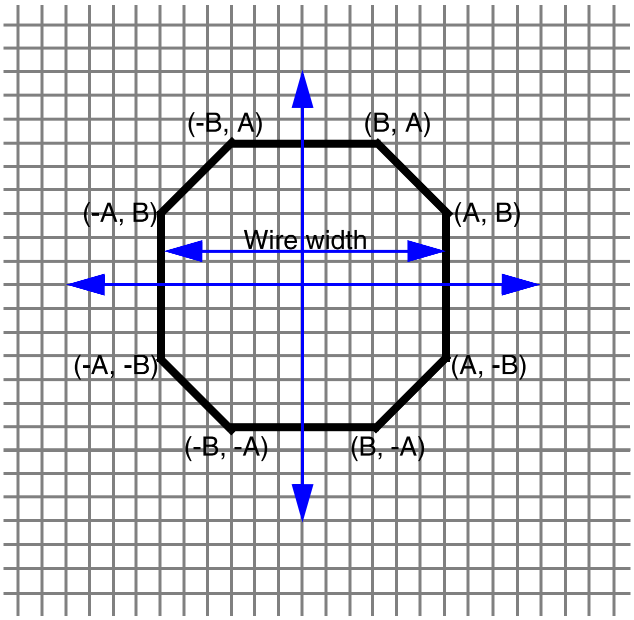
Figure 8-21



The octagon is always symmetric about the x and y axis. The coordinates are computed to be exactly the same wire width as equivalent horizontal or vertical wire widths, and as close as possible for the diagonal widths (they are always slightly bigger because of rounding of irrational values), while forcing the coordinates to remain on the manufacturing grid. The wire width must be an even multiple of the manufacturing grid in order to keep A and B on the manufacturing grid.

Assume the following rules:

- W = wire width
- M = manufacturing grid (mgrid). This is derived from the LEF `MANUFACTURINGGRID` statement.
- D = diagonal width
- ceiling = round up the result to the nearest integer

The octagon coordinates are computed as:

$$\mathrm{A} = W/2$$

$$\mathrm{B} = [\mathrm{ceiling}(W/(\mathrm{sqrt}(2) * M)) * M] - A$$

The derivation of B can be understood as:

$$\mathrm{D} = \mathrm{sqrt}((A + B)^2 + (A + B)^2) \text{ or } \mathrm{D} = \mathrm{sqrt}(2) * (A + B)$$

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The diagonal width (D) must be greater than or equal to the wire width (W), and B must be on the manufacturing grid, so D must be equal to W, which results in:

$D/\text{sqrt}(2) = A + B$

$B = D/\text{sqrt}(2) - A$ or $W/\text{sqrt}(2) - A$

To force B to be on the manufacturing grid, and keep the diagonal width greater than or equal to the wire width:

B on mgrid = $\text{ceiling}(B / M) * M$

Which results in the computation:

$\text{B} = [\text{ceiling}(W/(\text{sqrt}(2) * M)\ ) * M] - A$

The following table lists examples coordinate computations:

Table 8-1

W = Width (μm)	**M = mgrid (μm)**	**D = W/(sqrt(2)*M)**	**ceiling (D)**	**A (μm)**	**B (μm)**	**Diagonal width (μm)**
1.0	0.005	141.42	142	0.5	0.21	1.0041
0.5	0.005	70.71	71	0.25	0.105	0.5020
0.15	0.005	21.21	22	0.075	0.035	0.1556
0.155*	0.005	21.92	22	0.0775*	0.0325*	0.1556
* A width of 0.155 is an odd multiple of the manufacturing grid and is not allowed because it would create coordinates for A and B that are off the manufacturing grid. It is shown for completeness to illustrate how the result is off grid.						

The ‘octagon’ default style defined above applies only to 45-degree route segments; it does not apply to 90-degree route segments.

Note that point extension acts on both connected segments for internal DEF paths. In addition, diagonal segments and any orthogonal segments connected to them are not subject to point extension. If point extensions are specified for such segments, the application should ignore them. Finally, custom-style DEF paths are not subject of point extensions. If point extensions are specified for such segments, the application should ignore them.

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Routing Point Extensions (x y extValue)

If path point expression has *extValue*, then the resulting routing geometry should be created with the following rules:

- By default, the end extension of `NETS` segments is half-width.
- By default, the end extension of `SPECIALNETS` segments is zero.
- DEF path start and endpoint extensions will be applied to start and end path ends.
- Internal DEF path point extension acts on both the connected segments. If that is not desired, you need to use the `( * * <ext> )` method to make them different.
- Diagonal segments are not subject to endpoint extensions; if *extValue* exists, the application should ignore it.
- DEF path segments with `+ STYLE` are not subject to endpoint extensions; if *extValue* exists, the application should ignore it.

Example 1

In `SPECIALNETS`, points with extension `0` are equivalent to points without extensions. The following two routes produce identical routing shapes, as shown in Figure 8-22 on page 1084.

```
SPECIALNETS 1 ;
    - POWER (* power)
        + ROUTED metal3 80 ( 0 0 ) ( 100 0 ) ( 200 100 ) ( 300 100 ) ;
END SPECIALNETS

NETS 1 ;
    - mySignal ...
        + ROUTED metal3 ( 0 0 0 ) ( 100 0 0 ) ( 200 100 0 ) ( 300 100 0 ) ;
        #mySignal uses the default routing rule width of 80,
        #need to suppress half-width extensions in every point.
END NETS
```


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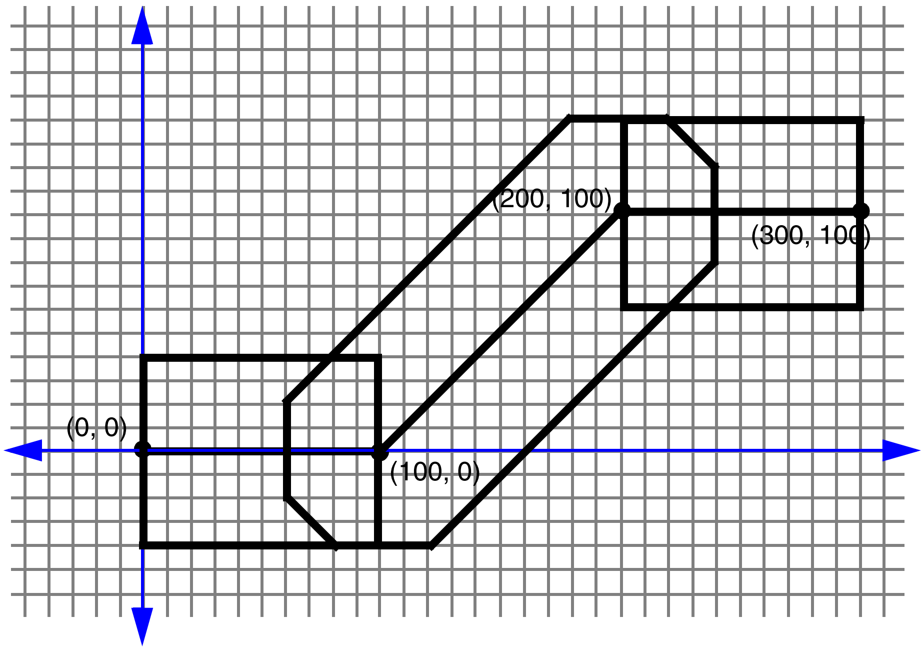
Figure 8-22



Example 2

The following `NETS` regular route definition, using the traditional default wire extension of 1/2 * *width*, produces the route shown in Figure 8-23 on page 1085.

```
NETS 1 ;
    - mySignal ... #mySignal uses the default routing rule with width of 80
        + ROUTED metal3 ( 0 0 ) ( 100 0 0 ) ( 200 100 0 ) ( 300 100 ) ;
           #The default extension is half-width (40) for NETS, need to set
           #the extension to 0 explicitly for the end connected to
           #a diagonal segment.
END NETS

SPECIALNETS 1;
    - POWER (* power)
        + ROUTED metal3 80 ( 0 0 40 ) ( 100 0 ) ( 200 100 ) ( 300 100 40 ) ;
             #Need to add half-width extensions to the start and
             #end points explicitly.
END SPECIALNETS
```


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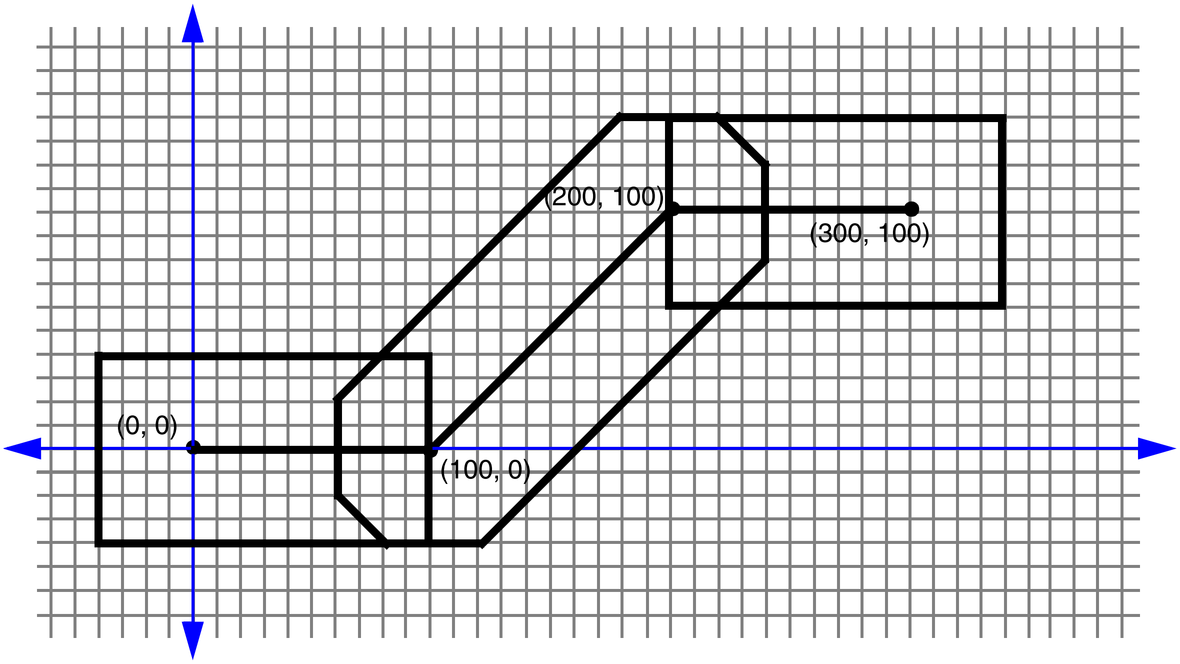
Figure 8-23



Example 3

The following definition, using the traditional default wire extension of 1/2 * *width* for all of the points, produces the route in Figure 8-24 on page 1086. In all cases, extensions to diagonal segments will be ignored.

```
SPECIALNETS 1 ;
    - POWER (* power)
        + ROUTED metal3 80 ( 0 0 40 ) ( 100 0 40 ) ( 200 100 40 ) ( 300 100 40 ) ;
           #The half-width extensions are given explicitly, they will be
           #ignored for diagonal segments.
END SPECIALNETS

NETS 1 ;
    - mySignal ... #mySignal uses the default routing rule width of 80
        + ROUTED metal3 ( 0 0 ) ( 100 0 ) ( 200 100 ) ( 300 100 ) ;
           #NETS segments by default have half-width end extensions.
END NETS
```


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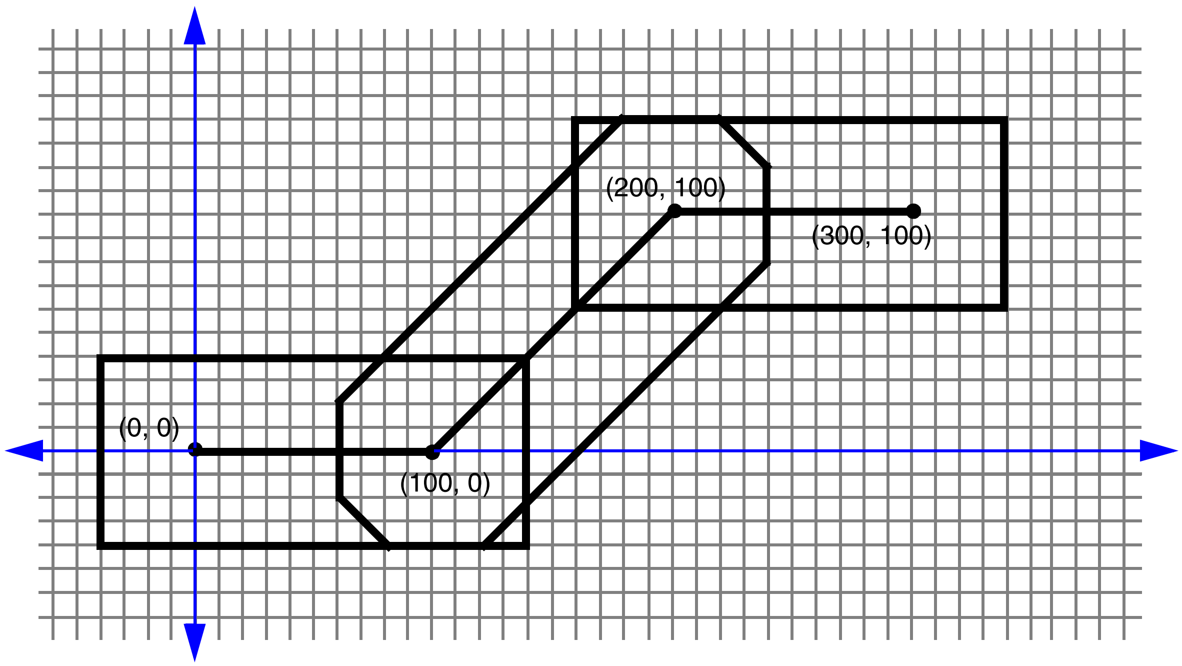
Figure 8-24



Technology

```
[TECHNOLOGY technologyName ;]
```

Specifies a technology name for the design in the database. In case of a conflict, the previous name remains in effect.

Tracks

```
[TRACKS
     [{X | Y} start DO numtracks STEP space
       [MASK maskNum [SAMEMASK]]
       [LAYER layerName ...]
     ;] ...]
```

Defines the routing grid for a standard cell-based design. Typically, the routing grid is generated when the floorplan is initialized. The first track is located at an offset from the placement grid set by the `OFFSET` value for the layer in the LEF file. The track spacing is the `PITCH` value for the layer defined in LEF.

`DO` *`numTracks`*

Specifies the number of tracks to create for the grid. You cannot specify 0 *`numtracks`*.

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`LAYER` *`layerName`*

Specifies the routing layer used for the tracks. You can specify more than one layer.

`MASK` *`maskNum`* `[SAMEMASK]`

Specifies which mask for double or triple patterning lithography to use for the first routing track. The *`maskNum`* variable must be a positive integer - most applications support values of 1, 2, or 3 only. The layer(s) must be declared as two or three mask layers in the `LEF LAYER` section.

By default, the tracks cycle through all the masks. So you will see alternating masks, such as, 1, 2, 1, 2, etc. for a two-mask layer and 1, 2, 3, 1, 2, 3, etc., for a three-mask layer.

If the `SAMEMASK` keyword is specified, then all the routing tracks are the same mask as the first track mask. Tracks without any defined mask do not have a mask set (that is, they are uncolored).
See Example 8-35 on page 1087.

`STEP` *`space`*

Specifies the spacing between the tracks.

`{X | Y}` *`start`*

Specifies the location and direction of the first track defined. `X` indicates vertical lines; `Y` indicates horizontal lines. *`start`* is the `X` or `Y` coordinate of the first line. For example, `X 3000` creates a set of vertical lines, with the first line going through (3000 0).

Example 8-35 Mask Assignments for Routing Tracks

- The following example shows a three-mask layer `M1` that has a first track of mask 2 with cycling mask numbers after that:

  ```
  TRACKS X 0 DO 20 STEP 5 MASK 2 LAYER M1 ;
  ```

  This statement will result in `M1` vertical tracks at `X` coordinates with mask assignments of 0 (mask 2), 5 (mask 3), 10 (mask 1), 15 (mask 2), etc., for 20 tracks.

- The following statement will result in `M1` vertical tracks at `X` coordinates with mask assignments of 0 (mask 1), 10 (mask 1), 20 (mask 1), 30 (mask 1), etc., for 20 tracks.

  ```
  TRACKS X 0 DO 20 STEP 10 MASK 1 SAMEMASK LAYER M1 ;
  ```

Track Properties

```
PROPERTYDEFINITIONS

DESIGN TRACKPROPERTIES STRING
```


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```
"{TRACKS [{X | Y} start DO numtracks STEP space
     [MASK maskNum [SAMEMASK]] [LAYER layerName ...]
     [WIDTH width] [NDR ruleName]}...
     ;...";
```

```
END PROPERTYDEFINITIONS
```

Defines the routing grid for a standard cell-based design with additional definition of either the width of the tracks and/or the NDR to which the tracks belong.

`DO` *`numTracks`*

Specifies the number of tracks to create for the grid. You cannot specify 0 *`numtracks`*.

`LAYER` *`layerName`*

Specifies the routing layer used for the tracks. You can specify more than one layer.

`MASK` *`maskNum`* `[SAMEMASK]`

Specifies which mask for double or triple patterning lithography to use for the first routing track. The *`maskNum`* variable must be a positive integer - most applications support values of 1, 2, or 3 only. The layer(s) must be declared as two or three mask layers in the `LEF LAYER` section.

By default, the tracks cycle through all the masks. So you will see alternating masks, such as, 1, 2, 1, 2, etc. for a two-mask layer and 1, 2, 3, 1, 2, 3, etc., for a three-mask layer.

If the `SAMEMASK` keyword is specified, then all the routing tracks are the same mask as the first track mask. Tracks without any defined mask do not have a mask set (that is, they are uncolored).
See Example 8-35 on page 1087.

`NDR` *`ruleName`*

Specifies that the tracks belong to the `NDR` with name *`ruleName`*. See Example 8-36 on page 1089.

`STEP` *`space`*

Specifies the spacing between the tracks.

`WIDTH` *`width`*

Specifies the width of the tracks as *`width`*. See Example 8-36 on page 1089.

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`{X | Y} start`

Specifies the location and direction of the first track defined. `X` indicates vertical lines; `Y` indicates horizontal lines. *`start`* is the `X` or `Y` coordinate of the first line. For example, `X 3000` creates a set of vertical lines, with the first line going through (3000 0).

Example 8-36 Width/NDR Specification for Routing Tracks

Consider the following example:

```
PROPERTYDEFINITIONS
DESIGN TRACKPROPERTIES STRING "
      TRACKS Y 560 DO 72 STEP 1120 LAYER M1 WIDTH 2400 ;
      TRACKS X 2640 DO 46 STEP 1980 LAYER M2 NDR NDR1 ; " ;
END PROPERTYDEFINITIONS
```

This means that the tracks on `M1` in the first statement has a width of `2400` and the tracks on `M2` in the second statement belong to the NDR with the name `NDR1`.

Units

`[UNITS DISTANCE MICRONS dbuPerMicron ;]`

Specifies the database units per micron (dbuPerMicron) to convert DEF distance units into microns.

LEF supports values of 100, 200, 400, 800, 1000, 2000, 4000, 8000, 10,000, and 20,000 for the LEF dbuPerMicron. The LEF dbuPerMicron must be greater than or equal to the DEF dbuPerMicron, otherwise you can get round-off errors. The LEF convert factor must also be an integer multiple of the DEF convert factor so no round-off of DEF database unit values is required (e.g., a LEF convert factor of 1000 allows DEF convert factors of 100, 200, 1000, but not 400, 800).

The following table shows the valid pairings of the LEF dbuPerMicron and the corresponding legal DEF dbuPerMicron values.

LEF dbuPerMicron	Legal DEF dbuPerMicron
100	100
200	100, 200
400	100, 200, 400
800	100, 200, 400, 800

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LEF dbuPerMicron	Legal DEF dbuPerMicron
1000	100, 200, 1000
2000	100, 200, 400, 1000, 2000
4000	100, 200, 400, 800, 1000, 2000, 4000
8000	100, 200, 400, 800, 1000, 2000, 4000, 8000
10,000	100, 200, 400, 1000, 2000, 10,000
20,000	100, 200, 400, 800, 1000, 2000, 4000, 10,000, 20,000

Using DEF Units

The following table shows examples of how DEF units are used:

Units	DEF Units	DEF Value Example	Real Value
Time	.001 nanosecond	1500	1.5 nanoseconds
Capacitance	.000001 picofarad	1,500,000	1.5 picofarads
Resistance	.ooo1 ohm	15,000	1.5 ohms
Power	.0001 milliwatt	15,000	1.5 milliwatts
Current	.0001 milliamp	15,000	1.5 milliamps
Voltage	.001 volt	1500	1.5 volts

The DEF reader assumes divisor factors such that DEF data is given in the database units shown below.

Unit	Database Precision
1 nanosecond	= 1000 DBUs
1 picofarad	= 1,000,000 DBUs
1 ohm	= 10,000 DBUs
1 milliwatt	= 10,000 DBUs
1 milliampere	= 10,000 DBUs
1 volt	= 1000 DBUs

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Version

```
[VERSION versionNumber ;]
```

Specifies which version of the DEF syntax is being used.

Note: The `VERSION` statement is not required in a DEF file; however, you should specify it, because it prevents syntax errors caused by the inadvertent use of new versions of DEF with older tools that do not support the new version syntax.

Vias

```
[VIAS numVias ;
     [– viaName
        [   + VIARULE viaRuleName
               + CUTSIZE xSize ySize
               + LAYERS botmetalLayer cutLayer topMetalLayer
               + CUTSPACING xCutSpacing yCutSpacing
               + ENCLOSURE xBotEnc yBotEnc xTopEnc yTopEnc
               [+ ROWCOL numCutRows NumCutCols]
               [+ ORIGIN xOffset yOffset]
               [+ OFFSET xBotOffset yBotOffset xTopOffset yTopOffset]
               [+ PATTERN cutPattern] ]
        | [ + RECT layerName [+ MASK maskNum] pt pt
                    | + POLYGON layerName [+ MASK maskNum] pt pt pt] ...]
     ;] ...
END VIAS]
```

Lists the names and geometry definitions of all vias in the design. Two types of vias can be listed: fixed vias and generated vias. All vias consist of shapes on three layers: a cut layer and two routing (or masterslice) layers that connect through that cut layer.

A fixed via is defined using rectangles or polygons, and does not use a `VIARULE`. The fixed via name must mean the same via in all associated LEF and DEF files.

A generated via is defined using `VIARULE` parameters to indicate that it was derived from a `VIARULE GENERATE` statement. For a generated via, the via name is only used locally inside this DEF file. The geometry and parameters are maintained, but the name can be freely changed by applications that use this via when writing out LEF and DEF files to avoid possible via name collisions with other DEF files.

`CUTSIZE` *`xSize ySize`*

Specifies the required width (*`xSize`*) and height (*`ySize`*) of the cut layer rectangles.
Type: Integer, specified in DEF database units

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`CUTSPACING` *`xCutSpacing yCutSpacing`*

Specifies the required x and y spacing between cuts. The spacing is measured from one cut edge to the next cut edge.
Type: Integer, specified in DEF database units

`ENCLOSURE` *`xBotEnc yBotEnc xTopEnc yTopEnc`*

Specifies the required x and y enclosure values for the bottom and top metal layers. The enclosure measures the distance from the cut array edge to the metal edge that encloses the cut array (see Figure 8-25 on page 1096).
Type: Integer, specified in DEF database units

`LAYERS` *`botMetalLayer cutLayer TopMetalLayer`*

Specifies the required names of the bottom routing/masterslice layer, cut layer, and top routing/masterslice layer. These layer names must be previously defined in layer definitions, and must match the layer names defined in the specified LEF *`viaRuleName`*.

`MASK` *`maskNum`*

Specifies the mask for double or triple patterning lithography that is to be applied to the shapes defined in `RECT` or `POLYGON` statements of the via master. The *`maskNum`* variable must be a positive integer; most applications support values of 1, 2, or 3 only. For a fixed via made up of `RECT/POLYGON` statements, the cut-shapes must be either colored or uncolored. It is an error to have partially colored cuts for one via. Uncolored cut shapes should be automatically colored by the reader if the layer is a multi-mask layer.
For a via layer with a single shape, the corresponding layer mask in the via definition (via-master) is ignored and only the *`viaMaskNum`* specified in the via instantiation is used.

For a via layer with multiple shapes, the colors of both via-master and via-instance masks are processed.
Some readers may, however, color them for internal consistency (see Example 8-39 on page 1100). So a writer may write out MASK 1 for metal shapes even if they were read in with no mask value.
For uncolored fixed vias, or parameterized vias (with `+ VIARULE` ...), the mask of the cuts are pre-defined as an alternating pattern starting with MASK 1 at the bottom-left. The mask cycles, from left-to-right and bottom-to-top, for the cuts are as shown in Figure 8-30 on page 1101.

`numVias`

Specifies the number of vias listed in the `VIA` statement.

`OFFSET` *`xBotOffset yBotOffset xTopOffset yTopOffset`*

Specifies the x and y offset for the bottom and top metal layers. These values allow each metal layer to be offset independently.

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By default, the `0,0` origin of the via is the center of the cut array, and the enclosing metal rectangles. After the non-shifted via is computed, the metal layer rectangles are shifted by adding the appropriate values—the *`x/y BotOffset`* values to the metal layer below the cut layer, and the *`x/y TopOffset`* values to the metal layer above the cut layer.

These offset values are in addition to any offset caused by the `ORIGIN` values. For an example and illustration of this syntax, see Example 8-37 on page 1095.
Default: `0`, for all values
Type: Integer, in DEF database units

`ORIGIN` *`xOffset yOffset`*

Specifies the x and y offset for all of the via shapes. By default, the `0,0` origin of the via is the center of the cut array, and the enclosing metal rectangles. After the non-shifted via is computed, all cut and metal rectangles are shifted by adding these values. For an example and illustration of this syntax, see Example 8-37 on page 1095.
Default: `0`, for both values
Type: Integer, in DEF database units

`PATTERN` *`cutPattern`*

Specifies the cut pattern encoded as an ASCII string. This parameter is only required when some of the cuts are missing from the array of cuts, and defaults to "all cuts are present," if not specified.

For information on and examples of via cut patterns, see "Creating Via Cut Patterns" on page 1101.

The *`cutPattern`* syntax is defined as follows:

```
numRows_rowDefinition
  [_numRows_rowDefinition] ...
```

`numRows`	Specifies a hexadecimal number that indicates how many times to repeat the following row definition. This number can be more than one digit.
`rowDefinition`	Defines one row of cuts, from left to right.

The *`rowDefinition`* syntax is defined as follows:

```
{[RrepeatNumber]hexDigitCutPattern} ...
```

`hexDigitCutPattern`

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Specifies a single hexadecimal digit that encodes a 4-bit binary value in which `1` indicates a cut is present, and `0` indicates a cut is not present.

repeatNumber Specifies a single hexadecimal digit that indicates how many times to repeat *hexDigitCutPattern*.

For parameterized vias (with `+ VIARULE` ...), the *cutPattern* has an optional suffix added to allow three types of mask color patterns. The default mask color pattern (no suffix) is a checker-board (see Figure 8-28 on page 1098). The other two patterns supported are alternating rows, and alternating columns (see Figure 8-29 on page 1099).

The optional suffixes are:

<*cut_pattern*>_MR alternating rows
<*cut_pattern*>_MC alternating columns

`POLYGON` *layerName pt pt pt*

Defines the via geometry for the specified layer. You must specify at least three points to generate the polygon, and the edges must be parallel to the x axis, the y axis, or at a 45-degree angle.
Type: (*x y*) Integer, specified in database units

Each `POLYGON` statement defines a polygon generated by connecting each successive point, and then the first and last points. The *pt* syntax corresponds to a coordinate pair, such as (*x y*). Specify an asterisk (`*`) to repeat the same value as the previous *x* or *y* value from the last point.

For example, `+ POLYGON ( 0 0 ) ( 10 10 ) ( 10 0 )` creates a triangle shape.

All vias consist of shapes on three layers: a cut layer and two routing (or masterslice) layers that connect through that cut layer. There should be at least one `RECT` or `POLYGON` on each of the three layers.

`RECT` *layerName pt pt*

Defines the via geometry for the specified layer. The points are specified with respect to the via origin. In most cases, the via origin is the center of the via bounding box. All geometries for the via, including the cut layers, are output by the DEF writer.
Type: (*x y*) Integer, specified in database units

All vias consist of shapes on three layers: a cut layer and two routing (or masterslice)

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layers that connect through that cut layer. There should be at least one `RECT` or `POLYGON` on each of the three layers.

`ROWCOL` *`numCutRows numCutCols`*

Specifies the number of cut rows and columns that make up the cut array.
Default: `1`, for both values
Type: Positive integer, for both values

`viaName`

Specifies the via name. Via names are generated by appending a number after the rule name. Vias are numbered in the order in which they are created.

`VIARULE` *`viaRuleName`*

Specifies the name of the LEF `VIARULE` that produced this via. This name must be specified before you define any of the other parameters, and must refer to a `VIARULE GENERATE` via rule. It cannot refer to a `VIARULE` without a `GENERATE` keyword.

Specifying the reserved via rule name of `DEFAULT` indicates that the via should use the previously defined `VIARULE GENERATE` rule with the `DEFAULT` keyword that exists for this routing-cut-routing (or masterslice-cut-masterslice) layer combination. This makes it possible for a tool that does not use the LEF `VIARULE` technology section to still generate DEF generated-via parameters by using the default rule.

Example 8-37 Via Rules

The following via rule describes a non-shifted via (that is, a via with no `OFFSET` or `ORIGIN` parameters). There are two rows and three columns of via cuts. Figure 8-25 on page 1096 illustrates this via rule.

```
- myUnshiftedVia
    + VIARULE myViaRule
    + CUTSIZE 20 20              #xCutSize yCutSize
    + LAYERS metal1 cut12 metal2
    + CUTSPACING 30 30           #xCutSpacing yCutSpacing
    + ENCLOSURE 20 50 50 20      #xBotEnc yBotEnc xTopEnc yTopEnc
    + ROWCOL 2 3 ;
```


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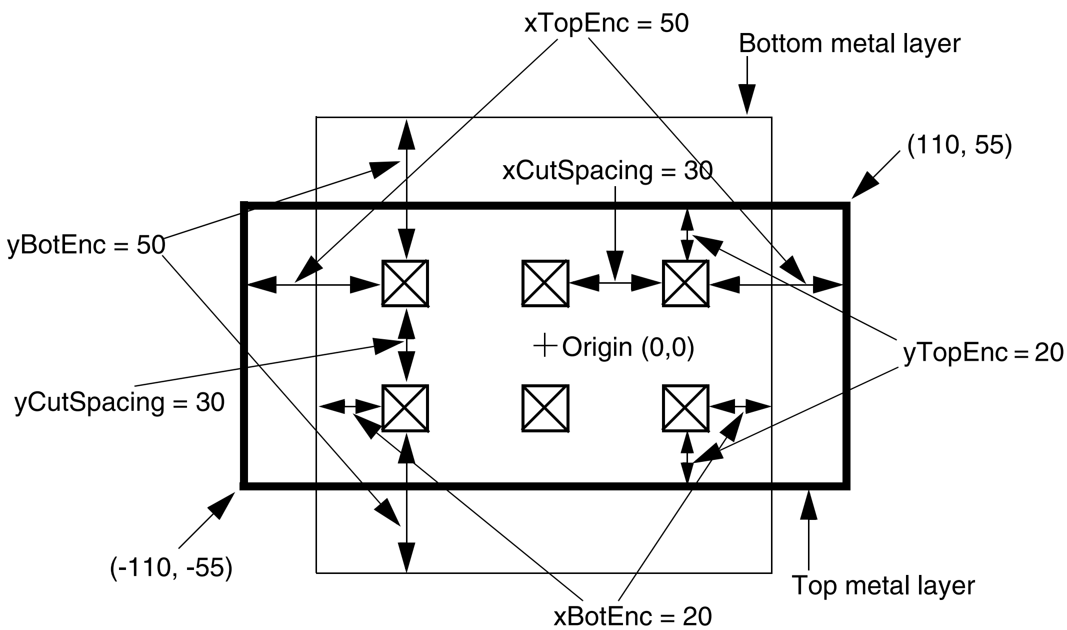
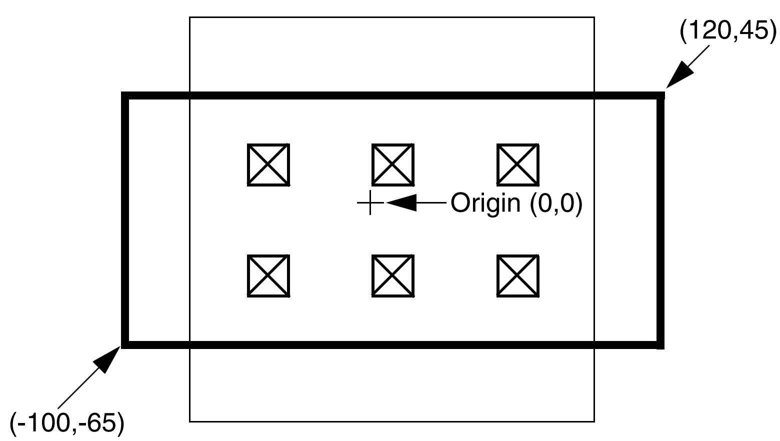
Figure 8-25 Via Rule



The same via rule with the following `ORIGIN` parameter shifts all of the metal and cut rectangles by 10 in the x direction, and by -10 in the y direction (see Figure 8-26 on page 1096):

```
+ ORIGIN 10 -10
```

Figure 8-26 Via Rule With Origin



If the same via rule contains the following `ORIGIN` and `OFFSET` parameters, all of the rectangles shift by 10, -10. In addition, the top layer metal rectangle shifts by 20, -20, which means that the top metal shifts by a total of 30, -30.

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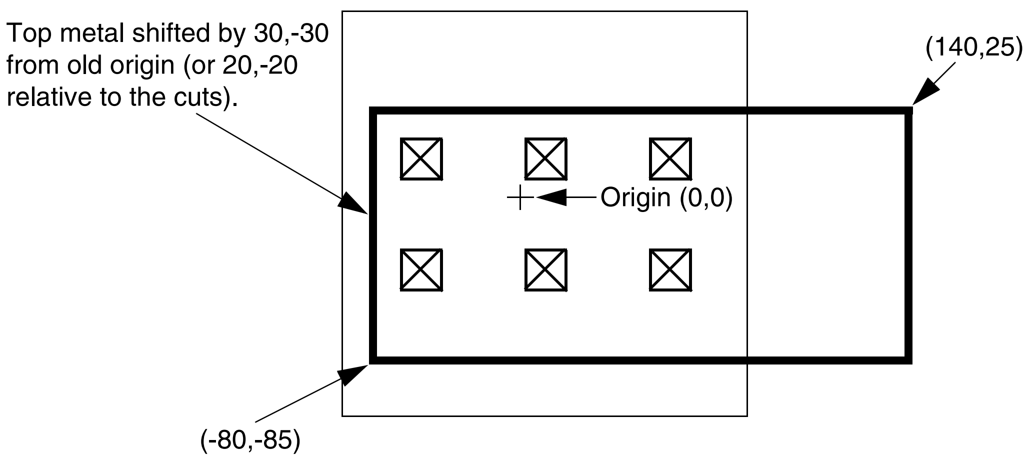
```
    + ORIGIN 10 -10
    + OFFSET 0 0 20 -20
```

Figure 8-27 Via Rule With Origin and Offset



Example 8-38 Multi-Mask Patterns for Parameterized Vias with Via Rule

The following via rule describes a via cut mask pattern for a parameterized via:

```
- myParamVia1
    + VIARULE myGenVia1         + CUTSIZE 40 40
    + LAYERS M1 VIA1 M2         + CUTSPACING 40 40
    + ENCLOSURE 40 0 0 40       + ROWCOL 3 4
    + PATTERN 2_F_1_D ;                            #1 cut in top row is missing
```


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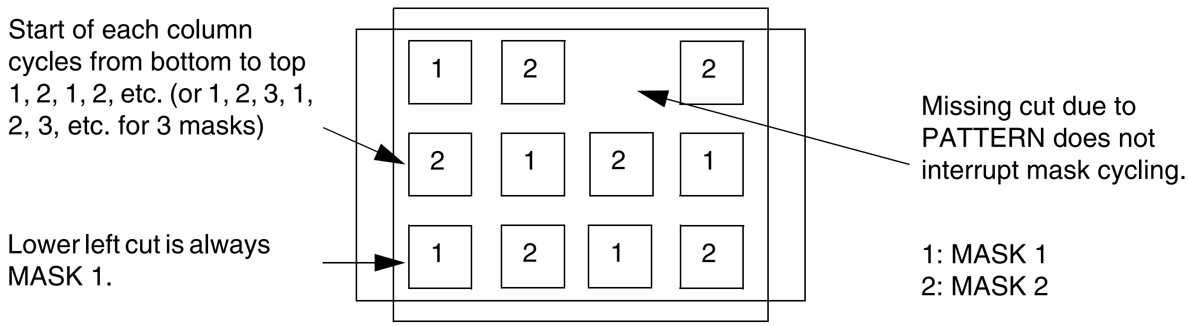
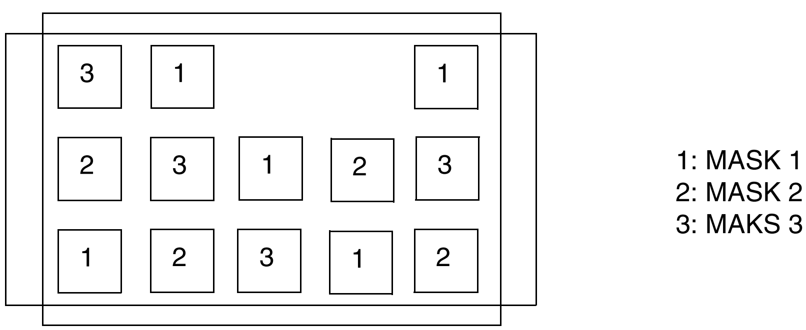
Figure 8-28 Multi-Mask Patterns for Parameterized Vias



Assuming VIA1 is a 2-mask cut-layer, then the mask for each cut-shape is shown above. The default checker-board pattern is: the bottom left cut starts at 1, and goes left to right, 1, 2, 1, 2. The next row up starts at 2, and goes 2, 1, 2, 1, etc. Missing cuts due to PATTERN do not change the mask assignments.



Example of a check-board parameterized via cut-mask pattern for a 3-mask layer with 2 missing cuts due to PATTERN statement.

- The following examples show a parameterized via with cut-mask patterns for a 3-mask layer and 2-mask layer using _MC and _MR suffixes:

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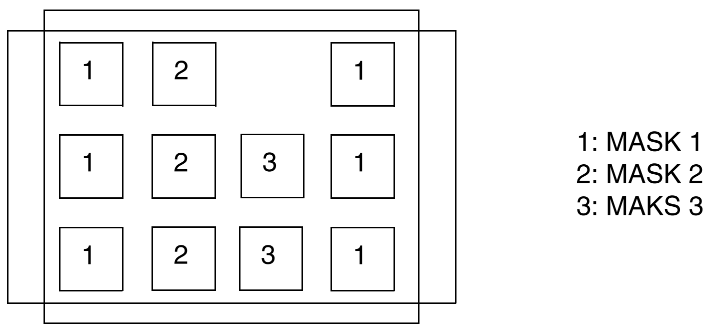
Figure 8-29 Multi-Mask Patterns for Parameterized Vias using Suffixes

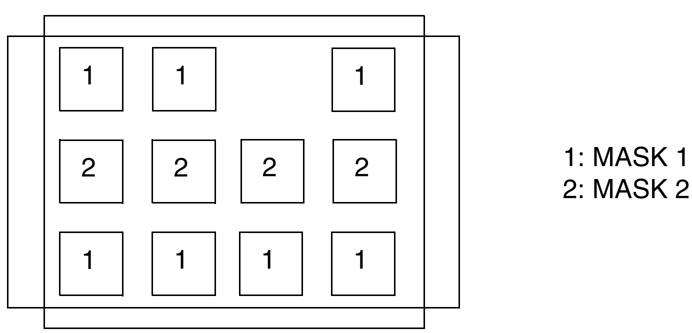
On a 3 mask cut layer

Mask colors for a parameterized via like:
- myParamVia1
+ VIARULE myGenVia1 + CUTSIZE 40 40
+ LAYERS M1 VIA1 M2 + CUTSPACING 40 40
+ ENCLOSURE 40 0 0 40 + ROWCOL 3 4
+ PATTERN 2_F_1_D_MC; #alternating mask color columns due to _MC suffix



On a 2 mask cut layer

Mask colors for a parameterized via like:
- myParamVia1
+ VIARULE myGenVia1 + CUTSIZE 40 40
+ LAYERS M1 VIA1 M2 + CUTSPACING 40 40
+ ENCLOSURE 40 0 0 40 + ROWCOL 3 4
+ PATTERN 2_F_1_D_MR ; #alternating mask color rows due to _MR suffix



For a fixed via specified using `RECT` or `POLYGON` statements, the cut shapes must either be all colored or uncolored. If the cuts are not colored, they will be automatically colored in a checkerboard pattern as shown in Figure 8-28 on page 1098. Each via cut with the same lower-left Y value is considered as one row, and each via in one row is a new column. For common "array" style vias with no missing cuts, this coloring is a good one. For vias that do

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not have a row and column structure or are missing cuts, then this coloring may not be good (see Figure 8-30 on page 1101). If the metal layers are not colored, some applications will color them to mask 1 for internal consistency, even though the via master metal shape colors are not really used by LEF or DEF via instances.

Example 8-39 Multi-Mask Patterns for Fixed Via

The following example shows a fixed-via with pre-colored cut shapes:

```
-  myVia1
    + RECT  M1 ( -40 -20 ) ( 120 20 )            #no mask, some readers set to 1
    + RECT VIA1 + MASK 1 ( -20 -20 ) ( 20 20 )       #first cut on mask 1
    + RECT VIA1 + MASK 2  ( 60 -20 ) ( 100 20 )      #second cut on mask 2
    + RECT ( -20 -40 ) ( 100 40 )                   #no mask, some readers set to 1
```


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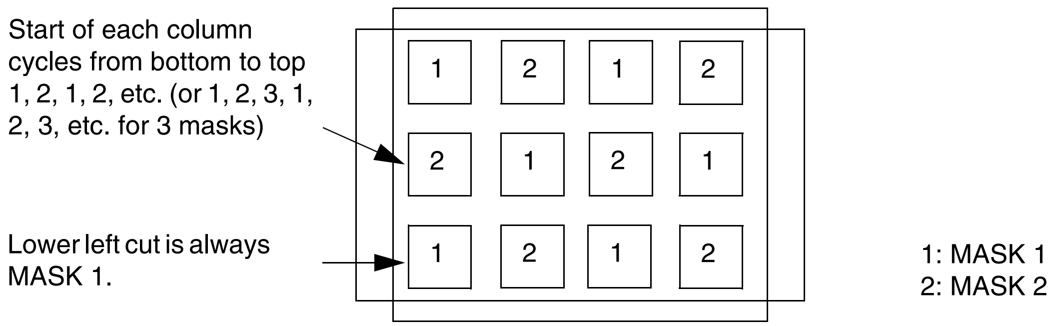
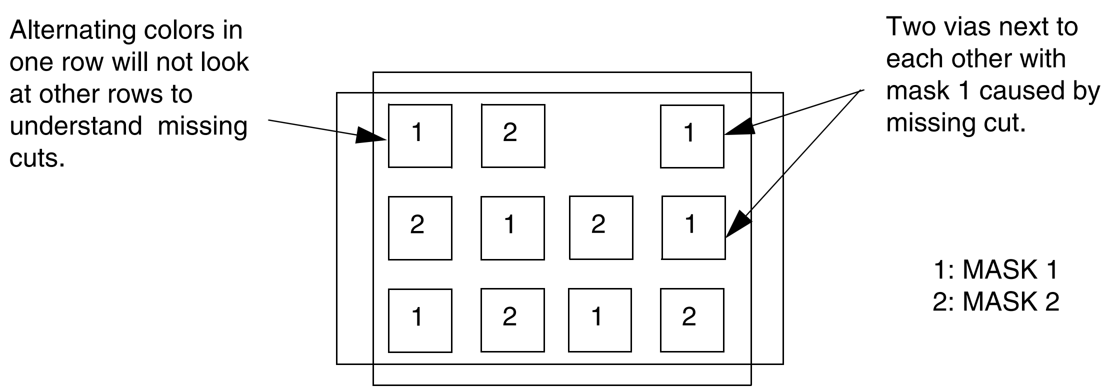
Figure 8-30 Multi-Mask Patterns for Fixed Via



Uncolored fixed vias (from RECT/POLYGON statements) will get similar coloring as a parameterized via. Each cut with the same Y value is one row. The cuts inside one row alternate. This works well for cut-arrays without missing cuts as shown here.



Unlike parameterized vias, there is no real row/column structure required in a fixed via, so the automatic coloring will not work well for a fixed via with missing cuts (or cuts that are not aligned). This type of via must be pre-colored to be useful.

See the Fills, Nets, and Special Nets routing statements to see how a via instance uses these via-master mask values.

Creating Via Cut Patterns

Via cuts are defined as a series of rows, starting at the bottom, left corner. Each row definition defines one row of cuts, from left to right, and rows are numbered from bottom to top.

The `PATTERN` syntax that defines rows uses the `ROWCOL` parameters to specify the cut array. If the row has more bits than the *`numCutCols`* value in the `ROWCOL` parameter for this via,

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the last bits are ignored. The number of rows defined must equal the *numCutRows* value in the ROWCOL parameter.

Figure 8-31 on page 1102 illustrates the following via cut pattern syntax:

```
- myVia
    + VIARULE myViaRule
    ...
    + ROWCOL 5 5
    + PATTERN 2_F0_2_F8_1_78 ;]
```

The last three bits of F0, F8, and 78 are ignored because only five bits are allowed in a row. Therefore, the following PATTERN syntax gives the identical pattern:

```
+ PATTERN 2_F7_2_FF_1_7F
```

Figure 8-31

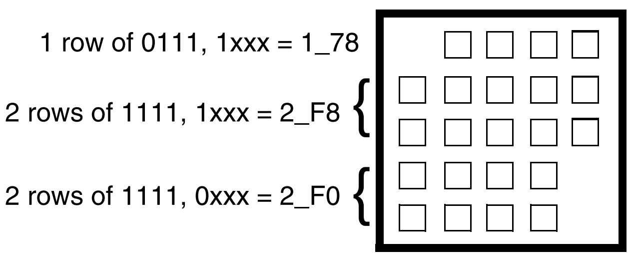

Figure 8-32 on page 1103 illustrates the following via cut pattern syntax:

```
- myVia
    + VIARULE myViaRule
    ...
    + ROWCOL 5 14
    + PATTERN 2_FFE0_3_R4F ;
```

The R4F value indicates a repeat of four Fs. The last two bits of each row definition are ignored because only 14 bits allowed in each row.

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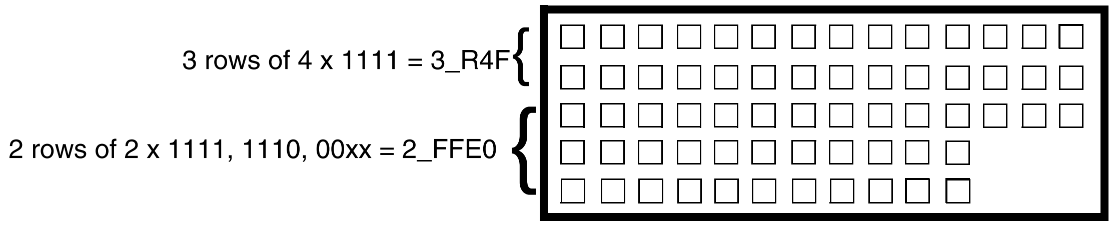
Figure 8-32



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A

Examples

This appendix contains information about the following topics.

- LEF on page 1105
- DEF on page 1116
- Scan Chain Synthesis Example on page 1121

LEF

```
VERSION 5.7 ;

# DEMO4 CHIP - 1280 ARRAY

     &alias &&area = (73600,74400) (238240,236400) &endalias
          &alias &&core = (85080,85500) (226760,224700) &endalias
          &alias &&m2stripes = sroute stripe net vss net vdd layer m2
               width
               320 count 2 pattern 87900 4200 218100
               area &&area core &&core &endalias
          &alias &&m3stripes = sroute stripe net vss net vdd layer m3
               width
               600 count 2 pattern 89840 6720 217520
               area &&area core &&core &endalias
          &alias &&powerfollowpins = sroute follow net vss net vdd layer
               m1 width 560
               area &&area core &&core &endalias
          &alias &&powerrepair = sroute repair net vss net vdd area
               &&area core &&core &endalias
     # PLACEMENT SITE SECTION
     SITE CORE1 SIZE 67.2 BY 6 ; # GCD of all Y sizes of Macros END
          CORE1
     SITE IOX SIZE 37.8 BY 444 ; # 151.2 / 4 = 37.8 , 4 sites per pad END IOX
     SITE IOY SIZE 436.8 BY 30 ; # 150 / 5 = 30 , 5 sites per pad END IOY
     SITE SQUAREBLOCK SIZE 268.8 BY 252 ; END SQUAREBLOCK
     SITE I2BLOCK SIZE 672 BY 504 ; END I2BLOCK
     SITE LBLOCK SIZE 201.6 BY 168 ; END LBLOCK
     SITE CORNER SIZE 436.8 BY 444 ; END CORNER
     LAYER POLYS TYPE MASTERSLICE ; END POLYS
```


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```
LAYER PW TYPE MASTERSLICE ; END PW
LAYER NW TYPE MASTERSLICE ; END NW
LAYER PD TYPE MASTERSLICE ; END PD
LAYER ND TYPE MASTERSLICE ; END ND
LAYER CUT01 TYPE CUT ; END CUT01
LAYER M1 TYPE ROUTING ; DIRECTION VERTICAL ; PITCH 5.6 ; WIDTH2.6 ;
     SPACING 1.5 ;
END M1
LAYER CUT12 TYPE CUT ;
SPACING 4.0 SAMENET LAYER CUT01 ;
END CUT12
LAYER M2 TYPE ROUTING ; DIRECTION HORIZONTAL ; PITCH 6.0 ;
     WIDTH 3.2 ;SPACING 1.6 ;
END M2
LAYER CUT23 TYPE CUT ;
SPACING 4.0 SAMENET LAYER CUT01 ;
END CUT23
LAYER M3 TYPE ROUTING ; DIRECTION VERTICAL ; PITCH 5.6 ; WIDTH 3.6;
     SPACING 1.6 ;
END M3
LAYER OVERLAP TYPE OVERLAP ; END OVERLAP
VIA C2PW DEFAULT LAYER PW ; RECT -2.0 -2.0 2.0 2.0 ;
     LAYER CUT01 ; RECT -0.6 -0.6 0.6 0.6 ;
     LAYER M1 ; RECT -2.0 -2.0 2.0 2.0 ;
END C2PW
VIA C2NW DEFAULT LAYER NW ; RECT -2.0 -2.0 2.0 2.0 ;
     LAYER CUT01 ; RECT -0.6 -0.6 0.6 0.6 ;
     LAYER M1 ; RECT -2.0 -2.0 2.0 2.0 ;
END C2NW
VIA C2PD DEFAULT LAYER PD ; RECT -2.0 -2.0 2.0 2.0 ;
     LAYER CUT01 ; RECT -0.6 -0.6 0.6 0.6 ;
     LAYER M1 ; RECT -2.0 -2.0 2.0 2.0 ;
END C2PD
VIA C2ND DEFAULT LAYER ND ; RECT -2.0 -2.0 2.0 2.0 ;
     LAYER CUT01 ; RECT -0.6 -0.6 0.6 0.6 ;
     LAYER M1 ; RECT -2.0 -2.0 2.0 2.0 ;
END C2ND
VIA C2POLY DEFAULT LAYER POLYS ; RECT -2.0 -2.0 2.0 2.0 ;
     LAYER CUT01 ; RECT -0.6 -0.6 0.6 0.6 ;
     LAYER M1 ; RECT -2.0 -2.0 2.0 2.0 ;
END C2POLY
VIA VIA12 DEFAULT LAYER M1 ; RECT -2.0 -2.0 2.0 2.0 ;
     LAYER CUT12 ; RECT -0.7 -0.7 0.7 0.7 ;
     LAYER M2 ; RECT -2.0 -2.0 2.0 2.0 ;
END VIA12
VIA VIA23 DEFAULT LAYER M3 ; RECT -2.0 -2.0 2.0 2.0 ;
     LAYER CUT23 ; RECT -0.8 -0.8 0.8 0.8 ;
     LAYER M2 ; RECT -2.0 -2.0 2.0 2.0 ;
END VIA23
VIA VIACENTER12 LAYER M1 ; RECT -4.6 -2.2 4.6 2.2 ;
     LAYER CUT12 ; RECT -3.1 -0.8 -1.9 0.8 ; RECT 1.9 -0.8 3.1 0.8 ;
```


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```
     LAYER M2 ; RECT -4.4 -2.0 4.4 2.0 ;
END VIACENTER12
VIA VIATOP12 LAYER M1 ; RECT -2.2 -2.2 2.2 8.2 ;
     LAYER CUT12 ; RECT -0.8 5.2 0.8 6.8 ;
     LAYER M2 ; RECT -2.2 -2.2 2.2 8.2 ;
END VIATOP12
VIA VIABOTTOM12 LAYER M1 ; RECT -2.2 -8.2 2.2 2.2 ;
     LAYER CUT12 ; RECT -0.8 -6.8 0.8 -5.2 ;
     LAYER M2 ; RECT -2.2 -8.2 2.2 2.2 ;
END VIABOTTOM12
VIA VIALEFT12 LAYER M1 ; RECT -7.8 -2.2 2.2 2.2 ;
     LAYER CUT12 ; RECT -6.4 -0.8 -4.8 0.8 ;
     LAYER M2 ; RECT -7.8 -2.2 2.2 2.2 ;
END VIALEFT12
VIA VIARIGHT12 LAYER M1 ; RECT -2.2 -2.2 7.8 2.2 ;
     LAYER CUT12 ; RECT 4.8 -0.8 6.4 0.8 ;
     LAYER M2 ; RECT -2.2 -2.2 7.8 2.2 ;
END VIARIGHT12
VIA VIABIGPOWER12 LAYER M1 ; RECT -21.0 -21.0 21.0 21.0 ;
     LAYER CUT12 ; RECT -2.4 -0.8 2.4 0.8 ;
          RECT -19.0 -19.0 -14.2 -17.4 ; RECT -19.0 17.4 -14.2
               19.0;
          RECT 14.2 -19.0 19.0 -17.4 ; RECT 14.2 17.4 19.0 19.0 ;
          RECT -19.0 -0.8 -14.2 0.8 ; RECT -2.4 -19.0 2.4 -17.4 ;
          RECT 14.2 -0.8 19 0.8 ; RECT -2.4 17.4 2.4 19.0 ;
     LAYER M2 ; RECT -21.0 -21.0 21.0 21.0 ;
END VIABIGPOWER12
VIARULE VIALIST12 LAYER M1 ; DIRECTION VERTICAL ; WIDTH 9.0 TO
     9.6;
     LAYER M2 ; DIRECTION HORIZONTAL ; WIDTH 3.0 TO 3.0 ;
          VIA VIACENTER12 ; VIA VIATOP12 ; VIA VIABOTTOM12 ;
          VIA VIALEFT12 ; VIA VIARIGHT12 ;
END VIALIST12
VIARULE VIAGEN12 GENERATE
     LAYER M1 ;
      ENCLOSURE 0.01 0.05 ;
     LAYER M2 ;
      ENCLOSURE 0.01 0.05 ;
     LAYER CUT12 ;
      RECT -0.06 -0.06 0.06 0.06 ;
      SPACING 0.14 BY 0.14 ;
END VIAGEN12
VIA VIACENTER23 LAYER M3 ; RECT -2.2 -2.2 2.2 2.2 ;
     LAYER CUT23 ; RECT -0.8 -0.8 0.8 0.8 ;
     LAYER M2 ; RECT -2.0 -2.0 2.0 2.0 ;
END VIACENTER23
VIA VIATOP23 LAYER M3 ; RECT -2.2 -2.2 2.2 8.2 ;
     LAYER CUT23 ; RECT -0.8 5.2 0.8 6.8 ;
     LAYER M2 ; RECT -2.2 -2.2 2.2 8.2 ;
END VIATOP23
VIA VIABOTTOM23 LAYER M3 ; RECT -2.2 -8.2 2.2 2.2 ;
```


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```
          LAYER CUT23 ; RECT -0.8 -6.8 0.8 -5.2 ;
          LAYER M2 ; RECT -2.2 -8.2 2.2 2.2 ;
     END VIABOTTOM23
     VIA VIALEFT23 LAYER M3 ; RECT -7.8 -2.2 2.2 2.2 ;
          LAYER CUT23 ; RECT -6.4 -0.8 -4.8 0.8 ;
          LAYER M2 ; RECT -7.8 -2.2 2.2 2.2 ;
     END VIALEFT23
     VIA VIARIGHT23 LAYER M3 ; RECT -2.2 -2.2 7.8 2.2 ;
          LAYER CUT23 ; RECT 4.8 -0.8 6.4 0.8 ;
          LAYER M2 ; RECT -2.2 -2.2 7.8 2.2 ;
     END VIARIGHT23
     VIARULE VIALIST23 LAYER M3 ; DIRECTION VERTICAL ; WIDTH 3.6 TO
          3.6 ;
               LAYER M2 ; DIRECTION HORIZONTAL ; WIDTH 3.0 TO 3.0 ;
                    VIA VIACENTER23 ; VIA VIATOP23 ; VIA VIABOTTOM23 ;
                    VIA VIALEFT23 ; VIA VIARIGHT23 ;
     END VIALIST23
     VIARULE VIAGEN23 GENERATE
          LAYER M2 ;
           ENCLOSURE 0.01 0.05 ;
          LAYER M3 ;
           ENCLOSURE 0.01 0.05 ;
          LAYER CUT23 ;
           RECT -0.06 -0.06 0.06 0.06 ;
           SPACING 0.14 BY 0.14 ;
     END VIAGEN23
     MACRO CORNER CLASS ENDCAP BOTTOMLEFT ; SIZE 436.8 BY 444 ; SYMMETRY X Y ; SITE
     CORNER ;
          PIN VDD SHAPE RING ; DIRECTION INOUT ;
               PORT LAYER M2 ; WIDTH 20 ; PATH 426.8 200 200 200 200 434 ;END
          END VDD
          PIN VSS SHAPE RING ; DIRECTION INOUT ;
               PORT LAYER M2 ; WIDTH 20 ; PATH 100 434 100 100 ; LAYER M1;
                    WIDTH 20 ; PATH 100 100 426.8 100 ;END
          END VSS
     END CORNER

MACRO IN1X class pad ; FOREIGN IN1X ; SIZE 151.2 BY 444 ;
          SYMMETRY X ; SITE IOX ;
          PIN Z DIRECTION OUTPUT ;
               PORT LAYER M1 ; PATH 61.6 444 72.8 444 ; END
          END Z
          PIN PO DIRECTION OUTPUT ;
               PORT LAYER M1 ; PATH 78.4 444 84.0 444 ; END
          END PO
          PIN A DIRECTION INPUT ;
               PORT LAYER M1 ; PATH 95.2 444 100.8 444 ; END
          END A
          PIN PI DIRECTION INPUT ;
               PORT LAYER M1 ; PATH 106.4 444 112 444 ; END
          END PI
```


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```
          PIN VDD DIRECTION INOUT ; SHAPE ABUTMENT ;
               PORT LAYER M2 ; WIDTH 20 ; PATH 10 200 141.2 200 ; END
          END VDD
          PIN VSS DIRECTION INOUT ; SHAPE ABUTMENT ;
               PORT LAYER M1 ; WIDTH 20 ; PATH 10 100 141.2 100 ; END
          END VSS
     END IN1X

MACRO IN1Y EEQ IN1X ; FOREIGN IN1Y ; class pad ;SIZE 436.8 BY 150 ;
          SYMMETRY Y ; SITE IOY ;
          PIN Z DIRECTION OUTPUT ;
               PORT LAYER M2 ; PATH 0 69 0 75 ; END
          END Z
          PIN PO DIRECTION OUTPUT ;
               PORT LAYER M2 ; PATH 0 81 0 87 ; END
          END PO
          PIN A DIRECTION INPUT ;
               PORT LAYER M2 ; PATH 0 51 0 57 ; END
          END A
          PIN PI DIRECTION INPUT ;
               PORT LAYER M2 ; PATH 0 39 0 45 ; END
          END PI
          PIN VDD DIRECTION INOUT ; SHAPE ABUTMENT ;
               PORT LAYER M2 ; WIDTH 20 ; PATH 236.8 10 236.8 140 ; END
          END VDD
          PIN VSS DIRECTION INOUT ; SHAPE ABUTMENT ;
               PORT LAYER M2 ; WIDTH 20 ; PATH 336.8 10 336.8 140 ; END
          END VSS
     END IN1Y

MACRO FILLER FOREIGN FILLER ; SIZE 67.2 BY 6 ; SYMMETRY X Y;
          SITE CORE1 ;
          PIN VDD DIRECTION INOUT ; SHAPE ABUTMENT ;
               PORT LAYER M1 ; RECT 45.8 0 55 6 ; END
          END VDD
          PIN VSS DIRECTION INOUT ; SHAPE ABUTMENT ;
               PORT LAYER M1 ; RECT 12.2 0 21.4 6 ; END
          END VSS
          OBS LAYER M1 ; RECT 24.1 1.5 43.5 4.5 ; END
     END FILLER

MACRO INV FOREIGN INVS ; SIZE 67.2 BY 24 ; SYMMETRY X Y ; SITE CORE1 ;
          PIN Z DIRECTION OUTPUT ;
               PORT LAYER M2 ; PATH 30.8 9 42 9 ; END
          END Z
          PIN A DIRECTION INPUT ;
               PORT LAYER M1 ; PATH 25.2 15 ; END
          END A
          PIN VDD DIRECTION INOUT ; SHAPE ABUTMENT ;
               PORT LAYER M1 ; WIDTH 5.6 ; PATH 50.4 4.6 50.4 13.4 ; END
          END VDD
          PIN VSS DIRECTION INOUT ; SHAPE ABUTMENT ;
```


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```
            PORT LAYER M1 ; WIDTH 5.6 ; PATH 16.8 4.6 16.8 13.4 ; END
        END VSS
        OBS LAYER M1 ; RECT 24.1 1.5 43.5 16.5 ; END
    END INV

MACRO BUF FOREIGN BUFS ; SIZE 67.2 BY 126 ; SYMMETRY X Y ; SITE
        CORE1 ;
        PIN Z DIRECTION OUTPUT ;
            PORT LAYER M2 ; PATH 25.2 39 42 39 ; END
        END Z
        PIN A DIRECTION INPUT ;
            PORT LAYER M1 ; PATH 30.8 33 ; END
        END A
        PIN VDD DIRECTION INOUT ; SHAPE FEEDTHRU ;
            PORT LAYER M1 ; WIDTH 5.6 ;
            PATH 50.4 4.6 50.4 10.0 56.0 10.0 56.0 115.8 50.4 115.8
                50.4 121.4 ; END
        END VDD
        PIN VSS DIRECTION INOUT ; SHAPE FEEDTHRU ;
            PORT LAYER M1 ; WIDTH 5.6 ;
            PATH 16.8 4.6 16.8 10.0 11.2 10.0 11.2 115.8 16.8 115.8
                16.8 121.4 ; END
        END VSS
        OBS LAYER M1 ; RECT 24.1 1.5 43.5 124.5 ; END
    END BUF

MACRO BIDIR1X FOREIGN BIDIR1X ; class pad ; SIZE 151.2 BY 444 ;
        SYMMETRY X ; SITE IOX ;
        PIN IO DIRECTION INOUT ;
            PORT LAYER M1 ; PATH 61.6 444 67.2 444 ; END
        END IO
        PIN ZI DIRECTION OUTPUT ;
            PORT LAYER M1 ; PATH 78.4 444 84.0 444 ; END
        END ZI
        PIN PO DIRECTION OUTPUT ;
            PORT LAYER M1 ; PATH 95.2 444 100.8 444 ; END
        END PO
        PIN A DIRECTION INPUT ;
            PORT LAYER M1 ; PATH 106.4 444 112.0 444 ; END
        END A
        PIN EN DIRECTION INPUT ;
            PORT LAYER M1 ; PATH 134.4 444 140.0 444 ; END
        END EN
        PIN TN DIRECTION INPUT ;
            PORT LAYER M1 ; PATH 28.0 444 33.6 444 ; END
        END TN
        PIN PI DIRECTION INPUT ;
            PORT LAYER M1 ; PATH 44.8 444 50.4 444 ; END
        END PI
        PIN VDD DIRECTION INOUT ; SHAPE ABUTMENT ;
            PORT LAYER M2 ; WIDTH 20 ; PATH 10 200 141.2 200 ; END
        END VDD
```


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```
          PIN VSS DIRECTION INOUT ; SHAPE ABUTMENT ;
               PORT LAYER M1 ; WIDTH 20 ; PATH 10 100 141.2 100 ; END
          END VSS
     END BIDIR1X

MACRO BIDIR1Y EEQ BIDIR1X ; class pad ; FOREIGN BIDIR1Y ; SIZE 436.8
          BY 150 ; SYMMETRY Y ; SITE IOY ;
          PIN IO DIRECTION INOUT ;
               PORT LAYER M2 ; PATH 0 69 0 75 ; END END IO
          PIN ZI DIRECTION OUTPUT ;
               PORT LAYER M2 ; PATH 0 93 0 99 ; END END ZI
          PIN PO DIRECTION OUTPUT ;
               PORT LAYER M2 ; PATH 0 81 0 87 ; END END PO
          PIN A DIRECTION INPUT ;
               PORT LAYER M2 ; PATH 0 15 0 21 ; END END A
          PIN EN DIRECTION INPUT ;
               PORT LAYER M2 ; PATH 0 27 0 33 ; END END EN
          PIN TN DIRECTION INPUT ;
               PORT LAYER M2 ; PATH 0 39 0 45 ; END END TN
          PIN PI DIRECTION INPUT ;
               PORT LAYER M2 ; PATH 0 51 0 57 ; END END PI
          PIN VDD DIRECTION INOUT ; SHAPE ABUTMENT ;
               PORT LAYER M2 ; WIDTH 20 ; PATH 236.8 10 236.8 140 ; END
          END VDD
          PIN VSS DIRECTION INOUT ; SHAPE ABUTMENT ;
               PORT LAYER M2 ; WIDTH 20 ; PATH 336.8 10 336.8 140 ; END
               END VSS
     END BIDIR1Y

MACRO OR2 FOREIGN OR2S ; SIZE 67.2 BY 42 ; SYMMETRY X Y ; SITE
          CORE1 ;
          PIN Z DIRECTION OUTPUT ;
               PORT LAYER M2 ; PATH 25.2 39 42 39 ; END
          END Z
          PIN A DIRECTION INPUT ;
               PORT LAYER M1 ; PATH 25.2 15 ; END
          END A
          PIN B DIRECTION INPUT ;
               PORT LAYER M1 ; PATH 25.2 3 ; END
          END B
          PIN VDD DIRECTION INOUT ; SHAPE FEEDTHRU ;
               PORT LAYER M1 ; WIDTH 5.6 ;
               PATH 50.4 4.6 50.4 10.0 ; PATH 50.4 27.4 50.4 37.4 ;
               VIA 50.4 3 C2PW ; VIA 50.4 21 C2PW ; VIA 50.4 33 C2PW ;
               VIA 50.4 39 C2PW ; END
          END VDD
          PIN VSS DIRECTION INOUT ; SHAPE FEEDTHRU ;
               PORT LAYER M1 ; WIDTH 5.6 ; PATH 16.8 4.6 16.8 10.0 ;
               PATH 16.8 27.4 16.8 37.4 ;
               VIA 16.8 3 C2NW ; VIA 16.8 15 C2NW ; VIA 16.8 21 C2NW ;
               VIA 16.8 33 C2NW ; VIA 16.8 39 C2NW ; END
```


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```
          END VSS
          OBS LAYER M1 ; RECT 24.1 1.5 43.5 40.5 ; END
     END OR2

MACRO AND2 FOREIGN AND2S ; SIZE 67.2 BY 84 ; SYMMETRY X Y ; SITE
          CORE1 ;
          PIN Z DIRECTION OUTPUT ;
               PORT LAYER M2 ; PATH 25.2 39 42 39 ; END
          END Z
          PIN A DIRECTION INPUT ;
               PORT LAYER M1 ; PATH 42 15 ; END
          END A
          PIN B DIRECTION INPUT ;
               PORT LAYER M1 ; PATH 42 3 ; END
          END B
          PIN VDD DIRECTION INOUT ; SHAPE ABUTMENT ;
               PORT LAYER M1 ; WIDTH 5.6 ; PATH 50.4 4.6 50.4 79.4 ; END
          END VDD
          PIN VSS DIRECTION INOUT ; SHAPE ABUTMENT ;
               PORT LAYER M1 ; WIDTH 5.6 ; PATH 16.8 4.6 16.8 79.4 ; END
          END VSS
          OBS LAYER M1 ; RECT 24.1 1.5 43.5 82.5 ; END
     END AND2

MACRO DFF3 FOREIGN DFF3S ; SIZE 67.2 BY 210 ; SYMMETRY X Y ; SITE
          CORE1 ;
          PIN Q DIRECTION OUTPUT ;
               PORT LAYER M2 ; PATH 19.6 99 47.6 99 ; END
          END Q
          PIN QN DIRECTION OUTPUT ;
               PORT LAYER M2 ; PATH 25.2 123 42 123 ; END
          END QN
          PIN D DIRECTION INPUT ;
               PORT LAYER M1 ; PATH 30.8 51 ; END
          END D
          PIN G DIRECTION INPUT ;
               PORT LAYER M1 ; PATH 25.2 3 ; END
          END G
          PIN CD DIRECTION INPUT ;
               PORT LAYER M1 ; PATH 36.4 75 ; END
          END CD
          PIN VDD DIRECTION INOUT ; SHAPE FEEDTHRU ;
               PORT LAYER M1 ; WIDTH 5.6 ; PATH 50.4 4.6 50.4 205.4 ;
          END
          END VDD
          PIN VSS DIRECTION INOUT ; SHAPE FEEDTHRU ;
               PORT LAYER M1 ; WIDTH 5.6 ; PATH 16.8 4.6 16.8 205.4 ;
          END
          END VSS
          OBS LAYER M1 ; RECT 24.1 1.5 43.5 208.5 ; PATH 8.4 3 8.4 123 ;
               PATH 58.8 3 58.8 123 ; PATH 64.4 3 64.4 123; END
     END DFF3
```


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```
MACRO NOR2 FOREIGN NOR2S ; SIZE 67.2 BY 42 ; SYMMETRY X Y ; SITE
          CORE1 ;
          PIN Z DIRECTION OUTPUT ;
               PORT LAYER M1 ; PATH 42 33 ; END
          END Z
          PIN A DIRECTION INPUT ;
               PORT LAYER M1 ; PATH 25.2 15 ; END
          END A
          PIN B DIRECTION INPUT ;
               PORT LAYER M1 ; PATH 36.4 9 ; END
          END B
          PIN VDD DIRECTION INOUT ; SHAPE FEEDTHRU ;
               PORT LAYER M1 ; WIDTH 5.6 ; PATH 50.4 4.6 50.4 37.4 ; END
          END VDD
          PIN VSS DIRECTION INOUT ; SHAPE FEEDTHRU ;
               PORT LAYER M1 ; WIDTH 5.6 ; PATH 16.8 4.6 16.8 37.4 ; END
          END VSS
          OBS LAYER M1 ; RECT 24.1 1.5 43.5 40.5 ; END
     END NOR2

MACRO AND2J EEQ AND2 ;FOREIGN AND2SJ ; SIZE 67.2 BY 48 ;
          SYMMETRY X Y ; ORIGIN 0 6 ; SITE CORE1 ;
          PIN Z DIRECTION OUTPUT ;
               PORT LAYER M2 ; PATH 25.2 33 42 33 ; END
          END Z
          PIN A DIRECTION INPUT ;
               PORT LAYER M1 ; PATH 42 15 ; END
          END A
          PIN B DIRECTION INPUT ;
               PORT LAYER M1 ; PATH 42 3 ; END
          END B
          PIN VDD DIRECTION INOUT ; SHAPE FEEDTHRU ;
               PORT LAYER M1 ; WIDTH 5.6 ; PATH 50.4 -1.4 50.4 37.4 ;
          END
          END VDD
          PIN VSS DIRECTION INOUT ; SHAPE FEEDTHRU ;
               PORT LAYER M1 ; WIDTH 5.6 ; PATH 16.8 -1.4 16.8 37.4 ;
          END
          END VSS
          OBS LAYER M1 ; RECT 24.1 1.5 43.5 34.5 ; END
     END AND2J

MACRO SQUAREBLOCK FOREIGN SQUAREBLOCKS ; CLASS RING ;SIZE 268.8
          BY 252 ; SITE SQUAREBLOCK ;
          PIN Z DIRECTION OUTPUT ;
               PORT LAYER M2 ; PATH 22.8 21 246.0 21 ; END
          END Z
          PIN A DIRECTION OUTPUT ;
               PORT LAYER M2 ; PATH 64.4 33 137.2 33 ;
               PATH 137.2 33 137.2 69 ; PATH 137.2 69 204.4 69 ; END
          END A
          PIN B DIRECTION INPUT ;
```


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```
            PORT LAYER M2 ; PATH 22.8 129 246.0 129 ; END
        END B
        PIN C DIRECTION INPUT ;
            PORT LAYER M2 ; PATH 70 165 70 153 ; PATH 70 153 126 153 ;
            END
        END C
        PIN D DIRECTION INPUT ;
            PORT LAYER M2 ; PATH 22.8 75 64.4 75 ; END
        END D
        PIN E DIRECTION INPUT ;
            PORT LAYER M2 ; PATH 22.8 87 64.4 87 ; END
        END E
        PIN F DIRECTION INPUT ;
            PORT LAYER M2 ; PATH 22.8 99 64.4 99 ; END
        END F
        PIN G DIRECTION INPUT ;
            PORT LAYER M2 ; PATH 22.8 111 64.4 111 ; END
        END G
        PIN VDD DIRECTION INOUT ; SHAPE RING ;
            PORT LAYER M1 ; WIDTH 3.6 ; PATH 4.0 3.5 4.0 248 ;
            PATH 264.8 100 264.8 248 ; PATH 150 3.5 150 100 ;
            LAYER M2 ; WIDTH 3.6 ; PATH 4.0 3.5 150 3.5 ;
            PATH 150 100 264.8 100 ; PATH 4.0 248 264.8 248 ; END
        END VDD
        PIN VSS DIRECTION INOUT ; SHAPE RING ;
            PORT LAYER M1 ; WIDTH 3.6 ; PATH 10 10 10 150 ;
            PATH 100 150 100 200 ; PATH 50 200 50 242 ;
            PATH 258.8 10 258.8 242 ; LAYER M2 ; WIDTH 3.6 ;
            PATH 10 150 100 150 ; PATH 100 200 50 200 ;
            PATH 10 10 258.8 10 ; PATH 50 242 258.8 242 ; END
        END VSS
        OBS LAYER M1 ; RECT 13.8 14.0 255.0 237.2 ; END
    END SQUAREBLOCK

MACRO I2BLOCK FOREIGN I2BLOCKS ; CLASS RING ; SIZE 672 BY 504 ;
        SITE I2BLOCK ;
        PIN Z DIRECTION OUTPUT ;
            PORT LAYER M2 ; PATH 22.8 21 649.2 21 ; END
        END Z
        PIN A DIRECTION OUTPUT ;
            PORT LAYER M2 ; PATH 22.8 63 154.0 63 ; PATH 154.0 63 154.0
            129;
            PATH 154.0 129 447.6 129 ; END
        END A
        PIN B DIRECTION INPUT ;
            PORT LAYER M2 ; PATH 137.2 423 447.6 423 ; END
        END B
        PIN C DIRECTION INPUT ;
            PORT LAYER M2 ; PATH 204.4 165 271.6 165 ; END
        END C
        PIN D DIRECTION INPUT ;
```


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```
            PORT LAYER M2 ; PATH 204.4 171 271.6 171 ; END
        END D
        PIN E DIRECTION INPUT ;
            PORT LAYER M1 ; PATH 204.4 213 204.4 213 ; END
        END E
        PIN F DIRECTION INPUT ;
            PORT LAYER M1 ; PATH 406 249 406 273 ; END
        END F
        PIN G DIRECTION INPUT ;
            PORT LAYER M1 ; PATH 338.8 249 338.8 273 ; END
        END G
        PIN H DIRECTION INPUT ;
            PORT LAYER M1 ; PATH 372.4 357 372.4 381 ; END
        END H
        PIN VDD DIRECTION INOUT ; SHAPE RING ;
            PORT LAYER M1 ; WIDTH 3.6 ; PATH 668 3.5 668 80.5 ;
            PATH 467 80.5 467 465.5 ; PATH 668 465.5 668 500.5 ;
            PATH 4 500.5 4 465.5 ; PATH 138 465.5 138 80.5 ;
            PATH 4 80.5 4 3.5 ; LAYER M2 ; WIDTH 3.6 ; PATH 4 3.5 668 3.5;
            PATH 668 80.5 467 80.5 ; PATH 467 465.5 668 465.5 ;
            PATH 668 500.5 4 500.5 ; PATH 4 465.5 138 465.5 ;
            PATH 138 80.5 4 80.5 ; END
        END VDD
        PIN VSS DIRECTION INOUT ; SHAPE RING ;
            PORT LAYER M1 ; WIDTH 3.6 ; PATH 662 10 662 74 ;
            PATH 461 74 461 472 ; PATH 662 472 662 494 ; PATH 10 494 10
                472;
            PATH 144 472 144 74 ; PATH 10 74 10 10 ;LAYER M2 ; WIDTH
                3.6 ;
            PATH 10 10 662 10 ; PATH 662 74 461 74 ; PATH 461 472 662
                472 ;
            PATH 662 494 10 494 ; PATH 10 472 144 472 ; PATH 144 74 10
                74 ;
            END
        END VSS
        OBS LAYER M1 ; RECT 14 14 658 70 ; RECT 14 476 658 490 ;
            RECT 148 14 457 490 ; # rectilinear shape description
            LAYER OVERLAP ; RECT 0 0 672 84 ; RECT 134.4 84 470.4 462 ;
            RECT 0 462 672 504 ; END
    END I2BLOCK

MACRO LBLOCK FOREIGN LBLOCKS ; CLASS RING ; SIZE 201.6 BY 168 ; SITE
        LBLOCK ;
        PIN Z DIRECTION OUTPUT ;
            PORT LAYER M2 ; PATH 2.8 15 198.8 15 ; END
        END Z
        PIN A DIRECTION OUTPUT ;
            PORT LAYER M2 ; PATH 2.8 81 137.2 81 ; PATH 137.2 81 137.2
                69 ;
            PATH 137.2 69 198.8 69 ; END
        END A
```


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```
          PIN B DIRECTION INPUT ;
               PORT LAYER M2 ; PATH 2.8 165 64.4 165 ; END
          END B
          PIN C DIRECTION INPUT ;
               PORT LAYER M1 ; PATH 2.8 93 2.8 105 ; END
          END C
          PIN D DIRECTION INPUT ;
               PORT LAYER M1 ; PATH 64.4 93 64.4 105 ; END
          END D
          PIN E DIRECTION INPUT ;
               PORT LAYER M1 ; PATH 198.8 39 198.8 39 ; END
          END E
          PIN F DIRECTION INPUT ;
               PORT LAYER M1 ; PATH 198.8 45 198.8 45 ; END
          END F
          PIN G DIRECTION INPUT ;
               PORT LAYER M1 ; PATH 2.8 111 2.8 111 ; END END G
               PORT LAYER M2 ; WIDTH 3.6 ; PATH 1.8 27 199.8 27 ; END
          END VDD
          PIN VSS DIRECTION INOUT ;
               PORT LAYER M2 ; WIDTH 3.6 ; PATH 1.8 57 199.8 57 ; END
          END VSS
          OBS LAYER M2 ; RECT 1.0 80 66.2 166.5 ; RECT 1.0 1.5 200.6 23 ;
               RECT 1.0 31 200.6 53 ; RECT 1.0 61 200.6 82.5 ;
               # rectilinear shape description
               LAYER OVERLAP ; RECT 0 0 201.6 84 ; RECT 0 84 67.2 168 ;
               END
     END LBLOCK

END LIBRARY
```

DEF

The following example shows a design netlist.

```
DESIGN DEMO4CHIP ;
     TECHNOLOGY DEMO4CHIP ;
     UNITS DISTANCE MICRONS 100 ;
     COMPONENTS 243 ;

- CORNER1 CORNER ; - CORNER2 CORNER ; - CORNER3 CORNER ;
     - CORNER4 CORNER ; - C01 IN1X ; - C02 IN1Y ; - C04 IN1X ;
     - C05 IN1X ; - C06 IN1Y ;
     - C07 IN1Y ; - C08 IN1Y ; - C09 IN1Y ; - C10 IN1X ; - C11 IN1X ;
     - C13 BIDIR1Y ; - C14 INV ; - C15 BUF ; - C16 BUF ; - C17 BUF ;
     - C19 BIDIR1Y ; - C20 INV ; - C21 BUF ; - C22 BUF ; - C23 BUF ;
     - C25 BIDIR1Y ; - C26 INV ; - C27 BUF ; - C28 BUF ; - C29 BUF ;
     - C31 BIDIR1Y ; - C32 INV ; - C33 BUF ; - C34 BUF ; - C35 BUF ;
     - C37 BIDIR1X ; - C39 INV ; - C40 BUF ; - C41 BUF ; - C42 BUF ;
     - C44 BIDIR1X ; - C45 INV ; - C46 BUF ; - C47 BUF ; - C48 BUF ;
```


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    - C50 BIDIR1Y ; - C51 INV ; - C52 BUF ; - C53 BUF ; - C54 BUF ;
    - C56 BIDIR1X ; - C57 INV ; - C58 BUF ; - C59 BUF ; - C60 BUF ;
    - D02 BIDIR1X ; - D03 INV ; - D04 BUF ; - D05 BUF ; - D06 BUF ;
    - D08 BIDIR1X ; - D09 INV ; - D10 BUF ; - D11 BUF ; - D12 BUF ;
    - D14 BIDIR1X ; - D15 INV ; - D16 BUF ; - D17 BUF ; - D19 BUF ;
    - D33 BIDIR1Y ; - D34 INV ; - D35 BUF ; - D36 BUF ; - D37 BUF ;
    - D39 BIDIR1Y ; - D40 INV ; - D41 BUF ; - D42 BUF ; - D43 BUF ;
    - D45 BIDIR1Y ; - D46 INV ; - D47 BUF ; - D48 BUF ; - D49 BUF ;
    - D82 OR2 ; - D83 OR2 ; - D84 OR2 ; - D85 OR2 ; - D86 OR2 ;
    - D87 OR2 ; - D88 OR2 ; - D89 OR2 ; - D90 OR2 ; - D91 OR2 ;
    - D92 OR2 ; - D93 OR2 ;
    - E01 AND3 ; - E02 AND3 ; - E03 AND3 ; - E04 AND3 ; - E05 AND3 ;
    - E06 AND3 ; - E07 AND3 ; - E08 AND3 ; - E09 AND3 ; - E10 AND3 ;
    - E11 AND3 ; - E12 AND3 ; - E13 AND3 ; - E14 AND3 ; - E15 AND3 ;
    - E16 AND3 ;
    - EE16 IN1X ; - E17 IN1X ; - E18 IN1X ; - E19 IN1X ; - E20 IN1X ;
    - E21 IN1X ; - E22 IN1X ; - E23 IN1Y ; - E24 IN1Y ; - E25 IN1Y ;
    - E26 INV ; - E27 AND2 ; - E28 AND2 ; - E29 AND2 ; - E30 AND2 ;
    - E31 AND2 ; - E32 AND2 ; - E33 OR2 ; - E34 OR2 ; - E35 OR2 ;
    - E36 OR2 ; - E37 IN1Y ; - E38A01 DFF3 ; - E38A02 DFF3 ;
    - E38A03 DFF3 ;- E38A04 DFF3 ; - E38A05 DFF3 ; - F01 I2BLOCK ;
    - F04 OR2 ; - F06 OR2 ; - F07 OR2 ; - F08 OR2 ; - F09 SQUAREBLOCK ;
    - F12 LBLOCK ;
    - Z14 INV ; - Z15 BUF ; - Z16 BUF ; - Z17 BUF ; - Z20 INV ;
    - Z21 BUF ; - Z22 BUF ; - Z23 BUF ; - Z26 INV ; - Z27 BUF ;
    - Z28 BUF ; - Z29 BUF ; - Z32 INV ; - Z33 BUF ; - Z34 BUF ;
    - Z35 BUF ; - Z39 INV ; - Z40 BUF ; - Z41 BUF ; - Z42 BUF ;
    - Z45 INV ; - Z46 BUF ; - Z47 BUF ; - Z48 BUF ; - Z51 INV ;
    - Z52 BUF ; - Z53 BUF ; - Z54 BUF ; - Z57 INV ; - Z58 BUF ; - Z59 BUF ;
    - Z60 BUF ; - Z103 INV ; - Z104 BUF ; - Z105 BUF ; - Z106 BUF ;
    - Z109 INV ; - Z110 BUF ; - Z111 BUF ; - Z112 BUF ; - Z115 INV ;
    - Z116 BUF ; - Z117 BUF ; - Z119 BUF ; - Z134 INV ; - Z135 BUF ;
    - Z136 BUF ; - Z137 BUF ; - Z140 INV ; - Z141 BUF ; - Z142 BUF ;
    - Z143 BUF ; - Z146 INV ; - Z147 BUF ; - Z148 BUF ; - Z149 BUF ;
    - Z182 OR2 ; - Z183 OR2 ; - Z184 OR2 ; - Z185 OR2 ; - Z186 OR2 ;
    - Z187 OR2 ; - Z188 OR2 ; - Z189 OR2 ; - Z190 OR2 ; - Z191 OR2 ;
    - Z192 OR2 ; - Z193 OR2 ; - Z201 AND3 ; - Z202 AND3 ; - Z203 AND3 ;
    - Z204 AND3 ; - Z205 AND3 ; - Z206 AND3 ; - Z207 AND3 ; - Z208 AND3 ;
    - Z209 AND3 ; - Z210 AND3 ; - Z211 AND3 ; - Z212 AND3 ; - Z213 AND3 ;
    - Z214 AND3 ; - Z215 AND3 ; - Z216 AND3 ; - Z226 INV ; - Z227 AND2 ;
    - Z228 AND2 ; - Z229 AND2 ; - Z230 AND2 ; - Z231 AND2 ; - Z232 AND2 ;
    - Z233 OR2 ; - Z234 OR2 ; - Z235 OR2 ; - Z236 OR2 ; - Z38A01 DFF3 ;
    - Z38A02 DFF3 ; - Z38A03 DFF3 ; - Z38A04 DFF3 ; - Z38A05 DFF3 ;
    END COMPONENTS

NETS 222 ;
    - VDD ( Z216 B ) ( Z215 B ) ( Z214 C ) ( Z214 B )
    ( Z213 C ) ( Z213 B ) ( Z212 C ) ( Z212 B ) ( Z211 C ) ( Z211 B )
    ( Z210 C ) ( E23 Z ) ( Z143 Z ) ( Z142 Z ) ( Z141 Z ) ( Z119 Z )
    ( Z117 Z ) ( Z116 Z ) ( Z106 Z ) ( Z105 Z ) ( Z104 Z ) ( Z34 Z )
    ( Z33 Z ) ( Z28 Z ) ( Z27 Z ) ( Z22 Z ) ( Z21 Z ) ( Z16 Z )
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( Z15 Z ) ( D45 PO ) ( D14 PO ) ( C01 PI ) ( D45 TN ) ( D39 TN )
( D33 TN ) ( D14 TN ) ( D08 TN ) ( D02 TN ) ( C56 TN ) ( C50 TN )
( C44 TN ) ( C37 TN ) ( C31 TN ) ( C25 TN ) ( C19 TN ) ( C13 TN ) ;
- VSS ( Z209 C ) ( Z208 C ) ( Z207 C ) ( Z206 C ) ( Z205 C )
( Z204 C ) ( Z203 C ) ( Z202 C ) ( Z201 C ) ( Z149 Z ) ( Z148 Z )
( Z147 Z ) ( Z137 Z ) ( Z136 Z ) ( Z135 Z ) ( Z112 Z ) ( Z111 Z )
( Z110 Z ) ( Z60 Z ) ( Z59 Z ) ( Z58 Z ) ( Z54 Z ) ( Z53 Z )
( Z52 Z ) ( Z47 Z ) ( Z46 Z ) ( Z41 Z ) ( Z40 Z ) ( E18 Z )
( D49 Z ) ( D43 Z ) ( D45 A ) ( D39 A ) ( D33 A ) ( D14 A )
( D08 A ) ( D02 A ) ( C56 A ) ( C50 A ) ( C44 A ) ( C37 A )
( C31 A ) ( C25 A ) ( C19 A ) ( C13 A ) ; - XX1001 ( Z38A04 G )
( Z38A02 G ) ; - XX100 ( Z38A05 G ) ( Z38A03 G ) ( Z38A01 G ) ;
- XX907 ( Z236 B ) ( Z235 B ) ; - XX906 ( Z234 B ) ( Z233 B ) ;
- XX904 ( Z232 B ) ( Z231 B ) ; - XX903 ( Z230 B ) ( Z229 B ) ;
- XX902 ( Z228 B ) ( Z227 B ) ;
- XX900 ( Z235 A ) ( Z233 A ) ( Z232 A ) ( Z230 A ) ( Z228 A ) ( Z226 A ) ;
- Z38QN4 ( Z38A04 QN ) ( Z210 B ) ; - COZ131 ( Z38A04 Q ) ( Z210 A ) ;
- Z38QN3 ( Z38A03 QN ) ( Z209 B ) ; - COZ121 ( Z38A03 Q ) ( Z209 A ) ;
- Z38QN2 ( Z38A02 QN ) ( Z208 B ) ; - COZ111 ( Z38A02 Q ) ( Z208 A ) ;
- Z38QN1 ( Z38A01 QN ) ( Z207 B ) ; - COZ101 ( Z38A01 Q ) ( Z207 A ) ;
- XX901 ( Z236 A ) ( Z234 A ) ( Z231 A ) ( Z229 A ) ( Z227 A ) ( Z226 Z )
     ( Z193 A ) ;
- X415 ( Z149 A ) ( Z148 A ) ( Z147 A ) ( Z146 Z ) ; - X413 ( Z143 A )
     ( Z142 A ) ( Z141 A ) ( Z140 Z ) ;
- X411 ( Z137 A ) ( Z136 A ) ( Z135 A ) ( Z134 Z ) ;
- X405 ( Z119 A ) ( Z117 A ) ( Z116 A ) ( Z115 Z ) ;
- X403 ( Z112 A ) ( Z111 A ) ( Z110 A ) ( Z109 Z ) ;
- X401 ( Z106 A ) ( Z105 A ) ( Z104 A ) ( Z103 Z ) ;
- X315 ( Z60 A ) ( Z59 A ) ( Z58 A ) ( Z57 Z ) ;
- X313 ( Z54 A ) ( Z53 A ) ( Z52 A ) ( Z51 Z ) ;
- DIS051 ( Z216 A ) ( Z48 Z ) ;
- X311 ( Z48 A ) ( Z47 A ) ( Z46 A ) ( Z45 Z ) ;
- DIS041 ( Z215 A ) ( Z42 Z ) ; - X309 ( Z42 A ) ( Z41 A ) ( Z40 A )
     ( Z39 Z ) ;
- X307 ( Z35 A ) ( Z34 A ) ( Z33 A ) ( Z32 Z ) ;
- DIS031 ( Z214 A ) ( Z35 Z ) ; - DIS021 ( Z213 A ) ( Z29 Z ) ;
- X305 ( Z29 A ) ( Z28 A ) ( Z27 A ) ( Z26 Z ) ;
- DIS011 ( Z212 A ) ( Z23 Z ) ;
- X303 ( Z23 A ) ( Z22 A ) ( Z21 A ) ( Z20 Z ) ;
- DIS001 ( Z211 A ) ( Z17 Z ) ;
- X301 ( Z17 A ) ( Z16 A ) ( Z15 A ) ( Z14 Z ) ;
- X1000 ( E38A05 G ) ( E38A03 G ) ( E38A01 G ) ( E37 Z ) ;
- CNTEN ( Z38A05 Q ) ( E38A05 Q ) ( E25 A ) ;
- VIH20 ( E37 PI ) ( E25 PO ) ; - X0907 ( E36 B ) ( E35 B ) ( E25 Z ) ;
- CCLK0 ( F09 A ) ( E24 A ) ; - VIH19 ( E25 PI ) ( E24 PO ) ;
- X0906 ( E34 B ) ( E33 B ) ( E24 Z ) ; - CATH1 ( F09 Z ) ( E23 A ) ;
- VIH18 ( E24 PI ) ( E23 PO ) ; - CRLIN ( F08 Z ) ( E22 A ) ;
- VIH17 ( E23 PI ) ( E22 PO ) ; - X0904 ( E32 B ) ( E31 B ) ( E22 Z ) ;
- NXLIN ( F07 Z ) ( E21 A ) ; - VIH16 ( E22 PI ) ( E21 PO ) ;
- X0903 ( E30 B ) ( E29 B ) ( E21 Z ) ; - RPT1 ( F06 Z ) ( E20 A ) ;
- VIH15 ( E21 PI ) ( E20 PO ) ; - X0902 ( E28 B ) ( E27 B ) ( E20 Z ) ;
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- AGISL ( F04 Z ) ( E19 A ) ; - VIH14 ( E20 PI ) ( E19 PO ) ;
- X0900 ( E35 A ) ( E33 A ) ( E32 A ) ( E30 A ) ( E28 A ) ( E26 A )
    ( E19 Z ) ;
- TSTCN ( Z38A05 QN ) ( E38A05 QN ) ( E18 A ) ;
- VIH13 ( E19 PI ) ( E18 PO ) ; - BCLK1 ( F01 A ) ( E17 A ) ;
- VIH12 ( E18 PI ) ( E17 PO ) ; - CLR0 ( F01 Z ) ( EE16 A ) ;
- VIH11 ( E17 PI ) ( EE16 PO ) ; - BCLKX1 ( Z216 C ) ( E17 Z )
    ( E16 C ) ; - CLRX0 ( Z38A05 CD ) ( Z38A03 CD ) ( Z38A01 CD )
    ( Z215 C ) ( E38A05 CD ) ( E38A03 CD ) ( E38A01 CD ) ( EE16 Z )
    ( E15 C ) ; - E38QN4 ( E38A04 QN ) ( E10 B ) ;
- CAX131 ( E38A04 Q ) ( E10 A ) ; - E38QN3 ( E38A03 QN ) ( E09 B ) ;
- CAX121 ( E38A03 Q ) ( E09 A ) ; - E38QN2 ( E38A02 QN ) ( E08 B ) ;
- CAX111 ( E38A02 Q ) ( E08 A ) ; - E38QN1 ( E38A01 QN ) ( E07 B ) ;
- CAX101 ( E38A01 Q ) ( E07 A ) ;
- SDD111 ( Z38A05 D ) ( Z205 Z ) ( E38A05 D ) ( E05 Z ) ;
- SDD121 ( Z38A04 D ) ( Z204 Z ) ( E38A04 D ) ( E04 Z ) ;
- X0901 ( E36 A ) ( E34 A ) ( E31 A ) ( E29 A ) ( E27 A ) ( E26 Z )
    ( D93 A ) ;
- VIH21 ( Z192 A ) ( E37 PO ) ( D92 A ) ;
- STRDENB0 ( Z206 B ) ( Z202 B ) ( Z201 B ) ( Z189 B ) ( Z188 B )
    ( F12 A ) ( E06 B ) ( E02 B ) ( E01 B ) ( D89 B ) ( D88 B ) ;
- STRDENA0 ( Z202 A ) ( Z201 A ) ( Z183 B ) ( Z182 B ) ( F12 Z )
    ( F01 H ) ( E02 A ) ( E01 A ) ( D83 B ) ( D82 B ) ;
- DAB151 ( F12 H ) ( D48 Z ) ; - DAA151 ( F08 B ) ( D47 Z ) ;
- X0415 ( D49 A ) ( D48 A ) ( D47 A ) ( D46 Z ) ;
- SDD151 ( Z38A01 D ) ( Z201 Z ) ( E38A01 D ) ( E01 Z ) ( D45 EN ) ;
- X0414 ( Z146 A ) ( D46 A ) ( D45 ZI ) ; - D151 ( E14 C ) ( D45 IO ) ;
- DAB141 ( F12 G ) ( D42 Z ) ; - DAA141 ( F08 A ) ( D41 Z ) ;
- X0413 ( D43 A ) ( D42 A ) ( D41 A ) ( D40 Z ) ;
- SDD141 ( Z38A02 D ) ( Z202 Z ) ( E38A02 D ) ( E02 Z ) ( D39 EN ) ;
- VIH60 ( D45 PI ) ( D39 PO ) ; - X0412 ( Z140 A ) ( D40 A ) ( D39 ZI ) ;
- D141 ( E13 C ) ( D39 IO ) ; - SDI131 ( E16 B ) ( D37 Z ) ;
- DAB131 ( F12 F ) ( D36 Z ) ; - DAA131 ( F07 B ) ( D35 Z ) ;
- X0411 ( D37 A ) ( D36 A ) ( D35 A ) ( D34 Z ) ;
- VIH58 ( Z193 Z ) ( D93 Z ) ( D33 PI ) ;
- SDD131 ( Z38A03 D ) ( Z203 Z ) ( E38A03 D ) ( E03 Z ) ( D33 EN ) ;
- VIH59 ( D39 PI ) ( D33 PO ) ; - X0410 ( Z134 A ) ( D34 A ) ( D33 ZI ) ;
- D131 ( E12 C ) ( D33 IO ) ; - SDI101 ( E15 B ) ( D19 Z ) ; ...
- X0315 ( C60 A ) ( C59 A ) ( C58 A ) ( C57 Z ) ;
- SDD071 ( Z211 Z ) ( E11 Z ) ( C56 EN ) ;
- VIH53 ( Z190 Z ) ( D90 Z ) ( D02 PI ) ( C56 PO ) ;
- X0314 ( Z57 A ) ( C57 A ) ( C56 ZI ) ;
- D071 ( E08 C ) ( C56 IO ) ; - SDI061 ( E11 B ) ( C54 Z ) ;
- DAB061 ( F09 H ) ( C53 Z ) ; - DAA061 ( F04 A ) ( C52 Z ) ;
- X0313 ( C54 A ) ( C53 A ) ( C52 A ) ( C51 Z ) ;
- SDD061 ( Z212 Z ) ( E12 Z ) ( C50 EN ) ;
- VIH52 ( Z189 Z ) ( D89 Z ) ( C56 PI ) ( C50 PO ) ;
- X0312 ( Z51 A ) ( C51 A ) ( C50 ZI ) ;
- D061 ( E07 C ) ( C50 IO ) ; - SDI051 ( E16 A ) ( C48 Z ) ;
- DAB051 ( F09 G ) ( C47 Z ) ; - DAA051 ( F01 G ) ( C46 Z ) ;
- X0311 ( C48 A ) ( C47 A ) ( C46 A ) ( C45 Z ) ;
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```
- SDD051 ( Z213 Z ) ( E13 Z ) ( C44 EN ) ;
- VIH51 ( Z188 Z ) ( D88 Z ) ( C50 PI ) ( C44 PO ) ;
- X0310 ( Z45 A ) ( C45 A ) ( C44 ZI ) ;
- D051 ( E06 C ) ( C44 IO ) ; - SDI041 ( E15 A ) ( C42 Z ) ;
- DAB041 ( F09 F ) ( C41 Z ) ; - DAA041 ( F01 F ) ( C40 Z ) ;
- X0309 ( C42 A ) ( C41 A ) ( C40 A ) ( C39 Z ) ;
- SDD041 ( Z214 Z ) ( E14 Z ) ( C37 EN ) ;
- VIH50 ( Z187 Z ) ( D87 Z ) ( C44 PI ) ( C37 PO ) ;
- X0308 ( Z39 A ) ( C39 A ) ( C37 ZI ) ;
- D041 ( E05 C ) ( C37 IO ) ; - SDI031 ( E14 A ) ( C35 Z ) ;
- DAB031 ( F09 E ) ( C34 Z ) ; - DAA031 ( F01 E ) ( C33 Z ) ;
- X0307 ( C35 A ) ( C34 A ) ( C33 A ) ( C32 Z ) ;
- SDD031 ( Z215 Z ) ( E15 Z ) ( C31 EN ) ;
- VIH49 ( Z186 Z ) ( D86 Z ) ( C37 PI ) ( C31 PO ) ;
- X0306 ( Z32 A ) ( C32 A ) ( C31 ZI ) ;
- D031 ( E04 C ) ( C31 IO ) ; - SDI021 ( E13 A ) ( C29 Z ) ;
- DAB021 ( F09 D ) ( C28 Z ) ; - DAA021 ( F01 D ) ( C27 Z ) ;
- X0305 ( C29 A ) ( C28 A ) ( C27 A ) ( C26 Z ) ;
- SDD021 ( Z216 Z ) ( E16 Z ) ( C25 EN ) ;
- VIH48 ( Z185 Z ) ( D85 Z ) ( C31 PI ) ( C25 PO ) ;
- X0304 ( Z26 A ) ( C26 A ) ( C25 ZI ) ;
- D021 ( E03 C ) ( C25 IO ) ; - SDI011 ( E12 A ) ( C23 Z ) ;
- DAB011 ( F09 C ) ( C22 Z ) ; - DAA011 ( F01 C ) ( C21 Z ) ;
- X0303 ( C23 A ) ( C22 A ) ( C21 A ) ( C20 Z ) ;
- SDD011 ( Z209 Z ) ( E09 Z ) ( C19 EN ) ;
- VIH47 ( Z184 Z ) ( D84 Z ) ( C25 PI ) ( C19 PO ) ;
- X0302 ( Z20 A ) ( C20 A ) ( C19 ZI ) ;
- D011 ( E02 C ) ( C19 IO ) ; - SDI001 ( E11 A ) ( C17 Z ) ;
- DAB001 ( F09 B ) ( C16 Z ) ; - DAA001 ( F01 B ) ( C15 Z ) ;
- X0301 ( Z14 A ) ( C17 A ) ( C16 A ) ( C15 A ) ( C14 Z ) ;
- VIH45 ( Z182 Z ) ( D82 Z ) ( C13 PI ) ;
- SDD001 ( Z210 Z ) ( E10 Z ) ( C13 EN ) ;
- VIH46 ( Z183 Z ) ( D83 Z ) ( C19 PI ) ( C13 PO ) ;
- X0300 ( C14 A ) ( C13 ZI ) ; - D001 ( E01 C ) ( C13 IO ) ;
- CCLKB0 ( Z234 Z ) ( Z189 A ) ( E34 Z ) ( D89 A ) ( C11 A ) ;
- VIH10 ( EE16 PI ) ( C11 PO ) ;
- STRAAA ( Z206 A ) ( E06 A ) ( C11 Z ) ;
- CCLKA0 ( Z233 Z ) ( Z188 A ) ( E33 Z ) ( D88 A ) ( C10 A ) ;
- VIH9 ( C11 PI ) ( C10 PO ) ;
- STRB00 ( Z192 B ) ( D92 B ) ( C10 Z ) ;
- CRLINB1 ( Z232 Z ) ( Z187 A ) ( E32 Z ) ( D87 A ) ( C09 A ) ;
- VIH8 ( C10 PI ) ( C09 PO ) ;
- STRA00 ( Z187 B ) ( D87 B ) ( C09 Z ) ;
- CRLINA1 ( Z231 Z ) ( Z186 A ) ( E31 Z ) ( D86 A ) ( C08 A ) ;
- VIH7 ( C09 PI ) ( C08 PO ) ;
- X10001 ( E38A04 G ) ( E38A02 G ) ( C08 Z ) ;
- NXLINB1 ( Z230 Z ) ( Z185 A ) ( E30 Z ) ( D85 A ) ( C07 A ) ;
- VIH6 ( C08 PI ) ( C07 PO ) ;
- CLRX00 ( Z38A04 CD ) ( Z38A02 CD ) ( E38A04 CD ) ( E38A02 CD )
    ( C07 Z ) ;
- NXLINA1 ( Z229 Z ) ( Z184 A ) ( E29 Z ) ( D84 A ) ( C06 A ) ;
```


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```
    - VIH5 ( C07 PI ) ( C06 PO ) ;
    - STRBB0 ( Z205 B ) ( Z193 B ) ( E05 B ) ( D93 B ) ( C06 Z ) ;
    - RPTB1 ( Z228 Z ) ( Z183 A ) ( E28 Z ) ( D83 A ) ( C05 A ) ;
    - VIH4 ( C06 PI ) ( C05 PO ) ;
    - STRAA0 ( Z205 A ) ( Z186 B ) ( E05 A ) ( D86 B ) ( C05 Z ) ;
    - RPTA1 ( Z227 Z ) ( Z182 A ) ( E27 Z ) ( D82 A ) ( C04 A ) ;
    - VIH3 ( C05 PI ) ( C04 PO ) ;
    - STRB0 ( Z204 B ) ( Z203 B ) ( Z191 B ) ( Z190 B ) ( E04 B )
        ( E03 B ) ( D91 B ) ( D90 B ) ( C04 Z ) ;
    - CNTENB0 ( Z236 Z ) ( Z191 A ) ( E36 Z ) ( D91 A ) ( C02 A ) ;
    - VIH2 ( C04 PI ) ( C02 PO ) ;
    - STRA0 ( Z204 A ) ( Z203 A ) ( Z185 B ) ( Z184 B ) ( E04 A )
        ( E03 A ) ( D85 B ) ( D84 B ) ( C02 Z ) ;
    - CNTENA0 ( Z235 Z ) ( Z190 A ) ( E35 Z ) ( D90 A ) ( C01 A ) ;
    - VIH1 ( C02 PI ) ( C01 PO ) ; - CALCH ( E37 A ) ( C01 Z ) ;

#
```

Scan Chain Synthesis Example

You define the scan chain in the `COMPONENTS` and `SCANCHAINS` sections in your DEF file.

```
COMPONENTS 100 ;
- SIN MUX ;
- SOUT PAD ;
- C1 SDFF ;
- C2 SDFF ;
- C3 SDFF ;
- C4 SDFF ;
- B1 BUF ;
- A1 AND ; ...
END COMPONENTS

NETS 150 ;
- N1 (C1 SO) (C3 SI) ;
- N2 (C3 SO) (A1 A) ; ...
END NETS
```

You do not need to define any scan nets in the `NETS` section. This portion of the `NETS` section shows the effect of the scan chain process on existing nets that use components you specify in the `SCANCHAINS` section.

```
SCANCHAINS 1 ;
- SC
     + COMMONSCANPINS (IN SI) (OUT SO)
     + START SIN Z2
     + FLOATING C1 C2 C3
     + ORDERED C4 B1 (IN A) (OUT Q) ;
     + STOP SOUT A ;
END SCANCHAINS
```


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Because components C1, C2, and C3 are floating, `TROUTE SCANCHAIN` can synthesize them in any order in the chain. `TROUTE` synthesizes ordered components (C4 and B1) in the order you specify.

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B

Optimizing LEF Technology for Place and Route

This appendix contains the following information.

Overview

This appendix provides guidelines for defining the optimized technology section in the LEF file to get the best performance using Cadence® place-and-route tools.

For the following guidelines, the preferred routing direction for `metal1` and all other odd metal layers is horizontal. The preferred routing direction for `metal2` and all other even metal layers is vertical. Standard cells are arranged in horizontal rows.

This appendix discusses the following LEF statements.

```
LAYER layerName
     TYPE ROUTING ;
     PITCH distance ;
     WIDTH defWidth ;
     SPACING minSpacing [RANGE minwidth maxwidth] ;
     WIREEXTENSION value ;

END layerName
```


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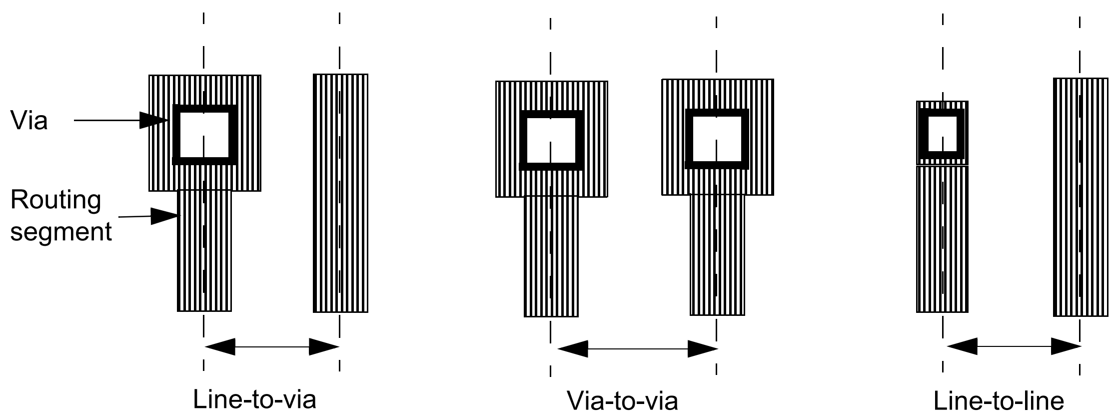
```
VIA viaName DEFAULT
     [TOPSTACKONLY]
     LAYER layerName RECT pt pt ; ...
END viaName
SPACING
     SAMENET
        layerName layerName minSpace [STACK] ;
END SPACING
```

Guidelines for Routing Pitch

The following is a summary for choosing the right pitch for an existing design library. For detailed information on determining routing pitch, refer to the *Cadence Abstract Generator User Guide*.

Pitch Measurement



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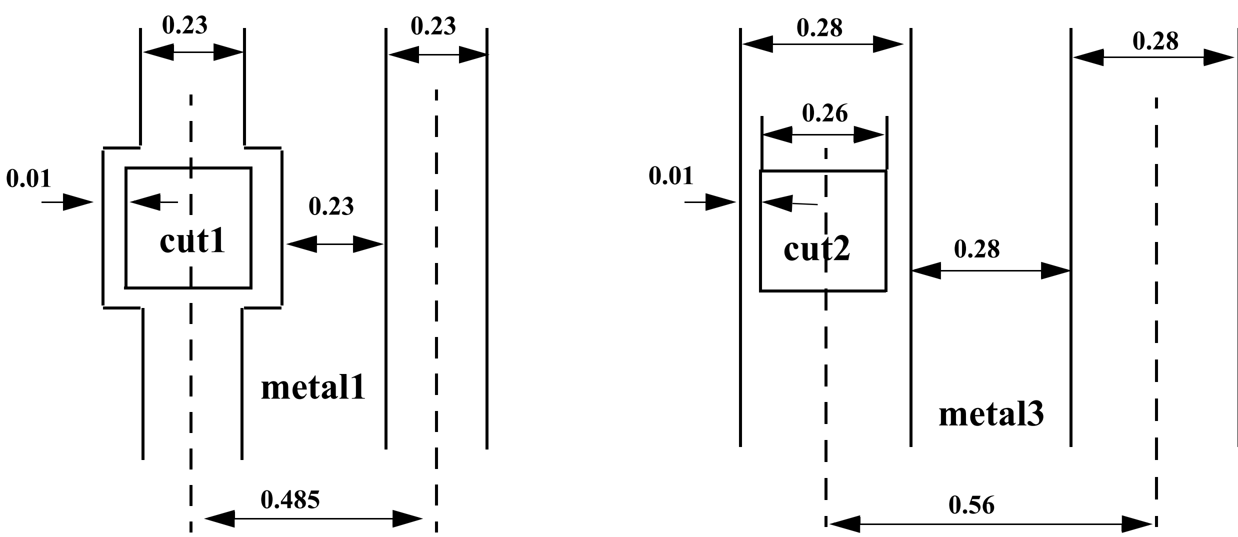
DESIGN RULE No. 1

W.1	Minimum width of `metal1` = 0.23 µm
S.1	Minimum space between two `metal1` regions = 0.23 µm
W.2	Minimum and maximum width of `cut1` = 0.26 µm
E.1	Minimum extension of `metal1` beyond `cut1` = 0.01 µm
W.3	Minimum width of `metal3` = 0.28 µm
S.2	Minimum space between two `metal3` regions = 0.28 µm
W.4	Minimum and maximum width of `cut2` = 0.26 µm
E.2	Minimum extension of `metal1` beyond `cut2` = 0.01 µm



Although the minimum `metal1` routing pitch is 0.485um from the design rule, you should use 0.56um instead, to match the `metal3` routing pitch in the same preferred direction.

LEF Construct No. 1

```
LAYER metal1
     TYPE ROUTING ;
     WIDTH 0.23 ;
     SPACING 0.23 ;
     PITCH 0.56 ;
     DIRECTION HORIZONTAL ;
```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```
END metal1


LAYER metal3
     TYPE ROUTING ;
     WIDTH 0.28 ;
     SPACING 0.28 ;
     PITCH 0.56 ;
     DIRECTION HORIZONTAL ;
END metal3
```

Recommendations

- Use line-to-via spacing for both the horizontal and vertical direction.
- Allow diagonal vias with the routing pitch.
- Align the routing pitch for `metal1` and `metal2`, with the pins inside the standard cells.
- Have uniform routing pitch in the same preferred direction. The pitch ratio should be 2 - 3 or 1 - 2. It is better to define the `metal1` pitch larger than necessary in order to achieve a 1 - 1 ratio because the `metal1` width is usually smaller the `metal2` and `metal3` widths.

Pitch Recommendations for Library Development

- All pins should be on the grid, and only those portions of the pins that are accessible to the router should be modeled as pins. For example, 45 degree pin geometry.
- The height of the cell should be the even multiple of the `metal1` pitch, and the width of the cell should be the even multiple of the `metal2` pitch.
- The blockage modeling, especially for `metal1`, should be simplified as much as possible. For example, it is very common for the entire area within the cell boundary to be obstructed in `metal1`, so use a single rectangular blockage instead of many small blockages.

Guidelines for Wide Metal Spacing

The `SPACING` statement in the LEF `LAYER` section is applied to both regular and special wires. You can use the Cadence® ultra router option `frouteUseRangeRule` to determine which objects to check against the `SPACING RANGE` statement. The default checks both pin and obstruction.

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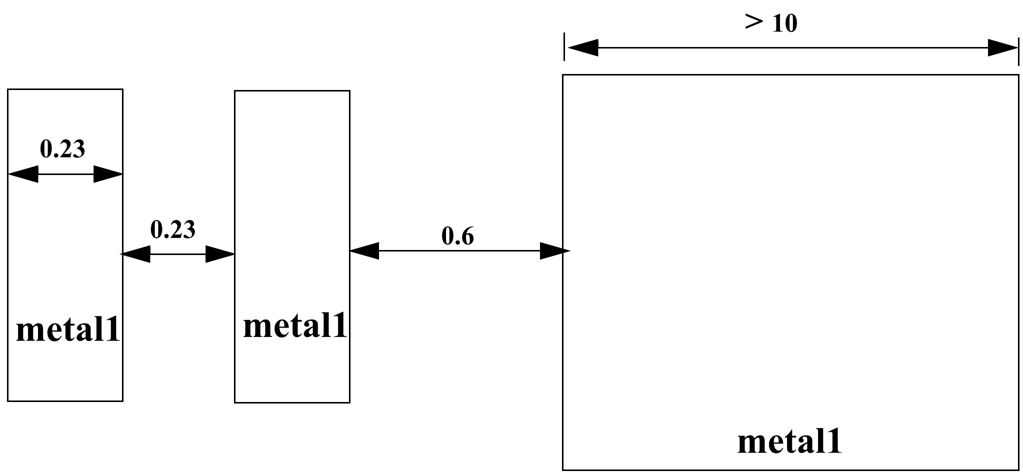
DESIGN RULE No. 2

S.1	Minimum space between two `metal1` regions = 0.23 µm
S.2	Minimum space between metal lines with one or both metal line width and length are greater than 10um = 0.6 µm



LEF CONSTRUCT No. 2

```
LAYER metal1
     WIDTH 0.23 ;
     SPACING 0.23 ;
     SPACING 0.6 RANGE 10.002 1000 ;

END metal1
```

Guidelines for Wire Extension at Vias

The following guidelines are for wire extension at vias.

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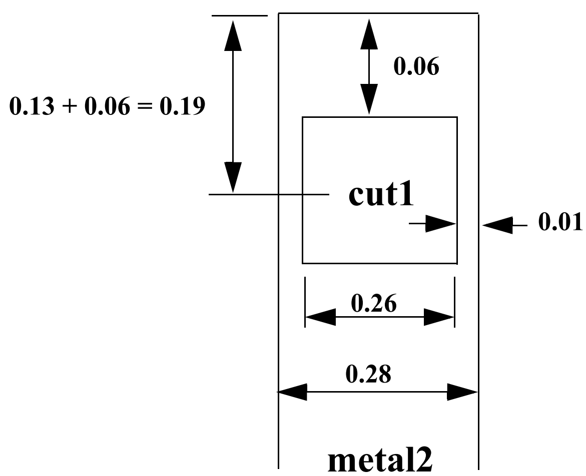
DESIGN RULE No. 3

W.1	Minimum and maximum width of `cut1` = 0.26 µm
W.2	Minimum width of `metal2` = 0.28 µm
E.1	Minimum extension of `metal2` beyond `cut1` = 0.01 µm
E.2	Minimum extension of `metal2` end-of-line region beyond `cut1` = 0.06 µm



LEF CONSTRUCT No. 3

```
LAYER metal2
     TYPE ROUTING ;
     WIDTH 0.28 ;
     SPACING 0.28 ;
     PITCH 0.56 ;
     WIREEXTENSION 0.19 ;
     DIRECTION VERTICAL ;

END metal2



VIA via23 DEFAULT
     LAYER metal2 ;
          RECT -0.14 -0.14 0.14 0.14 ;         # Use square via
     LAYER cut2 ;
          RECT -0.13 -0.13 0.13 0.13 ;
     LAYER metal3 ;
          RECT -0.14 -0.14 0.14 0.14 ;         # Use square via
```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```
END via23
```

Recommendations

- Use the `WIREEXTENSION` statement instead of defining multiple vias because the width of the `metal2` in `cut1` is the same as the default routing width of the `metal2` layer.
- The `WIREEXTENSION` statement only extends wires and not vias. For 65nm and below, `WIREEXTENSION` is no longer recommended because it may generate some advance rule violations if wires and vias have different widths.
- Define the `DEFAULT VIA` as a square via.

Guidelines for Default Vias

The following guidelines are for default vias.

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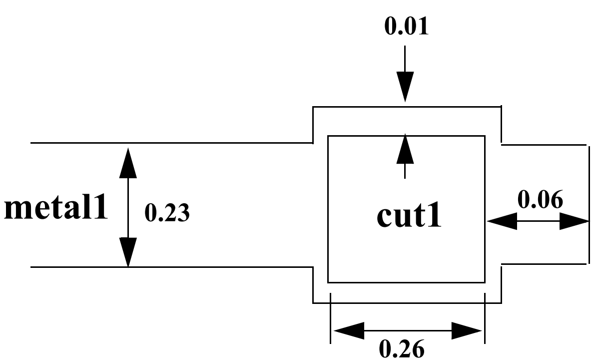
DESIGN RULE No. 4

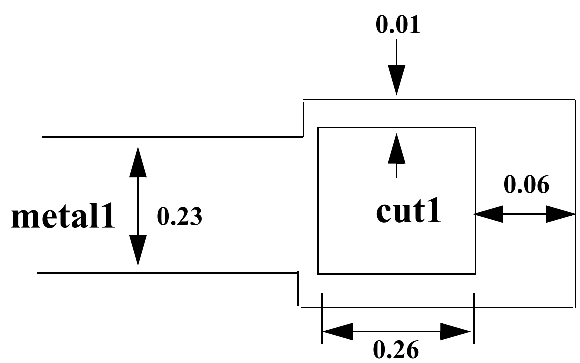
W.1	Minimum width of `metal1` = 0.23 µm
W.2	Minimum and maximum width of `cut1` = 0.26 µm
E.1	Minimum extension of `metal1` beyond `cut1` = 0.01 µm
E.2	Minimum extension of `metal1` end-of-line region beyond `cut1` = 0.06 µm

Case A:



Use `WIREEXTENSION` and square `DEFAULT VIA`

Case B:



Use Horizontal and Vertical `DEFAULT VIA`s

LEF CONSTRUCT No. 4 (Case B)

```
LAYER metal1
     TYPE ROUTING ;
     WIDTH 0.23 ;
     SPACING 0.23 ;
     PITCH 0.56 ;
     DIRECTION HORIZONTAL ;
END metal1


VIA via12_H DEFAULT
     LAYER metal1 ;
          RECT -0.19 -0.14 0.19 0.14 ;        # metal1 end-of-line
```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```
          extension 0.6 in both directions
     LAYER cut1 ;
          RECT -0.13 -0.13 0.13 0.13 ;
     LAYER metal2 ;
          RECT -0.14 -0.14 0.14 0.14 ;
END via12_H


VIA via12_V DEFAULT
     LAYER metal1 ;
          RECT -0.14 -0.19 0.14 0.19 ;        # metal1 end-of-line
          extension 0.6 in both directions
     LAYER cut1 ;
          RECT -0.13 -0.13 0.13 0.13 ;
     LAYER metal2 ;
          RECT -0.14 -0.14 0.14 0.14 ;
END via12_V
```

Recommendations

- If the width of the end-of-line metal extension is the same as the default metal routing width, as in Case A, use the `WIREEXTENSION` statement in the LEF `LAYER` section, and define a square via in the `DEFAULT VIA` section.
- If the width of the end-of-line metal extension is the same as the width of the via metal, as in Case B, define one horizontal `DEFAULT VIA` and one vertical `DEFAULT VIA` to cover the required metal extension area in both pregerred and non-preferred routing directions. Do not use the `WIREEXTENSION` statement in the LEF `LAYER` section.

Guidelines for Stack Vias (MAR Vias) and Samenet Spacing

The following guidelines are for stack vias (minimum area rule) and `SAMENET SPACING`.

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DESIGN RULE No. 5

W.1	Minimum width of `metal2` = 0.28 µm
W.2	Minimum and maximum width of `cut2` = 0.26 µm
E.1	Minimum extension of `metal2` beyond `cut2` = 0.01 µm
A.1	Minimum area of `metal2` = 0.2025 µm
C.1	`Cut2` can be fully or partially stacked on `cut1`, contact or any combination
W.1	Minimum width of `metal3` = 0.28 µm
W.2	Minimum and maximum width of `cut3` = 0.26 µm
E.1	Minimum extension of `metal2` beyond `cut3` = 0.01 µm
A.1	Minimum area of `metal3` = 0.2025 µm
C.1	`Cut3` can be fully or partially stacked on `cut2`, `cut1`, contact or any combination

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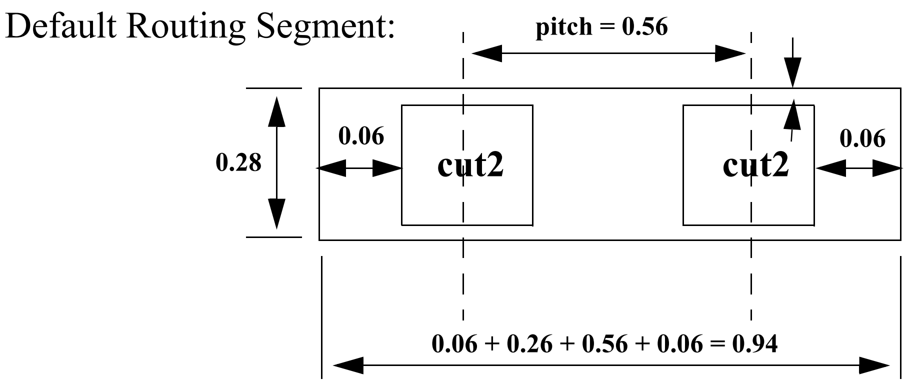
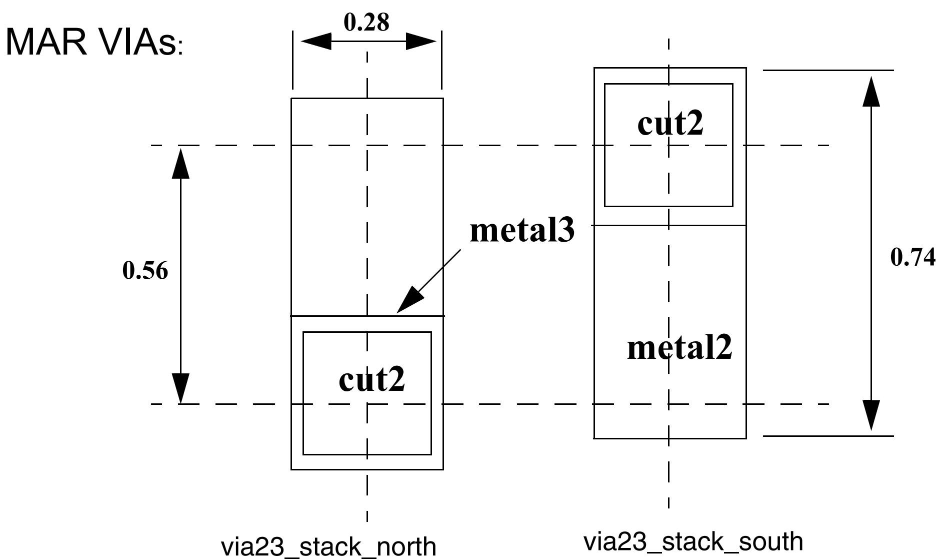


Minimum routing area of metal3 = 0.28 x 0.94 = 0.2632 > 0.2.25 (MAR)



LEF CONSTRUCT No. 5

```
VIA via23_stack_north DEFAULT
     LAYER metal2 ;
          RECT -0.14 -0.14 0.14 0.6 ;   # MAR = 0.28 x 0.74
     LAYER cut2 ;
```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```
          RECT -0.13 -0.13 0.13 0.13 ;
     LAYER metal3 ;
          RECT -0.14 -0.14 0.14 0.14 ;
END via23_stack_north


VIA via23_stack_south DEFAULT
     LAYER metal2 ;
          RECT -0.14 -0.6 0.14 0.14 ;   # MAR = 0.28 x 0.74
     LAYER cut2 ;
          RECT -0.13 -0.13 0.13 0.13 ;
     LAYER metal3 ;
          RECT -0.14 -0.14 0.14 0.14 ;
END via23_stack_south


VIA via34_stack_east DEFAULT
     LAYER metal3 ;
          RECT -0.14 -0.14 0.6 0.14 ;   # MAR = 0.28 x 0.74
     LAYER cut3 ;
          RECT -0.13 -0.13 0.13 0.13 ;
     LAYER metal4 ;
          RECT -0.14 -0.14 0.14 0.14 ;
END via34_stack_east


VIA via34_stack_west DEFAULT
     LAYER metal3 ;
          RECT -0.6 -0.14 0.14 0.14 ;   # MAR = 0.28 x 0.74
     LAYER cut3 ;
          RECT -0.13 -0.13 0.13 0.13 ;
     LAYER metal4 ;
          RECT -0.14 -0.14 0.14 0.14 ;
END via34_stack_west
```

Recommendations

- The minimum metal routing segment (two vias between one pitch grid) with or without end-of-line metal extension should automatically satisfy the minimum area rule.
- If vias are stackable, create the `TOPSTACKONLY` vias with a rectangular shape blocking only one neighboring grid for both sides of the preferred routing direction. In other words, one north oriented and one south oriented for vertical-preferred routing layers, and one east oriented and one west oriented for horizontal-preferred routing layers.

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- Use slightly larger dimensions for the via size to make them an even number, so they snap to the manufacturing grids.
- The `STACK` keyword in the `SAMENETSPACING` statements only allows vias to be fully overlapped (stacked) by SROUTE commands. To allow vias to be partially overlapped, set the environment variable `SROUTE.ALLOWOVERLAPINSTACKVIA` to `TRUE`.
- The `metal1` layer does not require a MAR via because all metal1 pins should satisfy the minimum area rules.

Example of an Optimized LEF Technology File

```
VERSION 5.8 ;

BUSBITCHARS "[]" ;


UNITS
     DATABASE MICRONS 100  ;

END UNITS



LAYER metal1
     TYPE ROUTING ;
     WIDTH 0.23 ;
     SPACING 0.23 ;
     SPACING 0.6 RANGE 10.02 1000 ;
     PITCH 0.56 ;
     DIRECTION HORIZONTAL ;

END metal1



LAYER cut1
     TYPE CUT ;

END cut1



LAYER metal2
     TYPE ROUTING ;
     WIDTH 0.28 ;
     SPACING 0.28 ;
     SPACING 0.6 RANGE 10.02 1000 ;
     PITCH 0.56 ;
     WIREEXTENSION 0.19 ;
     DIRECTION VERTICAL ;

END metal2
```


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```
LAYER cut2
     TYPE CUT ;
END cut2


LAYER metal3
     TYPE ROUTING ;
     WIDTH 0.28 ;
     SPACING 0.28 ;
     SPACING 0.6 RANGE 10.02 1000 ;
     PITCH 0.56 ;
     WIREEXTENSION 0.19 ;
     DIRECTION HORIZONTAL ;
END metal3


LAYER cut3
     TYPE CUT ;
END cut3


LAYER metal4
     TYPE ROUTING ;
     WIDTH 0.28 ;
     SPACING 0.28 ;
     SPACING 0.6 RANGE 10.02 1000 ;
     PITCH 0.56 ;
     WIREEXTENSION 0.19 ;
     DIRECTION VERTICAL ;
END metal4


LAYER cut4
     TYPE CUT ;
END cut4


LAYER metal5
     TYPE ROUTING ;
     WIDTH 0.28 ;
     SPACING 0.28 ;
     SPACING 0.6 RANGE 10.02 1000 ;
```


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```
     PITCH 0.56 ;
     WIREEXTENSION 0.19 ;
     DIRECTION HORIZONTAL ;
END metal5


LAYER cut5
     TYPE CUT ;
END cut5


LAYER metal6
     TYPE ROUTING ;
     WIDTH 0.44 ;
     SPACING 0.46 ;
     SPACING 0.6 RANGE 10.02 1000 ;
     PITCH 1.12 ;
     DIRECTION VERTICAL ;
END metal6
```

start DEFAULT VIA

```
VIA via12_H DEFAULT
     LAYER metal1 ;
          RECT -0.19 -0.14 0.19 0.14 ;      # metal1 end-of-line ext 0.6
     LAYER cut1 ;
          RECT -0.13 -0.13 0.13 0.13 ;
     LAYER metal2 ;
          RECT -0.14 -0.14 0.14 0.14 ;
END via12_H


VIA via12_V DEFAULT
     LAYER metal1 ;
          RECT -0.14 -0.19 0.14 0.19 ;      # metal1 end-of-line ext 0.6
     LAYER cut1 ;
          RECT -0.13 -0.13 0.13 0.13 ;
     LAYER metal2 ;
          RECT -0.14 -0.14 0.14 0.14 ;
END via12_V


VIA via23 DEFAULT
     LAYER metal2 ;
          RECT -0.14 -0.14 0.14 0.14 ;
     LAYER cut2 ;
```


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```
          RECT -0.13 -0.13 0.13 0.13 ;
     LAYER metal3 ;
          RECT -0.14 -0.14 0.14 0.14 ;
END via23


VIA via34 DEFAULT
     LAYER metal3 ;
          RECT -0.14 -0.14 0.14 0.14 ;
     LAYER cut3 ;
          RECT -0.13 -0.13 0.13 0.13 ;
     LAYER metal4 ;
          RECT -0.14 -0.14 0.14 0.14 ;
END via34


VIA via45 DEFAULT
     LAYER metal4 ;
          RECT -0.14 -0.14 0.14 0.14 ;
     LAYER cut4 ;
          RECT -0.13 -0.13 0.13 0.13 ;
     LAYER metal5 ;
          RECT -0.14 -0.14 0.14 0.14 ;
END via45


VIA via56_H DEFAULT
     LAYER metal5 ;
          RECT -0.24 -0.19 0.24 0.19 ;
     LAYER cut5 ;
          RECT -0.18 -0.18 0.18 0.18 ;
     LAYER metal6 ;
          RECT -0.27 -0.27 0.27 0.27 ;
END via56_H


VIA via56_V DEFAULT
     LAYER metal5 ;
          RECT -0.19 -0.24 0.19 0.24 ;
     LAYER cut5 ;
          RECT -0.18 -0.18 0.18 0.18 ;
     LAYER metal6 ;
          RECT -0.27 -0.27 0.27 0.27 ;
END via56_V
```

end DEFAULT VIA

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start STACK VIA

```
VIA via23_stack_north DEFAULT
     LAYER metal2 ;
          RECT -0.14 -0.14 0.14 0.6 ;   # MAR = 0.28 x 0.74
     LAYER cut2 ;
          RECT -0.13 -0.13 0.13 0.13 ;
     LAYER metal3 ;
          RECT -0.14 -0.14 0.14 0.14 ;
END via23_stack_north


VIA via23_stack_south DEFAULT
     LAYER metal2 ;
          RECT -0.14 -0.6 0.14 0.14 ;   # MAR = 0.28 x 0.74
     LAYER cut2 ;
          RECT -0.13 -0.13 0.13 0.13 ;
     LAYER metal3 ;
          RECT -0.14 -0.14 0.14 0.14 ;
END via23_stack_south


VIA via34_stack_east DEFAULT
     LAYER metal3 ;
          RECT -0.14 -0.14 0.6 0.14 ;   # MAR = 0.28 x 0.74
     LAYER cut3 ;
          RECT -0.13 -0.13 0.13 0.13 ;
     LAYER metal4 ;
          RECT -0.14 -0.14 0.14 0.14 ;
END via34_stack_east


VIA via34_stack_west DEFAULT
     LAYER metal3 ;
          RECT -0.6 -0.14 0.14 0.14 ;   # MAR = 0.28 x 0.74
     LAYER cut3 ;
          RECT -0.13 -0.13 0.13 0.13 ;
     LAYER metal4 ;
          RECT -0.14 -0.14 0.14 0.14 ;
END via34_stack_west


VIA via45_stack_north DEFAULT
     LAYER metal4 ;
          RECT -0.14 -0.14 0.14 0.6 ;   # MAR = 0.28 x 0.74
     LAYER cut4 ;
```


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```
          RECT -0.13 -0.13 0.13 0.13 ;
     LAYER metal5 ;
          RECT -0.14 -0.14 0.14 0.14 ;
END via45_stack_north


VIA via45_stack_south DEFAULT
     LAYER metal4 ;
          RECT -0.14 -0.6 0.14 0.14 ;    # MAR = 0.28 x 0.74
     LAYER cut4 ;
          RECT -0.13 -0.13 0.13 0.13 ;
     LAYER metal5 ;
          RECT -0.14 -0.14 0.14 0.14 ;
END via45_stack_south


VIA via56_stack_east DEFAULT
     LAYER metal5 ;
          RECT -0.19 -0.19 0.35 0.19 ;    # MAR = 0.38 x 0.54
     LAYER cut5 ;
          RECT -0.18 -0.18 0.18 0.18 ;
     LAYER metal6 ;
          RECT -0.27 -0.27 0.27 0.27 ;
END via56_stack_east


VIA via56_stack_west DEFAULT
     LAYER metal5 ;
          RECT -0.35 -0.19 0.19 0.19 ;    # MAR = 0.38 x 0.54
     LAYER cut5 ;
          RECT -0.18 -0.18 0.18 0.18 ;
     LAYER metal6 ;
          RECT -0.27 -0.27 0.27 0.27 ;
END via56_stack_west
```

end STACK VIA

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C

Calculating and Fixing Process Antenna Violations

This appendix describes process antenna violations and how you can use the router to correct them. It includes the following sections:

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Overview

During deep submicron wafer fabrication, gate damage can occur when excessive static charges accumulate and discharge, passing current through a gate. If the area of the layer connected directly to the gate or connected to the gate through lower layers is large relative to the area of the gate and the static charges are discharged through the gate, the discharge can damage the oxide that insulates the gate and cause the chip to fail. This phenomenon is called the *process antenna effect* (PAE).

To determine the extent of the PAE, the router calculates the area of the layer relative to the area of the gates connected to it, or connected to it through lower layers. The number it calculates is called the *antenna ratio*. Each foundry sets a maximum allowable antenna ratio for the chips it fabricates.

For example, assume a foundry sets a maximum allowable antenna ratio of 500. If a net has two input gates that each have an area of 1 micron^2, any metal layers that connect to the gates and have an area larger than 1,000 micron^2 have process antenna violations because they would cause the antenna ratio to be higher than 500:

$$\text{Antenna Ratio} = \frac{\text{Area of metal layer}}{\text{Area of gates}} \qquad 500 = \frac{1000}{1 + 1}$$

To tell the router the values to use when it calculates the antenna ratio, you set antenna keywords in the LEF and DEF files. The router measures potential damage caused by PAE by checking the ratio it calculates against the values specified by the antenna keywords. When it finds a net whose antenna ratio for a specified layer exceeds the maximum allowed value for that layer, it finds a *process antenna violation* and attempts to fix it using one or both of the following methods:

- Changing the routing so the routing layers connected to a gate or connected to a gate through lower layers are not so large that they build enough static charge to damage the gate
- Inserting diodes that protect the gate by providing an alternate path to discharge the static charge

LEF can specify several types of antenna ratios, including ratios for PAE damage on one layer only and ratios calculated by adding accumulated damage on several layers. In addition, LEF can specify ratios based on the area of the metal wires or the cut area of vias.

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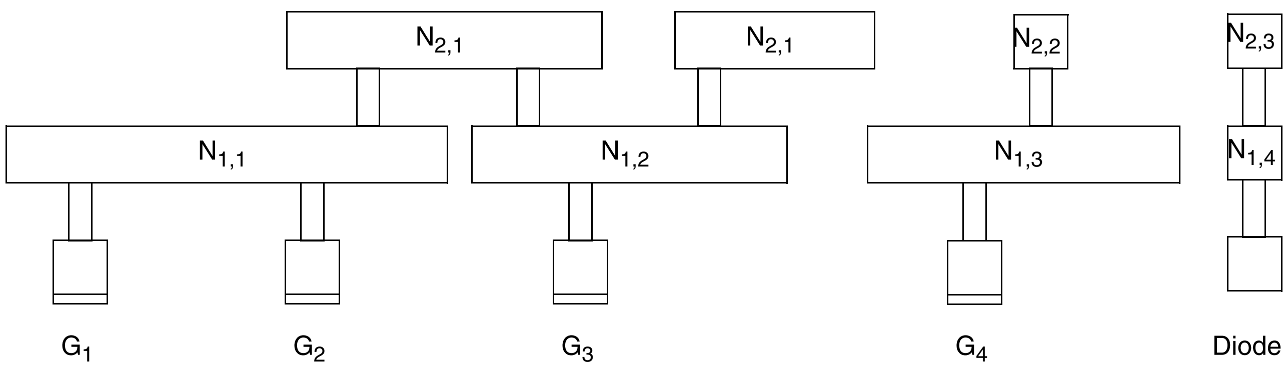
What Are Process Antennas?

In a chip manufacturing process, metal layers are built up, layer by layer, starting with the first-level metal layer (usually referred to as *metal1*). Next, the *metal1-metal2* vias are created, then the second-level metal layer, then *metal2-metal3* vias, and so on.

On each metal layer, metal is initially deposited so it covers the entire chip. Then, the unneeded portions of the metal are removed by etching, typically in plasma (charged particles).

Figure C-1 on page 1143 shows a section of an imaginary chip after the unneeded metal from *metal2* is removed.

Figure C-1



In the figure,

- Gate areas for transistors are labelled G_k, where k is a sequential number starting with 1.
- Wire segments are labelled $N_{i,j}$
  - N signifies that the wire segment is an electrically connected node
  - i specifies the metal layer to which the node belongs
  - j is a sequential number for the node on that metal layer
- Nodes are labelled so that all pieces of the metal geometry on layer *metal$_i$* that are electrically connected by conductors at layers below *metal$_i$* belong to the same node. For example, the two *metal2* wire segments that belong to node $N_{2,1}$ are electrically connected to gates G_1, G_2, and G_3 by a piece of wire on *metal1* (labelled $N_{1,2}$).

Thick oxide insulates the already-fabricated structures below *metal2*, preventing them from direct contact with the plasma. The *metal2* geometries, however, are exposed to the plasma,

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and collect charge from it. As the metal geometries collect charge, they build up voltage potential.

Because the metal geometries collect charge during the metallization process, they are referred to as process antennas. In general, the more area covered by the metal geometries that are exposed to the plasma (that is, the larger the process antennas), the more charge they can collect.

In Figure C-1 on page 1143, note the following:

- Node $N_{1,1}$ is electrically connected to gates G_1 and G_2.
- Node $N_{1,2}$ is electrically connected to gate G_3.
- Node $N_{2,1}$ (node $N_{2,1}$ has two pieces of metal) is electrically connected to gates G_1, G_2, and G_3.
- Node $N_{1,3}$ and node $N_{2,2}$ are electrically connected to gate G_4.
- Node $N_{1,4}$ and node $N_{2,3}$ are electrically connected to the diffusion (diode).

What Is the Process Antenna Effect (PAE)?

If the voltage potential across the gate oxide becomes large enough to cause current to flow across the gate oxide, from the process antennas to the gates to which the process antennas are electrically connected, the current can damage the gate oxide. The process antenna effect (PAE) is the term used to describe the build-up of charge and increase in voltage potential. The larger the total gate area that is electrically connected to the process antennas on a specific layer, the more charge the connected gates can withstand.

In the imaginary chip in Figure C-1 on page 1143, if the current were to flow, the following would happen, as a result of the node-gate connections:

- The charge collected by process antennas on nodes $N_{1,1}$, $N_{1,2}$, and $N_{2,1}$ would be discharged through one or more of gates G_1, G_2, and G_3.
- The charge collected by process antennas on nodes $N_{1,3}$ and $N_{2,2}$ would be discharged through gate G_4.
- The charge collected by process antennas on node $N_{1,4}$ and $N_{2,3}$ would be discharged through the diode.

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What Is the Antenna Ratio?

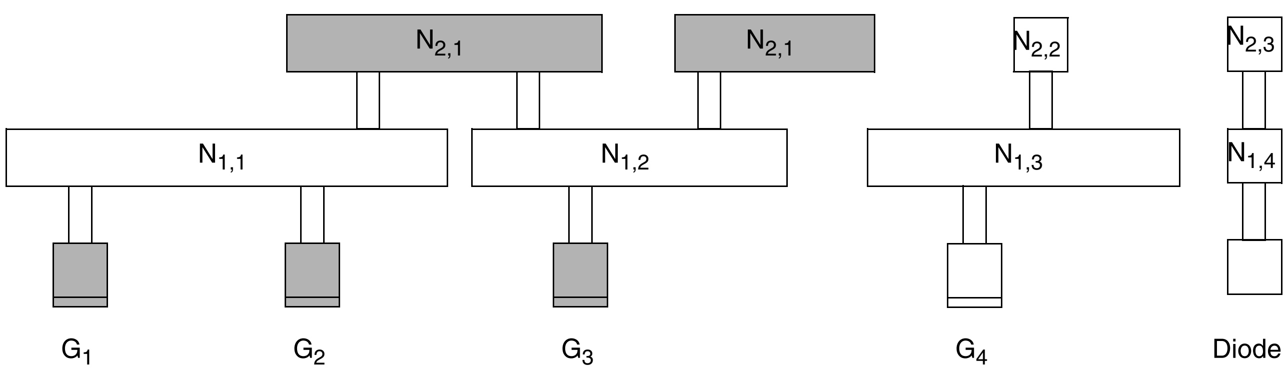
Because the total gate area that is electrically connected to a node (and therefore connected to the process antennas) determines the amount of charge from the process antennas the electrically connected gates can withstand, and because the size of the process antennas connected to the node determines how much charge the antennas collect, it is useful to calculate the ratio of the size of the process antennas on a node to the size of the gate area that is electrically connected to the node. This is the antenna ratio. The greater the antenna ratio, the greater the potential for damage to the gate oxide.

If you check a chip and obtain an antenna ratio greater than the threshold specified by the foundry, gate damage is likely to occur.

Figure C-2 on page 1145 shows the same section of the imaginary chip as the previous figure. The shaded areas in this figure represent the process antennas on node $N_{2,1}$ and the gates to which they connect: gates G_1, G_2, and G_3. The shaded gates discharge the electricity collected by the process antennas on node $N_{2,1}$.

Figure C-2

$$\text{Antenna Ratio} = \frac{\text{Area of process antennas on a node}}{\text{Area of gates electrically connected to the node}}$$



What Can Be Done to Improve the Antenna Ratio?

If there is an alternate path for the current to flow, the charge on the node can be discharged through the alternate path before the voltage potential reaches a level that damages the gate. For example, a Zener diode, which allows current to flow in the reverse direction when the reverse bias reaches a specified breakdown voltage, provides an alternate path, and helps avoid building up so much charge at the node that the charge is discharged through the gate

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oxide. Diffusion features that form the output of a logic gate (source and drain of transistors) can provide such an alternate discharge path.

Routers typically use two methods to decrease the antenna ratio:

- Changing the routing by breaking the metal layers into smaller pieces
- Inserting antenna diode cells to discharge the current

Both of these methods supply alternate paths for the current. For details about how to specify antenna diode cells, see "Using Antenna Diode Cells" on page 1176.

Using Process Antenna Keywords in the LEF and DEF Files

You tell the router the values to use for the gate, diffusion, and metal areas by setting values for process antenna keywords in the LEF and DEF files for your design. You also tell the router the values to use for the threshold process antenna ratios by setting the keywords.

The following table lists LEF version 5.5 antenna keywords.

If the keyword ends with ...	It refers to ...	Examples
`area`	Area of the gates or diffusion Measured in micron^2	`ANTENNADIFFAREA` `ANTENNAGATEAREA`

`factor`	Area multiplier used for the metal nodes	`ANTENNAAREAFACTOR`
		`ANTENNASIDEAREAFACTOR`
		
		Note: Use `DIFFUSEONLY` if you want the multiplier to apply only when connecting to diffusion. For more information, see "Using DiffUseOnly" on page 1177.

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If the keyword ends with ...	It refers to ...	Examples

`ratio`	Relationship the router is calculating
	
	`Cum` is used in keywords for cumulative antenna ratio.
		`ANTENNAAREARATIO`
		`ANTENNASIDEAREARATIO`
		`ANTENNADIFFAREARATIO`
		`ANTENNADIFFSIDEAREARATIO`
		`ANTENNACUMAREARATIO`
		`ANTENNACUMSIDEAREARATIO`
		`ANTENNACUMDIFFAREARATIO`
		`ANTENNACUMDIFFSIDEAREARATIO`

Calculating Antenna Ratios

Tools should calculate antenna ratios using one of the following models:

- The partial checking model

  Using this model, you calculate damage to gates by process antennas on one layer. For example, if you use the partial checking model to calculate the PAE referred to a gate from *metal3*, you do not consider any potential damages referred to that gate from metallization steps on *metal1* or *metal2*.

  You use this model to calculate a partial antenna ratio (PAR). A PAR tells you if any single metallization step is likely to inflict damage to a gate.

- The cumulative checking model

  This model is more conservative than the partial checking model. It adds damage to a gate caused by the PAE referred to the gate from each metallization step, starting from *metal1* up to the layer that is being checked. For example, if you use the cumulative checking model to calculate the PAE referred to a gate from *metal3*, you add the PAR from the relevant antenna areas on *metal1*, *metal2*, and *metal3*.

  You use this model to calculate a cumulative antenna ratio (CAR). A CAR adds the damages on successive layers together to accumulate them as the layers are built up.

Calculating the Antenna Area

The area used to model the charge-collecting ability of a node is called the *antenna area*. The router calculates the antenna area for one of the following areas:

- The drawn area (the top surface area of the metal shape)

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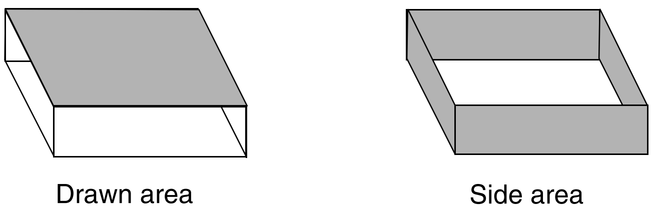
- The side area (the area of the sides of the metal shape)

The height of each side is taken from the THICKNESS statement for that layer.

Figure C-3 on page 1148 shows drawn and side areas.

Figure C-3



Antenna Area Factor

You can increase or decrease the calculated antenna area by specifying an antenna area factor in the LEF file.

- Use ANTENNAAREAFACTOR to adjust the calculation of the drawn area.
- Use ANTENNASIDEAREAFACTOR to adjust the calculation of the side area.

The default value of both factors is 1.

The final ratio check can be scaled (that is, made more or less pessimistic) by using the ANTENNAAREAFACTOR or ANTENNASIDEAREAFACTOR values that are used to multiply the final PAR and CAR values.

Note: The LEF and DEF ANTENNA values are always unscaled values; only the final ratio-check is affected by the scale factors.

Calculating a PAR

The general PAR(m_1) equation for a single layer is calculated as:

$$\mathrm{PAR}(m_i) = \frac{\{(\mathit{metalFactor} \times \text{metal\_area}) \times \mathit{diffMetalReduceFactor} - \mathit{minusDiffFactor} \times \text{diff\_area}\}}{(\text{gate\_area} + \mathit{plusDiffFactor} \times \text{diff\_area})}$$

The existing ANTENNAAREAFACTOR statement is shown as *metalFactor* for the metal area. It has no effect on the diff_area, gate_area, or cut_area shown. Likewise, the ANTENNAAREADIFFREDUCEPWL statement is shown as *diffMetalReduceFactor*, the

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ANTENNAAREAMINUSDIFF statement is shown as *minusDiffFactor*, and the ANTENNAGATEPLUSDIFF statement is shown as *plusDiffFactor*. For cut layer, the ratio equation illustrates the effect of an ANTENNAAREAFACTOR *cutFactor* statement as *metalFactor*. If there is no preceding ANTENNAAREAFACTOR statement, the *metalFactor* value defaults to 1.0.

For single layer rules, the PAR value is compared to ANTENNA[SIDE]AREARATIO and/or ANTENNADIFF[SIDE]AREARATIO, as appropriate. For cumulative layer rules, the CAR values is compared to ANTENNACUM[SIDE]AREARATIO and/or ANTENNACUMDIFF[SIDE]AREARATIO, as appropriate.

The following example uses a simplified formula to calculate a PAR, without including the various area factors:

$$\mathrm{PAR}(N_{i,j}, G_k) = \frac{\mathrm{Area}(N_{i,j})}{\sum_{G_k \in C(N_{i,j})} \mathrm{Area}(G_k)}$$

$\mathrm{PAR}(N_{i,j}, G_k)$ is the partial antenna ratio for node j on *metal*$_i$ with respect to gate G_k, where G_k is electrically connected to node $N_{i,j}$ by layer i or below.

$\mathrm{Area}(N_{i,j})$ is the drawn or side area of node $N_{i,j}$.

$C(N_{i,j})$ is the set of gates G_k that are electrically connected to $N_{i,j}$ through the layers below *metal*$_i$.

$\mathrm{Area}(G_k)$ is the drawn or side area of gate G_k. (The reason to include the G_k parameter for PAR is to maintain uniformity with the notation for CAR.)

Note: For a specified node $N_{i,j}$, the $\mathrm{PAR}(N_{i,j}, G_k)$ for all gates G_k that are connected to the node $N_{i,j}$ using *metal*$_i$ or below are identical.

Calculations for PAR on the First Metal Layer

Figure C-4 on page 1150 shows a section of an imaginary chip after the first metal layer is processed.

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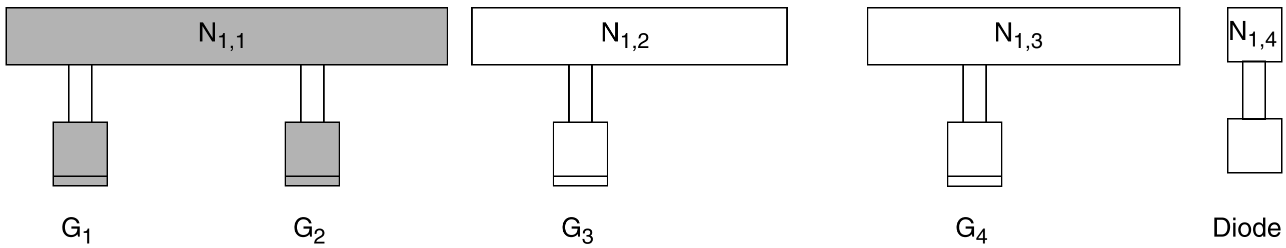
Figure C-4



The shaded areas in the figure represent the wire segment and the gates whose areas you must compute to evaluate the formula below.

To calculate $\mathrm{PAR}(N_{i,j}, G_k)$ for node $N_{1,1}$, a node on the first metal layer, with respect to gate G_1, use the following formula:

$$\mathrm{PAR}\ (N_{1,1},G_1) = \frac{\mathrm{Area}(N_{1,1})}{\mathrm{Area}(G_1) + \mathrm{Area}(G_2)}$$

Because gates G_1 and G_2 both connect to node $N_{1,1}$, the following statement is true:

$\mathrm{PAR}(N_{1,1},G_1) = \mathrm{PAR}(N_{1,1},G_2)$

To calculate PAR for node $N_{1,2}$, another node on the first metal layer, with respect to gate G_3, use the following formula:

$$\mathrm{PAR}\ (N_{1,2},G_3) = \frac{\mathrm{Area}(N_{1,2})}{\mathrm{Area}(G_3)}$$

To calculate $\mathrm{PAR}(N_{i,j}, G_k)$ for node $N_{1,3}$, another node on the first metal layer, with respect to gate G_4, use the following formula:

$$\mathrm{PAR}\ (N_{1,3},G_4) = \frac{\mathrm{Area}(N_{1,3})}{\mathrm{Area}(G_4)}$$

Calculations for PAR on the Second Metal Layer

Figure C-5 on page 1151 shows the chip after the second metal layer is processed.

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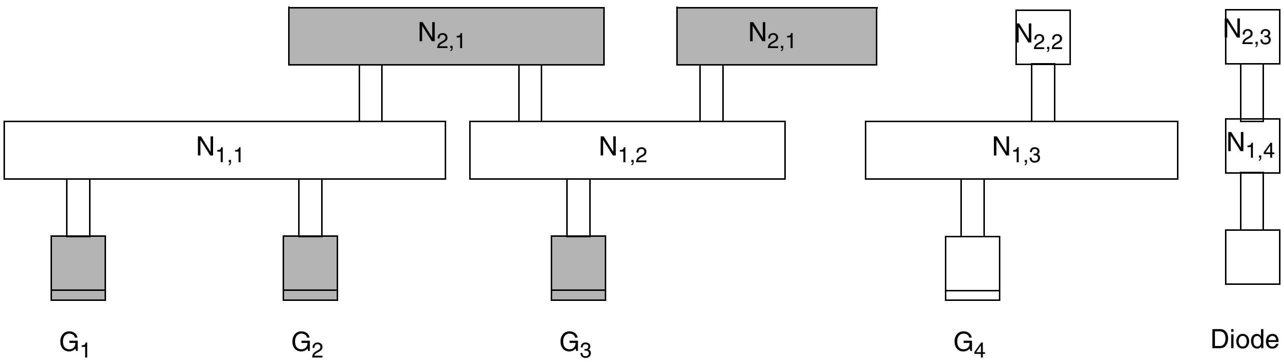
Figure C-5



The shaded areas in the figure represent the wire segments and the gates whose areas you must compute to evaluate the formula below.

$N_{2,1}$ consists of two pieces of metal on the second layer that are electrically connected at this step in the fabrication process. Therefore, to calculate $PAR(N_{2,1},G_1)$, you must add the area of both pieces together.

To calculate $PAR(N_{i,j},G_k)$ for node $N_{2,1}$, a node on the second metal layer, with respect to gate G_1, use the following formula:

$$PAR\ (N_{2,1},G_1) = \frac{Area(N_{2,1})}{Area(G_1) + Area(G_2) + Area(G_3)}$$

As on the first layer,

$PAR(N_{2,1},G_1) = PAR(N_{2,1},G_2) = PAR(N_{2,1},G_3)$

Calculations for PAR on the Third Metal Layer

Figure C-6 on page 1152 shows the chip after the third metal layer is processed.

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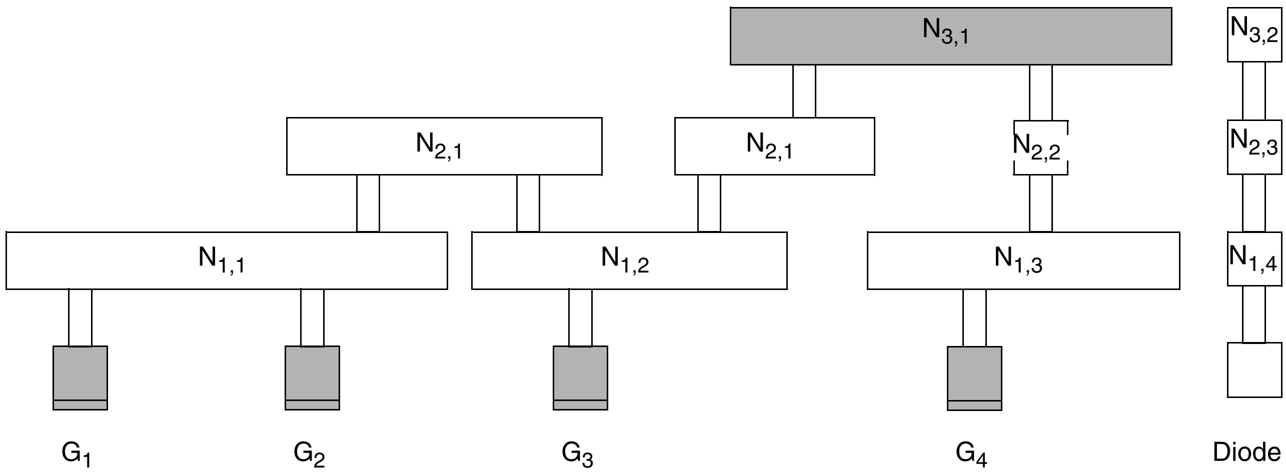
Figure C-6



The shaded areas in the figure represent the wire segment and the gates whose areas you must compute to evaluate the formula below.

To calculate `PAR(`$N_{i,j}$`, `G_k`)` for node $N_{3,1}$, a node on the third metal layer, with respect to gate G_1, use the following formula:

$$PAR\ (N_{3,1},G_1) = \frac{Area(N_{3,1})}{Area(G_1) + Area(G_2) + Area(G_3) + Area(G_4)}$$

As on the prior layers,

$PAR(N_{3,1},G_1) = PAR(N_{3,1},G_2) = PAR(N_{3,1},G_3) = PAR(N_{3,1},G_4)$

Calculations for PAR on the Fourth Metal Layer

Figure C-7 on page 1153 shows the chip after the fourth metal layer is processed.

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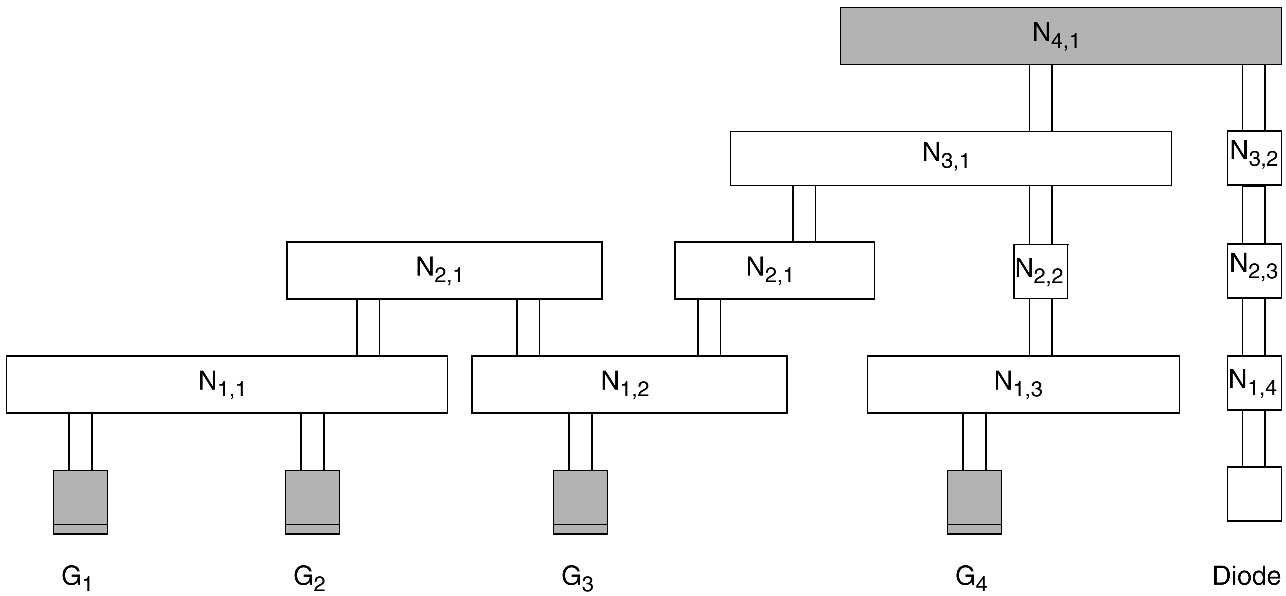
Figure C-7



The shaded areas in the figure represent the wire segment and the gates whose areas you must compute to evaluate the formula below.

To calculate `PAR(N`$_{i,j}$`, G`$_k$`)` for the fourth metal layer, use the following formula:

$$\text{PAR}(N_{4,1}, G_1) = \frac{\text{Area}(N_{4,1})}{\text{Area}(G_1) + \text{Area}(G_2) + \text{Area}(G_3) + \text{Area}(G_4)}$$

As on the prior layers,

$\text{PAR}(N_{4,1},G_1) = \text{PAR}(N_{4,1},G_2) = \text{PAR}(N_{4,1},G_3) = \text{PAR}(N_{4,1},G_4)$

Note: Node $N_{4,1}$ is connected to the diffusion layer through the output diode. After the router calculates the antenna ratio, it compares its calculations to the area of the diffusion, instead of the area of the gates.

Calculating a CAR

To calculate a CAR, the router adds the PARs for all the relevant nodes on the specified or lower metal layers that are electrically connected to a gate. Therefore, `CAR(N`$_{i,j}$`,G`$_k$`)` designates the cumulative damage to gate G_k by metallization steps up to the current level of metal, i.

To create a single accumulative model that combines both metal and cut damage into one model, specify the `ANTENNACUMROUTINGPLUSCUT` statement for the layer, so that:

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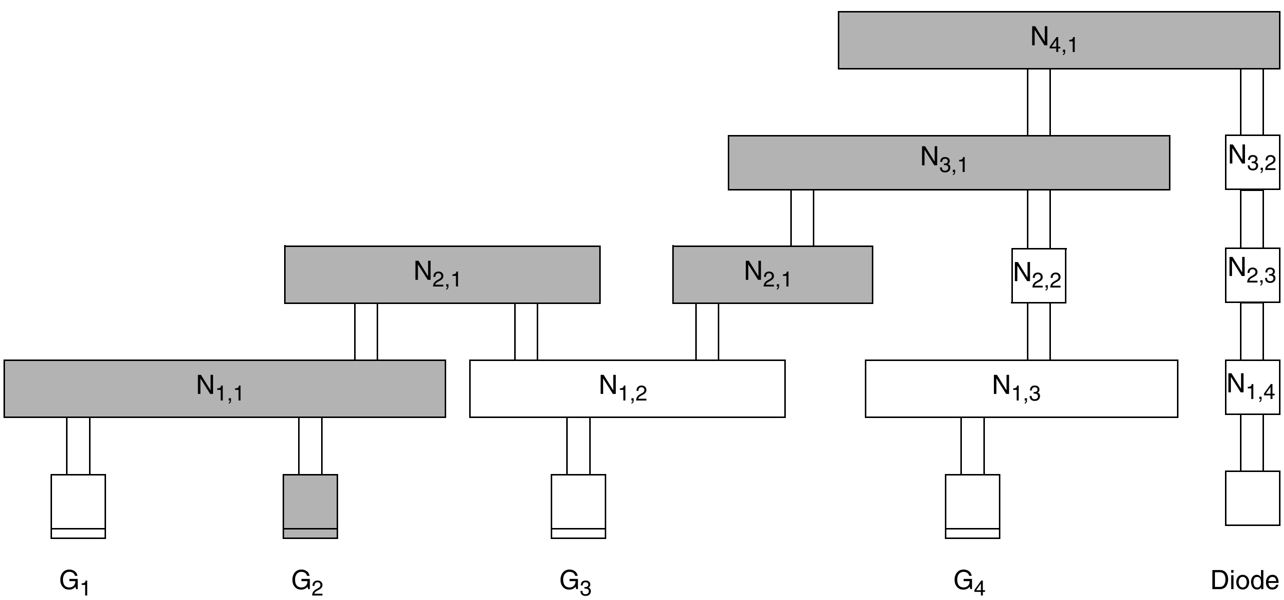
$CAR(m_i) = PAR(m_i) + CAR(v_{i-1})$

This means that the CAR from the cut layer below this metal layer is accumulated, instead of the CAR from the metal layer below this metal layer.

Note: In practice, the router only needs to keep track of the worst-case CAR; however, the CARs for all of the gates shown in Figure C-8 on page 1154 are described here.

The router calculates an antenna ratio with respect to a node-gate pair. To find the CAR for the node $N_{i,j}$ - gate G_k pair, you trace the path of the current between gate G_k and node $N_{i,j}$ and add the PAR with respect to gate G_k for the all nodes in the path between the first metal layer and layer i that you can trace back to G_k.

Figure C-8



The path of the current between gate G_2 and node $N_{4,1}$ is shaded.

Important

In Figure C-8 on page 1154, node $N_{1,2}$ is not shaded because it was not electrically connected to G_2 when *metal1* was processed. That is, because the charge accumulated on $N_{1,2}$ when *metal1* was processed cannot damage gate G_1, the router does not include it in the calculations for $CAR(N_{2,1},G_1)$.

Another way to explain this is to say that the PAE from node $N_{1,2}$ with respect to gate G_2 is `0`.

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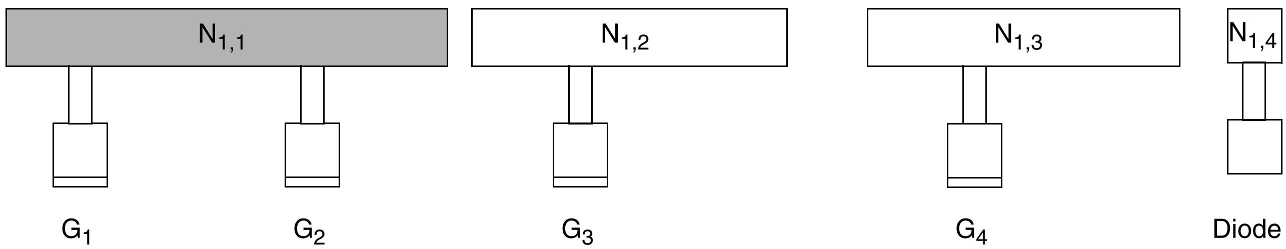
Calculations for CAR on the First Metal Layer

Figure C-9 on page 1155 shows the chip after the first metal layer is processed.

Figure C-9



In the figure above,

$CAR(N_{1,1},G_1) = PAR(N_{1,1},G_1)$

$CAR(N_{1,1},G_2) = PAR(N_{1,1},G_2)$

Because $PAR(N_{1,1},G_1)$ equals $PAR(N_{1,1},G_2)$, $CAR(N_{1,1},G_1)$ equals $CAR(N_{1,1},G_2)$.

Note: In general, $CAR(N_{i,j},G_k)$ equals $CAR(N_{i,j},G_{k'})$ if the two gates G_k and $G_{k'}$ are electrically connected to the same node on *metal1*, the lowest layer that is subject to the process antenna effect.

Calculations for CAR on the Second Metal Layer

Figure C-10 on page 1156 shows the chip after the second metal layer is processed.

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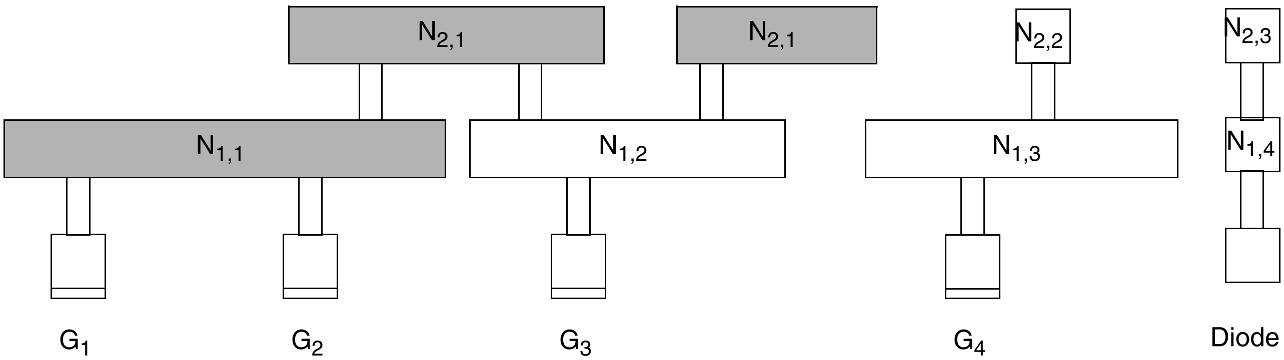
Figure C-10



The path of the current between gate G_1 and node $N_{2,1}$ is shaded.
Note that node $N_{2,1}$ comprises two pieces of metal.

Important

In the figure above, $N_{1,2}$ is not included in the calculations for $CAR(N_{2,1},G_1)$ because it was not electrically connected to G_1 when *metal1* was processed. That is, because the charge accumulated on $N_{1,2}$ when *metal1* was processed cannot damage gate G_1, the router does not include it in the calculations for $CAR(N_{2,1},G_1)$.

In the figure above,

$CAR(N_{2,1},G_1) = PAR(N_{1,1},G_1) + PAR(N_{2,1},G_1)$

$CAR(N_{2,1},G_2) = PAR(N_{1,1},G_2) + PAR(N_{2,1},G_2)$

Gates G_1 and G_2 have the same history with regard to PAE because they are connected to the same piece of *metal1*, so they have the same CAR for any node on a specified layer:

$CAR(N_{2,1},G_1) = CAR(N_{2,1},G_2)$

Calculations for CAR on the Third Metal Layer

Figure C-11 on page 1157 shows the chip after the third metal layer is processed.

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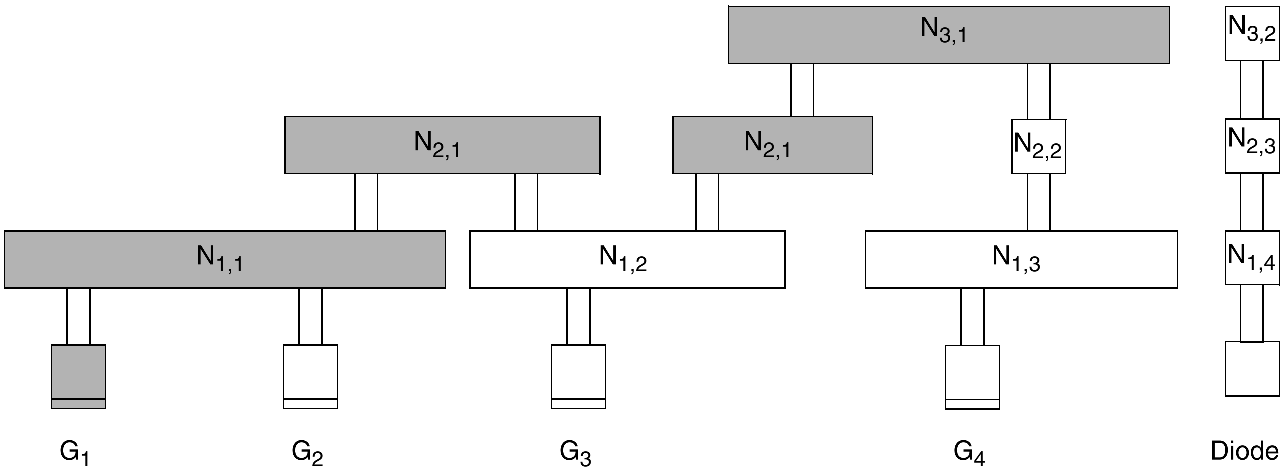
Figure C-11



The path of the current between gate G_1 and node $N_{3,1}$ is shaded.

Gate G_1

In the figure above,

$CAR(N_{3,1},G_1) = PAR(N_{1,1},G_1) + PAR(N_{2,1},G_1) + PAR(N_{3,1},G_1)$

Gate G_2

In the figure above,

$CAR(N_{3,1},G_2) = PAR(N_{1,1},G_2) + PAR(N_{2,1},G_2) + PAR(N_{3,1},G_2)$

$CAR(N_{3,1},G_1)$ equals $CAR(N_{3,1},G_2)$ because gates G_1 and G_2 are both electrically connected to the same node, $N_{1,1}$, on *metal1* and therefore have the same history with regard to PAE. Therefore, the formula for $CAR(N_{3,2},G_2)$ is $CAR(N_{3,1},G_1) = CAR(N_{3,1},G_2)$

Gates G_3 and G_4

Gates G_3 and G_4 are not connected to the same node on *metal1* and therefore do not have the same history with regard to PAE. Therefore, the $CAR(N_{3,1},G_3)$ and $CAR(N_{3,1},G_4)$ do not necessarily equal $CAR(N_{3,1},G_1)$ or $CAR(N_{3,1},G_2)$.

In Figure C-12 on page 1158, the relevant areas for calculating CAR for gate G_3 are shaded.

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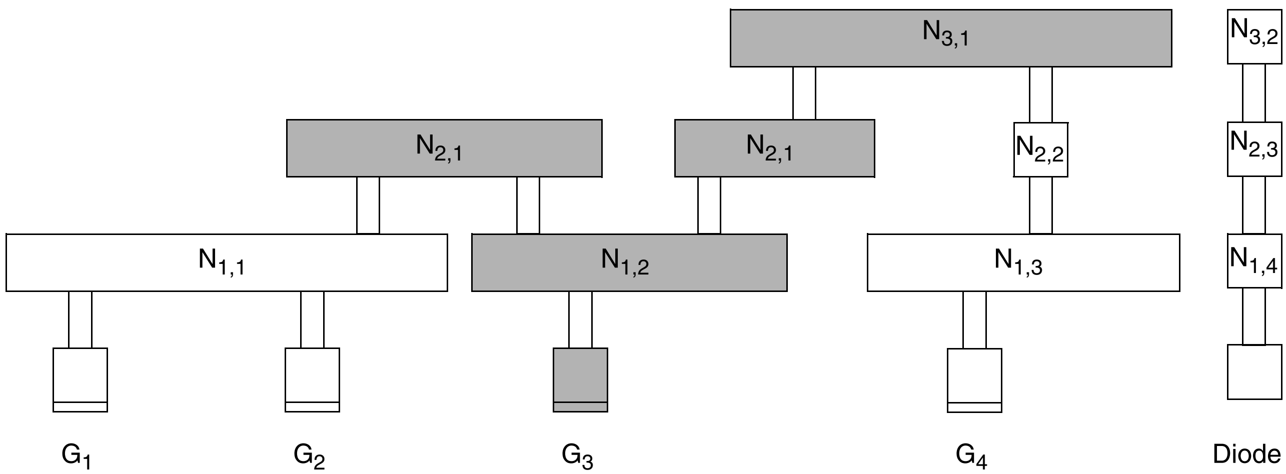
Figure C-12

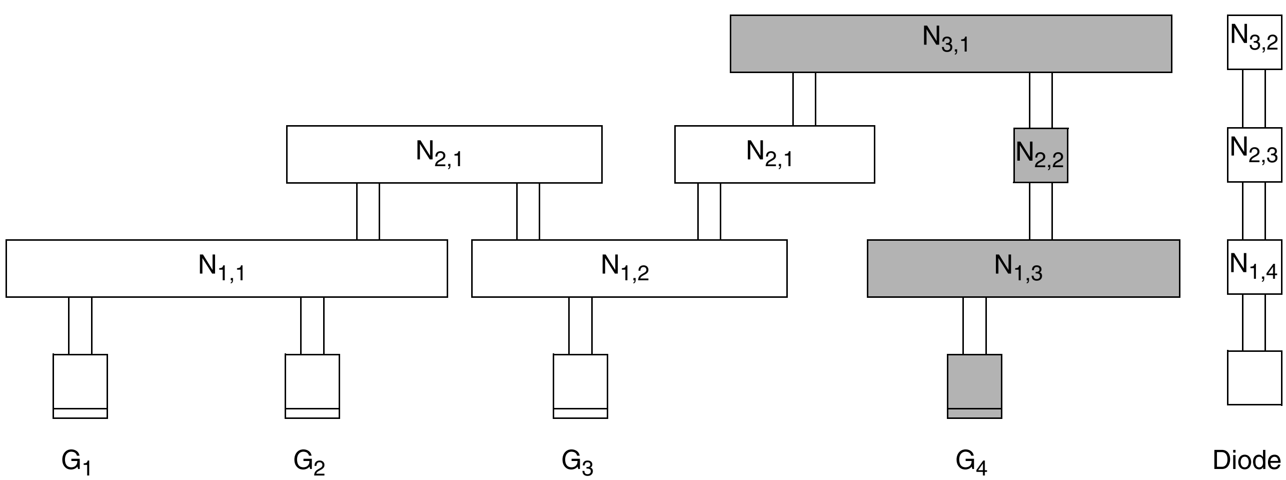


In the figure above,

$CAR(N_{3,1},G_3) = PAR(N_{1,2},G_3) + PAR(N_{2,1},G_3) + PAR(N_{3,1},G_3)$

In Figure C-13 on page 1158, the relevant areas for calculating CAR for gate G_4 are shaded.

Figure C-13



In the figure above,

$CAR(N_{3,1},G_4) = PAR(N_{1,3},G_4) + PAR(N_{2,2},G_4) + PAR(N_{3,1},G_4)$

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Calculations for CAR on the Fourth Metal Layer

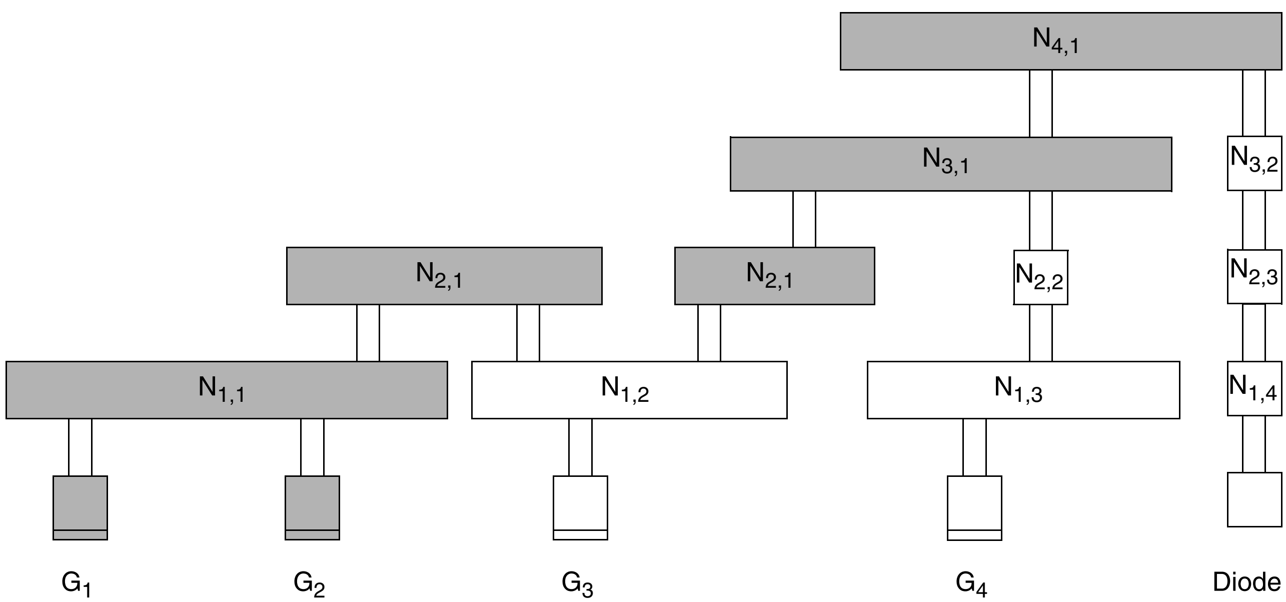
The following figure shows the chip after the fourth metal layer is processed.

Note: Node $N_{4,1}$ is connected to the diffusion layer through the output diode. After the router calculates the antenna ratio, it compares its calculations to the area of the diffusion, instead of the area of the gates.

Gates G_1 and G_2

In Figure C-14 on page 1159, the relevant areas for calculating $CAR(N_{4,1},G_1)$ and $CAR(N_{4,1},G_2)$ are shaded.

Figure C-14



In the figure above,

$$CAR(N_{4,1},G_1) = PAR(N_{1,1},G_1) + PAR(N_{2,1},G_1) + PAR(N_{3,1},G_1) + PAR(N_{4,1},G_1)$$

$$CAR(N_{4,1},G_2) = PAR(N_{1,1},G_2) + PAR(N_{2,1},G_2) + PAR(N_{3,1},G_2) + PAR(N_{4,1},G_2)$$

$$CAR(N_{4,1},G_1) = CAR(N_{4,1},G_2)$$

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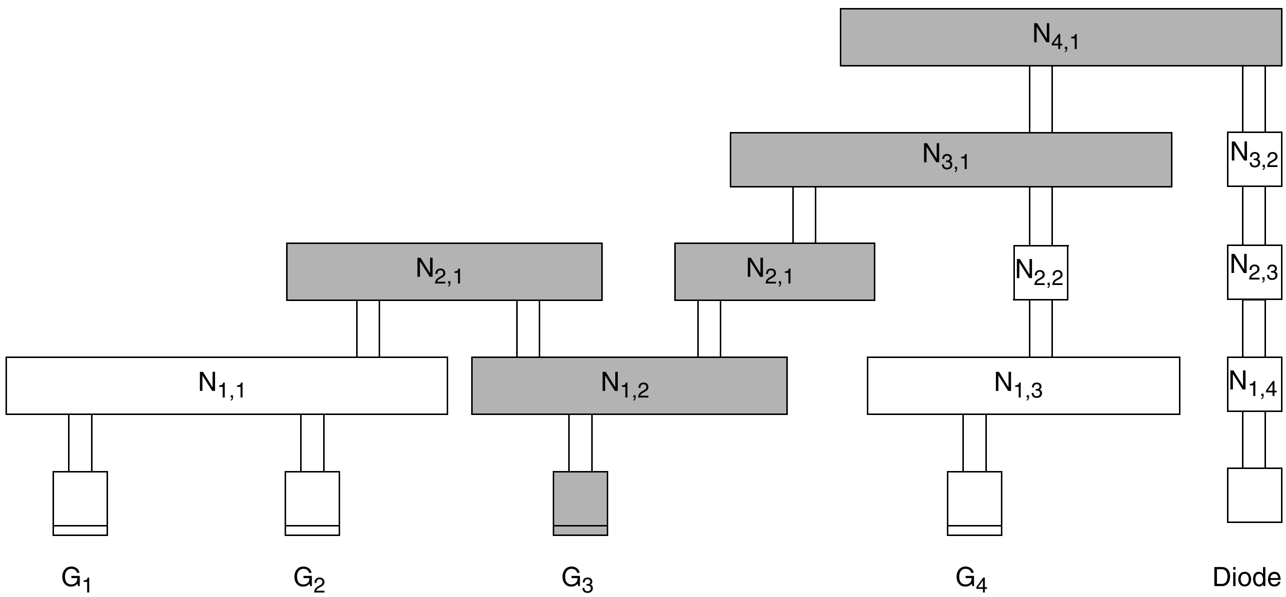
Gate G_3

In Figure C-15 on page 1160, the relevant areas for calculating `CAR(`$N_{4,1}$`,`G_3`)` are shaded.

Figure C-15



In the figure above,

$$CAR(N_{4,1},G_3) = PAR(N_{1,2},G_3) + PAR(N_{2,1},G_3) + PAR(N_{3,1},G_3) + PAR(N_{4,1},G_3)$$

`CAR(`$N_{4,1}$`,`G_3`)` does not equal `CAR(`$N_{4,1}$`,`G_1`)` or `CAR(`$N_{4,1}$`,`G_2`)` because it is not connected to the same node on *metal1*.

Gate G_4

In Figure C-16 on page 1161, the relevant areas for calculating `CAR(`$N_{4,1}$`,`G_4`)` are shaded.

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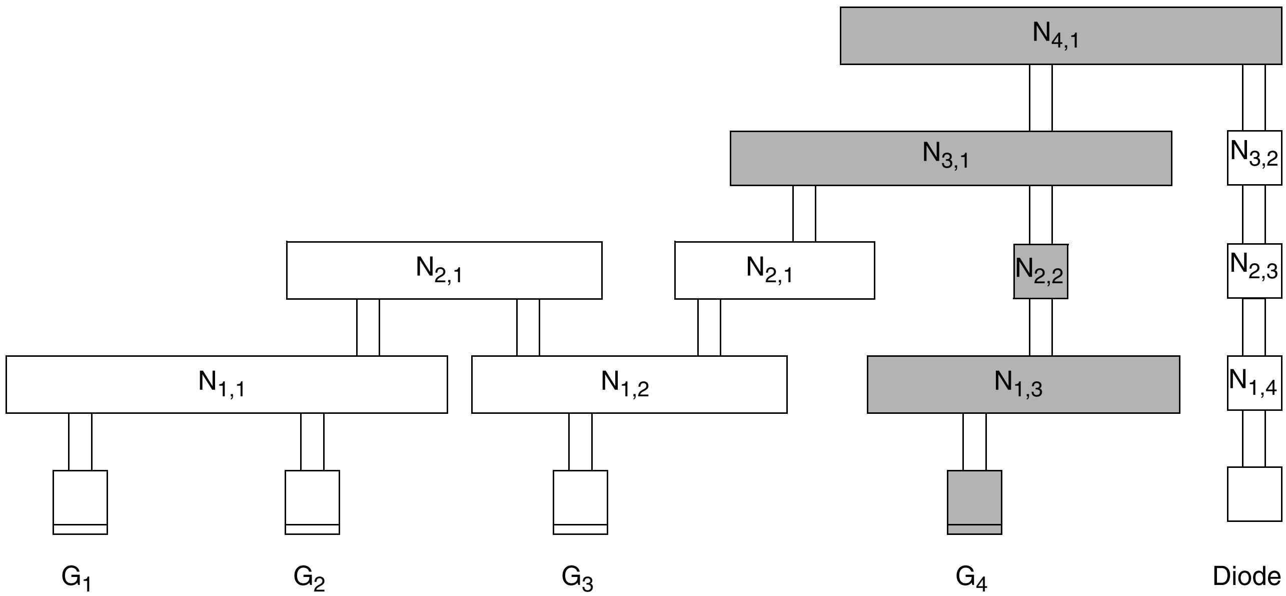
Figure C-16



In the figure above,

`CAR(`$N_{4,1}$`,`G_4`) = PAR(`$N_{1,3}$`,`G_4`) + PAR(`$N_{2,2}$`,`G_4`)`
`+ PAR(`$N_{3,1}$`,`G_4`) + PAR(`$N_{4,1}$`,`G_4`)`

`CAR(`$N_{4,1}$`,`G_4`)` does not equal `CAR(`$N_{4,1}$`,`G_1`)`, `CAR(`$N_{4,1}$`,`G_2`)`, or `CAR(`$N_{4,1}$`,`G_3`)` because it is not connected to the same node on *metal1*.

Calculating Ratios for a Cut Layer

The router calculates damage from a cut layer separately from damage from a metal layer. Calculations for the cut layers do not use side area modelling.

Calculating a PAR on a Cut Layer

The general PAR(c_i) equation for a single layer is calculated as:

$$PAR(c_i) = \frac{((\mathit{cutFactor} \text{ x cut\_area}) \text{ x } \mathit{diffAreaReduceFactor}) - (\mathit{minusDiffFactor} \text{ x diff\_area})}{\text{gate\_area} + (\mathit{plusDiffFactor} \text{ x diff\_area})}$$

The existing `ANTENNAAREAFACTOR` statement is shown as *`cutFactor`* for the metal area. Likewise, the `ANTENNAAREADIFFREDUCEPWL` statement is shown as *`diffAreaReduceFactor`*, the `ANTENNAAREAMINUSDIFF` statement is shown as

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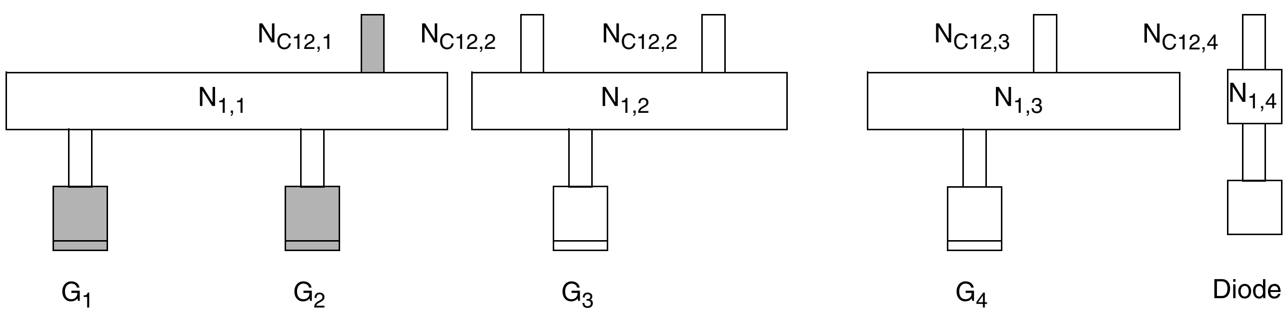
minusDiffFactor, and the ANTENNAGATEPLUSDIFF statement is shown as *plusDiffFactor*. For cut layer, the ratio equation illustrates the effect of an ANTENNAAREAFACTOR *cutFactor* statement as *metalFactor*. If there is no preceding ANTENNAAREAFACTOR statement, the *metalFactor* value defaults to 1.0.

In the figures and text that follow,

- *Cij* is the cut layer between $metal_i$ and $metal_j$.
- $N_{Cij,k}$ specifies an electrically connected node on *Cij*.
- The nodes are numbered sequentially, from left to right.

Figure C-17 on page 1162 shows the chip after the C12 process step.

Figure C-17



In the figure above,

$$PAR(N_{C12,1},G_1) = \frac{Area(N_{C12,1})}{Area(G_1) + Area(G_2)}$$

As in calculations on the metal layers,

$PAR(N_{C12,1},G_1) = PAR(N_{C12,1},G_2)$

Calculating a CAR on a Cut Layer

As explained in "Calculating Antenna Ratios":

$$CAR(c_i) = PAR(c_i) + CAR(c_{i-1})$$

To create a single accumulative model that combines both metal and cut damage into one model, specify the ANTENNACUMROUTINGPLUSCUT statement for the layer, so that:

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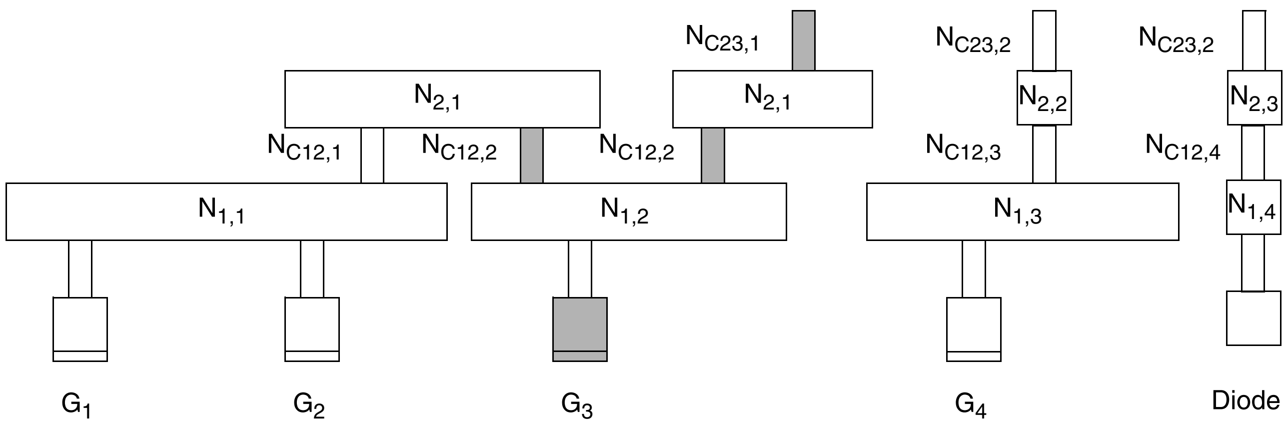
$\mathrm{CAR}(c_i) = \mathrm{PAR}(c_i) + \mathrm{CAR}(m_{i-1})$

This means that the CAR from the *metal* layer below this cut layer is accumulated, instead of the CAR from the cut layer below this cut layer.

Figure C-18 on page 1163 shows the chip after the `C23` process step.

Figure C-18



The router calculates the CAR with respect to gate G_3 after the cut `C23` process step as follows:

$$\mathrm{CAR}(N_{C23,1}, G_3) = \frac{\mathrm{Area}(N_{C12,2})}{\mathrm{Area}(G_3)} + \frac{\mathrm{Area}(N_{C23,1})}{\mathrm{Area}(G_1) + \mathrm{Area}(G_2) + \mathrm{Area}(G_3)}$$

Figure C-19 on page 1164 shows the chip after the `C34` process step.

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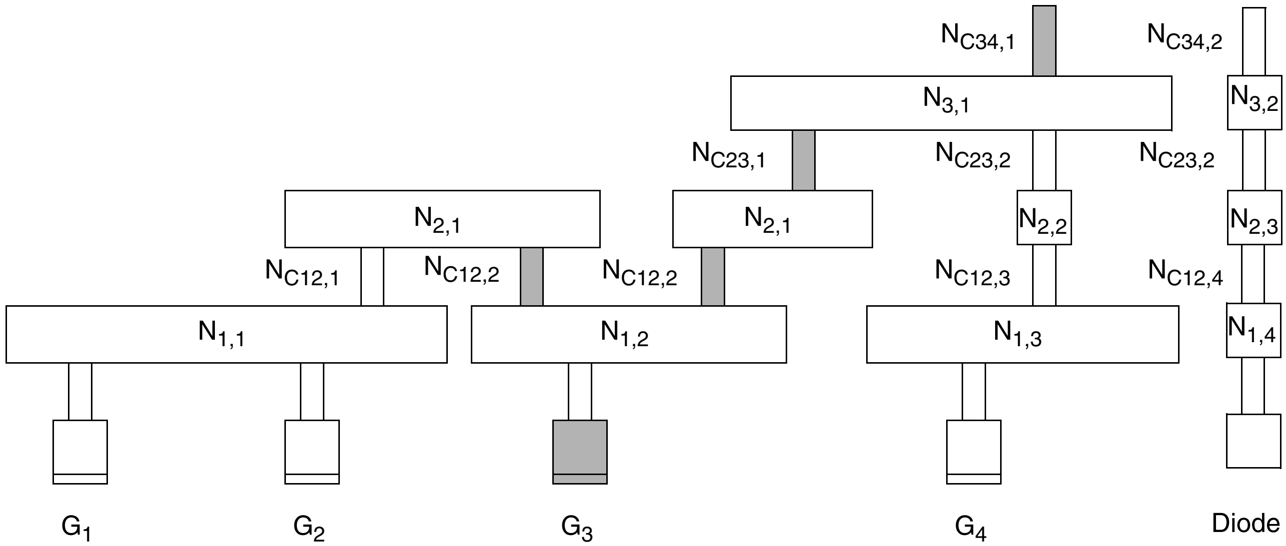
Figure C-19



The router calculates the CAR with respect to gate G_3 after the cut `C34` process step as follows:

$$CAR(N_{C34,1},G_3) =$$

$$\frac{Area(N_{C12,2})}{Area(G_3)} + \frac{Area(N_{C23,1})}{Area(G_1) + Area(G_2) + Area(G_3)} + \frac{Area(N_{C34,1})}{Area(G_1) + Area(G_2) + Area(G_3) + Area(G_4)}$$

Checking for Antenna Violations

For each metal layer, the router performs several antenna checks, using the keywords and values specified in the LEF or DEF file. The router can perform the following four types of antenna checks, depending on the keywords you set in the LEF file:

- Area Ratio Check
- Side Area Ratio Check
- Cumulative Area Ratio Check
- Cumulative Side Area Ratio Check

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Area Ratio Check

The area ratio check compares the PAR for each layer to the value of the ANTENNAAREARATIO or ANTENNADIFFAREARATIO.

The router calculates the PAR as follows:

$$\text{PAR}\ (\text{N}_{i,j},\ \text{G}_k) = \frac{\text{Drawn area of N}_{i,j}}{\Sigma\ \text{Area of gates connected below N}_{i,j}}$$

According to the formula above, the area ratio check finds the PAR for node $N_{i,j}$ with respect to gate G_k by dividing the drawn area of the node by the area of the gates that are electrically connected to it. The final PAR is multiplied by the ANTENNAAREAFACTOR (the default value for the factor is 1) and compared to the ANTENNAAREARATIO or ANTENNADIFFAREARATIO. If the PAR is greater than the ANTENNAAREARATIO or ANTENNADIFFAREARATIO specified in the LEF file, the router finds a process antenna violation and attempts to fix it.

The link between $PAR(N_{i,j}, G_k)$ and a PAE violation at node $N_{i,j}$ depends on whether node $N_{i,j}$ is connected to a piece of diffusion, as follows:

- If there is no connection from node $N_{i,j}$ to a diffusion area through the current and lower layers, a violation occurs when the PAR is greater than the ANTENNAAREARATIO.
- If there is a connection from node $N_{i,j}$ to a diffusion area through current and lower layers, a violation occurs when the PAR is greater than the ANTENNADIFFAREARATIO.
- If there is a connection from node $N_{i,j}$ to a diffusion area through current and lower layers, and ANTENNADIFFAREA is not specified for an output or inout pin, the value is 0.

Side Area Ratio Check

The side area ratio check compares the PAR computed based on the side area of the nodes for each layer to the value of the ANTENNASIDEAREARATIO or ANTENNADIFFSIDEAREARATIO.

The router calculates the PAR as follows:

$$\text{PAR}\ (\text{N}_{i,j}\ \text{G}_k) = \frac{\text{Side area of N}_{i,j}}{\Sigma\ \text{Area of gates connected below N}_{i,j}}$$

According to the formula above, the area ratio check finds the PAR for node $N_{i,j}$ with respect to gate G_k by dividing the side area of the node by the area of the gates that are electrically

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connected to $N_{i,j}$. The final PAR is multiplied by the `ANTENNASIDEAREAFACTOR` (the default value for the factor is `1`) and compared to the `ANTENNASIDEAREARATIO` or `ANTENNADIFFSIDEAREARATIO`. If the PAR is greater than the `ANTENNASIDEAREARATIO` or `ANTENNADIFFSIDEAREARATIO` specified in the LEF file, the router finds a process antenna violation and attempts to fix it.

The link between $PAR(N_{i,j}, G_k)$ and a PAE violation at node $N_{i,j}$ depends on whether node $N_{i,j}$ is connected to a piece of diffusion, as follows:

- If there is no connection to the diffusion area through the current and lower layers, a violation occurs when the PAR is greater than the `ANTENNASIDEAREARATIO`.
- If there is a connection to the diffusion area through current and lower layers, a violation occurs when the PAR is greater than the `ANTENNADIFFSIDEAREARATIO`.
- If there is a connection to the diffusion area through current and lower layers, and `ANTENNADIFFAREA` is not specified for an output or inout pin, the value is `0`.

Cumulative Area Ratio Check

The cumulative area ratio check compares the CAR to the value of `ANTENNACUMAREARATIO` or `ANTENNACUMDIFFAREARATIO`. The CAR is equal to the sum of the PARs of all nodes on the same or lower layers that are electrically connected to the gate.

Note: When you use CARs, you can ignore metal layers by not specifying the CAR keywords for those layers. For example, if you want to check *metal1* using a PAR and the remaining metal layers using a CAR, you can define `ANTENNAAREARATIO` or `ANTENNASIDEAREARATIO` for *metal1*, and `ANTENNACUMAREARATIO` or `ANTENNACUMSIDEAREARATIO` for the remaining metal layers.

The cumulative area ratio check finds the CAR for node $N_{i,j}$ with respect to gate G_k by adding the PARs for all layers of metal, from the current layer down to *metal1*, for all nodes that are electrically connected G_k. The final CAR is multiplied by the `ANTENNAAREAFACTOR` (the default value for the factor is `1`) and compared to the `ANTENNACUMAREARATIO` or `ANTENNACUMDIFFAREARATIO`. If the CAR is greater than the `ANTENNACUMAREARATIO` or `ANTENNACUMDIFFAREARATIO` specified in the LEF file, the router finds a process antenna violation and attempts to fix it.

The link between $CAR(N_{i,j}, G_k)$ and a PAE violation at node $N_{i,j}$ depends on whether node $N_{i,j}$ is connected to a piece of diffusion, as follows:

- If there is no connection to a diffusion area through the current and lower layers, a violation occurs when the CAR is greater than the `ANTENNACUMAREARATIO`.

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- If there is a connection to a diffusion area through current and lower layers, a violation occurs when the CAR is greater than the `ANTENNACUMDIFFAREARATIO`.
- If there is a connection to a diffusion area through current and lower layers, and `ANTENNADIFFAREA` is not specified for an output or inout pin, the value is `0`.

Cumulative Side Area Ratio Check

The cumulative side area ratio check compares the CAR to the value of the `ANTENNACUMSIDEAREARATIO` or `ANTENNACUMDIFFAREARATIO`.

Note: When you use CARs, you can ignore metal layers by not specifying the CAR keywords for those layers. For example, if you want to check *metal1* using a PAR and the remaining metal layers using a CAR, you can define `ANTENNAAREARATIO` or `ANTENNASIDEAREARATIO` for *metal1*, and `ANTENNACUMAREARATIO` or `ANTENNACUMSIDEAREARATIO` for the remaining metal layers.

The cumulative side area ratio check finds the CAR for node $N_{i,j}$ with respect to gate G_k by adding the PARs for all layers of metal, from the current layer down to *metal1*, for all nodes that are electrically connected G_k. The final CAR is multiplied by the `ANTENNASIDEAREAFACTOR` (the default value for the factor is `1`) and compared to the `ANTENNACUMSIDEAREARATIO` or `ANTENNACUMDIFFAREARATIO`. If the CAR is greater than the `ANTENNACUMSIDEAREARATIO` or `ANTENNACUMDIFFAREARATIO` specified in the LEF file, the router finds a process antenna violation and attempts to fix it.

- If there is no connection to a diffusion area through the current and lower layers, a violation occurs when the CAR is greater than the `ANTENNACUMSIDEAREARATIO`.
- If there is a connection to a diffusion area through current and lower layers, a violation occurs when the CAR is greater than the `ANTENNACUMSIDEAREARATIO`.
- If there is a connection to a diffusion area through current and lower layers, and `ANTENNACUMDIFFAREA` is not specified for an output or inout pin, the value is `0`.

Cut Layer Process Antenna Model Examples

- **Example 1**

  To create the following process antenna rule for a cut layer *via1*:

  cut_area / (gate_area + 2.0 x diff_area) <= 10

  Cut layers should include the following information:

  ```
  ANTENNAGATEPLUSDIFF 2.0 ;
  ANTENNADIFFAREARATIO 10 ;
  ```


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- **Example 2**

  Assume the following process antenna rule:

  cut_area x PWL(diff_area) / gate_area <= 10

  This rule uses a cumulative model with diffusion area reduction function, where:

  - PAR = (cut_area x diffReduceFactor) / gate_area <= 10
  - diffReduceFactor = 1.0 for diff_area < 0.1μm^2
  - diffReduceFactor = 0.2 for diff_area >= 0.1 μm^2

  Cut layers should include the following information:

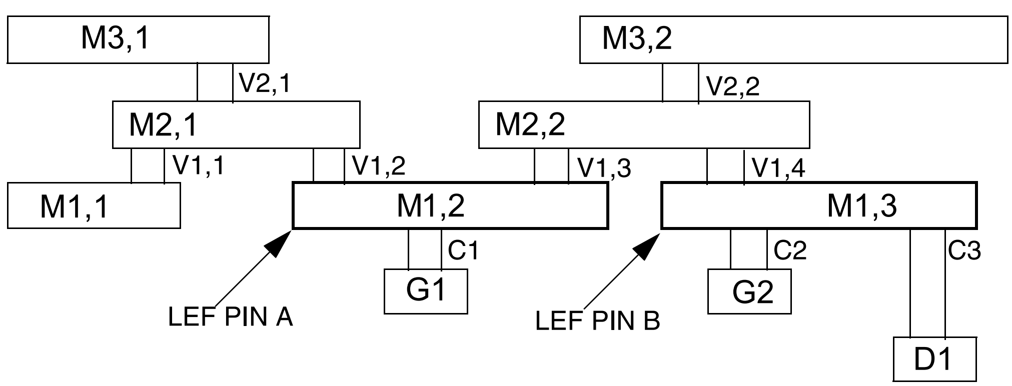
```
ANTENNAAREADIFFREDUCEPWL ( ( 0.0 1.0 ) ( 0.0999 1.0 ) ( 0.1 0.2 )
    ( 1000.0 0.2 ) ) ;
ANTENNACUMDIFFAREARATIO 10 ;
```

For examples of models that use the `ANTENNACUMROUTINGPLUSCUT` and the `ANTENNAAREAMINUSDIFF` rules, see the examples below in "Routing Layer Process Antenna Models."

Routing Layer Process Antenna Model Examples

The following process antenna rule examples use the topology shown in Figure C-20 on page 1168. In this figure, there are two polysilicon gates (G1, G2), one diffusion connection (D1), contacts (C), and via (*V1*, *V2*) and metal (*M1*, *M2*, *M3*) shapes. Note that *M1,2* is one LEF `PIN`, and *M1,3* is a different LEF `PIN`. The other metal is routing.

Figure C-20



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The following area values are also used for the examples:

G1 = 1.0	D1 = 0.5	M2,1 = 4.0
G2 = 2.0	M1,1 = 1.0	M2,2 = 5.0
All Cs = 0.1	M1,2 = 2.0	M3,1 = 6.0
All Vs = 0.1	M1,3 = 3.0	M3,2 = 9.0

Example 1

The following process antenna rule combines cut area and metal area into one cumulative rule:

ratio = (metal _area + 10 x cut_area) / gate_area

- The cumulative ratio <= 1000 for diffusion < 0.1, and <= 4000 for diffusion >= 0.1
- The single layer ratio <= 500 for diffusion < 0.1, and <= 1500 for diffusion >= 0.1

Every routing layer should include the following information:

```
ANTENNACUMROUTINGPLUSCUT ;
ANTENNACUMDIFFAREARATIO ( ( 0.0 1000 ) ( 0.0999 1000 ) ( 0.1 4000 )
    ( 1000.0 4000 ) ) ;
ANTENNADIFFAREARATIO ( ( 0.0 5000 ) ( 0.0999 500 ) ( 0.1 1500 )
    ( 1000.0 1500 ) ) ;
```

Every cut layer should include the following information:

```
ANTENNAAREAFACTOR 10 ; #10.0 x cut area
ANTENNACUMROUTINGPLUSCUT ;
ANTENNACUMDIFFAREARATIO ( ( 0.0 1000 ) ( 0.0999 1000 ) ( 0.1 4000 )
    ( 1000.0 4000 ) ) ;
ANTENNADIFFAREARATIO ( ( 0.0 5000 ) ( 0.0999 500 ) ( 0.1 1500 )
    ( 1000.0 1500 ) ) ;
```

Note: `ANTENNAAREARATIO` and `ANTENNACUMAREARATIO` are not required because the `*DIFFAREARATIO` statements are checked, even if diff_area is equal to 0.

For gate `G1`, the PARs and CARs are computed as follows:

1. CAR(C,G1) = 10 x area(C1) / area(G1) = 10 x 0.1 / 1.0 = 1.0

   The polysilicon and contact cut layer and shapes are not normally visible in LEF and DEF. If the contact cut area should be included, its CAR value should be included with LEF `PIN A`, using appropriate `ANTENNA` statements. The *M1* `PIN` area should not be

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included because *M1* area is a `PIN` shape in the LEF and will be added in by tools reading LEF. Therefore, there should be two antenna statements for LEF `PIN A`, either:

```
ANTENNAGATEAREA 1.0 LAYER M1 ;
ANTENNAMAXCUTCAR 1.0 LAYER C ;
```

or:

```
ANTENNAGATEAREA 1.0 LAYER M1 ;
ANTENNAMAXAREACAR 1.0 LAYER M1 ;
```

Because the *M1* `PIN` area is not included in the `MAXAREACAR` value, both of sets of statements give the same results. For more details, see "Calculations for Hierarchical Designs."

Similarly, the LEF `PIN B` should have values, such as either:

```
ANTENNAGATEAREA 2.0 LAYER M1 ;
ANTENNADIFFAREA 0.5 LAYER M1 ;
ANTENNAMAXCUTCAR 1.0 LAYER C ; #only C2 affects G2; C3 does not
```

or:

```
ANTENNAGATEAREA 2.0 LAYER M1 ;
ANTENNADIFFAREA 0.5 LAYER M1 ;
ANTENNAMAXAREACAR 1.0 LAYER M1 ; #only C2 affects G2; C3 does not
```

2. PAR(M1,G1) = area(M1,2) / area(G1) = 2 / 1 = 2.0
3. CAR(M1,G1) = PAR(M1,G1) + PIN A's CAR(C,G1)
   PIN A's CAR(C,G1) = ANTENNAMAXCUTCAR for LAYER C = 1.0
   = 2.0 + 1.0 = 3.0
4. diode_area = 0, single-layer PWL(0) = 500, check PAR(M1,G1) = 2.0 <= 500,
   cum-layer PWL(0) = 1000, therefore check CAR(M1,G1) = 3.0 <= 1000
5. PAR(V1,G1) = 10 x area(V1,2 + V1,3) / area(G1) = 10 x 0.2 / (1) = 2.0
6. CAR(V1,G1) = PAR(V1,G1) + CAR(M1,G1) = 2.0 + 3.0 = 5.0
7. diode_area = 0, single-layer PWL(0) = 500, check PAR(V1,G1) = 2.0 <= 500,
   cum_layer PWL(0) = 1000, therefore check CAR(V1,G1) = 5.0 <= 1000
8. PAR(M2,G1) = area(M2,1 + M2,2) / area(G1 + G2) = (4+5) / (1 + 2) = 3.0
9. CAR(M2,G1) = PAR(M2,G1) + CAR(V1,G1) = 3.0 + 5.0 = 8.0
10. diode_area = 0.5, single-layer PWL(0.5) = 1500, check PAR(M2,G1) = 3.0 <= 1500,
    cum_layer PWL(0.5) = 4000, therefore check CAR(M2, G1) = 8.0 <= 4000
11. PAR(V2,G1) = 10 x area(V2,1 + V2,2) / area(G1 + G2) = 10 x 0.2 / (1 + 2) = 0.67

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12. CAR(V2,G1) = PAR(V2,G1) + CAR(M2,G1) = 0.67 + 8.0 = 8.67

13. diode_area = 0.5, single-layer PWL(0.5) = 1500, check PAR(V2,G1) = 0.67 <= 1500, cum_layer PWL(0.5) = 4000, therefore check CAR(V2, G1) = 8.67 <= 4000

14. PAR(M3,G1) = area(M3,1 + M3,2) / area(G1 + G2) = (6 + 9) / (1 + 2) = 5

15. CAR(M3,G1) = PAR(M3,G1) + CAR(V2,G1) = 5 + 8.67 = 12.34

16. diode_area = 0.5, single-layer PWL(0.5) = 1500, check PAR(M3,G1) = 5 <= 1500, cum_layer PWL(0.5) = 4000, therefore check CAR(M3,G1) = 13.67 <= 4000

Example 2

The following cumulative rule is the same as the rule in Example 1, except it also subtracts the diff_area factor. Only the cumulative model is used.

ratio = [(metal_area + 10 x cut_area) - (100 x diff_area)] / gate_area

Every routing layer should include the following information:

```
ANTENNACUMROUTINGPLUSCUT ;
ANTENNAAREAMINUDIFF 100.0 ;
ANTENNACUMDIFFAREARATIO 1000 ;
```

Every cut layer should include the following information:

```
ANTENNAAREAFACTOR 10 ;  #10.0 x cut area
ANTENNACUMROUTINGPLUSCUT ;
ANTENNAAREAMINUDIFF 100.0 ;
ANTENNACUMDIFFAREARATIO 1000 ;
```

For gate `G1`, the PARs and CARs are computed as follows:

1. CAR(C,G1) = 10 x area(C1) / area(G1) = 10 x 0.1 / 1.0 = 2.0

   This value is on the LEF `PIN`, as mentioned in Example 1.

2. PAR(M1,G1) = area(M1,2) / area(G1) - (100 x diff_area) = (2 / 1) - (100 x 0) = 2.0

3. CAR(M1,G1) = PAR(M1,G1) + PIN A’s CAR(C,G1)
   PIN A’s CAR(M1) = ANTENNAMAXAREACAR for LAYER M1 = 1.0
   = 2.0 + 1.0 = 3.0

4. Check CAR(M1,G1) = 3.0 <= 1000

5. PAR(V1,G1) = [10 x area(V1,2 + V1,3) - (100 x diff_area)] / area(G1)
   = [(10 x .2) - (100 x 0)] / (1) = 2.0

6. CAR(V1,G1) = PAR(V1,G1) + CAR(M1,G1) = 2.0 + 3.0 = 5.0

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7. Check CAR(V1,G1) = 5.0 <= 1000
8. PAR(M2,G1) = [area(M2,1 + M2,2) - (100 x area(D1))] / area(G1 + G2)
   = [(4 + 5) - (100 x 0.5) / (1 + 2) = -13.67
9. CAR(M2,G1) = PAR(M2,G1) + CAR(V1,G1) = -13.67 + 5.0 = -8.67, truncate to 0
10. Check CAR(M2,G1) = 0 <= 1000
11. PAR(V2,G1) = [(10 x area(V2,1 + V2,2)) - (100 x area(D1)] / area(G1 + G2)
    = [(10 x 0.2) - (100 x 0.5)] / (1 + 2) = -16.0
12. CAR(V2,G1) = PAR(V2,G1) + CAR(M2,G1) = -16.0 + 0 = -16.0, truncate to 0
13. Check CAR(V2,G1) = 0 <= 1000
14. PAR(M3,G1) = [area(M3,1 + M3,2) - (100 x area(D1))] / area(G1 + G2)
    = [(6 + 9) - (100 x 0.5)] / (1 + 2) = -11.67
15. CAR(M3,G1) = PAR(M3,G1) + CAR(V2,G1) = -11.67 + 0 = -11.67, truncate to 0
16. Check CAR(M3,G1) = 0 <= 1000

Example 3

The following cumulative rule for metal layers includes a diffusion area factor added into the denominator of the ratio:

Single layer: metal_area / (gate_area + 2.0 x diff_area) <= 1000

Cumulative for the layer: metal_area / (gate_area + 2.0 x diff_area) <= 5000

Every metal layer should include the following information:

```
ANTENNAPLUSGATEDIFF 2.0 ;
ANTENNADIFFAREARATIO 1000 ;
ANTENNACUMDIFFAREARATIO 5000 ;
```

Note: The via area is ignored in this example. If an independent via model is needed, similar statements should be added to the via layers, which would be computed separately.

For gate `G1`, the PARs and CARs are computed as follows:

1. PAR(M1,G1) = area(M1,2) / area(G1) = 2.0 / 1 = 2
2. CAR(M1,G1) = PAR(M1,G1) = 2
3. Check PAR(M1,G1) = 2 <= 1000,
   check CAR(M1,G1) = 2 <= 5000

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4. PAR(M2,G1) = area(M2,1 + M2,2) / [area(G1 + G2) + 2 x area(D1)]
   = (4 + 5) / [(1 + 2) + 2 x 0.5] = 2.25
5. CAR(M2,G1) = CAR(M1,G1) + PAR(M2,G1) = 2 + 2.25 = 4.25
6. Check PAR(M1,G1) = 2.25 <= 1000,
   check CAR(M1,G1) = 4.25 <= 5000
7. PAR(M3,G1) = area(M3,1 + M3,2) / [area(G1 + G2) + 2 x area(D1)]
   = (6 + 9) / [(1 + 2) + 2 x 0.5] = 3.75
8. CAR(M3,G1) = PAR(M3,G1) + CAR(M2,G1) = 3.75 + 4.25 = 8.0
9. Check PAR(M1,G1) = 3.75 <= 1000,
   check CAR(M1,G1) = 8.0 <= 5000

Example 4

Assume a cumulative rule that includes a diffusion area reduction value and a routing ratio of 1000. The reduction value is 1.0 if the diff_area is less than 0.1, 0.2 if the diff_area equals 0.1, and decreases linearly to 0.1 if the diff_area equals 1.0. The reduction value remains 0.1 if the diff_area is greater than 1.0.

Every metal layer should include the following information:

```
ANTENNAAREADIFFREDUCEPWL ( ( 0.0 1.0 ) ( 0.0999 1.0 ) ( 0.1 0.2 )( 1.0 0.1 )
    ( 1000.0 0.1 ) ) ;" ;
ANTENNACUMDIFFAREARATIO 1000 ;
```

Note: The via area is ignored in this example. If an independent via model is needed, similar statements should be added to the via layers, which would be computed separately.

For gate `G1`, the PARs and CARs are computed as follows:

1. Initial PAR(M1,G1) = area(M1,2) / area(G1) = 2.0 / 1 = 2
2. diode_area = 0, PWL(0) = 1.0, therefore initial PAR(M1,G1) is multiplied by 1.0
   to give PAR(M1,G1) = 2 x 1 = 2
3. CAR(M1,G1) = PAR(M1,G1) = 2
4. Check CAR(M1,G1) <= 1000, therefore check 2 <= 1000
5. Initial PAR(M2,G1) = area(M2,1 + M2,2) / area(G1 + G2) = (4 + 5) / (1 + 2) = 3
6. diode_area = 0.5, PWL(0.5) = 0.155, therefore initial PAR(M2,G1) is multiplied by 1.0
   to give PAR(M2,G1) = 3 x 0.155 = 0.465
7. CAR(M2,G1) = CAR(M1,G1) + PAR(M2,G1) = 2 + 0.465 = 2.465

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8. Check CAR(M2,G1) <= 1000, therefore check 2.465 <= 1000
9. Initial PAR(M3,G1) = area(M3,1 + M3,2) / area(G1 + G2) = (6 + 9) / (1 + 2) = 5
10. diode_area = 0.5, PWL(0.5) = 0.155, therefore initial PAR(M3,G1) is multiplied by 0.155 to give PAR(M3,G1) = 5 x 0.1555 = 0.775
11. CAR(M3,G1) = PAR(M3,G1) + CAR(M2,G1) = 0.775 + 2.465 = 3.24
12. Check CAR(M3,G1) <= 1000, therefore check 3.24 <= 1000

Example Using the Antenna Keywords

The following example is a portion of a LEF file that shows the antenna keywords for a process that has cumulative area ratio damage for metal and cut layers.

Assume you have the following antenna rules for your process:

1. A maximum cumulative metal to gate area ratio of 1000
2. If a diode of diffusion area greater than `.1` micron^2 is connected to the metal, the maximum metal ratio is calculated as follows:
   - If *diff_area* > `0` and *diff_area* <= `0.099`, ANTENNACUMDIFFAREARATIO rule ratio `1000` is applied for the CAR check.
   - If *diff_area* >= `0.1`, the ANTENNACUMDIFFAREARATIO rule ratio can be calculated with the interpolate formula as:
     - `d =` *diff_axis*`[`*end*`] -` *diff_axis*`[`*start*`];`
     - `ratio =` *ratio_value*`[`*start*`] * (`*diff_axis*`[`*end*`] -` *pin's diff_area*`) /` *d* `+` *ratio_value*`[`*end*`] * (`*pin's diff_area* `-` *diff_axis*`[`*start*`]) /` *d*

     Inserting values in the formula:

     - `d = 100 - 0.1 = 99.9`
     - `ratio = 5200 * (100 -` *pin's diff_area*`) / 99.9 + 205000 * (pin's diff_area - 0.1) / 99.9`

       `= 5200 * 100 / 99.9 - (5200 *` *pin's diff_area*`) / 99.9 + (205000 *` *pin's diff_area*`) / 99.9 - 20500/99.9`

       `= 5000 + (205000x-5200x)/99.9`

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= `5000 + 2000 * pin's diff_area`

3. A maximum cumulative via to gate area ratio of 20
4. If a diode of diffusion area greater than .1 micron^2 is connected to the via, the maximum via ratio is: ratio = diode_area x 200 + 100

The corresponding LEF file would include:

```
LAYER M1
  TYPE ROUTING ;
  ...
  ANTENNACUMAREARATIO 1000 ;
  ANTENNACUMDIFFAREARATIO
       PWL ( ( 0 1000 ) ( 0.099 1000 ) ( 0.1 5200 ) ( 100 205000 ) ) ;
END M1

LAYER VIA1
  TYPE CUT ;
  ...
  ANTENNACUMAREARATIO 20 ;
  ANTENNACUMDIFFAREARATIO
       PWL ( ( 0 20 ) ( 0.099 20 ) ( 0.1 120 ) ( 100 20100 ) ) ;
END VIA1
```

A typical standard cell that has only *M1* pins and routing inside of it would have:

```
MACRO INV1X
  CLASS CORE ;
  ...
  PIN IN
    DIRECTION INPUT ;
    ANTENNAGATEAREA .5 LAYER M1 ; # connects to 0.5 µm^2 poly gate
    ANTENNAPARTIALMETALAREA 1.0 LAYER M1 ; # has 1.0 µm^2 M1 area.
           # Note that it should not include the M1 pin area, just the M1 routing
           # area that is not included in the PIN shapes. In many cases, all of the
           # M1 routing is included in the PIN, so this value is 0, and not in the
           # LEF at all.
    ANTENNAMAXAREACAR 10.0 LAYER M1 ; # has 10.0 cumulative ratio so far.
           # This value can include area from internal poly routing if poly routing
           # damage is accumulated with the metal layers. It does not include
           # the area of the M1 pin area, just the M1 routing area that is not
           # included in the PIN shapes. If poly damage is not included, and all
           # of the M1 routing is included in the PIN, this value will be 0, and
```


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```
          # not in the LEF at all.
    ...
  END IN
  PIN OUT
    DIRECTION OUTPUT ;
    ANTENNADIFFAREA .2 LAYER M1 ; # connects to 0.2 µm^2 difusion area
    ANTENNAPARTIALMETALAREA 1.0 LAYER M1 ; # has 1.0 µm^2 M1 area
    # No ANTENNAMAXAREACAR value because no internal poly gate is connected
    ...
  END OUT
END INV1X
```

Using Antenna Diode Cells

Routers generally use one of two methods to fix process antenna violations:

- Change the routing by breaking the metal layers into smaller pieces
- Insert antenna diode cells to discharge the current

Changing the Routing

One method routers use to fix antenna violations is to limit the charge that is collected through the metal nodes exposed to the plasma. To do this, it goes up one layer or pushes the routing down one layer whenever the process antenna ratio exceeds the ratio set in the LEF file.

The router changes the routing by disconnecting nets with antenna violations and making the connections to higher metal layers instead. It does not make the connections to lower layers. This method works because the top metal layer always completes the connection from the gate to the output drain area of the driver, which is a diode that provides a discharge path.

Inserting Antenna Diode Cells

The second method routers use to repair antenna violations is to insert antenna diode cells in the design. The electrical charges on the metal that connects to the diodes is then discharged through the diode diffusion layer and substrate. The router inserts the diode cells automatically.

The following example shows a LEF definition of an antenna diode cell, with the `CLASS CORE ANTENNACELL` and `ANTENNADIFFAREA` defined:

```
MACRO antenna1
```


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```
    CLASS CORE ANTENNACELL ;
    ...
    PIN ANT1
        AntennaDiffArea 1.0 ;
        PORT
        LAYER metal1 ;
        RECT  0.190 2.380 0.470 2.660 ;
        END
    END ANT1
END antenna1
```

Using DiffUseOnly

LEF defines only one value for ANTENNAAREAFACTOR and one value for ANTENNASIDEAREAFACTOR, with or without DIFFUSEONLY, per layer. If you specify more than one antenna area or side area factor for a layer, only the last one is used. The AREAFACTOR value lets you scale the value of the metal area. If you use the DIFFUSEONLY keyword, only metal attached to diffusion is scaled.

Suppose you have the following LEF file:

```
Antenna.lef
-----------
LAYER M3
TYPE ROUTING ;
PITCH 0.56 ;
DIRECTION HORIZONTAL ;
WIDTH 0.28 ; SPACING 0.28 ;
SPACING 0.36 RANGE 1.0 250.0 ;
CAPACITANCE CPERSQDIST 0.0009762 ;
RESISTANCE RPERSQ 0.129 ;
THICKNESS 0.60 ;
AntennaAreaRatio 10000 ;
AntennaDiffAreaRatio 10000 ;
AntennaAreaFactor 1.2 DiffUseOnly ;
AntennaSideAreaRatio 5000 ;
AntennaDiffSideAreaRatio 5000 ;
AntennaSideAreaFactor 1.4 DiffUseOnly ;
END M3
```


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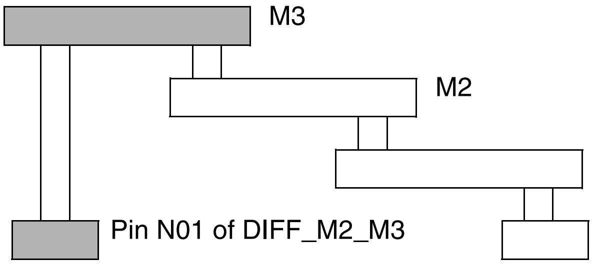
Figure C-21



In the figure,

- The input pin `H01` of `GATE_M2_M3` connects the metal wires to *metal1*, *metal2*, and *metal3* in sequence.
- The `ANTENNAAREAFACTOR 1.2 DIFFUSEONLY` and `ANTENNASIDEAREAFACTOR 1.4 DIFFUSEONLY` apply to *metal3* routing.
- Prior to *metal3* fabrication, there is no path to the diffusion diode. This causes the default factor of `1.0` to apply to the *metal1* and *metal2* segments shown when calculating PARs.

Calculations for Hierarchical Designs

The following section illustrates computation of antenna ratios for hierarchical designs.

LEF and DEF Keywords for Hierarchical Designs

If the keyword ends with ...	It refers to ...	Examples

`area`
`sideArea`
	Drawn area or side area of the metal wires. Measured in micron^2.	`ANTENNAPARTIALCUTAREA`
		`ANTENNAPARTIALMETALAREA`
		`ANTENNAPARTIALMETALSIDEAREA`
		`ANTENNAPINDIFFAREA`
		`ANTENNAPINGATEAREA`
		`ANTENNAPINPARTIALCUTAREA`

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If the keyword ends with ...	It refers to ...	Examples
CAR	Relationship the router is calculating CAR is used in keywords for cumulative antenna ratio.	ANTENNAMAXAREACAR ANTENNAMAXCUTCAR ANTENNAMAXSIDEAREACAR ANTENNAPINMAXAREACAR ANTENNAPINMAXCUTCAR ANTENNAPINMAXSIDEAREACAR

Design Example

Figure C-22 on page 1180 represents a macro block. This block can be a custom hard block or part of a bottom-up hierarchical flow. The resulting PAE values will be the same in either case. In the example,

- Gates G_1, G_2, G_3, and G_4 are the same size.
- Node $N_{1,3}$ is larger than node $N_{1,2}$.
- Vias (cuts) are all the same size.
- The I/O pin is on *metal3*.
- The area of diffusion for D_1 is $area(Diff_1)$.
- The area of diffusion for D_2 is $area(Diff_2)$.
- The area of the cut layer that connects node $N_{3,2}$ and node $N_{4,2}$ is $area(N_{34,2})$.
- Any damage from the poly layer or poly-to-metal1 via is ignored.

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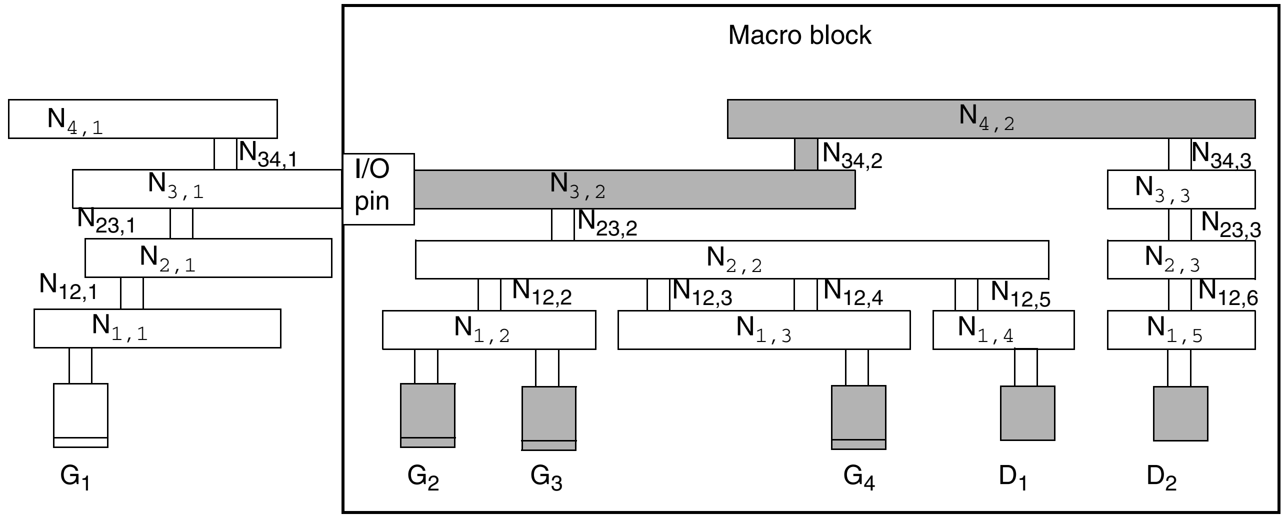
Figure C-22



Relevant Metal Areas

- The relevant metal area for PAE calculations is the partial metal drawn area and side area connected directly to the I/O pin on the inside of the macro on the specified layer.
- Only the same metal layer as the I/O pin or above is needed for PAR calculations in hierarchical designs.

Important

Do not include the drawn area or side area of the I/O pin in the area calculations for the block, because the router includes these areas in the calculations for the upper level. Only the internal routing area that is not part of the I/O pin should be included.

For the design in the figure above, you must specify values for the following metal areas in the LEF file:

```
ANTENNAPARTIALMETALAREA area(N3,2) LAYER Metal3 ;
ANTENNAPARTIALMETALAREA area(N4,2) LAYER Metal4 ;
ANTENNAPARTIALMETALSIDEAREA sideArea(N3,2) LAYER Metal3 ;
ANTENNAPARTIALMETALSIDEAREA sideArea(N4,2) LAYER Metal4 ;
```

You do not need to specify an ANTENNAPARTIALMETALAREA or ANTENNAPARTIALSIDEMETALAREA for any layer lower than *metal3* because the I/O pin is on *metal3*; that is, there is no connection outside the block until *metal3* is processed.

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Relevant Gate, Diffusion, and Cut Areas

- The relevant gate and diffusion areas are the gate and diffusion areas that connect directly to the I/O pin on the specified layer or are electrically connected to the pin through lower layers.
- The relevant partial cut area is above the current pin layer and inside the macro on the specified layer.

For the design in the figure above, you must specify values for the following gate, diffusion, and cut areas in the LEF file:

ANTENNAGATEAREA *area*($G_2 + G_3 + G_4$) LAYER Metal3 ;

ANTENNADIFFAREA *area*($Diff_1$) LAYER Metal3 ;

ANTENNADIFFAREA *area*($Diff_1 + Diff_2$) LAYER Metal4 ;

ANTENNAPARTIALCUTAREA *area*($N_{34,2}$) LAYER Via34 ;

Calculating the CAR

Use the following keywords to calculate the actual CAR on the I/O pin layer or above.

- The relevant maximum CAR value of the drawn and side areas are from the metal layer that is on or below the I/O pin layer.
- The relevant maximum CAR value of the cut layer is from the cut layer that is immediately above the I/O pin layer.

For the example in Figure C-22 on page 1180, the keywords and calculations for *metal3* and *via34* would be:

$$\text{ANTENNAMAXAREACAR}\left(\frac{N_{1,3}}{G_4}+\frac{N_{2,2}}{G_2+G_3+G_4}+\frac{N_{3,2}}{G_2+G_3+G_4}\right)\ \text{LAYER Metal3 ;}$$

$$\text{ANTENNAMAXSIDEAREACAR}\left(\frac{N_{1,3}}{G_4}+\frac{N_{2,2}}{G_2+G_3+G_4}+\frac{N_{3,1}}{G_2+G_3+G_4}\right)\ \text{LAYER Metal3 ;}$$

$$\text{ANTENNAMAXCUTCAR}\left(\frac{N_{12,3}+N_{12,4}}{G_4}+\frac{N_{23,2}}{G_2+G_3+G_4}+\frac{N_{34,2}}{G_2+G_3+G_4}\right)\ \text{LAYER Via34;}$$

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Sample LEF File for a Bottom-Up Hierarchical Design

For a macro block like that shown in Figure C-22 on page 1180, you should have the following pin information in your LEF file, ignoring `SIDEAREA` values:

```
PIN example
    ANTENNAGATEAREA 0.3 LAYER METAL3 ; # area of G2 + G3 + G4
    ANTENNADIFFAREA 1.0 LAYER METAL3 ; # area of D1
    ANTENNAPARTIALMETALAREA 10.0 LAYER METAL3 ; # area of N3,2
    ANTENNAMAXAREACAR 100.0 LAYER METAL3 ; # max CAR of N3,2

    ANTENNAPARTIALCUTAREA 0.1 LAYER VIA34 ; # area of N34,2
    ANTENNAMAXCUTCAR 5.0 LAYER VIA34 ; # max cut CAR of N34,2

    ANTENNAGATEAREA 0.3 LAYER METAL4 ; # area of G2 + G3 + G4
    ANTENNADIFFAREA 2.0 LAYER METAL4 ; # area of D1 + D2
    ANTENNAPARTIALMETALAREA 12.0 LAYER METAL4 ; # area of N4,2
    ANTENNAMAXAREACAR 130.0 LAYER METAL4 ; # max CAR of N4,2
END example
```


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